Class-11th

Kattar Advanced EaJEE NOTES *for* MATHEMATICS

By Tarun Khandelwal (T.K. Sir)
B. Tech IIT Delhi (AIR 289)

IIT JEE ki Jung Me Aapka sabse Important Hathiyar hota hai Well STRUCTURED NOTES

Class Notes in Hand Written Format for

JEE Main & Advanced

Detailed Notes and Short Notes /
Basic to Advanced Illustrations /
Tips & Tricks / Brain Teasers / PYQs

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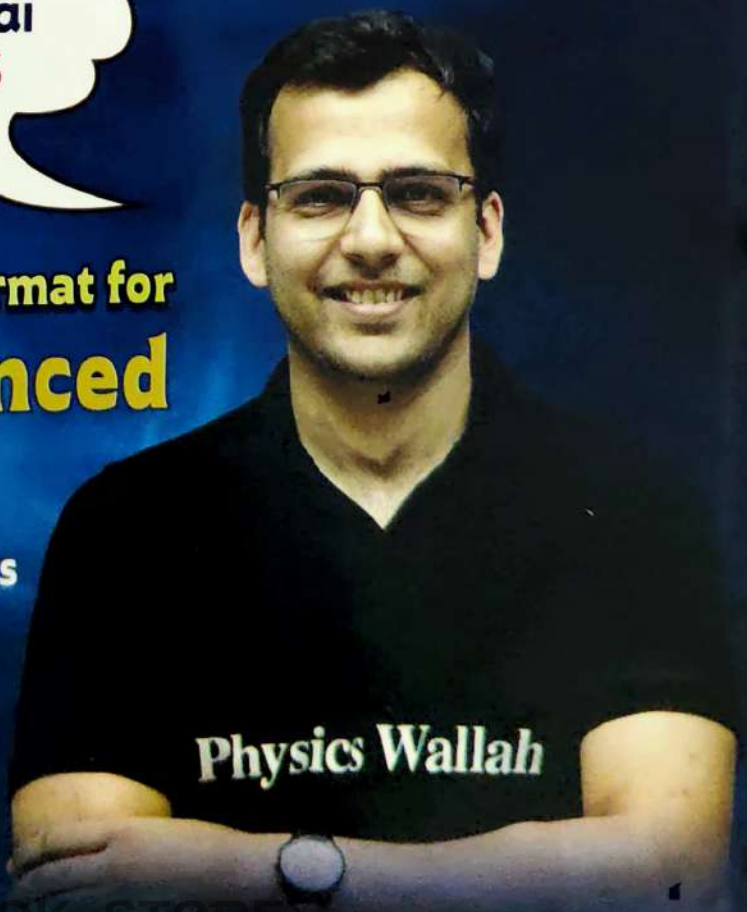

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Notes

1. Detailed Notes and Short Notes
2. Tips, Tricks & Shortcuts
3. Most Important PYQs
4. Brain Teasers to Develop Problem Solving Skills
5. Basic to Advanced illustrations

5 in 1 ▶ Kattar Advanced EaJEE NOTES for Mathematics

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Preface

Physics Wallah (PW) strongly believes in conceptual and fun-based learning. PW provides highly exam-oriented content to bring quality and clarity to the students.

A plethora of Competitive Exams Books is available in the market but PW professionals are continuously working to provide the most effective book for Competitive Exams to our students.

Books adopt a multi-faceted approach to master and understanding the concepts by having a rich diversity of questions asked in the examination and equip the students with the knowledge for the competitive exam.

The main objective of the book is to provide a detailed notes and short notes and a large number of quality problems with varying cognitive levels to facilitate teaching-learning of concepts that are presented through the book.

Every effort has been made to ensure that the content is thoroughly reviewed for clarity, ease of understanding, language quality, and comprehensive coverage of all relevant topics and chapters.

Mastering the Physics Wallah (PW) study material curated by the PW team, can help students qualify for the exam with good marks.

In each chapter, for better understanding detailed notes, short notes, tips & tricks, examples, PYQs and brain teasers are given.

- To strengthen the concept of the topic at the zenith level

This Book Consists of

- Detailed Notes and Short Notes
- Tips, Tricks & Shortcuts
- Most Important PYQs
- Brain Teasers to Develop Problem Solving Skills
- Basic to Advanced illustrations

A step to make our book even better for our students:

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About the Author

Tarun Khandelwal

B. Tech IIT Delhi (AIR 289)

"Tarun Khandelwal – Mathematician, Educator & IIT Delhi Alumnus, inspiring learners to master mathematics with clarity and confidence!"

Tarun Khandelwal (T.K sir) is a passionate educator and mentor, a graduate from IIT Delhi with an All India Rank of 289 in the prestigious IIT JEE Examination. For over 15 years, he has dedicated himself to guiding thousands of students in mastering mathematics—helping them transform fear of the subject into confidence and success. Under his mentorship, countless learners have achieved their dream of cracking IIT JEE, one of the most challenging examinations in the country. A passionate mathematician and author, T.K. Sir believes that numbers are not just symbols but stories waiting to be uncovered. With a sharp analytical mind and a creative spirit, he has dedicated his journey to making mathematics engaging, relatable, and meaningful for learners from all walks of life.

As a teacher and thinker, T.K. Sir blends logic with imagination, showing that mathematics is not confined to equations but extends into the patterns of life, nature, and thought itself. His writings capture this unique perspective—where precision meets creativity, and knowledge transforms into inspiration.

A firm believer in lifelong learning, T.K. Sir continues to explore, question, and create—living proof that mathematics is not just a subject but a way of thinking and a way of life. In this book the author has Uniquely blended creativity with precision to redefine the IIT JEE preparation, offering students the most effective and engaging guide for both JEE Main and Advanced.

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This book is dedicated to my beloved father

Late Mr. Dinesh Khandelwal

I lost him when I was only six years old, far too soon to truly know him, yet never too soon to feel his love and warmth. Though life gave me only a few years with him, his presence has lived on in my heart as a constant source of strength and inspiration.

Every step of my journey in learning, teaching, and now writing this book has been guided by the values he instilled and the blessings I still feel. Mathematics, to me, has always been more than just a subject—it is a way of thinking with clarity, discipline, and creativity. And in dedicating this book to him, I honor not only his memory but also the foundation of strength and perseverance he left within me.

This work is my humble tribute—a way of saying that even though he could not walk beside me, he has been with me in every chapter of my life...

Acknowledgement

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Chapter 1

Basic Mathematics

ALGEBRAIC IDENTITIES-1

1. $(a+b)^2 = a^2 + b^2 + 2ab$
2. $(a-b)^2 = a^2 + b^2 - 2ab$
3. $a^2 - b^2 = (a-b)(a+b)$
4. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

Ques. If $a + b = 10$ and $ab = 3$, find the value of
1. $a^2 + b^2$
2. $a^4 + b^4$

Ans. 1. $a + b = 10$
squaring both sides
$\Rightarrow (a+b)^2 = 100$
$\Rightarrow a^2 + b^2 + 2ab = 100$
$\Rightarrow a^2 + b^2 + 2 \times 3 = 100$ [$\because ab = 3$]
$\Rightarrow a^2 + b^2 = 100 - 6$
$\Rightarrow a^2 + b^2 = 94$

2. Squaring both sides
$\Rightarrow (a^2 + b^2)^2 = (94)^2$
$\Rightarrow (a^2)^2 + (b^2)^2 + 2a^2b^2 = (94)^2$
$\Rightarrow a^4 + b^4 + 2(ab)^2 = (94)^2$
$\Rightarrow a^4 + b^4 + 2(3)^2 = (94)^2 \Rightarrow a^4 + b^4 + 18 = (94)^2$
$\Rightarrow a^4 + b^4 = (94)^2 - 18 = 8818$

ALGEBRAIC IDENTITIES-2

1. $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$
2. $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$
3. $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$
4. $a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$

Proof:

$a^3 + b^3 = (a+b)^3 - 3ab(a+b)$
$= (a+b)[(a+b)^2 - 3ab]$
$= (a+b)[a^2 + b^2 + 2ab - 3ab]$
$= (a+b)(a^2 + b^2 - ab)$

5. $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$
6. $a^3 - b^3 = (a-b)(a^2 + b^2 + ab)$

Ques. If $a + b = 10$ and $a^2 + b^2 = 58$, find the value of $a^3 + b^3$.

Ans. $\because a + b = 10$

Squaring both sides
$\Rightarrow (a+b)^2 = 100$
$\Rightarrow a^2 + b^2 + 2ab = 100$
$\Rightarrow 58 + 2ab = 100$
$\Rightarrow 2ab = 42$
$\Rightarrow ab = 21$

Method 1: $a + b = 10$
Cube both sides
$a^3 + b^3 + 3ab(a+b) = 1000$
$a^3 + b^3 + 3 \times 21 \times 10 = 1000$
$a^3 + b^3 = 1000 - 3 \times 21 \times 10$
$= 1000 - 630 = 370$

Method 2: $a^3 + b^3 = (a+b)(a^2 + b^2 - ab)$
$= 10[58 - 21] = 10 \times 37 = 370$

Ques. If $x = 3 + 3^{1/3} + 3^{2/3}$, then the value of $x^3 - 9x^2 + 18x - 12$ is
(a) 0 (b) -1 (c) 1 (d) 2

Ans. $x - 3 = 3^{1/3} + 3^{2/3}$
cube both sides $\Rightarrow (x-3)^3 = (3^{1/3} + 3^{2/3})^3$
$\Rightarrow x^3 - 3^3 - 3 \cdot x \cdot 3(x-3) = (3^{1/3})^3 + (3^{2/3})^3 + 3.3^{1/3}.3^{2/3}(x-3)$
$\Rightarrow x^3 - 27 - 9x(x-3) = 3^1 + 3^2 + 3.3^1.(x-3)$
$\Rightarrow x^3 - 27 - 9x^2 + 27x = 12 + 9(x-3)$
$\Rightarrow x^3 - 27 - 9x^2 + 27x = 12 + 9x - 27$
$\Rightarrow x^3 - 27 - 9x^2 + 27x = 9x - 15$
$\Rightarrow x^3 - 27 - 9x^2 + 27x - 9x + 15 = 0$
$\Rightarrow x^3 - 9x^2 + 18x - 12 = 0$

ALGEBRAIC IDENTITIES-3

1. $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$
2. If $a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 - 3abc = 0$
$\Rightarrow a^3 + b^3 + c^3 = 3abc$

Ques. Simplify $\dfrac{(a^2-b^2)^3 + (b^2-c^2)^3 + (c^2-a^2)^3}{(a-b)^3 + (b-c)^3 + (c-a)^3}$

Ans. Let $a - b = \alpha$, $b - c = \beta$, $c - a = \gamma$
$\therefore \alpha + \beta + \gamma = 0$
Then $\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma = 3(a-b)(b-c)(c-a)$
Let $a^2 - b^2 = p$; $b^2 - c^2 = q$; $c^2 - a^2 = r$
$\therefore p + q + r = 0$
Then $p^3 + q^3 + r^3 = 3pqr$
$= 3(a^2 - b^2)(b^2 - c^2)(c^2 - a^2)$

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$$= \frac{3(a^2-b^2)(b^2-c^2)(c^2-a^2)}{3(a-b)(b-c)(c-a)}$$

$$= \frac{(a-b)(a+b)(b-c)(b+c)(c-a)(c+a)}{(a-b)(b-c)(c-a)}$$

$$= (a+b)(b+c)(c+a)$$

Ques. If $a + b + c = 10$, $ab + bc + ca = 5$ & $abc = 2$, find the value of

1. $a^2 + b^2 + c^2$
2. $a^3 + b^3 + c^3$

Ans. 1. $a + b + c = 10$

Squaring both sides

$\Rightarrow (a + b + c)^2 = 100$

$\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 100$

$\Rightarrow a^2 + b^2 + c^2 + 10 = 100$

$\Rightarrow a^2 + b^2 + c^2 = 90$

2. $a^3 + b^3 + c^3 - 3abc$

$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$= 10[90 - 5] = 10 \times 85 = 850$

$\Rightarrow a^3 + b^3 - c^3 - 3abc = 850$

$\Rightarrow a^3 + b^3 + c^3 - 3 \times 2 = 850$

$\Rightarrow a^3 + b^3 + c^3 = 850 + 6 = 856$

Brain Teaser

If $a, b, c \in R$ and $a, b, c \neq 0$ such that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 6$ and $\frac{b}{a} + \frac{c}{b} + \frac{a}{c} = 8$ then $\frac{a^3}{b^3} + \frac{b^3}{c^3} + \frac{c^3}{a^3} - 3$ is equal to

(a) 81 (b) 48 (c) 72 (d) 84

Ans. Let $\frac{a}{b} = p$, $\frac{b}{c} = q$, $\frac{c}{a} = r$.

$\Rightarrow p + q + r = 6$ ...(i)

& $pqr = \frac{a}{b}.\frac{b}{c}.\frac{c}{a} = 1$ ...(ii)

$\Rightarrow \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 8 \Rightarrow pq + qr + pr = 8$ ...(iii)

$\because p + q + r = 6$

Squaring both sides

$\Rightarrow p^2 + q^2 + r^2 + 2(pq + qr + pr) = 36$

$\Rightarrow p^2 + q^2 + r^2 + 16 = 36$

$\Rightarrow p^2 + q^2 + r^2 = 20$ ...(iv)

$p^3 + q^3 + r^3 - 3pqr = (p + q + r)(p^2 + q^2 + r^2 - pq - qr - pr)$

From equation (i), (ii), (iii) and (iv) we get

$\Rightarrow p^3 + q^3 + r^3 - 3 \times 1 = 6[20 - 8]$

$\Rightarrow p^3 + q^3 + r^3 - 3 = 72$

EXPONENTS

(Power) or (Indices)

(1) $a^x.a^y = a^{x+y}$, Example $\rightarrow 3^7.3^{12} = 3^{7+12} = 3^{19}$

(2) $\frac{a^x}{a^y} = a^{x-y}$, Example $\rightarrow \frac{5^{10}}{5^3} = 5^{10-3} = 5^7$

(3) $a^0 = 1 \quad (a \neq 0)$

(4) $(a^x)^y = a^{xy}$, Example $\rightarrow (3^5)^7 = 3^{35}$

(5) $a^{-x} = \left(\frac{1}{a}\right)^x = \frac{1}{a^x}$, Example $\rightarrow \frac{1}{3^{10}} = 3^{-10}$

(6) $\left(\frac{a}{b}\right)^x = \left(\frac{b}{a}\right)^{-x}$, Example $\rightarrow \left(\frac{3}{5}\right)^{10} = \left(\frac{5}{3}\right)^{-10}$

(7) $\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$, Example $\rightarrow \left(\frac{3}{5}\right)^{10} = \frac{3^{10}}{5^{10}}$

(8) $\sqrt{a} = a^{1/2}$

(9) $\sqrt{ab} = a^{\frac{1}{2}}b^{\frac{1}{2}}$

(10) $(ab)^x = a^x b^x$

(11) $\sqrt[3]{ab} = (ab)^{\frac{1}{3}} = a^{\frac{1}{3}}.b^{\frac{1}{3}}$

Example $\rightarrow 3^1 + 3^2 + 3^3 = 3^1[1 + 3 + 3^2]$

Example $\rightarrow 3^{10} + 3^{20} = 3^{10}[1 + 3^{10}]$

Ques. $\left[\left\{\left(\frac{1}{7^2}\right)^{-2}\right\}^{-1/3}\right]^{1/4} = 7^x$ then find value of $-3x$.

(JEE Arjuna Mathematics M-1)

Ans. $\because \frac{1}{7^2} = 7^{-2}$

$\left[\left[\left\{(7^{-2})^{-2}\right\}^{-1/3}\right]^{1/4}\right] = 7^x$

$\Rightarrow 7^{(-2)(-2)(-1/3) \times 1/4} = 7^x \Rightarrow 7^{4(-1/3)1/4} = 7^x$

$\Rightarrow 7^{-1/3} = 7^x$

$\Rightarrow x = -\frac{1}{3} \Rightarrow 3x = -1 \Rightarrow -3x = 1$

Ques. If $\left(\frac{x^6y^{-3}}{x^{-2}y^3}\right)^{-1/2} \div \left(\frac{x^{-1}y^2}{x^3y^{-2}}\right)^{-1/3} = x^a y^b$, then the value of $(a + b + 1)$ is

(a) 0 (b) 2 (c) -1 (d) -2

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Ans. $\left(\frac{x^8}{y^6}\right)^{1/2} \div \left(\frac{y^4}{x^4}\right)^{1/3}$ $\boxed{\left(\frac{x}{y}\right)^m = \left(\frac{y}{x}\right)^m \& \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}}$

$= \left(\frac{y^6}{x^8}\right)^{1/2} \div \left(\frac{x^4}{y^4}\right)^{1/3} = \frac{y^3}{x^4} \div \frac{x^{4/3}}{y^{4/3}}$

$= \frac{y^3}{x^4} \times \frac{y^{4/3}}{x^{4/3}} = \frac{y^{3+\frac{4}{3}}}{x^{4+\frac{4}{3}}} = \frac{y^{\frac{9+4}{3}}}{x^{\frac{12+4}{3}}} = \frac{y^{\frac{13}{3}}}{x^{\frac{16}{3}}} = x^{-16/3}.y^{13/3}$

$\Rightarrow a = -\frac{16}{3}$ & $b = \frac{13}{3}$

$\therefore a + b + 1 = -\frac{16}{3} + \frac{13}{3} + 1 = \frac{-16+13+3}{3} = 0$

Ques. If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is

(a) 3 (b) 2 (c) 4 (d) 1

Ans. $x^2 - 64x + 256 = 0$

α and β are roots of the given equations.

$\therefore \alpha+\beta = -\frac{(-64)}{1} = 64$ and $\alpha\beta = \frac{256}{1} = 256$

$\therefore \frac{\alpha^{3/8}}{\beta^{5/8}} + \frac{\beta^{3/8}}{\alpha^{5/8}} = \frac{\alpha^{3/8}.\alpha^{5/8} + \beta^{3/8}.\beta^{5/8}}{\alpha^{5/8}.\beta^{5/8}}$

$= \frac{\alpha^1 + \beta^1}{(\alpha\beta)^{5/8}} = \frac{\alpha+\beta}{(\alpha\beta)^{5/8}}$

$= \frac{64}{(256)^{5/8}} = \frac{64}{2^{8.\frac{5}{8}}} = \frac{64}{2^5} = \frac{64}{2^5} = 2$

Ques. If $5^{x-1} + 5(0.2)^{x-2} = 26$, then x may have the value.

(a) 25 (b) 1/3

(c) 3 (d) None of these

Ans. $\frac{5^x}{5^1} + 5\left(\frac{1}{5}\right)^{x-2} = 26$

$\Rightarrow \frac{5^x}{5^1} + 5\frac{1}{5^{x-2}} = 26$

$\Rightarrow \frac{5^x}{5} + \frac{5}{\frac{5^x}{5^2}} = 26$

$\Rightarrow \frac{5^x}{5} + \frac{5.5^2}{5^x} = 26$

Let $5^x = t$

$\Rightarrow \frac{t}{5} + \frac{125}{t} = 26$

$\Rightarrow \frac{t^2 + 625}{5t} = 26$

$\Rightarrow t^2 + 625 = 26 \times 5t$

$\Rightarrow t^2 + 625 = 130t$

$\Rightarrow t^2 - 130t + 625 = 0$

$\Rightarrow t^2 - 125t - 5t + 625 = 0$

$\Rightarrow (t - 125)(t-5) = 0$

$\Rightarrow t = 125$ or $t = 5$

$\therefore 5^x = 5^3$ or $5^x = 5^1$

$\Rightarrow x = 3$ or $x = 1$

SURDS

Example: $\frac{1}{2+\sqrt{3}} = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{2^2 - (\sqrt{3})^2} = \frac{2-\sqrt{3}}{4-3}$

Hence $\frac{1}{2+\sqrt{3}} = 2-\sqrt{3}$

Ques. Value of $\sqrt{\frac{1}{3-\sqrt{8}} - \frac{1}{\sqrt{8}-\sqrt{7}} + \frac{1}{\sqrt{7}-\sqrt{6}} - \frac{1}{\sqrt{6}-\sqrt{5}} + \frac{1}{\sqrt{5}-2}}$ is:

(a) 1 (b) 5

(c) $\sqrt{5}$ (d) $\sqrt{8}$

Ans. $\sqrt{\frac{1}{3-\sqrt{8}} - \frac{1}{\sqrt{8}-\sqrt{7}} + \frac{1}{\sqrt{7}-\sqrt{6}} - \frac{1}{\sqrt{6}-\sqrt{5}} + \frac{1}{\sqrt{5}-2}}$

$= \sqrt{(3+\sqrt{8}) - (\sqrt{8}+\sqrt{7}) + (\sqrt{7}+\sqrt{6}) - (\sqrt{6}+\sqrt{5}) + (\sqrt{5}+2)}$

$= \sqrt{3+2} = \sqrt{5}$

Ques. Find the number of Real roots of the equation **[01 Feb 2023 (Shift 1)]**

$(\sqrt{3}+\sqrt{2})^{x^2-4} + (\sqrt{3}-\sqrt{2})^{x^2-4} = 10$

(a) 2 (b) 4

(c) 6 (d) 0

Ans. $(\sqrt{3}+\sqrt{2})^{x^2-4} + (\sqrt{3}-\sqrt{2})^{x^2-4} = 10$

$\downarrow t$ $\quad$ $\downarrow \frac{1}{t}$

$\Rightarrow \boxed{t + \frac{1}{t} = 10}$

$\Rightarrow \frac{t^2+1}{t} = 10 \Rightarrow t^2 + 1 = 10t$

$\Rightarrow t^2 - 10t + 1 = 0$

$\Rightarrow t = \frac{10 \pm \sqrt{100-4\times1\times1}}{2\times1}$

$\Rightarrow t = \frac{10 \pm \sqrt{96}}{2}$

$\Rightarrow t = \frac{2(5 \pm 2\sqrt{6})}{2}$

$\Rightarrow t = 5 + 2\sqrt{6}, 5 - 2\sqrt{6}$

$\Rightarrow t = (\sqrt{3}+\sqrt{2})^2, (\sqrt{3}-\sqrt{2})^2$

$\Rightarrow (\sqrt{3}+\sqrt{2})^{x^2-4} = (\sqrt{3}+\sqrt{2})^2$ or $(\sqrt{3}-\sqrt{2})^2$

$\Rightarrow (\sqrt{3}+\sqrt{2})^{x^2-4} = (\sqrt{3}+\sqrt{2})^2$ or $(\sqrt{3}+\sqrt{2})^{-2}$

$\Rightarrow x^2 - 4 = 2$ or $x^2 - 4 = -2$

$\Rightarrow x^2 = 6$ or $x^2 = 2$

$\Rightarrow x = +\sqrt{6}$ or $x = \pm\sqrt{2}$

$\Rightarrow x = 2$ value or $x = 2$ value

$\Rightarrow \{x = \sqrt{6}, -\sqrt{6}, \sqrt{2}, -\sqrt{2}\}$

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COMPONENDO & DIVIDENDO (C & D)

If $\frac{a}{b} = \frac{c}{d}$ then according to C & D

$\Rightarrow \boxed{\frac{a+b}{a-b} = \frac{c+d}{c-d}}$

Example: $\frac{5}{3} = \frac{15}{9} \Rightarrow \frac{5+3}{5-3} = \frac{15+9}{15-9}$

$\Rightarrow \frac{8}{2} = \frac{24}{6} \Rightarrow 4 = 4$

Proof:

Given : $\frac{a}{b} = \frac{c}{d}$

$\frac{a}{b} + 1 = \frac{c}{d} + 1$

$\Rightarrow \frac{a+b}{b} = \frac{c+d}{d}$ ...(i)

Subtract

$\frac{a}{b} - 1 = \frac{c}{d} - 1 \Rightarrow \frac{a-b}{b} = \frac{c-d}{d}$ ...(ii)

Divide equation (i) by (ii)

$\Rightarrow \frac{\frac{a+b}{b}}{\frac{a-b}{b}} = \frac{\frac{c+d}{d}}{\frac{c-d}{d}}$

$\Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$

Hence proved.

Ques. If $\frac{a+3b+2c+6d}{-a-3b+2c+6d} = \frac{3a+b+6c+2d}{-3a-b+6c+2d}$, then the correct statement is

(a) $ad = bc$ ✓ (b) $ac = bd$

(c) $c = \frac{ab}{d}$ (d) $a + d = b + c$

Ans. Using C & D, we get

$\Rightarrow \frac{4c+12d}{2a+6b} = \frac{12c+4d}{6a+2b}$

$\Rightarrow \frac{2(c+3d)}{2(a+3b)} = \frac{2(3c+d)}{2(3a+b)}$

$\Rightarrow \frac{c+3d}{a+3b} = \frac{3c+d}{3a+b} \Rightarrow \frac{c+3d}{3c+d} = \frac{a+3b}{3a+b}$

Apply C & D again

$\Rightarrow \frac{4c+4d}{-2c+2d} = \frac{4a+4b}{-2a+2b}$

$\Rightarrow \frac{2(c+d)}{2(-c+d)} = \frac{2(a+b)}{2(-a+b)}$

Apply C & D again:

$\Rightarrow \frac{2d}{2c} = \frac{2b}{2a}$

$\Rightarrow \frac{d}{c} = \frac{b}{a} \Rightarrow ad = bc.$

INEQUATIONS/INEQUALITY

Any Expression involving $>$, $<$, $\geq$ OR $\leq$ sign is known as Inequality.

(i) $5 > 3 \rightarrow$ True (iv) $7 > 7 \rightarrow$ False

(ii) $7 \geq 4 \rightarrow$ True (v) $8 \leq 15 \rightarrow$ True

(iii) $7 \geq 7 \rightarrow$ True (vi) $8 \leq 8 \rightarrow$ True

INTERVALS

Closed Interval [ ]

Example: $3 \leq x \leq 7 \Rightarrow [3, 7]$

Open Interval ( )

Example: $3 < x < 7 \Rightarrow (3, 7)$

Semi open or semi closed

Example: $3 \leq x < 7 \Rightarrow [3, 7)$; $3 < x \leq 7 \Rightarrow (3, 7]$

Another examples are:

1. $x \geq 2 \Rightarrow x \in [2, \infty)$
2. $x > 2 \Rightarrow x \in (2, \infty)$

Important Note

$\infty \rightarrow$ larger than largest number you can think of

$\frac{1}{0} \rightarrow$ Not Defined

3. $x \leq 5 \Rightarrow \in (-\infty, 5]$
4. $x < 0 \Rightarrow x \in (-\infty, 0)$

Important Note

∞ or $-\infty$ will always be open bracket.

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5. $-3 < x \leq 5$

$x \in (-3, 5]$

↓ belongs to

6. $-1 \leq x < 0 \Rightarrow x \in [-1, 0)$

7. $-5 \leq x \leq 3 \Rightarrow x \in [-5, 3]$

Important Note

{ } ⟶ Curly bracket

Example: $x \in \{3, 5\} \Rightarrow x$ is only 3 or 5

Ques. Find the number of Integers in the following Intervals-

(a) (3, 9]

Ans. 4, 5, 6, 7, 8, 9

number of integers = 6

(b) (-4, 5)

Ans. -3, -2, -1, 0, 1, 2, 3, 4

number of integers = 8

(c) (1, 7]

Ans. 2, 3, 4, 5, 6, 7

number of integers = 6

(d) [-3, 5]

Ans. -3, -2, -1, 0, 1, 2, 3, 4, 5

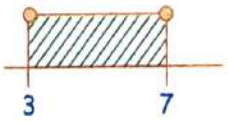
number of integers = 9

NUMBER LINE REPRESENTATION OF INTERVALS

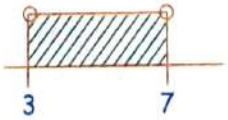
Open bracket ⟶ ○ (Hollow dot).

Closed bracket ⟶ ● (Solid dot).

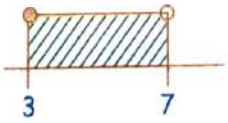
$3 \leq x \leq 7 \Rightarrow x \in [3, 7]$



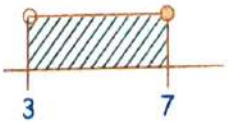
$3 < x < 7 \Rightarrow x \in (3, 7)$



$3 \leq x < 7 \Rightarrow x \in [3, 7)$

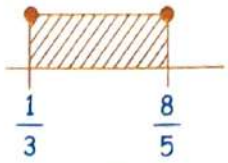


$3 < x \leq 7 \Rightarrow x \in (3, 7]$

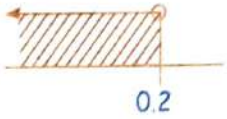


$x \in (3, 7)$ →

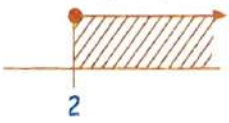
$x \in \left[\frac{1}{3}, \frac{8}{5}\right]$



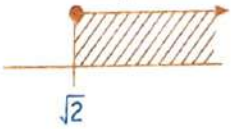
$x \in (-\infty, 0.2)$



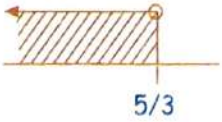
$x \in [2, \infty)$



$x \in [\sqrt{2}, \infty)$



$x \in (-\infty, 5/3)$



BASIC RULES FOR SOLVING INEQUATIONS

1. If we cross Multiply a positive constant, the sign of inequality remains unchanged.

   Example: $\frac{3x - 5}{7} > 1$

   $\Rightarrow 3x - 5 > 7 \Rightarrow 3x > 7 + 5 \Rightarrow 3x > 12 \Rightarrow x > 4$

2. If we cross Multiply a negative constant, the sign of inequality reverses.

   Example: $\frac{3x - 5}{-7} > 1$

   $\Rightarrow 3x - 5 < -7 \Rightarrow 3x < 5 - 7 \Rightarrow 3x < -2 \Rightarrow x < -2/3$

3. While solving Inequalities, we cannot cross Multiply a variable.

   Example: $\frac{3x - 5}{x - 2} > 1$ ⟶ cannot cross multiply

Ques. Solve the following Inequalities & write Solution in interval Form

1. $2 \leq 3x - 4 \leq 5$

Ans. $2 \leq 3x - 4 \leq 5$

add 4

$\Rightarrow 4 + 2 \leq 3x - 4 + 4 \leq 5 + 4 \Rightarrow 6 \leq 3x \leq 9$

divide by 3

$\Rightarrow 6/3 \leq x \leq 9/3 \Rightarrow 2 \leq x \leq 3$ or $x \in [2, 3]$

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2. $6 \le -3(2x-4) < 12$

Ans. **Ist Method:** $6 \le -3(2x-4) < 12$

$\Rightarrow \frac{6}{-3} \ge (2x-4) > \frac{12}{-3} \Rightarrow -2 \ge 2x-4 > -4$

$\Rightarrow -2+4 \ge 2x - \cancel{4} + \cancel{4} > -\cancel{4} + \cancel{4} \Rightarrow 2 \ge 2x > 0$

$\Rightarrow \frac{2}{2} \ge x > \frac{0}{2} \Rightarrow 1 \ge x > 0$

$x \in (0, 1]$

IInd Method:

$6 \le -3(2x-4) < 12$

$\Rightarrow 6 \le -6x + 12 < 12 \Rightarrow 6 - 12 \le -6x + 12 - 12 < 12 - 12$

$\Rightarrow -6 \le -6x < 0$

$\Rightarrow \frac{-6}{-6} \ge x > \frac{0}{-6} \Rightarrow 1 \ge x > 0$

or $x \in (0, 1]$

RATIONAL INEQUALITIES

Ques. $\frac{x-4}{x-7} < 0$

Ans. LHS is negative

Case 1:

$x > 7$

Say $x = 9$

LHS: $\frac{9-4}{9-7} = \frac{5}{2} = +ve$

Case 2:

$4 < x < 7$

Say $x = 5$

LHS: $\frac{5-4}{5-7} = \frac{1}{-2} = -ve$

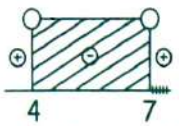
Case 3:

$x < 4$

Say: $x = 0$

LHS: $\frac{0-4}{0-7} = \frac{-4}{-7} = +ve$

$x \in (4, 7)$



$x - 4 = 0 \Rightarrow x = 4$

$x - 7 = 0 \Rightarrow x = 7$

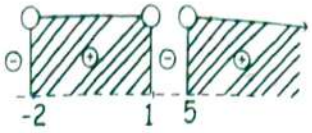
Ques. $\frac{(x+2)(x-1)}{(x-5)} > 0$

Ans. Critical pts

$x + 2 = 0 \Rightarrow x = -2,$

$x - 1 = 0 \Rightarrow x = 1,$

$x - 5 = 0 \Rightarrow x = 5$



$x \in (-2, 1) \cup (5, \infty)$

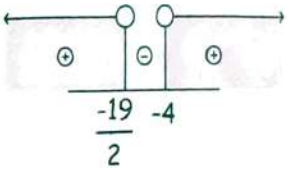
Ques. $\frac{2x-3}{x+4} < 4$

Ans. $\frac{2x-3}{x+4} - \frac{4}{1} < 0$

$\frac{2x-3-4(x+4)}{x+4} < 0$

$\frac{2x-3-4x-16}{x+4} < 0 \Rightarrow \frac{-2x-19}{x+4} < 0$

$\frac{-(2x+19)}{x+4} < 0 \Rightarrow \frac{2x+19}{x+4} > 0$



$x \in (-\infty, -19\backslash 2) \cup (-4, \infty)$

Wavy Curve Method

STEPS for Applying Wavy Curve Method

STEP-1: Create Zero in RHS and Simplify.

STEP-2: Convert LHS into Linear Factors.

STEP-3: Make the coefficient of x positive in all the linear Factors.

STEP-4: Find the critical points by equating each linear factor to zero and plot them on number line.

STEP-5: Start writing plus minus sign alternate from the right most end of the number line.

STEP-6: Shade the required region.

STEP-7: For > or < sign- all critical points are open bracket.

For ≥ or ≤ Sign, numerator critical points are closed bracket whereas denominator critical points are open bracket.

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Ques. $\frac{2x+7}{3x+1} > 2$

Ans. $\frac{2x+7}{3x+1} - 2 > 0$

$\Rightarrow \frac{2x+7-2(3x+1)}{3x+1} > 0$

$\Rightarrow \frac{2x+7-6x-2}{3x+1} > 0$

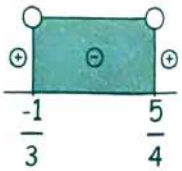
$\Rightarrow \frac{-4x+5}{3x+1} > 0$

$\Rightarrow \frac{-(4x-5)}{(3x+1)} > 0$

$\Rightarrow \frac{4x-5}{3x+1} < 0$

$4x - 5 = 0 \Rightarrow x = \frac{5}{4}$

Or $3x + 1 = 0 \Rightarrow x = -\frac{1}{3}$



$x \in \left(\frac{-1}{3}, \frac{5}{4}\right)$

Ques. The set of real values of x for which $\frac{x - x^2}{x^2 + 2x} > 0$, is

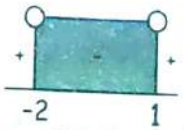
(a) $(-2, 1)$ (b) $(-2, -1)$

(c) $(-2, 1) - \{0\}$ (d) None of these

Ans. $\Rightarrow \frac{x(1-x)}{x(x+2)} > 0 \quad (x \neq 0)$

$\Rightarrow \frac{-(x-1)}{x+2} > 0$

$\Rightarrow \frac{x-1}{x+2} < 0$



$x \in (-2, 1) - \{0\}$

Ques. The set of real values of x for which $\frac{10x^2 + 17x - 34}{x^2 + 2x - 3} < 8$, is

(a) $\left(-\frac{5}{2}, 2\right)$ (b) $\left(-3, -\frac{5}{2}\right) \cup (1, 2)$

(c) $(-3, 1)$ (d) None of these

Ans. $\frac{10x^2 + 17x - 34}{x^2 + 2x - 3} - 8 < 0$

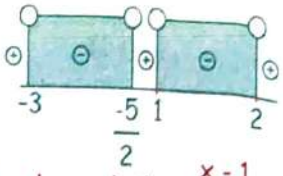
$\Rightarrow \frac{10x^2 + 17x - 34 - 8(x^2 + 2x - 3)}{x^2 + 2x - 3} < 0$

$\Rightarrow \frac{10x^2 + 17x - 34 - 8x^2 - 16x + 24}{x^2 + 2x - 3} < 0$

$\Rightarrow \frac{2x^2 + x - 10}{x^2 + 2x - 3} < 0 \Rightarrow \frac{2x^2 + 5x - 4x - 10}{x^2 + 3x - x - 3} < 0$

$\Rightarrow \frac{(2x+5)(x-2)}{(x+3)(x-1)} < 0$

$x \in \left(-3, -\frac{5}{2}\right) \cup (1, 2)$



Ques. If S is the set of all real x such that $\frac{x-1}{x} - \frac{x+1}{x-1} \leq 2$, then S is equal to

(a) $(-\infty, -1) \cup \left(0, \frac{1}{2}\right) \cup (1, \infty)$

(b) $(-\infty, -1] \cup \left(0, \frac{1}{2}\right] \cup (1, \infty)$

(c) $(-1, 0) \cup \left[\frac{1}{2}, 1\right)$

(d) None of these

Ans. $\frac{x-1}{x} - \frac{x+1}{x-1} - \frac{2}{1} \leq 0$

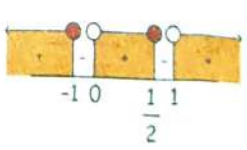
$\Rightarrow \frac{(x-1)^2 - x(x+1) - 2x(x-1)}{x(x-1)} \leq 0$

$\Rightarrow \frac{x^2 + 1 - 2x - x^2 - x - 2x^2 + 2x}{x(x-1)} \leq 0$

$\Rightarrow \frac{1 - x - 2x^2}{x(x-1)} \leq 0 \Rightarrow \frac{-(2x^2 + x - 1)}{x(x-1)} \leq 0$

$\Rightarrow \frac{2x^2 + x - 1}{x(x-1)} \geq 0$

$\Rightarrow \frac{2x^2 + 2x - x - 1}{x(x-1)} \geq 0$



$\Rightarrow \frac{(2x-1)\cdot(x+1)}{x(x-1)} \geq 0$

$x \in (-\infty, -1] \cup \left(0, \frac{1}{2}\right] \cup (1, \infty)$

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Ques. $1 + \frac{2}{x-1} \geq \frac{6}{x}$

Ans. $1 + \frac{2}{x-1} - \frac{6}{x} \geq 0$

$$\frac{x(x-1) + 2x - 6(x-1)}{x(x-1)} \geq 0$$

$$\frac{x^2 - x + 2x - 6x + 6}{x(x-1)} \geq 0$$

$$\frac{x^2 - 5x + 6}{x(x-1)} \geq 0$$

$$\frac{(x-3)(x-2)}{(x-1)} \geq 0$$

$x \in (-\infty, 0) \cup (1, 2] \cup [3, \infty)$

Union & Intersection of Intervals

UNION & INTERSECTION OF INTERVALS

- $\cup$ → OR
- $\cap$ → AND (Common)
- Always draw on number line

Ex: (I)(N)DI(A)

CH(I)(N)A

Intersection = {I, N, A}; Union = {INDACH}

Ex: $x = \{1, 2, 3, 4\}$; $y = \{2, 3, 5, 7, 8\}$

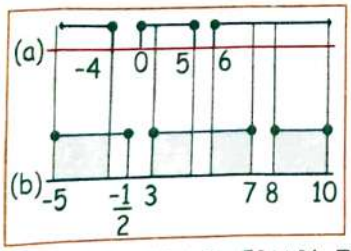
$x \cup y = \{1, 2, 3, 4, 5, 7, 8\}$; $x \cap y = \{2, 3\}$

Ques. Find intersection of the following Intervals

(a) $(-\infty, -4] \cup [0, 5] \cup [6, \infty)$

(b) $\left[-5, \frac{-1}{2}\right] \cup [3, 7] \cup [8, 10]$

Ans.



$A \cap B = [-5, -4] \cup [3, 5] \cup [6, 7] \cup [8, 10]$

Ques. $-1 < \frac{3x-1}{x-2} \leq 4$

Ans. $-1 < \frac{3x-1}{x-2}$

$$\Rightarrow \frac{3x-1}{x-2} > -1$$

$$\Rightarrow \frac{3x-1}{x-2} + 1 > 0 \Rightarrow \frac{3x-1+x-2}{x-2} > 0$$

$$\Rightarrow \frac{4x-3}{x-2} > 0$$

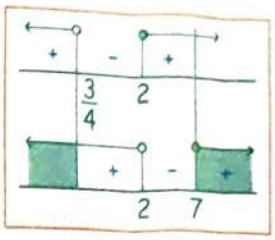
and, $\frac{3x-1}{x-2} \leq 4 \Rightarrow \frac{3x-1}{x-2} - 4 \leq 0$

$$\Rightarrow \frac{3x-1-4(x-2)}{x-2} \leq 0$$

$$\Rightarrow \frac{3x-1-4x+8}{x-2} \leq 0 \Rightarrow \frac{-x+7}{x-2} \leq 0$$

$$\Rightarrow \frac{-(x-7)}{x-2} \leq 0$$

$$\Rightarrow \frac{x-7}{x-2} \geq 0$$



$\left(-\infty, \frac{3}{4}\right] \cup [7, \infty)$

Ques. The solution set of $x^2 + 2 \leq 3x \leq 2x^2 - 5$, is

(a) ϕ

(b) $[1, 2]$

(c) $(-\infty, -1] \cup \left[\frac{5}{2}, \infty\right)$

(d) None of these

Ans. $x^2 + 2 \leq 3x$

$x^2 - 3x + 2 \leq 0$

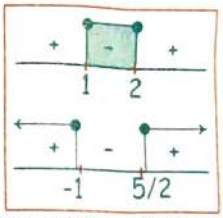
$(x-2)(x-1) \leq 0$

and, $3x \leq 2x^2 - 5$

$2x^2 - 5 \geq 3x$

$2x^2 - 3x - 5 \geq 0$

$(2x-5)(x+1) \geq 0$



Intersection → ϕ Phi (Nothing)

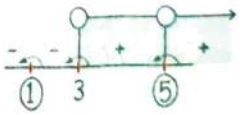
Wavy Curve method For repeated linear Factors)

Important Note

Sign about that critical point will not change whose power is even.

Ques. $\frac{(x-3)^3(x-1)^4}{(x-5)^2} > 0$

Ans. $x \in (3, 5) \cup (5, \infty)$



Ans can also be written as: $x \in (3, \infty) - \{5\}$

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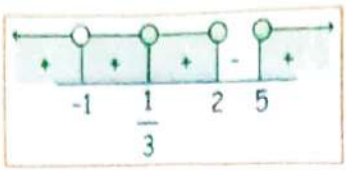
Ques. $\frac{(x-2)^3(x-5)(3x-1)^2}{(x+1)^4} \geq 0$

Ans.



$x \in (-\infty, 2] \cup [5, \infty) - \{-1\}$

Ques. Solution of $\frac{x^4(x+1)^2(x-2)}{(x-3)^3(x+4)} > 0$ is

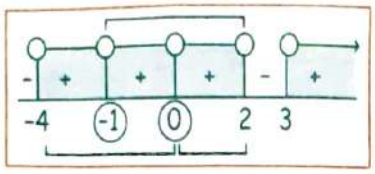
(a) $x \in (-4, 2) \cup (4, \infty)$

(b) $x \in (-4, -1) \cup (-1, 2) \cup (3, \infty) - \{0\}$

(c) $x \in (-4, 0) \cup (0, 2) \cup (3, \infty)$

(d) $x \in (-4, -1) \cup (-1, 0) \cup (0, 2) \cup (3, \infty)$

Ans.



Wavy Curve when Quadratic has D<0

Method-1: Complete the square

$0 = x^2 + x + 1$

$a = 1, b = 1, c = 1$

$D = b^2 - 4ac$

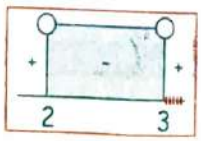
$= 1 - 4 \times 1 \times 1 = 1 - 4 = -3$

$D = -3$

$= -ve$

$\frac{x^2 - 5x + 6}{x^2 + x + 1} < 0$

$\frac{(x-2)(x-3)}{x^2 + x + 1} < 0$



Always +ve

$\Rightarrow (x-2)(x-3) < 0$

Method-2: By Graph of Quadratic

If $a > 0$ & $D < 0$

$\Rightarrow$ Quad will always be +ve

If $a < 0$ & $D < 0$

$\Rightarrow$ Quad is always -ve

Ques. $\frac{0.5}{x - x^2 - 1} < 0$ (+ve / -ve)

Ans. $-x^2 + x - 1$

$a = -1, b = 1, c = -1$

$D = 1^2 - 4(-1)(-1)$

$D = 1 - 4 = -3$

$D = -ve$

$x \in R$

Ques. $\frac{x^2 - 7x + 12}{2x^2 + 4x + 5} > 0$

Ans. $2x^2 + 4x + 5$

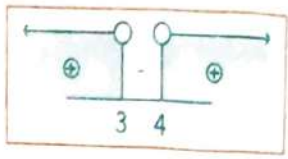
$a = 2, b = 4, c = 5$

$D = 16 - 4 \times 2 \times 5$

$D = -ve$

$\frac{(x-3)(x-4)}{2x^2 + 4x + 5} > 0$

always +ve



$\Rightarrow (x-3)(x-4) > 0$

$\Rightarrow x \in (-\infty, 3) \cup (4, \infty)$

Ques. $\frac{5x^2 - 2}{4x^2 - x + 3} < 1$ always +ve

Ans. $a > 0, D < 0 \Rightarrow$ always +ve

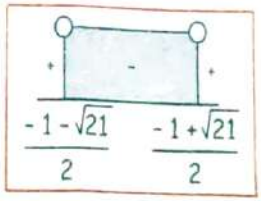
Cross multiply

$\Rightarrow 5x^2 - 2 < 4x^2 - x + 3$

$\Rightarrow x^2 + x - 5 < 0$

$\Rightarrow x = \frac{-1 \pm \sqrt{1^2 - 4(-5)}}{2}$

$\Rightarrow x = \frac{-1 \pm \sqrt{21}}{2}$



$x \in \left(\frac{-1-\sqrt{21}}{2}, \frac{-1+\sqrt{21}}{2}\right)$

Ques. $\frac{x^2 + 6x - 7}{x^2 + 1} \leq 2$

always +ve

Ans. $\Rightarrow x^2 + 6x - 7 \leq 2(x^2 + 1)$

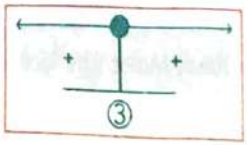
$\Rightarrow x^2 + 6x - 7 \leq 2x^2 + 2$

$\Rightarrow 2x^2 + 2 - x^2 - 6x + 7 \geq 0$

$\Rightarrow x^2 - 6x + 9 \geq 0$

$\Rightarrow (x-3)^2 \geq 0$

$x \in (-\infty, \infty)$



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MODULUS FUNCTION/ ABSOLUTE VALUE FUNCTION

$y = |x| = x$ if $x \geq 0$

$y = |x| = -x$ if $x < 0$

Example:

(i) $|-2| = -(-2) = 2$ (ii) $|3| = 3$

(iii) $|10| = 10$ (iv) $|1.5| = 1.5$

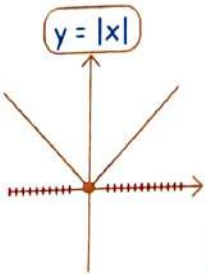
(v) $|\sqrt{2}| = \sqrt{2}$ (vi) $|0| = 0$

(vii) $|-3| = 3$

Graph of $y = |x|$

$y = |x| = x$ if $x \geq 0$

$y = -x$ if $x < 0$



Example:

$$|x^2 - 5| = \begin{cases} +(x^2 - 5) \text{ if } x^2 - 5 > 0 \\ -(x^2 - 5) \text{ if } x^2 - 5 < 0 \end{cases}$$

Graph of Modulus Function

Example: $y = |x - 2| = x - 2$ if $x - 2 \geq 0 \Rightarrow x \geq 2$

$y = |x - 2| = -(x - 2)$ if $x - 2 < 0 \Rightarrow x < 2$

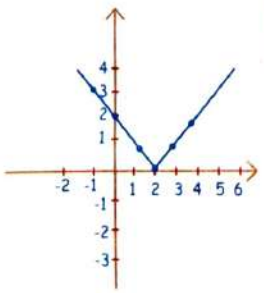
$y = x - 2$ if $x \geq 2$

x	2	3	4
y	0	1	2

$y = -(x - 2)$ if $x < 2$

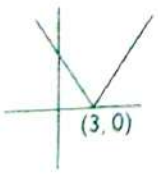
$y = 2 - x$ if $x < 2$

x	1	0	-1
y	1	2	3



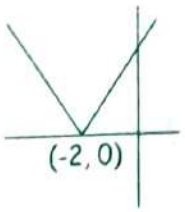
Ques. Make the graph of $y = |x - 3|$

Ans.



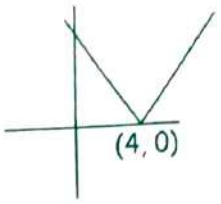
Ques. Make the graph of $y = |x + 2|$

Ans.



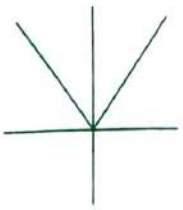
Ques. Make the graph of $y = |x - 4|$

Ans.



Ques. Make the graph of $y = |x|$

Ans.



Example: $y = |x| + 2$

$y = x + 2$ if $x \geq 0$

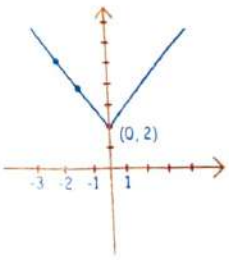
$y = -x + 2$ if $x < 0$

$y = x + 2$ if $x \geq 0$

x	0	1
y	2	3

$y = -x + 2$ if $x < 0$

x	-1	-2
y	3	4

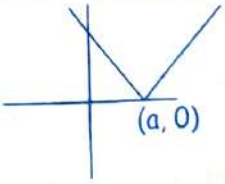


Graphical Transformations of Modulus Functions

Important Note

Note 1: To obtain the graph of $|(x + a)|$ shift the graph of $|(x)|$ 'a' units to left.

Note 2: To obtain the graph of $|(x - a)|$ shift the graph of $|(x)|$ 'a' units to right.



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Important Note

Note 3: To obtain the graph of $|(x)| + a$, shift the graph of $|(x)|$ 'a' units to upwards.

Note 4: To obtain the graph of $|(x)| - a$, shift the graph of $|(x)|$ 'a' units to downwards.

Important properties of Modulus

Property 1: $|-x| = |x|$

Property 2: $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$

Property 3: $|xy| = |x||y|$

Property 4: $\sqrt{x^2} = |x|$, $x \in R$, Also $x^2 = |x|^2$

Property 5: $|x| \geq 0$

Example: $\sqrt{25} = 5$

$\sqrt{-10}$ = Not real

$\sqrt{0} = 0$

Property 6: If $|x + y| = |x| + |y|$ both x & y are +ve or both are -ve.

Case 1: If x & y both are +ve

$\Rightarrow |x| = x, |y| = y \Rightarrow x + y = +ve$

$\Rightarrow |x + y| = x + y = LHS$

$RHS = x + y$

Case 2: If x & y are both are - ve.

$x + y = -ve \qquad |x| = -x, |y| = -y$

$|x + y| = -(x + y)$

$\Rightarrow \begin{cases} LHS = -(x + y) \\ RHS = -x - y \end{cases}$ equal

Summary of property 6:

If $|x + y| = |x| + |y| \Rightarrow$ either $x \geq 0$ & $y \geq 0$

$x \leq 0$ & $y \leq 0$

If $|x + y| = |x| + |y| \Rightarrow \boxed{xy \geq 0}$

$\boxed{|a| + |b| = |a + b| \Rightarrow ab \geq 0}$

$\boxed{|a| + |b| = |a - b| \Rightarrow ab \leq 0}$

Brain Teaser

Solve: $\underbrace{|x^2 + 2x - 3|}_{\alpha} + \underbrace{|1 - 2x|}_{\beta} = |x^2 - 2|$

Ans. $\alpha = x^2 + 2x - 3$

$\oplus \quad \beta = 1 - 2x$

$\alpha + \beta = x^2 - 2$

Given: $|\alpha| + |\beta| = |\alpha + \beta|$

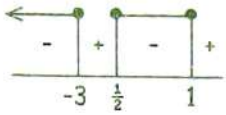
$\Rightarrow \alpha\beta \geq 0$

$\Rightarrow (x^2 + 2x - 3)(1 - 2x) \geq 0$

$\Rightarrow (x + 3)(x - 1)(1 - 2x) \geq 0$

$\Rightarrow -(x + 3)(x - 1)(2x - 1) \geq 0$

$\Rightarrow (x + 3)(x - 1)(2x - 1) \leq 0$



Final answer: $x \in (-\infty, -3] \cup [1/2, 1]$

Solving Modulus Equations

1. $|x| = 5 \Rightarrow x = \pm 5$
2. $|x| = 7 \Rightarrow x = \pm 7$
3. $|x| = \sqrt{2} \Rightarrow x = \pm\sqrt{2}$
4. $|x| = 0.5 \Rightarrow x = \pm 0.5$
5. $|x| = -3 \Rightarrow$ no solution

Solving Modulus equations- Having Single mod sign

Ques. (a) $|x| = 2$

Ans. $x = \pm 2$

(b) $|x - 2| = 3$

Ans. $x - 2 = \pm 3$

$x - 2 = 3$ or $x - 2 = -3$

$x = 5$ or $x = -1$

(c) $\left|\frac{x - 2}{2x - 1}\right| = 5$

Ans. $\frac{x - 2}{2x - 1} = \pm 5$

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Taking positive value

$$\frac{x-2}{2x-1} = 5$$

$x - 2 = 10x - 5$

$x - 10x = -5 + 2$

$-9x = -3$

$\Rightarrow x = 1/3$

Taking negative value

$$\frac{x-2}{2x-1} = -5$$

$\Rightarrow x - 2 = -5(2x - 1)$

$\Rightarrow x - 2 = -10x + 5$

$\Rightarrow x + 10x = 2 + 5$

$\Rightarrow 11x = 7$

$\Rightarrow x = \frac{7}{11}$

Ques. $||x - 2| - 3| - 4| = 1$

Ans. $\Rightarrow ||x - 2| - 3| - 4 = \pm 1$

$\Rightarrow ||x - 2| - 3| = 4 \pm 1$

$\Rightarrow ||x - 2| - 3| = 5 \text{ or } 3$

$\Rightarrow |x - 2| - 3 = \pm 5 \text{ or } \pm 3$

$\Rightarrow |x - 2| - 3 = \pm 5 \text{ or } |x - 2| - 3 = \pm 3$

$\Rightarrow |x - 2| = 3 \pm 5 \text{ or } |x - 2| = 3 \pm 3$

$\Rightarrow |x - 2| = 8 \text{ or } -2 \text{ or } |x - 2| = 6 \text{ or } |x - 2| = 0$

(i) $|x - 2| = 8$

$\Rightarrow x - 2 = \pm 8$

$x = 2 \pm 8 \Rightarrow x = 10, -6$

(ii) $|x - 2| = -2 \rightarrow$ Reject

(iii) $|x - 2| = 6$

(iv) $x - 2 = \pm 6 \Rightarrow x = 2 \pm 6$

$x = 8, -4$

(v) $|x - 2| = 0$

$x - 2 = \pm 0$

$x = 2 \pm 0$

$x = 2$

Hence, $x = 2, 8, -4, 10, -6$

Ques. $(x - 3)^2 + |x - 3| - 11 = 0$, the sum of the solutions of this equation is?

(a) 5 (b) 6 (c) 7 (d) 8

Ans. $|x - 3|^2 + |x - 3| - 11 = 0 \quad (\because (x-3)^2 = |x-3|^2)$

Let, $|x - 3| = t$

$t^2 + t - 11 = 0$

$D = b^2 - 4ac = (1)^2 - 4(1)(-11) = 1 + 44 = 45$

$$t = \frac{-b \pm \sqrt{D}}{2a}$$

$$t = \frac{-1 \pm \sqrt{45}}{2 \times 1}$$

$$t = \frac{-1 \pm 3\sqrt{5}}{2}$$

$$|x - 3| = t = \underbrace{\frac{-1 + 3\sqrt{5}}{2}}_{①} \text{ or } \underbrace{\frac{-1 - 3\sqrt{5}}{2}}_{②}$$

(+ve for ①, -ve for ②)

Reject

$$|x - 3| = \frac{3\sqrt{5} - 1}{2} = a$$

$|x - 3| = a$

$\Rightarrow x - 3 = \pm a$

$\Rightarrow x = 3 \pm a$

$\Rightarrow x_1 = 3 + a, x_2 = 3 - a$

$\Rightarrow x_1 + x_2 = 6$

$x_1 = 3 + a$
$\oplus \; x_2 = 3 - a$
$x_1 + x_2 = 6$

Solving Modulus Equations - Having Multiple Mod sign

Ques. Solve for x : $|x - 3| + 2|x - 4| = 11$

Ans. **Case 1:**

$x < 3$

$-(x - 3) - 2(x - 4) = 11$

$-x + 3 - 2x + 8 = 11$

$-3x + 11 = 11$

$\boxed{x = 0}$

accept

Case 2:

$3 \le x \le 4$

$(x - 3) - 2(x - 4) = 11$

$x - 3 - 2x + 8 = 11$

$-x + 5 = 11$

$-x = 6$

$\boxed{x = -6}$

Reject

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Case 3:

$x > 4$

$(x - 3) + 2(x - 4) = 11$

$x - 3 + 2x - 8 = 11$

$3x - 11 = 11$

$3x = 22$

$\boxed{x = 22/3}$

Accept

Ans $x = \{0, 22/3\}$

Ques. Find the number of real values of x satisfying.

$|5x + 2| + |3 - x| = 7x + 5$

Ans. **Case 1:**

$x < -2/5$

$-(5x + 2) + 3 - x = 7x + 5$

$-5x - 2 + 3 - x = 7x + 5$

$-6x + 1 = 7x + 5$

$-13x = 4$

$x = -4/13$

$-2/5 = -0.4$

$-4/13 = -0.30 \quad -0.4 < -0.3$ $\boxed{\text{Reject}}$

Case 2:

$-2/5 \le x < 3$

$5x + 2 + 3 - x = 7x + 5$

$4x + 5 = 7x + 5$

$4x = 7x$

$3x = 0$

$\Rightarrow x = 0$ $\boxed{\text{Accept}}$

Case 3:

$x \ge 3$

$5x + 2 - (3 - x) = 7x + 5$

$5x + 2 - 3 + x = 7x + 5$

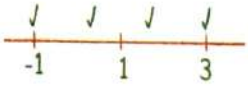
$6x - 1 = 7x + 5$

$-x = 6$

$x = -6$ $\boxed{\text{Reject}}$

Ques. Solve for x: $|x - 1| - 2|x - 3| + 3|x + 1| = 10$

Ans. **Case 1:**



$x \le -1$

$-x + 1 + 2(x - 3) - 3(x + 1) = 10$

$-x + 1 + 2x - 6 - 3x - 3 = 10$

$-2x + 1 - 9 = 10$

$-2x = 18$

$\boxed{x = -9}$ (Accept)

Case 2:

$-1 < x < 1$

$-x + 1 + 2(x - 3) + 3(x + 1) = 10$

$-x + 1 + 2x - 6 + 3x + 3 = 10$

$4x - 2 = 10$

$4x = 12$

$\boxed{x = 3}$ (Reject)

Case 3:

$1 \le x \le 3$

$x - 1 + 2(x - 3) + 3(x + 1) = 10$

$x - 1 + 2x - 6 + 3x + 3 = 10$

$6x - 4 + 10$

$6x = 14$

$x = 14/6 = 7/3$ (Accept)

Case 4:

$x > 3$

$x - 1 - 2(x - 3) + 3(x + 1) = 10$

$x - 1 - 2x + 6 + 3x + 3 = 10$

$2x + 8 + 10$

$2x = 2$

$\boxed{x = 1}$ (Reject)

Ques. Solve for x: $\boxed{|x| + |x + 2| = 2}$

Ans. **Case 1:** $x \le -2$

$-x - (x + 2) = 2$

$-2x - 2 = 2$

$-2x = 4$

$\boxed{x = -2}$

Accept

Case 2: $-2 < x < 0$

$-x + (x + 2) = 2$

$-x + x + 2 = 2$

$\boxed{2 = 2}$

always true

Case 3: $x \ge 0$

$x + x + 2 = 2$

$2x = 0$

$\boxed{x = 0}$

(Accept)

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Ans. $\boxed{-2 \le x \le 0}$

$x \in [-2, 0]$

always true ⇒ complete case is accepted as solution

Never true ⇒ complete case is rejected.

Ques. Solution set of $3^x(3^x - 1) + 2 = |3^x - 1|\,|3^x - 2|$ contains **[JEE Mains 2020]**

(a) 1 Element

(b) two elements

(c) at least four elements

(d) infinite elements

Ans. $\boxed{3^x = t}$

$t(t - 1) + 2 = |t - 1| + |t - 2|$

$\Rightarrow t^2 - t + 2 = |t - 1| + |t - 2|$

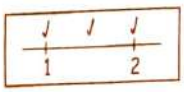
Case → 1

$\boxed{t \le 1}$

$t^2 - t + 2 = -2t + 3$

$\Rightarrow t^2 + t - 1 = 0$

$t = \dfrac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2 \times 1}$



$t = \dfrac{-1 \pm \sqrt{5}}{2}$

$t = \dfrac{-1 - \sqrt{5}}{2}$ or $t = \dfrac{-1 + \sqrt{5}}{2}$

$\boxed{3^x = \dfrac{-1 - \sqrt{5}}{2}}$ or $3^x = \dfrac{\sqrt{5} - 1}{2}$ +ve

Reject

Accept 1 solution

Case → 2

$1 < t < 2$

$t^2 - t + 2 = t - 1 - t + 2$

$\boxed{t^2 - t + 1 = 0}$

$D < 0$

no sol^n

Case → 3

$t \ge 2$

$t^2 - t + 2 = t - 1 + t - 2$

$t^2 - t + 2 = 2t - 3$

$\boxed{t^2 - 3t + 5 = 0}$

$\boxed{D < 0}$ (No Solution)

Ques. $|3x - 2| + 3|2x - 5| = 7$

Ans. **Case-1:** $x < 2/3$

$-(3x - 2) - 3(2x - 5) = 7$

$-3x + 2 - 6x + 15 = 7$

$-9x + 17 = 7$

$-9x = -10$

$\boxed{x = \dfrac{10}{9}}$ Reject

Caes-2: $2/3 \le x \le 5/2$

$(3x - 2) - 3(2x - 5) = 7$

$3x - 2 - 6x + 15 = 7$

$-3x + 13 = 7$

$-3x = 7 - 13$

$-3x = -6$

$\boxed{x = 2}$

Caes-3: $x >$ 5/2 (2.5)

$3x - 2 + 3(2x - 5) = 7$

$3x - 2 + 6x - 15 = 7$

$9x = 17 + 7$

$9x = 24$

$x = \dfrac{24}{9} = 8/3$ (Accepted)

Ans. $x = \{2, 8/3\}$

Ques. Solve for x: $\underbrace{|x - 2024|}_{\text{non -ve}} + \underbrace{|x - 2023|}_{\text{non -ve}} + \underbrace{|x - 2022|}_{\text{non -ve}} = 0$

Ans. $x - 2024 = 0$ & $x - 2023 = 0$ & $x - 2022 = 0$

$x = 2024$ & $x = 2023$ & $x = 2022$

no solution exists.

Ques. The sum of the roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is **(06 April 2023 Shift-I)**

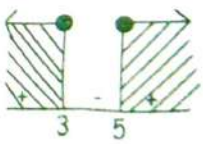
(a) $9 - \sqrt{3}$ (b) $9 + \sqrt{3}$

(c) $11 - \sqrt{3}$ (d) $11 + \sqrt{3}$

Ans. **Case 1:** $x^2 - 8x + 15 \ge 0$

$(x - 5)(x - 3) \ge 0$

$x^2 - 8x + 15 - 2x + 7 = 0$



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$\Rightarrow x^2 - 10x + 22 = 0$

$\Rightarrow x = \dfrac{10 \pm \sqrt{10^2 - 4 \times 1 \times 22}}{2 \times 1}$

$x = \dfrac{10 \pm \sqrt{100 - 88}}{2}$

$= \dfrac{10 \pm 2\sqrt{3}}{2}$

$= \boxed{5 \pm \sqrt{3}}$ $\quad 5 + \sqrt{3} \approx 6.73 \rightarrow$ Accept

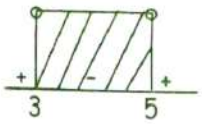
$5 - \sqrt{3} = 5 - 1.73 \rightarrow$ Reject

$= 3.2$

Case 2:

$x^2 - 8x + 15 < 0$

$(x - 3)(x - 5) < 0$



$-(x^2 - 8x + 15) - 2x + 7 = 0$

$\Rightarrow -x^2 + 8x - 15 - 2x + 7 = 0$

$\Rightarrow -x^2 + 6x - 8 = 0$

$\Rightarrow x^2 - 6x + 8 = 0$

$\Rightarrow (x - 4)(x - 2) = 0$

$x = 2$ or $x = 4$

Reject Accept

Sum $= 5 + \sqrt{3} + 4$

$= 9 + \sqrt{3}$

MOD INEQUALITIES

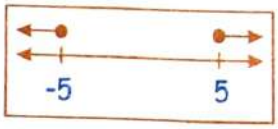
Single mod

If $|x| \le a \Rightarrow x \in [-a, a]$

Example: $|x| \le 3 \Rightarrow x \in [-3, 3]$

$|x| \ge a$

$\Rightarrow x \in (-\infty, -a] \cup [a, \infty)$



Example: If $|x| \ge 5$

$\Rightarrow x \in (-\infty, -5] \cup [5, \infty)$

Solving Modulus Inequtions Having Single Mod Sign

Ques. Set of values of x satisfying $\dfrac{|x| - 1}{|x| + 2} > 0$ is

(a) $(-\infty, -2) \cup (2, \infty)$

(b) $(-2, -1) \cup (1, 2)$

(c) $(-\infty, -1) \cup (1, \infty)$

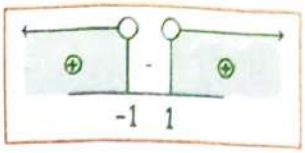
(d) $(-2, 2)$

Ans. $|x| + 2$

+ve + 2 = +ve

$\dfrac{|x| - 1}{|x| + 2} > 0$

always +ve



$\Rightarrow |x| - 1 > 0$

$|x| > 1$

$\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$

Ques. $1 < |x| < 5$

Ans. **Ist Method**

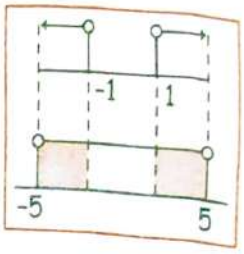
$|x| > 1$

$x \in (-\infty, -1) \cup (1, \infty)$

$|x| < 5$

$x \in (-5, 5)$

$(-5, -1) \cup (1, 5)$



IInd Method

$1 < |x| < 5$

$\Rightarrow x \in (-5, -1) \cup (1, 5)$

Ques. $\dfrac{3}{2} < |x| \le 9$

Ans. $x \in \left[-9, \dfrac{-3}{2}\right) \cup \left(\dfrac{3}{2}, 9\right]$

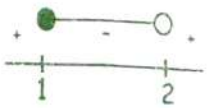
Ques. $\dfrac{|x| - 1}{(|x| - 2)(|x| + 3)} \le 0$

always + ve

Ans. $\dfrac{|x| - 1}{|x| - 2} \le 0$

$|x| = t$

$\dfrac{t - 1}{t - 2} \le 0$



$t \in [1, 2)$

$1 \le |x| < 2$

$x \in (-2, -1] \cup [1, 2)$

Ques. $1 < |x + 3| \le 10$

Ans. $1 < x + 3 \le 10$ OR $-10 \le x + 3 < -1$

Subtract 3

$1 - 3 < x \le 10 - 3$ or $-10 - 3 \le x < -1 - 3$

$-2 < x \le 7$ OR $-13 \le x < -4$

$x \in (-2, 7] \cup [-13, -4)$

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Ques. If $||x - 2| - 3| \le 2$ then complete set of values of x is

(a) $[-3, 7]$ (b) $[-3, 1] \cup [3, 7]$

(c) $[1, 7]$ (d) $[-3, 3] \cup [4, 7]$

Ans. $||x-2|-3| \le 2$

$-2 \le |x - 2| - 3 \le 2$

add 3

$3 - 2 \le |x - 2| \le 3 + 2$

$1 \le |x - 2| \le 5$

$1 \le x - 2 \le 5$ or $-5 \le x - 2 \le -1$

add 2 add 2

$3 \le x \le 7$ or $-3 \le x \le 1$

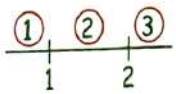
$x \in [-3, 1] \cup [3, 7]$

Solving Modulus Inequations - having Multiple mod sign

Ques. $|x - 1| + |x - 2| < 7$

Ans. $x - 1 = 0$

$x - 2 = 0$



Case 1:

$x \le 1$

$-(x - 1) - (x - 2) < 7$

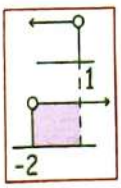
$-x + 1 - x + 2 < 7$

$-2x + 3 < 7$

$-2x < 4$

$x > -2$

$I_1 : (-2, 1]$



Case 2:

$1 < x \le 2$

$x - 1 - (x - 2) < 7$

$x - 1 - x + 2 < 7$

$1 < 7$

always true

$1 < x \le 2$

$I_2 : (1, 2]$

Case 3:

$x > 2$

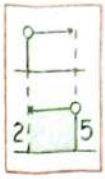
$x - 1 + x - 2 < 7$

$2x - 3 < 7$

$2x < 10$

$x < 5$

$I_3 : (2, 5)$



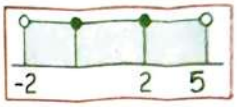
Final Answer: $I_1 \cup I_2 \cup I_3$

$I_1 : (-2, 1]$

$I_2 : (1, 2]$

$I_3 : (2, 5)$

$\Rightarrow x \in (-2, 5)$



Ques. $|x - 1| + 2|x + 3| \ge 7$

Ans. **Case 1:**

$x \le -3$

$-(x - 1) - 2(x + 3) \ge 7$

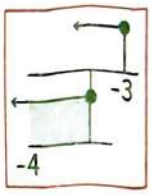
$-x + 1 - 2x - 6 \ge 7$

$-3x - 5 \ge 7$

$-3x \ge 12$

$x \le -4$

$I_1 : (-\infty, -4]$



Case 2:

$-3 < x < 1$

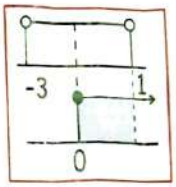
$-(x - 1) + 2(x + 3) \ge 7$

$-x + 1 + 2x + 6 \ge 7$

$x + 7 \ge 7$

$x \ge 0$

$I_2 = [0, 1)$



Case 3:

$x \ge 1$

$x - 1 + 2(x + 3) \ge 7$

$x - 1 + 2x + 6 \ge 7$

$3x + 5 \ge 7$

$3x \ge 2$

$x \ge \frac{2}{3}$

$I_3 : [1, \infty)$

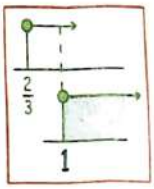
So,

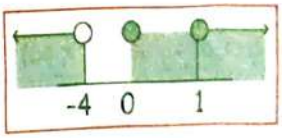
$I_1 : (-\infty, -4]$

$I_2 : [0, 1)$

$I_3 : [1, \infty)$

$x \in (-\infty, -4] \cup [0, \infty)$





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LOGARITHM DEFINITION

$\log_b N$ — Number, base

$N > 0$

$b > 0,\ b \neq 1$

If $\log_b N = y$

$\Rightarrow N = (b)^y$

$\Rightarrow$ Number = $(\text{base})^{RHS}$

Example: If $\log_2 16 = y$

$\Rightarrow 16 = 2^y$

$\Rightarrow 2^4 = 2^y$

$\Rightarrow y = 4$

Example: $\log_5 625 = y \Rightarrow 625 = (5)^y$

$\Rightarrow (5)^4 = (5)^y$

$\Rightarrow y = 4$

Example: $\log_{(-2)} 5$ = Not defined

$\log_2 0$ = Not defined

$\log_{\sqrt{3}} \sqrt{2}$ = defined

$\log_1 5$ = Not defined

$\log_5 1$ = Defined

Example: If $\log_1 25 = y$

$(1)^y = 25$ (Reject).

$\Rightarrow$ base is never equal 2

$\log_2 25 = y$

$25 = (2)^y$

$\Rightarrow y$ = Real (but we can find the value using log tables)

Important Note

Note 1: $\log_1 1$ = Not defined

Note 2: $\log_1 1 = y \Rightarrow 1 = (1)^y$

Not defined as we don't get a unique value of y.

Basic Properties of Logarithm

Property 1: $\log_a a = 1$

Proof → Let $\log_a (a) = y$

$(a)^1 = (a)^y$

$\Rightarrow y = 1$

Property 2: $\log_{1/a} a = -1$

Proof → $\log_a a = y$

$\Rightarrow a = \left(\frac{1}{a}\right)^y \Rightarrow a^1 = a^{-y} \Rightarrow y = -1$

Property 3: $\log_b 1 = 0$

Proof → $\log_b 1 = y$

$\Rightarrow 1 = b^y \Rightarrow y = 0$

Property 4: $\log_b xy = \log_b x + \log_b y$

Proof → $\log_b (xy) = \log_b x + \log_b y$ (p = q + r)

To Prove: $p = q + r$

$\log_b (xy) = p$ $\quad \log_b x = q$ $\quad \log_b y = r$

$\Rightarrow xy = (b)^p$ $\quad \Rightarrow x = (b)^q$ $\quad \Rightarrow y = (b)^r$

So, $(b)^q \cdot (b)^r = (b)^p$

$\Rightarrow (b)^{q+r} = (b)^p$

$\Rightarrow q + r = p$ (hence proved)

Similarly, $\log_b (xyz) = \log_b x + \log_b y + \log_b z$

Property 5: $\log_b (x/y) = \log_b x - \log_b y$

Generalize: $\log_b \frac{xy}{z} = \log_b x + \log_b y - \log_b z$

$\log_k \frac{ab}{cd} = \log_k a + \log_k b - \log_k c - \log_k d$

Property 6: $\log_b (x)^y = y \log_b x$

Proof → $\log (x)^y = \log (x.x.x.x.x...x)$

$= \log x + \log x + \log x \ldots$

$= y \log x$

Ques. Simplify the following

i. $\log_{10} 2 + \log_{10} 5$

(a) 2 (b) 1 (c) 5 (d) 9

Ans. $\log_{10} (2 \times 5) = \log_{10} 10 = 1$

ii. $\log_{10} 25 + \log_{10} 4 - \log_{10} 10$

(a) 1 (b) 3

(c) 5 (d) 4

Ans. $\log_{10}\left(\frac{25 \times 4}{10}\right) = \log_{10} 10 = 1$

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iii $\log_{39}\frac{15}{7} + \log_{39}\frac{13}{3} - \log_{39}\frac{5}{21}$

(a) 3

(b) 2

(c) 0

(d) None

Ans. $\log_{39}\left(\frac{\frac{15\times13}{7\times3}}{\frac{5}{21}}\right)$

$\log_{39}\left[\frac{15\times13}{7\times3}\times\frac{21}{5}\right]$

$= \log_{39} 39 = 1$

Ques. Find the value of $\log_2[\log_2\{\log_3(\log_3 27^3)\}]$

(a) 1 (b) 2

(c) 3 (d) 0

Ans. First, $\log_3 27^3$

$= \log_3 (27)^3$

$= \log_3 [(3)^3]^3$

$= \log_3 (3)^9$ $[\because (a^m)^n = a^{mn}]$

$= 9 \log_3 3$

$= 9$

Now, $\{\log_3 (\log_3 27^3)\} = \{\log_3 (9)\}$

$= \log_3 (3)^2 = 2(\log_3 3) = 2$

Now, $[\log_2\{\log_3 (\log_3 27^3)\}] = \log_2 2 = 1$

So, $\log_2[\log_2\{\log_3 (\log_3 27^3)\}] = \log_2 (1) = 0$

Ques.

	Column I		Column II
(A)	$\frac{\log_3 243}{\log_2\sqrt{32}}$	(p)	positive integer
(B)	$\frac{2\log 6}{(\log 12 + \log 3)}$	(p)	negative integer
(C)	$\log_{1/3}\left(\frac{1}{9}\right)^{-2}$	(r)	rational but not integer
(D)	$\frac{\log_5 16 - \log_5 4}{\log_5 128}$	(s)	prime

Ans. (A) $243 = (3)^5$

$\sqrt{32} = (32)^{1/2} = (2^5)^{1/2} = 2^{5/2}$

$$\frac{\log_3 243}{\log_2\sqrt{32}} = \frac{\log_3 (3^5)}{\log_2 (2^{5/2})}$$

$$= \frac{5\log_3 3}{5/2\log_2 2} \quad (\log_3 3 \to 1,\ \log_2 2 \to 1)$$

$$= \frac{5}{(5/2)} = \frac{5}{5} \times 2 = 2$$

Hence, (A) → (p), (s)

(B) $\frac{2\log 6}{\log 36} = \frac{\log 6^2}{\log 36}$

$= \frac{\log 36}{\log 36} = 1$

Hence, (B) → (p)

(C) $\log_{1/3}(1/3)^{2\times(-2)} = \log_{1/3}(1/3)^{-4}$

$= -4\log_{1/3} 1/3 = -4 \times 1 = -4$

Hence, (C) → q

(D) $\frac{\log_5 16 - \log_5 4}{\log_5 128}$

$= \frac{\log_5 (16/4)}{\log_5 (2)^7} = \frac{\log_5 (2)^2}{\log_5 (2)^7}$

$= \frac{2\log_5 2}{7\log_5 2} = 2/7$

Hence, (D) → r

Mistakes While Applying Log

1) $\log_b (x + y) = \log_b x + \log_b y$ (x)

2) If $\frac{\log x}{\log 2} = 3 \Rightarrow x/2 = 3$ (x)

$\Rightarrow x = 6$

3) $\frac{\log_b x}{\log y} = \log_b x - \log_b y$ (x)

4) $\log_b (x/y) = \log_b x - \log_b y$ (✓)

5) $(\log x)(\log y) = \log x + \log y$ (x)

6) $\log (x y) = \log x + \log y$ (✓)

Ques. The value of $7\log\frac{16}{15} + 5\log\frac{25}{24} + 3\log\frac{81}{80}$ is

(a) log 2 (b) log 3

(c) log 5 (d) None of these

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Ans. $\log\left(\frac{16}{15}\right)^7 + \log\left(\frac{25}{24}\right)^5 + \log\left(\frac{81}{80}\right)^3$

$= \log\left[\left(\frac{16}{15}\right)^7 \cdot \left(\frac{25}{24}\right)^5 \cdot \left(\frac{81}{80}\right)^3\right]$ [$\because \log x + \log y = \log(xy)$]

$\log\left[\left(\frac{2^4}{3.5}\right)^7 \cdot \left(\frac{5^2}{2^3.3}\right)^5 \cdot \left(\frac{3^4}{2^4.5}\right)^3\right]$

$\log\left[\frac{2^{28}}{3^7.5^7} \cdot \frac{5^{10}}{2^{15}.3^5} \cdot \frac{3^{12}}{2^{12}.5^3}\right]$

$= \log\left[\frac{2^{28} \cdot 5^{10} \cdot 3^{12}}{3^{12} \cdot 5^{10} \cdot 2^{27}}\right]$

$= \log \frac{2^{28}}{2^{27}} = \log 2^{28-27} = \log 2$

Examples on Basic Properties of Logarithm (Property-1 to Property-6)

Ques. The value of $\sqrt{(\log_{0.5}^2 4)}$ is

(a) -2 (b) $\sqrt{(-4)}$

(c) (2) (d) Name of these

Ans. $\sqrt{(\log_{0.5}^2 4)} = \sqrt{(\log_{0.5} 4)^2}$

Since, $\sqrt{x^2} = |x|$ & $\log^2 x = (\log x)^2$

$= |\log_{1/2} 4|$

$= |\log_{1/2} (2)^2|$

$= |2 \log_{1/2} 2|$ ($\log_{1/2} 2 \to -1$)

$= |2 \times (-1)| = |-2| = 2$

Ques. Simplify: $\log_{\frac{1}{3}} \sqrt[4]{729.\sqrt[3]{9^{-1}.27^{-\frac{4}{3}}}}$

Ans. We know, $\sqrt[3]{x} = (x)^{1/3}$

$\sqrt[4]{x} = (x)^{1/4}$

Let $N = \sqrt[4]{729.\sqrt[3]{9^{-1}.27^{-\frac{4}{3}}}}$

$\Rightarrow N = [3^6.(3^{-2}.3^{3(-4/3)})^{1/3}]^{1/4}$

$= [3^6 (3^{-2}.3^{-4})^{1/3}]^{1/4} = [3^6 (3^{-6})^{1/3}]^{1/4}$

$= [3^6.3^{-2}]^{1/4} = [3^{6-2}]^{1/4}$

$= [3^4]^{1/4} = 3$

Final Answer: $\log_{1/3} 3 = -1$

Two Most Common bases of Logarithm

(i) b = base = 10

Example: $\log_{10} N$

(ii) base = e = Napier's constant = 2.71

Example: $\log_e x \to$ Natural log

$\log_e x$ is same as $\ln x$

$\ln 2 = \log_e 2$

Solving Logarithm Equations

Ques. $\log_2 x = 5$

Ans. $x = 2^5 = 32$

Ques. $|\log_2 x| = 3$

Ans. $\log_2 x = \pm 3$

$x = (2)^{\pm 3}$

$x = 2^3$ or $x = 2^{-3}$

$x = 8$ or $x = \frac{1}{2^3} = \frac{1}{8}$

Ques. Find the value of x

$\log_2 (x + 1) - \log_2 (2x - 3) = 0$

(a) 1 (b) 3 (c) 4 (d) 8

Ans. $\log_2 \left[\frac{(x+1)}{(2x-3)}\right] = 0$

$\Rightarrow \frac{x+1}{2x-3} = 2^0$

$\frac{x+1}{2x-3} = 1 \Rightarrow x + 1 = 2x - 3$

$x - 2x = -3 - 1$

$-x = -4$

$\Rightarrow x = 4$

Ques. Solve for x: $\log_4 (2 \log_3 (1 + \log_2 (1 + 3 \log_3 x))) = 1/2$ ($\log_3 (\ldots) \to N$)

Ans. $\log_4 N = 1/2$

$\Rightarrow N = (4)^{1/2} = 2$

$2 \log_3 (1 + \log_2 (1 + 3 \log_3 x)) = 2$

$\log_3 (1 + \log_2 (1 + 3 \log 3 x)) = 1$ ($\log_2 (\ldots) \to N$)

$\log_3 N = 1$

$\Rightarrow N = 3^1 = 3$

$1 + \log_2 (1 + 3 \log_3 x) = 3$

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$\Rightarrow \log_2 (1 + 3 \log_3 x) = 2$

$\Rightarrow 1 + 3 \log_3 x = (2)^2 \Rightarrow 1 + 3 \log_3 x = 4$

$\Rightarrow 3 \log_3 x = 3 \Rightarrow \log_3 x = 1$

$\Rightarrow x = 3^1 \Rightarrow x = 3$

Ques. Solve for x: ${}_x(\log_{10} x + 1) = 10^6$

Ans. Taking log both sides

$\log_{10} (x)^{\log_{10} x + 1} = \log_{10}(10)^6$

[$\because \log (x)^y = y \log x$]

$\Rightarrow (\log_{10} x + 1) \log_{10} x = 6 \log_{10} 10$ (1)

$\Rightarrow (\log_{10} x + 1) \log_{10} x = 6$

$\Rightarrow$ Let $\log_{10} x = t$

$\Rightarrow (t + 1) t = 6 \Rightarrow t^2 + t - 6 = 0$

$\Rightarrow t^2 + 3t - 2t - 6 = 0 \Rightarrow (t + 3)(t - 2) = 0$

either $t + 3 = 0 \Rightarrow t = -3$

or $t - 2 = 0 \Rightarrow t = 2$

$\log_{10} x = 2$ or $\log_{10} x = -3$

$\Rightarrow x = 10^2$ or $x = 10^{-3}$

$\Rightarrow x = 100$ or $x = \frac{1}{1000}$

Advanced Properties of Logarithm

Property 7: Base Change Theorem

$$\log_b N = \frac{\log_c N}{\log_c b}$$

$$\text{B.C.T} = \frac{\text{log of Number}}{\text{Log of base}}$$

Example: $\log_2 5 = \frac{\log_3 5}{\log_3 2}$

Example: $\log_3 7 = \frac{\log_{10} 7}{\log_{10} 3}$

Ques. Solve: $(\log_3 7) \cdot (\log_{49} 3)$

Ans. $\frac{\log 7}{\log 3} \cdot \frac{\log 3}{\log 49} = \frac{\log 7}{\log (7)^2} = \frac{\log 7}{2 \log 7} = 1/2$

Ques. The value of $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$ is

(a) 1 (b) 2 (c) 3 (d) 4

Ans. $\frac{\log 4}{\log 3} \cdot \frac{\log 5}{\log 4} \cdot \frac{\log 6}{\log 5} \cdot \frac{\log 7}{\log 6} \cdot \frac{\log 8}{\log 7} \cdot \frac{\log 9}{\log 8}$

$= \frac{\log 9}{\log 3} = \frac{2 \log (3)}{\log 3} = 2$

Derived Properties of Base Change Theorem

Property 8: $\log_b a = \frac{1}{\log_a b}$

Proof → L.H.S = $\log_b a = \frac{\log_a a}{\log_a b}$ (1) $= \frac{1}{\log_a b} = \text{RHS}.$

Hence proved.

Example: (i) $\log_3 7 = \frac{1}{\log_7 3}$, (ii) $\log_{100} 5 = \frac{1}{\log_5 100}$,

(iii) $\log_2 x = \frac{1}{\log_x 2}$

Ques. Solve for x:

$1 + 2 \log_{(x+2)} 5 = \log_5 (x + 2)$

Ans. Let $\log_{(x+2)} 5 = t$

$1 + 2t = \frac{1}{t}$

$\Rightarrow t + 2t^2 = 1$

$\Rightarrow 2t^2 + t - 1 = 0$

$\Rightarrow 2t^2 + 2t - t - 1 = 0$

$\Rightarrow 2t(t + 1) - 1(t + 1) = 0$

$\Rightarrow (2t - 1)(t + 1) = 0$

$\Rightarrow t = \frac{1}{2}$

$\Rightarrow \log_{x+2} 5 = \frac{1}{2}$

$\Rightarrow 5 = (x + 2)^{1/2}$

Squaring both sides

$25 = x + 2$

$\Rightarrow \boxed{x = 23}$

or $t = -1$

or $\log_{x+2} 5 = -1$

$\Rightarrow 5 = (x + 2)^{-1}$

$\Rightarrow 5 = \frac{1}{x + 2}$

$\Rightarrow 5(x + 2) = 1$

$\Rightarrow 5x + 10 = 1$

$\Rightarrow 5x = -9$

$\Rightarrow \boxed{x = -9/5}$

Hence, the solutions for x are 23 and -9/5.

Ques. Find the exact value of $\frac{\log_3 135}{\log_{15} 3} - \frac{\log_3 5}{\log_{405} 3}$

Ans. $(\log_3 135)(\log_3 15) - (\log_3 5)(\log_3 405)$

$= \log_3 (3^3 5) \log_3 (3 \times 5) - (\log_3 5) \log_3 (3^4 5)$

$= (\log_3 3^3 + \log_3 5)(\log_3 3 + \log_3 5) - \log_3 5 (\log_3 3^4 + \log_3 5)$

(3) (1) (4)

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$= (3 + \log_3 5)(1 + \log_3 5) - \log_3 5 (4 + \log_3 5)$

Let $\log_3 5 = t$

$= (3 + t)(1 + t) - t(4 + t)$

$= (3 + 4t + t^2) - 4t - t^2 = 3$

Property 9: Base power formula $\log_{a^k} m = \frac{1}{k} \log_a m$

Proof → LHS: $\log_{(a)^k} m = \frac{\log_a m}{\log_a (a)^k}$

$= \frac{\log_a m}{k \log_a a}$ ($\log_a a \to 1$)

$= \frac{1}{k} \log_a m$ = RHS

General: $\boxed{\log_{(b)^k} (x)^y = \frac{y}{k} \log_b x}$

Example: $\log_{(2)^{100}} 2$

Ans. $\log_{(2)^{100}} 2 = \frac{1}{100} \log_2 2$ ($\log_2 2 \to 1$)

$= \frac{1}{100}$

Example: $\log_{\sqrt{2}} 4$

Ans. $\log_{\sqrt{2}} 4 = \log_{(2)^{1/2}} 4 = \frac{1}{1/2} \log_2 4$

$= 2 \log_2 (2)^2$

$= 2 \times 2 \log_2 2$ ($\log_2 2 \to 1$)

$= 4$

Ques. Logarithm of $32\sqrt[5]{4}$ to the base $2\sqrt{2}$ is

(a) 3.6 (b) 5

(c) 5.6 (d) None of these

Ans. $\log_{(2)^{3/2}} (2)^{27/5}$

$= \frac{27/5}{3/2} \log_2 2$ ($\log_2 2 \to 1$)

$= \frac{27 \times 2}{5 \times 3}$

$= \frac{18}{5} = 3.6$

Ques. If $\log_7 2 = m$, then $\log_{49} 28$ is equal to

(a) $2(1 + 2m)$ (b) $\frac{1 + 2m}{2}$

(c) $\frac{2}{1 + 2m}$ (d) $1 + m$

Ans. $\log_{7^2} 28 = \frac{1}{2} \log_7 (28)$

$= \frac{1}{2} \log_7 (7 \times 2^2)$

$= \frac{1}{2} [\log_7 7 + \log_7 2^2]$

$= \frac{1}{2} [1 + 2 \log_7 2]$

$= \frac{1}{2} [1 + 2m]$

Property 10: $N = a^{\log_a N}$

Proof → Let $a^{\log_a N} = x$

taking $\log_a$ both Sides

$\log_a a^{\log_a N} = \log_a x$

$(\log_a N) \log_a a = \log_a x$

$\log_a N = \log_a x \Rightarrow N = x$

Example

(i) $5^{\log_5 3} = 3$

(ii) $7^{\log_7 10} = 10$

(iii) $3^{2\log_3 5} = 3^{\log_3 5^2} = 3^{\log_3 25} = 25$

Important Note

If $\log_a x = \log_a y \Rightarrow x = y$

Ques. The value of $81^{(1/\log_5 3)} + 27^{\log_9 36} + 3^{4/\log_7 9}$ is equal to

(a) 49 (b) 625 (c) 216 (d) 890

Ans. $(81)^{\frac{1}{\log_5 3}} = (81)^{\log_3 5} = (3^4)^{\log_3 5} = 3^{\log_3 (5)^4}$

$= (5)^4 = 25^2 = 625$

$(27)^{\log_9 36} = (27)^{\log_{3^2} 6^2} = (3)^{3\log_3 6} = 3^{\log_3 6^3}$

$= (6)^3 = 216$

$3^{4/\log_7 9} = 3^{4 \log_9 7} = 3^{4\log_{3^2} 7} = 3^{2\log_3 7} = 3^{\log_3 7^2}$

$= 7^2 = 49$

Hence, $81^{(1/\log_5 3)} + 27^{\log_9 36} + 3^{4/\log_7 9}$

$= 625 + 216 + 49 = 890.$

Ques. $x^2 + 7^{\log_7 x} - 2 = 0$ then x is

(a) 1 (b) -2 (c) -1 (d) 2

Ans. $x^2 + x - 2 = 0, x > 0$

$\Rightarrow x^2 + 2x - x - 2 = 0$

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$\Rightarrow (x+2)(x-1) = 0$

$x = -2$ or $x = 1$

Reject Accept

Option (a) is correct.

Property 11: $a^{\log_b c} = c^{\log_b a}$

Example:

(i) $5^{\log_3 7} = 7^{\log_3 5}$

(ii) $10^{\log_7 5} = 5^{\log_7 10}$

Ques. If $a = \log_{24} 12$ and $b = \log_{36} 24$ and $c = \log_{48} 36$ then the value of abc = ? (DPP-7)

(a) $2bc - 1$ (b) $2bc + 1$

(c) $bc - 1$ (d) $bc + 1$

Ans. $abc = \frac{\log 12}{\log 24} \times \frac{\log 24}{\log 36} \times \frac{\log 36}{\log 48}$

$abc = \frac{\log_{12} 12}{\log_{12} 48} = \frac{1}{\log_{12} 48}$

Now, $bc = \frac{\log 24}{\log 36} \times \frac{\log 36}{\log 48} = \frac{\log_{12} 24}{\log_{12} 48} = bc$

$2bc = \frac{2\log_{12} 24}{\log_{12} 48} = \frac{\log(24\times 24)}{\log 48} = \frac{\log_{12}(48\times 12)}{\log_{12} 48}$

$2bc = \frac{\log_{12} 48 + \log_{12} 12}{\log_{12} 48} \rightarrow 1$

$2bc = 1 + \frac{1}{\log_{12} 48} \Rightarrow 2bc - 1 = \frac{1}{\log_{12} 48}$

Hence, $abc = 2bc - 1$

Brain Teaser

Solve for x : $25^{\log_{10} x} = 5 + 4 \cdot x^{\log_{10} 5}$

Ans. $5^{2\log_{10} x} = 5 + 4 \cdot (x)^{\log_{10} 5}$ → (P-11)

$\Rightarrow 5^{2\log x} = 5 + 4 \cdot (5)^{\log_{10} x}$

Let $5^{\log_{10} x} = t$

$\Rightarrow \boxed{t^2 = 5 + 4t}$

$\Rightarrow t^2 - 4t - 5 = 0$

$\Rightarrow (t-5)(t+1) = 0$

$\Rightarrow t = 5$ or $t = -1$

$5^{\log_{10} x} = 5^1$ or $5^{\log_{10} x} = -1$

$\Rightarrow \log_{10} x = 1$ | Reject

$\Rightarrow x = 10^1$ | 5^y is non -ve. ×

$\Rightarrow \boxed{x = 10}$

Ans. ✓

Brain Teaser

Solve the system of equations for x, y and z :

(a) $(\log_a x)\log_a(xyz) = 48$

(b) $(\log_a y)\log_a(xyz) = 12$

(c) $(\log_a z)\log_a(xyz) = 84$

$a > 0, a \neq 1$.

Ans. $(\log_a x)\log_a(xyz) = 48$ ...(i)

$(\log_a y)\log_a(xyz) = 12$ ...(ii)

$(\log_a z)\log_a(xyz) = 84$ ...(iii)

Adding equation (i), (ii) and (iii)

$\Rightarrow \log_a(xyz)[\log_a x + \log_a y + \log_a z] = 48 + 12 + 84$

$\Rightarrow \log_a(xyz) \cdot \log_a(xyz) = 144$

$[\log_a(xyz)]^2 = 144 \Rightarrow \log_a(xyz) = \pm 12$

Case 1:

$\log_a(xyz) = 12$

From (i), we get

$(\log_a x)12 = 48$

$\Rightarrow \log_a x = \frac{48}{12}$

$\Rightarrow \log_a x = 4$

$\Rightarrow x = a^4$

From (ii), we get

$(\log_a y) \times 12 = 12$

$\Rightarrow y = a^1$

From (iii), we get

$(\log_a z) \times 12 = 84$

$\Rightarrow \log_a z = 7 \Rightarrow z = a^7$

Case 2:

$\log_a(xyz) = -12$

$\begin{cases} x = a^{-4} \\ y = a^{-1} \\ z = a^{-7} \end{cases}$

Final Answer:

$(x, y, z) = (a^4, a, a^7)$

OR

(a^{-4}, a^{-1}, a^{-7})

Brain Teaser

The number of real solutions of the equation $\log(-x) = 2\log(x+1)$, is

(a) 0 (b) 1 (c) 2 (d) 4

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Ans. $\log(-x) = \log(x+1)^2 \Rightarrow -x = (x+1)^2$

$-x = x^2 + 1 + 2x$

$\Rightarrow x^2 + 3x + 1 = 0$

$\because x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\therefore x = \dfrac{-3 \pm \sqrt{9 - 4 \times 1 \times 1}}{2 \times 1}$

$x = \dfrac{-3 \pm \sqrt{5}}{2}$ → $\dfrac{-3+\sqrt{5}}{2}$ -ve, $\dfrac{-3-\sqrt{5}}{2}$ -ve

$x = \dfrac{-3-\sqrt{5}}{2}$

$\Rightarrow x + 1 = \dfrac{-3+\sqrt{5}}{2} + \dfrac{1}{1}$

$x + 1 = \dfrac{-3-\sqrt{5}+2}{2}$

$(x+1) = \dfrac{\sqrt{5}-1}{2}$ ✓ +ve

only 1 solutions

$x = \dfrac{-3-\sqrt{5}}{2}$

$x + 1 = \dfrac{-3-\sqrt{5}}{2} + 1$

$x + 1 = \dfrac{-3-\sqrt{5}+2}{2}$

$x + 1 = \dfrac{-1-\sqrt{5}}{2} = $ -ve

(Reject)

Characteristic & Mantissa of Logarithm

$\log_{10}101 = 2$ ....

$\log_{10}999 = 2.9$.....

$\log_5 5 = 1$

$\log_5 25 = 2$

$\log_5 125 = 3$

$\log_5 90$

Characteristic = 2

Characteristic

If $\log_{10}N = 1.201$

Mantissa

$\log_{10}100 = 2.0$

Characteristic Mantissa

$\log_{10}1000 = 3$

Ques. Find the number of positive integers which have the characteristic 3, if the base of the logarithm is 5.

Ans. $\log_5 125 = 3$

$\log_5 625 = 4$

Characteristic is 3 for {125, 126, 127, 128, ... 624}

$624 = a + (n-1)d$

$624 = 125 + (n-1) \times 1$

$624 = 124 + n$

$n = 500$

$\log_{10}10 = 1$

$\log_{10}100 = 2$

$\log_{10}1000 = 3$

$\log_{10}10{,}000 = 4$

Characteristic = 2 is for {100, 101, 102, 103, ... 999}.

Characteristic = 3 is for {1000, 1001, 1002, 1003, ... 9999}.

Important Note

If characteristic of $\log_{10}N$ is x then no. of digits in N is x + 1.

Example:

(i) If $\log_{10}N$ has characteristic = 3, then N is a 4 digit number.

(ii) Characteristic of $\log_{10}N$ is 20 then no. of digits in N is = 21.

Ques. Number of digits in the number 12^{300} are (Given $\log_{10}2 = 0.3010$, $\log_{10}3 = 0.4771$)

(a) 323 (b) 324 ✓

(c) 325 (d) None of these

Ans. $N = (12)^{300}$

$\log_{10}N = \log_{10}(12)^{300}$

$= 300 \log_{10}12$

$300 \log_{10}(2^2 . 3)$

$300 [2 \log_{10}2 + \log_{10}3]$

$= 300 [2 \times 0.301 + 0.4771]$

$= 300 \times (0.602 + 0.4771)$

$\log_{10}N = 323.7$ → characteristics

No. of digits = 323 + 1

= 324

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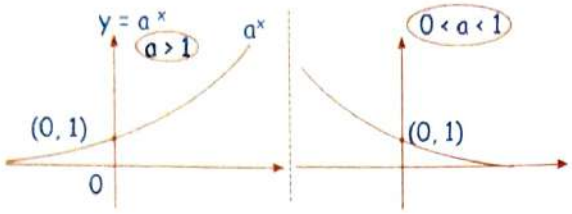
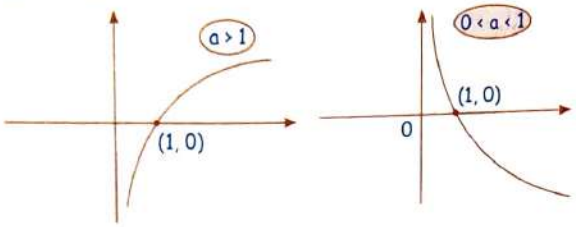
Graphs of exponential functions

(i) $y = a^x$



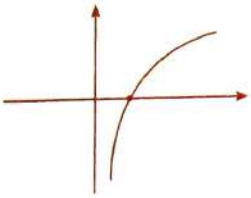
Graphs of LOGARITHM functions

(1) $y = \log_a x$



Important Property

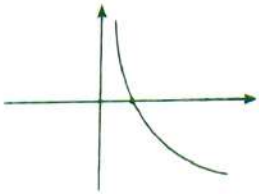
If number and base both are either less than 1 or greater than 1 then the value of log is positive.



$\log_2 10 > 0$

$\log_{0.05} 0.01 > 0$

If number and base are on opposite side of 1 on the number line then the value of log is negative.



$\log_{0.2} 10 < 0$

LOGARITHM INEQUALITIES

Case-1 : When base of log is constant.

Example: $\log_3(x - 2) \geq 0$

Case-2 : When base of log is variable.

Example: $\log_{3x}(x - 2) < 2$

Important Note

Always find domain and take its intersection with your answer before writing the final answer.

Logarithm Inequalities - CASE-1 (When Base is Constant)

Case 1 (A): When base of log is > 1

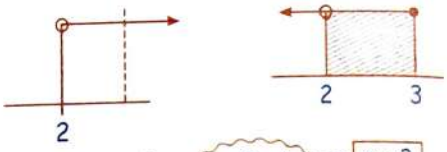
Important Note

Sign of inequality remains same.

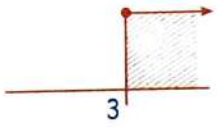
Example:

$\log_3(x - 2) \geq 0$

$\Rightarrow (x - 2 > 0) \Rightarrow \boxed{x > 2}$ ----- (i)



$\Rightarrow (x - 2) \geq 3^0 \Rightarrow (x - 2 \geq 1) \Rightarrow \boxed{x \geq 3}$ ---- (ii)



From equations (i) & (ii)

Intersection = $[3, \infty)$

Ques. Solve: $\log_3(2x - 3) \leq 2$.

Ans.

$\log_3(2x - 3) \leq 2.$

$\Rightarrow 2x - 3 \leq (3)^2$

$2x - 3 \leq 9$

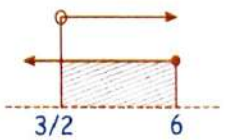
$2x \leq 12$

$x \leq 6$

Domain :

$2x - 3 > 0 \Rightarrow 2x > 3$

$\Rightarrow x > \frac{3}{2}$



Final Answer = $\left(\frac{3}{2}, 6\right]$

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Case 1 (B): When base of log is < 1

Important Note

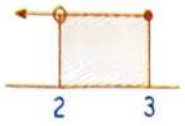
Sign of inequality reverses.

Example:

$\log_{1/3}(x-2) \geq 0$ Defined when

$\Rightarrow (x-2) \leq (1/3)^0$ $x-2>0 \Rightarrow x>2$ ...(i)

$\Rightarrow x-2 \leq 1 \Rightarrow x \leq 3$ ...(ii)



From equations (i) & (ii)

Intersection = (2, 3]

Ques. Solve: $\log_{1/2}[3x+2] \leq 3$.

Ans. $(3x+2) \geq \left(\frac{1}{2}\right)^3$

$3x+2 \geq \frac{1}{8}$

$3x \geq \frac{1}{8} - 2$

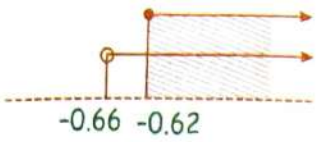
$3x \geq \frac{1-16}{8}$

$3x \geq \frac{-15}{8}$

$x \geq \frac{-5}{8}$

$\frac{-5}{8} = -0.62$

Domain : $3x+2>0 \Rightarrow x>-2/3 \Rightarrow x>-0.66$



Final Answer: $x \in \left[\frac{-5}{8}, \infty\right)$

Ques. $\log_2\left(\frac{2x^2-3x+1}{x^2-5x+6}\right) \geq 1$

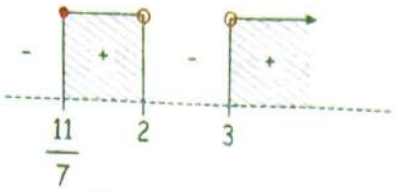
Ans. $\frac{2x^2-3x+1}{x^2-5x+6} \geq 2^1$

$\Rightarrow \frac{2x^2-3x+1}{x^2-5x+6} - 2 \geq 0$

$\Rightarrow \frac{2x^2-3x+1-2(x^2-5x+6)}{x^2-5x+6} \geq 0$

$\Rightarrow \frac{2x^2-3x+1-2x^2+10x-12}{x^2-5x+6} \geq 0$

$\Rightarrow \frac{7x-11}{(x-2)(x-3)} \geq 0$



$x \in \left[\frac{11}{7}, 2\right) \cup (3, \infty)$

Ques. $\log_{1/2}(x^2-1) > 0$

(a) $(-\sqrt{2}, -1) \cup (1, \sqrt{2})$

(b) $(-2, -1) \cup (1, 2)$

(c) $(-\infty, -1) \cup (1, \infty)$

(d) $(-2, 2)$

Ans. $x^2 - 1 < \left(\frac{1}{2}\right)^0$

$x^2 - 1 < 1$

$\Rightarrow x^2 < 2$

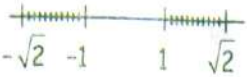
$1 < x^2 < 2$

$\Rightarrow x \in (1, \sqrt{2}) \cup (-\sqrt{2}, -1)$

Domain :

$x^2 - 1 > 0$

$x^2 > 1$



Logarithm Inequalities- CASE-2 (When Base is Variable)

Ques. $\log_{2x}(x^2-5x+6) \geq 1$.

Ans. Case 1: $2x>1 \Rightarrow x>\frac{1}{2}$

$\Rightarrow x^2-5x+6 \geq (2x)^1$

$\Rightarrow x^2-7x+6 \geq 0$

$\Rightarrow (x-6)(x-1) \geq 0$

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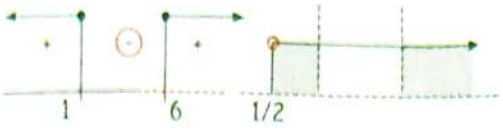


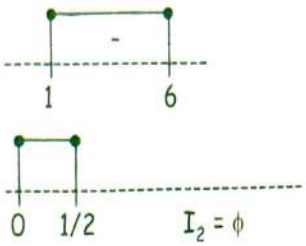
$I_1 : \left(\frac{1}{2}, 1\right] \cup [6, \infty)$

Case 2: $0 < 2x < 1$

$$\boxed{0 < x < \frac{1}{2}}$$

base is less than 1

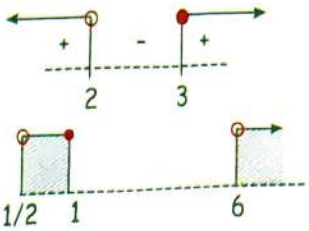
$x^2 - 5x + 6 \leq (2x)^1$



Domain: $x^2 - 5x + 6 > 0$

$(x - 3)(x - 2) > 0$

Ans : $I_1 \cup I_2 \cap$ Domain



Final Answer: $\left(\frac{1}{2}, 1\right] \cup [6, \infty)$

Solving Exponential Inequations

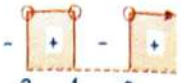
Case 1(A) : When base is > 1

Ques. $5^{x+3} > \left(\frac{1}{25}\right)^{1/x}$

Ans. $5^{x+3} > (5^{-2})^{1/x} \Rightarrow 5^{x+3} > 5^{-2/x} \Rightarrow x + 3 > \frac{-2}{x}$

$$\Rightarrow x + 3 + \frac{2}{x} > 0 \Rightarrow \frac{x^2 + 3x + 2}{x} > 0$$

$$\Rightarrow \frac{(x + 2)(x + 1)}{x} > 0$$



$x \in (-2, -1) \cup (0, \infty)$

Case 1(B) : When base is < 1

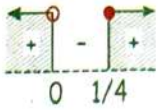
Ques. Solve $(0.5)^{1/x} \geq 0.0625$

Ans. $(0.5)^{1/x} > (0.5)^4$

$$\frac{1}{x} \leq 4$$

$$\frac{1}{x} - 4 \leq 0$$

$$\frac{1 - 4x}{x} \leq 0 \qquad \Rightarrow \frac{4x - 1}{x} \geq 0$$



$x \in (-\infty, 0) \cup \left[\frac{1}{4}, \infty\right)$

Ques. $\log_4 x - \log_x 4 \leq \frac{3}{2}$

Ans. Domain : $x > 0, x \neq 1$

Let $\log_4 x = t$

$\Rightarrow \log_x 4 = \frac{1}{t}$

$$t - \frac{1}{t} \leq \frac{3}{2}$$

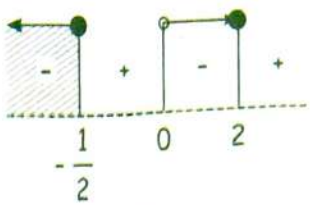
$$t - \frac{1}{t} - \frac{3}{2} \leq 0$$

$$\frac{2t^2 - 2 - 3t}{2t} \leq 0$$

$$\frac{2t^2 - 3t - 2}{2t} \leq 0$$

$$\frac{2t^2 - 4t + t - 2}{2t} \leq 0$$

$$\frac{(2t + 1)(t - 2)}{2t} \leq 0$$



$t \in \left(-\infty, -\frac{1}{2}\right] \cup (0, 2]$

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$\log_4 x \in \left(-\infty, -\frac{1}{2}\right] \cup (0, 2]$

$x \in \left(4^{-\infty}, 4^{-\frac{1}{2}}\right] \cup (4^0, 4^2]$

Final Answer:

$x \in \left(0, \frac{1}{2}\right] \cup (1, 16]$

$4^{-\infty} \to 0$

$4^{-x} \to \frac{1}{4^x} \to \frac{1}{\infty} \to 0$

$4^{-\frac{1}{2}} = (2)^{2\left(-\frac{1}{2}\right)} = 2^{-1}$

$= \frac{1}{2}$

Ques. $\log_2 x \leq \frac{2}{\log_2 x - 1}$

(a) $\left(0, \frac{1}{2}\right] \cup (2, 4)$

(b) $\left(0, \frac{1}{2}\right] \cup (2, 4]$

(c) $\left(\frac{1}{2}, 2\right) \cup [4, \infty)$

(d) None of these

Ans. Let $\log_2 x = t$

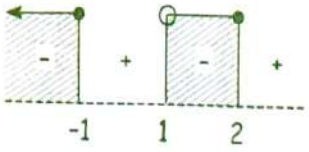
$t \leq \frac{2}{t-1}$

$t - \frac{2}{t-1} \leq 0$

$\frac{t(t-1)-2}{(t-1)} \leq 0$

$\frac{t^2 - t - 2}{(t-1)} \leq 0$

$\frac{(t-2)(t+1)}{(t-1)} \leq 0$



$\log_2 x \in (-\infty, -1] \cup (1, 2]$

$x \in (2^{-\infty}, 2^{-1}] \cup (2^1, 2^2]$

$\downarrow$

$x \in \left(0, \frac{1}{2}\right] \cup (2, 4]$

SQUARE ROOT INEQUATIONS

Important Note

We must Square Both Sides of an Inequality only if both RHS & LHS are Non-Negative.

Always find domain in sq. Root inequalities

L	R
+	+
-	+
+	-
-	-

Example:

(i) $\sqrt{x+2} > x$

(ii) $\sqrt{x-3} \geq \sqrt{7-x}$

(iii) $3 > 2 \Rightarrow 3^2 > 2^2$

(iv) $3 > -7$

Square $\Rightarrow 9 > 49$ (×)

(v) $3 > -2$

Square $9 > 4$ (✓)

Ques. $\sqrt{2x-5} > \sqrt{5-x}$

Non negative — Non negative

Ans. Squaring both sides

$2x - 5 > 5 - x$

$2x + x > 5 + 5$

$3x > 10$

$x > \frac{10}{3}$ ✓

✓ ⇒ LHS & RHS both are non negative ∴ We can S.B.S.

always take intersection with domain

Domain:

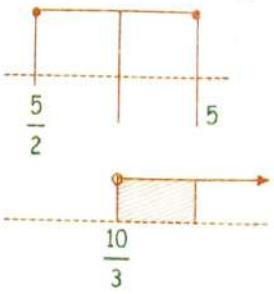
$2x - 5 \geq 0 \Rightarrow x \geq \frac{5}{2}$

&

$5 - x \geq 0 \quad -x \geq -5$

$x \leq 5$

Domain: $\left[\frac{5}{2}, 5\right]$ ✓



Ans. $\left(\frac{10}{3}, 5\right]$

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Ques. $\sqrt{7x-6} < (x) \rightarrow +ve$

Domain:

$7x - 6 \geq 0$

$7x \geq 6$

$x \geq \frac{6}{7}$ ✓

Ans. LHS = non -ve

RHS : x = +ve in the domain:

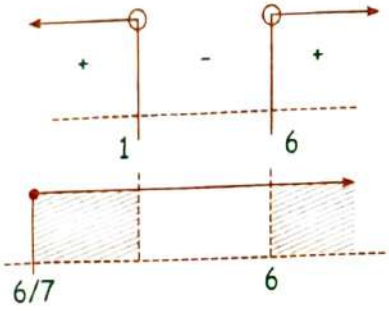
Now, we can square both sides

$7x - 6 < x^2$

$-x^2 + 7x - 6 < 0$

$x^2 - 7x + 6 > 0$

$(x - 6)(x - 1) > 0$



Final Answer: $x \in \left[\frac{6}{7}, 1\right) \cup (6, \infty)$

Ques. $\sqrt{8 + 2x - x^2} > 6 - 3x$

Ans. LHS = +ve

RHS = ?

Case 1:

$RHS \geq 0$ ✓

$6 - 3x \geq 0$

$6 \geq 3x$

$2 \geq x \Rightarrow x \leq 2$

$\sqrt{8 + 2x - x^2}$

Squaring both sides

$8 + 2x - x^2 > (6 - 3x)^2$

$8 + 2x - x^2 > 36 + 9x^2 - 36x$

$0 > 36 - 8 + 10x^2 - 38x$

$10x^2 - 38x + 28 < 0$

$5x^2 - 19x + 14 < 0$

$5x^2 - 5x - 14x + 14 < 0$

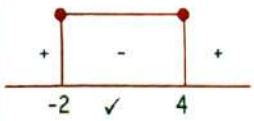
$(5x - 14)(x - 1) < 0$

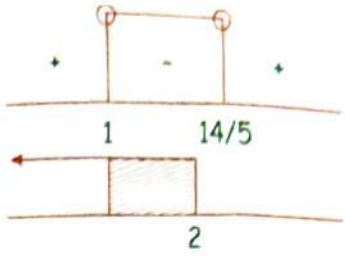
Domain:

$8 + 2x - x^2 \geq 0$

$x^2 - 2x - 8 \leq 0$

$(x - 4)(x + 2) \leq 0$





$I_1 : (1, 2]$

Case 2:

$RHS < 0$

$6 - 3x < 0 \Rightarrow 6 < 3x \Rightarrow x > 2$

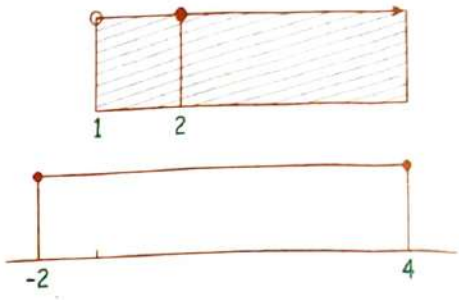
LHS = +ve

$\underbrace{LHS}_{+ve} > \underbrace{RHS}_{-ve}$

$\Rightarrow$ always true.

$x > 2$

$I_2 : x \in (2, \infty)$ ✓



$x \in (1, 4]$

Concluding Example

$|2x - 4| < x - 1$

LHS = +ve

RHS = ? = +ve

$\underbrace{LHS}_{+ve} < RHS$

$+ve < RHS$

$x - 1 > 0$ $\quad x > 1$ ✓

$|x|^2 = x^2$

Now, we can S.B.S.

$|2x - 4|^2 < (x - 1)^2$

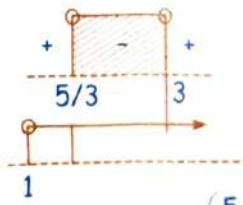
$(2x - 4)^2 < x^2 + 1 - 2x$

$4x^2 + 16 - 16x < x^2 + 1 - 2x$

$3x^2 - 14x + 15 < 0$

$3x^2 - 9x - 5x + 15 < 0$

$(3x - 5)(x - 3) < 0$



Final Answer: $x \in \left(\frac{5}{3}, 3\right)$

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Solving Exponential Equations

Brain Teaser

The number of solutions of the equation

$|x-3|^{\frac{x^2-8x+15}{x-2}} = 1$ is /are:

Ans.

$x = 3$	$x = 2$
Question $(0)^0$	Rejected as Denominator $(x-2)$ becomes zero.

0^0 is not defined

$$\log_{10}|x-3|^{\frac{x^2-8x+15}{x-2}} = \log_{10} 1$$

$$\left(\frac{x^2-8x+15}{x-2}\right)\log|x-3| = 0$$

$\because$ If $ab = 0$
$\Rightarrow$ either $a = 0$ or $b = 0$

either $\dfrac{x^2-8x+15}{x-2} = 0$

$\Rightarrow x^2-8x+15 = 0$

$\Rightarrow (x-3)(x-5) = 0 \Rightarrow x = 3$ or 5

OR $\log_{10}|x-3| = 0$

$|x-3| = 1$

$x - 3 = \pm 1$

$x = 3 \pm 1 \Rightarrow x = 4$ or 2

Hence, $x = 5$ or $x = 4$ are the only answers.

Important Note

If $|a|^b = 1$

$\Rightarrow b = 0$ or $|a| = 1$

Power = 0 or Base = 1

Brain Teaser

Find all the solutions of the equation

$|x-1|^{(\log x)^2 - \log x^2} = |x-1|^3$, where base of logarithm is (10).

Ans.

$$\frac{|x-1|^{(\log x)^2 - \log x^2}}{|x-1|^3} = 1$$

$$|x-1|^{(\log x)^2 - 2\log x - 3} = 1$$

$(\log x)^2 - 2\log x - 3 = 0$

Let $\log x = t$

$t^2 - 2t - 3 = 0$

$(t-3)(t+1) = 0$

$t = 3$ or $t = -1$

$\log_{10} x = 3$ or -1

$x = 10^3$ or 10^{-1}

$= 1000$ or $\dfrac{1}{10}$

Or $|x-1| = 1$

$x - 1 = \pm 1$

$x = 1 \pm 1$

$x = 2$ or $x = 0$ (Reject)

$x = 1000, \dfrac{1}{10}$ or 2

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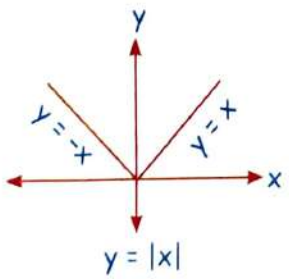
Short Notes

WAVY CURVE METHOD

- Points where denominator is zero will never be included in the answer.
- If we are asked to find the intervals where f(x) is non-negative or non-positive then make the intervals closed corresponding to the roots of the numerator and let it remain open corresponding to the roots of denominator.
- Normally we cannot cross-multiply while solving inequalities. But we cross multiply if we are sure that quantity in denominator is always positive.
- Normally we cannot square both sides of inequalities. But we can square if we are sure that both sides are non negative.
- We can multiply both sides with a negative number by changing the sign of inequality.
- We can add or subtract equal quantity to both sides of inequalities without changing the sign of inequality.

ABSOLUTE VALUE FUNCTION/ MODULUS FUNCTION

- $y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$



- **If $|x| < a$,** The solution set is $(-a, a)$.
- **If $|x| > a$,** The solution set is $(-\infty, -a) \cup (a, \infty)$

LOGARITHM

(P-1) $\log_a 1 = 0$

(P-2) $\log_N N = 1$

(P-3) $\log_{\frac{1}{N}} N = -1$

(P-4) $\log_a(MN) = \log_a M + \log_a N$

(P-5) $\log_a \frac{m}{n} = \log_a m - \log_a n$

(P-6) $\log_a m^x = x \log_a m$

(P-7) $\log_b m = \frac{\log_a m}{\log_a b}$

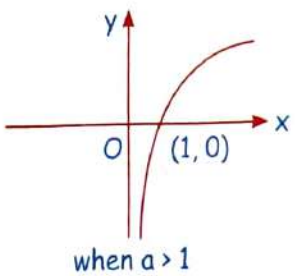
(P-8) $\log_b a . \log_a b = 1$

(P-9) $\log_{a^k} m = \frac{1}{k}\log_a m$

(P-10) $N = (a)^{\log_a N}$

(P-11) $a^{\log_b c} = c^{\log_b a}$

Graph of $y = \log_a x$:



- Whenever the number and the base are on the same side of unity then logarithm of that number to that base is positive.
- If the number and the base are located on different side of unity then logarithm of that number to that base is negative.
- The mantissa part of logarithm of a number is always positive $(0 \leq m < 1)$
- If the characteristic of $\log_{10} N$ be n, then the number of digits in N is $(n + 1)$
- If the characteristic of $\log_{10} N$ be $(-n)$, then there exist $(n - 1)$ zeros after decimal in N.

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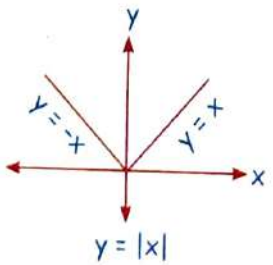

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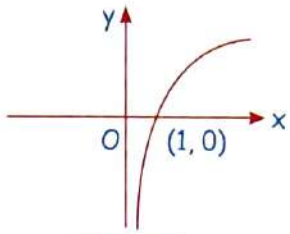
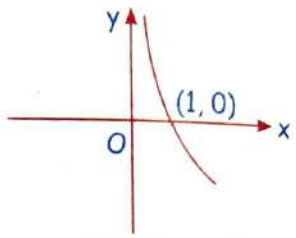


when $a > 1$



when $0 < a < 1$

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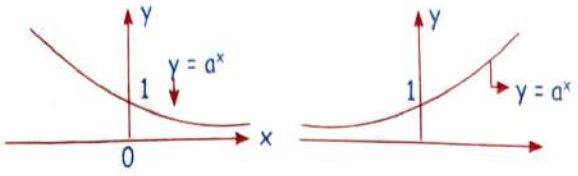
EXPONENTIAL FUNCTIONS

Graphs of $y = a^x$



$(0 < a < 1)$ $(a > 1)$

- When $a < 1$, a^x decreases as x increases and when $a > 1$, a^x increases as x increases.
- The values of a^x are always positive as the graphs are always above the x-axis for all x.

EXPONENTIAL INEQUALITIES

- $a^x > 1 \Rightarrow \begin{cases} x > 0 & \text{if } a > 1 \\ x < 0 & \text{if } a < 1 \end{cases}$

  & $a^x \geq 1 \Rightarrow \begin{cases} x \geq 0 & \text{if } a \geq 1 \\ x \leq 0 & \text{if } a \leq 1 \end{cases}$

- $a^x < 1 \Rightarrow \begin{cases} x < 0 & \text{if } a > 1 \\ x > 0 & \text{if } a < 1 \end{cases}$

  & $a^x \leq 1 \Rightarrow \begin{cases} x \leq 0 & \text{if } a \geq 1 \\ x \geq 0 & \text{if } a \leq 1 \end{cases}$

- $a^x > a^y \Rightarrow \begin{cases} x > y & \text{if } a > 1 \\ x < y & \text{if } 0 < a < 1 \end{cases}$

  & $a^x \geq a^y \Rightarrow \begin{cases} x \geq y & \text{if } a \geq 1 \\ x \leq y & \text{if } 0 < a \leq 1 \end{cases}$

- $a^x < a^y \Rightarrow \begin{cases} x < y & \text{if } a > 1 \\ x > y & \text{if } 0 < a < 1 \end{cases}$

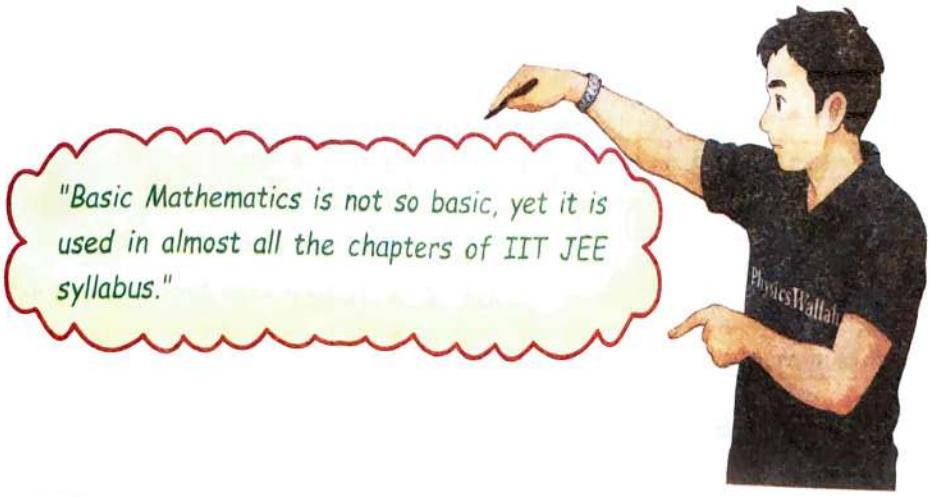
  & $a^x \leq a^y \Rightarrow \begin{cases} x \leq y & \text{if } a \geq 1 \\ x \geq y & \text{if } 0 < a \leq 1 \end{cases}$

- $a^x > b \Rightarrow \begin{cases} x > \log_a b & \text{if } a > 1 \\ x < \log_a b & \text{if } 0 < a < 1 \end{cases}$

  & $a^x \geq b \Rightarrow \begin{cases} x \geq \log_a b & \text{if } a \geq 1 \\ x \leq \log_a b & \text{if } 0 < a \leq 1 \end{cases}$

- $a^x < b \Rightarrow \begin{cases} x < \log_a b & \text{if } a > 1 \\ x > \log_a b & \text{if } 0 < a < 1 \end{cases}$

  & $a^x \leq b \Rightarrow \begin{cases} x \leq \log_a b & \text{if } a \geq 1 \\ x \geq \log_a b & \text{if } 0 < a \leq 1 \end{cases}$



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Chapter 2

Sets

SET

A well-defined collection of objects.

Example:

✓ The collection of vowels in English alphabets.
→ {a, e, i, o, u}

✓ The collection of first five prime natural number.
→ {2, 3, 5, 7, 11}

× The collection of Best cricket players of India..
→ not well defined

× The collection of difficult topics in mathematics.
→ not a set

✓ The integers whose squares are less than 20.
→ {0, 1, 2, 3, 4, -1, -2, -3, -4}

× The set of most Naughty students of Arjuna
→ not a set

NOTATIONS

N = Set of all natural number = {1, 2, 3, 4.....}

W = Set of all whole number = {0, 1, 2, 3,.....}

Z or I = Set of all integers = {...-3, -2, -1,0,1,2,3.... }

Z^+ = Set of all positive integers = {1, 2, 3,...} = N

Z^- = Set of all negative integers = {-1, -2, -3,...}

Q = The set of all rational number = $\left\{\frac{p}{q} : p, q \in I, q \neq 0\right\}$

R = The set of all real numbers

R - Q = The set of all irrational numbers

REPRESENTATION OF A SETS

Roster or Tabular Notation

Ques. Write the Following Sets in Roster form

(i) Set of all Natural numbers lying between 2 and 9.

(ii) Set of all letters in the word 'MATHEMATICS'.

(iii) Set of all ODD natural numbers.

(iv) B = {x : $x^2 + 7x - 8 = 0, x \in R$}

Ans. (i) Roster From → {3, 4, 5, 6, 7, 8}

(ii) Roster form → {M, A, T, H, E, I, C, S}

(iii) Roster form → {1, 3, 5, 7, 9....}

(iv) $(x + 8)(x - 1) = 0$

$x = -8, 1$

B = {-8, 1}

Set Builder Form

Ques. Write the following sets in set builder form

(i) A = {1,4,9,16,25,36,49}

(ii) B = {1,9,25,49,81,121}

(iii) C = {2,9,28,65,126}

Ans. (i) A : {x | x is sq of first 7 natural no's}

(ii) B : {x | x is sq of first 6 odd natural no's}

(iii) C : {x | $x = n^3 + 1$ where n is a natural number $n \leq 5$}

TYPES OF SETS

Empty Set or Null set or Void set (ϕ, { })

A set containing no element

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Singleton Set

Example: {3}, {5}, {a}, {0}

Finite Set and Infinite Set

Example:

(i) Let W be the set of the days of the week.

→ Finite Set

(ii) Let S be the set of solutions of the equation $x^2 - 16 = 0$.

$x^2 = 16$

$x = \pm 4$ $x = \{4, -4\}$

→ Finite Set

(iii) Let G be the set of points on a line.

→ Infinite Set

(iv) Set of Natural Numbers → Infinite Set

Equivalent Sets $A \approx B$

If no. of elements in both sets are same

Example:

A = {1, 2, 3, 4, 5}

B = {a, e, i, o, u}

C = {Ph, Ch, M, E, H}

Equivalent sets

Cardinal number = 5

$n(A) = n(B) = n(C)$

5 5 5

Equal Sets

Example:

A = {1, 2, 3, 4, 5}

B = {2, 1, 3, 5, 4}

C = {2, 2, 2, 1, 3, 3, 5, 4}

$A = B = C$

Ques. Which of the following pairs of sets are equal? Give reasons.

(i) A = {0, 3} and B = {x : x is root of the equation $x^2 - 3x = 0$}

(ii) A = {-4, -3, -2, -1, 0, 2, 3, 4} and B = {x : x ∈ Z and |x| < 4}

Ans. (i) A = {0, 3}, B = {x : x is root of the equation $x^2 - 3x = 0$}

$x(x - 3) = 0$

$x = 0$ or $x = 3$

B = {0, 3}

Hence, $A = B$ → Equal set

(ii) A = {-4, -3, -2, -1, 0, 2, 3, 4} and B

= {x : x ∈ Z and $|x| < 4$}

$-4 < x < 4$

B = {-3, -2, -1, 0, 1, 2, 3}

Hence, $A \neq B$ → Not equal set

Universal Set

Example:

A = {1, 2, 3}

B = {2, 3, 4, 5}

C = {5, 6, 7, 8}

U = {1, 2, 3, 4, 5, 6, 7, 8}

U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

U = N

U = Q

Ques. Which of the following may be considered as universal set(s) for all the three sets

X = {a, c}, Y = {a, e, i, o, u}, Z = {i, j, k, l}

(a) {a, b, c, d,.... m} → Not universal set

(b) {a, e, i, o, u, k, l} → Not universal set

(c) {x : x is a letter of English alphabet} → Universal set

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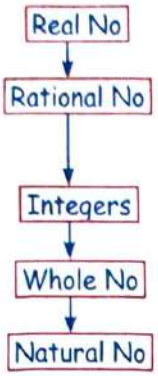
SUB-SET & SUPER-SET



Ex: A = {1, 2, 3, 4}

Subset of A are {1, 2}, {1, 2, 3,}, {2, 3, 4}, {2}, {1, 2, 3, 4}

Note: Null Set is subset of Every Set

Difference between ∈ & ⊂

∈ & ⊂

is an element of | Subset of

A = {1, 2, 3, 4}
1 ∈ A ✓
1 ⊂ A ×
{1} ⊂ A ✓

A = {a, b, c}
B = {a}
B ⊂ A
a ∈ A
b ∈ A
c ∈ A

Ques. Examine whether the following statements are ture or false:

(i) {a, b} ⊄ {b, c, a}

(ii) {a, e} ⊂ {x : x is a vowel in the English alphabet} ({a, e, i, o, u})

(iii) {1, 2, 3} ⊂ {1, 3, 5}

(iv) {a} ⊂ {a, b, c}

(v) {a} ∈ {a, b, c}

(vi) {x : x is an even natural number less than 6} ⊂ {x : x is a natural number which divides 36}

Ans. (i) False, (ii) True, (iii) False, (iv) True, (v) False, (vi) True

Ques. A = {x : x is a natural number and x < 8} and B = {1, 2, 3, 4, 5}. Which of the following is true?

(a) A ⊂ B (b) B ⊄ A (c) B ∈ A (d) B ⊂ A

Ans. A = {1, 2, 3, 4, 5, 6, 7}
B = {1, 2, 3, 4, 5}
⇒ B ⊂ A

Ques. Let X = {n ∈ N : 1 ≤ n ≤ 50}. A = {n ∈ x: n is a multiple of 2}; B = {n ∈ x : n is a multiple of 7}, then the number of elements in the smallest subset of X containing both A and B is

[JEE Main 2020 (Jan.)]

Ans. X = {1, 2, 3, 4, 5,.....50}

A = {2, 4, 6, 8, 10,....50}

B = {7, 14, 21, 28, 35, 42, 49}

⇒ A ∪ B = {2, 4, 6, 8, 10,...50, 7, 21, 35, 49}
(25) + (4)

= 29

PROPER SUBSET

A = {1, 2}

Subset ϕ
{1} → proper subsets = 3
{2}
{1, 2}

Total = (4) = 2^2

A = {a, b, c}

ϕ]
{a}]
{b}]
{c}] → 7 proper subsets
{a, b}]
{b, c}]
{c, a}]
{a, b, c} → not part of proper sub set

Total = 8 = 2^3

Number of Subsets

Total number of Subsets of a set A is 2^n

where (n) is the number of elements present in set A

no of proper subsets = $2^n - 1$

Example: A = {a, b, c, d}

no of subsets = 2^4 = 16

Ques. If A = {a, b, c, d, e} then the number of proper subsets are

(a) 32 (b) 33
(c) 31 (d) None of these

Ans. Proper subset = $2^n - 1 = 2^5 - 1 = 31$

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Ques. Let $A = \{1, 2, \{3, 4\}, 5\}$. State TRUE or FALSE?

(i) $\{3, 4\} \subset A$

(ii) $\{3, 4\} \in A$

(iii) $\{\{3, 4\}\} \subset A$

(iv) $1 \in A$

(v) $1 \subset A$

(vi) $\{1, 2, 5\} \subset A$

(vii) $\{1, 2, 5\} \in A$

(viii) $\{1, 2, 3\} \subset A$

(ix) $\phi \in A$

(x) $\phi \subset A$

(xi) $\{\phi\} \subset A$

$A = \{1, 2, \{3, 4\}, 5\}$

$\{\{3, 4\}\}$

Ans. (i) False, (ii) True, (iii) True, (iv) True, (v) False, (vi) True, (vii) False, (viii) False, (ix) False (x) True (xi) False

POWER SET

Set of all subsets of A is called power set of A, It is denoted by P(A)

$n(P(A)) = 2^n$

Example: → $A = \{1, 2, 3\}$

$P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2,3\}, \{3, 1\} \{1, 2, 3\}\}$

Ques. Find power set of the following sets

1. $A = \{a, b\}$

Ans. $P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$

2. $A = \{a, \{b, c\}\}$

Ans. $P(A) = \{\phi, \{a\}, \{\{b, c\}\}, \{a, \{b, c\}\}\}$

$\boxed{n = 3}$

Ques. If $A = \{1, 2, 4\}$ How many elements are present in

1. $P(A)$

Ans. $P(A) = 2^n = 2^3 = 8$

2. $P(P(A))$

Ans. $P(P(A)) = 2^n = 2^8$

Ques. How many element has P(A), if $A = \{\ \}$

(a) 0 (b) 1 (c) 2 (d) 4

Ans. $A = \phi$ only subset of A is ϕ $P(A) = \{\phi\}$

Ques. What is $P(P(\phi))$

(a) $\{\phi, \{\phi\}\}$ (b) $\{\phi\}$

(c) $\{\phi, \{\phi\}\ \{\{\phi\}\}\}$ (d) $\{\{\phi\}\}$

Ans. $P(\phi) = \{\phi\}$

$P\{P(\phi)\} = \{\phi, \{\phi\}\}$

Ques. What is $n(P(P(P(\phi))))$

(a) 1 (b) 2 (c) 4 (d) 8

Ans. $n(P(\phi)) \rightarrow 1$

$n(P(P(\phi))) = 2^1 = 2$

$n(P(P(P(\phi)))) = 2^2 = 4$

Operations of Sets

Ques. If $A = \{1, 2, 3\}$, $B = \{2, 3, 4, 5\}$

(i) Union of Sets (ii) Intersection of sets

Ans. (i) $A \cup B = \{1, 2, 3, 4, 5\}$ (ii) $A \cap B = \{2, 3\}$

UNION & INTERSECTION OF SETS

Note

$\boxed{A \cap B \subseteq A}$ ✓

$\boxed{A \cap B \subseteq B}$ ✓

→ $\boxed{A \cap \phi = \phi}$

$\boxed{A \cap U = A}$

Note

$A \subseteq (A \cup B)$

$B \subseteq (A \cup B)$

$A \cup \phi = A$

$A \cup U = U$

Ques. If $X = \{4^n - 3^n - 1; n \in N\}$ and $Y = \{9(n - 1): n \in N\}$ Where N is a set of natural numbers. Then $X \cup Y$ is equal to:

(a) X (b) Y (c) N (d) Y-X

Ans. $X = \{0, 9, 54, 243, ..\}$

$Y = \{0, 9, 18, 27, 36, ...\}$

$X \cup Y = \{0, 9, 18, 27, 36, ...\}$

$= Y$

DIFFERENCE OF SETS

A - B = {all those elements of A which do not belong to B}

B - A = {all those elements of B which do not belong to A}.

$B - A = B - (A \cap B)$

Ques. $A = \{x : x = 2n, n \leq 3, n \in N\}$

$B = \{x : x = 3n, n \leq 3, n \in N\}$

Find $A \cup B$ and A - B:

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Ans. $A = \{2, 4, 6\}$

$B = \{3, 6, 9\}$

$A \cup B = \{2, 4, 6, 3, 9\}$

$A - B = \{2, 4\}$

Ques. Find $A - (A - B)$

If $A = \{1, 4, 7, 10, 13\}$ and

$B = \{2, 4, 6, 8, 10, 12, 14, 16\}$

Ans. $A \cap B = \{4, 10\}$

$A - B = A - (A \cap B)$

$\Rightarrow A - B = \{1, 7, 13\}$

Now, $A - (A - B) = \{4, 10\}$

COMPLEMENT OF A SET

not A, $\overline{A}$, A^c, A'

Example: $U = \{1, 2, 3, 4\}$ and $A = \{1, 2\}$

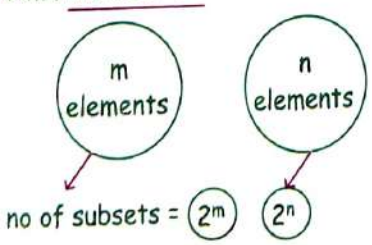
Then, $A' = U - A = \{3, 4\}$

Brain Teaser

Two finite sets have m and n elements respectively. The total number of subsets of first set is 992 more than the total number of subsets of the second set.

Find the value of m & n.

Ans. Method → 1



$\# 2^m = 992 + 2^n$

$\Rightarrow 2^m - 2^n = 992$

$\Rightarrow 2^{10} - 2^5 = 992$

Hence, $m = 10, n = 5$

Method → 2

$2^m - 2^n = 992$

$\Rightarrow 2^n \left[\frac{2^m}{2^n} - 1\right] = 992$

$\Rightarrow 2^n [2^{m-n} - 1] = 992$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

$2^5 = 32$

$2^6 = 64$

$2^7 = 128$

$2^8 = 256$

$2^9 = 512$

$2^{10} = 1024$

2	992
2	496
2	248
2	124
2	62
	31

$2^n [2^{m-n} - 1] = 2^5 \times 31$

$\Rightarrow 2^n = 2^5$ or $2^{m-n} - 1 = 31$

$\Rightarrow n = 5 \Rightarrow 2^{m-n} = 32 = 2^5$

$\Rightarrow m - n = 5 \Rightarrow m = 10$

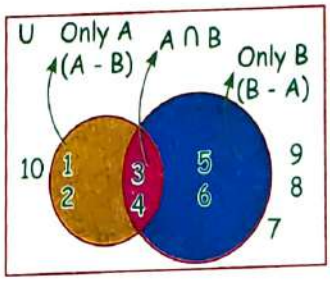
VENN DIAGRAM

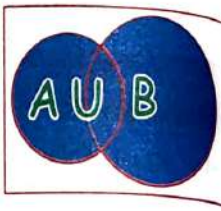
$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

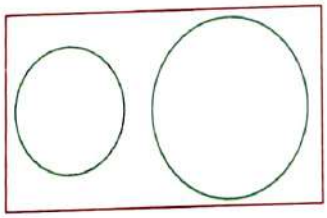
$A \cap B = \{3, 4\}$





DISJOINT SETS

$A \cap B = \phi$



LAWS OF ALGEBRA OF SETS

Commutative Law

$A \cap B = B \cap A$

$A \cup B = B \cup A$

Associative Law

$(A \cup B) \cup C = A \cup (B \cup C)$

$(A \cap B) \cap C = A \cap (B \cap C)$

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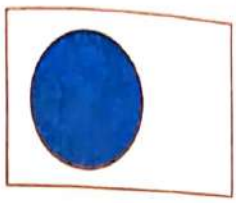
Identity Law

$A \cap U = A$

$A \cup \phi = A$



Complement Law

$A \cup A' = U$

$A \cap A' = \phi$

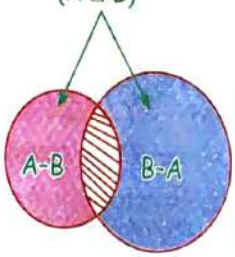
$(A') = A$

SYMMETRIC DIFFERENCE

$(A \Delta B)$

$A \Delta B = (A \cup B) - (A \cap B)$

Symmetric difference $(A \Delta B)$



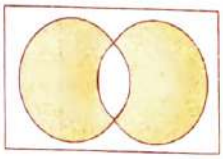
Ques. Which is equal to $(A - B) \cup (B - A)$?

(a) $(A \cup B) \cup (A - B)$

(b) $(A \cup B) \cup (A \cap B)$

(c) $(A \cup B) - (A \cap B)$

(d) $A - B$



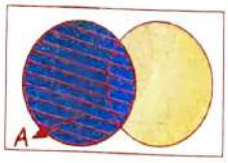
Ques. If A and B are two sets then $A \cap (A \cup B)$ equals

(a) A

(b) B

(c) ϕ

(d) $A \cup B$



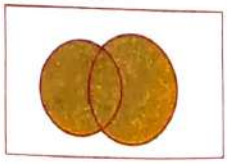
Ques. If A and B are two sets then $A \cup (A \cup B)$ equals

(a) A

(b) B

(c) ϕ

(d) $A \cup B$



CARDINAL NUMBER

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Ques. Let A and B be two sets such that $n(A) = 24$, $n(A \cup B) = 46$ and $n(A \cap B) = 8$.

(i) $n(B)$

(ii) $n(A - B) =$

(iii) $n(B - A)$

Ans. (i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$\Rightarrow 46 = 24 + n(B) - 8$

$\Rightarrow 46 = 16 + n(B)$

$\Rightarrow n(B) = 30$

(ii) $n(A-B) = n(A) - n(A \cap B)$

$= 24 - 8 = 16$

(iii) $n(B-A) = n(B) - n(A \cap B)$

$= 30 - 8 = 22$

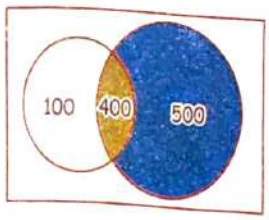
Ques. If $n(A \cup B) = 1000$, $n(A - B) = 100$ & $n(B - A) = 500$ then what is the value of $n(A \cap B)$?

(a) 600

(b) 400

(c) 500

(d) 900



Ans. $n(A \Delta B) = n(A-B) + n(B-A) = 600$

$\Rightarrow n(A \cup B) - n(A \cap B) = 600$

$\Rightarrow 1000 - n(A \cap B) = 600$

$\Rightarrow n(A \cap B) = 400$

DE-MORGAN'S LAW

$n(\overline{A \cup B}) = n(\bar{A} \cap \bar{B})$

$n(\overline{A \cap B}) = n(\bar{A} \cup \bar{B})$

Ques. If A, B and C are three sets and U is the universal set such that $n(U) = 700$, $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$. Find $n(A' \cap B')$.

(a) 600 (b) 400 (c) 500 (d) 300

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Ans. $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

Now, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 200 + 300 - 100 = 400$

$n(A' \cap B') = 700 - 400 = 300$

Ques. In a school there are 20 teachers who teach math's or chemistry. Out of these 12 teach math and 4 teach both math and chemistry. How many teach only chemistry?

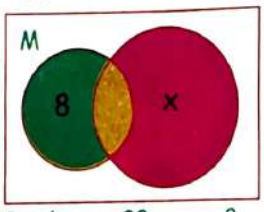
(a) 10 (b) 12

(c) 8 (d) None of these

Ans. $n(M \cup C) = 20$

$n(M) = 12$

$n(M \cap C) = 4$



$8 + 4 + x = 20 \Rightarrow x = 8$

Ques. There are 200 individuals with a skin disorder, out of them 120 have been exposed to chemical C_1, 50 to chemical C_2 and 30 to both chemical C_1 and C_2 find the number of individuals exposed to neither C_1 nor C_2.

(a) 60

(b) 40

(c) 50

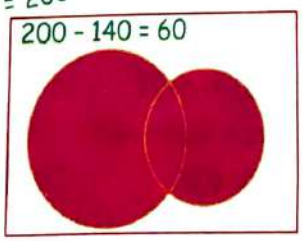
(d) 80

Ans. $n(C_1) = 120$

$n(C_2) = 50$

$n(C_1 \cup C_2) = 30$

$U = 200$



$n(C_1 \cup C_2) = n(C_1) + n(C_2) - n(C_1 \cap C_2))$

$= 120 + 50 - 30$

$\Rightarrow n(C_1 \cup C_2) = 140$

So, $n(\overline{C}_1 \cup \overline{C}_2) = U(n) - n(C_1 \cup C_2) = 200 - 140 = 60$

Ques. In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. How many people like coffee and tea?

Ans. $n(C) = 37, n(T) = 52$

$n(C \cup T) = 70$

Let, $n(C \cap T) = x$

$n(C \cup T) = n(C) + n(T) - n(C \cap T)$

$\Rightarrow 70 = 37 + 52 - x$

$\Rightarrow 70 = 89 - x$

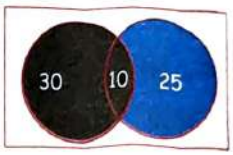
$\Rightarrow x = 19$

Ques. In a group of 65 people, 40 like cricket, 10 like both cricket and tennis. How many like tennis only and not cricket? How many like tennis?

Ans. $n(C) = 40,$

$n(C \cap T) = 10,$

$n(C \cup T) = 65,$



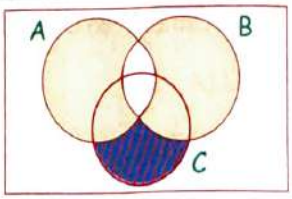
$\Rightarrow 65 = 40 + n(T) - 10$

$\Rightarrow 25 = n(T) - 10$

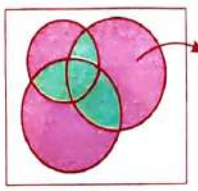
$\Rightarrow \boxed{n(T) = 35}$

$\Rightarrow n(\text{Tennis only}) = n(T) - n(T \cap C) = 35 - 10 = 25$

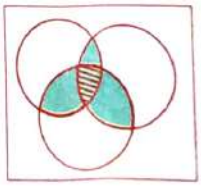
Cardinal Number for 3 Sets



$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$



exacty one



Ques. In a survey of 60 people, it was found that 25 people read newspaper H, 26 read newspaper T, 26 read newspaper I, 9 read both H and I, 11 read both H and T, 8 read both T and I, 3 read all three newspapers. Find:

(i) the number of people who read at least one of the newspapers.

(ii) the number of people who read exactly one newspaper.

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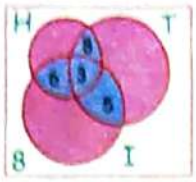
Ans. $n(U) = 60$



$n(H) = 25$ $n(H \cap I) = 9$

$n(T) = 26$ $n(H \cap T) = 11$ $n(H \cap T \cap I) = 3$

$n(I) = 26$ $n(T \cap I) = 8$

(i) $n(H \cup T \cup I) = 25 + 26 + 26 - 9 - 11 - 8 + 3$

$= 26 + 26 = 52$

(ii) n[people read exactly one newspaper]

$= 52 - [6 + 8 + 5 + 3]$

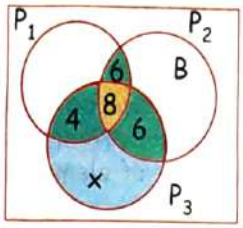
$= 52 - [22]$

$= 30$

Ques. In a survey it was found that 21 persons liked product P_1, 26 liked product P_2 and 29 liked product P_3. If 14 persons liked products P_1 and P_2; 12 persons liked product P_3 and P_1; 14 persons liked products P_2 and P_3, and 8 liked all the three products. Find how many liked product P_3 only.

(JEE Arjuna Mathematics M-1)

Ans.



From then diagram,

$n(P_1) = 21$
$n(P_2) = 26$
$n(P_3) = 29$
$n(P_1 \cap P_2) = 14$
$n(P_1 \cap P_3) = 12$
$n(P_2 \cap P_3) = 14$
$n(P_1 \cap P_2 \cap P_3) = 8$

$x + 4 + 8 + 6 = 29$

$\Rightarrow x + 18 = 29$

$\Rightarrow x = 11$

Ques. There are 70 students studying Math or English or German. 40 are studying Math, 30 studying German, 35 studying English and 15 in all three course. How many students are enrolled in exactly two of the courses?

(a) 10 (b) 5

(c) 15 (d) 20

Ans. $n(M \cup E \cup G) = 70$

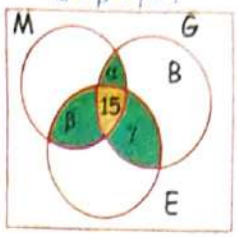
$n(M) = 40$, $n(M \cap G) = \alpha + 15$

$n(G) = 30$, $n(M \cap E) = 15 + \beta$

$n(E) = 35$, $n(E \cap G) = \gamma + 15$

$n(M \cap E \cap G) = 15$

$\alpha + \beta + \gamma = ?$



$n(M \cup E \cup G) = 40 + 30 + 35 - (\alpha + 15 + \beta + 15 + \gamma + 15) + 15$

$\Rightarrow 70 = 70 + 35 - (45 + \alpha + \beta + \gamma) + 15$

$\Rightarrow 50 - 45 - \alpha - \beta - \gamma = 0$

$5 = \alpha + \beta + \gamma$

Ques. An organization awarded 48 medals in event 'A', 25 in event 'B' and 18 in event 'C'. If these medals went to total 60 men and only five men got medals in all the three events, then, how many received medals in exactly two of the three events?

(11th April Shift 2023)

(a) 10 (b) 5 (c) 21 (d) 15

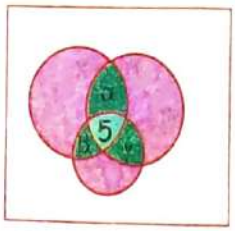
Ans. $n(A) = 48$

$n(B) = 25$

$n(C) = 18$

$n(A \cap B \cap C) = 5$

$n(A \cup B \cup C) = 60$



$\Rightarrow 60 = 48 + 25 + 18 - (\alpha + 5 + \beta + 5 + \gamma + 5) + 5$

$\Rightarrow 12 = 25 + 18 - (\alpha + \beta + \gamma + 15) + 5$

$\Rightarrow 12 = 30 + 3 - (\alpha + \beta + \gamma)$

$\Rightarrow \alpha + \beta + \gamma = 33 - 12 = 21$

Ques. In a class of 140 students numbered 1 to 140, all even numbered students opted Mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is

(JEE Main 2019 (Jan.))

(a) 102 (b) 42 (c) 1 (d) 38

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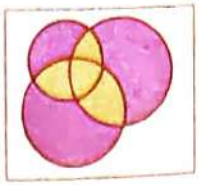
Ans. M → 2, 4, 6, 8, 10, ... 140 $n(M) = 70$

P → 3, 6, 9, 12, ... 138 $n(P) = 46$

C → 5, 10, 15, 20, ... 140 $n(C) = 28$



$(M \cap C)$: 10, 20, 30, ... 140, $n(M \cap C) = 14$

$(M \cap P)$: 6, 12, 18, 24, ... 138, $n(M \cap P) = 23$

$(P \cap C)$: 15, 30, 45, ... 135, $n(P \cap C) = 9$

$(P \cap C \cap M)$: {30, 60, 90, 120}, $n(P \cap C \cap M) = 4$

$(P \cup C \cup M) = (70 + 46 + 28) - (14 + 23 + 9) + 4$

$= 70 + 46 + 28 - 46 + 4 = 70 + 32$

$n(P \cup C \cup M) = 102$

students did not opt any courses. $= 140 - 102 = 38$

MAX & MIN VALUE OF $A \cap B$

Ques. In a survey it was found that 60% Indians like mangoes whereas 70% Like apples. If x % of Indians like both Apples and Mangoes then

Ans. $n(M) = 60$, $n(A) = 70$

Let, $n(M \cap A) = x$

Now, $n(M \cup A) = n(M) + n(A) - n(M \cap A)$

$\Rightarrow n(M \cup A) = 60 + 70 - x$

$\Rightarrow n(M \cup A) = 130 - x$

$\Rightarrow x = 130 - n(M \cup A)$

For $x_{min} \Rightarrow n(MUA)_{max} = 100$

$\Rightarrow x_{min} = 130 - 100 = 30\%$

For $x_{max} \Rightarrow n(MUA)_{min} = 70$

$\Rightarrow x_{max} = 60$

Ques. A survey shows that 63% of the people in a city read newspaper A whereas 76% read newspaper B. If x % of the people read both the newspapers, then a possible value of x can be:

(Main Sep.04, 2020 (I))

(a) 29 (b) 37 (c) 65 (d) 55

Ans. $n(A) = 63\%$, $n(B) = 76\%$ → $x_{max} = 63$

Let, $x = n(A \cap B)$

$n(A \cup B) = 63 + 76 - x$

$\Rightarrow n(A \cup B) = 139 - x$

$\Rightarrow x = 139 - n(A \cup B) \Rightarrow x_{min} = 39$

$39 \le x \le 63$

Ques. Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If K % of them are suffering from both ailments, then K can not belong to the set

(26th Aug 1st Shift 2021)

(a) {79, 81, 83, 85}

(b) {84, 87, 90, 93}

(c) {80, 83, 86, 89}

(d) {84, 86, 88, 90}

Ans. $n(A) = 89$

$n(B) = 98$

$n(A \cap B) = k$

$n(A \cup B) = 89 + 98 - k$

$\Rightarrow k = 89 + 98 - n(A \cup B)$

$\Rightarrow k_{min} = 89 + 98 - 100$

$= 87$

$K_{max} = 89$

$87 \le k \le 89$

Ques. If A and B be two sets containing 3 and 6 elements respectively, what can be the maximum number of elements in $(A \cup B)$?

(a) 9 (b) 6

(c) 3 (d) Can't say

Ans. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$\Rightarrow \underbrace{n(A \cup B)}_{max} = 3 + 6 - \underbrace{n(A \cap B)}_{min}$

$\Rightarrow (A \cup B)_{max} = 9$

LAWS OF ALGEBRA OF 3 SETS

Distributive Law

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

i.e. union is distributive over intersection.

(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

i.e Intersection is distributive over Union.

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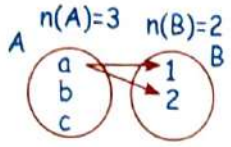
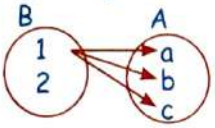
CARTESIAN PRODUCT OF 2 SETS

Example: (i)



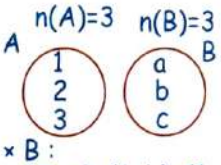
$A \times B$: {(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)}

$n(A \times B) = 6$



$B \times A$: {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)}

Example: (ii)



$A \times B$: {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c) (3, a), (3, b), (3, c)}

$n(A \times B) = 9$

$n(A \times B) = n(A).\ n(B)$

Note: If $n(A) = x$, $n(B) = y$ then total no. of subsets of $(A \times B)$ will be 2^{xy}

Property-1

In general $A \times B \neq B \times A$

but $n(A \times B) = n(B \times A)$

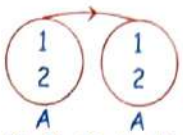
Property-2

$n(A \times B) = n(A).n(B)$

Property-3

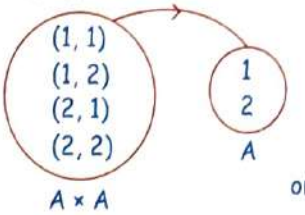
$A \times A \times A = \{(a, b, c) : a, b, c, \in A\}$.
Here (a, b, c) is called an ordered triplet.

$A = \{1, 2\}$



$A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$
ordered pair

$A \times A \times A =$



ordered pair

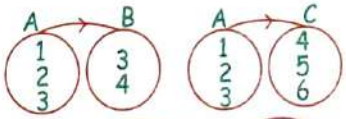
$A \times A \times A = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2) (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)\}$

Important Note

$n(A \times B \times C) = n(A).\ n(B)\ n(C)$

Ques. Let $A = \{1, 2, 3\}$, $B = (3, 4)$ and $C = \{4, 5, 6\}$. Find the set $(A \times B) \cap (A \times C)$

Ans.



$A \times B$: {(1, 3), (1, 4), (2, 3), (2, 4) (3, 3), (3, 4)}

$A \times C$: {(1, 4), (1, 5), (1, 6), (2, 4), (2, 5), (2, 6), (3, 4) (3, 5), (3, 6)}

$\therefore (A \times B) \cap (A \times C) = \{(1, 4), (2, 4), (3, 4)\}$

Ques. If $A = \{x : x^2 - 5x + 6 = 0\}$, $B = \{2, 4\}$, $C = \{4, 5\}$ then $A \times (B \cap C)$ is

(a) {(2, 4), (3, 4)}

(b) {(4, 2), (4, 3)}

(c) {(2, 4), (3, 4), (4, 4)}

(d) {(2, 2), (3, 3), (4, 4), (5, 5)}

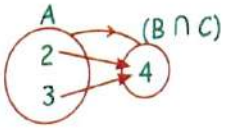
Ans.

$x^2 - 5x + 6 = 0 \Rightarrow 2, 3$

$A = \{2, 3\}$

$B = \{2, 4\}$

$C = \{4, 5\}$



$\Rightarrow B \cap C = \{4\}$

$\therefore A \times (B \cap C) = \{(2, 4), (3, 4)\}$

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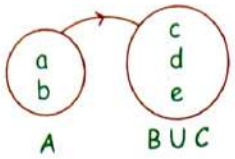
Ques. If A = {a, b}, B = {c, d}, C = {d, e} then {(a, c), (a, d), (a, e), (b, c), (b, d), (b, e)} is equal to

(a) $A \cap (B \cup C)$

(b) $A \cup (B \cap C)$

(c) $A \times (B \cup C)$

(d) $A \times (B \cap C)$



Ans. $\therefore A \times (B \cup C) = \{(a, c), (a, d), (a, e), (b, c), (b, d), (b, e)\}$

Ques. Let Z be the set of integers.
If $A = \{x \in Z: 2^{(x+2)(x^2-5x+6)} = 1\}$ and $B = \{x \in Z: -3 < 2x - 1 < 9\}$, then the number of subsets of the set $A \times B$, is:

(a) 2^{18} (b) 2^{15} (c) 2^{12} (d) 2^{10}

Ans. $2^{(x+2)(x^2-5x+6)} = 2^0$

$(x + 2)(x - 2)(x - 3) = 0$

$\Rightarrow x = -2, 2, 3$

$A = \{-2, 2, 3\}$

$n(A) = 3$

$n(A \times B) = 15$

Subsets = 2^{15}

B :

$-3 < 2x - 1 < 9$

$-2 < 2x < 10$

$-1 < x < 5$

$B = \{0, 1, 2, 3, 4\}$

$n(B) = 5$

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Short Notes

DEFINITION OF SET

A set is a well defined collection of objects which are distinct from each other

If a is an element of a set A, then we write $a \in A$ and say a belongs to A.

REPRESENTATION OF SETS

Roster Form

In this form a set is described by listing elements, separated by commas and enclose them by curly brackets.

Example- The set of vowels of English Alphabet may be described as

$A = \{a,e,i,o,u\}$

Set Builder Form

In this, we write down a property or rule Which gives us all the element of the set.

Example- $A = \{x : x \in N \text{ and } x = 2n \text{ for } n \in N\}$

TYPES OF SETS

Null set or Empty set

A set having no element in it is called an Empty set or a null set or void set it is denoted by ϕ or $\{\}$

Singleton

A set consisting of a single element is called a singleton set.

e.g. Then set {0}, is a singleton set.

Finite Set

A set which has only finite number of elements is called a finite set.

e.g. A = {a, b, c}

Order of a finite set

The number of elements in a finite set is called the order of the set A and is denoted O(A) or n(A). It is also called cardinal number of the set.

e.g. $A = \{a,b,c,d\} \Rightarrow n(A) = 4$

Infinite set

A set which has an infinite number of elements is called an infinite set.

e.g. $A = \{1,2,3,4,....\}$ is an infinite set.

Equal sets

Two sets A and B are said to be equal if every element of A is a member of B, and every element of B is a member of A.

If sets A and B are equal. We write A = B and A and B are not equal then $A \neq B$.

e.g. $A = \{1,2,6,7\}$ and $B = \{6,1,2,2,7\} \Rightarrow A = B$

Equivalent sets

Two finite sets A and B are equivalent if their number of elements are same ie. n(A) = n(B).

e.g. $A = \{1,3,5,7\}$, $B = \{a,b,c,d\}$

Equal sets are always equivalent but 2 equivalent sets may not always be equal.

Subsets

Let A and B be two sets if every element of A is an element B, then A is called a subset of B. If A is a subset of B. we write $A \subseteq B$.

Example: $A = \{1,2,3,4\}$ and $B = \{1,2,3,4,5,6,7\} \Rightarrow A \subseteq B$

Proper subset

If A is a subset of B and $A \neq B$ then A is a proper subset of B. and we write $A \subset B$

1. Every set is a subset of itself i.e. $A \subseteq A$ for all A.

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2. Empty set ϕ is a subset of every set.

3. The total number of subsets of a finite set containing n elements is 2^n

Universal set

A set consisting of all possible elements which occur in the discussion is called a Universal set and is denoted by U.

Example- If $A=\{1,2,3\}, B=\{2,4,5,6\}, C=\{1,3,5,7\}$ then $U=\{1,2,3,4,5,6,7\}$ can be taken as the Universal set.

Power set

Let A be any set. The set of all subsets of A is called power set of A and is denoted by P(A).

OPERATION ON SETS

Union of two sets

$A\cup B=\{x: x\in A \text{ or } x\in B\}$

e.g. $A=\{1,2,3\}$, $B=\{2,3,4\}$ then $A\cup B=\{1,2,3,4\}$

Intersection of two sets

$A\cap B=\{x: x\in A \text{ and } x\in B\}$

e.g. $A=\{1,2,3\}, B=\{2,3,4\}$ then $A\cap B=\{2,3\}$

Difference of two sets

$A-B=\{x: x\in A \text{ and } x\notin B\}$

e.g. $A=\{1,2,3\}, B=\{2,3,4\}; A-B=\{1\}$

Complement of a set

$A'=\{x: x\notin A \text{ but } x\in U\}=U-A$

e.g. $U=\{1,2,....10\}$, $A=\{1,2,3,4,5\}$

then $A'=\{6,7,8,9,10\}$

De-Morgan Law

$(A\cup B)'=A'\cap B'; (A\cap B)'=A'\cup B'$

Distributive Law

$A\cup(B\cap C)=(A\cup B)\cap(A\cup C)$

$A\cap(B\cup C)=(A\cap B)\cup(A\cap C)$

Commutative Law

$A\cup B=B\cup A; A\cap B=B\cap A$

Associative Law

$(A\cup B)\cup C=A\cup(B\cup C)$

$(A\cap B)\cap C=A\cap(B\cap C)$

$A\cap\phi=\phi; A\cap U=A \;\&\; A\cup\phi=A; A\cup U=U$

CARDINAL NUMBER

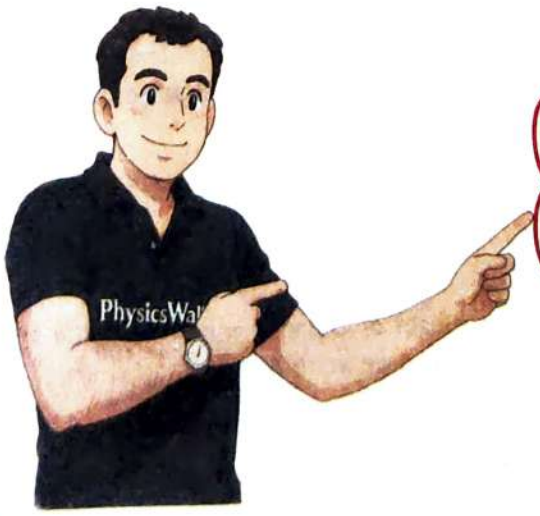
For 2 Sets

$n(A\cup B)=n(A)+n(B)-n(A\cap B)$

For 3 Sets

$n(A\cup B\cup C)=n(A)+n(B)+n(C)-n(A\cap B$

$n(B\cap C)-n(A\cap C)+n(A\cap B\cap C)$



"In mathematics, every relation builds a connection — just like every concept links to the next."

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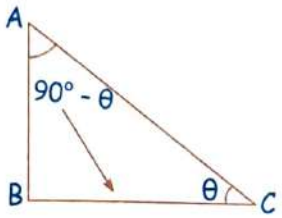
Chapter 3

Trigonometric Identities

TRIGONOMETRIC RATIOS OF COMPLEMENTARY ANGLES

1. $\sin(90° - \theta) = \cos\theta$
2. $\cos(90° - \theta) = \sin\theta$
3. $\tan(90° - \theta) = \cot\theta$
4. $\cot(90° - \theta) = \tan\theta$
5. $\sec(90° - \theta) = \operatorname{cosec}\theta$
6. $\operatorname{cosec}(90° - \theta) = \sec\theta$

Proof:



$$\sin\theta = \frac{AB}{AC}, \quad \text{...(i)}$$

$$\cos(90° - \theta) = \frac{AB}{AC} \quad \text{...(ii)}$$

By eqⁿ (i) and (iii):

$\sin\theta = \cos(90° - \theta)$

Ques. $\dfrac{2[\cos^2 20° + \cos^2 70°]}{[\sin^2 25° + \sin^2 65°]} - \tan 45° + \tan 13° \tan 23°$

$\tan 30° \tan 67° \tan 77° =$

Ans. $\dfrac{2[\cos^2 20° + \cos^2 70°]}{[\sin^2 25° + \sin^2 65°]} - \tan 45° + \tan 13° \tan 23°$

$\tan 30° \tan 67° \tan 77°$

$$= \frac{2[\cos^2 20° + \sin^2 20°]}{[\sin^2 25° + \cos^2 25°]} - 1 + \tan 13° \times \tan 23° \times \frac{1}{\sqrt{3}} \times \cot 13° \times \cot 23°$$

$$= 2 - 1 + \frac{1}{\sqrt{3}} = 1 + \frac{1}{\sqrt{3}}$$

Ques. $\left(\dfrac{\tan 20°}{\operatorname{cosec} 70°}\right)^2 + \left(\dfrac{\cot 20°}{\sec 70°}\right)^2 + 2\tan 15° \tan 37°$

$\tan 53° \tan 60° \tan 75° =$

Ans. $\left(\dfrac{\tan 20°}{\sec 20°}\right)^2 + \left(\dfrac{\cot 20°}{\operatorname{cosec} 20°}\right)^2 + 2\tan 15° \tan 37°$

$\cot 37° \times \sqrt{3} \times \cot 15°$

$$= \left(\frac{\sin 20°}{\cos 20° \sec 20°}\right)^2 + \left(\frac{\cos 20°}{\sin 20° \operatorname{cosec} 20°}\right)^2 + 2\sqrt{3}$$

$$= \sin^2 20° + \cos^2 20° + 2\sqrt{3} = 1 + 2\sqrt{3}$$

Ques. $\tan 1° \tan 2° \tan 3° \ldots\ldots \tan 89° =$

Ans. $\tan 1° \tan 2° \tan 3° \tan 4° \ldots\ldots \tan 88° \tan 89°$

$\because \tan\theta = \cot(90° - \theta)$

$= \tan 1° \tan 2° \tan 3° \tan 4° \ldots \tan 45° \ldots \cot 3° \cot 2° \cot 1° = \tan 45° = 1$

Ques. $\sin^2 5° + \sin^2 10° + \ldots + \sin^2 85° + \sin^2 90° = 9\dfrac{1}{2}$

Ans. $\sin^2 5° + \sin^2 10° + \sin^2 15° + \ldots\ldots \sin^2 40° + \sin^2 45° + \sin^2 50° + \ldots\ldots \sin^2 85° + \sin^2 90°$

$[\because \sin^2\theta = \cos^2(90° - \theta)]$

$= \sin^2 5° + \sin^2 10° + \ldots\ldots + \sin^2 40° + \sin^2 45° + \cos^2 40° + \ldots\ldots + \cos^2 10° + \cos^2 5° + \sin^2 90°$

$= 8 + \sin^2 45° + \sin^2 90°$

$$= 8 + \left(\frac{1}{\sqrt{2}}\right)^2 + (1)^2 = 9 + \frac{1}{2} = 9.5$$

Ques. If θ is an acute angle such that $5\cot\theta = 3$, then find the value of $\dfrac{5\sin\theta - 3\cos\theta}{(4\sin\theta + 3\cos\theta)}$

(a) $\dfrac{11}{18}$ (b) ✓ $\dfrac{16}{29}$

(c) $\dfrac{14}{27}$. (d) None of these

Ans. $\dfrac{5\sin\theta - 3\cos\theta}{(4\sin\theta + 3\cos\theta)}$

divide Numerator & Denominator with $\sin\theta$

$$= \frac{5 - \dfrac{3\cos\theta}{\sin\theta}}{4 + \dfrac{3\cos\theta}{\sin\theta}} = \frac{5 - 3 \times \dfrac{3}{5}}{4 + 3 \times \dfrac{3}{5}}$$

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$$= \frac{5 - \frac{9}{5}}{4 + \frac{9}{5}} = \frac{25 - 9}{20 + 9} = \frac{16}{29}$$

Ques. $(\text{cosec}^2 72° - \tan^2 18°) =$

(a) 0 (b) 1

(c) $\frac{3}{2}$ (d) None of these

Ans. $(\text{cosec}^2 72° - \tan^2 18°) = \sec^2 18° - \tan^2 18° = 1$

Ques. Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ for $k = 1, 2, 3 \ldots$.

Then for all $x \in R$, the value of $f_4(x) - f_6(x)$ is equal to:

(a) $\frac{1}{12}$ (b) $\frac{1}{4}$

(c) $\frac{-1}{12}$ (d) $\frac{5}{12}$

Ans. $f_4(x) = \frac{1}{4}[\sin^4 x + \cos^4 x] = \frac{1}{4}[1 - 2\sin^2 x \cos^2 x]$

$= \frac{1}{4} - \frac{1}{2}\sin^2 x \cos^2 x$

$f_6(x) = \frac{1}{6}[\sin^6 x + \cos^6 x] = \frac{1}{6}[1 - 3\sin^2 x \cos^2 x]$

$= \frac{1}{6} - \frac{1}{2}\sin^2 x \cos^2 x$

$f_4(x) - f_6(x) = \left[\frac{1}{4} - \frac{1}{2}\sin^2 x \cos^2 x\right]$

$-\left[\frac{1}{6} - \frac{1}{2}\sin^2 x \cos^2 x\right]$

$= \frac{1}{4} - \frac{1}{6} = \frac{3 - 2}{12} = \frac{1}{12}$

Ques. If $\sec\theta + \tan\theta = p$, then $\cos\theta =$

(a) $\frac{2p}{1 - p^2}$ (b) $\frac{2p}{1 + p^2}$

(c) $\frac{2p}{p^2 - 1}$ (d) $\frac{p}{p^2 + 1}$

Ans. If $\sec\theta + \tan\theta = p$ ...(i)

$\sec\theta - \tan\theta = \frac{1}{p}$ ...(ii)

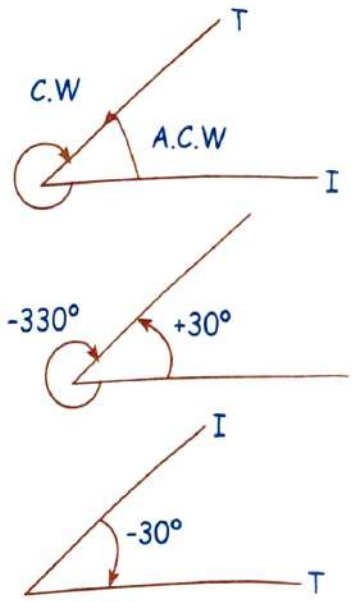
Eq^n. (i) + (ii)

$2\sec\theta = p + \frac{1}{p} \Rightarrow 2\sec\theta = \frac{p^2 + 1}{p}$

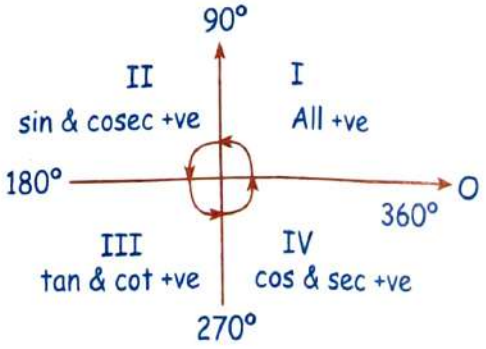
$\Rightarrow \sec\theta = \frac{p^2 + 1}{2p} \Rightarrow \cos\theta = \frac{2p}{p^2 + 1}$

SIGN CONVENTION IN TRIGONOMETRY

1. In trigonometry angles are always measure from initial side to terminal side
2. If rotation is A.C.W (Anti Clock were) then angl is +ve
   If rotation is C.W (clock wise) then angle is -v

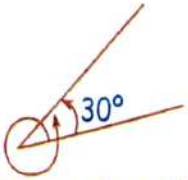


ASTC RULE



EX:

1. $\cos 190° < 0$
2. $\tan 95° < 0$
3. $\cot 195° > 0$
4. $\sec 330° > 0$
5. $\text{cosec}\ 330° < 0$
6. $\sin 170° > 0$
7. $\sin(800°) = \sin 80° > 0$
8. $\cos(850°) = \cos 130° < 0$
9. $\tan 2000° = \tan 200° > 0$



$30° = 360° + 30° = 390°$

$400° = 400° - 360° = 40°$

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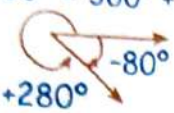
$30° = 360° + 30° = 390°$

$400° = 400° - 360° = 40°$

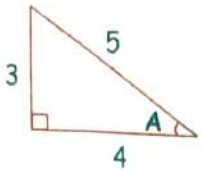


$-80° = +280°$

Ques. If $\sin A = 3/5$ and $A \in (90°, 180°)$ then find the value of

(a) $\tan A$

Ans. $\tan A = -\frac{3}{4}$



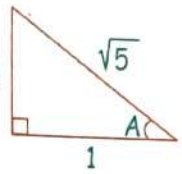
(b) $\cos A$

Ans. $\cos A = -\frac{4}{5}$

Ques. If $\tan A = 2$ and $A \in (180°, 270°)$ then find the value of $\sin A$ and $\cos A$.

Ans. $\tan A = \frac{2}{1}$, $A \in (180°, 270°)$

$\text{Sin } A = -\frac{2}{\sqrt{5}}$, $\cos A = -\frac{1}{\sqrt{5}}$



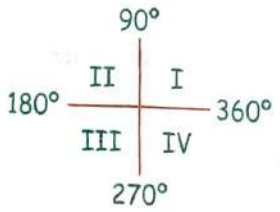
Ques. Find the Quadrant in which the following angles lie:

(a) 280°

(b) 380°

(c) 580°

(d) -300°

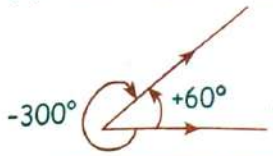


Ans. (a) $280° \rightarrow IV^{th}$ Quad

(b) $380° \rightarrow 380° - 360° = 20° = I^{st}$ Quad

(c) $580° \rightarrow 580° - 360° = 220° = III^{rd}$ Quad

(d) $-300° + 360° \rightarrow 60° \rightarrow I^{st}$ Quad



$-300° = +60°$

RADIAN MEASURE OF ANGLE

π Radians $= 180°$

$1 \text{ Radians} = \left(\frac{180}{\pi}\right)°$

$180° \rightarrow \pi$ Radians

$1° = \left(\frac{\pi}{180}\right)$ Radians

Ex: $30° \rightarrow \frac{\pi}{180} \cdot 30 = \frac{\pi}{6}$

Degree	Radians
0°	0
180°	π
90°	$\frac{\pi}{2}$
45°	$\frac{\pi}{4}$
30°	$\frac{\pi}{6}$
60°	$\frac{\pi}{3}$
360°	2π

Ques. $23\pi^c/4$ is equal to-

(a) 675° (b) 1080°

(c) 745° (d) 1035°

Ans. $\frac{23\pi^c}{4} = \frac{23}{4} \times 180°$

$= 23° \times 45° = 1035°$

Ques. Convert 10°30' into radians.

(a) $\frac{5\pi}{12}$ (b) $\frac{7\pi}{120}$

(c) $\frac{11\pi}{12}$ (d) $\frac{\pi}{12}$

Ans. $10° 30' = (10.5)°$

$\because 180° \rightarrow \pi$ or $1° \rightarrow \frac{\pi}{180}$

$1° \rightarrow 60'$
$0.5° \rightarrow 30'$
$1' \rightarrow 60''$

$\therefore 10° 30' = (10.5)° = \frac{\pi}{180} \times 10.5 = \frac{\pi}{180} \times \frac{105}{10} = \frac{7\pi}{120}$

Ques. 45° 15'30" change into degree

(a) $\left(\frac{5431}{120}\right)°$ (b) $\left(\frac{5400}{120}\right)°$

(c) 45° (d) $\left(\frac{543}{120}\right)°$

Ans. $15' 30'' = (15.5)'$

$\because 60' \rightarrow 1°$ or $1' \rightarrow \left(\frac{1}{60}\right)°$

$\therefore (15.5)' = \left(\frac{15.5}{60}\right)° = \left(\frac{155}{10 \times 60}\right)° = \left(\frac{31}{120}\right)°$

Hence, $45° 15'30'' = \left(45 + \frac{31}{120}\right)° = \left(\frac{5431}{120}\right)°$

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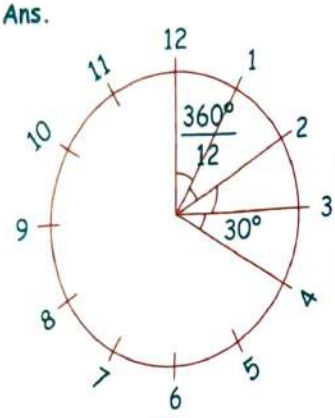
Brain Teaser

Assuming Hour hand to be initial side, Find the angle between the Hour hand & Minute hand of a clock at 3.30 PM (in Radians)

(a) $\frac{\pi}{2}$ (b) $-\frac{\pi}{2}$

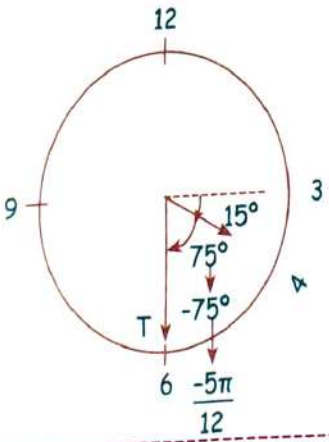
(c) $\frac{5\pi}{12}$ (d) $-\frac{5\pi}{12}$

Ans.



12 hr → 360°

1 hr → $\frac{360°}{12} = 30°$



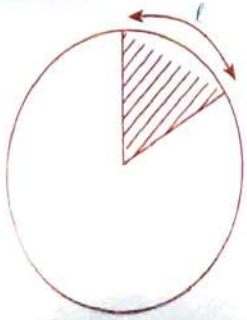
$180° \to \pi$

$1° \to \frac{\pi}{180}$

$75° \to \frac{\pi}{180} \times 75$

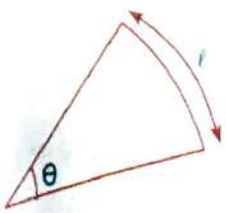
$= \frac{15\pi}{36} = \frac{5\pi}{12}$

AREA OF SECTOR & LENGTH OF ARC



Area sector = $\frac{1}{2} r(r\theta)$ (ℓ)

Area of sector = $\frac{1}{2} r\ell$



Area = $\frac{\theta \text{ in degree}}{360°} \pi r^2$

Area sector = $\frac{\theta}{2\pi} \cdot \pi r^2 = \frac{\theta}{2} r^2 = \frac{1}{2} r^2\theta$

Area sector = $\frac{1}{2} r^2\theta$ ($\theta \to$ radians)

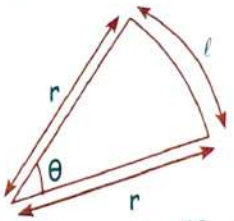
length of Arc.

$\ell = \frac{\theta}{360°} \cdot 2\pi r$ ($\theta \to$ degree)

$\ell = \frac{\theta}{2\pi} \cdot 2\pi r \Rightarrow \ell = r\theta$ ($\theta \to$ Radians)

Ques. If the perimeter of a certain sector is 10 units and radius of the circle is 3 units, find the area of the sector.

Ans.



$r = 3, r + r + l = 10 \Rightarrow 3 + 3 + l = 10$

$\Rightarrow l = 4$

$\Rightarrow l = 4 = r \times \theta$

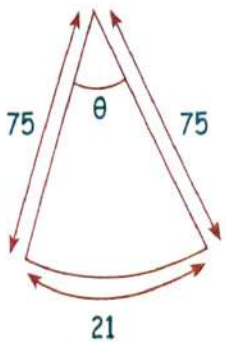
$\Rightarrow \theta = \frac{4}{r} = \frac{4}{3}$ Radians

Area of sector = $\frac{1}{2} r^2\theta = \frac{1}{2}(3)^2 \times \frac{4}{3} = \frac{12}{2} = 6$

Area = 6 (unit)^2

Ques. Find the angle in radian through which a pendulum swings if its length is 75 cm and the tip describes and arc of length 21 cm. use $\pi = 22/7$.

Ans.



$l = r\theta$

$\Rightarrow 21 = 75\theta \Rightarrow \theta = \frac{21}{75} = \frac{7}{25}$ Radians

Ques. A rail road curve is to be laid out on a circle. If the track is to change direction by 28° in a distance of 44 meters then find the radius of the curve. use $\pi = 22/7$.

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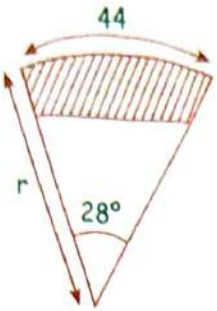


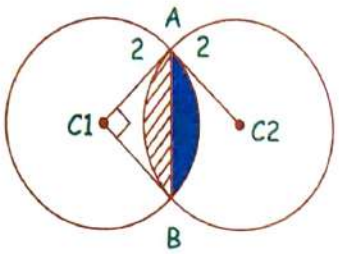
$\frac{\theta}{360^\circ} \cdot 2\pi r = 44$

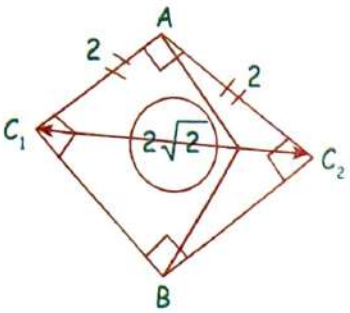
$\Rightarrow \frac{28^\circ}{360^\circ} \times 2 \times \frac{22}{7} r = 44$

$\Rightarrow \frac{4r}{360} = 1 \Rightarrow r = \frac{360}{4} \Rightarrow r = 90$ m

Ques. Two circles of radii 2 cm have their centers 2 $\sqrt{2}$ cm apart. Find the perimeter and area of the shaded region.

Ans.





$(C_1A)^2 + (C_2A)^2 = (C_1C_2)^2$

$\Rightarrow C_1AC_2B$ is a square

Blue area = Area of sector - ΔC_1AB

$= \frac{1}{2} r^2\theta - \frac{1}{2} \times 2 \times 2$

$= \frac{1}{2} (2)^2 \frac{\pi}{2} - 2 = (\pi - 2)$

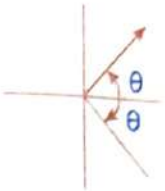
Area of shaped region = $2(\pi - 2)$ cm^2

perimeter = $2(r\theta)$

$= 2 \times 2 \times \frac{\pi}{2} = 2\pi$ cm

NEGATIVE ANGLE FORMULAS

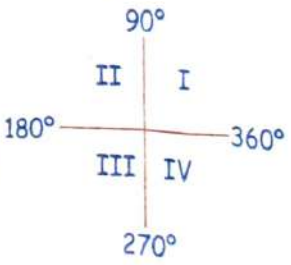
1. $\sin(-\theta) = -\sin\theta$
2. $\cos(-\theta) = \cos\theta$
3. $\tan(-\theta) = -\tan\theta$
4. $\cot(-\theta) = -\cot\theta$
5. $\sec(-\theta) = \sec\theta$
6. $\operatorname{cosec}(-\theta) = -\operatorname{cosec}\theta$



ALLIED ANGLE FORMULAS

If sum or difference of 2 angles is 90°, 180°, 270° or 360° such angles are called allied angles.

1. $\sin(180^\circ - \theta) = +\sin\theta$
2. $\cos(180^\circ + \theta) = -\cos\theta$
3. $\tan(270^\circ + \theta) = -\cot\theta$
4. $\sec(90^\circ + \theta) = -\operatorname{cosec}\theta$
5. $\tan(270^\circ - \theta) = +\cot\theta$
6. $\cos(360^\circ - \theta) = +\cos\theta$
7. $\sin(90^\circ + \theta) = +\cos\theta$



Steps to Derive Allied Angle formulas

Step-1 Assuming θ to be a Positive Acute angle find the Quadrant in which angle in LHS Lies.

Step-2 Find the sign of the trigonometric functions present in LHS in the quadrant found above

Step-3 If LHS has 180° or 360° then trigonometric functions remains same, if LHS has 90° or 270° then trigonometric function changes as sin ↔ cos, tan ↔ cot, sec ↔ cosec

Ques. The value of $\cos(270^\circ + \theta)\cos(90^\circ - \theta) - \sin(270^\circ - \theta)\cos\theta$ is

(a) 0 (b) -1

(c) 1/2 (d) 1

Ans. $\cos(270^\circ + \theta) = +\sin\theta$, $\sin(270^\circ - \theta) = -\cos\theta$

(IV) (3rd)

$\because \cos(90^\circ - \theta) = \sin\theta$

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$\therefore \cos(270° + \theta)\cos(90° - \theta) - \sin(270° - \theta)\cos\theta$
$= \sin^2\theta + \cos\theta \cdot \cos\theta = \sin^2\theta + \cos^2\theta = 1$

Ques. $\dfrac{\tan(90° - \theta)\sec(180° - \theta)\sin(-\theta)}{\sin(180° + \theta)\cot(360° + \theta)\operatorname{cosec}(90° - \theta)}$ is equals:

(a) 0

(c) −1

(b) 1

(d) 2

Ans. $\dfrac{\cot\theta \cdot (-\sec\theta) \cdot (-\sin\theta)}{(-\sin\theta)\cot\theta \cdot \sec\theta} = -1$

Ques. Simplify: $\dfrac{\sec\theta \quad \cot(90° + \theta) \to -\tan\theta}{\operatorname{cosec}(90° + \theta) + \cot(450° + \theta)} + \dfrac{\tan\theta \quad -\sec\theta}{\tan(180° + \theta) + \sec(180° - \theta)}$

$\sec\theta \leftarrow \operatorname{cosec}(90° - \theta) \quad \tan\theta \qquad \tan(360° + \theta) \quad -\sec(-\theta)$

$\operatorname{cosec}(450° - \theta) - \tan(180° + \theta)$ $\qquad \tan\theta \qquad \sec\theta$

Ans. $= \dfrac{\sec\theta - \tan\theta}{\sec\theta - \tan\theta} + \dfrac{\tan\theta - \sec\theta}{\tan\theta - \sec\theta} = 1 + 1 = 2$

TABLE FOR STANDARD ANGLES

θ	$0°$	$30°$	$45°$	$60°$	$90°$	$120°$	$135°$	$150°$	$180°$	$210°$	$225°$	$240°$	$270°$	$300°$	$315°$	$330°$	$360°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$+\frac{\sqrt{3}}{2}$	$\frac{+1}{\sqrt{2}}$	$\frac{+1}{2}$	0	$\frac{-1}{2}$	$\frac{-1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	−1	$-\frac{\sqrt{3}}{2}$	$\frac{-1}{\sqrt{2}}$	$\frac{-1}{2}$	0
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$\frac{-1}{2}$	$\frac{-1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	−1	$-\frac{\sqrt{3}}{2}$	$\frac{-1}{\sqrt{2}}$	$\frac{-1}{2}$	0	$\frac{+1}{2}$	$\frac{+1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	ND	$-\sqrt{3}$	−1	$\frac{-1}{\sqrt{3}}$	0	$\frac{+1}{\sqrt{3}}$	+1	$+\sqrt{3}$	ND	$-\sqrt{3}$	−1	$\frac{-1}{\sqrt{3}}$	0

Ques. $\cos 570° \sin 510° - \sin 330° \cos 390°$ equals:

(a) 0 (b) 1 (c) −1 (d) 2

Ans. $\cos 570° = \cos(570° - 360°) = \cos(210°)$

$= -\dfrac{\sqrt{3}}{2}$,

$\sin 510° = \sin(510° - 360°) = \sin 150° = \dfrac{1}{2}$

$\sin 330° = -\dfrac{1}{2}$

$\cos(390°) = \cos 30° = \dfrac{\sqrt{3}}{2}$

$\cos 570° \sin 510° - \sin 330° \cos 390°$

$= \left(-\dfrac{\sqrt{3}}{2}\right)\left(\dfrac{1}{2}\right) - \left(\dfrac{-1}{2}\right)\dfrac{\sqrt{3}}{2}$

$= -\dfrac{\sqrt{3}}{4} + \dfrac{\sqrt{3}}{4} = 0$

Ques. Find the value:

(a) $\cos\dfrac{19\pi}{3}$

Ans. $\cos\dfrac{19\pi}{3} = \cos\left(\dfrac{19\pi}{3} - 6\pi\right)$

$= \cos\left(\dfrac{19\pi - 18\pi}{3}\right)$

$= \cos\dfrac{\pi}{3} = \dfrac{1}{2}$

(b) $\tan\dfrac{-2023\pi}{4}$

Ans. $= \tan\left(\dfrac{-2023\pi}{4}\right) = -\tan\left(\dfrac{2023\pi}{4}\right)$

$= -\tan\left[\dfrac{2023\pi}{4} - 506\pi\right] = -\tan\left[\dfrac{2023\pi - 2024}{4}\right.$

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$= -\tan\left(\frac{-\pi}{4}\right) = \tan\left(\frac{\pi}{4}\right) = 1$

(c) $\cos\left(\frac{119\pi}{3}\right) = \left(\frac{119\pi}{3} - 40\pi\right)$

$= \cos\left(\frac{119\pi - 120\pi}{3}\right) = \cos\left(\frac{-\pi}{3}\right) = \cos\frac{\pi}{3} = \frac{1}{2}$

Important Concept

If = A + B = 180°

Then, cos A + cos B = 0

sin A = sin B

tan A + tan B = 0

Proof:

A + B = 180°

B = 180° - A

sin both sides

sin B = sin (180° - A)

sin B = sin A

cos both sidess

cos B = cos (180° - A)

cos B = -cos A

cos A + cos B = 0

tan both sides

tan B = tan(180° - A)

tan B = - tan A

tan A + tan B = 0

Ex: 20° & 160° ⇒ cos 20° + cos 160° = 0 and sin20° = sin160°

If = A + B = 90°

sin A = cos B

cos A = sin B

Ex → 10°, 80° ⇒ sin 10° = cos 80° and cos10° = sin80°

Ques. $\tan\frac{\pi}{11} + \tan\frac{2\pi}{11} + \tan\frac{4\pi}{11} + \tan\frac{7\pi}{11} + \tan\frac{9\pi}{11} + \tan\frac{10\pi}{11}$ equals:

(a) 0 (b) 1 (c) -1 (d) 2

Ans. $\frac{\pi}{11} + \frac{10\pi}{11} = \frac{11\pi}{11} = \pi, \frac{2\pi}{11} + \frac{9\pi}{11} = \pi, \frac{4\pi}{11} + \frac{7\pi}{11} = \pi$

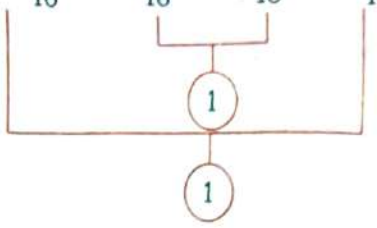
$\Rightarrow \tan\pi = 0$

Ques. $\cos^2\frac{\pi}{16} + \cos^2\frac{3\pi}{16} + \cos^2\frac{5\pi}{16} + \cos^2\frac{7\pi}{16}$

(a) 2 (b) 1

(c) 0 (d) None of these

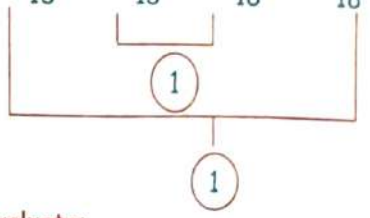
Ans. $\cos^2\frac{\pi}{16} + \cos^2\frac{3\pi}{16} + \cos^2\frac{5\pi}{16} + \cos^2\frac{7\pi}{16} = 1 + 1 = 2$



Ques. $\sin^2\frac{\pi}{18} + \sin^2\frac{\pi}{9} + \sin^2\frac{7\pi}{18} + \sin^2\frac{4\pi}{9}$ equal

(a) 0 (b) 1 (c) -1 (d) 2

Ans. $\sin^2\frac{\pi}{18} + \sin^2\frac{2\pi}{18} + \sin^2\frac{7\pi}{18} + \sin^2\frac{8\pi}{18} = 1 + 1 = 2$



Ques. Evaluate:

(a) $\frac{\sin\frac{10\pi}{17}\cos\frac{11\pi}{14}\tan\frac{\pi}{7}}{\cos\frac{3\pi}{14}\sin\frac{7\pi}{17}\tan\frac{6\pi}{7}}$ (-1, -1) = 1

Ans. $\frac{10\pi}{17} + \frac{7\pi}{17} = \frac{17\pi}{17} = \pi$

$\frac{11\pi}{14} + \frac{3\pi}{14} = \pi$

$\because \cos A + \cos B = 0$

or $\cos A = -\cos B$

$\Rightarrow \frac{\cos A}{\cos B} = -1$

(b) $\frac{\sin\frac{\pi}{2024}\cos\frac{17\pi}{90}\tan\frac{7\pi}{11}}{\cos\frac{73\pi}{90}\sin\frac{2023\pi}{2024}\tan\frac{4\pi}{11}}$ (-1) = 1

Ans. $\frac{\pi}{2024} + \frac{2023\pi}{2024} = \frac{2024\pi}{2024} = \pi$

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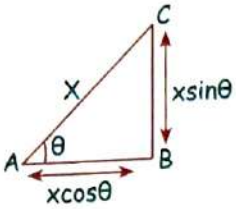
COMPOUND ANGLES

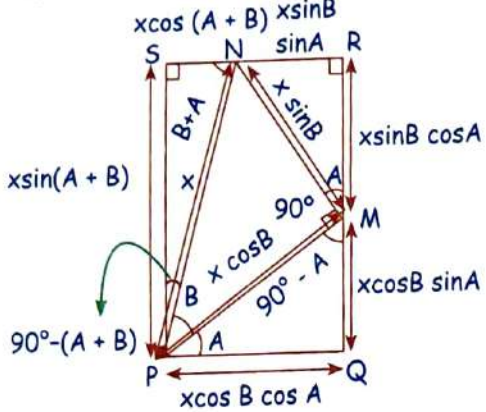
1. $\sin(A + B) = \sin A \cos B + \cos A \sin B$
2. $\sin(A - B) = \sin A \cos B - \cos A \sin B$
3. $\cos(A + B) = \cos A \cos B - \sin A \sin B$
4. $\cos(A - B) = \cos A \cos B + \sin A \sin B$

Proof:



$\cos\theta = \frac{AB}{x} \Rightarrow AB = x\cos\theta$

$\sin\theta = \frac{BC}{x} \Rightarrow BC = x\sin\theta$



PN = x, PM = xcosB

PQ = xcosB cosA, PS = xsin(A + B)

QR = xsinB cosA + xcosB sinA

Equate

$\because$ PS = QR

$\Rightarrow x\sin(A + B) = x\sin B\cos A + x\cos B\sin A$

$\Rightarrow \sin(A + B) = \sin B\cos A + \cos B\sin A$

and, PQ = SR

$\therefore x\cos(A + B) + x\sin B\sin A = x\cos B\cos A$

$\Rightarrow \cos(A + B) = \cos A\cos B - \sin A\sin B$

Important Values

1. $\sin(15°) = \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos(75°) = \cos\frac{5\pi}{12}$

Proof: $\sin 15° = \sin(60° - 45°)$

$= \sin 60°\cos 45° - \cos 60°\sin 45°$

$= \frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{2}} - \frac{1}{2}\cdot\frac{1}{\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}}$

2. $\sin(75°) = \sin\frac{5\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \cos(15°) = \cos\frac{\pi}{12}$

3. $\tan(15°) = \tan\frac{\pi}{12} = 2 - \sqrt{3} = \cot(75°) = \cot\frac{5\pi}{12}$

Proof: $\tan 15° = \frac{\sin 15°}{\cos 15°} = \frac{(\sqrt{3}-1)\times(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)}$

$= \frac{(\sqrt{3}-1)^2}{(\sqrt{3})^2-(1)^2} = \frac{(\sqrt{3}-1)^2}{2}$

$= \frac{3+1-2\sqrt{3}}{2} = \frac{4-2\sqrt{3}}{2} = \frac{2(2-\sqrt{3})}{2}$

4. $\tan(75°) = \tan\frac{5\pi}{12} = 2 + \sqrt{3} = \cot(15°) = \cot\frac{\pi}{12}$

Ques. If $\tan 15° + \frac{1}{\tan 75°} + \frac{1}{\tan 105°} + \tan 195° = 2a$, then the value of $\left(a + \frac{1}{a}\right)$ is:

(a) 4 (b) $4 - 2\sqrt{3}$

(c) 2 (d) $5 - \frac{3}{2}\sqrt{3}$

Ans. $\tan 15° + \frac{1}{\cot 15°} + \frac{1}{-\cot 15°} + \tan 15° = 2a$

$\Rightarrow 2\tan 15° = 2a \Rightarrow a = \tan 15°$

Hence, $a + \frac{1}{a} = \tan 15° + \frac{1}{\tan 15°}$

$= \tan 15° + \cot 15° = 2 - \sqrt{3} + 2 + \sqrt{3} = 4$

Ques. Prove that

$\underbrace{\cos(45° - A)}_{\phi}\ \underbrace{\cos(45° - B)}_{\theta} - \underbrace{\sin(45° - A)}_{\phi}\ \underbrace{\sin(45° - B)}_{\theta} = \sin(A+B)$

Ans. $\because \cos\theta\cos\phi - \sin\theta\cos\phi = \cos(\theta + \phi)$

LHS: $\cos(45° - A)\cos(45° - B) - \sin(45° - A)\sin(45° - B) = \cos[45° - A + 45° - B]$

$= \cos(90° - (A + B))$

$= \sin(A + B) = RHS$

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Ques. Prove that $1 + \tan A \tan\left(\frac{A}{2}\right) = \sec A$

Ans. LHS: $1 + \tan A \tan\left(\frac{A}{2}\right)$

$$= 1 + \frac{\sin A}{\cos A}\frac{\sin\frac{A}{2}}{\cos\frac{A}{2}}$$

$$= \frac{\cos A \cos\frac{A}{2} + \sin A \sin\frac{A}{2}}{\cos A \cos\frac{A}{2}}$$

$$= \frac{\cos(A - A/2)}{\cos A \cos A/2} = \frac{\cos A/2}{\cos A \cos A/2}$$

$$= \frac{1}{\cos A} = \sec A = \text{RHS, Hence Prove.}$$

Ques. Prove that $\tan A \cot\frac{A}{2} - 1 = \sec A$

Ans. LHS: $\frac{\sin A}{\cos A}\frac{\cos A/2}{\sin A/2} - 1$

$$= \frac{\sin A \cos A/2 - \cos A \sin A/2}{\cos A \sin A/2}$$

$$= \frac{\sin(A - A/2)}{\cos A \sin A/2} = \frac{\sin A/2}{\cos A \sin A/2}$$

$$= \frac{1}{\cos A} = \sec A = \text{RHS, Hence Prove.}$$

Ques. Prove that $\frac{\sin(A - B)}{\cos A \cos B} + \frac{\sin(B - C)}{\cos B \cos C} + \frac{\sin(C - A)}{\cos C \cos A} = 0$

Ans. Ist term $= \frac{\sin A \cos B - \cos A \sin B}{\cos A \cos B}$

$$= \frac{\sin A \cos B}{\cos A \cos B} - \frac{\cos A \cdot \sin A}{\cos A \cos B} = \tan A - \tan B$$

Similarly

2nd term = tan B − tan C

& 3rd term = tan C − tan A

Hence, $\frac{\sin(A - B)}{\cos A \cos B} + \frac{\sin(B - C)}{\cos B \cos C} + \frac{\sin(C - A)}{\cos C \cos A}$

$= \tan A - \tan B + \tan B - \tan C + \tan C - \tan A = 0$

5. $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

6. $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

7. $\cot(A + B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$

8. $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$

Ques. If $\tan A = \frac{n}{n+1}$ and $\tan B = \frac{1}{2n+1}$ find least positive value of $\tan(A + B)$.

(a) 30° (b) 60° (c) 90° (d) 45°

Ans. $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$= \frac{\frac{n}{n+1} + \frac{1}{2n+1}}{1 - \frac{n}{n+1}\cdot\frac{1}{2n+1}} = \frac{n(2n+1) + n + 1}{(n+1)(2n+1) - n}$$

$$= \frac{2n^2 + 2n + 1}{2n^2 + n + 2n + 1 - n} = \frac{2n^2 + 2n + 1}{2n^2 + 2n + 1} = 1$$

$\Rightarrow \tan(A + B) = 1 \Rightarrow A + B = 45°$

Ques. Prove that: $\tan 3A \cdot \tan 2A \cdot \tan A = \tan 3A - \tan 2A - \tan A$

Ans. Consider $3A = 2A + A$

tan both sides

$\Rightarrow \tan 3A = \tan(2A + A)$

$\Rightarrow \tan 3A = \frac{\tan 2A + \tan A}{1 - \tan 2A \tan A}$

$\Rightarrow \tan 3A (1 - \tan 2A \tan A) = \tan 2A + \tan A$

$\Rightarrow \tan 3A - \tan 3A \tan 2A \tan A = \tan 2A + \tan A$

$\Rightarrow \tan 3A \tan 2A \tan A = \tan 3A - \tan 2A - \tan A.$

Ques. Find the value of $\tan 125° + \tan 100° + \tan 100° \tan 125°$

Ans. Consider $125° + 100° = 225°$

$\Rightarrow \tan(125° + 100°) = \tan 225°$

$\Rightarrow \frac{\tan 125° + \tan 100°}{1 - \tan 125° \tan 100°} = 1$

$(\because \tan 225° = \tan(180° + 45°)$

$= \tan 45° = 1)$

$\Rightarrow \tan 125° + \tan 100° = 1 - \tan 125° \tan 100°$

$\Rightarrow \tan 125° + \tan 100° + \tan 125° \tan 100° = 1$

Ques. Prove that: $\tan 80° = \tan 10° + 2\tan 70°$

Ans. Consider $80 = 10° + 70°$

$\Rightarrow \tan 80° = \tan(10° + 70°)$

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$$\Rightarrow \tan 80^\circ = \frac{\tan 10^\circ + \tan 70^\circ}{1 - \tan 10^\circ \tan 70^\circ}$$

$$\Rightarrow \tan 80^\circ [1 - \tan 10^\circ \tan 70^\circ] = \tan 10^\circ + \tan 70^\circ$$

$$\Rightarrow \tan 80^\circ - \tan 80^\circ \tan 10^\circ \tan 70^\circ = \tan 10^\circ + \tan 70^\circ$$

$$\Rightarrow \tan 80^\circ - \frac{1}{\tan 10^\circ} \cdot \tan 10^\circ \cdot \tan 70^\circ = \tan 10^\circ + \tan 70^\circ$$

$$\left(\because \tan 80^\circ = \cot 10^\circ = \frac{1}{\tan 10^\circ}\right)$$

$$\Rightarrow \tan 80^\circ = 2\tan 70^\circ + \tan 10^\circ$$

DOUBLE ANGLE FORMULAS

(i) $\sin 2\theta = 2 \sin\theta \cos\theta = \dfrac{2\tan\theta}{1 + \tan^2\theta}$

(ii) $\cos 2\theta = \cos^2\theta - \sin^2\theta$

$= 2\cos^2\theta - 1$

$= 1 - 2\sin^2\theta$

$= \dfrac{1 - \tan^2\theta}{1 + \tan^2\theta}$

(iii) $\tan 2\theta = \dfrac{2\tan\theta}{1 - \tan^2\theta}$

Ques. $\cos A = \frac{1}{3}$, find $\cos 2A$

(a) $\frac{7}{9}$ (b) $\frac{-7}{9}$ ✓

(c) $\frac{1}{9}$ (d) None of these

Ans. $\cos 2A = 2\cos^2 A - 1$

$\Rightarrow \cos 2A = 2\left(\frac{1}{3}\right)^2 - 1 = \frac{2}{9} - 1 = \frac{2-9}{9} = \frac{-7}{9}$

Ques. $\sin A = \frac{2}{5}$, find $\cos 2A$

(a) $\frac{24}{25}$ (b) $\frac{17}{25}$ ✓

(c) $\frac{-17}{25}$ (d) None of these

Ans. $\cos 2A = 1 - 2\sin^2 A$

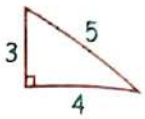
$= 1 - 2\left(\frac{2}{5}\right)^2 = 1 - \frac{2 \times 4}{25}$

$= 1 - \frac{8}{25} = \frac{17}{25}$

Ques. $\sin A = \frac{3}{5}$, find $\sin 2A$

(a) $\frac{24}{25}$ (b) $\frac{4}{25}$ (c) $\frac{-24}{25}$ (d) $\frac{3}{5}$

Ans. $\sin A = \frac{3}{5}$



$\cos A = \frac{4}{5}$

or

$\cos A = \frac{-4}{5}$

$\sin 2A = 2\sin A \cos A$

$= 2 \times \frac{3}{5} \times \cos A$

$= 2 \times \frac{3}{5} \times \frac{4}{5}$ or $2 \times \frac{3}{5} \times \left(\frac{-4}{5}\right) = \frac{24}{25}$ or $\frac{-24}{25}$

Ques. Exact value of $\tan 200^\circ (\cot 10^\circ - \tan 10^\circ)$ is

Ans. $\tan(200^\circ) = \tan(180^\circ + 20^\circ) = \tan 20^\circ$

$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta} \Rightarrow \tan 20^\circ = \frac{2\tan 10^\circ}{1 - \tan^2 10^\circ}$

Now, $\tan 200^\circ (\cot 10^\circ - \tan 10^\circ)$

$= \tan 20^\circ (\cot 10^\circ - \tan 10^\circ)$

$= \frac{2\tan 10^\circ}{1 - \tan^2 10^\circ} \cdot \left[\frac{1}{\tan 10^\circ} - \tan 10^\circ\right]$

$= \frac{2\tan 10^\circ}{1 - \tan^2 10^\circ} \frac{1 - \tan^2 10^\circ}{\tan 10^\circ} = 2$

Ques. The tangents of two acute angles are 3 and 2. The sine of twice their difference is

(a) 7/24 (b) 7/48

(c) 7/50 (d) 7/25 ✓

Ans. Given, $\tan A = 3$, $\tan B = 2$

$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{3 - 2}{1 + 3 \times 2} = \frac{1}{7}$

$\Rightarrow \tan(A - B) = \frac{1}{7}$

Now, $\sin 2(A - B) = \frac{2\tan(A - B)}{1 + \tan^2(A - B)}$

$= \frac{2 \times 1/7}{1 + (1/7)^2} = \frac{2/7}{1 + 1/49} = \frac{2/7}{50/49}$

$= \frac{2 \times 7}{50} = \frac{7}{25}$

Ques. If $\sin x + \cos x = \frac{4}{3}$ then find $\sin 2x$

(a) 7/9 ✓ (b) 7/16

(c) 16/9 (d) 7/25

Ans. $\sin x + \cos x = \frac{4}{3}$

$\Rightarrow (\sin x + \cos x)^2 = 16/3$

$\Rightarrow \sin^2 x + \cos^2 x + 2 \sin x \cos x = 16/9$

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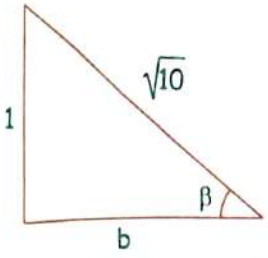
$\Rightarrow 1 + \sin 2x = 16/9$

$\Rightarrow \sin 2x = \frac{16}{9} - 1 = \frac{16-9}{9} = \frac{7}{9}$

Ques. $0 < \alpha, \beta < \frac{\pi}{2}$, $\tan\alpha = \frac{1}{7}$ and $\sin\beta = \frac{1}{\sqrt{10}}$. The value of $\alpha + 2\beta$ is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

Ans.



$b^2 = (\sqrt{10})^2 - 1^2$
$= 10 - 1 \Rightarrow b^2 = 9$
$\Rightarrow b = 3$

$\tan\alpha = \frac{1}{7}$ and $\tan\beta = \frac{1}{3}$

$\tan(\alpha + 2\beta) = \frac{\tan\alpha + \tan 2\beta}{1 - \tan\alpha \tan 2\beta}$

$\because \tan 2\beta = \frac{2\tan\beta}{1 - \tan^2\beta}$

$= \frac{2 \times 1/3}{1 - (1/3)^2} = \frac{2/3}{1 - 1/9}$

$= \frac{2/3}{8/9} = \frac{2 \times 3}{8} = \frac{3}{4}$

$\Rightarrow \tan(\alpha + 2\beta) = \frac{1/7 + 3/4}{1 - 1/7 \times 3/4}$

$= \frac{\frac{4+21}{28}}{\frac{28-3}{28}} = \frac{25}{25} = 1$

$\Rightarrow \tan(\alpha + 2\beta) = 1 \Rightarrow \alpha + 2\beta = 45°$

Brain Teaser

Simplify $\sqrt{3}$ cosec 20° - sec 20°

Ans. $\sqrt{3}\ \text{cosec}\ 20° - \sec 20° = \tan 60° \frac{1}{\sin 20°} - \frac{1}{\cos 20°}$

$= \frac{\sin 60°}{\cos 60°} \cdot \frac{1}{\sin 20°} - \frac{1}{\cos 20°}$

$= \frac{\sin 60° \cos 20° - \cos 60° \sin 20°}{\cos 60° \sin 20° \cos 20°}$ → $\sin(60° - 20°)$

$= \frac{\sin 40°}{\cos 60° \sin 20° \cos 20°}$

$= \frac{2 \sin 20° \cos 20°}{\cos 60° \sin 20° \cos 20°}$

$(\because \sin 40° = 2 \sin 20° \cos 20°)$

$= \frac{2}{\cos 60°} = \frac{2}{1/2} = 4$

Ques. The value of $\left(1 + \cos\frac{\pi}{8}\right)\left(1 + \cos\frac{3\pi}{8}\right)\left(1 + \cos\frac{5\pi}{8}\right)\left(1 + \cos\frac{7\pi}{8}\right)$ is

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{1}{8}$ (d) $\frac{3}{8}$

Ans. If $A + B = 180° \Rightarrow \cos B = -\cos A$

$\therefore \cos\frac{7\pi}{8} = -\cos\pi/8, \cos\frac{5\pi}{8} = -\cos 3\pi/8$

Now, $\left(1 + \cos\frac{\pi}{8}\right)\left(1 + \cos\frac{3\pi}{8}\right)\left(1 + \cos\frac{5\pi}{8}\right)\left(1 + \cos\frac{7\pi}{8}\right)$

$= \left(1 + \cos\frac{\pi}{8}\right)\left(1 + \cos\frac{3\pi}{8}\right)\left(1 - \cos\frac{3\pi}{8}\right)\left(1 - \cos\frac{\pi}{8}\right)$

$= (1^2 - \cos^2\pi/8)(1^2 - \cos^2 3\pi/8)$

$= \sin^2\pi/8\ \sin^2 3\pi/8$ $(\because 1 - \cos^2\theta = \sin^2\theta)$

$= \sin^2\pi/8\ \cos^2\pi/8$ $(\because \frac{\pi}{8} + \frac{3\pi}{8} = 90°)$

$= \left(\frac{2\sin\frac{\pi}{8}\cos\frac{\pi}{8}}{2}\right)^2$

$= \left[\frac{\sin(2\pi/8)}{2}\right]^2 = \left[\frac{\sin\pi/4}{2}\right]^2 = \left(\frac{1}{2\sqrt{2}}\right)^2 = \left(\frac{1}{8}\right)$

Brain Teaser

The difference $(\sin^8 75° - \cos^8 75°)$ is equal to

(a) 1 (b) $\frac{3\sqrt{3}}{8}$

(c) $\frac{3\sqrt{3}}{16}$ (d) $\frac{7\sqrt{3}}{16}$

Ans. $[\cos 15°]^8 - [\sin 15°]^8$

$= (\cos^4 15°)^2 - (\sin^4 15°)^2$

$= (\cos^4 15° - \sin^4 15°)(\cos^4 15° + \sin^4 15°)$

$= [(\cos^2 15°)^2 - (\sin^2 15°)^2][\cos^4 15° + \sin^4 15°]$

$= (\cos^2 15° - \sin^2 15°)(\cos^2 15° + \sin^2 15°)[\cos^4 15° + \sin^4 15°]$

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$= \cos(30°) \times [1 - 2\sin^2 15° \cos^2 15°]$

$(\because \cos^4\theta + \sin^4\theta = 1 - 2\sin^2\theta\cos^2\theta)$

$= \cos 30° \left[1 - \frac{4}{2}\sin^2 15° \cos^2 15°\right]$

$= \cos 30° \left[1 - \frac{(2\sin 15° \cos 15°)^2}{2}\right]$

$= \cos 30° \left[1 - \frac{(\sin 30°)^2}{2}\right]$

$= \frac{\sqrt{3}}{2}\left[1 - \frac{(1/2)^2}{2}\right] = \frac{\sqrt{3}}{2}\left[1 - \frac{1}{8}\right] = \frac{7\sqrt{3}}{16}$

TRIPLE ANGLE FORMULAS

(i) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

Proof: $\sin 3\theta = \sin(\theta + 2\theta)$

$= \sin\theta\cos 2\theta + \cos\theta \sin 2\theta$

$= \sin\theta(1 - 2\sin^2\theta) + \cos\theta\, 2\sin\theta\cos\theta$

$= \sin\theta(1 - 2\sin^2\theta) + 2\sin\theta(1 - \sin^2\theta)$

$= \sin\theta - 2\sin^3\theta + 2\sin\theta - 2\sin^3\theta$

$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

(ii) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

Proof: $\cos 3\theta = \cos(\theta + 2\theta)$

$= \cos\theta\cos 2\theta - \sin\theta\sin 2\theta$

$= \cos\theta(2\cos^2\theta - 1) - \sin\theta \cdot 2\sin\theta\cos\theta$

$= \cos\theta(2\cos^2\theta - 1) - 2\cos\theta(1 - \cos^2\theta)$

$= 2\cos^3\theta - \cos\theta - 2\cos\theta + 2\cos^3\theta$

$= 4\cos^3\theta - 3\cos\theta$

(iii) $\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

Ques. Prove that $\left(\frac{\sin 3\theta}{\sin\theta}\right)^2 - \left(\frac{\cos 3\theta}{\cos\theta}\right)^2 = 8\cos 2\theta$

Ans. LHS: $\left(\frac{3\sin\theta - 4\sin^3\theta}{\sin\theta}\right)^2 - \left(\frac{4\cos^3\theta - 3\cos\theta}{\cos\theta}\right)^2$

$= (3 - 4\sin^2\theta)^2 - (4\cos^2\theta - 3)^2$

$= (3 - 4\sin^2\theta + 4\cos^2\theta - 3)(3 - 4\sin^2\theta - 4\cos^2\theta + 3)$

$= 4[\cos^2\theta - \sin^2\theta][6 - 4(\sin^2\theta + \cos^2\theta]$

$= 4\cos 2\theta [6 - 4]$

$= 4\cos 2\theta \times 2 = 8\cos 2\theta = RHS$

Ques. $(4\cos^2 9° - 3)(4\cos^2 27° - 3) = \tan 9°$

Ans. LHS: $\left(\frac{4\cos^3 9° - 3\cos 9°}{\cos 9°}\right)\left(\frac{4\cos^3 27° - 3\cos 27°}{\cos 27°}\right)$

$= \frac{\cos(3 \times 9°)}{\cos 9°} \frac{\cos(27° \times 3)}{\cos 27°}$

$= \frac{\cos 27°}{\cos 9°} \cdot \frac{\cos 81°}{\cos 27°} = \frac{\sin 9°}{\cos 9°} = \tan 9° = RHS$

Ques. $\tan^6\frac{\pi}{9} - 33\tan^4\frac{\pi}{9} + 27\tan^2\frac{\pi}{9}$ is equal to?

Ans. $\frac{\pi}{9} = \frac{180°}{9} = 20°$

Let, $\theta = 20° = \frac{\pi}{9}$

$\Rightarrow 3\theta = 60° \Rightarrow \tan 3\theta = \tan 60°$

$\Rightarrow \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} = \sqrt{3}$

$\Rightarrow 3\tan\theta - \tan^3\theta = \sqrt{3}(1 - 3\tan^2\theta)$

$\Rightarrow (3\tan\theta - \tan^3\theta)^2 = 3(1 - 3\tan^2\theta)^2$

$\Rightarrow 9\tan^2\theta + \tan^6\theta - 6\tan^4\theta = 3[1 + 9\tan^4\theta - 6\tan^2\theta]$

$\Rightarrow 9\tan^2\theta + \tan^6\theta - 6\tan^4\theta = 3 + 27\tan^4\theta - 18\tan^2\theta$

$\Rightarrow \tan^6\theta - 33\tan^4\theta + 27\tan^2\theta = 3$

Ques. Prove that : $\tan 4x = \frac{4\tan x(1 - \tan^2 x)}{1 - 6\tan^2 x + \tan^4 x}$

Ans. $\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$

$\therefore$ L.H.S: $\tan 4x = \frac{2\tan 2x}{1 - \tan^2 2x} = \frac{2\tan 2x}{1 - (\tan 2x)^2}$

$= \frac{2 \cdot \frac{2\tan x}{1 - (\tan^2 x)^2}}{1 - \left(\frac{2\tan x}{1 - (\tan^2 x)^2}\right)^2} = \frac{\frac{4\tan x}{1 - \tan^2 x}}{\frac{(1 - \tan^2 x)^2 - 4\tan^2 x}{(1 - \tan^2 x)^2}}$

$= \frac{(4\tan x)(1 - \tan^2 x)}{1 + \tan^4 x - 2\tan^2 x - 4\tan^2 x} = RHS$

Ques. Prove that: $\cos 4x = 1 - 8\sin^2 x \cos^2 x$

Ans. $\because \cos 2\theta = 1 - 2\sin^2\theta$

$\therefore \cos 4x = 1 - 2\sin^2 2x$

$= 1 - 2(\sin 2x)^2 = 1 - 2(2\sin x \cos x)^2$

$= 1 - 2 \times 4\sin^2 x \cos^2 x = 1 - 8\sin^2 x \cos^2 x = RHS$

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Ques. If $\sin\theta + \cos\theta = \frac{1}{2}$ then $16(\sin(2\theta) + \cos(4\theta) + \sin(6\theta))$

(a) 23 (b) 27 (c) -23 (d) -27

Ans. $\sin\theta + \cos\theta = \frac{1}{2}$

$\Rightarrow (\sin\theta + \cos\theta)^2 = 1/4$

$\Rightarrow 1 + 2\sin\theta\cos\theta = 1/4$

$\Rightarrow 1 + \sin 2\theta = 1/4$

$\Rightarrow \sin 2\theta = \frac{-3}{4} = \frac{-12}{16}$

$\sin 6\theta = 3\sin 2\theta - 4\sin^3 2\theta$

$= 3(-3/4) - 4\left(\frac{-3}{4}\right)^3 = \frac{-9}{4} + \frac{27 \times 4}{4^3} = \frac{-9}{4} + \frac{27}{16}$

$= \frac{-36 + 27}{16} = \frac{-9}{16}$

$\cos 4\theta = 1 - 2\sin^2 2\theta$

$= 1 - 2\left(\frac{-3}{4}\right)^2 = 1 - 2 \times \frac{9}{16} = \frac{16 - 18}{16} = \frac{-2}{16}$

Hence, $16(\sin 2\theta + \cos 4\theta + \sin 6\theta)$

$= 16\left[\frac{-12}{16} - \frac{2}{16} - \frac{9}{16}\right]$

$= -12 - 2 - 9 = -12 - 11 = -23$

Ques. If $\sin\alpha - \cos\alpha = \frac{1}{5}$, then $\tan\left(\frac{\alpha}{2}\right) =$

(a) -3 or $\frac{1}{2}$ (b) 3 or $-\frac{1}{2}$

(c) -2 or $\frac{1}{3}$ (d) 2 or $-\frac{1}{3}$

Ans. $\because \sin 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$

$\therefore \sin\alpha = \frac{2\tan(\alpha/2)}{1 + \tan^2(\alpha/2)}$

Similarly, $\cos\alpha = \frac{1 - \tan^2\alpha/2}{1 + \tan^2\alpha/2}$

Given, $\sin\alpha - \cos\alpha = \frac{1}{5}$

$\Rightarrow \frac{2\tan\alpha/2}{1 + \tan^2\alpha/2} - \frac{(1 - \tan^2\alpha/2)}{1 + \tan^2\alpha/2} = \frac{1}{5}$

$\Rightarrow \frac{2\tan\alpha/2 - 1 + \tan^2\alpha/2}{1 + \tan^2\alpha/2} = \frac{1}{5}$

Let, $\tan\alpha/2 = x \Rightarrow \frac{2x - 1 + x^2}{1 + x^2} = \frac{1}{5}$

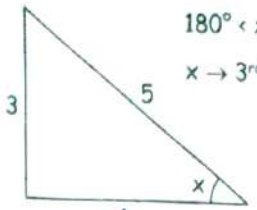
$\Rightarrow 10x - 5 + 5x^2 = 1 + x^2 \Rightarrow 4x^2 + 10x - 6 = 0$

$\Rightarrow 2x^2 + 5x - 3 = 0 \Rightarrow 2x^2 + 6x - x - 3 = 0$

$\Rightarrow (2x - 1)(x + 3) = 0 \Rightarrow x = 1/2$ or $x = -3$

Ques. If $\tan x = \frac{3}{4}, \pi < x < \frac{3\pi}{2}$, find the values of $\sin\frac{x}{2}, \cos\frac{x}{2}$ and $\tan\frac{x}{2}$.

Ans.



$180° < x < 270°$

$x \rightarrow 3^{rd}$ Quad

$\cos x = -4/5$

Consider $\cos 2\theta = 1 - 2\sin^2\theta$

$\Rightarrow \cos x = 1 - 2\sin^2 x/2$

$\Rightarrow -4/5 = 1 - 2\sin^2 x/2$

$\Rightarrow 2\sin^2 x/2 = 1 + 4/5$

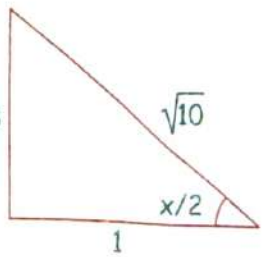
$\Rightarrow 2\sin^2 x/2 = 9/5 \Rightarrow \sin^2 x/2 = \frac{9}{10}$

$\frac{x}{2}$ in 2^{nd} Quad

$\sin\left(\frac{x}{2}\right) = +\sqrt{\frac{9}{10}} = \frac{3}{\sqrt{10}}$,

$\cos x/2 = -\frac{1}{\sqrt{10}}$

$\tan x/2 = -\frac{3}{1}$



Ques. If $\tan\frac{x}{2} = \frac{m}{n}$ find the value of $m\sin x + n\cos x$.

Ans. **Method - 1**

$\frac{m \cdot 2\tan x/2}{1 + \tan^2 x/2} + \frac{n \cdot (1 - \tan^2 x/2)}{1 + \tan^2 x/2}$

$= \frac{2m \cdot m/n}{1 + m^2/n^2} + \frac{n(1 - m^2/n^2)}{1 + m^2/n^2}$

$= \frac{2m^2 \cdot n}{m^2 + n^2} + \frac{n[n^2 - m^2]}{n^2 + m^2}$

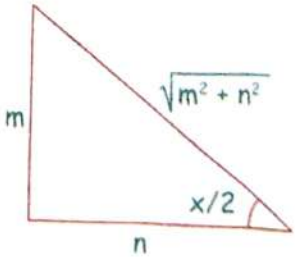
$= \frac{2m^2 n + n(n^2 - m^2)}{m^2 + n^2}$

$= n\frac{[2m^2 + n^2 - m^2]}{(m^2 + n^2)} = n\frac{[m^2 + n^2]}{(m^2 + n^2)} = n$

Method - 2

$\tan\left(\frac{x}{2}\right) = \frac{m}{n}$

$\frac{m}{\sqrt{m^2 + n^2}} = \sin\frac{x}{2}$



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$m = \sqrt{m^2 + n^2}\ \sin x/2$

$n = \sqrt{m^2 + n^2}\ \cos x/2$

Now, $m\sin x + n\cos x = \sqrt{m^2 + n^2}\ \sin x/2 \sin x + \sqrt{m^2 + n^2}\ \cos\frac{x}{2}\cos x$

$= \sqrt{m^2 + n^2}\ [\cos(x - x/2)] = \sqrt{m^2 + n^2}\ \cos\frac{x}{2} = n$

Ques. Find the value of $\sin 18°$

Ans.

If $\theta = 18°$

$\Rightarrow 5\theta = 90°$

$\Rightarrow 2\theta + 3\theta = 90°$

$\Rightarrow 2\theta = 90° - 3\theta$

Sin both sides

$\Rightarrow \sin 2\theta = \sin(90° - 3\theta)$

$\Rightarrow \sin 2\theta = \cos 3\theta$

$\Rightarrow 2\sin\theta\cos\theta = 4\cos^3\theta - 3\cos\theta$

$\Rightarrow 2\sin\theta = (4\cos^2\theta - 3)$

$\Rightarrow 2\sin\theta = 4(1 - \sin^2\theta) - 3$

let, $\sin\theta = t$

$\Rightarrow 2t = 4(1 - t^2) - 3$

$\Rightarrow 2t = 4 - 4t^2 - 3$

$\Rightarrow 2t = 1 - 4t^2$

$\Rightarrow 4t^2 + 2t - 1 = 0$

$\Rightarrow t = \frac{-2 \pm \sqrt{4 - 4(4)(-1)}}{2 \times 4}$

$\Rightarrow t = \frac{-2 \pm \sqrt{20}}{2 \times 4} = \frac{-2 \pm 2\sqrt{5}}{2 \times 4}$

$\Rightarrow t = 2\frac{[-1 \pm \sqrt{5}]}{2 \times 4}$

$\because\ t > 0$

$\therefore \sin 18° = \frac{-1 + \sqrt{5}}{4}$

$\Rightarrow \sin 18° = \frac{\sqrt{5} - 1}{4} = \cos 72°$

Ques. Find the value of $\cos 36°$

Ans. $\cos 2\theta = 1 - 2\sin^2\theta$

$\Rightarrow \cos 36° = 1 - 2\sin^2 18°$

$= 1 - 2\left(\frac{\sqrt{5} - 1}{4}\right)^2 = 1 - \frac{2}{16}(5 + 1 - 2\sqrt{5})$

$= 1 - \frac{1}{8}(6 - 2\sqrt{5}) = 1 - \frac{1}{4}(3 - \sqrt{5})$

$= \frac{4 - 3 + \sqrt{5}}{4} = \frac{1 + \sqrt{5}}{4}$

$\cos 36° = \frac{\sqrt{5} - 1}{4} = \sin 54°\ (\because \sin 54° = \cos 36°)$

Ques. Find the value of the expression

$\cot 9° + \cot 27° + \cot 63° + \cot 81°$

Ans. $\cot 9° + \cot 27° + \tan 27° + \tan 9°$

$= (\cot 9° + \tan 9°) + (\cot 27° + \tan 27°)$

$= \left(\frac{\cos 9°}{\sin 9°} + \frac{\sin 9°}{\cos 9°}\right) + \left(\frac{\cos 27°}{\sin 27°} + \frac{\sin 27°}{\cos 27°}\right)$

$= \left(\frac{2}{2\sin 9°\cos 9}\right) + \left(\frac{2}{2\sin 27°\cos 27}\right)$

$= \frac{2}{\sin 18°} + \frac{2}{\sin 54°}$

$= \frac{2 \times 4}{\sqrt{5} - 1} + \frac{2 \times 4}{\sqrt{5} + 1} = 8\left[\frac{\sqrt{5} + 1 + \sqrt{5} - 1}{(\sqrt{5} - 1)(\sqrt{5} + 1)}\right]$

$= \frac{8 \times 2\sqrt{5}}{5 - 1} = \frac{16\sqrt{5}}{4} = 4\sqrt{5}$

Ques. The value of

$36(4\cos^2 9° - 1)(4\cos^2 27° - 1)(4\cos^2 81° - (4\cos^2 243° - 1)$ is

(a) 54 (b) 18 (c) 27 (d) 36

Ans. $\because \cos^2 81° = \sin^2 9°$,

and $\cos^2 243° = \cos^2(270° - 27°) = \sin^2 27°$

$\therefore 36(4\cos^2 9° - 1)(4\cos^2 27° - 1)(4\cos^2 81° - (4\cos^2 243° - 1)$

$= 36(4\cos^2 9° - 1)(4\sin^2 9° - 1).$

$(4\cos^2 27° - 1)(4\sin^2 27° -$

$= 36[16\cos^2 9°\sin^2 9° - 4\sin^2 9° - 4\cos^2 9° + 1]$

$[16\cos^2 27°\sin^2 27° - 4\sin^2 27° - 4\cos^2 27° +$

$= 36[4(2\sin 9°\cos 9°)^2 - 4(1) + 1]$

$[4(2\sin 27°\cos 27°)^2 - 4(1) +$

$= 36[4(\sin 18°)^2 - 4 + 1][4(\sin 54°)^2 - 4 + 1]$

$= 36[4\sin^2 18° - 3][4\sin^2 54° - 3]$

$= 36\frac{(4\sin^3 18° - 3\sin 18°)}{\sin 18°}\left[\frac{4\sin^3 54° - 3\sin 54°}{\sin 54°}\right]$

$= 36\frac{(-\sin 54°)}{\sin 18°}\frac{(-\sin 54° \times 3)}{\sin 54°}$

$= \frac{36\sin 162°}{\sin 18°} = \frac{36\sin(180° - 18°)}{\sin 18} = \frac{36\sin 18°}{\sin 18°} = 3$

DE-FACTORIZATION FORMULAS

1. $\sin(A + B) = \sin A\cos B + \cos A\sin B$
2. $\sin(A - B) = \sin A\cos B - \cos A\sin B$
3. $\cos(A + B) = \cos A\cos B - \sin A\sin B$
4. $\cos(A - B) = \cos A\cos + B + \sin A\sin B$

(1) + (2)

5. $2\sin A\cos B = \sin(A + B) + \sin(A - B)$

(1) - (2)

6. $2\cos A\sin B = \sin(A + B) - \sin(A - B)$

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(3) + (4)

7. $2\cos A \cos B = \cos(A + B) + \cos(A - B)$

(3) - (4)

$-2\sin A \sin B = \cos(A + B) - \cos(A - B)$

8. $\boxed{2\sin A \sin B = \cos(A - B) - \cos(A + B)}$

Ques. Prove that

$$2\cos\frac{\pi}{13}\cos\frac{9\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13} = 0$$

Ans. LHS: $2\cos\frac{\pi}{13}\cos\frac{9\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13}$

$$= \cos\left[\frac{\pi}{13} + \frac{9\pi}{13}\right] + \cos\left[\frac{\pi}{13} - \frac{9\pi}{13}\right] + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13} \quad [\because 2\cos A\cos B = \cos(A+B) + \cos(A-B)]$$

$$= \cos\left(\frac{10\pi}{13}\right) + \cos\left(\frac{8\pi}{13}\right) + \cos\left(\frac{3\pi}{13}\right) + \cos\frac{5\pi}{13}$$

($\because$ If $A + B = \pi$ than $\cos A + \cos B = 0$)

$= 0 = R.H.S.$

Ques. $\sin(\beta - \gamma)\cos(\alpha - \delta) - \sin(\gamma - \delta)\cos(\beta - \delta) + \sin(\alpha - \beta)\cos(\delta - \beta) = 0$

Ans. LHS:

$$\frac{1}{2}\Big[\sin[\beta-\gamma+\alpha-\delta] + \sin[\beta-\gamma-\alpha+\delta] + \sin[\gamma-\alpha+\beta-\delta] + \sin[\alpha-\beta-\gamma+\delta] + \sin[\alpha-\beta+\gamma-\delta] + \sin[\alpha-\beta-\gamma+\delta]\Big]$$

$$= \frac{1}{2} \times 0 = RHS$$

FACTORIZATION FORMULAS

1. $\sin C + \sin D = 2\sin\left(\frac{C + D}{2}\right)\cos\left(\frac{C - D}{2}\right)$

2. $\sin C - \sin D = 2\cos\left(\frac{C + D}{2}\right)\sin\left(\frac{C - D}{2}\right)$

3. $\cos C + \cos D = 2\cos\left(\frac{C + D}{2}\right)\cos\left(\frac{C - D}{2}\right)$

4. $\cos D - \cos C = 2\sin\left(\frac{C + D}{2}\right)\sin\left(\frac{C - D}{2}\right)$

or $\cos C - \cos D = -2\sin\left(\frac{C + D}{2}\right)\sin\left(\frac{C - D}{2}\right)$

or $\cos C - \cos D = 2\sin\left(\frac{C + D}{2}\right)\sin\left(\frac{D - C}{2}\right)$

Ques. Prove that $\frac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} = \tan\theta$

Ans. LHS: $\frac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta}$

$$= \frac{2\cos\left(\frac{7\theta + 5\theta}{2}\right)\sin\left(\frac{7\theta - 5\theta}{2}\right)}{2\cos\left(\frac{7\theta + 5\theta}{2}\right)\cos\left(\frac{7\theta - 5\theta}{2}\right)}$$

$$= \frac{\cos 6\theta \sin\theta}{\cos 6\theta \cos\theta} = \tan\theta = R.H.S$$

Ques. Prove that $\frac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B} = \tan(A+B)\cot(A-B)$

Ans. L.H.S: $\frac{\sin 2A + \sin 2B}{\sin 2A - \sin 2B}$

$$= \frac{2\sin\left(\frac{2A + 2B}{2}\right)\cos\left(\frac{2A - 2B}{2}\right)}{2\cos\left(\frac{2A + 2B}{2}\right)\sin\left(\frac{2A - 2B}{2}\right)}$$

$$= \frac{\sin(A + B)\cos(A - B)}{\cos(A + B)\sin(A - B)} = \tan(A + B)\cot(A - B)$$

$= R.H.S$

Ques. Prove that $\cos 3A + \cos 5A + \cos 7A + \cos 15A = 4\cos 4A \cos 5A \cos 6A$

Ans. L.H.S : $\cos 3A + \cos 5A + \cos 7A + \cos 15A$

$$= 2\cos\left(\frac{3A + 5A}{2}\right)\cos\left(\frac{3A - 5A}{2}\right) + 2\cos\left(\frac{7A + 15A}{2}\right)\cos\left(\frac{7A - 15A}{2}\right)$$

$= 2\cos(4A)\cos(-A) + 2\cos(11A)\cos(-4A)$

$= 2\cos 4A \cos A + 2\cos 11A \cos 4A$

$= 2\cos 4A\,[\cos A + \cos 11A]$

$$= 2\cos 4A \cdot 2\cos\left(\frac{A + 11A}{2}\right)\cos\left(\frac{A - 11A}{2}\right)$$

$= 2\cos 4A \cdot 2\cos 6A \cos(-5A)$

$= 4\cos 4A \cos 6A \cos 5A$

Ques. Prove that $\sin 80° - \cos 70° = \cos 50°$

Ans. L.H.S: $\sin 80° - \sin 20°$

$$= 2\cos\left(\frac{80° + 20°}{2}\right)\sin\left(\frac{80° - 20°}{2}\right)$$

$$= 2\cos\left(\frac{100°}{2}\right)\sin\left(\frac{60°}{2}\right) = 2\cos 50° \sin 30°$$

$$= 2\cos 50° \times \frac{1}{2} = \cos 50° = R.H.S$$

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Ques. Prove that: $\sin51° + \cos81° = \cos21°$

Ans. L.H.S : $\cos39° + \cos81°$ $(\because \sin51° = \cos39°)$

$$= 2\cos\left(\frac{39° + 81°}{2}\right)\cos\left(\frac{39° - 81°}{2}\right)$$

$$= 2\cos\left(\frac{120°}{2}\right)\cos\left(\frac{42°}{2}\right) = 2\cos60° \cdot \cos21°$$

$$= 2 \times \frac{1}{2}\cos21° = \cos21° = \text{R.H.S}$$

Ques. Prove that $\sin\frac{\theta}{2}\sin\frac{7\theta}{2} + \sin\frac{3\theta}{2}\sin\frac{11\theta}{2}$

$= \sin 2\theta \sin 5\theta$

Ans. L.H.S: $\sin\frac{\theta}{2}\sin\frac{7\theta}{2} + \sin\frac{3\theta}{2}\sin\frac{11\theta}{2} =$

$$\frac{2\sin\theta/2 \sin 7\theta/2 + 2\sin 3\theta/2 \sin 11\theta/2}{2}$$

$$= \frac{1}{2}\left[\cos\left(\frac{\theta}{2} - \frac{7\theta}{2}\right) - \cos\left(\frac{\theta}{2} + \frac{7\theta}{2}\right) + \cos\left(\frac{3\theta}{2} - \frac{11\theta}{2}\right) - \cos\left(\frac{3\theta}{2} + \frac{11\theta}{2}\right)\right]$$

$$= \frac{1}{2}\left[\cos(-3\theta) - \cos4\theta + \cos(4\theta) - \cos7\theta\right]$$

$$= \frac{1}{2}\left[\cos 3\theta - \cos7\theta\right]$$

$$= \frac{1}{2}\left[2\sin\left(\frac{3\theta + 7\theta}{2}\right)\sin\left(\frac{7\theta - 3\theta}{2}\right)\right]$$

$= \sin 5\theta \cdot \sin2\theta = \text{R.H.S}$

Ques. Prove that $\frac{\sin A - \sin5A + \sin9A - \sin13A}{\cos A - \cos5A - \cos9A + \cos13A}$

$= \cot4A$

Ans. L.H.S : $\frac{\sin A - \sin5A + \sin9A - \sin13A}{\cos A - \cos5A - \cos9A + \cos13A}$

$$= \frac{2\cos\left(\frac{A + 5A}{2}\right)\sin\left(\frac{A - 5A}{2}\right) + 2\cos\left(\frac{9A + 13A}{2}\right)\sin\left(\frac{9A - 13A}{2}\right)}{2\sin\left(\frac{A + 5A}{2}\right)\sin\left(\frac{5A - A}{2}\right) - 2\sin\left(\frac{9A + 13A}{2}\right)\sin\left(\frac{13A - 9A}{2}\right)}$$

$$= \frac{\cos 3A \sin(-2A) + \cos(11A)\sin(-2A)}{\sin 3A \sin 2A - \sin(11A)\sin(2A)}$$

$$= \frac{-\sin 2A\,[\cos 3A + \cos 11A]}{\sin 2A\,[\sin 3A - \sin 11A]}$$

$$= \frac{-[\cos 3A + \cos 11A]}{\sin 3A - \sin 11A}$$

$$= \frac{-2\cos\left(\frac{3A + 11A}{2}\right)\cos\left(\frac{3A - 11A}{2}\right)}{2\cos\left(\frac{3A + 11A}{2}\right)\sin\left(\frac{3A - 11A}{2}\right)}$$

$$= \frac{-\cos 7A \cos 4A}{\cos 7A \sin(-4A)} = \frac{\cos 4A}{\sin 4A} = \cot 4A = \text{R.H.S}$$

Ques. $\frac{\cos 289° - \sin 289°}{\cos 289° + \sin 289°}$

(a) $\cot 26°$ (b) $\tan 26°$

(c) $-\tan 26°$ (d) $-\cot 26°$

Ans. $\cos289° = \cos(270° + 19°) = \sin19°$

$\sin289° = \sin(270° + 19°) = -\cos19°$

$$\therefore \frac{\cos 289° - \sin 289°}{\cos 289° + \sin 289°}$$

$$= \frac{\sin 19° + \cos 19°}{\sin 19° - \cos 19°} = \frac{\cos 71° + \cos 19°}{\cos 71° - \cos 19°}$$

$$= \frac{2\cos\left(\frac{71° + 19°}{2}\right)\cos\left(\frac{71° - 19°}{2}\right)}{2\sin\left(\frac{71° + 19°}{2}\right)\sin\left(\frac{19° - 71°}{2}\right)}$$

$$= \frac{\cos 45° \cdot \cos 26°}{\sin 45° \cdot \sin(-26°)} = \frac{\cos 26°}{-\sin 26°} = -\cot 26°$$

Brain Teaser

If $3\sin\alpha = 5\sin\beta \neq 0$, then the value of $\frac{\tan\left(\frac{\alpha + \beta}{2}\right)}{\tan\left(\frac{\alpha - \beta}{2}\right)}$ is:

Ans. Given, $3\sin\alpha = 5\sin\beta \Rightarrow \frac{\sin\alpha}{\sin\beta} = 5/3$

By componendo and dividendo

$$\Rightarrow \frac{\sin\alpha + \sin\beta}{\sin\alpha - \sin\beta} = \frac{5 + 3}{5 - 3}$$

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$$\Rightarrow \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}{2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)} = \frac{8}{2}$$

$$\Rightarrow \frac{\tan\left(\frac{\alpha+\beta}{2}\right)}{\tan\left(\frac{\alpha-\beta}{2}\right)} = 4$$

Ques. Let $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$ & $\tan\alpha = k\tan\beta$, then find the value of k. **[JEE Main 2024]**

(a) $-\frac{2}{3}$ (b) -5 (c) $\frac{2}{3}$ (d) 5

Ans. Given $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$

$$\Rightarrow \frac{\sin(\alpha+\beta)}{\sin(\alpha-\beta)} = 2/3$$

By componendo and dividendo

$$\Rightarrow \frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{\sin(\alpha+\beta)-\sin(\alpha-\beta)} = \frac{2+3}{2-3}$$

$$\Rightarrow \frac{2\sin\alpha\cos\beta}{2\cos\alpha\sin\beta} = \frac{5}{-1} \Rightarrow \frac{\tan\alpha}{\tan\beta} = -5$$

Ques. Find the value of the expression, $2\sin 12° - \sin 72°$ **[JEE Main 2022]**

Ans. $\sin12° + \sin12° - \sin72°$

$$\sin12° + 2\cos\left(\frac{12°+72°}{2}\right)\sin\left(\frac{12°-72°}{2}\right)$$

$= \sin12° + 2\cos42° \cdot \sin(-30°)$

$= \sin12° - 2\cos42° \sin 30°$

$= \sin12° - \cos42° = \sin12° - \sin48°$

$$= 2\cos\left(\frac{12°+48°}{2}\right)\sin\left(\frac{12°-48°}{2}\right)$$

$= 2\cos 30° \sin(-18°) = -2\cos 30° \sin 18°$

$$= -2 \times \frac{\sqrt{3}}{2}\left(\frac{\sqrt{5}-1}{4}\right) = -\sqrt{3}\left(\frac{\sqrt{5}-1}{4}\right)$$

Ques. Find the exact value of $\cos 24° - \cos 12° + \cos 48° - \cos 84°$.

Ans. $(\cos 24° - \cos 84°) + (\cos 48° - \cos 12°)$

$$= 2\sin\left(\frac{24°+84°}{2}\right)\sin\left(\frac{84°-24°}{2}\right) +$$

$$2\sin\left(\frac{48°+12°}{2}\right)\sin\left(\frac{12°-48°}{2}\right)$$

$= 2\sin 54° \cdot \sin 30° + 2\sin 30° \sin(-18°)$

$= 2\sin 30° [\sin 54° - \sin 18°]$

$$= 2 \times \frac{1}{2}\left[\frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4}\right] = \frac{2}{4} = \frac{1}{2}$$

Extra Formulas - Double Angles

(i) $\cos 2\theta = 2\cos^2\theta - 1 \Rightarrow 1 + \cos 2\theta = 2\cos^2\theta$

(ii) $\cos 2\theta = 1 - 2\sin^2\theta \Rightarrow 2\sin^2\theta = 1 - \cos 2\theta$

(iii) $\tan^2\theta = \frac{2\sin^2\theta}{2\cos^2\theta} \Rightarrow \tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$

(iv) $1 + \sin 2\theta = (\cos\theta + \sin\theta)^2$

(v) $1 - \sin 2\theta = (\cos\theta - \sin\theta)^2$

Ques. Prove that :

(i) $\frac{\sin 2A}{1-\cos 2A} = \cot A$ (ii) $\frac{\sin 2A}{1+\cos 2A} = \tan A$

Ans. (i) L.H.S : $\frac{2\sin A\cos A}{2\sin^2 A}$

$= \frac{\sin A\cos A}{\sin A \sin A}$

$= \cot A$ = R.H.S

(ii) L.H.S: $\frac{2\sin A\cos A}{2\cos^2 A}$

$= \frac{\sin A}{\cos A} = \tan A$

= R.H.S

Ques. $\frac{\sin\theta + \sin 2\theta}{1+\cos\theta+\cos 2\theta} = \tan\theta$

Ans. L.H.S: $\frac{\sin\theta + \sin 2\theta}{1+\cos 2\theta+\cos\theta}$

$= \frac{\sin\theta + 2\sin\theta\cos\theta}{2\cos^2\theta + \cos\theta}$ $(\because 1 + \cos 2\theta = 2\cos^2\theta)$

$= \frac{\sin\theta(1+2\cos\theta)}{\cos\theta(2\cos\theta+1)} = \tan\theta$ = R.H.S

Ques. $\frac{1-\cos A+\cos B-\cos(A+B)}{1+\cos A-\cos B-\cos(A+B)} = \tan\frac{A}{2}\cot\frac{B}{2}$

Ans. L.H.S: $\frac{(1-\cos A)+(\cos B-\cos(A+B))}{(1+\cos A)-(\cos B+\cos(A+B))}$

$$= \frac{2\sin^2 A/2 + 2\sin\left[\frac{B+A+B}{2}\right]\sin(A/2)}{2\cos^2 A/2 - 2\cos\left[\frac{B+A+B}{2}\right]\cos(A/2)}$$

$$= \frac{2\sin A/2\,[\sin A/2 + \sin(B+A/2)]}{2\cos A/2\,[\cos A/2 - \cos(B+A/2)]}$$

$$= \frac{\tan A/2\,[\sin A/2 + \sin(B+A/2)]}{[\cos A/2 - \cos(B+A/2)]}$$

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$$= \frac{\tan A/2 \cdot 2\sin\left(\frac{A+B}{2}\right)\cos(-B/2)}{2\sin\left(\frac{A+B}{2}\right)\sin(B/2)}$$

$$= \frac{\tan A/2 \cos(B/2)}{\sin(B/2)} = \tan A/2 \cdot \cot B/2 = RHS$$

Ques. Prove the identity $\cos^4 t = \frac{3}{8} + \frac{1}{2}\cos 2t + \frac{1}{8}\cos 4t$

Ans. L.H.S : $\cos^4 t = (\cos^2 t)^2$

$$= \left[\frac{1+\cos 2t}{2}\right]^2 = \frac{(1+\cos 2t)^2}{4}$$

$$= \frac{1}{4} + \frac{2\cos 2t}{4} + \frac{\cos^2 2t}{4}$$

$$= \frac{1}{4} + \frac{1}{2}\cos 2t + \frac{\cos^2 2t}{4}$$

$$\left[\because 2\cos^2\theta = 1 + \cos 2\theta \Rightarrow \cos^2\theta = \frac{1+\cos 2\theta}{2}\right]$$

$$\left[\because \cos^2 2\theta = \frac{1+\cos 4\theta}{2}\right]$$

$$= \frac{1}{4} + \frac{1}{2}\cos 2t + \frac{\frac{1+\cos 4t}{2}}{4}$$

$$= \frac{2}{8} + \frac{1}{2}\cos 2t + \frac{1}{8} + \frac{\cos 4t}{8}$$

$$= \frac{3}{8} + \frac{1}{2}\cos 2t + \frac{\cos 4t}{8} = R.H.S$$

Ques. Prove that $\cos^2 x + \cos^2\left(x + \frac{\pi}{3}\right) + \cos^2\left(x - \frac{\pi}{3}\right) = \frac{3}{2}$.

Ans. L.H.S : $\frac{1+\cos 2x}{2} + \frac{1+\cos 2(x+\pi/3)}{2} + \frac{1+\cos 2(x-\pi/3)}{2}$

$$= \frac{1}{2}\left[3 + \cos 2x + \cos(2x + 2\pi/3) + \cos(2x - \frac{2\pi}{3})\right]$$

Let,

$C = 2x + 2\pi/3$, $D = 2x - 2\pi/3$

$\Rightarrow C + D = 4x \Rightarrow \frac{C+D}{2} = 2x$

$C - D = 4\pi/3 \Rightarrow \frac{C-D}{2} = 2\pi/3$

$$= \frac{1}{2}\left[3 + \cos 2x + 2\cos(2x)\cos(2\pi/3)\right]$$

$$= \frac{1}{2}\left[3 + \cos 2x + 2\cos 2x\,(-1/2)\right]$$

$$\left(\because \cos 2\pi/3 = -\frac{1}{2}\right)$$

$$= \frac{1}{2}[3 + \cos 2x - \cos 2x] = \frac{3}{2} = R.H.S$$

Ques. If $\pi < 2\theta < \frac{3\pi}{2}$, then $\sqrt{2+\sqrt{2+2\cos 4\theta}}$ equal to

(a) $-2\cos\theta$ (b) $-2\sin\theta$

(c) $2\cos\theta$ (d) $2\sin\theta$

Ans. Since, $1 + \cos 2\theta = 2\cos^2\theta$

$\therefore 1 + \cos 4\theta = 2\cos^2 2\theta$,

$$\sqrt{2+\sqrt{2+2\cos 4\theta}} = \sqrt{2+\sqrt{2(1+\cos 4\theta)}}$$

$$= \sqrt{2+\sqrt{2\cdot 2\cos^2 2\theta}}$$

$$= \sqrt{2+\sqrt{4\cos^2 2\theta}}$$

$$= \sqrt{2+2\cos 2\theta}$$

$$= \sqrt{2-2\cos 2\theta} \quad \left(\because \pi < 2\theta < \frac{3\pi}{2} \Rightarrow 2\theta \to 3^{rd}\text{quad}\right.$$

$\left.\cos 2\theta \to (-)ve\right)$

$$= \sqrt{2(1-\cos 2\theta)}$$

$$= \sqrt{2 \times 2\sin^2\theta} \quad (\because 1 - \cos 2\theta = 2\sin^2\theta)$$

$$= \sqrt{4\sin^2\theta} = |2\sin\theta| = 2\sin\theta$$

$$\left[\because \pi < 2\theta < \frac{3\pi}{2} \Rightarrow \frac{\pi}{2} < \theta < \frac{3\pi}{4}\right.$$

$\left.\theta \to 2^{nd} \Rightarrow \sin\theta = +ve\right]$

Ques. If $\frac{2\sin\alpha}{\{1+\cos\alpha+\sin\alpha\}} = y$, then $\frac{\{1-\cos\alpha+\sin\alpha\}}{1+\sin\alpha}$

(a) $\frac{1}{y}$ (b) y (c) $1-y$ (d) $1+y$

Ans. $y = \frac{2\sin\alpha}{1+\cos\alpha+\sin\alpha}$

$$= \frac{2\cdot 2\sin\alpha/2\cos\alpha/2}{2\cos^2\alpha/2 + 2\sin\alpha/2\cos\alpha/2}$$

$$= \frac{4\sin\alpha/2\cos\alpha/2}{2\cos\alpha/2\,[\cos\alpha/2 + \sin\alpha/2]}$$

$$y = \frac{2\sin\alpha/2}{\cos\alpha/2 + \sin\alpha/2} \quad \ldots (i)$$

Let $x = \frac{\{1-\cos\alpha+\sin\alpha\}}{1+\sin\alpha}$

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$$x = \frac{2\sin^2 a/2 + 2\sin a/2 \cos a/2}{(\cos a/2 + \sin a/2)^2}$$

$$x = \frac{2\sin a/2\left[\sin\frac{a}{2} + \cos\frac{a}{2}\right]}{[\cos a/2 + \sin a/2]^2}$$

$$x = \frac{2\sin a/2}{\cos a/2 + \sin a/2} \Rightarrow x = y$$

Extra Formulas

$$\because \tan\left(\frac{\pi}{4} + \theta\right) = \frac{\tan \pi/4 + \tan\theta}{1 - \tan \pi/4 \tan\theta}$$

1. $\tan\left(\frac{\pi}{4} + \theta\right) = \frac{1 + \tan\theta}{1 - \tan\theta}$
2. $\tan\left(\frac{\pi}{4} - \theta\right) = \frac{1 - \tan\theta}{1 + \tan\theta}$
3. $\sin(A + B)\sin(A - B) = \sin^2 A - \sin^2 B$

Proof : $(\sin A \cos B + \cos A \sin B)(\sin A \cos B - \cos A \sin B)$

$= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B$

$= \sin^2 A (1 - \sin^2 B) - (1 - \sin^2 A)(\sin^2 B)$

$= \sin^2 A - \sin^2 A \sin^2 B - \sin^2 B + \sin^2 A \sin^2 B$

$= \sin^2 A - \sin^2 B = RHS$

4. $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B$

Proof:

$(\cos A \cos B - \sin A \sin B)(\cos A \cos B + \sin A \sin B)$

$= \cos^2 A \cos^2 B - \sin^2 A \sin^2 B$

$= \cos^2 A (1 - \sin^2 B) - (1 - \cos^2 A)\sin^2 B$

$= \cos^2 A - \cos^2 A \sin^2 B - \sin^2 B + \cos^2 A \sin^2 B$

$= \cos^2 A - \sin^2 B = R.H.S.$

5. $\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A\tan B - \tan B\tan C - \tan C\tan A}$

Ques. Find the value of $\cos^2 48° - \sin^2 12°$

Ans. $\cos^2 48° - \sin^2 12° = \cos(48° + 12°)\cos(48° - 12°)$

$= \cos 60° \cdot \cos 36° = \frac{1}{2} \cdot \frac{\sqrt{5} + 1}{4}$

$= \frac{\sqrt{5} + 1}{8}$

Ques. Find the value of $\frac{\cos 289° - \sin 289°}{\cos 289° + \sin 289°}$

(a) $\cot 26°$ (b) $\tan 26°$

(c) $-\tan 26°$ (d) $-\cot 26°$

Ans. $\frac{\cos 289° - \sin 289°}{\cos 289° + \sin 289°}$

divide by $\cos 289°$

$= \frac{1 - \tan 289°}{1 + \tan 289°}$

$= \tan(45° - 289°) = \tan(-244°)$

$= -\tan 244° = -\tan(270° - 26°) = -\cot 26°$

$(\because \tan(270° - \theta) = \cot\theta)$

6. $\sin\theta \sin(60° - \theta)\sin(60° + \theta) = \frac{1}{4}\sin 3\theta$ ...(i)
7. $\cos\theta \cos(60° - \theta)\cos(60° + \theta) = \frac{1}{4}\cos 3\theta$ ...(ii)
8. $\tan\theta \tan(60° - \theta)\tan(60° + \theta) = \tan 3\theta$ ...(iii)

Proof of 6: L.H.S : $\sin\theta \sin(60° - \theta)\sin(60 + \theta)$

$= \sin\theta[\sin^2 60° - \sin^2\theta] = \sin\theta\left[\frac{3}{4} - \sin^2\theta\right]$

$= \sin\theta \frac{[3 - 4\sin^2\theta]}{4}$

$= \frac{3\sin\theta - 4\sin^3\theta}{4} = \frac{\sin 3\theta}{4} = R.H.S$

Proof of 7: L.H.S: $\cos\theta \cos(60° - \theta)\cos(60° + \theta)$

$= \cos\theta[\cos^2 60° - \sin^2\theta] = \cos\theta\left[\frac{1}{4} - (1 - \cos^2\theta)\right]$

$= \cos\theta\left[\frac{1}{4} - 1 + \cos^2\theta\right] = \cos\theta\left[\frac{1 - 4 + 4\cos^2\theta}{4}\right]$

$= \cos\theta\left[\frac{-3 + 4\cos^2\theta}{4}\right] \frac{4\cos^3\theta - 3\cos\theta}{4}$

$= \frac{\cos 3\theta}{4} = R.H.S$

Ques. $\cos 20° \cos 40° \cos 60° \cos 80° =$

Ans. $\cos 60° \cos 20° \cos 40° \cos 80°$

$= \cos 60° \cos 20° \cos(60° - 20°)\cos(60° + 20°)$

$= \cos 60° \times \frac{1}{4}\cos(3 \times 20°)$

$= \cos 60° \times \frac{1}{4}\cos 60° = \frac{1}{2} \times \frac{1}{4} \times \frac{1}{2} = \frac{1}{16}$

Ques. $\sin 10° \sin 50° \sin 70° = ?$

Ans. $\sin 10° \sin(60° - 10°)\sin(60° + 10°)$

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$$= \frac{1}{4}\sin(3 \times 10^\circ) = \frac{1}{4}\sin 30^\circ$$

$$= \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

Ques. $\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ = ?$

Ans. $$\frac{\tan 6^\circ \tan(60^\circ + 6^\circ)\tan(60^\circ - 6^\circ)}{\tan 54^\circ} \tan 42^\circ \tan 78^\circ$$

$$= \frac{\tan 18^\circ}{\tan 54^\circ} \tan 42^\circ \tan 78^\circ$$

$$= \frac{\tan 18^\circ \tan(60^\circ - 18^\circ)\tan(60^\circ + 18^\circ)}{\tan 54^\circ}$$

$$= \frac{\tan(3 \times 18^\circ)}{\tan 54^\circ} = 1$$

Ques. Find the value of the continued product $\prod_{k=1}^{17} \sin \frac{k\pi}{18}$

Ans. $$\prod_{k=1}^{17} \sin\frac{k\pi}{18} = \sin\frac{\pi}{18}\sin\frac{2\pi}{18}\sin\frac{3\pi}{18}\sin\frac{4\pi}{18}\ldots\left(\sin\frac{9\pi}{18}\right)\ldots \sin\left(\frac{17\pi}{18}\right)$$

$$= \left(\sin\frac{\pi}{18}\right)^2\left(\sin\frac{2\pi}{18}\right)^2\left(\sin\frac{3\pi}{18}\right)^2\ldots\left(\sin\frac{8\pi}{18}\right)^2\left(\sin\frac{9\pi}{18}\right)^1$$

$(\because \sin(\pi-\theta) = \sin\theta)$

$$= \left[\sin\frac{\pi}{18}\sin\frac{2\pi}{18}\sin\frac{3\pi}{18}\ldots\sin\frac{7\pi}{18}\sin\frac{8\pi}{18}\right]^2$$

$$= \frac{1}{16^2}\left[2\sin\frac{\pi}{18}\cos\frac{\pi}{18}\,2\sin\frac{2\pi}{18}\cos\frac{2\pi}{18}\,2\sin\frac{3\pi}{18}\cos\frac{3\pi}{18}\,2\sin\frac{4\pi}{18}\cos\frac{4\pi}{18}\right]^2$$

$(\because \sin(\frac{\pi}{18}-\theta) = \cos\theta)$

$$= \frac{1}{(16)^2}\left[\sin\frac{2\pi}{18}\sin\frac{4\pi}{18}\sin\frac{6\pi}{18}\sin\frac{8\pi}{18}\right]^2$$

$$= \frac{1}{(16)^2}[\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ]^2$$

$$= \frac{1}{16^2}\left(\frac{3}{16}\right)^2 = \frac{9}{(16)^4} = \frac{9}{(256)^2}$$

Note: Summation $\sum_{k=1}^{10} (k)^2$

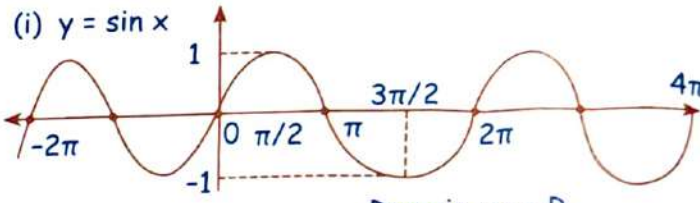
$$= 1^2 + 2^2 + 3^3 + 4^2 + \ldots + 10^2$$

Product $\prod_{k=1}^{10} (k)^2$

$$= (1)^2 \times (2)^2 \times (3)^3 \times (4)^2 \ldots (10)^2$$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

(i) $y = \sin x$



Domain : $x \in R$

Range : $y \in [-1, 1]$

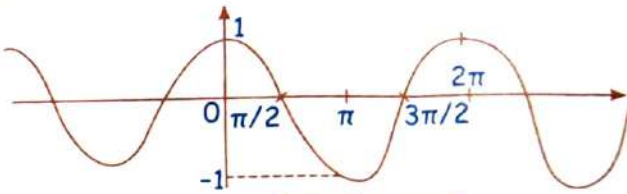
Period : $2n\pi$

Fundamental period = 2

$\sin x$ is ↑ (Increasing) for $x \in (0, \pi/2) \cup (3\pi/2, 2\pi)$

$\sin x$ is ↓ (decreasing) for $x \in (\pi/2, 3\pi/2)$

(ii) $y = \cos x$



Domain : $x \in R$

Range : $y \in [-1, 1]$

Period : $2n\pi$

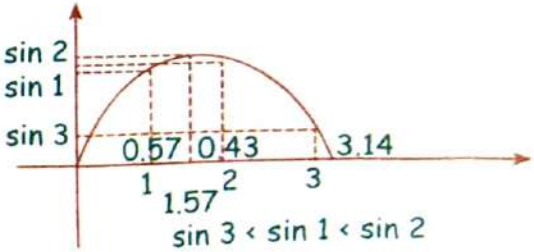
Fundamental period = 2π

$\cos x$ is decreasing for $x \in (0, \pi)$

$\cos x$ is increasing for $x \in (\pi, 2\pi)$

Ques. Arrange in Ascending order : sin 1, sin 2, si

Ans.



$\sin 3 < \sin 1 < \sin 2$

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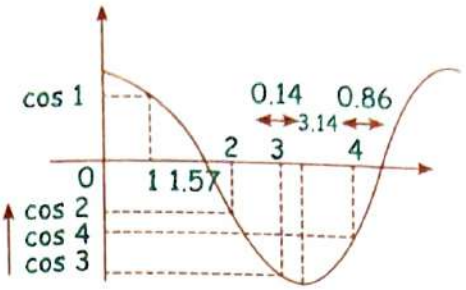
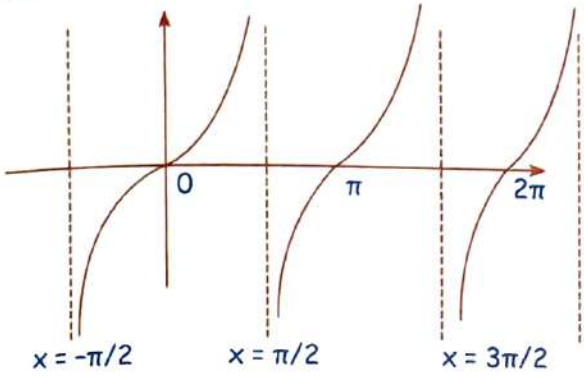
Ques. Arrange in Ascending Order : cos 1, cos 2, cos 3

Ans.



cos 3 < cos 2 < cos 1

(iii) y = tanx



Graph of tan x is increasing in each quandrant

$y = \frac{\sin x}{\cos x}$

$\cos x \neq 0$

$\Rightarrow x \neq \pi/2, 3\pi/2, 5\pi/2, 7\pi/2$ ..... .

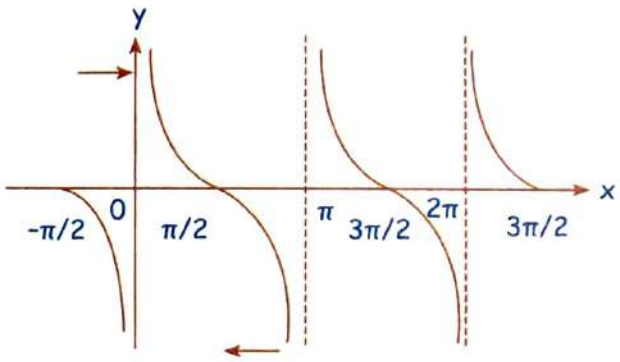
$\Rightarrow x \neq (2n + 1)\pi/2$

$\therefore$ Domain : $x \in R - (2n + 1)\pi/2$

Range : $y \in (-\infty, \infty)$ or $y \in R$

Fundamental period $\rightarrow \pi$

(iv) Graph of y = cot x



$y = \frac{\cos x}{\sin x}$

Domain : $x \in R - n\pi$

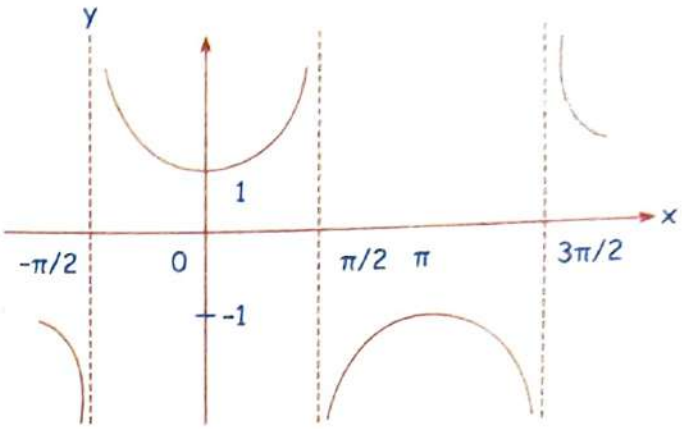
Range : $y \in R$

$\sin x \neq 0$

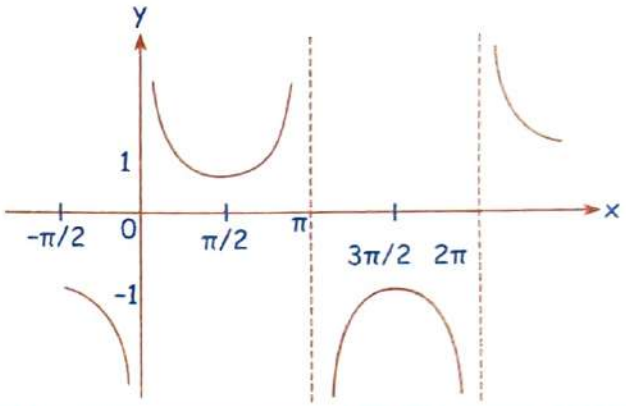
$x \neq 0, \pi, 2\pi, 3\pi, ...$

$x \neq n\pi$

(v) Graph of y = sec x



(vi) Graph of y = cosec x



DOMAIN AND RANGE SUMMARY

Domains, Ranges and Periodicity of Trigonometric Functions

T-Ratio	Domain	Range	Period
sin x	R	[-1, 1]	2π
cos x	R	[-1, 1]	2π
tan x	R - {(2n + 1) π/2; n ∈ I}	R	π
cot x	R - {nπ : n ∈ I}	R	π
sec x	R - {(2n + 1) π/2; n ∈I}	$(-\infty, -1] \cup [1, \infty]$	2π
cosec x	R - {nπ : n ∈ I}	$(-\infty, -1] \cup [1, \infty]$	2π

Ques. State whether True of False

(a) The domain of sin x is [-1, 1] → (F)
(b) The range of cos x is [-1, 1] → (T)
(c) The domain of tan x is all real numbers. → (F)
(d) The range of tan x is all real numbers. → (T)
(e) sin x is increasing function in $x \in [0, \pi]$ → (F)
(f) cos x is decreasing function in $x \in [0, \pi]$ → (T)

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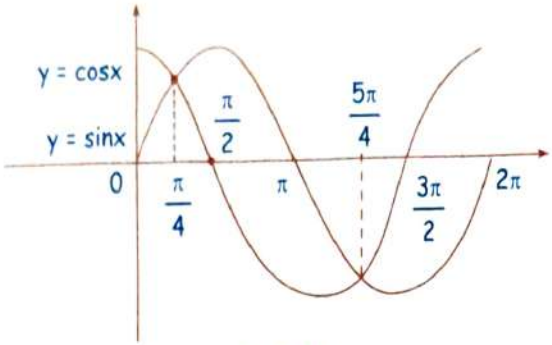
GRAPHS OF sinx VS cosx



1. $\sin x = \cos x \Rightarrow x = \pi/4, 5\pi/4$
2. $\cos x > \sin x \Rightarrow x \in [0, \pi/4) \cup \left(\frac{5\pi}{2}, 2\pi\right]$
3. $\sin x > \cos x \Rightarrow x \in (\pi/4, 5\pi/4)$s

Ques. State whether True of False

(a) $\sin 4^\circ < \cos 4^\circ \rightarrow$ (T)

(b) $\sin 46^\circ < \cos 46^\circ \rightarrow$ (F)

(c) $\sin 100^\circ > \cos 100^\circ \rightarrow$ (T)

(d) $\sin 200^\circ > \cos 200^\circ \rightarrow$ (T)

(e) $\sin 289^\circ > \cos 289^\circ \rightarrow$ (F)

Ques. find the value of $\sin 9^\circ - \cos 9^\circ$

Ans. Let $\sin 9^\circ - \cos 9^\circ = x$

SBS

$\sin^2 9^\circ + \cos^2 9^\circ - 2\sin 9^\circ \cos 9^\circ = x^2$

$\Rightarrow 1 - \sin 18^\circ = x^2 \Rightarrow 1 - \left(\frac{\sqrt{5}-1}{4}\right) = x^2$

$\Rightarrow \frac{4-\sqrt{5}+1}{4} = x^2 \Rightarrow x^2 = \frac{5-\sqrt{5}}{4}$

$\Rightarrow x = \pm\sqrt{\frac{5-\sqrt{5}}{4}}$ ($\because \sin 9^\circ < \cos 9^\circ$

$\sin 9^\circ - \cos 9^\circ < 0 \Rightarrow x < 0$)

$\therefore x = -\sqrt{\frac{5-\sqrt{5}}{4}}$

Ques. If x lies in 2nd Quadrant then $\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}$

(a) $\cot\frac{x}{2}$ (b) $\tan\frac{x}{2}$

(c) $\sec\frac{x}{2}$ (d) $\operatorname{cosec}\frac{x}{2}$

Ans $\sqrt{1+\sin x} = |\cos x/2 + \sin x/2| = (\cos x/2 + \sin x/2)$

$\sqrt{1-\sin x} = \sqrt{(\cos x/2 - \sin x/2)^2}$

$\Rightarrow \sqrt{1-\sin x} = |\cos x/2 - \sin x/2|$

$90^\circ < x < 180^\circ$
$\Rightarrow 45^\circ < \frac{x}{2} < 90^\circ$
$\Rightarrow \cos\frac{x}{2} < \sin\frac{x}{2}$

$\Rightarrow \sqrt{1-\sin x} = \sin x/2 - \cos x/2$

Now, $\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}$

$= \frac{\cos x/2 + \sin x/2 + \sin x/2 - \cos x/2}{\cos x/2 + \sin x/2 - (\sin x/2 - \cos x/2)}$

$= \frac{2\sin x/2}{2\cos x/2} = \tan(x/2)$

Ques. If $A = 340^\circ$ then $2\sin\frac{A}{2}$ is identical to

(a) $\sqrt{1+\sin A}+\sqrt{1-\sin A}$

(b) $-\sqrt{1+\sin A}-\sqrt{1-\sin A}$

(c) $\sqrt{1+\sin A}-\sqrt{1-\sin A}$

(d) $-\sqrt{1+\sin A}+\sqrt{1-\sin A}$

Ans. $\sqrt{1+\sin A} = |\cos A/2 + \sin A/2| = |\cos 170^\circ + \sin 170^\circ|$
$= |\sin 10^\circ - \cos 10^\circ|$

$\sqrt{1-\sin A} = |\cos A/2 - \sin A/2| = |\cos 170^\circ - \sin 170^\circ|$

($\because \cos 170^\circ < \sin 170^\circ \Rightarrow \cos 170^\circ - \sin 170^\circ < 0$)

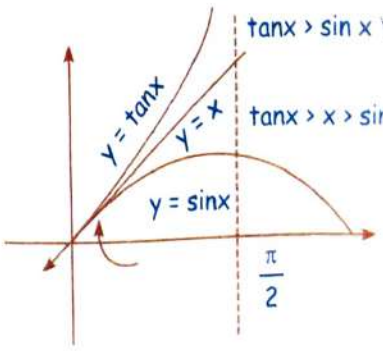
$\sqrt{1+\sin A} = -(\cos A/2 + \sin A/2) = -\cos A/2 - \sin A/2$

$\sqrt{1-\sin A} = -(\cos A/2 - \sin A/2) = -\cos A/2 + \sin A/2$

$\Rightarrow \sqrt{1+\sin A} - \sqrt{1-\sin A} = -2\sin A/2$

$\Rightarrow -\sqrt{1+\sin A} + \sqrt{1-\sin A} = 2\sin A/2$

GRAPHS OF sinx VS tanx



$\tan x > \sin x \ \forall\ x \in \left(0, \frac{\pi}{2}\right)$

$\tan x > x > \sin x \ \forall\ x \in \left(0, \frac{\pi}{2}\right)$

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IDENTITIES IN A TRIANGLE/ CONDITIONAL IDENTITIES

(1) If $A + B + C = 180°$ then

$\tan A + \tan B + \tan C = \tan A \tan B \tan C$

Proof: $A + B = 180° - C$

$\Rightarrow \tan(A + B) = \tan(180° - C)$

$\Rightarrow \dfrac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$

$\Rightarrow \tan A + \tan B = -\tan C (1 - \tan A \tan B)$

$\Rightarrow \tan A + \tan B = -\tan C + \tan A \tan B \tan C$

$\Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C$

or $\Sigma \tan A = \prod \tan A$

(2) $\tan\dfrac{A}{2}\tan\dfrac{B}{2} + \tan\dfrac{B}{2}\tan\dfrac{C}{2} + \tan\dfrac{C}{2}\tan\dfrac{A}{2} = 1$

or $\Sigma \tan\left(\dfrac{A}{2}\right)\tan\left(\dfrac{B}{2}\right) = 1.$

Proof: $A + B = 180° - C$

$A/2 + B/2 = 90° - C/2$

$\tan\left(\dfrac{A}{2} + \dfrac{B}{2}\right) = \tan\left(90° - \dfrac{C}{2}\right)$

$\Rightarrow \tan\left(\dfrac{A}{2} + \dfrac{B}{2}\right) = \cot\dfrac{C}{2}$

$\Rightarrow \dfrac{\tan A/2 + \tan B/2}{1 - \tan A/2 \tan B/2} = \dfrac{1}{\tan C/2}$

$\Rightarrow \tan C/2 (\tan A/2 + \tan B/2) = 1 - \tan A/2 \tan B/2$

$\Rightarrow \tan\dfrac{A}{2}\tan\dfrac{B}{2} + \tan\dfrac{B}{2}\tan\dfrac{C}{2} + \tan\dfrac{C}{2}\tan\dfrac{A}{2} = 1$

Ques. If in a triangle $\tan A + \tan B + \tan C = 6$ & $\tan A \tan B = 2$ then find the value of $\tan C$

Ans. $\tan A \tan B \tan C = 6$

$\Rightarrow 2 \times \tan C = 6 \Rightarrow \tan C = 6/2$

$\Rightarrow \tan C = 3$

Ques. If $A + B + C = 180°$ than

$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

Ans. L.H.S: $\sin 2A + \sin 2B + \sin 2C$

$= 2\sin(A + B)\cos(A - B) + 2\sin C \cos C$

$= 2\sin(180° - C)\cos(A - B) + 2\sin C \cos C$

$(\because A + B = 180° - C)$

$= 2\sin C \cos(A - B) + 2\sin C \cos C$

$= 2\sin C [\cos(A - B) + \cos C]$

$C = 180° - (A + B)$

$\cos C = -\cos(A + B)$

$= 2\sin C [\cos(A - B) - \cos(A + B)]$

$= 2\sin C . 2\sin A \sin B$

$= 4\sin A \sin B \sin C$ = R.H.S

Ques. If $\tan A = \dfrac{1}{\sqrt{x(x^2 + x + 1)}}$, $\tan B = \dfrac{\sqrt{x}}{\sqrt{(x^2 + x + 1)}}$ and $\tan C = (x^{-3} + x^{-2} + x^{-1})^{1/2}$, $0 < A, B, C < \dfrac{\pi}{2}$ then $A + B$ is equal to: [01 feb 2024(shift -1)]

(a) C (b) $\pi - C$

(c) $2\pi - C$ (d) $\dfrac{\pi}{2} - C$

Ans. $\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$

$$= \frac{\dfrac{1}{\sqrt{x}\sqrt{x^2 + x + 1}} + \dfrac{x}{\sqrt{x}\sqrt{x^2 + x + 1}}}{1 - \dfrac{1 . x}{x(x^2 + x + 1)}}$$

$$= \frac{(1 + x)(x^2 + x + 1)}{\sqrt{x}\sqrt{x^2 + x + 1}(x^2 + x + 1 - 1)}$$

$$= \frac{(1 + x)(x^2 + x + 1)}{\sqrt{x}\sqrt{x^2 + x + 1}\, x(x + 1)}$$

$$= \frac{\sqrt{x^2 + x + 1}}{\sqrt{x^3}} = \tan(A + B)$$

$$\tan C = \sqrt{\frac{1}{x^3} + \frac{1}{x^2} + \frac{1}{x}} = \sqrt{\frac{1 + x + x^2}{x^3}}$$

(3) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

(4) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

Proof: L.H.S: $2\cos(A + B)\cos(A - B) + 2\cos^2 C - 1$

$= 2\cos(\pi - C)\cos(A - B) + 2\cos^2 C - 1$

$= -2\cos C \cos(A - B) + 2\cos^2 C - 1$

$= -2\cos C [\cos(A - B) - \cos C] - 1$

$= -2\cos C [\cos(A - B) + \cos(A + B)] - 1$

$= -2\cos C . 2\cos A \cos B - 1$

$= -4\cos A \cos B \cos C - 1$

= R.H.S

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Ques. If $A + B + C = \pi$, then the value of expression $\frac{\cos A}{\sin B \sin C} + \frac{\cos B}{\sin C \sin A} + \frac{\cos C}{\sin A \sin B}$ is

Ans. $\frac{\cos A}{\sin B \sin C} + \frac{\cos B}{\sin C \sin A} + \frac{\cos C}{\sin A \sin B}$

$= \frac{2\sin A \cos A + 2\sin B \cos B + 2\sin C \cos C}{2\sin A \sin B \sin C}$

$= \frac{\sin 2A + \sin 2B + \sin 2C}{2\sin A \sin B \sin C}$

$= \frac{4\sin A \sin B \sin C}{2\sin A \sin B \sin C} = 2$

(5) $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

Proof: L.H.S: $2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\sin C/2 \cos C/2$

$\left[\because \sin\left(\frac{A+B}{2}\right) = \sin\left(\frac{180^\circ - C}{2}\right)\right.$

$\left. = \sin(90^\circ - C/2) = \cos C/2\right]$

$= 2\cos C/2 \cos\left(\frac{A-B}{2}\right) + 2\sin C/2 \cos C/2$

$= 2\cos C/2 \left[\cos\frac{A-B}{2} + \sin C/2\right]$

$\left[\because \sin C/2 = \sin\left(\frac{180^\circ - (A+B)}{2}\right)\right.$

$\left. = \sin\left(90^\circ - \frac{(A+B)}{2}\right) = \cos\frac{(A+B)}{2}\right]$

$= 2\cos\frac{C}{2}\left[\cos\frac{A-B}{2} + \cos\left(\frac{A+B}{2}\right)\right]$

$= 2\cos\frac{C}{2}\, 2\cos\frac{A}{2}\cos\frac{B}{2}$

$= 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$ = R.H.S

Ques. If $A + B + C = \pi$, then $\cos^2 A + \cos^2 B - \cos^2 C$ is equal to

(a) $1 - 2\sin A \sin B \cos C$

(b) $2\sin A \sin B \sin C$

(c) $1 - 2\cos A \sin B \sin C$

(d) $2\cos A \sin B \sin C$

Ans. $\cos^2 A + \cos^2 B - \cos^2 C$

$= \cos^2 A + 1 - \sin^2 B - \cos^2 C$

$= 1 + \cos(A + B)\cos(A - B) - \cos^2 C$

$(\because \cos^2 A - \sin^2 B = \cos(A + B)\cos(A - B))$

$= 1 - \cos C \cos(A - B) - \cos^2 C$

$= 1 - \cos C\,[\cos(A - B) + \cos C]$

$= 1 - \cos C\,[\cos(A - B) - \cos(A + B)]$

$= 1 - \cos C \cdot 2\sin A \sin B = 1 - 2\sin A \sin B \cos C$

Brain Teaser

If $xy + xz + zx = 1$ where $x, y, z \in R^+$ then prov that

$\frac{x}{1+x^2} + \frac{y}{1+y^2} + \frac{z}{1+z^2} = \frac{2}{\sqrt{(1+x^2)(1+y^2)(1+z^2)}}$

Ans. Let $x = \tan A/2$, $y = \tan B/2$, $z = \tan C/2$ } Given

$xy + yz + zx = 1$

$\Rightarrow \Sigma \tan A/2 \tan B/2 = 1 \Rightarrow A + B + C = 180^\circ$

L.H.S: $\frac{x}{1+x^2} + \frac{y}{1+y^2} + \frac{z}{1+z^2}$

$= \frac{2\tan A/2}{2(1+\tan^2 A/2)} + \frac{2\tan B/2}{2(1+\tan^2 B/2)} + \frac{2\tan C/2}{2(1+\tan^2 C/2)}$

$= \frac{\sin A}{2} + \frac{\sin B}{2} + \frac{\sin C}{2}$

$= \frac{\sin A + \sin B + \sin C}{2}$

$= \frac{4\cos A/2 \cos B/2 \cos C/2}{2}$

$= 2\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

R.H.S : $= \frac{2}{\sqrt{(1+x^2)(1+y^2)(1+z^2)}}$

$= \frac{2}{\sqrt{1+\tan^2 A/2}\sqrt{1+\tan^2 B/2}\sqrt{1+\tan^2 C/2}}$

$= \frac{2}{|\sec A/2||\sec B/2||\sec C/2|}$

$= \frac{2}{\sec A/2 \sec B/2 \sec C/2}$

$= 2\cos A/2 \cos B/2 \cos C/2$

L.H.S = R.H.S

Hence Proved.

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Important Inequalities in a Triangle

In an acute angled triangle ABC, $\tan A + \tan B + \tan C \geq 3\sqrt{3}$

In any triangle ABC, $\cos A + \cos B + \cos C \leq 3/2$

In any triangle ABC, $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \leq \frac{1}{8}$

SUM OF SINES/COSINES OF n ANGLES IN AP

Sum of sines of "n" angles in AP

$$\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots + \sin(\alpha+\overline{n-1}\beta) = \frac{\sin(n\beta/2)}{\sin(\beta/2)}\sin\left(\alpha+\frac{(n-1)}{2}\beta\right)$$

$\alpha \rightarrow$ first angle

$\beta \rightarrow$ c.d

$n \rightarrow$ no of terms

sum of cosines of "n" angles in AP

$$\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots + \cos(\alpha+\overline{n-1}\beta) = \frac{\sin(n\beta/2)}{\sin(\beta/2)}\cos\left(\alpha+\frac{(n-1)}{2}\beta\right)$$

Ques. The value of $\cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}$ is

(a) 0 (b) 1

(c) $\frac{1}{2}$ (d) None of these

Ans. $\cos\alpha + \cos(\alpha+\beta) + \ldots + \cos(\alpha+\overline{n-1}\beta) = \frac{\sin\frac{n\beta}{2}}{\sin\beta/2}\cos(\alpha+(n-1)\beta/2)$

$$\therefore \cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}$$

$$= \frac{\sin 5\pi/11}{\sin\pi/11}\cos\left[\frac{\pi}{11}+\frac{4\pi}{11}\right]$$

$$= \frac{2\sin\frac{5\beta}{11}\cos\frac{5\beta}{11}}{2\sin\pi/11} = \frac{\sin\left(\frac{10\pi}{11}\right)}{2\sin\left(\frac{\pi}{11}\right)} = \frac{1}{2}$$

Ques. Find the value of $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$

Ans. $\alpha = 2\pi/7, \beta = 2\pi/7 \Rightarrow \beta/2 = \pi/7, n = 3$

$$\text{Sum} = \frac{\sin n\beta/2}{\sin\beta/2}\cos(\alpha+(n-1)\beta/2)$$

$$= \frac{\sin 3\pi/7}{\sin\pi/7}\cos\left[\frac{2\pi}{7}+2\pi/7\right]$$

$$= \frac{2\sin 3\pi/7\cos 4\pi/7}{2\sin\pi/7}$$

$$= \frac{\sin\left[\frac{3\pi}{7}+\frac{4\pi}{7}\right]+\sin\left[\frac{3\pi}{7}-\frac{4\pi}{7}\right]}{2\sin\pi/7}$$

$$= \frac{\sin\pi+\sin[-\pi/7]}{2\sin\pi/7} = \frac{-\sin\pi/7}{2\sin\pi/7} = \frac{-1}{2}$$

Ques. $\sum_{r=0}^{n}\sin^2\frac{r\pi}{n}$ is equal to

(a) $\frac{n+1}{2}$ (b) $\frac{n-1}{2}$

(c) $\frac{n}{2}$ (d) None of these

Ans. $\sin^2\pi/n + \sin^2\frac{2\pi}{n} + \sin^2\frac{3\pi}{n} + \ldots\ldots \sin^2\frac{n\pi}{n}$

$$[\because \sin^2\theta = \frac{1-\cos 2\theta}{2}]$$

$$= \frac{1-\cos 2\pi/n}{2} + \frac{1-\cos 4\pi/n}{2} + \frac{1-\cos 6\pi/n}{2} + \ldots + \frac{(1-\cos 2n\pi/n)}{2}$$

$$= \frac{1}{2}[1-\cos 2\pi/n + 1-\cos 4\pi/n + 1-\cos 6\pi/n + \ldots + 1-\cos 2n\pi/n]$$

$$= \frac{1}{2}[n-(\cos 2\pi/n + \cos 4\pi/n + \cos 6\pi/n + \ldots + \cos 2n\pi/n]$$

$$= \frac{1}{2}\left[n-\frac{\sin n\beta/2}{\sin\beta/2}\cos(\alpha+(n-1)\frac{\beta}{2})\right]$$

Here, $\alpha = \frac{2\pi}{n}, \beta = \frac{2\pi}{n} \Rightarrow \frac{\beta}{2} = \frac{\pi}{n}$

no. of terms = n

$$= \frac{1}{2}\left[n-\frac{\sin\pi}{\sin\frac{\pi}{n}}\cos\left(\frac{2\pi}{n}+(n-1)\frac{\pi}{n}\right)\right]$$

$$= \frac{1}{2}[n-0] = n/2$$

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CONTINUED PRODUCT OF COSINE SERIES

$$\cos A \cos 2A \cos 4A \cos 8A \ldots \cos 2^{n-1}A = \frac{1}{2^n \sin A} \sin(2^n A)$$

Proof: $2\sin A \cos A = \sin 2A$

$2\sin 2A \cos 2A = \sin 4A$

$2\sin 4A \cos 4A = \sin 8A$

.....

.....

$2\sin 2^{n-1}A \cos 2^{n-1}A = \sin 2^n A$

Multiply all,

$\Rightarrow 2^n \sin A \cos A \cos 2A \cos 4A \ldots \cos 2^{n-1}A = \sin 2^n$

$\Rightarrow \cos A \cos 2A \cos 4A \ldots \cos 2^{n-1}A = \frac{\sin 2^n A}{2^n \sin A}$

Ques. Find the value of $\cos\frac{\pi}{7} \cos\frac{2\pi}{7} \cos\frac{4\pi}{7}$.

Ans. $A = \frac{\pi}{7}$, $n = 3$

$$\cos\frac{\pi}{7} \cos\frac{2\pi}{7} \cos\frac{4\pi}{7} = \frac{\sin 2^n A}{2^n \sin A}$$

$$= \frac{\sin 2^3 \pi/7}{2^3 \sin \pi/7} = \frac{\sin 8\pi/7}{8\sin \pi/7}$$

$$= \frac{\sin[\pi + \pi/7]}{8\sin \pi/7} = \frac{-\sin \pi/7}{8\sin \pi/7} = \frac{-1}{8}$$

Ques. Prove that: $\cos\frac{2\pi}{15} \cos\frac{4\pi}{15} \cos\frac{8\pi}{15} \cos\frac{16\pi}{15} = \frac{1}{16}$

Ans. $A = \frac{2\pi}{15}$, $n = 4$

L.H.S: $\cos\frac{2\pi}{15} \cos\frac{4\pi}{15} \cos\frac{8\pi}{15} \cos\frac{16\pi}{15}$

$$= \frac{\sin 2^n A}{2^n \sin A} = \frac{\sin 2^4 \times 2\pi/15}{2^4 \sin 2\pi/15}$$

$$= \frac{\sin(32\pi/15)}{16\sin(2\pi/15)} = \frac{\sin(2\pi + 2\pi/15)}{16\sin(2\pi/15)}$$

$$= \frac{1}{16} = \text{R.H.S}$$

Ques. $96 \cos\frac{\pi}{33} \cos\frac{2\pi}{33} \cos\frac{4\pi}{33} \cos\frac{8\pi}{33} \cos\frac{16\pi}{33}$ is equal is

(a) 3 (b) 1 (c) 4 (d) 2

Ans. $A = \pi/33$, $n = 5$

$$96 \cos\frac{\pi}{33} \cos\frac{2\pi}{33} \cos\frac{4\pi}{33} \cos\frac{8\pi}{33} \cos\frac{16\pi}{33}$$

$$= 96 \cdot \frac{\sin 2^5 \pi/33}{2^5 \sin \pi/33}$$

$$= \frac{96 \sin 32\pi/33}{32 \sin \pi/33} = \frac{96 \sin(\pi - \pi/33)}{32 \sin \pi/33}$$

$$= \frac{96}{32} = 3$$

Brain Teaser

Arithmetic mean of $2\sin 2°, 4\sin 4°, 6\sin 6$ ... upto 90 terms

(a) $\cot 1°$ (b) $\frac{\cot 44° - 1}{\cot 44° + 1}$

(c) $\frac{\cos^2 20° - \sin^2 19°}{\sin^2 26° - \sin^2 25°}$ (d) $\sqrt{\frac{1 - \cos 2°}{1 + \cos 2°}}$

Ans. A.M. $= 2[\sin 2° + \sin 4° + \sin 6° + \ldots \sin 88°] + 1$

$\alpha = 2°$, $\beta = 2° \Rightarrow \beta/2 = 1°$

$n = 44$

$$= 2 \cdot \frac{\sin n\beta/2}{\sin \beta/2} \sin[\alpha + (n-1)\beta/2] + 1$$

$$= 2\frac{\sin(44 \times 1°)}{\sin 1°} \sin[2° + 43 \times 1°] + 1$$

$$= \frac{2\sin 44° \sin 45°}{\sin 1°} + 1 = \frac{\cos 1° - \cos 89°}{\sin 1°} + 1$$

$$= \frac{\cos 1° - \sin 1°}{\sin 1°} + 1$$

$$= \frac{\cos 1° - \sin 1° + \sin 1°}{\sin 1°} = \frac{\cos 1°}{\sin 1°} = \cot 1°$$

(b) $\frac{\cot 44° - 1}{\cot 44 + 1} = \frac{1 - \tan 44°}{1 + \tan 44°}$

$(\because \tan(45° - \theta) = \frac{1 - \tan\theta}{1 + \tan\theta})$

$= \tan(45° - 44°) = \tan 1°$

(c) $\frac{\cos^2 20° - \sin^2 19°}{\sin^2 26° - \sin^2 25°}$

$$= \frac{\cos(20° + 19°)\cos(20° - 19°)}{\sin(26° + 25°)\sin(26° - 25°)}$$

$$= \frac{\cos 39° \cos 1°}{\sin 51° \sin 1°} = \cot 1°$$

(d) $\sqrt{\frac{1 - \cos 2°}{1 + \cos 2°}} = \sqrt{\frac{2\sin^2 1°}{2\cos^2 1°}} = \tan 1°$

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Ques. Find the range of the followings

1. $y = 3\cos x$

Ans. $y \in [-3, 3]$

2. $y = 3\sin x + 9$

Ans. $y \in [6, 12]$

3. $y = 3\sin x - 6$

Ans. $y_{max} = 3 - 6 = -3, y_{min} = -3 - 6 = -9$ $y \in [-9, -3]$

4. $y = 7 - 3\sin x$

Ans. $y_{min} = 7 - 3(1) = 4, y_{max} = 7 - 3(-1) = 7 + 3 = 10$
$y \in [4, 10]$

5. $y = 4 - 8\cos x$

Ans. $y_{min} = 4 - 8 \times 1 = -4, y_{max} = 4 - 8(-1) = 4 + 8 = 12$
$y \in [-4, 12]$

6. $y = 6 + 3\sin^2 x$

Ans. $y_{max} = 6 + 3(1)^2 = 9,$
for y_{min} $(\sin^2 x)$ must be min $\to 0, y_{min} = 6$
$\Rightarrow y \in [6, 9],$

7. $y = 8 - 5\cos^2 x$

Ans. $y_{max} = 8 - 5 \times 0 = 8, y_{min} = 8 - 5 \times 1 = 3$
$y \in [3, 8]$

Ques. Find the range of the following

1. $y = 5\sin^2 x + 8\cos^2 x$

Ans. $y = 5\sin^2 x + 5\cos^2 x + 3\cos^2 x$
$y = 5 + 3\cos^2 x$
$y_{max} = 5 + 3 = 8$
$y_{min} = 5 + 0 = 5$
$y \in [5, 8]$

2. $y = 5\sin^2 x + 9\cos^2 x - 2$

Ans. $y = 5 + 4\cos^2 x - 2 = 3 + 4\cos^2 x, y \in [3, 7]$

3. $y = 2\sin^2 x - 8\cos^2 x$

Ans. $y = 2 - 10\cos^2 x, y_{max} = 2, y_{min} = 2 - 10 = -8,$
$y \in [-8, 2]$

4. $y = \sin^6 x + \cos^6 x$

Ans. $1 - 3\sin^2 x\cos^2 x = 1 - 3(\sin x \cos x)^2 = 1 - 3/4\sin^2 2x$
$y_{max} = 1, y_{min} = 1 - 3/4 \times 1 = 1/4$
$y \in [1/4, 1]$

5. $y = 3\sin x + 4\cos z$

Ans. $y_{max} = 7, y_{min} = -7$ $y \in [-7, 7]$

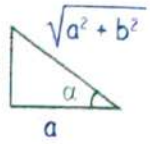
RANGE OF $y = a\sin x + b\cos x$

(Miximum and Minimum Values of Trigonometric Expressions)

The range of $y = a\cos x + b\sin x$ will be

$[-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$

Proof: $y = a\sin x + b\cos x$



divide by $\sqrt{a^2 + b^2}$

$$\frac{y}{\sqrt{a^2 + b^2}} = \frac{a}{\sqrt{a^2 + b^2}}\sin x + \frac{a}{\sqrt{a^2 + b^2}}\cos x$$

$$\frac{y}{\sqrt{a^2 + b^2}} = \cos\alpha \sin x + \sin\alpha \cos x$$

$y = \sqrt{a^2 + b^2}\ \sin(x + \alpha)$

$y_{max} = \sqrt{a^2 + b^2} \cdot 1 = \sqrt{a^2 + b^2}$

$y_{min} = -\sqrt{a^2 + b^2}$

Example $y = 5\cos 3x + 12\sin 3x$

$\sqrt{a^2 + b^2} = \sqrt{5^2 + 12^2} = 13$

$y \in [-13, 13]$

Ques. Find the Range of

1. $y = \cos 3x + \sin 3x$

Ans. $[-\sqrt{2}, \sqrt{2}]$

2. $y = \cos 2x - \sin 2x$

Ans. $[-\sqrt{2}, \sqrt{2}]$

3. $y = 3\cos x - 4\sin x + 7$

Ans. $3\cos x - 4\sin x \in [-5, 5] \Rightarrow y \in [7 - 5, 7 + 5] = [2, 12]$

4. $y = \sqrt{3}\sin x - \cos x$

Ans. $a = \sqrt{3}, b = -1$

$\sqrt{3 + 1} = \sqrt{a^2 + b^2} = 2$ $\quad y \in [-2, 2]$

5. $y = 6\sin x \cos x - 4\cos 2x$

Ans. $y = 3\sin 2x - 4\cos 2x \Rightarrow y \in [-5, 5]$

6. $y = \cos^2 x - 6\sin x \cos x + 3\sin^2 x + 2$

Ans. $3 + 2\sin^2 x - 6\sin x \cos x$
$y = 3 + 1 - \cos 2x - 3\sin 2x$

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$y = 4 - \cos 2x - 3\sin 2x$

$[\sqrt{1^2 + 3^2} = \sqrt{10}]$

$y_{max} = 4 + \sqrt{10}$

$y_{min} = 4 - \sqrt{10}$

Ques. Find the maximum value of the expression $\frac{1}{\sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta}$.

Ans. $D = \sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta$

$= 1 + 4\cos^2\theta + 3\sin\theta\cos\theta$

$= 1 + 2(1 + \cos 2\theta) + 3/2\sin 2\theta$

$D = 3 + 2\cos 2\theta + 3/2\sin 2\theta$

$D_{min} = 3 - \sqrt{2^2 + \left(\frac{3}{2}\right)^2} = 3 - \sqrt{4 + \frac{9}{4}}$

$= 3 - \sqrt{\frac{25}{4}} = 3 - 5/2 = 1/2$

$y_{max} = \frac{1}{D_{min}} = 2$

Ques. Prove that $5\cos\theta + 3\cos\left(\theta + \frac{\pi}{3}\right) + 3$ lies between -4 and 10.

Ans. $y = 5\cos\theta + 3[\cos\theta \cos \pi/3 - \sin\theta \sin \pi/3] + 3$

$y = 5\cos\theta + 3\left[\frac{1}{2}\cos\theta - \frac{\sqrt{3}}{2}\sin\theta\right] + 3$

$y = 5\cos\theta + 3/2\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta + 3$

$y = \cos\theta\left[5 + \frac{3}{2}\right] - \frac{3\sqrt{3}}{2}\sin\theta + 3$

$y = \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta + 3$

$y_{max} = 7 + 3 = 10$

$y_{min} = -7 + 3 = -4$

$\sqrt{a^2 + b^2} = \sqrt{\left(\frac{13}{2}\right)^2 + \left(\frac{3\sqrt{3}}{2}\right)^2}$

$= \left[\sqrt{\frac{196}{4}}\right]$

$= 7$

Range Using AM - GM Inequality

For +ve numbers AM ≥ GM

Equality holds (AM = GM) when given numbers are equal

Example: 3, 27

$A.m = \frac{3 + 27}{2} = \frac{30}{2} = 15$

$G.m = \sqrt{3 \times 27} = 9$

Here $15 > 9$

MAXIMUM & MINIMUM VALUES

1. Minimum value of $a^2\tan^2\theta + b^2\cot^2\theta$ is $2ab$ where $a, b > 0$

Proof: $y = a^2\tan^2\theta + b^2\cot^2\theta$

$a^2 + \tan^2\theta, b^2\cot^2\theta$ (+ve numbers)

$AM \geq GM$

$\Rightarrow \frac{a^2\tan^2\theta + b^2\cot^2\theta}{2} \geq \sqrt{a^2\tan^2\theta\, b^2\cot^2\theta}$

$\Rightarrow \frac{a^2\tan^2\theta + b^2\cot^2\theta}{2} \geq ab$

$y_{min} = 2ab$ or $y \in [2ab, \infty]$

2. Minimum value of $a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta$ is $(a+b)^2$ where $a, b > 0$

Proof: $y = a^2(1 + \tan^2\theta) + b^2(1 + \cot^2\theta)$

$= a^2 + a^2\tan^2\theta + b^2 + b^2\cot^2\theta$

$\Rightarrow y = a^2 + b^2 + a^2\tan^2\theta + b^2\cot^2\theta$

$[\because a^2\tan^2\theta + b^2\cot^2\theta \geq 2ab$

$y_{min} = a^2 + b^2 + 2ab \Rightarrow y_{min} = (a + b)^2$

Ques. Find range of $y = 16\tan^2\theta + 64\cot^2\theta$

Ans. $a^2 = 16 \Rightarrow a = 4$

$b^2 = 64 \Rightarrow b = 8$

Range = $y \in [2ab, \infty]$

$\Rightarrow y \in [2 \times 4 \times 8, \infty)$

$\Rightarrow y \in [64, \infty)$

Ques. Find the value of $y = 4\sec^2\theta + 9\operatorname{cosec}^2\theta$

Ans. $y_{min} = (a + b)^2 = (2 + 3)^2 = 25$

Ques. The minimum value of $3\tan^2\theta + 12\cot^2\theta$ is

(a) 6 (b) 15 (c) 24 (d) 12

Ans. $a^2 = 3 \Rightarrow a = \sqrt{3}$

$b^2 = 12 \Rightarrow b = \sqrt{12}$

$y_{min} = 2ab = 2\sqrt{3}\sqrt{12} = 2\sqrt{36} = 2 \times 6 = 12$

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Range Using Completing the Square Methods

Ques. $y = 2\sin^2 x - 3\sin x + 4$

Ans. $\sin x = t$

$y = 2t^2 - 3t + 4$

$= 2\left[t^2 - \frac{3t}{2} + 2\right] = 2\left[\left(t - \frac{3}{4}\right)^2 - \frac{9}{16} + 2\right]$

$= 2\left[\left(t - \frac{3}{4}\right)^2 + \frac{23}{16}\right]$

$\Rightarrow y = \frac{23}{8} + 2\left(\sin x - \frac{3}{4}\right)^2$

$y_{min} = \frac{23}{8} + 2 \times 0$ when $\sin x - 3/4 = 0$ {possible}

$y_{min} = \frac{23}{8}$

for y_{max} put $\sin x = -1$

$y_{max} = 2(-1)^2 - 3(-1) + 4$

$= 2 + 3 + 4 = 9$

$y \in \left[\frac{23}{8}, 9\right]$

Ques. $y = 2\sin^2 x - 5\sin x - 1$

Ans. $\sin x = t \Rightarrow y = 2t^2 - 5t - 1$

$\Rightarrow y = 2\left[t^2 - \frac{5t}{4} - \frac{1}{2}\right]$

$\Rightarrow y = 2\left[\left(t - \frac{5}{4}\right)^2 - \frac{25}{16} - \frac{8}{16}\right]$

$\Rightarrow y = 2\left[\left(t - \frac{5}{4}\right)^2 - \frac{33}{16}\right]$

$\Rightarrow y = \frac{-33}{8} + 2\left[\sin x - \frac{5}{4}\right]^2$

For y_{min}, $\sin x - 5/4 = 0 = \sin x = 5/4$ {not possible}

So, for y_{min}, we put $\sin x = 1$

y_{max} we put $\sin x = -1$

$y_{min} = 2 - 5 - 1 = -4$

$y_{max} = 2(-1)^2 - 5(-1) - 1 = 2 + 5 - 1 = 6.$

$y \in [-4, 6]$

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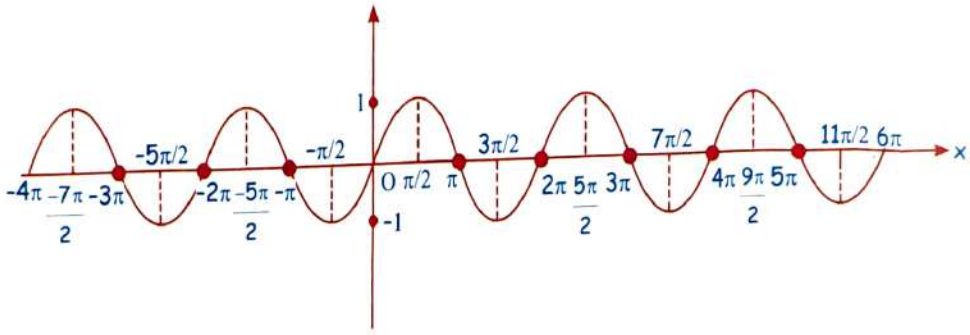
MEASURE OF AN ANGLE IN RADIANS

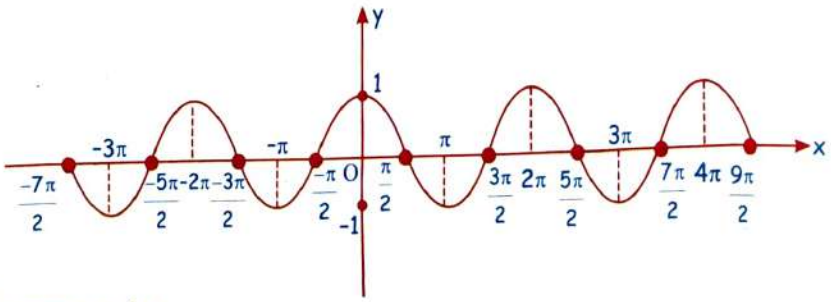
- π radians = 180°
- Length of an arc = $r\theta$
- Area of a sector = $\frac{1}{2} r^2\theta$

GRAPHS OF TRIGONOMETRIC FUNCTIONS

(i) y = sinx



(ii) y = cosx

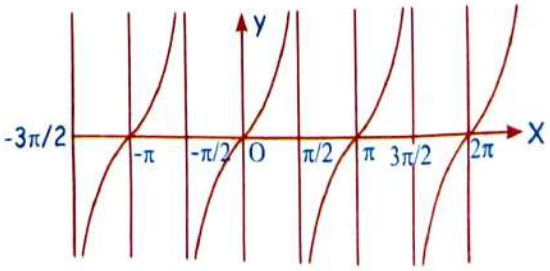


- If $x \in (0, \frac{\pi}{4})$, cos x > sin x,

  If $x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ sinx > cosx,

  If $x = \frac{\pi}{4}$, sin x = cos x

(iii) y = tanx



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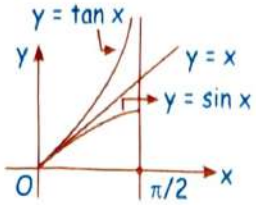
- If $x \in (0, \frac{\pi}{2})$, $\sin x < x < \tan x$



FORMULAS FOR NEGATIVE ANGLES

- $\sin(-\theta) = -\sin\theta$
- $\operatorname{cosec}(-\theta) = -\operatorname{cosec}\theta$
- $\cos(-\theta) = \cos\theta$
- $\sec(-\theta) = \sec\theta$
- $\tan(-\theta) = -\tan\theta$
- $\cot(-\theta) = -\cot\theta$

TRIGONOMETRIC RATIOS OF THE SUM & DIFFERENCE OF TWO ANGLES

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$.
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$.
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\cot(A + B) = \dfrac{\cot B \cot A - 1}{\cot B + \cot A}$
- $\cot(A - B) = \dfrac{\cot B \cot A + 1}{\cot B - \cot A}$
- $\sin(A + B) \cdot \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
- $\cos(A + B) \cdot \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$
- $\tan\left(\frac{\pi}{4} + x\right) = \dfrac{1 + \tan x}{1 - \tan x}$
- $\tan\left(\frac{\pi}{4} - x\right) = \dfrac{1 - \tan x}{1 + \tan x}$
- $\sin(15^\circ) = \sin\dfrac{\pi}{12}$

  $= \dfrac{\sqrt{3} - 1}{2\sqrt{2}} = \cos(75^\circ) = \cos\dfrac{5\pi}{12}$
- $\sin(75^\circ) = \sin\dfrac{5\pi}{12}$

  $= \dfrac{\sqrt{3} + 1}{2\sqrt{2}} = \cos(15^\circ) = \cos\dfrac{\pi}{12}$
- $\tan(15^\circ) = \tan\dfrac{\pi}{12}$

  $= 2 - \sqrt{3} = \cot(75^\circ) = \cot\dfrac{5\pi}{12}$
- $\tan(75^\circ) = \tan\dfrac{5\pi}{12}$

  $= 2 + \sqrt{3} = \cot(15^\circ) = \cot\dfrac{\pi}{12}$
- If $A + B = \dfrac{\pi}{4}$, then $(1 + \tan A) \cdot (1 + \tan B) = 2$
- If $A + B = 45^\circ$, then $(\cot A - 1)(\cot B - 1) = 2$
- $\tan(A + B + C)$

  $= \dfrac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - (\tan A \tan B + \tan A \tan C + \tan B \tan C)}$

FORMULAE TO TRANSFORM THE PRODUCT INTO SUM OR DIFFERENCE (DEFACTORIZATION FORMULAE)

- $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$.
- $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$.
- $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$
- $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

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FORMULAE TO TRANSFORM SUM OR DIFFERENCE INTO PRODUCT (FACTORIZATION FORMULAE)

- $\sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$
- $\sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right) \sin\left(\frac{C-D}{2}\right)$
- $\cos C + \cos D = 2 \cos\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$
- $\cos C - \cos D = 2 \sin\left(\frac{C+D}{2}\right) \sin\left(\frac{D-C}{2}\right)$

TRIGONOMETRIC IDENTITIES FOR MULTIPLE ANGLES

- $\sin 2\theta = 2 \sin\theta \cos\theta = \frac{2\tan\theta}{1+\tan^2\theta}$
- $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$
  $= 1 - 2\sin^2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$
- $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$
- $\tan(22.5^\circ) = \tan\frac{\pi}{8} = \sqrt{2}-1 = \cot(67.5^\circ) = \cot\frac{3\pi}{8}$
- $\tan(67.5^\circ) = \tan\frac{3\pi}{8} = \sqrt{2}+1 = \cot(22.5^\circ) = \cot\frac{\pi}{8}$
- $1 + \cos 2\theta = 2\cos^2\theta$
- $1 - \cos 2\theta = 2\sin^2\theta$

TRIGONOMETRIC IDENTITIES FOR TRIPLE ANGLE

- $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$
- $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$
- $\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$
- $\sin 18^\circ = \sin\frac{\pi}{10} = \frac{\sqrt{5}-1}{4} = \cos 72^\circ = \cos\frac{2\pi}{5}$
- $\cos 36^\circ = \cos\frac{\pi}{5} = \frac{\sqrt{5}+1}{4} = \sin 54^\circ = \sin\frac{3\pi}{10}$
- $\sin\theta \sin(60^\circ - \theta)\sin(60^\circ + \theta) = \frac{1}{4}\sin 3\theta$
- $\cos\theta . \cos(60^\circ - \theta) \cos(60^\circ + \theta) = \frac{1}{4}\cos 3\theta$
- $\tan\theta \tan(60^\circ - \theta) \tan(60^\circ + \theta) = \tan 3\theta$
- $\cot\theta \cot(60^\circ - \theta) \cot(60^\circ + \theta) = \cot 3\theta$
- $\tan\theta + \tan(60^\circ + \theta) + \tan(120^\circ + \theta) = 3\tan 3$
- $\cot\theta + \cot(60^\circ + \theta) + \cot(120^\circ + \theta) = 3\cot 3$

CONDITIONAL TRIGONOMETRIC IDENTITIES

If $A + B + C = 180^\circ$, then

- $\tan A + \tan B + \tan C = \tan A \tan B \tan C$
- $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$
- $\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$
- $\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}$
- $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$
- $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$
- $\sin A + \sin B + \sin C = 4 \cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$
- $\cos A + \cos B + \cos C = 1 + 4 \sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

IMPORTANT INEQUALITIES IN A TRIANGLE

- In an acute angled triangle ABC $\tan A + \tan B + \tan C \geq 3\sqrt{3}$
- In any triangle ABC, $\cos A + \cos B + \cos C \leq \frac{3}{2}$
- In any triangle ABC, $\sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2} \leq \frac{1}{8}$

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SUM OF SINES OR COSINES OF "n" ANGLES IN A.P.

- Sum of sines of "n" angles in AP

$$\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin(\alpha+\overline{n-1}\beta) = \frac{\sin n\beta/2}{\sin\beta/2}\sin\left(\alpha+\frac{n-1}{2}\beta\right)$$

- Sum of cosines of "n" angles in AP

$$\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots\ldots + \cos(\alpha+\overline{n-1}\beta) = \frac{\sin n\beta/2}{\sin\beta/2}\cos\left(\alpha+\frac{n-1}{2}\beta\right)$$

- Product of cosines of " n " angles in which are being doubled

$$\cos A\cos 2A\cos 4A\cos 8A \ldots\ldots \cos 2^{n-1}A = \frac{1}{2^n\sin A}\sin\left(2^n A\right)$$

MAXIMUM & MINIMUM VALUES OF TRIGONOMETRIC EXPRESSIONS

- $y = a\cos\theta + b\sin\theta$ will always lie in the interval $[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}]$ i.e. the maximum and minimum values of y are $\sqrt{a^2+b^2}$, $-\sqrt{a^2+b^2}$ respectively
- Minimum value of $a^2\tan^2\theta + b^2\cot^2\theta$ is $2ab$ where $a, b > 0$
- Minimum value of $a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta = (a+b)^2$ where $a, b > 0$

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Chapter

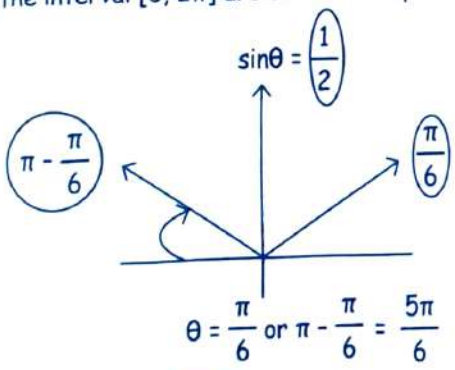
4 Trigonometric Equations

TRIGONOMETRIC EQUATION VS IDENTITY

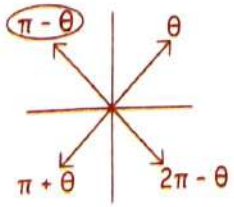
Identity	Equation
(1) $\sin^2\theta + \cos^2\theta = 1$	(1) $\sin\theta + \cos\theta = 1$
(2) $\sin 2\theta = 2\sin\theta\cos\theta$	(2) $\sin\theta = \frac{1}{2}$
Identities are true in the entire domain.	Equations are not true for all θ in domain. We will solve for θ

PRINCIPAL SOLUTIONS

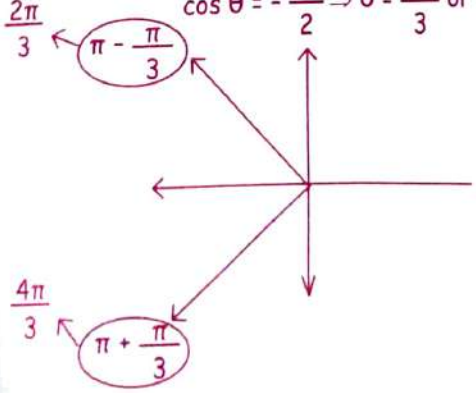
The solutions of the Trigonometric Equation which lie in the interval $[0, 2\pi]$ are called Principal Solutions.



$$\theta = \frac{\pi}{6} \text{ or } \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$



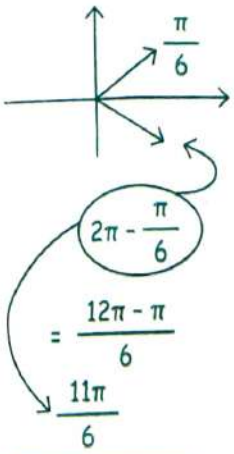
$$\cos\theta = -\frac{1}{2} \Rightarrow \theta = \frac{2\pi}{3} \text{ or } \frac{4\pi}{3}$$



Ques. Find the principal solutions for the following

1. $\cos\theta = \frac{\sqrt{3}}{2}$

Ans.



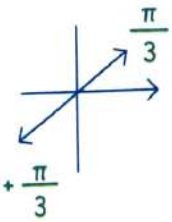
$$= \frac{12\pi - \pi}{6}$$

$$\frac{11\pi}{6}$$

$$\therefore \theta = \frac{\pi}{6}, \frac{11\pi}{6}$$

2. $\tan\theta = \sqrt{3}$

Ans.

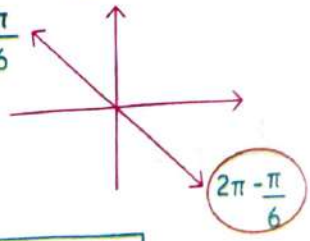


$$= \frac{4\pi}{3}$$

$$\theta = \frac{\pi}{3} \text{ or } \frac{4\pi}{3}$$

3. $\tan\theta = -\frac{1}{\sqrt{3}}$

Ans.



$$\therefore \theta = \frac{5\pi}{6} \text{ or } \frac{11\pi}{2}$$

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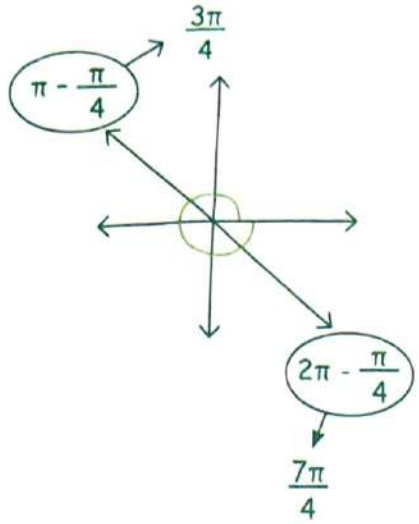
Ques. Find the principal solution of $\tan x = -1$

(a) $\frac{\pi}{4}, \frac{3\pi}{4}$ (b) $\frac{3\pi}{4}, \frac{5\pi}{4}$

(c) $\frac{5\pi}{4}, \frac{7\pi}{4}$ (d) $\frac{3\pi}{4}, \frac{7\pi}{4}$

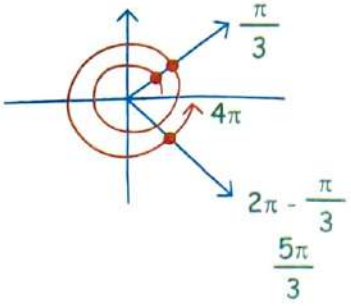
Ans.



Ques. Find the principal solutions for the following:

1. $\cos 2\theta = \frac{1}{2}$

Ans.



$2\theta = \frac{\pi}{3}$ or $\frac{5\pi}{3}, 2\pi + \frac{\pi}{3}, 2\pi + \frac{5\pi}{3}$

$2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

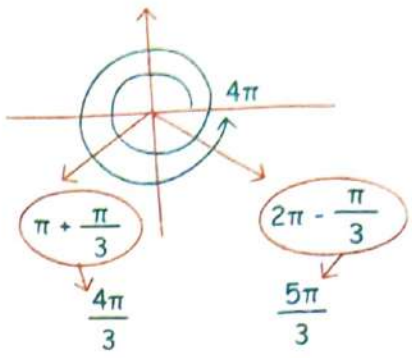
↓

4 Principal Solution.

2. $\sin 2\theta = -\frac{\sqrt{3}}{2}$

Ans. $2\theta = \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{4\pi}{3} + 2\pi, \frac{5\pi}{3} + 2\pi$

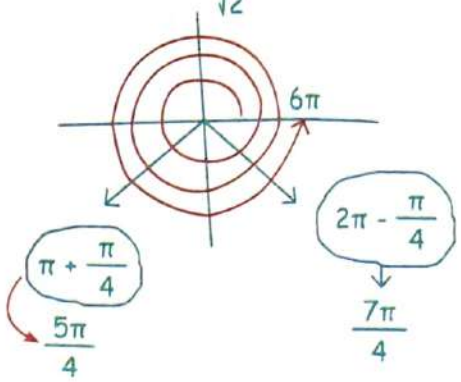
$2\theta = \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{10\pi}{3}, \frac{11\pi}{3}$



$\therefore \theta = \frac{4\pi}{6}, \frac{5\pi}{6}, \frac{10\pi}{6}, \frac{11\pi}{6}$

3. $\sin 3\theta = -\frac{1}{\sqrt{2}}$

Ans. $\sin 3\theta = -\frac{1}{\sqrt{2}}$



$3\theta = \frac{5\pi}{4}, \frac{7\pi}{4}, 2\pi + \frac{5\pi}{4}, 2\pi + \frac{7\pi}{4},$

$4\pi + \frac{5\pi}{4}, 4\pi + \frac{7\pi}{4}$

$3\theta = \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{13\pi}{4}, \frac{15\pi}{4}, \frac{21\pi}{4}, \frac{23\pi}{4}$

$\theta = \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{13\pi}{12}, \frac{15\pi}{12}, \frac{21\pi}{12}, \frac{23\pi}{12}$

Ques. Find the principal solution of $\sin 2x = -\frac{1}{2}$

(a) $\frac{7\pi}{6}, \frac{11\pi}{6}$

(b) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{15\pi}{12}, \frac{19\pi}{12}$

(c) $\frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$

(d) None of these

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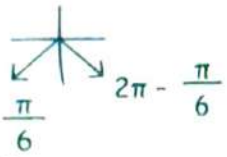
Ans.



$$2x = \frac{7\pi}{6}, \frac{11\pi}{6}, 2\pi + \frac{7\pi}{6}, 2\pi + \frac{11\pi}{6}$$

$$2x = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}, \frac{23\pi}{6}$$

$$x = \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$$

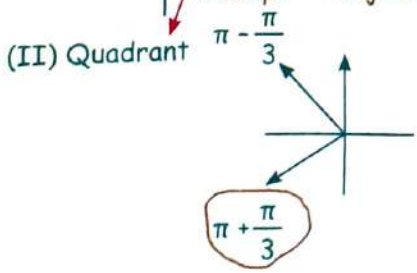
Ques. The number of solutions of the equation,

$$|\cot x| = \cot x + \frac{1}{\sin x} \quad (0 \le x \le 2\pi) \text{ is} \qquad \textbf{[PYQ]}$$

(a) 0 (b) 1 (c) 2 (d) 3

Ans.

$\cot x > 0$	$\cot x < 0$
$\cot x = \cot x + \frac{1}{\sin x}$	$-\cot x = \cot x + \frac{1}{\sin x}$
$0 = \frac{1}{\sin x}$	$-2\cot x = \frac{1}{\sin x}$
$\Rightarrow$ Not possible	$-\frac{2\cos x}{\sin x} = \frac{1}{\sin x}$
	$-2\cos x = 1$
	$\cos x = -\frac{1}{2}$
	$x = \frac{2\pi}{3}$ or $\frac{4\pi}{3}$
	Accept: $\frac{2\pi}{3}$ (II) Quadrant; Reject: $\frac{4\pi}{3}$ (IV) Quadrant



Ques. The values of x between 0 and 2π, which satisfy the equation $\sin x\sqrt{8\cos^2 x} = 1$ are in increasing A.P. with common difference? **[PYQ]**

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{8}$ (c) $\frac{3\pi}{8}$ (d) $\frac{5\pi}{8}$

Ans. $\sin x\sqrt{8}\,|\cos x| = 1$

$2\sqrt{2}\sin x\,|\cos x| = 1$

Case-1:

$\cos x \ge 0$

$2\sqrt{2}\sin x\cos x = 1$

$\sin 2x = \frac{1}{\sqrt{2}}$

$$2x = \frac{\pi}{4}, \frac{3\pi}{4}, 2\pi + \frac{\pi}{4}, 2\pi + \frac{3\pi}{4}$$

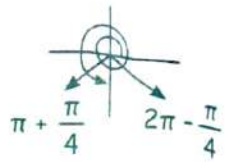
$$2x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}$$

$$x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$$

(I) Quadrant (III) Quadrant

Case-2:

$\cos x < 0$



$-2\sqrt{2}\sin x\cos x = 1$

$\sin 2x = -\frac{1}{\sqrt{2}}$

$$2x = \frac{5\pi}{4}, \frac{7\pi}{4}, 2\pi + \frac{5\pi}{4}, 2\pi + \frac{7\pi}{4}$$

$$2x = \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{13\pi}{4}, \frac{15\pi}{4}$$

$$x = \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$$

II II IV IV

Final value of $x = \left\{\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}\right\} \rightarrow$ A.P.

Common difference $= \frac{2\pi}{8}$

Ques. The number of distinct solutions of the equation $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$ in the interval $[0, 2\pi]$, is **[JEE Main 202**

Ans. $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$

$\log_{1/2}|\sin x| + \log_{1/2}|\cos x| = 2$

$\log_{1/2}(|\sin x|\,|\cos x|) = 2$ [$\because |a|\,|b| = |ab$

$|\sin x|\,|\cos x| = (1/2)^2$

$|\sin x.\cos x| = \frac{1}{4}$

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$|2 \sin x \cos x| = \frac{1}{2}$

$|\sin 2x| = \frac{1}{2}$

$\sin 2x = \pm \frac{1}{2}$

$\sin 2x = \frac{1}{2}$ → 4 Solutions

$\sin 2x = -\frac{1}{2}$ → 4 Solutions

Hence, the number of distinct solutions of the equations, = 8 Solutions

Ques. Find the principal solutions for the following:

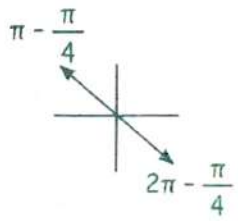
1. $\sin x + \cos x = 0$

Ans. $\sin x + \cos x = 0$

$\Rightarrow \sin x = -\cos x$

$\Rightarrow \frac{\sin x}{\cos x} = -1$

$\Rightarrow \boxed{\tan x = -1}$



$x = \frac{3\pi}{4}, \frac{7\pi}{4}$

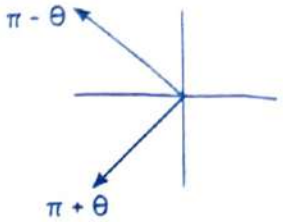
2. $\sin\theta = \frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{-1}{2}$

Ans. $\sin\theta = \frac{\sqrt{3}}{2}$

$\cos\theta = \frac{-1}{2}$

$\Rightarrow \theta = \frac{2\pi}{3}$

3. $\cos\theta = -\frac{1}{3}$



Ans. $\theta = \cos^{-1}\frac{1}{3}$

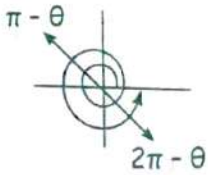
$\theta = \pi - \cos^{-1}\frac{1}{3}$

or

$\theta = \pi + \cos^{-1}\frac{1}{3}$

4. $\boxed{\tan 2\theta = -\sqrt{2}}$

Ans. $2\theta = \tan^{-1}(-\sqrt{2})$



$2\theta = \pi - \tan^{-1}\sqrt{2},\ 2\pi - \tan^{-1}\sqrt{2},$

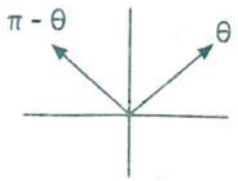
$3\pi - \tan^{-1}\sqrt{2},\ 4\pi - \tan^{-1}\sqrt{2}$

$\theta = \frac{\pi - \tan^{-1}\sqrt{2}}{2}, \frac{2\pi - \tan^{-1}\sqrt{2}}{2},$

$\frac{3\pi - \tan^{-1}\sqrt{2}}{2}, \frac{4\pi - \tan^{-1}\sqrt{2}}{2}$

Ques. Find Principal solution of $\sin\theta = \frac{1}{5}$

Ans. $\theta = \sin^{-1}\frac{1}{5}$



$\sin^{-1}\frac{1}{5}$ & $\pi - \sin^{-1}\frac{1}{5}$

Ques. The number of elements in the set $S = \{\theta \in [-4\pi, 4\pi]\}$:

$3\cos^2 2\theta + 6\cos 2\theta - 10\cos^2\theta + 5 = 0$ is _____.

[JEE Main 2022 (29 june shift 2)]

Ans. $3\cos^2 2\theta + 6\cos 2\theta - 5[2\cos^2\theta - 1] = 0$

$\Rightarrow 3\cos^2 2\theta + 6\cos 2\theta - 5\cos 2\theta = 0$

$\Rightarrow 3\cos^2 2\theta + \cos 2\theta = 0$

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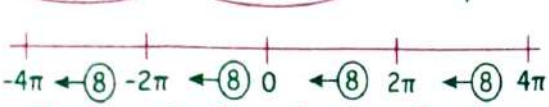
$\Rightarrow \cos 2\theta\,[3\cos 2\theta + 1] = 0$

either $\cos 2\theta = 0$ or $\cos 2\theta = -\frac{1}{3}$

4 Solutions + 4 Solutions = 8 Principal sol^n



Total = $8 \times 4 = 32$

Ques. If for $x \in \left(0, \frac{\pi}{2}\right)$, $\log_{10} \sin x + \log_{10} \cos x = -1$ and $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, $n > 0$ then the value of n is equal to:

[JEE Main 2021]

(a) 20 (b) 12 (c) 9 (d) 16

Ans. $\log_{10} \sin x + \log_{10} \cos x = -1$

$\Rightarrow \log_{10}(\sin x \cos x) = -1$

$[\because \log x + \log y = \log_{10} xy]$

$\Rightarrow \sin x \cos x = 10^{-1}$

$\Rightarrow \sin x \cos x = \frac{1}{10}$

$\because (\sin x + \cos x)^2 = 1 + 2\sin x \cos x$

$(\sin x + \cos x)^2 = 1 + 2 \times \frac{1}{10} = 1 + \frac{1}{5} = \frac{6}{5}$

$(\sin x + \cos x)^2 = 6/5$

$\therefore \log_{10}(\sin x + \cos x) = \frac{1}{2}[\log_{10} n - 1]$

$\Rightarrow \log_{10}(\sin x + \cos x)^2 = [\log_{10} n - 1]$

$\Rightarrow \log_{10}(6/5) = (\log n - 1)$

$\Rightarrow \log_{10}\left(\frac{12}{10}\right) = \log_{10} 12 - \log_{10} 10 = (\log_n - 1)$

$\Rightarrow \log 12 = \log n$

Therefore, n = 12

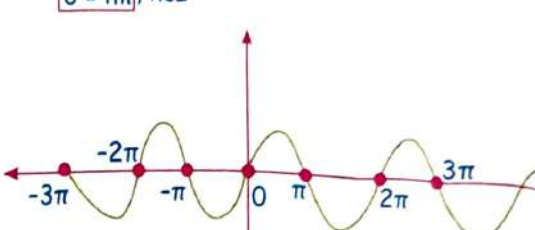
GENERAL SOLUTIONS

The solution consisting of all possible solutions of a trigonometric equation in general form is called general solution.

$\sin\theta = 0$

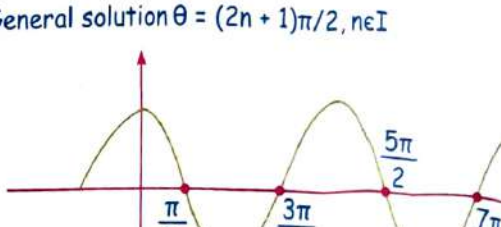
$\Rightarrow \theta = 0, \pi, 2\pi, 3\pi, 4\pi, \ldots$

$\theta = n\pi$, $n \in I$



$\cos\theta = 0 \Rightarrow \theta = \pi/2, 3\pi/2, 5\pi/2, 7\pi/2 \ldots$

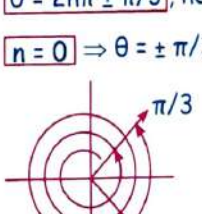
General solution $\theta = (2n+1)\pi/2$, $n \in I$



$\cos\theta = 1/2 \rightarrow$ general sol?

$\theta = 2n\pi \pm \pi/3$, $n \in I$

$n = 0 \Rightarrow \theta = \pm \pi/3$



$n = 1 \Rightarrow \theta = 2\pi \pm \pi/3$ → $2\pi + \pi/3$, $2\pi - \pi/3$

$n = 2 \Rightarrow \theta = 4\pi \pm \pi/3$ → $4\pi - \pi/3$

General solution of $\cos\theta = \cos\alpha$

$\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$, where $n \in I$ here α is any one sol^n

Proof: $\cos\theta = \cos\alpha$

$\cos\theta - \cos\alpha = 0$

$-2\sin\left(\frac{\theta + \alpha}{2}\right)\sin\left(\frac{\theta - \alpha}{2}\right) = 0$

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$\sin\left(\frac{\theta+\alpha}{2}\right)\sin\left(\frac{\theta-\alpha}{2}\right)=0$

either $\sin\left(\frac{\theta+\alpha}{2}\right)=0 \Rightarrow \frac{\theta+\alpha}{2}=n\pi \Rightarrow \theta+\alpha=2n\pi$

$\Rightarrow \boxed{\theta=2n\pi-\alpha}$

OR

$\sin\left(\frac{\theta-\alpha}{2}\right)=0 \Rightarrow \frac{\theta-\alpha}{2}=n\pi \Rightarrow \boxed{\theta=2n\pi+\alpha}$

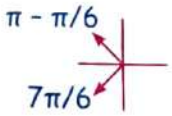
Therefore $\boxed{\theta=2n\pi\pm\alpha}$

Example:

1) $\boxed{\cos\theta=1/2}$

$\cos\theta=\cos\pi/3$

$\boxed{\theta=2n\pi\pm\pi/3}$, $n\in I$



2) $\cos\theta=\frac{1}{\sqrt{2}}=\cos\pi/4$

$\boxed{\theta=2n\pi\pm\pi/4}$

3) $\cos\theta=-\frac{\sqrt{3}}{2}$

$\boxed{\theta=2n\pi\pm5\pi/6}$, $n\in I$

or $n=0 \Rightarrow \theta=\pm5\pi/6$ (correct)

$\boxed{\theta=2n\pi\pm7\pi/6}$

$n=1 \Rightarrow \theta=2\pi-7\pi/6=\frac{12\pi-7\pi}{2}=\frac{5\pi}{6}$

$2\pi+7\pi/6$

Ques. Find the general solutions for the following

1. $\cos2\theta=\frac{\sqrt{3}}{2}$

Ans. $\cos2\theta=\cos\pi/6$

$2\theta=2n\pi\pm\pi/6$

$\boxed{\theta=n\pi\pm\pi/12}$

2. $\sec3\theta=-\sqrt{2}$

$\cos3\theta=\frac{-1}{\sqrt{2}}$

$\cos3\theta=\cos(3\pi/4)$

$3\theta=2n\pi\pm3\pi/4$

$\boxed{\theta=\frac{2n\pi}{3}\pm\pi/4}$

Ques. The general solution of the following equation:

$\cos3\theta=-\frac{1}{2}$

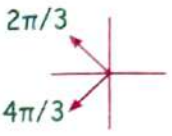
(a) $2n\pi\pm\frac{2\pi}{3}$ (b) $2n\pi\pm\frac{\pi}{3}$

(c) $\frac{2n\pi}{3}\pm\frac{2\pi}{9}$ (d) $\frac{2n\pi}{3}\pm\frac{4\pi}{3}$

Ans. $\cos3\theta=\cos2\pi/3$

$\Rightarrow 3\theta=2n\pi\pm2\pi/3$

$\Rightarrow \theta=\frac{2n\pi}{3}\pm\frac{2\pi}{9}$



General Solutions of $\tan\theta=\tan\alpha$

$\tan\theta=\tan\alpha \Rightarrow \theta=n\pi+\alpha$, $n\in I$

proof $\rightarrow \frac{\sin\theta}{\cos\theta}=\frac{\sin\alpha}{\cos\alpha}$

$\sin\theta\cos\alpha=\cos\theta\sin\alpha$

$\sin\theta\cos\alpha-\cos\theta\sin\alpha=0$

$\sin(\theta-\alpha)=0$

$(\theta-\alpha)=n\pi$

$\theta=n\pi+\alpha$

Example:

(i) $\tan\theta=\sqrt{3}$

$\tan\theta=\tan\pi/3$

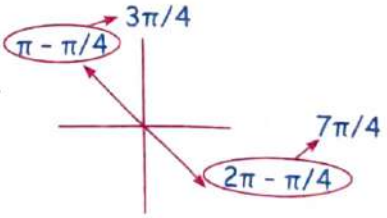
$\theta=n\pi+\pi/3$

(ii) $\tan\theta=-1$

$\tan\theta=\tan3\pi/4$

$\theta=n\pi+3\pi/4$

or $\theta=n\pi+7\pi/4$



OR

$\tan\theta=\tan(-\pi/4)$

$\theta=n\pi-\pi/4$ Ans

Best

Ques. The general solution of $\tan\left(\frac{2\theta}{3}\right)=\sqrt{3}$ is ____.

(a) $\frac{3n\pi}{2}+\frac{\pi}{2}$; $n\in I$ (b) $\frac{n\pi}{2}+\frac{\pi}{2}$; $n\in I$

(c) $n\pi\pm\frac{\pi}{2}$; $n\in I$ (d) None of these

Ans. $\tan\left(\frac{2\theta}{3}\right)=\tan\pi/3$

$\frac{2\theta}{3}=n\pi+\pi/3$

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$2\theta = 3n\pi + 3\pi/3$

$$\theta = \frac{3n\pi}{2} + \frac{\pi}{2}$$

General Solutions of $\sin\theta = \sin\alpha$

$\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n\alpha$, where $n \in I$

$(-1)^{even} = +1$

$(-1)^{odd} = -1$

proof: $\sin\theta - \sin\alpha = 0$

$$2\cos\left(\frac{\theta+\alpha}{2}\right)\sin\left(\frac{\theta-\alpha}{2}\right) = 0$$

$$\cos\left(\frac{\theta+\alpha}{2}\right) = 0 \Rightarrow \frac{\theta+\alpha}{2} = (2n+1)\pi/2$$

$\theta = (2n+1)\pi - \alpha$

OR

$$\sin\left(\frac{\theta-\alpha}{2}\right) = 0$$

$$\Rightarrow \frac{\theta-\alpha}{2} = n\pi$$

$\theta = 2n\pi + \alpha$

OR

union: $\theta = n\pi + (-1)^n\alpha$

$\sin\theta = 1/2$

$\sin\theta = \sin\pi/6 \Rightarrow \theta = n\pi + (-1)^n\pi/6$

$n = 0 \Rightarrow \theta = \pi/6$

$n = 1 \Rightarrow \theta = \pi + (-1)^1\pi/6$

$= \pi - \pi/6 = 5\pi/6$

$n = 2 \Rightarrow \theta = 2\pi + (-1)^2\pi/16 = 2\pi + \pi/6$

Ques. Find the general solutions for the following

1. $\sin 2\theta + \cos 2\theta = 0$

Ans. $\sin 2\theta = -\cos 2\theta \Rightarrow \tan 2\theta = -1$

$\tan 2\theta = \tan(-\pi/4)$

$2\theta = n\pi + (-\pi/4)$

$2\theta = n\pi - \pi/4$

$\theta = n\pi/2 - \pi/8$

2. $\sin 3\theta = -\frac{1}{2}$

Ans. $\sin 3\theta = \sin(-\pi/6)$

$3\theta = n\pi + (-1)^n(-\pi/6)$

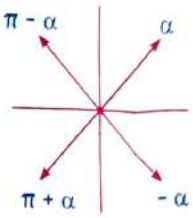
$\theta = n\pi/3 + (-1)^n(-\pi/18)$

General Solutions of $\sin^2\theta = \sin^2\alpha$, $\cos^2\theta = \cos^2\alpha$, $\tan^2\theta = \tan^2\alpha$

$\sin^2\theta = \sin^2\alpha$, $\cos^2\theta = \cos^2\alpha$, $\tan^2\theta = \tan^2\alpha$,

$\Rightarrow \theta = n\pi \pm \alpha$, $n \in I$

Proof:



$\tan^2\theta = 3$

$\tan\theta = \pm\sqrt{3}$

$\theta = n\pi \pm \alpha$

$n = 0 \Rightarrow \theta = \pm\alpha$

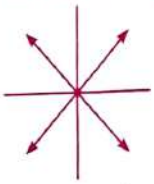
$n = 1 \Rightarrow \theta = \pi \pm \alpha$

$n = 2 \Rightarrow \theta = 2\pi \pm \alpha$

Ques. Find the general solutions for the following

1. $\sin^2\theta = 1/4$

Ans. $\sin^2\theta = 1/4 \Rightarrow \sin\theta = \pm 1/2$ → soln in all 4th quad



$\sin^2\theta = \sin^2\pi/6$

$\theta = n\pi \pm \pi/6$

2. $\cos^2\theta = 1/2 = \left(\frac{1}{\sqrt{2}}\right)^2 = \cos^2\pi/4$

Ans. $\theta = n\pi \pm \pi/4$

Ques. If $2\tan^2\theta = \sec^2\theta$, then the general value of θ is

(a) $n\pi + \frac{\pi}{4}$; $n \in I$ (b) $n\pi - \frac{\pi}{4}$; $n \in I$

(c) $n\pi \pm \frac{\pi}{4}$; $n \in I$ (d) $2n\pi \pm \frac{\pi}{4}$; $n \in I$

Ans. $2\tan^2\theta = 1 + \tan^2\theta$

$\tan^2\theta = 1$

$\tan^2\theta = \tan^2\pi/4$

$\theta = n\pi \pm \pi/4$

Ques. General solution of $\tan 5\theta = \cot 2\theta$ is

(a) $\theta = \frac{n\pi}{7} + \frac{\pi}{14}$ (b) $\theta = \frac{n\pi}{7} + \frac{\pi}{5}$

(c) $\theta = \frac{n\pi}{7} + \frac{\pi}{2}$ (d) $\theta = \frac{n\pi}{7} + \frac{\pi}{3}$, $n \in Z$

Ans. **Method 1:**

$$\frac{\sin 5\theta}{\cos 5\theta} = \frac{\cos 2\theta}{\sin 2\theta}$$

$\sin 5\theta \sin 2\theta = \cos 5\theta \cos 2\theta$

$\cos 5\theta \cos 2\theta - \sin 5\theta \sin 2\theta = 0$

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$\cos(5\theta + 2\theta) = 0$

$\cos 7\theta = 0$

$7\theta = (2n + 1)\pi/2$

$\theta = (2n + 1)\frac{\pi}{14}$

$\theta = n\pi/7 + \pi/14$

Method 2:

$\tan 5\theta = \cot 2\theta$

$\tan(5\theta) = \tan(\pi/2 - 2\theta)$

$\tan\theta = \tan\alpha$ ($\alpha = \pi/2 - 2\theta$)

$\theta = n\pi + \alpha$

$5\theta = n\pi + (\pi/2 - 2\theta)$

$7\theta = n\pi + \pi/2$

$\theta = n\pi/7 + \pi/14$

Ques. If $\tan\theta = \cot 2\theta$ then the value of θ is

(a) $(2n + 1)\frac{\pi}{6}$; $n \in I$

(b) $(2n + 1)\frac{\pi}{6}$; $n \in I$; $n \neq 3m + 1$; $m \in I$

(c) $(2n + 1)\frac{\pi}{6}$; $n \in I$; $n \neq 2m + 1$; $m \in I$

(d) None

Ans. $\tan\theta = \tan(\pi/2 - 2\theta)$

$\theta = n\pi + \pi/2 - 2\theta$

$3\theta = n\pi + \pi/2$

$3\theta = \pi[n + 1/2]$

$3\theta = (2n + 1)\pi/2$

$\theta = (2n + 1)\pi/6$, $n \in I$, $n \neq 1, 4, 7, 10, \ldots\ldots$

$n = 0 \Rightarrow \theta = \pi/6$

$n = 1 \Rightarrow \theta = 3\pi/6 = \pi/2$ → Ques is N.D as $\tan\theta$ is N.D

$n = 2 \Rightarrow \theta = 5\pi/6$

$n = 3 \Rightarrow \theta = 7\pi/6$

$n = 4 \Rightarrow \theta = 9\pi/6 = 3\pi/2$

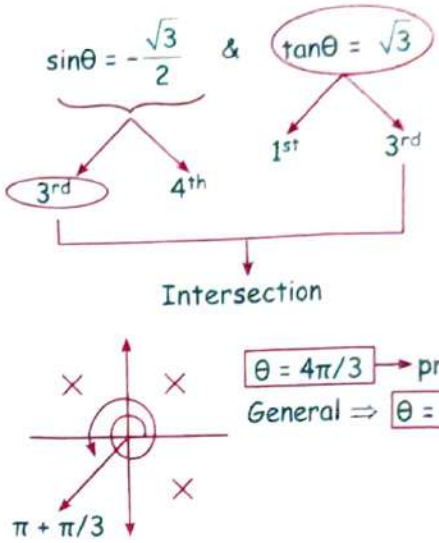
$n \neq 1, 4, 7, 10, \ldots\ldots \Rightarrow n \neq 3m + 1, n \in I$

Ques. The most general value of θ which satisfies both the equations $\tan\theta = \sqrt{3}$ and $\operatorname{cosec}\theta = -\frac{2}{\sqrt{3}}$ is

(a) $n\pi + \frac{4\pi}{3}$; $n \in I$ (b) $n\pi + \frac{2\pi}{3}$; $n \in I$

(c) $2n\pi + \frac{4\pi}{3}$; $n \in I$ (d) $2n\pi + \frac{2\pi}{3}$; $n \in I$

Ans.



Ques. The number of solution of the equation

$\cos\left(x + \frac{\pi}{3}\right)\cos\left(\frac{\pi}{3} - x\right) = \frac{1}{4}\cos^2 2x$, $x \in [-3\pi, 3\pi]$ is **[JEE Main 2022]**

(a) 8 (b) 5 (c) 6 (d) 7

Ans. $2\cos(x + \frac{\pi}{3})\cos(\frac{\pi}{3} - x) = \frac{1}{2}\cos^2 2x$

$\Rightarrow [\cos(\frac{2\pi}{3}) + \cos 2x] = \frac{1}{2}\cos^2 2x$

$\Rightarrow -\frac{1}{2} + \cos 2x = \frac{1}{2}\cos^2 2x$

Let $\cos 2x = t$

$\frac{-1}{2} + t = \frac{1}{2}t^2$

$\frac{-1 + 2t}{2} = \frac{t^2}{2}$

$\Rightarrow t^2 - 2t + 1 = 0 \Rightarrow (t - 1)^2 = 0$

$\Rightarrow t - 1 = 0 \Rightarrow t = 1$

$\cos 2x = 1$

$\cos 2x = \cos 0$

$\Rightarrow 2x = 2n\pi \pm 0 \Rightarrow 2x = 2n\pi$

$\therefore x = n\pi$

$n = 0, x = 0$

$n = 1, x = \pi$

$n = 2, x = 2\pi$

$n = 3, x = 3\pi$

$n = -1, x = -\pi$

$n = -2, x = -2\pi$

$n = -3, x = -3\pi$

Total solutions = 7

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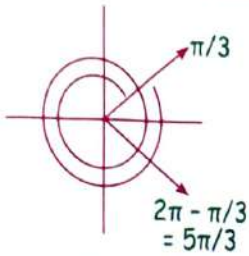
FINDING PRINCIPLE SOLUTION FROM GENERAL SOLUTION

Ques. Find solutions of $\cos 2x = 1/2$ in the interval $[0, 2\pi]$

Ans. **Method 1:** jalebi method



$2x = \pi/3, 5\pi/3, 2\pi + \pi/3, 2\pi + 5\pi/3$

$2x = \pi/3, 5\pi/3, 7\pi/3, 11\pi/3$

$x = \pi/6, 5\pi/6, 7\pi/6, 11\pi/6$

Method 2: By general solution

$\cos 2x = 1/2 = \cos \pi/3$

$2x = 2n\pi \pm \pi/3$

$x = n\pi \pm \pi/6$

$x \in [0, 2\pi]$

$n = 0 \Rightarrow x = \pm \pi/6 \Rightarrow \boxed{x = \pi/6}$

$n = 1 \Rightarrow x = \pi \pm \pi/6 \begin{cases} 5\pi/6 \\ 7\pi/6 \end{cases}$

$n = 2 \Rightarrow x = 2\pi - \pi/6 = 11\pi/6$

$n = 3 \Rightarrow x = 3\pi \pm \pi/6 \rightarrow$ ignore

Ques. Find the number of solution of $\tan 3x = 1$ in the interval $x \in \left[-\frac{2\pi}{3}, \frac{5\pi}{6}\right]$, $x \in [-120°, 150°]$

Ans. $\tan 3x = \tan \pi/4$

$3x = n\pi + \pi/4$

$\boxed{x = \frac{n\pi}{3} + \frac{\pi}{12}}$

$n = 0 \Rightarrow x = \pi/12 = (15°)$ ✓

$n = 1 \Rightarrow x = \pi/3 + \pi/12 = 60° + 15° = (75°)$ ✓

$n = 2 \Rightarrow x = 2\pi/3 + \pi/12 = 120° + 15° = (135°)$ ✓

$n = 3 \Rightarrow x = 3\pi/3 + \pi/12 = 180° + 15° = 195°$ (X)

$n = -1 \Rightarrow x = -\pi/3 + \pi/12 = -60° + 15° = (-45°)$ ✓

$n = -2 \Rightarrow x = -2\pi/3 + \pi/12 = -120° + 15° = (-105°)$ ✓

$n = -3 \Rightarrow x = -\pi + \pi/12 = -180° + 15° = -165°$ (X)

Exact Answer:

$x = 15°, 75°, 135°, -46°, -105°$

TYPES OF TRIGONOMETRIC EQUATIONS

$\sin x = 0 \Rightarrow x = n\pi$

$\cos x = 0 \Rightarrow x = (2n + 1)\frac{\pi}{2}$

Type-1:

Trigonometric equations which can be solved b factorization.

Example: Solve $2\sin^2 x \cos x - \cos x = 0$, where $0 \le x \le 2$

Method 1: By general solution $\cos x [2\sin^2 x - 1] = 0$

either $\cos x = 0$ or $2\sin^2 x - 1 = 0 \Rightarrow 2\sin^2 x = 1$

$x = (2n + 1)\frac{\pi}{2}$

$n = 0 \Rightarrow x = (\pi/2)$ ✓

$n = 1 \Rightarrow x = (3\pi/2)$ ✓

$n = 2 \Rightarrow x = 5\pi/2$

$\Rightarrow \sin^2 x = 1/2 = \left(\frac{1}{\sqrt{2}}\right)^2$

$\Rightarrow \sin^2 x = \sin^2 \pi/4$

$\Rightarrow \boxed{x = n\pi \pm \pi/4}$

$n = 0 \Rightarrow x = (\pi/4)$ ✓

$n = 1 \Rightarrow x = \pi - \pi/4 = (3\pi/4)$ ✓

$\Rightarrow \pi + \pi/4 = (5\pi/4)$ ✓

$n = 2 \Rightarrow x = 2\pi - \pi/4$

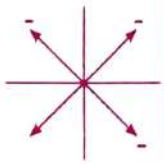
$= (7\pi/4)$ ✓

Method 2: Jalebi method

$\cos x = 0$

$\sin^2 x = 1/2$

$\sin x = \pm \frac{1}{\sqrt{2}}$



OR

$\therefore 2\sin^2 x - 1 = 0$

$\Rightarrow -(1 - 2\sin^2 x) = 0$

$\Rightarrow \cos 2x = 0$

$\Rightarrow 2x = (2n + 1)\pi/2$

$\Rightarrow x = (2n + 1)\pi/4$

Important Note

1. $x^2 = x \Rightarrow x^2 - x = 0 \Rightarrow x(x - 1) = 0$

   $\Rightarrow x = 0$ or $x = 1$

   Alternate method: $x^2 = x$

   $\Rightarrow x = 1$ or $x = 0$

2. $(x - 1)^2 = 5(x - 1) \Rightarrow x - 1 = 5 \Rightarrow x = 6$

   or $x - 1 = 0 \Rightarrow x = 1$

3. $(\ln x)^2 = 5(\ln x)$

   $\ln x = 5$ or $\ln x = 0$

   $\Rightarrow x = e^5$ or $x = e^0 = 1$

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Type-2:

Trigonometric equation which can be solved by reducing then into quadratic equation.

solve: $16^{\sin^2 x} + 16^{\cos^2 x} = 10$, where $0 \le x \le 2\pi$

$$16^{\sin^2 x} + 16^{1-\sin^2 x} = 10$$

$$16^{\sin^2 x} + \frac{16^1}{16^{\sin^2 x}} = 10$$

Let $16^{\sin^2 x} = t$

$$t + \frac{16}{t} = 10$$

$$\frac{t^2 + 16}{t} = 10$$

$$t^2 - 10t + 16 = 0$$

$$(t-8)(t-2) = 0$$

$t = 8$ or $t = 2$

$16^{\sin^2 x} = 8$ or 2

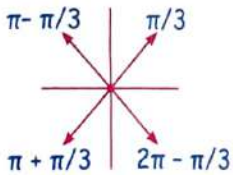
$2^{4\sin^2 x} = 2^3$ or 2^1

$\Rightarrow 4\sin^2 x = 3$ or $4\sin^2 x = 1$

$\Rightarrow \sin^2 x = 3/4$ or $\sin^2 x = 1/4$

$\sin x = \pm \sqrt{3}/2$ or $\sin x = \pm 1/2$

$x = \pi/6, \pi - \pi/6, \pi + \pi/6, 2\pi - \pi/6$



Ques. The number of solutions of the equation $\sin x = \cos^2 x$ in the interval $(0, 10)$ is _____.

[JEE Main 29 june 2022 shift 2]

Ans. $\sin x = 1 - \sin^2 x$

Let, $\sin x = t$

$t = 1 - t^2$

$t^2 + t - 1 = 0$

$$t = \frac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2 \times 1}$$

$$t = \frac{-1 \pm \sqrt{5}}{2}$$

$$\sin x = \frac{-1 \pm \sqrt{5}}{2} \rightarrow \text{H.W}$$

Type-3:

sum/diff → product

Trigonometric equations which can be solved by transforming a sum or difference of trigonometric ratios into their product.

Example: Find the values of x, $0 \le x \le 2\pi$ such that

$\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x$

$\sin x + \sin 3x + \sin 2x = \cos x + \cos 3x + \cos 2x$

$2\sin 2x \cos x + \sin 2x = 2\cos 2x \cos x + \cos 2x$

$\sin 2x (2\cos x + 1) = \cos 2x (2\cos x + 1)$

$\sin 2x = \cos 2x$ or $2\cos x + 1 = 0$

$\sin 2x = \cos 2x$ | or $2\cos x = -1$

$\frac{\sin 2x}{\cos 2x} = 1$ | $\cos x = -1/2$

$\tan 2x = 1$ | $x = 2n\pi \pm 2\pi/3$

$2x = n\pi + \pi/4$

$x = n\pi/2 + \pi/8$

Ques. If $0 \le x < \frac{\pi}{2}$, then the number of values of x for which $\sin x - \sin 2x + \sin 3x = 0$, is

(a) 2 (b) 1 (c) 3 (d) 4

Ans. $\sin x + \sin 3x - \sin 2x = 0$

$2\sin 2x \cos x - \sin 2x = 0$

$\sin 2x (2\cos x - 1) = 0$

$\sin 2x = 0 \Rightarrow 2x = n\pi \Rightarrow x = n\pi/2 \rightarrow x = 0$

or $2\cos x - 1 = 0 \Rightarrow x = \pi/2$ (Reject)

$\cos x = 1/2 \rightarrow x = \pi/3$

$\Rightarrow x = 60°$

$\Rightarrow x = \pi/3$

Type-4:

Trigonometric equations which can be solved by transforming a product of trigonometric ratios into their sum or difference.

Example: $\sin 5x \cos 3x = \sin 6x \cos 2x$

$2\sin 5x \cos 3x = 2\sin 6x \cos 2x$

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Using de-factorization formulas, we get

$\sin 8x + \sin 2x = \sin 8x + \sin 4x$

$\sin 2x = \sin 4x$

Method 1:

$\sin 4x - \sin 2x = 0$

$\Rightarrow 2 \cos 3x \sin x = 0$

either

$\cos 3x = 0$ or $\sin x = 0$

$3x = (2n + 1)\pi/2$ or $x = n\pi$

$x = (2n + 1)\pi/6$ or $x = n\pi$

$n = \checkmark$

$x = \checkmark$

Method 2:

$\sin 2x = \sin 4x$

$\sin 2x = 2\sin 2x \cos 2x$

$\sin 2x = 0$ or $2\cos 2x = 1$

$2x = n\pi$ or $\cos 2x = 1/2 \Rightarrow 2x = 2n\pi \pm \pi/3$

$x = n\pi/2$ or $x = n\pi \pm \pi/6$

Brain Teaser

Find all value of θ, between 0 and π, which satisfy the equation, $\cos\theta \cdot \cos 2\theta \cdot \cos 3\theta = \frac{1}{4}$

Ans. $2 \cos\theta \cos 2\theta \cos 3\theta = 1/2$

$(2 \cos\theta \cos 3\theta) \cos 2\theta = 1/2$

$[\cos 4\theta + \cos 2\theta] \cos 2\theta = 1/2$

$\cos 4\theta \cos 2\theta + \cos^2 2\theta = 1/2$

$2 \cos 4\theta \cos 2\theta + 2\cos^2 2\theta - 1 = 0$

$2\cos 4\theta . \cos 2\theta + \cos 4\theta = 0$ [$\because \cos 2\theta = 2\cos^2 A - 1$]

$\cos 4\theta [2\cos 2\theta + 1] = 0$

$\cos 4\theta = 0$ or $\cos 2\theta = -1/2$

Type-5:

Trigonometric equations of the form $a\sin x + b\cos x = c$ where $a, b, c \in R$, can be solved by 2 methods.

Method 1: Dividing both sides of the equation by $\sqrt{a^2 + b^2}$ & "enjoy".

Method 2: By converting into $\tan(x/2)$

Ques. Solve $\sin x + \cos x = 1$

Ans. $a = 1 = b$

$\sqrt{a^2 + b^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$

divide the equation by $\sqrt{2}$

$\frac{\sin x}{\sqrt{2}} + \frac{\cos x}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

$\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = \frac{1}{\sqrt{2}}$

$\sin\frac{\pi}{4}$ $\quad$ $\cos\frac{\pi}{4}$

$\cos(x - \pi/4) = \frac{1}{\sqrt{2}} = \cos \pi/4$

$x - \pi/4 = 2n\pi \pm \pi/4$

$x = 2n\pi \pm \pi/4 + \pi/4$

$\oplus$ $x = 2n\pi + \pi/4 + \pi/4$

$\boxed{x = 2n\pi + \pi/2}$

OR

$\ominus$ $x = 2n\pi - \pi/4 + \pi/4$

$\boxed{x = 2n\pi}$

Ques. Solve $3 \cos x + 4 \sin x = 5$

Ans. $\because \sin x = \frac{2\tan(x/2)}{1 + \tan^2(x/2)}$,

$\cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)}$

$\therefore \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} + \frac{2\tan(x/2)}{1 + \tan^2(x/2)} = 1$

put $\tan(x/2) = t$

$\Rightarrow \frac{1 - t^2}{1 + t^2} + \frac{2t}{1 + t^2} = 1 \Rightarrow \frac{1 - t^2 + 2t}{1 + t^2} = 1$

$\Rightarrow 1 - t^2 + 2t = 1 + t^2$

$\Rightarrow 2t^2 = 2t$

$\Rightarrow t = 0$ or $t = 1$

$\therefore \tan(x/2) = 0$ or $\tan\frac{x}{2} = 1$

$x/2 = n\pi$ or $x/2 = n\pi + \frac{\pi}{4}$

$x = 2n\pi$ or $x = 2n\pi + \frac{\pi}{2}$

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Ques. The number of solutions of the equation $\sin x = \cos^2 x$ in the interval $(0, 10)$ is _____.

$\approx (0, 3\pi)$

$3\pi \approx 3 \times 3.14 \approx 10$

[JEE Main 29 june 2022 shift 2]

Ans. $\sin x = 1 - \sin^2 x$

Let, $\sin x = t$

$\therefore t = 1 - t^2$

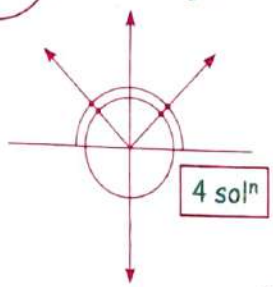
$t^2 + t - 1 = 0$

$$t = \frac{-1 \pm \sqrt{1 - 4(1)(-1)}}{2 \times 1}$$

$$\sin x = \frac{-1 \pm \sqrt{5}}{2}$$

$$\sin x = \frac{-1 + \sqrt{5}}{2} \text{ or } \frac{-1 - \sqrt{5}}{2} \text{ out of range}$$

$$\sin x = \underbrace{\frac{\sqrt{5} - 1}{2}}_{+ve}$$



Ques. Let $S = \left\{x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) : 9^{1-\tan^2 x} + 9^{1-\tan^2 x} = 10\right\}$

and $\beta = \sum_{x \in S} \tan^2\left(\frac{x}{3}\right)$ then $\frac{1}{6}(\beta - 14)^2$

[JEE Main 2023]

(a) 16 (b) 8 (c) 64 (d) 32

Ans. $\frac{9^1}{9^{\tan^2 x}} + 9^{\tan^2 x} = 10$

put $9^{\tan^2 x} = t$

$\frac{9}{t} + t = 10$

$9 + t^2 = 10t$

$t^2 - 10t + 9 = 0$

$(t - 9)(t - 1) = 0$

$t = 9, 1$

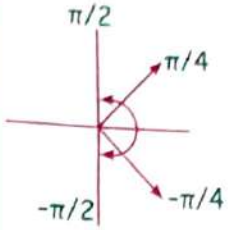
$9^{\tan^2 x} = 9^1$ or $9^{\tan^2 x} = 1 = 9^0$

$\tan^2 x = 1$ or $\tan^2 x = 0$

$\tan x = \pm 1$ $\tan x = 0$

$x = \pi/4, -\pi/4$ $\Rightarrow x = n\pi$

$n = 0, x = 0$



$S = \{0, \pi/4, -\pi/4\}$

$\beta = \tan^2\left(\frac{0}{3}\right) + \tan^2\frac{\pi/4}{3} + \tan^2\frac{(-\pi/4)}{3}$

$\beta = \tan^2 \pi/12 + \tan^2(-\pi/12)$

$\therefore \beta = 2\tan^2 \pi/12$ [$\because \tan^2(-\theta) = (-\tan\theta)^2 = \tan^2\theta$]

$\beta = 2\tan^2 15°$

$= 2(2 - \sqrt{3})^2$ [$\because \tan 15° = 2 - \sqrt{3}$]

$\beta = 2[4 + 3 - 4\sqrt{3}]$

$\beta = 2[7 - 4\sqrt{3}]$

$\beta = 14 - 8\sqrt{3}$

$\beta - 14 = -8\sqrt{3}$

S.B.S

$(\beta - 14)^2 = 64 \times 3$

$\frac{1}{6}(\beta - 14)^2 = \frac{1}{\cancel{6}_2} \times 64 \times \cancel{3} = 32$

Ques. If m and n respectively are the numbers of positive and negative value of θ in the internal $[-\pi, \pi]$ that satisfy the equation $\cos 2\theta \cos\frac{\theta}{2} = \cos 3\theta \cos\frac{9\theta}{2}$, then mn is equal to _____

[JEE Main 2023]

Ans. $2\cos 2\theta \cos\frac{\theta}{2} = 2\cos 3\theta \cos\frac{9\theta}{2}$

$\cos(2\theta + \theta/2) + \cos(2\theta - \theta/2) = \cos(3\theta + 9\theta/2) + \cos(3\theta - 9\theta/2)$

$\cos(5\theta/2) + \cos(3\theta/2) = \cos(15\theta/2) + \cos(3\theta/2)$

$\cos 5\theta/2 = \cos(15\theta/2)$

$\cos 5\theta/2 - \cos 15\theta/2 = 0$

$2\sin(10\,\theta/2)\sin(5\theta/2) = 0$

$\sin 5\theta \sin(5\theta/2) = 0$

$\sin 5\theta = 0$ or $\sin 5\theta/2 = 0$

$5\theta = n\pi$ or $5\theta/2 = n\pi$

$\theta = n\pi/5$ or $\theta = 2n\pi/5$

$\theta = 0, \underbrace{\pi/5, 2\pi/5, 3\pi/5, 4\pi/5, 5\pi/5}_{+ve}$

$\theta = \underbrace{-\pi/5, -2\pi/5, -3\pi/5, -4\pi/5, -5\pi/5}_{-ve}$

here, $m = 5$ and $n = 5$

$\therefore mn = 5 \times 5 = 25$

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Ques. Consider the following lists:

List-1		List-2	
I.	$\left\{x\in\left[-\frac{2\pi}{3},\frac{2\pi}{3}\right]: \cos x+\sin x=1\right\}$	P.	has two elements
II.	$\left\{x\in\left[-\frac{5\pi}{18},\frac{5\pi}{18}\right]: \sqrt{3}\tan 3x=1\right\}$	Q.	has three elements
III.	$\left\{x\in\left[-\frac{6\pi}{5},\frac{6\pi}{5}\right]: 2\cos(2x)=\sqrt{3}\right\}$	R.	has four elements
IV.	$\left\{x\in\left[-\frac{7\pi}{4},\frac{7\pi}{4}\right]: \sin x-\cos x=1\right\}$	S.	has five elements
		T.	has six elements

The correct options is: **[JEE Advance 2022]**

(a) (I) → (P); (II) → (S); (III) → (P); (IV) → (S)

(b) (I) → (P); (II) → (P); (III) → (T); (IV) → (R)

(c) (I) → (Q); (II) → (P); (III) → (T); (IV) → (S)

(d) (I) → (Q); (II) → (S); (III) → (P); (IV) → (R)

Ans.

(I) $x = 2n\pi$ or $x = 2n\pi + \pi/2$

$n = 0$ $\quad n = 0 \Rightarrow x = \pi/2$

$x = 0$ $\quad n = -1 \Rightarrow x = -2\pi + \pi/2 = -3\pi/2 = -270°$

(II) $\tan 3x = \frac{1}{\sqrt{3}}$

$3x = n\pi + \pi/6$

$x = n\pi/3 + \pi/18$

$x = \frac{6n\pi}{18} + \frac{\pi}{18}$

$n = 0 \Rightarrow x = \pi/18$

$n = 1 \Rightarrow x = 7\pi/18$ (×)

$n = -1 \Rightarrow x = \frac{-6\pi}{18} + \frac{\pi}{18} = \frac{-5\pi}{18}$

2 soln

Type-6:

Trigonometric equations of the form $P(\sin x \pm \cos x, \sin x \cos x) = 0$ (where p(y,z) is a polynomial) can be solved by using the substitution $\sin x \pm \cos x = t$.

Example: Solve $\sin x + \cos x = 1 + \sin x.\cos x$

Let, $\sin x + \cos x = t$

Squaring both sides

$(\sin x + \cos x)^2 = t^2$

$1 + 2\sin x \cos x = t^2$

$\sin x \cos x = \frac{t^2-1}{2}$

Now, $\sin x + \cos x = 1 + \sin x.\cos x$

$\therefore t = 1 + \left(\frac{t^2-1}{2}\right)$

$t = \frac{2 + t^2 - 1}{2}$

$2t = t^2 + 1$

$t^2 + 1 - 2t = 0$

$(t-1)^2 = 0$

$t = 1$

$\sin x + \cos x = 1$

Ans: $x = 2n\pi$ or $x = 2n\pi + \pi/2$

$\sin x + \cos x = 2$
no soln

SOME IMPORTANT GENERAL SOLUTIONS

S. No.	Trigonometric Equation	Principal Solution	General Equation
1.	$\sin\theta = 0$	$\theta = 0, \pi$	$\theta = n\pi, n \in I$
2.	$\sin\theta = 1$	$\theta = \frac{\pi}{2}$	$\theta = (4n+1)\frac{\pi}{2}, n\in I$
3.	$\sin\theta = -1$	$\theta = \frac{3\pi}{2}$	$\theta = (4n-1)\frac{\pi}{2}, n\in I$
4.	$\cos\theta = 0$	$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$	$\theta = (2n+1)\frac{\pi}{2}, n\in I$
5.	$\cos\theta = 1$	$\theta = 0$	$\theta = 2n\pi, n \in I$
6.	$\cos\theta = -1$	$\theta = \pi$	$\theta = (2n+1)\pi, n \in I$
7.	$\tan\theta = 0$	$\theta = 0, \pi$	$\theta = n\pi, n \in I$

$\cos\theta = 0$

$\cos\theta = \cos\pi/2$

$\theta = 2n\pi \pm \pi/2$

$\sin\theta = 1 = \sin\pi/2$

$\theta = n\pi + (-1)^n \pi/2$

$\sin\theta = 1 \Rightarrow \theta = (4n+1)\pi/2$

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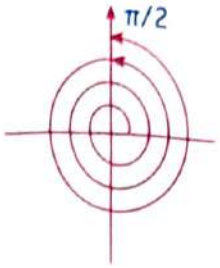
$\sin\theta = 1$

$\Rightarrow \theta = \pi/2 + 2n\pi$

$\theta = \frac{\pi + 4n\pi}{2}$

$\theta = \frac{\pi(4n+1)}{2}$

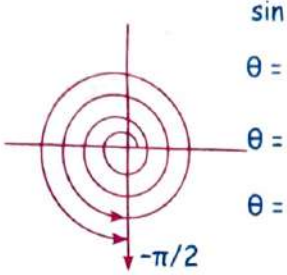


$\sin\theta = -1$

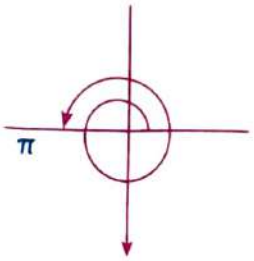
$\theta = -\pi/2 + 2n\pi$

$\theta = \frac{-\pi + 4n\pi}{2}$

$\theta = \frac{\pi(4n-1)}{2}$



$\cos\theta = -1 \Rightarrow \theta = \pi + 2n\pi = \pi(2n+1)$



Type 7(a):

Trigonometric equation (having multiple variables) which can be solved by the use of boundness of the trigonometric functions.

Solved using range (max/min value)

Ques. If $\theta \in [0, 5\pi]$ and $r \in R$ such that $2\sin\theta = r4 - 2r2 + 3$ then the maximum number of values of the pair (r, θ) is

Ans. LHS: $2\sin\theta$

$[-2, 2]$

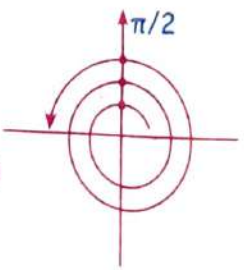
$\Rightarrow$ LHS = RHS = 2

$2\sin\theta = 2$ & $r^2 - 1 = 0$

$\sin\theta = 1$ & $r^2 = 1$

$\theta = (4n+1)\pi/2$ & $r = \pm 1$

$\theta = \pi/2, 5\pi/2, 9\pi/2$

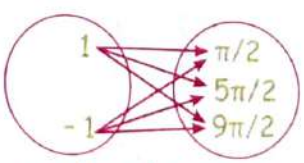


RHS: $r^4 - 2r^2 + 3$

$= (r^2 - 1)^2 - 1 + 3$

$= (r^2 - 1)^2 + 2$

RHS: $[2, \infty)$



number of pairs = 6

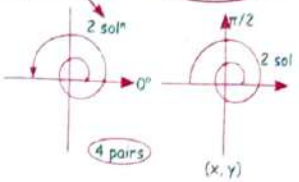
Ques. If $0 \le x \le 3\pi$, $0 \le y \le 3\pi$ and $\cos x \sin y = 1$ then the possible number of values of the ordered pair (x, y) is

Ans.

$\cos x \sin y = 1$

Case 1:

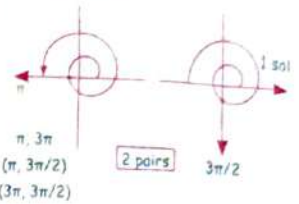
$\cos x = 1$ & $\sin y = 1$



Case 2:

$\cos x = -1$ & $\sin y = -1$

x = 2 values



Important Concept

$x + 1/x \ge 2$ for any $x > 0$

$(x, 1/x)$ (+ve)

AM $\ge$ GM

$\frac{x + 1/x}{2} \ge \sqrt{x \cdot \frac{1}{x}}$

$\frac{x + \frac{1}{x}}{2} \ge 1$

$x + \frac{1}{x} \ge 2$

If $x < 0$ $\boxed{x + \frac{1}{x} \le -2}$

$-x, -\frac{1}{x}$ (+ve)

$\frac{-x - 1/x}{2} \ge \sqrt{1}$

$\frac{-\left(x + \frac{1}{x}\right)}{2} \ge 1$

$-\left(x + \frac{1}{x}\right) \ge 2$

$x + \frac{1}{x} \le -2$

Ques. If $\sin x + \cos x = \sqrt{\left(y + \frac{1}{y}\right)}$, $x \in [0, \pi]$, then

(a) $x = \frac{\pi}{4}$, $y = 1$ (b) $x \in R$, $y = 1$

(c) $x = \frac{\pi}{4}$, $y = 2$ (d) $x = \frac{3\pi}{4}$, $y \in R$

Ans. $-\sqrt{2} \le \sin x + \cos x \le \sqrt{2}$

LHS : $[-\sqrt{2}, \sqrt{2}]$

$y + \frac{1}{y} \in [2, \infty)$

$\sqrt{y + \frac{1}{y}} \in [\sqrt{2}, \infty)$

R.H.S : $[\sqrt{2}, \infty)$

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LHS = RHS = $\sqrt{2}$

$\sin x + \cos x = \sqrt{2}$ & $y + \frac{1}{y} = 2$

$x = \frac{\pi}{4}$ $\quad$ $y = 1$

Type-7(b):

Trigonometric equations (having single variable) which can be solved by the use of boundness of the trigonometric ratios.

Example: $\cos^{50} x - \sin^{50} x = 1$

$(\cos x)^{50} = 1 \oplus (\sin x)^{50}$

$LHS_{max} = 1 \quad RHS_{min} = 1$

$\Rightarrow LHS = RHS = 1$

$(\cos x)^{50} = 1$ & $(\sin x)^{50} = 0$

$(\cos x)^{50} = 1$

$(\cos x)^2 = 1$

$\cos^2 x = 1$

$\cos^2 x = \cos^2 0$

$x = n\pi \pm \alpha$ (here $\alpha = 0$)

$x = n\pi$

$(\sin x)^{50} = 0$

$\sin x = 0$

$x = n\pi$

Intersection

$x = n\pi$

Ques. The number of solutions of the equation

$$\sin\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4 \text{ is}$$

Ans. LHS: $[-1, 1]$

LHS = RHS = 1

$\sin \frac{\pi x}{2\sqrt{3}} = 1$ &

Check, $x = \sqrt{3}$

$\sin\left(\frac{\pi\sqrt{3}}{2\sqrt{3}}\right) = 1$

Ans: $x = \sqrt{3}$ ✓

RHS: $(x - \sqrt{3})^2 - 3 + 4$

RHS: $(x - \sqrt{3})^2 + 1$

RHS: $[1, \infty]$

RHS = $1 \Rightarrow x = \sqrt{3}$

Ques. The number of solution of the equation

$$1 + \sin^4 x = \cos^2 3x, \; x \in \left[-\frac{5\pi}{2}, \frac{5\pi}{2}\right] \text{ is}$$

[JEE Main 2019]

(a) 7 (b) 4 (c) 3 (d) 5

Ans. $1 + \sin^4 x = \cos^2 3x$

min = 1 max = 1

$\Rightarrow \sin^4 x = 0$ & $\cos^2 3x = 1$

$\sin x = 0$

$x = n\pi$

$x = 0, \pi, 2\pi, -\pi, -2\pi$

All are common

$\cos^2 3x = \cos^2 0$

$3x = n\pi \pm 0$

$x = n\pi/3$

$x = 0, \pi/3, 2\pi/3, \pi, 2\pi$

$x = -\pi, -2\pi$

Ques. Find the general solution for

1. $\cos x + \cos 2x + \cos 3x + \cos 4x = 5$

(1 + 1 + 1 + 1)

Ans. No sol^n

2. $\cos x + \cos 2x + \cos 3x + \cos 4x = 4$

(1 + 1 + 1 + 1)

Ans. $\cos x = 1 \Rightarrow x = 2n\pi \rightarrow x = 0, 2\pi, 4\pi, 6\pi, \ldots\ldots$

$\cos 2x = 1 \Rightarrow 2x = 2n\pi \Rightarrow x = n\pi \rightarrow x = 0, \pi, 2\pi, 3\pi, 4\pi, \ldots\ldots$

$\cos 3x = 1 \Rightarrow 3x = 2n\pi \Rightarrow x = 2n\pi/3 \rightarrow x = 0, 2\pi/3, 4\pi/3, 6\pi/3, 8\pi/3 \ldots\ldots 12\pi/3$

$\cos 4x = 1 \Rightarrow 4x = 2n\pi \Rightarrow x = n\pi/2 \rightarrow x = 0, \pi/2, 2\pi/2, 3\pi/2, 4\pi/2, 5\pi/2, 6\pi/2 \ldots\ldots 8\pi/2$

$\Rightarrow$ common sol$^n \rightarrow x = 0, 2\pi, 4\pi, 6\pi \ldots\ldots$

$x = 2n\pi, n \in I$

Ques. The number of solutions of the equation

$4\sin^2 x - 4\cos^3 x + 8 - 4\cos x = 0; \; x \in [-2\pi, 2\pi]$ is **[JEE Main 2024 (Feb shift 2)]**

Ans. $4(1 - \cos^2 x) - 4\cos^3 x - 4\cos x + 8 = 0$

$12 = 4\cos^2 x + 4\cos^3 x + 4\cos x$

$\cos^2 x = 1$ & $\cos^3 x = 1$ & $\cos x = 1$ $\rightarrow$ $\cos x = 1$

$\cos x = 1$

$x = 2n\pi$

$x = 0, 2\pi, -2\pi$

The number of solution = 3.

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Brain Teaser

Solve for x: $\sin x\left(\cos\frac{x}{4} - 2\sin x\right) + \left(1 + \sin\frac{x}{4} - 2\cos x\right)\cos x = 0$

Ans. $\sin x\cos\frac{x}{4} - 2\sin^2 x + \cos x + \cos x\sin\frac{x}{4} - 2\cos^2 x = 0$

$\boxed{\sin x\cos\frac{x}{4} + \cos x\sin\frac{x}{4}} + \cos x = 2(\sin^2 x + \cos^2 x)$

$\sin(x + x/4) + \cos x = 2$

$\sin(5x/4) + \cos x = 2$

$\sin\frac{5x}{4} = 1 \Rightarrow \frac{5x}{4} = (4n+1)\pi/2$

$\Rightarrow x = (4n+1)2\pi/5$

&

$\cos x = 1 \Rightarrow x = 2n\pi \Rightarrow x = 0, 2\pi, 4\pi, 6\pi, \ldots\ldots$

$n = 1 \Rightarrow x = 2\pi$

$n = 6 \Rightarrow x = 25 \times 2\pi/5 = 10\pi$

$n = 11 \Rightarrow x = 45 \times 2\pi/5 = 18\pi$

common sol^n $2\pi, 10\pi, 18\pi, 26\pi, \ldots$

$a = 2\pi, d = 8\pi \Rightarrow T_n = a + (n-1)d$

$= 2\pi + (n-1).8\pi$

$= 2\pi + 8n\pi - 8\pi = 8n\pi - 6\pi, n\epsilon I$

Final Answer: $8n\pi - 6\pi, n\epsilon I$

Ques. Solve: $\sin(\cos\theta) = \cos(\sin\theta)$ **[Miscellaneous Type]**

Ans. $\cos(\sin\theta) = \cos(\pi/2 - \cos\theta)$

$\boxed{\sin\theta = 2n\pi \pm (\pi/2 - \cos\theta)}$

$\oplus\ \sin\theta = 2n\pi \pm (\pi/2 - \cos\theta)$

$\boxed{\sin\theta + \cos\theta = 2n\pi + \pi/2}$

$n = 0 \Rightarrow \sin\theta + \cos\theta = \pi/2$

$\boxed{\sin\theta + \cos\theta = 1.57}$ → no sol^n

$n = 1 \Rightarrow \sin\theta + \cos\theta = 2\pi + \pi/2$ → no sol^n

$n = -1 \Rightarrow$ No sol^n

2nd case

$\ominus$

$\sin\theta = 2n\pi - \pi/2 + \cos\theta$

$\boxed{\sin\theta - \cos\theta = 2n\pi - \pi/2}$

$\boxed{\pi/2 = \frac{3.14}{2} \approx 1.57}$

$\boxed{\sqrt{2} \approx 1.41}$

Answer: No Solution exists.

Ques. Solve: $(2+\sqrt{3})\cos x = 1 - \sin x$

Ans. $(\cot\pi/12)\cos x = 1 - \sin x$ $[\because \cot 15^0 = 2+\sqrt{3}]$

$\frac{\cos\pi/12}{\sin\pi/12}\cos x = 1 - \sin x$

$\cos\pi/12\cos x = \sin\pi/12 - \sin x\sin\pi/12$

$\boxed{\cos x\cos\frac{\pi}{12} + \sin x\sin\frac{\pi}{12}} = \sin\frac{\pi}{12}$

$\cos\left(x - \frac{\pi}{12}\right) = \sin\frac{\pi}{12}$ → $\cos\frac{5\pi}{12}$

$\cos(x - \pi/12) = \cos\frac{5\pi}{12}$

$\oplus\ x - \pi/12 = 2n\pi \pm 5\pi/12$

$x - \pi/12 = 2n\pi + 5\pi/12$

$x = 2n\pi + \pi/2$

$\ominus\ x - \pi/12 = 2n\pi - 5\pi/12$

$x = 2n\pi - \frac{5\pi}{12} + \frac{\pi}{12} \Rightarrow x = 2n\pi - \frac{4\pi}{12}$

or $x = 2n\pi - \pi/3$

Brain Teaser

If $2\tan^2 x - 5\sec x = 1$ for exactly 7 distinct value of $x \epsilon \left[0, \frac{n\pi}{2}\right]$, $n \epsilon N$, then the greatest value of 'n' is

(a) 6 (b) 12 (c) 13 (d) 15

Ans. $2(\sec^2 x - 1) - 5\sec x = 1$

Let, $\sec x = t$

$2(t^2 - 1) - 5t = 1$

$2t^2 - 5t - 3 = 0$

$2t^2 - 6t + t - 3 = 0$

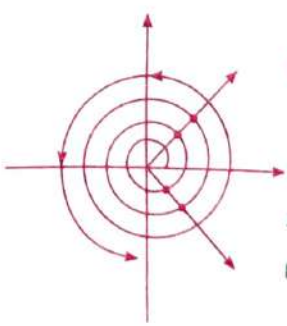
$(2t+1)(t-3) = 0$

$t = -1/2$ or $t = 3$

$\sec x = -1/2$ or $\sec x = 3$

$\cos x = -2$ (Rejected) or $\cos x = 1/3$

$\cos x = 1/3$



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$x \in [0, 3\pi/2]$

For n = 12

$\Rightarrow x \in [0, 6\pi]$

$\Rightarrow$ 6 solution

$n = 13 \Rightarrow x \in [0, 13\pi/2] \rightarrow$ 7 solution

$n_{max} = 15 \rightarrow x \in [0, 15\pi/2] \rightarrow$ 7 solution

Ques. Solve $\frac{3}{2} - \sin 2x = \sqrt{9 + 10\sin 2x}$

Ans. Let, $\sin 2x = t$

$3/2 - t = \sqrt{9 + 10t}$

$(\sqrt{t}^2) = t$

$\sqrt{(t^2)} = |t|$

S.B.S

$(3/2 - t)^2 = 9 + 10t$

$9/4 + t^2 - 3t = 9 + 10t$

$t^2 - 13t + 9/4 - 9 = 0$

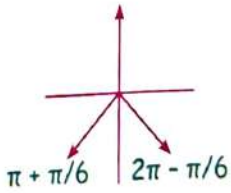
$t^2 - 13t + 9(-3/4) = 0$

$4t^2 - 52t - 27 = 0$

$4t^2 - 54t + 2t - 27 = 0$

$(2t + 1)(2t - 27) = 0$

$t = -1/2$ or $t = 27/2$



$\sin 2x = -1/2$ or $\sin 2x = 27/2$ (Rejected)

$2x = n\pi + (-1)^n(-\pi/6)$

$x = n\pi/2 + (-1)^n(-\pi/12)$

Brain Teaser

Find the solution set of the equation,

$$\log_{\frac{-x^2-6x}{10}}(\sin 3x + \sin x) = \log_{\frac{-x^2-6x}{10}}(\sin 2x)$$

Ans. $\sin 3x + \sin x = \sin 2x$

$2\sin 2x \cos x = \sin 2x$

$2\cos x = 1$ or $\sin 2x = 0$

$\cos x = 1/2$ or $2x = n\pi$

$x = 2n\pi \pm \pi/3$ or $x = n\pi/2$

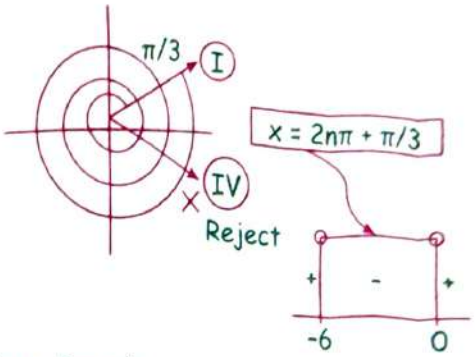
$\cos x = 1/2$

$\sin 2x = 2\sin x (\cos x)$

$= 2\sin x \, 1/2$

$\sin 2x = (\sin x) = -ve$

$-ve$



Base: Domain

$\frac{-x^2 - 6x}{10} > 0$

$-(x^2 + 6x) > 0$

$x^2 + 6x < 0$

$x(x + 6) < 0$

$x = 2n\pi + \pi/3$

$n = 0 \Rightarrow x = \pi/3 = \frac{3.14}{3} \approx 1$

$n = 1, 2, 3$ ..... (X)

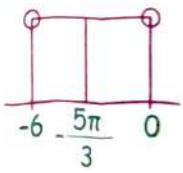
$n = -1 \Rightarrow x = -2\pi + \pi/3$

$x = -5\pi/3 \approx -5$

$n = -2 \Rightarrow x = -4\pi + \pi/3$

$x = -11/3 \approx -11$ (X)

Answer: $-5\pi/3$ only



Brain Teaser

$\cos^2\left(\frac{\pi}{4}(\sin x + \sqrt{2}\cos^2 x)\right) - \left(\tan^2\left(x + \frac{\pi}{4}\right)\tan^2 x\right) = 1$

Ans. $\cos^2\left(\frac{\pi}{4}(\sin x + \sqrt{2}\cos^2 x)\right) = 1 + \tan^2\left(x + \frac{\pi}{4}\right)\tan^2 x$

$\cos^2\left(\frac{\pi}{4}(\sin x + \sqrt{2}\cos^2 x)\right) = 1$ & $\tan^2\left(x + \tan^2 x\right) = 0$

$\frac{\pi}{4}(\sin x + \sqrt{2}\cos^2 x) = n\pi$

$\sin x + \sqrt{2}\cos^2 x = 4n$

$n = 0$ only

$\sin x + \sqrt{2}\cos^2 x = 0$

$\sin x + \sqrt{2}(1 - \sin^2 x) = 0$

Let, $\sin x = t$

$t + \sqrt{2}(1 - t^2) = 0$

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$\sqrt{2}t^2 - t - \sqrt{2} = 0$

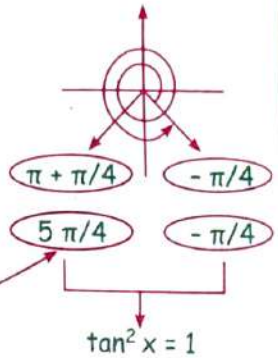
$\sqrt{2}t^2 - 2t + t - \sqrt{2} = 0$

$\sqrt{2}t(t - \sqrt{2}) + 1(t - \sqrt{2}) = 0$

$(\sqrt{2}t + 1)(t - \sqrt{2}) = 0$

$t = \frac{-1}{\sqrt{2}}$ or $t = \sqrt{2} \approx 1.414$ (Rejected)

$\sin x = \frac{-1}{\sqrt{2}}$



$\tan^2\left(x + \frac{\pi}{4}\right)\tan^2 x = 0$ (1)

$\tan^2\left(x + \frac{\pi}{4}\right) = 0$

$\tan^2 x = 1$

$x = \frac{5\pi}{4}$

$\tan^2(5\pi/4 + \pi/4) = 0$

$\tan^2(6\pi/4) = 0$

$\tan^2(3\pi/2) = 0$

Not defined [Reject]

$x = -\pi/4$ satisfies

Answer: $x = -\pi/4 + 2n\pi$

Ques. The number of distinct solution of the equation $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is

[JEE Advance 2015]

Ans. $\frac{5}{4}\cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$

$\frac{5}{4}\cos^2 2x + 1 - 2\sin^2 x \cos^2 x + 1 - 3\sin^2 x \cos^2 x = 2$

$5/4\cos^2 2x - 5\sin^2 x \cos^2 x = 0$

$5/4\cos^2 2x - 5/4(2\sin x \cos x)^2 = 0$

$5/4\cos^2 2x - 5/4\sin^2 2x = 0$

$5/4(\cos^2 2x - \sin^2 2x) = 0$

$\cos 4x = 0$

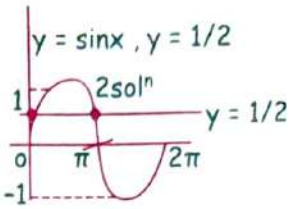
$4x = (2n + 1)\pi/2$

$x = (2n+1)\pi/8$

GRAPHICAL APPROACH

Ques. Find the number of solutions of equation $\sin x = \frac{1}{2}$ using graph in $x \in [0, 2\pi]$

Ans.



Ques. Find the number of solutions of equation $\sin x = \frac{x}{10}$

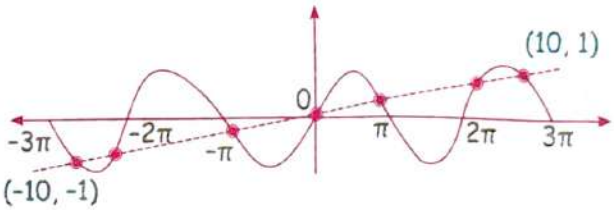
Ans. $y = \sin x$ & $y = x/10$

$10 \approx 3\pi$

$3\pi = 3 \times 3.14 = 9.42$

x	0	10	-10
y	0	1	-1

$-1 \le x/10 \le 1 \Rightarrow -10 \le x \le 10$



Hence, number of solutions = 7.

Ques. The number of solutions of the equation, $x + 2\tan x = \frac{\pi}{2}$ in the interval $[0, 2\pi]$ is:

[JEE Main 2021 (march)]

(a) 3 (b) 4 (c) 2 (d) 5

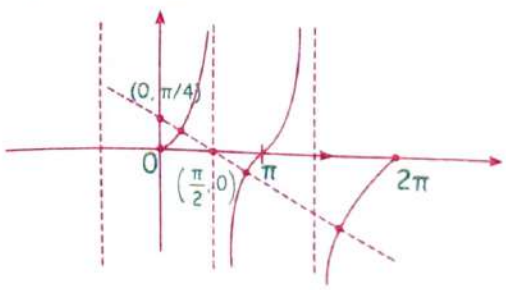
Ans. $2\tan x = \pi/2 - x$

$\tan x = \pi/4 - x/2$

$y = \tan x$

$y = \pi/4 - x/2$

x	0	π/2
y	π/4	0



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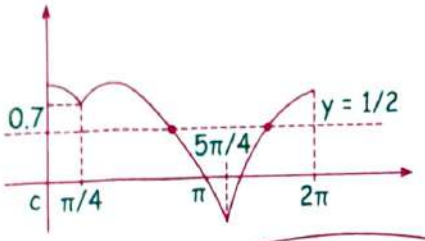
Brain Teaser

Number of solutions of the equation: max {sinx, cosx} = $\frac{1}{2}$ in $x \in (-2\pi, 5\pi)$, is equal to

Ans. $y = \max\{\sin x, \cos x\}$ & $y = 1/2$



$\sin\pi/4 = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} = \frac{1.4}{2} \approx 0.7$

2 sol^n → $[0, 2\pi)$

2 sol^n → $[2\pi, 4\pi)$

2sol^n → $(-2\pi, 0)$

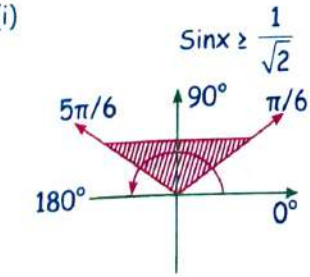
1 sol^n → $(4\pi, 5\pi)$

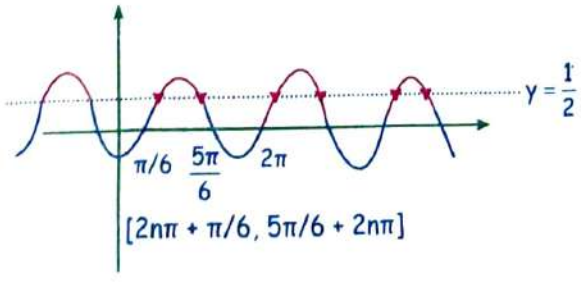
Hence, number of solution = 7.

SOLVING TRIGONOMETRIC INEQUATIONS FOR sinx & cosx

Example:

(i)





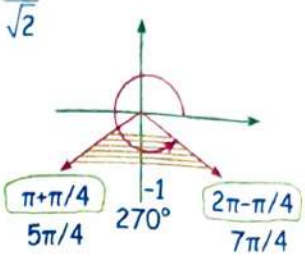
$[2n\pi + \pi/6, 5\pi/6 + 2n\pi]$

$x \in [\pi/6, 5\pi/6]$

add $2n\pi$ to this interval

$x \in [2n\pi + \pi/6, 2n\pi + 5\pi/6]$

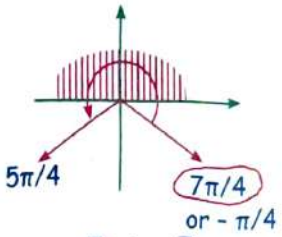
(ii) $\sin x \leq -\frac{1}{\sqrt{2}}$



$x \in [5\pi/4, 7\pi/4]$

$x \in [2n\pi + 5\pi/4, 2n\pi + 7\pi/4]$

(iii) $\sin x \geq -\frac{1}{\sqrt{2}}$



$\left[\frac{7\pi}{4}, \frac{5\pi}{4}\right]$

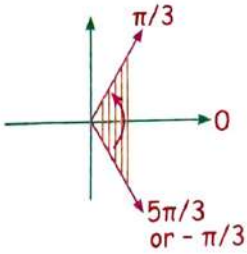
$x \in [-\pi/4, 5\pi/4]$ or $x \in [0, 5\pi/4] \cup [7\pi/4, 2\pi]$

Final Answer: $x \in [2n\pi - \pi/4, 2n\pi + 5\pi/4]$

Ques. Solve:

1. $\cos x \geq \frac{1}{2}$

Ans.



$[-\pi/3, \pi/3]$

$[2n\pi - \pi/3, 2n\pi + \pi/3]$

2. $|\cos x| < \frac{\sqrt{3}}{2}$

Ans. $|x| < 1 \Rightarrow -1 < x < 1$

$\frac{-\sqrt{3}}{2} < \cos x < \frac{\sqrt{3}}{2}$

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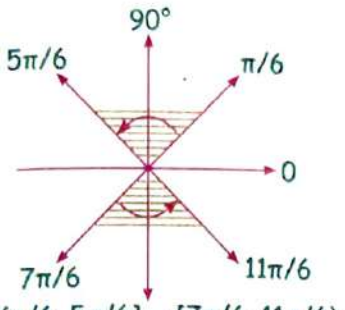
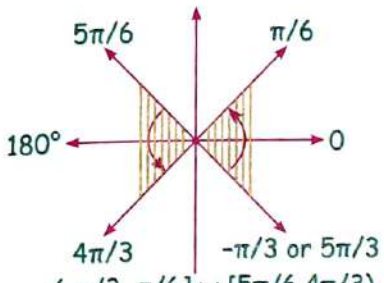


add $2n\pi$ → $(\pi/6, 5\pi/6] \cup [7\pi/6, 11\pi/6)$

3. $-\frac{\sqrt{3}}{2} < \sin x \le \frac{1}{2}$

Ans.



add $2n\pi$ → $(-\pi/3, \pi/6] \cup [5\pi/6, 4\pi/3)$

Ques. If $0 < \theta < 2\pi$, then the interval of values of θ for which $2\sin^2\theta - 5\sin\theta + 2 > 0$, is

[JEE Advance 2006]

(a) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (b) $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$

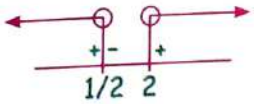
(c) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ (d) $\left(\frac{41\pi}{48}, \pi\right)$

Ans. Let $\sin\theta = t$

$2t^2 - 5t + 2 > 0$

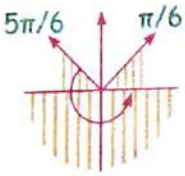
$2t^2 - 4t - t + 2 > 0$

$(2t - 1)(t - 2) > 0$



$t < 1/2$ or $t > 2$ (×)

$\boxed{\sin\theta < 1/2}$



Ques. Find the exhaustive set of values of α lying between 0 and π which satisfies the inequality, $\sin\alpha\cos^3\alpha > \sin^3\alpha\cos\alpha$

(a) $\alpha \in \left(0, \frac{\pi}{4}\right)$ (b) $\alpha \in \left(0, \frac{\pi}{2}\right)$

(c) $\alpha \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (d) None of these

Ans. $\sin\alpha\cos^3\alpha - \sin^3\alpha\cos\alpha > 0$

$\sin\alpha\cos\alpha[\cos^2\alpha - \sin^2\alpha] > 0$

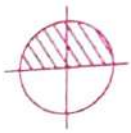
$\left(\frac{2\sin\alpha\cos\alpha}{2}\right)\cos 2\alpha > 0$

$\frac{2\sin 2\alpha\cos 2\alpha}{2} > 0$

$\sin 4\alpha > 0$

$4\alpha \in (0, \pi) \cup (2\pi, 3\pi)$

$\alpha \in (0, \pi/4) \cup \left(\frac{2\pi}{4}, \frac{3\pi}{4}\right)$



$\alpha \in (0, \pi/4) \cup (\pi/2, 3\pi/4)$

OR

$4\alpha \in (0, \pi)$

$4\alpha \in (2n\pi, 2n\pi + \pi)$

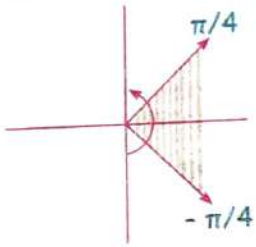
$\alpha \in \left(\frac{2n\pi}{4}, \frac{2n\pi + \pi}{4}\right)$

Ques. If $\cos x - \sin x \ge 1$ and $0 \le x \le 2\pi$ then the solution set for x is

(a) $\left[0, \frac{\pi}{4}\right] \cup \left[\frac{7\pi}{4}, 2\pi\right]$ (b) $\left[\frac{3\pi}{2}, \frac{7\pi}{4}\right] \cup \{0\}$

(c) $\left[\frac{3\pi}{2}, 2\pi\right] \cup \{0\}$ (d) None of these.

Ans. $\cos x - \sin x \ge 1$



$\frac{\cos x}{\sqrt{2}} - \frac{\sin x}{\sqrt{2}} \ge \frac{1}{\sqrt{2}}$

$\cos(x + \pi/4) \ge \frac{1}{\sqrt{2}}$

$x + \pi/4 \in [-\pi/4, \pi/4]$

Subtract $\pi/4$

$x \in [-\pi/4 - \pi/4, \pi/4 - \pi/4]$

$x \in [-\pi/2, 0]$

$x \in [2n\pi - \pi/2, 2n\pi]$

$n = 0 \Rightarrow x \in [-\pi/2, 0]$

$n = 1 \Rightarrow x \in [3\pi/2, 2\pi]$

SOLVING TRIGONOMETRIC INEQUATIONS FOR tanx

Period of $\tan x \to \pi$

Solve in 1/2 cycle & add $n\pi$

Example: (i) $\tan x \ge 1$

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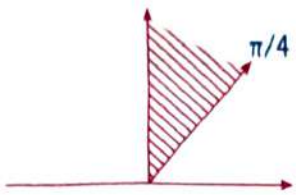
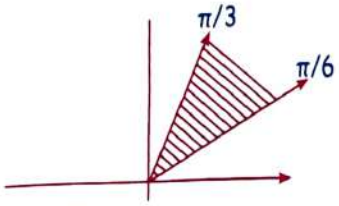


$x \in [\pi/4, \pi/2)$

Final Answser: $x \in [n\pi + \pi/4, n\pi + \pi/2)$

$n = 1 \Rightarrow x \in \left[5\pi/4, \frac{3\pi}{2}\right)$

(ii) $\frac{1}{\sqrt{3}} < \tan x \leq \sqrt{3}$

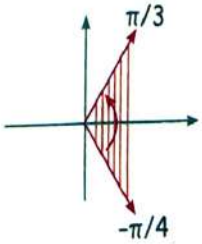


$x \in (\pi/6, \pi/3]$

Final Answer: $x \in (n\pi + \pi/6, n\pi + \pi/3]$

Ques. Solve: $-1 < \tan x \leq \sqrt{3}$

Ans.



$(-\pi/4, \pi/3]$

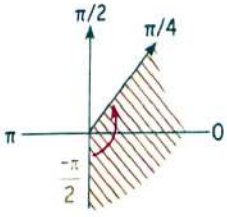
$x \in [n\pi - \pi/4, n\pi + \pi/3]$

Ques. Solve: $\tan x \leq 1$

Ans. $[0, \pi/4] \cup (\pi/2, \pi]$

or

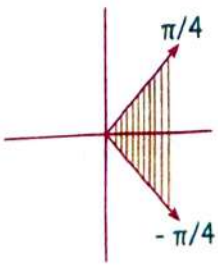
$(-\pi/2, \pi/4]$



Final Answer: $(n\pi - \pi/2, n\pi + \pi/4]$

Ques. Solve: $|\tan x| \leq 1$

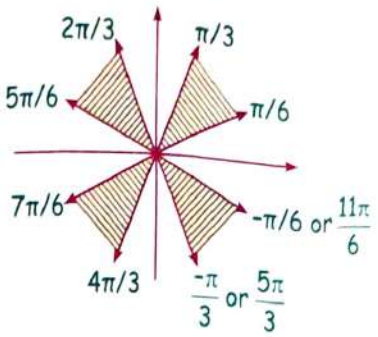
Ans. $-1 \leq \tan x \leq 1$



$[-\pi/4 + n\pi, \pi/4 + n\pi]$

Ques. Solve: $\frac{1}{4} \leq \sin^2 x \leq \frac{3}{4}$

Ans.



Let $\sin x = t$

$1/4 \leq t^2 \leq 3/4$

$t \in \left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right] \cup \left[\frac{-\sqrt{3}}{2}, \frac{-1}{2}\right]$

$\sin x \in \left[\frac{1}{2}, \frac{\sqrt{3}}{2}\right] \cup \left[\frac{-\sqrt{3}}{2}, \frac{-1}{2}\right]$

$x \in [n\pi + \pi/6, \pi/3 + n\pi] \cup [2\pi/3 + n\pi, 5\pi/6 + n\pi]$ → Better

$x \in \left[\frac{n\pi}{2} + \frac{\pi}{6}, \frac{\pi}{3} + \frac{n\pi}{2}\right]$ → Best

Ques. All possible values of $\theta \in [0, 2\pi]$ for which $\sin 2\theta + \tan 2\theta > 0$ lie in: [JEE Main 2021(Feb

(a) $\left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

(b) ✓ $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{5\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$

(c) $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

(d) $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right)$

Ans. $\sin 2\theta + \frac{\sin 2\theta}{\cos 2\theta} > 0 \Rightarrow \sin 2\theta \left[1 + \frac{1}{\cos 2\theta}\right] > 0$

$\sin 2\theta \cdot \frac{[\cos 2\theta + 1]}{\cos 2\theta} > 0$

$\tan 2\theta \cdot (2\cos^2\theta) > 0$

+ve

$\tan 2\theta > 0$

$2\theta \in (0, \pi/2) \Rightarrow 2\theta \in (n\pi, n\pi + \pi/2)$

$\Rightarrow \theta \in (n\pi/2, n\pi/2 + \pi/4)$

$n = 0 \Rightarrow \theta \in (0, \pi/4)$

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Short Notes

PRINCIPAL SOLUTIONS

- The solutions of the Trigonometric Equation which lie in the interval $[0, 2\pi)$ are principal solutions.

GENERAL SOLUTION

The solution consisting of all possible solution of a trigonometric equation in general form is called General Solution.

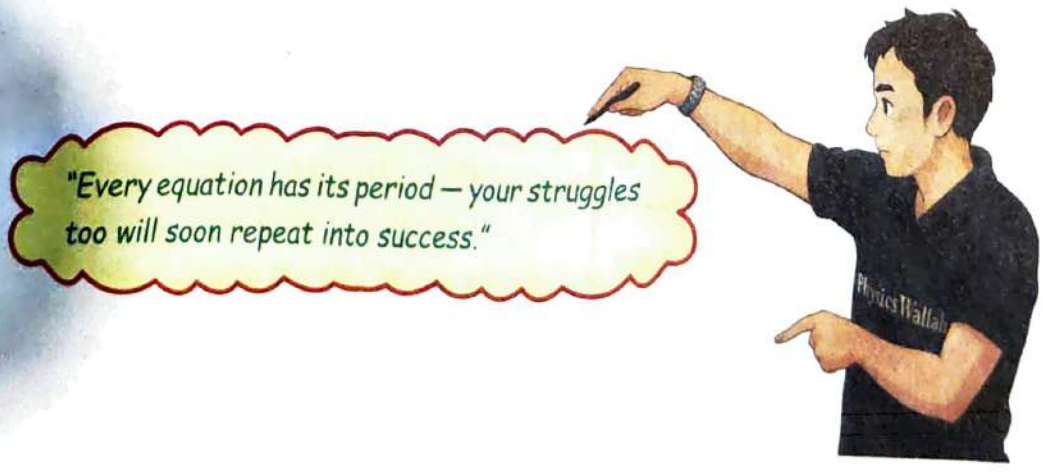
- Some common general solutions

Sl.	Trigonometric Equation	Principal Angle	Principal Solution	General Equation
1.	$\sin\theta = 0$	$\theta = 0$	$\theta = 0, \pi$	$\theta = n\pi, n \in I$
2.	$\sin\theta = 1$	$\theta = \frac{\pi}{2}$	$\theta = \frac{\pi}{2}$	$\theta = (4n+1)\frac{\pi}{2}, n \in I$
3.	$\sin\theta = -1$	$\theta = -\frac{\pi}{2}$	$\theta = \frac{3\pi}{2}$	$\theta = (4n-1)\frac{\pi}{2}, n \in I$
4.	$\cos\theta = 0$	$\theta = \frac{\pi}{2}$	$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$	$\theta = (2n+1)\frac{\pi}{2}, n \in I$
5.	$\cos\theta = 1$	$\theta = 0$	$\theta = 0$	$\theta = 2n\pi, n \in I$
6.	$\cos\theta = -1$	$\theta = \pi$	$\theta = \pi$	$\theta = (2n+1)\pi, n \in I$
7.	$\tan\theta = 0$	$\theta = 0$	$\theta = 0, \pi$	$\theta = n\pi, n \in I$

- $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$ where $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], n \in I$
- $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$ where $\alpha \in [0, \pi], n \in I$
- $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha$ where $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], n \in I$
- $\sin^2\theta = \sin^2\alpha$, $\cos^2\theta = \cos^2\alpha$, $\tan^2\theta = \tan^2\alpha$ $\Rightarrow \theta = n\pi \pm \alpha$

IMPORTANT POINTS TO KEEP IN MIND WHILE SOLVING TRIGONOMETRIC EQUATIONS

- Squaring the equation at any step should be avoided as far as possible.
- Never cancel terms containing unknown terms on the two sides, which are in product.
- Domain should not be changed
- Check that denominator is not zero at any stage while solving equations
- While solving trigonometric equations, you may get same set of solution repeated in your answer. It is necessary for you to exclude these repetitions
- Sometimes the two solution set consist partly of common values. In all such cases the common part must be presented only once
- If $|c| > \sqrt{a^2 + b^2}$, then the equation $a\cos\theta + b\sin\theta = c$ has no solution
- If the problem involves only one equation consisting of more than one variable or equation involves variable of different natures then the boundary conditions of Trigonometric functions is generally used



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Chapter 5

Complex Numbers-I

COMPLEX NUMBER (INTRODUCTION)

$z = x + iy$

1) Real Part: $Re(z) = x$

Imaginary Part: $Im(z) = y$

i = iota = $\sqrt{-1}$ $\quad i^2 = -1$

Example: $z = 2 + 3i$

$Re(z) = 2$

$Im(z) = 3$

2) If $y = 0 \Rightarrow z = x \rightarrow$ Purely Real C.N.

3) If $x = 0 \Rightarrow z = iy \rightarrow$ Purely Imaginary C.N.

Note: Both real & imaginary parts are actually real.

Complex Number
↓
Real No
↓
Rational
↓
Integers
↓
Whole No
↓
Natural No

Ques. If $\alpha, \beta \in R$ are such that $1 - 2i$ (here $i^2 = -1$) is root of $z^2 + \alpha z + \beta = 0$, then $(\alpha - \beta)$ is equal to

(JEE Main 2021 Feb.)

(a) - 7 (b) 7 (c) 3 (d) - 3

Ans. ∵ α, β are real.

∴ The roots of given eqⁿ are $1-2i, 1+2i$.

Sum of roots = $1 - 2i + 1 + 2i = 2$

Product of Roots = $(1 - 2i)(1 + 2i)$

$= 1^2 - 4i^2$

$= 1^2 + 4$

$= 5$

Sum = $-\alpha$

Product = β

$\Rightarrow \beta = 5$

And $-\alpha = 2 \Rightarrow \alpha = -2$

So, $\alpha - \beta = -2 - 5 = -7$

Ques. If $x = 3 + i$, then $x^3 - 3x^2 - 8x + 15 =$

(UPSEAT 2023)

(a) 6 (b) 10

(c) - 18 (d) - 15

Ans. $x = 3 + i$

$\Rightarrow x - 3 = i$ .....(i)

S.B.S

$(x - 3)^2 = i^2$

$\Rightarrow x^2 - 6x + 9 = -1$

$\Rightarrow x^2 - 6x = -10$ .....(ii)

$\Rightarrow x^2 = 6x - 10$

$\therefore x^3 + 3x^2 - 8x + 15$

$= x^2(x - 3) - 8x + 15$

By eqn (i) and (ii):

$= (6x - 10)i - 8x + 15$

$= 6xi - 10i - 8x + 15$

$= x(6i - 8) - 10i + 15$

$= (3 + i)(6i - 8) - 10i + 15$

$= 18i + 6i^2 - 24 - 8i - 10i + 15$

$= 18i - 6 - 24 - 18i + 15 = -15$

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HIGHER POWERS OF IOTA (i)

$i^1 = i$ | $w^0 = 1$

$i = \sqrt{-1}$ $i^2 = -1$ | $w^1 = w$

$i^2 = -1$ | $w^2 = w^2$

$i^3 = i^2 . i = -i$ | $w^3 = 1$

$i^4 = i^2 . i^2 = (-1)^2 = 1$ | $w^4 = w$

$i^5 = i^4 \; i = i$ | $w^5 = w^2$

$i^6 = i^4 . i^2 = -1$ | $w^6 = 1$

$i^7 = i^4 \; i^3 = i^3 = -i$

$i^8 = i^4 . i^4 = 1$

- $i^{4n} = 1$
- $i^{4n+1} = i$
- $i^{4n+2} = -1$
- $i^{4n+3} = -i$
- $i^{4n} + i^{4n+1} i^{4n+2} + i^{4n+3} = 0$

Ex. (i) $i^{2056} = 1$

(ii) $i^{289} = i^{288} \; i = i$

(iii) $i^{1002} = i^{1000} \; i^2 = -1$

Ques. $i^{437} + i^{2021} + i^{2026} + i^{446}$

$i + i + i^2 + i^2$

Ans. $2i - 1 - 1$

$= 2i - 2$

Ques. $i^{2027} + i^{2029} =$

$i^3 + i$

Ans. $-i + i = 0$

Ques. $\sum_{r=1}^{57} i^r =$

Ans. $= i + i^2 + i^3 + i^4 + \text{--------} + i^{57}$

$i^{56} \, i = (i)$

Ques. If $i^2 = -1$, then sum $i^1 + i^2 + i^3 + \ldots$ to 1000 terms is equal to

(a) 1 (b) -1

(c) i (d) 0

Ans. $i + i^2 + i^3 + i^4 + \text{------} + i^{1000} = 0$

Equality: If $x_1 + iy_1 = x_2 + iy_2$, implies $x_1 = x_2$ & $y_1 = y_2$

Ex. If $x + iy = 7 + 3i$

$x = 7$ & $y = 3$

Ques. Find real values of x and y satisfying equation:-

$(x^4 + 2xi) - (3x^2 + yi) = (3 - 5i) + (1 + 2yi)$

Ans. $(x^4 - 3x^2) + i(2x - y) = 4 + i(2y - 5)$

$\Rightarrow x^4 - 3x^2 - 4 = 0$ & $2x - y = 2y - 5$

$x^4 - 3x^2 - 4 = 0$

$x^2 = t$

$t^2 - 3t - 4 = 0$

$(t - 4)(t + 1) = 0$

$t = 4$ or $-1 \rightarrow x^2 = -1$ (x)

$x^2 = 4 \Rightarrow x = \pm 2$

$2x + 5 = 3y$

Case → 1: $x = 2$, $4 + 5 = 3y \Rightarrow y = 3$

Case → 2: $x = -2$, $-4 + 5 = 3y \Rightarrow y = 1/3$

Ans: $(x, y) = (2, 3)$ or $\left(-2, \frac{1}{3}\right)$

Inequality of Complex Number

$x_1 + iy_1 > x_2 + iy_2$

$\Rightarrow y_1 = y_2 = 0$

$\Rightarrow x_1 > x_2$

Ex. $3 + i > 2 + i$

(x)

Two non real numbers can never be compared

ALGEBRA OF COMPLEX NUMBER

Algebraic Operations with Complex Numbers

(i) **Addition:** $[x_1 + iy_1] + [x_2 + iy_2] = [x_1 + x_2] + i[y_1 + y_2]$

Example: $3 + 4i + 7 + 8i = 10 + 12i$

(ii) **Subtraction:** $(x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2)$

(iii) **Multiplication:** $(x_1 + iy_1)(x_2 + iy_2)$

(iv) **Division:** $\frac{x_1 + iy_1}{x_2 + iy_2}$ ⇒ "Always rationalize denominator"

Example: $\frac{3 + 4i}{3 - i} = \frac{3 + 4i}{3 - i} \times \frac{3 + i}{3 + i}$

$= \frac{(3 + 4i)(3 + i)}{3^2 - i^2}$

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$$= \frac{(3+4i)(3+i)}{10} = \frac{9+3i+12i+4i^2}{10}$$

$$= \frac{9+15i-4}{10}$$

$$= \frac{5+15i}{10} = \frac{5(1+3i)}{10} = \frac{(1+3i)}{2}$$

Ques. $z_1 = 1 - 3i$, $z_2 = 2 + i$ then find the value of $iz_1 + z_2$. Also find Im $(2z_1 - z_2)$ = ?

Ans. $z_1 = 1 - 3i$

$i z_1 = i(1 - 3i) = i - 3i^2$

$\boxed{\begin{matrix} i z_1 = i + 3 \\ z_2 = 2 + i \end{matrix}}$

add

$\Rightarrow i z_1 + z_2 = 2i + 5$

Now, $2z_1 = 2 - 6i$

$\therefore 2z_1 - z_2 = (2 - 6i) - (2 + i) = -6i - i = -7i$

$\Rightarrow$ Im $(2z_1 - z_2) = -7$

Ques. let $\left(-2 - \frac{1}{3}i\right)^3 = \frac{x+iy}{27}$, where x and y are real numbers, then y - x equals:

(JEE Main 2019 Jan.)

(a) 91 (b) - 85

(c) 85 (d) - 91

Ans. $\frac{(-6-i)^3}{27} = \frac{x+iy}{27}$

$\Rightarrow (-1)^3 (6+i)^3 = x + iy$

$\Rightarrow -[6^3 + i^3 + 3 \times 6i (6+i)] = x + iy$

$\Rightarrow -[216 - i + 18i (6+i)] = x + iy$

$\Rightarrow -[216 - i + 108i - 18)] = x + iy$

$\Rightarrow -[198 + 107i] = x + iy$

$\Rightarrow -198 = x$ and $-107 = y$

$\Rightarrow y - x = -107 + 198 = 91$

Brain Teaser

Dividing f [z] by z - i, we obtain the remainder i and diving it by z + i, we get the remainder 1 + i, then remainder upon the division of f [z] by $z^2 + 1$ is

(a) $\frac{1}{2}(z+1) + i$ (b) $\frac{1}{2}(iz+1) + i$

(c) $\frac{1}{2}(iz-1) + i$ (d) $\frac{1}{2}(z+1) + i$

Ans. $f(i) = i$

$f(-i) = 1 + i$

when dividing by $z^2 + 1$

let remainder by $az + b$

$\boxed{f(z) = (z^2+1) Q(z) + az + b}$

Put $z = i$

$f(i) = (i^2+1) Q(z) + ai + b$

$\boxed{i = ai + b} \rightarrow (1)$

Put $z = -i$

$f(-i) = (i^2+1) Q(z) - ai + b$

$\boxed{1 + i = -ai + b} \rightarrow (2)$

(1) + (2)

$1 + 2i = 2b$

$b = \frac{1}{2} + i$

(1) - (2)

$i - (i+1) = 2ai$

$-1 = 2ai$

$a = \frac{-1}{2i} \cdot \frac{i}{i} = \frac{-i}{2i^2} = \frac{i}{2}$

$\therefore$ Remainder $= \frac{i}{2}z + \frac{1}{2} + i$

Ques. The value of $(1+i)^6 + (1-i)^6$ is **(RPET 2002)**

(a) 0 (b) 2^7

(c) 2^6 (d) None of these

Ans. $(1+i)^2 = 1^2 + i^2 + 2i$

$= 0 + 2i = 2i$

$(1+i)^6 = [(1+i)^2]^3$

$= (2i)^3 = 8i^3 = -8i$

$(1-i)^6 = [(1-i)^2]^3$

$= (-2i)^3 = -8i^3 = 8i$

$\boxed{(1+i)^6} + (1-i)^6$

$-8i \quad 8i$

$\Rightarrow (1+i)^6 + (1-i)^6 = 0$

Ques. The Smallest positive integer n for which $(1+i)^{2n} = (1-i)^{2n}$ is: **(Karnataka CET 2004)**

(a) 1 (b) 2

(c) 3 (d) 4

Ans. $(1+i)^{2n} = (1-i)^{2n} \Rightarrow [(1+i)^2]^n = [(1-i)^2]^n$

$\Rightarrow [1+i^2+2i]^n = [1+i^2-2i]^n \Rightarrow (2i)^n = (-2i)^n$

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$\Rightarrow 2^n \cdot i^n = 2^n (-i)^n \Rightarrow i^n = (-i)^n$

$n = 1 \Rightarrow i = -i$

$n = 2 \Rightarrow i^2 = (-i)^2$

$\therefore$ Smallest value of $n = 2$

Ques. The Sum of all possible values of $\theta \in [\pi, 2\pi]$, for which $\frac{1 + i\cos\theta}{(1 - 2i\cos\theta)}$ is purely imaginary, is equal

(JEE Main - 2024(Apr. shift 2))

(a) 3π (b) 2π

(c) 5π (d) 4π

Ans. $z = \frac{1 + i\cos\theta}{1 - 2i\cos\theta} \times \frac{1 + 2i\cos\theta}{1 + 2i\cos\theta}$

$= \frac{(1 + i\cos\theta)(1 + 2i\cos\theta)}{1^2 - (2i\cos\theta)^2}$

$= \frac{1 + 2i\cos\theta + i\cos\theta + 2i^2\cos^2\theta}{1 + 4\cos^2\theta}$

$z = \frac{1 - 2\cos^2\theta + 3i\cos\theta}{1 + 4\cos^2\theta}$

$\because$ Re (Z) = 0

$\Rightarrow \frac{1 - 2\cos^2\theta}{1 + 4\cos^2\theta} = 0$

$\Rightarrow 1 - 2\cos^2\theta = 0 \Rightarrow 2\cos^2\theta - 1 = 0$

$\Rightarrow \cos 2\theta = 0 \Rightarrow 2\theta = (2n + 1)\frac{\pi}{2}$

$\theta = (2n + 1)\frac{\pi}{4}$

$n = 0$

$\Rightarrow \theta = \frac{\pi}{4}$

$n = 1$

$\theta = \frac{3\pi}{4}$

} Net in $[\pi, 2\pi]$

$n = 2$

$\theta = \frac{5\pi}{4}$

$n = 3$

$\theta = \frac{7\pi}{4}$

} $\Rightarrow$ Valid values

$\therefore$ sum of possible values $= \frac{5\pi}{4} + \frac{7\pi}{4} = 3\pi$

Ques. The real part of $(1 - \cos\theta + 2i\sin\theta)^{-1}$ is

(IIT 1978, 86)

(a) $\frac{1}{3 + 5\cos\theta}$ (b) $\frac{1}{5 - 3\cos\theta}$

(c) $\frac{1}{3 - 5\cos\theta}$ (d) $\frac{1}{5 + 3\cos\theta}$

Ans. $\frac{1}{(1 - \cos\theta) + 2i\sin\theta} \times \frac{(1 - \cos\theta) - 2i\sin\theta}{(1 - \cos\theta) - 2i\sin\theta}$

$= \frac{1 - \cos\theta - 2i\sin\theta}{(1 - \cos\theta)^2 - (2i\sin\theta)^2}$

$= \frac{1 - \cos\theta - 2i\sin\theta}{1 + \cos^2\theta - 2\cos\theta + 4\sin^2\theta}$

$= \frac{1 - \cos\theta - 2i\sin\theta}{1 + \cos^2\theta - 2\cos\theta + 4 - 4\cos^2\theta}$

$\text{Re}(z) = \frac{1 - \cos\theta}{5 - 3\cos^2\theta - 2\cos\theta}$

$= \frac{-(1 - \cos\theta)}{3\cos^2\theta + 2\cos\theta - 5}$

$= \frac{-(1 - \cos\theta)}{3\cos^2\theta + 5\cos\theta - 3\cos\theta - 5}$

$= \frac{(\cos\theta - 1)}{(3\cos\theta + 5) + (\cos\theta - 1)}$

$= \frac{1}{3\cos\theta + 5}$

SQUARE ROOT OF A COMPLEX NUMBER

Ques. Find the square root of $5 + 12i$

Ans. Let $\sqrt{5 + 12i} = x + iy$

S.B.S.

$5 + 12i = x^2 + (iy)^2 + 2ixy$

$5 + 12i = x^2 - y^2 + (2xy)i$

$x^2 - y^2 = 5 \rightarrow (1)$

$2xy = 12 \rightarrow (2)$

Consider $(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2$

$(x^2 + y^2)^2 = 5^2 + 12^2$

$(x^2 + y^2)^2 = 169$

$\Rightarrow x^2 + y^2 = 13 \rightarrow (3)$

$(1) + (3)$

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$2x^2 = 5 + 13$

$\Rightarrow 2x^2 = 18$

$x^2 = 9 \Rightarrow y^2 = 4$

$\Rightarrow x = \pm 3 \quad y = \pm 2$

$2xy = 12$

$\Rightarrow xy = 6$

$\Rightarrow x = 3, y = 2$ only

or $x = -3, y = -2$ only

$x + iy = 3 + 2i$ or $-3 - 2i$

Ques. Find the square root of $z = 9 - 40i$

Ans. $\sqrt{9 - 40i} = x + iy$

S.B.S.

$9 - 40i = x^2 - y^2 + 2ixy$

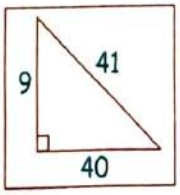
$\boxed{x^2 - y^2 = 9} \rightarrow (1)$

$\boxed{2xy = -40} \rightarrow (2)$

$(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2$

$(x^2 + y^2)^2 = (9)^2 + (40)^2$

$\boxed{x^2 + y^2 = 41} \rightarrow (3)$



(1) + (3)

$2x^2 = 50$

$\Rightarrow x^2 = 25 \Rightarrow x = \pm 5$

$y^2 = 16 \Rightarrow y = \pm 4$

$2xy = -40 \Rightarrow xy = -20$

$x = 5 \Rightarrow y = -4$

$x = -5 \Rightarrow y = 4$

$\therefore \sqrt{9 - 40i} = \quad 5 - 4i$ or $-5 + 4i$

Important Note

$\sqrt{ab} = \sqrt{a}\sqrt{b} \quad a \geq 0 \ \& \ b \geq 0$

Ques. Find the square root of $3 - 4i$

(a) $\pm [2 - i]$ (b) $\pm [2 + i]$

(c) $\pm [\sqrt{3} - 2i]$ (d) $\pm [\sqrt{3} + 2i]$

Ans. (c) $[\pm (\sqrt{3} - 2i)]^2 = 3 + (2i)^2 - 2\sqrt{3}\,2i$

$= 3 - 4 - 4\sqrt{3}i$

$= -1 - 4\sqrt{3}i \neq 3 - 4i$

(a) $[\pm (2 - i)]^2 = (2 - i)^2 = 4 + i^2 - 4i$

$= 4 - 1 - 4i$

$= 3 - 4i$

Important Note

Three important terms for a complex number:

(i) Conjugate of a complex number $(\bar{z})$

(ii) Modulus of a complex number $(|z|)$

(iii) Argument of a complex number $(\arg(z))$

CONJUGATE OF A COMPLEX NUMBER (z)

If $z = x + iy$ then $\bar{z} = x - iy$

Ex → $z = 3 + 4i$

$\bar{z} = 3 - 4i$

Ex → $z = -3i$

$\bar{z} = 3i$

Ex → $z = 4$

$\bar{z} = 4$

Ex → $z = \dfrac{3 - 4i}{7 + 5i}$

$\bar{z} = \dfrac{3 + 4i}{7 - 5i}$

Ques. The Conjugate of $\left(\dfrac{(2 + i)^2}{3 + i}\right)^z$ in the form of $a + ib$, is

(a) $\dfrac{13}{2} + i\left(\dfrac{15}{2}\right)$ (b) $\dfrac{13}{10} + i\left(\dfrac{-15}{2}\right)$

(c) $\dfrac{13}{10} + i\left(\dfrac{-9}{10}\right)$ (d) $\dfrac{13}{10} + i\left(\dfrac{9}{10}\right)$

Ans. $\bar{z} = \dfrac{(2 - i)^2}{3 - i}$

$= \dfrac{4 + i^2 - 4i}{3 - i}$

$= \dfrac{3 - 4i}{(3 - i)} \cdot \dfrac{(3 + i)}{(3 + i)}$

$= \dfrac{(3 - 4i)(3 + i)}{3^2 - i^2}$

$= \dfrac{9 + 3i - 12i - 4i^2}{10}$

$= \dfrac{13 - 9i}{10} = \dfrac{13}{10} - \dfrac{9}{10}i$

Properties of Conjugate of a complex Number

(1) If Z is purely real then $\bar{Z} = Z$

(2) If Z is purely imaginary then $\bar{Z} = -Z$

(3) $\overline{Z_1 + Z_2} = \bar{Z}_1 + \bar{Z}_2$

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(4) $\overline{Z_1 - Z_2} = \bar{Z}_1 - \bar{Z}_2$

(5) $\overline{Z_1 Z_2} = (\bar{Z}_1)(\bar{Z}_2)$

(6) $\left(\frac{\bar{Z}_1}{Z_2}\right) = \frac{\bar{Z}_1}{\bar{Z}_2}\ [Z_2 \neq 0]$

(7) $Z_1\bar{Z}_2 + \bar{Z}_1 Z_2 = 2\,\text{Re}\,(Z_1\bar{Z})$

(8) $Z_1\bar{Z}_2 - \bar{Z}_1 Z_2 = 2iI_m\,(Z_1\bar{Z}_2)$

(9) $(Z^n) = (\bar{Z})^n$

(10) $Z + \bar{Z} = 2\,\text{Re}\,(Z)$

(11) $Z - \bar{Z} = 2i\,\text{Im}\,(Z)$

(12) $\overline{(\bar{Z})} = (Z)$

Ques. If $x + iy = \frac{a + ib}{a - ib}$ then show that $x^2 + y^2 = 1$

Ans. $x + iy = \frac{a + ib}{a - ib} \rightarrow (1)$

Conjugate both sides

$\Rightarrow x - iy = \frac{a - ib}{a + ib} \rightarrow (2)$

eqn (1) × (2)

$\Rightarrow (x + iy)(x - iy) = \left(\frac{a + ib}{a - ib}\right)\left(\frac{a - ib}{a + ib}\right)$

$\Rightarrow x^2 - (iy)^2 = 1$

$\Rightarrow x^2 + y^2 = 1$

Ques. If the conjugate of $(x + iy)(1 - 2i)$ be $1 + i$, then

(a) $y = \frac{1}{5}$

(b) $x = \frac{3}{5}$

(c) $x + iy = \frac{1 - i}{1 - 2i}$

(d) $x - iy = \frac{1 + i}{1 + 2i}$

Ans. $(x - iy)(1 + 2i) = 1 + i$

$x - iy = \frac{1 + i}{1 + 2i}$

Conjugate both sides

$x + iy = \frac{1 - i}{1 - 2i} \times \frac{1 + 2i}{1 + 2i}$

$\Rightarrow x + iy = \frac{(1 - i)(1 + 2i)}{1^2 - (2i)^2} = \frac{1 - i + 2i - 2i^2}{1 + 4}$

$\Rightarrow x + iy = \frac{3 + i}{5}$

$\Rightarrow x = \frac{3}{5}$ and $y = \frac{1}{5}$

Ques. Solve for z:

(i) $z^2 + \bar{z} = 0;$

Ans. Let $Z = x + iy$

$(x + iy)^2 + x - iy = 0$

$\Rightarrow x^2 - y^2 + 2ixy + x - iy = 0$

$\Rightarrow (x^2 - y^2 + x) + iy(2x - 1) = 0 + 0i$

$x^2 - y^2 + x = 0$ & $y(2x - 1) = 0$

$\Rightarrow$ either $y = 0$ or $2x - 1 = 0$

$\Rightarrow x = \frac{1}{2}$

Case → 1

$y = 0$

$x^2 - y^2 + x = 0$

$\Rightarrow x^2 + x = 0$

$\Rightarrow x(x + 1) = 0$

$\Rightarrow x = 0, x = -1$

Case → 2 $\Rightarrow x = \frac{1}{2}$

$\therefore x^2 - y^2 + x = 0$

$\frac{1}{4} - y^2 + \frac{1}{2} = 0$

$y^2 = \frac{3}{4}$

$y = \pm\frac{\sqrt{3}}{2}$

$y = 0 \Rightarrow x = 0, -1$

$x = \frac{1}{2} \Rightarrow y = \pm\frac{\sqrt{3}}{2}$

$z = 0$

$z = -1$

$z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$

$z = \frac{1}{2} - \frac{\sqrt{3}}{2}i$

MODULUS OF A COMPLEX NUMBER |z|

If $z = x + iy$, then modulus

$|z| = \sqrt{x^2 + y^2}$

$|z| \rightarrow$ real no

$|z| \geq 0$

Ex. (i) $z = 1 - i$

$|z| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$

(ii) $z = 3 + 4i$

$|z| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

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(iii) $z = 2i$

$|z| = \sqrt{0^2 + 2^2} = \sqrt{4} = 2$

(iv) $z = 5$

$|z| = \sqrt{5^2 + 0^2} = 5$

(v) $z = -3$

$|z| = 3$

Ques. Solve for z: $z^2 + |z| = 0$

Ans. Let $z = x + iy$

$\Rightarrow (x + iy)^2 + \sqrt{x^2 + y^2} = 0$

$\Rightarrow x^2 - y^2 + 2ixy + \sqrt{x^2 + y^2} = 0$

$\Rightarrow (x^2 - y^2 + \sqrt{x^2 + y^2}) + i(2xy) = 0$

$\Rightarrow x^2 - y^2 + \sqrt{x^2 + y^2} = 0$ & $2xy = 0 \Rightarrow x = 0$ or $y = 0$

Case → 1: $x = 0$

$x^2 - y^2 + \sqrt{x^2 + y^2} = 0$

$\Rightarrow -y^2 + \sqrt{y^2} = 0$

$-y^2 + |y| = 0$

$y \geq 0$	$y < 0$
$-y^2 + y = 0$	$-y^2 - y = 0$
$y(-y + 1) = 0$	$-y(y + 1) = 0$
$y = 0$ or $y = 1$	$y = 0$ or $y = -1$
accept	accept

Case → 2: $y = 0$

$x^2 - y^2 + \sqrt{x^2 + y^2} = 0$

$= x^2 + |x| = 0$

0 0

$\Rightarrow x = 0$ only nothing new

Therefore, $z = 0 + 0i$, $z = 0 + i$, $z = 0 - i$

Ques. If $z = x + iy$, $xy \neq 0$, satisfies the equation $z^2 + i\bar{z} = 0$, then $|z^2|$ is equal to

(JEE Main-2024 (Jan. Shift I))

(a) 9 (b) 1

(c) 4 (d) $\frac{1}{4}$

Ans. $(x + iy)^2 + i(x - iy) = 0$

$\Rightarrow x^2 - y^2 + 2ixy + ix + y = 0$

$(x^2 - y^2 + y) + ix(2y + 1) = 0$

$x^2 - y^2 + y = 0$ or $x(2y + 1) = 0$

$x = 0$ or $2y + 1 = 0$

Net accept ($\because x \neq 0$)

$y = \frac{-1}{2}$

$\Rightarrow x^2 - \frac{1}{4} - \frac{1}{2} = 0$

$x^2 = \frac{3}{4}$

$|z^2| = |z|^2$

$= (\sqrt{x^2 + y^2})^2$

$= x^2 + y^2$

$|z^2| = \frac{3}{4} + \frac{1}{4} = \frac{4}{4} = 1$

Properties of |z|

(i) $\boxed{|z| = |\bar{z}| = |-z| = |-\bar{z}|}$

Proof: $z = x + iy$

$\bar{z} = x - iy$

$|z| = \sqrt{x^2 + y^2}$

$|\bar{z}| = \sqrt{x^2 + (-y)^2}$

(ii) If $|z| = 1$ $\Rightarrow z\bar{z} = 1$ $\Rightarrow \bar{z} = 1/z$

$|z|^2 = z\bar{z}$

Proof: $z = x + iy$

LHS: $(\sqrt{x^2 + y^2})^2 = x^2 + y^2$

RHS: $(x + iy)(x - iy)$

$= x^2 - (iy)^2$

$= x^2 + y^2$

LHS = RHS

$|x|^2 = x^2$ If x is real

$|z|^2 = z^2$

Real Non Real

(iii) $|z_1 z_2| = |z_1||z_2|$

Proof: LHS. $|(x_1 + iy_1)(x_2 + iy_2)|$

$= |(x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2)|$

$= \sqrt{(x_1 x_2 - y_1 y_2)^2 + (x_1 y_2 + y_1 x_2)^2}$

$= \sqrt{x_1^2 x_2^2 + y_1^2 y_2^2 + x_1^2 y_2^2 + y_1^2 x_2^2}$

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RHS: $\sqrt{x_1^2 + y_1^2}\sqrt{x_2^2 + y_2^2}$

$= \sqrt{(x_1^2 + y_1^2)(x_2^2 + y_2^2)}$

LHS = RHS

(iv) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$

(v) In general $|z_1 + z_2| \neq |z_1| + |z_2|$

Proof: Let $Z_1 = x_1 + iy_1$

$Z_2 = x_2 + iy_2$

RHS: $\sqrt{x_1^2 + y_1^2} + \sqrt{x_2^2 + y_2^2}$

LHS: $|(x_1 + x_2) + i(y_1 + y_2)|$

$= \sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2}$

LHS ≠ RHS

(vi) $|z^n| = |z|^n$

Example: $|\sqrt{3 + 4i}|$

$|(3 + 4i)^{\frac{1}{2}}| = |3 + 4i|^{\frac{1}{2}}$

$= (5)^{\frac{1}{2}} = \sqrt{5}$

(vii) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + z_2\bar{z}_1$

$= |z_1|^2 + |z_2|^2 + 2\,Re(z_1\bar{z}_2)$

Proof: $|z_1 + z_2|^2 =$

$= (z_1 + z_2)(\overline{z_1 + z_2})$

$= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2)$

$= z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2$

$= |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + z_2\bar{z}_1$

(viii) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 - z_1\bar{z}_2 - z_2\bar{z}_1$

Proof: $|z_1 - z_2|^2$

$= (z_1 - z_2)(\overline{z_1 - z_2})$

$= (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$

$= z_1\bar{z}_1 - z_1\bar{z}_2 - z_2\bar{z}_1 + z_2\bar{z}_2$

$= |z_1|^2 - (z_1\bar{z}_2 \times z_2\bar{z}_1) + |z_2|^2$

Ques. Find the value of: $\left|\frac{(3 - 4i)(7 + 24i)}{(4 - 3i)}\right|$

(a) 1 (b) 2 (c) 25 (d) 625

Ans. $\left|\frac{(3 - 4i)(7 + 24i)}{(4 - 3i)}\right| \quad \frac{|3 - 4i|\,|7 + 24i|}{|4 + 3i|}$

$= |7 + 24i| = \sqrt{7^2 + (24)^2} = 25$

Ques. Find the value of: $\left|\frac{1}{(4 + 3i)} + \frac{1}{(3 - 2i)}\right|$

(a) 1

(b) $\frac{\sqrt{2}}{\sqrt{13}}$

(c) $\frac{\sqrt{50}}{\sqrt{13}}$

(d) None of the above

Ans. $\left|\frac{1}{(4 + 3i)} + \frac{1}{(3 + 2i)}\right| = \left|\frac{3 - 2i + 4 + 3i}{(4 + 3i)(3 - 2i)}\right|$

$= \left|\frac{(7 + i)}{(4 + 3i)(3 - 2i)}\right|$

$= \frac{|7 + i|}{|4 + 3i|\,|3 - 2i|} = \frac{\sqrt{7^2 + 1^2}}{\sqrt{4^2 + 3^2}\sqrt{3^2 + 2^2}}$

$= \frac{\sqrt{50}}{5\sqrt{13}} = \frac{5\sqrt{2}}{5\sqrt{13}} = \frac{\sqrt{2}}{\sqrt{13}}$

Ques. if z_1 and z_2 are unimodular then find $|3z_1 - 4z_2|^2 + |4z_1 + 3z_2|^2$

Ans. Given $|z_1| = |z_2| = 1$

$|3z_1 - 4z_2|^2 + |4z_1 + 3z_2|^2$

$|3z_1|^2 + |4z_2|^2 - 24|2z_1 + 1z_2|^2$

$+ |4z_1|^2 + |3z_2|^2 - 24|2_1 2_2|$

$= 2[|3z_1|^2 + |4z_2|^2]$

$= 2[9|z_1|^2 + 16|z_2|^2]$

$= 2[9 + 16] = 2 \times 25 = 50$

Ques. Prove that: $|1 - \bar{z}_1 z_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$

Ans. $|1 - \bar{z}_1 z_2|^2 - |z_1 - z_2|^2 = (1 - |z_1|^2)(1 - |z_2|^2)$

LHS: $(1 - \bar{z}_1 z_2)(1 - \bar{z}_1 z_2) - [|z_1|^2 + |z_2|^2 - z_1\bar{z}_2 - \bar{z}_1 z_2]$

$= (1 - \bar{z}_1 z_2)(1 - z_1\bar{z}_2) - [|z_1|^2 + |z_2|^2 - z_1\bar{z}_2 - \bar{z}_1 z_2]$

$= (1 - z_1\bar{z}_2 - \bar{z}_1 z_2 + (z_1\bar{z}_1)(z_2\bar{z}_2)) - |z_1|^2 - |z_2|^2 + z_1\bar{z}_2 + \bar{z}_1 z_2$

$= (1 + |z_1|^2|z_2|^2 - |z_1|^2 - |z_2|^2)$

$= -|z_1|^2(1 - |z_2|^2) + (1 - |z_2|^2)$

$(1 - |z_2|^2)(1 - |z_1|^2)$

= RHS

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Ques. Let $\bar{z}$ denote the complex conjugate of a complex number z and let $i = \sqrt{-1}$. In the set of complex numbers, the number of distinct roots of the equation. $\bar{z} - z^2 = i(\bar{z} + z^2)$ is .....

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Ans. $\bar{z} - z^2 = i\bar{z} + iz^2$

$\Rightarrow \bar{z}(1 - i) = z^2(1 + i)$

$\Rightarrow \bar{z}\dfrac{(1 - i)}{1 + i} = z^2$

$\Rightarrow \bar{z}\dfrac{(1 - i)^2}{1^2 - i^2} = z^2$

$\Rightarrow \bar{z}\dfrac{(1^2 + i^2 - 2i)}{2} = z^2$

$-\bar{z}i = z^2$

$-i[x - iy] = (x + iy)^2$

$-ix - y = x^2 - y^2 + 2ixy$

$x^2 - y^2 = -y$ & $2xy = -x$

$2xy + x = 0$

$x(2y + 1) = 0 \Rightarrow x = 0$ or $x = -\dfrac{1}{2}$

Case 1: $x = 0$

$x^2 - y^2 + y = 0$

$\Rightarrow y^2 = y$

$\Rightarrow y = 0, 1$

Case 2: $y = -\dfrac{1}{2}$

$x^2 - \dfrac{1}{4} - \dfrac{1}{2} = 0$

$\Rightarrow x^2 = \dfrac{3}{4} \Rightarrow x = \pm\dfrac{\sqrt{3}}{4}$

Hence, 4 roots

Ques. If $x + iy = \left(\dfrac{a + ib}{a - ib}\right)$, then show that $x^2 + y^2 = 1$

Ans. mod both sides

$\Rightarrow |x + iy| = \left|\dfrac{a + ib}{a - ib}\right|$ $\quad (\because |a + ib| = \sqrt{a^2 + b^2})$

$\Rightarrow \sqrt{x^2 + y^2} = \left|\dfrac{a + ib}{a - ib}\right| \Rightarrow x^2 + y^2 = 1$

Ques. Prove that $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$ if and only if $\frac{z_1}{z_2}$ is purely imaginary given $z_1 \neq 0, z_2 \neq 0$

Ans. Given, $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$

$\because |z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + \bar{z}_1z_2$

$\therefore z_1\bar{z}_2 + \bar{z}_1z_2 = 0$

$\Rightarrow z_1\bar{z}_2 = -\bar{z}_1z_2$

$\Rightarrow \dfrac{z_1}{z_2} = \dfrac{-\bar{z}_1}{\bar{z}_2}$

$\Rightarrow \left(\dfrac{z_1}{z_2}\right) = -\left(\overline{\dfrac{z_1}{z_2}}\right)$

$\left(\dfrac{z_1}{z_2}\right) \rightarrow z$, $\left(\overline{\dfrac{z_1}{z_2}}\right) \rightarrow \bar{z}$

$\Rightarrow z = -\bar{z}$

$\Rightarrow z$ is purely Im

$\Rightarrow \dfrac{z_1}{z_2}$ is purely Im.

Ques. If $|z_1| = 2$ & $|z_2| = 1$ and $|z_1 + z_2| = 3$, then fi

$|4z_2 + z_1| = ?$

(a) 4 (b) 6

(c) 1 (d) 2

Ans. $|z_1 + z_2| = 3$

S.B.S

$\Rightarrow |z_1 + z_2|^2 = 9$

$\Rightarrow |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + \bar{z}_1z_2 = 9$

$\Rightarrow 4 + 1 + z_1\bar{z}_2 + \bar{z}_1z_2 = 9$

$\Rightarrow z_1\bar{z}_2 + \bar{z}_1z_2 = 4$

Let $x = |4z_2 + z_1|$

$\Rightarrow x^2 = |4z_2 + z_1|^2$

$\Rightarrow x^2 = 16|z_2|^2 + |z_1|^2 + 4z_2\bar{z}_1 + 4\bar{z}_2z_1$

$\Rightarrow x^2 = 16 \times 1 + 4 + 4[z_2\bar{z}_1 + \bar{z}_2z_1]$

$\Rightarrow x^2 = 16 + 4 + 16$

$\Rightarrow x^2 = 36$

$\Rightarrow x = 6$

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Ques. If $|z_1| = |z_2| = \ldots = |z_n| = 1$ Prove that $|z_1 + z_2 + \ldots + z_n| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \ldots + \frac{1}{z_n}\right|$

$z_1\bar{z}_2 = 1 = z_2\bar{z}_2 = \ldots = z_n\bar{z}_n$

Ans. $z_1 = \frac{1}{\bar{z}_1}$

LHS : $|z_1 + z_2 + z_3 + \ldots + z_n|$

$$\frac{\left|\frac{1}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3} + \ldots + \frac{1}{\bar{z}_n}\right|}{\left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} + \ldots + \frac{1}{z_n}\right|}$$

$$\left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} + \ldots + \frac{1}{z_n}\right| \quad (\because |\bar{z}| = |z|)$$

= R.H.S

Ques. If $|z_1| = 2, |z_2| = 3, |z_3| = 4$ and $|z_1 + z_2 + z_3| = 5$, then find $|4 z_2z_3| + 9z_1z_3 + 16z_1z_2| = ?$

(a) 40 (b) 60

(c) 120 (d) 180

Ans. $= \left||z_1|^2 z_2z_3 + |z_2|^2 z_1z_3 + |z_3|^2 z_1z_2\right|$

$= |z_1\bar{z}_1z_2z_3 + z_2\bar{z}_2z_1z_3 + z_3\bar{z}_3z_1z_2|$

$= |z_1z_2z_3[\bar{z}_1 + \bar{z}_2 + \bar{z}_3]|$

$= |z_1||z_2||z_3|\,|\bar{z}_1 + \bar{z}_2 + \bar{z}_3|$

$= |z_1||z_2||z_3|\,|z_1 + z_2 + z_3|$

$= 2 \times 3 \times 4 \times 5$

$= 120$

Ques. If α & β are different complex numbers with $|\beta| = 1$, then $\left|\frac{\beta - \alpha}{1 - \bar{\alpha}\beta}\right|$ is equal to

(a) 0 (b) 1/2

(c) 1 (d) 2

Ans. $\beta\bar{\beta} = 1 \Rightarrow \beta = \frac{1}{\bar{\beta}}$

$$\left|\frac{\beta - \alpha}{1 - \bar{\alpha}\beta}\right| = \left|\frac{\beta - \alpha}{1 - \bar{\alpha}\cdot\frac{1}{\bar{\beta}}}\right|$$

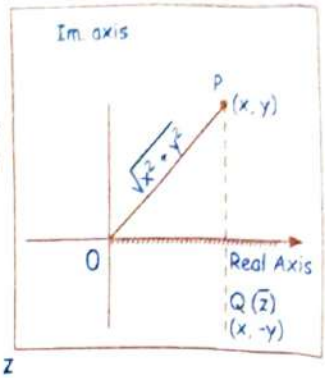
$$= \left|\frac{(\beta - \alpha)\bar{\beta}}{(\bar{\beta} - \bar{\alpha})}\right| = \frac{|\beta - \alpha||\bar{\beta}|}{(\bar{\beta} - \bar{\alpha})} = |\bar{\beta}| = |\beta| = 1 \ (\because |z| = \bar{z})$$

ARGAND PLANE

$z = x + iy$

$OP = \sqrt{x^2 + y^2} = |z|$

$|z|$ represents distance of origin from the point representing complex number in argand plane



$\bar{z} = x - iy$

$(x, -y)$

$\bar{z}$ represents image of z in real axis

Ex: $z = 2 + 3i$

$(2,3) \rightarrow$ Point in argent plane

$z = 3i \rightarrow (0, 3)$

$z = -2 \rightarrow (-2, 0)$

Ex: $z = -2-5i$ $(-2, -5)$

$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

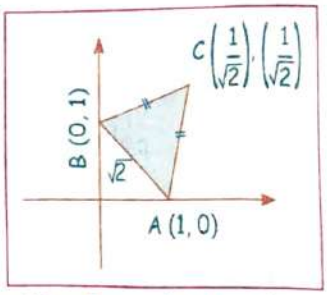
(1,0) (0,1)

Ques. The triangle formed by the points (1), $\frac{1+i}{\sqrt{2}}$ and (i) as vertices in the argand diagram is

(a) Scalene (b) Equilateral

(c) Isosceles (d) Right-angled

Ans.



$\because$ BC = AC $\Rightarrow$ Isosceles triangle

Ques. If the four complex number $z, \bar{z}, \bar{z} - 2\text{Re}(\bar{z})$ and $z - 2\text{Re}(z)$ represent the vertices of the square of side 4 units in the argand plane, then $|z|$ is equal to **(JEE Main 2020)**

(a) $4\sqrt{2}$ (b) $2\sqrt{2}$ (c) 2 (d) 4

Ans. Let, $z = x + iy \rightarrow (x, y)$,

$\bar{z} = x - iy \rightarrow (x, -y)$,

$\bar{z} - 2\text{Re}(z) = -x - iy - 2x$

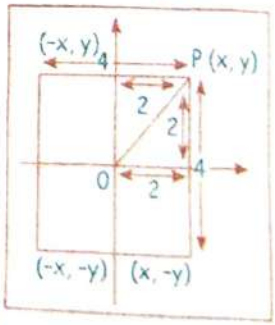
$= -x - iy$ $(-x, -y)$

$z - 2\text{Re}(z) = x + iy - 2x$

$= -x + iy \rightarrow (-x, y)$

$OP^2 = 2^2 + 2^2 \Rightarrow OP^2 = 8$

$|z| = OP = \sqrt{8} = 2\sqrt{2}$



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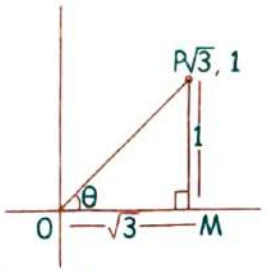
ARGUMENT OF A COMPLEX NUMBER

Definition

The angle made by the line joining origin to the point representing complex number in the argand plane with positive direction of X axis (in any direction) is called Argument of a Complex Number.
It is represent by arg (Z)

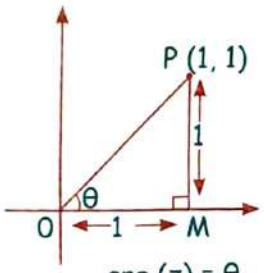
Example 1: $z = \sqrt{3} + i$



$\tan\theta = \frac{1}{\sqrt{3}}$

$\theta = \frac{\pi}{6}$

Example 2: $z = 1 + i$



$\tan\theta = \frac{1}{1}$

$\tan\theta = 1$

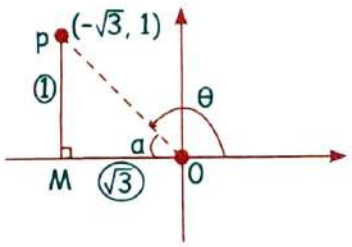
$\theta = \frac{\pi}{4}$

$\arg(z) = \theta$

Example 3: $z = -\sqrt{3} + i$

Ans. $\tan\alpha = \frac{1}{\sqrt{3}}$

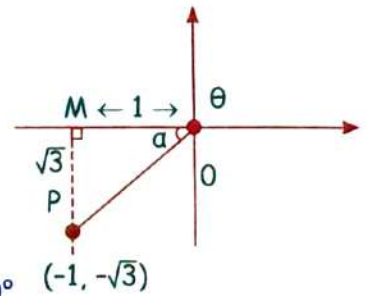
$\alpha = \frac{\pi}{6} \Rightarrow \theta = \frac{5\pi}{6}$



Example 4: $z = -1 - \sqrt{3}i$

Ans. $\tan\alpha = \frac{\sqrt{3}}{1}$

$\Rightarrow \alpha = 60° = \frac{\pi}{3}$



$\arg(z) = 180° + 60°$

$= 240° = \frac{4\pi}{3}$

or $\arg(z) = -\frac{2\pi}{3} \rightarrow$ Principle argument

Principal Argument: The value of θ such that $-\pi < \theta < \pi$ is called Principal argument. It is represented by amp (Z).

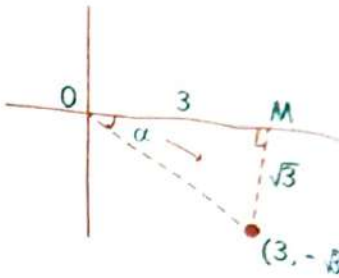
Ques. Find amplitude of

1. $z = \sqrt{3} - 3i$

Ans. $\tan\alpha = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$

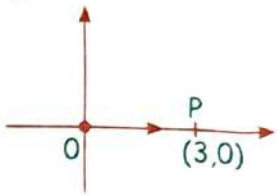
$\alpha = \frac{\pi}{6}$

$\Rightarrow \text{amp}(z) = -\frac{\pi}{6}$



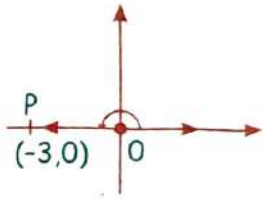
2. $z = 3$

Ans. amp (z) = 0



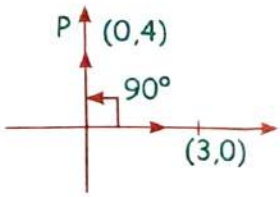
3. $z = -3$

Ans. amp (z) = π



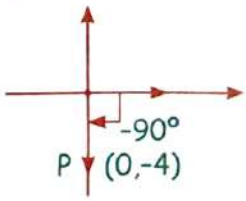
4. $z = 4i$

Ans. $\arg(z) = \frac{\pi}{2}$



5. $z = -4i$

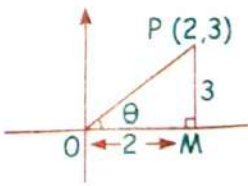
Ans. $\arg(z) = -\frac{\pi}{2}$



6. $z = 2 + 3i$

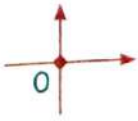
Ans. $\tan\theta = \frac{3}{2}$

$\theta = \tan^{-1}(3/2)$



7. $z = 0$

Ans. arg (z) = N.D.



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Ques. If $\frac{3+i\sin\theta}{4-i\cos\theta}$, $\theta \in [0,2\pi]$ is a real number, then an argument of $\sin\theta + i\cos\theta$ is:

(JEE Main Sep. 05, 2020(II))

(a) $\pi - \tan^{-1}\left(\frac{3}{4}\right)$ (b) $\tan^{-1}\left(\frac{4}{3}\right)$

(c) $\pi - \tan^{-1}\left(\frac{4}{3}\right)$ (d) $-\tan^{-1}\left(\frac{3}{4}\right)$

Ans. $z = \frac{3+i\sin\theta}{4-i\cos\theta} \Rightarrow \bar{z} = \frac{3-i\sin\theta}{4+i\cos\theta}$

$\because z = \bar{z}$ (For purely Real)

$\therefore \frac{3+i\sin\theta}{4-i\cos\theta} = \frac{3-i\sin\theta}{4+i\cos\theta}$

$\Rightarrow \frac{3+i\sin\theta}{3-i\sin\theta} = \frac{4-i\cos\theta}{4+i\cos\theta}$

Componendo and dividendo

$\frac{6}{2i\sin\theta} = \frac{8}{-2i\cos\theta} \Rightarrow \frac{3}{\sin\theta} = \frac{4}{-\cos\theta}$

$\Rightarrow \boxed{\tan\theta = -\frac{3}{4}}$ — 2nd Quad, 4th Quad

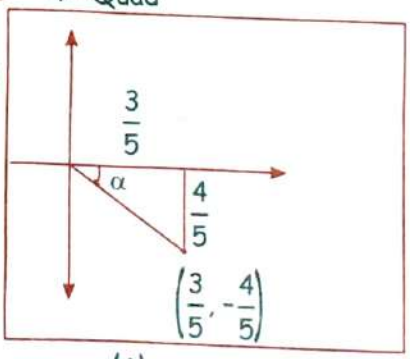
Case 1:

θ in 2nd Quad

$\tan\theta = -\frac{3}{4}$

$\sin\theta = \frac{3}{5}$

$\cos\theta = -\frac{4}{5}$



$z = \frac{3}{5} - \frac{4i}{5} \Rightarrow \theta = -\tan^{-1}\left(\frac{4}{3}\right)$

Case 2:

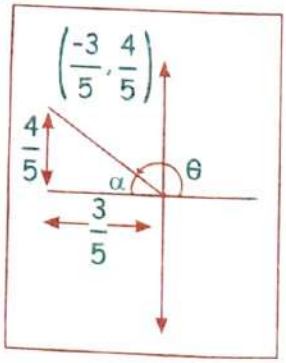
θ in 4th Quad

$\sin\theta = -\frac{3}{5}$

$\cos\theta = \frac{4}{5} = -\frac{3}{5} + \frac{4}{5}i$

$\tan\alpha = \frac{4}{3} \Rightarrow \alpha = \tan^{-1}\frac{4}{3}$

$\arg(z) = \pi - \alpha = \pi - \tan^{-1}\frac{4}{3}$



Ques. Let $z = 1 + i$ and $z_1 = \frac{1+i\bar{z}}{\bar{z}(1-z)+\frac{1}{z}}$. Then $\frac{12}{\pi}$ $\arg(z_1)$ is equal to?

(JEE Main-2023)

Ans. $|z| = \sqrt{1^2+1^2} = \sqrt{2}$

$|z|^2 = 2$

$z_1 = \frac{(1+i\bar{z})z}{z\bar{z}(1-z)+1}$

$= \frac{z + iz\bar{z}}{z\bar{z}(1-z)+1}$

$= \frac{z+i\times 2}{2(1-z)+1}$

$= \frac{1+i+2i}{2(1-1-i)+1} = \frac{1+3i}{1-2i}$

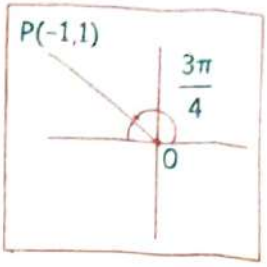
$z_1 = \frac{1+3i}{1-2i}\cdot\frac{1+2i}{1+2i}$

$= \frac{1+5i+6i^2}{1^2-(2i)^2} = \frac{-5+5i}{5}$

$\boxed{z_1 = -1+i}$

$\therefore \arg(z_1) = \tan^{-1}(-1) = 3\pi/4$

$\Rightarrow \frac{12}{\pi}\arg(z_1) = \frac{12}{\pi}\times\frac{3\pi}{4} = 9$



Properties of Argument of a Complex Number

(i) $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

In general $\arg(z_1 z_2 z_3 \ldots z_n) = \arg(z_1) + \arg(z_2) + \arg(z_3) \ldots + \arg(z_n)$

(ii) $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$

(iii) $\arg x$ → 0 if $x > 0$; π if $x < 0$

(iv) $\arg(iy) = \frac{\pi}{2}$ if $y > 0$

$= -\frac{\pi}{2}$ if $y < 0$

(v) $\arg(\bar{z}) = -\arg(z) = \arg\left(\frac{1}{z}\right)$

(vi) $\arg(z^n) = n\arg(z)$

Ques. If $\boxed{\arg(z_1) = \frac{2\pi}{3}}$ and $\boxed{\arg(z_2) = -\frac{\pi}{4}}$ then find the following:

Ans. (i) $\arg(z_1 . z_2) = \arg z_1 + \arg z_2 = \frac{2\pi}{3} - \frac{\pi}{4} = \frac{8\pi - 3\pi}{12}$

$= \frac{5\pi}{12}$

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(ii) $\arg\left(\frac{z_1}{7}\right) = 7 \arg(z_1) = 7 \times \frac{2\pi}{3} = \frac{14\pi}{3} - 4\pi$

$= \frac{14\pi - 12\pi}{3} = \frac{2\pi}{3}$

(iii) $\arg(3z_2^5) = \arg 3 + 5 \arg z_2 = 5\left(\frac{-\pi}{4}\right) = \frac{-5\pi}{4} +$

$2\pi = \frac{5\pi + 8\pi}{4} = \frac{3\pi}{4}$

(iv) $\arg\left(-\frac{2iz_2}{z_1^3}\right) = \arg(-2i) + \arg z_2 - 3 \arg z_1$

$\frac{-\pi}{2} - \frac{\pi}{4} - 3\left(\frac{2\pi}{3}\right) = -\frac{\pi}{2} - \frac{\pi}{4} - 2\pi + 2\pi = \frac{-3\pi}{4}$

Ques. If $z = \frac{(1-i)^{100}(-3+\sqrt{3}i)^{51}}{i^{50}(3i+\sqrt{3})^{60}}$. Find $|z|$ and $\arg(z)$

Ans. $|z^n| = |z|^n$

$|z| = \frac{|1-i|^{100}\,|-3+\sqrt{3}i|^{51}}{|i|^{50}\,|3i+\sqrt{3}|^{60}}$

$|1-i| = \sqrt{2}$

$|-3+\sqrt{3}i| = \sqrt{3^2+3} = \sqrt{12}$

$|i| = \sqrt{0^2+1^2} = 1$

$|3i+\sqrt{3}| = \sqrt{3^2+3} = \sqrt{12}$

$\Rightarrow |z| = \frac{(\sqrt{2})^{100}(\sqrt{12})^{51}}{(1)^{50}(\sqrt{12})^{60}}$

$= \frac{2^{50}}{(\sqrt{12})^9} = \frac{2^{50}}{(2\sqrt{3})^9} = \frac{2^{50}}{2^9(\sqrt{3})^9} = \frac{2^{41}}{(\sqrt{3})^9}$

$\arg z = 100\,\boxed{\arg(1-i)} + 51\,\boxed{\arg(-3+\sqrt{3}i)} -$

$50\,\boxed{\arg(i)} - 60\,\boxed{\arg(3i+\sqrt{3})}$

$\Rightarrow \arg(z) = 100\left(\frac{-\pi}{4}\right) + 51 \times \frac{5\pi}{6} - 50 \times \frac{\pi}{2} - 60 \times \frac{\pi}{3}$

$= -25\pi + \frac{85\pi}{2} - 25\pi - 20\pi$

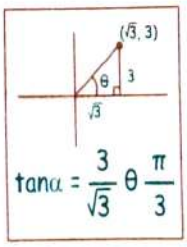
$= -70\pi + \frac{85\pi}{2}$

$= \frac{-140\pi + 85\pi}{2}$

$= \frac{-55\pi}{2}$

$= \frac{-55\pi}{2} + 28\pi$

$= \frac{-55\pi + 56\pi}{2} = \frac{\pi}{2}$



$\tan\alpha = \frac{3}{\sqrt{3}}$ θ $\frac{\pi}{3}$

Ques. Let z_1 and z_2 be two complex number such that $\overline{z}_1 = i\overline{z}_2$ and $\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$. Then

(JEE Main 2020)

(a) $\arg z_2 = \frac{\pi}{4}$ (b) $\arg z_2 = -\frac{3\pi}{4}$

(c) $\arg z_1 = \frac{\pi}{4}$ (d) $\arg z_1 = -\frac{3\pi}{4}$

Ans. $\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$

$\arg \overline{z}_1 - \arg \overline{z}_2 = \pi$

$\Rightarrow \arg z_1 + \arg z_2 = \pi \rightarrow (1)$

$\because \overline{z}_1 = i\overline{z}_2$

$\Rightarrow \arg \overline{z}_1 = \arg i + \arg \overline{z}_2$

$\Rightarrow -\arg z_1 = \frac{\pi}{2} - \arg z_2$

$\Rightarrow \boxed{\arg z_2 - \arg z_1 = \pi/2} \rightarrow (2)$

(1) + (2)

$\Rightarrow 2 \arg z_2 = \pi + \frac{\pi}{2} \Rightarrow \arg(z_2) = \frac{3\pi}{4}$

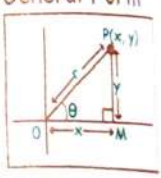
$\Rightarrow \arg z_1 + \frac{3\pi}{4} = \pi$

$\Rightarrow \arg z_1 = \frac{\pi}{4}$

POLAR FORM OF A COMPLEX NUMBER

(r, θ), $r \rightarrow |z|$ & $\theta \rightarrow \arg(z)$

General Form



Trigonometric Form

$z = (r\cos\theta) + i\,r\sin\theta$

$\boxed{z = r(\cos\theta + i\sin\theta)}$

$\boxed{z = r\,\text{cis}\theta}$

Ex: $z = 3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right) = 3\text{cis}\left(\frac{\pi}{6}\right)$

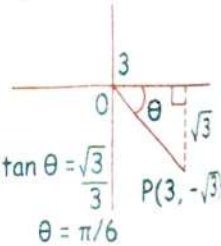
$\Rightarrow |z| = 3$ & $\arg(z) = \frac{\pi}{6}$

Ques. Convert $z = 3 - \sqrt{3}i$ into trigonometric form

Ans. $(|z| = r = \sqrt{3^2+3} = \sqrt{12})$

$\theta = \arg(z) = -\frac{\pi}{6}$

$z = r(\cos\theta + i\sin\theta)$



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$= \sqrt{12}\left[\cos\left(\frac{-\pi}{6}\right) + i\sin\left(\frac{-\pi}{6}\right)\right]$

$= \sqrt{12}\,\text{Cis}\left(\frac{-\pi}{6}\right)$

Ques. If $z = 1 + \text{Cos}\frac{6\pi}{5} + i\sin\frac{6\pi}{5}$ (r) Find $|z|$ and θ (amp z).

Ans. $2\cos^2\frac{3\pi}{5} + i \times 2\text{Sin}\frac{3\pi}{5}\text{Cos}\frac{3\pi}{5}$

$z = \left(2\,\text{Cos}\frac{3\pi}{5}\right)\left[\text{Cos}\frac{3\pi}{5} + i\,\text{Sin}\frac{3\pi}{5}\right]$

(-ve)

$= -2\cos\frac{2\pi}{5}\left[\text{Cos}\frac{3\pi}{5} + i\,\text{Sin}\frac{3\pi}{5}\right]$

$= \left(2\,\text{Cos}\frac{2\pi}{5}\right)\left[-\text{Cos}\frac{3\pi}{5} - i\,\text{Sin}\frac{3\pi}{5}\right]$

(+ve)

$= 2\cos\frac{2\pi}{5}\left[\text{Cos}\frac{2\pi}{5} - i\,\text{Sin}\frac{2\pi}{5}\right]$

$\left(\because \text{Cos}\frac{3\pi}{5} = -\text{ve}\right)$

$\therefore \text{Cos}\frac{3\pi}{5} = -\text{Cos}\frac{2\pi}{5}$

$\text{Sin}\frac{3\pi}{5} = \text{Sin}\frac{2\pi}{5}$

$z = \left(2\,\text{Cos}\frac{2\pi}{5}\right)\left[\text{Cos}\frac{-2\pi}{5} + i\,\text{Sin}\frac{-2\pi}{5}\right]$

$r\,[\cos\theta + i\sin\theta]$

$\Rightarrow r = |z| = 2\cos\frac{2\pi}{5}$

and $\theta = \frac{-2\pi}{5} = \text{amp}\,(z)$

Ques. Let r and θ respectively be the modulus and amplitude of the complex number $z = 2 - i\left(2\tan\frac{5\pi}{8}\right)$, Then (r, θ) is equal to

(JEE MAIN 29 Jan 2024 Shift 2)

(a) $\left(2\text{Sec}\frac{3\pi}{8}, \frac{3\pi}{8}\right)$ (b) $\left(2\text{Sec}\frac{3\pi}{8}, \frac{5\pi}{8}\right)$

(c) $\left(2\text{Sec}\frac{5\pi}{8}, \frac{3\pi}{8}\right)$ (d) $\left(2\text{Sec}\frac{11\pi}{8}, \frac{11\pi}{8}\right)$

Ans. $z = 2 - i\left(2\frac{\text{Sin}\frac{5\pi}{8}}{\text{Cos}\frac{5\pi}{8}}\right)$

$= \frac{2}{\text{Cos}\frac{5\pi}{8}}\left[\text{Cos}\frac{5\pi}{8} - i\,\text{Sin}\frac{5\pi}{8}\right]$

(-ve)

$= 2\,\text{Sec}\frac{5\pi}{8}\left[\text{Cos}\frac{5\pi}{8} - i\,\text{Sin}\frac{5\pi}{8}\right]$

$= 2\,\text{Sec}\frac{3\pi}{8}\left[-\text{Cos}\frac{5\pi}{8} + i\,\text{Sin}\frac{5\pi}{8}\right]$

$= \left(2\,\text{Sec}\frac{3\pi}{8}\right)\left[\text{Cos}\frac{3\pi}{8} + i\,\text{Sin}\frac{3\pi}{8}\right]$

(+ve)

$r = 2\,\text{Sec}\frac{3\pi}{8}$

$\theta = \frac{3\pi}{8}$

Exponential Form

$z = re^{i\theta}$, is Known as exponential form of a complex number.

$z = re^{i\theta}$

$z = r\,[\text{Cos}\theta + i\,\text{Sin}\theta]$

$\Rightarrow r\,(\text{Cos}\theta + i\,\text{Sin}\theta) = re^{i\theta}$

$\Rightarrow \text{Cos}\theta + i\,\text{Sin}\theta = e^{i\theta}$

Put $\theta = 0 \Rightarrow z = r\,[\text{Cos}0 + i\,\text{Sin}0]$

$\Rightarrow z = r$

$z = r\,(\text{Cos}\theta + i\,\text{Sin}\theta)$

$\Rightarrow \frac{dz}{d\theta} = r\,[-\text{Sin}\theta + i\text{Cos}\theta]$

$= r\,[i^2\,\text{Sin}\theta + i\,\text{Cos}\theta]$

$\Rightarrow \frac{dz}{d\theta} = i\,(r\,[\text{Cos}\theta + i\,\text{Sin}\theta]) \rightarrow z$

$r \rightarrow |z|$

$\theta \rightarrow \arg(z)$

$e \rightarrow$ Napier's count

$\Rightarrow \frac{dz}{d\theta} = iz \Rightarrow \int_r^z \frac{dz}{z} = \int_0^\theta i\,d\theta$

$\Rightarrow \ln z\Big|_r^z = i\theta\Big|_0^\theta \quad \left(\because \int \frac{dz}{d\theta} = iz\right)$

$\Rightarrow \ln z - \ln r = i(\theta - 0)$

$\Rightarrow \ln\left(\frac{z}{r}\right) = i\theta \Rightarrow \frac{z}{r} = e^{i\theta} \Rightarrow z = re^{i\theta}$

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Exponential Form

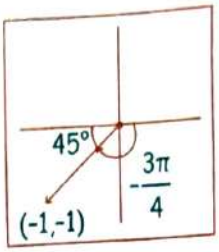
Ques. Express $z = -1 - i$ into Euler's Form

Ans. $z = re^{i\theta}$

$r = |z| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\theta = \frac{-3\pi}{4}$

$\Rightarrow z = \sqrt{2}\, e^{i\frac{3\pi}{4}}$



Ques. Find the value of $[1 + i]^{100}$.

(a) $2^{100}(\cos 100\pi + i \sin 100\pi)$

(b) $2^{100}(\cos 25\pi + i \sin 25\pi)$

(c) $2^{50}(\cos 100\pi + i \sin 100\pi)$

(d) ✓ $2^{50}(\cos 25\pi + i \sin 25\pi)$

Ans. $r = |1 + i| = \sqrt{2}$

$\theta = \frac{\pi}{4}$

$1 + i = re^{i\theta}$

$\Rightarrow (1 + i) = \sqrt{2}\, e^{\frac{i\pi}{4}}$

$\Rightarrow (1 + i)^{100} = \left[\sqrt{2}\, e^{\frac{i\pi}{4}}\right]^{100}$

$= (\sqrt{2})^{100}\left[e^{\frac{i100\pi}{4}}\right]$

$= 2^{50}\,[e^{i25\pi}]$

$= 2^{50}\,[\cos 25\pi + i \sin 25\pi]$

Ques. If $(\sqrt{3} + i)^{100} = 2^{99}(p + iq)$ then p and q are roots of the equation **(JEE Main(Aug) 2021)**

(a) ✓ $x^2 - (\sqrt{3}-1)x - \sqrt{3} = 0$

(b) $x^2 - (\sqrt{3}+1)x + \sqrt{3} = 0$

(c) $x^2 + (\sqrt{3}-1)x - \sqrt{3} = 0$

(d) $x^2 - (\sqrt{3}+1)x + \sqrt{3} = 0$

Ans. $z = \sqrt{3} + i$

$|z| = \sqrt{3 + 1} = 2$

$\Rightarrow (\sqrt{3} + i)^{100} = (2e^{\frac{i\pi}{6}})^{100}$

$= 2^{100}\, e^{i\frac{100\pi}{6}}$

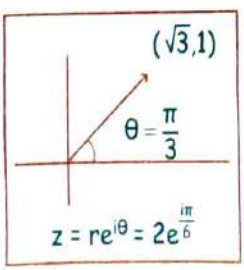
$= 2^{100}\, e^{i\frac{50\pi}{3}}$

$= 2^{100}\,[\cos\frac{50\pi}{3} + i \sin\frac{50\pi}{3}]$

$\because \frac{50\pi}{3} - 16\pi = \frac{50\pi - 48\pi}{3} = \frac{2\pi}{3}$

$= 2^{100}\,[\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}]$

$= 2^{100}\,[\frac{-1}{2} + \frac{\sqrt{3}}{2}] = 2^{99}\,[-1 + \sqrt{3}i]$



$p = -1,\ q = \sqrt{3}$

sum $= \sqrt{3} - 1$,

product $= -\sqrt{3}$

Ques. In algebraic form $z = i^i$ simplifies to

(a) $-e^{-\frac{\pi}{2}}$ (b) $e^{-\frac{i\pi}{2}}$

(c) ✓ $e^{-\frac{\pi}{2}}$ (d) $ie^{-\frac{\pi}{2}}$

Ans. $z = re^{i\theta}$

We know that $i = e^{\frac{i\pi}{2}}$

$i^i = (e^{\frac{i\pi}{2}})^i = e^{\frac{i^2\pi}{2}} = e^{-\frac{\pi}{2}}$

Ques. If $(\cos\theta + i\sin\theta)(\cos 2\theta + i\sin 2\theta)\ldots(\cos n\theta + i\sin n\theta) = 1$ then the value of θ is

(a) $4m\pi$ (b) $\frac{2m\pi}{n(n+1)}$

(c) ✓ $\frac{4m\pi}{n(n+1)}$ (d) $\frac{m\pi}{n(n+1)}$

Ans. $\Rightarrow e^{i\theta}.\, e^{i2\theta}.\, e^{i3\theta}.\ldots.\, e^{in\theta} = 1$

$\Rightarrow e^{i\theta + i2\theta + i3\theta \ldots + in\theta} = 1$

$\Rightarrow e^{i\theta\,[1 + 2 + 3 + \ldots + n]} = 1$

$\Rightarrow e^{\left[i\theta\frac{n(n+1)}{2}\right]} = 1$

$\Rightarrow \cos\left(\frac{n(n+1)\theta}{2}\right) + i\sin\frac{n(n+1)\theta}{2} = 1$

$\Rightarrow \cos\left(\frac{n(n+1)\theta}{2}\right) = 1$

$\Rightarrow \frac{n(n+1)\theta}{2} = 2m\pi \quad (\because \cos x = 1 \therefore x = 2m\pi)$

$\Rightarrow n(n+1)\theta = 4m\pi$

$\Rightarrow \theta = \frac{4m\pi}{n(n+1)}$

DE MOIVRE'S THEOREM

The only value of $(\cos\theta + i\sin\theta)^n$ is $\cos(n\theta) + i\sin(n\theta)$ if 'n' is integer and it is one of the values of $(\cos\theta + i\sin\theta)^n$ if n is a non integral rational number

Proof: LHS: $(\cos\theta + i\sin\theta)^n$

$(e^{i\theta})^n = e^{in\theta}$

$= \cos n\theta + i\sin n\theta$ = RHS

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Ex. (1) $(\cos\theta + i\sin\theta)^{\frac{2}{3}} = \cos\frac{2\theta}{3} + i\sin\frac{2\theta}{3}$

one of the values

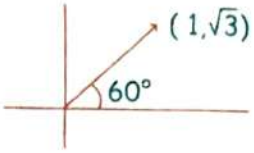
(2) $(\cos\theta + i\sin\theta)^{\frac{1}{5}} = \cos\frac{\theta}{5} + i\sin\frac{\theta}{5}$

one of the values

Ques. If $z = 1 + i\sqrt{3}$ then find $z^4 = ?$

Ans. $|z| = r = 2$,

$\arg z = \theta = \frac{\pi}{3}$



$z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$z^4 = 2^4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^4$

$= 2^4\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right) \rightarrow$ D.M.T

$= 16\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$

$= 8(-1 - \sqrt{3}i) = -8[1 + \sqrt{3}i]$

Ques. If $\frac{1}{x} + x = 2\cos A$, then $x^n + \frac{1}{x^n}$ is equal to

(a) $2\cos nA$ (b) $2\sin nA$

(c) $\cos nA$ (d) $\sin nA$

Ans. $x + \frac{1}{x} = 2\cos A$ $\qquad \frac{1}{x^n} = x^{-n}$

$\Rightarrow x^2 + 1 = 2x\cos A$

$\Rightarrow x^2 - (2\cos A)x + 1 = 0$

$\Rightarrow x = \frac{2\cos A \pm \sqrt{4\cos^2 A - 4}}{2}$

$\Rightarrow x = \frac{2\cos A \pm 2\sqrt{\cos^2 A - 1}}{2}$

$(\because 1 - \cos^2 A = \sin^2 A)$

$\Rightarrow x = \cos A \pm \sqrt{-\sin^2 A}$

$\Rightarrow x = \cos \pm i\sin A$

$\oplus$ $x = \cos A + i\sin A$

$\Rightarrow \boxed{x^n = \cos nA + i\sin nA} \rightarrow$ ①

& $x^{-n} = \cos(-nA) + i\sin(-nA)$

$\Rightarrow \boxed{x^{-n} = \cos nA - i\sin nA} \rightarrow$ ②

By adding eqn (i) and (ii)

$\Rightarrow x^n + x^{-n} = 2\cos nA$

Ques. $\left(\frac{1 + \sin\theta + i\cos\theta}{1 + \sin\theta - i\cos\theta}\right)^n =$

(a) $\cos\left(\frac{n\pi}{2} - n\theta\right) + i\sin\left(\frac{n\pi}{2} - n\theta\right)$

(b) $\cos\left(\frac{n\pi}{2} + n\theta\right) + i\sin\left(\frac{n\pi}{2} + n\theta\right)$

(c) $\sin\left(\frac{n\pi}{2} - n\theta\right) + i\cos\left(\frac{n\pi}{2} - n\theta\right)$

(d) $\cos n\left(\frac{n\pi}{2} + n\theta\right) + i\sin n\left(\frac{n\pi}{2} + n\theta\right)$

Ans. $= \left(\frac{1 + \cos\left(\frac{\pi}{2} - \theta\right) + i\sin\left(\frac{\pi}{2} - \theta\right)}{1 + \cos\left(\frac{\pi}{2} - \theta\right) - i\sin\left(\frac{\pi}{2} - \theta\right)}\right)^n$

$= \frac{2\cos^2\left(\frac{\pi}{4} - \frac{\theta}{2}\right) + i\,2\sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right)\cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right)}{2\cos^2\left(\frac{\pi}{4} - \frac{\theta}{2}\right) - i\,2\sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right)\cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right)}$

$= \left(\frac{\cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right) + i\sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right)}{\cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right) - i\sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right)}\right)^n$

$= \left[\left(\cos\frac{\pi}{4} - \frac{\theta}{2}\right) + i\sin\left(\frac{\pi}{4} - \frac{\theta}{2}\right)^{2n}\right]$

$= \cos 2n\left(\frac{\pi}{4} - \frac{\theta}{2}\right) + i\sin 2n\left(\frac{\pi}{4} - \frac{\theta}{2}\right)$

$= \cos n\left(\frac{\pi}{2} - \theta\right) + i\sin n\left(\frac{\pi}{2} - \theta\right)$

Ques. The value of $\left(\frac{1 + \sin\frac{2\pi}{9} + i\cos\frac{2\pi}{9}}{1 + \sin\frac{2\pi}{9} - i\cos\frac{2\pi}{9}}\right)^3$ is:

(JEE Main 2 sept. 2020)

(a) $-\frac{1}{2}(1 - i\sqrt{3})$ (b) $\frac{1}{2}(1 - i\sqrt{3})$

(c) $\frac{1}{2}(\sqrt{3} - i)$ (d) $-\frac{1}{2}(\sqrt{3} - i)$

Ans. $\left(\frac{1 + \sin 40^\circ + i\cos 40^\circ}{1 + \sin 40^\circ - i\cos 40^\circ}\right)^3$

$= \left(\frac{1 + \cos 50^\circ + i\sin 50^\circ}{1 + \cos 50^\circ - i\sin 50^\circ}\right)^3$

$\frac{\pi}{9} = \frac{180^\circ}{9} = 20^\circ$

$\sin\frac{2\pi}{9} = \sin 40^\circ$

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$$= \left(\frac{2\cos^2 25^\circ + i.2\sin 25^\circ \cos 25^\circ}{2\cos^2 25^\circ - i.2\sin 25^\circ \cos 25^\circ}\right)^3$$

$$= \left(\frac{\cos 25^\circ + i \sin 25^\circ}{\cos 25^\circ - i \sin 25^\circ}\right)^3 = (\cos 25^\circ + i \sin 25^\circ)^6$$

$$= \cos 150^\circ + i \sin 150^\circ = -\frac{\sqrt{3}}{2} + \frac{i}{2}$$

Ques. The value of $i^{\frac{1}{3}}$ is

(a) $\frac{\sqrt{3}+i}{2}$ (b) $\frac{\sqrt{3}-i}{2}$ (c) $\frac{1+i\sqrt{3}}{2}$ (d) $\frac{1-i\sqrt{3}}{2}$

Ans. $i = \cos\frac{\pi}{2} + i \sin\frac{\pi}{2}$

$$\Rightarrow i^{\frac{1}{3}} = \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right)\right)^{\frac{1}{3}}$$

$$= \left(\cos\left(2m\pi + \frac{\pi}{2}\right) + i \sin\left(2m\pi + \frac{\pi}{2}\right)\right)^{\frac{1}{3}}$$

$$= \left(\cos\left(\frac{2m\pi + \frac{\pi}{2}}{3}\right) + i \sin\left(\frac{2m\pi + \frac{\pi}{2}}{3}\right)\right)$$

$m = 0 \Rightarrow \text{cis}\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} + \frac{i}{2}$

$m = 1 \Rightarrow \text{cis}\left(\frac{5\pi}{6}\right) = \cos\frac{5\pi}{6} + i \sin\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} + \frac{i}{2}$

$m = 2 \Rightarrow \text{cis}\left(\frac{9\pi}{6}\right) = \cos\frac{3\pi}{2} + i \sin\frac{3\pi}{2} = -i$

$$m = 3 \Rightarrow \text{cis}\left(\frac{6\pi + \frac{\pi}{2}}{3}\right) = \text{cis}\left(2\pi + \frac{\pi}{6}\right) = \text{cis}\left(\frac{\pi}{6}\right)$$

Finding Cube Roots of Unity Using D.M.T

$z = (1)^{\frac{1}{3}} = (\cos 0 + i \sin 0)^{\frac{1}{3}}$

$= (\cos 2m\pi + i \sin 2m\pi)^{\frac{1}{3}}$

$$z = \left(\cos\frac{2m\pi}{3} + i \sin\frac{2m\pi}{3}\right)$$

$m = 0 \Rightarrow z = \cos 0 + i \sin 0 = 1$

$$m = 1 \Rightarrow z = \cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3} = \frac{-1}{2} + \frac{\sqrt{3}}{2}i + \frac{-1+\sqrt{3}i}{2} = \omega$$

$$m = 2 \Rightarrow z = \cos\frac{4\pi}{3} + i \sin\frac{4\pi}{3} - \frac{-1}{2} + \frac{-\sqrt{3}}{2}i = \frac{-1-\sqrt{3}i}{2} = \omega^2$$

$m = 3 \Rightarrow z = \cos 2\pi + i \sin 2\pi = 1$

4th roots of Unity

$z = (1)^{\frac{1}{4}}$

$\Rightarrow z = (\cos 0 + i \sin 0)^{\frac{1}{4}}$

$$\Rightarrow z = \cos\frac{2m\pi}{4} + i \sin\frac{2m\pi}{4}$$

$m = 0 \Rightarrow z_1 = \cos 0 + i \sin 0 = 1$

$m = 1 \Rightarrow z_2 = \cos\frac{2\pi}{4} + i \sin\frac{2\pi}{4} = i$

$m = 2 \Rightarrow z_3 = \cos\frac{4\pi}{4} + i \sin\frac{4\pi}{4} = -1$

$m = 3 \Rightarrow z_4 = \cos\frac{6\pi}{4} + i \sin\frac{6\pi}{4} = -i$

4th roots of unity $\pm 1, \pm i$

Ques. Find 5th Root of $(1 + i)$

Ans. $z = (1+i)^{\frac{1}{5}}$

$$z = \left(\sqrt{2}\left(\cos\frac{\pi}{4} + i \sin\frac{\pi}{4}\right)\right)^{\frac{1}{5}}$$

$$= (2)^{\frac{1}{10}}\left(\cos\frac{\pi}{4} + i \sin\frac{\pi}{4}\right)^{\frac{1}{5}}$$

$$= (2)^{\frac{1}{10}}\left(\cos\left(2m\pi + \frac{\pi}{4}\right) + i \sin\left(2m\pi + \frac{\pi}{4}\right)\right)^{\frac{1}{5}}$$

$$z = 2^{\frac{1}{10}}\left[\cos\frac{2m\pi + \frac{\pi}{4}}{5} + i \sin\frac{2m\pi + \frac{\pi}{4}}{5}\right]$$

$m = 0 \Rightarrow z = 2^{\frac{1}{10}}\,\text{cis}\left(\frac{\pi}{20}\right)$

$m = 1 \Rightarrow z = 2^{\frac{1}{10}}\,\text{cis}\left(\frac{9\pi}{20}\right)$

$m = 2 \Rightarrow z = 2^{\frac{1}{10}}\,\text{cis}\left(\frac{17\pi}{20}\right)$

$m = 3 \Rightarrow z = 2^{\frac{1}{10}}\,\text{cis}\left(\frac{25\pi}{20}\right)$

$m = 4 \Rightarrow z = 2^{\frac{1}{10}}\,\text{cis}\left(\frac{33\pi}{20}\right)$

Ques. Let $A = \left\{\frac{1967 + 1686i \sin\theta}{7 - 3i\cos\theta} : \theta \in R\right\}$. If A Contains exactly one positive integer n, then value of n is (JEE Main 2023)

Ans. $\frac{1967 + 1686i \sin\theta}{7 - 3i\cos\theta} = n$

$\Rightarrow 1967 + 1686i \sin\theta = 7n - 3in\cos\theta$

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Comparing real parts we get

$1967 = 7n$

$\Rightarrow n = \frac{1967}{7} = 281$

Ques. Using DMT prove $\sin 3\theta$ and $\cos 3\theta$ formula.

Ans. Proof: $(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i \sin 3\theta$

$\Rightarrow \cos^3\theta + (i\sin\theta)^3 + 3\cos\theta\, i \sin\theta\, [\cos\theta + i\sin\theta]$
$= \cos 3\theta + i \sin 3\theta$

$\Rightarrow \cos^3\theta - i\sin^3\theta + 3\sin\theta \cos^2\theta\, i + 3\cos \sin^2\theta\, i^2$
$= \cos 3\theta + i \sin 3\theta$

$\Rightarrow (\cos^3\theta - 3\cos\theta \sin^2\theta) + i\,[3\sin\theta \cos^2\theta - \sin^3\theta]$
$= \cos 3\theta + i \sin 3\theta$

$\Rightarrow [\cos^3\theta - 3\cos\theta\,(1 - \cos^2\theta)] + i\,[3\sin\theta\,(1 - \sin^2\theta) - \sin^3\theta] = \cos 3\theta + i \sin 3\theta$

$\Rightarrow (4\cos^3\theta - 3\cos\theta) + i\,[3\sin\theta - 4\sin^3\theta] = \cos 3\theta + i \sin 3\theta$

Comparing real and imaginary parts, we get

$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

Ques. If α, β be the roots of the equation $\mu^2 - 2\mu + 2 = 0$ and if $\cot\theta = x + 1$, then $[(x + \alpha)^n - (x + \beta)^n] / [\alpha - \beta]$ is equal to

(a) $\frac{\sin n\theta}{\sin^n\theta}$ ✓ (b) $\frac{\cos n\theta}{\cos^n\theta}$

(c) $\frac{\sin n\theta}{\cos^n\theta}$ (d) $\frac{\cos n\theta}{\sin^n\theta}$

Ans. $\mu = \frac{2 \pm i\sqrt{4 - 4 \times 2}}{2 \times 1} = \frac{2 \pm 2i}{2} = 1 \pm i$

$\alpha = 1 + i$ & $\beta = 1 - i$

$\alpha - \beta = 2i$

$x + \alpha = \cot\theta - 1 + 1 + i$

$= \cot\theta + i$

$x + \alpha = \frac{\cos\theta + i\,\sin\theta}{\sin\theta} \Rightarrow$

$x + \beta = \cot\theta - 1 + 1 - i$

$= \cot\theta - i$

$x + \beta = \frac{\cos\theta - i\sin\theta}{\sin\theta}$

$(x + \alpha)^n = \frac{(\cos\theta + i\sin\theta)^n}{(\sin\theta)^n} = \frac{\cos n\theta + i \sin n\theta}{(\sin\theta)^n}$

$(x + \beta)^n = \frac{\cos n\theta - i \sin n\theta}{(\sin\theta)^n}$

$\frac{(x + \alpha)^n - (x + \beta)^n}{\alpha - \beta} = \frac{2i \sin n\theta}{2i(\sin\theta)^n} = \frac{\sin n\theta}{\sin^n\theta}$

Cube Roots of Unity Using D.M.T

$1, \omega, \omega^2 \rightarrow$ cube roots

$\omega = \frac{-1 + \sqrt{3}\,i}{2}$, $\omega^2 = \frac{-1 - \sqrt{3}\,i}{2}$,

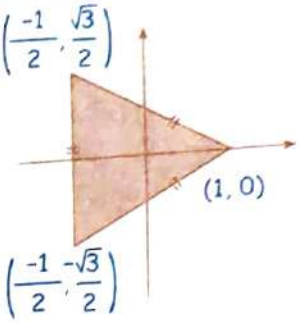
$1 + \omega + \omega^2 = 0$

$\omega^3 = 1$

$\omega^{3n} = 1$

$\omega^{3n+1} = \omega$

$\omega^{3n+2} = \omega^2$



Geometrical Significance of cube roots. They represent vertices of equilateral triangle.

$x^2 + x + 1 = 0 \Rightarrow \omega, \omega^2$

$x^2 - x + 1 = 0 \Rightarrow -\omega, -\omega^2$

Fourth Roots of Unity Using D.M.T

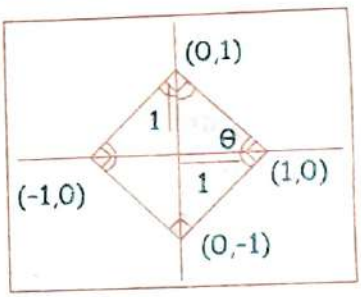
$z = (1)^{\frac{1}{4}}$ $\quad i = 0 + i$

$z^4 = 1$ $\quad -i = 0 - i$

$z = 1, -1, i, -i$

4^{th} roots of unity lie on square"

$\tan\theta = \frac{1}{1} \Rightarrow \theta = 45°$



Ques. If ω is an imaginary cube root of Unity, then simplify $\frac{2 + 3\omega + \omega^2}{1 + 2\omega + 3\omega^2} + \frac{1 + 2\omega + 3\omega^2}{3 + 2\omega + \omega^2}$

(a) -1 (b) 1 (c) ω ✓ (d) None

Ans. $= \frac{1 + 2\omega}{\omega + 2\omega^2} + \frac{\omega + 2\omega^2}{2 + \omega^2}$ $\quad [\because 1 + \omega + \omega^2 = 0]$

$= \frac{(1 + 2\omega)}{\omega(1 + 2\omega)} + \frac{\omega(1 + 2\omega)}{2 + \omega}$

$= \frac{1}{\omega} + \frac{\omega(1 + 2\omega)}{2 + \omega}$

$= \frac{2 + \omega + \omega^2(1 + 2\omega)}{\omega(2 + \omega)} = \frac{2 + \omega + \omega^2 + 2\omega^3}{\omega(2 + \omega)}$

$= \frac{1 + 2}{\omega(2 + \omega)} = \frac{3}{\omega(2 + \omega)} = \frac{3}{2\omega + \omega^2}$

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Ques. The sum of 162^{th} power of the roots of the equation $x^3 - 2x^2 + 2x - 1 = 0$ is

(JEE Main 26 Feb 2020 I)

Ans. $(x^3 - 1) - 2x(x - 1) = 0$

$(x - 1)(x^2 + x + 1) - 2x(x - 1) = 0$

$(x - 1)[x^2 + x + 1 - 2x] = 0$

$\boxed{(x - 1) + (x^2 - x + 1) = 0}$

$\boxed{x = 1}$ or $\boxed{x^2 - x + 1 = 0}$

$x^2 - x + 1 = 0 \Rightarrow x = \frac{1 \pm \sqrt{1 - 4}}{2 \times 1} = \frac{1 \pm \sqrt{3}i}{2}$

$\frac{1 + \sqrt{3}\,i}{2}$, $\frac{1 - \sqrt{3}\,i}{2}$

$\downarrow$ $-\omega^2$ $\quad$ $\downarrow$ $-\omega$

$1, -\omega, -\omega^2$ are the roots

$\alpha^{162} + \beta^{162} + \gamma^{162} = (1)^{162} + (-\omega)^{162} + (-\omega^2)^{162}$

$= 1 + \omega^{162} + \omega^{324} = 1 + 1 + 1 = 3$

Brain Teaser

If $z^2 - 2z + 4 = 0$, then the value of $\left(\frac{z}{2} + \frac{2}{z}\right)^2 + \left(\frac{z^2}{4} + \frac{4}{z^2}\right)^2 + \left(\frac{z^3}{8} + \frac{8}{z^3}\right)^2 + \ldots + \left(\frac{z^{24}}{8^{24}} + \frac{2^{24}}{z^{24}}\right)^2$ is equal to

(a) 24 (b) 32

(c) ~~48~~ (d) None of these

Ans. $\left(\frac{z}{2}\right)^2 - \frac{z}{2} + 1 = 0$

$\frac{z}{2} = -\omega$ or $-\omega^2 \Rightarrow z = -2\omega$ or $-2\omega^2$

$\frac{z}{2} = -\omega$

$\left(\frac{z}{2} + \frac{2}{z}\right)^2$	$\left(\omega^2 + \frac{1}{\omega^2}\right)^2$	$\left(-\omega^3 - \frac{1}{\omega^3}\right)^2$
$\left(-\omega - \frac{1}{\omega}\right)^2$	$\left(\frac{\omega^4 + 1}{\omega^2}\right)^2$	$(-1 - 1)^2$
$\left(\frac{-\omega^2 - 1}{\omega}\right)^2$	$\left(\frac{\omega + 1}{\omega^2}\right)^2$	$(-2)^2$
$\left(\frac{\omega}{\omega}\right)^2$	$\left(\frac{-\omega^2}{\omega^2}\right)^2 = 1$	④

$= 1 = (1 + 1 + 4) \times 8 = 6 \times 8 = 48$

Ques. If a and b are real numbers such that $(2 + \omega)^4 = a + b\omega$, where $= \frac{-1 + i\sqrt{3}}{2}$, then a + b equal to

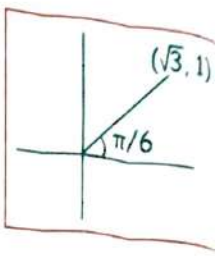
[JEE Main 4 Sept 2020 II]

(a) 33 (b) ~~9~~ (c) 24 (d) 57

Ans. $(2 + \omega)^4 = \left[2 - \frac{1 + \sqrt{3}i}{2}\right]^4$

$= \left[\frac{4 - 1 + \sqrt{3}i}{2}\right]^4$

$= \left[\frac{3 + \sqrt{3}i}{2}\right]^4$



$= \frac{(\sqrt{3})^4}{2^4}(\sqrt{3} + i)^4$

$= \frac{3^2}{16}\left[2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^4$ $\quad |\sqrt{3} + i| = 2$

$\quad \arg(\sqrt{3} + i) = \frac{\pi}{6}$

$= 3^2\left[\cos\frac{4\pi}{6} + i\sin\frac{4\pi}{6}\right]$

$= 9\left[\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right] = 9\left[\frac{-1}{2} + \frac{\sqrt{3}i}{2}\right] = 9\omega$

$\boxed{\therefore b = 9}$

$\boxed{a = 0}$

$a + b = 9$

Important Note

1) The roots of $x^3 = 1$ are $1, \omega, \omega^2$

2) The roots of $x^4 = 1$ are $\pm 1, \pm i$

3) The roots of $x^3 = 27$ are

$3, 3\omega, 3\omega^2$

4) The roots of $x^3 = -64$ are

$-4, -4\omega, -4\omega^2$

5) The roots of $(x - 1)^3 = 8$ are

$x - 1 = 2, 2\omega, 2\omega^2 \Rightarrow x = 3, 1 + 2\omega, 1 + 2\omega^2$

6) The roots of $(2x + 1)^3 + 27 = 0$ are

$\Rightarrow (2x + 1)^3 = -27$

$\Rightarrow 2x + 1 = -3, -3\omega, -3\omega^2$

$\Rightarrow 2x = -1 - 3, -1 - 3\omega, -1 - 3\omega^2$

$x = -2, \frac{-1 - 3\omega}{2}, \frac{-1 - 3\omega^2}{2}$

7) The roots of $x^4 = 16$ are

$\Rightarrow x = 2, -2, +2i, -2i$

8) The roots of $(x - 2)^4 = 81$ are

$x - 2 = 3, -3, 3i, -3i$

$x = 5, -1, 2 + 3i, 2 - 3i$

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log OF COMPLEX NUMBER

$z = re^{i\theta}$

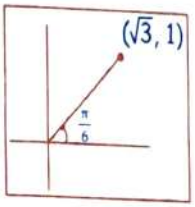
$\ln z = \ln r + \ln e^{i\theta}$

$\ln z = \ln r + i\theta \ln e$

$\boxed{\ln z = \ln r + i\theta}$ → Particular value

$\boxed{\ln z = \ln r + i(\theta + 2m\pi)}$ → General value

$\boxed{\ln i = i\left(\frac{\pi}{2} + 2m\pi\right)}$



Ex: $z = \sqrt{3} + i$, Find $\ln z$

$\ln z = \ln r + i\theta$

$r = |z| = 2$

$\theta = \frac{\pi}{6}$

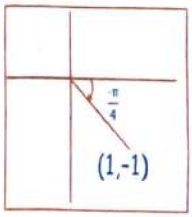
$\ln z = \ln 2 + i\frac{\pi}{6} \Rightarrow \ln(\sqrt{3} + i) = \ln 2 + i\frac{\pi}{6}$

Ex: $\ln i = \ln 1 + i\frac{\pi}{2}$

$\boxed{\ln i = i\frac{\pi}{2}}$ → one of the values

Ex: $\ln(1 + i) = \ln\sqrt{2} + i\frac{\pi}{4}$

Ques. If $z = 1 - i$. Find Re $(\ln z)$ & Im $(\ln z)$



Ans. $r = \sqrt{2}$

$\ln z = \ln r + i\theta$

$\ln z = \underbrace{\ln\sqrt{2}}_{\text{Real part}} + i\underbrace{\left(\frac{-\pi}{4}\right)}_{\text{Imaginary part}}$

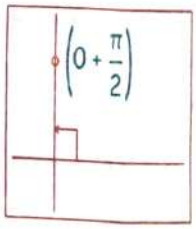
Ques. $\ln(\ln i) = ?$

(a) $\ln\frac{\pi}{2} + i\frac{\pi}{2}$

(b) $\ln\frac{\pi}{2} - i\frac{\pi}{2}$

(c) $\ln\frac{\pi}{2}$

(d) $\frac{\pi}{2}\ln i$



Ans. $\because \ln i = i\frac{\pi}{2}$

$\ln z = \ln\left(\frac{i\pi}{2}\right) = \ln i + \ln\frac{\pi}{2} = i\frac{\pi}{2} + \ln\frac{\pi}{2}$

EULER'S IDENTITIES

$\cos x = \frac{e^{ix} + e^{-ix}}{2}$ and $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$

$x \to i\theta$

$\cos(i\theta) = \frac{e^{i\cdot i\theta} + e^{-i\cdot i\theta}}{2}$

$\cos(i\theta) = \frac{e^{-\theta} + e^{\theta}}{2}$

$\cos(i\theta) = \frac{e^{\theta} + e^{-\theta}}{2}$ → Real & + ve

$x \to i\theta$

$\sin i\theta = \frac{e^{-\theta} - e^{\theta}}{2i}$ → Purely Imaginary

Proof: Consider

$e^{ix} = \cos x + i\sin x$

$\oplus\ e^{-ix} = \cos x - i\sin x$

$e^{ix} + e^{-ix} = 2\cos x$

$\Rightarrow \cos x = \frac{e^{ix} + e^{-ix}}{2}$

$\ominus\ e^{ix} - e^{-ix} = 2i\sin x$

$\Rightarrow \sin x = \frac{e^{ix} - e^{-ix}}{2i}$

Ques. Find the value of the following -

(1) $\cos[3i]$

Ans. $\boxed{\cos i\theta = \frac{e^{\theta} - e^{-i\theta}}{2}}$

Put $\theta = 3$

$\cos 3i = \frac{e^{3} + e^{-3}}{2}$ → Real

(2) $\cos[i \ln 2]$

Ans. $\boxed{e^{\ln x} = x}$

Put $\theta = \ln 2$

$\cos(i \ln 2) = \frac{e^{\ln 2} + e^{-\ln 2}}{2}$

$= \frac{2 + \frac{1}{2}}{2} = \frac{5}{4}$

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Short Notes

DEFINITION

- Any number of the form x + iy (where x and y are real numbers) is called a complex number.
- x is called the real part of the complex number and y is called the imaginary part of the complex number z. 'x' is denoted as Re(z) and 'y' is denoted as Im(z).
- For any complex number both the real parts and the imaginary parts are always real.
- In general $i^{4n} = 1$, $i^{4n+1} = i$, $i^{4n+2} = -1$, $i^{4n+3} = -i$, where $n \in I$

CONJUGATE OF A COMPLEX NUMBER i.e., $\bar{z} = x - iy$

- $\bar{z}$ is the mirror image of z along real axis.
- If $z = \bar{z} \Leftrightarrow z$ is purely real.
- If $z = -\bar{z} \Leftrightarrow z$ is purely imaginary
- $z + \bar{z} = 2Re(z)$ & $z - \bar{z} = 2i\,Im(z)$
- $\overline{z_1 + z_2} = \bar{z_1} + \bar{z_2}$
- $\overline{z_1 z_2} = \left(\bar{z_1}\right)\left(\bar{z_2}\right)$
- $\left(\overline{\frac{z_1}{z_2}}\right) = \frac{\bar{z_1}}{\bar{z_2}} (z_2 \neq 0)$
- $z_1\bar{z}_2 + \bar{z}_1 z_2 = 2Re(z_1\bar{z}_2)$
- $\overline{z^n} = \left(\bar{z}\right)^n$

MODULUS OF A COMPLEX NUMBER i.e., $|z| = \sqrt{x^2 + y^2}$

- Distance of a complex number z from origin is called the modulus of the complex number z and it is denoted by |z|
- $z\bar{z} = |z|^2$
- $|z_1 z_2| = |z_1||z_2|$
- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
- $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + z_2\bar{z}_1$
  $= |z_1|^2 + |z_2|^2 + 2Re(z_1\bar{z}_2)$
- $|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - z_1\bar{z}_2 - z_2\bar{z}_1$
  $= |z_1|^2 + |z_2|^2 - 2Re(z_1\bar{z}_2)$
- $|z|^n = |z^n|$, $n \in N$

ARGUMENT OF A COMPLEX NUMBER

- The angle made by the line joining origin with the point representing complex number in the argand plane, with the positive direction of X-axis is called argument of a complex number.
- Value of θ in the range $-\pi < \theta \leq \pi$, is called the Principal Argument of z. It is also known as amplitude and represented by amp (z).
- $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
- $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$
- $\arg(z)^n = n \arg(z)$

REPRESENTATION OF A COMPLEX NUMBER IN VARIOUS FORMS

- Cartesian Form $z = x + iy$
- Trigonometric Form $z = r(\cos\theta + i\sin\theta)$
- Exponential Form $z = re^{i\theta}$
- Euler's Formula $e^{ix} = \cos x + i\sin x$
  & $e^{-ix} = \cos x - i\sin x$

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ROTATION OF COMPLEX NUMBERS

- (Final Complex number) = (Initial Complex number) $e^{i\theta}$ if rotation is anticlockwise.
- (Final Complex number) = (Initial Complex number) $e^{-i\theta}$ if rotation is clockwise.
- If Complex numbers z_1, z_2, z_3 respectively represent the vertices of an isosceles right angled triangle ABC with right angle at C then $z_1^2 + z_2^2 + 2z_3^2 = 2z_3(z_1 + z_2)$.
- If z_1, z_2, z_3 represent the vertices of an equilateral triangle then

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$$

DE'MOIVRE'S THEOREM & CUBE ROOTS OF UNITY

- The value of $(\cos\theta + i\sin\theta)^n$ is $\cos(n\theta) + i\sin(n\theta)$ if 'n' is integer
- $\cos(n\theta) + i\sin(n\theta)$ is one of the values of $(\cos\theta + i\sin\theta)^n$ if n is a rational number of the form p/q, where p & q are co-prime.
- The 3 cube roots of unity are 1, $\frac{-1+i\sqrt{3}}{2}(\omega)$, $\frac{-1-i\sqrt{3}}{2}(\omega^2)$.
- $1 + \omega + \omega^2 = 0$
- $\omega^3 = 1$
- $1 + \omega^n + \omega^{2n} = 3$ (if n is multiple of 3)
- $1 + \omega^n + \omega^{2n} = 0$ (if n is not a multiple of 3).
- The three cube roots of unity when plotted on the argand plane constitute the vertices of an equilateral triangle.

"When problems get complex, learn to see beyond the real and embrace the bigger picture."

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Chapter 6

Quadratic Equations

EQUATION VS IDENTITY

$ax^2 + bx + c = 0$

- $a \neq 0 \Rightarrow$ Quad eqⁿ → 2 Roots
- $a = 0$ & $b \neq 0 \Rightarrow$ linear eqⁿ → 1 Root
  $bx + c = 0$
- $a = 0, b = 0, c \neq 0$
  → Contradiction → no root
  ex → $0x^2 + 0x + 5 = 0$
- $a = 0, b = 0, c = 0$ → Identity
  → having Infinite Solⁿ

  Ex: $0x^2 + 0x + 0 = 0$

Example: $(x - 1)^2 = x^2 + 1 - 2x$ → Identity

Ques. The values of λ for which $(\lambda^2 - 3\lambda + 2)x^2 + (\lambda^2 - 5\lambda + 6)x + (\lambda^2 - 4) = 0$ is an identity in x is **[(AIEEE 2011)]**

(a) 1 (b) 2 (c) -2 (d) 0

Ans. $\lambda^2 - 3\lambda + 2 = 0$

$(\lambda - 2)(\lambda - 1) = 0 \Rightarrow \lambda = 2, 1$

&

$\lambda^2 - 5\lambda + 6 = 0 \Rightarrow (\lambda - 3)(\lambda - 2) = 0 \Rightarrow \lambda = 2, 3$

&

$\lambda^2 - 4 = 0 \Rightarrow (\lambda - 2)(\lambda + 2) = 0 \Rightarrow \lambda = 2, -2$

Hence, $\lambda = 2$

DIFFERENCE OF ROOTS OF A QUADRATIC EQUATION

If α, β are roots of equation $ax^2 + bx + c = 0$

then, $|\alpha - \beta| = \frac{\sqrt{D}}{a}$

Proof: $\alpha + \beta = \frac{-b}{a}, \alpha\beta = \frac{c}{a}$

Consider $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$$\Rightarrow (\alpha - \beta)^2 = \left(-\frac{b}{a}\right)^2 - \frac{4c}{a} = \frac{b^2}{a^2} - \frac{4c}{a}$$

$$\Rightarrow (\alpha - \beta)^2 = \frac{b^2 - 4ac}{a^2}$$

$$\Rightarrow (\alpha - \beta)^2 = \frac{D}{a^2}$$

$$\Rightarrow (\alpha - \beta) = \pm\frac{\sqrt{D}}{a}$$

$$\Rightarrow |\alpha - \beta| = \frac{\sqrt{D}}{a}$$

Ques. If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals

(a) -2 (b) 3 (c) 2 (d) 1

Ans. ∵ Roots are consecutive integers

$\therefore \beta = \alpha + 1$

$$\Rightarrow |\beta - \alpha| = 1 \;\&\; |\beta - \alpha| = \frac{\sqrt{D}}{a}$$

$$\Rightarrow 1 = \frac{\sqrt{b^2 - 4ac}}{a} \Rightarrow 1 = \frac{\sqrt{b^2 - 4c}}{1} \Rightarrow 1 = b^2 - 4c$$

Ques. If α, β are the roots of $ax^2 + bx + c = 0$ $\alpha + h, \beta + h$ are the roots of $px^2 + qx + r = 0$ and D_1, D_2 the respective discriminants of these equations, then $D_1 : D_2$ equals **(AIEEE- 2005)**

(a) $\frac{a^2}{p^2}$ (b) $\frac{b^2}{q^2}$

(c) $\frac{c^2}{r^2}$ (d) None of these

Ans. $ax^2 + bx + c = 0$ < α, β

& $px^2 + qx + r = 0$ < $\alpha + h$, $\beta + h$

Difference of roots of both the eqⁿ s is same

$$\frac{\sqrt{D_1}}{a} = \frac{\sqrt{D_2}}{p} \Rightarrow \frac{\sqrt{D_1}}{\sqrt{D_2}} = \frac{a}{p} \Rightarrow \frac{D_1}{D_2} = \frac{a^2}{p^2}$$

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Ques. Let $a \in R$ and let α, β be the roots of the equation $x^2 + 60^{\frac{1}{4}}x + a = 0$. if $\alpha^4 + \beta^4 = -30$, then the product of all possible values of a is ____. **[JEE Main 2023 (25 Jan., Shift II)]**

Ans. $\alpha + \beta = -(60)^{\frac{1}{4}}$, $\alpha\beta = \frac{a}{1} = a$

Squaring both sides

$\Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta = \left((60)^{\frac{1}{4}}\right)^2$

$\Rightarrow \alpha^2 + \beta^2 + 2a = (60)^{\frac{1}{2}} = \sqrt{60}$

$\Rightarrow \alpha^2 + \beta^2 = \sqrt{60} - 2a$

Squaring both sides

$\Rightarrow \alpha^4 + \beta^4 + 2\alpha^2\beta^2 = (\sqrt{60} - 2a)^2$

$\Rightarrow -30 + 2(\alpha\beta)^2 = 60 + 4a^2 - 4\sqrt{60}a$

$\Rightarrow -30 + 2a^2 = 60 + 4a^2 - 4\sqrt{60}a$

$\Rightarrow 2a^2 - 4\sqrt{60}a + 90 = 0$

$\therefore$ Product $= \frac{90}{2} = 45$

Ques. The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is:

(JEE Main - 2022)

(a) 4 (b) 5 (c) 6 (d) 8

Ans. $x^2 + (3 - a)x + 1 - 2a = 0$ (roots α, β)

$\alpha + \beta = \frac{-(3 - a)}{1} = (a - 3)$,

$\alpha\beta = \left(\frac{(1 - 2a)}{1}\right) = 1 - 2a$

$\because \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (a - 3)^2 - 2(1 - 2a)$

$= a^2 + 9 - 6a - 2 + 4a = a^2 - 2a + 7$

$= (a - 1)^2 - 1 + 7 = (a - 1)^2 + 6$

$\Rightarrow (\alpha^2 + \beta^2)_{min} = 6$

Ques. α & β are the roots of the equation $ax^2 + bx + c = 0$. Find a relation between a, b & c such that

(i) $\beta = 2\alpha$

Ans. $\alpha + \beta = -\frac{b}{a} \Rightarrow \alpha + 2\alpha = -\frac{b}{a}$ $[\because \beta = 2\alpha]$

$\Rightarrow 3\alpha = -\frac{b}{a}$

$\Rightarrow \alpha = \frac{-b}{3a}$

$\alpha\beta = \alpha \cdot 2\alpha = \frac{c}{a}$

$\Rightarrow 2\alpha^2 = \frac{c}{a}$

$\Rightarrow 2\left[-\frac{b}{3a}\right]^2 = \frac{c}{a} \Rightarrow \frac{2b^2}{9a^2} = \frac{c}{a} \Rightarrow \boxed{2b^2 = 9ac}$

(ii) $\beta = 3\alpha$

$\alpha + \beta = -\frac{b}{a}$

$\Rightarrow \alpha + 3\alpha = -\frac{b}{a}$

$\Rightarrow \alpha = \frac{-b}{4a}$

$\alpha \cdot 3\alpha = \frac{c}{a}$

$\Rightarrow 3\alpha^2 = \frac{c}{a}$

$\Rightarrow 3\left[-\frac{b}{4a}\right]^2 = \frac{c}{a} \Rightarrow \frac{3b^2}{16a^2} = \frac{c}{a} \Rightarrow 3b^2 = 16ac$

(iii) $\beta = \alpha^2$

$\Rightarrow \boxed{\alpha + \alpha^2 = \frac{-b}{a}} \rightarrow (1)$

$\alpha \cdot \alpha^2 = \frac{c}{a}$

$\boxed{\alpha^3 = \frac{c}{a}} \rightarrow (2)$

Cube both sides

$(\alpha + \alpha^2)^3 = \left(\frac{-b}{a}\right)^3$

$\Rightarrow \alpha^3 + (\alpha^2)^3 + 3\alpha \cdot \alpha^2(\alpha + \alpha^2) = \frac{-b^3}{a^3}$

$\Rightarrow \alpha^3 + \alpha^6 + 3\alpha^3(\alpha + \alpha^2) = -\frac{b^3}{a^3}$

$\Rightarrow \frac{c}{a} + \left(\frac{c}{a}\right)^2 + \frac{3c}{a}\left(\frac{-b}{a}\right) = \frac{-b^3}{a^3}$

$\Rightarrow \frac{c}{a} + \frac{c^2}{a^2} - \frac{3bc}{a^2} = \frac{-b^3}{a^3}$

$\Rightarrow \frac{ac + c^2 - 3bc}{a^2} = -\frac{b^3}{a^3}$

$\Rightarrow a^2c + ac^2 - 3abc = -b^3$

$\Rightarrow a^2c + ac^2 + b^3 = 3abc$

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FORMATION OF A QUADRATIC EQUATION

The quadratic equation whose roots are α and β is given by

$$K(x^2 - (\alpha + \beta)x + \alpha\beta) = 0$$

OR

$$K[x^2 - (\text{Sum of roots})\,x + \text{Product of roots}] = 0$$

Ex: $\alpha = 2$ & $\beta = 5$

sum = 7, Product = 10

Then equation is $x^2 - (\text{Sum})x + \text{Product} = 0$

$\Rightarrow K(x^2 - 7x + 10 = 0)$

Ques. If the sum of the squares of the reciprocals of the roots α and β of the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to:

[JEE Main (24 June 2022, Shift-1)]

(a) 18 (b) 24 (c) 36 (d) 96

Ans. $\alpha + \beta = -\frac{\lambda}{3}, \alpha\beta = -\frac{1}{3}$

Given, $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 15$

$$\Rightarrow \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = 15$$

$$\Rightarrow \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2} = 15$$

$$\Rightarrow \frac{\left(-\frac{\lambda}{3}\right)^2 - 2\left(-\frac{1}{3}\right)}{\frac{1}{9}} = 15$$

$$\Rightarrow \frac{\frac{\lambda^2}{9} + \frac{2}{3}}{\frac{1}{9}} = 15$$

$\Rightarrow \lambda^2 + 6 = 15$

$\Rightarrow \boxed{\lambda^2 = 9}$

$\Rightarrow \lambda = \pm 3$

$\boxed{\lambda = 3}$

$6(\alpha^3 + \beta^3)^2$

$= 6((\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta))^2$

$= 6\left(\left(-\frac{\lambda}{3}\right)^3 - 3\left(-\frac{1}{3}\right)\left(-\frac{\lambda}{3}\right)\right)^2$

$= 6((-1)^3 - 1)^2$

$= 6(-2)^2 = 6 \times 4 = 24$

$= 6 \times 4 = 24$

EQUATIONS REDUCIBLE TO QUADRATIC EQUATIONS

Ques. Solve the equation $\sqrt{x+1} + \sqrt{2x-5} = 3$

Ans. $\sqrt{2x-5} = 3 - \sqrt{x+1}$

Whenever squaring both sides always check the solution

Squaring both sides

$\Rightarrow 2x - 5 = (3 - \sqrt{x+1})^2$

$\Rightarrow 2x - 5 = 9 + x + 1 - 6\sqrt{x+1}$

$\Rightarrow 2x - 5 = 10 + x - 6\sqrt{x+1}$

$\Rightarrow 6\sqrt{x+1} = 15 - x$

Squaring both sides

$\Rightarrow 36(x+1) = (15 - x)^2$

$\Rightarrow 36x + 36 = 225 + x^2 - 30x$

$\Rightarrow x^2 - 66x + 189 = 0$

$\Rightarrow x^2 - 63x - 3x + 189 = 0$

$\Rightarrow (x - 63)(x - 3) = 0$

$\Rightarrow x = 63$ or $x = 3$

Check	Check
$x = 63$	$x = 3$
L.H.S: $\sqrt{64} + \sqrt{2 \times 63 - 5}$	L.H.S: $\sqrt{3+1} + \sqrt{6-5}$
$\Rightarrow 8 + 11 = 19$	$\Rightarrow 2 + 1 = 3$
R.H.S: 3	R.H.S = 3
L.H.S $\neq$ R.H.S	L.H.S = R.H.S (Satisfies)
Reject	

Ques. The number of real roots of the equation $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$, is:

(31 Jan 2023 Shift 1)

(a) 0 (b) 1

(c) 3 (d) 2

Ans. $4x^2 - 14x + 6 \Rightarrow 4x^2 - 2x - 12x + 6$

$\Rightarrow (2x - 6)(2x - 1) \Rightarrow 2(x - 3)(2x - 1)$

Now,

$\sqrt{(x-3)(x-1)} + \sqrt{(x-3)(x+3)} = \sqrt{2(x-3)(2x-1)}$

$\Rightarrow \sqrt{x-3} = 0 \Rightarrow x = 3$

or $\sqrt{x-1} + \sqrt{x+3} = \sqrt{2(2x-1)}$

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Squaring both sides

$x - 1 + x + 3 + 2\sqrt{(x-1)(x+3)} = 4x - 2$

$\Rightarrow 2\sqrt{(x-1)(x+3)} = 2x - 4 \Rightarrow \sqrt{(x-1)(x+3)} = x - 2$

Squaring both sides

$\Rightarrow x^2 + 2x - 3 = x^2 + 4 - 4x$

$\Rightarrow 6x = 7 \Rightarrow x = 7/6$

Check, $x = 7/6$

$\Rightarrow \sqrt{1/6} + \sqrt{19/6} = \sqrt{2 \times (4/3)}$

$\Rightarrow 1 + \sqrt{19} = 4$ (Reject)

Ques. $9\left(x^2 + \frac{1}{x^2}\right) - 27\left(x + \frac{1}{x}\right) + 8 = 0,\ x \in R$

Ans. Let, $x + \frac{1}{x} = t$

Squaring both sides

$\Rightarrow x^2 + \frac{1}{x^2} + 2x\frac{1}{x} = t^2$

$\Rightarrow x^2 + \frac{1}{x^2} + 2 = t^2$

$\Rightarrow x^2 + \frac{1}{x^2} = t^2 - 2$

Now, $9\left(x^2 + \frac{1}{x^2}\right) - 27\left(x + \frac{1}{x}\right) + 8 = 0$

$\Rightarrow 9(t^2 - 2) - 27t + 8 = 0$

$\Rightarrow 9t^2 - 18 - 27t + 8 = 0 \Rightarrow 9t^2 - 27t - 10 = 0$

$\Rightarrow 9t^2 - 30t + 3t - 10 = 0$

$\Rightarrow (3t - 10)(3t + 1) = 0$

$\Rightarrow t = \frac{10}{3}$ or $t = -\frac{1}{3}$

$\Rightarrow x + \frac{1}{x} = \frac{10}{3}$ or $x + \frac{1}{x} = -\frac{1}{3}$

$\Rightarrow \frac{x^2 + 1}{x} = \frac{10}{3}$ Rejected

$\Rightarrow 3x^2 + 3 = 10x$

$\Rightarrow 3x^2 - 10x + 3 = 0$

$\Rightarrow 3x^2 - 9x - x + 3 = 0$

$\Rightarrow (3x - 1)(x - 3) = 0$

$\Rightarrow x = 3$ or $\frac{1}{3}$

$x + \frac{1}{x} \geq 2$ for $x > 0$

$x + \frac{1}{x} \leq -2$ for $x < 0$

Ques. The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^{x} + 1 = 0$ is _____

[(28 June 2022 Shift 1)]

Ans. Let $e^x = t,\ t > 0$

$\Rightarrow t^4 + 4t^3 - 58t^2 + 4t + 1 = 0$

divide by t^2

$\Rightarrow t^2 + 4t - 58 + \frac{4}{t} + \frac{1}{t^2} = 0$

$\Rightarrow \left(t^2 + \frac{1}{t^2}\right) + 4\left(t + \frac{1}{t}\right) - 58 = 0$

Let $t + \frac{1}{t} = z$

$\Rightarrow (z^2 - 2) + 4z - 58 = 0$

$\Rightarrow z^2 + 4z - 60 = 0$

$\Rightarrow z^2 + 10z - 6z - 60 = 0$

$\Rightarrow (z - 6)(z + 10) = 0$

$\Rightarrow z = 6$ or -10

$\Rightarrow t + \frac{1}{t} = 6$ or $t + \frac{1}{t} = -10$

$\Rightarrow \frac{t^2 + 1}{t} = 6$ $t > 0$ or $\frac{1}{t} > 0$

$\Rightarrow t^2 - 6t + 1 = 0$ (Reject)

$\Rightarrow t = \frac{6 \pm \sqrt{36 - 4}}{2}$

$\Rightarrow t = \frac{6 \pm \sqrt{32}}{2} = \frac{6 \pm 4\sqrt{2}}{2}$

$\Rightarrow t = 3 \pm 2\sqrt{2}$

$\Rightarrow e^x = 3 + 2\sqrt{2}$ or $e^x = 3 - 2\sqrt{2}$

$\Rightarrow x = \log_e(3 + 2\sqrt{2})$ or $x = \log_e(3 - 2\sqrt{2})$

Ques. Find number of integer roots of equation $x(x+1)(x+2)(x+3) = 120$

Ans. $(x^2 + 3x)(x^2 + 3x + 2) = 120$

Let $x^2 + 3x = t$

$\Rightarrow t[t + 2] = 120 \Rightarrow t^2 + 2t = 120$

$\Rightarrow t^2 + 2t - 120 = 0$

$\Rightarrow t^2 + 12t - 10t - 120 = 0$

$\Rightarrow (t + 12)(t - 10) = 0$

$\Rightarrow t = -12$ or $t = 10$

$x^2 + 3x = -12$ or $x^2 + 3x = 10$

$\Rightarrow x^2 + 3x + 12 = 0$ $x^2 + 3x - 10 = 0$

$D < 0$ $x^2 + 5x - 2x - 10$

$(x + 5)(x - 2) = 0$

$x = -5$ or 2

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Brain Teaser

Solve the equation $(x+2)(x+3)(x+8)(x+12) = 4x^2$

Ans. $(x+2)(x+12)(x+3)(x+8) = 4x^2$

divide by x^2

$$\Rightarrow \left(\frac{x^2+14x+24}{x}\right)\left(\frac{x^2+11x+24}{x}\right) = \frac{4x^2}{x^2}$$

$$\Rightarrow \left(x+14+\frac{24}{x}\right)\left(x+11+\frac{24}{x}\right) = 4$$

Let, $x + \frac{24}{x} + 11 = t$

$\Rightarrow (t+3)\,t = 4 \Rightarrow t^2 + 3t - 4 = 0$

$\Rightarrow t^2 + 4t - t - 4 = 0$

$\Rightarrow (t+4)(t-1) = 0 \Rightarrow t = -4 \text{ or } 1$

$\Rightarrow x + \frac{24}{x} + 11 = -4$ or $x + \frac{24}{x} + 11 = 1$

$\Rightarrow x + \frac{24}{x} = -15$ $\quad x + \frac{24}{x} = -10$

$\Rightarrow x^2 + 24 = -15x$ $\quad x^2 + 10x + 24 = 0$

$\Rightarrow x^2 + 15x + 24 = 0$ $\quad (x+6)(x+4) = 0$

$D = 15^2 - 4 \times 24$ $\quad \boxed{x = -6, -4}$

$= 225 - 96 = 129$

$$\Rightarrow x = \frac{-15 \pm \sqrt{129}}{2} < 0$$

Ques. Evaluate $\sqrt{6+\sqrt{6+\sqrt{6+\ldots\ldots\ldots}}} = x$

Ans. $\sqrt{6+x} = x$

Squaring both sides

$\Rightarrow 6 + x = x^2$ | $\sqrt{x}$ = non negative

$\Rightarrow x^2 - x - 6 = 0$ | $x = 3$ or $x = -2$ **Reject**

$\Rightarrow (x-3)(x+2) = 0$

$\Rightarrow x = 3 \text{ or } -2$

Ques. The value of $3 + \cfrac{1}{4 + \cfrac{1}{3 + \cfrac{1}{4 + \cfrac{1}{3 + \ldots\infty}}}} = x$ is

[JEE Main 2021]

(a) $1.5 + \sqrt{3}$ (b) $2 + \sqrt{3}$

(c) $3 + 2\sqrt{3}$ (d) $4 + \sqrt{3}$

Ans. $3 + \cfrac{1}{4 + \cfrac{1}{x}} = x$

$$\Rightarrow 3 + \frac{x}{4x+1} = x$$

$$\Rightarrow \frac{x}{4x+1} = x - 3$$

$\Rightarrow x = (x-3)(4x+1)$

$\Rightarrow x = 4x^2 + x - 12x - 3$

$\Rightarrow 4x^2 - 12x - 3 = 0$

$$\Rightarrow x = \frac{12 \pm \sqrt{(12)^2 - 4 \times 4 (-3)}}{2 \times 4}$$

$$\Rightarrow x = \frac{12 \pm 4\sqrt{3^2+3}}{2 \times 4} = \frac{12 \pm 4 \times 2\sqrt{3}}{8}$$

$$\Rightarrow x = \frac{4(3 \pm 2\sqrt{3})}{8} = \frac{3+2\sqrt{3}}{2}, \frac{3-2\sqrt{3}}{2} \quad (x > 3)$$

$$\Rightarrow x = \frac{3}{2} + \sqrt{3}$$

NEWTON'S FORMULA

If α, β are roots of $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$ or $T_n = \alpha^n - \beta^n$

Then $aS_{n+2} + bS_{n+1} + cS_n = 0 \;\forall\, n \in N$

or $aT_{n+2} + bT_{n+1} + cT_n = 0 \;\forall\, n \in N$

Proof:

$ax^2 + bx + c = 0$

put $x = \alpha$ and multiply with α^n

$\Rightarrow \alpha^n [a\alpha^2 + b\alpha + c = 0]$

$\Rightarrow \boxed{a\alpha^{n+2} + b\alpha^{n+1} + c\alpha^n = 0} \longrightarrow ①$

put $x = \beta$ and multiply with β^n

$\Rightarrow \beta^n [a\beta^2 + b\beta + c = 0]$

$\Rightarrow \boxed{a\beta^{n+2} + b\beta^{n+1} + c\beta^n = 0} \longrightarrow ②$

① + ②

$\Rightarrow a(\alpha^{n+2} + \beta^{n+2}) + b(\alpha^{n+1} + \beta^{n+1}) + c(\alpha^n + \beta^n) = 0$

$\Rightarrow aS_{n+2} + bS_{n+1} + cS_n = 0$

① - ②

$\Rightarrow aT_{n+2} + bT_{n+1} + cT_n = 0$

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Ques. Let α and β be the roots of $x^2 - 6x - 2 = 0$ with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is **[IIT-2011]**

(a) 1 (b) 2

(c) 3 (d) 4

Ans. By using Newton's Formula,

$a_{10} - 6a_9 - 2a_8 = 0$

$\Rightarrow a_{10} - 2a_8 = 6a_9$

$\Rightarrow \frac{a_{10} - 2a_8}{2a_9} = 3$

Ques. Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n$, $n = 1, 2, 3, \ldots$, then: **[JEE Advance (Sep.) 2020]**

(a) $5S_6 + 6S_5 = 2S_4$

(b) $6S_6 + 5S_5 + 2S_4 = 0$

(c) $6S_6 + 5S_5 = 2S_4$

(d) $5S_6 + 6S_5 + 2S_4 = 0$

Ans. By using Newton's Formula,

$5S_6 + 6S_5 - 2S_4 = 0$

$\Rightarrow 5S_6 + 6S_5 = 2S_4$

Ques. The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0, x \in R$ has: **[JEE Main (31 Jan. 2023 Shift 2)]**

(a) two solutions and both are negative

(b) no solution

(c) four solutions two of which are negative

(d) two solutions and only one of them is negative

Ans. Let $e^x = t$ $(t > 0)$

$\Rightarrow t^4 + 8t^3 + 13t^2 - 8t + 1 = 0$

divide by t^2

$\Rightarrow t^2 + 8t + 13 - \frac{8}{t} + \frac{1}{t^2} = 0$

$\Rightarrow \left(t^2 + \frac{1}{t^2}\right) + 8\left(t - \frac{1}{t}\right) + 13 = 0$

Let, $\left(t - \frac{1}{t}\right) = z$

$\Rightarrow t^2 + \frac{1}{t^2} - 2 = z^2 \Rightarrow t^2 + \frac{1}{t^2} = z^2 + 2$

Now, equation become

$(z^2 + 2) + 8z + 13 = 0 \Rightarrow z^2 + 8z + 15 = 0$

$\Rightarrow (z + 5)(z + 3) = 0 \Rightarrow z = -5$ or $z = -3$

$t - \frac{1}{t} = -3$ or $t - \frac{1}{t} = -5$

$\Rightarrow t^2 - 1 = -3t$ $\Rightarrow t^2 - 1 = -5t$

$\Rightarrow t^2 + 3t - 1 = 0$ $\Rightarrow t^2 + 5t - 1 = 0$

$\Rightarrow t = \frac{-3 \pm \sqrt{9 + 4}}{2}$ $\Rightarrow t = \frac{-5 \pm \sqrt{25 + 4}}{2}$

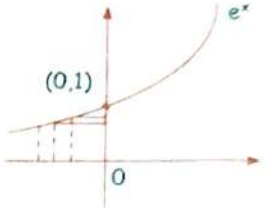
$\Rightarrow t = \frac{-3 \pm \sqrt{13}}{2}$ $\Rightarrow t = \frac{-5 \pm \sqrt{29}}{2}$

$\Rightarrow e^x = \frac{\sqrt{13} - 3}{2} < 1$ $\Rightarrow e^x = \frac{\sqrt{29} - 5}{2} < 1$

$e^x = e^0 = 1 \Rightarrow x = 0$

$e^x > 1 \Rightarrow x > 0,$

$e^x < e^0 \Rightarrow x < 0$



$\Rightarrow$ 1 soln $\Rightarrow$ 1 soln

- ve soln - ve soln

Ques. If for some $p, q, r \in R$ not all have same sign, one of the roots of the equation $(p^2 + q^2)x^2 - 2q(p + r)x + q^2 + r^2 = 0$ is also a root of the equation $x^2 + 2x - 8 = 0$ then $\frac{q^2 + r^2}{p^2}$ is equal to: **[(26 July 2022 Shift 1)]**

Ans. $x^2 + 2x - 8 = 0 \Rightarrow x^2 + 4x - 2x - 8 = 0$

$\Rightarrow (x + 4)(x - 2) = 0 \Rightarrow x = -4$ or 2

$p^2x^2 + q^2x^2 - 2pqx - 2qrx + q^2 + r^2 = 0$

$\Rightarrow (px - q)^2 + (qx - r)^2 = 0$

$\Rightarrow px - q = 0$ & $qx - r = 0$

$\Rightarrow x = \frac{q}{p}$ & $x = \frac{r}{q}$ or $x = \frac{q}{p} = \frac{r}{q}$

$\frac{q}{p} = \frac{r}{q} = 2$

$q = 2p$

$r = 2q$

If $p > 0$, $q > 0$, $r > 0$

$\frac{q}{p} = \frac{r}{q} = -4$

$q = -4p$ Accept

$r = -4q$

Reject because p, q, r not have same sign

Hence, $\frac{q^2 + r^2}{p^2} = \frac{q^2 + 16q^2}{p^2}$

$= \frac{17q^2}{p^2} = 17(-4)^2 = 17 \times 16 = 272$

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Ques Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$ then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ is equal to _____.

[06 Apr 2024 (Shift 2)]

Ans. By Newton's formula

$U_{n+2} + \sqrt{2}U_{n+1} - 8U_n = 0$

$n = 8 \Rightarrow U_{10} + 2U_9 - 8U_8 = 0$

$\Rightarrow \frac{U_{10} + \sqrt{2}U_9}{2U_8} = \frac{8U_8}{2U_8} \Rightarrow \frac{U_{10} + \sqrt{2}U_9}{2U_8} = 4$

Ques. If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ each positive integer n, then the value of $\left(\frac{P_{17}P_{20} + 5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}P_{18}^2}\right)$ is equal to **[(JEE Mains, July 2021)]**

Ans. Using Newton's formula,

$P_{n+2} + 5\sqrt{2}\,P_{n+1} + 10P_n = 0$

$\Rightarrow P_{20} + 5\sqrt{2}\,P_{19} + 10P_{18} = 0$

& $P_{19} + 5\sqrt{2}\,P_{18} + 10P_{17} = 0$

Now, $\left(\frac{P_{17}P_{20} + 5\sqrt{2}\,P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}\,P_{18}^2}\right) =$

$\frac{P_{17}(P_{20} + 5\sqrt{2}\,P_{19})}{P_{18}(P_{19} + 5\sqrt{2}\,P_{18})}$ ($P_{20} + 5\sqrt{2}\,P_{19} = -10P_{18}$, $P_{19} + 5\sqrt{2}\,P_{18} = -10P_{17}$)

$= \frac{P_{17}(-10\,P_{18})}{P_{18}(-10\,P_{17})} = 1$

Ques. Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $P_n = (\alpha)^n + (\beta)^n$, $P_{n-1} = 11$ and $P_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of P_n^2 is: **[(JEE Mains, Feb 2021)]**

Ans. $\because x^2 - (\text{Sum})\,x + \text{product} = 0$

Sum $= \alpha + \beta = 1$, Product $= \alpha\beta = -1$

$\therefore x^2 - x - 1 = 0$

Using Newton's formula

$P_{n+2} - P_{n+1} - P_n = 0$

$x \to n - 1 \Rightarrow P_{n+1} - P_n - P_{n-1} = 0$

$\Rightarrow 29 - P_n - 11 = 0 \Rightarrow P_n = 29 - 11$

$\Rightarrow P_n = 18 \Rightarrow P_n^2 = 18^2 = 324$

Ques. Let α and β be roots of $x^2 - 6(t^2 - 2t + 2)x - 2 = 0$ with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then find the minimum value of $\frac{a_{100} - 2a_{98}}{a_{99}}$ (where $t \in R$)

Ans. Using Newton's formula

$a_{100} - 6(t^2 - 2t + 2)a_{99} - 2a_{98} = 0$

$\Rightarrow a_{100} - 2a_{98} = 6(t^2 - 2t + 2)a_{99}$

$\Rightarrow \frac{a_{100} - 2a_{98}}{a_{99}} = 6(t^2 - 2t + 2) = 6((t-1)^2 + 1)$

Hence, min value of $\left(\frac{a_{100} - 2a_{98}}{a_{99}}\right) = 6(0 + 1) = 6$

FINDING NON REAL ROOTS

Ex: $x^2 + 3x + 5 = 0$

$D = 9 - 4 \times 1 \times 5$

$= 9 - 20 = -11$ (-ve) $\rightarrow$ no real roots

$x = \frac{-3 \pm \sqrt{-11}}{2 \times 1} = \frac{-3 \pm \sqrt{11}\sqrt{-1}}{2}$ (iota)

$\Rightarrow x = \frac{-3 \pm \sqrt{11}\,i}{2}$ or $x = \frac{-3 - \sqrt{11}\,i}{2}, \frac{-3 + \sqrt{11}\,i}{2}$

Important Note

For a Quadratic Equation defined over real Coefficients, the non real roots will always occur in Conjugate pairs.

Example: $x^2 + 2x + 4 = 0$

$x = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times 4}}{2 \times 1}$

$x = \frac{-2 \pm \sqrt{4 - 16}}{2} = \frac{-2 \pm \sqrt{12}i}{2} = \frac{-2 \pm 2\sqrt{3}i}{2}$

$= -1 + \sqrt{3}i$ or $-1 - \sqrt{3}i$

Ques. Form a Quadratic Equation with real coefficients whose one of the root is $5 - 2i$.

Ans. $\alpha = 5 - 2i \quad \Rightarrow$ other root $(\beta) = 5 + 2i$

$+\ \beta = 5 + 2i$

$\alpha + \beta = 10$

$\alpha\beta = (5 - 2i)(5 + 2i) = 5^2 - (2i)^2$

$= 25 - 4i^2 = 25 + 4 = 29$

Hence, Quadratic equations is $x^2 - (10)x + 29 = 0$

Ques. If one root of the equation $2x^2 - 6x + c = 0$ is $\frac{3 + 5i}{2}$, then the real value of c will be-

(a) 7 (b) -7

(c) 17 (d) -17

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Ans. $\alpha = \frac{3+5i}{2} \Rightarrow \beta = \frac{3-5i}{2}$

Product of roots $= \left(\frac{3+5i}{2}\right)\left(\frac{3-5i}{2}\right) = \frac{c}{2}$

$\Rightarrow \frac{3^2-(5i)^2}{2} = c \Rightarrow \frac{9+25}{2} = c$

$\Rightarrow c = 17$

Ques. Form a Quadratic with rational coefficients whose one of the root is $(4-\sqrt{5})$.

Ans. $\alpha = 4-\sqrt{5}$

$\beta = 4+\sqrt{5}$

$\Rightarrow \alpha + \beta = 8$

$\alpha\beta = (4-\sqrt{5})(4+\sqrt{5}) = 4^2 - 5 = 11$

Hence, equation is $x^2 - 8x + 11 = 0$

SOLVING CUBIC EQUATIONS

Example: Factorise $x^3 + 4x^2 + x - 6 = 0$

1st Root by (Hit & trial)

$x = 1$

$\Rightarrow 13 + 4 \times 1 + 1 - 6 = 6 - 6 = 0$

$\Rightarrow x = 1$ is a root

$\Rightarrow (x-1)$ is a factor

Now, $x^3 + 4x^2 + x - 6 = 0$

$\Rightarrow x^2(x-1) + 5x(x-1) + 6(x-1) = 0$

$\Rightarrow (x-1)(x^2 + 5x + 6) = 0$

$\Rightarrow (x-1)(x+3)(x+2) = 0 \Rightarrow x = 1, -3, -2$

Ques. Factorize $x^3 - 23x^2 + 142x - 120$

Ans. $\boxed{x = 1}$

$= 1 - 23 + 142 - 120 = 143 - 143 = 0$

So, $(x-1)$ is a factor.

Now, $x^3 - 23x^2 + 142x - 120$

$= x^2(x-1) - 22x(x-1) + 120(x-1)$

$= (x-1)(x^2 - 22x + 120)$

$= (x-1)(x-10)(x-12)$

THEORY OF EQUATION

(i) For Cubic Polynomial

$\overset{+}{a}x^3 \overset{-}{+} bx^2 \overset{+}{+} cx \overset{-}{+} d = 0 \Leftarrow \begin{matrix}\alpha\\ \beta\\ \gamma\end{matrix}$

$\alpha + \beta + \gamma = -\frac{b}{a}$

$\alpha\beta + \beta\gamma + \gamma\alpha = +\frac{c}{a}$

$\alpha\beta\gamma = -\frac{d}{a}$

Example: $3x^3 - 4x^2 + 7x - 1 = 0 \Leftarrow \begin{matrix}\alpha\\ \beta\\ \gamma\end{matrix}$

$\alpha + \beta + \gamma = \frac{4}{3}$

$\alpha\beta + \beta\gamma + \gamma\alpha = +\frac{7}{3}$

$\alpha\beta\gamma = \frac{1}{3}$

(ii) For Quadratic Polynomial

$\overset{+}{a}x^4 \overset{-}{+} bx^3 \overset{+}{+} cx^2 \overset{-}{+} dx \overset{+}{+} e = 0 \Leftarrow \begin{matrix}\alpha\\ \beta\\ \gamma\\ \delta\end{matrix}$

$\alpha + \beta + \gamma + \delta = -\frac{b}{a}$

$\alpha\beta + \beta\gamma + \gamma\delta + \ldots = +\frac{c}{a}$

$\alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \ldots = -\frac{d}{a}$

$\alpha\beta\gamma\delta = +\frac{e}{a}$

Example: $\overset{+}{3}x^4 \overset{-}{-} 4x^3 \overset{+}{+} 7x^2 \overset{-}{+} 3x \overset{+}{-} 1 = 0$

$\alpha + \beta + \gamma + \delta = \frac{4}{3}$, $\alpha\beta + \beta\gamma \ldots = +\frac{7}{3}$

$\alpha\beta\gamma + \beta\gamma\delta + \ldots = -\frac{3}{3}$, $\alpha\beta\gamma\delta = -\frac{1}{3}$

Ques. If α, β, γ are the roots of the equation $\overset{+}{x}^3 \overset{-}{-} 3x^2 \overset{+}{+} 5x \overset{-}{-} 1 = 0$ then find the value of

1. $\alpha^2 + \beta^2 + \gamma^2$

$\alpha + \beta + \gamma = 3$
$\alpha\beta + \beta\gamma + \gamma\alpha = 5$
$\alpha\beta\gamma = 1$

Ans. $\alpha + \beta + \gamma = 3$

Squaring both sides

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 + 2\underbrace{(\alpha\beta + \beta\gamma + \gamma\alpha)}_{5} = 9$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = -1$

2. $\alpha^3 + \beta^3 + \gamma^3$

Ans. $\alpha^3+\beta^3+\gamma^3-3\alpha\beta\gamma = (\alpha+\beta+\gamma)(\alpha^2+\beta^2+\gamma^2-(\alpha\beta+\beta\gamma+\gamma\alpha))$

$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 - 3 \times 1 = 3 \times (-1-5)$

$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 - 3 = -18$

$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 = -15$

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Ques. If α, β, γ are the roots of the equation $x^3 + ax + b = 0$, then $\frac{\alpha^3 + \beta^3 + \gamma^3}{\alpha^2 + \beta^2 + \gamma^2} =$

(a) $\frac{3b}{2a}$ (b) $-\frac{3b}{2a}$

(c) $3b$ (d) $2a$

Ans. $\alpha + \beta + \gamma = 0$

$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{a}{1} = a$

$\alpha\beta\gamma = -b$

Squaring both sides

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 + 2a = 0 \Rightarrow \alpha^2 + \beta^2 + \gamma^2 = -2a$

$$\frac{\alpha^3 + \beta^3 + \gamma^3}{\alpha^2 + \beta^2 + \gamma^2} = \frac{3\,\alpha\beta\gamma}{-2a} = \frac{3(-b)}{-2a} = \frac{3b}{2a}$$

Ques. The imaginary roots of the equation $(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$ are

(a) $1 \pm i$ (b) $2 \pm i$

(c) $-1 \pm i$ (d) None of these

Ans. $(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$

divide by x^2

$$\Rightarrow \left(\frac{x^2 + 2}{x}\right)^2 + 8 = \frac{6x(x^2 + 2)}{x^2}$$

$$\Rightarrow \left(\frac{x^2 + 2}{x}\right)^2 + 8 = 6\left(\frac{x^2 + 2}{x}\right)$$

Let $\frac{x^2 + 2}{x} = t$

$\Rightarrow t^2 + 8 = 6t \Rightarrow t^2 - 6t + 8 = 0 \Rightarrow t = 2, 4$

$\Rightarrow \frac{x^2 + 2}{x} = 2$

$x^2 - 2x + 2 = 0$

$x = \frac{2 \pm \sqrt{4 - 8}}{2}$

$= \frac{2 \pm 2i}{2} = 1 \pm i$

or

$\frac{x^2 + 2}{x} = 4$

$x^2 - 4x + 2 = 0$

Real $D > 0$

Ques. If α, β and γ are the roots of the equation $x^3 - px^2 + qx - r = 0$, then the value of $\alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta$ is equal to

(a) $pq + 3r$ (b) $pq + r$

(c) $pq - 3r$ (d) $\frac{q^2}{r}$

Ans. $\alpha + \beta + \gamma = p \Rightarrow \beta + \gamma = p - \alpha$,

$\alpha\beta + \beta\gamma + \gamma\alpha = q$,

$\alpha\beta\gamma = r$

Now, $\alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta$

$= \alpha^2(\beta + \gamma) + \beta^2(\alpha + \gamma) + \gamma^2(\alpha + \beta)$

$= \alpha^2(p - \alpha) + \beta^2(p - \beta) + \gamma^2(p - \gamma)$

$= -[(\alpha^3 - p\alpha^2) + (\beta^3 - p\beta^2) + (\gamma^3 - p\gamma^2)]$

$(\because \alpha^3 - p\alpha^2 + q\alpha - r = 0 \Rightarrow \alpha^3 - p\alpha^2 = r - q\alpha)$

$= -[r - q\alpha + r - q\beta + r - q\gamma]$

$= -[3r - q(\alpha + \beta + \gamma)] = -[3r - qp] = pq - 3r$

Important Note

- A cubic equation will have either 1 real and 2 imaginary roots or all 3 real roots.
- A polynomial equation with degree odd will have atleast one real root.
- $ax^4 + bx^3 + cx^2 + dx + e = 0$ ← All 4 real / 2 Real & 2 non real / 0 Real & 4 non real

Method of Substitution

Example: If α & β are the roots of the equation $x^2 - 7x + 2 = 0$. Find a quadratic Equation whose roots are $\alpha + 2$ & $\beta + 2$

Method of Substitution

Let old quadratic is in 'x' and let new quadratic is in 'y'.

$x^2 - 7x + 2 = 0$

$\Rightarrow x = \alpha$ or $x = \beta$

also $y = \alpha + 2$ or $y = \beta + 2$

So, $y = x + 2 \Rightarrow x = y - 2$

$\Rightarrow (y - 2)^2 - 7(y - 2) + 2 = 0$

$\Rightarrow y^2 + 4 - 4y - 7y + 14 + 2 = 0$

Hence, new quadratic equations is $y^2 - 11y + 20 = 0$

Ques. If α, β, γ are the roots of the equation $x^3 - 2x^2 + x - 5 = 0$ then obtain a cubic equation whose roots are:

1. $2/\alpha, 2/\beta, 2/\gamma$

Ans. $\alpha + \beta + \gamma = 2$, $\alpha\beta + \beta\gamma + \gamma\alpha = 1$, $\alpha\beta\gamma = 5$

Let old Quad is in 'x' and let New Quad is in 'y'.

So, $y = \frac{2}{x} \Rightarrow x = \frac{2}{y}$

Hence, equation is $\left(\frac{2}{y}\right)^3 - 2\left(\frac{2}{y}\right)^2 + \frac{2}{y} - 5 = 0$

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2. $(\beta+\gamma), (\gamma+\alpha), (\alpha+\beta)$

$2-\alpha \quad 2-\beta \quad 2-\gamma$

Ans. $\boxed{y = 2 - x}$

$\Rightarrow x = 2 - y$

Hence, equation is $(2-y)^3 - 2(2-y)^2 + (2-y) - 5 = 0$

3. $\alpha\beta + 1/\gamma\ ,\ \beta\gamma + 1/\alpha\ ,\ \alpha\gamma + 1/\beta$

Ans. $\frac{\alpha\beta\gamma+1}{\gamma}, \frac{\alpha\beta\gamma+1}{\alpha}, \frac{\alpha\beta\gamma+1}{\beta} = \frac{6}{\gamma}, \frac{6}{\alpha}, \frac{6}{\beta}$

So, $y = \frac{6}{x} \Rightarrow x = \frac{6}{y}$

Hence, equation is $\left(\frac{6}{y}\right)^3 - 2\left(\frac{6}{y}\right)^2 + \frac{6}{y} - 5 = 0$

Ques. If α, β, γ are the roots of $x^3 - px^2 + qx - r = 0$ then the equation whose roots are $\beta+\gamma, \gamma+\alpha, \alpha+\beta$ is:

(a) $x^3 - 2px^2 + (p^2+q)x + (r - qp) = 0$

(b) $x^3 + 2px^2 + (p^2 - q)x + (r + qp) = 0$

(c) $x^3 + 2px^2 + (p^2+q)x + (r + qp) = 0$

(d) $x^3 + 2px^2 - (p^2+q)x - (r + qp) = 0$

Ans. $\alpha+\beta+\gamma = P \Rightarrow \beta+\gamma = P - \alpha$

$y = p - x \Rightarrow x = p - y$

Hence, required equation is

$(p-y)^3 - p(p-y)^2 + q(p-y) - r = 0$

$\Rightarrow p^3 - y^3 - 3py(p-y) - p(p^2+y^2-2py) + qp - qy - r = 0$

$\Rightarrow -y^3 - 3p^2y + 3py^2 - py^2 + 2p^2y + qp - qy - r = 0$

$\Rightarrow -y^3 + 2py^2 - p^2y - qy + qp - r = 0$

$\Rightarrow y^3 - 2py^2 + (p^2+q)y + r - pq = 0$

Ques. If α, β, γ are the roots of $ax^3 + 2x^2 + 3x + 2 = 0$, then the transformed equation whose roots are $\alpha(\beta+\gamma), \beta(\gamma+\alpha), \gamma(\alpha+\beta)$ is obtained by the substitution $x =$

(a) $\frac{-2}{y-3}$ (b) $\frac{2}{y+3}$ (c) $\frac{-2}{y+3}$ (d) $\frac{2}{y-3}$

Ans. $\alpha+\beta+\gamma = -2, \alpha\beta+\beta\gamma+\gamma\alpha = 3, \alpha\beta\gamma = -2$

Now, $y = \alpha\beta + \alpha\gamma + \beta\gamma - \beta\gamma$

$\Rightarrow y = 3 - \beta\gamma \Rightarrow y = 3 - \frac{\alpha\beta\gamma}{\alpha}$

$\Rightarrow y = 3 - \frac{(-2)}{\alpha} = 3 + \frac{2}{\alpha}$

So, $y = 3 + \frac{2}{x} \Rightarrow y - 3 = \frac{2}{x} \Rightarrow x = \frac{2}{y-3}$

PROBLEMS ON FACTOR & REMAINDER THEOREM

Factor Theorem

If $x = a$ is a root of the polynomial $P(x) = 0$ then $P(\alpha) = 0$ & $(x - \alpha)$ is a factor of $P(x)$

Remainder Theorem

If a polynomial $P(x)$ is divided by $x - \alpha$ then then remainder obtained is $P(\alpha)$

Proof: Dividend = Divisor × Quotient + Remainder

$\Rightarrow P(x) = (x - a)Q + R$

Put $x = \alpha$

$P(\alpha) = 0 + R \Rightarrow R = P(\alpha)$

Example: $P(x) = x^3 - 3x^2 + 2x - 5$ Divide karo $P(x)$ Ko by $(x - 2)$

$\Rightarrow R = P(2)$

Remainder $= (2)^3 - 3(2)^2 + 2 \times 2 - 5$

$= 8 - 12 + 4 - 5 = -5$

Ques. If $x + 1$ is a factor of the expression $x^4 + (p-3)x^3 - (3p-5)x^2 + (2p-9)x + 6$ then the value of p is-

(a) 1 (b) 2

(c) 3 (d) 4

Ans. $x = -1$ is a root

$\Rightarrow P(-1) = (-1)^4 + (p-3)(-1)^3 - (3p-5)(-1)^2 + (2p-9)(-1) + 6 = 0$

$\Rightarrow 1 - (p-3) - (3p-5) - (2p-9) + 6 = 0$

$\Rightarrow 1 - p + 3 - 3p + 5 - 2p + 9 + 6 = 0$

$\Rightarrow -6p + 24 = 0 \Rightarrow 6p = 24 \Rightarrow p = 4$

Ques. The polynomial $ax^3 + 3x^2 - 3$ and $2x^3 - 5x + a$ when divided by $(x - 4)$ leaves remainder R_1 and R_2 respectively then value of a if $2R_1 - R_2 = 0$

(a) $-\frac{18}{127}$ (b) $\frac{18}{127}$

(c) $\frac{17}{127}$ (d) $-\frac{17}{127}$

Ans. $f(4) = R_1 = a(4)^3 + 3(4)^2 - 3 = 64a + 45$

$g(4) = R_2 = 2(4)^3 - 5(4) + a = 128 - 20 + a = 108 + a$

$\because 2R_1 = R_2$

$\Rightarrow 2(64a + 45) = 108 + a \Rightarrow 128a + 90 = 108 + a$

$\Rightarrow 127a = 18 \Rightarrow a = \frac{18}{127}$

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Brain Teaser

If P(x) is a real polynomial and when divided by x - 1 the remainder is 10 & when it is divided by x + 1 then also the remainder is 10, then the remainder when it is divided by $x^2 - 1$ will be ax + b then $\frac{3a+b}{2}$ is

Ans. P (1) = 10, P (- 1) = 10

Let remainder be ax + b

$(x^2 - 1) Q(x) + ax + b = P(x)$

Put x = 1

$\Rightarrow 0 \times Q(x) + a + b = P(1)$

$\Rightarrow a + b = 10$ ...(i)

Put x = - 1

$\Rightarrow 0 \times Q(x) + a + b = P(-1)$

$\Rightarrow -a + b = 10$ ...(ii)

By adding eqn (i) and (ii):

$2b = 20 \Rightarrow b = 10 \Rightarrow a = 0$

Remainder

$ax + b = 0x + 10 = 10$

Hence, $\frac{3a+b}{2} = \frac{3 \times 0 + 10}{2} = 5$

Brain Teaser

Let p(x) be a 4° polynomial with leading Coefficient 1 such that P (1) = 1, P (2) = 8, P (3) = 27, P (4) = 64 then find P (10)/2012.

Ans. **Pattern Based Problems**

$Q(x) = (x - 1)(x - 2)(x - 3)(x - 4)$

Consider a new polynomial Q (x)

Pattern

$\boxed{Q(x) = P(x) - x^3}$

$Q(1) = P(1) - 1^3 = 1 - 1 = 0$

$Q(1) = 0 \Rightarrow (x - 1)$ is a factor of Q (x)

$Q(2) = P(2) - 2^3 = 8 - 8 = 0$

$Q(2) = 0 \Rightarrow (x - 2)$ is a factor of Q (x)

$Q(3) = 0 \Rightarrow (x - 3)$ is a factor of Q (x)

$Q(4) = 0 \Rightarrow (x - 4)$ is a factor of Q (x)

$(x - 1)(x - 2)(x - 3)(x - 4) = P(x) - x^3$

$\Rightarrow P(x) = (x - 1)(x - 2)(x - 3)(x - 4) + x^3$

Put, x = 10

$P(10) = 9 \times 8 \times 7 \times 6 + 10^3$

$= 72 \times 42 + 1000 = 4024$

Hence, $\frac{P(10)}{2012} = \frac{4024}{2012} = 2$

COMMON ROOTS

Case 1: Both the roots are common.

$a_1 x^2 + b_1 x + c_1 = 0 \begin{matrix}\alpha \\ \beta\end{matrix}$

$a_2 x^2 + b_2 x + c_2 = 0 \begin{matrix}\alpha \\ \beta\end{matrix}$

$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Case 2: Exactly one roots is common.

$a_1 x^2 + b_1 x + c_1 = 0 \begin{matrix}\alpha \\ \beta\end{matrix}$

$a_2 x^2 + b_2 x + c_2 = 0 \begin{matrix}\alpha \\ \gamma\end{matrix}$

Step 1: Eliminate x^2 from both equations

Step 2: Get the value of $x \rightarrow$ common root

$\boxed{x = \alpha}$

Step 3: Put $x = \alpha$ in any one of the two given equations

condition for exactly one root common:

$(c_1 a_2 - c_2 a_1)^2 = (a_1 b_2 - a_2 b_1)(b_1 c_2 - b_2 c_1)$

Ques. If $x^2 + 3x + 5 = 0$ and $ax^2 + bx + c = 0$ have c common root and a, b, c ∈ N, then minimum value of a + b + c is equal to

(a) 3 (b) 9

(c) 6 (d) 12

Ans. $x^2 + 3x + 5 = 0 \rightarrow D < 0$, Non real roots

So, roots will be like p + iq, p - iq

"here both roots will be common"

$\Rightarrow \frac{a}{1} = \frac{b}{3} = \frac{c}{5} = k$

$\Rightarrow a = k, b = 3k, c = 5k$

Hence, a + b + c = 9k

$k_{min} = 1 \Rightarrow (a + b + c)_{min} = 9$

Ques. If a non zero root of the equation $x^2 + 2x +$ = 0 and $2x^2 + 3x + 5\lambda = 0$ is common, then t value of λ will be:

(a) 2 (b) 1

(c) -1 (d) 0

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Ans. $2(x^2 + 2x + 3\lambda = 0) \rightarrow (1)$

$\ominus \quad 2x^2 + 3x + 5\lambda = 0 \rightarrow (2)$

$\Rightarrow 4x + 6\lambda - 3x - 5\lambda = 0$

$\Rightarrow x + \lambda = 0 \Rightarrow x = -\lambda$

Put $x = -\lambda$ in equation $x^2 + 2x + 3\lambda = 0$

$\Rightarrow (-\lambda)^2 + 2(-\lambda) + 3\lambda = 0 \Rightarrow \lambda^2 - 2\lambda + 3\lambda = 0$

$\Rightarrow \lambda^2 + \lambda = 0 \Rightarrow \lambda(\lambda + 1) = 0$

$\Rightarrow \lambda = 0$ or $\lambda = -1$

Reject Common root must be non zero

Ques. If the quadratic equation $x^2 + bx + c = 0$ (roots 1, β) and $x^2 + cx + b = 0$ ($b \neq c$) (roots γ, 1) have a common root then prove that their uncommon roots are the roots of the equation $x^2 + x + bc = 0$.

Ans. $x^2 + bx + c = 0$

$x^2 + cx + b = 0$

$\ominus \quad - \quad -$

$(b - c)x + c - b = 0$

$\Rightarrow (b - c)x = (b - c) \times 1 \qquad (\because b \neq c)$

$\Rightarrow x = 1 \rightarrow$ common root

put $x = 1$ in equation $x^2 + bx + c = 0$

$\Rightarrow 1^2 + b + c = 0 \Rightarrow b + c = -1$

Now in equations $x^2 + cx + b = 0$, and $x^2 + bx + c = 0$

$1 \times \beta = \frac{c}{1}$ and $1 \times \gamma = \frac{b}{1}$

$\Rightarrow \beta = c$ and $\gamma = b$,

Sum $= b + c = -1$, Product $= bc$

Hence, equation is, $x^2 - (-1)x + bc = 0$

$\Rightarrow x^2 + x + bc = 0$

Ques. Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation $x^2 - x + 2\lambda = 0$ (roots 1/3, β) and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to: (α = 1/3)

[JEE Main-2020]

(a) 18 (b) 9 (c) 27 (d) 36

Ans. $3x^2 - 3x + 6\lambda = 0$

$3x^2 - 10x + 27\lambda = 0$

$- \quad + \quad -$

$\Rightarrow 7x - 21\lambda = 0 \Rightarrow x = 3\lambda$

Put $x = 3\lambda$ in eq^n $x^2 - x + 2\lambda = 0$

$\Rightarrow (3\lambda)^2 - 3\lambda + 2\lambda = 0$

$\Rightarrow 9\lambda^2 - \lambda = 0 \Rightarrow \lambda(9\lambda - 1) = 0$

$\Rightarrow \lambda = 0, \lambda = \frac{1}{9}$ (but $\lambda \neq 0$)

$\Rightarrow \lambda = \frac{1}{9} \quad \therefore x = 3 \times \frac{1}{9} = \frac{1}{3} = \alpha$

$\frac{1}{3} + \gamma = \frac{10}{3}$ | $\frac{1}{3} + \beta = 1$

$\Rightarrow \gamma = 3$ | $\Rightarrow \beta = \frac{2}{3}$

Hence, $\frac{\beta\gamma}{\lambda} = \frac{\frac{2}{3} \times 3}{\frac{1}{9}} = 2 \times 9 = 18$

Ques. If the equations $x^2 + 4ax + 3 = 0$ & $2x^2 + 3x - 9 = 0$ have exactly one root common, Then the un - common roots are

(a) $-1, \frac{3}{2}$ (b) 2, - 3

(c) - 1, 2 (d) None of these

Ans. Given, $2x^2 + 3x - 9 = 0 \Rightarrow 2x^2 + 6x - 3x - 9 = 0$

$\Rightarrow (2x - 3)(x + 3) = 0 \Rightarrow x = \frac{3}{2}$ or -3

Case 1:

If $x = \frac{3}{2}$ is common

then, $x^2 + 4ax + 3 = 0$ (roots 3/2, β) $\Rightarrow \frac{3}{2} \cdot \beta = 3 \Rightarrow \beta = 2$

$2x^2 + 3x - 9 = 0$ (roots 3/2, γ = -3)

(2, -3) $\rightarrow$ un-common roots

Case 2:

If $x = -3$ is common root then,

$x^2 + 4ax + 3 = 0$ (roots -3, β) $\Rightarrow -3 \times \beta = \frac{3}{1} \Rightarrow \beta = -1$

$2x^2 + 3x - 9 = 0$ (roots -3, 3/2, γ)

$\left(-1, \frac{3}{2}\right) \rightarrow$ un-common roots

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CUBE ROOTS OF UNITY

$x = (1)^{\frac{1}{3}}$

Cube both sides

$\Rightarrow x^3 = 1 \Rightarrow x^3 - 1^3 = 0 \Rightarrow (x-1)(x^2+x+1) = 0$

either $x - 1 = 0$ or

$x^2 + x + 1 = 0$

$\Rightarrow x = 1,\ x^2 + x + 1 = 0$

$\Rightarrow x = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}i}{2}$

$\Rightarrow x = \frac{-1+\sqrt{3}i}{2}, \frac{-1-\sqrt{3}i}{2}$

ω $\quad$ ω^2

$\omega = \frac{-1+\sqrt{3}i}{2}$

Squaring both sides

$\omega^2 = \frac{(-1+\sqrt{3}i)^2}{4}$

$= \frac{1+3i^2-2\sqrt{3}i}{4}$

$= \frac{1-3-2\sqrt{3}i}{4}$

$= \frac{-2-2\sqrt{3}i}{4}$

$\omega^2 = \frac{-1-\sqrt{3}i}{2}$

1, ω, ω^2 are 3 cube roots of Unity

$x^2 + x + 1 = 0 \rightarrow \omega, \omega^2$

So, $\omega + \omega^2 = -1$

$\Rightarrow \boxed{1 + \omega + \omega^2 = 0}$

$\omega \cdot \omega^2 = \frac{1}{1}$

$\Rightarrow \boxed{\omega^3 = 1}$

Ques. The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The others roots of the first and second are integers in the ratio 4:3. Then the common root is **[(AIEEE 2008)]**

α, 4β $\quad$ α, 3β

(a) 4 $\quad$ (b) 3

(c) 2 $\quad$ (d) 1

Ans. $\alpha + 4\beta = 6$ $\quad$ $\alpha + 3\beta = c$

$\alpha \cdot 4\beta = a$ $\quad$ $\alpha \cdot 3\beta = 6$

$\alpha\beta = 2$

$\Rightarrow 2 \times 4 = a \Rightarrow a = 8$

$x^2 - 6x + 8 = 0 \rightarrow 2, 4$

Case → 1	Case → 2
$\alpha = 2$	$\alpha = 4$
$4\beta = 4$	$4\beta = 2$
$\beta = 1$	$\beta = \frac{1}{2}$ (x)
	$3\beta = \frac{3}{2}$ (Not a integer)

$\therefore$ Common roots (α) = 2

Ques. If ω is a cube of unity, then $(3 + 5\omega + 3\omega^2)^2 + (3 + 3\omega + 5\omega^2)^2$ is equal to:

(a) 4 $\quad$ (b) 0

(c) -4 $\quad$ (d) none of these

Ans. $(3 + 5\omega + 3\omega^2)^2 + (3 + 3\omega + 5\omega^2)^2$

$= (3 + 3\omega + 3\omega^2 + 2\omega)^2 + (3 + 3\omega + 3\omega^2 + 2\omega^2)^2$

$= (2\omega)^2 + (2\omega^2)^2$

$= 4\omega^2 + 4\omega^4$ ($\omega^4 = \omega$)

$= 4[\omega^2 + \omega] = 4(-1) = -4$

$[1 + \omega + \omega^2 = 0$
$3 + 3\omega + 3\omega^2 = 0$
$\because \omega^4 = \omega^3 \omega$ ($\omega^3 = 1$)
$\Rightarrow \omega^4 = \omega]$

Higher Powers of ω

1. $\omega^{3n} = 1 \quad n \in I$
2. $\omega^{3n+1} = \omega$
3. $\omega^{3n+2} = \omega^2$

Ques. If $i = \sqrt{-1}$ then

$4 + 5\left(\frac{-1}{2} + i\frac{\sqrt{3}}{2}\right)^{334} - 3\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{365}$ is equal to:

ω $\quad$ $-\omega^2$

(a) $1 - i\sqrt{3}$ $\quad$ (b) $-1 + i\sqrt{3}$

(c) $4\sqrt{3}i$ $\quad$ (d) $-i\sqrt{3}$

Ans. $4 + 5\omega^{334} - 3(-\omega^2)^{365}$

$= 4 + 5\omega^{334} + 3\omega^{730}$ (ω, ω)

$= 4 + 5\omega + 3\omega$

$= 4 + 8\omega = 4 + 8\left(\frac{-1+\sqrt{3}i}{2}\right) = \frac{8 - 8 + 8\sqrt{3}i}{2} = 4\sqrt{3}i$

$[\omega^2 = \frac{-1}{2} - \frac{\sqrt{3}i}{2}$
$\Rightarrow -\omega^2 = \frac{1}{2} + \frac{\sqrt{3}i}{2}]$

Ques. Let α be a root of the equation $1 + x^2 + x^4 = 0$. Then the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to: **[JEE Main - 2023]**

(a) 1 $\quad$ (b) α

(c) $1 + \alpha$ $\quad$ (d) $1 + 2\alpha$

Ans. Let $x^2 = t \Rightarrow \boxed{t^2 + t + 1 = 0} \rightarrow \omega, \omega^2$

$\Rightarrow t = \omega$ or $\omega^2 \Rightarrow x^2 = \omega$ or ω^2

$\Rightarrow \alpha^2 = \omega$ or ω^2

$\Rightarrow \boxed{\alpha = \omega}$ (one of the value)

$\omega^{1011} + \omega^{2022} - \omega^{3033} = 1 + 1 - 1 = 1$

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Ques. If α satisfies the equation $x^2 + x + 1 = 0$ and $(1 + \alpha)^7 = A + B\alpha + C\alpha^2$, $A, B, C \geq 0$, then $5(3A - 2B - C)$ is equal to ... **[JEE Main-2024]**

Ans. Let $\alpha = \omega$

$(1 + \omega)^7 = A + B\omega + C\omega^2$

$\Rightarrow (-\omega^2)^7 = A + B\omega + C\omega^2 \quad (\because 1 + \omega = -\omega^2)$

$\Rightarrow -\omega^{14} = A + B\omega + C\omega^2 \Rightarrow -\omega^2 = A + B\omega + C\omega^2$

$\Rightarrow 1 + \omega = A + B\omega + C\omega^2 \Rightarrow A = 1, B = 1, C = 0$

Hence, $5(3A - 2B - C)$

$= 5(3 \times 1 - 2 \times 1 - 0) = 5 \times 1 = 5$

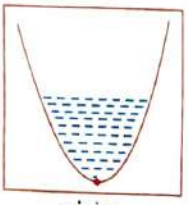
GRAPH OF QUADRATIC EXPRESSION

$y = ax^2 + bx + c \rightarrow$ Parabolic in nature

(i) if $a > 0 \rightarrow$ Concave up

mouth opening upward

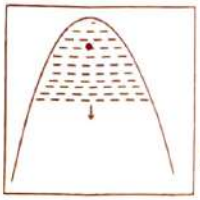
Filling water



minima

(ii) If $a < 0 \rightarrow$ Concave down

Maxima

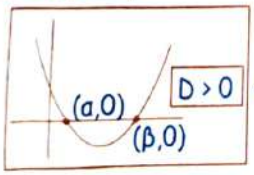


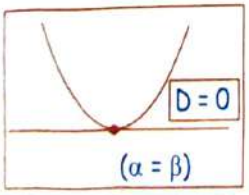
Mouth opening downward

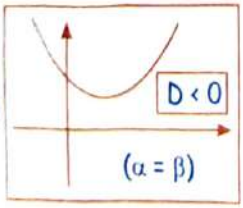
Spilling water

6 Possible graphs for a Quadratic Expression

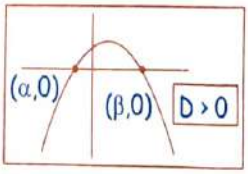
Case - I $a > 0$

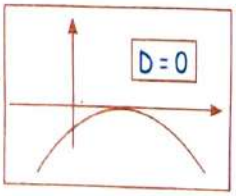
1) 

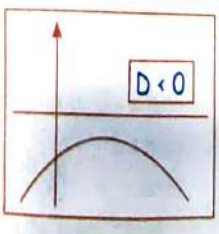
2) 

3) 

Case - II $a < 0$

4) 

5) 

6) 

Important Points to Remember While Plotting Graph of Quadratic Expression

(1) If $a > 0 \Rightarrow$ concave upwards, $a < 0 \Rightarrow$ concave downwords

(2) Graph cuts x-axis at $(\alpha, 0)$ & $(\beta, 0)$

(3) Graph cuts y-axis at $[0,c]$

(4) Vertex of parabola is $\left[-\frac{b}{2a}, \frac{-D}{4a}\right]$

where, $D \rightarrow$ Discriminant

proof: $y = ax^2 + bx + c$

$\frac{dy}{dx} = 2ax + b = 0$

For Vertex, $\frac{dy}{dx} = 0$

$\Rightarrow 2ax = -b \Rightarrow x = -\frac{b}{2a}$

$y = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c = \frac{b^2}{4a} - \frac{b^2}{2a} + c$

$= \frac{b^2 - 2b^2 + 4ac}{4a} = \frac{-b^2 + 4ac}{4a} = -\frac{(b^2 - 4ac)}{4a} = -\frac{D}{4a}$

Example: Find Vertex of $y = x^2 - 3x + 1$

$a = 1, b = -3, c = 1$

$D = b^2 - 4ac = 9 - 4 \times 1 \times 1 = 5$

Vertex $= \left[\frac{-b}{2a}, \frac{-D}{4a}\right] = \left[\frac{-(-3)}{2 \times 1}, \frac{-5}{4 \times 1}\right] = \left[\frac{3}{2}, \frac{-5}{4}\right]$

Ques. Plot the graph of $y = x^2 - 5x - 6$

Ans. $a = 1, b = -5, c = -6$

$D = b^2 - 4ac$

$= (-5)^2 - 4 \times 1 (-6)$

$= 25 + 24 = 49 > 0$

$x^2 - 5x - 6 = 0$

$\Rightarrow (x - 6)(x + 1) = 0$

$\Rightarrow x = -1, 6$

x - axis $(\alpha, 0)$ $(\beta, 0) = (-1, 0)$ & $(6, 0)$

y - axis $(0,c) = (0, -6)$

Vertex $= \left[\frac{-b}{2a}, \frac{-D}{4a}\right] = \left[\frac{5}{2 \times 1}, \frac{-49}{4}\right]$

$= \left[\frac{5}{2}, \frac{-49}{4}\right] \rightarrow$ IVth Quad

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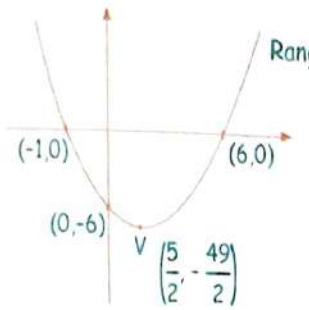


Range: $y \in \left(-\frac{49}{4}, \infty\right)$

Ques. Plot the graph of the following:

(i) $y = 3x^2 + x + 2$

Ans. $a = 3\ b = 1, c = 2$

$D = 1^2 - 4 \times 3 \times 2 = 1 - 24 = -23 < 0$

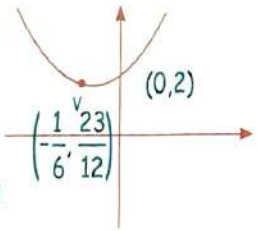
does not cut x - axis

y - axis (0,2)

Vertex $\left[-\frac{b}{2a}, -\frac{D}{4a}\right]$

$= \left[\frac{-1}{2 \times 3}, \frac{-(-23)}{4 \times 3}\right]$

$= \left[\frac{-1}{6}, \frac{23}{12}\right]$ = 2nd Quad



Range: $y \in \left(\frac{23}{12}, \infty\right)$

(ii) $y = -x^2 + 6x - 9$

Ans. $a = -1, b = 6, c = -9$

$D = b^2 - 4ac = 36 - 4 \times (-1)(-9) = 36 - 36 = 0$

Now, $-x^2 + 6x - 9 = 0$

$\Rightarrow x^2 - 6x + 9 = 0$

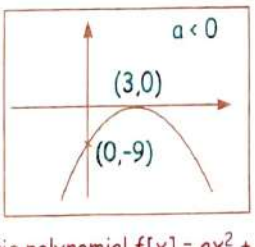
$\Rightarrow (x - 3)^2 = 0 \Rightarrow x = 3$

x - axis $\rightarrow$ (3, 0)

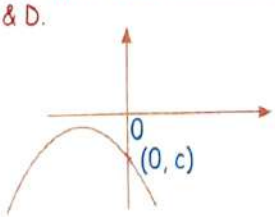
y - axis $\rightarrow$ (0, -9)

Vertex = [3, 0]

Range: $y \in (-\infty, 0)$



Ques. The graph of quadratic polynomial $f[x] = ax^2 + bx + c$ is shown below. Comment on the sign of a, b, c & D.



Ans. $a < 0$

$D < 0$

$c < 0$

$X_v = -\frac{b}{2a}$

But Vertex is in 3rd Quad

$\Rightarrow x_v < 0 \Rightarrow -\frac{b}{2a} < 0$

$\Rightarrow \frac{b}{2a} > 0 \Rightarrow b < 0$

(-ve)

Ques. The graph of quadratic polynomial $f(x) = ax^2 + bx + c$ is shown below .Comment on the sign of a, b, c & D.

Ans. $a > 0$

$D > 0$

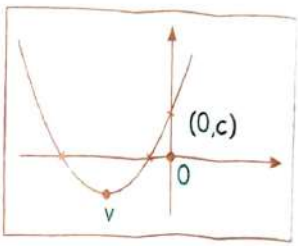
$c > 0$

and $-\frac{b}{2a} < 0$

$\Rightarrow \frac{b}{2a} > 0$

(2a) + ve

$\Rightarrow b > 0$



FORMULA FOR RANGE OF QUADRATIC EXPRESSION

(i) if $a > 0 \Rightarrow y \in \left[-\frac{D}{4a}, \infty\right)$ (ii) if $a < 0 \Rightarrow y \in \left(-\infty, \frac{-D}{4a}\right]$

Example: $y = x^2 - 3x + 1$

Ans. $D = 9 - 4 = 5,\ a = 1 > 0$

So, Range, $y \in \left[-\frac{D}{4a}, \infty\right)$

$\Rightarrow y \in \left[-\frac{5}{4 \times 1}, \infty\right) \Rightarrow y \in \left[-\frac{5}{4}, \infty\right)$

Ques. The least value of 'n' such that $(n-2)x^2 + 8x + n + 4 > 0\ \forall x \in R$, where $n \in N$ is

(a) 3 (b) 5

(c) 4 (d) None

Ans. $a = n - 2, b = 8, c = n + 4$

$D < 0$

$\Rightarrow 64 - 4(n - 2)(n + 4) < 0$

$\Rightarrow 16 - (n - 2)(n + 4) < 0$

$\Rightarrow 16 - (n^2 + 2n - 8) < 0$

$\Rightarrow n^2 + 2n - 8 > 16$

$\Rightarrow n^2 + 2n - 24 > 0$

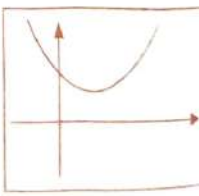
$\Rightarrow (n + 6)(n - 4) > 0$

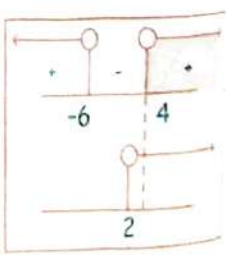
and $a > 0 \Rightarrow n - 2 > 0$

$\Rightarrow n > 2$

Hence, $n \in (4, \infty)$

$\therefore$ least value of n = 5





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Ques. Find the possible value of 'k' for which the inequality $(k-4)x^2 - 2kx + 2k - 6 < 0$ is satisfied for all $x \in R$.

Ans. $a = k-4,\ b = 2k,\ c = 2k-6$

$a < 0$

$\Rightarrow k - 4 < 0 \Rightarrow k < 4$

and $D < 0$

$\Rightarrow 4k^2 - 4(k-4)(2k-6) < 0$

$\Rightarrow k^2 - (2k^2 - 6k - 8k + 24) < 0$

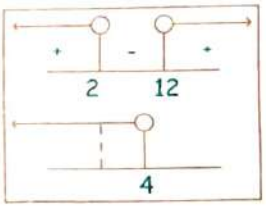
$\Rightarrow k^2 - 2k^2 + 14k - 24 < 0$

$\Rightarrow -k^2 + 14k - 24 < 0$

$\Rightarrow k^2 - 14k + 24 > 0$

$\Rightarrow (k-12)(k-2) > 0$

Hence, $k \in (-\infty, 2)$

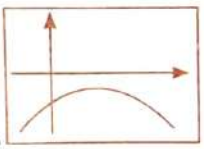


Ques. If $ax^2 + bx + c = 0$ does not have real roots & $c < 0$, then prove that

(i) $a + b + c < 0$ (ii) $a - b + c < 0$ (iii) $9a - 3b + c < 0$

Ans. $y = ax^2 + bx + c$

y - axis $(0, c)$



$\Rightarrow ax^2 + bx + c < 0$ for all $x \in R$

(i) $x = 1 \Rightarrow a + b + c < 0$

(ii) $x = -1 \Rightarrow a(-1)^2 + b(-1) + c < 0$

$\Rightarrow a - b + c < 0$

(iii) $x = -3 \Rightarrow 9a - 3b + c < 0$

Ques. If $ax^2 + bx + 5 = 0$ does not have distinct real roots then find minimum value of $(5a + b)$

Ans. $y = ax^2 + bx + 5$

y - axis $[0, 5]$

$\Rightarrow ax^2 + bx + 5 \geq 0$ for all $x \in R$

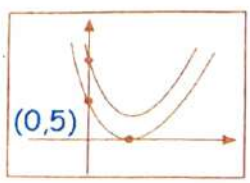
Put, $x = 5$

$\Rightarrow 25a + 5b + 5 \geq 0$

$\Rightarrow 5(5a + b + 1) \geq 0$

$\Rightarrow 5a + b \geq -1$

minimum value of $(5a + b) = -1$



Ques. $ax^2 + 2bx - 3c = 0$ has no real root and $a + b > \frac{3c}{4}$ then

(a) $a > 0, c < 0$ (b) $a < 0, c > 0$

(c) $a > 0, c > 0$ (d) None of these

Ans. Given, $a + b > \frac{3c}{4}$

$\Rightarrow 4a + 4b > 3c$

$\Rightarrow 4a + 4b - 3c > 0$

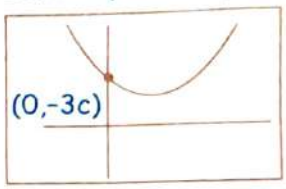
$f(x) = ax^2 + 2bx - 3c$

Put $x = 2$

$f(2) = 4a + 4b - 3c > 0$

$f(2) > 0$

So, the possible graph is



$a > 0$

$-3c > 0$

$\Rightarrow c < 0$

Brain Teaser

If $x^2 + (a-b)x + (1-a-b) = 0$ where $a, b, \in R$ then find the value of a for which equation has unequal real roots for all value of b

Ans. $D > 0$ (for unequal real roots)

$\Rightarrow (a-b)^2 - 4 \times 1 \times (1-a-b) < 0$

$\Rightarrow a^2 + b^2 - 2ab - 4 + 4a + 4b > 0\ \forall\ b \in R$

$\Rightarrow b^2 + b(4 - 2a) + a^2 + 4a - 4 > 0$

$a = 1,\ b \to 4 - 2a,\ c \to a^2 + 4a - 4$

$D < 0$ (for above eqn to be always +ve.)

$\Rightarrow (4-2a)^2 - 4 \times 1 \times (a^2 + 4a - 4) < 0$

$\Rightarrow 4(2-a)^2 - 4(a^2 + 4a - 4) < 0$

$\Rightarrow 4 + a^2 - 4a - a^2 - 4a + 4 < 0$

$\Rightarrow 8 - 8a < 0 \Rightarrow 8 < 8a \Rightarrow a > 1$

Range of Rational Expressions

1) $y = \frac{\text{linear}}{\text{linear}}$

Ex: $y = \frac{②x - 3}{④x + 7}$

Range, $y \in R - \left\{\frac{2}{4}\right\}$

$\Rightarrow y \in R - \left\{\frac{1}{2}\right\}$

$y = \frac{\text{Quad}}{\text{linear}}$ or $\frac{\text{linear}}{\text{Quad}}$ or $\frac{\text{Quad}}{\text{Quad}}$

same method

Ques. Find the range of y for real x if $y = \frac{x+2}{2x^2 + 3x + 6}$

Ans. $y = \frac{x+2}{2x^2 + 3x + 6}$

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Cross multiply

$\Rightarrow 2x^2y + 3xy + 6y = x + 2$

$\Rightarrow 2y\,x^2 + x(3y - 1) + 6y - 2 = 0\ ,\ x \in R$

$D \geq 0$

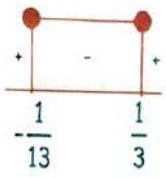
$\Rightarrow (3y - 1)^2 - 4 \times 2y\,(6y - 2) \geq 0$

$\Rightarrow (3y - 1)^2 - 16y\,(3y - 1) \geq 0$

$\Rightarrow (3y - 1)\,[(3y - 1) - 16y] \geq 0$

$\Rightarrow (3y - 1)\,[-13y - 1] \geq 0$

$\Rightarrow (3y - 1)\,(13y + 1) \leq 0$



$\Rightarrow y \in \left[\frac{-1}{13}, \frac{1}{3}\right]$

Ques. Find the range of y for real x if $y = \frac{x^2 - 3x + 2}{x^2 + x - 6}$

Ans. $y = \frac{x^2 - 3x + 2}{x^2 + x - 6}$

$y = \frac{(x - 2)(x - 1)}{(x + 3)(x - 2)}$ Domain: $x \in R - \{2, -3\}$

$\Rightarrow x - 2 \neq 0 \Rightarrow x \neq 2$

put $x = 2$

$y = \frac{2 - 1}{2 + 3} = \frac{1}{5}$

$\Rightarrow y \neq \frac{1}{5}$

and $y = \frac{x - 1}{x + 3} \Rightarrow y \in R - \left\{\frac{1}{1}\right\}$

Hence, $y \in R - \left\{1, \frac{1}{5}\right\}$

Ques. Let $f: R - \{2,6\} \to R$ be real valued function defined as $f(x) = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}$ Then range of f is **[JEE Main - 2023]**

(a) $\left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$

(b) $\left(-\infty, -\frac{21}{4}\right) \cup (0, \infty)$

(c) $\left(-\infty, -\frac{21}{4}\right] \cup \left[\frac{21}{4}, \infty\right)$

(d) $\left(-\infty, -\frac{21}{4}\right] \cup [1, \infty]$

Ans. $y = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}$

$\Rightarrow yx^2 - 8xy + 12y = x^2 + 2x + 1$

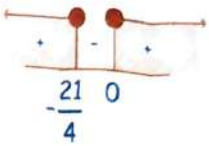
$\Rightarrow x^2 (y - 1) - 2x\,[4y + 1] + 12y - 1 = 0$

$D \geq 0$

$\Rightarrow 4\,(4y + 1)^2 - 4\,(y - 1)\,(12y - 1) \geq 0$

$\Rightarrow 16y^2 + 1 + 8y - (12y^2 - 13y + 1) \geq 0$

$\Rightarrow 4y^2 + 21y \geq 0$



$\Rightarrow y\,(4y + 21) \geq 0$

$\Rightarrow y \in \left(-\infty, -\frac{21}{4}\right] \cup [0, \infty]$

Ques. Range of the function $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}; x \in R$ is

(a) $(1, \infty)$ (b) $\left(1, \frac{11}{7}\right)$

(c) $\left(1, \frac{7}{3}\right]$ (d) $\left(1, \frac{7}{5}\right)$

Ans. $y = \frac{x^2 + x + 2}{x^2 + x + 1}$

$\Rightarrow yx^2 + yx + y = x^2 + x + 2$

$\Rightarrow x^2 (y - 1) + x\,(y - 1) + y - 2 = 0$

$D \geq 0$

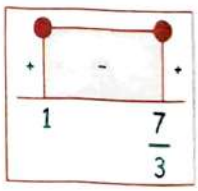
$\Rightarrow (y - 1)^2 - 4\,(y - 1)\,(y - 2) \geq 0$

$\Rightarrow (y - 1)\,[y - 1 - 4(y - 2)] \geq 0$

$\Rightarrow (y - 1)\,[y - 1 - 4y + 8] \geq 0$

$\Rightarrow (y - 1)\,(-3y + 7) \geq 0$

$\Rightarrow (y - 1)\,(3y - 7) \leq 0$



$y \in \left[1, \frac{7}{3}\right]$

Check for $x = 1$,

$y = 1 = \frac{x^2 + x + 2}{x^2 + x + 1}$

$x^2 + x + 1 = x^2 + x + 2 \Rightarrow 1 = 2\ (\times)$

$\Rightarrow$ open at $y = 1$

Hence, $y \in \left(1, \frac{7}{3}\right]$

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AM - 2

$$y = \frac{x^2 + x + 2}{x^2 + x + 1}$$

$$= \frac{(x^2 + x + 1) + 1}{x^2 + x + 1}$$

$$y = 1 + \frac{1}{\underbrace{x^2 + x + 1}_{t}}$$

$$y = 1 + \frac{1}{t}$$

$t = x^2 + x + 1$

$$t = \left(x + \frac{1}{2}\right)^2 - \frac{1}{4} + 1$$

$$\boxed{t = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} = t_{min}}$$

for $y_{max} = 1 + \frac{1}{t_{min}}$

$$y_{max} = 1 + \frac{1}{\frac{3}{4}}$$

$$= 1 + \frac{4}{3} = \frac{7}{3}$$

for y_{min}, $t \to \max$

$y \to 1 + \frac{1}{\infty}$ ($\frac{1}{\infty} \to 0$)

$y \to 1$

Hence, $y \in \left(1, \frac{7}{3}\right]$

Brain Teaser

If x is real, then $\frac{x^2 - x + c}{x^2 + x + 2c}$ can take all real values if:

(a) $c \in [0, 6]$ (b) $c \in [-6, 0]$

(c) $c \in (-\infty, -6) \cup (0, \infty)$ (d) $c \in (-6, 0)$

Ans. $y = \frac{x^2 - x + c}{x^2 + x + 2c}$

$\Rightarrow x^2y + xy + 2cy = x^2 - x + c$

$\Rightarrow x^2 (y - 1) + x (y + 1) + 2cy - c = 0$

$D \geq 0$

$\Rightarrow (y + 1)^2 - 4 (y - 1) (2cy - c) \geq 0$

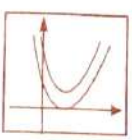
$\Rightarrow y^2 + 1 + 2y - 4 [2cy^2 - cy - 2cy + c] \geq 0$

$\Rightarrow y^2 + 1 + 2y - 8cy^2 + 12cy - 4c \geq 0$

$\Rightarrow y^2(1 - 8c) + 2y (6c + 1) + 1 - 4c \geq 0 \to \forall\, y \in R$

$a > 0, D \leq 0$

$1 - 8c > 0 \Rightarrow 8c < 1 \Rightarrow c < \frac{1}{8}$



and $4 (6c + 1)^2 - 4 (1 - 8c) (1 - 4c) \leq 0$

$\Rightarrow 36c^2 + 1 + 12c - [1 - 12c + 32c^2] \leq 0$

$\Rightarrow 36c^2 + 1 + 12c - 1 + 12c - 32c^2 \leq 0$

$\Rightarrow 4c^2 + 24c \leq 0 \Rightarrow 4c (c + 6) \leq 0$

Consider $c = 0$

$$y = \frac{x^2 - x}{x^2 + x}$$

$$y = \frac{x (x - 1)}{x (x + 1)}, x \neq 0$$

$$y = \frac{x - 1}{x + 1}$$

$$\Rightarrow y \neq \frac{0 - 1}{0 + 1} \Rightarrow y \neq -1$$

But y takes all real values.

So, $c \in (-6, 0)$

Consider $c = -6$

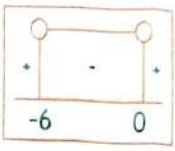
$$y = \frac{x^2 - x - 6}{x^2 + x - 12}$$

$$y = \frac{(x - 3) (x + 2)}{(x + 4) (x - 3)}$$

$$y = \frac{x + 2}{x + 4}, x \neq 3$$

$$y \neq \frac{3 + 2}{3 + 4}$$

$$y \neq \frac{5}{7}$$



Brain Teaser

Complete set of values of 'a' such that $\frac{x^2 - x}{1 - ax}$ attains real values is

(a) [1, 4] (b) (0, 4]

(c) (0, 1) (d) $(1, \infty)$

Ans. $y = \frac{x^2 - x}{1 - ax}$

$\Rightarrow y - axy = x^2 - x$

$\Rightarrow x^2 + (ay - 1)x - y = 0$

$D \geq 0$

$\Rightarrow (ay - 1)^2 - 4 \times 1 (-y) \geq 0$

$\Rightarrow a^2y^2 + 1 - 2ay + 4y \geq 0$

$\Rightarrow a^2y^2 + 2y (2 - a) + 1 \geq 0\ \forall\, y \in R$

$D \leq 0$

$\Rightarrow 4 (2 - a)^2 - 4a^2 (1) \leq 0$

$\Rightarrow 4 + a^2 - 4a - a^2 \leq 0$

$\Rightarrow 4 \leq 4a \Rightarrow a \geq 1$

put $a = 1$

$$y = \frac{x^2 - x}{1 - x} = \frac{x (x - 1)}{-(x - 1)} = -x, x \neq 1$$

$y \neq -1$, But y takes all real values.

Hence, $a \in (1, \infty)$

RATIONAL ROOTS/ INTEGRAL ROOTS

$ax^2 + bx + c = 0$

$$\alpha, \beta = \frac{-b \pm \sqrt{D}}{2a}$$

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D → Perfect squares

&

a, b, c → Rational

if $\boxed{a = 1}$

$\alpha, \beta = \frac{-b \pm \sqrt{D}}{2}$ → Integer

$D = (P)^2$

$\alpha, \beta = \frac{-b \pm P}{2}$

Case: 1

b → even

$D = b^2 - 4ac$

⇒ D → Even

$\Rightarrow \sqrt{D}$ → Even

Case: 2

b = odd

$D = b^2 - 4ac$ = odd

$\sqrt{D}$ → odd

Ques. If a and b are the odd integers, then the roots of the equation $2ax^2 + (2a + b)x + b = 0$, $a \neq 0$, will be

(a) Rational ✓ (b) Irrational

(c) Non - real (d) Equal

Ans. $a \to 2a, b \to (2a + b), c \to b$

$D = b^2 - 4ac = (2a + b)^2 - 4 \cdot 2a \cdot b$

$= 4a^2 + b^2 + 4ab - 8ab = 4a^2 + b^2 - 4ab$

$D = (2a - b)^2$ D → Perfect sq.

Brain Teaser

Find the number of the value of 'a' such that the equation $x^2 - x - a = 0$ has integral roots $a \in N, a \in [6, 100]$

Ans. $x^2 - x - a = 0$ (roots α, β)

$\alpha + \beta = 1$ & $\alpha\beta = -a$

(α, β)

~~(-1, 2)~~

(- 2, 3) → a = 6

(- 3, 4) → a = 12

(- 4, 5) → a = 20

(- 5, 6) → a = 30

(- 6, 7) → a = 42

(- 7, 8) → a = 56

(- 8, 9) → a = 72

(- 9, 10) → a = 90

~~(- 10, 11) → a = 110~~

Number of value of a = 8

Brain Teaser

Find the integral of 'a' for which the equation $(x + 2024)(x + a) + 1 = 0$ has integral roots.

Ans. $\underbrace{(x + 2024)}_{I_1}\underbrace{(x + a)} = -1$ $\quad a \in I \quad x \in I$

$\Rightarrow \boxed{I_1 I_2 = -1}$

Case: 1

$I_1 = 1$

$x + 2024 = 1$

$\Rightarrow x = -2023$

$I_2 = -1$

$x + a = -1$

$-2023 + a = -1$

$\boxed{a = 2022}$

Case: 2

$I_1 = -1$

$\Rightarrow x + 2024 = -1$

$\Rightarrow x = -2025$

$I_2 = 1$

$x + a = 1$

$-2025 + a = 1$

$\Rightarrow a = 2026$

LOCATION OF ROOTS

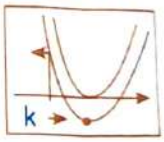
Step 1: Draw possible graphs (assuming a > 0)

Step 2: Write "necessary & sufficient conditions for the graph drawn above

Step 3: Solve all the conditions & take Intersection

Case 1

Both the roots are greater then as given number 'k'



1. $D \geq 0$
2. $x_v > k \Rightarrow \frac{-b}{2a} > k$
3. $f(k) > 0$

Ques. Find all the values of the parameter 't' for which both roots of the equation $x^2 - 6tx + [2 - 2t + 9t^2] = 0$ exceed the number 3.

Ans. $f(x) = x^2 - 6t\,x + (2 - 2t + 9t^2)$

(1) $D \geq 0$

$\Rightarrow (-6t)^2 - 4(2 - 2t + 9t^2) \geq 0$

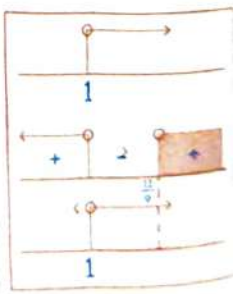
$\Rightarrow 36t^2 - 4(2 - 2t + 9t^2) \geq 0$

$\Rightarrow 36t^2 - 8 + 8t - 36t^2 \geq 0$

$\Rightarrow 8t \geq 8 \Rightarrow t \geq 1$

(2) $\frac{-b}{2a} > 3 \Rightarrow \frac{6t}{2 \times 1} > 3$

$\Rightarrow t > 1$



(3) $f(3) > 0$

$\Rightarrow 9 - 6t \times 3 + 2 - 2t + 9t^2 > 0$

$\Rightarrow 9 - 18t + 2 - 2t + 9t^2 > 0$

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$\Rightarrow 9t^2 - 20t + 11 > 0$

$\Rightarrow 9t^2 - 9t - 11t + 11 > 0$

$\Rightarrow (9t - 11)(t - 1) > 0$

$\therefore t \in \left(\frac{11}{9}, \infty\right)$

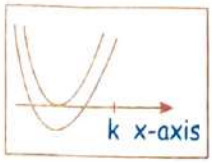
Case 2

Both the roots are less then a given number 'k'

(1) $D \geq 0$

(2) $\frac{-b}{2a} < k$

(3) $f(k) > 0$



Ques. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less then 3 then

(a) $a < 2$ (b) $2 \geq a \geq 3$

(c) $3 < a \geq 4$ (d) $a > 4$

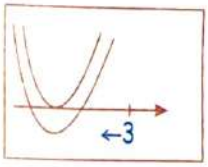
Ans. $f(x) = x^2 - 2ax + (a^2 + a - 3)$

(1) $D \geq 0$

$\Rightarrow 4a^2 - 4(a^2 + a - 3) \geq 0$

$\Rightarrow a^2 - a^2 - a + 3 \geq 0$

$\Rightarrow -a + 3 \geq 0 \Rightarrow a \leq 3$



(2) $\frac{-b}{2a} < 3$

$\Rightarrow \frac{2a}{2 \times 1} < 3 \Rightarrow a < 3$

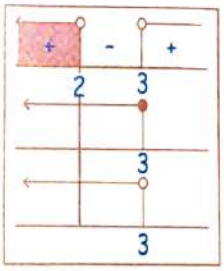
(3) $f(3) > 0$

$\Rightarrow 9 - 6a + a^2 + a - 3 > 0$

$\Rightarrow a^2 - 5a + 6 > 0$

$\Rightarrow (a - 2)(a - 3) > 0$

$\therefore a < 2$

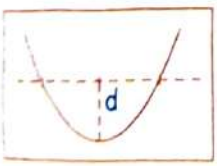


Case 3

Both roots lie on either side of a fixed number say 'd'. Alternatively one root is greater than 'd' and other less than 'd' or 'd' lies between the roots of the given equation.

Case: 1

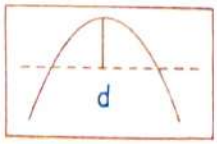
$a > 0$



$f(d) < 0$

Case: 2

$a < 0$



$f(d) > 0$

$of\ (d) < 0 \Rightarrow$ sufficient condition

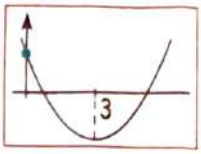
Important Note

For mouth opening Upwards graph , whenever we solve $f(k) < 0$, no need to solve $D > 0$. Similarly, for mouth opening down word graph, when even we solve $f(k) > 0$ no need to solve $D > 0$.

Ques. Find the set of value of 'a' for which zeroes of the quadratic polynomial $(a^2 + a + 1)x^2 + (a - 1)x + a^2$ are located on either side of 3.

Ans. $f(x) = \underbrace{(a^2 + a + 1)}_{+ve}x^2 + (a - 1)x + a^2 = 0$

$f(3) < 0$ is sufficient



$\Rightarrow (a^2 + a + 1) \times 9 + (a - 1) \times 3 + a^2 < 0$

$\Rightarrow 9a^2 + 9a + 9 + 3a - 3 + a^2 < 0$

$\Rightarrow 10a^2 + 12a + 6 < 0$

$\Rightarrow \boxed{5a^2 + 6a + 3 < 0} \rightarrow D = -ve,\ a = 5$

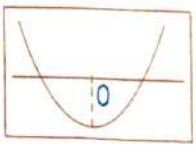
$\Rightarrow$ always + ve

No solution $a \in \phi$

Important Note

The necessary and sufficient condition for a quadratic $ax^2 + bx + c = 0$ to have roots of opposite in sign is $c/a < 0$.

Proof: $f(x) = x^2 + \frac{b}{a}x + \frac{c}{a} = 0$



sufficient condition, $f(0) < 0$

$\Rightarrow \frac{c}{a} < 0$

Important Note

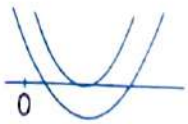
The necessary and sufficient condition for a quadratic equation $ax^2 + bx + c = 0$ to have roots positive.

(i) Product of roots > 0 or $\frac{c}{a} > 0 \rightarrow$ necessary not sufficient.

(ii) $D \geq 0$

(iii) $f(0) > 0$ } sufficient

(iv) $-\frac{b}{2a} > 0$



Ques. The equation $x^2 + ax - a^2 + 1 = 0$ will have roots of opposite sign if

(a) $a \in (-\infty, \infty)$ (b) $a \in (-1, 1)$

(c) $a \in (-\infty, -1) \cup (1, \infty)$ (d) None of these

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Ans. $x^2 + ax + 1 - a^2 = 0$

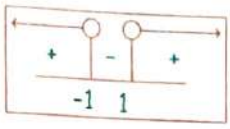
condition is $\frac{c}{a} < 0$

$\Rightarrow \frac{1 - a^2}{1} < 0$

$\Rightarrow (1 - a)(1 + a) < 0$

$\Rightarrow (a - 1)(a + 1) > 0$

$\therefore a \in (-\infty, -1) \cup (1, \infty)$

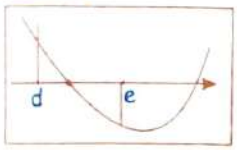


Case 4

Exactly one roots in the interval (d, e) when $d < e$.

Case: A $(a > 0)$

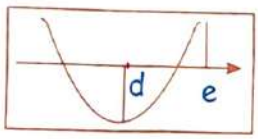
Case: 1



$f(d) > 0$

& $f(e) < 0$

Case: 2



$f(d) < 0$

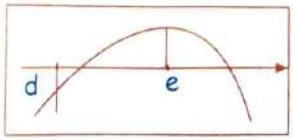
&

$f(e) > 0$

$f(d) \cdot f(e) < 0 \rightarrow$ sufficient

Case: B $(a < 0)$

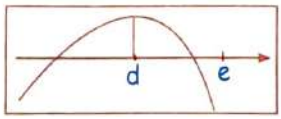
Case: 1



$f(e) > 0$

& $f(d) < 0$

Case: 2



$f(d) > 0$

$f(e) < 0$

$f(d) \cdot f(e) < 0 \rightarrow$ sufficient

Ques. Find the set of values of m for which exactly one root of the equation $x^2 + mx + (m^2 + 6m) = 0$ lie in (-2,0).

Ans. $f(x) = x^2 + mx + (m^2 + 6m)$

$f(0) = m^2 + 6m$

& $f(-2) = (-2)^2 + m(-2) + m^2 + 6m$

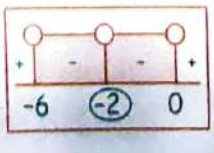
$= 4 - 2m + m^2 + 6m = m^2 + 4m + 4$

Condition: $f(d)\, f(e) < 0$

$\Rightarrow f(-2)\, f(0) < 0 \Rightarrow (m^2 + 4m + 4)(m^2 + 6m) < 0$

$\Rightarrow (m + 2)^2\, m\, (m + 6) < 0$

$m \in (-6, 0) - \{-2\}$



Brain Teaser

Find all possible values of 'a' for exactly one root of the quadratic equation $x^2 - (a + 1)x + 2a = 0$ lie in the interval (0,3).

Ans. $f(x) = x^2 - (a + 1)x + 2a = a$

$f(0) = 2a$

& $f(3) = 3^2 - 3(a + 1) + 2a$

$= 9 - 3a - 3 + 2a = 6 - a$

Condition: $f(0)\, f(3) < 0$

$\Rightarrow 2a(6 - a) < 0$

$\Rightarrow a(a - 6) > 0$

Now, If

$f(0) = 0$

$2a = 0$

$\Rightarrow a = 0$

$\Rightarrow$ Quad becomes $x^2 - x = 0$

$\Rightarrow x(x - 1) = 0 \Rightarrow x = 0, 1$

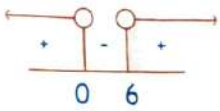
$1 \in (0,3) \Rightarrow a = 0$ is correct

and, If

$f(3) = 0$

$\Rightarrow 6 - a = 0$

$\Rightarrow a = 6$



$\Rightarrow$ Quad becomes

$x^2 - 7x + 12 = 0 \Rightarrow x = 4, 3$

and $4 \notin (0, 3)$ So, $a = 6$ is rejected

Hence, $a \in (-\infty, 0] \cup (6, \infty)$

Case 5

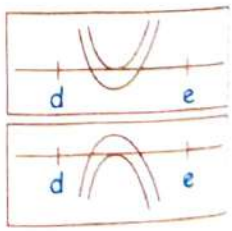
Both the roots lie in [d, e].

(1) $D \geq 0$

(2) $d < -\frac{b}{2a} < e$

(3) $af(d) > 0$

(4) $af(e) > 0$



Ques. If α, β are the roots of the quadratic equation $x^2 + 2(k - 3)x + 9 = 0$ $(\alpha \neq \beta)$. If $\alpha, \beta \in (-6, 1)$ then find the value of k.

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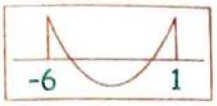
Ans. (1) $D > 0$

$\Rightarrow 4(k-3)^2 - 4 \times 1 \times 9 > 0$

$\Rightarrow k^2 - 6k + 9 - 9 > 0$

$\Rightarrow k(k-6) > 0$



(2) $-6 < \frac{-b}{2a} < 1$

$\Rightarrow -6 < \frac{-2(k-3)}{2 \times 1} < 1$

$\Rightarrow 6 > k - 3 > -1 \Rightarrow 9 > k > 2$

(3) $f(-6) > 0$

$\Rightarrow 36 - 2(k-3) \times 6 + 9 > 0$

$\Rightarrow 36 - 12(k-3) + 9 > 0$

$\Rightarrow 36 - 12k + 36 + 9 > 0 \Rightarrow 81 > 12k$

$\Rightarrow k < \frac{81}{12} \Rightarrow k < \frac{27}{4}$

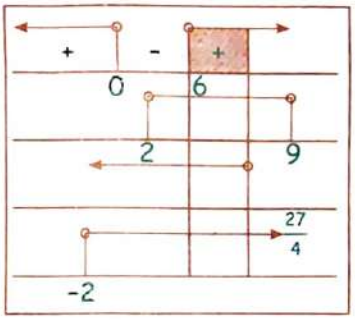
(4) $f(1) > 0$

$\Rightarrow 1 + 2(k-3) + 9 > 0$

$\Rightarrow 2k - 6 + 10 > 0$

$\Rightarrow 2k + 4 > 0$

$\Rightarrow k > -2$



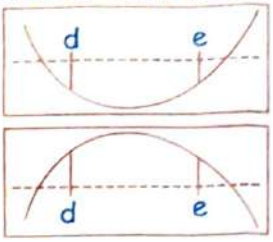
Hence, $k \in \left(6, \frac{27}{4}\right)$

Case 6

One root is greater than e and the other roots is less than d.

1. ~~$D \geq 0$~~
2. $af(d) < 0$
3. $af(e) < 0$

→ Sufficient



Ques. Find the Value of k for which one root of the equation of $[k-5]x^2 + 2kx + k - 4 = 0$ is smaller than 1 and the other root not exceed 2.

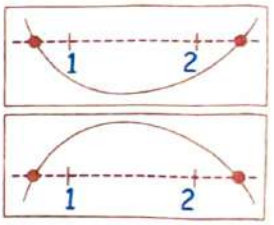
Ans. (1) $af(1) < 0$

$\Rightarrow (k-5)[(k-5)+2k+k-4] < 0$

$\Rightarrow (k-5)(4k-9) < 0$

$\Rightarrow (k-5)\left(k - \frac{9}{4}\right) < 0$

$k \in \left(\frac{9}{4}, 5\right)$



(2) $af(2) < 0$

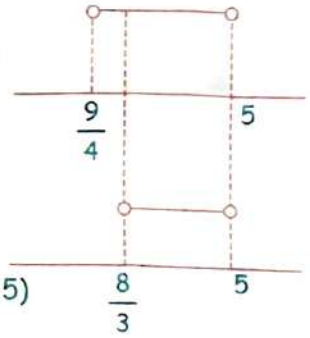
$\Rightarrow (k-5)[4k-20+4k+k-4] < 0$

$\Rightarrow (k-5)[9k-24] < 0$

$\Rightarrow (k-5)\left(k - \frac{8}{3}\right) < 0$

$k \in \left(\frac{8}{3}, 5\right)$

Final Answer: $k \in \left(\frac{8}{3}, 5\right)$



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Short Notes

ROOTS AND COEFFICIENTS

- An equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$ and a, b, c are any real numbers, is called a quadratic equation.
- The roots of the quadratic equation are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- Let α and β be two roots of the given quadratic equation. Then $\alpha+\beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$.
- A quadratic equation, whose roots are α and β can be written as $(x - \alpha)(x - \beta) = 0$
- i.e., $ax^2 + bx + c \equiv a(x - \alpha)(x - \beta)$.

NATURE OF ROOTS

- If the quadratic equation is satisfied by more than two distinct numbers (real or complex), then it becomes an identity, i.e. $a = b = c = 0$.
- The quadratic equation has real and equal roots if and only if $D = 0$ i.e. $b^2 - 4ac = 0$.
- The quadratic equation has real and distinct roots if and only if $D > 0$ i.e. $b^2 - 4ac > 0$
- The quadratic equation has complex roots with non-zero imaginary parts if and only if $D < 0$ i.e. $b^2 - 4ac < 0$.
- If $p + iq$ (p and q being real) is a root of the quadratic equation where $i = \sqrt{-1}$, then $p - iq$ is also a root of the quadratic equation.
- If $p + \sqrt{q}$ is an irrational root of the quadratic equation, then $p - \sqrt{q}$ is also a root of the quadratic equation provided that all the coefficients are rational.
- The quadratic equation has rational roots if D is a perfect square and a, b, c are rational.
- If $a = 1$ and b, c are integers and the roots of the quadratic equation are rational, then the roots must be integers.

CONDITION FOR COMMON ROOT(S)

The condition for the quadratic equations $ax^2 + bx$ $c = 0$ and $dx^2 + ex + f = 0$ to have

- Exactly one common root

  $\Rightarrow (dc - af)^2 = (bf - ce)(ae - bd)$

- Both the roots to be common is $\frac{a}{d} = \frac{b}{e} = \frac{c}{f}$

NEWTON'S FORMULA

$S_n = \alpha^n + \beta^n$, $T_n = \alpha^n - \beta^n$, $U_n = p\alpha^n + q\beta^n$,

$V_n = p\alpha^n - q\beta^n$

Let α and β are the roots of the equation $ax^2 + bx$ $c = 0$, then

- $a S_{n+2} + b S_{n+1} + c S_n = 0$
- $a T_{n+2} + b T_{n+1} + c T_n = 0$
- $a U_{n+2} + b U_{n+1} + c U_n = 0$
- $a V_{n+2} + b V_{n+1} + c V_n = 0$

If α, β, γ are roots of equation $ax^3 + bx^2 + cx + d =$ and $T_n = p\alpha^n + q\beta^n + r\gamma^n$, then

- $a T_{n+3} + b T_{n+2} + c T_{n+1} + d T_n = 0$

GRAPH OF QUADRATIC EXPRESSION

- The expression $ax^2 + bx + c$ is said to b real quadratic expression in x where a, b are real and $a \neq 0$.
- Let $f(x) = ax^2 + bx + c$ where a, b, c, $\in R$ (0). Then $y = f(x)$ represents a parabola who axis is parallel to the y-axis, with vertex $A\left(-\frac{b}{2a}, \frac{-D}{4a}\right)$.
- Depending upon the values of 'a' and D th parabola will have following different shapes

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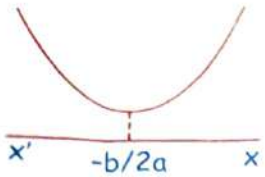
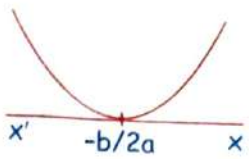


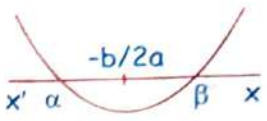
$a > 0, D < 0$

$f(x) \geq -D/4a \quad \forall x \in R$



$a > 0, D = 0$

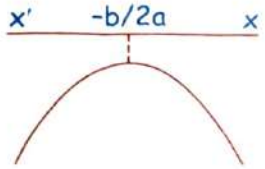
$f(x) \geq 0 \quad \forall x \in R$



$a > 0, D > 0$

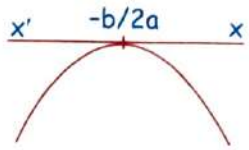
$-\frac{D}{4a} \leq f(x) < 0 \quad \forall x \in (\alpha, \beta)$

$f(x) > 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$



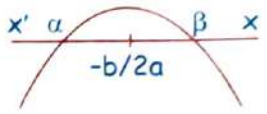
$a < 0, D < 0$

$f(x) \leq -D/4a \quad \forall x \in R$



$a < 0, D = 0$

$f(x) \leq 0 \quad \forall x \in R$



$a < 0, D > 0$

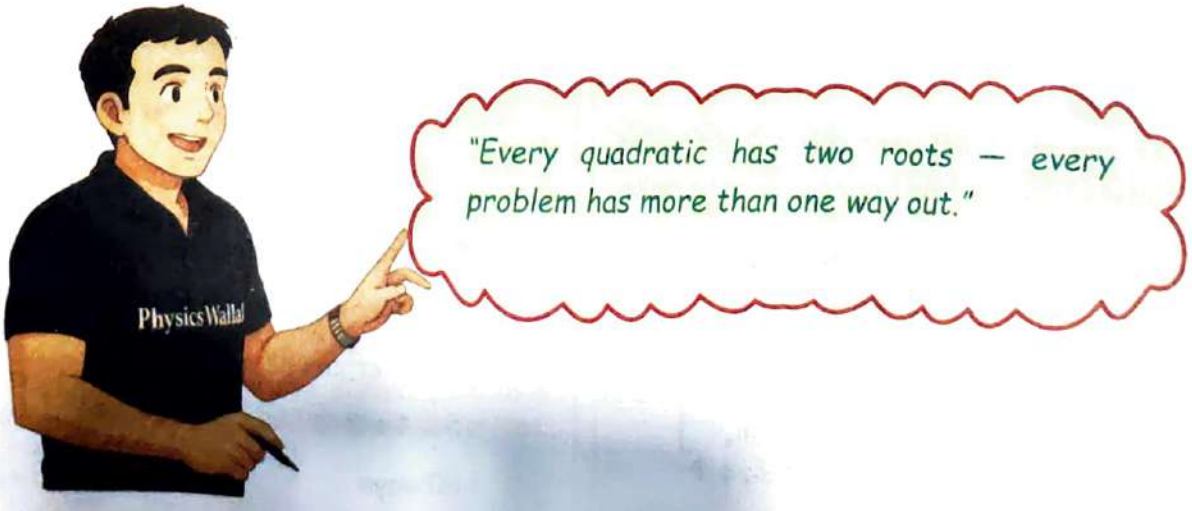
$-\frac{D}{4a} \geq f(x) > 0 \quad \forall x \in (\alpha, \beta)$

$f(x) < 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$

LOCATION OF ROOTS

Let $f(x) = ax^2 + bx + c$, where $a, b, c \in R$ and $a \neq 0$. Suppose $d, e \in R$ and $d < e$. Then the following results hold good;

- Both roots of $f(x) = 0$ will be greater than d, if $D \geq 0$, $af(d) > 0$ and $d < -b/2a$.
- Both roots of $f(x) = 0$ will be less than d, if $D \geq 0$, $af(d) > 0$ and $-b/2a < d$.
- If d lies between the roots of $f(x) = 0$ then $af(d) < 0$.
- Both roots of $f(x) = 0$ will belong to the interval (d,e) if $D \geq 0$, $af(d) > 0$, $af(e) > 0$ and $d < -\frac{b}{2a} < e$.
- Exactly one root of $f(x) = 0$ will lie in (d,e) if $f(d).f(e) < 0$.
- Interval (d, e) will be contained between the roots of $f(x) = 0$ if $af(d) < 0$, $af(e) < 0$.
- 'd' will be the repeated root of $f(x) = 0$ if $f(d) = 0$ and $f'(d) = 0$.



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Chapter 7 Permutation & Combination

INTRODUCTION TO P & C

Permutation & Combination

Arrangement & Selection

Order is important — Order is not Important

{1, 2 3} Combination — 123

{3, 2, 1} — 321

{2, 1, 3} — 132 | Permutation

231

FUNDAMENTAL PRINCIPLE OF COUNTING

Illustration-1

Suppose a person has 3 different shirts and 2 different Jeans. How many pairs can he make.

$S_1\ S_2\ S_3$ → (3)

(X)

$J_1\ J_2$ → (2)

(6)

$S_1\ J_1$
$S_1\ J_2$
$S_2\ J_1$
$S_2\ J_2$
$S_3\ J_1$
$S_3\ J_2$

→ (6) pairs

Illustration-2

Suppose a person has 3 different shirts, 2 different Jeans & 4 different shoes. How many pairs can he make.

$S_1\ S_2\ S_3$ → (3)

$J_1\ J_2$ → (2)

$Sh_1\ Sh_2\ Sh_3\ Sh_4$ → 4

$S_1\ J_1\ Sh_1$
$S_1\ J_1\ Sh_2$
$S_1\ J_1\ Sh_3$
$S_1\ J_1\ Sh_4$

Total ways = 3 x 2 x 4 = 24

Ques. A boy has 6 marbles, find number of ways in which he can put the marbles in 3 pockets.

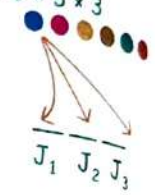
Ans. 1st Marble can be put in 3 ways

Total no .of ways = 3 × 3 × 3 × 3 × 3 × 3

$= 3^6$

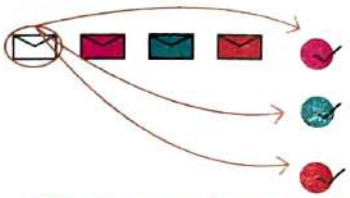
$= (27)^2$

$= 729$



Ques. Number of ways of posting 4 letters in 3 letters boxes?

Ans.



1st Letter can be posted in 3 ways

2nd Letter can be posted in 3 ways

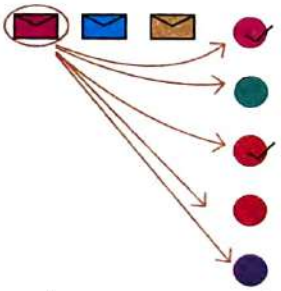
3rd Letter can be posted in 3 ways

4th Letter can be posted in 3 ways

∵ Number of ways = 3 x 3 x 3 x 3 = 81

Ques. Number of ways of posting 3 letters in 5 letters boxes such that no box has more than 1 letter.

Ans.



1st Letter can be posted in 5 ways

2nd Letter can be posted in 4 ways

3rd Letter can be posted in 3 ways

= 5 x 4 x 3

= 60 ways

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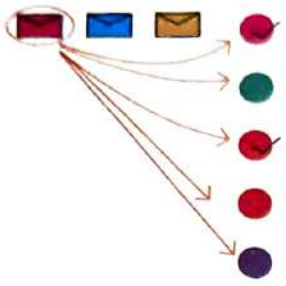
Ques. Number of ways of posting 3 letters in 5 letters boxes Such that not all letters are put in 1 box

Ans



Total ways = 5 x 5 x 5 = 125

= 125 - (when all letters go to 1 box)^{5}

= 125 - 5 = 120

Ques. The number of possible outcomes when a coin is tossed 6 times is

(a) 36 (b) 64

(c) 12 (d) 32

Ans. 1st coin → 2 outcomes

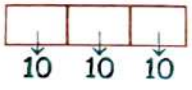
2nd coin → 2 outcomes

= 2 x 2 x 2 x 2 x 2 x 2

= 2^6 = 64

Ques. A suitcase full of money has a 3 digit lock with each digit being from 0 to 9. A thief wants to open it, how many max unsuccessful attempts can be made by the thief.

Ans.



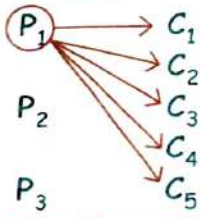
Total no of passwords

= 10 x 10 x 10 = 1000

Unsuccessful = 1000 - 1 = 999

Ques. In a singing competition, in how many ways 3 different prizes can be distributed to 5 children.

Ans.



5 X 5 X 5 = 125

Ques. In a singing competition, in how many ways 3 different prizes can be distributed to 5 children such that no child gets all the prizes.

Ans. 125 - 5 = 120

Principle

If the total number of ways of occurrence of event A is "m" ways and the total number of ways of occurrence of another event B is "n" ways. Then, the total number of ways of simultaneous occurrence of both the events is "m.n"

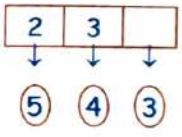
If the total number of ways of occurrence of event A is "m" ways and the total number of ways of occurrence of another event B is "n" ways & the total number of ways of occurrence of another event C is "p" ways then the total number of ways of simultaneous occurrence 3 events is "m.n.p"

Example: Let there are 25 MCQs in JEE Main, Maths Paper. In how many ways can a student answer all of them if each question has exactly one correct option.

All ques can be answered in 4 ways:

∴ Total ways = $(4)^{25}$

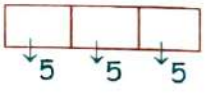
Example: Find number of 3 digit numbers formed using the digits 1, 2, 3, 4, 5 without repetition.



5 x 4 x 3 = 60

Ques. Find number of 3 digit numbers formed using the digits 1,2,3,4,5 with repetition allowed.

Ans.



= 5^3 = 125

Ques. There are 10 students in Arjuna Batch, in how many ways we can send them to 4 countries of Pluto.

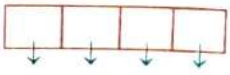
Ans. Number of ways = $(4)^{10}$

Ques. The number of different four digit numbers that can be formed with the digits 2, 3, 4, 7 and using each digit only once is

(a) 120 (b) 96

(c) 24 (d) 100

Ans.



4 x 3 x 2 x 1 = 24

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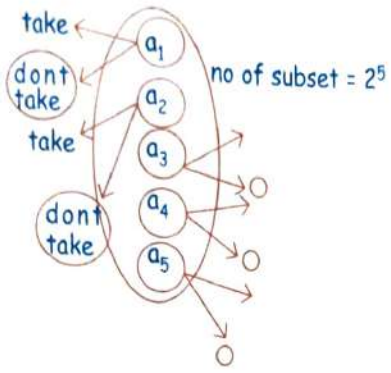
Important Note

Number of subsets of a set containing n elements is 2^n.

Example:



Number of subsets = $2 \times 2 \times 2 \ldots . 2 = 2^n$

Proper subsets = $2^n - 1$

Ques. Find the number of 5 digit integers which can be formed such that

1. Repetition of digits is allowed

Ans.

↓	↓	↓	↓	↓
9	10	10	10	10

$9 \times 10^4 = \boxed{90,000}$

2. Repetition of digits is NOT allowed.

Ans.

2	1			
↓	↓	↓	↓	↓
9	9	8	7	6

$= 9 \times 9 \times 8 \times 7 \times 6 = 27216$

3. At least 2 digit is repeated

Ans. Repetition of digits is allowed - Repetition of digits is NOT allowed = 90,000 - 27,216 = 62,784

Ques. Find the number of 5 digit integers greater than 50,000 which can be formed such that

1. Repetition of digits is allowed

Ans.

5	0	0	0	0
↓	↓	↓	↓	↓
5	10	10	10	10

$5 \times 10 \times 10 \times 10 \times 10$

$= 50,000 = 50,000 - 1 = 49,999$

2. Repetition of digits is NOT allowed.

Ans.

6				
↓	↓	↓	↓	↓

$5 \times 9 \times 8 \times 7 \times 6 = 15,120$

Ques. The number of 5-digit ATM passwords having at least one of their digits repeated is

(a) 90,000 (b) 10,000

(c) 30,240 (d) 69,760

Ans.

↓	↓	↓	↓	↓
10	10	10	10	10

→ Total

↓	↓	↓	↓	↓

$10 \times 9 \times 8 \times 7 \times 6$

→ No digit is repeated

Total - no digit is repeated

$10^5 - 10 \times 9 \times 8 \times 7 \times 6$

$= 100000 - 30240 = 69760$

Ques. Find the number of 5 digit integers divisible by 3 which can be formed using the digits {0, 1, 2, 3, 4, 5} without repetition.

Ans. **Case 1:** Zero is not in our number {1, 2, 3, 4, 5}

↓	↓	↓	↓	↓
5	4	3	2	1

$= 5 \times 4 \times 3 \times 2 \times 1 = 120$

Case 2: Zero in our number

{0, 1, 2, 4, 5}

2				
↓	↓	↓	↓	↓

$4 \times 4 \times 3 \times 2 \times 1$

$16 \times 6 = 96$

Ques. Using the digits 1,2,3,4,5,6,7, a number of 4 different digits is formed. Find

(1) How many numbers are formed? (A) 840

(2) How many numbers are exactly divisible by 2? (B) 200

(3) How many numbers are exactly divisible by 25? (C) 360

(4) How many numbers are exactly divisible by 4? (D) 40

(E) 180

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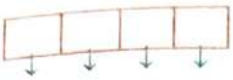
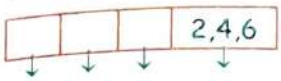
Ans. (1) 

$7 \times 6 \times 5 \times 4$

Total numbers = $7 \times 5 \times 6 \times 4 = 42 \times 20 = 840$

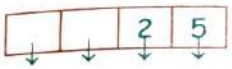
Hence, (1) → (A)

(2) 

$4 \times 5 \times 6 \times 3$

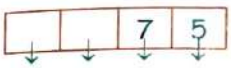
Total numbers = $20 \times 18 = 360$

Hence, (2) → (C)

(3) 

$4 \times 5 = 20$

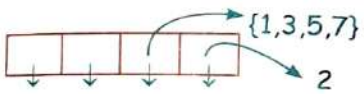
or



$5 \times 4 = 20$

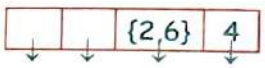
Total numbers = 40

Hence, (3) → (D)

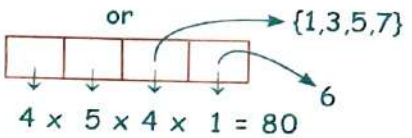
(4) 

$4 \times 5 \times 4 \times 1 = 80$

or



$4 \times 5 \times 2 \times 1 = 40$

or 

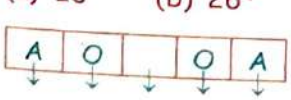
$4 \times 5 \times 4 \times 1 = 80$

∴ Total numbers = 80 + 40 + 80 = 200

Hence, (4) → (B).

Ques. The number of 5 lettered palindromes (with repetition) which can be formed using the letters from the English alphabets, is

(a) 26^5 (b) 26^4 (c) 26^3 (d) 26^2

Ans. 

26 26 26 1 1

Total = $26 \times 26 \times 26 \times 1 \times 1$

$= 26^3$

Ques. Let S be the set of all passwords which are six to eight character long, where each character is either an alphabet from {A, B, C, D, E} or a number from with {1, 2, 3, 4, 5} the repetition of characters allowed, if the number of passwords in S whose at least one character is a number from {1,2,3,4,5} is $\alpha \times 5^6$, then α is equal to _____. **[JEE Main-2022, July-I]**

Ans.

Total =	Six	7	or	8
	10^6	10^7		10^8

Total Passwords

$= (10^6 + 10^7 + 10^8)$

When numbers {1, 2, 3, 4, 5} are absent

$= (5^6 + 5^7 + 5^8)$

When at least one number is present

$= (10^6 + 10^7 + 10^8) - (5^6 + 5^7 + 5^8)$

$= 10^6[1 + 10 + 100] - 5^6[1 + 5 + 5^2]$

$= 10^6 \times 111 - 5^6.31$

$= 5^6 [2^6 \times 111 - 31]$

comparing with $\alpha \times 5^6$

$\alpha = 2^6 \times 111 - 31$

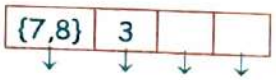
$\alpha = 64 \times 111 - 31$

Ques. The number of integers, greater than 7000 that can be formed, using the digits 3,5,6,7,8 without repetition, is **[JEE Main-2023]**

(a) 120 (b) 168 (c) 220 (d) 48

Ans. Case 1

4 digit no

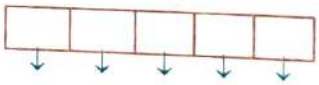


$2 \times 4 \times 3 \times 2$

$= 48$

Case 2

5 digit no



$5 \times 4 \times 3 \times 2 \times 1$

$= 5! = 120$

∴ Total = 48 + 120 = 168

Ques. The number of five-digit number, greater than 40000 and divisible by 5, which can be formed using the digits 0, 1, 3, 5, 7 and 9 without repetition, is equal to

(a) 132 (b) 120 (c) 72 (d) 96

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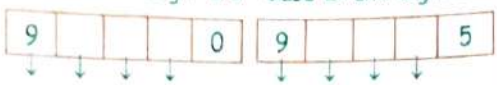
Ans. Case 1: Unit digit is 0 Case 2: unit digit is 5.



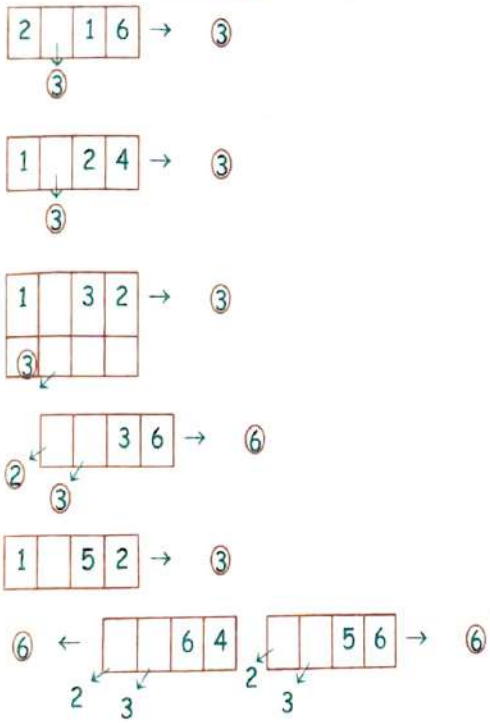
$3 \times 4 \times 3 \times 2 = 72$ $2 \times 4 \times 3 \times 2 = 48$

Total = 72 + 48 = 120

Ques. Numbers are to be formed between 1000 and 3000, which are divisible by 4, using the digits 1, 2, 3, 4, 5 and 6 without repetition of digits. Then the total number of such numbers is ____.

[JEE Main-2022]

Ans. Last 2 digits ? 1 2 → Reject



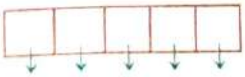
3 + 3 + 3 + 6 + 3 + 6 + 6

= 12 + 18

= 30

Ques. If digits are not allowed to repeat in any number formed by using the digits 0, 2, 4, 6, 8, then the numbers of all numbers greater than 10,000 is equal to. **[JEE Main 22 July-2021]**

Ans.



$4 \times 4 \times 3 \times 2 \times 1$

$= 4 \times 4!$

$= 4 \times 24$

$= 96$

FACTORIAL NOTATION

$4! = 4 \times 3 \times 2 \times 1$ $n! = 1 \times 2 \times 3 \times \ldots n$

$0! = 1$

$n! = n(n-1)!$ = $n(n-1)(n-2)!$

Example:

(i) $10! = 10 \times 9! = 10 \times 9 \times 8!$

(ii) $(4n-1)! = (4n-1)(4n-2)!$

(iii) $2r! = 2r(2r-1)!$

(iv) $(2r-1)! = (2r-1)(2r-2)!$

$0! = 1 = 1!$
$2! = 2$
$3! = 6$
$4! = 24$
$5! = 120$
$6! = 720$

Important Note

Show that $1! + 2! + 3! + 4! + \ldots\ldots\ldots\ldots n!$ Can not be a perfect square for any $n > 3$

s! onwards, last digits is always 0

$n = 1$ $s = 1! = (1) \rightarrow$ Perfect square

$n = 2$ $s = 1! + 2! = 3$

$n = 3$, $s = 1! + 2! + 3! = 1 + 2 + 6 + (9) \rightarrow$ Perfect sq

$n = 4$, $s = 1! + 2! + 3! + 4! = 9 + 24 = 33$

$n = 5$ $s = 1! + 2! + 3! + 4! + 5! \rightarrow$ last digit 3

$n = 6$ 33+ ..... 0 $\rightarrow$ last digit 3

FORMULA FOR ${}^nP_r$

$${}^nP_r = \frac{n!}{(n-r)!} \quad r \le n$$

Ques: Find the value of:

1. ${}^{10}P_3$

Ans. $= \frac{10!}{(10-3)!} = \frac{10!}{7!} = \frac{10 \times 9 \times 8 \times 7!}{7!} = 720$

2. ${}^5P_1$

Ans. $= \frac{5!}{4!} = \frac{5 \times 4!}{4!} = 5$

3. ${}^{10}P_0$

Ans. $= \frac{10!}{10!} = 1$

4. ${}^5P_7$

Ans. $= \frac{5!}{-2!} \rightarrow$ Not defined

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5. ${}^{10}P_{10}$

Ans. $\frac{10!}{0!} = 10!$

6. ${}^{n}P_{2}$

Ans. $\frac{n!}{(n-2)!} = n(n-1)$

7. ${}^{o}P_{o}$

Ans. $\frac{0!}{0!} = 1$

Ques. If ${}^{2n}P_4 = 396 \times {}^{n}P_2$, then the value of n, is

(a) 4 (b) 5 (c) 6 (d) 7

Ans. ${}^{2n}P_4 = 396.\ {}^{n}P_2 \Rightarrow \frac{(2n)!}{(2n-4)!} = 396.\ \frac{n!}{(n-2)!}$

$$\frac{2n\cdot(2n-1)(2n-2)(2n-3)\cdot(2n-4)!}{(2n-4)!}$$

$= 396\ n(n-1)$

$\Rightarrow 2n(2n-1)(2n-2)(2n-3) = 396\ n(n-1)$

$\Rightarrow 4(2n-1)(n-1)(2n-3) = 396(n-1)$

$\Rightarrow 4(2n-1)(2n-3) = 396$

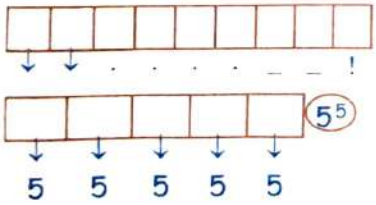
$\Rightarrow (2n-1)(2n-3) = 9 \times 11$

$2n-1 = 11$	$2n-3 = 9$
$2n = 12$	$2n = 12$
$n = 6$	$n = 6$

THEOREMS OF PERMUTATION OF DISTINCT OBJECTS

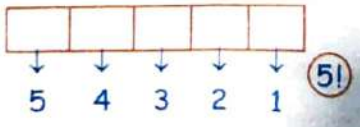
Theorem 1

The total number of permutations of n distinct objects taken all at a time with repetition is n^n



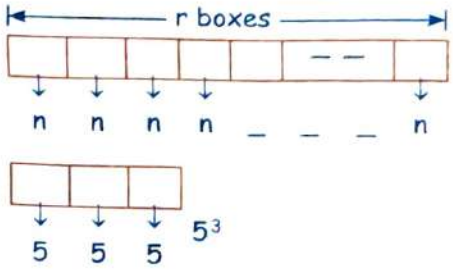
Theorem 2

The total number of permutations of n distinct objects taken all at atime without repetition is = n!



Theorem 3

The total numbers of permutations of n distinct objects taken "r" at a time with repetition is n^r

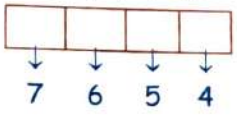


Theorem 4

The total numbers of permutations of r objects,out of n distinct objects without repetition is $\frac{n!}{(n-r)!}$, $1 \le r \le n$. This number is denoted as ${}^{n}P_{r}$.

Example: n = 7, r = 4

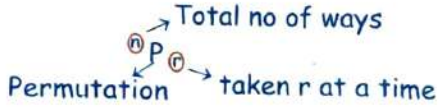
1, 2, 3, 4, 5, 6, 7



$\frac{7 \times 6 \times 5 \times 4 \times 3!}{3!}$

$= \frac{7!}{3!} = {}^{7}P_4$

$n = n, \quad r \to r$



Ques. A class has 12 student. In how many ways we can arrange six of them in a line

Ans. ${}^{12}P_6 = \frac{12!}{6!}$

Ques. How many 5 lettered words can be formed from DAUGHTER?

Ans. ${}^{8}P_5 = \frac{8!}{3!}$

Ques. How many 8 lettered words can be formed from DAUGHTER?

Ans. ${}^{8}P_8 = \frac{8!}{0!} = 8!$

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Ques. Total number of words that can be formed using "all letters" of the word MACHINE in which repetition is not allowed such that:

(i) No restriction → 7!

(ii) "Relative order" of vowels & consonants remains same

(iii) Word starts with "A"

(iv) Word ends with a vowel

(v) Vowel occupies even places

Ans. & → (x) Multiply

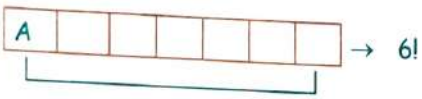
OR → (+) Add

(i) 7!

(ii) (4! x 3!)

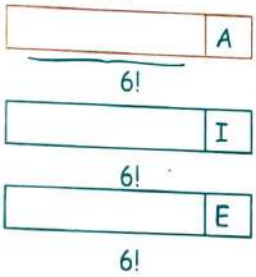
3! x 4! = 6 x 24

(iii) 6!



→ 6!

(iv) 3 x (6!)



6! + 6! + 6!

= 720 x 3

(v) (4! x 3!)]

Ques. Find the number of ways in which 5 boys and 5 girls can be arranged in a line such that

(1) All boys are together

(2) All boys are together & all girls are together

(3) All boys are not together

(4) No 2 boys are together

Ans. (1) G_1, G_2, G_3, G_4, G_5 B_1, B_2, B_3, B_4, B_5

$G_1, G_2, G_3, G_4, G_5, \alpha$

$6! \times 5!$

5 girls & α arrangement — Internal arrangement of α

(2) G_1, G_2, G_3, G_4, G_5

β, α

$2! \times 5! \times 5!$

arrangement of α & β — internal boys arrangement of β

(3) Total - 6! x 5! = 10! - 6! x 5!

(4) Method of Gaps

$\downarrow G_1 \downarrow G_2 \downarrow G_3 \downarrow G_4 \downarrow G_5 \downarrow$

$B_1 \quad B_2 \quad B_3 \quad B_4 \quad B_5$

$5! \times {}^6P_5$

THEOREMS OF PERMUTATION

Theorem 1

The total number of permutations of n distinct objects taken all at a time with repetition is n^n

Theorem 2

The total number of permutations of n distinct objects taken all at a time without repetition is n!

Theorem 3

The total numbers of permutations of n distinct objects taken "r" at a time with repetition is n^r

Theorem 4

The total numbers of permutations of r objects out of n distinct objects without repetition is $\frac{n!}{(n-r)!}$, $r \le n$. This number is denoted as ${}^nP_r$.

Ques. Find number of words that can be formed using all the letters of the word EQUATION in which

(i) Vowels are always together

(ii) Consonants are always together

(iii) Vowels consonants are together

(iv) No 2 consonants are together

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Ans. (i) A E I O U Q, T, N

α, Q, T, N

$= 4! \times 5!$

(ii) A, E, I, O, U Q T N $\rightarrow \beta$

$= 6! \times 3!$

(iii) A E I O U

$\alpha, \beta,$

$= 2! \times 5! \times 3!$

(iv) | A | E | I | O | U |

$^6P_3 \times 5!$

Ques. Eleven books consisting of 5 Mathematics, 4 Physics and 2 Chemistry are placed on a shelf. The number of possible ways of arranging them on the assumption that the books of the same subjects are all together is

(a) 4! 2! (b) 11!

(c) 5! 4! 3! 2! (d) None of these

Ans. $M_1\ M_2\ M_3\ M_4\ M_5$

α, β, γ

$3! \times 5! \times 4! \times 2!$

Ques. The number of six letter words (with or without meaning), formed using all the letters of the words 'Vowels', so that all the consonants never come together, is

(a) 576 (b) 140

(c) 356 (d) 209

Ans. Total - (when all consonants are together)

$6! - 3! \times 4!$

V, W, L, S $\quad 4! [6 \times 5] - 6$

α, O, E $\quad 6 \times 4! [5 - 1]$

$= 6 \times 4! \times 4$

$= 24 \times 24 = (24)^2 = 576.$

Ques. The number of words that can be formed out of the letters of the word ARTICLE so that the vowels occupy even places is

(a) 36 (b) 574

(c) 144 (d) 754

Ans. R A I I C E L

$3! \times 4! = 6 \times 24 = 144$

PERMUTATION OF ALIKE OBJECTS

Theorem 5

(Permutations of objects not all distinct)

Ques. Find the number of arrangement:

1. Let we have 3 identical A's and 4 identical B's and 6 identical C's then find total number of arrangements of these 13 alphabets.

Ans. $\underbrace{A\,A\,A}_{3}\ \underbrace{B\,B\,B\,B}_{4}\ \underbrace{C\,C\,C\,C\,C\,C}_{6}$

$= \dfrac{13!}{3! \cdot 4! \cdot 6!}$

2. INDEPENDENCE

Ans. Number of arrangement $= \dfrac{12!}{3!2!4!}$

3. TEACHER

Ans. Number of arrangement $= \dfrac{7!}{2!}$

Ques. Number of arrangements of all the letters of the word MISSISSIPPI

Ans. S = 4

I = 4

P = 2

M = 1

(11)

Total letters = 11

$\therefore$ Total arrangements $= \dfrac{11!}{4!\,4!\,2!}$

Ques. Number of words with or without meaning that can be formed using all the letters of the words ASSASSINATION such that:

(i) starts & ends with 'S'.

Ans. $\dfrac{11!}{2!3!2!2!} = \dfrac{11!}{2!4!}$

(After fixing s in starts & ends.)

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(ii) ends with vowel.

Ans. Case → 1 Ends with O = $\frac{12!}{4!3!2!2!}$

_ _ _ _ _ _ _ _ _ _ _ O

Case → 2 Ends with I = $\frac{12!}{4!3!2!}$

_ _ _ _ _ _ _ _ _ _ _ I

Case → 3 Ends with A = $\frac{12!}{4!2!2!2!}$

_ _ _ _ _ _ _ _ _ _ _ A

Ques. The number of words ,with or without meaning,that can be formed using all the letters of the word ASSASSINATION so that the vowels occur together, is _____.

[JEE Main 2023 (Feb.,Shift - 1)]

Ans. S = 4, A = 3, I = 2, N = 2, O = 1, T = 1

(A A A I I O) → α

α, S, S, S, S, N, N, T.

α with all other consonants: $\frac{8!}{4!\,2!} \times \frac{6!}{3!\,2!}$ ← arrangement inside 'α'

$= \frac{8 \times 7 \times 6 \times 5 \times 4!}{4! \times 2} \times \frac{6 \times 5 \times 4}{2}$

$= 56 \times (30)^2 = 56 \times 900 = 50400$

Ques. Find all possible arrangements of the letters of the word INTEGRATION such that

(a) All the vowels are together

Ans. I = 2, N = 2, T = 2, E = 1, G = 1, R = 1, A = 1, D = 1

(I I E A O)

α, N, N, T, T, G, R, → $\frac{7!}{2!\,2!} \times \frac{5!}{2!}$

(b) All Consonants are together.

Ans. (N N T T G R)

β, I, I, R, A, O

$= \frac{6!}{2!} \times \frac{6!}{2!\,2!}$

(c) Relative position of vowels and consonants do not change.

Ans. I _ _ E _ _ A _ I O _

$= \frac{5!}{2!} \times \frac{6!}{2!\,2!}$

(d) There are exactly 6 alphabets between G & R

Ans. G _ _ _ _ _ _ R _ _ _

$= 4 \times \frac{9!}{2!\,2!\,2!} \times (2)$ (GR & RG)

Ques. The number of seven digits odd numbers, that can be formed using all the seven digits 1, 2, 2, 2, 3, 3, 5 is _____.

[JEE Main 2023 (Jan., Shift - II)]

Ans. _ _ _ _ _ _ 5 = $\frac{6!}{3!\,2!}$

_ _ _ _ _ _ 3 = $\frac{6!}{3!}$

_ _ _ _ _ _ 1 = $\frac{6!}{3!2!}$

$= \frac{6!}{3!2!} + \frac{6!}{3!} + \frac{6!}{3!\,2!} = \frac{5!}{2} + (5!) + \frac{5!}{2}$

$= (60) + 120 + 60 = 240$

Ques. In how many ways 6 Boys & 4 Girls can be arranged in a row such that

1. Girls occur in increasing order of Height.

Ans. Let $G_1 = G_2 = G_3 = G_4 = G$

$B_1, B_2, B_3, B_4, B_5, B_6, G\,G\,G\,G$

$= 10!/4!$

2. Boys occur in increasing order of Height.

Ans. $\frac{10!}{6!}$

3. Girls occur in increasing order of Height as well as boys also.

Ans. B B B B B B G G G G

$\frac{10!}{6!\,4!}$

Ques. Find all possible arrangements of the letters of the word ABCDEF such that

1. A comes before B.

Ans. $\frac{6!}{2!}$

2. A comes just before B.

Ans. (AB) → α, C, D, E, F = 5!

3. A comes before B & B comes before C.

Ans. A > B > C = $\frac{6!}{3!}$

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4. A comes before B, C comes before D & E comes before F.

Ans. $A > B,\ C > D,\ E > F = \dfrac{6!}{2!\,2!\,2!}$

5. A comes before B, B comes before C & D comes before E, E comes before F.

Ans. $A > B > C \qquad D > E > F$

$$= \frac{6!}{3!\,3!}$$

Ques. Find the number of 7 digit integers whose sum of digits is 59.

Ans. **Case 1:**

$$\underline{9}\,\underline{9}\,\underline{9}\,\underline{9}\,\underline{9}\,\underline{9}\,\underline{5} = \frac{7!}{6!} = \frac{7 \times 6!}{6!} = 7$$

Case 2:

$$\underline{9}\,\underline{9}\,\underline{9}\,\underline{9}\,\underline{9}\,\underline{8}\,\underline{6} \rightarrow = \frac{7!}{5!} = 7 \times 6 = 42$$

Case 3:

$$\underline{9}\,\underline{9}\,\underline{9}\,\underline{9}\,\underline{9}\,\underline{7}\,\underline{7} \rightarrow = \frac{7!}{5!\,2!} = \frac{7 \times 6}{2!} = 21$$

Case 4: $\underline{9}\,\underline{9}\,\underline{9}\,\underline{9}\,\underline{8}\,\underline{8}\,\underline{7} \rightarrow = \dfrac{7!}{4!\,2!} = \dfrac{7 \times \overset{3}{\cancel{6}} \times 5}{\cancel{2}} = 105$

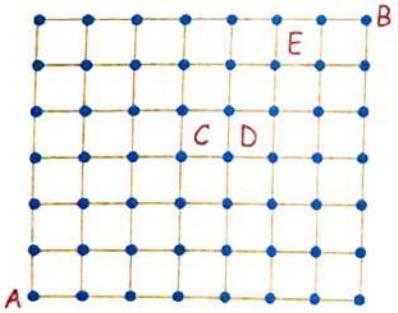
Case 5:

$$\underline{9}\,\underline{9}\,\underline{9}\,\underline{8}\,\underline{8}\,\underline{8}\,\underline{8} \rightarrow = \frac{7!}{4!\,3!} = \frac{7 \times 6 \times 5}{6} = 35$$

Total $= 7 + 40 + 21 + 105 + 35 = \boxed{210}$

Brain Teaser

Consider a network as shown in the figure. Paths from A to B consists of the horizontal or vertical line segments. No diagonal movements is allowed.



(i) How may paths are there from A to B?

Ans. $\underbrace{x\,x\,x\,x\,x\,x\,x}_{\text{7 steps in x-direction.}}\ \underbrace{y\,y\,y\,y\,y\,y}_{\text{6 steps in y-direction.}} \rightarrow \dfrac{13!}{6!\,7!}$

(ii) How many paths go via C?

Ans. $(A \rightarrow C)\ \&\ (C \rightarrow B)$

$$\frac{8!}{4!\,4!} \times \left(\frac{5!}{3!\,2!}\right)$$

(iii) How many paths go via CD?

Ans. $(A \rightarrow C)\ \&\ (C \rightarrow D)\ \&\ (D \rightarrow B)$

$$= \frac{8!}{4!\,4!} \times 1 \times \frac{4!}{2!\,2!}$$

(iv) Paths via C.E.

Ans. $(A \rightarrow C)\ \&\ (C \rightarrow E)\ \&\ (E \rightarrow B)$

$$= \frac{8!}{4!\,4!} \times 2 \times \frac{3!}{2!\,1!} = \frac{8 \times 7 \times 6 \times 5}{4!} \times 31$$

$= 2 \times 7 \times 6 \times 5 = 14 \times 30 = 420$

FINDING RANK ACCORDING TO DICTIONARY

Ques. If all the letters of the word "VIJAY" are arranged in all possible manner as they are in a dictionary, then find the rank of the word "VIJAY"

Ans. Arrange in order = A, I, J, V, Y

A					$\rightarrow 4! = 24$
I					$\rightarrow 4! = 24$
J					$\rightarrow 24$
V	A				$\rightarrow 3! = 6$
V	I	A			$\rightarrow 2! = 2$
V	I	J	A	Y	$\rightarrow 1$

$= 24 + 24 + 24 + 6 + 2 + 1 = 81$st word is VIJAY

Ques. All words, with or without meaning, are made using all the letters of the word MONDAY. These words are written as in a dictionary with serial numbers. The serial number of the word MONDAY is

[JEE Main 2023 (13 April Shift II)]

(a) 327 ✓ (b) 328

(c) 324 (d) 326

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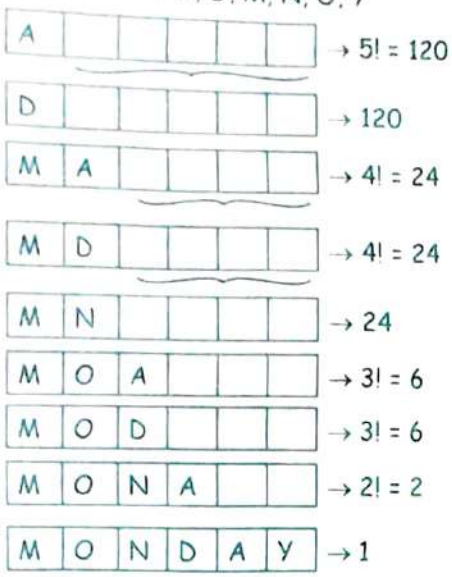
Ans. MONDAY → A, D, M, N, O, Y

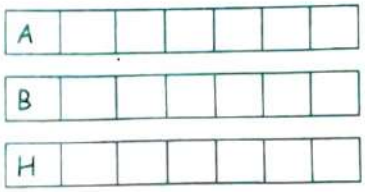


Rank = 240 + 72 + 12 + 3

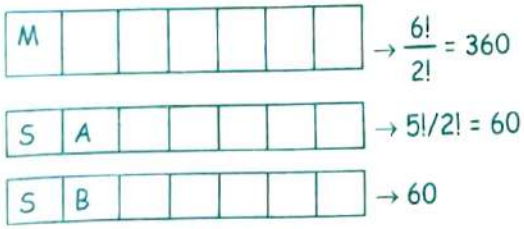
= 240 + 72 + 15 = 240 + 87 = 327

Ques. If the letters of the word 'SHUBHAM' are arranged in all possible manner as they are in a dictionary, then find the rank of the word 'SHUBHAM'.

Ans. A, B, H, H, M, S, U → Two same letters



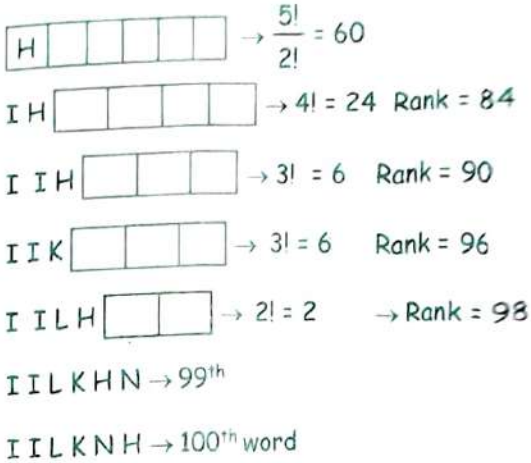
After putting H in first place other all letters are different. So, we don't divide by 2!



Hence, rank of SHUBHAM = 2025th.

Ques. Find 100th word in N I K H I L

Ans. H, I, I, K, L, N

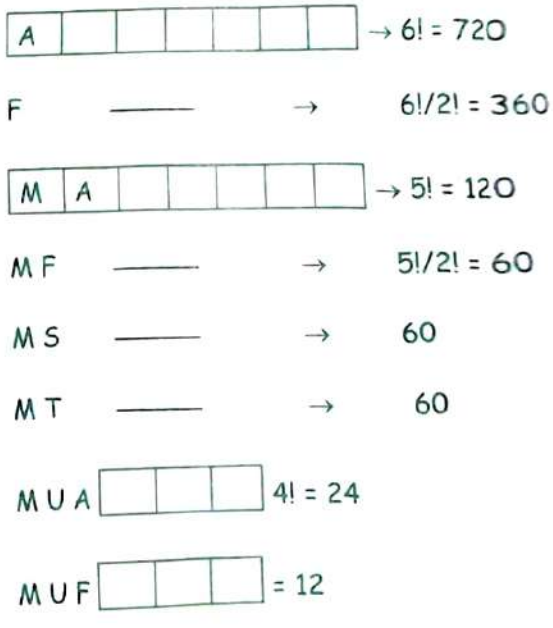


I I L K H N → 99th

I I L K N H → 100th word

Ques. If all the letters of the word 'MUSTAFA' are arranged in all possible manner as they are in a dictionary, then find the rank of the word 'MUSTAFA'.

Ans. A, A, F, M, S, T, U



1427th ward

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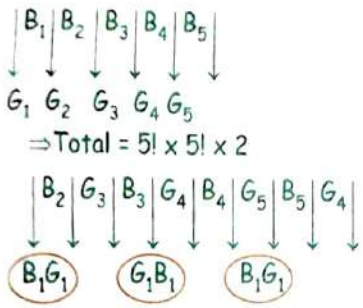
Ques. In how many ways 5 boys and 5 girls can be arranged in a line such that boys and girls are alternate & B_1 G_1 are never together.

Ans.



$\Rightarrow$ Total = 5! x 5! x 2

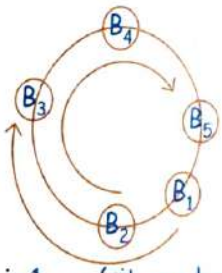
B_1G_1 together = 4! x 4! x 2 x 9

Required ways = 5! x 5! x 2 - when B_1G_1 are always together

CIRCULAR PERMUTATION OF DISTINCT OBJECTS

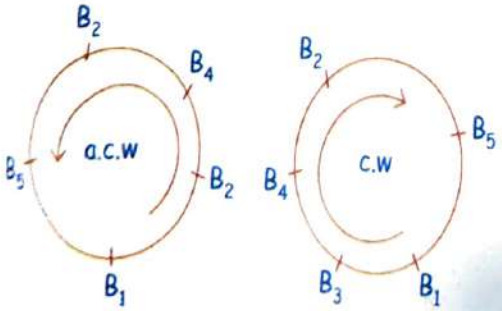
Example: B_1 B_2 B_3 B_4 B_5

5!



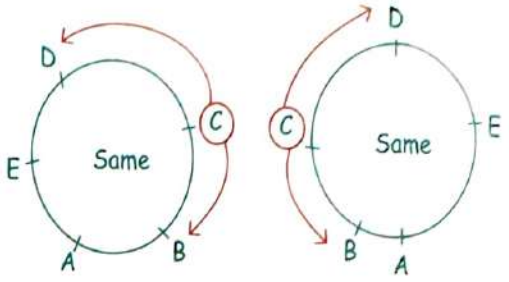
B_1 Can sit in 1 way (sit anywhere)
B_2 Can sit in 4 ways
B_3 Can sit in 3 ways
B_4 can sit in 2 ways
B_5 can sit in 1 way
Total no. of arrangements = 4!

- The total number of circular arrangement of n distinct objects is (n - 1)!
- If clockwise and anti-clockwise arrangements are considered to be same then the number of arrangements is $\frac{(n-1)!}{2}$



Ques. In how many ways 5 boys can be seated around a circular table such that they don't have same neighbours?

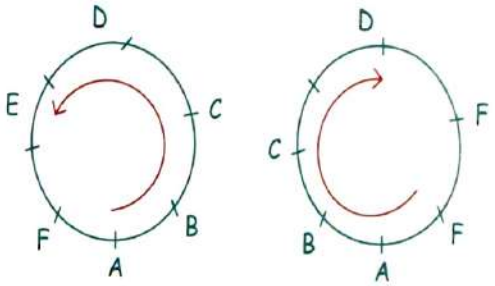
Ans.



No. of ways = $\frac{(n-1)!}{2}$ = 4!/2 = 24/2 = 12

Ques. How many garlands can be made using all of 6 different flowers?

Ans.

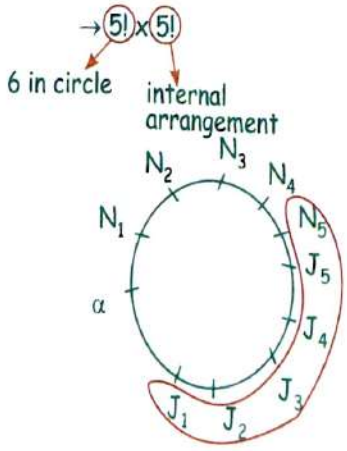


5!/2 = 60

Ques. In how many ways 5 NEET & 5 JEE Aspirants can be seated round a table such that.

(a) All JEE aspirants are together?

Ans. Total no. of ways =



(b) All NEET aspirants are together & all JEE Aspirants are together?

Ans. Total no. of ways =

$(2 - 1)! \times 5! \times 5! = (5!)^2$

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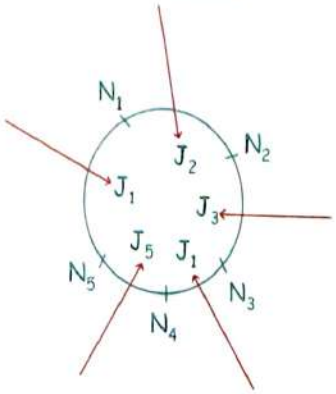
(c) Not all JEE Aspirants are together

Ans. Total no. of ways = 9! - (5!)2

(d) No 2 JEE aspirants are together

Ans. Total no. of ways = $\boxed{4! \times 5!}$



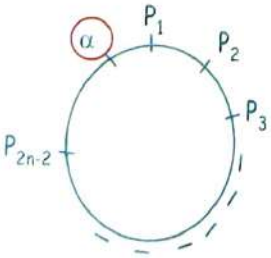
Ques. 2n chairs are arranged symmetrically around a table. There are 2n people, including A and B, whose wish to occupy the chairs. Find the number of seating arrangements, if:

(i) A and B are next to each other

Ans. P_1 can sit in 1 way and P_2 can sit in 2 ways.

Remaining = (2n - 2)!

Total number of ways = 2 x (2n - 2)!



(ii) A and B are diametrically opposite.

Ans. A can sit in 1 way

B can sit in 1 way (2n - 2)!

Remaining →

No. of ways = 1 x 1 x (2n - 2)! = (2n - 1)!

COMBINATIONS (SELECTIONS)

$$^nC_r = \frac{^nP_r}{r!} = \frac{n!}{(n-r)!r!}$$

Example: (i) $^{10}C_3 = \frac{10!}{7!\,3!} = \frac{10 \times 9 \times 8 \times 7!}{7! \times 3 \times 2}$

$= 10 \times 3 \times 4 = 120$

(ii) $^5C_2 = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times 2} = 10$

Ques. Number of ways in which four seats can be occupied by 7 persons.

Ans. $^7C_4 \rightarrow$ If 4 seats are alike.

$^7P_4 \rightarrow$ If 4 seats are different.

Ques. 10 students have gathered for JEE Advance Result celebrations, find total number to Hand-shakes assuming that each child shakes hand with every other.

Ans. no. of handshakes = no. of ways of selecting 2 students out of 10

$= {}^{10}C_2 = \frac{10!}{8!\,3!} = \frac{10 \times 9 \times 8!}{8! \times 2!}$

$= \frac{10 \times 9}{2} = 45$

Ques. Assuming IPL is played Between 10 teams. how many total matched are played in in league stage if each team plays with every other team twice.

Ans. Total Matches = $^{10}C_2 \times 2 = 45 \times 2 = 90$

Ques. If $^{2n}C_3 : {}^nC_3 = 10 : 1$, then the ratio $(n^2 + 3n) : (n^2 - 3n + 4)$ is **(JEE Main-2023)**

(a) 65 : 37 (b) 2 : 1

(c) 35 : 16 (d) 27 : 11

Ans. $\frac{^{2n}C_3}{^nC_3} = \frac{10}{1} \Rightarrow \frac{(2n)!}{(2n-3)!\,3!} \times \frac{(n-3)!3!}{n!} = \frac{10}{1}$

$\Rightarrow \frac{2n(2n-1)(2n-2)}{n(n-1)(n-2)} = \frac{10}{1}$

$\Rightarrow \frac{4(2n-1)(n-1)}{(n-1)(n-2)} = \frac{10}{1} \Rightarrow \frac{4(2n-1)}{n-2} = \frac{10}{1}$

$\Rightarrow 2(2n-1) = 5(n-2)$

$\Rightarrow 4n - 2 = 5n - 10 \Rightarrow n = 8$

$\therefore \frac{n^2 + 3n}{n^2 - 3n + 4} = \frac{8^2 + 24}{8^2 - 24 + 4} = \frac{64 + 24}{64 - 24 + 4} = \frac{88}{44} = 2$

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Ques. In how many ways can a committee of 6 be selected from a class of 5 Boys & 6 Girls such that it has

(a) Exactly 3 boys

Ans. (Select 3B & 3G)

No. of Committee = ${}^5C_3 \times {}^6C_3$

$= \frac{5!}{3!\,2!} \times \frac{6!}{3!\,3!} = \frac{5 \times 4}{2} \times \frac{6 \times 5 \times 4}{3!}$

$= 10 \times 20 = 200$

(b) At least 3 boys

Ans. (3B & 3G) OR (4B & 2G) or (5B & 1G)

No. of Committee

${}^5C_3 \times {}^6C_3 + {}^5C_4 \cdot {}^6C_2 + {}^5C_5 \; {}^6C_1$

$= 200 + 5 \times 15 + 1 \times 6 = 200 + 75 + 6 = \boxed{281}$

(b) At most 3 boys

Ans. (3B & 3G) OR (2B & 4G) OR (1B & 5G) OR (0B & 6G)

No. of Committee

$= 200 + {}^5C_2 \times {}^6C_4 + {}^5C_1 \times {}^6C_5 + {}^5C_0 \times {}^6C_6$

Master Question

A delegation of four students is to be selected from a total of 12 students.

In how many ways can be delegation be selected.

(i) if all the students are equally willing?

Ans. No. of Teams = ${}^{12}C_4 = \frac{12!}{8!\,4!} = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2}$

$= 11 \times 5 \times 9$

$= 99 \times 5 = 495$

(ii) if two particular students have to be included in the delegation?

Ans. No. of teams = ${}^{10}C_2 = 45$

(iii) if two particular students do not wish to be together in the delegation?

Ans. No. of Teams = Total — when those 2 go together

$= 495 - {}^{10}C_2 = 495 - 45 = 450$

(iv) if two particular students wish to be included together only in the delegation?

Ans. No. of Teams (A & B giving with 2 others) + (out of 10 any 4 are going)

$= {}^{10}C_2 + {}^{10}C_4$

(v) if two particular students refuse to be together and two other particular students wish to be together only in the delegation?

Ans. No. of Team = B_3 & $\bar{B}_4$ & 2 other and ($\bar{B}_3$ & $\bar{B}_4$)

$= ({}^{10}C_2 - 1) + {}^{10}C_4$ - (when B_1 & B_2 all selected together)

$= ({}^{10}C_2 - 1) + ({}^{10}C_4 - {}^8C_2) = 70$

Ques. There are ten boys $B_1, B_2 \ldots B_{10}$ and five girls $G_1, G_2, \ldots G_5$ in a class. Then the number of ways of forming a group consisting of three boys and three girls, if both B_1 and B_2 together should not be the numbers of a group is _____

[JEE Main June 2022]

Ans. Total ways - when B_1 & B_2 are going together

$= {}^{10}C_3 \; {}^5C_3 - {}^8C_1 \; {}^5C_3$

$= {}^5C_3 [{}^{10}C_3 - {}^8C_1]$

$= 10\left[\frac{10 \times 9 \times 8}{3 \times 2} - 8\right]$

$= 10\,[3 \times 5 \times 8 - 8] = 80\,[14] = 1120$

Ques. A scientific committee is to formed from 6 Indians and 8 foreigners, which includes at least 2 Indians and double the number of foreigners as Indians. Then the number of ways, the committee can be formed is:

[JEE Main, 24 Feburary 2021 (I)]

(a) 560 (b) 1050

(c) 1625 (d) 575

Ans. The no. of committee = (2I & 4F) or (3I & 6F) or (4I & 8F)

$= {}^6C_2 \cdot {}^8C_4 + {}^6C_3 \cdot {}^8C_6 + {}^6C_4 \cdot {}^8C_8$

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PROPERTIES OF ${}^nC_r$

1. ${}^nC_0 = {}^nC_n = 1$

Proof: ${}^nC_0 = \frac{n!}{0!\,(n-0)!} = \frac{n!}{n!} = 1$

${}^nC_0 = \frac{n!}{n!\,(n-n)!} = \frac{n!}{n!} = 1$

2. ${}^nC_1 = n$

Proof: ${}^nC_1 = \frac{n!}{1!\,(n-1)!} = \frac{n(n-1)!}{(n-1)!} = n$

3. ${}^nC_r = {}^nC_{n-r}$

4. ${}^nC_2 = \frac{n(n-1)}{2}$

Proof: ${}^nC_2 = \frac{n!}{(n-1)!\,2!} = \frac{n(n-1)\,(n-2)!}{(n-2)\,2!}$

$= \frac{n(n-1)}{2}$

5. ${}^nP_r = {}^nC_r \cdot r!$

6. ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

Proof → LHS:

$$\frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!\,(n-r+1)!}$$

$$= \frac{n!}{(r-1)!\,(n-r)!}\left[\frac{1}{r} + \frac{1}{n-r+1}\right]$$

$$= \frac{n!}{(r-1)!\,(n-r)!}\left[\frac{n-r+1+r}{r\,(n-r+1)}\right]$$

$$= \frac{n!\,(n+1)}{(r-1)!\,(n-r)\,r\,(n-r+1)}$$

$$= \frac{(n+1)!}{r!\,(n-r+1)!} = {}^{n+1}C_r = \text{RHS}$$

Ques. ${}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3$

(a) ${}^{51}C_4$ (b) ${}^{52}C_4$ ✓

(c) ${}^{53}C_4$ (d) None of these

Ans. Here, ${}^{47}C_4 + \sum_{r=1}^{5} {}^{52-r}C_3$

$= {}^{47}C_4 + {}^{51}C_3 + {}^{50}C_3 + {}^{49}C_3 + {}^{48}C_3 + {}^{47}C_3$

$= ({}^{47}C_4 + {}^{47}C_3) + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3$ (using ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$)

$= ({}^{48}C_4 + {}^{48}C_3) + {}^{49}C_3 + {}^{50}C_3 + {}^{51}C_3$

$= ({}^{49}C_4 + {}^{49}C_3) + {}^{50}C_3 + {}^{51}C_3$

$= ({}^{50}C_4 + {}^{50}C_3) + {}^{51}C_3 = ({}^{51}C_4 + {}^{51}C_3) = {}^{52}C_4$

Ques. If ${}^nC_8 = {}^nC_6$, then find ${}^nC_2$.

Ans. If ${}^nC_p = {}^nC_q \Rightarrow p = q$ or $p + q = n$

$\because {}^nC_8 = {}^nC_6$

$\therefore 8 + 6 = n \Rightarrow n = 14$

Now, ${}^nC_2 = {}^{14}C_2 = \frac{14 \times 13}{2} = 91$

If ${}^nC_p = {}^nC_q \Rightarrow p = q$ or $p + q = n$

Ques. A boy needs to select five courses from 12 available courses, out of which 5 courses are language courses. If he can choose at most two language courses, then the number of ways he can choose five courses is

[JEE Main Jan. 2023 (I)]

Ans.

5 language 7 non lang (core)

2L & 3C OR (1L & 4C) OR (0L & 5C).

the number of ways he can choose five courses is

${}^5C_2\ {}^7C_3 + {}^5C_1\ {}^7C_4 + {}^5C_0\ {}^7C_5$

$= 10.\ {}^7C_3 + 5.\ {}^7C_3 + 1 \times 21$

$= 10 \times 35 + 5 \times 35 + 21 = 546$

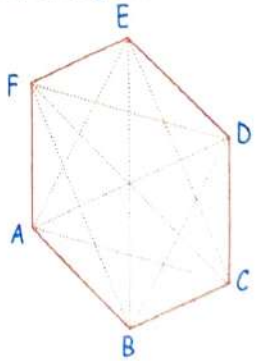
Important Note

The total number of diagonals in a "n" sided Polygon is $\boxed{\frac{n(n-3)}{2} = {}^nC_2 - n}$

Joining 2 ventix will give side or diagonal

$\Rightarrow {}^6C_2$ = no of side + no. of diagonal

$\Rightarrow {}^6C_2 = 6$ + non of diagonal



$\Rightarrow$ no of diagonal $= {}^6C_2 - 6$

generalize : no of diagonal $= {}^nC_2 - n = \frac{n(n-1)}{2} - n$

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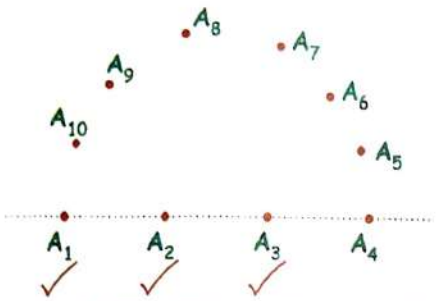
Ques. Given 10 points in a plane out of which exactly 4 are collinear. Find the maximum number of
(1) Straight Line (2) Triangles
that can be drawn passing through these points.

Ans.



(1) no of Straight Line = $^{10}C_2$ — when 2 collinear points are selected

$= {}^{10}C_2 - {}^4C_2 + 1$

$= 45 - 6 + 1 = 40$

(2) no of triangles = $^{10}C_3 - {}^4C_3$

$= \frac{10 \times 9 \times 8}{3 \times 2} - 4$

$= 120 - 4 = 116$

Ques. Find the maximum number of points of Intersection that can be obtained using 5 distinct Circles & 6 different straight lines drawn in a plane.

Ans.

Line Vs Line	Circle Vs Circle	Circle Vs Line
No. of points $= {}^6C_2 \times 1 +$	$^5C_2 \times 2$ +	$^5C_1 \cdot {}^6C_1 \times 2$

$= 15 + 10 \times 2 + 5 \times 6 \times 2$

$= 15 + 20 + 60 = 95$

Ques. Let the number of elements in sets A and B be five and two respectively. Then the number of subsets of A x B each having at least 3 and at most 6 elements is **(JEE Main 2023)**

(a) 752 (b) 772

(c) 782 (d) 792

Ans. no of subsets with 3 elements = $^{10}C_3$

Similar No. of subsets with 4 elements = $^{10}C_4$

No. of subsets with 5 elements = $^{10}C_5$

No. of subsets with 6 elements = $^{10}C_6$

the number of subsets of A x B each having at least 3 and at most 6 elements is

$= {}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 + {}^{10}C_6$

$= 120 + 210 + 252 + 210 = 792$

Ques. If the sides AB, BC and CA of a triangle ABC have 3, 5 and 6 interior points respectively, then the total number of triangles that can be constructed using these points as vertices, is equal to **[JEE Main March 2021 (II)]**

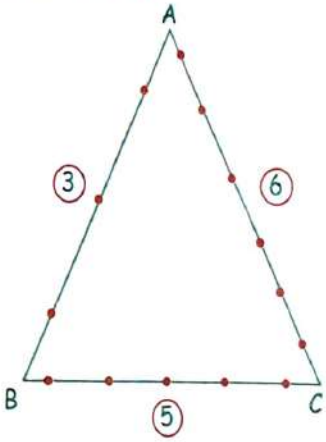
(a) 364 (b) 240

(c) 333 (d) 360

Ans. Total Triangles $^{14}C_3 - [{}^3C_3 + {}^5C_3 + {}^6C_3]$

$= {}^{14}C_3 - [1 + 10 + 20] = \frac{14 \times 13 \times 12}{3 \times 2} - 31$

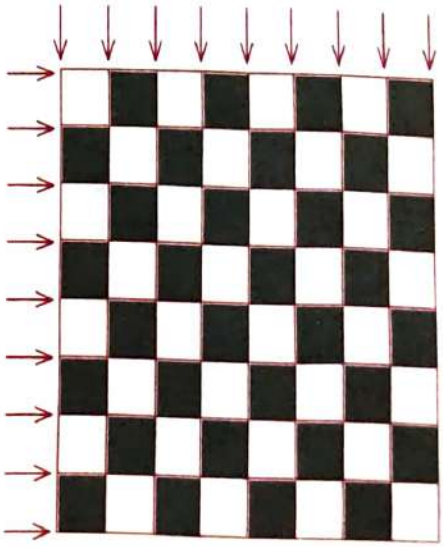
$= 14 \times 26 - 31 = 333$



Brain Teaser

Find the numbers of rectangles & the number of squares present in a chess board.

Ans.



To obtain a rectangle we need any 2 horizontal lines & any 2 vertical lines

∴ No. of rectangles = $({}^9C_2 \times {}^9C_2)$

no of squares = $8^2 + 7^2 + 6^2 + 5^2 + 4^2 + \dots + 1^2$

$= \frac{8 \times 9 \times 17}{6} = 4 \times 3 \times 17 = 204$

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Important Note

Note 1: The number of triangles formed by joining the vertices of a n sided polygon is ${}^nC_3$.

Note 2: The number of triangles having exactly 2 sides common to the sides of the polygon is "n".

Note 3: The number of triangles having exactly 1 side common to the sides of the polygon is "n(n - 4)".

Note 4: The number of triangles having no side common to the side of the polygon is $\frac{n(n-4)\,n(n-5)}{6}$.

Ques. Find the maximum number of points of intersection of 7 straight lines and 5 circles when 3 straight lines are parallel and 2 circles are concentric.

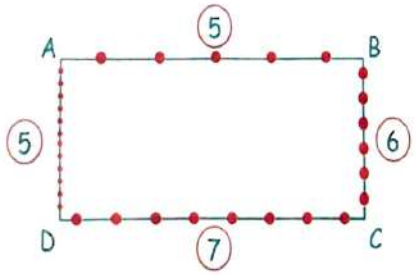
Ans. Total ways = ${}^7C_2 \cdot 1 + {}^5C_2 \cdot 2 + {}^7C_1 \cdot {}^5C_1 \cdot 2 -$ $[{}^5C_2 \times 1 + {}^2C_2 \times 2] = 21 + 20 + 70 - [3 + 2]$

$= 111 - 5 = 106$

Ques. Consider a rectangle ABCD having 5, 7, 6, 9 points in the interior of the line segments AB, CD, BC, DA respectively. Let a be the number of triangles having these points from 4 different sides as vertices and β be the number of quadrilaterals having these points from all 4 different sides as vertices. The $(\beta - \alpha)$ is equal to **[JEE Main 2021, (16 March Shift-II)]**

(a) 795 (b) 1173 (c) 1890 (d) 717

Ans.



$\beta = {}^5C_1 \cdot {}^7C_1 \cdot {}^6C_1 \cdot {}^9C_1$

$\Rightarrow \beta = 5 \cdot 7 \cdot 6 \cdot 9 = 1890$

$\alpha = 5 \cdot 7 \cdot 6 + 6 \cdot 7 \cdot 9 + 7 \cdot 9 \cdot 5 + 9 \cdot 5 \cdot 6 = 1173$

$\beta - \alpha = 717$

Ques. Find the number of 6 digit integers which can be formed such that the digits occur in

(1) Ascending order

Ans. $= {}^9C_6 \times 1 = {}^9C_6$

(2) Descending Order

Ans. ${}^{10}C_6 \times 1 = {}^{10}C_6$

Ques. Out of 10 consonants and 4 vowels, how many words can be formed each containing 3 consonants and 2 vowels?

Ans. No. of words = ${}^{10}C_3 \cdot {}^4C_2 \times 5!$

Ques. Total number of 6-digit numbers in which all the odd digits and only odd digits appears, is

(a) $\frac{5}{2}(6!)$ (b) 6!

(c) $\frac{1}{2}(6!)$ (d) none

Ans. $\underbrace{1, 3, 5, 7, 9.}$

1 3 5 7 9 1

Out of 5 digits one should repeat

Total Number = ${}^5C_1 \times \frac{6!}{2!} = 5 \cdot \frac{6!}{2!}$

Ques. A person invites a group of 10 friends at dinner and sits

(i) 5 on a round table and 5 more on another round table.

(ii) 4 on one round table and 6 on the other round table.

Find the number of ways in each case in which he can arrange the guests.

Ans. (i) No. of ways

$= {}^{10}C_5 \cdot {}^5C_5\, 4! \cdot 4! = {}^{10}C_5 \times 4! \cdot 4!$

$= \frac{10!}{25}$

Ans. (ii) No. of Ways = ${}^{10}C_4 \cdot 3! \cdot 5! = \frac{10!}{4!\,6!} \cdot 3! \cdot 5!$

$= \frac{10!}{4!3!} \times \frac{3! \cdot 5!}{6 \times 5!} = \frac{10!}{24}$.

Ques. In how many of the permutations of n things take r at a time will three given things always occur?

Ans. $\underbrace{T_1, T_2, T_3,}\ \underbrace{T_4, - - - - T_n}_{n-3}$

$\underbrace{T_1\ T_2\ T_3}$ $\underbrace{- - - - - -}$

3 things (r - 3) places

No. of total permutations = ${}^{n-3}C_{r-3} \cdot r!$

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PERMUTATION OF ALIKE OBJECTS (TAKEN SOME AT A TIME)

Ques. Find the total number of ways of selecting five letters from the word "INDEPENDENCE".

Ans. I = 1, N = 3, E = 4, D = 2 P = 1, C = 1

Cases	Selections
4 Alike & 1 diff	$1 \times {}^5C_1 = 5$
3 Alike & 2 diff	$2 \times {}^5C_2 = 20$
3 Alike & 2 other Alike	$2 \times 2 = 4$
2 Alike & 3 diff	${}^3C_1 \cdot {}^5C_3 = 30$
2 Alike, 2 others alike & 1 diff	${}^3C_1 \cdot {}^2C_1 \times {}^4C_1 = {}^3C_2 \cdot {}^4C_1 = 12$
All 5 different	${}^6C_5 = 6$

Total selection = 77

Arrangement	Total Arrangement
$\frac{5!}{4!}$	$5 \times \frac{5!}{4!} = 25$
5!/3! = 5 x 4	= 20 x 20 = 400
5!/3!2! = 10	= 40
5!/2! = 60	60 x 30 = 1800
5!/2!2! = 30	30 x 12 = 360
5!	6 x 5! = 720
	Total = 3345

Ques. Find the total number of ways of selecting five letters from the word "INDEPENDENT".

Ans.

I = 1 P = 1

N = 3 T = 1

E = 3

D = 2

3 alike & 2 different	$2 \times {}^5C_2 = 20$
3 alike & 2 other alike	$2 \times 2 = 4$
2 alike & 3 different	$3 \times {}^5C_3 = 30$
2 alike & 2 alike & 1 different	${}^3C_2 \times {}^4C_1 = 12$
All 5 different	${}^6C_5 = 6$

Total Selection = 72

Ques. The number of ways five alphabets can be chosen from the alphabets of the word MATHEMATICS, where the chosen alphabets are not necessarily distinct, is equal to:

[JEE main 2024 (Apr. Shift-II)]

(a) 179 (b) 177 (c) 181 (d) 175

Ans. M = 2, A = 2, T = 2, H = 1, E = 1, I = 1, C = 1, S = 1

2 A & 3 diff	$3 \times {}^7C_3 = 3 \times 35 = 105$
2 A & 2 A & 1 diff	${}^3C_2 \cdot {}^6C_1 = 3 \times 6 = 18$
All 5 diff.	${}^8C_5 = 56$

Total Selection = 179

Ques. Find the number of 6 lettered words which can be formed using letters of the word INTEGRATION such that each word contains 3 vowels & 3 Consonants

Ans. I = 2 N = 2, T = 2

E = 1 G = 1, R = 1

A = 1

O = 1

2 alike & 1 different vowels & 2 alike & 1 different consonants	$1 \times {}^3C_1 \cdot 2 \cdot {}^3C_2 \cdot \frac{6!}{2!\,2!}$
(2 alike & 1 different) vowels & 3 different consonants	$1 \cdot {}^3C_1 \cdot {}^4C_3 \cdot 6!/2!$
3 different vowels & 2 A & different consonants	${}^4C_3 \cdot 2 \cdot {}^3C_1 \cdot 6!/2!$
3 different vowels & 3 different consonants	${}^4C_3 \cdot {}^4C_3 \cdot 6!$

EXPONENT OF a PRIME "p" in n!

Exponent of a prime 'p' n (α) =

$$\alpha = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \text{----}$$

Example: $100! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \text{ - - - - - } 100$

(a) Exponent of 5: $\alpha = \left[\frac{100}{5}\right] + \left[\frac{100}{25}\right] + \left[\frac{100}{125}\right] + \text{- - - -}$

$\alpha = 20 + 4 = 24$

(b) Exponent of 2:

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$$\alpha = \left[\frac{100}{2}\right] + \left[\frac{100}{4}\right] + \left[\frac{100}{8}\right] + \left[\frac{100}{16}\right] + \left[\frac{100}{32}\right] + \left[\frac{100}{64}\right] + \left[\frac{100}{128}\right] + \ldots$$

$\alpha = 50 + 25 + 12 + 6 + 3 + 1 = 97$

Ques. Find no of zero's at the end of 200!

Ans. Exponents of 2 $\rightarrow \alpha = \left[\frac{200}{2}\right] + \left[\frac{200}{4}\right] + \left[\frac{200}{8}\right] + \ldots$

Exponents of 5 $\rightarrow \beta = \left[\frac{200}{5}\right] + \left[\frac{200}{25}\right] + \left[\frac{200}{125}\right] + \ldots$

$\beta = 40 + 8 + 1$

Power of 10 = 49

Ques. Find the exponent of 80 in 200!

Ans. $\because 80 = 16 \times 5 = 2^4 \times 5$

Exponent of 2

$(\alpha) = \left[\frac{200}{2}\right] + \left[\frac{200}{4}\right] + \left[\frac{200}{8}\right] + \left[\frac{200}{16}\right] + \left[\frac{200}{32}\right] + \left[\frac{200}{64}\right] + \left[\frac{200}{128}\right] + \ldots$

$\alpha = 100 + 50 + 25 + 12 + 6 + 3 + 1 = 197$

Exponent 5 $(\beta) = \left[\frac{200}{5}\right] + \left[\frac{200}{25}\right] + \left[\frac{200}{125}\right] + \ldots$

$= 40 + 8 + 1 = 49$

For Exponent of 80 $\rightarrow 2^{197} \cdot 5^{49} = 2.2^{196} \cdot 5^{49}$

$= 2(2^4)^{49} \cdot (5)^{49} = 2(16)^{49} \cdot (5)^{49} = 2(80)^{49}$

So, the exponent of 80 = 49

FORMATION OF GROUPS

Theorems

Theorem-1 The number of ways in which (m + n) different things can be divided into two groups such that one of them contains m things and other than has n things, is $\frac{(m+n)!}{m!n!}$. $(m \neq n)$.

If these groups are the be "distributed" bet two persons then the number of ways $\frac{(m+n)!}{m!n!}$.

Theorem-2 The number of ways in which different things can be divided into two groups that EACH of them contains n things is $\frac{(2n)!}{n!\,n!\,2!}$

If these groups are the be "distributed" be two persons then the number of ways $\frac{(2n)!}{n!\,n!\,2!}$.

Theorem-3 Number of ways in which (m + n different things can be divided into three g containing m, n & p things respectively, is: $\frac{(m+n}{m!\,n}$ $(m \neq n \neq p)$.

If these groups are the be "distributed" be three persons then the number of ways $\frac{(m+}{m!}$ x 3!

Theorem-4 If m = n = p then the number of g $\frac{(3n)!}{n!\,n!\,n!\,3!}$.

If these groups are the be "distributed" be three persons then the number of ways $\frac{(}{n!\,n}$ x 3!

Theorem-5 Number of ways in which (m different things can be divided into three containing m, n & p things respectively, is: $\frac{(m}{m}$ $(m \neq n)$.

If these groups are the be "distributed" t persons then the number of ways $\frac{(m+2n)!}{m!\,n!n!2!}$ x

Conclusion

✓ Jitne Equal Groups divide by utne factori

✓ Always use groups to distribute distinct

✓ Read question carefully and see if we DIVIDE or DISTRIBUTE.

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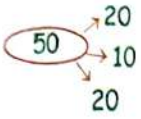
Ques. In how many ways can 50 volunteers can be sent for election campaigning to 3 districts having a requirement of 20, 10, 20 volunteers respectively.

Ans.

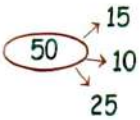


A	B	C
20	10	20

$$\text{division} = \frac{50!}{20!10!20!\,2!} \times (2!) \rightarrow \text{distribution in } A \& C$$

Ques. In how many ways can 50 volunteers can be sent for election campaigning to 3 districts having a requirement of 15, 10, 25 volunteers respectively.

Ans.

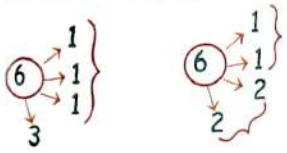


A	B	C
15	10	25

$$= \frac{50!}{20!10!25}$$

Ques. In how many ways can 6 distinct toys be distributed to 4 children such that each child gets at least 1 toy.

Ans.



$$\text{Total ways} = \frac{6!}{(1!)^3\,3!}\,\frac{1}{3!} \times 4! + \frac{6!}{(1!)^2\,(2!)^2}\,\frac{1}{2!2!} \times 4!$$

$$= \frac{6 \times 5 \times 4}{3!} \times 4 \times 3! + \frac{6!}{2 \times 2 \times 2 \times 2} \times 4 \times 3 \times 2!$$

$$= 6 \times 5 \times 16 + \frac{6 \times 5 \times 4 \times 3 \times 2 \times 3 \times 2}{2 \times 2}$$

$$= 30 \times 16 + 30 \times 36$$

$$= 30\,[16 + 36]$$

$$= 30 \times 52 = 1560$$

Ques. In how many ways can 10 distinct computers be distributed to 5 schools such that each school gets at least 1 computer and at most 5 computers.

Ans.

Case-1: 1 1 1 1 6 more than 5 (X)

Case-2: 1 1 1 2 5 $\frac{10!}{2!\,5!\,3!} \times 5! + \frac{10!}{3!\,4!\,3!} \times 5!$

Case-3: 1 1 1 3 4

Case-4: 1 1 2 2 4 $\frac{10!}{(2!)^2\,4!\,2!\,2!} \times 5! + \frac{10!}{2!\,3!\,3!\,2!\,2!} \times 5!$

Case-5: 1 1 2 3 3

Case-6: 1 2 2 2 3 $\frac{10!}{3!\,(2!)^3\,3!} \times 5! + \frac{10!}{(2!)^5\,5!} \times 5!$

Case-7: 2 2 2 2 2

Ques. The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is **[IIT JEE Adv 2012]**

(a) 75 (b) 150

(c) 210 (d) 243

Ans. 1 1 3 $\rightarrow \frac{5!}{3!\,2!} \times 3!$

1 2 2 $\rightarrow \frac{5!}{2!\,2!\,2!} \times 3!$

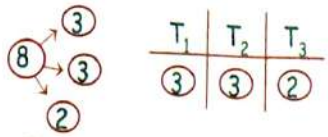
$$\text{Total ways} = \frac{5!}{3!\,2!} \times 3! + \frac{5!}{2!\,2!\,2!} \times 3!$$

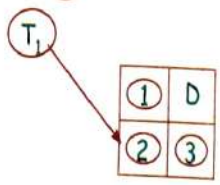
$$= 60 + \frac{120}{2 \times 2 \times 2} \times 3 \times 2 = 60 + 30 \times 3 = 60 + 90$$

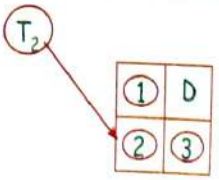
Ques. In how many ways 8 people can be seated in 3 different taxis each having 3 duly numbered seats such that

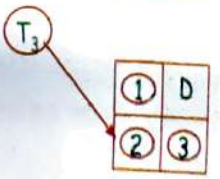
(1) Arrangement inside the taxi is immaterial

Ans.









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$$\text{div} = \frac{8!}{3!\,3!\,2!\,2!}$$

$$\text{Distribution} = \frac{8!}{3!\,3!\,2!\,2!} \times 3!$$

(2) Internal arrangement also matters

Ans. M-1 Total ways $= \frac{8!}{3!\,3!\,2!\,2!} \times 3! \times 3! \times 3! \times {}^3P_2$

$= \frac{8!}{2!\,2!} \cdot 3! \cdot 3! = 8!.3.3 = 9!$

M-2 ${}^9P_8 = \frac{9!}{1!} = 9!$

Ques. In a hotel, four rooms are available. Six perosns are to be accommodated in these four rooms in such a way that each of these rooms contains at least one person and at most two persons. Then the number of all possible ways in which this can be done _____ .

[JEE Adv. 2020]

Ans. R_1 R_2 R_3 R_4

1 1 2 2

$$\text{Total ways} = \frac{6!}{2!\,2!\,2!\,2!} \times 4!$$

$$= \frac{720 \times 3 \times 2}{2 \times 2} = 360 \times 3 = 1080$$

Ques. Find the sum of all 5 digit numbers which can be formed using the digit 1, 3, 5, 7, 9 (without repetition)?

(a) 66,666,00 (b) 67,666,00

(c) 66,666,60 (d) 66,666,66

Ans. 1, 3, 5, 7, 9

Total number = 5! = 120

Method-1 Carry Method

3, 5, 7, 9

				1

4!

last digit will be 1 in 24 no's

				3

4!

last digit is 3 in 24 no's

Sum of last digit = 24 × 1 + 24 × 3 + 24 × 5 + 24 × 7 + 24 × 9

= 24 [1 + 3 + 5 + 7 + 9]

24 × 25 = 600

66 66 66 60

66, 6 6,6 00

			1	

4! = 24

Sum of second last digit = 600 and 60 will be added which comes from last digit.

= 600 + 60 = 660

Similarly third last digit = 600 + 66 = 666

Hence, total sum = 66,666,00

Method-2 Place Value Method

Contribution of 1 in the entires sum

$= 1 \times 10^0 \times 4! + 1 \times 10^1 \times 4! + 1 \times 10^2 \times 4! + 1 \times 10^3 \times 4! + 1 \times 10^4 \times 4!$

$= 1 \times 4! \underbrace{[10^0 + 10^1 + 10^2 + 10^3 + 10^4]}_{\Delta} = 24\Delta$

Contribution of 3 in the entires sum

$= 3 \times 4! \underbrace{[10^0 + 10^1 + 10^2 + 10^3 + 10^4]}_{\Delta} = 3 \times 24\Delta$

5 ⟶ $= 5 \times 24\Delta$

7 ⟶ $= 7 \times 24\Delta$

9 ⟶ $= 9 \times 24\Delta$

∵ 10,000 + 1000 + 100 + 10 + 1 = Δ = 1 1 1 1 1

Total Sum = 24 Δ [1 + 3 + 5 + 7 + 9]

= 24 Δ . 25 = 600 Δ

= 600 × 11111 = 6666600

METHOD-3 FORMULA METHOD

When given digits are in 'AP' ⟶ {1,3,5,

$$\text{Sum} = \frac{n}{2}[I + L]$$

$$= \frac{120}{2}[13579 + 97531] \longrightarrow [\because n = 5!]$$

= 6666600

Proof → .....

S = 13579 + 13597 + - - - - - - + 97513 + 97

+ S = 97531 + 97513 + - - - - - - + 13597 + 135

2S = 111110 + 111110 + - - - - - - + 111110 + 11

2S = 111110 [120]

$$S = \frac{120}{2}[111110] = 6666600$$

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Ques. A Group of 9 Students, $S_1, S_2, ..., S_9$ is to be divided into 3 teams X, Y, Z of size 2, 3, 4 respectively. Suppose that S_1 cant be selected for the team X, S_2 cant be selected for team Y then number of ways to form such teams is?
(JEE Adv. 2024)

Ans.

(2) X	(3) Y	(4) Z
S_2		

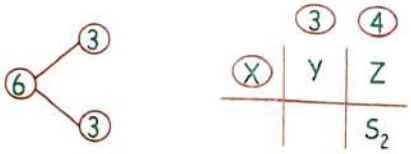
Case → 1

S_2 goes to team X

$${}^7C_1 \times \frac{7!}{3!\,4!} = \frac{7 \times 7 \times 6 \times 5}{3 \times 2} = 49 \times 5 = 245$$

no. of ways of selecting 1 student out of $\{S_3, S_4, S_5, ... S_9\}$ for team X

Case → 2 : S_2 goes to Z



$$\text{Groups} = {}^7C_1 \times \frac{6! \times 2!}{3!3! \times 2!}$$

$$= 21 \times \frac{6 \times 5 \times 4}{3 \times 2}$$

$$= 21 \times 20 = \boxed{420}$$

Total Groups = 245 + 420 = 665

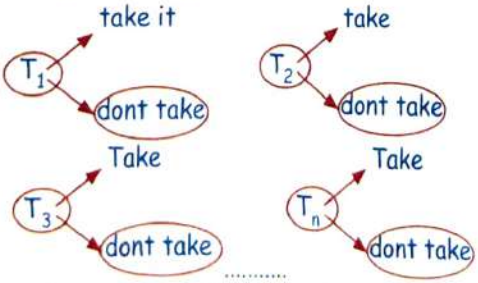
SELECTION OF ATLEAST ONE OBJECT

Case-1 From n Distinct objects

$T_1, T_2, T_3, T_4, T_5, ..., T_n$

M-1 $${}^nC_1 + {}^nC_2 + {}^nC_3 + {}^nC_4 + ... + {}^nC_n = 2^n - 1$$

M-2



Total no. of ways of selecting at least one object = $2^n - 1$

Ques. From a library containing 4 different physics books. 5 different chemistry books & 6 different maths books. Find the number of ways in which we can select

$P_1, P_2, P_3, P_4, C_1, C_2, C_3, C_4, C_5, M_1, M_2, M_3, M_4, M_5, M_6$

(a) Atleast 1 book

Ans. $(2^{15} - 1)$

(b) Atleast 1 book of each Subject

Ans. $(2^4 - 1) \cdot (2^5 - 1) \cdot (2^6 - 1) = 15 \times 31 \times 63$

(c) Atleast 2 books

Ans. ${}^{15}C_2 + {}^{15}C_3 + {}^{15}C_4 - - - {}^{15}C_{15} = 2^{15} - 1 - {}^{15}C_1$
$= 2^{15} - 1 - 15 = (2^{15} - 16)$

(d) Atleast 2 book of each Subject

Ans. $(2^4 - 1 - {}^4C_1)(2^5 - 1 - {}^5C_1)(2^6 - 1 - {}^6C_1)$

(e) Exactly 2 book

Ans. ${}^{15}C_2$

(f) Exactly 2 book of each Subject

Ans. ${}^4C_2 \times {}^5C_2 \times {}^6C_2$

Ques. From a class of 5 Boys and 5 Girls, in how many ways can a teacher take at-least one boy and at-least 1 girl to a Sports event.

(a) 32^2 (b) $10^2 - 1$

(c) ✓ 31^2 (d) 62

Ans. Total ways = $(2^5 - 1) \times (2^5 - 1) = 31 \cdot 31 = (31)^2$

Ques. In JEE ADVANCED Exam, in How many ways can a student answer a Question which has one or more than 1 correct option out of 4 given options.

(a) 14 (b) ✓ 15

(c) 11 (d) None

Ans. $2^4 - 1 = 15$

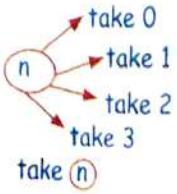
Ques. In JEE ADVANCED Exam in How many ways can a student answer a Question which has More than 1 correct options out of 4 given options.

(a) 14 (b) 15

(c) ✓ 11 (d) None

Ans. Total ways = $2^4 - 1 - {}^4C_1$
$= 16 - 1 - 4 = 16 - 5 = 11$

Case-2 From n Identical objects. $\underbrace{T, T, T, T, T, ..., T}_{n \text{ things}}$



Total no. of ways of dealing with n identical objects = (n + 1)

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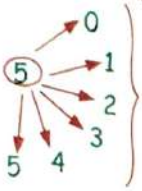
Ques. From a BASKET having 5 identical Apples, in how many ways can a Child take atleast one fruit.

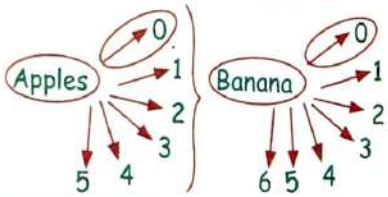
(a) 5 (b) 6 (c) 32 (d) 31

Ans. 

Total way = (5 + 1) - 1 = 6 - 1 = 5

Ques. From a BASKET having 5 identical Apples & 6 Identical Bananas, in how many ways can a Child take atleast one fruit.

(a) 10 (b) 30 (c) 29 (d) 41

Ans. 

no. of ways deating with Apples = 6
no. of ways deating with bananas = 7
Total ways = 6 × 7 = 1 = 41

Ques. From a library containing 4 identical physics books. 5 identical chemistry books & 6 identical maths books. Find the number of ways in which we can select

(a) Atleast 1 book

Ans. (5 x 6 x 7 - 1) = 210 - 1 = 209

(b) At-least 1 book of each Subject

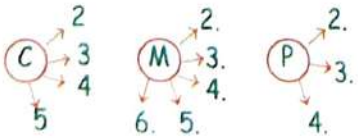
Ans. (5-1)(6-1)(7-1) = 4 x 5 x 6 = 20 x 6 = 120

(c) At-least 2 books

Ans. 209 - [when 1 book was selected]
= 209 - 3 = 206

(d) At-least 2 book of each Subject

Ans. 3 x 4 x 5 = 3 x 20 = 60



(e) Exactly 2 book

Ans. PP or CC or MM or PC or CM or MP
1 + 1 + 1 + 1 + 1 + 1 = 6 ways

(f) Exactly 2 book of each Subject

Ans. 1 × 1 × 1 = 1

Case-3 From "n" mixed objects.

Ques. From a library containing 4 different physics books. 5 identical chemistry books & 6 identical maths books. Find the number of ways in which we can select

P_1, P_2, P_3, P_4, CCCCC, MMMMMM

(a) Atleast 1 book

Ans. $2^4 . 6 . 7 - 1 = 671$

(b) Atleast 1 book of each Subject

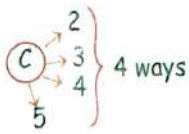
Ans. $(2^4 - 1) \times 5 \times 6 = 15 \times 5 \times 6 = 15 \times 30 = 450$

(c) Atleast 2 books

Ans. 671 - [when exactly 1 book selected]
= 671 - 6 = 665

(d) Atleast 2 book of each Subject

Ans. $(2^4 - 1 - 4) \times 4 \times 5 = 11 \times 20 = 220$



(e) Exactly 2 book

Ans. PP + CC + MM + PC + PM + CM
$= {}^4C_2 + 1 + 1 + {}^4C_1 \times 1 + {}^4C_1 \times 1 + 1 = 17$

(f) Exactly 2 book of each Subject

Ans. ${}^4C_2 \times 1 \times 1 = 6$

FINDING NUMBER OF DIVISORS

$N = (P_1)^a (P_2)^b (P_3)^c$ - - - .

Where, P_1, P_2, P_3,.... are Prime numbers.

no. of divisors = (a + 1) (b + 1) (c + 1) - -

Ques. Find the total number of Divisors of 9600

(a) 60 (b) 58 (c) 48 (d) 46

Ans. 96 x 100 = 24 x 4 x 100
$= 8 \times 3 \times 4 \times 4 \times 25 = 2^3 . 3^1 \times 2^4 . 5^2$
$= 2^7 . 3^1 . 5^2$
No. of divisors = 8 x 2 x 3

Note: 1 & number it self are called improp divisors

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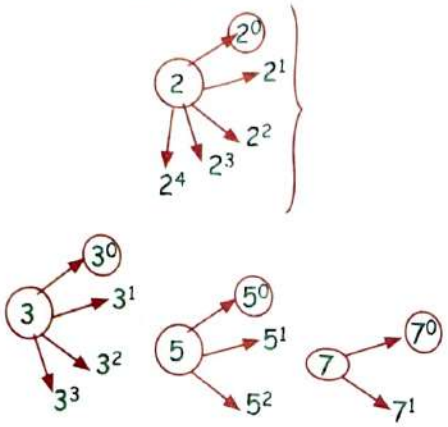
Ques. Find the number of Positive integral Divisors of the number N = 75600.

Ans. $N = 2^4 . 3^3 . 5^2 . 7^1$



no of divisors = 5 x 4 x 3 x 2 = 120

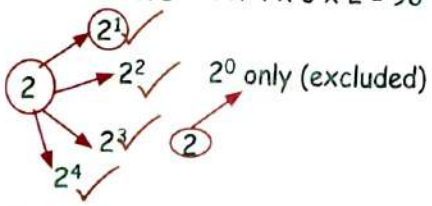
no. of ways of taking power of 2

Ques. In the previous example, out these 120 divisors of the number N = 75600,

(a) How many divisors are even.

Ans. $N = 2^4 . 3^3 . 5^2 . 7^2$

even divisors = 4 x 4 x 3 x 2 = 96



(b) How many divisors are odd

Ans. odd divisons = Total - even 120 - 96 = 24

(c) How many are proper divisors

Ans. 1 and the number itself

120 - 2 = 118

(d) How many divisors are perfect squares.

Ans. Total perfect squares divisors = 3 x 2 x 2 x 1 = 12

(e) Sum of all the divisors

Ans. $(2^0 + 2^1 + 2^2 + 2^3 + 2^4) . (3^0 + 3^1 + 3^2 + 3^3) . (5^0 + 5^1 + 5^2) . (7^0 + 7^1)$

$= 31 \times 40 \times 31 \times 8 = 307520$

(f) Sum of all even Divisors

Ans. $(2^1 + 2^2 + 2^3 + 2^4) . (3^0 + 3^1 + 3^2 + 3^3) . (5^0 + 5^1 + 5^2) . (7^0 + 7^1)$

$= 31 . 40 . 31 . 8$

(g) Sum of all odd Divisors

Ans. $2^0 (3^0 + 3^1 + 3^2 + 3^3) . (5^0 + 5^1 + 5^2) . (7^0 + 7^1) = 1.40.31.8$

(h) Sum of all proper Divisors

Ans. 31.40.31.8 — (75601)

(i) The number of ways in which the number 75600 can be expressed as a Product of two divisor

Ans. 75600 = 756 x 100 <--> 75600 = 100 x 756

75600 = 10 x 7560

75600 = 1 x 75600

The number of ways in which the number 75600 can be expressed as a Product of two divisor = 120/2 = 60

Ques. Find the sum of all Divisors of 960.

(a) 3048 (b) 3087

(c) 3047 (d) 2180

Ans. $96 = 2^6 . 3^1 . 5^1$

Sum of all divisors $(2^0 + 2^1 + 2^2 + - - + 2^6) . (3^0 + 3^1) (5^0 + 5^1)$

$= \left(\frac{2^7 - 1}{2 - 1}\right) . 4.6 = (128 - 1) . 4.6$

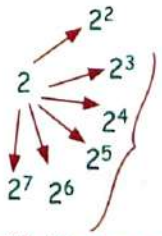
$= 127 \times 24 = 3048$

Ques. If $N = 2^4 . 3^3 . 5^2 . 6^3 = 2^4 . 3^3 . 5^2 . 2^3 . 3^3 = 2^7 . 3^6 . 5^2$.

(i) Number of divisors of the form

(4n) = divisible by 4

Ans.



(4n) = divisible by 4 6 x 7 x 3

= 42 x 3

= 126

(ii) Number of divisors of the form

(4n + 2) → 6, 10, 14, 18, - .

Ans. 2, 4, 6, 8, 10, 12, 14, 16.,

Number of divisors of the form (4n + 2) = no. of even divisors - 126

= 7 x 7 x 3 - 126 = 147 - 126 = 21

Ques. Find the number of ways in which the number $N = 2^4 3^2 5^6 7^8$ can be expressed as a product of two divisors.

Ans. N - Perfect sq

$N = x . x$

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$N = (\underbrace{2^2 . 3^1 . 5^3 . 7^4}_{x}) \times (\underbrace{2^2 . 3^1 . 5^3 . 7^4}_{x})$

$5 . 3 . 7 . 9 = 945 =$ No. of divisors

$= \left(\frac{945 + 1}{2}\right) = 473$

Brain Teaser

Find the number of lattice points on the hyperbola $xy = 30030$

Hint- (a, b) is a Lattice point if both a & b are integers.

Ans. $xy = 30030$

$xy = 2^1 . 3^1 . 5^1 . 7^1 . 11^1 . 13^1$

Total divisors $= 2 \times 2 \times 2 \times 2 \times 2 \times 2$

$= 2^6 = \boxed{64} \rightarrow$ Total laltice points

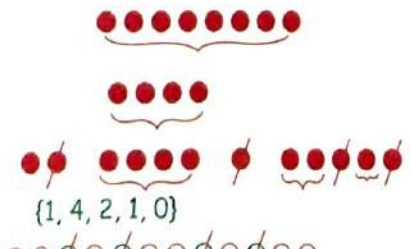
$xy = 3003 \times 10$ (3003, 10)

$xy = 10 \times 3003$ (10, 3003) $= 2 \times 64 = 128$

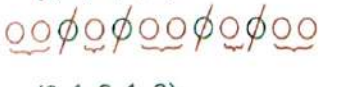
DISTRIBUTION OF IDENTICAL OBJECTS (BEGGAR'S METHOD)

Illustration-1 In how many ways can we distribute 8 identical coins to 5 beggars such that each beggar can receive any number of coins.

Ans. $^{12}C_4 = \frac{12!}{8!4!}$

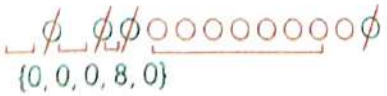


{1, 4, 2, 1, 0}



(2, 1, 2, 1, 2)

{8, 0, 0, 0, 0}



{0, 0, 0, 8, 0}

{0, 0, 0, 0, 8}

Method

STEP-1 Introduce fake identical coins, 1 less th the number of beggars.

STEP-2 Arrange these fake coins along with th real coins to get the number of ways in which th distribution can be done.

Each arrangement of these (fake coins + real coin corresponds to 1 distribution pattern.

Logic-2 The fake coins act as partition between th number of coins received by each beggar.

Ex → 10 coins → to 6 beggars

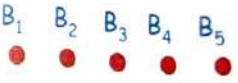
10 → Real
5 → Fake
$\Big\} \frac{15!}{10!5!} = {}^{15}C_5$

Illustration-2 In how many ways can we distribut 12 identical coins to 5 beggars such that each begga receives at least one coin.

Ans. $1 \times {}^{11}C_4$

7 coins

4 Fake



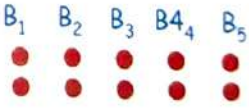
$\frac{11!}{7!4!} = {}^{11}C_4$

Illustration-3 In how many ways can we distribut 12 identical coins to 5 beggars such that each begga receives at least 2 coin.

Ans.

2 real

$\frac{11!}{7!4!} = {}^{11}C_4$



Theorems

Theorem-1 Number of ways to distribute identical objects among "r" person so that each m get any number of things is equal to

$^{n+r-1}C_{r-1} = {}^{n+r-1}C_{n}$

Real = n
Fake = r - 1
$\Big\} \rightarrow$ Total = n + r - 1

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Theorem-2 Number of ways to distribute "n" identical objects among "r" person so that each gets at least 1 thing is equal to. ${}^{n-1}C_{r-1}$

$n - r \rightarrow$ real

$n - r \rightarrow$ Fake ${}^{n-r+r-1}C_{r-1} = {}^{n-1}C_{r-1}$

Ques. There are infinite number of Identical BLUE, RED, GREEN AND YELLOW balls available for selections. In How many ways can we select 10 Balls.

(a) ${}^{14}C_3$ (b) ${}^{13}C_{10}$

(c) ${}^{12}C_{10}$ (d) ${}^{14}C_4$

Ans. $B_1 \rightarrow \underline{OOOO}$ - - - ∞

$B_2 \rightarrow \underline{OOOO}$ - - - ∞

$B_3 \rightarrow OOOO$ - - - ∞

$B_4 \rightarrow OOOO$ - - ∞

$\rightarrow \boxed{B_1 + B_2 + B_3 + B_4 = 10}$

${}^{10+3}C_3 = {}^{13}C_3 = {}^{13}C_{10}$

FINDING NUMBER OF SOLUTIONS

Number of Non-Negative Integral Solutions

Ques. Find the number of Non negative Integral solutions for the following:

1. $x_1 + x_2 + x_3 + x_3 + x_4 + x_5 = 25$

Ans. $x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0$

Real = 25, Fake = 4 $\Rightarrow {}^{29}C_4$

2. $x_1 + x_2 + x_3 = 30$

Ans. Real = 30

Fake = 2

$\Rightarrow {}^{32}C_2$

Ques. Find the number of Natural solutions for the following:

1. $x_1 + x_2 + x_3 + x_3 + x_4 + x_5 = 25$

Ans. $x_1 \geq 1, x_2 \geq 1, x_3 \geq 1, x_4 \geq 1, x_5 \geq 1$

Real = 20

Fake = 4 $\Rightarrow {}^{24}C_4$

2. $x_1 + x_2 + x_3 = 30$

Ans. $27 \rightarrow$ Real

Fake $\rightarrow 2 \Rightarrow {}^{29}C_2$

Ques. Find the number of ways in which Rahul Ji can distribute 40 identical blankets to 4 beggars having a family of 4, 5, 6, & 3 members respectively such that each beggar gets at least as many blankets as the number of members in his family.

Ans. $x_1 + x_2 + x_3 + x_4 + = 40$

$x_1 \geq 4, x_2 \geq 5, x_3 \geq 6, x_4 \geq 3$

$4 + 5 + 6 + 3 = 18$

$22 \rightarrow$ real

Fake $\rightarrow 3$

Total = ${}^{25}C_3$

Number of Integral Solutions

Ques. Find the number of Integral solutions for the following:

1. $x_1 + x_2 + x_3 + x_4 = 20,$

$x_1 \geq 0, x_2 \geq 1, x_3 \geq 2, x_4 \geq 4$

Ans. Real = 23, Fake = 3 ${}^{16}C_3$

2. $x_1 + x_2 + x_3 = 15$

$x_1 \geq 4, x_2 \geq -3, x_3 \geq -1$

Ans. Real = 15 = ${}^{17}C_2$

3. $x_1 + x_2 + x_3 + x_4 \leq 12, x_1 \geq 0$

Ans. Introduce a fake beggar

$x_1 + x_2 + x_3 + x_4 + x_5 = 12$

$x_5 \geq 0$

Real = 12

Fake = 4 $\Rightarrow {}^{16}C_4$

4. $x_1 + x_2 + x_3 < 15, x_1 \geq 0, x_2 \geq 2, x_3 \geq -1$

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Ans. $x_1 + x_2 + x_3 \le 14$

$x_1 \ge 0, x_2 \ge 2, x_3 \ge -1, x_4 \ge 0$

Real = 13

Fake = 3

Total = $^{16}C_3$

5. $x_1 + x_2 + x_3 \ge 20, x_1 \ge 0$

Ans. (x)

6. $x_1 + x_2 + x_3 + 3x_4 = 20, x_1 \ge 0$

Case→1

$x_4 = 0 \Rightarrow x_1 + x_2 + x_3 = 20$	$= {}^{22}C_2$
$x_4 = 1 \Rightarrow x_1 + x_2 + x_3 = 17$	$= {}^{19}C_2$
$x_4 = 2 \Rightarrow x_1 + x_2 + x_3 = 14$	${}^{16}C_2$
$x_4 = 3$	${}^{13}C_2$
$x_4 = 4$	${}^{10}C_2$ Add
$x_4 = 5$	${}^{7}C_2$
$x_4 = 6$	${}^{4}C_2$

Ques. In how many ways can Rs. 16 (in 16 one Rupee identical coins) be divided into 4 person when none of them get less than Rs.3.

(a) 70 (b) 35

(c) 64 (d) 192

Ans. $x_1 + x_2 + x_3 + x_4 = 16$

$x_i \ge 3$

Real = 16 - 12 = 4

Fake = 3

Total way = $^{7}C_3 = \frac{7 \times 6 \times 5}{3!} = 35$

Ques. The total number of ways of selecting six coins out of 20 identical one rupee coins, 10 Identical two rupee coins and 7 identical five rupee coins is

(a) 28 (b) 56

(c) $^{37}C_6$ (d) None of these

Ans. $x_1 + x_2 + x_3 = 6$

$x_i \ge 0$

Total no. = $^{8}C_2 = \frac{8 \times 7}{2!} = 28$

Ques. Find number of integral soln of $35 < x + y + z \le 52$ such that (x, y, z are positive integers)

Ans. $x + y + z \le 52 \Rightarrow x + y + z + w = 52$

Real = 49
Fake = 3 $\rightarrow {}^{52}C_3$

$x + y + z \le 35$

$x + y + z + k = 35$

$z \ge 1$ $k \ge 0$

$y \ge 1$

$z = 1$

real 35 - 3 = 32 $\rightarrow {}^{35}C_3$

Fake = 3

Total No. = $^{52}C_3 - {}^{35}C_3$

Ques. $(x_1 + x_2 + x_3) \cdot (y_1 + y_2) = 77$, Find number Natural solutions.

Ans.

$x_1 + x_2 + x_3$	$y_1 + y_2$
11	7
7	11

Case-1

$x_1 + x_2 + x_3 = 11$	$y_1 + y_2 = 7$
Real = 8	Real = 5
Fake = 2	Fake = 1
Sol = $^{10}C_2$	Sol = $^{6}C_1$

Total Soln = $^{10}C_2 \times {}^{6}C_1 = 45 \times 6 = 270$

Case-2

$x_1 + x_2 + x_3 = 7$

Real = 4

Fake = 2

$= {}^{6}C_2$

$y_1 + y_2 = 11$

Real = 9

Fake = 1

$= {}^{10}C_1$

Total = $^{6}C_2 \times {}^{10}C_1$

$15 \times 10 = 150$

Final = 270 + 150 = 420

Number of Integral Solutions Theorems

Theorem-1 The number of non-negative Integ solutions of the equation $x_1 + x_2 + x_3 + \ldots + x_r = n$ $^{n+r-1}C_{r-1}$, $x_i \ge 0$

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Theorem-2 The number of Natural solutions for the equation $x_1 + x_2 + x_3 + \ldots + x_r = n$ is ${}^{n-1}C_{r-1}$

Theorem-3 The number of non-negative integral solutions of $x_1 + x_2 + x_3 + \ldots + x_r \le n$ is ${}^{n+r}C_r$

Ques. Find the number of Lattice points laying inside the triangle formed by lines $x = 0$, $y = 0$ & $x + y = 2024$.

Ans. By Beggar's Method

$x + y = 2024$

$y = 0 \Rightarrow x = 2024$

$x > 0$, $y > 0$

& $x + y < 2024$

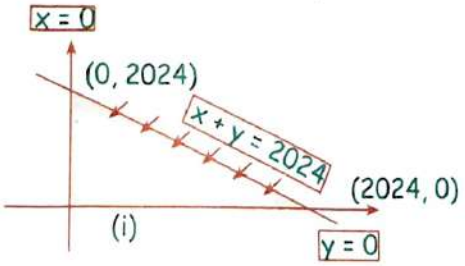
$\Rightarrow$ $x + y \le 2023$ $\quad x + y + z = 2023$

$x \ge 1, y \ge 1$ $\quad x \ge 1$ Real = 2021

$y \ge 1$

$z \ge 0$ Fake = 2

Total no. of soln = ${}^{2023}C_2$ = no of lattice pts



Brain Teaser

Find the number of Lattice points laying inside the triangle formed by lines $x + 0$, $y + 0$ & $x + y = 2024$.

Ans. By Beggar's Method

$x = 1, \Rightarrow y = 1, 2, 3, - - - 2022 = 2022$

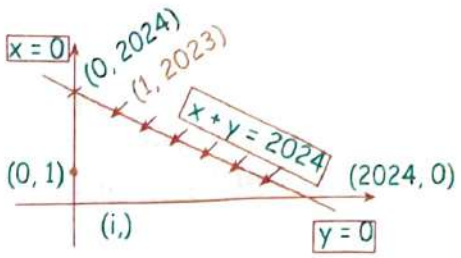
$x = 2, \Rightarrow y = 1, 2, 3, - - - 2021 = 2021$

$x = 3, \Rightarrow y = 1, 2, 3, - - - 2020 = 2020$

$x = 2022, y = 1 \longrightarrow = 1$ pt

Add = Total No. = $1 + 2 + 3 + - - + 2022$

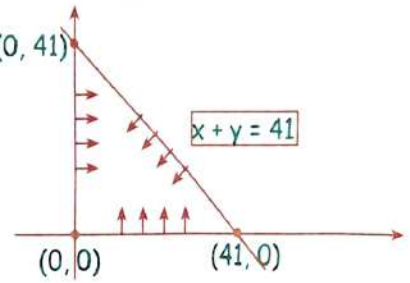
$= \dfrac{2022 \times 2023}{2}$



Ques. The number of points, having both co-ordinates as integers, that lie in the interior of the triangle with vertices (0, 0), (0, 41) and (41, 0) is :

(a) 820 (b) 780 (c) 901 (d) 816

Ans.



$x + y \le 40$

$x \ge 1, y \ge 1$

$x + Y + z = 40$

$x \ge 1, y \ge 1, z \ge 0$

Total no. = ${}^{40}C_2 = \dfrac{40 \times 39}{2} = 20 \times 39 = 780$

Brain Teaser

In a unique Cricket series between India & Pakistan, they decide to play on till a team wins 5 matches. The number of ways in which the series can be won by India, if no match ends in a draw is:

Ans.

W W W W W

X_1 X_2 X_3 X_4 X_5

$X_1 \rightarrow$ no. of Losses before Ist W

$\rightarrow X_1 + X_2 + X_3 + X_4 + X_5 \le 4$

WWWWW $\leftarrow X_1 = X_2 = X_3 = X_4 = X_5 = 0$

$X_1 = 1$ $\quad X_1 + X_2 + X_3 + X_4 + X_5 + W = 4$

$X_2 = 1$ $\quad X_1 \ge 0, W \ge 0$

$X_3 = 2$ $\quad = {}^9C_5$

$X_4 = 0$

$X_5 = 0$ $\quad$ LWLWLLWWW

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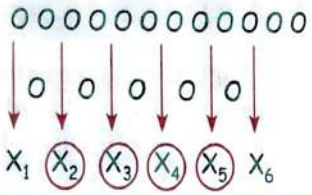
Ques. The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can be placed in a row so that between any two red cubes, there should be at least 2 blue cubes, is _____.

(JEE MAIN 27 JUNE 2022 (Shift-1))

Ans.



$X_1 + X_2 + X_3 + X_4 + X_5 + X_6 = 11$

$X_1 \geq 0 \quad X_2 \geq 2$

$X_6 \geq 0 \quad X_3$

X_4

X_5

Real = 3

Fake = 5

Total ways = ${}^8C_3 = 56$

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Short Notes

PERMUTATIONS

- **Without Repetition:** arranging n objects, taken r at a time is equivalent to filling r places from n things. $\frac{n!}{(n-r)!} = {}^nP_r$
- The number of arrangements of n different objects taken all at a time $= {}^nP_n = n!$
- **With Repetition:** The number of permutations (arrangements) of n different objects, taken r at a time, when each object may occur once, twice, thrice .... upto r times in any arrangement $= (n)^r$
- The number of arrangements that can be formed using n objects out of which p are identical (and of one kind), q are identical (and of another kind), r are identical (and of another kind) and the rest are distinct is $\frac{n!}{p!q!r!}$.
- EXPONENT "α" OF A PRIME "p" in n!

$$\alpha = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \ldots$$ where [ ] stands for greatest integer function.

CIRCULAR PERMUTATIONS

- The total number of circular arrangements of n persons is $\frac{n!}{n} = (n-1)!$.
- If no distinction is made between the clockwise and the anti-clockwise arrangements of n different objects around a circle, then the number of arrangements is $\frac{1}{2}(n-1)!$

COMBINATIONS (SELECTION OF DISTINCT OBJECTS)

- The number of selections (combinations or groups) that can be formed from n different objects taken r $(0 \le r \le n)$ at a time is

$${}^nC_r = \frac{n!}{r!(n-r)!}.$$

- The number of ways in which r objects can be selected from n different objects if k particular objects are
  - (i) always included $= {}^{n-k}C_{r-k}$
  - (ii) never included $= {}^{n-k}C_r$
- The number of arrangements of n distinct objects taken r at a time so that k particular objects are
  - (i) always included $= {}^{n-k}C_{r-k} \cdot r!$
  - (ii) never included $= {}^{n-k}C_r \cdot r!$

SELECTION OF ATLEAST 1 OBJECT

Selection from Distinct Objects

- The number of selections from n different objects, taken at least one $= {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots + {}^nC_n = 2^n - 1$.

Selection from Identical Objects

- The number of selections of r objects out of n identical objects is 1.
- Total number of selections of zero or more objects from n identical objects is n + 1.
- The total number of selections of at least one out of $a_1 + a_2 + a_3 + \ldots + a_n$ objects, where a_1 are alike (of one kind), a_2 are alike (of second kind) and so on .... a_n are alike (of nth kind), is $[(a_1+1)(a_2+1)(a_3+1)\ldots(a_n+1)] - 1$.

Selection When both Identical and Distinct Objects are Present

- The number of selections taking at least one out of $a_1 + a_2 + a_3 + \ldots + a_n + k$ objects, where a_1 are alike (of one kind), a_2 are alike (of second kind) and so on ... a_n are alike (of nth kind), and k are distinct

$= [(a_1+1)(a_2+1)(a_3+1)\ldots(a_n+1)]\,2^k - 1$.

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FORMATION OF GROUPS

Groups of Unequal Size (Different Number of Objects in Each Group)

- The number of ways in which n distinct objects can be divided into r unequal groups containing a_1 objects in the 1st group, a_2 objects in the second group and so on

$$= {}^{n}C_{a_1} \cdot {}^{n-a_1}C_{a_2} \cdot {}^{n-a_1-a_2}C_{a_3} \cdots {}^{a_r}C_{a_r} = \frac{n!}{a_1!a_2!a_3!\cdots a_r!}.$$

Here $a_1 + a_2 + a_3 + \ldots + a_r = n$.

- The number of ways in which n distinct objects can be distributed among r persons such that first person gets a_1 objects, 2nd person gets a_2 objects ... , rth person gets a_r objects $= \frac{n!r!}{a_1!a_2!a_3!\cdots a_r!}$.

Groups of Equal Size (Each Group Containing Same Number of Objects)

- The number of ways in which m × n distinct objects can be divided equally into n groups (unmarked)

$$= \frac{(mn)!}{(m!)^n n!}$$

- The number of ways in which m × n different objects can be distributed equally among n persons (or numbered groups)
= (number of ways of dividing into groups) × (number of groups)! $= \frac{(mn)!n!}{(m!)^n n!} = \frac{(mn)!}{(m!)^n}$.

DISTRIBUTION OF IDENTICAL OBJECTS

- **Theorem 1**: Number of ways to distribute "n" identical objects among "r" persons so that each may get any number of things is equal to ${}^{n+r-1}C_{r-1}$
- **Theorem 2**: Number of ways to distribute "n" identical objects among "r" persons so that each gets at least 1 thing is equal to ${}^{n-1}C_{r-1}$

PRINCIPLE OF INCLUSION AND EXCLUSION

- $n(A_1 \cup A_2 \cup \ldots\ldots \cup An)$

$$= \sum n(A_i) - \sum_{i \neq j} n(A_i \cap A_j) + \sum_{i \neq j \neq k} n(A_i \cap A_j \cap A_k)$$

$$+ \ldots\ldots + (-1)^n \sum n(A_1 \cap A_2 \cap \ldots \cap A_n)$$

DE-ARRANGEMENT

- A derangement of 1, 2, ... , n is a permutation of the numbers such that no number occupies its natural position.
- $D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!}\right]$

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Chapter 8

Binomial Theorem

PASCAL'S TRIANGLE

$[x+y]^1$	$x+y$
$[x+y]^2$	$x^2+2xy+y^2$
$[x+y]^3$	$x^3+3x^2y+3xy^2+y^3$
$[x+y]^4$	$x^4+4x^3y+6x^2y^2+4xy^3+y^4$
$[x+y]^5$	$x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5$
$[x+y]^6$	$x^6+6x^5y+15x^4y^2+20x^3y^3+15x^2y^4+6xy^5+y^6$

Coefficient

$^1C_0$ (1) (1) $^1C_1$

$^2C_0$ (1) (2) $^2C_1$ (1) $^2C_2$

$^3C_0$ (1) (3) $^3C_1$ (3) $^3C_2$ (1) $^3C_3$

1 4 6 4 1

1 5 10 10 5 1

1 6 15 20 15 6 1

BINOMIAL THEOREM (FOR NATURAL INDEX)

$$(x+y)^n = {}^nC_0x^n + {}^nC_1x^{n-1}y + {}^nC_2x^{n-2}y^2 + \dots {}^nC_ny^n$$

Some Important Expansions

(i) $(x-y)^n = {}^nC_0\,x^n - {}^nC_1\,x^{n-1} + C_2\,x^{n-2}y^2 - \dots + (-1)^n\,{}^nC_ny^n$

(ii) $(1+x)^n = {}^nC_0 + {}^nC_1\,x + {}^nC_2\,x^2 + \dots + {}^nC_nx^n$

(iii) $(1-x)^n = {}^nC_0 - {}^nC_1\,x + {}^nC_2\,x^2 + \dots + (-1)^n.\,{}^nC_n\,x^n$

ex → 1. $(1+x)^{20} = {}^{20}C_0 + {}^{20}C_1\,x + {}^{20}C_2\,x^2 + {}^{20}C_3x^3 + \dots + {}^{20}C_{19}\,x^{19} + {}^{20}C_{20}\,x^{20}$

ex → 2. $(1-3x)^{15} = {}^{15}C_0 - {}^{15}C_1\,(3x) + {}^{15}C_2\,(3x)^2 - {}^{15}C_2(3x)^3 + \dots - {}^{15}C_{15}\,(3x)^{15}$

3. $(x+1)^{20} = {}^{20}C_0x^{20} + {}^{20}C_1\,x^{19} + {}^{20}C_2\,x^{18} + {}^{20}C_3x^{17} + \dots + {}^{20}C_{20}$

Important Note

Note 1: Number of terms in the expansion of $(x+y)^n$ is $(n+1)$ i.e.

Note 2: Binomial Expansion is a Homogeneous expansion i.e the sum of indices of x and y for each and every terms is n.

Note 3: The binomial coefficients of the terms $({}^nC_0, {}^nC_1 \dots)$ equidistant from the beginning and the end are equal i.e. ${}^nC_r = {}^nC_{r-1}$

Ques. Using Theorem, Find the value of $\left(\sqrt{2}+1\right)^6 + \left(\sqrt{2}-1\right)^6$

Ans. $\left(\sqrt{2}+1\right)^6 = {}^6C_0\left(\sqrt{2}\right)^6 + {}^6C_1\left(\sqrt{2}\right)^5 + {}^6C_2\left(\sqrt{2}\right)^4 + {}^6C_3\left(\sqrt{2}\right)^3 + {}^6C_4\left(\sqrt{2}\right)^2 + {}^6C_5\left(\sqrt{2}\right)^1 + {}^6C_6$

$\left(\sqrt{2}-1\right)^6 = {}^6C_0\left(\sqrt{2}\right)^6 - {}^6C_1\left(\sqrt{2}\right)^5 + {}^6C_2\left(\sqrt{2}\right)^4 - {}^6C_3\left(\sqrt{2}\right)^3 \dots + {}^6C_6$

$\left(\sqrt{2}+1\right)^6 + \left(\sqrt{2}-1\right)^6 = 2[{}^6C_0\left(\sqrt{2}\right)^6 + {}^6C_2\left(\sqrt{2}\right)^4 + {}^6C_4\left(\sqrt{2}\right)^2 + {}^6C_6]$

$= 2\,[2^3 + 15.2^2 + 15.(2) + 1]$

$= 2\,[8 + 60 + 30 + 1]$

$= 99 \times 2 = 198$

Ques. If the 4th term in the expansion $\left(px + \frac{1}{x}\right)^m$ is $\frac{5}{2}\ \forall x \in R$ then find the value of m and p.

Ans. $\left(px+\frac{1}{x}\right)^m = \underbrace{{}^mC_0\,(px)^m}_{T_1} + \underbrace{{}^mC_1\,(px)^{m-1}\left(\frac{1}{x}\right)}_{T_2} + \underbrace{{}^mC_2\,(px)^{m-2}\left(\frac{1}{x}\right)^2}_{T_3} + \underbrace{{}^mC_3\,(px)^{m-3}\left(\frac{1}{x}\right)^3}_{T_4}$

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$T_4 = {}^mC_3 (px)^{m-3} \frac{1}{x^3} = \frac{5}{2} \forall x \in R$

$\Rightarrow {}^mC_3 p^{m-3} \frac{x^{m-3}}{x^3} = \frac{5}{2}$ for all $x \in R$

$\Rightarrow m - 3 = 3$

$\Rightarrow m = 6$

${}^mC_3 (p)^{m-3} = \frac{5}{2}$

$\Rightarrow {}^6C_3 (p)^{6-3} = \frac{5}{2}$

$\Rightarrow \frac{6 \times 5 \times 4}{3!} (p)^3 = \frac{5}{2}$

$\Rightarrow 5 \times 4\, p^3 = \frac{5}{2}$

$\Rightarrow p^3 = \frac{1}{8} \Rightarrow p = \frac{1}{2}$

GENERAL TERM

The general term or the $(r + 1)^{th}$ in the expansion of $(x + y)^n$ is given by $T_{r+1} = {}^nC_r x^{n-r} y^r$

General terms for the expansion of $(1 + x)^n$ is given by

$T_{r+1} = {}^nC_r x^r$

Ex → $\left(2x + \frac{7}{y}\right)^{10}$ Find 5^{th} terms

$T_{r-1} = {}^{10}C_r (2x)^{10-r} \left(\frac{7}{y}\right)^r$

Put $r = 4$

$T_5 = {}^{10}C_4 (2x)^6 \left(\frac{7}{y}\right)^4$

Ques. The coefficient Of x^{-6} in the expansion of $\left(\frac{4x}{5} + \frac{5}{2x^2}\right)^9$, is ..... .

[JEE Main 31 Jan. 2023 Shift II]

Ans. $T_{r+1} = {}^nC_r (x)^{n-r} (y)^r$

$= {}^9C_r \left(\frac{4x}{5}\right)^{9-r} \left(\frac{5}{2x^2}\right)^r$

$= {}^9C_r \left(\frac{4}{5}\right)^{9-r} \left(\frac{5}{2}\right)^r \frac{x^{9-r}}{x^{2r}}$

$T_6 = {}^9C_5 \left(\frac{4}{5}\right)^4 \frac{5^5}{2^5} \cdot x^{9-3.5}$

put $9 - 3r = -6$

$\Rightarrow 15 = 3r \Rightarrow r = 5$

Coeff $= {}^9C_5 \left(\frac{4}{5}\right)^4 \left(\frac{5}{2}\right)^5$

$= \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2} \cdot \left(\frac{4}{5}\right)^4 \frac{5^5}{2^5}$

$= 9 \times 7 \times 2 \times 5 \cdot \frac{2^8}{2^5} = 9 \times 7 \times 10 \times 2^3$

$= 63 \times 10 \times 8 = 5040$

Ques. The natural number m, for which the coffiecent of x in binomial expansion of $\left(x^m + \frac{1}{x^2}\right)^{22}$ is 1540 is ..... **[JEE Main Sept. 2020]**

Ans. $T_{r+1} = {}^{22}C_r (x^m)^{22-r} \left(\frac{1}{x^2}\right)^r$

$= {}^{22}C_r \frac{(x)^{m(22-r)}}{x^{2r}}$

$= {}^{22}C_r x^{m(22-r)-2r}$

$\because 1540 = {}^{22}C_r$

$\Rightarrow {}^{22}C_3 = {}^{22}C_r$

and

$m(22 - r) - 2r = 1$

$r = 3$

$m(22 - r) - 2r = 1$

$\Rightarrow m(19) - 6 = 1$

$\Rightarrow 19m = 7$

$\Rightarrow m = \frac{7}{19}$

$m \in N$

or $r = 19$

$m(22 - r) - 2r = 1$

$\Rightarrow m(3) - 38 = 1$

$\Rightarrow 3m = 39$

$\Rightarrow m = 13$

Ques. If the fourth terms in the expansion of $(x + x\log_2 x)^7$ is 4480, then the value of x where $x \in N$ is equal to: **[JEE Main March 2021]**

(a) 2 (b) 4 (c) 3 (d) 1

Ans. $T_{r+1} = {}^7C_r (x)^{7-r} [x\log_2 x]^r$

$r = 3, T_4 = {}^7C_3 (x)^4 [x\log_2 x]^3$

$\Rightarrow 35 (x)^4 x^3 \log_2 x = 4480$

$\Rightarrow (x)^4 x^3 \log_2 x = 128 = 2^7$

$\Rightarrow x^4 x^3 \log_2 x = 2^4 \cdot 2^3$

$\Rightarrow x = 2$

TERM INDEPENDENT OF x

Term independent of x means that there is a term present which has no exponent of x.

Hence, to find the terms independent of x, find the value of r for which the combined exponent of x is zero.

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Ques. If the term independent of x in the expansion of $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ is 105, then a^2 is equal to:

[JEE Main April 2024 Shift II]

(a) 2 (b) 4 (c) 6 (d) 9

Ans. $T_{r+1} = {}^{10}C_r\left(\sqrt{a}x^2\right)^{10-r}\left(\frac{1}{2x^3}\right)^r$

$= {}^{10}C_r \cdot \frac{\left(\sqrt{a}\right)^{10-r}}{2^r} \cdot \frac{x^{20-2r}}{x^{3r}}$

$= \frac{{}^{10}C_r\left(\sqrt{a}\right)^{10-r}}{2^r} x^{20-5r} \Rightarrow 20 - 5r = 0 \Rightarrow 20 = 5r \Rightarrow r = 4$

$T_5 = \frac{{}^{10}C_4\left(\sqrt{a}\right)^6}{2^4} = \frac{10\times9\times8\times7}{4\times3\times2\times2^4} \cdot (a)^3 = 105$

$\Rightarrow \frac{10\times21\times a^3}{16} = 105 \Rightarrow \frac{10a^3}{16} = 5$

$\Rightarrow \frac{2a^3}{16} = 1 \Rightarrow a^3 = 8 \Rightarrow a = 2$

Hence, $a^2 = 4$

Ques. If the coefficients of x^4, x^5 and x^6 in the expansion of $(1 + x)^n$ are the arithmetic progression, then the maximum value of n is:

[JEE Main April 2024 Shift II]

(a) 7 (b) 21 (c) 28 (d) 14

Ans. $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \cdots + {}^nC_nx^n$

coeff of x^4: ${}^nC_4$

coeff of x^5: ${}^nC_5$ → AP

coeff of x^6: ${}^nC_6$

$\Rightarrow 2\,{}^nC_5 = {}^nC_4 + {}^nC_6$

$\Rightarrow 2\frac{n!}{5!(n-5)!} = \frac{n!}{4!(n-4)!} + \frac{n!}{6!(n-6)!}$

$\Rightarrow \frac{2}{5(n-5)} = \frac{1}{(n-4)(n-5)} + \frac{1}{6\times5}$

$\Rightarrow \frac{2}{5(n-5)} = \frac{30 + (n-4)(n-5)}{30(n-4)(n-5)}$

$\Rightarrow 2 \times 6(n - 4) = 30 + n^2 - 9n + 20$

$\Rightarrow 12n - 48 = 50 + n^2 - 9n$

$\Rightarrow n^2 - 21n + 98 = 0 \Rightarrow n = 14, 7$

Brain Teaser

Let the coefficients of three consecutive terms in the binomial expansion of $(1 + 2x)^r$ be the ratio 2:5:8. Then the coefficient of the terms, which is in the middle of these three terms, is

Ans. $T_{r+1} = {}^nC_r(2x)^r$

$= {}^nC_r2^r(x)^r$

${}^nC_{r-1}2^{r-1}, {}^nC_r2^r, {}^nC_{r+1}2^{r+1}$

2 : 5 : 8

$\frac{{}^nC_r2^r}{{}^nC_{r-1}2^{r-1}} = \frac{5}{2}$

$\Rightarrow \frac{(n-r+1)}{r}.2^1 = \frac{5}{2}$

$\Rightarrow \frac{(n-r+1)2}{r} = \frac{5}{2}$

$\Rightarrow 4(n - r + 1) = 5r$

$\Rightarrow 4n + 4 = 9r$ ...(i)

$\frac{{}^nC_{r+1}2^{r+1}}{{}^nC_r2^r} = \frac{8}{5}$

$\Rightarrow \left(\frac{n-r}{r+1}\right).2^1 = \frac{8}{5}$

$\Rightarrow \frac{n-r}{r+1} = \frac{4}{5}$

$\Rightarrow 5n - 5r = 4r + 4$

$\Rightarrow 5n = 9r + 4$ ...(ii)

By eqn (i) and (ii)

$\Rightarrow n = 8, r = 4$

MIDDLE TERM

Theorem (i) If n is even, there is only one middle term which is given by $T_{\left(\frac{n}{2}+1\right)}$

Theorem (ii) If n is odd, there are two middle terms which are $T_{(n+1)/2}$ and $T_{[(n+1)/2]+1}$

Ques. Find the middle term is the expansion of $\left(2x^2 - \frac{1}{3x}\right)^{11}$ (11 odd)

Ans. $T_{r+1} = {}^{11}C_r(2x^2)^{11-r}\left(-\frac{1}{3x}\right)^r$

for T_6 Put r = 5

$T_6 = {}^{11}C_5(2x^2)^6\left(-\frac{1}{3x}\right)^5$

For T_7 Put r = 6

$T_7 = {}^{11}C_6(2x^2)^5\left(-\frac{1}{3x}\right)^6$

2 middle terms

$T_{\frac{n+1}{2}}$ & $T_{\frac{n+3}{2}}$

$= T_{\frac{11+1}{2}}, T_{\frac{11+3}{2}}$

$= T_6$ & T_7

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Ques. The middle term in the expansion of $\left(x-\frac{1}{x}\right)^{18}$ is

(a) ${}^{18}C_9$ (b) $-{}^{18}C_9$ (c) ${}^{18}C_{10}$ (d) $-{}^{18}C_{10}$

Ans. $T_{r+1} = 18_{C_r}(x)^{18-r}\left(-\frac{1}{x}\right)^r$ | 1 middle terms

for T_{10}, put r = 9 | $T_{(\frac{18}{2}+1)} = T_{10}$

$$T_{10} = {}^{18}C_9 (x)^9\left(-\frac{1}{x}\right)^9 = -{}^{18}C_9 x^9 \frac{1}{x^9} = -{}^{18}C_9$$

Ques. 5^{th} terms the end in the expansion of $\left(\frac{x^3}{2}-\frac{2}{x^2}\right)^{12}$ is

(a) $-7920x^{-4}$ (b) $x+7920x^4$

(c) $7920x^{-4}$ (d) $x-7920x^4$

Ans. r^{th} term from end for $(x+y)^n$ is same as r^{th} term from beginning for $(y+x)^n$

5^{th} Terms from end $= {}^{12}C_4\left(\frac{-2}{x^2}\right)^{12-4}\left(\frac{x^3}{2}\right)^4$

$$= {}^{12}C_4\frac{(-2)^8}{x^{16}}\frac{x^{12}}{2^4} = \frac{{}^{12}C_4 2^8}{x^4.2^4} = \frac{{}^{12}C_4 2^4}{x^4} = 7920x^{-4}$$

Ques. If in the expansion of $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n$, the ratio of the 7^{th} term from the beginning to the 7^{th} terms from end is equal to 1/6 then n is equqal to

Ans. $T_{r+1} = {}^nC_r\left(2^{1/3}\right)^{n-r}\left(\frac{1}{\sqrt[3]{3}}\right)^r$

put r = 6

$$T_7 = {}^nC_6\left(2^{1/3}\right)^{n-6}\left(\frac{1}{\sqrt[3]{3}}\right)^6 \Rightarrow T_7 = {}^nC_6\left(2^{\frac{1}{3}}\right)^{n-6}\frac{1}{3^2}$$

$$T'_{r+1} = {}^nC_r\left(\frac{1}{\sqrt[3]{3}}\right)^{n-r}\left(\sqrt[3]{2}\right)^r$$

put r = 6

$$T'_7 = {}^nC_6\left(\frac{1}{\sqrt[3]{3}}\right)^{n-6}\left(\sqrt[3]{2}\right)^6 \Rightarrow T'_7 = {}^nC_6\left(\frac{1}{\sqrt[3]{3}}\right)^{n-6}.2^2$$

$$\frac{T_7}{T'_7} = \frac{1}{6} \Rightarrow \frac{{}^nC_6\left(2^{1/3}\right)^{n-6}}{3^2\,{}^nC_6\,2^2}\left(\sqrt[3]{3}\right)^{n-6} = \frac{1}{6}$$

$$\Rightarrow \frac{\left(\frac{1}{2^3}\right)^{n-6}\left(\frac{1}{3^3}\right)^{n-6}}{3^2\,2^2} = \frac{1}{3\times 2}$$

$$\Rightarrow \left(\frac{1}{2^3.3^3}\right)^{n-6} = 6 \Rightarrow (2.3)^{\frac{n-6}{3}} = 6^1$$

$$\Rightarrow (6)^{\frac{n-6}{3}} = 6^1 \Rightarrow \frac{n-6}{3} = 1 \Rightarrow n = 9$$

Important Note

Note 1: The r^{th} term from end is $(T_{n-r+2})^{th}$ term from beginning.

Note 2: Better Approach r^{th} term from end in $(x+y)^n$ is same as r^{th} from beginning in $(y+x)^n$

Ques. In the expansion of $(5^{1/2}+7^{1/8})^{1024}$ the numb if integral term is

(a) 128 (b) 129 (c) 130 (d) 131

Ans. $T^{r+1} = {}^{1024}C_r\left(5^{\frac{1}{2}}\right)^{1024-r}\left(7^{\frac{1}{8}}\right)^r$

for integral terms, 1024 - r is even & r is multiple of 8

$\Rightarrow$ r = multiple of 8

no of terms, r = 0, 8, 16, 24,---, 1024

1024 = 0 +(n - 1).8

$$\Rightarrow \frac{1024}{8} = n-1 \Rightarrow 128 = n-1 \Rightarrow n = 129$$

Ques. The number of integral terms in the expansio of $(3^{1/2}+5^{1/4})^{680}$ is equal to

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Ans. $T_{r+1} = {}^{680}C_r\left(3^{\frac{1}{2}}\right)^{680-r}\left(5^{\frac{1}{4}}\right)^r$

680 - r = even & r $\rightarrow$ multiple of 4

$\Rightarrow$ r = even & r$\rightarrow$multiple of 4

$\Rightarrow$ r = 4k

r = 0 4, 8, 12,---- 680 (680 = 4 × 170)

Hence, total terms = 171

Important Note

Note 1: $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$

Note 2: $\frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n-r}{r+1}$

Note 3: In the expansion $(1+x)^n$

$$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r}|x| \Rightarrow \frac{T_{r+1}}{T_r} = \frac{n-r+1}{r}|x|$$

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NUMERICALLY GREATEST TERM

Ques. Find numerically greatest term in the expansion of $(1 + 4x)^8$ if $x = \frac{1}{3}$.

Ans. $(1 + 4x)^8 = {}^8C_0 + {}^8C_1 4x + {}^8C_2 (4x)^2 + \ldots + {}^8C_8 (4x)^8$

Consider:

$$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r}|4x|$$

$x = \frac{1}{3}, n = 8$

$$\Rightarrow \frac{T_{r+1}}{T_r} = \frac{8-r+1}{r}\cdot\left(\frac{4}{3}\right)$$

$$\Rightarrow \frac{T_{r+1}}{T_r} = \frac{4(9-r)}{3r}$$

For Numerically greatest term,

$$\frac{T_{r+1}}{T_r} = \frac{4(9-r)}{3r} \geq 1$$

$\Rightarrow 36 - 4r \geq 3r \Rightarrow 36 \geq 7r \Rightarrow 7r \leq 36 \Rightarrow r \leq \frac{36}{7}$

$r_{max} = 5$

Numerically greatest term = $T_{5+1} = T_6$

$$= {}^8C_5\left(\frac{4}{3}\right)^5$$

Ques. Find the greatest term in the expansion of $(3 - 2x)^9$ when $x = 1$.

Ans. $$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r}\left|\frac{-2x}{3}\right|$$

$\because n = 9$ & $x = 1$

$$\frac{T_{r+1}}{T_r} = \left(\frac{9-r+1}{r}\right)\left(\frac{2}{3}\right)$$

$$\Rightarrow \frac{T_{r+1}}{T_r} = \frac{2}{3}\left(\frac{10-r}{r}\right) \geq 1$$

$\Rightarrow 2(10 - r) \geq 3r \Rightarrow 20 \geq 5r \Rightarrow r \leq 4$

$r = 4, \frac{T_5}{T_4} = \frac{2}{3}\left(\frac{6}{4}\right) = 1$

$\Rightarrow \frac{T_5}{T_4} = 1 \Rightarrow T_5 = T_4$

$r = 5, \frac{T_6}{T_5} < 1 \Rightarrow T_6 < T_5$

$T_1 - T_2 + T_3 - T_4 + T_5 \ldots\ldots$

Numerically greatest $\rightarrow T_4 = T_5 \rightarrow$ both

Greatest term (with sign) $\rightarrow T_5$ only

Ques. The largest terms in the expansion of $(3 + 2x)^{50}$, where, $x = \frac{1}{5}$.

Ans. $$(3 + 2x)50 = 3^{50}\left[1 + \frac{2x}{3}\right]^{50}$$

Considers $$\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r}\cdot\left|\frac{2x}{3}\right|$$

$\because n = 50, x = 1/5$

$$\Rightarrow \frac{T_{r+1}}{T_r} = \left(\frac{51-r}{r}\right)\frac{2}{15} \geq 1$$

$\Rightarrow 2(51 - r) \geq 15r \Rightarrow 102 \geq 17r$

$\Rightarrow r \leq \frac{102}{17} \Rightarrow r \leq 6$

$r = 6, \frac{T_7}{T_6} = 1 \Rightarrow T_7 = T_6$

$T_{r+1} = {}^{50}C_r(3)^{50-r}(2x)^r$

2 largest terms, T^6 and T_7

$$T_6 = T_7 = {}^{50}C_6 (3)^{44}\left(\frac{2}{5}\right)^6$$

Ques. Find the remainder when

1. 5^{99} is divided by 8

Ans. $5^{99} = (1 + 4)^{99}$

$= {}^{99}C_0 + {}^{99}C_1(4) + {}^{99}C_2(4)^2 + {}^{99}C_3(4)^3 + \ldots {}^{99}C_{99}4^{99}$

$$\frac{1+99\times4+16I}{8} = \frac{1+99\times4}{8} = \frac{397}{8} = \frac{49\times8+5}{8}$$

Remainder = 5

2. 5^{99} is divided by 13

Ans. $5^{99} = 5.5^{98} = 5.(25)^{49} = 5\,[26 - 1]^{49}$

$= 5[{}^{49}C_0(26)^{49} - {}^{49}C_1(26)^{48} + \ldots\ldots + 1]$

$$= \frac{5(26I-1)}{13} = \frac{5\times26I-5}{13}$$

Remainder = $-5 + 13 = 8$

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3. 3^{2033} is divided by 28

Ans. $3^{2023} = 3 \times 3^{2022} = 3 \times (3^3)^{674} = 3 \times (27)^{674}$

$$= \frac{3.(28-1)^{674}}{28} = \frac{3[28I+1]}{28} = \frac{3 \times 28I + 3}{28}$$

Remainder = 3

Ques. Find last 2 digits of $(3)^{400}$

Ans. $(1 \pm 10k)^n \rightarrow$ Break

$(3^2)^{200} = (9)^{200} = (10-1)^{200}$

$= {}^{200}C_0(10)^{200} - {}^{200}C_1(10)^{199} + {}^{200}C_2(10)^{198} -$

$\ldots -200 \times 10\, C_{199}(10)^1 + {}^{200}C_{200}(1)$

$= 1000I + 1$

Hence, last 2 digits = 01

Problem on GIF & Fractional Function

Add or Subtract the two equations depending on how irrational terms get cancelled.

Ques. If $X = \left(3+\sqrt{7}\right)^n$, $n \in N$. Show that [X] is always an odd Integer & hence prove that $X(1-\{X\}) = 2^n$

Ans. $X = \left(3+\sqrt{7}\right)^n \nearrow I \; \searrow f$

Consider $I + f = \left(3+\sqrt{7}\right)^n$

$= {}^nC_0 3^n + {}^nC_1 3^{n-1}\left(\sqrt{7}\right)^1 + {}^nC_2 3^{n-2}\left(\sqrt{7}\right)^2 + \ldots$ ....(i)

$\left(3-\sqrt{7}\right)^n \approx (0.4)^n \Rightarrow 0 < \left(3-\sqrt{7}\right)^n < 1 \Rightarrow 0 < f' < 1$

$f' = \left(3-\sqrt{7}\right)^n$

$= {}^nC_0 3^n - {}^nC_1 3^{n-1}\left(\sqrt{7}\right)^1 + {}^nC_2 3^{n-2}\left(\sqrt{7}\right)^2$

$- {}^nC_3 3^{n-3}\left(\sqrt{7}\right)^3 + \ldots$ ....(ii)

Equation (i) + (ii),

$I + f + f' = 2[{}^nC_0 3^n + {}^nC_2 3^{n-2}\left(\sqrt{7}\right)^2$

$+ {}^nC_4 3^{n-4}\left(\sqrt{7}\right)^4 + \ldots.]$

$\Rightarrow I + f + f'$ = Even Integer

$\Rightarrow f + f'$ = Even Integer - I $\Rightarrow f + f'$ = Integer

We know

$0 < f' < 1$

$0 < f < 1$

$0 < f + f' < 2$

$\Rightarrow f + f' = 1 \Rightarrow f' = 1 - f$

$I + f + f'$ = Even

$\Rightarrow I + 1$ = Even

$\Rightarrow I$ = odd, Hence proved

To Prove: $X(1-\{x\}) = 2$

LHS: $(I+f)(1-f)$

$= (I+f)f'$

$= \left(3+\sqrt{7}\right)^n \cdot \left(3-\sqrt{7}\right)^n$

$= \left[\left(3+\sqrt{7}\right)\left(3-\sqrt{7}\right)\right]^n$

$= \left[3^2 - \left(\sqrt{7}\right)^2\right]^n$

$= (2)^n$

Ques. Let $R = \left(5\sqrt{5}+11\right)^{2n+1}$ and $f = R - [R]$ where [ ] denotes the greatest integer function. The value of R.f is 4^{2n+1}

Ans. $I + f = \left(5\sqrt{5}+11\right)^{2n+1} = {}^{2n+1}C_0\left(5\sqrt{5}\right)^{2n+1} +$

${}^{2n+1}C_1\left(5\sqrt{5}\right)^{2n}.(11)^1 + {}^{2n+1}C_2\left(5\sqrt{5}\right)^{2n-1}(11)^2 + \ldots$

$f' = \left(5\sqrt{5}-11\right)^{2n+1} = {}^{2n+1}C_0\left(5\sqrt{5}\right)^{2n+1} -$

${}^{2n+1}C_1\left(5\sqrt{5}\right)^{2n}(11)^1 + {}^{2n+1}C_2\left(5\sqrt{5}\right)^{2n-1}(11)^2 - \ldots$

$\ominus$

$I + f - f'$ = Even

$\Rightarrow f - f'$ = Even Integer - I

$\Rightarrow f - f'$ = Integer

we know,

$0 < f < 1$,

$0 < f' < 1 \Rightarrow -1 < -f' < 0$

$\Rightarrow -1 < f - f' < 1$

$\Rightarrow f - f' = 0 \Rightarrow f = f' \Rightarrow I$ = Even Integer

Now, the value of Rf =

$\left(5\sqrt{5}+11\right)^{2n+1}.\left(5\sqrt{5}-11\right)^{2n+1}$

$= (125-121)^{2n+1} = 4^{2n+1}$

Important Note

Note 1: If we add $I + f$ & f' to cancel out irrational terms then always $I + f + f'$ = even & $f + f'$ = 1

Note 2: If we subtract $I + f$ & f' to cancel out the irrational terms then always $I + f - f'$ = even & $f - f' = 0$

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Ques. Let $x = \left(2+\sqrt{3}\right)^n$, then the value of $x - x^2 + x[x]$, where [.] denotes the greatest integer function, is equal to

(a) 1 (b) 2

(c) 2^{2n} (d) 2^n

Ans. $I + f = \left(2+\sqrt{3}\right)^n = {}^nC_0\, 2^n + {}^nC_1\, 2^{n-1}\left(\sqrt{3}\right)^1 + \ldots$

$f' = \left(2-\sqrt{3}\right)^n = {}^nC_0\, 2^n - {}^nC_1\, 2^{n-1}\left(\sqrt{3}\right)^1 + \ldots$

$+$

$I + f + f' = \text{Even}$ & $f + f' = 1$

Now, $x - x^2 + x[x]$

$= x - x(x - [x]) = x - xf$

$= x(1 - f)$

$= \left(2+\sqrt{3}\right)^n\left(2-\sqrt{3}\right)^n = (4-3)^n = 1$

Ques. If $\left(9+\sqrt{80}\right)^n = I + f$ where I, n are Natural numbers and $0 < f < 1$, then

(a) I is an odd integer

(b) I is an even integer

(c) $(I + f)(1 - f) = 1$

(d) $1 - f = \left(9-\sqrt{80}\right)^n$

Ans. $I + f = \left(9+\sqrt{80}\right)^n$

$f' = \left(9-\sqrt{80}\right)^n$

$+$

$I + f + f' = \text{Even}$ & $f + f' = 1$

$\Rightarrow I = \text{odd}$

$\Rightarrow I + f = \left(90+\sqrt{80}\right)^n$

and $1 - f = f' = \left(90-\sqrt{80}\right)^n$

Hence, $(I + f)(1 - f) = (81 - 80)^n = 1$

Ques. Let $x = \left(8\sqrt{3}+\sqrt{13}\right)^{13}$, and $y = \left(7\sqrt{2}+\sqrt{13}\right)^9$. If [t] denotes the greatest integer $\le t$, then **[JEE Main Jan. 2023 Shift II]**

(a) [x] + [y] is even

(b) [x] is odd but [y] is even

(c) [x] is even but [y] is odd

(d) [x] and [y] are both odd

Ans. For x:

$I + f = \left(8\sqrt{3}+\sqrt{13}\right)^{13}$

$\ominus$ $f' = \left(8\sqrt{3}-\sqrt{13}\right)^{13}$

$I + f - f' = \text{Even}$ & $f - f' = 0$

$\Rightarrow I = \text{Even} \Rightarrow [x] = \text{Even}$

For y:

$I + f = \left(7\sqrt{2}+9\right)^9$

$f' = \left(7\sqrt{2}-9\right)^9$

$-$

$I + f - f' = \text{Even}$ & $f - f' = 0$

$\Rightarrow I = \text{Even}$

Hence, [x] + [y] is even

Brain Teaser

Let $x = \left(3\sqrt{6}+7\right)^{89}$. If {x} denotes the fractional part of 'x' then find the remainder when $x\{x\} + (x\{x\})^2 + (x\{x\})^3$ is divided by 31.

Ans. $I + f = \left(3\sqrt{6}+7\right)^{89}$

$f' = \left(3\sqrt{6}-7\right)^{89}$

$-$

$I + f - f' = \text{Even}$ & $f - f' = 0$

$\Rightarrow I = \text{Even}$ & $f = f'$

$xf = \left(3\sqrt{6}+7\right)^{89}\left(3\sqrt{6}-7\right)^{89}$

$= (54 - 49)^{89} = (5)^{89}$

Now, $xf + (xf)^2 + (xf)^3$

$= (5)^{89} + (5^{89})^2 + (5^{89})^3$

$= \dfrac{5^{89} + 5^{178} + 5^{267}}{31}$

$= \dfrac{5^2.125^{29} + 5.125^{59} + (125)^{89}}{31}$

$= \dfrac{25(1+31I) + 5(1+31I) + (1+31I}{31}$

$= \dfrac{25+5+1}{31} = 1$

Hence, Remainder is 0.

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PROPERTIES OF BINOMIAL COEFFICIENTS

1. Sum of all the binomial coefficients in $(1 + x)^n$ is equal 2^n i.e.

$C_0 + C_1 + C_2 + C_3 \ldots\ldots C_n = 2^n$

Proof: $\rightarrow (1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \ldots\ldots + {}^nC_nx^n$

Put $x = 1$, $2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots\ldots + {}^nC_n$

2. Sum of Even coefficient equals the sum of Odd coefficient and is equal to 2^{n-1}

$C_0 + C_2 + C_4 + \ldots\ldots = C_1 + C_3 + C_5 + \ldots\ldots = 2^{n-1}$

Proof:

Put $x = -1$

$(1 - 1)^n = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 \ldots\ldots = 0$

${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots\ldots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots\ldots = x$

$\Rightarrow x + x = 2^n \Rightarrow x = \frac{2n}{2} = 2^{n-1}$

Ex: Sum of all coefficient

$(1 - 3x)^n = {}^{10}C_0 - {}^{10}C_1(3x) + {}^{10}C_2(3x)^2 - {}^{10}C_3(3x)^3$ .....

Put $x = 1$

Sum $= [1 - 3 \times 1]^{10} = (-2)^{10} = (2)^{10}$

Ex: $(2x - 3y)^{20} = {}^{20}C_0(2x)^{20} - {}^{20}C_1(2x)^{19}(3y)^1 + \ldots$

Put $x = 1, y = -1$

$\Rightarrow$ Sum of all coeff $= (2 + 3)^{20} = (5)^{20}$

To obtain sum of all coeff Put all variables equal to 1

Ques. The value of ${}^{10}C_2 + {}^{10}C_4 + {}^{10}C_6 + {}^{10}C_8 + {}^{10}C_{10}$ is.

(a) 2^{10} (b) 2^9 (c) 2^8 (d) 5^{11}

Ans. $[{}^{10}C_0 + {}^{10}C_2 + {}^{10}C_4 + \ldots. + {}^{10}C_{10}] - {}^{10}C_0$

$= 2^{n-1} - 1 = 2^{10-1} - 1 = 2^9 - 1 = 512 - 1 = 511$

Ques. The value of $\frac{1}{1!50!} + \frac{1}{3!48!} + \frac{1}{5!46!} + \ldots + \frac{1}{49!2!} + \frac{1}{51!0!}$ is

[JEE Main Feb. 2023 Shift II]

(a) $\frac{2^{50}}{50!}$ (b) $\frac{2^{50}}{51!}$ (c) $\frac{2^{51}}{51!}$ (d) $\frac{2^{51}}{50!}$

Ans. $\frac{1}{1!50!} + \frac{1}{3!48!} + \frac{1}{5!46!} + \ldots + \frac{1}{49!2!} + \frac{1}{51!0!}$

$= \frac{1}{51!}\left[\frac{51!}{1!50!} + \frac{51!}{3!48!} + \frac{51!}{5!46!} + \ldots + \frac{51!}{51!0!}\right]$

$= \frac{1}{51!}\left[{}^{51}C_1 + {}^{51}C_3 + {}^{51}C_5 + \ldots + {}^{51}C_{51}\right]$

$= \frac{1}{51!}\left[2^{51-1}\right] = \frac{1}{51!}.2^{50}$

$1 + 1$

Ques. If $1 + [(2) + {}^{49}C_1 + {}^{49}C_2 + \ldots. {}^{49}C_{49}][{}^{50}C_2 + {}^{50}C_4 + \ldots. {}^{50}C_{50}]$ is equal to $2^n \cdot m$, where m is odd, then $n + m$ is equal to _____.

[JEE Main Jul. 2022 Shift II]

Ans. $1 + [1 + {}^{49}C_0 + {}^{49}C_1 + {}^{49}C_2 + \ldots. + {}^{49}C_{49}]$

$[{}^{50}C_0 + {}^{50}C_2 + \ldots. + {}^{50}C_{50} - 1]$

$= 1 + (1 + 2^{49})(2^{49} - 1)$

$= 1 + (2^{49})^2 - 1^2 = 2^{98} = 2^n.m$

$\because n = 98, m = 1 \Rightarrow n + m = 99$

3. ${}^{n+1}C_{r+1} = \left(\frac{n+1}{r+1}\right){}^nC_r$

proof: ${}^{n+1}C_{r+1} =$

$\frac{(n+1)!}{(r+1)!(n-r)!} = \frac{(n+1)\,n!}{(r+1)r!(n-r)!} = \frac{n+1}{r+1}{}^nC_r$

Example 1: ${}^{100}C_{20} = \frac{100}{20}{}^{99}C_{19}$

Example 2: ${}^{200}C_{40} = \frac{200}{40}{}^{199}C_{39}$

4. ${}^{n+1}C_{r+1} = \left(\frac{n+1}{r+1}\right)\left(\frac{n}{r}\right){}^{n-1}C_{r-1}$

Example: ${}^{100}C_{20} = \frac{100}{20}{}^{99}C_{19} = \frac{100}{20}.\frac{99}{19}.{}^{98}C_{18}$

Ques. If A denotes the sum of all the coefficients in the expansion of $(1 - 3x + 10x^2)^n$ and B denotes the sum of all the coefficients in the expansion of $(1 + x^2)^n$, then:

[JEE Main Jan. 2024 Shift I]

(a) $A = B^3$ (b) $3A = B$

(c) $B = 3^3$ (d) $A = 3B$

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Ans. $A = (1 - 3 + 10)^n = 8^n = 2^{3n}$

$B = (2)^n$

$A = 2^{3n} = (2^n)^3 \Rightarrow A = B^3$

TYPES OF BINOMIAL SERIES

Type 1

Series in which the coefficients of binomial Coefficients are in AP

Type 2

Series in which the coefficients of binomial Coefficients are in HP

Type 3

Series in which the each term is a product of 2 or more binomial Coefficients.

Type 4

Series in which the binomial Coefficients occur with fixed gaps.

Binomial Series Type-1

When coefficients of binomial Coefficients are in A.P.

Ques. Prove that $1.C_1 + 2.C_2 + 3.C_3 + ... + n.C_n = n.2^{n-1}$

Method-1

Ans.

$S = 0.C_0 + 1.C_1 + 2.C_2 + 3.C_3 + ... + nC_n$

$S = n.C_0 + (n-1)C_1 + (n-2)C_2 + ... + 0.C_n$

$\Rightarrow 2S = n.C_0 + n.C_1 + n.C_2 + n.C_3 + ... + nC_n$

$\Rightarrow 2S = n[C_0 + C_1 + C_2 + C_3 + ... + C_n]$

$\Rightarrow 2S = n.2^n$

$\Rightarrow S = \frac{n.2^n}{2^1} = n.2^{n-1}$

Method-2

By differentiation

$(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ... + {}^nC_n x^n$

diff wr.t 'x'

$\Rightarrow n(1 + x)^{n-1} = 0 + {}^nC_1.1 + {}^nC_2.2x + ... + {}^nC_n.n x^{n-1}$

Put $x = 1$

$n\,2^{n-1} = 1.C_1 + 2C_2 + 3.C_3 + ... nC_n$

Ques. $1.C_0 + 2.C_1 + 3.C_2 + ... + (n+1)C_n = 2^{n-1}(n+2)$.

Ans.

$S = 1.C_0 + 2.C_1 + 3.C_2 + ... + (n+1)C_n$

$S = (n+1)C_0 + n.C_1 + {}^{(n-2)}C_2 + ... + 1.C_n$

$\Rightarrow 2S = C_0[n+2] + C_1[n+2] + C_2(n+2) + ... + (n+2)C_n$

$\Rightarrow 2S = (n+2)[C_0 + C_1 + C_2 + ... + C_n]$

$\Rightarrow 2S = (n+2).2^n$

$\Rightarrow S = (n+2)2^{n-1}$

Ques. $1.C_0 + 3.C_1 + 5.C_2 + 7.C_3 + ... + [2n+1].C_n = [n+1]2^n$.

Ans. Consider

$(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ... + {}^nC_n x^n$

In general c.d = d

$x \to x^d$

In given series c.d. = 2

$x \to x^2$

$\Rightarrow (1 + x^2)^n = {}^nC_0 + {}^nC_1 x^2 + {}^nC_2 x^4 + ... + {}^nC_n x^{2n}$

Multipy with x

$\Rightarrow x(1 + x^2)^n = {}^nC_0 + {}^nC_1 x^3 + {}^nC_2 x^5 + ... + {}^nC_n x^{2n+1}$

diff. wrt x

$\Rightarrow x(1 + x^2)^n.1 + x.n(1 + x^2)^{n-1}.2x = 1.C_0 + 3.C_1 x^2 + 5.C_2 x^4 + ... + (2n+1).C_n x^{2n}$

Put $x = 1$

$\Rightarrow 2^n + n.2^{n-1}.2 = 1.C_0 + 3.C_1 + 5.C_2 + ... + (2n+1).C_n$

$= 2^n[1 + n] = RHS$

Ques. $1.2.C_1 + 2.3.C_2 + 3.4.C_3 + ... + n[n+1]C_n = n[n+3]2^{n-2}$.

Ans. $(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + ... + {}^nC_n x^n$

diff. wrt x

$\Rightarrow n(1 + x)^{n-1} = 0 + 1.{}^nC_1 + 2.{}^nC_2 x + 3.{}^nC_3 x^2 + ... + n.{}^nC_n x^{n-1}$

Multipy with x^2

$\Rightarrow n x^2(1 + x)^{n-1} = 1.{}^nC_1 x^2 + 2.{}^nC_2 x^3 + 3.C_3 x^4 + ... + n.C_n x^{n+1}$

diff. wrt x

$\Rightarrow n\,2x(1 + x)^{n-1} + n.x^2(n-1)(1 + x)^{n-2} = 1.2.C_1 x + 2.3.C_2 x^2 + 3.4.C_3 x^3 + ... + n(n+1)C_n x^n$

Put $x = 1$

$\Rightarrow n.2.2^{n-1} + n.(n-1).2^{n-2} = S$

$\Rightarrow n.2^{n-2}[2.2 + n - 1] = S$

$\Rightarrow n.2^{n-2}(n+3) = S$

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Ques. $3C_0 - 8C_1 + 13C_2 - 18C_3 + \ldots$ upto $[n+1]$ terms $= 0$.

Ans. $(1+x^5)^n = {}^nC_0 + {}^nC_1 x^5 + {}^nC_2 x^{10} + {}^nC_3 x^{15} + \ldots + {}^nC_n x^{5n}$

Multiply with x^3

$\Rightarrow x^3(1+x^5)^n = {}^nC_0 x^3 + {}^nC_1 x^8 + {}^nC_2 x^{13} + \ldots$

differentiate wrt x

$\Rightarrow 3x^2(1+x^5)^n + x^3 n(1+x^5)^{n-1} 5x^4 = 3C_0 x^2 + 8C_1 x^7 + 13C_2 x^{12} + \ldots$

Put $x = -1$

$\Rightarrow 0 + 0 = 3.C_0 - 8C_1 + 13.C_2 - 18C_3 \ldots$

$\Rightarrow 0 = S$

Binomial Series Type-2

When coefficients of Binomial Coefficients are in H.P.

Ques. Show that: $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$

Ans. $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots + {}^nC_n x^n$

$$\int_0^1 (1+x)^n dx = \int_0^1 C_0 dx + \int_0^1 C_1 x dx + \int_0^1 C_2 x^2 dx + \ldots + \int_0^1 C_n x^n dx$$

$$\Rightarrow \int_0^1 (1+x)^n dx = C_0 x\Big|_0^1 + C_1 \frac{x^2}{2}\Big|_0^1 + C_2 \frac{x^3}{3}\Big|_0^1 + \ldots + C_n \frac{x^{n+1}}{n+1}\Big|_0^1$$

$$\Rightarrow \frac{(1+x)^{n+1}}{1(n+1)}\Big|_0^1 = C_0 + \underbrace{\frac{C_1}{2} + \frac{C_2}{3} + \ldots + \frac{C_n}{n+1}}_{S}$$

$$\Rightarrow S = \frac{2^{n+1}}{n+1} - \frac{1^{n+1}}{n+1}$$

Ques. Show that:

$$\frac{C_0}{2} + \frac{C_1}{3} + \frac{C_2}{4} + \ldots + \frac{C_n}{n+2} = \frac{n.2^{n+1}+1}{(n+1)(n+2)}$$

Ans. $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots + {}^nC_n x^n$

$$\int_0^1 x(1+x)^n = \underbrace{\int_0^1 C_0 x + \int_0^1 C_1 x^2 + \ldots + \int_0^1 C_n x^{n+1}}_{S}$$

$$\Rightarrow \int_0^1 x(1+x)^n dx = S \Rightarrow \int_0^1 (x+1-1)(1+x)^n dx = S$$

$$\Rightarrow \int_1^0 (1+x)^{n+1} - \int_0^1 (1+x)^n = S$$

$$\Rightarrow \frac{(1+x)^{n+2}}{n+2}\Big|_0^1 - \frac{(1+x)^{n+1}}{n+1}\Big|_0^1 = S$$

$$\Rightarrow S = \left(\frac{2^{n+2}-1}{n+2}\right) - \left(\frac{2^{n+1}-1}{n+1}\right)$$

$$\Rightarrow s = \frac{n.2^{n+1}+1}{(n+1)(n+2)}$$

Ques. Show that:

$$\frac{C_0}{1} + \frac{C_2}{3} + \frac{C_4}{5} + \ldots = \frac{2^n}{n+1}$$

Ans.

$(1+x)^n = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \ldots + {}^nC_n x^n$

$(1-x)^n = C_0 - C_1 x + C_2 x^2 - C_3 x^3 + \ldots$

$\oplus$

$$\Rightarrow \int_0^1 \frac{(1+x)^n + (1-x)^n}{2} = \underbrace{\int_0^1 C_0 + \int_0^1 C_2 x^2 + \int_0^1 C_4 x^4 + \int_0^1 C_6 x^6 + \ldots}_{S}$$

$$\Rightarrow S = \frac{1}{2}\int_0^1 \left[(1+x)^n + (1+x)^n\right] dx$$

$$\Rightarrow S = \frac{1}{2}\left\{\int_0^1 (1+x)^n dx + \int_0^1 (1+x)^n dx\right\}$$

$$\Rightarrow S = \frac{1}{2}\left[\frac{(1+x)^{n+1}}{n+1}\Big|_0^1 + \frac{(1-x)^{n+1}}{(-1)(n+1)}\Big|_0^1\right]$$

$$\Rightarrow S = \frac{1}{2}\left[\frac{2^{n+1}-1}{n+1} - \left\{\frac{0-1}{n+1}\right\}\right]$$

$$\Rightarrow S = \frac{1}{2}\left[\frac{2^{n+1}-1+1}{n+1}\right] = \frac{1}{2}\frac{2^{n+1}}{n+1} = \frac{2^n}{n+1}$$

Ques. If $\frac{1}{n+1}{}^nC_n + \frac{1}{n}{}^nC_{n-1} + \ldots + \frac{1}{2}{}^nC_1 + {}^nC_0 = \frac{1023}{10}$ then n is equal to

(a) 9 (b) 8 (c) 7 (d) 6

Ans. ${}^nC_0 + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} + \ldots + \frac{{}^nC_n}{n+1} = \frac{2^{n+1}-1}{n+1} = \frac{2^{10}-1}{10}$

$\Rightarrow n+1 = 10 \Rightarrow n = 9$

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Concluding Question

If $S=3.\frac{^{10}C_0}{1}+3^2.\frac{^{10}C_1}{2}+3^3.\frac{^{10}C_2}{3}+\ldots+3^{11}\frac{^{10}C_{10}}{11}$. Find S.

Ans. $\int_0^3(1+x)^{10}=\int_0^3{}^{10}C_0+\int_0^3{}^{10}C_1x+\int_0^3{}^{10}C_2x^2+\int_0^3{}^{10}C_3x^3$

$+\ldots+\int_0^3{}^{10}C_{10}x^{10}$

$=^{10}C_0x\Big|_0^3+\frac{^{10}C_1x^2}{2}\Big|_0^3+\frac{^{10}C_2x^3}{3}\Big|_0^3+\ldots+\frac{^{10}C_{10}x^{11}}{11}\Big|_0^3$

$\Rightarrow\frac{(1+x)^{11}}{11}\Big|_0^3=^{10}C_0.3^1+\frac{^{10}C_13^2}{2}+\frac{^{10}C_23^3}{3}+\ldots+\frac{^{10}C_{10}3^{11}}{11}$

$\Rightarrow\frac{4^{11}-1}{11}=S$

Method-2

Ques. Prove that $1.C_1+2.C_2+3.C_3+\ldots+n.C_n=n.2^{n-1}$

Ans. $T_r=r.{}^nC_r$

$\Rightarrow S_n=\sum_{r=1}^{n}T_r\quad(S_n=\sum_{r=1}^{n}T_r=T_1+T_2+T_3+\cdots+T_n)$

$=\sum_{r=1}^{n}r\,{}^nC_r$

$=\sum_{r=1}^{n}r.\frac{n}{r}{}^{n-1}C_{r-1}\qquad{}^nC_r=\frac{n}{r}{}^{n-1}C_{r-1}$

$=\sum_{r=1}^{n}n\,{}^{n-1}C_{r-1}$

$=n\sum_{r=1}^{n}{}^{n-1}C_{r-1}$

$=n\,[{}^{n-1}C_0+{}^{n-1}C_1+{}^{n-1}C_2+\cdots+{}^{n-1}C_{n-1}]$

$=n.2^{n-1}$

Ques. $1.C_0+2.C_1+3C_2+\ldots+(n+1)C_n=2^{n-1}(n+2)$.

Ans. $T_r=r.C_{r-1}$

$r\to1$ to $n+1$

$S_n=\sum_{r=1}^{n+1}T_r=\sum_{r=1}^{n}r\,C_{r-1}$

$=\sum\left(\{r-1\}+1\right)C_{r-1}$

$\Rightarrow S_n=\sum_{r=1}^{n+1}[(r-1)\,{}^nC_{r-1}+C_{r-1}]$

$=\sum\left[(r-1)\frac{n}{(r-1)}{}^{n-1}C_{r-2}+{}^nC_{r-1}\right]$

$=\sum[n.{}^{n-1}C_{r-2}+{}^nC_{r-1}]$

$=\sum_{r=2}^{n+1}n\,{}^{n-1}C_{r-2}+\sum_{r=1}^{n+1}{}^nC_{r-1}$

$=n\sum_{r=2}^{n+1}{}^{n-1}C_{r-2}+2^n$

$=n\,[{}^{n-1}C_0+C_1+{}^{n-1}C_2+\ldots.{}^{n-1}C_{n-1}]+2^n$

$=(n\,2^{n-1}+2^n)$

Method-3

Ques. Show that: $C_0+\frac{C_1}{2}+\frac{C_2}{3}+\ldots..+\frac{C_n}{n+1}=\frac{2^{n+1}-1}{n+1}$

Ans. $S_n=\sum_{r=0}^{n}\frac{^nC_r}{(r+1)}$

$\because\frac{1}{(n+1)}{}^{n+1}C_{r+1}=\frac{1}{r+1}.{}^nC_r$

$=\frac{1}{n+1}\sum_{r=0}^{n}{}^{n+1}C_{r+1}$

$=\frac{1}{n+1}[{}^{n+1}C_1+{}^{n+1}C_2+{}^{n+1}C_3+\ldots{}^{n+1}C_{n+1}]$

$=\frac{1}{n+1}\left(2^{n+1}-1\right)$

Binomial Series Type-3

When each term f the series is expressed as a product of two or more Binomial Coefficent.

TYPE-3 (A) When difference of lower suffixes is constant in each terms.

Step 1 write $(1+x)^n$

Step 2replce $x\to\frac{1}{x}$

Step 3 coeff of x^d in product $(1+x)^n\left(1+\frac{1}{x}\right)$

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Ques. $C_0^2 + C_1^2 + C_2^2 + \ldots + C_n^2 = \frac{(2n)!}{n!n!} = {}^{2n}C_n$

Ans. $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \ldots + C_nx^n$

$$\left(1+\frac{1}{x}\right)^n = C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \frac{C_3}{x^3} + \ldots + \frac{C_n}{x^n}$$

$(1+x)^n\left(1+\frac{1}{x}\right)^n \rightarrow$ coeff of x^0 in this product is our Ans

Ans: coeff of x^0 is $\frac{(1+x)^{2n}}{x^n}$

= coeff of x^n in $(1+x)^{2n}$

$= {}^{2n}C_n$

Ques. $C_0C_2 + C_1C_3 + C_2C_4 + \ldots + C_{n-2}C_n = \frac{(2n)!}{(n+2)!(n-2)!}$

d = 2

Ans. Coeff of x^2 in the product $\frac{(1+x)^{2n}}{x^n}$ is our Ans

$\Rightarrow$ Coeff of x^{n+2} in $(1+x)^{2n}$

$= {}^{2n}C_{n+2}$ or ${}^{2n}C_{n-2}$

TYPE-3(B) When sum of lower suffxes is constant in each term.

Trick method:

${}^mC_r\,{}^nC_0 + {}^mC_{r-1}\,{}^nC_1 + {}^mC_{r-2}\,{}^nC_2 + \ldots {}^mC_0\,{}^nC_r = {}^{m+n}C_r$

m boys, m girls → r student select → ${}^{m+n}C_r$

Ques. Prove that:

${}^mC_r\,{}^nC_0 + {}^mC_{r-1}\,{}^nC_1 + {}^mC_{r-2}\ldots + {}^mC_0\,{}^nC_r = {}^{m+n}C_r$

Ans. sum = r

$(1+x)^m = {}^mC_0 + {}^mC_1x + {}^mC_2x^2 + \ldots {}^mC_rx^r + \ldots + {}^mC_{m-1}x^{m-1} + {}^mC_mx^m$

$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_1x + {}^nC_2x^2 + \ldots + {}^4C_rx^r + \ldots {}^nC_nx^n$

coeff of x^r in the product $(1+x)^m(1+x)^n$

= coeff of x^r in $(1+x)^{m+n} = {}^{m+n}C_r$

Ques. The value of $\sum_{r=0}^{6} ({}^6C_r\,{}^6C_{6-r})$ is equal to:

(a) 1124 (b) 1324 (c) 1024 (d) 924

Ans. ${}^6C_0\,{}^6C_6 + {}^6C_1\,{}^6C_5 + {}^6C_2\,{}^6C_4 + \ldots + {}^6C_6\,{}^6C_0$

sum = 6

6 boys, 6 girls → Select 6 students

$= {}^{12}C_6 = 924$

Ques. The value $\sum_{r=0}^{22} {}^{22}C_r\,{}^{23}C_r$ is

[JEE Main Jan. 2023 Shift

(a) ${}^{45}C_{23}$ (b) ${}^{44}C_{23}$ (c) ${}^{45}C_{24}$ (d) ${}^{44}C_{22}$

Ans. $\sum_{r=0}^{22} {}^{22}C_r\,{}^{23}C_r = \sum_{r=0}^{22} {}^{22}C_r\,{}^{23}C_{23-r}$

$= {}^{22}C_0\,{}^{23}C_{23} + {}^{22}C_1\,{}^{23}C_{22} + {}^{22}C_2\,{}^{23}C_{21} + \ldots + {}^{22}C_{22}\,{}^{23}C_1$

22 boys, 23 girls → Select 23 students

$= {}^{45}C_{23}$

Ques. If $\sum_{k=1}^{31} ({}^{31}C_k)({}^{31}C_{k-1}) - \sum_{k=1}^{31} ({}^{30}C_k)({}^{30}C_{k-1}) = \frac{\alpha(60!)}{(30!)(31!}$

Where $\alpha \in R$. Then the value of 16α is equal

[JEE Main June 2022 Shift

(a) 1411 (b) 1320 (c) 1615 (d) 1855

Ans. $\sum_{k=1}^{31} {}^{31}C_k\,{}^{31}C_{32-k}$ $\left(\because {}^nC_r = {}^nC_{n-r}\right)$

$= {}^{31}C_1\,{}^{31}C_{31} + {}^{31}C_2\,{}^{31}C_{30} + \ldots + {}^{31}C_{31}\,{}^{31}C_1 = {}^{62}C_{32}$

31 B, 31 G → 32 students

$\sum_{k=1}^{30} {}^{30}C_k\,{}^{30}C_{31-k} = {}^{60}C_{31}$

30 B, 30 G → 31 students

$= {}^{62}C_{32} - {}^{60}C_{31}$

$= \frac{62!}{32!30!} - \frac{60!}{31!29!}$

$= \frac{60!}{31!}\left[\frac{62\times61}{32.30!} - \frac{30.}{30.29!}\right]$

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$$= \frac{60!}{31!30!}\left[\frac{31\times 61}{16} - 30\right] = \frac{\alpha(60!)}{(30!)(31!)}$$

$$\alpha = \left[\frac{31\times 61}{16} - 30\right] \Rightarrow \alpha = \frac{31\times 61 - 480}{16}$$

$\Rightarrow 16\alpha = 31 \times 61 - 480$

$\Rightarrow 16\alpha = 1411$

Misc. Series

Ques. $C_1^2 + 2C_2^2 + 3.C_3^2 + \ldots + n.C_n^2 = \dfrac{(2n-1)!}{\{(n-1)!\}^2}$

Method-1

Ans. Consider, $(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \ldots + C_nx^n$

Differentiate w.r.t. x, we get

$n(1 + x)^{n-1} = 1.C_1 + 2.C_2x + 3.C_3x^2 + \ldots + n.C_nx^{n-1}$

$$x \to \frac{1}{x}\left(1+\frac{1}{x}\right)^n = C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \frac{C_3}{x^3} + \ldots + \frac{C_1}{x^n}$$

$1.C_1^2 + 2.C_2^2 + 3.C_3^2 + \ldots + n.C_n^2$

= coefficient of x^{-1} in $n(1 + x)^{n-1}\left(1+\frac{1}{x}\right)^n$

= coefficient of x^{-1} in

$$\frac{n(1+x)^{n-1}.(1+x)^n}{x^n} = \frac{n(1+x)^{2n-1}}{x^n}$$

= coefficient of x^{-1} $\dfrac{n(1+x)^{2n-1}}{x^n}$

= coefficient of x^{n-1} in $n(1 + x)^{2n-1}$

$$= n.\ {}^{2n-1}C_{n-1} = n.\ \frac{(2n-1)!}{(n-1)!n!} = \frac{(2n-1)!}{(n-1)(n-1)!}$$

Method-2

Ans. $\Sigma r(C_r)^2 = \sum_{r=1}^{n} r\ {}^nC_r.{}^nC_r$

$$= \sum r\frac{n}{r}\ {}^{n-1}C_{r-1}\ {}^nC_r$$

$$= n\sum {}^{n-1}C_{r-1}\ {}^nC_r$$

$$= n\sum_{r=1}^{n} {}^{n-1}C_{n-r}\ {}^nC_r$$

$= n.{}^{n-1}C_{n-1}\ {}^nC_1 + {}^{n-1}C_{n-2}\ {}^nC_2 + \ldots$

(n-1) boys, n girls → Select n students

$= n.{}^{2n-1}C_n$

$$= n.\frac{(2n-1)!}{(n-1)!n!} = \frac{(2n-1)!}{(n-1)!(n-1)!}$$

Ques. Suppose $\sum_{r=0}^{2023} r^2\ {}^{2023}C_r = 2023 \times \alpha \times 2^{2022}$. Then the value of α is .....

[JEE Main Jan. 2023 Shift I]

Ans. $$\sum r^2\frac{2023}{r}\ {}^{2022}C_{r-1} = 2023\sum(r-1+1)\ {}^{2022}C_{r-1}$$

$$= 2023\left\{\sum_{r=1}^{2023}(r-1)\ {}^{2022}C_{r-1} + \sum_{r=1}^{2023}{}^{2022}C_{r-1}\right\}$$

$$= 2023\left\{\sum_{r=2}^{2023}\frac{2022}{r-1}(r-1)\ {}^{2021}C_{r-2} + 2^{2022}\right.$$

$$= 2023\left[2022\sum_{r=2}^{2023}{}^{2021}C_{r-2} + 2^{2022}\right]$$

$$= 2023\left[2022\times 2^{2021} + 2^{2022}\right]$$

$$= 2023\left[1011\times 2^{2022} + 2^{2022}\right]$$

$$= 2023.2^{2022}\left[1011+1\right] = 2023.2^{2022}(1012)$$

Hence, $\alpha = 1012$

Ques. If $\dfrac{1}{n+1}{}^nC_n + \dfrac{1}{n}{}^nC_{n-1} + \ldots + \dfrac{1}{2}{}^nC_1 + {}^nC_0 = \dfrac{1023}{10}$ then n is equal to

[JEE Main April 2023 Shift I]

(a) 9 (b) 8 (c) 7 (d) 6

Ans. $$\sum_{r=0}^{n}\frac{{}^nC_r}{r+1}$$

$$= \sum_{r=0}^{n}\frac{{}^{n+1}C_{r+1}}{n+1} = \frac{1}{n+1}\left(\sum_{r=0}^{n}{}^{n+1}C_{r+1}\right)$$

$$= \frac{1}{n+1}\left[{}^{n+1}C_1 + {}^{n+1}C_2 + \ldots + {}^{n+1}C_{n-1}\right]$$

$$= \frac{1}{n+1}\left(2^{n+1}-1\right) = \left(\frac{2^{10}-1}{10}\right) \Rightarrow n+1 = 10$$

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Ques. The value of $\binom{50}{0}\binom{50}{1}+\binom{50}{1}\binom{50}{2}+\ldots+\binom{50}{49}\binom{50}{50}$ is, where ${}^{n}C_r=\binom{n}{r}$

(a) $\binom{100}{50}$ (b) $\binom{100}{51}$ (c) $\binom{50}{25}$ (d) $\binom{50}{25}^2$

Ans. ${}^{50}C_0 \cdot {}^{50}C_1 + {}^{50}C_1 \cdot {}^{50}C_2 + \ldots + {}^{50}C_{49} \cdot {}^{50}C_{50}$

$= \sum_{r=0}^{49} {}^{50}C_r \, {}^{50}C_{r+1}$

$= \sum_{r=0}^{49} {}^{50}C_r \, {}^{50}C_{49-r}$

50B
50G → 49 students

$= {}^{100}C_{51}$ or ${}^{100}C_{49}$

Binomial Series Type-5

When binomial Coefficients occur with fixed gaps

Examples:

$C_0 + C_3 + C_6 + C_9 + \ldots$
$C_1 + C_4 + C_7 + C_{10} + \ldots$ → Gap of 3 (use cube root of unity)

$C_0 + C_4 + C_8 + C_{12} + \ldots$
$C_1 + C_5 + C_9 + C_{13} + \ldots$ → Gap of 4 (use 4th root of unity (i))

$$1+i=\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$$

Ques. Find the value of

(i) $C_1 + C_5 + C_9 \ldots\ldots$

(ii) $C_0 + C_4 + C_8 \ldots\ldots$

Ans. $(1+x)^n = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \ldots\ldots + C_n x^n$

Put $x = i$

$(1+i)^n = C_0 + C_1 i + C_2 i^2 + C_3 i^3 + C_4 i^4 + \ldots\ldots + C_n i^n$

$\Rightarrow (1+i)^n = C_0 + C_1 i - C_2 - i C_3 + C_4 + \ldots\ldots$

$\Rightarrow (1+i)^n = (C_0 - C_2 + C_4 - C_6 \ldots) + i(C_1 - C_3 + C_5 - C_7 \ldots)$

$\Rightarrow \left[\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)\right]^n$

$= (C_0 - C_2 + C_4 - C_6 \ldots\ldots) + i(C_1 - C_3 + C_5 - C_7 \ldots\ldots)$

$\Rightarrow \left(\sqrt{2}\right)^n \left[\cos\frac{n\pi}{4} + i\sin\frac{n\pi}{4}\right.$

$= (C_0 - C_2 + C_4 - C_6 \ldots\ldots) + i(C_1 - C_3 + C_5 - C_7 \ldots\ldots)$

(i) $\left(\sqrt{2}\right)^n \sin\frac{n\pi}{4} = C_1 - C_3 + C_5 - C_7 + C_9 \ldots\ldots$

$\oplus \quad 2^{n-1} = C_1 + C_3 + C_5 + C_7 + C_9 \ldots\ldots$

$$\frac{\left(\sqrt{2}\right)^n \sin\left(\frac{n\pi}{4}\right) + 2^{n-1}}{2} = C_1 + C_5 + C_9 + \ldots\ldots$$

$$\left(\sqrt{2}\right)^n \cos\frac{n\pi}{4} = C_0 - C_2 + C_4 - C_6 \ldots\ldots$$

(ii) $\oplus \quad 2^{n-1} = C_0 + C_2 + C_4 + C_6 \ldots\ldots$

$$\frac{\left(\sqrt{2}\right)^n \cos\frac{n\pi}{4} + 2^{n-1}}{2} = [C_0 + C_4 + C_8 + C_{12} + \ldots.]$$

$$\Rightarrow \frac{2^{\frac{n}{2}} \cos n\frac{\pi}{4} + 2^{n-1}}{2} = [C_0 + C_4 + C_8 + C_{12} + \ldots.]$$

Ques. Find the value of $C_0 + C_3 + C_6 + \ldots\ldots$

Ans. $(1+x)^n = C_0 + C_1 x + C_2 + C_3 x^2 + C_3 x^3 + \ldots. C_n x^n$

Put $x = 1$, $2^n = C_0 + C_1 + C_2 + C_3 + C_4 + \ldots. C_n$

Put $x = w$, $(1+w)^n = C_0 + C_1 w + C_2 w^2 + C_3 w^3 + C_4 w^4 + \ldots. C_n w^n$

Put $x = w^2$ $(1+w^2)^n = C_0 + C_1 w^2 + C_2 w^4 + C_3 w^6 + C_4 w^8 + \ldots. + C_n w^{2n}$

After adding these 3 equations we get,

$2^n + (1+w)^n + (1+w^2)^n = 3[C_0 + C_3 + C_6 + C_9 + \ldots]$

$\Rightarrow 2^n + 2\,\mathrm{Re}\,(1+w)^n = 3[C_0 + C_3 + C_6 + C_9 + \ldots\ldots]$

$$\Rightarrow \frac{2^n + 2\cos\frac{n\pi}{3}}{2} = [C_0 + C_3 + C_6 + C_9 + \ldots\ldots]$$

BINOMIAL THEOREM FOR RATIONAL INDEX

$$(x+y)^n = x^n + nx^{n-1}y + \frac{n(n-1)}{2!}x^{n-2}y^2 + \frac{n(n-1)(n-2)}{3!}x^{n-3}y^3 + \ldots$$

This expression is valid only if $\left|\frac{y}{x}\right| < 1 \,\&\, n \in Q$

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- $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots\ldots\infty$

  This expression is valid only if $|x| < 1$

- If x is very small, we can ignore higher power of x

  $\Rightarrow (1+x)^n \approx 1+nx$

Ques. If $|x| < 1$, then $(1+x)^{-2} =$

(a) $1 - 2x + 3x^2 - \ldots$

(b) $1 + 2x + 3x^2 + \ldots$

(c) $1 - \frac{2}{x} + \frac{3}{x^2} - \ldots$

(d) $\frac{1}{x^2} - \frac{2}{x^3} + \frac{3}{x^4} - \ldots$

Ans. $(1+x)^n = 1 + nx + \frac{(n-1)}{2}x^2 + \frac{n(n-1)(n-1)}{3!}x^3 + \ldots\infty$

$n = -2$

$\Rightarrow (1+x)^{-2} = 1 - 2x + \frac{-2(-3)}{2}x^2 - \ldots.$

$= 1 - 2x + 3x^2 - \frac{2(-3)(-4)}{3!}x^3 + \ldots.$

$= 1 - 2x + 3x^2 - 4x^3 + 5x^4 \ldots.$

SOME IMPORTANT EXPANSIONS

- $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \ldots\ldots\infty$
- $(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \ldots\ldots\infty$
- $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots\ldots\infty$
- $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots\ldots\infty$
- $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \ldots\ldots\infty$
- $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \ldots\ldots\infty$

Ques. The coefficient of x^3 in the expansion of $\frac{(1+3x)^2}{1-2x}$ will be

(a) 8 (b) 32

(c) 50 (d) None of these

Ans. $(1+3x)^2 (1-2x)^{-1}$

$= (1 + 9x^2 + 6x)\left(1 + 2x + (2x)^2 + (2x)^3 + \ldots\ldots\right)$

coefficient of $x^3 \Rightarrow 8 + 18 + 24 = 50$

Ques. If x is so small that x^2 & higher powers can be neglected and $\frac{\left(1+\frac{2x}{5}\right)^{-5} + \sqrt{4+2x}}{(4+x)^{\frac{3}{2}}} = a + bx.$

Find (a + b)

Ans. $\frac{\left[1 + \frac{2x}{5}(-5)\right] + (4+2x)^{\frac{1}{2}}}{(4+x)^{\frac{3}{2}}}$ $(\because (1+x)^n \approx 1 + nx)$

$= \frac{(1-2x) + 2\left(1+\frac{2x}{4}\right)^{\frac{1}{2}}}{4^{\frac{3}{2}}\left[1+\frac{x}{4}\right]^{\frac{3}{2}}}$

$= \frac{(1-2x) + 2\left[1 + \frac{2x}{4} \times \frac{1}{2}\right]}{2^3\left[1 + \frac{x}{4}.\frac{3}{2}\right]}$

$= \frac{(1-2x) + 2\left(1+\frac{x}{4}\right)}{8\left[1+\frac{3x}{8}\right]} = \frac{3 - 2x + \frac{x}{2}}{8\left[1+\frac{3x}{8}\right]} = \frac{3 - \frac{3x}{2}}{8\left[1+\frac{3x}{8}\right]}$

$= \frac{3\left(1-\frac{x}{2}\right)}{8}\left(1+\frac{3x}{8}\right)^{-1}$

$= \frac{3}{8}\left(1-\frac{x}{2}\right)\left(1-\frac{3x}{8}\right)$

$= \frac{3}{8}\left[1 - \frac{3x}{8} - \frac{x}{2} + 0\right]$

$= \frac{3}{8}\left[1 - \frac{3x}{8} - \frac{4x}{8}\right]$

$= \frac{3}{8}\left[1 - \frac{7x}{8}\right] = a + bx$

$\therefore a = \frac{3}{8}, b = \frac{-21}{64}$

Hence, $a + b = \frac{3}{8} - \frac{21}{64} = \frac{3}{64}$

Ques. If x is so small that x^3 and higher powers of x may be neglected, then $\frac{(1+x)^{\frac{3}{2}} - \left(1+\frac{1}{2}x\right)^3}{(1-x)^{\frac{1}{3}}}$ may be approximated as

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(a) $\frac{3}{8}x^2$ (b) $\frac{x}{2}-\frac{3}{8}x^2$

(c) $1-\frac{3}{8}x^2$ (d) $3x+\frac{3}{8}x^2$

Ans. $= \dfrac{\left[1+\frac{3x}{2}+\frac{\frac{3}{2}\left(\frac{3}{2}-1\right)}{2}x^2\right]-\left[1+\frac{3x}{2}+\frac{3(3-1)}{2}\left(\frac{x}{2}\right)^2\right]}{1-\frac{x}{2}+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2}(x)^2}$

$$=\frac{\frac{3}{8}x^2-\frac{3x^2}{4}}{1-\frac{x}{2}-\frac{1}{8}x^2}=-\frac{3x^2}{8}\left[1-\frac{x}{2}-\frac{x^2}{8}\right]^{-1}$$

$$=\frac{-3x^2}{8}\left(1-\frac{x}{2}\right)^{1}=\frac{-3x^2}{8}\left[1+\frac{x}{2}\right]=\frac{-3x^2}{8}$$

Ques. Simplify: $S=1-\frac{1}{2}.\frac{1}{2}+\frac{1.3}{2.4.2^2}-\frac{1.3.5}{2.4.6}.\frac{1}{2^3}+.....\infty$

Ans. $(1+x)^n=1+nx+\frac{n(n-1)}{2}x^2+\frac{n(n-1)(n-2)}{3!}x^3+$ .... ∞

Compare RHS

$nx=\frac{-1}{4}\Rightarrow x=\frac{-1}{4n}$ ...(i)

$\frac{n(n-1)x^2}{2}=\frac{3}{32}$

$\Rightarrow \frac{n(n-1)}{2}\frac{1}{16n^2}=\frac{3}{32}$

$\Rightarrow \frac{n-1}{n}=3\Rightarrow n-1=3n$

$\Rightarrow 2n=-1\Rightarrow n=-\frac{1}{2}$

By eqn (i)

$\left(-\frac{1}{2}\right)x=-\frac{1}{4}\Rightarrow x=\frac{1}{2}$

Compare LHS

$S=(1+x)^n$

$=\left(1+\frac{1}{2}\right)^{-\frac{1}{2}}$

$=\left(\frac{3}{2}\right)^{-\frac{1}{2}}=\left(\frac{2}{3}\right)^{\frac{1}{2}}$

$=\sqrt{\frac{2}{3}}$

Ques. $1-\frac{1}{8}+\frac{1}{8}.\frac{3}{16}-\frac{1.3.5}{8.16.24}+....=S$

(a) $\frac{2}{5}$ (b) $\frac{\sqrt{2}}{5}$

(c) $\frac{2}{\sqrt{5}}$ (d) None of t

Ans. $(1+x)^n=1+nx+\frac{n(n-1)}{2}x^2+....$

Compare RHS

$nx=\frac{-1}{8}\Rightarrow x=\frac{-1}{8n}$ ...(i)

$\frac{3}{8.16}=\frac{n(n-1)}{2}x^2$

$\Rightarrow \frac{3}{8.16}=\frac{n(n-1)}{2}.\left(\frac{-1}{8n}\right)^2$

$\Rightarrow \frac{3}{8\times16}=\frac{n(n-1)}{2}.\frac{1}{64n^2}$

$\Rightarrow 3=\frac{n-1}{n}$

$\Rightarrow 3n=n-1$

$\Rightarrow 2n=-1$

$\Rightarrow n=-\frac{1}{2}$

By eqn (i)

$\left(\frac{-1}{2}\right)x=\frac{-1}{8}\Rightarrow x=\frac{1}{4}$

Compare LHS

$S=(1+x)^n$

$=\left(1+\frac{1}{4}\right)^{-\frac{1}{2}}$

$=\left(\frac{5}{4}\right)^{-\frac{1}{2}}$

$=\left(\frac{4}{5}\right)^{\frac{1}{2}}$

$=\sqrt{\frac{4}{5}}=\frac{2}{\sqrt{5}}$

Ques. If $x=\frac{1}{3}+\frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}+.....\infty$, then find the value of x^2+2x

Ans. Add 1 both sides

$(1+x)=1+\frac{1}{3}+\frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}+.....\infty$

$(1+y)^n=1+ny+\frac{n(n-1)}{2}y^2+.....\infty$

$\frac{1}{3}=ny\Rightarrow y=\frac{1}{3n}$ ...(i)

$\frac{1}{6}=\frac{n(n-1)}{2}y^2$

$\Rightarrow \frac{1}{6}=\frac{n(n-1)}{2}.\frac{1}{9n^2}\Rightarrow 1=\frac{(n-1)}{3n}$

$\Rightarrow 3n=n-1\Rightarrow n=-\frac{1}{2}$

By eqn (i)

$1=\left(\frac{-1}{2}\right)y\Rightarrow y=-\frac{2}{3}$

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(a) $-\frac{3}{8}x^2$ (b) $\frac{x}{2}-\frac{3}{8}x^2$

(c) $1-\frac{3}{8}x^2$ (d) $3x+\frac{3}{8}x^2$

Ans. $= \dfrac{\left[1+\frac{3x}{2}+\frac{\frac{3}{2}\left(\frac{3}{2}-1\right)}{2}x^2\right]-\left[1+\frac{3x}{2}+\frac{3(3-1)}{2}\left(\frac{x}{2}\right)^2\right]}{1-\frac{x}{2}+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2}(x)^2}$

$$=\frac{\frac{3}{8}x^2-\frac{3x^2}{4}}{1-\frac{x}{2}-\frac{1}{8}x^2}=-\frac{3x^2}{8}\left[1-\frac{x}{2}-\frac{x^2}{8}\right]^{-1}$$

$$=\frac{-3x^2}{8}\left(1-\frac{x}{2}\right)^{-1}=\frac{-3x^2}{8}\left[1+\frac{x}{2}\right]=\frac{-3x^2}{8}$$

Ques. Simplify: $S=1-\frac{1}{2}\cdot\frac{1}{2}+\frac{1.3}{2.4.2^2}-\frac{1.3.5}{2.4.6}\cdot\frac{1}{2^3}+\ldots..\infty$

Ans. $(1+x)^n=1+nx+\frac{n(n-1)}{2}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\ldots.\infty$

Compare RHS

$nx=\frac{-1}{4}\Rightarrow x=\frac{-1}{4n}$ ...(i)

$\frac{n(n-1)x^2}{2}=\frac{3}{32}$

$\Rightarrow \frac{n(n-1)}{2}\frac{1}{16n^2}=\frac{3}{32}$

$\Rightarrow \frac{n-1}{n}=3\Rightarrow n-1=3n$

$\Rightarrow 2n=-1\Rightarrow n=-\frac{1}{2}$

By eqn (i)

$\left(-\frac{1}{2}\right)x=-\frac{1}{4}\Rightarrow x=\frac{1}{2}$

Compare LHS

$s=(1+x)^n$

$=\left(1+\frac{1}{2}\right)^{-\frac{1}{2}}$

$=\left(\frac{3}{2}\right)^{\frac{1}{2}}=\left(\frac{2}{3}\right)^{\frac{1}{2}}$

$=\sqrt{\frac{2}{3}}$

Ques. $1-\frac{1}{8}+\frac{1}{8}\cdot\frac{3}{16}-\frac{1.3.5}{8.16.24}+\ldots.=S$

(a) $\frac{2}{5}$ (b) $\frac{\sqrt{2}}{5}$

(c) $\frac{2}{\sqrt{5}}$ (d) None of these

Ans. $(1+x)^n=1+nx+\frac{n(n-1)}{2}x^2+\ldots.$

Compare RHS

$nx=\frac{-1}{8}\Rightarrow x=\frac{-1}{8n}$ ...(i)

$\frac{3}{8.16}=\frac{n(n-1)}{2}x^2$

$\Rightarrow \frac{3}{8.16}=\frac{n(n-1)}{2}\cdot\left(\frac{-1}{8n}\right)^2$

$\Rightarrow \frac{3}{8\times16}=\frac{n(n-1)}{2}\cdot\frac{1}{64n^2}$

$\Rightarrow 3=\frac{n-1}{n}$

$\Rightarrow 3n=n-1$

$\Rightarrow 2n=-1$

$\Rightarrow n=-\frac{1}{2}$

By eqn (i)

$\left(\frac{-1}{2}\right)x=\frac{-1}{8}\Rightarrow x=\frac{1}{4}$

Compare LHS

$S=(1+x)^n$

$=\left(1+\frac{1}{4}\right)^{-\frac{1}{2}}$

$=\left(\frac{5}{4}\right)^{-\frac{1}{2}}$

$=\left(\frac{4}{5}\right)^{\frac{1}{2}}$

$=\sqrt{\frac{4}{5}}=\frac{2}{\sqrt{5}}$

Ques. If $x=\frac{1}{3}+\frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}+\ldots.\infty$, then find the value of x^2+2x

Ans. Add 1 both sides

$(1+x)=1+\frac{1}{3}+\frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}+\ldots.\infty$

$(1+y)^n=1+ny+\frac{n(n-1)}{2}y^2+\ldots.\infty$

$\frac{1}{3}=ny\Rightarrow y=\frac{1}{3n}$ ...(i)

$\frac{1}{6}=\frac{n(n-1)}{2}y^2$

$\Rightarrow \frac{1}{6}=\frac{n(n-1)}{2}\cdot\frac{1}{9n^2}\Rightarrow 1=\frac{(n-1)}{3n}$

$\Rightarrow 3n=n-1\Rightarrow n=-\frac{1}{2}$

By eqn (i)

$\frac{1}{3}=\left(\frac{-1}{2}\right)y\Rightarrow y=-\frac{2}{3}$

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$\because 1+x=(1+y)^n$

$\Rightarrow 1+x=\left(1-\frac{2}{3}\right)^{\frac{1}{2}}=\left(\frac{1}{3}\right)^{\frac{1}{2}}=\sqrt{3}$

$\Rightarrow (1+x)^2=3 \Rightarrow x^2+2x+1=3$

$\Rightarrow x^2+2x=3-1=2$

FINDING NUMBER OF TERMS IN EXPANSION

$(x+y)^n \rightarrow$ no. of terms $=(n+1)$

$(x+y)^n = {}^nC_0x^n + C_1x^{n-1}y^1 + {}^nC_2x^{n-2}y^2 + \ldots {}^nC_nx^0y^n$

$(x+y+z)^n \rightarrow$

$(x+y+z)^n = ()\,x^0y^0z^n + ()\,x^1y^0z^{n-1} + ()\,x^0y^1z^{n-1} + ()\,x^1y^1z^{n-2} + \ldots..$

no . of terms in $(x+y+z)^n$ is equal to no of ways in which n identical things can be distributed to 3 beggars.

No. of term in $(x+y+z)^n = {}^{n+2}C_2$

$(a+b+c+d)^n \Rightarrow$ no of terms $= {}^{n+3}C_3$

Important Note

- Number of terms in the expansion of $(x+y+z)^n = {}^{n+2}C_2$
- The number of distinct terms in the expansion of $(x_1+x_2+\ldots+x_r)^n$ are ${}^{n+r-1}C_{r-1}$
- Sum of all the coefficients is obtained by substituting all the variables terms as unity i.e. 1.

TRINOMIAL THEOREM (FOR NATURAL INDEX)

Trinomial Expansion

$$(x+y+z)^n = \sum \frac{n!}{\alpha_1!\alpha_2!\alpha_3!}(x)^{\alpha_1}(y)^{\alpha_2}(z)^{\alpha_3}$$

where $\alpha_1, \alpha_2, \alpha_3$ are non negative integers such that $\alpha_1+\alpha_2+\alpha_3=n$

Examples:

$$(x+y+z)^{10} = \sum \frac{10!}{\alpha_1!\alpha_2!\alpha_3!}(x)^{\alpha_1}(y)^{\alpha_2}(z)^{\alpha_3}$$

where, $\alpha_1+\alpha_2+\alpha_3=10$

Ques. Find the coefficient of $a^3b^4c^5$ in the expansion of $(bc+ca+ab)^6$

Ans. $(bc+ca+ab)^6 = \sum \frac{6!}{\alpha_1!\alpha_2!\alpha_3!}(bc)^{\alpha_1}.(ca)^{\alpha_2}.(ab)^{\alpha_3}$

$= \sum \frac{6!}{\alpha_1!\alpha_2!\alpha_3!}(b)^{\alpha_1+\alpha_3}(c)^{\alpha_1+\alpha_2}(a)^{\alpha_2+\alpha_3}$ $\quad \alpha_1+\alpha_2+\alpha_3=6$

Coefficient

$\alpha_1+\alpha_3=4 \Rightarrow 6-\alpha_2=4 \Rightarrow \alpha_2=2$

$\alpha_1+\alpha_2=5 \Rightarrow 6-\alpha_3=5 \Rightarrow \alpha_3=1$

$\alpha_2+\alpha_3=3 \Rightarrow 6-\alpha_1=3 \Rightarrow \alpha_1=3$

Coefficient $= \frac{6!}{2!3!1!} = \frac{6\times5\times\cancel{4}^{2}}{\cancel{2}} = 60$

Ques. Find coefficient of $x^8y^8z^4$ in $(xy+yz+zx)^{10}$.

Ans. $(xy+yz+zx) = \frac{10!}{\alpha_1!\alpha_2!\alpha_3!}(xy)^{\alpha_1}(yz)^{\alpha_2}(zx)^{\alpha_3}$ $\quad \alpha_1+\alpha_2+\alpha_3=10$

$= \frac{10!}{\alpha_1!\alpha_2!\alpha_3!}X^{\alpha_1+\alpha_3}Y^{\alpha_1+\alpha_2}Z^{\alpha_1+\alpha_3}$

$\left.\begin{aligned}\alpha_1+\alpha_3=8 &\Rightarrow \alpha_2=2\\ \alpha_1+\alpha_2=8 &\Rightarrow \alpha_3=2\\ \alpha_2+\alpha_3=4 &\Rightarrow \alpha_1=6\end{aligned}\right\}$

coefficient $= \frac{10!}{6!2!2!} = 1260$

Ques. Find the coefficient of x^3 in the expansion of $(2-x+3x^2)^5$

M-1 by T.T.

Ans. $\sum \frac{5!}{\alpha_1!\alpha_2!\alpha_3!}(2)^{\alpha_1}(-x)^{\alpha_2}(3x^2)^{\alpha_3}$

$= \frac{5!}{\alpha_1!\alpha_2!\alpha_3!}(2)^{\alpha_1}(-1)^{\alpha_2}(3)^{\alpha_3}(x)^{\alpha_2+2\alpha_3}$

α_1	α_2	α_3
2	3	0
3	1	1
	-1	2
	?	

$\alpha_1+\alpha_2+\alpha_3=5$

$\alpha_2+2\alpha_3=3$

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Coeff of $x^3 = \frac{5!}{2!3!0!}(2)^2(-1)^3(3)^0 + \frac{5!}{3!1!1!}(2)^2(-1)^1(3)^1$

$= 10 \times 4 - 20 \times 8 \times 3$

$= 40 - 480 = -520$

M-2 by B.T.

$T_{r+1} = {}^5C_r (2)^{5-r}(-x + 3x^2)^r$

$r = 1 \quad {}^5C_1 (2)^4 (-x + 3x^2)^1$

$r = 2 \quad {}^5C_2 (2)^3 (-x + 3x^2)^2$

$r = 3 \quad {}^5C_3 (2)^2 (-x + 3x^2)^3$

$r = 4 \quad {}^5C_2 (2)^1 (-x + 3x^2)^4$

Coefficient of x^3 in

${}^5C_2 (2)^3 (-x + 3x^2)^2 + {}^5C_3 (2)^2 (-x + 3x^2)^3$

$= {}^5C_2 (2)^3 x^2 (-1 + 3x)^2 + {}^5C_3 2^2 x^3 (-1 + 3x)^3$

$= 80$ coeff of x^1 in $(\underbrace{-1 + 3x}_{-1})^2 + 40$ coeff of x^0 in $(\underbrace{-1 + 3x}_{-6})^3$

$= 80(-6) + 40(-1)$

$= -480 - 40 = -520$

MULTINOMIAL THEOREM (FOR NATURAL TNDEX)

The formula by which any power of an algebraic expression consisting of more than two terms separated by an operator {+, -} is known as a Multinomial.

$$(x_1 + x_2 + x_3 + \ldots\ldots x_r)^n = \sum \frac{n!}{(\alpha_1!)(\alpha_2!)(\alpha_3!)\ldots\ldots(\alpha_r!)} x_1^{\alpha_1} x_2^{\alpha_2} x_3^{\alpha_3} \ldots\ldots x_r^{\alpha_r}$$

where a_1, a_2, a_3 are all non-negative integers and subject to a condition that $a_1 + a_2 + a_3 + \ldots\ldots a_r = n$

Important Note

The coefficient of x^r in the expansion of $(1-x)^{-n}$, where $n \in \mathbb{N}$ is ${}^{n+r-1}C_r$

Ques. (i) Find coefficient of x^{17} in $(1-x)^{-2}$

$r = 17 \quad n = 2$

Ans. Coefficient: ${}^{n+r-1}C_r$

$= {}^{2+17-1}C_{17} = {}^{18}C_{17} = {}^{18}C_1 = 18$

(ii) Find coefficient of x^5 in $(1-x)^{-6}$.

$r = 5 \quad n = 6$

Ans. Coefficient: ${}^{n+r-1}C_r = {}^{6+5-1}C_5 = {}^{10}C_5$

Ques. If $|x| < 1$ then the coefficient of x^n in the expansion of $(1 + x + x^2 + \ldots\ldots)^2$ will be

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(a) 1 (b) n

(c) n + 1 (d) None of these

Ans. $\left(\frac{1}{1-x}\right)^2 = \frac{1}{(1-x)^2} = (1-x)^{-2}$

$n = 2, r \to n$

Coefficient ${}^{n+n-1}C_r$

$= {}^{2+n-1}C_n = {}^{n+1}C_n$

$= {}^{n+1}C_1 = n + 1$

P&C QUESTIONS USING MULTINOMIAL THEOREM

Ques. There are infinite number of identical red, white, green & blue balls available.

(a) select 10 balls;

Ans. **M-1**

$x_1 + x_2 + x_3 + x_4 = 10 \Rightarrow {}^{13}C_3$

M-2 by B.T.

Coeff of x^{10} in $(x^0 + x^1 + x^2 + x^3 \ldots\ldots)(x^0 + x^1 + x^2 + \ldots\ldots)(x^0 + x^1 + x^2 + x^3 \ldots\ldots)(x^0 + x^1 + x^2 + \ldots\ldots)$

Power of x in Ist GP represent no. of Red balls

Power of x in 2nd GP represent no. of white balls

Power of x in 3rd GP represent no. of Green balls

Power of x in 4th GP represent no. of Blue balls

coeff of x^{10} in $\left(\frac{1}{1-x}\right)^4 = (1-x)^{-4}$

$r = 10, n = 4$

${}^{n+r-1}C_r = {}^{4+10-1}C_{10} = {}^{13}C_{10} = {}^{13}C_3$

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(b) select 10 balls such that at least 1 ball of each colour is there.

Ans. **M-1**

$x_1 + x_2 + x_3 + x_4 = 10 \Rightarrow {}^9C_3$

M-2 by B.T

Coeff of x^{10} in $(x^1 + x^2 + x^3 .....)(x^1 + x^2 + ...)(x^1 + x^2 .....)(x^1 + x^2 + ...)$

Power of x in Ist GP represent no. of Red balls

Power of x in 2nd GP represent no. of white balls

Power of x in 3rd GP represent no. of Green balls

Power of x in 4th GP represent no. of Blue balls

Coeff of x^{10} in $\left(\frac{x}{1-x}\right)^4 = x^4(1-x)^{-4}$

Coeff of x^6 in $(1-x)^{-4}$, $n = 4, r = 6$

${}^{n+r-1}C_r = {}^9C_6 = {}^9C_3$

Ques. Find the number of integral solutions for the system $x_1 + x_2 + x_3 = 10, 0 \le x_1 \le 6, 0 \le x_2 \le 7, 0 \le x_3 \le 8$.

Ans. Coeff of x^{10} in

$(x^0 + x^1 + ... + x^6)(x^0 + x^1 + ... + x^7)(x^0 + x^1 + ... + x^8)$

$= \left(\frac{1-x^7}{1-x}\right)\left(\frac{1-x^8}{1-x}\right)\left(\frac{1-x^9}{1-x}\right)$

$= (1 - x^7)(1 - x^8)(1 - x^9)(1 - x)^{-3}$

Coeff of x^{10} in $(1 - x^7 - x^8 + x^{15})(1 - x^9)(1 - x)^{-3}$

$= (1 - x^7 - x^8)(1 - x^9)(1 - x)^{-3}$

$\Rightarrow$ Coeff of x^{10} in $(1 - x^7 - x^8 - x^9)(1 - x)^{-3}$

$= (1 - x)^{-3} - x^7(1 - x)^{-3} - x^8(1 - x)^{-3} - x^9(1 - x)^{-3}$

$r = 10$	$r = 3$	$r = 2$	$r = 1$
$n = 3$	$n = 3$	$n = 3$	$n = 3$

$= {}^{3+10-1}C_{10} - {}^{3+3-1}C_3 - {}^{3+2-1}C_2 - {}^{3+1-1}C_1$

$= {}^{12}C_{10} - {}^5C_3 - {}^4C_2 - {}^3C_1$

$= \frac{12 \times 11}{2} - 10 - 6 - 3$

$= 66 - 16 - 3 = 50 - 3 = 47$

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Short Notes

BINOMIAL THEOREM FOR NATURAL INDEX

- $(a + x)^n = {}^nC_0a^n + {}^nC_1a^{n-1}x + {}^nC_2a^{n-2}x^2 + \ldots + {}^nC_r a^{n-r}x^r + \ldots + {}^nC_nx^n$
- $(a - x)^n = {}^nC_0a^n - {}^nC_1a^{n-1}x + {}^nC_2a^{n-2}x^2 - \ldots + (-1)^r {}^nC_r a^{n-r}x^r + \ldots + (-1)^n {}^nC_nx^n$
- $(a + x)^n + (a - x)^n = \sum_{r=0}^{n} {}^nC_rx^r\left[1+(-1)^r\right] = 2({}^nC_0 a^n + {}^nC_2 a^{n-2}x^2 \ldots)$
- $(a + x)^n - (a - x)^n = \sum_{r=0}^{n} {}^nC_rx^r\left[1-(-1)^r\right] = 2({}^nC_1 a^{n-1}x + {}^nC_3 a^{n-3}x^3 \ldots..)$
- $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \ldots + {}^nC_rx^r + \ldots + {}^nC_nx^n = \sum_{r=0}^{n} {}^nC_rx^r$
- $(1-x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 + \ldots + (-1)^r {}^nC_rx^r + \ldots + (-1)^n {}^nC_nx^n = \sum_{r=0}^{n}(-1)^r {}^nC_rx^r$
- Number of terms in the expansion of $(x+y)^n$ is $(n + 1)$ i.e. one more than the index n.
- Binomial Expansion is a Homogeneous expansion i.e. the sum of indices of x and y for each and every term is n.
- The binomial coefficients of the terms $({}^nC_0, {}^nC_1 \ldots)$ equidistant from the beginning and the end are equal. i.e. ${}^nC_r = {}^nC_{n-r}$
- The r^{th} term from end is $(T_{n-r+2})^{th}$ term from beginning.
- r^{th} term from end in $(x + y)^n$ is same as r^{th} term from beginning in $(y + x)^n$

GENERAL TERM

- The general term or the $(r + 1)^{th}$ term in the expansion of $(x + y)^n$ is given by $T_{r+1} = {}^nC_rx^{n-r}y^r$
- General term for the expansion of $(1 + x)^n$ is given by $T_{r+1} = {}^nC_rx^r$

MIDDLE TERM

- If n is even, there is only one middle term which is given by $T_{[(n)/2]+1} = {}^nC_{n/2} \cdot x^{n/2} \cdot y^{n/2}$
- If n is odd, there are two middle terms which are $T_{(n+1)/2}$ and $T_{[(n+1)/2]+1}$

In Greatest Integer & Fractional Part Questions

- If we add I + f & f' to cancel out irrational terms then always I + f + f' = even & f + f' = 1
- If we subtract I + f & f' to cancel out the irrational terms then always I + f - f' = even & f - f' = 0

BASIC PROPERTIES OF BINOMIAL EXPANSION OF $(a + x)^n$

For convenience, the coefficients ${}^nC_0, {}^nC_1, \ldots, {}^nC_r, \ldots, {}^nC_n$ are usually denoted by $C_0, C_1, \ldots, C_r, \ldots, C_n$ respectively.

- $C_0 + C_1 + C_2 + \ldots + C_n = 2^n$
- $C_0 - C_1 + C_2 - \ldots + C_n = 0$
- $C_0 + C_2 + C_4 + \ldots = C_1 + C_3 + C_5 + \ldots = 2^{n-1}$
- ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
- $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$
- $r\,{}^nC_r = n\,{}^{n-1}C_{r-1}$
- $\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$.
- There are (n+1) terms in expansion.
- Binomial coefficients of terms equidistant from beginning and end are equal.
- The general term of the expansion is ${}^nC_r a^{n-r}$ this is in fact the $(r+1)^{th}$ term from the beginning

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- If n is even, there is only one middle term namely $\left(\frac{n}{2}+1\right)^{th}$ and is equal to ${}^{n}C_{\frac{n}{2}}a^{\frac{n}{2}}.x^{\frac{n}{2}}$.
- If n is odd, there are two middle terms namely $\left(\frac{n+1}{2}\right)^{th}$ and $\left(\frac{n+3}{2}\right)^{th}$ terms respectively.
- Binomial coefficient of middle term is the greatest binomial coefficient occurring in the expansion.
- Ratio of two consecutive terms in the expansion of $(x+a)^n$ is $=\left(\frac{n-r+1}{r}\right)\left|\frac{a}{x}\right| = \left(\frac{n+1}{r}-1\right)\left|\frac{a}{x}\right|$

BINOMIAL THEOREM FOR ANY NEGATIVE OR RATIONAL INDEX

- $(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3 +\ldots+\frac{n(n-1)(n-2)\ldots(n-(r-1))}{r!}x^r+\ldots\infty$

  The above expansion is valid if and only if $|x| < 1$.
- If n is a negative integer or a fraction, then ${}^{n}C_r$ will be written as $\frac{(n)(n-1)\ldots(n-(r-1))}{r!}$
- If n is a negative integer or a fraction, then the number of terms in the series are infinite. Hence, the series never terminates.
- If first term is not 1 (unity), then make first term unity in the following way $(x+y)^n = x^n\left(1+\frac{y}{x}\right)^n$ in such a way that $\left|\frac{y}{x}\right| < 1$.
- If x is so small that square and higher powers of x can be neglected, then $(1+x)^n = 1+nx$
- $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \ldots\infty$
- $(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \ldots\infty$
- $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots\infty$
- $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots\infty$
- $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \ldots\infty$
- $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \ldots\infty$

MULTINOMIAL THEOREM FOR NATURAL INDEX

- $(x_1+x_2+x_3+\ldots+x_r)^n = \sum\frac{n!}{(\alpha_1!)(\alpha_2!)(\alpha_3!)\ldots(\alpha_r!)}x_1^{\alpha_1}x_2^{\alpha_2}x_3^{\alpha_3}\ldots x_r^{\alpha_r}$

  where, $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_r$ are all non-negative integers and subject to a condition that $\alpha_1+\alpha_2+\alpha_3+\ldots+\alpha_r = n$
- The number of distinct terms in the expansion of $(x_1+x_2+x_3+\ldots+x_r)^n$ are ${}^{n+(r-1)}C_{(r-1)}$.
- Sum of all the coefficients is obtained by substituting all the variables terms equal to unity i.e. 1.

"Small terms add up to powerful results – just like your consistent efforts."

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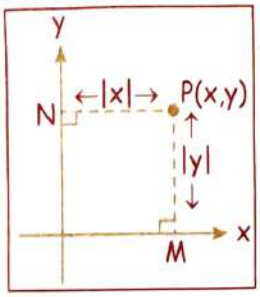
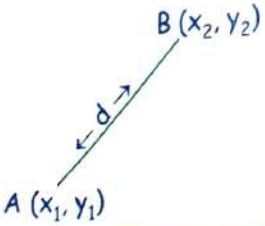
Chapter 10

Straight Lines

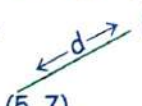
INTRODUCTION TO CO-ORDINATE GEOMETRY





$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example:



$$d = \sqrt{(5-(-1))^2 + (7-3)^2}$$

$$= \sqrt{6^2 + 4^2} = \sqrt{36 + 16} = \sqrt{52}$$

$x \rightarrow$ Abscissa

$y \rightarrow$ ordinate

x - axis (a, o)

y - axis (o, b)

Ques. Find the coordinates of a point on y-Axis which is equidistant from the points $[-1, 2]$ & $[3, 4]$.

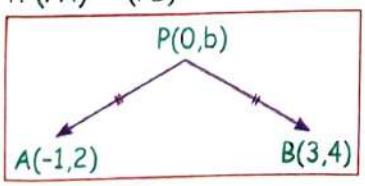
(a) (5, 0) (b) (0, 4)

(c) (0, 5) (d) None of these

Ans. Let the co-ordinate of a point on y-axis = P(0,b)

$\because$ P is equidistant from the points A and B.

$\therefore (PA)^2 = (PB)^2$



$\Rightarrow 1^2 + (b-2)^2 = 3^2 + (b-4)^2$

$\Rightarrow 1 + b^2 + 4 - 4b = 9 + b^2 + 16 - 8b$

$\Rightarrow 5 - 4b = 25 - 8b \Rightarrow 8b - 4b = 25 - 5$

$\Rightarrow 4b = 20 \Rightarrow b = 5$

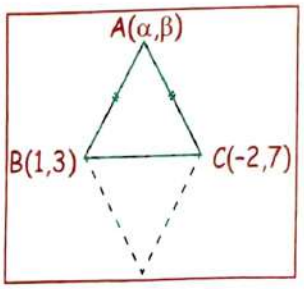
Ques. ABC is an isosceles triangle. If the coordinates of the base are B (1, 3) and C(- 2, 7), the coordinates of vertex A can be

(a) (1, 6) (b) $\left(-\frac{1}{2}, 5\right)$

(c) $\left(\frac{5}{6}, 6\right)$ (d) None of these

Ans. Let A (α, β)

$\because$ ABC is an isosceles triangle.



$\therefore (AB)^2 = (AC)^2$

$\Rightarrow (\alpha - 1)^2 + (\beta - 3)^2 = (\alpha + 2)^2 + (\beta - 7)^2$

$\Rightarrow \alpha^2 - 2\alpha + 1 + \beta^2 - 6\beta + 9 = \alpha^2 + 4 + 4\alpha + \beta^2 - 14\beta + 49$

$\Rightarrow -2\alpha + 10 - 6\beta = 4\alpha - 14\beta + 53$

$\Rightarrow -6\alpha + 8\beta = 43$

Now, check the given options:

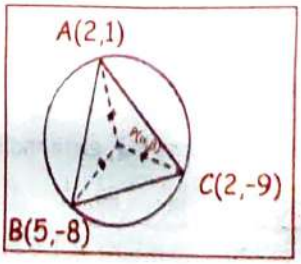
(A) $\alpha = 1, \beta = 6 \Rightarrow -6 + 48 \neq 43$

(B) $\alpha = -\frac{1}{2}, \beta = 5 \Rightarrow 3 + 40 = 43$

(C) $\alpha = \frac{5}{6}, \beta = 6 \Rightarrow -5 + 48 = 43$

Ques. Find the coordinates of the center of a circle passing through the points A (2, 1), B (5, - 8) and C (2, - 9). Also, find the radius of this circle.

Ans. Let P (α, β) is the circumcenter of the circle.



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$PA = PB = PC$

$(\alpha - 2)^2 + (\beta - 1)^2 = (\alpha - 5)^2 + (\beta + 8)^2$

$\Rightarrow \alpha^2 + 4 - 4\alpha + \beta^2 + 1 - 2\beta = \alpha^2 + 25 - 10\alpha + \beta^2 + 64 + 16\beta$

$\Rightarrow 5 - 4\alpha - 2\beta = 89 - 10\alpha + 16\beta$

$\Rightarrow 6\alpha - 18\beta = 89 - 5$

$\Rightarrow \alpha - 3\beta = 14$ ...(i)

$PA = PC$

$(\alpha - 2)^2 + (\beta - 1)^2 = (\alpha - 2)^2 + (\beta + 9)^2$

$\Rightarrow \beta^2 + 1 - 2\beta = \beta^2 + 81 + 18\beta$

$\Rightarrow -20\beta = 80 \Rightarrow \beta = -4$

By equation (i):

$\alpha - 3(-4) = 14 \Rightarrow \alpha = 2$

$\therefore P(2, -4)$

Therefore, $PA = \sqrt{0^2 + 5^2} = 5 = r$

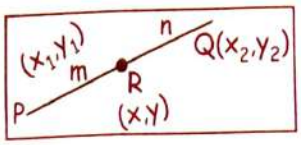
RECTANGULAR CO-ORDINATE SYSTEM

Section Formula

The co-ordinates of a point dividing a line joining the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the ratio $m : n$ is given by:

a) For internal division:

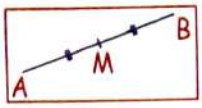
R divides line segment PQ, internally.



$$(x, y) = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}\right)$$

Mid Point Formula

$$\text{Mid point} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$



b) For external division:

R divides line segment PQ, externally.

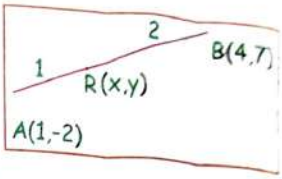
$$(x, y) = \left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}\right)$$

Ques. Find the co-ordinates of the point dividing the join of A (1, - 2) and B (4, 7)

(a) Internally in the ratio 1 : 2

(b) Externally in the ratio 1 : 2

Ans. (a) $\frac{RA}{RB} = \frac{1}{2}$

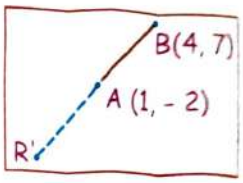


$$x = \frac{1 \times 4 + 2 \times 1}{1 + 2} = \frac{6}{3} = 2$$

$$y = \frac{1 \times 7 + 2(-2)}{1 + 2} = \frac{7 - 4}{3} = \frac{3}{3} = 1$$

$\therefore R(2, 1)$

(b) $\frac{R'A}{R'B} = \frac{1}{2}$



$$R'\left(\frac{4 \times 1 - 2 \times 1}{1 - 2}, \frac{1 \times 7 - 2(-2)}{1 - 2}\right)$$

$$R'\left(\frac{2}{-1}, \frac{7 + 4}{-1}\right)$$

$\Rightarrow R'(-2, -11)$

Ques. If the line $2x + y = k$ passes through the point which divides the line segment joining the points (1, - 1) and (2, 4) in the ratio 3 : 2, then k is equal to

(a) $\frac{26}{5}$ (b) 5

(c) 6 (d) $\frac{11}{15}$

Ans. Let, the point divides the line externally

$$D = \left(\frac{3 \times 2 - 2(1)}{3 - 2}, \frac{3 \times 4 - 2(-1)}{3 - 2}\right)$$

$\Rightarrow D = (4, 14)$

put the point in the given line

$2 \times 4 + 14 = k$

$\Rightarrow 8 + 14 = k \Rightarrow k = 22$ (Which is not an option)

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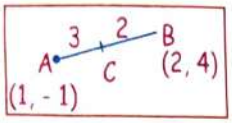
$$C = \left(\frac{3 \times 2 + 2 \times (1)}{3+2}, \frac{3 \times 4 + 2 \times (-1)}{3+2}\right)$$

Now, let the point divides the line internally

$$C\left(\frac{8}{5}, \frac{10}{5}\right)$$



Satisfies $2x + y = k$

$$\therefore \frac{2 \times 8}{5} + \frac{10}{5} = k$$

$$\Rightarrow \frac{26}{5} = k$$

Important Notes

Harmonic Conjugate Points:

If P divides AB internally in the ratio m : n & Q divides AB externally in the ratio m : n then P : Q are said to be harmonic conjugate points of each other with respect to line AB.

Mathematically: $\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$ i.e. AP, AB & AQ are in H.P.

Ques. If P (1, 2), Q (4, 6), R (5, 7) and S (a, b) are the vertices of a parallelogram PQRS, then

(a) a = 2, b = 4

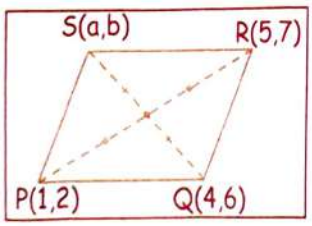
(b) a = 4, b = 4

(c) a = 2, b = 3

(d) a = 3, b = 5

Ans. ∵ PQRS is a parallelogram

∴ Mid point of QS is same as Mid point of PR



$$\frac{a+4}{2} = \frac{1+5}{2}$$

$$\Rightarrow a + 4 = 6 \Rightarrow a = 2$$

$$\frac{b+6}{2} = \frac{2+7}{2}$$

$$\Rightarrow b + 6 = 9 \Rightarrow b = 3$$

Area of Triangle

Let A (x_1, y_1), B (x_2, y_2) and C (x_3, y_3) are vertices of a triangle, then

$$\text{Area of } \Delta ABC = \left|\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$$

$$= \frac{1}{2}\left|[x_1 (y_2 - y_3) + x_2 (y_3 - y_1) + x_3 (y_1 - y_2)]\right|$$

If A, B & C are collinear pts. Then,

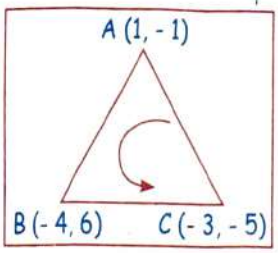
$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Ques. Find the area of triangle formed by A (1, - 1), B (- 4, 6) & C (- 3, - 5)

Ans. $\text{Area} = \left|\frac{1}{2}[1[6+5] - 4[-5+1] - 3[-1-6]]\right|$

$$= \frac{1}{2}[11 + 16 + 21]$$

$$= \frac{1}{2}[48] = 24$$



Ques. Show that the points A (a, b + c), B (b, c + a) & C (c, a + b) are collinear.

Ans. $a[c - b] + b[a - c] + c[b - a]$

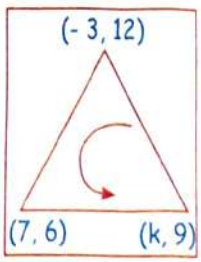
$= ac - ab + ab - bc + bc - ac = 0$

Hence, A, B and C are collinear

Ques. Find the value of k such that the points A (- 3, 12), B (7, 6) & C (k, 9) are collinear.

Ans. ∵ A, B and C are collinear

$$\therefore \begin{vmatrix} -3 & 12 & 1 \\ 7 & 6 & 1 \\ k & 9 & 1 \end{vmatrix} = 0$$



$$\Rightarrow -3[6 - 9] - 12[7 - k] + 1[63 - 6k] = 0$$

$$\Rightarrow 9 - 84 + 12k + 63 - 6k = 0 \Rightarrow 6k = 12 \Rightarrow k = 2$$

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Important Note

All the co-ordinates of the vertices of an equilateral triangle cannot be rational.

Proof: Let us assume that all the co-ordinates are rational.

$$\text{Area} = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \text{Rational}$$

But Area $= \frac{\sqrt{3}}{4}(S)^2$

$\because S^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 \rightarrow$ Rational

and $\sqrt{3}$ is irrational

$\therefore$ Area = Irrational × Rational = Irrational

Hence our assumption is incorrect.

Ques. A and B are two points (3, 4) and (5, - 2). Find the coordinates of a point P such that PA = PB and the area of $\Delta PAB = 10$.

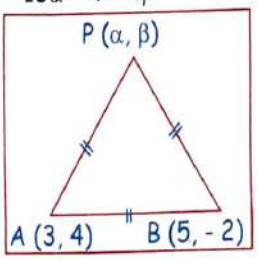
Ans. $\because PA = PB$

$\Rightarrow (\alpha - 3)^2 + (\beta - 4)^2 = (\alpha - 5)^2 + (\beta + 2)^2$

$\Rightarrow 9 - 6\alpha + 16 - 8\beta = 25 - 10\alpha + 4 + 4\beta$

$\Rightarrow 4\alpha - 12\beta = 4$

$\Rightarrow \alpha - 3\beta = 1 \quad ...(i)$



$\because$ Area of $\Delta PAB = 10$

$$\left|\frac{1}{2}\begin{vmatrix} \alpha & \beta & 1 \\ 3 & 4 & 1 \\ 5 & -2 & 1 \end{vmatrix}\right| = 10$$

$$\Rightarrow \begin{vmatrix} \alpha & \beta & 1 \\ 3 & 4 & 1 \\ 5 & -2 & 1 \end{vmatrix} = \pm 20$$

$\Rightarrow \alpha(4 + 2) - \beta(3 - 5) + 1(- 6 - 20) = \pm 20$

$\Rightarrow 6\alpha + 2\beta - 26 = \pm 20$

$\Rightarrow 3\alpha + \beta = 13 \pm 10$

$3\alpha + \beta = 23$ or $3\alpha + \beta = 3$

put the value of $\alpha = 1 + 3\beta$ from eq. (i)

$3(3\beta + 1) + \beta = 23$

$\Rightarrow 10\beta = 20$

$\Rightarrow \beta = 2$ and $\alpha = 7$

$3(3\beta + 1) + \beta = 3$

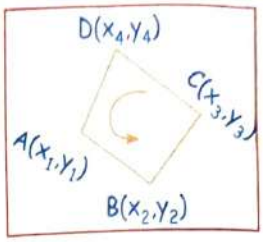
$\Rightarrow 10\beta = 0$

$\Rightarrow \beta = 0$ and $\alpha = 1$

Hence, coordinates of point P are (7, 2) or (1, 0).

Area of Quadrilateral

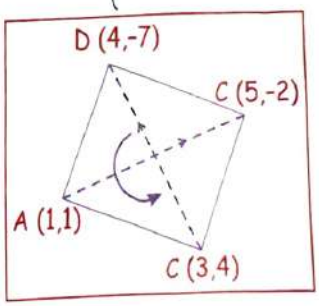
Let A (x_1, y_1), B (x_2, y_2) and C (x_3, y_3) and D (x_4, y_4) are vertices of a Quadrilateral then area is given by



$$\text{Area} = \left|\frac{1}{2}\left\{\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \begin{vmatrix} x_4 & y_4 \\ x_1 & y_1 \end{vmatrix}\right\}\right|$$

Ques. Find the area of a quadrilateral whose vertices in order are (1, 1), (3, 4), (5, - 2) & (4, - 7).

Ans. $$\text{Area} = \frac{1}{2}\left\{\begin{vmatrix} 1 & 1 \\ 3 & 4 \end{vmatrix} + \begin{vmatrix} 3 & 4 \\ 5 & -2 \end{vmatrix} + \begin{vmatrix} 5 & -2 \\ 4 & -7 \end{vmatrix} + \begin{vmatrix} 4 & -7 \\ 1 & 1 \end{vmatrix}\right\}$$



$$= \frac{1}{2}[(4 - 3) + (- 6 - 20) + (- 35 + 8) + (4 + 7)]$$

$$= \frac{1}{2}[1 - 26 - 27 + 11] = \left|\frac{1}{2}[- 14 - 27]\right|$$

$$= \frac{1}{2} \times 41 = \frac{41}{2} \text{ Sq. units}$$

CENTERS OF A TRIANGLE

Centroid (G)

The point where 3 medians of a triangle meet is call Centroid.

Centriod always lies inside the Δ.

$$G\left[\frac{X_1 + X_2 + X_3}{3}, \frac{Y_1 + Y_2 + Y_3}{3}\right]$$

Proof →

$\because$ D is the mid point of BC.

$$\therefore D\left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}\right)$$

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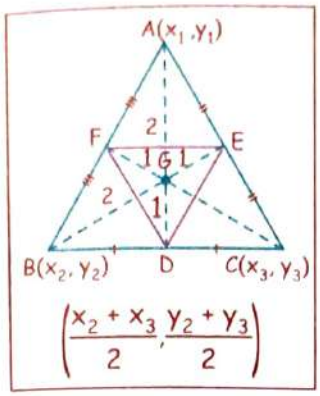
Now AG : GD = 2 : 1

$$x = \frac{2\left(\frac{x_2 + x_3}{2}\right) + 1 \times x_1}{3}$$

$$\therefore x = \frac{x_1 + x_2 + x_3}{3}$$

Similarly, $y = \frac{y_1 + y_2 + y_3}{3}$

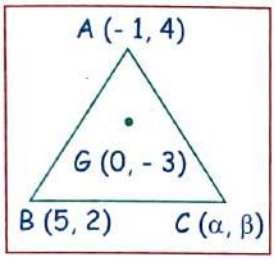


Ques. Two vertices of ΔABC are A(- 1, 4) and B (5, 2) and its centroid is G (0, - 3). The coordinates of C are

(a) (4, 3) (b) (4, 15)

(c) (- 4, - 15) (d) (- 15, - 4)

Ans. Let, the coordinates of C are (α, β).



$$\therefore \frac{\alpha - 1 + 5}{3} = 0$$

$$\Rightarrow \alpha + 4 = 0 \Rightarrow \alpha = -4$$

$$\text{and } \frac{\beta + 4 + 2}{3} = -3$$

$$\Rightarrow \beta + 6 = -9$$

$$\Rightarrow \beta = -15$$

Hence, the coordinates of C are (-4,-15).

Important Note

The centroid of Δ formed by joining mid points of the 3 sides of a triangle is same as the centroid of original Δ.

Ques. If the co-ordinates of the mid - points of the sides of a triangle are (1, 2) (0, - 1) and (2, - 1), find the co-ordinates of the vertices.

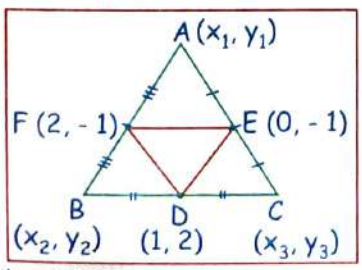
Ans. $\frac{x_2 + x_3}{2} = 1$

$\Rightarrow x_2 + x_3 = 2$ ... (i)

Similarly,

$x_1 + x_2 = 4$ ... (ii)

$x_1 + x_3 = 0$ ... (iii)



By adding eq (i), (ii) and (iii)

$2(x_1 + x_2 + x_3) = 6 \Rightarrow x_1 + x_2 + x_3 = 3$

By using above equations

We get $x_1 = 1, x_2 = 3, x_3 = -1$

Similarly,

$y_1 + y_2 = -2$ ...(iv)

$y_2 + y_3 = 4$ ...(v)

$y_1 + y_3 = -2$ ...(vi)

By Solving Equations (iv), (v) and (vi)

We get $y_1 = -4, y_2 = 2, y_3 = 2$

Hence, the coordinates of the vertices are:

A(1,-4), B(3,2), C(-1,2)

Ques. A triangle has a vertex at (1, 2) and the mid-points of the two sides through it are (- 1, 1) and (2, 3). Then the centroid of this triangle is

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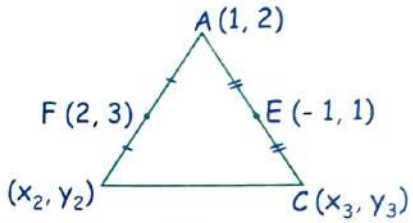
(a) $\left(1, \frac{7}{3}\right)$ (b) $\left(\frac{1}{3}, 1\right)$ (c) $\left(\frac{1}{3}, 2\right)$ (d) $\left(\frac{1}{3}, \frac{5}{3}\right)$

Ans.



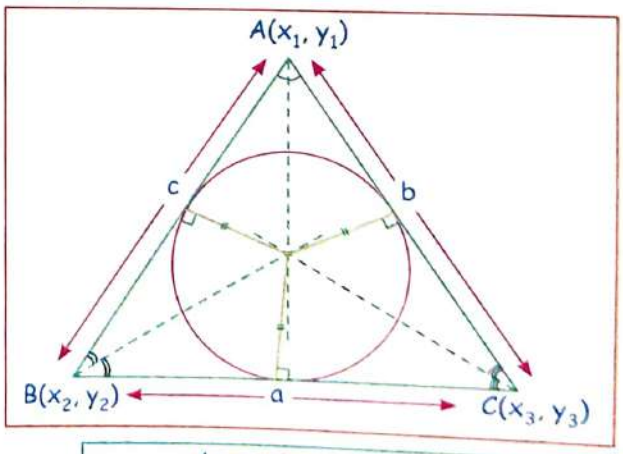
$$\left(\frac{x_2 + 1}{2}, \frac{y_2 + 2}{2}\right) = (2, 3) \Rightarrow (x_2, y_2) = (3, 4)$$

$$\text{and } \left(\frac{x_3 + 1}{2}, \frac{y_3 + 2}{2}\right) = (-1, 1) \Rightarrow (x_3, y_3) = (-3, 0)$$

$$\text{Hence, } G = \left(\frac{1 + 3 - 3}{3}, \frac{2 + 4 + 0}{3}\right) = \left(\frac{1}{3}, 2\right)$$

Incentre (I)

The Point where 3 angle bisectors of a triangle meet is called Incentre. OR Centre of the Inscribed Circle (Incircle).



Formula: $\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}\right)$

Point which is equidistant from 3 sides of a 'Δ'.

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Ques. The incentre of the triangle whose vertices are $(-36, 7)$, $(20, 7)$ and $(0, -8)$ is:

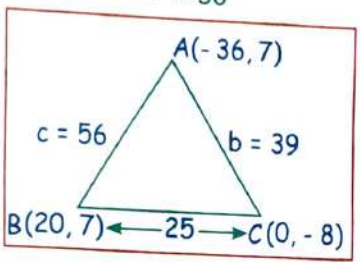
(a) $(0, -1)$ (b) $(-1, 0)$

(c) $(1, 1)$ (d) $\left(\frac{1}{2}, 1\right)$

Ans. $BC = a = \sqrt{20^2 + 15^2} = 5\sqrt{4^2 + 3^2} = 25$

$AC = b = \sqrt{(36)^2 + 15^2} = 3\sqrt{12^2 + 5^2} = 39$

$AB = c = \sqrt{56^2 + 0^2} = 56$



$$x = \frac{ax_1 + bx_2 + cx_3}{a + b + c}$$

$$x = \frac{-36 \times 25 + 39 \times 20 + 56 \times (0)}{56 + 39 + 25}$$

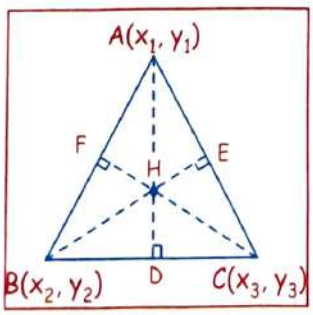
$$\Rightarrow x = \frac{-9 \times 100 + 39 \times 20}{120} = -1$$

$$\text{and } y = \frac{25 \times 7 + 39 \times 7 + 56 \times (-8)}{120} = 0$$

Hence, I = (-1,0)

Orthocentre (H)

The Point where 3 altitudes of a triangle meet is called orthocentre.



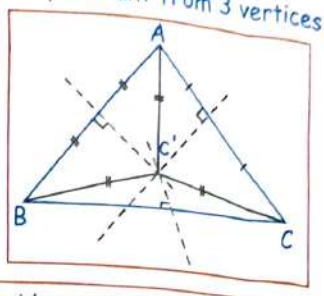
Co-ordinates of Orthocentre (H) =

$$\left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C}\right)$$

Circumcentre (C')

The Point where 3 perpendicular Bisectors of a triangle meet is called Circumcentre. OR Centre of the circumscribed circle.

Point which is equidistant from 3 vertices of a Δ.



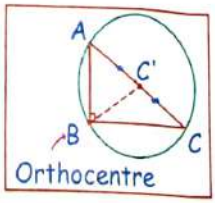
C' can be inside or outside the Δ.

Co-ordinates of circum-centre are given by

$$\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right)$$

Important Note

- If the triangle is equilateral, then centroid, in-centre, Ortho-centre, circum-centre, coincide.
- For right angled triangle, circum-centre is the midpoint of hypotenuse and ortho-centre is the right-angled vertex.

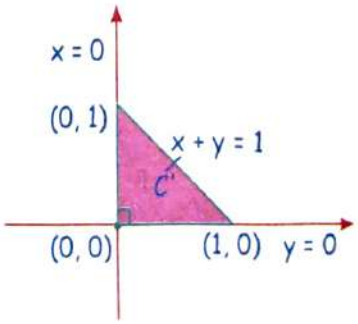


Ques. The orthocentre of triangle formed by the lines $xy = 0$ and $x + y = 1$ is

(a) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (b) $\left(\frac{1}{3}, \frac{1}{3}\right)$

(c) $(0, 0)$ (d) $\left(\frac{1}{4}, \frac{1}{4}\right)$

Ans. For $xy = 0$; $x = 0$ or $y = 0$



So, the triangle is a right angled triangle

$\therefore$ Orthocentre (0, 0)

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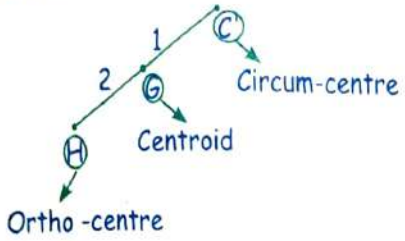
Important Note

Euler's Line

- In a non-equilateral triangle, Ortho-centre, centroid and circum-centre are always collinear. Further G divides the line joining H and C in the ratio 2 : 1.

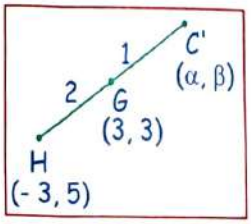


Ques. If the orthocentre and centroid of a triangle are (- 3, 5) and (3, 3) then its circumcentre is:

(a) (6, 2) (b) (3, - 1)

(c) (- 3, 5) (d) (- 3, 1)

Ans.



$$\frac{2\alpha + 1(-3)}{2+1} = 3 \qquad \frac{2\beta + 1 \times 5}{2+1} = 3$$

$$\Rightarrow 2\alpha - 3 = 9 \qquad \Rightarrow 2\beta + 5 = 9$$

$$\Rightarrow \alpha = 6 \qquad \Rightarrow \beta = 2$$

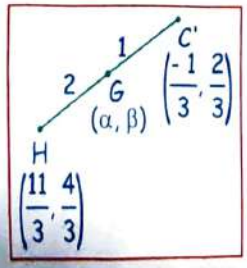
Ques. If in a triangle A = (1, 10), circumcentre = $\left(-\frac{1}{3}, \frac{2}{3}\right)$ and orthocentre $\left(\frac{11}{3}, \frac{4}{3}\right)$, then the co - ordinate of mid-point of side opposite to A is =

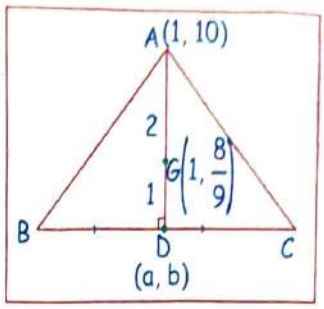
(a) (1, 6) (b) (1, 5)

(c) (1, - 3) (d) $\left(1, -\frac{11}{3}\right)$

Ans. $\alpha = \dfrac{2\left(\frac{-1}{3}\right) + 1 \times \frac{11}{3}}{3} = 1$

$$\beta = \frac{2 \times \frac{2}{3} + 1 \times \frac{4}{3}}{3} = \frac{8}{9}$$





$$\frac{2a + 1 \times 1}{3} = 1 \qquad \& \qquad \frac{2b + 10 \times 1}{3} = \frac{8}{9}$$

$$\Rightarrow 2a = 2 \qquad \Rightarrow 2b = \frac{8}{3} - 10$$

$$\Rightarrow a = 1 \qquad \Rightarrow b = \frac{4}{3} - 5 = \frac{-11}{3}$$

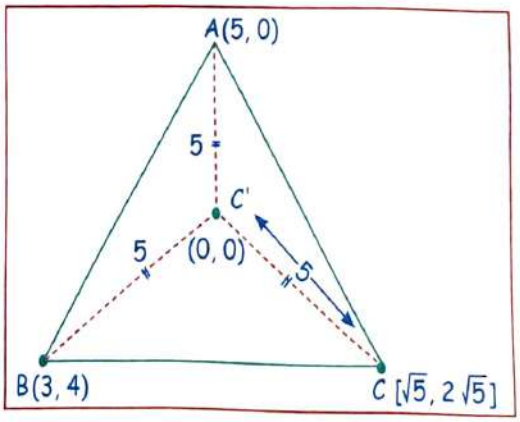
Hence, the co-ordinates of mid-point of side A is $\left(1, \frac{-11}{3}\right)$

Ques. Find ortho-centre of the triangle formed by the Points (5, 0), (3, 4) and $(\sqrt{5}, 2\sqrt{5})$.

Ans. $G\left(\frac{8+\sqrt{5}}{3}, \frac{4+2\sqrt{5}}{3}\right)$

$\because$ All the vertices are equidistant from the origin (0, 0)

$\therefore$ (0, 0) is the circumcentre



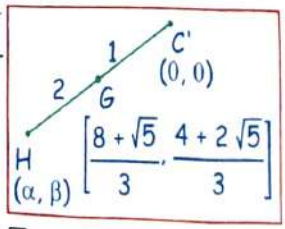
$$\frac{2 \times 0 + 1 \times \alpha}{3} = \frac{8+\sqrt{5}}{3}$$

$$\Rightarrow \alpha = 8 + \sqrt{5}$$

and

$$\frac{2 \times 0 + \beta \times 1}{3} = \frac{4 + 2\sqrt{5}}{3}$$

$$\Rightarrow \beta = 4 + 2\sqrt{5}$$



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Ques. Let A and B two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point P (m, n) from the point Q (- 2, - 3) is

[JEE Mains 27 Jan 2024-II]

Ans. ∵ Total number of subsets of set A = 2^m

∴ $2^m = 2^n + 56$

∵ m = 6, n = 3 satisfy the equation

So, P = (6,3) and Q = (-2,-3)

Thus, PQ = $\sqrt{8^2 + 6^2} = 10$

LOCUS

The path traced by a moving point under the given geometrical conditions is known as locus of that point.

Equation of Locus

The equation to a locus is the relation which exists between the coordinates of any point on the path, and which holds for no other point except those lying on the path.

Steps to Find the Equation of Locus

Step-1: If we are finding the equation of the locus of a point P, let P be (h, k).

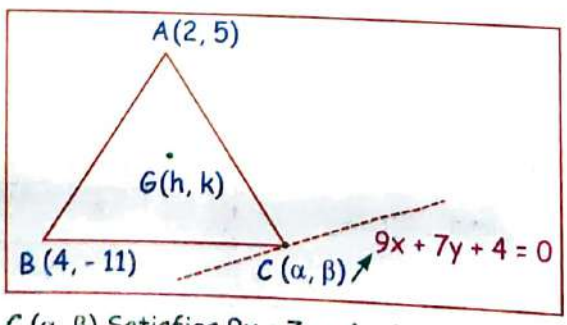
Step-2: Apply the given geometrical conditions.

Step-3: Get a relation between h & k by eliminating the variables assumed (if any).

Step-4: Replace (h, k) → (x, y). The resulting equation would be the equation of the locus of P.

Ques. ABC is a triangle in which A = (2, 5) and B = (4, - 11). Vertex C moves on the straight line 9x + 7y + 4 = 0. Prove that the centroid of ΔABC moves on 27x + 21y - 8 = 0.

Ans.



C (α, β) Satisfies 9x + 7y + 4 = 0

$9\alpha + 7\beta + 4 = 0$ ...(i)

Let centroid G [h, k]

$\therefore h = \frac{4 + 2 + \alpha}{3}$, $k = \frac{5 - 11 + \beta}{3}$

$\Rightarrow h = \frac{6 + \alpha}{3}$, $\Rightarrow k = \frac{\beta - 6}{3}$

$\alpha = 3h - 6$ ...(ii) $\beta = 3k + 6$

put the values of α and β in equation (i)

$9(3h - 6) + 7(3k + 6) + 4 = 0$

$\Rightarrow 27h - 54 + 21k + 42 + 4 = 0 \Rightarrow 27h + 21k$

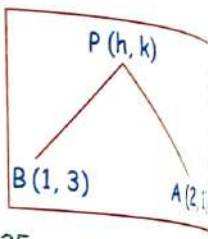
(h, k) → (x, y) ⇒ 27x + 21y = 8

Ques. If the locus of the point, whose distances from the point (2, 1) and (1, 3) are in the ratio 5 : 4, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to

(a) 37 (b) 437 (c) - 27 (d) 5

Ans. $PA = \sqrt{(h - 2)^2 + (k - 1)^2}$

$PB = \sqrt{(h - 1)^2 + (k - 3)^2}$



$\because \frac{PA}{PB} = \frac{5}{4}$

$\Rightarrow \frac{(PA)^2}{(PB)^2} = \frac{25}{16}$

$\Rightarrow \frac{(h - 2)^2 + (k - 1)^2}{(h - 1)^2 + (k - 3)^2} = \frac{25}{16}$

$\Rightarrow 16(h^2 - 4h + 4 + k^2 - 2k + 1) = 25(h^2 - 2h + 1 + k^2 - 6k + 9)$

$\Rightarrow 9h^2 + 9k^2 + 14h - 118k + 170 = 0$

(h,k) → (x,y)

$\Rightarrow 9x^2 + 9y^2 + 14x - 118y + 170 = 0$

Comparing with $ax^2 + by^2 + cxy + dx + ey + 170 = 0$

We get, a = b = 9, c = 0, d = 14, e = - 118

Therefore, $a^2 + 2b + 3c + 4d + e$

= 81 + 18 + 0 + 56 - 118 = 37

Ques. If the co-ordinates of a variable point P be (a cosθ, a sinθ), where θ is a parameter, find the locus of P.

Ans. $h = a\cos\theta$ ...(i)

$k = a\sin\theta$ ...(ii)

Squaring the equations and adding them

$h^2 + k^2 = a^2\cos^2\theta + a^2\sin^2\theta \Rightarrow h^2 + k^2 = a^2$

(h, k) → (x, y)

$x^2 + y^2 = a^2$

Hence, the locus of P is a circle.

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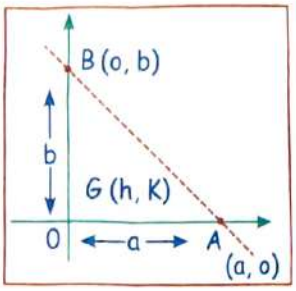
Ques. A variable line cuts the x and y axes at A and B respectively where OA = a, OB = b (O the origin) such that $a^2 + b^2 = 27$, find the locus of the centroid of ΔOAB.

Ans.



$$h = \frac{0 + a + 0}{3}$$

$$\Rightarrow a = 3h$$

$$k = \frac{0 + 0 + b}{3}$$

$$\Rightarrow b = 3k$$

$\because a^2 + b^2 = 27$

$(3h)^2 + (3k)^2 = 27$

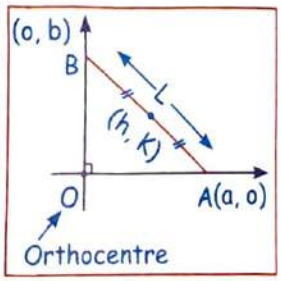
$\Rightarrow h^2 + k^2 = 3$

$x^2 + y^2 = 3 \rightarrow$ Circle (centre (0, 0) & $r = \sqrt{3}$)

Ques. A rod of lenght 'L' slides with its ends A and B always on the co-ordinate axes. Find the locus of

1. Midpoints of AB
2. Centroid of ΔOAB
3. Orthocentre of ΔOAB (O is origin)

Ans. 1) $h = \frac{a + o}{2}$, $k = \frac{0 + b}{2}$



$\Rightarrow 2h = a$ & $b = 2k$

$\because a^2 + b^2 = L^2$

$\therefore (2h)^2 + (2k)^2 = L^2$

$\Rightarrow h^2 + k^2 = \frac{L^2}{4}$

$\Rightarrow x^2 + y^2 = \frac{L^2}{4}$

2) $h = \frac{0 + a + o}{3} \Rightarrow a = 3h$

$k = \frac{0 + 0 + b}{3} \Rightarrow b = 3k$

$\because a^2 + b^2 = L^2$

$\therefore 9h^2 + 9k^2 = L^2 \Rightarrow x^2 + y^2 = \frac{L^2}{9}$

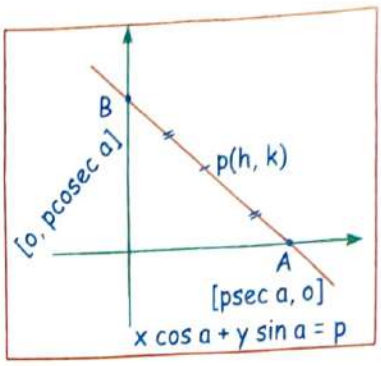
3) Orthocentre of ΔOAB is origin (0, 0)

Ques. Locus of midpoint of the point of the portion of $x \cos a + y \sin a = p$ intercepted between the coordinate axes, where p is a constant, is

(a) $x^2 + y^2 = \frac{4}{p^2}$ (b) $x^2 + y^2 = 4p^2$

(c) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{2}{p^2}$ (d) ✓ $\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$

Ans.



Put y = 0 to get A; $x \cos a = p \Rightarrow x = \frac{p}{\cos a}$

For B, put x = 0; $y = \frac{p}{\sin a} = p \text{ cosec } a$

$h = \frac{p \sec a + 0}{2} \Rightarrow 2h = \frac{p}{\cos a} \Rightarrow \cos a = \frac{p}{2h}$

$k = \frac{p \text{ cosec } a + 0}{2} \Rightarrow 2k = \frac{p}{\sin a} \Rightarrow \sin a = \frac{p}{2k}$

$\because \cos^2 a + \sin^2 a = 1$

$\Rightarrow \left(\frac{p}{2h}\right)^2 + \left(\frac{p}{2k}\right)^2 = 1 \Rightarrow \frac{p^2}{4h^2} + \frac{p^2}{4k^2} = 1$

$\Rightarrow \frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$

Ques. Vertices of a variable triangle are (3, 4), (5 cosθ, 5 Sinθ) and (5 sinθ, - 5 cosθ). Then locus of its ortho-center is: **[JEE Mains 2017]**

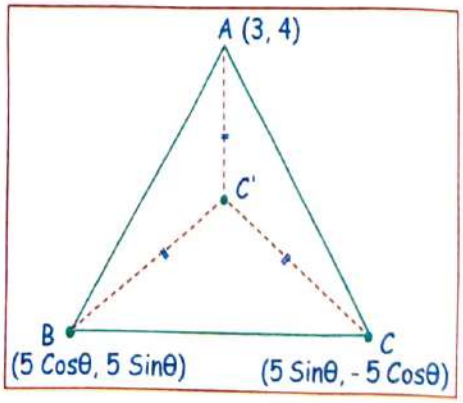
(a) $[x + y - 1]^2 + [x - y - 7]^2 = 100$

(b) $[x + y - 7]^2 + [x - y - 1]^2 = 100$

(c) $[x + y - 7]^2 + [x + y - 1]^2 = 100$

(d) ✓ $[x + y - 7]^2 + [x - y + 1]^2 = 100$

Ans.



$\because$ All the points (A, B and C) are equidistant from the origin (0, 0).

$\therefore$ co-ordinates of circumcentre (C') = (0, 0)

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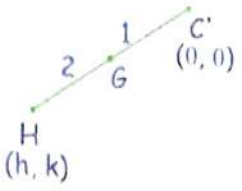


$$G = \left(\frac{5\cos\theta + 5\sin\theta + 3}{3}, \frac{5\sin\theta - 5\cos\theta + 4}{3}\right)$$

$\because GH : GC' = 2 : 1$

$$\therefore \frac{2 \times 0 + 1 \times h}{3} = \frac{5\cos\theta + 5\sin\theta + 3}{3}$$

$h - 3 = 5\cos\theta + 5\sin\theta$ ...(i)

Similarly

$K - 4 = 5\sin\theta - 5\cos\theta$ ...(ii)

by Adding equation (i) and (ii)

$h + k - 7 = 10\sin\theta$ ...(iii)

by Subtracting equation (ii) from (i)

$h - k + 1 = 10\cos\theta$ ...(iv)

Adding the squares of equation (iii) and (iv)

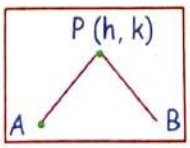
$(h + k - 7)^2 + (h - k + 1)^2 = 100$

$(h, k) \rightarrow (x, y)$

$(x + y - 7)^2 + (x - y + 1)^2 = 100$

Important Note

- **Parabola:** The locus of a point which moves such that it is equidistant from a fixed point and a fixed line.
- **Ellipse:** The locus of a point which moves such that the sum of its distances from 2 fixed points is a constant.

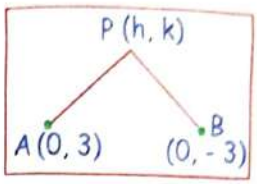


PA + PB = const

- **Hyperbola:** The locus of a point which moves such that the difference of its distances from 2 fixed point is a constant. $|PA - PB| = \text{Const}$

Ques. Find the equation of the locus of a points which moves so that the sum of its distances from the points (0, 3) and (0, - 3) is 8.

Ans.



$PA + PB = 8$

$\Rightarrow \sqrt{h^2 + (k - 3)^2} + \sqrt{h^2 + (k + 3)^2} = 8$

$\Rightarrow \sqrt{h^2 + (k - 3)^2} = 8 - \sqrt{h^2 + (k + 3)^2}$

Squaring both sides

$h^2 + (k - 3)^2 = 64 + h^2 + (k + 3)^2 - 2 \times 8\sqrt{h^2 + (k + 3)^2}$

$k^2 + 9 - 6k = 64 + k^2 + 9 + 6k - 16\sqrt{h^2 + (k + 3)^2}$

$\Rightarrow 16\sqrt{h^2 + (k + 3)^2} = 12k + 64$

$\Rightarrow 4\sqrt{h^2 + (k + 3)^2} = 3k + 16$

$\Rightarrow 16(h^2 + k^2 + 9 + 6k) = 9k^2 + 256 + 96k$

$\Rightarrow 16h^2 + 16k^2 + 144 + 96k = 9k^2 + 256 + 96k$

$\Rightarrow 16h^2 + 7k^2 = 256 - 144$

$$\Rightarrow \frac{16h^2 + 7k^2}{112} = \frac{112}{112}$$

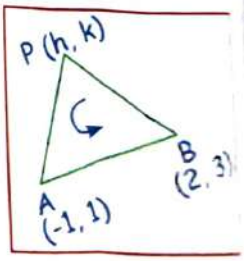
$$\Rightarrow \frac{h^2}{7} + \frac{k^2}{16} = 1 \rightarrow \frac{x^2}{7} + \frac{y^2}{16} = 1$$

Ques. Let A (- 1, 1) and B (2, 3) be two points and P be a variable point above the line AB such that the area of ΔPAB is 10. If the locus of P is $ax + by = 15$, then $5a + 2b$ is **[JEE Mains 5 April, 2024-II]**

(a) 6 (b) $\frac{-6}{5}$ (c) 4 (d) $\frac{-12}{5}$

Ans. $\because$ area of $\Delta PAB = 10$

$$\left|\frac{1}{2}\begin{vmatrix} h & k & 1 \\ -1 & 1 & 1 \\ 2 & 3 & 1 \end{vmatrix}\right| = 10$$



$$\Rightarrow \frac{1}{2}[h(1 - 3) - k(-1 - 2) + (-3 - 2)] = 10$$

$\Rightarrow -2h + 3k - 5 = 20 \Rightarrow -2x + 3y = 25$

$$\Rightarrow \frac{-2x}{5} + \frac{-3y}{5} = 5 \Rightarrow \frac{-6x}{5} + \frac{9y}{5} = 15$$

$$\therefore a = \frac{-6}{5}, b = \frac{9}{5}$$

$$5a + 2b = -6 + \frac{18}{5} = \frac{-30 + 18}{5} = \frac{-12}{5}$$

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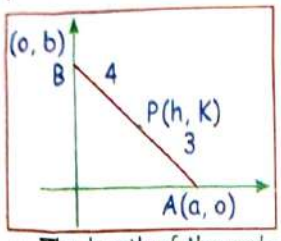
Ques. A fixed rod of length 5 units sliding between coordinates axes than find equation of locus of point which divides rod in the ratio 3 : 4.

Ans.



$\because$ The length of the rod = 5

$\therefore a^2 + b^2 = 25$ ...(i)

and $(h, k) = \left(\frac{4a+0}{4+3}, \frac{0+3b}{4+3}\right)$

$\Rightarrow h = \frac{4a}{7}, k = \frac{3b}{7} \Rightarrow a = \frac{7h}{4}, b = \frac{7k}{3}$

put the values of a & b is equation (i)

$\left(\frac{7h}{4}\right)^2 + \left(\frac{7k}{3}\right)^2 = 25 \Rightarrow \frac{h^2}{16} + \frac{k^2}{9} = \frac{25}{49}$

$\Rightarrow \frac{h^2}{25 \times 16} + \frac{k^2}{25 \times 9} = \frac{1}{49}$

$\Rightarrow \frac{h^2}{(20)^2} + \frac{k^2}{(15)^2} = \frac{1}{49}$

$(h, k) \rightarrow (x, y)$

$\Rightarrow \frac{x^2}{(20)^2} + \frac{y^2}{(15)^2} = \frac{1}{49}$

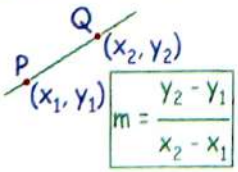
IMPORTANT TERMS FOR A STRAIGHT LINE

1. Inclination and Slope of a Line

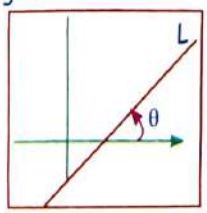
Let a line L make angle θ with the positive x - axis in anti-clockwise direction such that $0 \le \theta < \pi$. θ is called the inclination of the line.

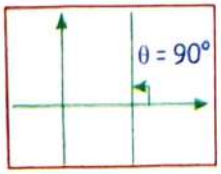
2. Slope (m)

$m = \tan\theta$, where θ is inclination angle.



$$m = \frac{y_2 - y_1}{x_2 - x_1}$$





$\Rightarrow$ $m = \tan 90°$

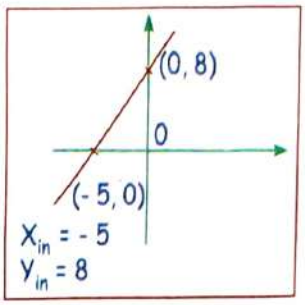
m = Not Defined

or

m is ∞

3. Intercepts

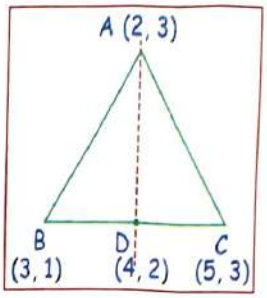
X-co-ordinates of the point where the line cuts X-axis is called X-intercept. Y-co-ordinate of the point where the line cuts Y-axis is called Y-intercept. Intercepts can be +ve or -ve.



Ques. If A (2, 3), B (3, 1) and C (5, 3) are three points, then the Slope of the line passing through A and bisecting BC is -

(a) $\frac{1}{2}$ (b) - 2

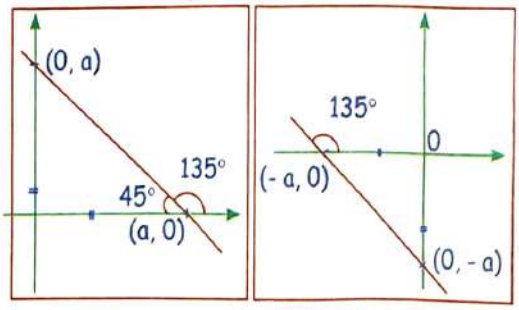
(c) $-\frac{1}{2}$ (d) 2



Ans. $m_{AD} = \frac{3-2}{2-4} = \frac{1}{-2}$

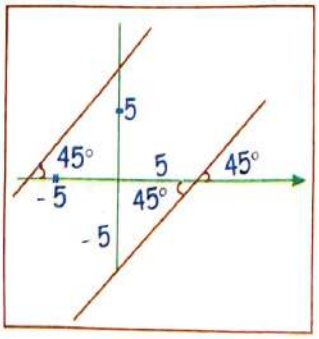
Important Note

1. If a line cuts off equal intercepts on co - ordinate axes then its slope is - 1.



$m = \tan 135° = -1$

2. If a line cuts off intercepts equal in magnitude but opposite in sign then its slope is 1.



$m = \tan 45° = 1$

3. If a line cuts off equal distances on co - ordinate axes then its slope is ± 1

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EQUATION OF STRAIGHT LINES

Slope Intercept Form

$y = mx + c$

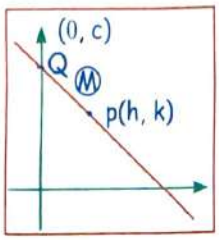
Slope — y - intercept

$m = \frac{k - c}{h - 0} \Rightarrow mh = k - c$

$(h, k) \to (x, y)$

$mx = y - c$

$\boxed{y = mx + c}$



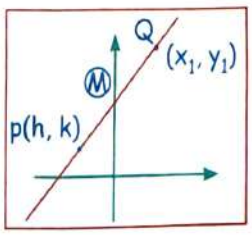
Slope Point Form

$y - y_1 = m[x - x_1]$

$\frac{k - y_1}{h - x_1} = m$

$\Rightarrow \frac{y - y_1}{x - x_1} = m$

$\boxed{y - y_1 = m(x - x_1)}$



Ques. Find the equation to the straight line cutting off an intercept 3 from the negative direction of the axis of y and inclined at 120° to the positive direction of x-axis.

(a) $y = \sqrt{3}x + 3$ (b) $y = \sqrt{3}x - 3$

(c) $y = -\sqrt{3}x - 3$ (d) $y = -\sqrt{3}x + 3$

Ans. $\theta = 120°$

$\Rightarrow m = \tan 120° = -\sqrt{3}$ and $c = -3$

$y = mx + c$, $y = -\sqrt{3}x - 3$

Ques. Find the equation of the line passing through intersection of the lines $x - 2y = 5$ & $2x + 3y + 4 = 0$ and

Ans. Put $x = 2y + 5$ in $2x + 3y + 4 = 0$

$\therefore 2(2y + 5) + 3y + 4 = 0 \Rightarrow 7y + 14 = 0$

$\Rightarrow y = -2$ and $x = 1$

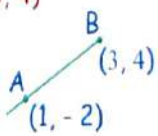
$\therefore P(1, -2)$

(1) Whose slope is 5

$\therefore y + 2 = 5(x - 1) \Rightarrow y + 2 = 5x - 5 \Rightarrow y + 7 = 5x$

(2) Which also passes through (3, 4)

$m = \frac{4 - (-2)}{3 - 1} = \frac{6}{2} = 3$



$\therefore y + 2 = 3(x - 1)$

$\Rightarrow y + 2 = 3x - 3 \Rightarrow y = 3x - 5$

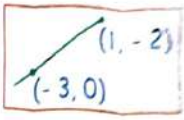
(3) Which cuts off Equal intercepts on co-ordinate axes.

$m = -1 \Rightarrow y + 2 = -1(x - 1) \Rightarrow y + 2 = -x + 1$

$\Rightarrow y + x + 1 = 0$

(4) Whose x intecept is - 3

$\therefore m = \frac{0 - (-2)}{-3 - 1} = \frac{-1}{2}$



thus, $(y + 2) = \frac{-1}{2}(x - 1)$

(5) Whose inclination angle is 75°

$m = \tan 75° = 2 + \sqrt{3} \Rightarrow y + 2 = m(x - 1)$

$y + 2 = (2 + \sqrt{3})(x - 1)$

Ques. Find the equation to the straight line passing through the point, (3, - 4) and cutting off intercepts, equal in magnitude but of opposite in signs from the two axes.

(a) $x + y + 1 = 0$ (b) $x - y - 7 = 0$

(c) $y = x + 7$ (d) $x + y + 7 = 0$

Ans. Point (3, - 4), $m = 1$

$y - (-4) = 1(x - 3)$

$\Rightarrow y + 4 = 1(x - 3)$

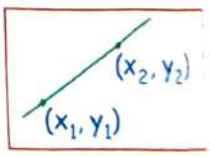
$\Rightarrow y = x - 7$

Two Point Form

$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)$

Where, $m = \frac{y_2 - y_1}{x_2 - x_1}$

$y - y_1 = m(x - x_1)$



Ques. Find the equation of the line passing through origin and parallel to the line $3x + 5y - 1 = 0$.

Ans. $5y = -3x + 1 \Rightarrow y = \frac{-3}{5}x + \frac{1}{5}$

$\therefore m = \frac{-3}{5}$

equation of new Line:

$y - 0 = \frac{-3}{5}(x - 0) \Rightarrow y = \frac{-3}{5}$

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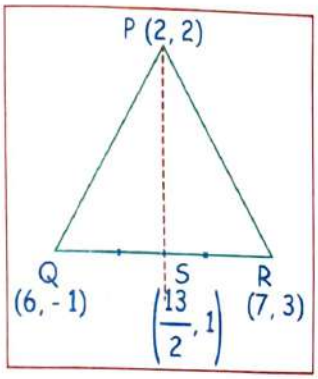
Ques. Let PS be the median of the triangle with vertices P (2, 2), Q (6, - 1) and R (7, 3). The equation of the line passing through (1, - 1) and parallel to PS is

(a) $2x - 9y - 7 = 0$ (b) $2x - 9y - 11 = 0$

(c) $2x + 9y - 11 = 0$ (d) $2x + 9y + 7 = 0$

Ans. $m = \dfrac{2 - 1}{2 - \frac{13}{2}} = \dfrac{1}{\frac{-9}{2}} = \dfrac{-2}{9}$

$m = \dfrac{-2}{9}$



Point (1, - 1)

$y + 1 = m(x - 1)$

$y + 1 = \dfrac{-2}{9}(x - 1)$

$9y + 9 = -2x + 2$

$9y + 2x + 7 = 0$

Ques. The line joining two points A (2, 0) and B (3, 1) is rotated about A in anticlockwise direction through an angle of 15°. Find the equation of the line in the new position.

Ans. $m_{AB} = \dfrac{1 - 0}{3 - 2} = 1$

$\Rightarrow \tan\theta = 1$

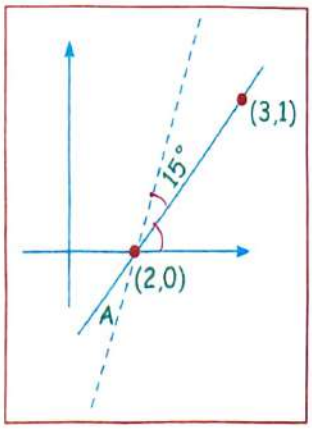
$\therefore \theta = 45°$

$m_{New} = \tan(60°)$

$= \sqrt{3}$

$y - 0 = m(x - 2)$

$y = \sqrt{3}(x - 2)$



Ques. A line passing through the point A (9, 0) makes an angle of 30° with the positive direction of x - axis. If this line is rotated about A through an angle of 15° in the clockwise direction, then its equation in the new position is

[JEE Mains 30 Jan, 2024-I]

(a) $\dfrac{y}{\sqrt{3} - 2} + x = 9$ (b) $\dfrac{x}{\sqrt{3} - 2} + y = 9$

(c) $\dfrac{x}{\sqrt{3} + 2} + y = 9$ (d) $\dfrac{y}{\sqrt{3} + 2} + x = 9$

Ans. $m_{New} = \tan 15° = 2 - \sqrt{3}$

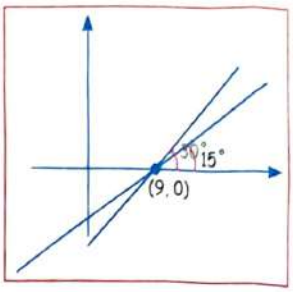
$y - 0 = m(x - 9)$

$\Rightarrow y = (2 - \sqrt{3})(x - 9)$

$\Rightarrow \dfrac{y}{2 - \sqrt{3}} = x - 9$

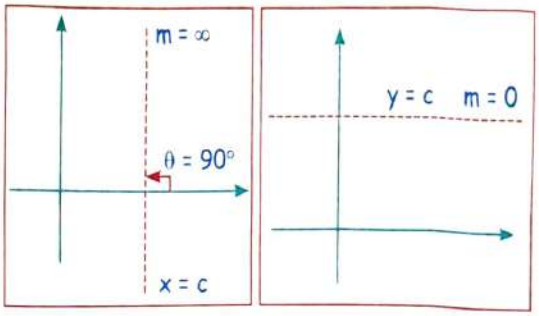
$\Rightarrow 9 = x - \dfrac{y}{2 - \sqrt{3}}$

$\Rightarrow 9 = x + \dfrac{y}{\sqrt{3} - 2}$



Important Note

Horizontal & Vertical Lines



x - y Intercept Form

$\dfrac{x}{a} + \dfrac{y}{b} = 1$

Proof: $m = \dfrac{b - 0}{0 - a} = \dfrac{-b}{a}$

$m = \dfrac{-b}{a}$

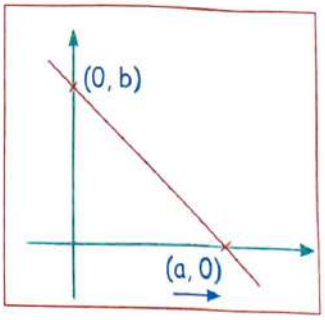
$y - 0 = m(x - a)$

$\Rightarrow y = \dfrac{-b}{a}(x - a)$

$\Rightarrow ay = -bx + ab$

$\Rightarrow \dfrac{bx + ay}{ab} = \dfrac{ab}{ab}$

$\Rightarrow \dfrac{x}{a} + \dfrac{y}{b} = 1$



Ques. Find equation of straight line through (1, 2) whose x - intercept is twice of y - intercept.

Ans. Equation of Straight Line:

$\dfrac{x}{a} + \dfrac{y}{b} = 1$

$x_{in} = a, y_{in} = b$

$a = 2b$ ...(i)

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and $(x, y) = (1, 2)$

$$\frac{1}{2b} + \frac{2}{b} = 1$$

$$\Rightarrow \frac{1+4}{2b} = 1 \Rightarrow b = \frac{5}{2} \text{ and } a = 5$$

$$\frac{x}{5} + \frac{y}{\frac{5}{2}} = 1 \Rightarrow \frac{x}{5} + \frac{2y}{5} = 1 \Rightarrow x + 2y = 5$$

Ques. Find the equation of a line passing through P (4, 3) & for which portion of its intercept between the co - ordinate axes is bisected at the point P.

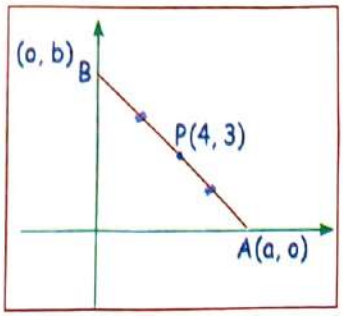
Ans. $\because 4 = \frac{a+0}{2}$

$\Rightarrow a = 8$

and $3 = \frac{b+0}{2}$

$\Rightarrow b = 6$

$\therefore \frac{x}{8} + \frac{y}{6} = 1$



Ques. A variable line is drawn passing through P (4, 3). Find the locus of mid point of its portion intercepted between co - ordinate axes.

Ans. $\frac{a+0}{2} = h \Rightarrow a = 2h$

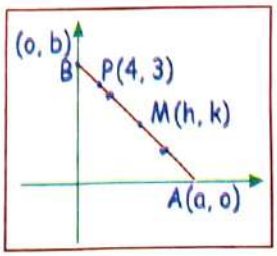
$\frac{b+0}{2} = k \Rightarrow b = 2k$

$\because \frac{x}{a} + \frac{y}{b} = 1$

$\Rightarrow \frac{x}{2h} + \frac{y}{2k} = 1$

$\therefore \frac{4}{2h} + \frac{3}{2k} = 1$

$\Rightarrow \frac{2}{x} + \frac{3}{2y} = 1$



Ques. A straight line through a fixed point (2, 3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus R is **[JEE Main 2018]**

(a) $3x + 2y = 6$ (b) $2x + 3y = xy$

(c) $3x + 2y = xy$ (d) $3x + 2y = 6xy$

Ans. $\because$ OPRQ is a rectangle

So, OP = QR and

OQ = PR

$\therefore$ Q = (0, k) and

P = (h, 0)

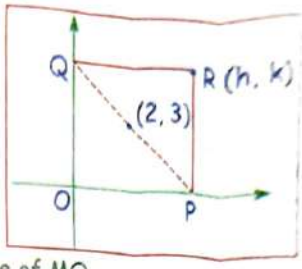
Now,

Slope of PM = slope of MQ

$$\frac{0-3}{h-2} = \frac{3-k}{2}$$

$\Rightarrow 6 = hk - 3h - 2k + 6 \Rightarrow 3h + 2k = hk$

$\therefore$ locus of R : $3x + 2y = xy$



Ques. Find number of straight lines passing through (2, 4) & forming a triangle of 16 sq. cm with the co - ordinate axes.

Ans. $\frac{x}{a} + \frac{y}{b} = 1$

$\frac{2}{a} + \frac{4}{b} = 1$ ...(i)

$\frac{1}{2}|a.b| = 16$

$\Rightarrow |ab| = 32$

$ab = \pm 32$ ...(ii)

When ab = 32

$b = \frac{32}{a}$

$\therefore \frac{2}{a} + \frac{4 \cdot a}{32} = 1 \Rightarrow \frac{2}{a} + \frac{a}{8} = 1 \Rightarrow \frac{16 + a^2}{8a} = 1$

$\Rightarrow a^2 + 16 = 8a \Rightarrow a^2 - 8a + 16 = 0$

$\therefore (a-4)^2 = 0 \Rightarrow a = 4$ and $b = \frac{32}{4} = 8$

When ab = - 32

$b = -\frac{32}{a} \Rightarrow \frac{2}{a} + \frac{4 \times a}{-32} = 1 \Rightarrow \frac{2}{a} - \frac{a}{8} = 1$

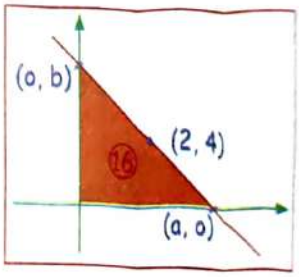
$\Rightarrow \frac{16 - a^2}{8a} = 1 \Rightarrow 16 - a^2 = 8a$

$\Rightarrow a^2 + 8a - 16 = 0$

$\because D > 0$

$\therefore$ It gives 2 values of a

Hence, total solutions are 3



Ques. Let A (- 2, - 1), B (1, 0), C (α, β) and D (γ, δ) be the vertices of a parallelogram ABCD. If the point C lies on $2x - y = 5$ and the point D lies on $3x - 2y = 6$, then the value of $|\alpha + \beta + \gamma + \delta|$ is equal to _____.

[JEE Main, 31 Jan 2024-II]

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Ans. $2\alpha - \beta = 5$ ...(i)

$3\gamma - 2\delta = 6$ ...(ii)

$\because$ Mid points of A & C and B & C will be same

$\frac{\gamma + 1}{2} = \frac{\alpha - 2}{2} \Rightarrow \alpha = \gamma + 3$ ...(iii)

$\frac{\delta + 0}{2} = \frac{\beta - 1}{2} \Rightarrow \beta = \delta + 1$ ...(iv)

Putting the value of α and β in equation (i)

$2(\gamma + 3) = 5 + \delta + 1$

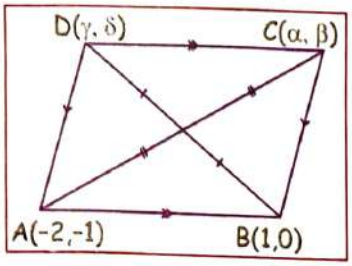
$\Rightarrow 2\gamma + 6 = 6 + \delta$

$\Rightarrow \delta = 2\gamma$

$3\gamma - 2(2\gamma) = 6$

$\Rightarrow 3\gamma - 4\gamma = 6$

$\gamma = -6$ and $\delta = -12$



Similarly, $\beta = -11$ and $\alpha = -6 + 3 = -3$

$\therefore |-11 - 3 - 6 - 12| = 32$

General Form

$$ax + by + c = 0$$

$\Rightarrow by = -ax - c \Rightarrow y = \frac{-a}{b}x - \frac{c}{b}$

$\Rightarrow$ Slope $(m) = \frac{-a}{b}$

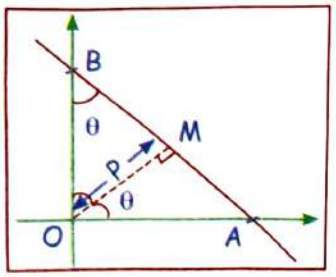
$m = \frac{-\text{coeff of } x}{\text{coeff of } y}$

Normal Form

$$x\cos\theta + y\sin\theta = P \quad [P > 0]$$

'P' $\rightarrow$ Length of $\perp$ from (0, 0) upon the line

'θ' $\rightarrow$ inclination angle of this perpendicular



In ΔOMA

$\cos\theta = \frac{P}{OA}$

$\Rightarrow OA = P\sec\theta$

In ΔOMB

$\sin\theta = \frac{P}{OB}$

$\Rightarrow OB = P\,\text{cosec}\,\theta$

$\frac{x}{x_{in}} + \frac{y}{y_{in}} = 1 \Rightarrow \frac{x}{P\sec\theta} + \frac{y}{P\,\text{cosec}\,\theta} = 1$

$\Rightarrow x\cos\theta + y\sin\theta = P$

Ques. Convert the line $3x - 4y = -10$ into,

(1) x - y intercept Form

Ans. Divide by -10

$\frac{3x}{-10} - \frac{4y}{-10} = 1$

$\frac{x}{\frac{-10}{3}} + \frac{y}{\frac{10}{4}} = 1$

(2) Normal Form

Ans. $3x - 4y = -10$

divide by $\sqrt{3^2 + (-4)^2}$

$\frac{3x}{5} - \frac{4y}{5} = \frac{-10}{5}$

$\Rightarrow \frac{3x}{5} - \frac{4y}{5} = -2 \Rightarrow \frac{-3x}{5} + \frac{4y}{5} = 2$

Ques. A straight line cuts off the intercepts $OA = a$ and $OB = b$ on the positive direction of x - axis and y - axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of y - axis and the area of ΔOAB is $\frac{98}{3}\sqrt{3}$, then $a^2 - b^2$ is equal to: **[JEE Main Jan 2023-I]**

(a) $\frac{392}{3}$ (b) 196 (c) $\frac{196}{3}$ (d) 98

Ans. Let OM = P

equation of Line AB:

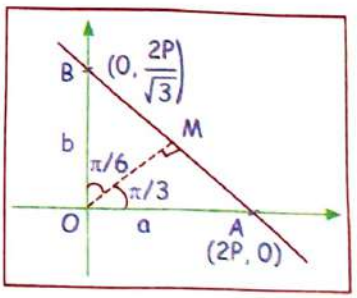
$x\cos 60° + y\sin 60° = P$

$\frac{x}{2} + \frac{y\sqrt{3}}{2} = P$

$\Rightarrow x + \sqrt{3}y = 2P$

$X_{in} = 2P$

$Y_{in} = \frac{2P}{\sqrt{3}}$



$\therefore \left|\frac{1}{2} \cdot 2P \cdot \frac{2P}{\sqrt{3}}\right| = \frac{98}{\sqrt{3}}$

$\Rightarrow |2P^2| = 98$

$\Rightarrow P^2 = 49 \Rightarrow P = 7$

$a = 2P = 14 \Rightarrow a^2 = 14^2$

and $b = \frac{2P}{\sqrt{3}} = \frac{14}{\sqrt{3}} \Rightarrow b^2 = \frac{14^2}{3}$

$\Rightarrow a^2 - b^2 = 14^2 - \frac{14^2}{3} = \frac{196 \times 2}{3}$

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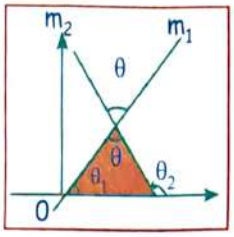
ANGLE BETWEEN LINES

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

$\theta \rightarrow$ is acute angle b/w the 2 lines

Proof: $m_1 = \tan\theta_1$ & $m_2 = \tan\theta_2$



$\theta_2 = \theta + \theta_1$

$\theta = |\theta_2 - \theta_1|$

taking tan both sides

$\tan\theta = |\tan(\theta_2 - \theta_1)|$

$$\Rightarrow \left|\frac{\tan\theta_2 - \tan\theta_1}{1 + \tan\theta_2 \tan\theta_1}\right| = \left|\frac{m_2 - m_1}{1 + m_2 m_1}\right|$$

Ques. Find the acute angle between lines given by $x - y = 7$ & $2x - 3y = 5$

(a) $\tan^{-1}\frac{2}{3}$ (b) $\tan^{-1}\frac{1}{3}$

(c) $\tan^{-1}\frac{1}{5}$ (d) $\tan^{-1}\frac{2}{5}$

Ans. $\because m_1 = 1$ and $m_2 = \frac{2}{3}$

$$\therefore \tan\theta = \left|\frac{1 - \frac{2}{3}}{1 + 1 \times \frac{2}{3}}\right| = \left|\frac{\frac{1}{3}}{\frac{5}{3}}\right| = \frac{1}{5}$$

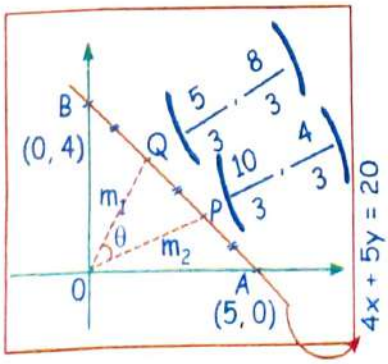
$$\Rightarrow \theta = \tan^{-1}\left(\frac{1}{5}\right)$$

Ques. The portion of the line $4x + 5y = 20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin. The tangent of an angle between the lines L_1 and L_2 is:

[JEE Main 27 Jan 2024-II]

(a) $\frac{8}{5}$ (b) $\frac{25}{1}$ (c) $\frac{2}{5}$ (d) $\frac{30}{41}$

Ans.



$$\therefore P = \left(\frac{1\times 0 + 2\times 5}{1+2}, \frac{1\times 4 + 2\times 0}{1+2}\right) = \left(\frac{10}{3}, \frac{4}{3}\right)$$

$$Q = \left(\frac{1\times 5 + 2\times 0}{1+2}, \frac{1\times 0 + 2\times 4}{1+2}\right) = \left(\frac{5}{3}, \frac{8}{3}\right)$$

Now, slope of OQ $= m_1 = \frac{\left(\frac{8}{3}\right)}{\left(\frac{5}{3}\right)} = \frac{8}{5}$

and slope of OP $= m_2 = \frac{\left(\frac{4}{3}\right)}{\left(\frac{10}{3}\right)} = \frac{2}{5}$

thus,

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{\left(\frac{8}{5} - \frac{2}{5}\right)}{1 + \left(\frac{8}{5}\right)\left(\frac{2}{5}\right)}\right|$$

$$= \frac{\left(\frac{6}{5}\right)}{\left(\frac{41}{25}\right)} = \frac{6}{5} \times \frac{25}{41} = \frac{30}{41}$$

Ques. If the line segment joining the points (5, 2) and (2, a) subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible value of a is:

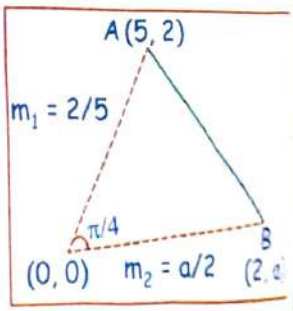
[JEE Main 08 Apr. 2024-I]

(a) 6 (b) 8 (c) 2 (d) 4

Ans. $\tan\frac{\pi}{4} = \left|\frac{\frac{2}{5} - \frac{a}{2}}{1 + \frac{2}{5}\cdot\frac{a}{2}}\right|$

$$\Rightarrow 1 = \left|\frac{\frac{4 - 5a}{10}}{\frac{10 + 2a}{10}}\right|$$

$$\Rightarrow \pm 1 = \frac{4 - 5a}{10 + 2a}$$



$10 + 2a = 4 - 5a$ or $-10 - 2a = 4 - 5a$

$\Rightarrow 7a = -6$ $\Rightarrow 3a = 14$

$\Rightarrow a = \frac{-6}{7}$ $\Rightarrow a = \frac{14}{3}$

Product $= \left|\frac{-6}{7} \times \frac{14}{3}\right| = 4$

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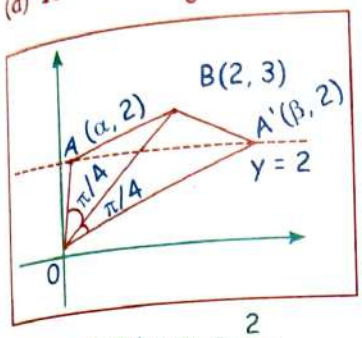
Ques. The distance between the two points A and A' which lie on $y = 2$ such that both the line segments AB and A'B (where B is the point (2, 3)) subtend angle $\frac{\pi}{4}$ at the origin, is equal to

[JEE Main June 2022-I)]

(a) 10 (b) $\frac{48}{5}$ (c) $\frac{52}{5}$ (d) 3

Ans.



Slope of $OA = m_1 = \frac{2}{\alpha}$

Slope of $OB = m_2 = \frac{3}{2}$

Slope of $OA' = M_3 = \frac{2}{\beta}$

$$\text{So, } \tan\frac{\pi}{4} = \left|\frac{\left(\frac{3}{2} - \frac{2}{\alpha}\right)}{1 + \left(\frac{3}{2}\right)\left(\frac{2}{\alpha}\right)}\right|$$

$$\Rightarrow 1 = \frac{3\alpha - 4}{2\alpha + 6} \Rightarrow 2\alpha + 6 = 3\alpha - 4 \Rightarrow \alpha = 10$$

Similarly

$$\tan\frac{\pi}{4} = \left|\frac{\left(\frac{2}{\beta} - \frac{3}{2}\right)}{1 + \left(\frac{3}{2}\right)\left(\frac{2}{\beta}\right)}\right| = \left|\frac{4 - 3\beta}{2\beta + 6}\right|$$

$$\Rightarrow 2\beta + 6 = 4 - 3\beta \Rightarrow \beta = \frac{-2}{5}$$

Hence, the distance between A and A'

$$= \sqrt{\left(10 + \frac{2}{5}\right)^2 + (2-2)^2} = \frac{52}{5}$$

whenever degree of Eqⁿ in m gets reduced due to cancellation, always take one slope as ∞

Important Note

1. If line are parallel $\Rightarrow \theta = 0 \Rightarrow \boxed{m_1 = m_2}$
2. If line are perpendicular $\Rightarrow \theta = 90^\circ \Rightarrow \boxed{m_1 m_2 = -1}$

Ques. In a triangle PQR, the co-ordinates of the point P and Q are (-2, 4) and (4, -2) respectively. If the equation of the perpendicular bisector of PR is $2x - y + 2 = 0$, Then the circumcenter of the ΔPQR is

[JEE Main 2021]

(a) (-1, 0) (b) (1, 4) (c) (0, 2) (d) (-2, -2)

Ans. $M\left(\frac{-2+4}{2}, \frac{4-2}{2}\right) = (1, 1)$

$m_{PQ} = \frac{-4-2}{4+2} = -1$

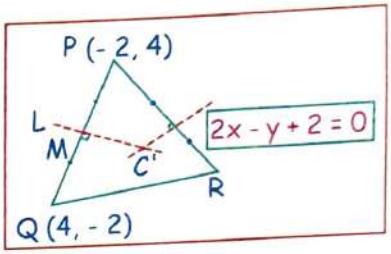
$\therefore m_L = 1$

eqⁿ of L:

$y - 1 = 1(x - 1)$

$\Rightarrow y - 1 = x - 1$

$\Rightarrow y = x \rightarrow \perp$ bisector



Solve both ⊥ bisectors to get circumcentre

$2x - x + 2 = 0$

$\Rightarrow x = -2$ and $y = -2$

Hence, circumcentre of ΔPQR is (-2, -2)

Ques. If the perpendicular bisector of the line segment joining the points P (1, 4) & Q (k, 3) has y intercept equal to - 4 then a value of k is equal to:

(a) $\sqrt{14}$ (b) $\sqrt{15}$ (c) - 4 (d) - 2

Ans. $m_{PQ} = \frac{4-3}{1-k} = \frac{1}{1-k}$

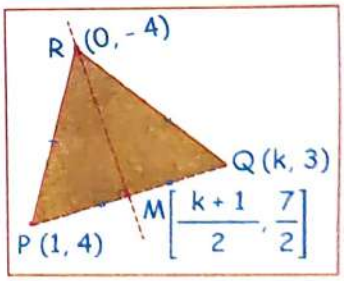
$$m_{RM} = \frac{\frac{7}{2} - (-4)}{\frac{k+1}{2} - 0} = \frac{15}{k+1}$$

$m_{PQ} \cdot m_{RM} = -1$

$\Rightarrow \frac{1}{1-k} \cdot \frac{15}{k+1} = -1$

$\Rightarrow 15 = (k+1)(k-1)$

$\Rightarrow 15 = k^2 - 1 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$



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Ques. If the orthocentre of the triangle, whose vertices are (1, 2), (2, 3) and (3, 1) is (α, β) then the quadratic equation whose roots are $\alpha + 4\beta$ and $4\alpha + \beta$, is

(a) $x^2 + 19x + 90 = 0$

(b) $x^2 - 18x + 80 = 0$

(c) $x^2 - 22x + 120 = 0$

(d) $x^2 - 20x + 99 = 0$

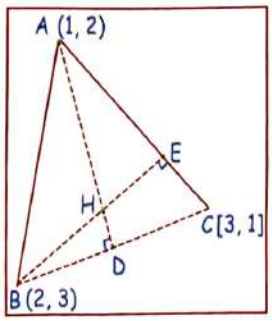
Ans. $m_{BC} = \frac{3-1}{2-3} = \frac{2}{-1} = -2$

$m_{AD} = \frac{1}{2}$

eqⁿ of AD: $y - 2 = \frac{1}{2}(x - 1)$

$2y - 4 = x - 1$

$2y = x + 3$ ...(i)



$m_{AC} = \frac{2-1}{1-3} = \frac{-1}{2}$

$\therefore m_{BE} = 2$

eqⁿ of BE:

$y - 3 = 2(x - 2)$

$\Rightarrow y - 3 = 2x - 4$

$\Rightarrow y = 2x - 1$ ...(ii)

By using equation (i) and (ii)

$2(2x - 1) = x + 3$

$\Rightarrow 4x - 2 = x + 3$

$\Rightarrow x = \frac{5}{3}$

and $y = \frac{10}{3} - 1 = \frac{7}{3}$

thus, orthocentere = $H\left[\frac{5}{3}, \frac{7}{3}\right]$

$\alpha = \frac{5}{3}, \beta = \frac{7}{3}$

$\therefore \alpha + 4\beta = 11$ and $4\alpha + \beta = 9$

therefore, the quadratic equation is:

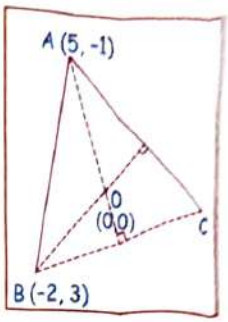
$x^2 - (11 + 9)x + 11 \times 9 = 0$

$\Rightarrow x^2 - 20x + 99 = 0$

Ques. Two vertices of a triangle are (5,-1) and (-2, 3). If origin is the orthocentre of the triangle, then the third vertex is

(a) (4,7) (b) (-4,7) (c) (4,-7) (d) (-4,-7)

Ans. $M_{AO} = \frac{-1-0}{5-0} = -\frac{1}{5}$



$\therefore M_{BC} = 5$

$M_{OB} = \frac{3-0}{-2-0} = \frac{-3}{2}$

$\therefore M_{AC} = \frac{2}{3}$

eqⁿ of AC:

$y + 1 = \frac{2}{3}(x - 5)$ ...(i)

eqⁿ of BC:

$y - 3 = 5(x + 2)$ ...(ii)

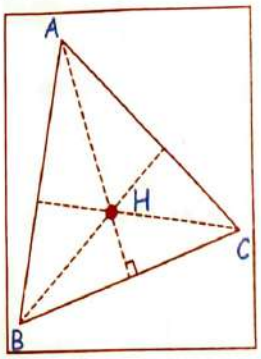
by solving equation (i) and (ii):

We get $x = -4, y = -7$

Important Note

Orthocentric System

In a Triangle ABC, the point A, B, C & H from a Orthocentric System, which means that the orthocenter of triangle formed by any 3 of these points is the 4th point.



	Orthocentre
ΔABC	H
AHB	C
AHC	B
BHC	A

Ques. Two vertices of a triangle ABC are A(3, -1) and B (- 2, 3), and its orthocentre is P(1, 1) .If the coordinates of the points C are (α, β) and the centre of the of the circle circumscribing the triangle PAB is (h, K), then the value of $(\alpha + \beta) + 2(h + k)$ equals

[JEE Main 9 Apr. 2024-II)]

Ans. $M_{PC} \cdot M_{AB} = -1$

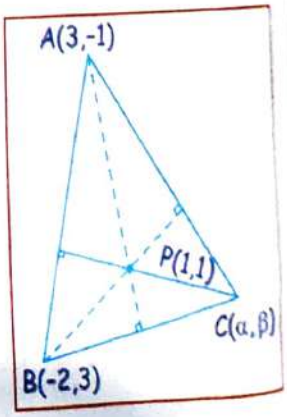
$\frac{\beta - 1}{\alpha - 1}\frac{4}{-5} = -1$

$4(\beta - 1) = 5(\alpha - 1)$

$4\beta = 5\alpha - 1$ ...(i)

$M_{AC} \cdot M_{BP} = -1$

$\left(\frac{\beta + 1}{\alpha - 3}\right)\left(\frac{3 - 1}{-2 - 1}\right) = -1$



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$2(\beta + 1) = 3(\alpha - 3)$

$2\beta = 3\alpha - 11$ ...(ii)

$2(3\alpha - 11) = 5\alpha - 1$

$6\alpha - 22 = 5\alpha - 1$

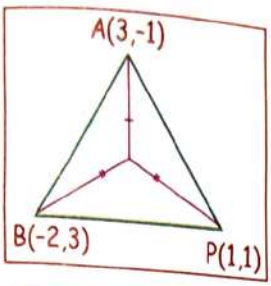
$\alpha = 21$

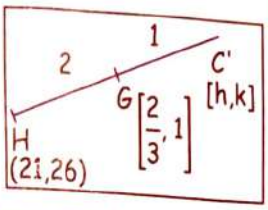
$2\beta = 3 \times 21 - 11$

$2\beta = 63 - 11$

$2\beta = 52$

$\beta = 26$





$\Rightarrow$ Orthocentre of ΔPAB is (21, 26)

and $G = \left[\frac{2}{3}, 1\right]$

$\frac{2k + 26}{3} = 1$ | $\frac{2h + 21}{3} = \frac{2}{3}$

$k = \frac{-23}{2}$ | $h = \frac{-19}{2}$

Hence, $\alpha + \beta + 2h + 2k$

$21 + 26 - 19 - 23 = 2 + 3 = 5$

Important Note

1. Equation of a line parallel to $ax + by + c = 0$ is given by $ax + by + k = 0$
2. Equation of a line Perpendicular to $ax + by + c = 0$ is given by $bx - ax + k = 0$

Example:

Line	Perpendicular Line
$2x + 3y + 7 = 0$	$3x - 2y + k = 0$
$4x - 7y + 10 = 0$	$-7x - 4y + k = 0$
$-5x + 8y + 3 = 0$	$8x + 5y + k = 0$

Ques. A line perpendicular to the line $ax + by + c = 0$ and passes through (a,b). The equations of the line is

(a) $bx - ay + (a^2 - b^2) = 0$

(b) $bx - ay - (a^2 - b^2) = 0$

(c) $bx - ay = 0$

(d) None of these

Ans. The perpendicular line is:

$bx - ay + k = 0$

at $x = a$, $y = b$

$ba - ab + k = 0 \Rightarrow k = 0$

$\Rightarrow$ eqn of Line $bx - ay = 0$

Ques. The equations to the straight line passing through the point of intersection of the lines $5x - 6y - 1 = 0$ and $3x + 2y + 53 = 0$ and perpendicular to the line $3x - 5y + 11 = 0$ is

(a) $5x + 3y + 8 = 0$

(b) $3x - 5y + 8 = 0$

(c) $5x + 3y + 11 = 0$

(d) $3x - 5y + 11 = 0$

Ans. Let the line be

$-5x - 3y + k = 0$

The point of intersection will be

$5x - 6y - 1 = 0$

$9x + 6y + 15 = 0$

$14x + 14 = 0$

$x = -1, y = -1$

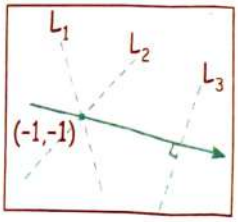
$\therefore 5 + 3 + k = 0$

$\Rightarrow k = -8$

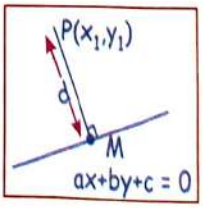
Thus

$-5x - 3y - 8 = 0$

$\Rightarrow 5x + 3y + 8 = 0$



PERPENDICULAR DISTANCE OF A POINT FROM A LINE



$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$ N^r → |point ko line me put| ; D^r → $\sqrt{(\text{coeff of } x)^2 + (\text{coeff of } y)^2}$

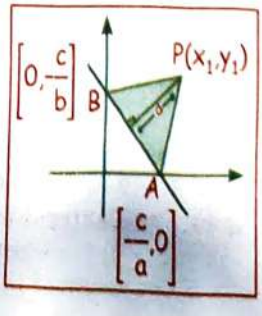
Proof:

ΔPAB:

Area of $\Delta PAB = \frac{1}{2}\sqrt{\frac{c^2}{a^2} + \frac{c^2}{b^2}}\, d$

$= \frac{1}{2} c \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}\, d$

$\Rightarrow \Delta = \frac{1}{2} \frac{cd}{ab} \sqrt{a^2 + b^2}$ ...(i)



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Area of $\Delta PAB = \left|\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ 0 & -\frac{c}{b} & 1 \\ -\frac{c}{a} & 0 & 1 \end{vmatrix}\right|$

$= \frac{1}{2}\left[\left|x_1\left[-\frac{c}{b}\right] - y_1\left[+\frac{c}{a}\right] + 1\left(\frac{-c^2}{ab}\right)\right|\right]$

$= \frac{1}{2}\left[\frac{cx_1}{b} + \frac{cy_1}{a} + \frac{c^2}{ab}\right] = \frac{1}{2}\left|\frac{acx_1 + bcy_1 + c^2}{ab}\right|$

$= \frac{1}{2}\frac{c}{ab}|ax_1 + by_1 + c| \ldots (ii)$

By equation (i) and (ii)

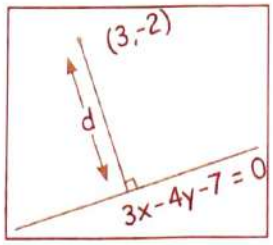
$= \frac{1}{2}\frac{cd}{ab}\sqrt{a^2+b^2} = \frac{1}{2}\frac{c}{ab}|ax_1 + by_1 + c|$

$\Rightarrow d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2+b^2}}$

Ques. Find distance of (3,-2) from line $3x - 7 = 4y$.

Ans. $d = \left|\frac{3\times 3 - 4(-2) - 7}{\sqrt{3^2+4^2}}\right|$

$= \left|\frac{9+8-7}{5}\right| = \frac{10}{5} = 2$



Ques. Let R be the point (3, 7) and let P and Q be two point on the line $x + y = 5$ such that PQR is an equilateral triangle. Then the area of ΔPQR is:

[JEE Main 1 June 2022-I]

(a) $\frac{25}{4\sqrt{3}}$ (b) $\frac{25\sqrt{3}}{2}$ (c) $\frac{25}{\sqrt{3}}$ (d) $\frac{25}{2\sqrt{3}}$

Ans. $RM = \left|\frac{3+7-5}{\sqrt{2}}\right| = \frac{5}{\sqrt{2}}$

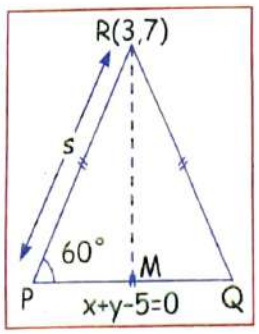
In ΔRPM

$\sin 60 = \frac{RM}{S}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{5}{\sqrt{2}s}$

$\Rightarrow S = \frac{10}{\sqrt{3}\sqrt{2}}$

$A = \frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}\frac{(10)10}{3\times 2} = \frac{25}{2\sqrt{3}}$



Ques. The Centroid of an equilateral triangle is (2, - 1) the equation of its base in $x + 2y = 1$. The length of its sides is

(a) $\frac{1}{\sqrt{5}}$ (b) $\frac{2\sqrt{5}}{\sqrt{3}}$ (c) $\frac{\sqrt{3}}{\sqrt{5}}$ (d) $\frac{2\sqrt{3}}{\sqrt{5}}$

Ans. $X = GM = \left|\frac{2-2-1}{\sqrt{1^2+2^2}}\right| = \frac{1}{\sqrt{5}}$

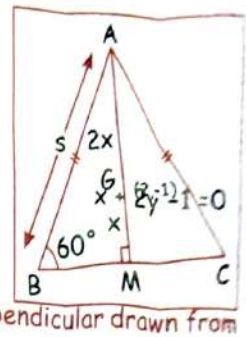
$AM = x + 2x = 3x = \frac{3}{\sqrt{5}}$

In ΔABM

$\frac{AM}{S} = \sin 60°$

$\Rightarrow AM = \frac{\sqrt{3}}{2}S$

$\Rightarrow \frac{3}{\sqrt{5}} = \frac{\sqrt{3}}{2}S \Rightarrow S = \frac{2\sqrt{3}}{\sqrt{5}}$



Ques. If the length of the perpendicular drawn from the origin to the line whose intercepts on the axes are a and b be p, then

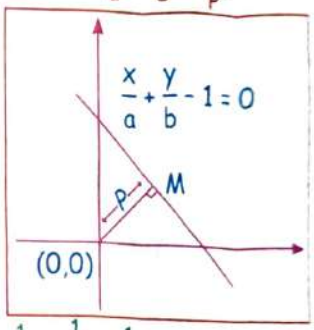
(a) $a^2 + b^2 = p^2$ (b) $a^2 + b^2 = \frac{1}{p^2}$

(c) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2}$ (d) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2}$

Ans. $P = \left|\frac{\frac{0}{a}+\frac{0}{b}-1}{\sqrt{\left(\frac{1}{a}\right)^2+\left(\frac{1}{b}\right)^2}}\right|$

$\Rightarrow p = \frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}$

$\Rightarrow p^2 = \frac{1}{\frac{1}{a^2}+\frac{1}{b^2}} \Rightarrow \frac{1}{a^2}+\frac{1}{b^2} = \frac{1}{p^2}$



Ques. A Straight line make variable a and b on the co-ordinate axes if $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$ (c is a constant) the equations of the path described by the foot of the perpendicular from the origin to the line, is

(a) $x^2 + y^2 = c^2$ (b) $x^2 + y^2 = \frac{c^2}{2}$

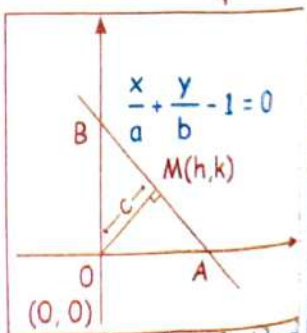
(c) $x^2 + y^2 = 2c^2$ (d) $x^2 + y^2 = \frac{c^2}{4}$

Ans. $OM = \left|\frac{-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}\right|$

$\Rightarrow OM = \left|\frac{1}{\sqrt{\frac{1}{c^2}}}\right|$

$OM = |c| \rightarrow$ Const. $\Rightarrow h^2 + k^2 = |c| \Rightarrow h^2 + k^2 = c^2$

$\therefore x^2 + y^2 = c^2$



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DISTANCE BETWEEN PARALLEL LINES

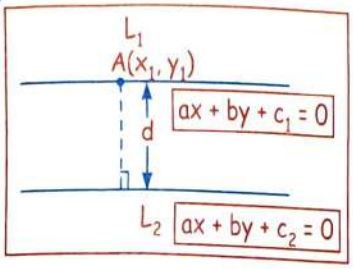
$$d = \frac{|c_2 - c_1|}{\sqrt{a^2 + b^2}}$$

Proof → Let (x_1, y_1) be a point on L_1

$ax_1 + by_1 + c_1 = 0$

$ax_1 + by_1 = -c_1$ ...(i)

$$d = \left|\frac{ax_1 + by_1 + c_2}{\sqrt{a^2 + b^2}}\right|$$



By equation (i)

$$d = \left|\frac{-c_1 + c_2}{\sqrt{a^2 + b^2}}\right|$$

$$\Rightarrow d = \left|\frac{c_2 - c_1}{\sqrt{a^2 + b^2}}\right|$$

Ques. Two tangents to a circle are given by $3x + 4y = 5$ and $6x + 8y + 15 = 0$. Find the radius of this circle.

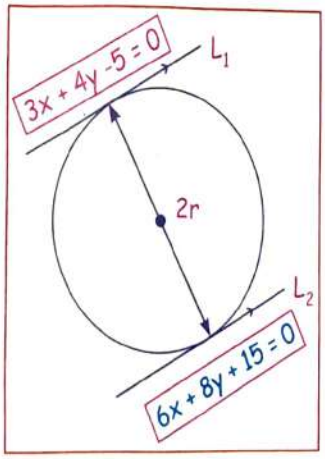
Ans. $\because L_1$ and L_2 are parallel.

$$\therefore \left|\frac{\frac{15}{2} - (-5)}{\sqrt{3^2 + 4^2}}\right| = 2r$$

$$\Rightarrow \left|\frac{\frac{25}{2}}{5}\right| = 2r$$

$$\Rightarrow \frac{5}{2} = 2r$$

$$\Rightarrow r = \frac{5}{4}$$



Ques. The equations of the lines through the point of intersection of the lines $x - y + 1 = 0$ and $2x - 3y + 5 = 0$ and whose distance from the point [3, 2] is $\frac{7}{5}$, is **[IIT JEE 1963]**

(a) $3x - 4y - 6 = 0$ and $4x + 3y + 1 = 0$

(b) $3x - 4y + 6 = 0$ and $4x - 3y - 1 = 0$

(c) $3x - 4y + 6 = 0$ and $4x - 3y + 1 = 0$

(d) None of these

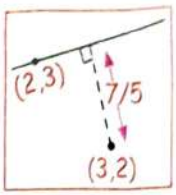
Ans. $x + 1 = y$ ...(i)

$2x + 5 = 3y$ ...(ii)

$2x + 5 = 3(x + 1) \Rightarrow 2 = x, y = 3$

Point of intersection of the lines $x - y + 1 = 0$ and $2x - 3y + 5 = 0$ is $(2, 3)$

Let slope be 'm'



eqn of Line: $y - 3 = m(x - 2)$

$y = mx + 3 - 2m$

$mx - y + 3 - 2m = 0$

and $\left|\frac{3m - 2 + 3 - 2m}{\sqrt{m^2 + 1}}\right| = \frac{7}{5}$

$\Rightarrow \left|\frac{m + 1}{\sqrt{m^2 + 1}}\right| = \frac{7}{5} \Rightarrow \frac{(m + 1)^2}{m^2 + 1} = \frac{49}{25}$

$\Rightarrow 25[m^2 + 1 + 2m] = 49(m^2 + 1)$

$\Rightarrow 25m^2 + 25 + 50m = 49m^2 + 49$

$\Rightarrow 24m^2 - 50m + 24 = 0$

$\Rightarrow 24m^2 - 32m - 18m + 24 = 0$

$\Rightarrow 8m(3m - 4) - 6(3m - 4) = 0$

$\Rightarrow (8m - 6)(3m - 4) = 0$

$\therefore m = \frac{6}{8}$ or $\frac{4}{3} \Rightarrow m = \frac{3}{4}$ or $\frac{4}{3}$

Hence, $y - 3 = \frac{3}{4}(x - 2)$ and $y - 3 = \frac{4}{3}(x - 2)$

$\Rightarrow 3x - 4y + 6 = 0$ and $4x - 3y + 1 = 0$

Brain Teaser

Find the equations of the straight lines, passing through (-2, -7) and cutting an intercept of 3 units between $4x + 3y = 12$ and $4x + 3y = 3$.

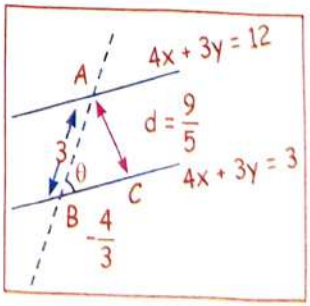
Ans. $d = \left|\frac{12 - 3}{\sqrt{4^2 + 3^2}}\right| = \frac{9}{5}$

In ΔABC:

$\sin\theta = \frac{AC}{AB} = \frac{9/5}{3} = \frac{3}{5}$

$\tan\theta = \frac{3}{4}$

Let slope of line AB be 'm'



$$\left|\frac{m - \left(-\frac{4}{3}\right)}{1 + m\left(-\frac{4}{3}\right)}\right| = \frac{3}{4}$$

$$\Rightarrow \left|\frac{3m + 4}{3 - 4m}\right| = \frac{3}{4}$$

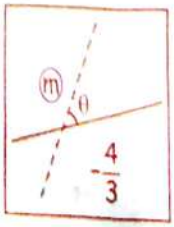
$$\Rightarrow \frac{3m + 4}{3 - 4m} = \pm\frac{3}{4}$$

$\Rightarrow 12m + 16 = \pm(9 - 12m)$

$\Rightarrow m = -\frac{7}{24}$ or ∞

$\therefore y + 7 = -\frac{7}{24}(x + 2)$

or $x = -2$



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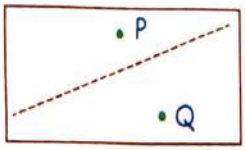
POSITION OF 2 POINTS WITH RESPECT TO A LINE

Same side ⇒ Same Sign

Opposite side ⇒ Opposite sign

Ques. Check point (2, 1) & (-3, 4) lines on same or opposite side of line $2y = 3x + 5$.

Ans. $3x - 2y + 5 = 0$



Put (2, 1) in the line

E_1: $3 \times 2 - 2 \times 1 + 5 = 9$

Put (-3, 4) in the line

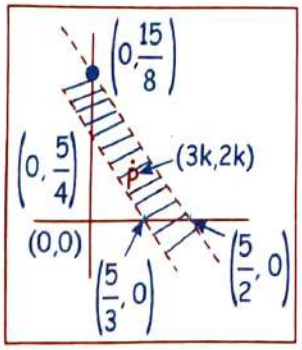
E_2: $-3 \times 3 - 2 \times 4 + 5 = -12$

$\because E_1 > 0, E_2 < 0$

$\therefore$ P and Q lies on opposite sides.

Ques. Find the range of k such that the point P (3k, 2k) lies between the Lines $3x + 4y = 5$ and $6x + 8y = 15$.

Ans. L_1: $3x + 4y - 5 = 0$



L_2: $3x + 4y - \frac{15}{2} = 0$

For L_2:

O & P are on same side

$\therefore 3(3k) + 4(2k) - \frac{15}{2} < 0$

$\Rightarrow 9k + 8k < \frac{15}{2} \Rightarrow 17k < \frac{15}{2} \Rightarrow k < \frac{15}{34}$

For L_1: O & P are on opp side

$3 \times 3k + 4(2k) - 5 > 0$

$\Rightarrow 9k + 8k > 5 \Rightarrow k > \frac{5}{17}$

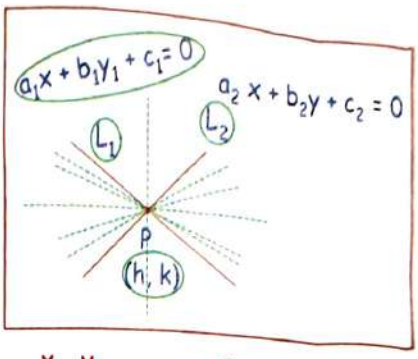
Hence, $k \in \left(\frac{5}{17}, \frac{15}{34}\right)$

FAMILY OF LINES

$$L_1 + \lambda L_2 = 0$$

$\lambda \in R$ represents the equations of straight lines passing through the points of intersection of L_1 and L_2.

It is used to avoid Finding P.O.I Of 2 Lines



Ques. $L_1 : \frac{x}{3} + \frac{y}{2} = 1$ and $L_2 : \frac{x}{2} + \frac{y}{3} = 1$.

A variable line passes through the intersection point of L_1 and L_2 and cuts the axes in A, B. Show that the mid -point of AB describes the curve, $5xy = 3(x + y)$.

Ans. L_1: $2x + 3y = 6$

L_2: $3x + 2y = 6$

Family

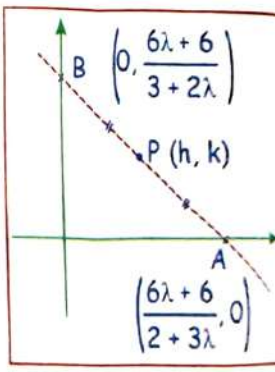
$(2x + 3y - 6) + \lambda(3x + 2y - 6) = 0$

$\Rightarrow (2 + 3\lambda)x + (3 + 2\lambda)y - 6\lambda - 6 = 0$

At $y = 0 \Rightarrow x = \frac{6\lambda + 6}{2 + 3\lambda}$

$2h = \frac{6\lambda + 6}{2 + 3\lambda}$ ...(i)

Similarly,



$2k = \frac{6\lambda + 6}{3 + 2\lambda}$ ...(ii)

$\because 2h = \frac{6\lambda + 6}{2 + 3\lambda}$

$\Rightarrow h = \frac{3\lambda + 3}{2 + 3\lambda}$

$\Rightarrow 2h + 3h\lambda = 3\lambda + 3 \Rightarrow \lambda(3h - 3) = 3 - 2h$

$\Rightarrow \lambda = \frac{3 - 2h}{3h - 3}$

So, $2k = \frac{6\lambda + 6}{2 + 3\lambda}$

$\Rightarrow k = \frac{3\lambda + 3}{2\lambda - 3} \Rightarrow k = \frac{3\left[\frac{3 - 2h}{3h - 3}\right] + 3}{2\left[\frac{3 - 2h}{3h - 3}\right] + 3}$

$\Rightarrow k = \frac{9 - 6h + 9h - 9}{6 - 4h + 9h - 9}$

$\Rightarrow k = \frac{3h}{5h - 3} \Rightarrow 5kh - 3k = 3h \Rightarrow 5hk = 3(h + k)$

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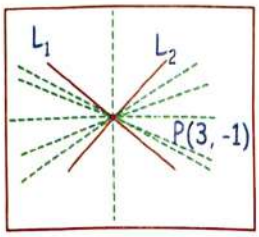
CONVERSE OF FAMILY OF LINES

If a line can be expressed in the form of a family then it always passes through a fixed point whose coordinates can be obtained by finding the point of intersection of the two lines L_1 & L_2.

Ques. p, q are parameters. All the lines of the family $(p + q)x + (2p + q)y = p + 2q$ pass through the point (x_1, y_1). The find the value of $x_1 + y_1$?



Ans. $p(x + 2y - 1) + q(x + y - 2) = 0$

divide by 'p'

$$(x + 2y - 1) + \frac{p}{q}(x + y - 2) = 0$$

$$\Rightarrow (x + 2y - 1) + \lambda(x + y - 2) = 0$$

$\therefore$ Point of intersection of $x + 2y = 1$ and $x + y = 2$ is $(3, -1)$

Ques. Line $ax + by + c = 0$ passes through fixed point for following condition, find that fixed point for every condition.

(i) a, b, c $\rightarrow$ A.P

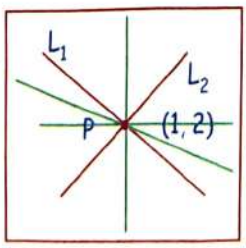
(ii) a b & c are 1st ,5th & 11th terms of an A. P

(iii) $3a = 2b + 5c$

(iv) $a^2 + 9b^2 = 6ab + 4c^2$

Ans. $ax + by + c = 0$

(i) a, b, c $\rightarrow$ AP



$\therefore 2b = a + c$

$$\Rightarrow b = \frac{a + c}{2}$$

$$\therefore ax + \left(\frac{a + c}{2}\right)y + c = 0$$

$$\Rightarrow 2ax + (a + c)y + 2c = 0$$

$$\Rightarrow a[2x + y] + c[y + 2] = 0$$

divide by a

$$(2x + y) + \frac{c}{a}(y + 2) = 0$$

Comparing with $L_1 + \lambda L_2 = 0$

L_1: $2x + y = 0$

L_2: $y + 2 = 0$

Point of intersection of L_1 and L_2 = $(1, -2)$

(ii) $a \rightarrow a$, $b = a + 4d$, $c = a + 10d$

$$\therefore d = \frac{b - a}{4}$$

$$\Rightarrow c = a + 10\left(\frac{b - a}{4}\right) \Rightarrow c = a + \frac{5(b - a)}{2}$$

$$\Rightarrow c = \frac{2a + 5b - 5a}{2} \Rightarrow c = \frac{5b - 3a}{2}$$

$$\Rightarrow \frac{3a}{2} - \frac{5b}{2} + c = 0$$

$$\therefore \text{Fixed point} = \left(\frac{3}{2}, \frac{-5}{2}\right)$$

(iii) $3a = 2d + 5c \Rightarrow \frac{3a}{5} = \frac{2b}{5} + c$

$$\therefore \text{Fixed point} = \left(\frac{-3}{5}, \frac{2}{5}\right)$$

(iv) $a^2 + 9b^2 - 6ab = 4c^2$

$(a - 3b)^2 = (2c)^2$

$a - 3b = \pm 2c$

$$\Rightarrow \frac{a}{\pm 2} - \frac{3b}{\pm 2} + c = 0$$

$$\therefore \text{Fixed point} = \left(\frac{-1}{2}, \frac{3}{2}\right) \text{ or } \left(\frac{1}{2}, \frac{-3}{2}\right)$$

FOOT OF PERPENDICULAR & IMAGE OF A POINT

Ques. Find image and foot of perpendicular (4, - 1) w.r.t the line $y = 2x + 6$.

Ans. $\because m_L = 2$

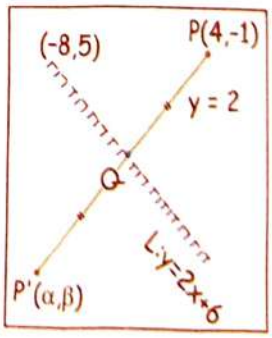
$$\Rightarrow m_{PQ} = \frac{-1}{2}$$

$\therefore$ eqn of PQ:

$$y + 1 = \frac{-1}{2}(x - 4)$$

$2y + 2 = -x + 4$

$x + 2y = 2 \rightarrow 2$



To find the intersection point of L and PQ:

$x + 2(2x + 6) = 2$

$\Rightarrow 5x + 12 = 2 \Rightarrow 5x = -10$

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$\therefore x = -2, y = 2$

Now, $\frac{\alpha + 4}{2} = -2 \Rightarrow \alpha = -8$

And $\frac{\beta - 1}{2} = 2 \Rightarrow \beta = 5$

$Q \rightarrow$ Foot of $\perp \rightarrow (-2, 2)$

$P' \rightarrow$ image of P in L $(-8, 5)$

FORMULA- FOOT OF PERPENDICULAR & IMAGE OF A POINT

Foot of Perpendicular & Reflection

(i) The foot of perpendicular drawn to $Ax + By + C = 0$ From the point (x_1, y_1) is by,

$$\frac{x - x_1}{A} = \frac{y - y_1}{B} = -\frac{Ax_1 + By_1 + C}{A^2 + B^2}$$

(ii) The image of a point (x_1, y_1) about the line $Ax + By + C = 0$ is given by,

$$\frac{x - x_1}{2A} = \frac{y - y_1}{2B} = -\frac{Ax_1 + By_1 + C}{A^2 + B^2}$$

Ques. Find image of the point $(4, -1)$ w.r.t the line $y = 2x + 6$ using Formula.

Ans. $y = 2x + 6$

$\Rightarrow 2x - y + 6 = 0$

$$\frac{x - 4}{2 \times 2} = \frac{y - (-1)}{2 \times (-1)} = \frac{-(2 \times 4 - (-1) + 6)}{2^2 + 1^2}$$

$$\Rightarrow \frac{x - 4}{4} = \frac{y - 1}{-2} = \frac{-15}{5}$$

$$\Rightarrow \frac{x - 4}{4} = \frac{y - 1}{-2} = -3$$

$\frac{x - 4}{4} = -3$ or $\frac{y + 1}{-2} = -3$

$\Rightarrow x - 4 = -12$ $\Rightarrow y + 1 = 6$

$\Rightarrow x = -8$ $\Rightarrow y = 5$

Ques. The image of a point A $(3, 8)$ in the line $x + 3y - 7 = 0$ is

(a) $(1, -4)$ (b) $(1, 2)$

(c) $(-1, -4)$ (d) $(3, 8)$

Ans. $$\frac{x - 3}{2 \times 1} = \frac{y - 8}{2 \times 3} = -\left[\frac{3 + 3 \times 8 - 7}{1^2 + 3^2}\right]$$

$$\Rightarrow \frac{x - 3}{2} = \frac{y - 8}{6} = -\left(\frac{20}{10}\right)$$

$$\Rightarrow \frac{x - 3}{2} = \frac{y - 8}{6} = -2$$

$\Rightarrow \frac{x - 3}{2} = -2$ or $y - 8 = -12$

$\Rightarrow x - 3 = -4$ $\Rightarrow y = -4$

$\Rightarrow x = -1$

Important Note

Image of the point (a, b) in the mirror $y = x$ is (b, a)

Proof:

$$\frac{x - a}{2 \times 1} = \frac{y - b}{2(-1)} = -\left[\frac{a - b}{1^2 + 1^2}\right]$$

$$\Rightarrow \frac{x - a}{2} = \frac{y - b}{-2} = \frac{-(a - b)}{2}$$

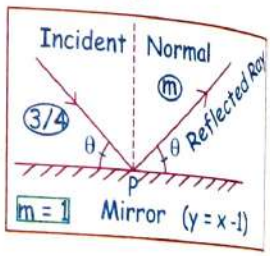
$\Rightarrow \frac{x - a}{2} = \frac{-(a - b)}{2}$ and $\frac{y - b}{-2} = \frac{-(a - b)}{2}$

$\Rightarrow x - a = -a + b \Rightarrow y - b = a - b$

$\Rightarrow x = b$ $\Rightarrow y = a$

Ques. If a incident ray is along the line $[3x - 4y - 5 = 0]$ get reflected by $(y = x - 1)$ then find equation of reflected ray & normal ray.

Ans. **Method - 1**



Get P:

$3x = 4y + 5$

$3x = 4(x - 1) + 5$

$3x = 4x - 4 + 5$

$-x = 1$

$x = -1, y = -2$

Let slope of reflected Ray be m

$$\left|\frac{1 - \left(\frac{3}{4}\right)}{1 + \left(\frac{3}{4}\right) \times 1}\right| = \left|\frac{m - 1}{1 + m \times 1}\right|$$

$$\frac{1}{7} = \left|\frac{m - 1}{1 + m}\right| \Rightarrow m + 1 = 7m - 7$$

$\Rightarrow 8 = 6m \Rightarrow m = \frac{4}{3}$

$\therefore$ Eqⁿ of reflected Ray: $y + 2 = \frac{4}{3}(x + 1)$

$\Rightarrow 3y + 6 = 4x + 4 \Rightarrow 3y + 2 = 4x$

Eqⁿ of normal Ray: $y + 2 = -1(x + 1)$

$\Rightarrow x + y + 3 = 0$

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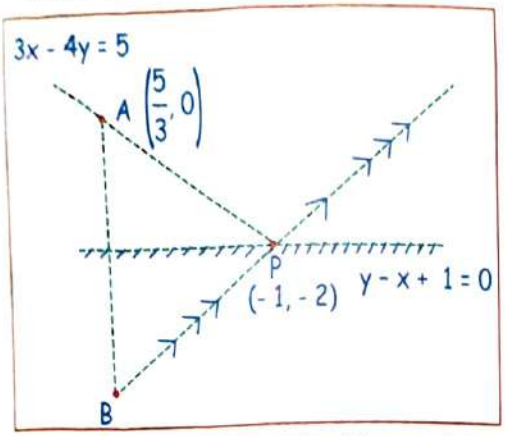
Method - 2

Get P [- 1, - 2]

Consider any point A on incident Ray $A\left[\frac{5}{3}, 0\right]$

Get image of A in mirror Say B.

Now B Will lie on Reflected Ray.



for B, we use the formula of image of a point-

$$\frac{(x - 5/3)}{-2 \times 1} = \frac{y - 0}{2 \times 1} = -\left(\frac{0 - 5/3 + 1}{(1)^2 + (-1)^2}\right)$$

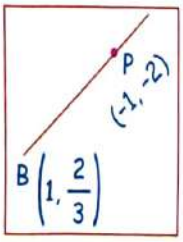
$$\Rightarrow \frac{(x - 5/3)}{-2} = \frac{y}{2} = \frac{1}{3}$$

$$\therefore x = \frac{-2}{3} + \frac{5}{3} = 1 \text{ and } y = \frac{2}{3}$$

So, Eqⁿ of reflected Ray will be-

$$(Y + 2) = \left(\frac{2/3 + 2}{1 + 1}\right)(x + 1)$$

$$\Rightarrow (y + 2) = \frac{4}{3}(x + 1)$$



Ques. A ray of light coming from the point P (1, 2) gets reflected from the point Q on the x - axis and then passes through the point R(4,3). if point S(h, k) is such that PQRS is a parallelogram, them hk^2 is equal to:

[JEE Main 9 Apr. 2004-I]

(a) 70 (b) 80 (c) 60 (d) 90

Ans. P' is the image of P with respect to x - axis

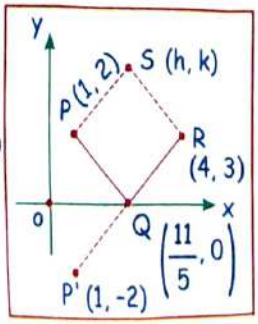
So, P' = (1, - 2)

Now, Eqⁿ of PR-

$$(y + 2) = \left(\frac{3 + 2}{4 - 1}\right)(x - 1)$$

$$\Rightarrow y + 2 = \frac{5}{3}(x - 1)$$

at $y = 0, x = \frac{11}{5}$



∵ PQRS is a parallelogram.

So, its diagonals bisect each other

Then, $\left(\frac{h + \frac{11}{5}}{2}\right) = \frac{4+1}{2} \Rightarrow h = 5 - \frac{11}{5} = \frac{14}{5}$

And $\frac{k}{2} = \frac{3 + 2}{2} \Rightarrow k = 5$

Hence, $hk^2 = \frac{14}{5} \times 25 = 70$

Brain Teaser

The equation of perpendicular bisectors of the sides AB and AC of a triangle ABC are x - y + 5 = 0 and x + 2y = 0 respectively. If the point A is (1, - 2), then the equation of line BC is

(a) 23x + 14y - 40 = 0

(b) 14x - 23y + 40 = 0

(c) 23x - 14y + 40 = 0

(d) 14x + 23y - 40 = 0

Ans. Image of A in the mirror L_1 is point B:

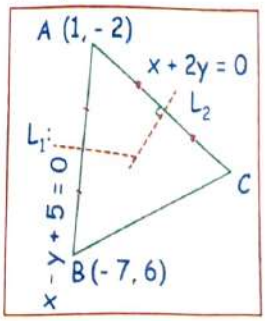
Image of A in mirror L_2 is C.

For B:

$$\frac{x - 1}{2 \times 1} = \frac{y - 2}{-2 \times 1} = -\left[\frac{1 + 2 + 5}{2}\right]$$

$$\Rightarrow \frac{x - 1}{2} = \frac{y + 2}{-2} = -4$$

$$\therefore x = -7, y = 6$$



For C:

$$\frac{x - 1}{2 \times 1} = \frac{y + 2}{2 \times 2} = \frac{-[1 - 4]}{1^2 + 2^2}$$

$$\Rightarrow \frac{x - 1}{2} = \frac{y + 2}{4} = \frac{3}{5}$$

$$\therefore C\left[\frac{11}{5}, \frac{2}{5}\right]$$

$$\therefore \text{Equation of BC: } (y - 6) = \frac{(6 - 2/5)}{(-7 - 11/5)}(x + 7)$$

$$\Rightarrow y - 6 = \frac{-28}{46}(x + 7)$$

$$\Rightarrow 23y - 138 = -14x - 98$$

$$\Rightarrow 14x + 23y - 40 = 0$$

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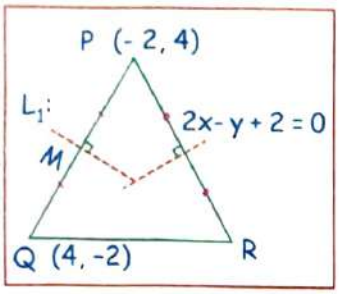
Ques. In a triangle PQR, the co-ordinates of the point P and Q are (- 2, 4) and (4, - 2) respectively If the equation of the perpendicular bisector of PR is 2x - y + 2 = 0, then the circumcenter of the ΔPQR is

(a) (- 1, 0) (b) (1, 4)

(c) (0, 2) (d) (- 2, - 2)

Ans.



Mid point of P and Q:

$$M = \left(\frac{4-2}{2}, \frac{-2+4}{2}\right) = (1, 1)$$

Now, slope of PQ:

$$m = \frac{-2-4}{4+2} = -1$$

$$\therefore \text{ slope of } L_1 = \frac{-1}{m} = 1$$

Then, Eqⁿ of perpendicular bisector of PQ:

$L_1 : y = 1 \times x + c$

Put (1, 1) in it

We get, y = x

Thus, circumcentre will be intersection of L_1 (y = x) and 2x - y + 2 = 0

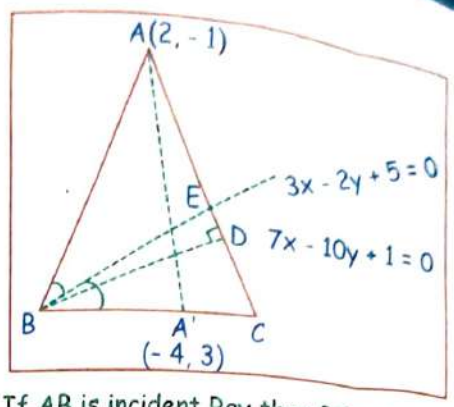
Hence, C' = (-2, -2)

Brain Teaser

A (2, - 1) B, C are the vertices of a triangle. The equations of the bisector of ∠ABC of the altitude from B, are 3x - 2y + 5 = 0 and 7x - 10y + 1 = 0, respectively. Find the equations of BC.

Ans. ∵ B is the point of intersection of 3x - 2y + 5 = 0 and 7x - 10y + 1 = 0

∴ B = (-3, -2)



If AB is incident Ray then BC is reflected ray wrt the line mirror BE.

$$\frac{x-2}{2\times 3} = \frac{y+1}{2\times(-2)} = -\left[\frac{6+2+5}{3^2+2^2}\right]$$

$$\Rightarrow \frac{x-2}{6} = \frac{y+1}{-4} = -1$$

$\therefore x = -4$ or $y = 3$

Equation of BC using (-3, -2) and (-4, 3) is 5x + y + 17 = 0

Ques. In a ΔABC, suppose y = x is equation of the bisector of the angle B and the equation of the side AC is 2x - y = 2. If 2AB = BC and the point A and B are respectively (4, 6) and (α, β), then α + 2β is equal to

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(a) 42 (b) 39 (c) 48 (d) 45

Ans. ∵ B, A' and C are collinear

$\therefore \beta = \alpha$

∵ point C lies on the line 2x - y = 2

$\therefore$ 2a - b = 2

$\Rightarrow$ b = 2a - 2 ...(i)

$$D\left(\frac{1\times a + 2\times 4}{3}, \frac{1\times b + 2\times 6}{3}\right)$$

$$D\left(\frac{a+8}{3}, \frac{b+12}{3}\right)$$

∵ point D lies on the line y = x

$$\therefore \frac{a+8}{3} = \frac{b+12}{3}$$

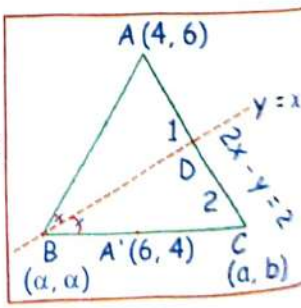
$\Rightarrow$ a = b + 4 ...(ii)

By using eqⁿ (i) & (ii)

a - 4 = 2a - 2

$\Rightarrow$ a = - 2

$\Rightarrow$ b = - 6



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point B lies on the line $y = x$

$$\begin{vmatrix} \alpha & \alpha & 1 \\ 6 & 4 & 1 \\ -2 & -6 & 1 \end{vmatrix} = 0$$

$\Rightarrow \alpha[4+6] - \alpha[6+2] + 1[-36+8] = 0$

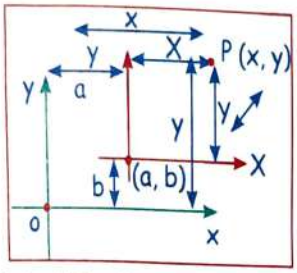
$\Rightarrow 10\alpha - 8\alpha = 28$

$\Rightarrow \alpha = 14$

SHIFTING OF ORIGIN (TRANSLATION OF AXES)

x & y → old co-ordinate system

X & Y → new co-ordinate system



$y = Y + b$

$\Rightarrow Y = y - b$

$Y_{new} = Y_{old} - b$

$X = x - a$

$X_{new} = x_{old} - a$

(a, b) is point where origin is shifted to.

Important Note

Slope of a line remains unchanged if we shift origin.

Ques. If the origin is shifted to (2, -3), Find:

1. The new co- ordinates of the point (3, 7)

Ans. $X = x - a$ & $Y = y - b$

$\Rightarrow X = x - 2$ & $Y = y + 3$

Given $x = 3$ & $y = 7$

$X = 3 - 2, Y = 7 + 3$

$\Rightarrow X = 1, Y = 10 \Rightarrow (1, 10)$

2. The new Equations of the line $3x = 4x + 10$

Ans. Old eqⁿ was a relation between x & y

new eqⁿ is a relation between X & Y

$\therefore$ put $x = X + 2$ and $y = Y - 3$

$3(X + 2) = 4(Y - 3) + 10$

$\Rightarrow 3X + 6 = 4Y - 12 + 10$

$\Rightarrow 3x = 4y - 8$

Ques. At what point the origin must be shifted to such that the equation $x^2 + y^2 - 6x + 8y - 1 = 0$ has no 1 degree term.

Ans. Let origin is shifted to (a, b)

$X = x - a$ & $Y = y - b$

$\Rightarrow x = X + a$ & $y = Y + b$

$(X + a)^2 + (Y + b)^2 - 6(X + a) + 8(Y + b) - 1 = 0$

$\Rightarrow X^2 + a^2 + 2aX + Y^2 + b^2 + 2bY - 6x - 6a + 8Y + 8b - 1 = 0$

$\Rightarrow X^2 + Y^2 + x(2a - 6) + Y(2b + 8) + a^2 + b^2 - 6a + 8b - 1 = 0$

$\therefore$ put 1 degree terms equal to zero

$\Rightarrow 2a - 6 = 0$ & $2b + 8 = 0$

$\Rightarrow a = 3$ $\quad b = -4$

Hence, the origin must be shifted to $(3, -4)$

ROTATION OF AXES

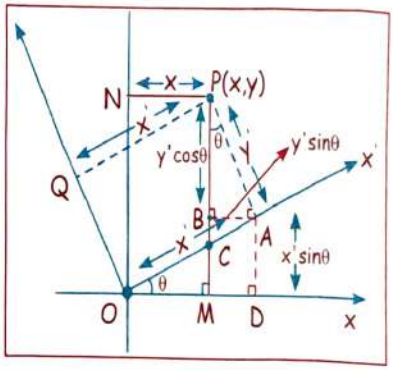
$OA = x'$

$PM = PB + AD$

$y = y'\cos\theta + x'\sin\theta$ ...(i)

$PN = OD - AB$

$x = x'\cos\theta - y'\sin\theta$ ...(ii)



[Eqⁿ (i)] × $\sin\theta$ + [Eqⁿ (ii)] × $\cos\theta$:

$\sin\theta$ ($y = y'\cos\theta + x'\sin\theta$)

$\cos\theta$ ($x = x'\cos\theta - y'\sin\theta$)

$y\sin\theta + x\cos\theta = x'\sin^2\theta + x'\cos^2\theta$

$\Rightarrow x' = x\cos\theta + y\sin\theta$

Similarly, [Eqⁿ (i)] × $\cos\theta$ − [Eqⁿ (ii)] × $\sin\theta$:

$\cos\theta$ ($y = y'\cos\theta + x'\sin\theta$)

$\sin\theta$ ($x = x'\cos\theta - y'\sin\theta$)

$y\cos\theta - x\sin\theta = y'$

$\Rightarrow y' = y\cos\theta - x\sin\theta$

$$\boxed{\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}}$$

and

$$\boxed{\begin{aligned} x' &= x\cos\theta + y\sin\theta \\ y' &= -x\sin\theta + y\cos\theta \end{aligned}}$$

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Ques. When the axes are turned through an angle $\tan^{-1}2$, The equations $4xy - 3x^2 = a^2$ becomes?

(a) $x^2 + 4x^2 = a^2$ (b) $x^2 - 4y^2 = a^2$

(c) $4x^2 + y^2 = a^2$ (d) $4x^2 - y^2 = a^2$

Ans. $4xy - 3x^2 = a^2$

Put $x = x'\cos\theta - y'\sin\theta$ and $y = x'\sin\theta + y'\cos\theta$

$4(x'\cos\theta - y'\sin\theta)(x'\sin\theta + y'\cos\theta) - 3(x'\cos\theta - y'\sin\theta)^2 = a^2$

$\Rightarrow 4\cos^2\theta[x' - y'\tan\theta][x'\tan\theta + y'] - 3\cos^2\theta(x' - y'\tan\theta)^2 = a^2$

$\because \theta = \tan^{-1}2 \Rightarrow \tan\theta = 2$

$4(x' - 2y')(2x' + y') - 3[x' - 2y']^2 = a^2\sec^2\theta$

$\because \sec^2\theta = 1 + \tan^2\theta = 1 + 2^2 = 5$

$\Rightarrow 4[2x'^2 + x'y' - 4x'y' - 2y'^2] - 3[x^2 + 4y'^2 - 4x'y'] = a^2(1 + 2^2)$

$\Rightarrow 8x'^2 - 12x'y' - 8y'^2 - 3x'^2 - 12y'^2 + 12x'y' = 5a^2$

$\Rightarrow 5x'^2 - 20y'^2 = 5a^2$

$\Rightarrow x'^2 - 4y'^2 = a^2$

$\Rightarrow x^2 - 4y^2 = a^2$

PARAMETRIC FORM OF A STRAIGHT LINE

Indirect relation b/w x & y

$r \rightarrow$ Parameter (variable for point on Line)

$x = x_1 \pm r\cos\theta$ and $y = y_1 \pm r\sin\theta$

Parametric co-ordinates of any point (x, y) lying at a distance of 'r' form Fixed pt (x_1, y_1)

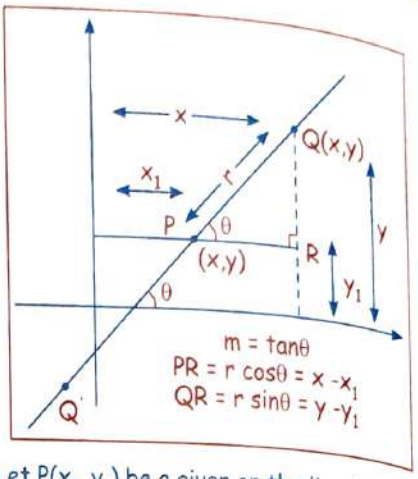
$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r \rightarrow$ Parametric form of Straight Line

$(x_1, y_1) \rightarrow$ Fixed point

$(x, y) \rightarrow$ variable point

$r \rightarrow$ distance

$m = \tan\theta$



Let $P(x_1, y_1)$ be a given on the line L and θ be the inclination of line L.

Let $Q(x,y)$ be a moving point it. Let $PQ = r$, then $x = x_1 \pm r\cos\theta$ and $y = y_1 \pm r\sin\theta$ is called parametric form of the equation of the line L.

Important Note

Whenever the distance from a fixed point is asked or is given then we can try to use parametric form of a line.

Ques. Find distance of (2, 3) form line $x - 3y + 5 = 0$ measured parallel to the line $y = x$.

Ans. Write parametric co-ordinates through point P (2, 3)

$x = x_1 + r\cos\theta$, $y = y_1 + r\sin\theta$

$\because$ distance is measured parallel to the line $y = x$

$\therefore \theta = 45°$

$Q(x = 2 + r\cos 45°, y = 3 + r\sin 45°)$

Put the coordinates in the given line:

$$\Rightarrow \left(2 + \frac{r}{\sqrt{2}}\right) - 3\left(3 + \frac{r}{\sqrt{2}}\right) + 5 = 0$$

$$\Rightarrow 2 + \frac{r}{\sqrt{2}} - 9 - \frac{3r}{\sqrt{2}} + 5 = 0$$

$$\Rightarrow -2 = \frac{2r}{\sqrt{2}} \Rightarrow r = -\sqrt{2}$$

Ques. ABCD represents a square if $A(3, 5)$ & $C(5, -3)$. Find the coordinates of the other 2 Vertices.

Ans. $r = PA = \sqrt{1^2 + 4^2} = \sqrt{17}$

$D[x_1 + r\cos\theta, y_1 + r\sin\theta]$

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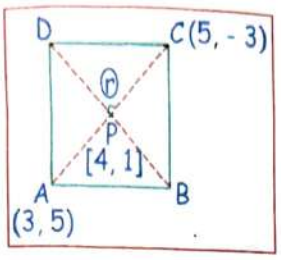
$M_{AC} = \frac{8}{3-5} = -4$

$\Rightarrow M_{BD} = \frac{1}{4}$

$\therefore \tan\theta = \frac{1}{4}$



$\cos\theta = \frac{4}{\sqrt{17}}, \sin\theta = \frac{1}{\sqrt{17}}$

$(x_1, y_1) = (4, 1)$

$D\ [4 + \sqrt{17}\ \cos\theta, 1 + \sqrt{17}\ \sin\theta]$

$D\ [4 + \sqrt{17} \times \frac{4}{\sqrt{17}}, 1 + \sqrt{17} \times \frac{1}{\sqrt{17}}$

$D\ [4 + 4, 1 + 1] = [8, 2]$

$B\ [4 - 4, 1 - 1] = (0, 0)$

Ques. The distance of the point (2,3) from the line $2x - 3y + 28 = 0$, measured parallel to the line $\sqrt{3}x - y + 1 = 0$ is equal to

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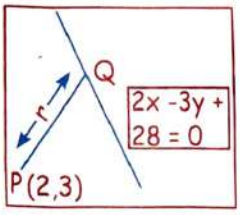
(a) $4\sqrt{2}$ (b) $6\sqrt{3}$

(c) $3 + 4\sqrt{3}$ (d) $4 + 6\sqrt{3}$

Ans. $\because$ The distance is parallel to $\sqrt{3}x - y + 1 = 0$



$\therefore M = \sqrt{3} = \tan\theta \Rightarrow \theta = \frac{\pi}{3}$

$Q[2 + r\cos 60°, 3 + r\sin 60°]$

$\therefore 2(2 + r\cos 60°) - 3(3 + r\sin 60°) + 28 = 0$

$\Rightarrow 4 - 9 + 28 = 3r\sin 60° - 2r\cos 60°$

$\Rightarrow 23 = r\left[\frac{3\sqrt{3}}{2} - \frac{2}{2}\right]$

$\Rightarrow r = \frac{46}{3\sqrt{3}-2} \times \frac{3\sqrt{3}+2}{3\sqrt{3}+2} = \frac{46(3\sqrt{3}+2)}{27-4} = 4 + 6\sqrt{3}$

Ques. A line through A(-5, -4) meets the line $x + 3y + 2 = 0$, $2x + y + 4 = 0$ & $x - y - 5 = 0$ at B, C & D respectively.

If $\left(\frac{15}{AB}\right)^2 + \left(\frac{10}{AC}\right)^2 = \left(\frac{6}{AD}\right)^2$ then find is equation.

Ans. $\because AB = r_1, AC = r_2, AD = r_3$

$\therefore \left(\frac{15}{r_1}\right)^2 + \left(\frac{10}{r_2}\right)^2 = \left(\frac{6}{r_3}\right)^2$

$B[-5 + r_1\cos\theta, -4 + r_1\sin\theta]$

$C\ [-5 + r_2\cos\theta, -4 + r_2\sin\theta]$

$D[-5 + r_2\cos\theta, -4 + r_3\sin\theta]$

$\therefore (-5 + r_1\cos\theta) + 3(-4 + r_1\sin\theta) + 2 = 0$

$\Rightarrow -5 + r_1\cos\theta - 12 + 3r_1\sin\theta + 2 = 0$

$\Rightarrow r_1(\cos\theta + 3\sin\theta) = 15$

$\Rightarrow \cos\theta + 3\sin\theta = \frac{15}{r_1}$ ...(i)

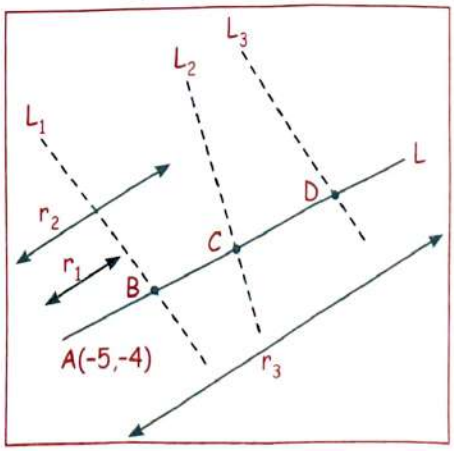
and $2(-5 + r_2\cos\theta + (-4 + r_2\sin\theta) + 4 = 0$

$\Rightarrow -10 + 2r_2\cos\theta - 4 + r_2\sin\theta + 4 = 0$

$\Rightarrow r_2[2\cos\theta + \sin\theta] = 10$

$\Rightarrow \frac{10}{r_2} = 2\cos\theta + \sin\theta$ ...(ii)

Similarly, $\frac{6}{r_3} = \cos\theta - \sin\theta$ ...(iii)



By adding the squares of Eq^n (i), (ii) and (iii)

$(\cos\theta + 3\sin\theta)^2 + (2\cos\theta + \sin\theta)^2 = (\cos\theta - \sin\theta)^2$

$\therefore \tan\theta = m$

$\Rightarrow [1 + 3m]^2 + (2 + m)^2 = (1 - m)^2$

We get, $m = -\frac{2}{3}$

eq^n of line: $y + 4 = m(x + 5)$

$\Rightarrow y + 4 = -\frac{2}{3}(x + 5)$

$\Rightarrow 2x + 3y + 22 = 0$

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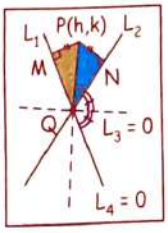
EQUATION OF ANGLE BISECTOR

Let two lines L_1: $a_1x + b_1y + c_1 = 0$ and L_2: $a_2x + b_2y + c_2 = 0$ be given. Then, the equations of angle bisectors, L_3 and L_4 are given by

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

Proof: drop perpendiculars PM & PN on L_1 & L_2 respectively



$\Delta PMQ \cong \Delta PNQ$

$\therefore$ PM = PN (CPCT)

$$\left|\frac{a_1h + b_1k + c_1}{\sqrt{a_1^2 + b_1^2}}\right| = \left|\frac{a_2h + b_2k + c_1}{\sqrt{a_2^2 + b_2^2}}\right|$$

$(h,k) \to (x,y)$

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

Ques. For the straight line $4x - 3x + 6 = 0$ and $5x + 12y - 9 = 0$, find the equation of he Angle bisectors.

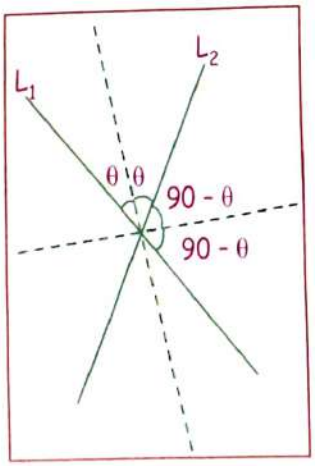
Ans. Eq^n of angle bisectors:

$$\frac{4x - 3y + 6}{\sqrt{4^2 + 3^2}} = \pm \frac{5x + 12y - 9}{\sqrt{5^2 + 12^2}}$$

$\Rightarrow 13(4x - 3y + 6) = \pm 5(5x + 12y - 9)$

$\Rightarrow 52x - 39y + 78 = 25x + 60y - 45$

$\Rightarrow 27x - 99y + 123 = 0$



$$\therefore m_1 = \frac{27}{99} = \frac{3}{11}$$

or $52x - 39y + 78 = -25x - 60y + 45$

$\Rightarrow 77x + 21y + 33 = 0$

$$\therefore m_2 = \frac{77}{21} = -\frac{11}{3}$$

$\because m_1m_2 = -1$

$\therefore$ Two bisectors are always perpendicular

ISOLATION OF ACUTE/OBTUSE ANGLE BISECTORS

Equation of bisector of acute /obtuse angle

Steps to find the equations of the bisector of the acute of obtuse angle

Step 1: In the equations of lines, make the constant terms c_1 & c_2 positive.

Step 2: Determine the sign of $a_1a_2 + b_1b_2$

Step 3: If $a_1a_2 + b_1b_2 < 0$, then + sign gives acute angle bisector

Hence $\frac{a_1x + b_1 + yc_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is acute angle bisector.

If sign of $a_1b_1 + b_1b_2$	For obtuse angle bisector	For acute angle bisector
+	use +sign	Use - sign
-	use - sign	use + sign

Ques. Find Obtuse angle bisector for the lines $4x - 2y + 7 = 0$ and $y = 13 - 2x$.

(a) $8x - 19y = 0$ (b) $8x - 19 = 0$

(c) $4y - 33 = 0$ (d) $4y - 99x = 0$

Ans. **Step 1:**

$4x - 2y + 7 = 0$ & $-2x - y + 13 = 0$

Step 2:

$a_1a_2 + b_1b_2 = -8 + 2 = -6 < 0$

Step 3:

Obtuse angle bisector is given by - ve sign.

$$\therefore \frac{4x - 2y + 7}{\sqrt{4^2 + 2^2}} = -\left[\frac{-2x - y + 13}{\sqrt{5}}\right]$$

$$\Rightarrow \frac{4x - 2y + 7}{2\sqrt{5}} = \frac{2x + y - 13}{\sqrt{5}}$$

$\Rightarrow 4x - 2y + 7 = 4x + 2y - 26$

$\Rightarrow 4y = 26 + 17$

$\Rightarrow 4y = 33$

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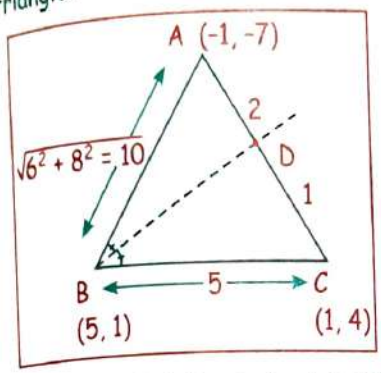
Ques. A (- 1, -7), B (5,1) and C(1,4) form a triangle. What is the equation of the bisector of $\angle ABC$?

Ans. Isolation of Internal & External angle bisector a triangle.



Get $D\left[\frac{2\times1 + 1\times(-1)}{3}, \frac{2\times4 + 1\times(-7)}{3}\right]$

$= \left(\frac{1}{3}, \frac{1}{3}\right)$

$\therefore$ Eqn of BD:

$y - 1 = \left(\frac{1 - 1/3}{5 - 1/3}\right)(x - 5)$

$\Rightarrow x - 7y + 2 = 0$

Important Note

Bisector of Angle Containing the Origin

If the equations of the lines are written with constant terms c_1 and c_2 positive, then the equations of the bisectors of the angle containing the origin is always obtained by taking positive sign

CONDITION OF CONCURRENCY

Three lines $a_1x + b_1y + c_1 = 0$; $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent, if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$ This implies

I) Lines are concurrent or

II) Lines are parallel

Ques. Line $x + y = 1$, $kx + y = 3$, $x + ky = 5$ are concurrent for $k = 7$ only. Given statement is True or False?

Ans. $\begin{vmatrix} 1 & 1 & -1 \\ k & 1 & -3 \\ 1 & k & -5 \end{vmatrix} = 0$

$\Rightarrow 1[-5 + 3k] - 1[-5k + 3] - 1[k^2 - 1] = 0$

$\Rightarrow -5 + 3k + 5k - 3 - k^2 + 1 = 0$

$\Rightarrow -7 + 8k - k^2 = 0 \Rightarrow k^2 - 8k + 7 = 0$

$\Rightarrow (k - 7)(k - 1) = 0$

$\Rightarrow k = 7$ or 1

if $k = 1 \Rightarrow$ lines are $\begin{cases} x + y = 1 \\ x + y = 3 \\ x + y = 5 \end{cases}$

These are parallel lines

$\therefore k = 1$ is Rejected

If $k = 7 \Rightarrow$ Lines are $\begin{cases} x + y = 1 \\ 7x + y = 3 \\ x + 7y = 5 \end{cases}$

These are concurrent lines

Hence, the given statement is true

Ques. For all real values a and b, lines $(2a + b)x + (a + 3b)y + (b - 3a) = 0$ and $mx + 2y + 6 = 0$ are concurrent. Them $|m|$ is equal to _____.

[JEE Main 27 Jan. 2024 Shift]

Ans. $a(2x + y - 3) + b(x + 3y + 1) = 0$

$\Rightarrow (2x + y - 3) + \lambda(x + 3y + 1) = 0$

$L_1: 2x + y - 3 = 0$, $L_2: x + 3y + 1 = 0$

$L_3: mx + 2y + 6 = 0$

$\because L_1, L_2$ and L_3 are concurrent

$\therefore \begin{vmatrix} 2 & 1 & -3 \\ 1 & 3 & 1 \\ m & 2 & 6 \end{vmatrix} = 0$

$\Rightarrow 2[18 - 2] - 1[6 - m] - 3[2 - 3m] = 0$

$\Rightarrow 32 - 6 + m - 6 + 9m = 0$

$\Rightarrow 10m = 12 - 32$

$\Rightarrow 10m = -20 \Rightarrow m = -2$

Hence, $|m| = 2$

AREA OF PARALLELOGRAM

$\because DM = P_1$, $BN = P_2$

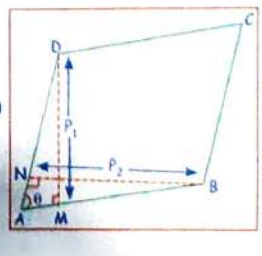
In ΔADM

$\sin\theta = \frac{P_1}{AD} \Rightarrow AD = P_1 \operatorname{Cosec}\theta$

Area = base × height

$= P_1 \operatorname{cosec}\theta \cdot P_2$

$= P_1P_2 \operatorname{cosec}\theta$



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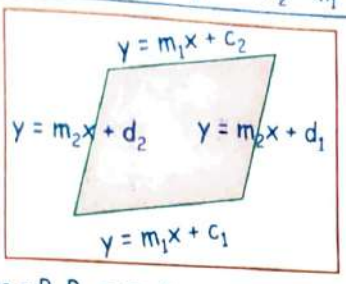
Area of parallelogram $= \left|\frac{(c_2 - c_1)(d_2 - d_1)}{m_2 - m_1}\right|$

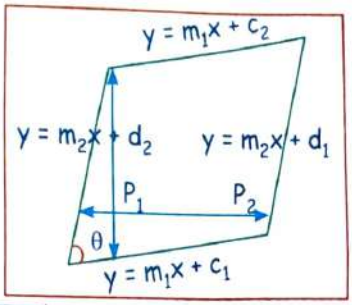


Proof: Area $= P_1 P_2 \operatorname{cosec}\theta$

$P_1 = \left|\frac{c_2 - c_1}{\sqrt{1^2 + m_1^2}}\right|$

$P_2 = \left|\frac{d_2 - d_1}{\sqrt{1^2 + m_2^2}}\right|$

$\tan\theta = \frac{|m_2 - m_1|}{|1 + m_1 m_2|}$



$\operatorname{cosec}\theta = \left|\frac{\sqrt{(1 - m_1)(1 - m_2)}}{m_2 - m_1}\right|$

Area of parallelogram $= \left|\frac{(c_2 - c_1)(d_2 - d_1)}{m_2 - m_1}\right|$

Ques. Find area of parallelogram formed by line $3x - 4y + 5 = 0$, $3x - 4y - 2 = 0$, $x + y - 1 = 0$ & $x + y - 5 = 0$

Ans. $4y = 3x + 5$

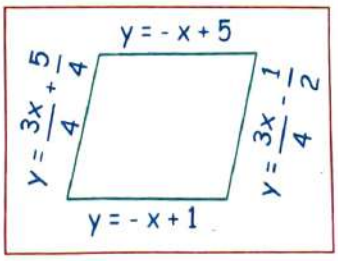
$\Rightarrow y = \frac{3x}{4} + \frac{5}{4}$

$4y = 3x - 2$

$\Rightarrow y = \frac{3x}{4} - \frac{2}{4}$

Also, $y = -x + 1$

and $y = -x + 5$



$\therefore$ Area $= \frac{(c_2 - c_1)|d_2 - d_1|}{|m_2 - m_1|}$

$m_1 = -1$ & $m_2 = \frac{3}{4}$

$c_1 = 1, c_2 = 5$

$d_1 = \frac{-1}{2}, d_2 = \frac{5}{4}$

$\therefore$ Area $= \frac{(5 - 1)\left(\frac{5}{4} + \frac{1}{2}\right)}{\left|\frac{3}{4} + 1\right|} = \frac{4 \times \frac{7}{4}}{\frac{7}{4}} = 4$

OPTIMIZATION PROBLEM-1

Two points A and B are on same side of line L, Point P on line L is such that PA + PB is minimum when point P is common point of line L and AC when C is image of point B in line L.

For three point P, A, B: $|PA + PB| \geq AB$ equality holds when P, A, B are collinear points.

Ques. Given points A(3, 4), B(7, 13), point P on the line $y = x$ such that PA + PB is minimum is

(a) $\left(\frac{31}{7}, \frac{31}{7}\right)$ (b) (6, 6)

(c) (5, 5) (d) none

Ans. eqⁿ of AC: $y - 4 = \left(\frac{7 - 4}{13 - 3}\right)(x - 3)$

$\Rightarrow y - 4 = \frac{3}{10}(x - 3)$

$\Rightarrow 10y - 40 = 3x - 9$

$\Rightarrow 10y = 3x + 31$

$\because$ P is the intersection point of AC and $y = x$

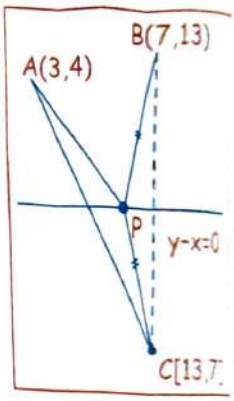
Put $y = x$

$\therefore 10x = 3x + 31$

$\Rightarrow x = \frac{31}{7}$

and $y = \frac{31}{7}$

Hence, $P = \left(\frac{31}{7}, \frac{31}{7}\right)$



OPTIMIZATION PROBLEM-2

Two points A and B are on same side of line L, point P on line L is such that |PA - PB| is maximum then point P is common point of line and AB.

For three points, P, A, B; $|PA - PB| \leq AB$, equality holds when P, A, B are collinear points.

Ques. A person starts from A (1, 5) and touches line $x + y = 2$ anywhere at P and reaches point B (7, 9). Then find coordination of point P if

(i) (PA + PB) is minimum

(ii) |PA - PB| is maximum

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Ans. $x + y - 2 = 0$

Put (1, 5) on it -

$1 + 5 - 2 = 4 > 0$

and put (2, 9) on it

$2 + 9 - 2 = 9 > 0$

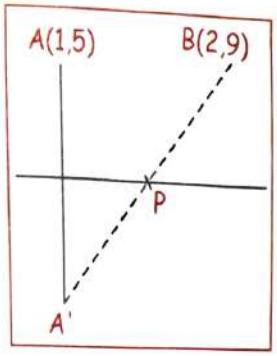
So, both points lie on the same side

$$\frac{x-1}{2\times1} = \frac{y-5}{2\times1} = -\left[\frac{1+5-2}{1^2+1^2}\right]$$

$$\frac{x-1}{2} = \frac{y-5}{2} = -2$$

$x = -3$ $y = 1$

$A'[-3, 1]$



(i) eqⁿ of A'B:

$$y - 1 = \frac{9-1}{2+3}(x+3)$$

$$\Rightarrow y - 1 = \frac{8}{5}(x+3)$$

$\Rightarrow 5y - 5 = 8x + 24$

$\Rightarrow 5y = 8x + 29$ ...(i)

$\because$ P will be the intersection point of A'B and $x + y = 2$

$\therefore 5(2 - x) = 8x + 29$

$10 - 5x = 8x + 29$

$13x = -19$

$$\therefore x = -\frac{19}{13} \text{ and } y = 2 + \frac{19}{13} = \frac{45}{13}$$

$$\Rightarrow P = \left(-\frac{19}{13}, \frac{45}{13}\right)$$

(ii) eqⁿ of AB:

$$(y - 5) = \frac{9-5}{2-1}(x-1)$$

$\Rightarrow (y - 5) = 4(x - 1)$

$\Rightarrow y - 5 = 4x - 4 \Rightarrow y = 4x + 1$

Put $y = 2 - x$

$\Rightarrow 2 - x = 4x + 1$

$\Rightarrow 5x = 1$

$$\Rightarrow x = \frac{1}{5}$$

$$\text{and } y = 2 - x = 2 - \frac{1}{5} = \frac{9}{5}$$

$$\Rightarrow P = \left(\frac{1}{5}, \frac{9}{5}\right)$$

Ques. A is (4, 1) B is (0, 4), point P on the line $y = 3x - 1$ is such that $|PA - PB|$ is maximum is

(a) (5, 2) (b) (2, 5) (c) (3, 5) (d) none

Ans. $\because$ A and B lies in the opposite directions

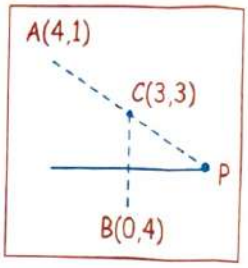
$\therefore$ Image point of B:

$$\frac{x-0}{2\times3} = \frac{y-4}{2\times(-1)} = -\left[\frac{0-4-1}{10}\right]$$

$$\Rightarrow \frac{x}{6} = \frac{y-4}{-2} = \frac{1}{2}$$

$\Rightarrow x = 3$ & $y = 3$

$\therefore C = (3, 3)$



eqⁿ of AC:

$$y - 1 = \frac{3-1}{3-4}(x-4)$$

$$\Rightarrow y - 1 = \frac{2}{-1}(x-4)$$

$\Rightarrow y - 1 = -2x + 8$

$\Rightarrow y + 2x = 9$

$\because$ P is the intersection point of $y = 3x - 1$ and $y + 2x = 9$

$\therefore 3x - 1 + 2x = 9 \Rightarrow x = 2$ and $y = 5$

Hence, P = (2, 5)

PAIRS OF STRAIGHT LINES

Ques. Let PQR be right angled isosceles triangle, right angle angled at p (2,1). If the equations of the line QR is $2x + y = 3$, then the equations representing the pair of lines PQ and PR is **[JEE Mains IIT 1999]**

(a) $3x^2 - 3y^2 + 8xy + 20x + 10y + 25 = 0$

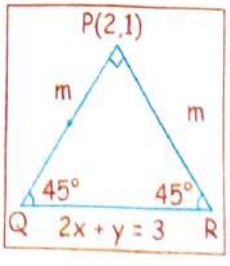
(b) $3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$

(c) $3x^2 - 3y^2 + 8xy + 10x + 5y + 20 = 0$

(d) $3x^2 - 3y^2 + 8xy - 10x - 15y - 20 = 0$

Ans. $2x + y = 3$

$\therefore m_1 = -2$



$$\tan 45° = \left|\frac{m - (-2)}{1 + m(-2)}\right|$$

$$\Rightarrow \pm 1 = \frac{m+2}{1-2m}$$

$\Rightarrow m + 2 = \pm(2m - 1)$

$\Rightarrow m = 3 \text{ or } -\frac{1}{3}$

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$\therefore y = 3x + c$ and $y = -\frac{1}{3}x + c$

Put (2, 1) in these lines

We get, $y = 3x - 5$ and $y = -\frac{1}{3}x + \frac{5}{3}$

$\therefore (y - 3x + 5)(3y + x - 5) = 0$

$\Rightarrow 3x^2 - 3y^2 + 8xy - 20x - 10y + 25 = 0$

GENERAL 2 DEGREE EQUATION

The general equations of 2 degree:

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines, if

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

i.e. $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

Ques. Determine the lines represented by $6x^2 + 11xy - 10y^2 + x + 31y - 15 = 0$. Find their point of intersection & Angle between the lines.

Ans. Factorize 2 degree terms:

$6x^2 + 11xy - 10y^2$

$= 6x^2 + 15xy - 4xy - 10y^2$

$= 3x[2x + 5y] - 2y[2x + 5y]$

$= (2x + 5y)(3x - 2y)$

Let the Lines be:

$(2x + 5y + c_1)(3x - 2y + c_2) = 0$

Coeff of x : $2c_2 + 3c_1 = 1$ ...(i)

Coeff of y: $5c_2 - 2c_1 = 31$ ...(ii)

By Eq^n (i) and (ii):

We get $c_1 = -3$, $c_2 = 5$

& $c_1 c_2 = -15$

eq^n of Lines

$2x + 5y - 3 = 0$

$3x - 2y + 5 = 0$

$\therefore$ point of intersection = (-1, 1)

and $\theta = \tan^{-1}\left|\dfrac{\frac{3}{2} - \left(\frac{-2}{5}\right)}{1 + \frac{3}{2}\left(\frac{-2}{5}\right)}\right| = \tan^{-1}\frac{19}{4}$

Important Note

- The point of Intersection of pair of straight lines can be directly obtained by partial differentiation.

Ex: f: $6x^2 + 11xy - 10y^2 + x + 3y - 15 = 0$

$\frac{\partial f}{\partial x} = 0 \Rightarrow 12x + 11y - 0 + 1 + 0 - 0 = 0$

$\Rightarrow 12x + 11y + 1 = 0$

$\frac{\partial f}{\partial y} = 0 \Rightarrow \frac{\partial f}{\partial y} = 0 + 11x - 30y + 31 = 0$

$\Rightarrow 11x - 20y + 31 = 0$

Solve simultaneoly to get P.O.I

HOMOGENEOUS 2 DEGREE EQUATION

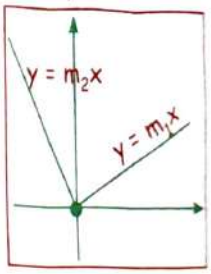
f: $ax^2 + 2hxy + by^2 = 0$

Homogeneous 2 degree equation always represents pair of straight lines intersecting at the origin.

a, h, b, g = f = c = 0 in Homogeneous 2 degree Eq^n

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ (always)

For P.O.I



$\left(\frac{\partial f}{\partial x}\right) = 0 \Rightarrow 2ax + 2hy + 0 = 0$

$\Rightarrow 2ax + 2hy = 0$ ...(i)

$\frac{\partial f}{\partial y} = 0 \Rightarrow 2hx + 2by = 0$ ...(ii)

Solving Eq^n (i) and (ii) Gives (0,0)

Angle Between Pair of Lines

If θ be the angle between the POSL, then

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}$$

Where, a = coeff of x^2

b = coefficient of y^2

$h = \dfrac{\text{coefficient of } xy}{2}$

(i) If $h^2 - ab = 0$

$\Rightarrow \theta = 0 \Rightarrow$ lines are parellel.

(ii) If $|a + b| = 0$

$\Rightarrow a + b = 0 \Rightarrow$ Lines are perpendicular

coeff. of x^2 + coeff. of $y^2 = 0 \Rightarrow$ POSL is perpendicular

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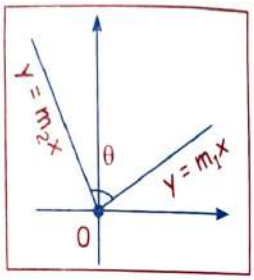
Proof: Consider Homogeneous POSL

$ax^2 + by^2 + 2hxy = 0$

divide by x^2

$$a + b\left(\frac{y}{x}\right)^2 + 2h\left(\frac{y}{x}\right) = 0$$

But $\frac{y}{x} = m_1$ or m_2



put $\frac{y}{x} = m$

$a + bm^2 + 2hm = 0$

$\therefore m_1 + m_2 = \frac{2h}{b}, m_1 m_2 = \frac{a}{b}$

$|m_2 - m_1| = \sqrt{(m_1 + m_2)^2 - 4 m_1 m_2}$

$$= \sqrt{\left(\frac{-2h}{b}\right)^2 - 4\left(\frac{a}{b}\right)} = \sqrt{\frac{4h^2 - 4ab}{b^2}}$$

$$\therefore \tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

$$\Rightarrow \tan\theta = \left|\frac{\frac{\sqrt{4h^2 - 4ab}}{b}}{1 + \frac{a}{b}}\right|$$

$$\Rightarrow \tan\theta = \left|\frac{\sqrt{4h^2 - 4ab}}{a + b}\right|$$

$\tan\theta$ Formula is valid for genral 2° POSL also.

Ques. If the angle between the pair of straight lines represented by the equation $x^2 - 3xy + \lambda y^2 + 3x - 5y + 2 = 0$ is $\tan^{-1}\left(\frac{1}{3}\right)$, where λ is a non-negative real number, then λ is

(a) 2 (b) 0 (c) 3 (d) 1

Ans. $a = 1,\ b = \lambda,\ h = -\frac{3}{2}$

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}$$

$$\Rightarrow \frac{1}{3} = \frac{2\sqrt{\frac{9}{4} - \lambda}}{|1 + \lambda|}$$

$$\Rightarrow \frac{|\lambda + 1|}{3} = 2\sqrt{\frac{9}{4} - \lambda}$$

$$\Rightarrow (\lambda + 1)^2 = 36\left[\frac{9}{4} - \lambda\right]$$

$\Rightarrow \lambda^2 + 1 + 2\lambda = 81 - 36\lambda$

$\Rightarrow \lambda^2 + 38\lambda - 80 = 0$

$\Rightarrow \lambda^2 + 40\lambda - 2\lambda - 80 = 0$

$\Rightarrow (\lambda + 40)(\lambda - 2) = 0$

$\Rightarrow \lambda = -40$ or $\lambda = +2$

$\because \lambda$ is a non-negative real number

$\therefore \lambda = 2$

Important Note

- These line will be
  - (i) Real and different, if $h^2 - ab > 0$
  - (ii) Real and Parallel, if $h^2 - ab = 0$
  - (iii) Imaginary, if $h^2 - ab < 0$
- The condition that these lines are :
  - (1) Perpendicular to each other is $a + b = 0$ i.e. coefficient of x^2 + coefficient of $y^2 = 0$
  - (2) Parallel is $h^2 = ab$
  - (3) Equally inclined to the axes of x if $h = 0$ i.e. coefficient of $xy = 0$.

Ques. If the lines $(p - q)x^2 + 2(p + q)xy + (q - p)y^2 = 0$ are mutually perpendicular, then

(a) $p = q$

(b) $q = 0$

(c) $p = 0$

(d) p and q may have any value

Ans. $\because a = p - q,\ b = q - p$

So, $a + b = 0 \Rightarrow p - q + q - p = 0$

Which is always true

Hence, p and q may have any value

Ques. The equation $8x^2 + 8xy + 2y^2 = 0$ represents a pair of _____ Straight Lines?

(a) Perpendicular (b) Parallel

(c) Coincident (d) None of these

Ans. $a = 8,\ h = 4,\ b = 2$

$\because h^2 = ab \therefore$ POSL intersect at (0,0)

$\Rightarrow$ Coincident

$8x^2 + 8xy + 2y^2 = 0$

$\Rightarrow 4x^2 + 4xy + y^2 = 0$

$\Rightarrow (2x + y)(2x + y) = 0$

Both lines coincident and intersect at (0, 0)

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Ques. The slope of one of the lines represented by the equation $3x^2 + 4xy + \lambda y^2 = 0$ is 2 times the slope of the other line. Find λ.

Ans. Divide by x^2

$$3 + \frac{4y}{x} + \lambda\left(\frac{y}{x}\right)^2 = 0$$

Put $\frac{y}{x} = m$

$$\lambda m^2 + 4m + 3 = 0$$

Roots are α and 2α

$$\alpha + 2\alpha = -\frac{4}{\lambda} \Rightarrow 3\alpha = -\frac{4}{\lambda} \Rightarrow \alpha = -\frac{4}{3\lambda} \quad ...(i)$$

$$\alpha.2\alpha = \frac{3}{\lambda} \Rightarrow 2\alpha^2 = \frac{3}{\lambda}$$

$$\Rightarrow 2\left(-\frac{4}{3\lambda}\right)^2 = \left(\frac{3}{\lambda}\right) \Rightarrow \frac{2 \times 16}{9\lambda^2} = \frac{3}{\lambda} \Rightarrow 32 = 27\lambda$$

$$\lambda = \frac{32}{27}$$

Ques. If the ratio of gradients of the lines represented by $ax^2 + 2hxy + by^2 = 0$ is 1 : 3, then the value of the ratio $h^2 : ab$ is

(a) $\frac{1}{3}$ (b) $\frac{3}{4}$ (c) $\frac{4}{3}$ (d) 1

Ans.

$$a + 2h\left(\frac{y}{x}\right) + b\left(\frac{y}{x}\right)^2 = 0$$

$$bm^2 + 2hm + a = 0$$

Roots are α and 3α

$$\therefore 4\alpha = -\frac{2h}{b} \Rightarrow \alpha = -\frac{h}{2b}$$

$$\text{and } 3\alpha^2 = \frac{a}{b} \Rightarrow 3\left(-\frac{h}{2b}\right)^2 = \frac{a}{b}$$

$$\Rightarrow 3h^2 = 4ab \Rightarrow \frac{h^2}{ab} = \frac{4}{3}$$

Ques. Show that the equation $Bx^2 - 2Hxy + Ay^2 = 0$ represents a pair of straight lines which are at right angle to the pair given by the equation $Ax^2 + 2Hxy + By^2 = 0$

Ans. $Bm^2 + 2Hm + A = 0$

$m \to -\frac{1}{m}$ (To find the line which are at right angles)

$$B\left(-\frac{1}{m}\right) + 2H\left(-\frac{1}{m}\right) + A = 0 \Rightarrow \frac{B}{m^2} - \frac{2H}{m} + A = 0$$

$$\Rightarrow Am^2 - 2Hm + B = 0$$

Put $m = \frac{y}{x}$

$$A\frac{y^2}{x^2} - 2H\frac{y}{x} + B = 0$$

$$\Rightarrow Ay^2 - 2Hxy + Bx^2 = 0$$

Distance Between Pair of Lines

First check for parallel lines $\Rightarrow h^2 = ab$

Distance between them is $2\sqrt{\frac{g^2 - ac}{a(a+b)}}$

Ques. The Equation $8x^2 + 8xy + 2y^2 + 26x + 13y + 15 = 0$ represent a pair of straight lines. The distance between them is

(a) $\frac{7}{\sqrt{5}}$ (b) $\frac{7}{2\sqrt{5}}$

(c) $\frac{\sqrt{7}}{5}$ (d) None of these

Ans. $a = 8, h = 4, b = 2$

$\because h^2 = ab, \Rightarrow$ Lines are parallel

$g = 13, C = 15$

$$\therefore d = 2\sqrt{\frac{g^2 - ac}{a(a+b)}}$$

$$= 2\sqrt{\frac{169 - 8 \times 15}{8(8+2)}} = 2\sqrt{\frac{49}{8 \times 10}}$$

$$= 7\sqrt{\frac{1}{2 \times 10}} = \frac{7}{\sqrt{20}} = \frac{7}{2\sqrt{5}}$$

Equation of Pair of Angle Bisector for POSL

The combined equation of angle bisectors between the lines represented by homogeneous equation of 2nd degree is given by $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$, $a \neq b$, $h \neq 0$.

Ques. The equation of the bisectors of the angle between lines represented by equation $4x^2 - 16xy - 7y^2 = 0$ is

(a) $8y^2 + 11xy - 8y^2 = 0$

(b) $8x^2 - 11xy - 8y^2 = 0$

(c) $16x^2 + 11xy - 16y^2 = 0$

(d) $16x^2 - 11xy - 16y^2 = 0$

Ans. $a = 4, b = -7, h = -8$

$$\because \frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

$$\therefore \frac{x^2 - y^2}{4 - (-7)} = \frac{xy}{-8} \Rightarrow \frac{x^2 - y^2}{11} = \frac{xy}{-8}$$

$$\Rightarrow -8x^2 + 8y^2 = 11xy \Rightarrow 8x^2 - 8y^2 + 11xy = 0$$

Product of perpendicular Distance of a Point from POSL

The product of Perpendicular distance of the point (x_1, y_1) from POSL $ax^2 + 2hxy + by^2 = 0$ is given by

$$\frac{|ax_1^2 + by_1^2 + 2hx_1y_1|}{4h^2 + (a - b)^2}$$

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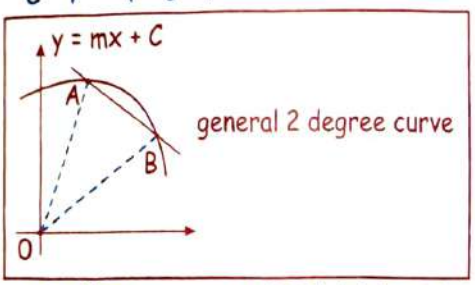
HOMOGENIZATION

Homogenized equation represents the POSL joining origin to the 2 points common to the curve (C) & the Line L w.r.t which the homogenization was carried out.

Genral curve → $ax^2 + by^2 + 2hxy + 2gx + 2fy + C = 0$

With respect to a line $y = mx + c \rightarrow \frac{y - mx}{C} = 1$

Homegenized Eqⁿ: $ax^2 + by^2 + 2hxy + 2gx\left(\frac{y - mx}{C}\right) + 2fy\left(\frac{y - mx}{C}\right) + C\left(\frac{y - mx}{C}\right)^2 = 0$



Homogenized Eqⁿ represents POSL joining origin with points common to curve & line L.

Ques. The equation of pair of lines joining origin to the points of intersection of $x^2 + y^2 = 27$ and $x + y = 3$ is

(a) $x^2 + y^2 + 6xy = 0$ (b) $x^2 + y^2 + 3xy = 0$ ✓

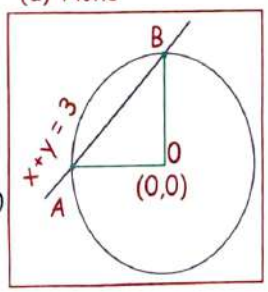
(c) $xy = 0$ (d) None

Ans. $\because \frac{x + y}{3} = 1$

$x^2 + y^2 - 27 = 0$

Homogenization

$x^2 + y^2 - 27\left(\frac{x + y}{3}\right)^2 = 0$



$x^2 + y^2 = \frac{3(x + y)^2}{9}$

$x^2 + y^2 = 3(x + y)^2$

$x^2 + y^2 = 3(x^2 + y^2 + 2xy)$

$2x^2 + 2y^2 + 6xy = 0 \Rightarrow x^2 + y^2 + 3xy = 0$

Ques. Find the equation of the line pair joining origin and the point of intersections of the line $2x - y = 3$ and the curve $x^2 - y^2 - xy + 3x - 6y + 18 = 0$. Also find the angle between these two lines.

Ans. $\frac{2x - y}{3} = 1$

Homogenized Eqⁿ

$x^2 - y^2 - xy + 3x\left[\frac{2x - y}{3}\right] - 6y\left[\frac{2x - y}{3}\right] + 18\left[\frac{2x - y}{3}\right]^2 = 0$

$\Rightarrow x^2 - y^2 - xy + x(2x - y) - 2y(2x - y) + 2(4x^2 + y^2 - 4xy) = 0$

$\Rightarrow x^2 - y^2 - xy + 2x^2 - xy - 4xy + 2y^2 + 8x^2 + 2y^2 - 8xy = 0 \Rightarrow 11x^2 + 3y^2 - 14xy = 0$

$\because a = 11, b = 3, h = -7$

$\therefore \tan\theta = \left|\frac{\sqrt{h^2 - ab}}{a + b}\right| = \left|\frac{2\sqrt{49 - 33}}{14}\right| = \frac{2\sqrt{16}}{14}$

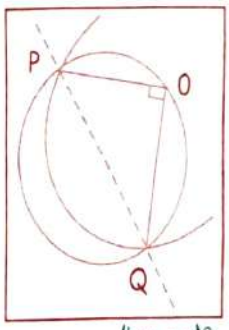
$\Rightarrow \tan\theta = \frac{4}{7}$

Ques. The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q. The circle on PQ as diameter passes through the origin. Prove that $n^2(a + b) = l^2 + m^2$

Ans. $lx + my = -n$

$\frac{lx + my}{-n} = 1$

Homogenization



$ax^2 + 2hxy + by^2 - 1\left(\frac{lx + my}{-n}\right)^2 = 0$

$a x^2 + 2hxy + by^2 - \frac{l^2x^2 + m^2y^2 + 2lmxy}{n^2}$

For $\theta = 90°$

Coeff of x^2 + coeff of $y^2 = 0$

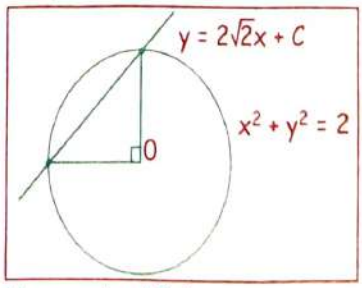
$\therefore a + b - \frac{l^2}{n^2} - \frac{m^2}{n^2} = 0 \Rightarrow a + b = \frac{l^2 + m^2}{n^2}$

Ques. The pair of straight lines joining the origin to the points of intersection of the line $y = 2\sqrt{2}x + c$ and the circle $x^2 + y^2 = 2$ are at right angles, if

(a) $c^2 - 4 = 0$ (b) $c^2 - 8 = 0$

(c) $c^2 - 9 = 0$ ✓ (d) $c^2 - 10 = 0$

Ans. $\frac{y - 2\sqrt{2}x}{C} = 1$



Homogenized Eqⁿ

$x^2 + y^2 - 2\left[\frac{y - 2\sqrt{2}x}{C}\right]^2 = 0$

$x^2 + y^2 - \frac{2}{C^2}[y^2 + 8x^2 - 4\sqrt{2}xy] = 0$

For 90° : Coeff of x^2 + coeff of $y^2 = 0$

$1 + 1 - \frac{2}{C^2} - \frac{16}{C^2} = 0 \Rightarrow 2 = \frac{18}{C^2} \Rightarrow c^2 = 9 \Rightarrow c^2 - 9 = 0$

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Short Notes

SECTION FORMULA

The co-ordinates of a point dividing a line joining the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the ratio m : n is given by:

(a) For internal division:

R divides line segment PQ, internally.

$$(x, y) \equiv \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

(b) For external division:

R divides line segment PQ, externally.

$$(x, y) \equiv \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right)$$

HARMONIC CONJUGATE POINTS

If P divides AB internally in the ratio m : n and Q divides AB externally in the ratio m : n, then P and Q are said to be harmonic conjugate points of each other with respect to line AB.

Mathematically: $\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$ i.e., AP, AB and AQ are in H.P.

AREA OF TRIANGLE

Let $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ are vertices of a triangle, then

$$\text{Area of } \Delta ABC = \left|\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$$

$$= \frac{1}{2}\left|\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]\right|$$

All the co-ordinates of the vertices of an equilateral triangle cannot be rational.

CENTRES OF A TRIANGLE

Centroid (G)

The point where 3 medians of a triangle meet is called Centroid.

$$G\left[\frac{X_1 + X_2 + X_3}{3}, \frac{Y_1 + Y_2 + Y_3}{3}\right]$$

The centroid of Δ formed by joining mid points of the 3 sides of a triangle is same as the centroid of original Δ.

Incentre (I)

The point where 3 angle bisectors of a triangle meet is called Incentre.

OR Centre of the Inscribed Circle (Incircle).

Formula: $\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$

Orthocentre (H)

The point where 3 altitudes of a triangle meet is called orthocentre.

Co-ordinates of Orthocentre:

$$\left(\frac{x_1\tan A + x_2\tan B + x_3\tan C}{\tan A + \tan B + \tan C}, \frac{y_1\tan A + y_2\tan B + y_3\tan C}{\tan A + \tan B + \tan C}\right)$$

Circumcentre (C)

The point where 3 perpendicular bisectors of a triangle meet is called Circumcentre. OR Centre the circumscribed circle.

Co-ordinates of circum-centre are given by:

$$\left(\frac{x_1\sin 2A + x_2\sin 2B + x_3\sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1\sin 2A + y_2\sin 2B + y_3\sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right)$$

- For Equilateral triangle, the centroid, in-centre, Ortho-centre, circum-centre, coincide.
- For Right angled triangle, circum-centre is the midpoint of hypotenuse and ortho-centre is the right-angled vertex.

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Euler's Line.

- In a non-equilateral triangle, Ortho-centre, centroid and circum-centre are always collinear. Further G divides the line joining H and C in the ratio 2 : 1.

ORTHOCENTRIC SYSTEM

In a Triangle ABC, the points A, B, C & H form a Orthocentric System, which means that the orthocenter of triangle formed by any 3 of these points is the 4th point.

In a Triangle ABC, the image of the orthocenter in any side of the triangle, always lies on the circumcircle of the triangle.

STRAIGHT LINES

1. If a line cuts off equal intercepts on co-ordinate axes then its slope is -1
2. If a line cuts off intercepts equal in magnitude but opposite in sign, then its slope is 1.
3. If a line cuts off equal distances on co-ordinate axes then its slope is 1 or -1.

VARIOUS FORMS OF STRAIGHT LINES

1. Slope Intercept Form $\rightarrow y = mx + c$
2. Slope Point Form $\rightarrow y - y_1 = m(x - x_1)$
3. Two Point Form $\rightarrow y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
4. X-Y Intercept Form $\rightarrow \frac{x}{a} + \frac{y}{b} = 1$
5. General Form $\rightarrow ax + by + c = 0$
6. Normal form: $x\cos\theta + y\sin\theta = P,\ (P > 0)$
7. Parametric Form of a Straight Line

   Let $P(x_1, y_1)$ be a given point on the line l and θ be the inclination of line l.

   Let $Q(x, y)$ be a moving point on it. Let $PQ = r$, then

   $x = x_1 \pm r\cos\theta$ and $y = y_1 \pm r\sin\theta$

   is called parametric form of the equation of the line l.

   Whenever the distance from a fixed point is asked or is given then we can try to use parametric form of a line.

ANGLE BETWEEN 2 LINES

Angle between two lines l_1 and l_2 having slopes m_1 and m_2 respectively.

Let θ be the acute angle between l_1 and l_2 then

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

If Lines are Parallel $m_1 = m_2$

If Lines are Perpendicular $m_1 m_2 = -1$

1. Equation of a line parallel to $ax + by + c = 0$ is given by $ax + by + k = 0$
2. Equation of a line Perpendicular to $ax + by + c = 0$ is given by $bx - ay + k = 0$

PERPENDICULAR DISTANCE OF A POINT FROM A LINE

$$p = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

DISTANCE BETWEEN PARALLEL LINES

$$d = \frac{|c_2 - c_1|}{\sqrt{a^2 + b^2}}$$

LOCUS

The path traced by a moving point under the given geometrical conditions is known as locus of that point.

Steps to Find the Equation of Locus

1. If we are finding the equation of the locus of a point P, let P be (h, k).
2. Apply the given geometrical conditions.
3. Get a relation between h & k by eliminating the variables assumed (if any).
4. Replace (h, k) $\rightarrow$ (x, y). The resulting equation would be the equation of the locus of P.

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Important Locus

Parabola:

The locus of a point which moves such that it is equidistant from a fixed point and a fixed line.

Ellipse:

The locus of a point which moves such that the sum of its distances from 2 fixed points is a constant.

Hyperbola:

The locus of a point which moves such that the difference of its distances from 2 fixed points is a constant.

FAMILY OF LINES

$L_1 + \lambda L_2 = 0, \lambda \in R$ represents the equation of straight lines passing through point of intersection of L_1 and L_2.

If a line can be expressed in the form of a family, then it always passes through a fixed point whose coordinates can be obtained by finding the point of intersection of the two lines L_1 and L_2.

Formula- Foot of perpendicular and Image of a Point

(i) The foot of perpendicular drawn to $Ax + By + C = 0$ from the point (x_1, y_1) is given by,

$$\frac{x - x_1}{A} = \frac{y - y_1}{B} = -\frac{Ax_1 + By_1 + C}{A^2 + B^2}$$

(ii) The image of a point (x_1, y_1) about the line $Ax + By + C = 0$ is given by,

$$\frac{x - x_1}{2A} = \frac{y - y_1}{2B} = -\frac{Ax_1 + By_1 + C}{A^2 + B^2}$$

Image of point (a, b) in the mirror $y = x$ is (b, a)

POSITION OF 2 POINTS WITH RESPECT TO A LINE

Same side ⇒ Same Sign

Opposite side ⇒ Opposite sign

EQUATION OF ANGLE BISECTOR

Let two lines $l_1 : a_1x + b_1y + c_1 = 0$ and $l_2 : a_2x + b_2y + c_2 = 0$ be given. Then, the equations of angle bisectors, l_3 and l_4, are given by

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

Isolation of Acute / Obtuse Angle Bisectors

Equation of bisector of acute/obtuse angles

Steps to find the equation of the bisector of the acute or obtuse angle

Step 1 In the equation of lines, make the constant terms c_1 & c_2 positive.

Step 2 Determine the sign of $a_1a_2 + b_1b_2$

Step 3 If $a_1a_2 + b_1b_2 < 0$, then + sign gives acute angle bisector

Hence bisector. $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ is acute angle

If sign of $a_1a_2 + b_1b_2$	For obtuse angle bisector	For acute andgle bisector
+	use + sign	use - sign
-	use - sign	use + sign

Bisector Containing the Origin

If the equation of the lines are written with constant terms c_1 and c_2 positive, then the equation of the bisectors of the angle containing the origin is always obtained by taking positive sign.

CONDITION OF CONCURRENCY

Three lines $a_1x + b_1y + c_1 = 0; a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent, if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

Optimization Problem-1

Two points A and B are on same side of line L, point P on line L is such that PA + PB is minimum when point P is common point of line L and AC when C is image of point B in line L.

For three point, P, A, B; $|PA + PB| \geq AB$, equality holds when P, A, B are collinear points.

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Optimization Problem-2

Two points A and B are on same side of line L, point P on line L is such that |PA – PB| is maximum then point P is common point of line L and AB.

For three points, P, A, B; $|PA - PB| \le AB$, equality holds when P, A, B are collinear points.

GENERAL 2° EQUATION

The general equation of 2 degree

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

represents a pair of straight lines, if

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \text{ i.e. } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

- The point of Intersection of POSL can be directly obtained by partial differentiation.

Homogeneous 2°Equation

$ax^2 + 2hxy + by^2 = 0$

Homogeneous 2° equation always represents pair of straight lines Intersecting at the origin.

Angle Between Pair of Lines

If θ be the angle between the POSL, then

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{|a+b|}$$

- These line will be :
  - (i) Real and different, if $h^2 - ab > 0$
  - (ii) Real and coincident, if $h^2 - ab = 0$
  - (iii) Imaginary, if $h^2 - ab < 0$
- The condition that these lines are :
  - (1) Perpendicular to each other is $a + b = 0$. i.e. coefficient of x^2 + coefficient of $y^2 = 0$.
  - (2) Parallel is $h^2 = ab$.
  - (3) Equally inclined to the axes of x if $h = 0$. i.e. coefficient of $xy = 0$.
- Distance between pair of Parallel Lines is $2\sqrt{\frac{g^2 - ac}{a(a+b)}}$

Equation of Pair of Angle Bisector for POSL

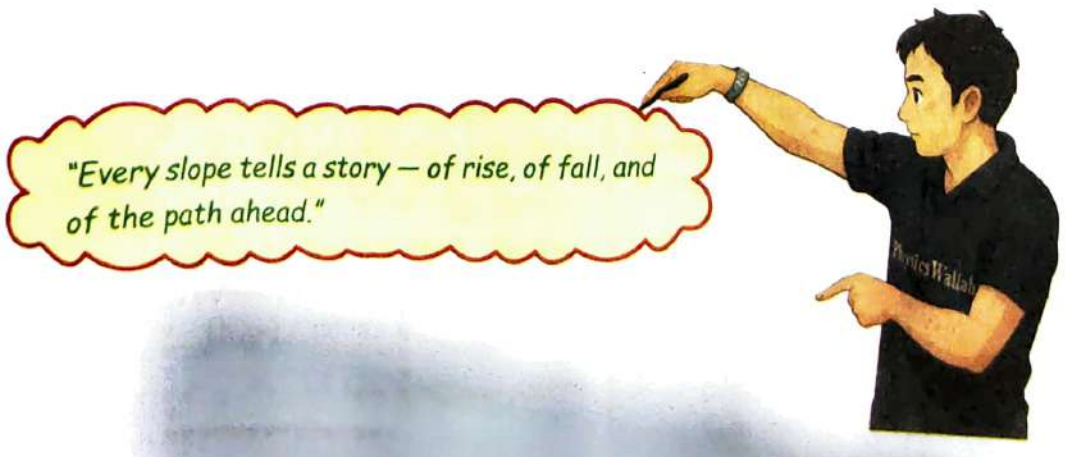
- The combined equation of angle bisectors between the lines represented by homogeneous equation of 2nd degree is given by $\frac{x^2 - y^2}{a-b} = \frac{xy}{h}, a \ne b, h \ne 0$.

Product of Perpendicular Distance of a Point from P.O.S.L.

- The product of Perpendicular distance of the point (x_1, y_1) from POSL $ax^2 + 2hxy + by^2 = 0$ is given by $\frac{|ax_1^2 + by_1^2 + 2hx_1y_1|}{\sqrt{4h^2 + (a-b)^2}}$

HOMOGENIZATION

Homogenized equation represents the POSL joining origin to the 2 points common to the curve (C) & the Line L w.r.t which the homogenization was carried out.



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Chapter 11

Circle

CIRCLE

$\because (h-a)^2 + (k-b)^2 = r^2$

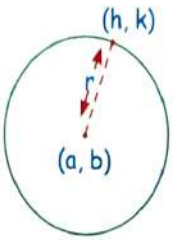
$(h, k) \rightarrow (x, y)$

Circle radius form:

$(x-a)^2 + (y-b)^2 = r^2$

Where, $(a, b) \rightarrow$ Centre

and $r \rightarrow$ radius



Example: $(3x-1)^2 + (3y+5)^2 = 9$

$$3^2\left[\left(x-\frac{1}{3}\right)^2+\left(y+\frac{5}{3}\right)^2\right]=9$$

$$\left(x-\frac{1}{3}\right)^2+\left(y+\frac{5}{3}\right)^2=1$$

$$\therefore \text{centre} = \left(\frac{1}{3}, -\frac{5}{3}\right), \text{Radius}(r) = 1$$

CENTRE-RADIUS FORM OF A CIRCLE

$(x-a)^2 + (y-b)^2 = r^2$

$(a,b) \rightarrow$ Centre

$r \rightarrow$ radius

if $C = (0, 0) = (a, b)$

Standard form of a circle: $x^2 + y^2 = r^2$

Ques. Find the equation of a circle for which the lines $3x + 4y + 5 = 0$ & $3x + y = 1$ acts as diameters and for which the perimeter of semi-circle is 36cm.

Ans. $\because$ Centre will be the intersection point of both lines

$\therefore C = (1, -2)$ $\left(\pi = \frac{22}{7}\right)$

Also, $\pi R + 2R = 36$

$$\Rightarrow R\left[\frac{22}{7}+2\right]=36$$

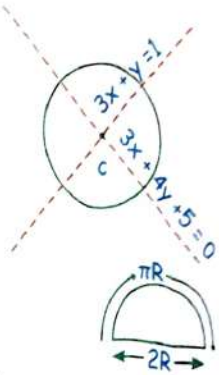
$$\Rightarrow R\left[\frac{22+14}{7}\right]=36$$

$$\Rightarrow R\,\frac{36}{7} = 36$$

$\Rightarrow R = 7$

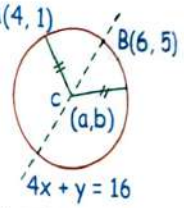
$\therefore$ eqⁿ of Circle

$(x-1)^2 + (y+2)^2 = 7^2$



Ques. Find the equation of the circle which passes through the points (4, 1), (6, 5) and has its center on the line $4x + y = 16$.

Ans. Let center be (a, b)



$\therefore 4a + b = 16$ ...(i)

$\because CB = CA$

$\Rightarrow (a-4)^2 + (b-1)^2 = (a-6)^2 + (b-5)^2$

$\Rightarrow a^2 + 16 - 8a + b^2 + 1 - 2b = a^2 + 36 - 12a + b^2 + 25 - 10b$

$\Rightarrow 4a + 8b = 36 + 25 - 16 - 1$

$\Rightarrow 4a + 8b = 44$

$\Rightarrow a + 2b = 11$ ...(ii)

By eqⁿ (i) & (ii):

We get, $(a,b) \equiv (3,4)$

GENERAL FORM OF A CIRCLE

$x^2 + y^2 + 2gx + 2fy + c = 0$

$\Rightarrow x^2 + 2gx + y^2 + 2fy + c = 0$

$\Rightarrow (x+g)^2 - g^2 + (y+f)^2 - f^2 + c = 0$

$\Rightarrow (x+g)^2 + (y+f)^2 = g^2 + f^2 - c$

center = $(-g, -f)$

$r = \sqrt{g^2 + f^2 - c}$

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$$\text{center} = \left[\frac{-\text{Coeff of } x}{2}, \frac{-\text{Coeff of } y}{2}\right]$$

Example: $x^2 + y^2 - 3x + 4y - 1 = 0$

$$\text{center} = \left[\frac{3}{2}, \frac{-4}{2}\right] = \left[\frac{3}{2}, -2\right]$$

$$r = \sqrt{\frac{9}{4} + 4 + 1} = \sqrt{\frac{29}{4}}$$

Ques. Find the centre and Radius of the Circle $4x^2 + 4y^2 - 12x + 4y + 1 = 0$

Ans. $x^2 + y^2 - 3x + y + \frac{1}{4} = 0$

$$\text{Center} = \left(\frac{3}{2}, \frac{-1}{2}\right)$$

$$r = \sqrt{\frac{9}{4} + \frac{1}{4} - \frac{1}{4}} = \frac{3}{2}$$

Important Note

(i) If $(g^2 + f^2 - c) > 0$, then r is real and positive and the circle is a real circle.

(ii) If $(g^2 + f^2 - c) = 0$, then radius $r = 0$ and circle is a point circle.

(iii) If $(g^2 + f^2 - c) < 0$, then r is imaginary $\Rightarrow$ circle is also imaginary.

(iv) A unique circle passes through three non collinear points.

Example: $x^2 + y^2 + 4x + 6y + 13 = 0$

$(x + 2)^2 + (y + 3)^2 = 0$

$g = 2, f = 3$

$g^2 + f^2 - c = 4 + 9 - 13 = 0$

$\therefore$ The circle is a point circle and the point is $(-2, -3)$

Example: $x^2 + y^2 + 4x + 2y + 7 = 0$

$g = 2, f = 1, c = 7$

$g^2 + f^2 - c = 4 + 1 - 7 = -2 < 0$

$\therefore$ The circle is not a real circle

Ques. Find the equation of the circle concentric with the circle $x^2 + y^2 - 8x + 6y - 5 = 0$ and passing through $(-2, -7)$.

Ans. Let the Circle be

$x^2 + y^2 - 8x + 6y + k = 0$

$\because (-2, -7)$ satisfes

$\therefore 4 + 49 + 16 - 42 + k = 0$

$\Rightarrow 20 + 7 + k = 0$

$\Rightarrow k = -27$

$\therefore$ The circle will be

$x^2 + y^2 - 8x + 6y - 27 = 0$

CONDITION FOR GENERAL 2 DEGREE CURVE TO REPRESENT CIRCLE

General 2 degree curve:

$ax^2 + by^2 + 2h\,xy + 2gx + 2fy + c = 0$

Circle $\rightarrow x^2 + y^2 + 2gx + 2fy + c = 0$

$\therefore$ condition to represent circle:

$h = 0 \Rightarrow$ coeff of xy

& $a = b \neq 0$

Ques. If the equations $2x^2 + \lambda xy + 2y^2 + (\lambda - 4)x + 6y - 5 = 0$ represents a circle then find λ and hence find the center and radius.

Ans. $\therefore \lambda$ is the co-effiient of xy

$\therefore \lambda = 0$

Thus, $2x^2 + 2y^2 - 4x + 6y - 5 = 0$

$\Rightarrow x^2 + y^2 - 2x + 3y - \frac{5}{2} = 0$

$\therefore$ centre $\left(1, \frac{-3}{2}\right)$

and $r = \sqrt{1 + \frac{9}{4} + \frac{5}{2}} = \sqrt{\frac{4 + 9 + 10}{4}} = \frac{\sqrt{23}}{2}$

Ques. Consider a circle $C_1 : x^2 + y^2 - 4x - 2y = a - 5$ Let its mirror image in the line $y = 2x + 1$ be another circle $C_2 : 5x^2 + 5y^2 - 10fx - 10gy + 36 = 0$. Let r be the radius of C_2. Then $a + r$ is equal to

[JEE Mains 2023 (Apr-1)]

Ans. $\because$ Centre of circle $C_1 = (2, 1)$ and

$(r_1) = \sqrt{2^2 + 1^2 + a - 5} = \sqrt{a}$

Centre of C_2 will be the image of centre of C_1 in the line $2x - y + 1 = 0$

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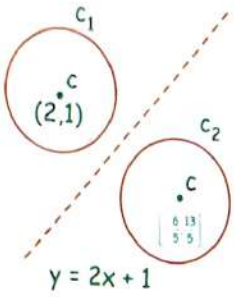
$$\therefore \frac{x-2}{2\cdot 2} = \frac{y-1}{2(-1)} = -\left(\frac{4-1+1}{5}\right)$$

$$\frac{x-2}{4} = \frac{y-1}{-2} = \frac{-4}{5}$$

$$\Rightarrow x = 2 - \frac{16}{5} = \frac{-6}{5}$$

and

$$y = \frac{13}{5}$$



$$C_2 : 5x^2 + 5y^2 - 10fx - 10gy + 36 = 0$$

$$\Rightarrow x^2 + y^2 - 2fx - 2gy + \frac{36}{5} = 0$$

$\because$ Centre $[f, g]$

$$\therefore f = \frac{-6}{5}, g = \frac{13}{5}$$

$$r_2 = \sqrt{\left(\frac{6}{5}\right)^2 + \left(\frac{13}{5}\right)^2 - \frac{36}{5}}$$

$$= \sqrt{\frac{205}{25} - \frac{36}{5}}$$

$$\Rightarrow r_2 = \sqrt{\frac{41}{5} - \frac{36}{5}} = 1 = r_1 = \sqrt{a}$$

$$\therefore \sqrt{a} = 1 \Rightarrow a = 1$$

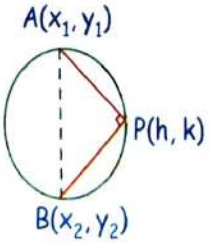
Hence, $a + r = 1 + 1 = 2$

DIAMETRIC FROM OF CIRCLE

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$ is the equation of a circle with diametric end points as (x_1, y_1) & (x_2, y_2)

Proof: $\because m_{PA}\, m_{PB} = -1$

$$\Rightarrow \left(\frac{k-y_1}{h-x_1}\right)\left(\frac{k-y_2}{h-x_2}\right) = -1$$



$(h, k) \to (x, y)$

$$\therefore \frac{(y-y_1)(y-y_2)}{(x-x_1)(x-x_2)} = -1$$

$$\Rightarrow (y - y_1)(y - y_2) = -(x - x_1)(x - x_2)$$

$$\Rightarrow (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

eqⁿ of Circle with AB as dia

Ques. (1) Find the equations of Circle passing through the points (0, 0), (a, 0) & (0, b).

(2) Find equations of the Smallest circle passing through (-1, 2) and (3, -4).

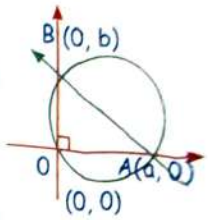
Ans. (i) $\because \angle AOB = 90°$

$\therefore$ AB must be a diame

$(x - a)(x - 0) + (y - 0)(y - b) = 0$

$\Rightarrow x^2 - ax + y^2 - by = 0$

$\Rightarrow x^2 + y^2 - ax - by = 0$



(ii) Circle will be smallest if (-1, 2) & (3, 4) will become end points of a diameter

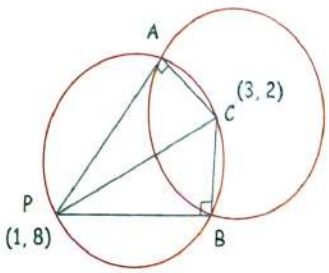
$\therefore (x + 1)(x - 3) + (y - 2)(y + 4) = 0$

Brain Teaser

Tangents drawn from the point P (1, 8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. Find the equations of the circumcircle of the triangle PAB.

Ans. Since $\angle A + \angle B = 180°$

$\therefore$ PACB is a cyclic quadrilateral



Further PC is diameter for this circle.

{Since Angle in semi circle is 90°}

Apply diametric form →

$(x - 1)(x - 3) + (y - 8)(y - 2) = 0$

$\Rightarrow x^2 - 4x + 3 + y^2 - 10y + 16 = 0$

$\Rightarrow x^2 + y^2 - 4x - 10y + 19 = 0$

Ques. Let O be the origin and OP and OQ be the tangents to the circle $x^2 + y^2 - 6x + 4y + 8 = 0$ at the points P and Q on it. If the circumcircle of the triangle OPQ passes through the points $\left(\alpha, \frac{1}{2}\right)$, then a value of α is **[JEE Main 2023]**

(a) $\frac{3}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{5}{2}$ (d) 1

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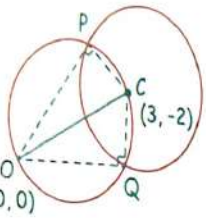
Ans. eqⁿ of Circumcircle of ΔOPQ:

$\because$ OC is the diameter

$\therefore (x - 0)(x - 3) + (y - 0)(y + 2) = 0$

$\Rightarrow x^2 - 3x + y^2 + 2y = 0$

$\Rightarrow x^2 + y^2 - 3x + 2y = 0$



$\because \left(\alpha, \frac{1}{2}\right)$ lies on the circle

$\therefore \alpha^2 + \frac{1}{4} - 3\alpha + 2 \times \frac{1}{2} = 0$

$\alpha^2 - 3\alpha + \frac{5}{4} = 0$

$\Rightarrow 4\alpha^2 - 12\alpha + 5 = 0$

$\Rightarrow 4\alpha^2 - 10\alpha - 2\alpha + 5 = 0$

$\Rightarrow 2\alpha[2\alpha - 5] - 1[2\alpha - 5]$

$\Rightarrow (2\alpha - 1)(2\alpha - 5) = 0$

$\therefore \alpha = \frac{1}{2}, \frac{5}{2}$

Important Note

Diametric form can also be obtained by adding 2 quadratic eqⁿ s whose roots are (x_1, x_2) & (y_1, y_2) Respectively.

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

(A quadratic with roots x_1 & x_2) + (a quadratic with roots y_1 & y_2) = 0

Before adding always ensure: (coeff of x^2 = coeff of y^2)

Ques. The x coordinates of two points A and B are roots of equation $x^2 + 2x - a^2 = 0$ and y coordinate are roots of equation $y^2 + 4y - b^2 = 0$ then equation of the circle which has diameter AB is-

(a) $(x - 1)^2 + (y - 2)^2 = 5 + a^2 + b^2$

(b) $(x + 1)^2 + (y + 2)^2 = \sqrt{(5 + a^2 + b^2)}$

(c) $(x + 1)^2 + (y + 2)^2 = (a^2 + b^2)$

(d) $(x + 1)^2 + (y + 2)^2 = 5 + a^2 + b^2$

Ans. $x^2 + 2x - a^2 = 0 \Rightarrow x_1, x_2$

$y^2 + 4y - b^2 = 0 \Rightarrow y_1, y_2$

$\therefore x^2 + y^2 + 2x + 4y - a^2 - b^2 = 0$

$\Rightarrow (x + 1)^2 - 1 + (y + 2)^2 - 4 - a^2 - b^2 = 0$

$\Rightarrow (x + 1)^2 + (y + 2)^2 = 5 + a^2 + b^2$

Ques. Let the abscissae of the two points P and Q a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2a + 2by + c = 0$, then the value of $(a + b - c)$ is: **[JEE Main 2022 July]**

(a) 12 (b) 13 (c) 14 (d) 16

Ans. $\therefore$ The eqn of the circle will be:

$(x^2 - 4x - 6) + (y^2 + 2y - 7) = 0$

$\Rightarrow x^2 + y^2 - 4x + 2y - 13 = 0$

Comparing with $x^2 + y^2 + 2ax + 2by + c = 0$

We get, $2a = -4$, $2b = 2$, $c = -13$

$\Rightarrow a = -2, b = 1, c = -13$

$\therefore a + b - c = -2 + 1 + 13 = 12$

Brain Teaser

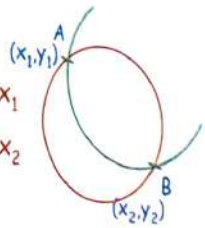
The parabola $y^2 = 4ax$ & the line $y = mx + c$ meet at two distinct points A & B. Find the equation of circle with AB as diameter.

Ans. To find the x-coordinates of end points of the diameter

$(mx + c)^2 = 4ax$

$\Rightarrow m^2x^2 + c^2 + 2mxc = 4ax$

$\Rightarrow m^2x^2 + (2mc - 4a)x + c^2 = 0 \Rightarrow x_1, x_2$ (I)



for y-coordinates

$y^2 = 4a\left(\frac{y - c}{m}\right)$

$my^2 - 4ay + 4ac = 0 \Rightarrow y_1, y_2$ (II)

$\therefore$ The eqⁿ of circle:

$m^2x^2 + m^2y^2 + (2mc - 4a)x - 4amy + c^2 + 4acm = 0$

Ques. Consider a triangle whose two sides lie on the x-axis and the line $x + y + 1 = 0$. If the orthocenter of Δ is (1, 1), then the equation of the circle passing through the vertices of the triangle Δ is **[JEE Adv. 2021]**

(a) $x^2 + y^2 - 3x + y = 0$

(b) $x^2 + y^2 + x + 3y = 0$

(c) $x^2 + y^2 + 2y - 1 = 0$

(d) $x^2 + y^2 + x + y = 0$

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Ans. $\because (\alpha, \beta)$ lies on the line AB

$\therefore \alpha + \beta = -1 \quad ...(i)$

$\because m_{AB} = -1$

$\therefore m_{CH} = 1$

$\frac{1-0}{1-a} = 1 \Rightarrow 1 = 1 - a$

$a = 0$

$\because m_{BE} . m_{AC} = -1$

$\Rightarrow \frac{1}{2} \cdot \left(\frac{+\beta}{\alpha}\right) = -1$

$\Rightarrow \beta = -2\alpha \quad ...(ii)$

So, $(\alpha, \beta) = (1, -2)$

Let Circle be

$x^2 + y^2 + 2gx + 2fy = 0$

put (1, -2) on it

$\Rightarrow 1^2 + (-2)^2 + 2g(1) + 2f(-2) = 0$

$\Rightarrow 5 + 2g - 4f = 0 \rightarrow 1$

put (-1, 0) on it

$(-1)^2 + 0^2 + 2g(-1) + 0 = 0$

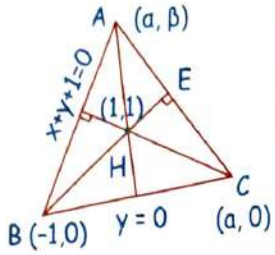
$\Rightarrow 1 = 2g$

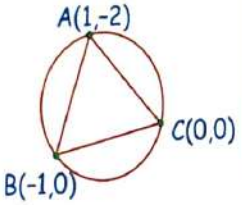
$\Rightarrow g = \frac{1}{2}$

and $5 + 1 = 4f \Rightarrow f = \frac{3}{2}$

Hence, Equation of circle is

$x^2 + y^2 + x + 3y = 0$





INTERCEPTS MADE BY A CIRCLE

X-intercept: The distance between the point where the circle cute X axis is called its X intercept.

Y-intercept: The distance between the point where the circle cuts Y axis is called its Y intercept.

$x^2 + y^2 + 2gx + 2fy + c = 0$

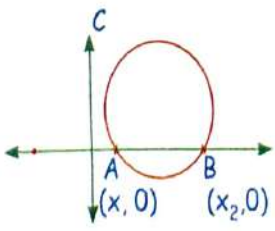
For X_{in} put $y = 0$

$x^2 + 2gx + c = 0 \begin{matrix} x_1 \\ x_2 \end{matrix}$

$AB = |x_2 - x_1| = \frac{\sqrt{D}}{a}$

$\frac{\sqrt{4g^2 - 4c}}{1} = 2\sqrt{g^2 - c}$

$x^2 + y^2 + 2gx + 2fy + C = 0$

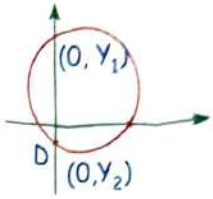


For Y_{in}, Put $x = 0$

$y^2 + 2fy + C = 0 \begin{matrix} y_1 \\ y_2 \end{matrix}$

$\Rightarrow Y_{in} = |y_1 - y_2| = \frac{\sqrt{D}}{a} = \frac{2\sqrt{f^2 - c}}{1}$

$= 2\sqrt{f^2 - c}$



Important Points to Remember

$x \text{ intercept} = |x_2 - x_1| = 2\sqrt{g^2 - c}$

$y \text{ intercept} = |y_2 - y_1| = 2\sqrt{f^2 - c}$

(i) Intercepts made by a circle are always non-negative

(ii) Intercepts made by a circle may be non-real

Example: $x^2 + y^2 + 8x + 4y + 5 = 0$

Centre [- 4,- 2]
3rd Quad

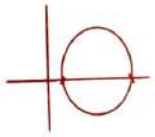
$2g = 8, 2f = 4, c = 5$

$g = 4, f = 2, c = 5$

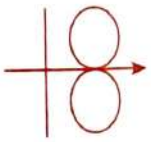
$X_{in} = 2\sqrt{g^2 - c}$
$= 2\sqrt{16 - 5} = 2\sqrt{11}$

$Y_{in} = 2\sqrt{f^2 - c}$
$= 2\sqrt{4 - 5} = 2\sqrt{-1} \rightarrow$ non real

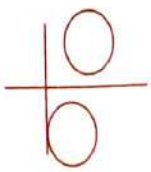
1. If $g^2 - C > 0 \rightarrow X_{in} \rightarrow$ non zero



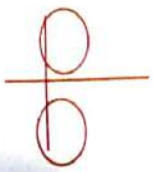
2. If $g^2 - c = 0 \rightarrow X_{in} =$ zero $\Rightarrow$ Circle tenches x - axis



3. If $g^2 - c < 0 \rightarrow X_{in} \rightarrow$ non real



4. $Y_{in} \rightarrow$ If $f^2 - c > 0$



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5. If $f^2 - c = 0$

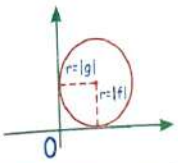
6. If $f^2 - c < 0$

- When circle touches x-axis $\Rightarrow g^2 = c$

$\Rightarrow r = |f|$ OR radius = |ordinate of centre|

- When circle touches y axis $\Rightarrow f^2 = c$

$\therefore r = \sqrt{g^2 + f^2 - C} = |g|$



$\Rightarrow r = |g|$ OR radius = |abscissa of centre|

- When circle touches both x and y axes

$\Rightarrow r = |g| = |f|$

Ques. Equation of the larger circle passing through (2, 1) and touching the coordinate axes is

(a) $x^2 + y^2 - 10x - 10y + 25 = 0$

(b) $x^2 + y^2 + 10x + 10y - 35 = 0$

(c) $x^2 + y^2 - 10x + 10y + 5 = 0$

(d) $x^2 + y^2 + 10x - 10y - 15 = 0$

Ans. circle touches in Ist Quad centre (r, r)

radius $\rightarrow$ r

$\therefore (x - r)^2 + (y - r)^2 = r^2$

put (2, 1) on it

$\therefore (2 - r)^2 + (1 - r)^2 = r^2$

$\Rightarrow 4 + r^2 - 4r + 1 + r^2 - 2r = r^2$

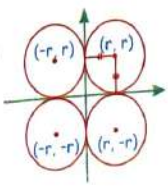
$\Rightarrow r^2 - 6r + 5 = 0$

$\Rightarrow (r - 5)(r - 1) = 0$

$r = 5$ or $r = 1$
✓ (✗)

$\therefore (x - 5)^2 + (y - 5) = 5^2$

$\Rightarrow x^2 + y^2 - 10x - 10y + 25 = 0$



Ques. Find the equation of the circle which touches both the coordinate-axes in 3rd Quadrant and also touches the line $3x - 4y + 8 = 0$

Ans. let centre be $(-r, -r)$

$CP = r$

$\Rightarrow \left|\frac{3(-r) - (4r) + 8}{5}\right| = r$

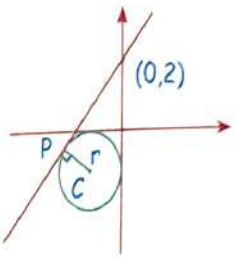
$\Rightarrow |-7r + 8| = 5r$

$\Rightarrow -7r + 8 = \pm 5r$

$\therefore 8 = 12r \Rightarrow r = \frac{2}{3}$

or $-7r + 8 = -5r$

$\Rightarrow 8 = 2r \Rightarrow r = 4$



Eq^n of circle

$(x + r)^2 + (y + r)^2 = r^2$

$r = \frac{2}{3}$ or $r = 4$

$\left(x + \frac{2}{3}\right)^2 + \left(y + \frac{2}{3}\right)^2 = \frac{4}{9}$

or

$(x + 4)^2 + (y + 4)^2 = 16$

Ques. The equation to the circle whose radius is 4 and which touches the x-axis at a distance -3 from the origin is-

(a) $x^2 + y^2 - 6x + 8y - 9 = 0$

(b) $x^2 + y^2 \pm 6x - 8y + 9 = 0$

(c) $x^2 + y^2 + 6x \pm 8y + 9 = 0$

(d) $x^2 + y^2 \pm 6x - 8y + 9 = 0$

Ans. Centre = (-3,4) or (-3,-4)

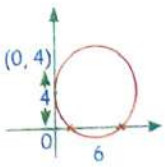
$\Rightarrow (x + 3)^2 + (y \pm 4)^2 = 4^2$

$x^2 + y^2 + 6x \pm 8y + 9 = 0$

Ques. Find the equation of the circle which touches the positive Y-axis at a distance of 4 units from origin and makes an intercept of length 6 on the x-axis.

Ans. **Method-I**

$\rightarrow$ Using formula



$2\sqrt{g^2 - c} = 6$

$g^2 - c = 9$ ... (i)

$f^2 - c = 0$ ... (ii)

$x^2 + y^2 + 2gx + 2fy + c = 0$

(0, 4) satisfies

$\therefore 16 + 2f \times 4 + c = 0$

$\Rightarrow 16 + 8f + c = 0$ ... (iii)

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By eqⁿ (ii)

$\Rightarrow 16 + 8f + f^2 = 0$

$\Rightarrow (f + 4)^2 = 0$

$\therefore f = -4$ and $c = 16$

Using eqⁿ (i)

$\Rightarrow g^2 - 16 = 9$

$\Rightarrow g^2 = 25$

$\Rightarrow g = \pm 5$

$\therefore x^2 + y^2 \pm 10x - 8y + 16 = 0$

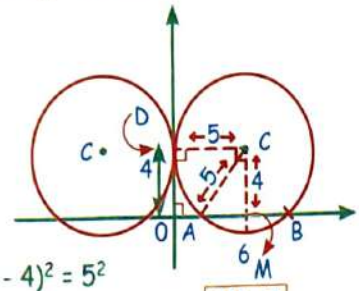
Method-II

$r = \sqrt{3^2 + 4^2}$

$\Rightarrow r = 5$

$C\,[5, 4]$

$C'[-5, 4]$



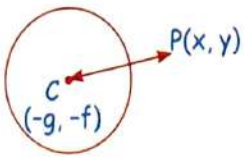
$\therefore (x \pm 5)^2 + (y - 4)^2 = 5^2$

$x^2 + y^2 \pm 10x - 8y + 16 = 0$

POSITION OF A POINT W.R.T A CIRCLE

Power of a Point: S_1

$S_1 : x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + C$



For Outside: $S_1 > 0$

For Inside: $S_1 < 0$

For on the circle $S_1 = 0$

Proof:

$PC > r$

$\Rightarrow \sqrt{(x_1 + g)^2 + (y_1 + f)^2} > \sqrt{g^2 + f^2 - C}$

$\Rightarrow x_1^2 + g^2 + 2gx_1 + y_1^2 + f^2 + 2fy_1 > g^2 + f^2 - C$

$\Rightarrow x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + C > 0$

$\Rightarrow S_1 > 0$

LENGTH OF TANGENT

Length of Tangent Drawn from an exterior Point (x_1, y_1)

$L = \sqrt{S_1}$, where S_1 is the power of the point (x_1, y_1)

Proof:

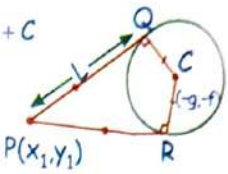
$L^2 = (PC)^2 - (QC)^2$

$\Rightarrow L^2 = (x_1 + g)^2 + (y_1 + f)^2 - (g^2 + f^2 - C)$

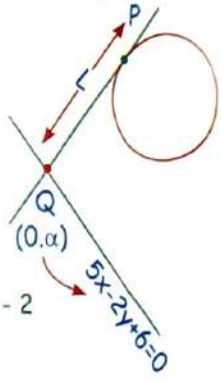
$\Rightarrow L^2 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + C$

$\Rightarrow L^2 = S_1$

$\Rightarrow L = \sqrt{S_1}$



Ques. If the tangent at the point P to the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis then find the length of PQ.



Ans. $5 \times 0 - 2\alpha + 6 = 0$

$\Rightarrow 2\alpha = 6$

$\Rightarrow \alpha = 3$

$\therefore Q\,(0, 3)$

Power of Q:

$S_1 = 0^2 + 3^2 + 6 \times 0 + 6 \times 3 - 2$

$S_1 = 9 + 18 - 2 = 25$

$\therefore L = \sqrt{S_1} = \sqrt{25} = 5$

Ques. If the length of the tangent drawn from the point (1, 2) to the circles $x^2 + y^2 + x + y - 4 = 0$ and $3x^2 + 3y^2 - x - y + k = 0$ be in the ratio 4 : 3, then k =

(a) $\frac{21}{2}$ (b) $\frac{7}{2}$ (c) $\frac{-21}{4}$ (d) $\frac{7}{4}$

Ans. $S_1 : 1 + 2^2 + 1 + 2 - 4$

$S_1 = 4$

$\therefore L_1 = 2$

$x^2 + y^2 - \frac{x}{3} - \frac{y}{3} + \frac{k}{3}$

$S_1' = 1^2 + 2^2 - \frac{1}{3} - \frac{2}{3} + \frac{k}{3}$

$S_1' = 4 + \frac{k}{3}$

$\therefore L_2 = \sqrt{4 + \frac{k}{3}}$

Thus, $\frac{\sqrt{4}}{\sqrt{4 + \frac{k}{3}}} = \frac{4}{3}$

$\Rightarrow \frac{4}{4 + \frac{k}{3}} = \frac{16}{9} \Rightarrow 4\left(4 + \frac{k}{3}\right) = 9$

$\Rightarrow 16 + \frac{4k}{3} = 9 \Rightarrow \frac{4k}{3} = -7 \Rightarrow k = \frac{-21}{4}$

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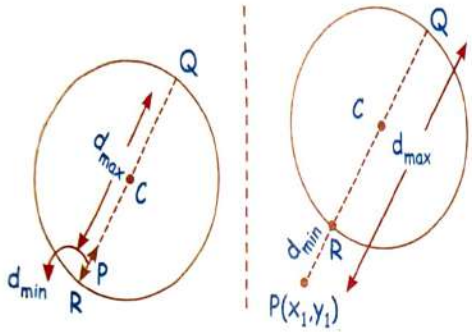
MAX & MIN DISTANCE OF A POINT FROM A CIRCLE

$d_{max} = PC + r$

$d_{min} = |PC - r|$



$d_{min} = r - PC$ $\qquad$ $d_{max} = PC + r$

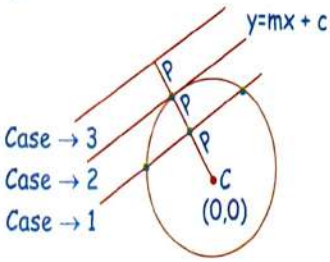
LINE AND A CIRCLE

Condition of tangency

$$CP = \left|\frac{c}{\sqrt{1+m^2}}\right| = r$$

$$\Rightarrow \frac{c}{\sqrt{1+m^2}} = \pm r$$

$$\Rightarrow c = \pm r\sqrt{1+m^2}$$



$CP < r \rightarrow$ Case $\rightarrow 1$

$CP = r \rightarrow$ Case $\rightarrow 2$

$CP > r \rightarrow$ Case $\rightarrow 3$

Ques. L_1: $5x + 12y - 10 = 0$ & L_2: $5x - 12y - 40 = 0$. The circle C_1 (radius = 3) touches L_1 and L_2 and has its center in 1st quadrant. The circle C_2 is concentric with C_1 and cuts off intercept of length 8 on the lines. what is the equation of C_2.

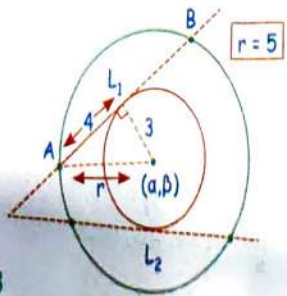
Ans. $\alpha > 0$

$\beta > 0$

$m_1 = \frac{-5}{12}$

$m_2 = \frac{5}{12}$

$$\left|\frac{5\alpha + 12\beta - 10}{13}\right| = 3$$



$\Rightarrow 5\alpha + 12\beta - 10 = \pm 39$

$\Rightarrow 5\alpha + 12\beta = \pm 39 + 10$

$\Rightarrow 5\alpha + 12\beta = 49$ or -29 (not possible)

$$\left|\frac{5\alpha - 12\beta - 40}{13}\right| = 3$$

$\Rightarrow 5\alpha - 12\beta - 40 = \pm 39$

$\Rightarrow 5\alpha - 12\beta = 79$ or 1

$5\alpha + 12\beta = 49$	$5\alpha + 12\beta = 49$
$5\alpha - 12\beta = 79$	$5\alpha - 12\beta = 1$
add	add
$10\alpha = 128$	$10\alpha = 50 \Rightarrow \alpha = 5$
$\Rightarrow \alpha = \frac{128}{10} = \frac{64}{5}$	$25 + 12\beta = 49$
$\frac{5 \times 64}{5} + 12\beta = 49$	$\Rightarrow 12\beta = 24$
	$\Rightarrow \beta = 2$

$\beta = -\frac{5}{4}$ (not possible)

eqn of C_2

$(x - 5)^2 + (y - 2)^2 = 5^2$

Ques. The equations of the tangents to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the straight line $4x + 3y + 5 = 0$ are

(a) $3x - 4y - 19 = 0, 3x - 4y + 31 = 0$

(b) $4x + 3y - 19 = 0, 4x + 3y + 31 = 0$

(c) $4x + 3y + 19 = 0, 4x + 3y - 31 = 0$

(d) $3x - 4y + 19 = 0, 3x - 4y + 31 = 0$

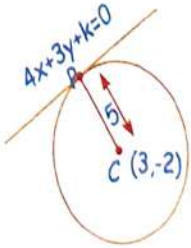
Ans. $x^2 + y^2 - 6x + 4y - 12 = 0$

Centre $[3, -2]$

$r = \sqrt{3^2 + 2^2 - (-12)} = \sqrt{25} = 5$

$\therefore PC = r$

$$\because \left|\frac{4(3) - 3 \times 2 + k}{5}\right| = 5$$



$$\Rightarrow \left|\frac{k+6}{5}\right| = 5$$

$\Rightarrow |k + 6| = 25$

$\Rightarrow k + 6 = \pm 25$

$k = 19$ or -31

$\therefore 4x + 3y + 19 = 0, 4x + 3y - 31 = 0$

Ques. A circle touch $x = 0, y = 0, 3x + 4y = 120$, centre of the circle in first quadrant is:

(a) (10, 10) (b) (60, 60)

(c) (24, 24) (d) None of these

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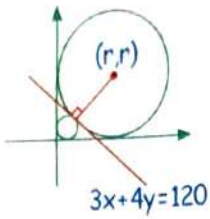
Ans. Let Centre be (r, r)

$$\frac{|3r + 4r - 120|}{\sqrt{3^2 + 4^2}} = r$$



$\Rightarrow |7r - 120| = 5r$

$\Rightarrow 7r - 120 = \pm 5r$

$2r = 120$ or $12r = 120$

$r = 60$ or $r = 10$

centre = (10, 10) or (60, 60)

Since the center is in the first quadrant and the circle touches the line $3x + 4y = 120$, the radius must be such that the circle is on the origin side of the line. For $r = 60$, the center is (60, 60), and $3(60) + 4(60) = 180 + 240 = 420$, which is greater than (120). This means the center is on the side of the line opposite to the origin.

For $r = 10$, the center is (10, 10), and $3(10) + 4(10) = 30 + 40 = 70$, which is less than (120). This indicates the circle is on the origin side of the line, which is consistent with touching the axes in the first quadrant and the given line. Therefore, $r = 10$ is the correct radius.

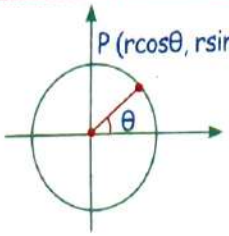
Hence centre of the circle in first quadrant = (10, 10).

PARAMETRIC COORDINATES ON CIRCLE

$P(r\cos\theta, r\sin\theta)$

$x^2 + y^2 = r^2$

$$\boxed{\begin{matrix} x = r\cos\theta \\ y = r\sin\theta \end{matrix} \quad \theta \in R}$$



OR

$x^2 + y^2 + 2gx + 2fy + c = 0$

$\Rightarrow (x + g)^2 + (y + f)^2 = g^2 + f^2 - c$

$\therefore x + g = r\cos\theta \Rightarrow x = -g + r\cos\theta$

and $y + f = r\sin\theta$, $y = -f + r\sin\theta$

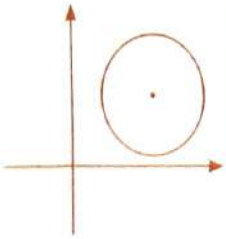
Where, $r = \sqrt{g^2 + f^2 - c}$

$x - x_1 = r\cos\theta$ & $y - y_1 = r\sin\theta$

$$\boxed{x = x_1 + r\cos\theta \ \& \ y = y_1 + r\sin\theta}$$

$\theta \rightarrow$ Variable

$r \rightarrow$ Fixed.



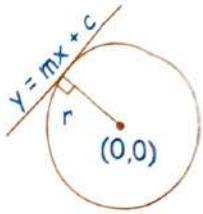
VARIOUS FORMS OF TANGENTS TO $x^2 + y^2 = r^2$

(1) Slope form of tangent

$$\boxed{y = mx + r\sqrt{1 + m^2}}$$

because these are always 2 parallel tangents to a circle

Proof: $\because y = mx + c$



$$\therefore \left|\frac{-c}{\sqrt{1 + m^2}}\right| = r$$

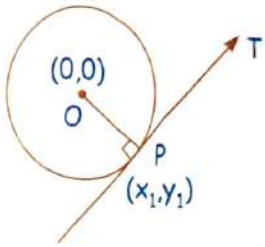
$\Rightarrow |c| = r\sqrt{1 + m^2}$

$\Rightarrow c = \pm r\sqrt{1 + m^2}$ (Condition of tangency)

$\therefore y = mx \pm r\sqrt{1 + m^2}$

(2) Cartesian Form →

$$\boxed{xx_1 + yy_1 = r^2}$$



Proof: $m_{op} = \frac{y_1 - 0}{x_1 - 0} = \frac{y_1}{x_1}$

$\therefore m_T = \frac{-x_1}{y_1}$

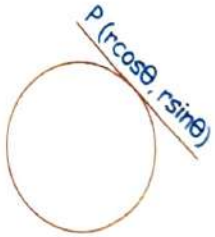
eqⁿ of Tangents: $y - y_1 = \frac{-x_1}{y_1}(x - x_1)$

$\Rightarrow yy_1 - y_1^2 = -xx_1 + x_1^2$

$\Rightarrow xx_1 + yy_1 = x_1^2 + y_1^2$

$\Rightarrow xx_1 + yy_1 = r^2$

(3) Parametric Form



$$\begin{cases} x_1 = r\cos\theta \\ y_1 = r\sin\theta \end{cases}$$

$xx_1 + yy_1 = r^2$

$x(r\cos\theta) + y(r\sin\theta) = r^2$

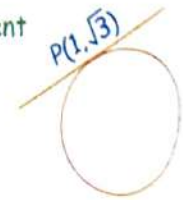
$$\Rightarrow \boxed{x\cos\theta + y\sin\theta = r}$$

Ques. Find the equation of tangent to the circle $x^2 + y^2 = 4$ which pass through the point

(i) $(1, \sqrt{3})$

Ans. Satisfies eqⁿ of circle

$\therefore$ There will be only one tangent



$x_1 = 1$, $y_1 = \sqrt{3}$

$\therefore xx_1 + yy_1 = r^2$

$\Rightarrow x + \sqrt{3}y = 4$

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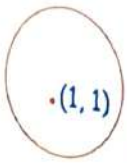
(ii) (1, 1)

Ans. $S: x^2 + y^2 - 4$

$S_1: 1^2 + 1^2 - 4 = -2$

$\Rightarrow S_1 < 0 \Rightarrow P(1, 1)$ lies inside

$\Rightarrow$ no tangent is possible



(iii) (2, 3)

Ans. $S_1: 2^2 + 3^2 - 4 = 9$

$S_1 > 0 \rightarrow P(2, 3)$ lies outside

Let the tangent be

$y = mx \pm r\sqrt{1+m^2}$

$\Rightarrow y = mx \pm 2\sqrt{1+m^2}$

$\because$ (2, 3) satisfies

$\therefore 3 = 2m \pm 2\sqrt{1+m^2}$

$\Rightarrow 3 - 2m = \pm 2\sqrt{1+m^2}$

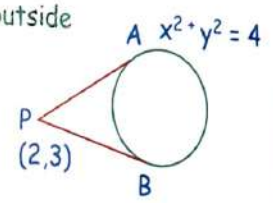
$\Rightarrow 9 + 4m^2 - 12m = 4 + 4m^2$

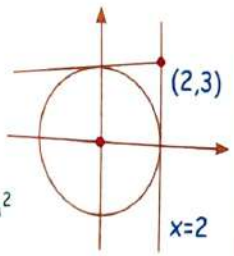
$\Rightarrow 5 = 12m$

$m = \frac{5}{12}$ or $m = \infty$

eqⁿ of tangent

$y - 3 = \frac{5}{12}(x - 2)$ or $x = 2$





Ques. Find the equation of tangent to the circle $x^2 + y^2 = 25$, which are perpendicular to the line $3x + 4y = 7$

Ans. Let the eqⁿ be

$y = mx \pm 5\sqrt{1+m^2}$

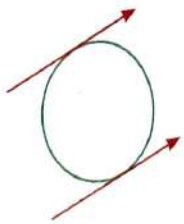
Slpe of given line $= -\frac{3}{4}$

$\therefore m = \frac{4}{3}$

$y = \frac{4}{3}x \pm 5\sqrt{1+\left(\frac{4}{3}\right)^2}$

$y = \frac{4}{3}x \pm \frac{5\times 5}{3}$

$\Rightarrow 3y = 4x \pm 25$



Ques. If the line $y = mx + c$ be a tangent to be circle $x^2 + y^2 = a^2$, then the point of contact is

(a) $\left(\frac{-a^2}{c}, a^2\right)$ (b) $\left(\frac{a^2}{c}, \frac{-a^2m}{c}\right)$

(c) $\left(\frac{-a^2m}{c}, \frac{a^2}{c}\right)$ (d) $\left(\frac{-a^2c}{m}, \frac{a^2}{c}\right)$

Ans. Let the tangent be

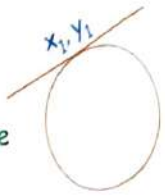
$xx_1 + yy_1 = a^2$ ...(i)

$-mx + y = c$ ...(ii)

Eqⁿ (i) & (ii) represent Same line

$\therefore \frac{x_1}{-m} = \frac{y_1}{1} = \frac{a^2}{c}$

$\therefore x_1 = \frac{-a^2m}{c}$ and $y_1 = \frac{a^2}{c}$



Ques. A variable tangent to the circle $x^2 + y^2 = 1$, meets the coordinate axes at points A & B. Find the locus of mid point of AB.

Ans. Let tangent be $x\cos\theta + y\sin\theta = 1$

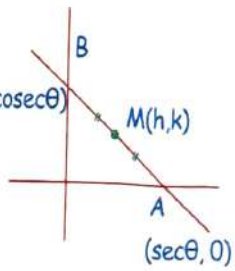
$2h = \sec\theta \Rightarrow 2h = \frac{1}{\cos\theta}$ $\Rightarrow \cos\theta = \frac{1}{2h}$

$2k = \operatorname{cosec}\theta \Rightarrow 2k = \frac{1}{\sin\theta}$ $\Rightarrow \sin\theta = \frac{1}{2k}$

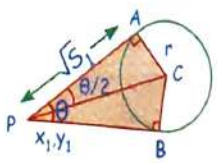
$\because \sin^2\theta + \cos^2\theta = 1$

$\therefore \left(\frac{1}{2h}\right)^2 + \left(\frac{1}{2k}\right)^2 = 1$

$\Rightarrow \frac{1}{x^2} + \frac{1}{y^2} = 4$



More About Tangents



(1) Area of quadrilateral formed by joining pair of tangents drawn from a point P (x_1, y_1) and the two radii of circle is $\boxed{r\sqrt{S_1}}$.

Proof: $\Delta PAC = \frac{1}{2} r\sqrt{S_1}$

Area of Quad PACB $= 2 \times \frac{1}{2} r\sqrt{S_1} = r\sqrt{S_1}$

(2) The angle between pair of tangents drawn from an exterior point (x_1, y_1) upon a circle is given by $\tan\theta = \frac{2r\sqrt{S_1}}{S_1 - r^2}$. Where S_1 represent power of point (x_1, y_1) and r is the radius of this circle.

Proof: In ΔPAC

$\tan\left(\frac{\theta}{2}\right) = \frac{r}{\sqrt{S_1}}$

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$$\because \tan\theta = \frac{2\tan\left(\frac{\theta}{2}\right)}{1-\tan^2\left(\frac{\theta}{2}\right)} = \frac{\left(\frac{2r}{\sqrt{S_1}}\right)}{1-\frac{r^2}{S_1}} = \frac{2r\sqrt{S_1}}{S_1 - r^2}$$

Ques. The angle between the two tangents from the origin to the circle $(x-7)^2 + (y+1)^2 = 25$ is

(a) 0 (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ ~~(d)~~ $\frac{\pi}{2}$

Ans. $S_1: 7^2 + 1^2 - 25 = 25$

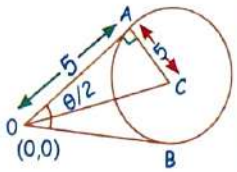
$OA = \sqrt{S_1} = \sqrt{25} = 5$

In Δ OAC

$\tan\left(\frac{\theta}{2}\right) = \frac{5}{5} = 1$

$\Rightarrow \frac{\theta}{2} = 45°$

$\therefore \theta = 90°$



Ques. Let the tangents at two points A and B on the circle $x^2 + y^2 - 4x + 3 = 0$ meet at origin O (0, 0). Then the area of the triangle OAB is:

[JEE Mains, 28 July 2022-II]

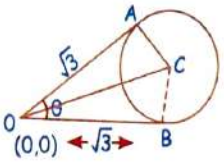
(a) $\frac{3\sqrt{3}}{2}$ ~~(b)~~ $\frac{3\sqrt{3}}{4}$ (c) $\frac{2}{2\sqrt{3}}$ (d) $\frac{3}{4\sqrt{3}}$

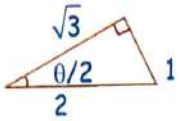
Ans. $S_1 = \sqrt{3}, r = \sqrt{(-2)^2 - 3} = 1$

Area $= \frac{1}{2}\sqrt{3}\sqrt{3}\sin\theta$

$\because \sin\frac{\theta}{2} = \frac{1}{2} \Rightarrow \frac{\theta}{2} = 30° \Rightarrow \theta = 60°$

$\therefore$ Area $= \frac{3\sqrt{3}}{4}$ then $\sin\theta = \frac{\sqrt{3}}{2}$





DIRECTOR CIRCLE

The locus of points of intersection of Perpendicular tangents drawn to a given circle is known as the director circle of the given circle.

Proof:

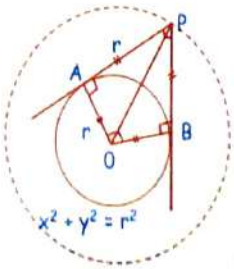
Quad PAOB is a square

$PA = OA$

$\sqrt{S_1} = r \Rightarrow \sqrt{h^2 + k^2 - r^2} = r$

$\Rightarrow h^2 + k^2 - r^2 = r^2$

$\Rightarrow h^2 + k^2 = 2r^2$



$(h, k) \rightarrow (x, y)$

Director Circle ← $x^2 + y^2 = 2r^2$ → Concentric, radius $= \sqrt{2}r$ $\left\{x^2 + y^2 = \left(\sqrt{2}r\right)^2\right.$

Important Note

Director circle is concentric with given circle & its radius is $\sqrt{2}$ the radius of the given circle.

Ques. Equation of the director circle of the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ is?

(a) $x^2 + y^2 - 2x + 4y - 8 = 0$

(b) $x^2 + y^2 - 2x + 4y - 4\sqrt{2} = 0$

(c) $x^2 + y^2 - 2x + 4y - 16 = 0$

~~(d)~~ $x^2 + y^2 - 2x + 4y - 13 = 0$

Ans. Let D.C be

$x^2 + y^2 - 2x + 4y + k = 0$

$r = \sqrt{1^2 + 2^2 - (-4)} = 3$

$R = \sqrt{2}\,.$

$\Rightarrow R = 3\sqrt{2}$

$\Rightarrow \sqrt{1^2 + 2^2 - k} = 3\sqrt{2}$

$\Rightarrow 5 - k = 18$

$\Rightarrow k = 5 - 18$

$\Rightarrow k = -13$

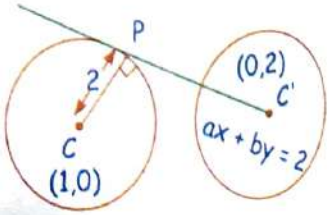
Hence, eqⁿ of the director circle is $x^2 + y^2 - 2x + 4y - 13 = 0$.

NORMAL TO A CIRCLE

Any line passing through center is called normal to circle.

Ques. If the straight line $ax + by = 2$; $a, b \neq 0$ touches the circle $x^2 + y^2 - 2x = 3$ and is normal to the circle $x^2 + y^2 - 4y = 6$, then find the value of a and b.

Ans.



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$CP = 2$

$$\Rightarrow \left|\frac{1\times a + 0\times b - 2}{\sqrt{a^2+b^2}}\right| = 2$$

$$\Rightarrow \left|\frac{a-2}{\sqrt{a^2+b^2}}\right| = 2$$

$\Rightarrow (a-2)^2 = 4(a^2+b^2)$

$(0, 2)$ Satisfies $\because ax + by = 2$

$\therefore a \times 0 + 2b = 2$

$\Rightarrow b = 1$

$\therefore (a-2)^2 = 4(a^2+1)$

$\Rightarrow a^2 + 4 - 4a = 4a^2 + 4$

$\Rightarrow 3a^2 + 4a = 0$

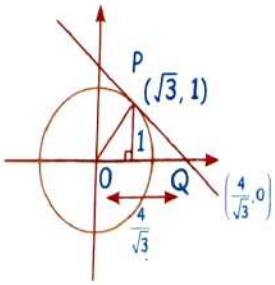
$$\Rightarrow a = \frac{-4}{3}$$

Hence, $a = \frac{-4}{3}, b = 1$

Ques. The tangents and the normal lines at the point $(\sqrt{3}, 1)$ to the circle $x^2 + y^2 = 4$ and the X-axis from a triangle. The area of this triangle (in square units) is **[JEE Main April 2019-II)]**

(a) $\frac{1}{3}$ (b) $\frac{4}{\sqrt{3}}$ (c) $\frac{2}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{3}}$

Ans. eqⁿ of tangent:



$xx_1 + yy_1 = 4$

$\Rightarrow \sqrt{3}x + y = 4$

at $y = 0, x = \frac{4}{\sqrt{3}}$

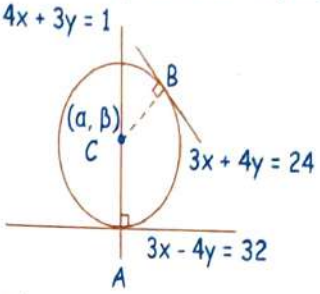
$$\therefore OQ = \frac{4}{\sqrt{3}}$$

Area of $\Delta POQ = \frac{1}{2} \times \frac{4}{\sqrt{3}} \times 1 = \frac{2}{\sqrt{3}}$

Ques. Let the center of a circle C be (α, β) and its radius $r < 8$. Let $3x + 4y = 24$ and $3x - 3y = 32$ be two tangents and $4x + 3y = 1$ be a normal to C. Then $(\alpha - \beta + r)$ is equal to

(a) 7 (b) 5 (c) 6 (d) 9

Ans. By solving $4x + 3y = 1$ and $3x - 4y = 32$



We get,

$A = (4, -5)$ and (α, β) lie on $4x + 3y = 1$

$$\therefore \beta = \frac{1-4\alpha}{3}$$

and $CA = CB$

$$\Rightarrow \left|\frac{3\alpha - 4\left(\frac{1-4\alpha}{3}\right) - 32}{5}\right| = \left|\frac{3\alpha + 4\left(\frac{1-4\alpha}{3}\right) - 24}{5}\right|$$

We get, $\alpha = 1$ and $\frac{28}{3}$

for $\alpha = 1$, centre $= (1, 1), r = 5$

for $\alpha = \frac{28}{3}$, centre $= \left(\frac{28}{3}, \frac{-109}{3}\right)$

and $r = 49.78$ (Rejected)

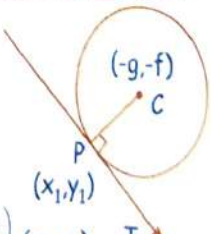
Hence, $\alpha = 1, \beta = -1, r = 5$

$\therefore \alpha - \beta + r = 7$

CARTESIAN FROM OF TANGENT TO $x^2 + y^2 + 2gx + 2fy + c = 0$

$$m_{CP} = \frac{y_1 + f}{x_1 + g}$$

$$m_T = -\left(\frac{x_1 + g}{y_1 + f}\right)$$



eqⁿ of Tangent : $y - y_1 = -\left(\frac{x_1 + g}{y_1 + f}\right)(x - x_1)$

$\Rightarrow (y - y_1)(y_1 + f) = -(x_1 + g)(x - x_1)$

$\Rightarrow yy_1 + fy - y_1^2 - fy_1 = -(xx_1 - x_1^2 + gx - gx_1)$

$\Rightarrow (xx_1 + y_1 + gx + fy) + gx_1 + fy_1 + c = (x_1^2 + y_1^2 + gx_1 + fy_1) + gx_1 fy_1 + c$

T: $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

$\Rightarrow T = 0$

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Equation of Cartesian tangent

$T: xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

Eqⁿ of circle: $x^2 + y^2 + 2gx + 2fy + c = 0$

∴ Cartesian tangent can be obtained by replacing

Learn
- $x^2 \to xx_1$,
- $y^2 \to yy_1$
- $x \to \frac{x + x_1}{2}$
- $y \to \frac{y + y_1}{2}$
- $c \to c$

$x^2 + y^2 = r^2$ → $\boxed{xx_1 + yy_1 = r^2}$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ → $\boxed{\frac{xx^1}{a^2} + \frac{yy^1}{b^2} = 1}$

SLOPE FORM OF TANGENT TO $(x - a)^2 + (y - b)^2 = r^2$

$x^2 + y^2 = r^2$ is

$y = mx \pm r\sqrt{1 + m^2}$

Tangent of $(x - a)^2 + (y - b)^2 = r^2$ will be:

$y - b = m(x - a) \pm r\sqrt{1 + m^2}$ Slope form

LENGTH OF A CHORD

Ques. Find the length of the chord intercepted by the line $x - y + 1 = 0$ on the circle $x^2 + y^2 - 2x + 4y - 4 = 0$.

Ans. $CP = \frac{1 + 2 + 1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$

$r = \sqrt{1^2 + 2^2 + 4} = 3$

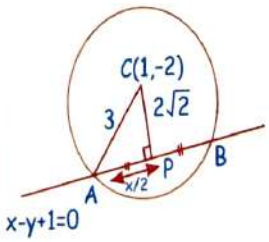
In ΔCAP:

$(CA)^2 = (CP)^2 + (AP)^2$

$\Rightarrow 3^2 = \left(2\sqrt{2}\right)^2 + \left(\frac{x}{2}\right)^2$

$\Rightarrow 1 = \left(\frac{x}{2}\right)^2$

$\Rightarrow \frac{x}{2} = 1 \Rightarrow x = 2$

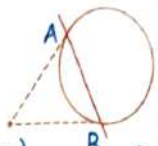


CHORD OF CONTACT

The Equation of Chord of contact for $x^2 + y^2 = r^2$ is given by $\boxed{xx_1 + yy_1 = r^2}$

AB is C.O.C with respect to point P

Equation of AB: $\boxed{xx_1 + yy_1 = r^2}$



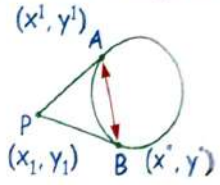
here (x_1, y_1) is Point of Intersection of tangents

eqⁿ of PA: $xx' + yy' = r^2$

eqⁿ of PB: $xx'' + yy'' = r^2$

$x_1x' + y_1y' = r^2$ ...(i)

$x_1x'' + y_1y'' = r^2$ ...(ii)



Consider an eqⁿ of a SL: $xx_1 + yy_1 = r^2$

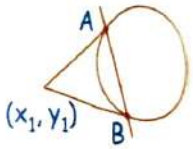
note that this straight line passes through A & B but C.O.C also passes through A & B.

Important Note

In Cartesian tangent $P(x_1, y_1)$ represent point of tangency. In chord of contact $P(x_1, y_1)$ represent point of intersection of tangents.

The Equation of Chord of contact for $x^2 + y^2 + 2gx + 2fy + c = 0$ is given by $T = 0$

AB: $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$



Ques. Tangents are drawn to the circle $x^2 + y^2 = 25$ at the points where it is met by line $3x + 4y = 5$. Find the point of intersection of these tangents.

Ans. Let $P(x_1, y_1)$

AB is Chord of contract for P:

eqⁿ of AB: $xx_1 + yy_1 = 25$ ...(i)

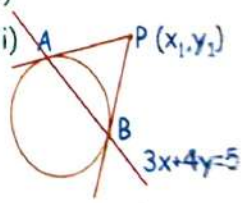
eqⁿ of AB: $3x + 4y = 5$ ...(ii)

By comparing eqⁿ (i) and (ii)

$\frac{x_1}{3} = \frac{y_1}{4} = \frac{25}{5}$

We get, $x_1 = 15$, $y_1 = 20$

Hence, point of intersection = (15, 20)



Ques. The chord of contact of tangents drawn from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$. Given $a > b > c > 0$. Show that a, b, c are in GP.

Ans. $x_1^2 + y_1^2 = a^2$ ...(i)

eqⁿ of AB: $xx_1 + yy_1 = b^2$

$OQ = c$

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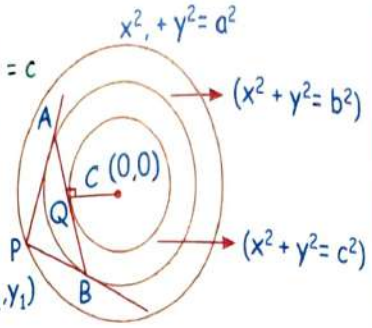
$$\therefore \left|\frac{-b^2}{\sqrt{x_1^2+y_1^2}}\right| = c$$



By eqⁿ (i)

$$\left|\frac{b^2}{\sqrt{a^2}}\right| = c$$

$$\left|\frac{b^2}{a}\right| = c \Rightarrow b^2 = ac$$

Hence, a, b, c are in GP

Ques. The distance between the chords of contact of the tangents to the circle $x^2+y^2+2gx+2fy+c=0$ from the origin and from the point (g, f) is

(a) $g^2 + f^2$ (b) $\frac{1}{2}(g^2 + f^2 + c)$

(c) $\frac{1}{2}\frac{g^2+f^2+c}{\sqrt{g^2+f^2}}$ (d) $\frac{1}{2}\frac{(g^2+f^2-c)}{\sqrt{g^2+f^2}}$

Ans. AB: $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

$x_1 = y_1 = 0$ (at P)

$\therefore gx + fy + c = 0$ ...(i)

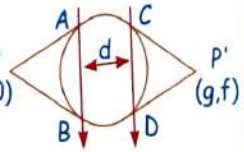
CD: $x_1 = g, y_1 = f$ (at P')

$\therefore xg + yf + g(x + g) + f(y + f) + c = 0$

$\Rightarrow 2xg + 2fy + g^2 + f^2 + c = 0$

$$\Rightarrow gx + fy + \frac{g^2+f^2+c}{2} = 0 \quad ...(ii)$$

$$\therefore d = \left|\frac{\frac{g^2+f^2+c}{2} - c}{\sqrt{g^2+f^2}}\right|$$



$$\therefore d = \frac{1}{2}\frac{(g^2+f^2-c)}{\sqrt{g^2+f^2}}$$

Ques. A circle with centre (2, 3) and radius 4 intersects the line $x + y = 3$ at the points P and Q. If the tangents at P and Q intersects at the point S (α, β) then $4\alpha - 7\beta$ is equal to

[JEE Mains, Jan 2023-II]

Ans. Eqⁿ of circle: $(x - 2)^2 + (y - 3)^2 = (4)^2$

$\therefore S: x^2 + y^2 - 4x - 6y - 3 = 0$

PQ is chord of contract for Points (α, β)

$\therefore$ PQ: $x\alpha + y\beta - 2(x + \alpha) - 3(y + \beta) - 3 = 0$

$\Rightarrow x(\alpha - 2) + y(\beta - 3) = 2\alpha + 3\beta + 3$

Comparing with

$x + y = 3$

$$\frac{\alpha-2}{1} = \frac{\beta-3}{1} = \frac{2\alpha+3\beta+3}{3}$$

$\Rightarrow \alpha = \beta - 1$

and $\beta - 3 = \frac{2\alpha+3\beta+3}{3}$

$\Rightarrow 3\beta - 9 = 2\alpha + 3\beta + 3$

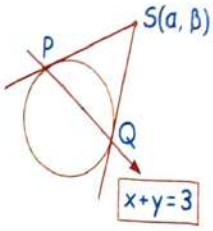
$\Rightarrow 2\alpha = -12 \Rightarrow \alpha = -6$

$\therefore \alpha = \beta - 1$

$\Rightarrow -6 = \beta - 1 \Rightarrow \beta = -5$

Hence, $4\alpha - 7\beta$

$= -24 + 35 = 11$

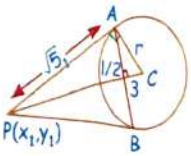


Important Note

The length of chord of contact drawn with respect to a point P (x_1, y_1) is given by

$$\frac{2r\sqrt{S_1}}{\sqrt{r^2+S_1}}$$

Proof: AB = L



Area of $\Delta PAC = \frac{1}{2}r\sqrt{S_1}$

Area of $\Delta PAC = \frac{1}{2}.PC.\frac{L}{2}$

$$\therefore \frac{1}{2}PC.\frac{L}{2} = \frac{1}{2}r\sqrt{S_1}$$

$$\Rightarrow L = \frac{2r\sqrt{S_1}}{PC} = \frac{2r\sqrt{S_1}}{\sqrt{S_1+r^2}} \quad [\because (PC)^2 = S_1 + r^2]$$

Ques. The area of the traingle formed by the tangents from the points (h, k) to the circle $x^2 + y^2 = a^2$ and the line joining their points of contact is-

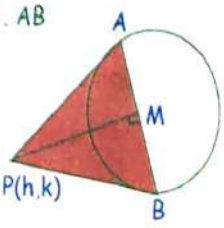
(a) $a\frac{\left(h^2+k^2-a^2\right)^{\frac{3}{2}}}{h^2+k^2}$ (b) $a\frac{\left(h^2+k^2-a^2\right)^{\frac{1}{2}}}{h^2+k^2}$

(c) $\frac{\left(h^2+k^2-a^2\right)^{\frac{3}{2}}}{h^2+k^2}$ (d) $\frac{\left(h^2+k^2-a^2\right)^{\frac{1}{2}}}{h^2+k^2}$

Ans. Area of $\Delta PAB = \frac{1}{2}.PM.AB$

eq of AB: $hx + ky = a^2$

$$PM = \left|\frac{\left(h^2+k^2-a^2\right)}{\sqrt{h^2+k^2}}\right|$$



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and $AB = \frac{2r\sqrt{S_1}}{\sqrt{S_1 + r^2}} = \frac{2a\sqrt{h^2+k^2-a^2}}{\sqrt{h^2+k^2-a^2+a^2}}$

$= \frac{2a\sqrt{h^2+k^2-a^2}}{\sqrt{h^2+k^2}}$

$\therefore$ Area $= \frac{a\left(h^2+k^2-a^2\right)^{\frac{3}{2}}}{h^2+k^2}$

CHORD WITH A GIVEN MID POINT

$\because PC \perp AB$ and $m_{PC} = \frac{y_1 + f}{x_1 + g}$

$\therefore m_{AB} = \frac{-(x_1+g)}{(y_1+f)}$

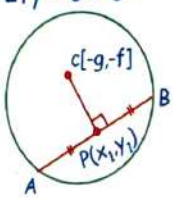
eqn of AB: $y - y_1 = \frac{-(x_1+g)}{y_1+f}(x - x_1)$

$\Rightarrow (y - y_1)(y_1 + f) = -(x_1 + g)(x - x_1)$

$\Rightarrow xx_1 + yy_1 + gx + fy + gx_1 + fy_1 + C = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + C$

$\therefore T = S_1 \rightarrow$ Chord with a given mid pt

$x^2 + y^2 + 2gx + 2fy + C = 0$



Ques. The equation of the chord of the circle $x^2 + y^2 = a^2$ having (x_1, y_1) as its mid point is

(a) $xy_1 + yx_1 = a^2$ (b) $x_1 + y_1 = a$

(c) $xx_1 + yy_1 = x_1^2 + y_1^2$ (d) $xx_1 + yy_1 = a^2$

Ans. T: $xx_1 + yy_1 - a^2$

S_1: $x_1^2 + y_1^2 - a^2$

$\therefore T = S_1$

$\Rightarrow xx_1 + yy_1 - a^2 = x_1^2 + y_1^2 - a^2$

$\Rightarrow xx_1 + yy_1 = x_1^2 + y_1^2$

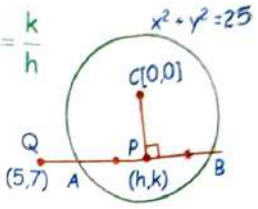
Important Note

While solving problems on this concept we prefer Using geometry ($PC \perp AB$) instead of using the formula $T = S_1$.

However, this formula is very useful in Problems on chord with a given midpoint for CONICS.

Ques. Find the locus of Mid point of all the chords of the circle $x^2 + y^2 = 25$ which pass through the point (5, 7)

Ans. **Method-1:** $M_{PC} = \frac{k-0}{h-0} = \frac{k}{h}$



and $M_{PQ} = \frac{k-7}{h-5}$

$PC \perp PQ$

$\therefore \frac{k}{h}\left(\frac{k-7}{h-5}\right) = -1$

$\Rightarrow k(k-7) = -h(h-5)$

$\Rightarrow h(h-5) + k(k-7) = 0$

$\Rightarrow h^2 + k^2 - 5h - 7k = 0$

$\therefore x^2 + y^2 - 5x - 7y = 0$

Method-2:

Equation of PQ by $T = S_1$

$\therefore hx + ky - 25 = h^2 + k^2 - 25$

$\Rightarrow 5h + 7k = h^2 + k^2$

Ques. The equation of the locus of the mid-points of chords of the circle $4x^2 + 4y^2 - 12x + 4y + 1 = 0$, which subtend angle 120° at the centre, is the circle

(a) $x^2 + y^2 - 3x + y + \frac{31}{16} = 0$

(b) $x^2 + y^2 - 3x + y - \frac{1}{16} = 0$

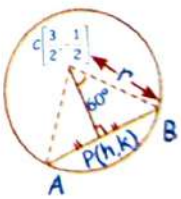
(c) $x^2 + y^2 - 3x + y + \frac{1}{4} = 0$

(d) $x^2 + y^2 - 3x + y - \frac{1}{4} = 0$

Ans. $x^2 + y^2 - 3x + y + \frac{1}{4} = 0$

$\therefore C = \left(\frac{3}{2}, \frac{-1}{2}\right)$

and $r = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2 - \frac{1}{4}}$



$\Rightarrow r = \frac{3}{2}$

$\because PC = r\cos 60°$

$\Rightarrow PC = \frac{3}{2} \times \frac{1}{2} = \frac{3}{4}$

So, the required eqn will be circle with

Centre $\left(\frac{3}{2}, \frac{-1}{2}\right)$ and radius $\frac{3}{4}$

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$$\therefore \left(h-\frac{3}{2}\right)^2+\left(k+\frac{1}{2}\right)^2=\left(\frac{3}{4}\right)^2$$

$$\Rightarrow h^2-3h+k^2+k+\frac{9}{4}+\frac{1}{4}=\frac{9}{16}$$

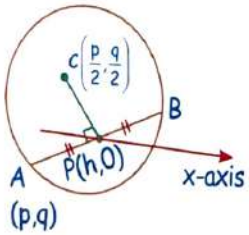
$(h,k) \to (x,y)$

$$\Rightarrow x^2+y^2-3x+y+\frac{31}{16}=0$$

Ques. If two distinct chords drawn from (p, q) on the circle $x^2 + y^2 = px + qy$, $pq \neq 0$ are bisected by the x - axis, then prove that $p^2 > 8q^2$

Ans. Let Mid pt be (h, 0)

$\because PC \perp PA$



$$\Rightarrow \left(\frac{\frac{q}{2}-0}{\frac{p}{2}-h}\right)\left(\frac{q-0}{p-h}\right)=-1$$

$$\Rightarrow \frac{q^2}{2}=-\left(\frac{p}{2}-h\right)(p-h)$$

$$\Rightarrow \frac{q^2}{2}+\left(\frac{p}{2}-h\right)(p-h)=0$$

$$\Rightarrow \frac{q^2}{2}+\frac{p^2}{2}-\frac{ph}{2}-ph+h^2=0$$

$$\Rightarrow h^2-\frac{3ph}{2}+\frac{p^2+q^2}{2}=0$$

$\Rightarrow 2h^2 - 3ph + p^2 + q^2 = 0$

'h' has 2 distinct values

$\therefore D > 0$

$(3p)^2 - 4 \times 2 \times (p^2 + q^2) > 0$

$\Rightarrow 9p^2 - 8p^2 > 8q^2$

$\Rightarrow p^2 > 8q^2$

Ques. For the circle $x^2 + y^2 = 50$, find the equation and the length of the shortest chord which pass through the point (3, 4)

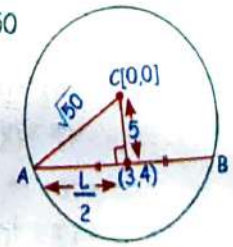
Ans. eqⁿ of AB: $T = S_1$

$xx_1 + yy_1 - 50 = x_1^2 + y_1^2 - 50$

$\Rightarrow 3x + 4y = 3^2 + 4^2$

$\Rightarrow 3x + 4y = 25$

and $\left(\frac{L}{2}\right)^2 = 50 - 25$



$\Rightarrow \left(\frac{L}{2}\right)^2 = 25$

$\Rightarrow \frac{L}{2} = 5 \Rightarrow L = 10$

Important Note

Of all the chords passing through a given point (x_1, y_1) lying inside a given circle, the shortest chord is the one whose "midpoint" is (x_1, y_1)

Proof:

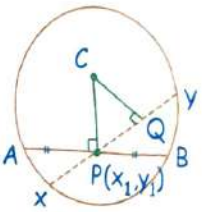
"Shortest chord wil be farthest from centre"

$\therefore CP > CQ$

$\Rightarrow$ Chord AB is farther from centre

$\Rightarrow$ Chord AB is shortest.

Which is possible if P is mid point of AB



Ques. The locus of the middle points of those chords of the circle $x^2 + y^2 = 4$ which subtend a right angle at the origin is

(a) $x^2 + y^2 - 2x - 2y = 0$

(b) $x^2 + y^2 = 4$

(c) $x^2 + y^2 = 2$

(d) $(x - 1)^2 + (y - 1)^2 = 5$

Ans. $OP = 2 \cos 45°$

$\Rightarrow OP = 2 \times \frac{1}{\sqrt{2}}$

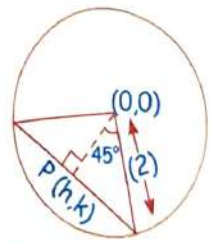
$\Rightarrow OP = \sqrt{2}$

$\Rightarrow \sqrt{h^2+k^2} = \sqrt{2}$

$\Rightarrow h^2 + k^2 = 2$

$(h,k) \to (x,y)$

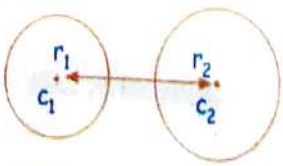
$x^2 + y^2 = 2$



NATURE OF 2 CIRCLE

Let two circles having C_1 and C_2 and radii, r_1 and r_2 and C_1C_2 is the distance between their centres then:

CASE-I: WHEN CIRCLES ARE APART FROM EACH OTHER



$C_1C_2 > r_1 + r_2$

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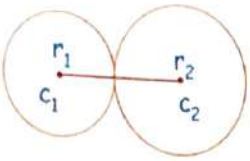
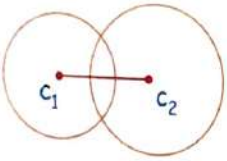
CASE-II: WHEN THE CIRCLE ARE TOUCHING EXTERNALLY



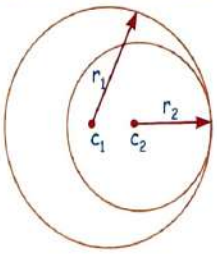
$C_1C_2 = r_1 + r_2$

CASE-III: WHEN THE CIRCLES ARE INTERSECTING AT 2 POINTS



$|r_1 - r_2| < C_1C_2 < r_1 + r_2$

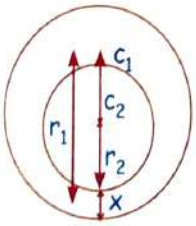
CASE-IV: WHEN THE CIRCLES ARE TOUCHING INTERNALLY



$C_1C_2 + r_2 = r_1$

$\Rightarrow C_1C_2 = |r_1 - r_2|$

CASE-V: WHEN ONE CIRCLE LIES COMPLETELY INSIDE THE OTHER



$r_1 = C_1C_2 + r_2 + x$

$\Rightarrow x + C_1C_2 = r_1 - r_2$

$\Rightarrow C_1C_2 < |r_1 - r_2|$

Ques. Circles $x^2 + y^2 - 2x - 4y = 0$ and $x^2 + y^2 - 8y - 4 = 0$

(a) Touch each other internally

(b) Touch each other externally

(c) Cuts each other at two points

(d) None of these

Ans. $C_1(1, 2), C_2(0, 4)$

$r_1 = \sqrt{1^2 + 2^2 - 0} \Rightarrow r_1 = \sqrt{5}$

and $r_2 = \sqrt{0^2 + 4^2 - (-4)} \Rightarrow r_2 = 2\sqrt{5}$

$\Theta\ C_1C_2 = \sqrt{1^2 + 2^2} = \sqrt{5}$

$C_1C_2 = \sqrt{5} = r_2 - r_1$

Hence, both circles touch each other internally

Ques. If the circles $x^2 + y^2 - 16x - 20y + 164 = r^2$ and $(x - 4)^2 + (y - 7)^2 = 36$ intersect at two distinct points, then

(a) $1 < r < 11$ (b) $r > 11$

(c) $r = 11$ (d) $0 < r < 11$

Ans. $(x - 8)^2 + (y - 10)^2 = r^2$

$\therefore C_1[8, 10], r_1 = r$

$(x - 4)^2 + (y - 7)^2 = 6^2$

$\therefore C_2[4, 7], r_2 = 6$

$\because$ Both circles intersect at two distinct points

$\therefore |r_1 - r_2| < C_1C_2 < r_1 + r_2$

$\because C_1C_2 = \sqrt{4^2 + 3^2} = 5$

$\therefore |r - 6| < 5 < r + 6$

$\therefore |r - 6| < 5$ or $r + 6 > 5$

$\Rightarrow -5 < r - 6 < 5$ or $r + 1 > 0$ always true

$\Rightarrow 6 - 5 < r < 11$

$\Rightarrow 1 < r < 11$

Ques. The centres of those circles which touch the circle, $x^2 + y^2 - 8x - 8y - 4 = 0$, externally and also touch the x-axis, lie on

(a) A circle

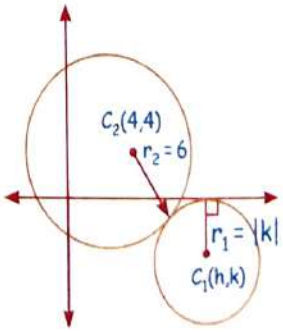
(b) An ellipse which is not a circle

(c) A hyperbola

(d) A parabola

Ans. $C_2 = (4,4), r = \sqrt{4^2 + 4^2 + 4} = \sqrt{36}$

$\Rightarrow r = 6$



$\because$ circles touches externally

$\therefore C_1C_2 = r_1 + r_2$

$\Rightarrow \sqrt{(h-4)^2 + (k-4)^2} = |k| + 6$

$\Rightarrow (h - 4)^2 + (k - 4)^2 = (|k| + 6)^2$

$\Rightarrow h^2 + 16 - 8h + k^2 + 16 - 8k = k^2 + 36 + 12|k|$

$\Rightarrow (h - 4)^2 = 20(k + 1)$ which is a parabola

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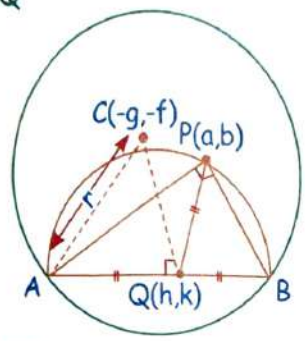
Brain Teaser

Find the locus of mid points of all the chords of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ which subtend a right angle at a given point (a, b)

Ans. $\because \angle APB = 90°$

$\therefore$ AB will be diameter of the circle having centre Q



In Δ CAQ

$(CA)^2 = g^2 + f^2 - C$

$(CQ)^2 = (h + g)^2 + (k + f)^2$

$(AQ)^2 = (PQ)^2 = (h - a)^2 + (k - b)^2$

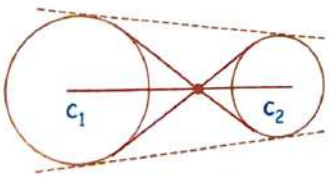
$\because (CA)^2 = (CQ)^2 + (AQ)^2$

$\Rightarrow g^2 + f^2 - c = (h + g)^2 + (k + f)^2 + (h - a)^2 + (k - b)^2$

NATURE OF 2 CIRCLE

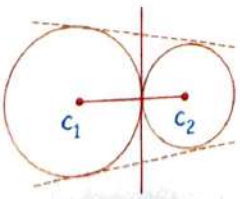
Let two circles having centre C_1 and C_2 and radii, r_1 and r_2 and C_1C_2 is the distance between their centres then:

CASE-I: WHEN CIRCLES ARE APART FROM EACH OTHER



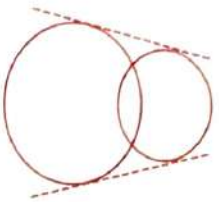
2 Direct common tangents + 2 transverse common langents = 4 common tangents

CASE-II: WHEN THE CIRCLES ARE TOUCHING EXTERNALLY



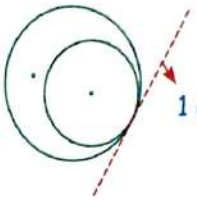
2 direct common tangents + 1 transverse common tangents = 3 common tangents

CASE-III: WHEN THE CIRCLES ARE INTERSECTING AT 2 POINTS



2 direct common tangents

CASE-IV: WHEN THE CIRCLES ARE TOUCHING INTERNALLY



1 direct common tangents

CASE-V: WHEN ONE CIRCLES COMPLETELY INSIDE THE OTHER

No common tangent

Ques. Let the centre of a circle, passing through the points (0, 0), (1, 0) and touching the circle $x^2 + y^2 = 9$, be (h, k). Then for all possible values of the coordinates of the centre (h, k), $4(h^2 + k^2)$ is equal to **[JEE Main 2024]**

Ans. $r = OC = 3 - r \Rightarrow 3 - r = r$

$r = \frac{3}{2}$

$\therefore \frac{9}{4} = h^2 + k^2$

$\Rightarrow 9 = 4(h^2 + k^2)$

$\therefore r^2 = (h - 0)^2 + (k - 0)^2 = (h - 1)^2 + k^2$

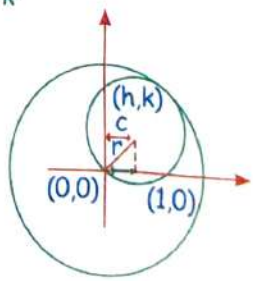
$\Rightarrow r^2 = h^2 + k^2 = (h - 1)^2 + k^2$

$\Rightarrow h^2 = h^2 + 1 - 2h$

$\Rightarrow h = \frac{1}{2}$

and $\left(\frac{1}{2}\right)^2 + k^2 = \frac{9}{4}$

$\Rightarrow k^2 = 2 \quad \Rightarrow k = \pm\sqrt{2}$



Hencer, possible values of (h,k) are

$\left(\frac{1}{2}, \sqrt{2}\right), \left(\frac{1}{2}, -\sqrt{2}\right)$

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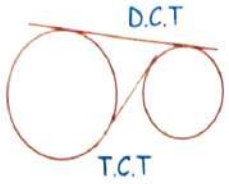
COMMON TANGENTS



Common Tangents

- Direct Common Tangents (D.C.T.) → Centres lie on same side of Tangent
- Transverse C.T. (T.C.T.) → Centres lie on opposite side of Tangent

Ques. The number of common tangents to the circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$ is **[JEE Main 2015]**

(a) 1 (b) 2 (c) 3 (d) 4

Ans. $C_1 [2, 3]$ | $C_2 [-3, -9]$

$r_1 = \sqrt{2^2 + 3^2 - (-12)}$ | $r_2 = \sqrt{9 + 81 - 26}$

$r_1 = \sqrt{13 + 12} = \sqrt{25} = 5$ | $r_2 = \sqrt{90 - 26} = 8$

$\therefore C_1C_2 = \sqrt{5^2 + 12^2} = 13$

and $r_1 + r_2 = 5 + 8 = 13$

$\because C_1C_2 = r_1 + r_2$

Hence, there will be 3 C.T.

Ques. The number of common tangents, to the circles $x^2 + y^2 - 18x - 15y + 131 = 0$ and $x^2 + y^2 - 6x - 6y - 7 = 0$, is **[JEE Main Apr. 2023-I]**

(a) 3 (b) 1 (c) 4 (d) 2

Ans. $C_1 \left(9, \frac{15}{2}\right)$ | $C_2 [3, 3]$

$r_1 = \sqrt{9^2 + \left(\frac{15}{2}\right)^2 - 131}$ | $r_2 = \sqrt{3^2 + 3^2 + 7}$

$= \sqrt{\frac{324 + 225 - 524}{4}}$ | $= \sqrt{25} = 5$

$= \sqrt{\frac{549 - 524}{4}} = \frac{5}{2}$

$\because C_1C_2 = \sqrt{6^2 + \left(\frac{15}{2} - 3\right)^2} = \sqrt{6^2 + \left(\frac{9}{2}\right)^2} = \frac{15}{2}$

and $r_1 + r_2 = 5 + \frac{15}{2} = \frac{15}{2}$

$\because C_1C_2 = r_1 + r_2$

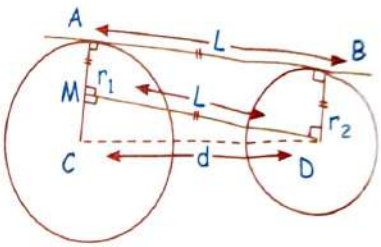
So, there will be 3 common tangents

Length of Direct Common Tangent

If length of D.C.T is L then $\Rightarrow L^2 = d^2 - (r_2 - r_1)^2$

$d \rightarrow$ distance b/w centres

Proof:



ΔCDM

$CM = r_1 - r_2$

$DM = L$

$CD = d$

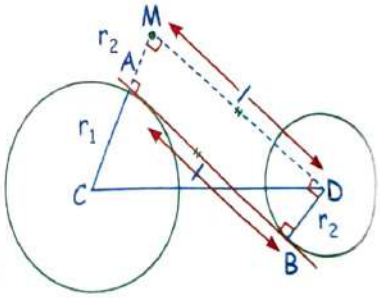
$d^2 = L^2 + (r_1 - r_2)^2$

$\Rightarrow L^2 = d^2 - (r_1 - r_2)^2$

Length of Transverse Common Tangent

If length of T.C.T is l then $\rightarrow l^2 = d^2 - (r_1 + r_2)^2$

Proof: Δ CDM



$CM = r_1 + r_2$

$CD = d$

$DM = l$

$d^2 = l^2 + (r_1 + r_2)^2$

$l^2 = d^2 - (r_1 + r_2)^2$

Ques. Find the sum of the lengths of all the common tangents to the circles

$x^2 + y^2 + 14x - 4y + 28 = 0$

$x^2 + y^2 - 14x + 4y - 28 = 0$

Ans.

$C_1 = (-7, 2), r_1 = \sqrt{7^2 + 2^2 - 28} = \sqrt{49 - 24} = \sqrt{25} = 5$

$C_2 = (7, -2), r_2 = \sqrt{7^2 + 2^2 + 28} = \sqrt{49 + 32} = 9$

$d = \sqrt{14^2 + 4^2} = \sqrt{196 + 16} = \sqrt{212}$

$\therefore L^2 = d^2 - (r_1 - r_2)^2$ | $l^2 = d^2 - (r_1 + r_2)^2$

$L^2 = 212 - 4^2 = 196$ | $l^2 = 212 - (14)^2 = 16$

$\Rightarrow L = 14$ | $l = 4$

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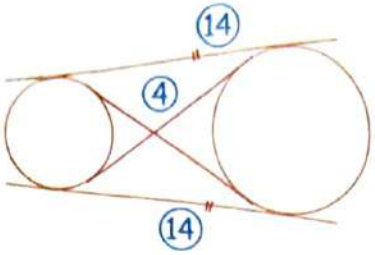


$\therefore$ the sum of the lengths = 2 (14 + 4) = 36

Finding Equation Of D.C.T & T.C.T

Ques. Determine the equations of all the common tangents to the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ and $x^2 + y^2 + 6x - 2y + 1 = 0$.

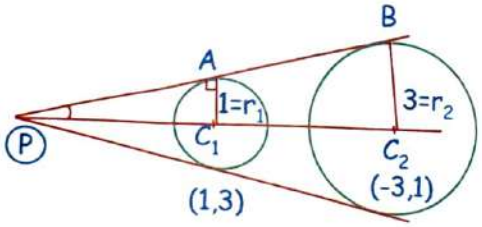
Ans. C_1 [1, 3] | C_2 [-3, 1]

$r_1 = \sqrt{1^2 + 3^2 - 9} = 1$ | $r_2 = \sqrt{3^2 + 1^2 - 1} = 3$

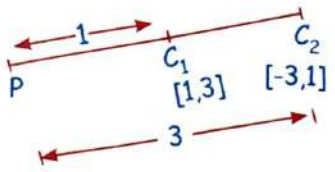
$C_1C_2 = \sqrt{4^2 + 2^2} = 2\sqrt{5}$

$r_1 + r_2 = 4$

$C_1C_2 > r_1 + r_2$



$\Delta PAC_1 \sim \Delta PBC_2$



$\Rightarrow \frac{PC_1}{PC_2} = \frac{r_1}{r_2} = \frac{1}{3}$

$P\left[\frac{1\times 3 - 1\times(-3)}{3-1}, \frac{3\times 3 - 1\times 1}{3-1}\right]$

P [3, 4]

Let slope of D.C.T be m

y - 4 = m (x - 3)

y = mx + 4 - 3m

$\because C_1A \perp PA$

and $C_1A = 1$

$\therefore \left|\frac{1\times m + 4 - 3m - 3}{\sqrt{1+m^2}}\right| = 1$

$\Rightarrow \left|\frac{(1-2m)}{\sqrt{1+m^2}}\right| = 1$

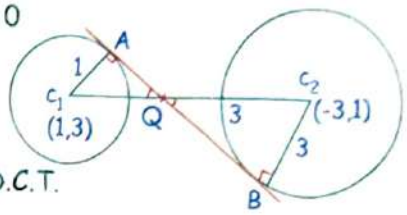
$\Rightarrow (1 - 2m)^2 = 1 + m^2$

$\Rightarrow 1 + 4m^2 - 4m = 1 + m^2$

$\Rightarrow 3m^2 - 4m = 0$

$\Rightarrow m (3m - 4) = 0$

$\Rightarrow m = 0, \frac{4}{3}$



Hence, Eqn of D.C.T.

$y = 4$ or $y = \frac{4}{3}x$

$\Delta AC_1Q \sim \Delta BC_2Q$

for T.C.T

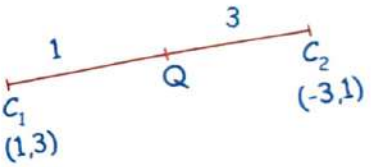
$\frac{C_1Q}{C_2Q} = \frac{r_1}{r_2} = \frac{1}{3}$

$\therefore Q = \left[\frac{1\times 3 + 1\times(-3)}{3+1}, \frac{3\times 3 + 1\times 1}{3+1}\right]$

$\Rightarrow Q = \left[0, \frac{5}{2}\right]$

Let slope of T.C.T be 'm'

$\therefore$ eqn of T.C.T



$y - \frac{5}{2} = m(x - 0)$

$\Rightarrow y = mx + \frac{5}{2}$

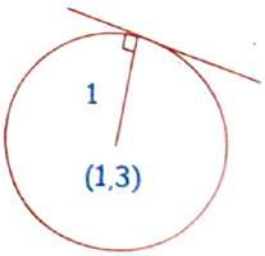
$\because C_1A \perp AQ$ and $C_1A = 1$

$\therefore \left|\frac{m - 3 + \frac{5}{2}}{\sqrt{1+m^2}}\right| = 1$

$\Rightarrow (m - \frac{1}{2})^2 = m^2 + 1$

$\Rightarrow m^2 + \frac{1}{4} - m = m^2 + 1$

$\Rightarrow m = \frac{-3}{4}$ & $m = \infty$



$\therefore$ eqn of T.C.T:

$y = \frac{-3}{4}x + \frac{5}{2}$ or $x = 0$

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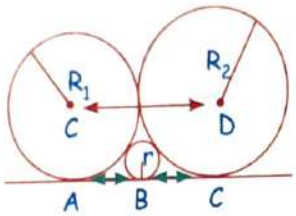
Brain Teaser

In the figure shown r is the radius of the smallest circle, R_1 and R_2 are the radii of the other two circles. Prove that $r = \dfrac{R_1R_2}{\left(\sqrt{R_1}+\sqrt{R_2}\right)^2}$.



Ans. $d = CD = R_1 + R_2$

$(AC)^2 = d^2 - (R_1 - R_2)^2$

$\Rightarrow (AC)^2 = (R_1 + R_2)^2 - (R_1 - R_2)^2$

$\Rightarrow (AC)^2 = 4R_1R_2$

$\Rightarrow AC = 2\sqrt{R_1R_2}$

Similarly $AB = 2\sqrt{R_1r}$

and $BC = 2\sqrt{R_2r}$

$\because AC = AB + BC$

$\Rightarrow 2\sqrt{R_1R_2} = 2\sqrt{R_1r} + 2\sqrt{R_2r}$

$\Rightarrow \sqrt{R_1R_2} = \sqrt{r}\left[\sqrt{R_1}+\sqrt{R_2}\right]$

$\Rightarrow \sqrt{r} = \dfrac{\sqrt{R_1R_2}}{\sqrt{R_1}+\sqrt{R_2}}$

$\Rightarrow r = \dfrac{R_1R_2}{\left(\sqrt{R_1}+\sqrt{R_2}\right)^2}$

Ques. Let the point B be the reflection of the point A(2, 3) with respect to the line $8x - 6y - 23 = 0$. Let Γ_A and Γ_B be circles of radii 2 and 1 with centers A and B respectively. Let T be a common tangents to the circles Γ_A and Γ_B such that both the circles are on the same side of T. If C is the point of intersection of T and the line passing through A and B, then the length of the segment AC is **[JEE Adv. 2019]**

Ans. $\Delta APC \sim \Delta BQC$

$\dfrac{AC}{BC} = \dfrac{2}{1}$

$AC = 2BC$

Let $BC = x \Rightarrow AB = x$

$\therefore AC = 2x$

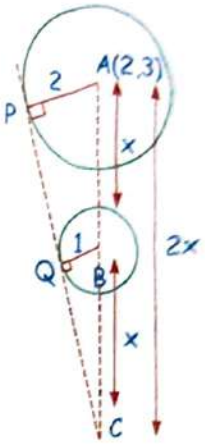
$\Rightarrow$ distance of A (2, 3) from mirror $= \dfrac{x}{2}$

$\Rightarrow \left|\dfrac{8\times2 - 6\times3 - 23}{\sqrt{8^2+6^2}}\right| = \dfrac{x}{2}$

$\Rightarrow \dfrac{25}{10} = \dfrac{x}{2} \Rightarrow x = 5$

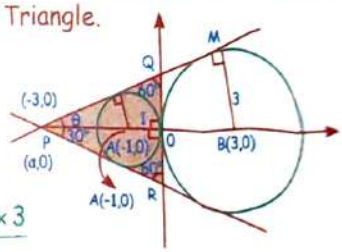
$\therefore AC = 2x = 10$

$\Rightarrow AC = 10$ units



Ques. Show that the triangle formed by the common tangents to $x^2 + y^2 + 2x = 0$ and $x^2 + y^2 - 6x = 0$ is an Equilateral Triangle.

Ans. $\Delta PAL \sim \Delta PBM$



$\dfrac{PA}{PB} = \dfrac{1}{3}$

$\therefore -1 = \dfrac{\alpha\times2 + 1\times3}{1+2}$

$\Rightarrow -3 = 2\alpha + 3$

$\Rightarrow -6 = 2\alpha \Rightarrow \alpha = -3$

$AL = 1$

$PA = 2$

In ΔPAL

$\sin\theta = \dfrac{AL}{PA} = \dfrac{1}{2} \Rightarrow \theta = 30°$

$\angle QPR = 60°$ and $\angle PQR = \angle QRP = 60°$

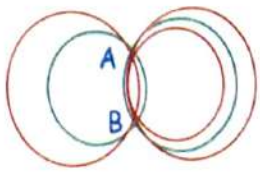
Hence, ΔPQR is eqⁿ an equilateral triangle.

FAMILY OF CIRCLES

CASE-I: Family of circles passing through point of intersection of 2 given circles.

$S_1 + \lambda S_2 = 0$

$\lambda \in R - \{-1\}$



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Example: The equation of the circle passing through the points of intersection of the circles $x^2 + y^2 - 8x - 2y + 7 = 0$ and $x^2 + y^2 - 4x + 10y + 8 = 0$, & whose center lies on y-axis.

Let Circle be $x^2 + y^2 - 8x - 2y + 7 + \lambda [x^2 + y^2 - 4x + 10y + 8] = 0$

$\Rightarrow x^2 (1 + \lambda) + y^2 (1 + \lambda) - x (8 + 4\lambda) + y (-2 + 10\lambda) + 7 + 8\lambda = 0$

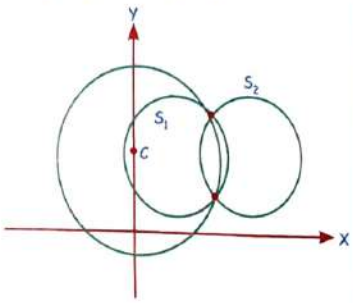
$$x^2 + y^2 - x\frac{(8+4\lambda)}{\lambda+1} + y\frac{(-2+10\lambda)}{\lambda+1} + \frac{7+8\lambda}{\lambda+1} = 0$$

$X_c = 0$

$$\Rightarrow \frac{8+4\lambda}{\lambda+1} = 0$$

$\Rightarrow 8 + 4\lambda = 0$

$\Rightarrow \lambda = -2$



$\therefore x^2 + y^2 - 8x - 2y + 7 - 2(x^2 + y^2 - 4x + 10y + 8) = 0$

$\Rightarrow x^2 + y^2 + 22y + 9 = 0$

Ques. The equation of the circle through the points of intersection of $x^2 + y^2 - 1 = 0$, $x^2 + y^2 - 2x - 4y + 1 = 0$ and touching the line $x + 2y = 0$, is

(a) $x^2 + y^2 + x + 2y = 0$

(b) $x^2 + y^2 - x + 20 = 0$

(c) $x^2 + y^2 - x - 2y = 0$

(d) $2(x^2 + y^2) - x - 2y = 0$

Ans. $x^2 + y^2 - 2x - 4y + 1 + \lambda (x^2 + y^2 - 1) = 0$

$\Rightarrow x^2 (1 + \lambda) + y^2 (1 + \lambda) - 2x - 4y + 1 - \lambda = 0$

$$\Rightarrow x^2 + y^2 - \frac{2x}{1+\lambda} - \frac{4y}{1+\lambda} + \frac{1-\lambda}{1+\lambda} = 0$$

$$\therefore C\left[\frac{1}{1+\lambda}, \frac{2}{1+\lambda}\right]$$

and $r = \sqrt{\left(\frac{1}{1+\lambda}\right)^2 + \frac{4}{(1+\lambda)^2} - \left(\frac{1-\lambda}{1+\lambda}\right)}$

$$= \sqrt{\frac{5-(1-\lambda)(1+\lambda)}{(1+\lambda)^2}} = \sqrt{\frac{4+\lambda^2}{(1+\lambda)^2}}$$

$\because$ This circle touches the line $x + 2y = 0$

$$\therefore \left|\frac{\frac{1}{1+\lambda} + \frac{2\times 2}{1+\lambda}}{\sqrt{5}}\right| = r$$

$$\Rightarrow \left|\frac{5}{\sqrt{5}(1+\lambda)}\right| = r$$

$$\Rightarrow \left|\frac{\sqrt{5}}{1+\lambda}\right| = \sqrt{\frac{4+\lambda^2}{(1+\lambda)^2}}$$

$\Rightarrow \sqrt{5} = \sqrt{4 + \lambda^2} \Rightarrow 5 = 4 + \lambda^2$

$\Rightarrow \lambda^2 = 1 \Rightarrow \lambda = \pm 1$

$\Rightarrow \lambda = -1$

(Reject because it will become line)

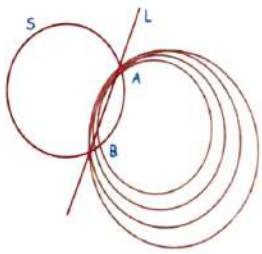
$\therefore \lambda = 1$

$$\Rightarrow x^2 + y^2 - \frac{2x}{2} - \frac{4}{2}y + 0 = 0$$

$\Rightarrow x^2 + y^2 - x - 2y = 0$

CASE-II: Family of circles passing though point of intersection of a given line and a given circle is:

$S + \lambda L = 0, \lambda \in R$



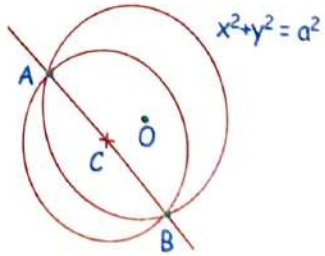
Ques. Given the circle $x^2 + y^2 = a^2$ and the line $x \cos\alpha + y \sin\alpha = p$ intersect at 2 district points A & B. Find the equation of circle with AB as diameter.

Ans. eqⁿ of Circle be: $S + \lambda L = 0$

$x^2 + y^2 - a^2 + \lambda (x \cos\alpha + y \sin\alpha - p) = 0$

$\Rightarrow x^2 + y^2 + (\lambda \cos\alpha) x + (\lambda \sin\alpha) y - a^2 - \lambda P = 0$

$$\therefore \text{Centre} \left[-\frac{\lambda\cos\alpha}{2}, -\frac{\lambda\sin\alpha}{2}\right]$$



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$\because$ centre will lie on the line

$\therefore \frac{-\lambda\cos\alpha}{2}\cdot\cos\alpha - \frac{\lambda\sin\alpha}{2}\cdot\sin\alpha = P$

$\Rightarrow \frac{-\lambda}{2}\left[\cos^2\alpha + \sin^2\alpha\right] = P$

$\Rightarrow \lambda = -2P$

Ques. The circle C: $x^2 + y^2 + kx + (1+k)y - (k+1) = 0$ passes through two fixed points for every real number k.

(i) Find the coordinates of these two points

(ii) Find the minimum value of the radius of a circle C.

Ans. C: $x^2 + y^2 + y - 1 + k(x + y - 1) = 0$

$\therefore$ S: $x^2 + y^2 + y - 1 = 0$ and L : $x + y = 1$

put $x = 1 - y$

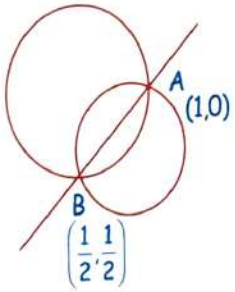
$\therefore (1-y)^2 + y^2 + y - 1 = 0$

$\Rightarrow 1 + y^2 - 2y + y^2 + y - 1 = 0$

$\Rightarrow 2y^2 - y = 0$

$\Rightarrow y(2y - 1) = 0$

$y = 0$, $x = 1$ or $y = \frac{1}{2}$, $x = \frac{1}{2}$



$\therefore A(1, 0)$ & $B\left(\frac{1}{2}, \frac{1}{2}\right)$

Min^m value of Radius $= \frac{AB}{2}$

$\because (AB)^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2$

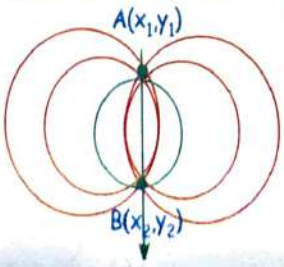
$\Rightarrow AB = \frac{1}{\sqrt{2}}$

CASE-III: Family of circles passing through 2 given points (x_1, y_1) and (x_2, y_2).

L: 2 point Form

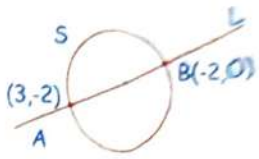
S: Diametric Form

then, $\boxed{S + \lambda L = 0}$, $\lambda \in R$



Ques. Find the equation of the circle passing through (3, -2) and (-2, 0) and having its centre on the line $2x - y = 3$.

Ans. $M = \frac{-2-0}{3+2} = \frac{-2}{5}$



$y - 0 = \frac{-2}{5}(x + 2)$

$\Rightarrow 5y = -2x - 4$

$\Rightarrow 2x + 5y + 4 = 0$

S: $(x - 3)(x + 2) + (y + 2)(y - 0)$

S: $x^2 + y^2 - x + 2y - 6$

L: $2x + 5y + 4$

Now we will write family $S + \lambda L = 0$

$\Rightarrow (x^2 + y^2 - x + 2y - 6) + \lambda(2x + 5y + 4) = 0$

$\Rightarrow x^2 + y^2 + (2\lambda - 1)x + y(5\lambda + 2) + 4\lambda - 6 = 0$

Centre $\left(\frac{1-2\lambda}{2}, \frac{-(5\lambda+2)}{2}\right)$

$\because$ centre lie on the line $2x - y = 3$

$\therefore \frac{2(1-2\lambda)}{2} + \frac{5\lambda+2}{2} = 3$

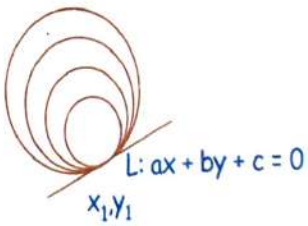
$\Rightarrow 2(1 - 2\lambda) + 5\lambda + 2 = 6$

$\Rightarrow 2 - 4\lambda + 5\lambda + 2 = 6$

$\Rightarrow \lambda = 2$

Hence, Eqn of the circle: $x^2 + y^2 + 3x + 12y + 2 = 0$

CASE-IV: Family of circles touching a given line L at a given point (x_1, y_1).



S: $(x - x_1)^2 + (y - y_1)^2$

L: $ax + by + c$

Family $\boxed{S + \lambda L = 0}$

Ques. Find the equation of a circle passing through Q(1, 2) and touching $2x + y = 5$ at P (2, 1).

Ans. **Method 1:**

S: $(x - 2)^2 + (y - 1)^2 = 0$

L: $2x + y - 5 = 0$

Family: $(x - 2)^2 + (y - 1)^2 + \lambda(2x + y - 5) = 0$

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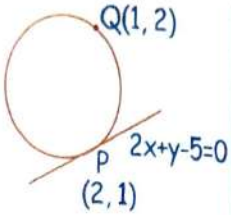
(1, 2) satisfies



$1^2 + 1^2 + \lambda(2 \times 1 + 2 - 5) = 0$

$\Rightarrow 2 + \lambda(-1) = 0$

$\Rightarrow \lambda = 2$

$\therefore (x-2)^2 + (y-2)^2 + 2(2x - y - 5) = 0$

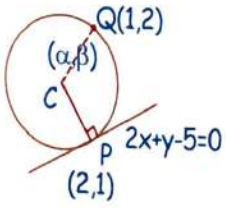
$\Rightarrow x^2 + y^2 - 2y - 2 = 0$

Method 2:

Let $C[\alpha, \beta]$

$(\alpha - 2)^2 + (\beta - 1)^2 = (\alpha - 1)^2 + (\beta - 2)^2 \rightarrow 1$ ...(i)

$M_{pc} = \frac{\beta - 1}{\alpha - 2}$



$M_T = -2 = \frac{-1}{M_{pc}}$

$\therefore M_{pc} = \frac{1}{2} = \frac{\beta - 1}{\alpha - 2}$ ...(iii)

calculate (α, β) and find the eqⁿ of circle.

Ques. A circle touching the x-axis at (3, 0) and making an intercept of length 8 on the y-axis passes through the point: **[JEE Main April 2019-II)]**

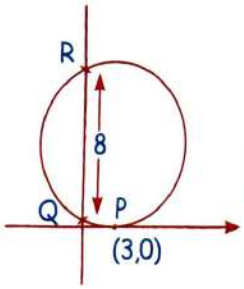
~~(a)~~ (3, 10) (b) (3, 5) (c) (2, 3) (d) (1, 5)

Ans. Method - 1

S: $(x - 3)^2 + (y - 0)^2 = 0$

L: $y = 0$

Family:



$(x - 3)^2 + y^2 + \lambda y = 0$

$x^2 + y^2 - 6x + \lambda y + 9 = 0$

$\because 2\sqrt{f^2 - c} = 8$

$\Rightarrow f^2 - c = 16$

$\because 2f = \lambda \Rightarrow f = \frac{\lambda}{2}$

$\therefore c = 9$

$\Rightarrow \left(\frac{\lambda}{2}\right)^2 - 9 = 16 \Rightarrow \left(\frac{\lambda}{2}\right)^2 = 25$

$\frac{\lambda}{2} = \pm 5 \Rightarrow \lambda = \pm 10$

$\therefore$ Eqⁿ of circle:

$x^2 + y^2 - 6x \pm 10y + 9 = 0$

$\therefore$ (3, 10) satisfy the eqⁿ

Method - 2

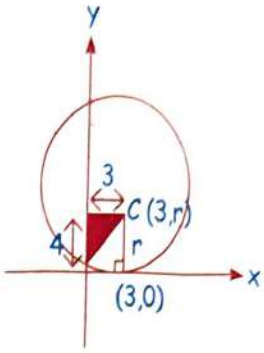
$r = \sqrt{3^2 + 4^2} = 5$

$\therefore C[3, 5]$ or $[3, -5]$

Eqⁿ of circle

$(x - 3)^2 + (y \pm 5)^2 = 5^2$

(3, 10) satisfy

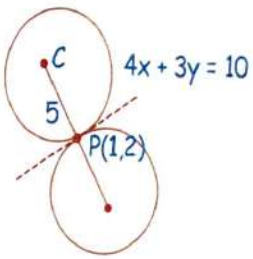


Brain Teaser

Two circles each of radius 5, touch other at (1, 2). If $4x + 3y = 10$ is common tangent, find the equations of the circles

Ans. $CP = 5 = r$

$\therefore x = 1 \pm 5\cos\theta$ and $y = 2 \pm 5\sin\theta$



$\because m = \frac{-4}{3}$

$\therefore m_{PC} = \frac{3}{4} = \tan\theta$

$\sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

So, $x = 1 \pm \frac{5 \times 4}{5} = 1 \pm 4$

and $y = 2 \pm \frac{5 \times 3}{5} = 2 \pm 3$

(5, 5) or (-3, -1)

$\therefore$ Equation of circles are -

$(x - 5)^2 + (y - 5)^2 = 5^2$

or $(x + 3)^2 + (y + 1)^2 = 5^2$

CASE-V: Equation of circle circumscribing a triangle whose sides are given

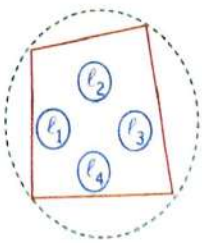
Family of curves circumscribing a triangle whose sides are given by $L_1 = 0$; $L_2 = 0$ and $L_3 = 0$ is given by;

$$L_1 L_2 + \lambda L_2 L_3 + \mu L_3 L_1 = 0$$

To Get λ & μ put coefficient of $xy = 0$

and coefficient of x^2 = coefficient of y^2.

CASE-VI: Equation of circle circumscribing a quadrilateral



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Equation of circle circumscribing a quadrilateral whose side in order are represented by the lines $L_1 = 0, L_2 = 0, L_3 = 0$ & $L_4 = 0$ are $\boxed{L_1 L_3 + \lambda L_2 L_4 = 0}$

where value of λ can be found out by using condition that coefficient of x^2 = coefficient of $y^2 \neq 0$ and coefficient of $xy = 0$.

Ques. Find the equation of Circumcircle of the Quadrilateral formed by the lines the order $2x + 3y = 2, 3x - 2y = 3, x + 2y = 3,$ & $2x - y = 1$

Ans. $(2x + 3y - 2)(x + 2y - 3) + \lambda (3x - 2y - 3)(2x - y - 1) = 0$

coeff of x^2: $2 + 6\lambda$
Coeff of y^2: $6 + 2\lambda$
→ equate: $2 + 6\lambda = 6 + 2\lambda \Rightarrow 4\lambda = 4 \Rightarrow \lambda = 1$

Coeff of xy: $7 - 7\lambda = 0$

$\lambda = 1$

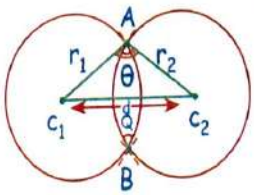
$\therefore (2x + 3y - 2)(x + 2y - 3) + (3x - 2y - 3)(2x - y - 1 = 0)$

ANGLE BETWEEN 2 CIRCLES

Angle between 2 curves is defined as angle between tangents at their Point of intersection.

Angle between 2 circles is same at both points A & B.

In ΔAC_1C_2 apply cosine rule



$$\boxed{\text{Cos}\theta = \left|\frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2}\right|}$$

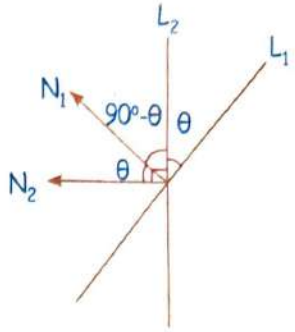
where $r_1 \rightarrow$ Radius of circle C_1

$r_2 \rightarrow$ Radius of circle C_2

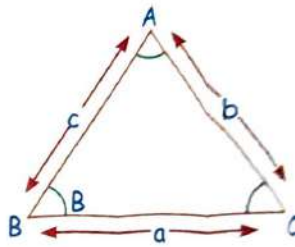
$d \rightarrow$ dis between centers

If 2 Curves dont intersect, then angle b/w them is not define.

"Angle b/w 2 Line is same as angle b/w their normals"



COSINE RULE



$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Ques. Find the angle between the circles $x^2 + y^2 = 16$ and $x^2 + y^2 - 4x - 4y - 8 = 0$

Ans. $\because \cos\theta = \left|\left(\frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2}\right)\right|$

$C_1[0,0]$ and $C_2[2,2]$

$r_1 = 4$ and $r_2 = \sqrt{2^2 + 2^2 - (-8)} = 4$

$d = C_1C_2 = \sqrt{2^2 + 2^2} = \sqrt{8} \Rightarrow d^2 = 8$

$\therefore \cos\theta = \left|\frac{4^2 + 4^2 - 8}{2 \times 4 \times 4}\right| = \frac{3}{4}$

$\Rightarrow \cos\theta = \frac{3}{4}$

ORTHOGONAL CIRCLES

"Perpendicular" $\Rightarrow \theta = 90°$

Definition: Two circles are said to be orthogonal if angle between them is 90°.

$\because \cos\theta = \left|\frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2}\right|$

and $\theta = 90° \Rightarrow r_1^2 + r_2^2 = d^2 \rightarrow$ For orthogonal circles

$\Rightarrow (g_1^2 + f_1^2 - c_1) + (g_2^2 + f_2^2 - c_2) = (g_1 - g_2)^2 + (f_1 - f_2)^2$

$\Rightarrow g_1^2 + f_1^2 - c_1 + g_2^2 + f_2^2 - c_2 = g_1^2 + g_2^2 - 2g_1g_2 + f_1^2 + f_2^2 - 2f_1f_2$

$\Rightarrow \boxed{2g_1g_2 + 2f_1f_2 = c_1 + c_2} \rightarrow$ Condition for orthogonality

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Ques. Two given circles $x^2 + y^2 + ax + by + c = 0$ and $x^2 + y^2 + dx + ey + f = 0$ will intersect each orthogonally, only when

(a) $a + b + c = d + e + f$

(b) $ad + be = c + f$

(c) $ad + be = 2c + 2f$

(d) $2ad + 2be = c + f$

Ans. $g_1 \to \frac{a}{2}$ | $g_2 \to \frac{d}{2}$

$f_1 \to \frac{b}{2}$ | $f_2 \to \frac{e}{2}$

$C_1 \to c$ | $C_2 \to f$

Condition for orthogonality

$2g_1 g_2 + 2f_1 f_2 = c_1 + c_2$

$\Rightarrow 2\frac{a}{2}\frac{d}{2} + 2.\frac{b}{2}.\frac{e}{2} = c + f$

$\Rightarrow ad + be = 2c + 2f$

Ques. Find the equation of the circle through the points of intersection of the circles $x^2 + y^2 - 4x - 6x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ and cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

Ans. $S_1 + \lambda S_2 = 0$

$\Rightarrow x^2 + y^2 - 4x - 6y - 12 + \lambda(x^2 + y^2 + 6x + 4y - 12) = 0$

$\Rightarrow (1 + \lambda) x^2 + (1 + \lambda) y^2 + (6\lambda - 4) x + (4\lambda - 6) y - 12\lambda - 12 = 0$

$\Rightarrow x^2 + y^2 + \left(\frac{6\lambda - 4}{1 + \lambda}\right)x + \left(\frac{4\lambda - 6}{\lambda + 1}\right)y - \frac{12(\lambda + 1)}{(\lambda + 1)} = 0$

$\because g_1 = \frac{3\lambda - 2}{1 + \lambda}, f_1 = \frac{2\lambda - 3}{\lambda + 1}, C_1 = -12$

$g_2 = -1, \quad f_2 = 0, \quad C_2 = -4$

Apply condition for orthogonality

$\Rightarrow \frac{2.(3\lambda - 2)}{\lambda + 1}(-1) = -12 - 4$

$\Rightarrow \frac{2.(3\lambda - 2)}{\lambda + 1} = 16$

$\Rightarrow 3\lambda - 2 = 8(\lambda + 1)$

$\Rightarrow 3\lambda - 2 = 8\lambda + 8$

$\Rightarrow 5\lambda = -10$

$\Rightarrow \lambda = -2$

Hence, eqn of the circle:

$x^2 + y^2 + 16x + 14y - 12 = 0$

Ques. The locus of centre of a circle passing through (a, b) and cuts orthogonally to circle $x^2 + y^2 = p^2$, is [IIT 1988]

(a) $2ax + 2by - (a^2 + b^2 + p^2) = 0$

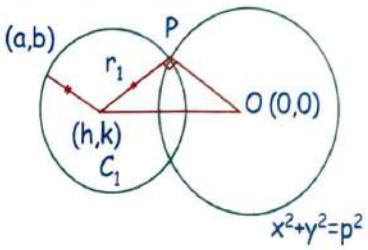
(b) $2ax + 2by - (a^2 - b^2 + p^2) = 0$

(c) $x^2 + y^2 - 3ax - 4by + (a^2 + b^2 - p^2) = 0$

(d) $x^2 + y^2 - 2ax - 3by + (a^2 - b^2 - p^2) = 0$

Ans. C_1PO is a ritht angled triangle

$\therefore h^2 + k^2 = r_1^2 + p^2$



$\Rightarrow h^2 + k^2 = (h - a)^2 + (k - b)^2 + p^2$

$\Rightarrow 0 = -2ah + a^2 + b^2 - 2bk + p^2$

$\Rightarrow 2ah + 2bk = a^2 + b^2 + p^2$

$(h, k) \to (x, y)$

$\Rightarrow 2ax + 2by = a^2 + b^2 + p^2$

Brain Teaser

C_1 and C_2 are circles such that

(i) they both pass through (0, a) and (0, -a)

(ii) they both touch $y = mx + c$;

(iii) they cut orthogonally.

Prove that $c^2 = a^2(2 + m^2)$

Ans. Let circle be $x^2 + y^2 + 2gx + 2fy + c' = 0$

(i) (0, a) satfisfes

$\therefore 0 + a^2 + 2fa + c' = 0$

$\Rightarrow a^2 + 2fa + c' = 0$ ...(i)

$\because$ (0, - a) satisfies

$\therefore a^2 - 2fa + c' = 0$ ...(ii)

eqn (i) - eqn (ii)

$4fa = 0$

$\because a \neq 0, \quad \therefore f = 0$

then $a^2 + c' = 0 \quad \Rightarrow c' = -a^2$

$\therefore$ eqn of Circle is $x^2 + y^2 + 2gx - a^2 = 0$

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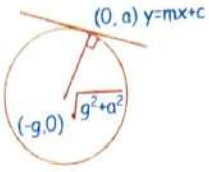
(ii) $\left|\frac{-mg+c}{\sqrt{1+m}}\right| = \sqrt{g^2+a^2}$



$\Rightarrow (-mg+c)^2 = (g^2+a^2)(1+m^2)$

$\Rightarrow m^2g^2 + c^2 - 2mgc = g^2 + a^2 + g^2m^2 + a^2m^2$

$\Rightarrow g^2 + 2mc\,g + (a^2 + a^2m^2 - c^2) = 0$

$\therefore g_1g_2 = a^2 + a^2m^2 - c^2$

Let g_1 and g_2 are roots of the quadratic eq^n

Thus, $x^2 + y^2 + 2g_1x - a^2 = 0$

or $x^2 + y^2 + 2g_2x - a^2 = 0$

(iii) $\because C_1$ and C_2 cuts orthogonally

$\therefore 2g_1g_2 + 2f_1f_2 = C_1 + C_2$

$2g_1g_2 = -a^2 - a^2$

$\Rightarrow g_1g_2 = -a^2$

So, $a^2 + a^2m^2 - c^2 = -a^2$

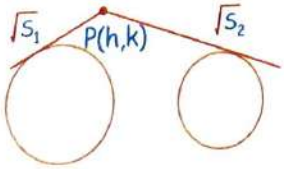
$\Rightarrow c^2 = 2a^2 + a^2m^2 \Rightarrow c^2 = a^2(m^2+2)$

RADICAL AXIS

Definition: The locus of a point, which moves in such a way that its power w.r.t the two given circles is same, is called the radical axis.

$\Rightarrow 2x(g_1 - g_2) + 2y(f_1 - f_2) + c_1c_2 = 0$

$S_1 - S_2 = 0$, is the equation of radical axis.



$\because \sqrt{S_1} = \sqrt{S_2}$

$\Rightarrow h^2 + k^2 + 2g_1h + 2f_1k + C_1 = h^2 + k^2 + 2g_2h + 2f_2k + c_2$

$\Rightarrow 2(g_1 - g_1)h + 2(f_1 - f_2)k + c_1 - c_2 = 0$

$(h, k) \to (x, y)$

$2(g_1 - g_2)x + 2(f_1 - f_2)y + C_1 - C_2 = 0 \to$ **Radical Axis**

$\Rightarrow \boxed{S_1 - S_2 = 0}$

Ques. Equation of radical axis of the circles $x^2 + y^2 - 3x - 4y + 5 = 0$, $2x^2 + 2y^2 - 10x - 12y + 12 = 0$ is

(a) $2x + 2y - 1 = 0$

(b) $2x + 2y + 1 = 0$

(c) $x + y - 7 = 0$

(d) $x + y - 7 = 0$

Ans. S_1: $x^2 + y^2 - 3x - 4y + 5 = 0$

S_2: $x^2 + y^2 - 5x - 6y + 6 = 0$

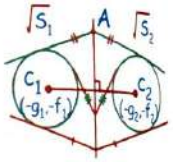
$- \quad - \quad + \quad + \quad -$

$S_1 - S_2$: $2x + 2y - 1 = 0$

$\Rightarrow x + y = \frac{1}{2}$

PROPERTIES OF RADICAL AXIS

PROPERTY-1: The length of two tangents drawn from a point on radical axis is always equal. (if defined).



$\sqrt{S_1} = \sqrt{S_2}$

PROPERTY-2: Line joining the centers of the two circles is always perpendicular to their Radical axis.

Proof: $m_{RA} = \frac{-2(g_1 - g_2)}{2(f_1 - f_2)}$ and $m_{C_1C_2} = \frac{f_1 - f_2}{g_1 - g_2}$

$m_{RA}\, m_{C_1C_2} = -1$

PROPERTY-3: For externally touching circles radical axis is same as tranverse common tangent

TCT = RA

Proof:

Consider a point A on T.C.T.

To Prove AM = AN

$\because$ AT = AN

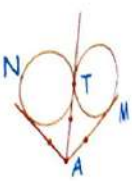
also, AT = AM

$\Rightarrow$ AM = AN

$\Rightarrow \sqrt{S_1} = \sqrt{S_2}$

$\Rightarrow S_1 = S_2$

$\therefore$ Point A lies on R.A also



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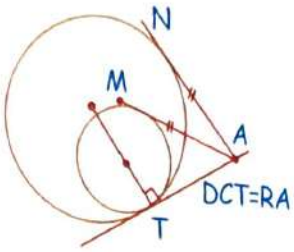
PROPERTY-4: For internally touching circles radical axis is same as direct common tangent.

Proof: To prove AM = AN



AT = AN

AT = AM

$\Rightarrow$ AM = AN

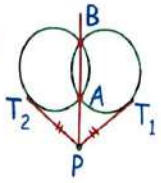
$\Rightarrow$ A lies on R.M also

PROPERTY-5: For intersecting circles radical axis is same as line along common chord.

Proof: Let P is a Point on common chord

To prove $PT_1 = PT_2$

"Using Tangent Secant theorem"



$PT_2^2 = PA.PB$ ...(i)

$PT_1^2 = PA.PB$ ...(ii)

$\therefore PT_1 = PT_2$

Ques. The length of the common chord of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$ is **[MP PET 2000]**

(a) $\frac{9}{2}$ (b) $2\sqrt{2}$ (c) $3\sqrt{2}$ (d) $\frac{3}{2}$

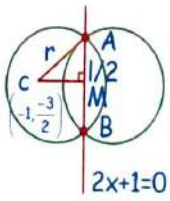
Ans. $r^2 = 1^2 + \left(\frac{3}{2}\right)^2 - 1$

$\Rightarrow r^2 = \left(\frac{3}{2}\right)^2$

Eqn of Common Chord:

$-2x - 1 = 0 \Rightarrow 2x + 1 = 0$

$\therefore CM = \left|\frac{2\times(-1)+1}{\sqrt{2^2+0^2}}\right| = \frac{1}{2}$



$\because r^2 = (CM)^2 + \left(\frac{L}{2}\right)^2$

$\Rightarrow \left(\frac{3}{2}\right)^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{L}{2}\right)^2$

$\Rightarrow 9 = 1 + L^2 \Rightarrow L^2 = 8 \Rightarrow L = 2\sqrt{2}$

Ques. Find the equation of the circle on the common chord of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$ as a diameter.

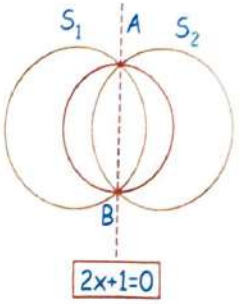
Ans. AB: 2x + 1 = 0

Family of Circles passing through A & B is: $S + \lambda L = 0$

$x^2 + y^2 + 2x + 3y + 1 + \lambda(2x + 1) = 0$

$\Rightarrow x^2 + y^2 + 2x(\lambda + 1) + 3y + \lambda + 1 = 0$

Center $\left[-(\lambda+1), -\frac{3}{2}\right]$



$\because$ Centre lies on AB

$\therefore -2(\lambda + 1) + 1 = 0$

$\Rightarrow -2\lambda - 2 + 1 = 0$

$\Rightarrow 2\lambda = -1$

$\Rightarrow \lambda = \frac{-1}{2}$

$\therefore$ Eqn of the circle:

$x^2 + y^2 + 2x + 3y + 1 - x - \frac{1}{2} = 0$

$\therefore x^2 + y^2 + x + 3y + \frac{1}{2} = 0$

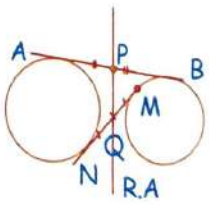
PROPERTY-6: For concentric circles radical axis is not defined.

PROPERTY-7: Radical axis bisects direct common tangent & transverse common tangent.

Proof: To Prove PA = PB

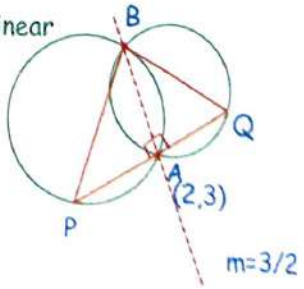
Length of tangent from P (which lies on RA) is same for both circle.

$\therefore$ PA = PB



Ques. Two circles interest at A (2, 3) and B. The slope of their radical axis is $\frac{3}{2}$. BP, BQ are diameter of the circles. What is the equation of PQ?

Ans. Here P, A, Q are collinear



$\because M_{AB} = \frac{3}{2}$

$\therefore M_{PQ} = \frac{-2}{3}$

Eqn of PQ:

$y - 3 = \frac{-2}{3}(x - 2)$

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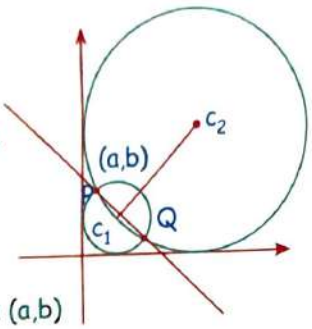
Ques. a, b > 0. Two circles pass through (a, b) and touch both the co-ordinate axes. The equation of the radical axis of the circles is: **[RPET 2003]**

(a) $x + y = a + b$ (b) $bx = ay$

(c) $x - y = a - b$ (d) $\frac{x}{a} + \frac{y}{b} = 2$

Ans. $C_1(r_1, r_1)$

$C_2(r_2, r_2)$



$\therefore M_{C_1C_2} = \frac{r_2 - r_1}{r_2 - r_1} = 1$

So, $M_{RA} = -1$

eqⁿ of R.A:

$\because$ it passes through (a,b)

$\therefore y - b = -1(x - a)$

$\Rightarrow x + y = a + b$

Brain Teaser

The circles C_1 and C_2 are such that

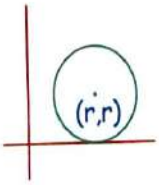
(i) they both pass through the point (a, b) of the first quadrant:

(ii) they both touch the co-ordinate axes.

(iii)they are orthogonal.

Prove that $\frac{a}{b} + \frac{b}{a} = 4$.

Ans. Equation $(x - r)^2 + (y - r)^2 = r^2$



(a, b) satisfies the eqⁿ:

$\therefore (a - r)^2 + (b - r)^2 = r^2$

$\Rightarrow r^2 - 2ar - 2br + a^2 + b^2 = 0$

$\Rightarrow r^2 - 2r(a + b) + a^2 + b^2 = 0 \begin{matrix} r_1 \\ r_2 \end{matrix}$

$C_1[r_1, r_1]$

$C_2[r_2, r_2]$

$d^2 = r_1^2 + r_2^2$ $\because$ Both circles are orthogonal.

$\therefore r_1^2 + r_2^2 = (C_1C_2)^2 \Rightarrow r_1^2 + r_2^2 = (r_1 - r_2)^2 + (r_1 - r_2)^2$

$\Rightarrow r_1^2 + r_2^2 - 4r_1r_2 = 0 \quad \Rightarrow r_1^2 + r_2^2 = 4_1 r_1r_2$

$\Rightarrow (r_1 + r_2)^2 = 6 r_1r_2$

$\Rightarrow (2(a + b))^2 = 6(a^2 + b^2)$

$\Rightarrow 4(a^2 + b^2 + 2ab) = 6(a^2 + b^2)$

$\Rightarrow 8ab = 2a^2 + 2b^2$

$\Rightarrow 4ab = a^2 + b^2$

$\Rightarrow 4 = \frac{a^2 + b^2}{ab}$

$\Rightarrow 4 = \frac{a}{b} + \frac{b}{a}$

Ques. The circle ω_1: $x^2 + y^2 = 16$ intersects a circle ω_2 (radius = 5) such that the common chord has a maximum length and the slope is $\frac{3}{4}$. The centre of ω_2 is

(a) $\left(\frac{9}{5}, \frac{-12}{5}\right)$ or $\left(\frac{-9}{5}, \frac{12}{5}\right)$

(b) $\left(\frac{12}{5}, \frac{-9}{5}\right)$ or $\left(\frac{-12}{5}, \frac{9}{5}\right)$

(c) $\left(\frac{5}{3}, \frac{-5}{4}\right)$ or $\left(\frac{5}{4}, \frac{-5}{3}\right)$

(d) None of these

Ans. **Method-1:**

$\because M_{AB} = \frac{3}{4}$

$\Rightarrow M_{OC} = \frac{-4}{3}$

$\because (OC)^2 = 5^2 - 4^2$

$\Rightarrow OC = 3$

$\therefore \tan\theta = \frac{-4}{3}$

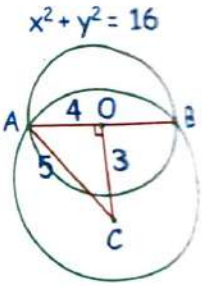
$\Rightarrow \sin\theta = \frac{4}{5}$ and $\cos\theta = \frac{-3}{5}$

$x = 0 \pm r\cos\theta, y = 0 \pm r\sin\theta$

$\Rightarrow x = 3\left(\frac{-3}{5}\right), y = 3\left(\frac{4}{5}\right)$

$\Rightarrow x = \frac{-9}{5}, y = \frac{12}{5}$

or $x = \frac{9}{5}, y = \frac{-12}{5}$



Method-2:

Equation of AB:

$y = \frac{3}{4}x$

$3x - 4y = 0$

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$S + \lambda L = 0$

$x^2 + y^2 - 16 + \lambda(3x - 4y) = 0$

$x^2 + y^2 + 3\lambda x - 4\lambda y - 16 = 0$

$\therefore C\left[\frac{-3\lambda}{2}, 2\lambda\right]$

and $r = \sqrt{\frac{9\lambda^2}{4} + 4\lambda^2 + 16} = 5$

$\therefore \lambda = \pm\frac{6}{5}$

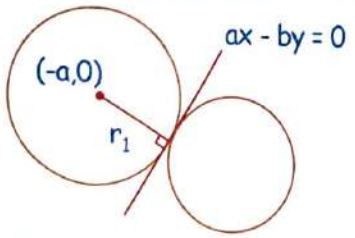
$\therefore c = \left(\pm\frac{6}{5}\times\frac{3}{2}, \pm\frac{6}{5}\times\left(\frac{-4}{2}\right)\right)$

$= \left(\frac{9}{5}, \frac{-12}{5}\right), \left(\frac{-9}{5}, \frac{12}{5}\right)$

Ques. Prove that the circle $x^2 + y^2 + 2ax + c^2 = 0$ and $x^2 + y^2 + 2by + c^2 = 0$ touch each other, if $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$.

Ans. Since Circle touch each other:

Radical axes → touches both circles



$\therefore S_1 - S_2 = 0$

$\Rightarrow 2ax - 2by = 0$

$\Rightarrow ax - by = 0$

$\because$ Perpendicular distance from (-a, 0) to Radical axes is r_1:

$\Rightarrow \left|\frac{a(-a)+0}{\sqrt{a^2+b^2}}\right| = r_1$

$\Rightarrow \left|\frac{a^2}{\sqrt{a^2+b^2}}\right| = \sqrt{a^2-c^2}$

$\Rightarrow a^4 = (a^2 + b^2)(a^2 - c^2)$

$\Rightarrow a^4 = a^4 - a^2c^2 + a^2b^2 - b^2c^2$

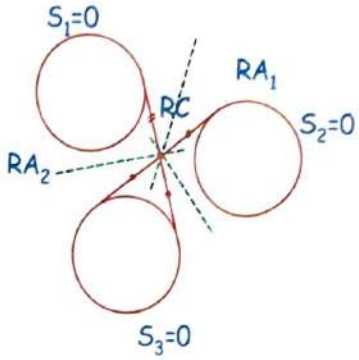
$\Rightarrow b^2c^2 + a^2c^2 = a^2b^2$

$\Rightarrow c^2(a^2 + b^2) = a^2b^2$

$\Rightarrow \frac{a^2+b^2}{a^2b^2} = \frac{1}{c^2} \Rightarrow \frac{1}{b^2} + \frac{1}{a^2} = \frac{1}{c^2}$

RADICAL CENTRE

Definition: The point of concurrency of the 3 radical axes defined between 3 given circle is known as Radical centre of those 3 circles



Ques. The radical centre of the circles $x^2 + y^2 + 4x + 6y = 19$, $x^2 + y^2 = 9$ and $x^2 + y^2 - 2x - 2y = 5$ will be

(a) (1,1) (b) (- 1, 1) (c) (1, -1) (d) (0, 1)

Ans. RA_1: $S_1 - S_2 = 0 \Rightarrow 4x + 6y = 10$

$\Rightarrow 2x + 3y = 5$ ...(i)

RA_2: $S_1 - S_3 = 0 \Rightarrow 6x + 8y = 14$

$\Rightarrow 3x + 4y = 7$ ...(ii)

$\because$ (1, 1) is POI of eq^n (i) & (ii)

$\therefore$ Radical Centre = (1, 1)

Ques. The line L_1: $a_1x + b_1y + c_1 = 0$ intersects the circle c_1: $x^2 + y^2 + A_1x + B_1y + C_1 = 0$ at P, Q. The line L_2: $a_2x + b_2y + c_2 = 0$ intersects the circle c_2: $x_2 + y_2 + A_2x + B_2y + C_2 = 0$ at R,S. The circle are not concentric and P, Q, R, S are concyclic.

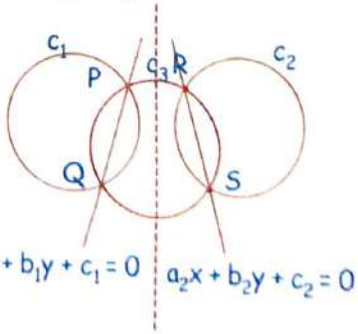
Prove that $\begin{vmatrix} A_1 - A_2 & B_1 - B_2 & C_1 - C_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$

Ans. C_1 & C_2 → RA: $(A_1 - A_2)x + (B_1 - B_2)y + C_1C_2 = 0$

C_1& C_3 → RA : $a_1x + b_1y + C_1 = 0$

C_2 & C_3 → RA : $a_2x + b_2y + C_2 = 0$

R.A. of C_1 & C_3



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R.A. of C_1 & C_3

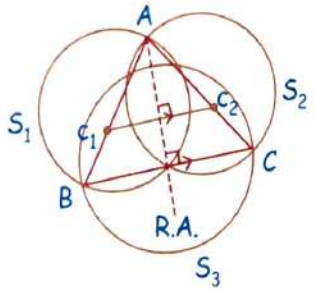
3 RA are always Con current

$$\therefore \begin{vmatrix} A_1 - A_2 & B_1 - B_2 & C_1 - C_2 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0.$$

Important Note

Property-1: The length of tangents drawn from radical centre upon the 3 circles is same (if defined).

Property-2: The radical centre of 3 circles defined on three sides of a triangle as diameter is same as the orthocentre of that triangle



Proof: $C_1C_2 \parallel BC$ (mid pt theorem)

but $C_1C_2 \perp RA$

$\Rightarrow RA \perp BC$

$\Rightarrow$ RA is same as altitude

Ques. Radical centre of the circles drawn on the sides as diameter of triangle formed by line $3x - 4y + 6 = 0$, $x - y + 2 = 0$ and $4x + 3y - 17 = 0$.

(a) (3, 2) (b) (3, -2) (c) (2, -3) (d) (2, 3)

Ans. $3x - 4y + 6 = 0$ ...(i)

$4x + 3y - 17 = 0$ ...(ii)

[Eqⁿ (ii)] × 4 + [Eqⁿ (i)] × 3

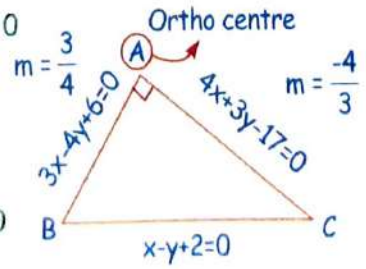
$25x + 18 - 68 = 0$

$\Rightarrow 25x = 50$

$\Rightarrow x = 2$

$\therefore 6 - 4y + 6 = 0$

$\Rightarrow y = 3$



Hence, Radical Centre = (2, 3)

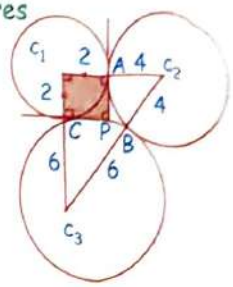
Ques. 3 Circles of radius 2, 4, 6 touch each other externally at points A, B and C respectively. If P is their Radical Centre. Find distance PA?

Ans. $\left.\begin{matrix} C_1C_3 = 8 \\ C_1C_2 = 6 \\ C_2C_3 = 10 \end{matrix}\right\} \rightarrow$ Pythagores

and $\angle C_1 = 90°$

also, $AC_1CP \Rightarrow$ Square

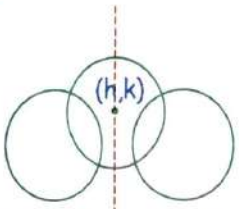
Hence, PA = 2



Property-3: The locus of centre of a variable circle which cuts two given circles orthogonally is same as the radical axis between the two given circles.

Let variable Circle by

$x^2 + y^2 + 2gx + 2fy + c = 0$



$x^2y^2+2g_1x+2f_1y+c_1=0$ $x^2y^2+2g_2x+2f_2y+c_2=0$

$-g = h, -f = k$ (for variable circle)

$\therefore 2gg_1 + 2ff_1 = C + C_1$ ...(i)

and $2gg_2 + 2ff_2 = C + C_2$ ...(ii)

eqⁿ (1) - eqⁿ (2)

$\Rightarrow 2g(g_1 - g_2) + 2f(f_1 - f_2) = C_1 - C_2$

$\Rightarrow -2h(g_1 - g_2) - 2k(f_1 - f_2) = C_1 - C_2$

$\Rightarrow 2h(g_1 - g_2) + 2k(f_1 - f_2) + C_1 - C_2 = 0$

$\Rightarrow 2(g_1 - g_2)x + 2(f_1 - f_2)y + C_1 - C_2 = 0$

$\Rightarrow S_1 - S_2 = 0 \rightarrow$ Radical axis

Ques. If locus of centre of circle which cuts the circle $x^2 + y^2 - 2x + 4y = 3$ and $x^2 + y^2 + 7x - 3y - 1 = 0$ orthogonally, is $lx + my + 4 = 0$, then $(l + m)$ is equal to:

(a) 0 (b) 2 (c) 4 (d) -2

Ans. $S_1 - S_2 = 0 \rightarrow$ locus

$\Rightarrow -2x - 7x + 4y + 3y - 3 + 1 = 0$

$\Rightarrow -9x + 7y - 2 = 0$

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$\Rightarrow 9x - 7y + 2 = 0$

$\Rightarrow 18x - 14y + 4 = 0$

Comparing with $lx + my + 4 = 0$

We get, $l = 18, \quad m = -14$

$\therefore l + m = 4$

Property-4: The Centre of a circle which cuts Three given circles orthogonally is same as the radical centre of the 3 given circles & Radius is equal to the length of tangent drawn from radical centre.

Ques. The equation of the circle which intersects circles $x^2 + y^2 + x + 2y + 3 = 0$, $x^2 + y^2 + 2x + 4y + 5 = 0$ and $x^2 + y^2 - 7x - 8y - 9 = 0$ at right angle, will be

(a) $x^2 + y^2 - 4x - 4y - 3 = 0$

(b) $3(x^2 + y^2) + 4x - 4y - 3 = 0$

(c) $x^2 + y^2 + 4x + 4y - 3 = 0$

(d) $3(x^2 + y^2) + 4(x + y) - 3 = 0$

Ans. Centre is same as R.C.

RA_1: $-x - 2y - 2 = 0 \Rightarrow (x + 2y + 2 = 0)$ ...(i)

RA_2: $8x + 10y + 12 = 0 \Rightarrow 4x + 5y + 6 = 0$ ...(ii)

By eqn (i) & (ii):

$$C\left[\frac{-2}{3}, \frac{-2}{3}\right] = (-g, -f)$$

and $r = \sqrt{S_1}$

$$\therefore \sqrt{\left(\frac{-2}{3}\right)^2 + \left(\frac{-2}{3}\right)^2 - c}$$

$$= \sqrt{\left(\frac{-2}{3}\right)^2 + \left(\frac{-2}{3}\right)^2 \frac{-2}{3} - \frac{4}{3} + 3} \Rightarrow c = -1$$

Hence, the circle will be-

$$x^2 + y^2 + \frac{4}{3}x + \frac{4}{3}y - 1 = 0$$

$$\Rightarrow 3(x^2 + y^2) + 4(x + y) - 3 = 0$$

Ques. Two circles intersect at the points (1, 1) and (3, -1) The y-axis is a common tangent. (See the adjoining figure). The length of AB is:

(a) $\sqrt{6}$

(b) $2\sqrt{6}$

(c) 6

(d) cannot be determined as data is insufficient

Ans. L [1, 1], M [3, -1]

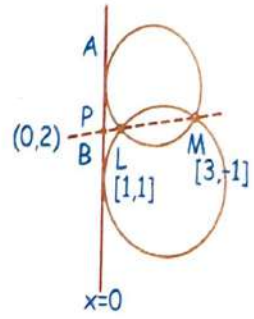
Eqn of RA: $x + y = 2$

at $x = 0 \Rightarrow y = 2$

$\therefore P = (0, 2)$

$\because PA = PB$

and $(PA)^2 = PL.PM$



$\because PL = \sqrt{2}$ and $PM = 3\sqrt{2}$

$\Rightarrow (PA)^2 = \sqrt{2}.3\sqrt{2} = 6$

$\Rightarrow PA = \sqrt{6} = PB$

Brain Teaser

Through a given point P(5, 2), secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$.

Find the value of $\sum_{i=1}^{5} PA_i^2 + \sum_{i=1}^{5} PB_i^2$

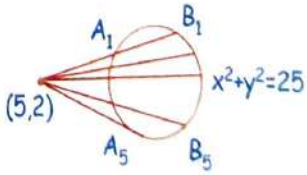
[Note: $A_r(B_r)$ denotes that the line passing through P(5, 2) meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r].

Ans. $PA_1.PB_1 = S_1$

$\Rightarrow PA_1 PB_1$

$= 5^2 + 2^2 - 25$

$\Rightarrow PA_1.PB_1 = 4$



$\therefore (PA_1^2 + PB_1^2) + (PA_2^2 + PB_2^2) + ...$

$= (PA_1 + PB_1)^2 - 2PA_1 PB_2 + (PA_2 + PB_2)^2 - 2PA_2.PB_2$

+-----

$= 5^2 - 2 \times 4 + 6^2 - 2.4 + 7^2 - 2.4 + 8^2 - 2.4 + 9^2 - 2.4$

$= (5^2 + 6^2 + 7^2 + 8^2 + 9^2) - 2 \times 4 \times 5 = 215$

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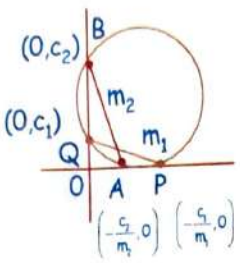
Important Note

If the lines with slopes m_1 and m_2 intersect co-ordinate axes at 4 concyclic points then $m_1m_2 = 1$



Proof: Let eqⁿ of PQ: $y = m_1x + c_1$

Let eqⁿ of AB: $y = m_2x + c_1$

$\because OA.OP = OQ.OB$

$\Rightarrow \left(-\frac{c_2}{m_2}\right)\left(-\frac{c_1}{m_1}\right) = c_1c_2$

$\Rightarrow \frac{1}{m_1m_2} = 1 \Rightarrow m_1m_2 = 1$

Ques. If the line $2x + 3y = 6$ and $ax + by = ab$ meet the coordinate axes in 4 distinct concylic points. then show that $3b = 2a$.

Ans. $\because m_1 = -\frac{2}{3}$ and $m_2 = -\frac{a}{b}$

$\because m_1.m_2 = 1 \Rightarrow \frac{2}{3}.\frac{a}{b} = 1$

$\Rightarrow 2a = 3b$

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Short Notes

EQUATIONS OF THE CIRCLE IN VARIOUS FORMS

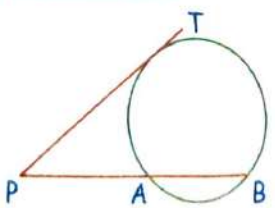
- The simplest equation of the circle is $x^2 + y^2 = r^2$ whose centre is (0, 0) and radius r.
- The equation $(x - a)^2 + (y - b)^2 = r^2$ represents a circle with centre (a, b) and radius r.
- The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ is the general equation of a circle with centre (-g, -f) and radius $\sqrt{g^2 + f^2 - c}$
- Equation of the circle with points $P(x_1, y_1)$ and $Q(x_2, y_2)$ as extremities of a diameter is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.
- Parametric Equation of a Circle: The equations $x = a\cos\theta$, $y = a\sin\theta$ are called parametric equations of the circle $x^2 + y^2 = a^2$ and θ is called a parameter. The point $(a\cos\theta, a\sin\theta)$ is also referred to as point θ. The parametric coordinates of any point on the circle $(x - h)^2 + (y - k)^2 = a^2$ are given by $(h + a\cos\theta, k + a\sin\theta)$ with $0 \le \theta < 2\pi$.

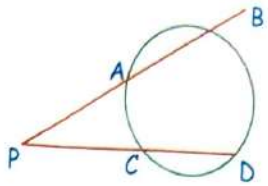
THE POSITION OF A POINT WITH RESPECT TO A CIRCLE

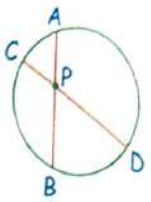
The point $P(x_1, y_1)$ lies outside, on, or inside a circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, according as $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c > = \text{or} < 0$.

TANGENT SECANT THEOREM

- $(PT)^2 = PA \cdot PB$
- $PA.PB = PC \cdot PD$







- If the lines with slopes m_1 & m_2 intersect co-ordinate axes at 4 con-cyclic points then $m_1 m_2 = 1$

TANGENT AND NORMAL TO A CIRCLE

- $T = 0$ is the equation which is obtained by substituting $x^2 = xx_1$, $y^2 = yy_1$, $2xy = xy_1 + yx_1$, $2x = x + x_1$, $2y = y + y_1$ in the equation $S = 0$.
- Normal to a circle always passes through its centre.
- Area of quadrilateral formed by joining pair of tangents drawn from a point $P(x_1, y_1)$. & the two radii of circle is $r\sqrt{S_1}$ where r is the radius of the circle & S_1 is the power of the point $P(x_1, y_1)$.
- The angle between pair of tangents drawn from an exterior point (x_1, y_1) upon a circle is given by $\tan\theta = \frac{2r\sqrt{S_1}}{S_1 - r^2}$. Where S_1 represent power of point (x_1, y_1) & r is the radius of this circle.

PAIR OF TANGENTS

The joint equation of a pair of tangents drawn from the point $A(x_1, y_1)$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $T^2 = SS_1$. $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ then $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$, and $T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$.

LENGTH OF TANGENT

The length of the tangent drawn from a point (x_1, y_1) outside the circle $S \equiv 0$ is $\sqrt{S_1}$.

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DIRECTOR CIRCLE

- Locus of point of intersection of two mutually perpendicular tangents to the circle is called director circle.
- For circle $x^2 + y^2 = a^2$, equation of director circle is given by $x^2 + y^2 = 2a^2$.
- The director circle is always concentric with given circle & its radius is $\sqrt{2}$ times the radius of the given circle.

CHORD OF CONTACT

- From a point $P(x_1, y_1)$ two tangents PA and PB can be drawn to the circle. The chord AB joining the points of contact A and B of the tangents from P is called the chord of contact of $P(x_1, y_1)$ with respect to the circle.
- The equation of chord of contact is given by $T = 0$.
- The length of chord of contact drawn w.r.t. a point $P(x_1, y_1)$ is given by $\frac{2r\sqrt{S_1}}{\sqrt{r^2 + S_1}}$

CHORD WITH A GIVEN MID-POINT

- The equation of the chord of the circle $S \equiv 0$, with midpoint (x_1, y_1) is $T = S_1$.
- Of all the chords passing through a given point (x_1, y_1) lying inside a given circle, the shortest chord is the one whose "midpoint" is (x_1, y_1).

FAMILY OF CIRCLES

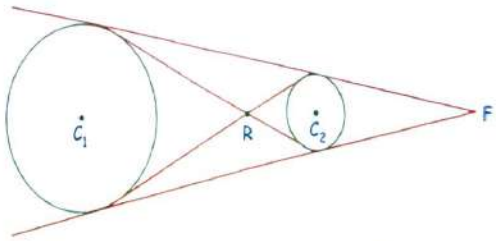
- If $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ and $S' \equiv x^2 + y^2 + 2g'x + 2f'y + c' = 0$ are two intersecting circles, then $S + \lambda S' = 0$, $\lambda \neq -1$, is the equation of the family of circles passing through the points of intersection of $S = 0$ and $S' = 0$.
- If $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ is a circle which is intersected by the straight line $\mu \equiv ax + by + c = 0$ at two real and distinct points, then $S + \lambda\mu = 0$ is the equation of the family of circles passing through the points of intersection of $S = 0$ and $\mu = 0$. If $\mu = 0$ touches $S = 0$ at P, then $S + \lambda\mu = 0$ is the equation of the family of circles, each touching $\mu = 0$ at P.
- The equation of the family of circles which touch the line $y - y_1 = m(x - x_1)$ at (x_1, y_1) for any value of m is $(x - x_1)^2 + (y - y_1)^2 + \lambda[(y - y_1) - m(x - x_1)] = 0$. If m is infinite, the equation is $(x - x_1)^2 + (y - y_1)^2 + \lambda(x - x_1) = 0$.

NATURE OF TWO CIRCLES

- When two circles do not intersect or touch, 4 common tangents can be drawn.
- When two circles touch each other externally, 3 common tangents can be drawn to the circles.
- When two circles intersect each other, two common tangents can drawn to the circles.
- When two circles touch each other internally 1 common tangent can be drawn to the circles.
- When a circle lie completely inside the other circle, no any common tangents can be drawn.

COMMON TANGENTS TO TWO CIRCLES

- The direct common tangents to two circles meet on the line of centres and divide it externally in the ratio of the radii.



- The transverse common tangents also meet on the line of centres and divide it internally in the ratio of the radii.

LENGTH OF DIRECT & TRANSVERSE COMMON TANGENT

- $L^2 = d^2 - (r_2 - r_1)^2$ and $l^2 = d^2 - (r_1 + r_2)^2$
- The length of Direct common tangent is always greater than the length of Transverse common tangent.

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RADICAL AXIS

- The radical axis of two circles is the locus of a point from which the tangent segments to the two circles are of equal length.
- In general $S - S' = 0$ represents the equation of the Radical Axis to the two circles . i.e. $2x(g - g') + 2y(f - f') + c - c' = 0$ where $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ and $S' \equiv x^2 + y^2 + 2g'x + 2f'y + c' = 0$
- Line joining the centres of the two circles is always $\perp$ to their Radical axis.
- For externally touching circles radical axis is same as TCT.
- For internally touching circles radical axis is same as DCT.
- For intersecting circles radical axis is same as line joining common chord.
- For concentric circles radical axis is not defined.
- Radical axis bisects DCT & TCT.

RADICAL CENTRE

- The point of concurrency of the 3 radical axes defined between 3 given circles is known as Radical centre of those 3 circles.
- The length of tangents drawn from radical centre upon the 3 circles is same (if defined).
- The radical centre of 3 circles defined on three sides of a Δ as diameter is same as the orthocentre of that Δ.

ANGLE BETWEEN TWO CIRCLES

$$\cos\theta = \left|\left(\frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}\right)\right|$$

ORTHOGONAL CIRCLES

- Two circles are said to be orthogonal if angle between their tangents is $90°$ at their point of intersection.
- For orthogonal circles

  $r_1^2 + r_2^2 = d^2$ or $2g_1g_2 + 2f_1f_2 = c_1 + c_2$
- The locus of centre of a variable circle which cuts two given circles orthogonally is same as the radical axis between the two given circles
- The circle which cuts the three given circles orthogonally will have centre as the radical centre of three given circles and radius equal to the length of tangent from radical centre to the three given circles.

"Life is a circle – what you give comes back around."

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Chapter 12

Parabola

CONIC SECTION

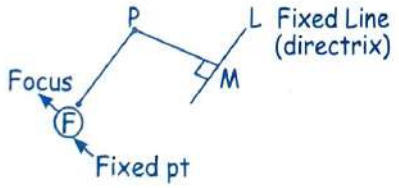
Fundamental Principle of Conic Section

The locus of a point which moves in a plane such that the ratio of its distance from a fixed point to its distance from a fixed straight line is always a constant.

(a) The fixed point is called the focus.

(b) The fixed straight line is called the directrix.

(c) The constant ratio is called the eccentricity denoted by e.



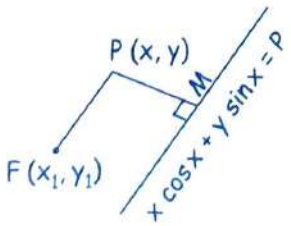
$\frac{PF}{PM}$ = const

$\frac{PF}{PM} = e$

If $e = 1 \Rightarrow$ Parabola

If $e > 1 \Rightarrow$ Hyperbola

If $0 < e < 1 \Rightarrow$ Ellipse



$PF = \sqrt{(x-x_1)^2 (y-y_1)^2}$

$PM = \left|\frac{x\cos x + y\sin x - P}{\sqrt{\cos^2 x + \sin^2 x}}\right|$

$PM = |x\cos x + y\sin x - P|$

$\frac{PF}{PM} = e \Rightarrow PF = e\,PM$

$\sqrt{(x-x_1)^2 + (y-y_1)^2} = e\,|x\cos x + y\sin x - P|$

$\Rightarrow (x-x_1)^2 + (y-y_1)^2 = e^2 (x\cos x + y\sin x - P)^2$

$x^2 + x_1^2 - 2xx_1 + y^2 + y_1^2 - 2yy_1 =$

$e^2 (x^2\cos^2 x + y^2\sin^2 x + p^2 + 2xy\sin x\cos x - 2px\cos x - 2py\sin x)$

$\Rightarrow x^2 (1 - e^2\cos^2 x) + y^2 (1 - e^2\sin^2 x) + xy$

$(-e^2\sin 2x) + ()\,x + (\;)\,y + \text{const} = 0$

General 2 degree Curve

$ax^2 + by^2 + 2h\,xy + 2g\,x + 2fy + C = 0$

$a = 1 - e^2\cos^2 x,\ b = 1 - e^2\sin^2 x$

$2h = -e^2\sin 2x \Rightarrow h = -e^2\sin x\cos x$

$\Delta = abc + 2fgh - af^2 - ch^2 - bg^2$

$\Delta = (x_1\cos x + y_1\sin x - P)^2$

$h^2 - ab = (-e^2\sin x\cos x)^2 - (1 - e^2\cos^2 x)(1 - e^2\sin^2 x)$

$= -[1 - e^2\cos^2 x - e^2\sin^2 x] - [1 - e^2]$

$\Rightarrow \boxed{h^2 - ab = e^2 - 1}$

For conic $\Delta \neq 0 \Rightarrow$ Fixed pt must not lie on Fixed Line

Parabola	Ellipse	Hyperbola
$e = 1$	$e < 1$	$e > 1 \Rightarrow e^2 > 1$
$\Rightarrow h^2 - ab = 0$	$\Rightarrow e^2 < 1$	$\Rightarrow h^2 - ab > 0$
	$\Rightarrow e^2 - 1 < 0$	
	$\Rightarrow h^2 - ab < 0$	

IF $\Delta = 0$

$\Rightarrow$ Locus is Pair of straight line

If $h^2 > ab$	$h^2 = ab$	If $h^2 < ab$
$\Rightarrow$ Real and distinct	$\Rightarrow$ Real and Parallel POSL	$\Rightarrow$ non real Pair of straight line

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Ques. The equation $x^2 - 2xy + y^2 + 3x + 2 = 0$ represents

(a) A parabola (b) An ellipse

(c) A hyperbola (d) A circle

Ans. $\begin{cases} a = 1 & 2g = 3 \\ b = 1 & 2f = 0 \\ h = -1 & c = 2 \end{cases}$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$= 1 \times 1 \times 2 - 1 \times (3/2)^2 - 2(-1)^2$

$\Delta = 2 - 9/4 - 2$

$\Delta = -9/4$

$\Delta \neq 0$ and $h^2 = ab \Rightarrow$ Parabola

General Conic Section

Case (i): When the focus does not lie on the directrix ($\Delta \neq 0$)

The conic represents:

A Parabola	An Ellipse	A Hyperbola
$e = 1; \Delta \neq 0$	$0 < e < 1; \Delta \neq 0$	$\Delta \neq 0; e > 1$
$h^2 = ab$	$h^2 < ab$	$h^2 > ab$

Case (ii): When the focus lies on the directrix

In this case $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

And we know that when $\Delta = 0$, the general 2° curve represents a pair of straight lines and if: $e^2 - 1 = h^2 - ab$

(a) $e > 1 \Rightarrow h^2 - ab > 0$ the lines will be real and distinct intersecting at S.

(b) $e = 1 \Rightarrow h^2 - ab = 0$ the lines will be Parallel.

(c) $e < 1 \Rightarrow h^2 < ab$ the lines will be imaginary

Ques. Curve $16x^2 + 8xy + y^2 - 74x - 78y + 212 = 0$ represents.

(a) Parabola (b) Hyperbola

(c) Ellipse (d) None of these

Ans. Comparing with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$a = 16; b = 1; h = 4$

$g = -37; f = -39; c = 212$

$\therefore \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Rightarrow \Delta = 16 \times 1 \times 212 + 2 \times 4 \times (-37) \times (-39) - 16(-39)^2 - 1(-37)^2 - 212(4)^2$

$\Rightarrow \Delta \neq 0$

also, $h^2 = 4^2 = 16$

and $ab = 16$

$\therefore h^2 = ab$

Hence, the curve represents parabola.

Trick for checking parabola:

$\Delta \neq 0, h^2 = ab \Rightarrow h = \pm \sqrt{ab}$

$a x^2 + by^2 + 2hxy + \ldots$

$ax^2 + by^2 \pm 2\sqrt{ab}\, xy + \ldots$

$\left(\sqrt{a} \times \pm\sqrt{by}\right)^2$

2 terms always form a perfect square in eqⁿ of parabola

Example: $x^2 + y^2 - 2xy - 8x - 7y + 1 = 0$

Perfect square now check $\Delta \neq 0$

Important Note

In the equation of a parabola, 2° terms will always form a complete square.

$y = ax^2 + bx + c \rightarrow$ Parabola

Ques. Which of the following equations will represent equation of parabola.

1. $x^2 = 4y - 3x + 1$
2. $y^2 + 4x^2 - 4xy = 5$
3. $2y^2 + 4x^2 - 4x = 7y$
4. $y^2 + 4x^2 + 2y + 4x + 1 = 0$
5. $\sqrt{(x-2)^2 + (y-5)^2} = \frac{1}{5}|3x + 4y - 7|$

Ans.

1. $x^2 = 4y - 3x + 1; \Delta \neq 0$

   x^2 is a complete square.

   $\therefore$ It is parabola
2. $y^2 + 4x^2 - 4xy = 5$

   Perfect square. Now check Δ.

   $\therefore 4x^2 + y^2 - 4xy - 5 = 0$

   $a = 4, b = 1, h = 2, g = f \equiv 0, c = -5$

   $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

   $= 4 \times 1 \times (-5) + 5(-2)^2$

   $\Rightarrow \Delta = -20 + 20 = 0 \Rightarrow$ Locus is parallel POSL.
3. $2y^2 + 4x^2 - 4x = 7y$

   $\because 2y^2 + 4x^2$ is not a perfect square

   $\therefore$ It is not a parabola
4. $y^2 + 4x^2 - 4xy + 2y + 4x + 1 = 0$;

   $\because y^2 + 4x^2 - 4xy = (2x - y)^2$ and $\Delta \neq 0$

   $\therefore$ It is parabola

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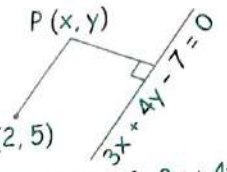
5. $\sqrt{(x-2)^2+(y-5)^2} = \frac{1}{5}|3x+4y-7|$



Since (2, 5) does not satisfy $3x + 4y = 7$

and $PF = PM \Rightarrow \frac{PF}{PM} = 1 \Rightarrow e = 1$

$\therefore$ It is a parabola

Ques. If equation $\sqrt{(3x-2)^2+(3x+5)^2} = \lambda|3x - 4y + 7|$ represents parabola then find λ.

Ans. $3\sqrt{\left(x-\frac{2}{3}\right)^2+\left(y+\frac{5}{3}\right)^2} = 5\lambda\left|\frac{3x-4y+7}{5}\right|$

$\Rightarrow \sqrt{\left(x-\frac{2}{3}\right)^2+\left(y+\frac{5}{3}\right)^2} = \frac{5\lambda}{3}\frac{|3x-4y+7|}{5}$

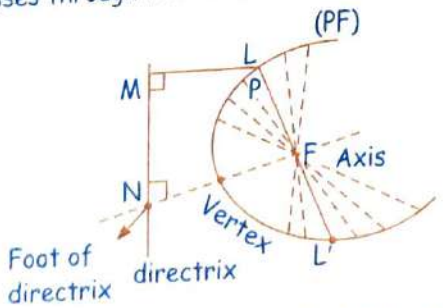
$\because PF = e\,PM$

$\therefore e = 1 \Rightarrow \frac{5\lambda}{3} = 1$

$\Rightarrow \lambda = \frac{3}{5}$

Important Definitions Related Conic Section

1. **Axis of a conic:** The line passing through the focus and perpendicular to the directrix is called the Axis of the Conic.
2. **Vertex of a conic:** Point of intersection of a conic with its axis is called a vertex
3. **Focal distance:** The distance of a point on the parabola from the focus is called the focal distance of the point
4. **Focal chord:** A chord of the parabola, which passes through the focus is called a focal chord.



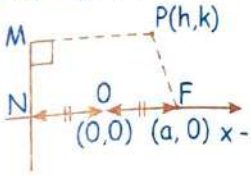
5. **Double ordinate:** A chord of the parabola perpendicular to the axis of the symmetry is called a double ordinate.
6. **Latus rectum:** A double ordinate passing through the focus or a focal chord perpendicular to the axis of parabola is called the Latus Rectum.
7. **Foot of directrix:** The point of intersection of axis and directrix which is called Foot of Directrix.

PARABOLA

Standard Equation-1

Let focus is (a, 0) and directrix is $\perp$ to x-axis $(x = -a)$

Also, Parabola passes through (0, 0)



$x = -a$

$\Rightarrow x + a = 0$

Equation of parabola

$PF = PM$

$\because PF = \sqrt{(h-a)^2+k^2}$ ...(i)

and $PM = \left|\frac{h+a}{\sqrt{1^2+0^2}}\right|$

$\Rightarrow PM = |h + a|$ ...(ii)

$PF = PM$

From eqn. (i) and (ii)

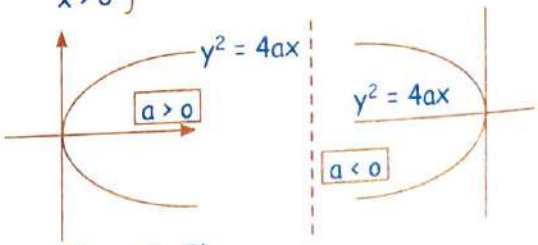
$\sqrt{(h-a)^2+k^2} = |h + a|$

$(h-a)^2 + k^2 = (h + a)^2$

$h^2 + a^2 - 2ah + k^2 = h^2 + a^2 + 2ah$

$\Rightarrow k^2 = 4ah \Rightarrow y^2 = 4ax$

If $\left.\begin{matrix} a > 0 \\ x > 0 \end{matrix}\right\} \Rightarrow ax > 0 \Rightarrow 4ax > 0 \Rightarrow y^2 > 0$



Focus (a, 0)

Directrix $x = -a$

Ques. The equation of the parabola with focus (0, 0) and directrix $x + y = 4$ is

(a) $x^2 + y^2 - 2xy + 8x + 8y - 16 = 0$

(b) $x^2 + y^2 - 2xy + 8x + 8y = 0$

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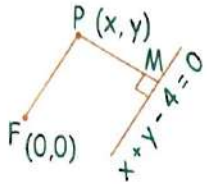
(c) $x^2 + y^2 + 8x + 8y - 16 = 0$

(d) $x^2 - y^2\ 8x + 8y - 16 = 0$

Ans. PF = PM



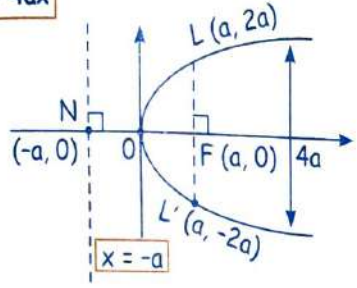
$\Rightarrow \sqrt{x^2 + y^2} = \left|\frac{x+y-4}{\sqrt{2}}\right|$

$\Rightarrow 2(x^2 + y^2) = (x + y - 4)^2$

$\Rightarrow 2x^2 + 2y^2 = x^2 + y^2 + 16 + 2xy - 8x - 8y$

$\Rightarrow x^2 + y^2 - 2xy + 8x + 8y - 16 = 0$

$y^2 = 4ax$



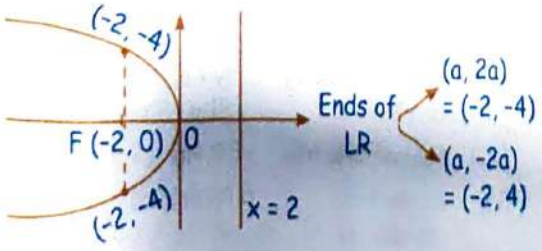
For L and L': put $x = a$ in y^2

$\Rightarrow 4ax \Rightarrow y^2 = 4a \cdot a \Rightarrow \boxed{y = \pm 2a}$

	Important Terms	For $y^2 = 4ax$	$x^2 = 4ay$
(i)	Vertex	(0, 0)	(0, 0)
(ii)	Focus	(a, 0)	(0, a)
(iii)	Directrix	$x = -a$	$y = -a$
(iv)	Axis	$y = 0$	$y = 0$
(v)	Tangent at the vertex	$x = 0$	$y = 0$
(vi)	L.R. (a) *Equation*	$\boxed{x = a}$	$y = a$
	(b) *Ends*	(a, 2a) and (a, -2a)	(2a, a) and (-2a, a)
	(c) *Length*	$\|4a\|$	$\|4a\|$
(vii)	Focal Distance of (x, y)	$\|x + a\|$	$\|y + a\|$
(viii)	Foot of directrix	(-a, 0)	(0, -a)
(ix)	Parametric coordinates	$p(at^2, 2at)$	$p(2at, at^2)$

Example: $y^2 = 8x \Rightarrow -8 = 4a \Rightarrow a = -2$

$\Rightarrow$ Focus (-2, 0) and directrix: $x = -a \Rightarrow x = 2$



Ques. Find every thing for the parabola $y^2 = 4x + 4y$.

	Important Terms	For $y^2 = 4ax$
(i)	Vertex	(-1, 2)
(ii)	Focus	(0, 2)
(iii)	Directrix	$x = -2$
(iv)	Axis	$y = 2$
(v)	Tangent at the vertex	$x = -1$
(vi)	L.R. (a) *Equation*	$x = 0$
	(b) *Ends*	(0, 0) and (0, 4)
	(c) *Length*	$4a = 4$
(vii)	Foot of directrix	$x = -a, y = 0$
		$x + 1 = 1, y - 2 = 0$
		$x = 2, y = 2$

Ans. $y^2 = 4x + 4y$

$\Rightarrow y^2 - 4y = 4x$

$\Rightarrow (y - 2)^2 - 4 = 4x \Rightarrow (y - 2)^2 = 4(x + 1)$

Comparing with $Y^2 = 4aX$

We get, $4a = 4 \Rightarrow a = 1$

$X = x + 1$

and $Y = y - 2$

Vertex: $X = 0$ and $Y = 0$

$\Rightarrow x + 1 = 0$ and $y - 2 = 0$

$\Rightarrow x = -1$ and $y = 2$

Focus: $X = a, Y = 0$

$\Rightarrow x + 1 = 1, y - 2 = 0$

$\Rightarrow x = 0, y = 2$

Directrix: $X = -a$

$\Rightarrow x + 1 = -1 \Rightarrow x = -2$

Axis $Y = 0, y - 2 = 0$

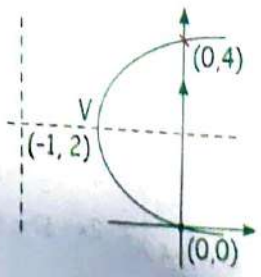
Ends of L.R

$X = a, Y = \pm 2a$

$x + 1 = 1, y - 2 = \pm 2$

$x = 0, y = 4$ or $y = 0$

L (0, 0) and L' (0, 4)



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Ques. Vertex of the parabola $y^2 + 2y + x = 0$ lies in the quadrant

(a) First (b) Second

(c) Third (d) fourth ✓

Ans. $(y + 1)^2 - 1 + x = 0$

$\Rightarrow (y + 1)^2 = -x + 1$

$\Rightarrow (y + 1)^2 = -(x - 1)$

comparing with $y^2 = -x$

$X = x - 1$ and $Y = y + 1$

Vertex: $X = 0 \Rightarrow x - 1 = 0 \Rightarrow x = 1$

$Y = 0 \Rightarrow y + 1 = 0 \Rightarrow y = -1$

$\therefore V = (1, -1) \Rightarrow$ fourth quadrant

Ques. The axis of the parabola $9y^2 - 16x - 12y - 57 = 0$ is

(a) $3y = 2$ ✓ (b) $x + 3y = 3$

(c) $2x = 3$ (d) $y = 3$

Ans. $9y^2 - 12y = 16x + 57$

$$\Rightarrow 9\left[y^2 - \frac{4}{3}y\right] = 16x + 57$$

$$\Rightarrow 9\left\{\left(y - \frac{2}{3}\right)^2 - \frac{4}{9}\right\} = 16x + 57$$

$$\Rightarrow 9\left(y - \frac{2}{3}\right)^2 - 4 = 16x + 57$$

$$\Rightarrow 9\left(y - \frac{2}{3}\right)^2 = 16x + 61$$

$$\Rightarrow 9\left(y - \frac{2}{3}\right)^2 = 16\left(x + \frac{61}{16}\right)$$

Comparing with $9Y^2 = 16X$

$$\Rightarrow y^2 = \frac{16}{9}X$$

Axis: $Y = 0$ $y - \frac{2}{3} = 0$; $y = \frac{2}{3}$

Ques. The length of latus rectum $4y^2 + 2x - 20y + 17 = 0$ of the parabola is?

(a) 3 (b) 6

(c) $\frac{1}{2}$ ✓ (d) 9

Ans. $4(y^2 - 5y) = -2x - 17$

$$\Rightarrow 4\left(\left(y - \frac{5}{2}\right)^2 - \frac{25}{4}\right) = -2x - 17$$

$$\Rightarrow 4\left(y - \frac{5}{2}\right)^2 - 25 = 2x - 17$$

$$\Rightarrow 4\left(y - \frac{5}{2}\right)^2 = -2x + 8$$

$$\Rightarrow 4\left(y - \frac{5}{2}\right)^2 = -2(x - 4)$$

$$\Rightarrow \left(y - \frac{5}{2}\right)^2 = -\frac{2}{4}(x - 4)$$

Comparing with

$$y^2 = -\frac{1}{2}X$$

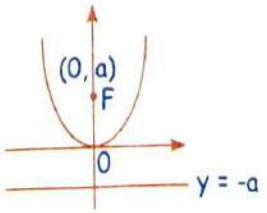
$$4a = \frac{1}{2}$$

$$\therefore L.R = |4a| = \frac{1}{2}$$

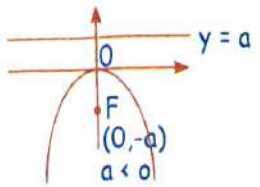
Standard Equation-2

$x^2 = 4ay$

(i) $a > 0$:



(ii) $a < 0$:



	Important Terms	For $x^2 = 4ay$
(i)	Vertex	(0,0)
(ii)	Focus	(0, a)
(iii)	Directrix	$y + a = 0$
(iv)	Axis	$x = 0$
(v)	Tangent at the vertex	$y = 0$
(vi)	L.R. (a) Equation	$y - a = 0$
	(b) Ends	L(2a, a) and L' (-2a, a)
	(c) Length	$\|4a\|$
(vii)	Focal Distance of (x, y)	$\|y + a\|$
(viii)	Foot of directrix	(0, -a)
(ix)	Parametric coordinates	$P(2at, at^2)$

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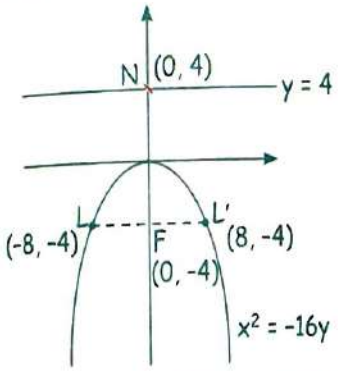
Ques. Find everything for the parabola $x^2 = -16y$.

Important Terms		For $x^2 = 4ay$
(i)	Vertex	$(0, 0)$
(ii)	Focus	$(0, a) = (0, -4)$
(iii)	Directrix	$y = -a \Rightarrow y = 4$
(iv)	Axis	$x = 0$
(v)	Tangent at the vertex	$y = 0$
(vi)	L.R. (a) Equation	$y = a \Rightarrow y = -4$
	(b) Ends	$(2a, a)$ or $(-2a, a) = (-8, -4)$ or $(8, -4)$
	(c) Length	$\|4a\| = 16$
(vii)	Foot of directrix	$(0, -a) = (0, 4)$

Ans. $4a = -16 \Rightarrow a = -4$



Ques. Focus and directrix of the parabola $x^2 = -8ay$ are

(a) $(0, 2a)$ and $y = 2a$

(b) $(0, 2a)$ and $y = 2a$

(c) $(2a, 0)$ and $x = -2a$

(d) $(-2a, 0)$ and $x = 2a$

Ans. Comparing with $x^2 = 4Ay$

$\Rightarrow 4A = -8a \Rightarrow A = -2a$

Focus $(0, A) = (0, -2a)$

Directrix: $y = -A \Rightarrow y = 2a$

Ques. The directrix of the parabola $x^2 - 4x - 8y + 12 = 0$ is

(a) $x = 1$ (b) $y = 0$

(c) $x = -1$ (d) $y = -1$

Ans. $x^2 - 4x = 8y - 12$

$\Rightarrow (x - 2)^2 - 4 = 8y - 12$

$\Rightarrow (x - 2)^2 = 8y - 8$

$\Rightarrow (x - 2)^2 = 8(y - 1)$

Comparing with $X^2 = 8Y$

$X = x - 2$; $Y = y - 1$ and $a = 2$

Directrix $Y = -a$

$\Rightarrow y - 1 = -2 \Rightarrow y = -1$

Ques. The length of the latus rectum of the parabola $9x^2 - 6x + 36y + 19 = 0$

(a) 36 (b) 9

(b) 6 (d) 4

Ans. $9\left(x^2 - \frac{2}{3}x\right) = -36y - 19$

$\Rightarrow 9\left(\left(x - \frac{1}{3}\right)^2 - \frac{1}{9}\right) = -36y - 19$

$\Rightarrow 9\left(x - \frac{1}{3}\right)^2 - 1 = -36y - 19$

$\Rightarrow 9\left(x - \frac{1}{3}\right)^2 = 36y - 18$

$\Rightarrow 9\left(x - \frac{1}{3}\right)^2 = -36\left(y + \frac{1}{2}\right)$

$\Rightarrow \left(x - \frac{1}{3}\right)^2 = -4\left(y + \frac{1}{2}\right)$

Comparing with $X^2 = -4aY$

$4a = -4$

$\therefore$ LR $= |4a| = 4$

Ques. If the line $x = 1$ is the directrix of $y^2 - kx + 8 = 0$, then find the value of k. **[JEE 2000]**

Ans. $y^2 = kx - 8$

$\Rightarrow y^2 = k\left(x - \frac{8}{k}\right)$

Comparing with $y^2 = 4ax$

$Y = y$, $X = x - \frac{8}{k}$

$4a = k, \Rightarrow a = \frac{k}{4}$

Directrix: $X = -a \Rightarrow x - \frac{8}{k} = -\frac{k}{4}$

$\therefore \frac{8}{k} - \frac{k}{4} = 1 \Rightarrow 32 - k^2 = 4k$

$\Rightarrow k^2 + 4k - 32 = 0 \Rightarrow (k + 8)(k - 4) = 0$

$\therefore k = -8$ or 4

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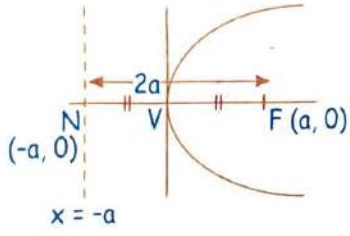
Important Note

(i) Perpendicular distance between focus and directrix is equal to semi latus rectum.

(ii) Vertex is middle point of the line joining Focus and the Foot of directrix.

(ii) Two parabolas are said to be equal if they have the same length of latus rectum.



Ques. If equation $\sqrt{(3x-2)^2+(3y+5)^2} = \lambda|3x-4y+7|$ represents parabola then find λ and also find the length of latus rectum.

Ans. $\sqrt{\left(x-\frac{2}{3}\right)^2+\left(y+\frac{5}{3}\right)^2} = \frac{5\lambda}{3}\left|\frac{3x-4y+7}{5}\right|$

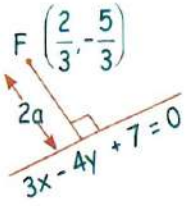
$PF = \frac{5\lambda}{3}PM \Rightarrow \frac{5\lambda}{3}=1 \Rightarrow \lambda = \frac{3}{5}$

$\therefore$ Focus $\left(\frac{2}{3},-\frac{5}{3}\right)$

and Directrix: $3x-4y+7=0$

$\therefore 2a = \left|\frac{3\cdot\frac{2}{3}+4\times\frac{5}{3}+7}{5}\right|$

$= \left|\frac{9+\frac{20}{3}}{5}\right|$

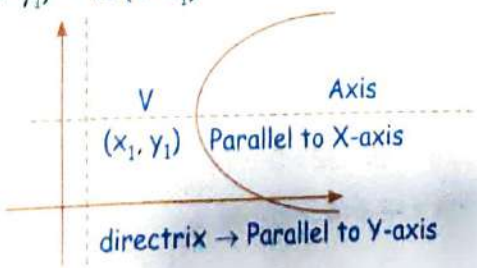


$\Rightarrow 2a = \frac{47}{15} \Rightarrow 4a = \frac{47\times 2}{15}$

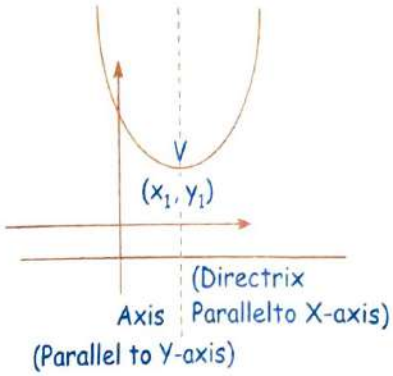
Hence, $\lambda = \frac{3}{5}$, length of latus rectum $= \frac{94}{15}$ units

SHIFTED FORMS OF PARABOLA

1. $(y-y_1)^2 = 4a(x-x_1)$



2. $(x-x_1)^2 = 4a(y-y_1)$



Ques. The equation of the parabola with its vertex at the origin, axis on the y-axis and passing through the point (6, - 3) is **(MP PET 2001)**

(a) $y^2 = 12x + 6$ (b) $x^2 = 12y$

(c) $x^2 = -12y$ (d) $y^2 = -12x + 6$

Ans. Let Parabola: $x^2 = 4ay$

$\because$ (6,-3) lies it on

$\therefore (6)^2 = 4a(-3)$

$\Rightarrow 36 = -12a \Rightarrow a = -3$

$\therefore x^2 = -12y$

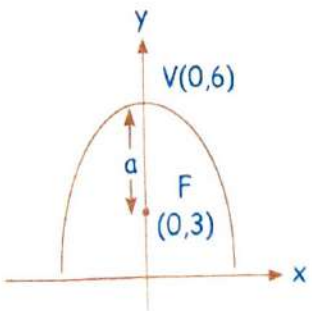
Ques. If (0, 6) and (0, 3) are respectively the vertex and focus of a parabola, then its equation is. **(Karnataka CET 2004)**

(a) $x^2 + 12y = 72$ (b) $x^2 - 12y = 72$

(c) $y^2 - 12x = 72$ (d) $y^2 + 12x = 72$

Ans. $VF = 3 = |a|, \Rightarrow a = -3$

$\therefore x^2 = -12(y-6), \Rightarrow x^2 = -12y + 72$



Ques. Axis of a parabola lies along x-axis. If its vertex and focus are at distance 2 and 4 respectively form the origin on the positive x-axis then which of the following points does not lie on it?

(a) (4, -4) (b) $(5, 2\sqrt{6})$

(c) $(6, 4\sqrt{2})$ (d) (8, 6)

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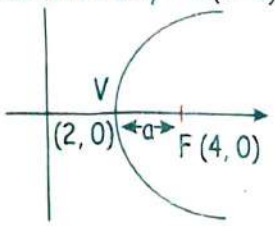
Ans. Let Parabola be $(y - y_1)^2 = 4a(x - x_1)$

$\because$ a = +2 and (h, k) = (2, 0)

$\therefore y^2 = 4a(x - 2) \Rightarrow y^2 = 8(x - 2)$

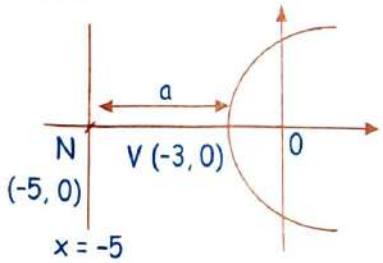
From the given options:

(8, 6) does not lie on $y^2 = 8(x - 2)$



Ques. If the vertex of a parabola is the point (-3, 0) and the directrix is the line x + 5 = 0, then find its equation.

Ans.



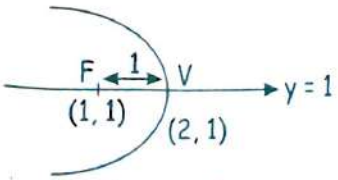
$(y - y_1)^2 = 4a(x - x_1)$

$\because$ a = 2, and (h,k) = (-3,0)

$\therefore (y - 0)^2 = 8(x + 3) \Rightarrow y^2 = 8x + 24$

Ques. Find the equation of the parabola whose focus is (1, 1) and whose vertex is (2, 1).

Ans.



$\because |a| = 1 \Rightarrow a = -1$

Let the equation be

$(y - y_1)^2 = 4a(x - x_1)$

$\Rightarrow (y - 1)^2 = 4a(x - 2) \Rightarrow (y - 1)^2 = -4(x - 2)$

Ques. Find the equation of a parabola whose axis is parallel to Y-axis and which passes through the points (0, 0), (1, 0) and (-2, 4)

(a) $2x^2 + 2x = 3y$ (b) $2x^2 - 2x = 3y$

(c) $2x^2 + 2x = y$ (d) $2x^2 - 2x = y$

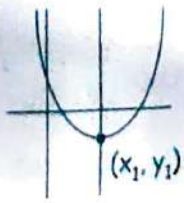
Ans. Let Parabola be: $y = ax^2 + bx + c$

Put (0, 0) on it

We get c = 0

$\therefore y = ax^2 + bx$

at (1, 0) and (-2, 4)



We get; a + b = 0 and 4a - 2b = 4

Thus, $a = \frac{2}{3}, b = \frac{2}{3}$

Hence, $3y = 2x^2 - 2x$

Ques. If vertex of a parabola is (2, - 1) and the equation of its directrix is 4x - 3y = 21, then the length of its latus rectum is.

(JEE Main 2022)

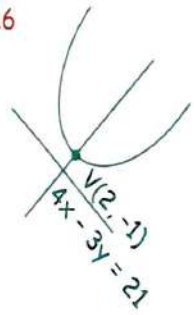
(a) 2 (b) 8

(c) 12 (d) 16

Ans. $a = \left|\frac{4 \times 2 - 3(-1) - 21}{\sqrt{4^2 + 3^2}}\right|$

$\Rightarrow a = \left|\frac{8 + 3 - 21}{5}\right|$

$\Rightarrow a = \left|\frac{-10}{5}\right| \Rightarrow a = 2$



$\therefore$ latus rectum = 4a = 8

Ques. If the length of the latus rectum of a parabola, whose focus is (a, a) and the tangent at its vertex is x + y = a, is 16, then |a| is equal to:

(JEE Main 2022)

(a) $2\sqrt{2}$ (b) $2\sqrt{3}$

(c) $4\sqrt{2}$ (d) 4

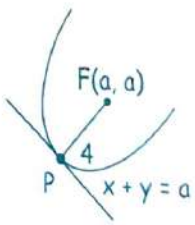
Ans. L.R = 4A = 16

$\Rightarrow A = 4$

and $\left|\frac{a + a - a}{\sqrt{2}}\right| = 4$

$\Rightarrow \left|\frac{a}{\sqrt{2}}\right| = 4$

$\Rightarrow a = \pm 4\sqrt{2} \Rightarrow |a| = 4\sqrt{2}$



Ques. Find the equation of a projectile fired from origin which attains a maximum height of 10 m after a horizontal displacement of 4 m.

Ans. Let the equation of parabola be

$(x - 4)^2 = 4a(y - 10)$

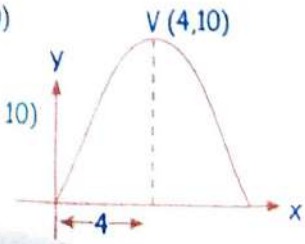
$\because$ (0, 0) satisfies

$\therefore (0 - 4)^2 = 4a(0 - 10)$

$\Rightarrow 16 = 4a(-10)$

$\Rightarrow 4a = -\frac{16}{10} = -\frac{8}{5}$

$\therefore (x - 4)^2 = -\frac{8}{5}(y - 10)$



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Ques. Find the equation of a parabola whose ends of the latus rectum are given by (3, 4) and (4, 3)

(a) $\left(x-\frac{7}{2}\right)^2+\left(y-\frac{7}{2}\right)^2=\left(\frac{x+y-6}{2}\right)^2$

(b) $\left(x-\frac{7}{2}\right)^2+\left(y-\frac{7}{2}\right)^2=\left(\frac{x+y-8}{\sqrt{2}}\right)^2$

(c) $\left(x-\frac{7}{2}\right)^2+\left(y-\frac{7}{2}\right)^2=\left(\frac{x+y-4}{2}\right)^2$

(d) None of these

Ans. Focus $=\left[\frac{3+4}{2},\frac{4+3}{2}\right]=\left[\frac{7}{2},\frac{7}{2}\right]$

$4a=\sqrt{1^2+1^2}$

$\because$ Eq^n of LL' : $x+y=7$

$\therefore$ Eqn of directrix : $x+y=k$

Now, NF = 2a

$4a=\sqrt{2} \Rightarrow 2a=\frac{1}{\sqrt{2}}$

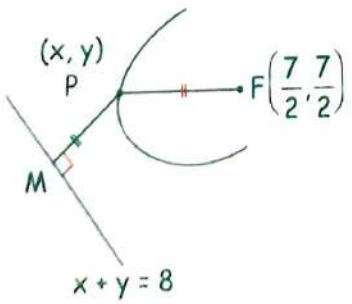
$\Rightarrow \left|\frac{k-7}{\sqrt{2}}\right|=\frac{1}{\sqrt{2}} \Rightarrow k-7=\pm 1$

$\Rightarrow k=8$ or 6

$\therefore x+y=8$ or $x+y=6$

Equation of Axis: $y-\frac{7}{2}=1\left(x-\frac{7}{2}\right)$

$y=x$

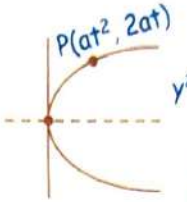


PF = PM

$\Rightarrow \sqrt{\left(x-\frac{7}{2}\right)^2+\left(y-\frac{7}{2}\right)^2}=\left|\frac{x+y-8}{\sqrt{2}}\right|$

$\Rightarrow \left(x-\frac{7}{2}\right)^2+\left(y-\frac{7}{2}\right)^2=\left(\frac{x+y-8}{\sqrt{2}}\right)^2$

PARAMETRIC COORDINATES



$\left.\begin{matrix}x=at^2\\y=2at\end{matrix}\right\}, t\in R$

Parameter

$x=at^2, y=2at$ are the parametric co-ordinates

Ques. The locus of the mid-point of the line segment joining the focus of the of the parabola $y^2=4ax$ to a moving point of the parabola, is another parabola whose directrix is: **(JEE Main 2021)**

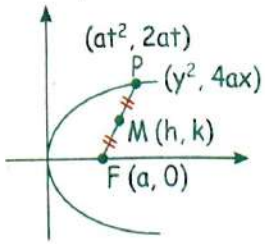
(a) $x=a$

(b) $x=0$

(c) $x=-\frac{a}{2}$

(d) $\frac{a}{2}$

Ans. $\because$ M is the mid point of PF.



$\therefore \frac{a+at^2}{2}=h, \frac{2at}{2}=K$

$\Rightarrow 2h=at^2+a$ and $2k=2at+0$

$\Rightarrow 2h=at^2+a$ and $t=\frac{k}{a}$

$\Rightarrow 2h=a\left(\frac{k}{a}\right)^2+a \Rightarrow 2h=\frac{k^2}{a}+a$

$2ax=y^2+a^2$

$\Rightarrow y^2=2ax-a^2 \Rightarrow y^2=2a\left(x-\frac{a}{2}\right)$

Comparing with $y^2=4aX$

$y^2=2ax$

Directrix: $X=-\frac{a}{2} \Rightarrow x-\frac{a}{2}=-\frac{a}{2} \Rightarrow x=0$

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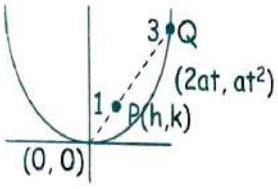
Ques. Let O be the vertex and Q be any point on the parabola $x^2 = 8y$. If the point P divides the line segment OQ internally in the ratio 1:3, then the locus of P is **(2015)**

(a) $x^2 = y$ (b) $y^2 = y$

(c) $y^2 = 2x$ (d) $x^2 = 2y$

Ans. $\because 8 = 4a \Rightarrow a = 2$

$\therefore Q = (4t, 2t^2)$



$$h = \frac{3\times 0 + 1\times 4t}{4}$$

$$\Rightarrow h = \frac{4t}{4} \Rightarrow h = t$$

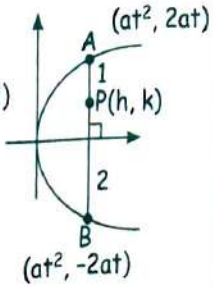
$$\text{and } k = \frac{3\times 0 + 1\times 2t^2}{4} \Rightarrow k = \frac{2t^2}{4} \Rightarrow k = \frac{t^2}{2}$$

$$\therefore k = \frac{h^2}{2} \Rightarrow x^2 = 2y$$

Ques. Find the locus of point of trisection of all the double ordinates of the parabola $y^2 = 4ax$.

Ans. $$h = \frac{1\times at^2 + 2\times at^2}{3}$$

$$\Rightarrow h = \frac{3at^2}{3} \Rightarrow h = at^2 \quad ...(i)$$



$$\text{and } k = \frac{1\times(-2at) + 2(2at)}{1+2}$$

$$\Rightarrow k = \frac{-2at + 4at}{3}$$

$$\Rightarrow k = \frac{2at}{3} \quad ...(ii)$$

$$\Rightarrow t = \frac{3k}{2a}$$

By eqⁿ (i) and (ii)

$$h = a\left(\frac{3k}{2a}\right)^2 \Rightarrow h = \frac{9k^2}{4a} \Rightarrow 9y^2 = 4ax$$

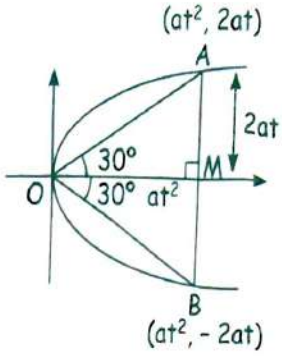
Ques. An equilateral Triangle is inscribed in a parabola $y^2 = 4ax$ such that one vertex of the TRIANGLE is same as vertex of the parabola. Find the area of this equilateral Triangle.

Ans. In ΔOAM; $\tan 30° = \frac{AM}{OM}$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{2at}{at^2}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{2}{t}$$

$$\Rightarrow t = 2\sqrt{3}$$



$\therefore A\left[12a, 4\sqrt{3}a\right]$

and $B\left[12a, -4\sqrt{3}a\right]$

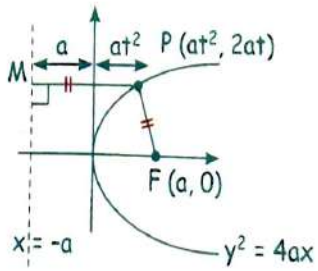
Area of ΔOAB: $\frac{\sqrt{3}}{4}s^2$

$\because s = 8\sqrt{3}a$

$\Rightarrow A = 48\sqrt{3}a^2$

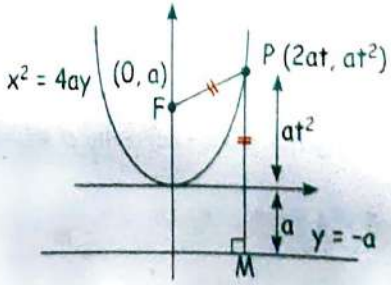
FOCAL DISTANCE

$PF = PM = |a + at^2|$



$PF = |a + x_1|$

$PM = PF = |a + at^2| = |a + y|$



$\therefore$ **Focal distance = PF = $|a + at^2|$**

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POWER OF A POINT

$P(x_1, y_1) \rightarrow$

For $y^2 = 4ax$

$S: y_1^2 - 4ax_1$

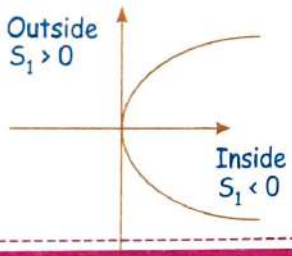
If $y_1^2 - 4ax_1 > 0 \Rightarrow P(x_1, y_1)$ lies outside

If $y_1^2 - 4ax_1 < 0 \Rightarrow P(x_1, y_1)$ lies inside

If $y_1^2 - 4ax_1 = 0 \Rightarrow P(x_1, y_1)$ lies on the parabola

For $x^2 = 4ay$

$S_1: x_1^2 - 4ay_1$



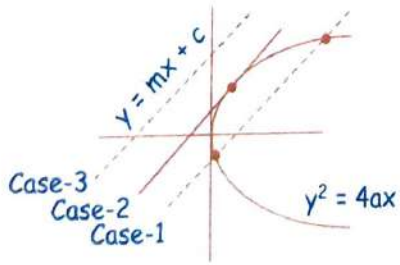
LINE AND A PARABOLA

Put $y = mx + c$ in the parabola $y^2 = 4ax$

$(mx + c)^2 = 4ax$

$\Rightarrow m^2x^2 + 2mxc + c^2 - 4ax = 0$

$\Rightarrow m^2x^2 + 2x(mc - 2a) + c^2 = 0$



Case-1: $D > 0$

Case-2: $D = 0 \rightarrow$ For condition of tangency

Ccase-3: $D < 0$

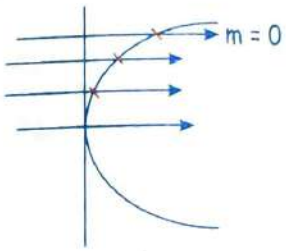
For tangency $4(mc - 2a)^2 - 4m^2c^2 = 0$

$\Rightarrow 4a^2 = 4acm \Rightarrow a = cm \Rightarrow c = \frac{a}{m}$

$\therefore y = mx + \frac{a}{m}$ is a Tangent

Important Note

1. Parallel tangents can't be drawn to parbola
2. Parabola to axis of parabola, no tangent can be drawn



EQUATION OF TANGENTS TO $y^2 = 4ax$

1. Slope form: $y = mx + \frac{a}{m}$
2. Cartesian form:

   $x^2 \rightarrow xx_1, y^2 \rightarrow yy_1$

   $x \rightarrow \left(\frac{x + x_1}{2}\right), y \rightarrow \frac{y + y_1}{2}$

   $\therefore yy_1 = 4a\left(\frac{x_1 + x_1}{2}\right)$

   $\Rightarrow \boxed{yy_1 = 2a(x + x_1)}$
3. Parametric form:

   put $x_1 = at^2, y_1 = 2at$

   in $yy_1 = 2a(x + x_1)$

   $\therefore y(2at) = 2a(x + at^2)$

   $\Rightarrow ty = x + at^2$

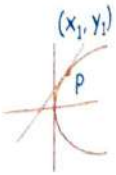
Proof: $y^2 = 4ax$

"Differentiate w.r.t.x"

$2y\frac{dy}{dx} = 4a \times 1$

$\Rightarrow \frac{dy}{dx} = \frac{4a}{2y} = \frac{2a}{y}$

$\Rightarrow m_T = \frac{2a}{y_1}$



$\therefore$ Equation of tangent: $y - y_1 = \frac{2a}{y_1}(x - x_1)$

Also: $y_1^2 = 4ax_1$

$yy_1 - 4ax_1 = 2ax - 2ax_1$

$\Rightarrow yy_1 = 2ax + 2ax_1 \Rightarrow yy_1 = 2a(x + x_1)$

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Important Note

Equation of Tangents to $y^2 = 4ax$

1. Slope form of Tangent: $y = mx + \frac{a}{m}$ at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$
2. Cartesian form of Tangent at (x_1, y_1): $yy_1 = 2a(x + x_1)$
3. Parametric form of Tangents: $ty = x + at^2$

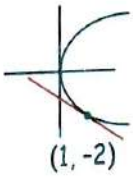
$\rightarrow$ slope $(m) = \frac{1}{t}$

Ques. If the parabola $y^2 = 4ax$ passes through the point $(1, -2)$, then the tangent at this point is

(a) $x + y - 1 = 0$ (b) $x - y - 1 = 0$

(c) $x + y + 1 = 0$ (d) $x - y + 1 = 0$

Ans. Put $(1,-2)$ in the parabola



$\therefore (-2)^2 = 4a \times 1$

$\Rightarrow 4 = 4a, \Rightarrow a = 1, y^2 = 4x$

Eqⁿ of tangent

$yy_1 = 2a(x + x_1)$

$\Rightarrow y(-2) = 2(x + 1)$

$\Rightarrow -y = x + 1$

Ques. The equation of the tangent to the parabola $y^2 = 16x$, which is perpendicular to the line $y = 3x + 7$ is

(a) $y - 3x + 4 = 0$ (b) $3y - x + 36 = 0$

(c) $3y + x - 36 = 0$ (d) $3y + x + 36 = 0$

Ans. $\because m = 3$

$\therefore m_T = -\frac{1}{3}$

$\Rightarrow y = mx + \frac{a}{m}$

$\Rightarrow y = -\frac{1}{3}x + \frac{4}{-\frac{1}{3}} \Rightarrow y = -\frac{x}{3} - 12$

$\Rightarrow 3y = -x - 36 \Rightarrow 3y + x + 36 = 0$

Ques. Find the equation of the tangents to the parabola $y^2 = 9x$ which pass through the point $(4, 10)$. Also find the point of tangency.

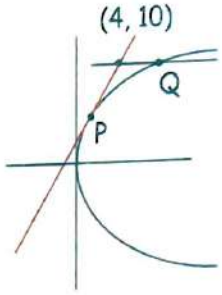
Ans. $y^2 - 9x = 0$

$S_1: 10^2 - 9 \times 4 = 64 > 0$

$\because S_1 > 0 \Rightarrow (4, 10)$ outside

Let the tangent by $ty = x + at^2$

$\Rightarrow ty = x + \frac{9t^2}{4}$



$\because (4, 10)$ satisfies

$\Rightarrow 10t = 4 + \frac{9t^2}{4}$

$\Rightarrow 40t = 16 + 9t^2$

$\Rightarrow 9t^2 - 40t + 16 = 0$

$\Rightarrow 9t^2 - 36t - 4t + 16 = 0$

$\Rightarrow (9t - 4)(t - 4) = 0$

$\Rightarrow t = \frac{4}{9}$ or $t = 4$

$\therefore y = \frac{x}{\left(\frac{4}{9}\right)} + \frac{9}{4} \times \frac{4}{9} \quad \left(\text{at } t = \frac{4}{9}\right)$

$\Rightarrow y = \frac{9}{4}x + 1$

$\Rightarrow y - 10 = \frac{9}{4}(x - 4)$

or $y = x + \frac{9}{4} \quad (\text{at } t = 1)$

Ques. For the parabola $y^2 = 8x$, find the angle between the tangents drawn form the point $(-1, -3)$.

Ans. $y^2 - 8x = 0$

$S_1: (-3)^2 - 8(-1)$

$S_1: 9 + 8 > 0 \Rightarrow$ outside

Let tangent be $y = mx + \frac{a}{m}$

$\Rightarrow y = mx + \frac{2}{m}$

at $(-1, -3)$

$-3 = -m + \frac{2}{m}$

$\Rightarrow -3 = \frac{-m^2 + 2}{m} \Rightarrow -3m = -m^2 + 2$

$\Rightarrow m^2 - 3m - 2 = 0 \rightarrow m_1, m_2$

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$$|m_1 - m_2| = \frac{\sqrt{D}}{a} = \frac{\sqrt{9+8}}{1} = \sqrt{17}$$

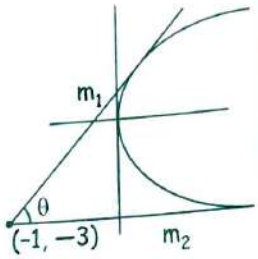
$m_1 + m_2 = 3$

$m_1 m_2 = -2$

$$\therefore \tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

$$\Rightarrow \tan\theta = \left|\frac{\sqrt{17}}{1 + (-2)}\right|$$

$$\Rightarrow \tan\theta = \sqrt{17}$$



Ques. Show locus of the point from which two tangents are drawn to $y^2 = 4ax$ such that slope of one is twice of other is a parabola. Find its length of latus rectum.

Ans. $y = mx + \frac{a}{m}$

To find the locus: $(x,y) \Rightarrow (h,k)$

$$\therefore k - mh = \frac{a}{m}$$

$$\Rightarrow km - m^2h = a$$

$$m^2h - km + a = 0 \quad (\alpha, 2\alpha)$$

$$\alpha + 2\alpha = \frac{k}{h} \Rightarrow 3\alpha = \frac{k}{h} \Rightarrow \alpha = \frac{k}{3h}$$

and $2\alpha^2 = \frac{a}{h}$

$$\Rightarrow 2\left(\frac{k}{3h}\right)^2 = \frac{a}{h}$$

$$\Rightarrow 2k^2 = 9ah \Rightarrow 2y^2 = 9ax$$

$\rightarrow$ parabola $y^2 = \frac{9a}{2}x$

Length of latus rectum $= \frac{9a}{2}$

Equation of Common Tangents

Ques. Find the equation of common tangents of the parabola $y^2 = 8ax$ and the circle $x^2 + y^2 = 2a^2$.

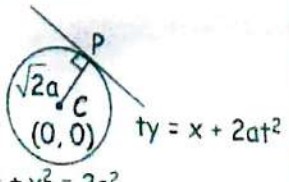
Ans. Eqⁿ of tangent of the parabola:

$ty = x + At^2$

$\because A = 2a$

$\Rightarrow ty = x + 2at^2$

$CP = \sqrt{2}a$



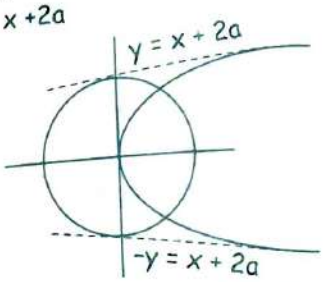
$x^2 + y^2 = 2a^2$

$$\Rightarrow \left|\frac{2at^2}{\sqrt{t^2+1}}\right| = \sqrt{2}a \Rightarrow \left(\sqrt{2}t^2\right)^2 = t^2 + 1,$$

$\Rightarrow 2t^4 - t^2 - 1 = 0$

Hence, $y = x + 2a$

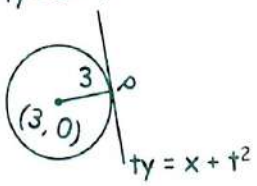
or $-y = x + 2a$

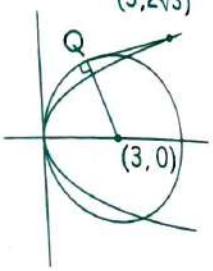


Ques. A line is a common tangent to the circle $(x-3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$. If the two points of contact (a, b) and (c, d) are distinct and lie in the first quadrant, then $2(a + c)$ is equal to **(JEE Main 2021)**

Ans. Eqⁿ of parabola:

$ty = x + t^2$





$$\therefore \left|\frac{3 + t^2}{\sqrt{1 + t^2}}\right| = 3$$

$\Rightarrow (3 + t^2)^2 = 9(1 + t^2) \Rightarrow 9 + t^4 + 6t^2 = 9 + 9t^2$

$\Rightarrow t^4 = 3t^2 \Rightarrow t = 0$ or $t^2 = 3 \Rightarrow t = \pm\sqrt{3}$

$\because t \neq 0, -\sqrt{3}$

$\therefore t = \sqrt{3} \Rightarrow P(at^2, 2at) = (3, 2\sqrt{3})$

Thus, $\sqrt{3}y = x + 3 \Rightarrow y = \frac{x+3}{\sqrt{3}}$

$$\therefore (x-3)^2 + \frac{(x+3)^2}{3} = 9$$

$$\Rightarrow x^2 + 9 - 6x + \frac{x^2 + 9 + 6x}{3} = 9$$

$\Rightarrow 3x^2 - 18x + x^2 + 9 + 6x = 0$

$\Rightarrow 4x^2 - 12x + 9 = 0 \Rightarrow (2x-3)^2 = 0 \Rightarrow x = \frac{3}{2}$

Hence, $(a,b) = \left(\frac{3}{2}, \frac{9}{2\sqrt{3}}\right)$ and $(3, 2\sqrt{3})$

$\Rightarrow 2(a + c) = 9$

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EQUATION OF TANGENTS TO $x^2 = 4ay$

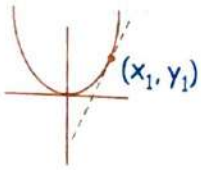
1. Slope Form of Tangent: $\boxed{y = mx - am^2}$

2. Cartesian form of Tangent (x_1, y_1):

$\because x^2 = 4ay$

$\Rightarrow xx_1 = 4a\frac{(y + y_1)}{2}$

$\Rightarrow xx_1 = 2a(y + y_1)$



3. Parametric form of Tangent:

Put $x_1 = 2at, y_1 = at^2$

$x(2at) = 2a(y + at^2)$

$\Rightarrow \boxed{tx = y + at^2}$

$m = t$ (slope)

Ques. Find the equation of the common tangent to the parabola $y^2 = 4x$ and $x^2 = 4y$.

Ans. $y = mx + \frac{1}{m}$ ...(i)

$y = mx - m^2$ ...(ii)

By eqⁿ (i) & (ii)

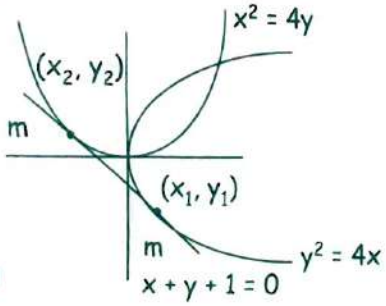
$\frac{1}{m} = -m^2$

$\Rightarrow m^3 = -1$

$\Rightarrow m = -1$

$\therefore y = -x - 1$

$\Rightarrow y + x + 1 = 0$



Ques. The distance of the point $(6, -2\sqrt{2})$ from the common tangent $y = mx + c$, $m > 0$, of the curves $x = 2y^2$ and $x = 1 + y^2$ is **(JEE Main 2023)**

(a) $\frac{1}{3}$ (b) 5

(c) $\frac{14}{3}$ (d) $5\sqrt{3}$

Ans. $y^2 = \frac{x}{2}$;

$\therefore 4a = \frac{1}{2}, \Rightarrow a = \frac{1}{8}$

Equation of Tangent: $y = mx + \frac{a}{m}$

$\Rightarrow y = mx + \frac{1}{8m}$ ...(i)

Also $y^2 = (x - 1)$,

$\therefore 4a = 1 \Rightarrow a = \frac{1}{4}$

$\therefore y = m(x - 1) + \frac{a}{m}$

$\Rightarrow y = m(x - 1) + \frac{1}{4m}$

$\Rightarrow y = mx - m + \frac{1}{4m}$ ...(ii)

By eqⁿ (i) and (ii)

$\frac{1}{8m} = -m + \frac{1}{4m}$,

$\Rightarrow \frac{1}{8m} = -m + \frac{2}{8m} \Rightarrow m = \frac{1}{8m}$

$\Rightarrow m^2 = \frac{1}{8} \Rightarrow m = \pm\frac{1}{2\sqrt{2}}$

Since, $m > 0$

$\therefore m = \frac{1}{2\sqrt{2}}$

Common tangent,

$y = \frac{1}{2\sqrt{2}}x + \frac{\sqrt{2}}{4}$

Distance of the point $(6, -2\sqrt{2})$ from the common tangent, $= \left|\frac{\frac{6}{2\sqrt{2}} + 2\sqrt{2} + \frac{\sqrt{2}}{4}}{\sqrt{\frac{1}{8} + 1}}\right| = 5$ unit.

Ques. The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the points P, Q and the parabola at the points R, S. Then the area of the quadrilateral PQRS is

(a) 3 (b) 6

(c) 9 (d) 15

Ans. $ty = x + at^2 \Rightarrow ty = x + 2t^2$

$\because OP = r = \sqrt{2}$

$\therefore \left|\frac{2t^2}{\sqrt{1 + t^2}}\right| = \sqrt{2} \Rightarrow \left|\frac{\sqrt{2}t^2}{\sqrt{1 + t^2}}\right| = 1$

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$\Rightarrow 2t^4 = 1 + t^2$

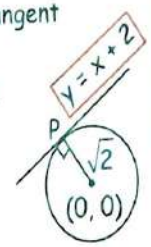
$\Rightarrow 2t^4 - t^2 - 1 = 0 \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$

$\therefore$ R $[at^2, 2at] \Rightarrow$ R $[2, 4]$

$\because$ Eqⁿ of the tangent:

$xx_1 + yy_1 = 2$

and $y - x = 2$ is also the eqⁿ of tangent



$\therefore \frac{x_1}{-1} = \frac{y_1}{1} = \frac{2}{2} \Rightarrow x_1 = -1, y_1 = 1$

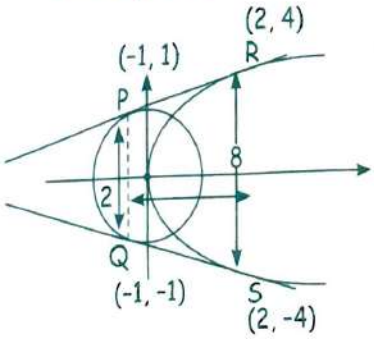
$m_{OP} = -1,$

$\Rightarrow \tan\theta = -1 \Rightarrow \theta = 135°$

$\therefore P\left[0 + \sqrt{2}\cos 135°, 0 + \sqrt{2}\sin 135°\right]$

$\Rightarrow P\left[\sqrt{2}\left(-\frac{1}{\sqrt{2}}\right), \sqrt{2} \times \frac{1}{\sqrt{2}}\right] \Rightarrow P[-1, 1]$

Similarly, Q $[-1, -1]$



Area of trap = $\frac{1}{2}(b_1 + b_2) \cdot h$

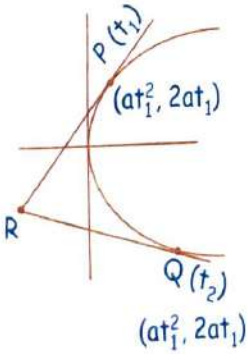
$= \frac{1}{2}(8 + 2) \times 3$

$= \frac{10}{2} \times 3$

$= 15$

Point of Intersection of Tangents Drawn at $P(t_1)$ and $Q(t_2)$

R $[at_1 t_2, a(t_1 + t_2)]$



Equation of PR: $t_1 y = x + at_1^2$ ...(i)

Equation of QR: $t_2 y = x + at_2^2$ ...(ii)

By eqⁿ (i) and (ii)

$y(t_1 - y_2) = a(t_1^2 - t_2^2)$

$\Rightarrow y(t_1 - t_2) = a(t_1 - t_2)(t_1 + t_2) \Rightarrow y = a(t_1 + t_2)$

OR

$X_R = \sqrt{at_1^2 \cdot at_2^2} = |at_1 t_2|$

and $Y_R = \frac{2at_1 + 2at_2}{2} = a(t_1 + t_2)$

Important Note

Point of intersection of tangents drawn at $P(t_1)$ and $Q(t_2)$ is given by R $[at_1 t_2, a(t_1 + t_2)]$
The abscissa of R is GM of abscissa of P and Q
The ordinate of R is AM of ordinates of P and Q.

Ques. The point of intersection of tangents at the ends of the latus-rectum of the parabola $y^2 = 4x$ is equal to

(a) (1, 0) (b) (-1, 0)

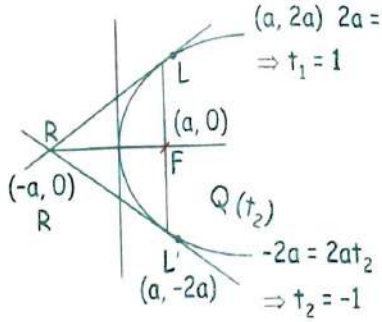
(c) (0, 1) (d) (0, -1)

Ans. $\because a = 1$

$\therefore$ R $[at_1 t_2, a(t_1 + t_2)]$

$=$ R $[a(1)(-1), a(1 - 1)]$

$=$ R $[-a, 0] =$ R $[-1, 0]$



Ques. Two tangents of the parabola $y^2 = 8x$, meet the tangent at its vertex in the points P and Q. If PQ = 4 then the locus of the point of intersection of the two tangents.

Ans. $h = at_1 t_2 = 2t_1 t_2$ ...(i)

$k = a(t_1 + t_2) = 2(t_1 + t_2)$ ...(ii)

P $[0, at_1]$

$= [0, 2t_1]$

Q $[0, at_2]$

$= [0, 2t_2]$

PQ = 4

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$\Rightarrow 2|t_1 - t_2| = 4$

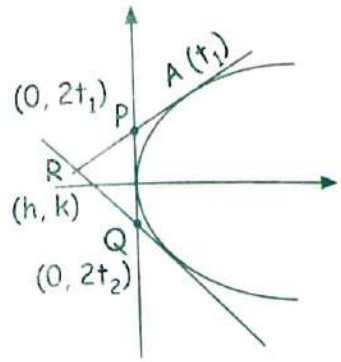
$\Rightarrow |t_1 - t_2| = 2$

$\Rightarrow (t_1 - t_2)^2 = 4$

$\Rightarrow (t_1 + t_2)^2 - 4t_1t_2 = 4$

By eqn (i) & (ii)

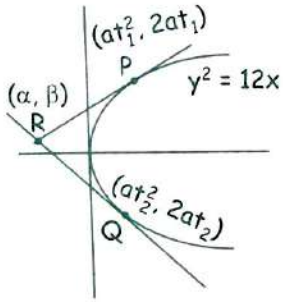
$\left(\frac{k}{2}\right)^2 - \frac{4h}{2} = 4$



$\Rightarrow \frac{k^2}{4} - 2h = 4 \Rightarrow k^2 - 8h = 16 \Rightarrow y^2 - 8x = 16$

Ques. The ordinates of the points P and Q on the parabola with focus (3, 0) and directrix $x = -3$ are in the ratio 3 : 1. If R (α, β) is the point of intersection of the tangents to the parabola at P and Q, then $\frac{\beta^2}{\alpha}$ is equal to **(JEE Main 2023)**

Ans. $\frac{2at_1}{2at_2} = \frac{3}{1} \Rightarrow t_1 = 3t_2$



$\therefore \alpha = at_1t_2, \quad \beta = a(t_1 + t_2)$

$\Rightarrow \alpha = 3.\ 3t_2 \cdot t_2$ and $\quad \beta = 3(3t_2 + t_2)$

$\Rightarrow \alpha = 9t_2^2 \quad \beta = 12t_2$

$$\frac{\beta^2}{\alpha} = \frac{(12t_2)^2}{9t_2^2} = \frac{(12)^2}{9} = \left(\frac{12}{3}\right)^2 = 16$$

CHORD JOINING 2 POINTS

Chord joining $P(t_1)$ and $Q(t_2)$ is given by

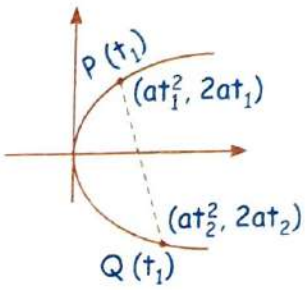
$2x - (t_1 + t_2)y + 2at_1t_2 = 0$

Proof: $m_{PQ} = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2} = \frac{2a(t_2 - t_1)}{a(t_2 - t_1)(t_2 + t_1)}$

$\Rightarrow m = \frac{2}{t_1 + t_2}$

Equation of chord: $y - 2at_1 = \frac{2}{t_1 + t_2}(x - at_1^2)$

$\Rightarrow 2x - (t_1 + t_2)y + 2at_1t_2 = 0$



Condition for Focal Chord

If the Chord Joining $P(t_1)$ and $Q(t_2)$ is a focal Chord then $t_1t_2 = -1$

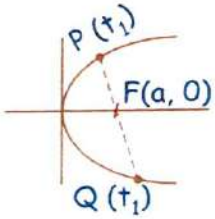
Proof: $2x - (t_1 + t_2)y + 2at_1t_2 = 0$

$\because$ (a, 0) satisfies

$\therefore 2a + 2at_1t_2 = 0$

$\Rightarrow 2a(1 + t_1t_2) = 0$

$\Rightarrow t_1t_2 = -1$



Important Note

Tangent drawn at the ends of any focal chord are always Perpendicular and they always meet on the Directrix.

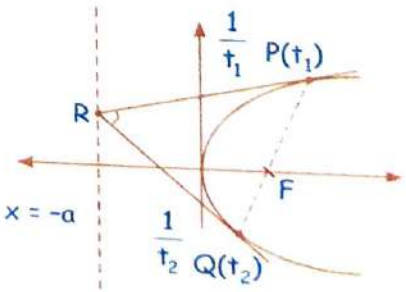
Proof: Given: $t_1t_2 = -1$ and $R[at_1t_2, (at_1 + t_2)]$

$\therefore X_R = -a$

$m_{PR} = \frac{1}{t_1}$, and $m_{QR} = \frac{1}{t_2}$

$m_{PR} \cdot m_{QR} = \frac{1}{t_1t_2} = \frac{1}{-1} = -1$

$\therefore$ PR and QR are perpendicular to each other



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Ques. If the line $y = 3x + c$ touches the parabola $y^2 = 12x$ at point p, then find the equation of the tangent at point Q, where PQ is a focal chord.

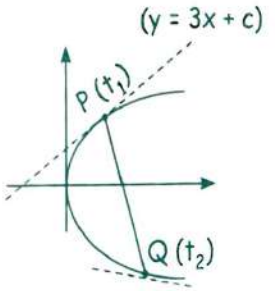
Ans. $t_1 t_2 = -1$

$\because t_1 = \frac{1}{m} = \frac{1}{3}$

$\Rightarrow \frac{1}{3} \times t_2 = -1 \Rightarrow t_2 = -3$

Equation of tangent: $t_2 y = x + at_2^2 \Rightarrow -3y = x + 3(-3)^2$

$\Rightarrow -3y = x + 27$



EQUATION OF NORMALS TO $y^2 = 4ax$

1. Cartesian form of Normal at (x_1, y_1):

$\because$ Equation of Tangent: $yy_1 = 2a(x + x_1)$

$m_T = \frac{2a}{y_1} \Rightarrow m_n = -\frac{y_1}{2a}$

$\therefore$ Equation of normal: $y - y_1 = -\frac{y_1}{2a}(x - x_1)$

2. Parametric form of Normal:

$\because x_1 = at^2, y_1 = 2at$

$\therefore y - 2at = -\frac{2at}{2a}(x - at^2)$

$\Rightarrow y - 2at = -t(x - at^2)$

$\Rightarrow y - 2at = -tx + at^3$

$\Rightarrow y + tx = 2at + at^3$

3. Slope Form of Normal:

$\because m = -t \Rightarrow t = -m$

$\therefore y + tx = 2at + at^3$

$\Rightarrow y - mx = 2a(-m) + a(-m)^3$

$\Rightarrow y = mx - 2am - am^3$

Ques. Find the equation of Normal to $y^2 = 16x$ at the lower end of the latus rectum

Ans. $\because$ y-coordinate of the lower end of latus rectum: $-2a = 2at$

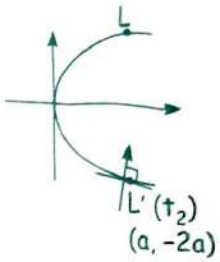
$\Rightarrow t = -1$

$y + tx = 2at + at^3$

$y - x = 2 \times 4(-1) + 4(-1)^3$

$\Rightarrow y = x - 8 - 4$

$\Rightarrow y = x - 12$



Ques. Find the equation of Normal to $y^2 = 8x$ at the upper end of the double ordinate whose length is 16.

Ans. $4at = 16$

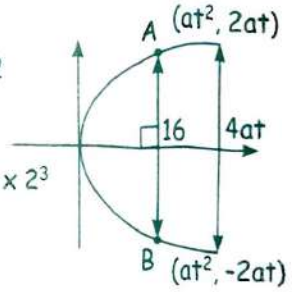
$\Rightarrow 4 \times 2t = 16 \Rightarrow t = 2$

$\therefore y + tx = 2at + at^3$

$y + 2x = 2 \times 2 \times 2 + 2 \times 2^3$

$\Rightarrow y + 2x = 8 + 16$

$\Rightarrow y + 2x = 24$



Important Note

For parabola maximum of 3 normals can be drawn passing a given pt.

Ques. The normal to the parabola $y^2 = 4ax$ from the point (5a, 2a) are

Ans. The normal to the parabola $y^2 = 4ax$ from the point (5a, 2a) are

$y + tx = 2at + at^3$

$\Rightarrow 2a + t(5a) = 2at + at^3$

$\Rightarrow 2a + 5at = 2at + at^3$

$\Rightarrow 2 + 5t = 2t + t^3$

$\Rightarrow t^3 - 3t - 2 = 0$

$\therefore t = -1, -1, 2$

$\therefore y - x = -2a - a$ [at $t = -1$]

$\Rightarrow y = x - 3a$

and $y + 2x = 4a + 8a$ [at $t = 2$]

$\Rightarrow y + 2x = 12a$

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EQUATION OF NORMALS TO $x^2 = 4ay$

1. Cartesian form of Normal at (x_1, y_1):

$\therefore$ Eqn of the tangent: $xx_1 = 2a(y + y_1)$

$\Rightarrow y + y_1 = \frac{xx_1}{2a}$; $\therefore m = \frac{x_1}{2a}$

$\Rightarrow m_N = \frac{-2a}{x_{-2a}}$

2. Parametric form of Normal:

put $x_1 = 2at$, $y_1 = at^2$

$x + ty = 2at + at^3$

3. Slope Form of Normal:

$y = mx + 2a + \frac{a}{m^2}$

$\boxed{y - y_1 = -\frac{2a}{x_1}(x - x_1)}$

Ques. Let a line perpendicular to $2x - y = 10$ touches the parabola $y^2 = 4(x - 9)$ at the point P, the distance of the point P from the centre of the circle $x^2 + y^2 - 14x - 8y + 56 = 0$ is

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Ans. $y = m(x - 9) + \frac{1}{m}$

$\therefore y = -\frac{1}{2}(x - 9) + \frac{1}{-\frac{1}{2}}$

$\Rightarrow y = -\frac{1}{2}(x - 9) - 2 \Rightarrow y = -\frac{x}{2} + \frac{9}{2} - 2$

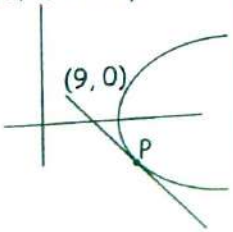
$\Rightarrow y = \frac{-x}{2} + \frac{5}{2} \Rightarrow 2y + x = 5 \Rightarrow x = 5 - 2y$

Thus, $y^2 = 4(5 - 2y - 9)$

$\Rightarrow y^2 = 4(-2y - 4)$

$\Rightarrow y^2 + 8y + 16 = 0$

$\Rightarrow (y + 4)^2 = 0$



$\Rightarrow y = -4$ and $x = 13$

$\therefore P(13, -4)$ and $C(7, 4)$

Therefore, $PC = \sqrt{(13-7)^2 + (-4-4)^2} = 10$

Ques. Prove that the two parabolas $y^2 = 4ax$ and $y^2 = 4c(x - b)$ cannot have common normal, other than the X axis unless $b/(a - c) > 2$.

Ans. Slope form of normals:

$y = mx - 2am - am^3$ ...(i)

and $y = m(x - b) - 2cm - cm^3$

$\Rightarrow y = mx - bm - 2cm - cm^3$ ...(ii)

Compare (i) and (ii)

$-2am - am^3 = -bm - 2cm - cm^3$

$\Rightarrow 2am + am^3 = bm + 2cm + cm^3$

$\because m = 0$ is the solution of the eqn.

$\therefore$ x-axis is the common normal.

$\Rightarrow 2a + am^2 = b + 2c + cm^2$ $\boxed{m = 0 \Rightarrow y = 0 \Rightarrow \text{x- axis}}$

$am^2 - cm^2 = b + 2c - 2a$

$\Rightarrow m^2 = \frac{b + 2c - 2a}{a - c}$

$\Rightarrow \frac{b + 2c - 2a}{a - c} > 0 \Rightarrow \frac{b - 2(a - c)}{a - c} > 0 \left[\because m^2 > 0\right]$

$\Rightarrow \frac{b}{a - c} - 2 > 0 \Rightarrow \frac{b}{a - c} > 2$

Ques. The normal $y = mx - 2am - am^3$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex if

(a) $m = 1$ (b) $m = \sqrt{2}$

(c) $m = -\sqrt{2}$ (d) $m = \frac{1}{\sqrt{2}}$

Ans. $y^2 = 4ax$ $\boxed{\text{Homogenized equation}}$

$y^2 - 4ax\left(\frac{y - mx}{-(2am + am^3)}\right) = 0$

$\Rightarrow y^2 + \frac{4ax(y - mx)}{a(2m + m^3)} = 0$

$\Rightarrow (2m + m^3)y^2 + 4x(y - mx) = 0$

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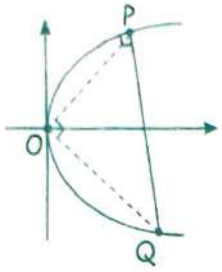
The normal subtends to a right angle

sum of multiples of x^2 and y^2 should be zero

$\therefore 2m + m^3 - 4m = 0 \Rightarrow m^3 = 2m \Rightarrow m^2 = 2$

$\Rightarrow m = \pm\sqrt{2}$



NORMAL CHORD

If the normal to the parabola $y^2 = 4ax$ at the point t_1, meets the parabola again at point t_2 then $t_2 = -t_1 - \frac{2}{t_1}$.

Proof: $m_{normal} = -t_1$ ...(i)

Equation of chord: $2x - (t_1 + t_2)y + 2at_1t_2 = 0$

$\therefore m_{chord} = \frac{2}{t_1 + t_2}$ ...(ii)

By equation (i) and (ii)

$$-t_1 = \frac{2}{t_1 + t_2} \Rightarrow t_1 + t_2 = \frac{2}{-t_1} \Rightarrow t_2 = -t_1 - \frac{2}{t_1}$$

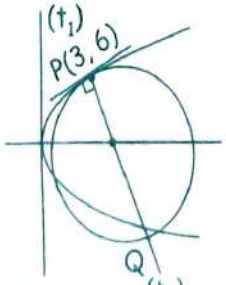
Ques. If the normal to $y^2 = 12x$ at P (3, 6) meets the parabola again at a point Q. Find the equation of circles with PQ as diameter.

Ans. $\because a = 3$ and $2at_1 = 6 \Rightarrow 6t_1 = 6 \Rightarrow t_1 = 1$

$\therefore t_2 = -t_1 - \frac{2}{t_1}$

$\Rightarrow t_2 = -1 - \frac{2}{1}$

$\Rightarrow t_2 = -3$



$\therefore Q[at_2^2, 2at_2] = [3(-3)^2, 2 \times 3(-3)]$

$= [27, -18]$

$\therefore$ Eqn of circle:

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

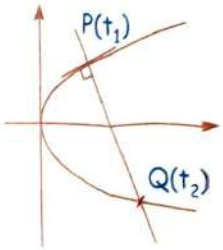
$\Rightarrow (x - 3)(x - 27) + (y - 6)(y + 18) = 0$

Important Note

If the normal to the parabola $y^2 = 4ax$ at point t_1 cuts the parabola again at point t_2 then prove that $t_2^2 \geq 8$.

Proof:

$$\boxed{t_2 = -t_1 - \frac{2}{t_1}}$$



$t_2^2 = \left(t_1 + \frac{2}{t_1}\right)^2$

$\Rightarrow t_2^2 = t_1^2 + \frac{4}{t_1^2} + 2t_1\frac{2}{t_1}$

$\Rightarrow t_2^2 = t_1^2 + \frac{4}{t_1^2} + 4$ ....(i)

$\because AM >, GM$

$$\Rightarrow \frac{t_1^2 + \frac{4}{t_1^2}}{2} \geq \sqrt{t_1^2\frac{4}{t_1^2}}$$

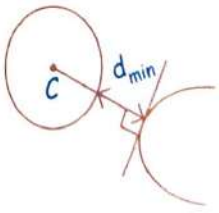
$$\Rightarrow \frac{t_1^2 + \frac{4}{t_1^2}}{2} \geq 2 \Rightarrow t_1^2 + \frac{4}{t_1^2} \geq 4$$

add 4: $t_1^2 + \frac{4}{t_1^2} + 4 \geq 8$

By eqn (i) $t_2^2 \geq 8$

Minimum Distance between 2 Curves

Minimum distance between 2 Curves always occurs along the line joining Common Normal.



Ques. The shortest distance between the curves $y^2 = 4x$ and $x^2 + y^2 - 24y + 128 = 0$, is $(a\sqrt{b} - c)$.

Find $a + c - b$.

Ans. Equation of normal at p(t)

$y + tx = 2at + at^3$

$\Rightarrow y + tx = 2t + t^3$

$\because$ Centre of the circle (0, 12) satifies

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$\Rightarrow 12 + t \times 0 = 2t + t^3 \Rightarrow t^3 + 2t - 12 = 0$

$\therefore t = 2$

$\therefore P[at^2, 2at] = [4, 4]$

Now, $d_{min} = PC - r$

$\because r = 4$

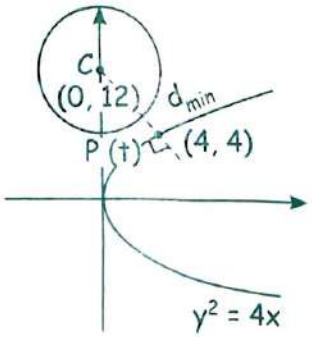
and $PC = \sqrt{4^2 + 8^2}$

$= \sqrt{80} = 4\sqrt{5}$

$\therefore d_{min} = 4(\sqrt{5} - 1)$

$\therefore a = 4, b = 5, c = 1$

$a + c - b = 0$



Ques. If the point (h, k) on the line $x + y = 4$ is nearest to the parabola $y^2 = 4(x - 9)$, then find the value of h and k.

Ans. $\because 2y\frac{dy}{dx} = 4 \Rightarrow \frac{dy}{dx} = \frac{2}{y}$

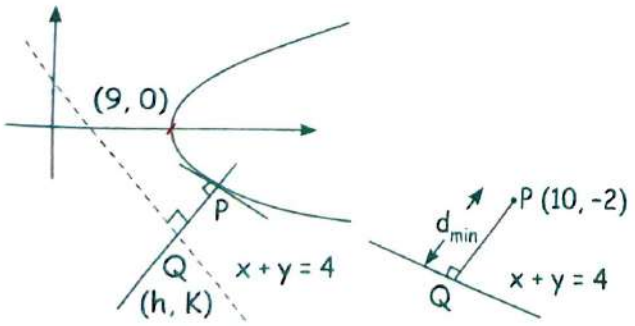
m_T at point P = -1

$\therefore \frac{dy}{dx} = -1 \Rightarrow \frac{2}{y} = -1 \Rightarrow y = -2$

$\therefore (-2)^2 = 4(x - 9)$

$\Rightarrow 1 = x - 9 \Rightarrow x = 10$

$\therefore P[10, -2]$



$\because$ Q is the foot of perpendicular

$$\therefore \frac{x-10}{1} = \frac{y+2}{1} = -\left|\frac{10-2-4}{1^2+1^2}\right|$$

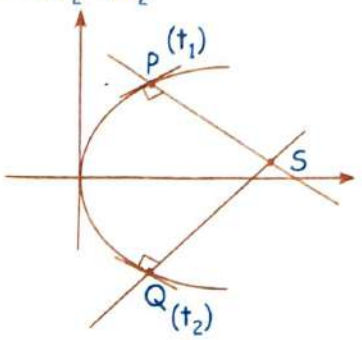
$\Rightarrow x - 10 = y + 2 = -2$

$\Rightarrow$ point = (8, -4)

Point of Intersection of Normals Drawn at $P(t_1)$ and $Q(t_2)$ to $y^2 = 4ax$

$y + t_1 x = 2at_1 + at_1^3$ ....(i)

$y + t_2 x = 2at_2 + at_2^3$ ....(ii)



By subtracting eqⁿ (ii) from eqⁿ (i)

$(t_1 - t_2)x = 2a(t_1 - t_2) + a(t_1^3 - t_2^3)$

$\Rightarrow (t_1 - t_2)x = 2a(t_1 - t_2) + a(t_1 - t_2)(t_1^2 + t_2^2 + t_1 t_2)$

$\Rightarrow x = 2a + a(t_1^2 + t_2^2 + t_1 t_2)$

$\Rightarrow x = a[t_1^2 + t_2^2 + t_1 t_2 + 2]$

Ques. For $y^2 = 8x$. Find locus of point of intersection of normals at the end points fo those chord which

(a) are also focal chord

(b) Subtend right angle at vertex.

Ans. (a) $h = a(t_1^2 + t_2^2 + t_1 t_2 + 2)$

$\because t_1 t_2 = -1$ ...(i)

$\therefore h = a(t_1^2 + t_2^2 + 1)$ ...(ii)

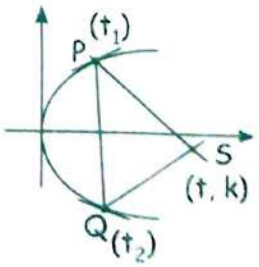
$k = -at_1 t_2 (t_1 + t_2)$

$\Rightarrow k = 2(t_1 + t_2) \Rightarrow t_1 + t_2 = \frac{k}{2}$ ...(iii)

By eqⁿ (ii)

$\Rightarrow \frac{h}{2} = t_1^2 + t_2^2 + 1$

$\frac{h}{2} = (t_1 + t_2)^2 - 2t_1 t_2 + 1$



By using eqⁿ. (i) & (ii)

$\frac{h}{2} = \left(\frac{k}{2}\right)^2 + 2 + 1$

$\Rightarrow \frac{h}{2} = \frac{k^2}{4} + 3 \Rightarrow \frac{h}{2} = \frac{k^2 + 12}{4} \Rightarrow 2h = k^2 + 12$

$2x = y^2 + 12$

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(b) $t_1t_2 = -4$

$\Rightarrow h = 2(t_1^2 + t_2^2 + t_1t_2 + 2)$

$\Rightarrow h = 2(t_1^2 + t_2^2 - 2)$

and $k = -at_1t_2(t_1 + t_2)$

$k = 4 \times 2(t_1 + t_2)$

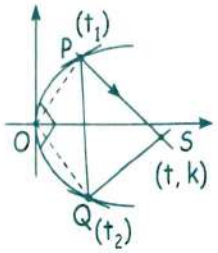
$\Rightarrow t_1 + t_2 = \frac{k}{8}$

$\therefore \frac{h}{2} = (t_1 + t_2)^2 - 2t_1t_2 - 2$

$\Rightarrow \frac{h}{2} = \left(\frac{k}{8}\right)^2 + 8 - 2$

$\Rightarrow \frac{h}{2} = \left(\frac{k}{8}\right)^2 + 6$

$\Rightarrow \frac{x}{2} = \left(\frac{y}{8}\right)^2 + 6$

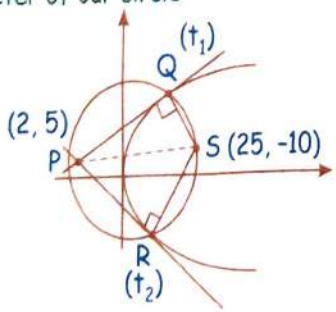


Brain Teaser

Tangent PQ and PR are drawn to $y^2 = 4x$ from the point P(2, 5) meeting parabola at Q and R respectively. Find the equation of circum-circle of triangle PQR.

Ans. Construct normals at points Q and R which meet at S,

Now PQSR is a cyclic quadulateral $\Rightarrow$ PS is diameter of our circle



$\because at_1t_2 = 2 \Rightarrow t_1t_2 = 2$

and $a(t_1 + t_2) = 5 \Rightarrow t_1 + t_2 = 5$

$\therefore S[(t_1^2 + t_2^2 + t_1t_2 + 2) - at_1t_2(t_1 + t_2)]$

$\Rightarrow S[25, -10]$

$(\because X_S = a(t_1^2 + t_2^2 + 2 + 2)((t_1 + t_2)^2 - 2t_1t_2 + 4 = 5^2 - 4 + 4) = 25$ and $y_s = at_1t_2(t_1 - t_2) = -2(3) = -10)$

$\therefore$ Eq^n the circle:

$(x - 2)(x - 25) + (y + 10)(y - 5) = 0$

Brain Teaser

The shortest distance between the parabola $y^2 = x - 1$ and $x^2 = y - 1$ is $\frac{3\sqrt{2}}{k}$, then the value of numercial quantity 'k' must be.

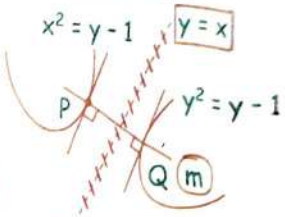
Ans. $y^2 = (x - 1)$; $a = \frac{1}{4}$

Two parabolas have symmetry about $y = x$

$\Rightarrow m_T = 1$

$\therefore y = m(x - 1) + \frac{a}{m}$

$\Rightarrow y = m(x - 1) + \frac{1}{4m}$



$\Rightarrow y = x - 1 + \frac{1}{4} \Rightarrow y = x - \frac{3}{4}$ ...(i)

also, $y - 1 = mx - am^2$

$y - 1 = 1 \times x - \frac{1}{4} \times 1^2 \Rightarrow y = x + \frac{3}{4}$ ...(ii)

$\therefore d = \left|\frac{\frac{3}{4} - \left(-\frac{3}{4}\right)}{\sqrt{1^2 + 1^2}}\right| = \frac{\frac{3}{2}}{\sqrt{2}} = \frac{3}{2\sqrt{2}} = \frac{3\sqrt{2}}{4}$

$\therefore k = 4$

$\because m_T = 1$

$y - 1 = x^2$

$\Rightarrow \frac{dy}{dx} = 2x = 1 \Rightarrow x = \frac{1}{2} \Rightarrow y - 1 = \left(\frac{1}{2}\right)^2 \Rightarrow y = \frac{5}{4}$

$\therefore P\left[\frac{1}{2}, \frac{5}{4}\right]$ and By symmetry $Q\left[\frac{5}{4}, \frac{1}{2}\right]$

Co-Normal Points/Concurrent Normals

The feet of 3 Concurrent normals are called co-normal Points

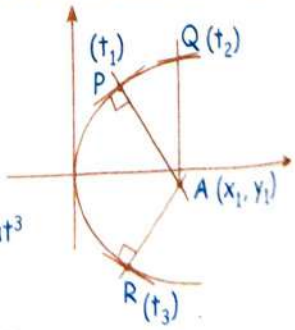
$y^2 = 4ax$

$y + tx = 2at + at^3$

(x_1, y_1) satisfies

$\therefore y_1 + tx_1 = 2at + at^3$

Cubic in 't'



P, Q and R are called co-normal points.

$at^3 + (2a - x_1)t - y_1 = 0$ $\rightarrow t_1, t_2, t_3$

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$m_1 = -t_1$

$m_2 = -t_2$

$+ \quad m_3 = -t_3$

$m_1 + m_2 + m_3 = -(t_1 + t_2 + t_3) = 0$

$t_1 + t_2 + t_3 = 0$

Properties of Co-Normal Points

1. If $P(t_1)$, $Q(t_2)$ and $R(t_3)$ are co-normal points of a parabola $y^2 = 4ax$ then $t_1 + t_2 + t_3 = 0$.
2. Algebraic sum of slopes of the three concurrent normal is zero.
3. Algebraic sum of ordinates of the three co-normal points on the parabola is zero.

   $\Rightarrow y_1 + y_2 + y_3 = 0$
4. Centroid of the Δ formed by three co-normal points lies on the axis of parabola (x-axis).

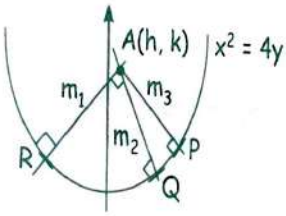
$$y_G = \frac{y_1 + y_2 + y_3}{3} = \frac{0}{3} = 0 \Rightarrow y_G = 0 \Rightarrow G \text{ has on X-axis}$$

Ques. Find the locus of the point of concurrency of 3 normals to the parabola $x^2 = 4y$ such that 2 of them are always at right angles to one another.

Ans. Slope form of Normal:

$$y = mx + 2a + \frac{a}{m^2}$$

$$y = mx + 2 + \frac{1}{m^2}$$



$$\therefore k = mh + 2 + \frac{1}{m^2}$$

$$\Rightarrow k = \frac{m^3h + 2m^2 + 1}{m^2} \Rightarrow m^3h + 2m^2 + 1 = km^2$$

$$\Rightarrow m^3h + (2 - k)m^2 + 1 = 0$$

$$m_1 m_2 m_3 = -\frac{1}{h}$$

$\because m_1 m_2 = -1$

$$\Rightarrow (-1)m_3 = -\frac{1}{h} \quad \Rightarrow m_3 = \frac{1}{h}$$

$$\therefore \left(\frac{1}{h}\right)^3 h + (2-k)\frac{1}{h^2} + 1 = 0 \Rightarrow \frac{1 + 2 - K + h^2}{h^2} = 0$$

$$\Rightarrow h^2 = k - 3$$

Ques. The parabola $y^2 = 4ax$ and the line $lx + my = n$ meet at two distinct points A and B. Now a normal is drawn to the parabola at these points A and B intersect at a point P. Now a third normal is drawn passing through the point P. Find the coordinates of feet of this 3rd normal.

Ans. $\because y_1 + y_2 + y_3 = 0 \Rightarrow y_3 = -(y_1 + y_2)$

$y^2 = 4ax$ and $lx + my = n \Rightarrow x = \left(\frac{n - my}{l}\right)$

$$y^2 = 4ax \Rightarrow \quad y^2 = 4a\left(\frac{n - my}{l}\right)$$

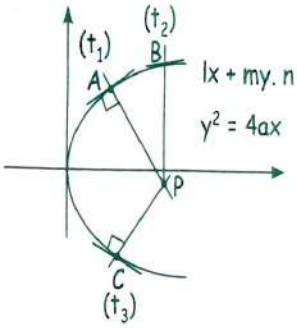
$ly^2 + 4amy - 4an = 0$ $\quad y_1, y_2$

$$y_1 + y_2 = -\frac{4am}{l} \Rightarrow y_3 = \frac{4am}{l},$$

Now $y_3^2 = 4ax_3$

$$\Rightarrow \left(\frac{4am}{l}\right)^2 = 4ax_3 \Rightarrow \frac{4am^2}{l^2} = x_3;$$

$$\therefore Q\left(\frac{4am^2}{l^2}, \frac{4am}{l}\right)$$



Important Note

Normals are drawn from (x_1, y_1) to $y^2 = 4ax$

(a) If these normals are real and distinct, then $x_1 > 2a$

Ques. If the three normals drawn to the parabola. $y^2 = 2x$ pass through the point $(a, 0)$ $a \neq 0$, then 'a' must be greater than:

(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$

(c) -1 (d) 1

Ans. $A = \frac{1}{2}$

$x_1 > 2A$

$\Rightarrow x_1 > 2 \times \frac{1}{2}$

$\Rightarrow x_1 > 1, a > 1$

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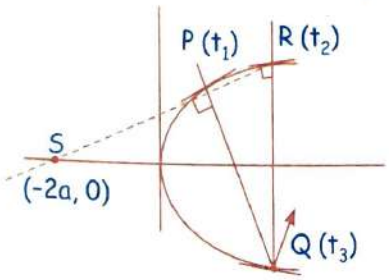
Important Note

If P and R are the points t_1 and t_2 on the parabola $y^2 = 4ax$ such that the normal to the parabola at P and R meet at Q (a point on the parabola) then $t_1t_2 = 2$ and the line PR passes through a fixed point on the axis of the curve.

Proof: PQ is a normal chord: $t_3 = -t_1 - \frac{2}{t_1}$ ...(i)



RQ is also a normal chord: $t_3 = -t_2 + \frac{2}{t_2}$ ...(ii)

By eqⁿ (i) & (ii)

$$-t_1 - \frac{2}{t_1} = -t_2 - \frac{2}{t_2} \Rightarrow t_1 + \frac{2}{t_1} = t_2 + \frac{2}{t_2}$$

$$\Rightarrow t_1 - t_2 = \frac{2}{t_2} - \frac{2}{t_1}$$

$$\Rightarrow (t_1 - t_2) = 2\,\frac{(t_1 - t_2)}{t_1t_2}$$

$$\Rightarrow t_1 t_2 = 2$$

$$\Rightarrow \frac{2}{t_1} = t_2$$

Put in equation (i): $t_3 = -t_1 - \frac{2}{t_1} \Rightarrow t_3 = t_1 - t_2$

$$\Rightarrow t_1 + t_2 + t_3 = 0$$

Equation of PR: $2x - (t_1 + t_2)y + 2at_1t_2 = 0$

$$\Rightarrow 2x + t_3y + 4a = 0$$

$$(2x + 4a) + t_3(y - 0) = 0$$

L_1 $\quad \lambda$ $\quad L_2 = 0$

L_1: $2x + 4a = 0$

L_2: $y = 0$

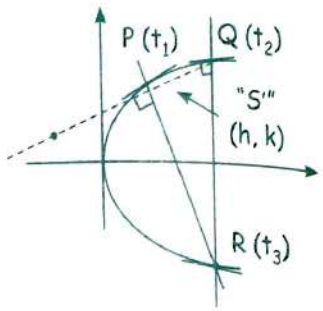
$\Big\}$ $(-2a, 0)$

Ques. P, Q are the points t_1, t_2 on the parabola $y^2 = 4ax$. The normal at P, Q meet on the parabola. Show that the middle point of PQ lies on the parabola $y^2 = 2a(x + 2a)$.

Ans. $t_1 t_2 = 2$

$2h = at_1^2 + at_2^2$, and $2k = 2at_1 + 2at_2$

$$\Rightarrow t_1 + t_2 = \frac{k}{a}$$



and $\frac{2h}{a} = t_1^2 + t_2^2$

$$\Rightarrow \frac{2h}{a} = (t_1 + t_2)^2 - 2t_1 t_2$$

$$\Rightarrow \frac{2h}{a} = \left(\frac{k}{a}\right)^2 - 2 \times 2$$

$$\Rightarrow \frac{2h}{a} = \frac{k^2}{a^2} - 4 \Rightarrow \frac{2h}{a} = \frac{k^2 - 4a^2}{a^2} \Rightarrow k^2 - 4a^2 = 2ha$$

Brain Teaser

The normal at $P(t_1)$ and $Q(t_2)$, to $y^2 = 4ax$, meet on the curve. The tangents at P and Q meet at R. Show that the circum-centre of ΔPQR, as P and Q very, traces a parabola.

Ans. $\because t_1 t_2 = 2$, $t_1 + t_2 + t_3 = 0$

Quad PRSQ is cyclic, RS is diameter of circle passing through RPSQ.

$R[at_1 t_2, a(t_1 + t_2)]$

$R[2a, -at_3]$ and $S[at_3^2, 2at_3]$

$\because$ (h,k) is the mid point of RS

$2h = 2a + at_3^2$ ...(i)

$2K = -at_3 + 2at_3$ ...(ii)

$2k = at_3$

$t_3 = \frac{2k}{a}$

By eqⁿ (i)

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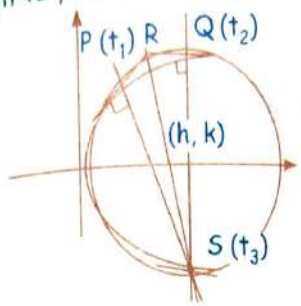
$$h = a + \frac{a}{2}\left(\frac{2k}{a}\right)^2 \Rightarrow h = a + \frac{2k^2}{a}$$

Hence, it vary as 'a' will change.

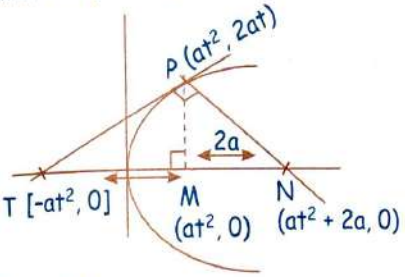


SUBTANGENT AND SUBNORMAL TO PARABOLA

Subtangent: Projection of PT on X-axis = TM = $|2at^2|$

Subnormal: Projection of PN on X-axis = MN = $|2a|$

Equation of PT: $ty = x + at^2$



For T: Put y = 0

$0 = x + at^2 \Rightarrow x = -at^2$

Equation of PN: $y + tx = 2at + at^3$, $y = 0 \Rightarrow tx = 2at + at^3$

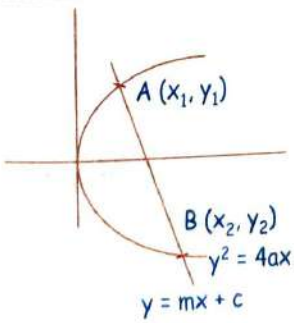
$x = at^2 + 2a$

The subtangent = twice the abscissa of the point P

The subnormal = 2a = constant

The subnormal is therefore constant for all point on the parabola and is equal to the semi-latus rectum.

LENGTH OF A CHORD



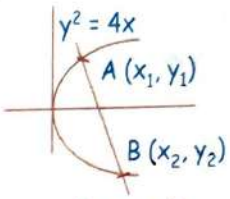
$$l = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$l = |x_2 - x_1|\sqrt{1 + \left(\frac{y_2 - y_1}{x_2 - x_1}\right)^2}$$

$$l = |x_2 - x_1|\sqrt{1 + m^2}$$

$$\boxed{l = \frac{\sqrt{D}}{m^2}\sqrt{1 + m^2}}$$

Ques. Find length AB of the chord.



$2x + y = 3$

Ans. Put $y = -2x + 3$ in the given parabola

$\therefore (-2x + 3)^2 = 4x$

$\Rightarrow 4x^2 + 9 - 12x = 4x$

$\Rightarrow 4x^2 - 16x + 9 = 0$

$\because \sqrt{D} = \sqrt{16^2 - 4 \times 4 \times 9}\ , = 4\sqrt{16 - 9} = 4\sqrt{7}$

$\therefore AB = |x_2 - x_1|\sqrt{1 + m^2} = |x_2 - x_1|\sqrt{1 + (-2)}$

$= |x_2 - x_1|\sqrt{5} = \frac{\sqrt{D}}{m^2}\sqrt{5} = \frac{4\sqrt{7} \times \sqrt{5}}{4} = \sqrt{35}$

CHORDS OF PARABOLA

CHORD OF CONTACT

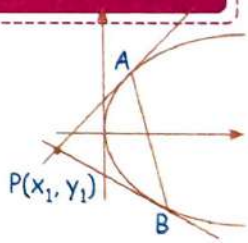
AB is C.O.C for point P

Equation of AB:

$yy_1 = 2a(x + x_1)$

For $x^2 = 4ay$

C.O.C: $\boxed{xx_1 = 2a(y + y_1)}$



Ques. From the point (-1, 2) tangent lines are drawn to the parabola $y^2 = 4x$, then the equation of chord of contact is

(a) $y = x + 1$ (b) $y = x - 1$

(c) $y + x = 1$ (d) None of these

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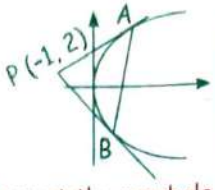
Ans. $yy_1 = 2(x + x_1)$

$\Rightarrow y(2) = 2(x - 1)$

$\Rightarrow y = x - 1$



Ques. If the line $x - y - 1 = 0$ intersect the parabola $y^2 = 8x$ at P and Q, then find the point of intersection of tangents at P and Q.

Ans. Let $R(x_1, y_1)$

Equation of PQ: $yy_1 = 4(x + x_1)$

Compare with: $y = x - 1$

Brain Teaser

Let P be a point on the parabola $y^2 = 4ax$, where $a > 0$. The normal to the parabola at P meets x-axis at a point Q. The area of the triangle PFQ, where F is the focus of the parabola, is 120. If the slope m of the normal and a are both positive integers, then the pair (a, m) is

(JEE ADV, 2023)

(a) (2, 3) (b) (1, 3)

(c) (2, 4) (d) (3, 4)

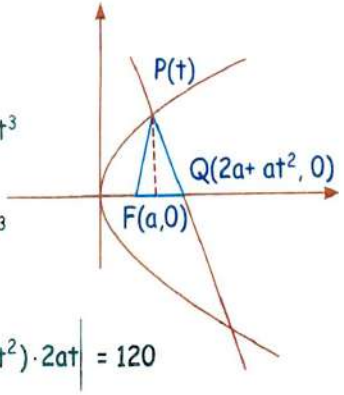
Ans. $\because m = -t$

$y + tx = 2at + at^3$

$y = 0$ (For Q)

$\therefore tx = 2at + at^3$

$\Rightarrow x = 2a + at^2$



Area: $= \left|\frac{1}{2}(a + at^2) \cdot 2at\right| = 120$

$\Rightarrow |a^2t(1 + t^2)| = 120 \Rightarrow |a^2(-m)(1 + m^2)| = 120$

$\Rightarrow a^2m(1 + m^2) = 4 \times 3 \times 10$

$\therefore a = 2, m = 3$

$\Rightarrow (a, m)\ (2, 3)$

Ques. Find the locus of point whose chord of contact with respect to the parabola $y^2 = 4bx$ is the tangent of the parabola $y^2 = 4ax$.

Ans. Let $P(h, k) \Rightarrow C.O.C : yk = 2b(x + h)$ ...(i)

Tangent of the parabola $y^2 = 4ax$ is:

$ty = x + at^2$ ...(ii)

Compare eq^n (i) & (ii)

$\frac{k}{t} = \frac{2b}{1} = \frac{2bh}{at^2}$

$\therefore t = \frac{k}{2b}, \frac{h}{at^2} = 1 \Rightarrow t = \frac{k}{2b}, h = at^2$

Thus $at^2 = a\left[\frac{k}{2b}\right]^2 = h \Rightarrow ak^2 = 4b^2h$

$\Rightarrow ay^2 = 4b^2x$

Important Note

The area of the triangle formed by the tangents from the point (x_1, y_1) and the chord of contact is

$\frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$ i.e. $\frac{(S_1)^{3/2}}{2a}$

Chord with a Given Mid-point

$T = S_1$

Where T: $yy_1 - 2a(x + x_1)$

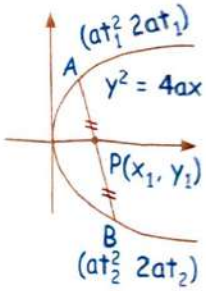
S_1: $y^2 - 4ax$

Equate to get equation of AB

$yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$

$\Rightarrow yy_1 - 2ax - 2ax_1 = y_1^2 - 4ax_1$

$\therefore$ AB: $yy_1 - 2ax = y_1^2 - 2ax_1$



Proof: $\because$ P is the mid point of AB

$\therefore 2x_1 = at_1^2 + at_2^2$,

$2y = 2a(t_1 + t_2)$

$\Rightarrow y_1 = a(t_1 + t_2) \Rightarrow t_1 + t_2 = \frac{y_1}{a}$,

$\therefore m_{AB} = \frac{2}{t_1 + t_2} = \frac{2}{\frac{y_1}{a}} = \frac{2a}{y_1}$

Equation of AB: $y - y_1 = \frac{2a}{y_1}(x - x_1)$

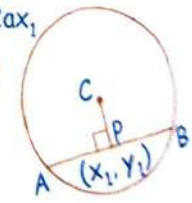
$yy_1 - y_1^2 = 2ax - 2ax_1$

$\Rightarrow yy_1 - 2ax - 2ax_1 = y_1^2 - 2ax_1 - 2ax_1$

$\Rightarrow yy_1 - 2a(x + x_1) = (y_1^2 - 4ax_1)$

$\Rightarrow T = S_1$

$\therefore$ Equation of Chord: $T = S_1$



Ques. Let PQ be a chord of the parabola $y^2 = 12x$ and the midpoint of PQ be at (4, 1). Then, which of the following point lies on the line passing through the points P and Q?

(JEE Main 4 April 2024-II)

(a) (3, 3) (b) (2, -9)

(c) $\left(\frac{3}{2}, -16\right)$ (d) $\left(\frac{1}{2}, -20\right)$

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Ans. T: $y \times 1 - 6(x + 4)$

S_1: $1^2 - 12 \times 4 = -47$

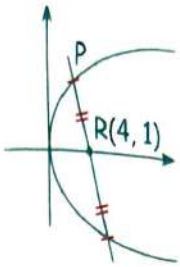
Equation of PQ: $T = S_1$

$\Rightarrow y - 6(x + 4) = -47$

$\Rightarrow y - 6x - 24 = -47$

$\Rightarrow y - 6x = 24 - 47$

$\Rightarrow y - 6x = -23$



From the given option $\left(\frac{1}{2}, -20\right)$ lies on the line

Ques. Find the locus of mid point of chord of parabola $y^2 = 4x$ which

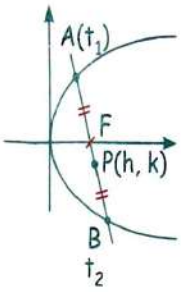
(a) are also focal chords

(b) are also normal chords

(c) subtend right angle at origin

(d) subtend 45° at origin

Ans.



(a) $t_1 t_2 = -1$ ...(i)

$\because$ P is the mid point of A and B.

$\therefore h = \frac{at_1^2 + at_2^2}{2} \Rightarrow t_1^2 + t_2^2 = 2h$ ...(ii)

$k = \frac{2at_1 + 2at_2}{2} \Rightarrow t_1 + t_2 = k$ ...(iii)

$t_1^2 + t_2^2 = 2h \Rightarrow (t_1 + t_2)^2 - 2t_1 t_2 = 2h$

By eqn (i), (ii) and (iii):

$(k)^2 - 2(-1) = 2h \Rightarrow k^2 = 2h - 2$

$\Rightarrow y^2 = 2x - 2$

(b) Equation of normal at A (t)

$y + tx = 2t + t^3$ ...(i)

Equation of AB: $T = S_1$

$yk - 2x = k^2 - 2h$ ...(ii)

By comparing both equations:

$\frac{1}{k} = \frac{t}{-2} = \frac{2t + t^3}{k^2 - 2h}$

$t = \frac{-2}{k}$

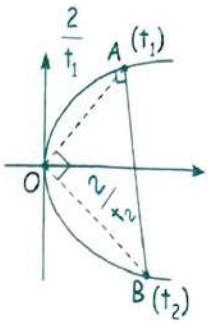
and $\frac{t}{-2} = \frac{2t + t^3}{k^2 - 2h} \Rightarrow -\frac{1}{2} = \frac{2 + t^2}{k^2 - 2h}$

$\Rightarrow \frac{-(k^2 - 2h)}{2} = 2 + t^2 = 2 + \left(\frac{-2}{k}\right)^2$

$\Rightarrow -k^2 (k^2 - 2h) = 4k^2 + 8$

$\Rightarrow -y^4 + 2xy^2 = 4y^2 + 8$

(c) $y^2 - 4x = 0$



AB: $yk - 2x = k^2 - 2h$

$\Rightarrow \frac{yk - 2x}{k^2 - 2h} = 1$

Homogenized equation:

$y^2 - 4x\left(\frac{yk - 2x}{k^2 - 2h}\right) = 0$

$\Rightarrow y^2(k^2 - 2h) - 4x(yk - 2x) = 0$

$\Rightarrow y^2(k^2 - 2h) - 4kxy + 8x^2 = 0$

$\because$ It subtend right angle at origin

$\therefore a + b = 0 \Rightarrow k^2 - 2h + 8 = 0 \Rightarrow y^2 = 2x - 8$

(d) $\tan\theta = \frac{2\sqrt{H^2 - ab}}{|a + b|}$

We know that $y^2 (k^2 - 2h) - 4kxy + 8x^2 = 0$

$\therefore H = -2k$,

$b = k^2 - 2h$

$a = 8$

$1 = \tan 45° = \frac{2\sqrt{(-2k)^2 - ab}}{|a + b|}$

$2\sqrt{16h - 4k^2} = |k^2 - 2h + 8|$

$\Rightarrow 4(16x - 4y^2) = (y^2 - 2x + 8)^2$

Chord Joining 2 Points

- Chord joining $P(t_1)$ and $Q(t_2)$ is given by $2x - (t_1 + t_2)y + 2at_1 t_2 = 0$
- For this to be a focal chord, then $t_1 t_2 = -1$

Important Note

Length of Focal Chord = $a(t_1 - t_2)^2 = 4a\,\text{cosec}^2\theta$

Where θ is the inclination of the focal chord.

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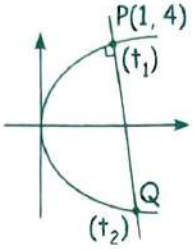
Ques. If one end of a focal chord of the parabola $y^2 = 16x$ is at $(1, 4)$ then the length of this focal chord is. **(JEE Main 2019)**

(a) 24 (b) 20

(c) 22 (d) 25

Ans. $4 = 2at_1 \Rightarrow 4 = 2 \times 4t_1$

$\therefore t_1 = \frac{1}{2}$



Thus, $t_1 t_2 = -1 \Rightarrow t_2 = -2$

$l = a(t_1 - t_2)^2$

$= 4\left(\frac{1}{2} + 2\right)^2 = 4 \times \left(\frac{5}{2}\right)^2 = 25$

Ques. Suppose AB is a focal chord of the parabola $y^2 = 12x$ of lenght l and slope $m < \sqrt{3}$. If the distance of the chord AB from the origin is d, then $l\,d^2$ is equal to.

(JEE Main 5 April 2024-I)

Ans. $a = 3$

$\tan\theta < \sqrt{3} \Rightarrow \theta < 60°$

$l = 4a\,\text{cosec}^2\theta$

$l = 4a \cdot \frac{a^2}{d^2}$

$d^2 l = 4a^3 = 4(3)^3$

$= 27 \times 4 = 108$

Ques. If the x-intercept of a focal chord of the parabola $y^2 = 8x + 4y + 4$ is 3, then the length of this chord is equal to.

Ans. $l = 4a\,\text{cosec}^2\theta$

$y^2 - 4y = 8x + 4$

$\Rightarrow (y - 2)^2 - 4 = 8x + 4$

$\Rightarrow (y - 2)^2 = 8(x + 1)$

$\therefore 4a = 8$

Focus: $x + 1 = a \Rightarrow x = 1$

$y - 2 = 0, \Rightarrow y = 2$

$\therefore F[1, 2]$

$\because$ point on focal chard = $(3, 0)$

$m = \frac{2-0}{1-3} = \frac{2}{-2} = -1$

$\Rightarrow \tan\theta = -1, \Rightarrow \theta = 135°$

$\therefore l = 4a\,\text{cosec}^2 135°$

$= 4 \times 2 \times \left(\sqrt{2}\right)^2, = 4 \times 2 \times 2$

$= 16$

DIAMETER OF PARABOLA

- Locus of Mid point of system of parallel chords is known as diameter of parabola.

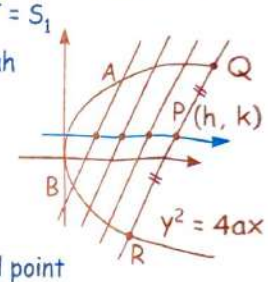
Proof: Equation of QR: $T = S_1$

$\Rightarrow yk - 2a(x + h) = k^2 - 4ah$

$\Rightarrow yk - 2ax = k^2 - 2ah$

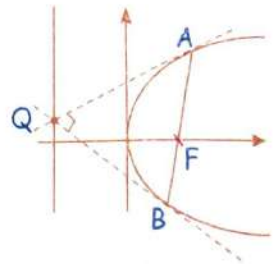
$\therefore m = \frac{2a}{k} \Rightarrow k = \frac{2a}{m}$

$\Rightarrow y = \frac{2a}{m} \rightarrow$ locus of mid point



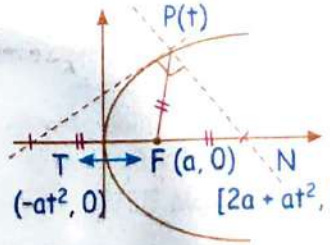
PROPERTIES OF PARABOLA

PROPERTY-1: Line joining the centers of the two Tangent drawn at the ends of any focal chord are always Perpenducular and they always meet on the Diretrix.



PROPERTY-2: If the tangent and normal drawn at any point P to a parabola meets the axis of Parabola T and N respectively then PF = TF = FN, where F is the focus of the parabola.

Proof: $FN = TF = PF = a + at^2$



PROPERTY-3: The foot of the perpendiculars from the focus, to any tangent, lies on the tangent at the vertex.

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Proof: To prove:

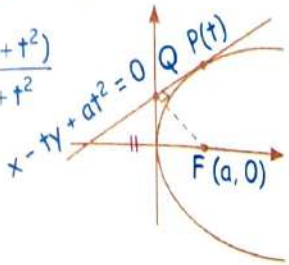
$$\frac{x-a}{1} = \frac{y-0}{-t} = -\left[\frac{a-0+at^2}{1^2+t^2}\right]$$

$$\Rightarrow \frac{x-a}{1} = \frac{y}{-t} = -\frac{a(1+t^2)}{1+t^2}$$

$$\Rightarrow x-a = \frac{y}{-t} = -a$$

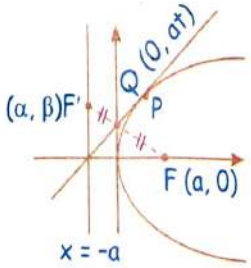
$\therefore x = 0, y = at$

$Q(0, at)$

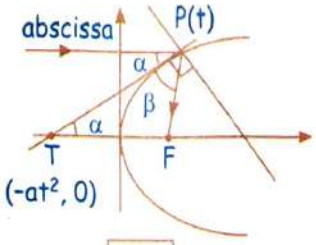


Which is a point lies on the tangent of the vertex

PROPERTY-4: The reflection of the focus F, in the tangent, T, at a point P on the curve lies on the directrix and it coincides with the foot of the perpenducualr from p to the directrix, d.



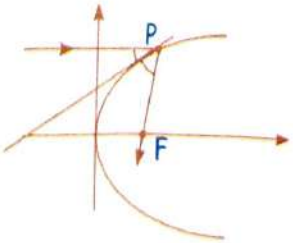
PROPERTY-5: The tangent and the normal at a point P bisect the angles between the focal radius of P and the abscissa of P.



$\alpha = \beta$

PROPERTY-6: (Reflection Property of Parabola)

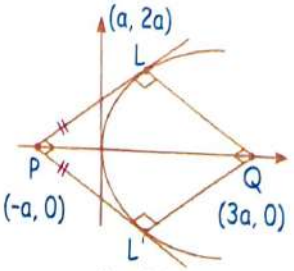
If incident ray is parallel to axis of the parabola then after getting reflected from the surface of the parabola, reflected ray passes through focus of the parabola.



PROPERTY-7: Tangent and normal drawn at the ends of the latus rectum enclose a square of side length $2\sqrt{2}a$.

Proof: $PL = \sqrt{(2a)^2 + (2a)^2} = 2a\sqrt{2}$

$PL' = \sqrt{(2a)^2 + (2a)^2} = 2a\sqrt{2} \Rightarrow PLQL'$ is a square

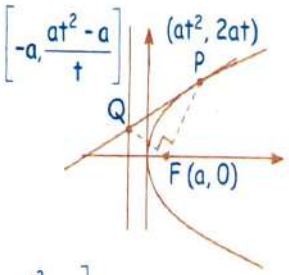


PROPERTY-8: The portion of a tangent to a parabola cut off between the directrix and the point of tangency subtends a right angle at the focus.

Proof: $ty = x + at^2$

$\rightarrow at\ x = -a$

$$ty = -a + at^2, \Rightarrow y = \frac{at^2 - a}{t}$$



$$\therefore Q\left[-a, \frac{at^2-a}{t}\right]$$

$$\therefore m_{PF} = \frac{2at-0}{at^2-a} = \frac{2t}{t^2-1}$$

$$\text{and } m_{QF} = \frac{\frac{at^2-a}{t}-0}{-2a} = \frac{t^2-1}{-2t}$$

$\Rightarrow M_{PF} \cdot M_{QF} = -1$

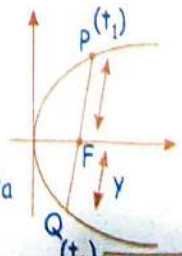
PROPERTY-9: Semi Latus Rectum is the Harmonic Mean between the 2 segments of any focal chord.

Proof:

$\because x = a + at_1^2$

and $y = a + at_2^2$

We need to prove: $\frac{2xy}{x+y} = 2a$



$t_1 t_2 = -1$

$$\text{LHS: } \frac{2(a+at_1^2)(a+at_2^2)}{a+at_1^2+a+at_2^2}$$

$$= \frac{2a^2(1+t_1^2)(1+t_2^2)}{a(t_1^2+t_2^2+2)} = \frac{2a(1+t_1^2+t_2^2+(t_1t_2)}{(t_1^2+t_2^2+2)} = 2a$$

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PROPERTY-10:

(a) Circle with focal distance as diameter touches the tangent at vertex

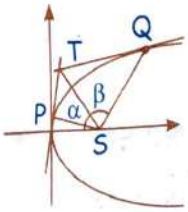
(b) Circle with Focal Chord as diameter touches the directrix

PROPERTY-11: If the tangents at P and Q meet in T, then

(a) TP and TQ subtend equal angles at the focus S.

(b) $ST^2 = SP.SQ$

(c) The triangles SPT and STQ are similar



PROPERTY-12: T is the triangle formed by three tangents to a parabola.

(a) The circumcircle of T passes through the focus;

(b) The orthocenter of T lies on the directrix.

PROPERTY-13: The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents these points.

PROPERTY-14: If P (t_1). Q(t_2), R(t_3), S(t_4) are 4 concyclic points on parabola $y^2 = 4ax$ then $t_1 + t_2 + t_3 + t_4 = 0$

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Short Notes

PARABOLA

Form	$y^2 = 4ax$	$x^2 = 4ay$
Vertex	(0, 0)	(0, 0)
Focus	(a, 0)	(0, a)
Equation of the Directrix	$x = -a$	$y = -a$
Equation of the Axis	$y = 0$	$x = 0$
Tangent at the Vertex	$x = 0$	$y = 0$
Foot of directrix	(-a, 0)	(0, -a)
Ends of Latus Rectum	(a, 2a) & (a,-2a)	(2a, a) & (-2a,a)

- Let the vertex be (α, β) and the axis be parallel to the x - axis. Then the equation of parabola is given by $(y - \beta)^2 = 4a(x - \alpha)$ which is equivalent to $x = Ay^2 + By + C$.
- When the axis is parallel to the y - axis, the equation of the parabola is $y = A'x^2 + B'x + C'$.
- The general equation of second degree i.e. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a parabola if $\Delta \neq 0$ and $h^2 = ab$. ($\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$).
- The focal distance of any point P (x, y) on $y^2 = 4ax$ is $|x + a|$.
- Parametric equations of the parabola $y^2 = 4ax$ are $x = at^2$, $y = 2at$.

EQUATION OF TANGENTS OF $y^2 = 4ax$

- Cartesian form of tangent is $yy_1 = 2a(x + x_1)$
- Parametric Form of tangent is $ty = x + at^2$.
- Slope form of tangent is $y = mx + a/m$ and the point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$
- Point of intersection of tangents drawn at $P(t_1)$ and $Q(t_2)$ is given by $[at_1t_2, a(t_1 + t_2)]$
- The abscissa of R is GM of abscissa of P and Q
- The ordinate of R is AM of ordinates of P and Q.

EQUATION OF TANGENTS TO $x^2 = 4ay$

- Cartesian form of tangent is $xx_1 = 2a(y + y_1)$
- Parametric Form of tangent is $tx = y + at^2$.
- Slope form of tangent is $y = mx - am^2$ and the point of contact is $(2am, am^2)$.

EQUATION OF NORMALS TO $y^2 = 4ax$

- Cartesian form of normal $y - y_1 = -\frac{y_1}{2a}(x - x_1)$
- Parametric Form of normal is $tx + y = at^3 + 2at$.
- Slope form of normal is $y = mx - 2am - am^3$ and the foot of normal is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$.

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EQUATION OF NORMALS TO $x^2 = 4ay$

- Cartesian form of normal $y - y_1 = -\frac{2a}{x_1}(x - x_1)$
- Parametric Form of normal is $ty + x = at^3 + 2at$.
- Slope form of normal is at $y = mx + 2a + \frac{a}{m^2}$ at $\left(-\frac{2a}{m}, \frac{a}{m^2}\right)$

MORE ABOUT NORMALS

- If the normal to the parabola $y^2 = 4ax$ at the point t_1, meets the parabola again at the point t_2 then $t_2 = -t_1 - \frac{2}{t_1}$
- A maximum of three normals can be drawn to a parabola from a given point (x_1, y_1) and the feet of these 3 normals are called co-normal points.
- If $P(t_1)$, $Q(t_2)$ and $R(t_3)$ are co-normal points of a parabola $y^2 = 4ax$ then $t_1 + t_2 + t_3 = 0$
- Algebraic sum of slopes of the three concurrent normal is zero.
- Algebraic sum of ordinates of the three co-normal points on the parabola is zero.
- Centroid of the Δ formed by three co-normal points lies on the axis of parabola (x-axis).
- Point of intersection of Normals at $P(t_1)$ & $Q(t_2)$ is $(a(t_1^2 + t_2^2 + t_1t_2 + 2), -at_1t_2(t_1 + t_2))$

CHORDS OF PARABOLA

- Equation of the chord joining 't_1' and 't_2' is $(t_1 + t_2)y = 2x + 2at_1t_2$. This is a focal chord if $t_1t_2 = -1$.
- Chord of Contact: Equation to the chord of contact for the parabola $y^2 = 4ax$ is $T = 0$, i.e. $yy_1 - 2a(x + x_1) = 0$.
- The area of the triangle formed by the tangents from the point (x_1, y_1) & the chord of contact is $\frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$ i.e. $\frac{(S_1)^{3/2}}{2a}$.
- Chord with a given mid point: Equation of the chord of the parabola $y^2 = 4ax$ whose middle point is (x_1, y_1) is $T = S_1$, where

  $T \equiv yy_1 - 2a(x + x_1)$ & $S_1 \equiv y_1^2 - 4ax_1$.
- If $P(t_1)$, $Q(t_2)$, $R(t_3)$ and $S(t_4)$ are concyclic points on the parabola then $t_1 + t_2 + t_3 + t_4 = 0$.
- Equation of the Tangents from an External Point: $SS_1 = T^2$, where $S = y^2 - 4ax$, $S_1 = y_1^2 - 4ax_1$, $T = yy_1 - 2a(x + x_1)$.
- DIAMETER OF PARABOLA: The locus of the middle point of a system of parallel chords of a parabola is called a Diameter and is given by $y = \frac{2a}{m}$

PROPERTIES OF PARABOLA

- The foot of the perpendicular from the focus, to any tangent, lies on the tangent at the vertex.
- The tangent and the normal at a point P bisect the angles between the focal radius of P and the abscissa of P.
- (Reflection Property)

  A ray of light, emanating from the focus, F, and incident on the mirror at P, is reflected from the mirror parallel to the axis.
- The reflection of the focus F, in the tangent, T, at a point P on the curve lies on the directrix and it coincides with the foot of the perpendicular from P to the directrix, d.

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- Tangents and Normals at the extremities of the latus rectum of a parabola $y^2 = 4ax$ constitute a square of side length $2\sqrt{2}a$
- The portion of a tangent to a parabola cut off between the directrix and the point of tangency subtends a right angle at the focus.
- Semi Latus Rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord of the parabola
- If the tangents at P and Q meet in T, then
  - I. TP and TQ subtend equal angles at the focus S,
  - II. $ST^2 = SP.SQ$, and
  - III. The triangles SPT and STQ are similar.
- The Ortho-centre of any triangle formed by three tangents to a parabola always lies on the directrix.
- The circle circumscribing the triangle formed by any three tangents to a parabola passes through the focus.
- Circle drawn with a focal chord as diameter always touches the directrix.
- Circle drawn with a focal distance as diameter always touches the tangent at vertex.

"The focus of a parabola is like your focus in studies – it decides the path of success."

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Chapter 13

Ellipse

ELLIPSE

The locus of a point which moves such that the ratio of its distance from a fixed point to its distance from a fixed straight line is always a constant (e) provided $0 < e < 1$.

For Ellipse:

$0 < e < 1$; $\Delta \neq 0$

$h^2 < ab$

Ques. Interpret the locus $4x^2 + y^2 - 8x - 2y + 1 = 0$.

Ans. $a = 4, b = 1, h = 0, g = -4, f = -1, c = 1$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$

and $h^2 - ab = 0 - 4 \times 1 = -4 < 0 \Rightarrow$ Ellipse

Ques. The equation of an ellipse whose focus is $(-1, 1)$, directrix is $x - y + 3 = 0$ and eccentricity is $1/2$, is given by

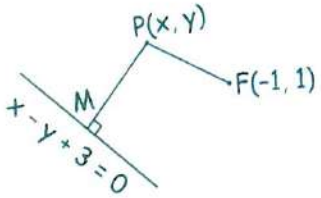
(a) $7x^2 + 2xy + 7y^2 + 10x - 10y + 7 = 0$

(b) $7x^2 - 2xy + 7y^2 - 10x + 10y + 7 = 0$

(c) $7x^2 - 2xy + 7y^2 - 10x - 10y - 7 = 0$

(d) $7x^2 - 2xy + 7y^2 + 10x + 10y - 7 = 0$

Ans.



$\because e = \frac{PF}{PM} = \frac{1}{2} \Rightarrow PF = \frac{1}{2}PM$

$\Rightarrow \sqrt{(x+1)^2 + (y-1)^2} = \frac{1}{2}\left|\frac{x - y + 3}{\sqrt{2}}\right|$

$\Rightarrow (x + 1)^2 + (y - 1)^2 = \frac{1}{4 \times 2}(x - y + 3)^2$

$\Rightarrow 8x^2 + 16x + 8 + 8y^2 - 16y + 8$

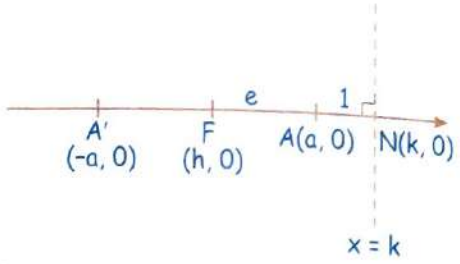
$= x^2 + y^2 + 9 - 2xy - 6y + 6x$

$\Rightarrow 7x^2 + 7y^2 + 2xy + 10x - 10y + 7 = 0$

STANDARD EQUATION-1

Let ellipse passes through 2-symmetric points on x-axis $(a, 0)$ and $(-a, 0)$ then ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Proof:



As we know,

$e = \frac{\text{Distance from any point on ellipse to focus}}{\text{Distance from that point to directrix}}$

$\therefore \frac{AF}{AN} = \frac{e}{1} \Rightarrow \frac{|a - h|}{|k - a|} = e$

$\Rightarrow ek + h = ae + a$ ...(i)

also, $\frac{A'F}{A'N} = \frac{e}{1} \Rightarrow \frac{(a + h)}{(a + k)} = e$

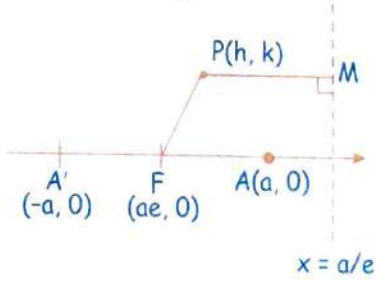
$\Rightarrow ek - h = -ae + a$ ...(ii)

By adding equation (i) and (ii)

$2ek = 2a \Rightarrow k = \frac{a}{e}$

By subtracting equation (ii) from (i)

$2h = 2ae \Rightarrow h = ae$



$PF = \sqrt{(h - ae)^2 + k^2}$

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and $PM = \left|\dfrac{h - \frac{a}{e}}{1}\right|$

$\because \dfrac{PF}{PM} = e \Rightarrow PF = ePM$

$\Rightarrow \sqrt{(h-ae)^2 + k^2} = e\left|h - \dfrac{a}{e}\right|$

$\Rightarrow (h - ae)^2 + k^2 = e^2 (h - a/e)^2$

$\Rightarrow h^2 + a^2e^2 - 2hae + k^2 = e^2h^2 + a^2 - 2hea$

$\Rightarrow h^2 (1 - e^2) + k^2 = a^2 - a^2e^2$

$(h, k) \rightarrow (x, y)$

$\therefore \dfrac{x^2(1-e^2)+y^2}{a^2(1-e^2)} = \dfrac{a^2(1-e^2)}{a^2(1-e^2)}$

$\Rightarrow \dfrac{x^2}{a^2} + \dfrac{y^2}{a^2(1-e^2)} = 1$

$\Rightarrow \boxed{\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1}$; where $b^2 = a^2 (1 - e^2)$ and $a > b$.

Put $x = 0$

$\dfrac{y^2}{b^2} = 1$

$y = \pm b$

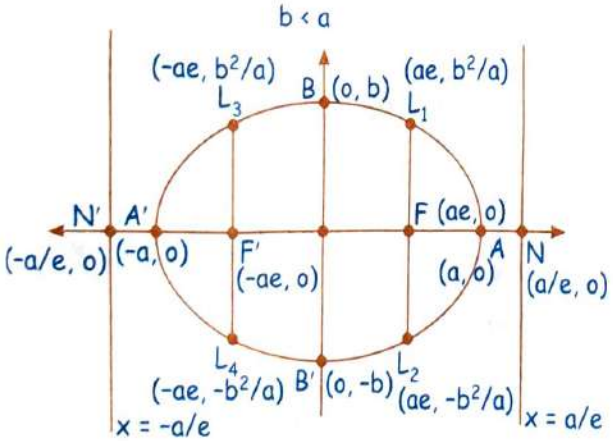
Put $y = 0$

$\Rightarrow \dfrac{x^2}{a^2} = 1$

$\Rightarrow x = \pm a$

$b^2 < a^2$

$b < a$



For $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ $(a > b)$

Major Axis	x - Axis (y = 0)
Minor Axis	y - axis (x = 0)
Ends of Major Axis/Vertices	(a, o) and (-a, o)
Ends of Minor Axis	(o, b) and (o, -b)

Centre	(o, o)
Foci	(ae, o) and (-ae, o)
Directrices	$x = \pm a/e$
Feet of Directrices	$(\pm a/e, o)$
Equation of L.R.	$x = ae$ or $x = -ae$
Ends of L.R.	$(ae, \pm b^2/a)$; $(-ae, \pm b^2/a)$
Length of L.R	$2b^2/a$
Relation (a, b and e)	$b^2 = a^2(1 - e^2)$
Area of Ellipse	πab

length of Major Axis = 2a

length of Minor Axis = 2b

For Finding ends of L.R

Put $x = ae$ in eqn $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$

$\dfrac{a^2e^2}{a^2} + \dfrac{y^2}{b^2} = 1$

$\Rightarrow \dfrac{y^2}{b^2} = 1 - e^2 \Rightarrow \dfrac{y^2}{b^2} = \dfrac{b^2}{a^2} \Rightarrow y^2 = \dfrac{b^4}{a^2}$

$\Rightarrow y = \pm \dfrac{b^2}{a}$

Ques. Find everything for the Ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{12} = 1$

Ans. $a^2 = 16$, $b^2 = 12 \Rightarrow a = 4$; $b = 2\sqrt{3}$

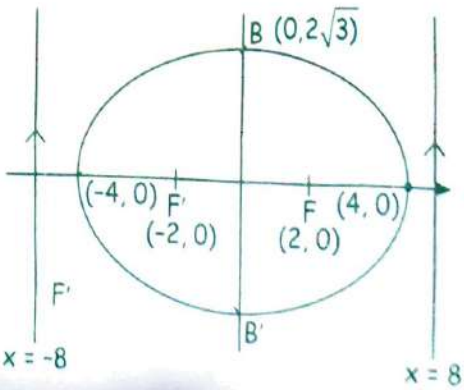
$\because a > b$

$\therefore b^2 = a^2 (1 - e^2)$

$\Rightarrow 12 = 16 (1 - e^2) \Rightarrow 3 = 4 (1 - e^2)$

$\Rightarrow 1 - e^2 = 3/4 \Rightarrow e^2 = 1/4 \Rightarrow e = \dfrac{1}{2}$

$F (\pm ae, o) = F\left[\pm 4 \times \dfrac{1}{2}, 0\right] = (\pm 2, 0)$



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Ques. If distance between the directrices be thrice the distance between the foci, then eccentricity of ellipse is-

(a) 1/2 (b) $\sqrt{2}/\sqrt{3}$

(c) $1/\sqrt{3}$ (d) 4/5

Ans. ∵ distance b/w directrices = 2a/e

and distance b/w foci = 2ae

$\therefore \frac{2a}{e} = 3.\ 2ae$

$\Rightarrow \frac{1}{e} = 3e \Rightarrow e^2 = \frac{1}{3} \Rightarrow e = \frac{1}{\sqrt{3}}$

Ques. If distance between foci of an ellipse is 6 and distance directrices is 12 then find the length of its L.R.

(a) $3\sqrt{2}$ (b) 4

(c) 9 (d) $2\sqrt{2}$

Ans. $2ae = 6$

$\Rightarrow ae = 3$ ...(i)

$2a/e = 12$

$\Rightarrow a/e = 6$ ...(ii)

By multiplying equation (i) and (ii)

$a^2 = 3 \times 6 \Rightarrow a = 3\sqrt{2}$

By equation (i)

$3\sqrt{2}e = 3 \Rightarrow e = \frac{1}{\sqrt{2}}$

$\therefore b^2 = a^2 (1 - e^2)$

$\Rightarrow b^2 = 18\ (1 - 1/2)$

$\Rightarrow b^2 = 9 \Rightarrow b = 3$

$\therefore L.R = \frac{2b^2}{a} = \frac{2(9)}{3\sqrt{2}} = \sqrt{2} \times 3 = 3\sqrt{2}$

Ques. If the length of the minor axis of ellipse is equal to half of the distance between the foci, then the eccentricity of the ellipse is:

[30 Jan. 2024]

(a) $\frac{\sqrt{5}}{3}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{5}}$

Ans. $2b = \frac{1}{2} 2ae$

$\Rightarrow 2b = ae$

$\Rightarrow 4b^2 = a^2 e^2$

$\Rightarrow 4a^2 (1 - e^2) = a^2 e^2$

$\Rightarrow 4 - 4e^2 = e^2$

$\Rightarrow 4 = 5e^2$

$\Rightarrow e^2 = \frac{4}{5} \Rightarrow e = \frac{2}{\sqrt{5}}$

Ques. Find everything for the ellipse

$4x^2 + 9y^2 - 8x - 36y + 4 = 0.$

Ans. $4x^2 + 9y^2 - 8x - 36y + 4 = 0.$

$\Rightarrow 4x^2 - 8x + 9y^2 - 36y + 4 = 0$

$\Rightarrow 4(x^2 - 2x) + 9(y^2 - 4y) + 4 = 0$

$\Rightarrow 4((x-1)^2 - 1) + 9((y-2)^2 - 4) + 4 = 0$

$\Rightarrow 4(x-1)^2 - 4 + 9(y-2)^2 - 36 + 4 = 0$

$\Rightarrow \frac{4(x-1)^2}{36} + \frac{9(y-2)^2}{36} = \frac{36}{36} = 1$

$\therefore \frac{(x-1)^2}{9} + \frac{(y-2)^2}{4} = 1$

$X = x - 1; \quad Y = y - 2; \quad a = 3, b = 2$

$\Rightarrow \frac{X^2}{9} + \frac{Y^2}{4} = 1$

∵ $b^2 = a^2 (1 - e^2)$

$\Rightarrow 4 = 9\ (1 - e^2)$

$\Rightarrow 9e^2 = 5 \Rightarrow e = \frac{\sqrt{5}}{3}$

∴ Centre (0, 0):

$X = 0$	$Y = 0$
$\Rightarrow x - 1 = 0$	$y - 2 = 0$
$\Rightarrow x = 1$	$y = 2$

Vertices:

$X = a$	$Y = 0$
$\Rightarrow x - 1 = 3$	$y - 2 = 0$
$\Rightarrow x = 4$	$y = 2$

(4, 2) and

$x = -a$	$y = 0$
$x - 1 = -3,$	$y - 2 = 0$
$x = -2,$	$y = 2$

(-2, 2)

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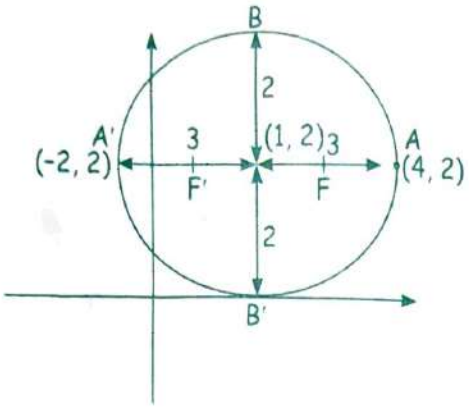
Foci: $X = \pm ae$ | $y = 0$

$\Rightarrow x - 1 = \pm \frac{\sqrt[3]{5}}{3}$ | $y - 2 = 0$

$\Rightarrow x = 1 \pm \sqrt{5}$ | $y = 2$



Ques. The coordinates of the foci of the ellipse $3x^2 + 4y^2 - 12x - 8y + 4 = 0$

(a) (1, 2), (3, 4) (b) (1, 4), (3, 1)

(c) (1, 1), (3, 1) (b) (2, 3), (5, 4)

Ans. $3(x^2 - 4x) + 4(y^2 - 2y) + 4 = 0$

$\Rightarrow 3((x-2)^2 - 4) + 4((y-1)^2 - 1) + 4 = 0$

$\Rightarrow 3(x-2)^2 - 12 + 4(y-1)^2 - 4 + 4 = 0$

$\Rightarrow 3(x-2)^2 + 4(y-1)^2 = 12$

$\Rightarrow \frac{(x-2)^2}{4} + \frac{(y-1)^2}{3} = 1$

$\because a^2 = 4, \ b^2 = 3$

$\therefore b^2 = a^2(1 - e^2)$

$\Rightarrow 3 = 4(1 - e^2) \Rightarrow 1 = 4e^2 \Rightarrow e = \frac{1}{2}$

$\therefore$ Foci:

$x = \pm ae, \ y = 0$

$\Rightarrow x - 2 = \pm \frac{2 \times 1}{2}, \ y - 1 = 0$

$\Rightarrow x - 2 = \pm 1, \quad y = 1 \Rightarrow x = 3 \text{ or } 1, \quad y = 1$

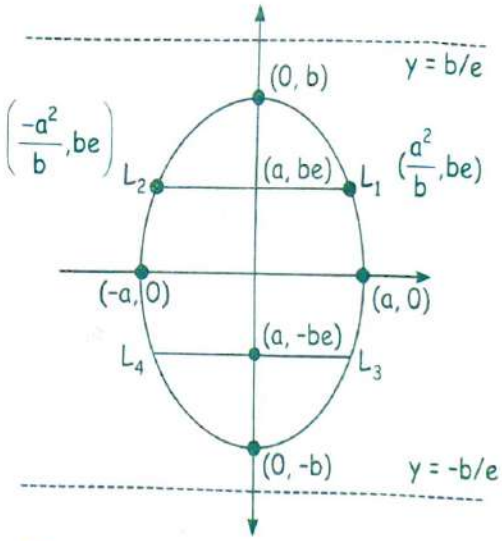
$\therefore$ Foci (3, 1) or (1, 1)

STANDARD EQUATION-2

It is given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a^2 = b^2(1 - e^2), (a < b)$

For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \ (a < b)$

Major axis	$x = 0$
Minor axis	$y = 0$
End of major axis/ vertices	(0, b) and (0, -b)
Ends of minor axis	(a, 0) and (-a, 0)
Centre	(0, 0)
Foci	(0, be) and (0, -be)
Directrices	$y = \pm b/e$
Feet of directrices	$(0, \pm b/e)$
Equation of L.R	$y = be$ or $y = -be$
Ends of L.R	$\left(\pm\frac{a^2}{b}, be\right)$ or $\left(\pm\frac{a^2}{b}, -be\right)$
Length of L.R	$2a^2/b$
Relation (a, b and e)	$a^2 = b^2(1 - e^2)$
Area of ellipse	πab



Ques. The eccentricity of the ellipse $9x^2 + 5y^2 - 30y = 0$ is.

(a) 1/3 (b) 2/3

(c) 3/4 (d) None of these

Ans. $9x^2 + 5(y^2 - 6y) = 0$

$\Rightarrow 9x^2 + 5((y-3)^2 - 9) = 0$

$\Rightarrow 9x^2 + 5(y-3)^2 = 45 \Rightarrow \frac{x^2}{5} + \frac{(y-3)^2}{9} = 1$

$a^2 = 5, \quad b^2 = 9$

$\because a < b$

$\therefore a^2 = b^2(1 - e^2) \Rightarrow 5 = 9(1 - e^2) \Rightarrow 5 = 9 - 9e^2$

$\Rightarrow 9e^2 = 4 \Rightarrow e = \frac{2}{3}$

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Ques. An ellipse, with foci at (0, 2) and (0, -2) and minor axis of length 4, passes through which of the following points?

[JEE Main 2019, Apr.11]

(a) $(\sqrt{2}, 2)$ (b) $(2, \sqrt{2})$

(c) $(2, 2\sqrt{2})$ (d) $(1, 2\sqrt{2})$

Ans. $\because be = 2 \Rightarrow b^2e^2 = 4$

$2a = 4 \Rightarrow a = 2$

$\because a^2 = b^2(1 - e^2)$

$\Rightarrow a^2 = b^2 - b^2e^2 \Rightarrow 4 = b^2 - 4 \Rightarrow b^2 = 8$

$\therefore \frac{x^2}{4} + \frac{y^2}{8} = 1$

Put $(\sqrt{2}, 2)$ in it:

$\Rightarrow \frac{(\sqrt{2})^2}{4} + \frac{2^2}{8} = 1$

$\Rightarrow \frac{2}{4} + \frac{4}{8} = 1$

Hence, the ellipse passes through $(\sqrt{2}, 2)$

Ques. In an ellipse, with centre at the origin, if the difference of the lengths of major axis and minor axis is 10 and one of the foci at $(0, 5\sqrt{3})$, then the length of its latus rectum is

[JEE Main 2019, Apr.1]

(a) 5 (b) 10 (c) 8 (d) 6

Ans. $be = 5\sqrt{3}$ ...(i)

$2b - 2a = 10$

$\Rightarrow b - a = 5$ ...(ii)

$\because a^2 = b^2(1 - e^2)$

$\Rightarrow a^2 = b^2 - b^2e^2$

$\Rightarrow a^2 = b^2 - 75$

$\Rightarrow b^2 - a^2 = 75$

$\Rightarrow (b - a)(b + a) = 75$

By equation (ii)

$5(b + a) = 75$

$\Rightarrow b + a = 15$ ...(iii)

By adding equation (ii) and (iii)

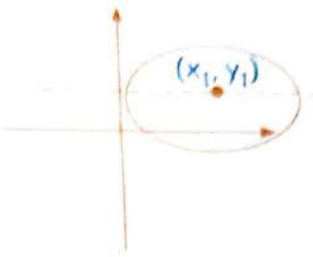
$2b = 25 \Rightarrow b = 10$

and $a = 5$

$\therefore L.R = \frac{2a^2}{b} = \frac{2(5)^2}{10} = 5$

SHIFTED FORMS OF ELLIPSE

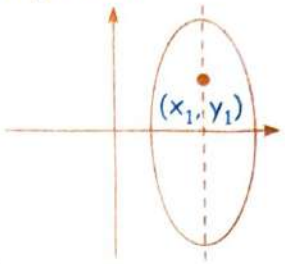
$a > b$



$\frac{(x - x_1)^2}{a^2} + \frac{(y - y_1)^2}{b^2} = 1$

major axis: $y = const.$

$b > a$



$\frac{(x - x_1)^2}{a^2} + \frac{(y - y_1)^2}{b^2} = 1$

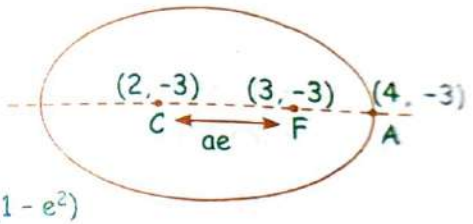
major axis: $x = const.$

Ques. Find the equation of the ellipse its centre at the point (2, -3), one focus at (3, -3) and one vertex at (4, -3).

Ans. $CF = ae = 1$

$CA = a = 2$

$\therefore e = \frac{1}{2}$



and $b^2 = a^2(1 - e^2)$

$\Rightarrow b^2 = 4(1 - \frac{1}{4}) \Rightarrow b^2 = 4 \times \frac{3}{4} \Rightarrow b^2 = 3$

$\therefore$ eqn of ellipse: $\frac{(x - x_1)^2}{a^2} + \frac{(y - y_1)}{b^2} = 1$

$\Rightarrow \frac{(x - 2)^2}{4} + \frac{(y + 3)^2}{3} = 1$

major axis is $y = -3$ which is parallel to x axis

$\Rightarrow$ Horizontal ellipse $a > b$

Ques. Find the equation of the ellipse having centre at (1, 2), one focus at (6, 2) and passing through the point (4, 6).

Ans. $b^2 = a^2(1 - e^2)$

$b^2 = a^2 - a^2e^2 \Rightarrow b^2 = a^2 - 25, ae = 5$

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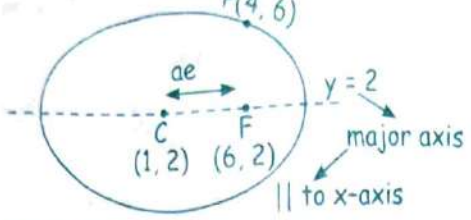
$\because \frac{(x-1)^2}{a^2} + \frac{(y-2)^2}{b^2} = 1$



Put (4, 6) in it

$\therefore \frac{3^2}{a^2} + \frac{4^2}{b^2} = 1 \Rightarrow \frac{9}{a^2} + \frac{16}{a^2 - 25} = 1$

$\Rightarrow \frac{9(a^2-25) + 16a^2}{a^2(a^2-25)} = 1$

$\Rightarrow 25a^2 - 225 = a^2(a^2 - 25)$

$\Rightarrow 25a^2 - 225 = a^4 - 25a^2 \Rightarrow a^4 - 50a^2 + 225 = 0$

$\Rightarrow (a^2 - 45)(a^2 - 5) = 0$

$a^2 = 45$ or $a^2 = 5$

$\because a^2 \neq 5$ because b^2 can not be -ve

$\therefore a^2 = 45, b^2 = 20$

$\Rightarrow$ eqⁿ of ellipse $\frac{(x-1)^2}{45} + \frac{(y-2)^2}{20} = 1$

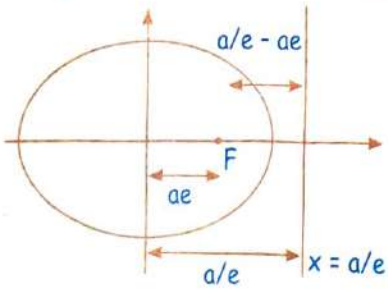
Important Note

The length of latus Rectum of Ellipse is $2b^2/a = 2e \times$ (distance between focus and Corresponding Directrix)

Proof: LR: $\frac{2b^2}{a} = \frac{2a^2(1-e^2)}{a} = 2a(1 - e^2)$

$= 2ae\left[\frac{1}{e} - e\right] = 2a\left[\frac{a}{e} - ae\right]$

$= 2e \times$ (dis b/w Focus and directrix)



Ques. Ellipse: Focus (1, 2) Directrix $3x + 4y - 1 = 0$ and $e = \frac{1}{3}$. Find L.R.

Ans. dis b/w focus and directrix:

$\frac{3\times1 + 4\times2 - 1}{\sqrt{3^2 + 4^2}} = \frac{3 + 8 - 1}{5} = \frac{10}{5} = 2$

L.R = $2e \times$ (dis b/w Focus and directrix)

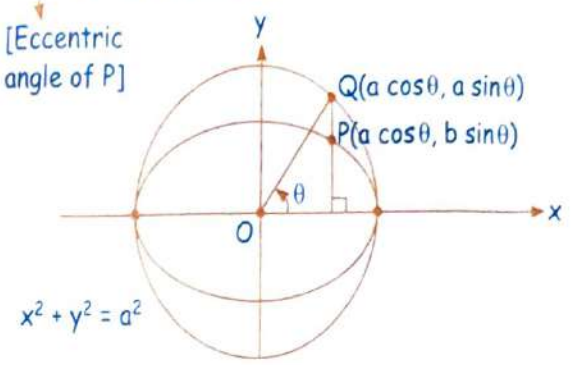
$= 2 \times \frac{1}{3} \times 2 = \frac{4}{3}$

PARAMETRIC COORDINATES AND AUXILIARY CIRCLE

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\theta \rightarrow$ Parameter

[Eccentric angle of P]



$\Rightarrow \angle QOX = \theta$

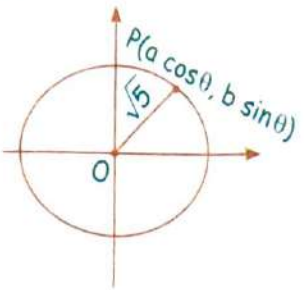
Q is called corresponding pt of P

Ques. Eccentric angle of the points on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ whose distance from the centre is $\sqrt{5}$ is

(a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

(c) $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ (d) $\frac{\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{15\pi}{8}$

Ans. $OP = \sqrt{5}$



$a^2\cos^2\theta + b^2\sin^2\theta = 5$

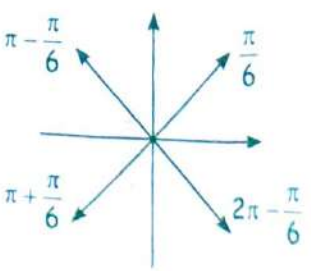
$6\cos^2\theta + 2\sin^2\theta = 5$

$4\cos^2\theta + 2 = 5$

$4\cos^2\theta = 3$

$\cos^2\theta = \frac{3}{4}$

$\cos\theta = \pm\frac{\sqrt{3}}{2}$



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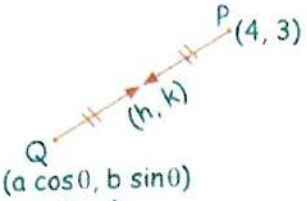
Ques. The locus of the mid point the line segment joining the point (4, 3) and the points on the ellipse $x^2 + 2y^2 = 4$ is ellipse with eccentricity:

[JEE Main-2022 (June-11)]

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{2\sqrt{2}}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{2}$

Ans. $\frac{x^2}{4} + \frac{y^2}{2} = 1$

$a = 2, \quad b = \sqrt{2}$



$\therefore 2h = a\cos\theta + 4$

$\Rightarrow \cos\theta = \frac{2h-4}{a}$ ...(i)

$2k = b\sin\theta + 3$

$\Rightarrow \sin\theta = \frac{2k-3}{b}$ ...(ii)

Square and add:

$\frac{(2h-4)^2}{a^2} + \frac{(2k-3)^2}{b^2} = 1$

$\Rightarrow \frac{(2h-4)^2}{4} + \frac{(2k-3)^2}{2} = 1$

$(h, k) \to (x, y)$

$\Rightarrow \frac{(2x-4)^2}{4} + \frac{(2y-3)^2}{2} = 1$

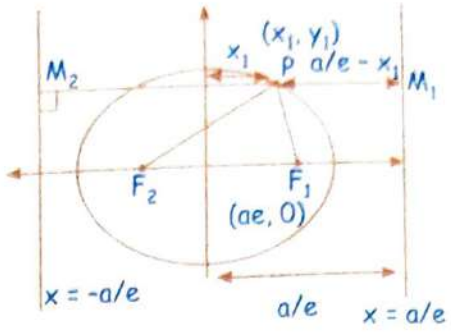
$\Rightarrow \frac{(x-2)^2}{1} + \frac{(y-3/2)^2}{1/2} = 1$

$A^2 = 1, B^2 = \frac{1}{2} \quad (A > B)$

$\because B^2 = A^2(1 - E^2)$

$\Rightarrow \frac{1}{2} = 1 - E^2 \Rightarrow E^2 = \frac{1}{2}$

FOCAL DISTANCES OF ANY POINT P ON ELLIPSE



$\frac{PF_1}{PM_1} = e \Rightarrow PF_1 = ePM_1$

$\because PM_1 = \frac{a}{e} - x_1 = \frac{a}{e} - a\cos\theta$

$\therefore PF_1 = e\left[\frac{a}{e} - a\cos\theta\right]$

$\Rightarrow PF_1 = a - ae\cos\theta$ ...(i)

$\because \frac{PF_2}{PM_2} = e \Rightarrow PF_2 = ePM_2$

$\because PM_2 = \frac{a}{e} + a\cos\theta$

$\Rightarrow PF_2 = e\left[\frac{a}{e} + a\cos\theta\right]$

$\Rightarrow PF_2 = a + ae\cos\theta$ ...(ii)

By adding equation (i) and (ii)

$\boxed{PF_1 + PF_2 = 2a}$

Ques. If the co-ordinates of two points A and B are $(\sqrt{7}, 0)$ and $(-\sqrt{7}, 0)$ respectively and P is any point on the conic, $9x^2 + 16y^2 = 144$, then PA + PB is equal to:

(a) 9 (b) 16 (c) 6 (d) 8

Ans. $\frac{x^2}{16} + \frac{y^2}{9} = 1, a^2 = 16, b^2 = 9$

$\because b^2 = a^2(1 - e^2)$

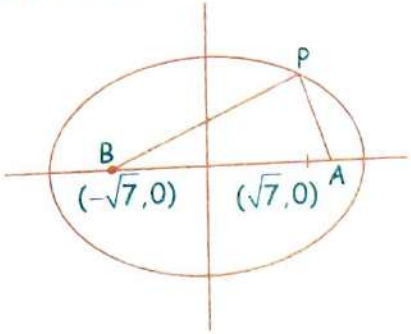
$\Rightarrow 9 = 16(1 - e^2) \Rightarrow 9 = 16 - 16e^2$

$\Rightarrow 16e^2 = 7 \Rightarrow e = \frac{\sqrt{7}}{4}$

$\therefore F[ae, 0] = \left[4 \times \frac{\sqrt{7}}{4}, 0\right] = [\sqrt{7}, 0]$

That means, $(\sqrt{7}, 0)$ and $(-\sqrt{7}, 0)$ are coordinates of focus.

Therefore, PA + PB = 2a = 8



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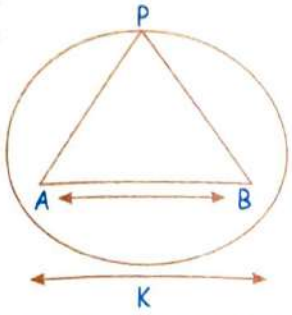
Important Note

The sum of focal distance of any point P lying on the ellipse is always equal to the length of Major Axis.

Conversely, if a point P moves such that sum of its distance from two fixed points is always a constant (k) then locus of P is an ellipse provided K > AB.



If PA + PB = K → Length of Major Axis
K < AB → no locus

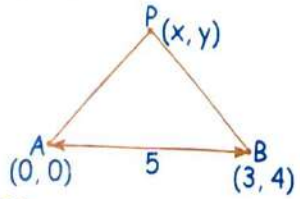
Example: Interpret the locus

1. $\sqrt{x^2+y^2}+\sqrt{(x-3)^2+(y-4)^2}=6$

PA + PB = 6 ⇒ K = 6

∵ AB = 5,

∴ K > AB



⇒ locus of P is ellipse

2. $\sqrt{x^2+y^2}+\sqrt{(x-3)^2+(y-4)^2}=3$

PA + PB = 3 ⇒ K = 3, AB = 5

K < AB ⇒ no locus.

3. $\sqrt{x^2+y^2}+\sqrt{(x-3)^2+(y-4)^2}=5$

PA + PB = 5

⇒ K = 5, AB = 5 ⇒ K = AB

Locus is Line Segment AB

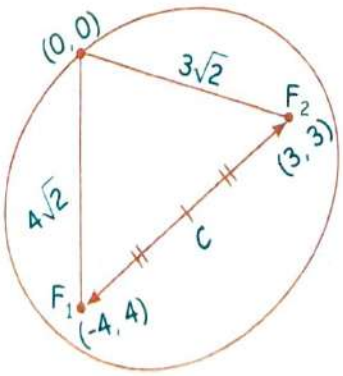
Ques. An ellipse having foci at (3, 3) and (-4, 4) and passing through the origin has eccentricity equal to

(a) $\frac{3}{7}$

(b) $\frac{2}{7}$

(c) $\frac{5}{7}$ ✓

(d) $\frac{3}{7}$



Ans. $F_1F_2 = 2ae = \sqrt{7^2+1^2} \Rightarrow 2ae = \sqrt{50}$

$\Rightarrow 2ae = 5\sqrt{2}$ ...(i)

$OF_1 + OF_2 = 2a$

$\Rightarrow 4\sqrt{2} + 3\sqrt{2} = 2a, \Rightarrow 2a = 7\sqrt{2}$ ...(ii)

By eqn (i) and (ii):

$e \cdot 7\sqrt{2} = 5\sqrt{2} \Rightarrow e = \frac{5}{7}$

POSITION OF A POINT W.R.T. AN ELLIPSE

Position of a given point (x_1, y_1) relative to a given ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Let $S(x, y) = \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \Rightarrow S_1: \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1$

1. If $S(x_1, y_1) > 0$, then (x_1, y_1) lies outside the ellipse.
2. If $S(x_1, y_1) < 0$, then (x_1, y_1) lies inside the ellipse.
3. If $S(x_1, y_1) = 0$, then (x_1, y_1) lies on the ellipse.

LINE AND ELLIPSE

$$\frac{x^2}{a^2}+\frac{(mx+C)^2}{b^2}=1$$

$\Rightarrow b^2x^2 + a^2[m^2x^2 + c^2 + 2mxc] = a^2b^2$

$\Rightarrow x^2(b^2 + a^2m^2) + 2ma^2cx + a^2c^2 - a^2b^2 = 0,$

$D = 0$

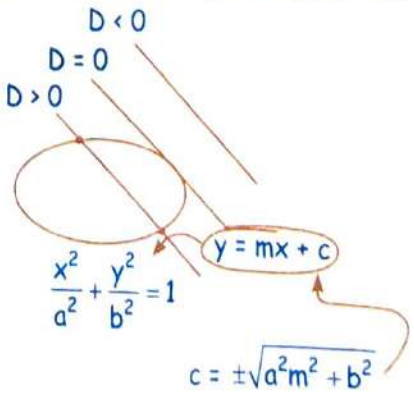
$\therefore 4m^2a^4c^2 - 4(b^2 + a^2m^2)\,a^2(c^2 - b^2) = 0$

$\Rightarrow m^2a^2c^2 - [b^2c^2 - b^4 + a^2m^2c^2 - a^2m^2b^2] = 0$

$\Rightarrow -b^2c^2 + b^4 + a^2m^2b^2 = 0$

$\Rightarrow -c^2 + b^2 + a^2m^2 = 0$

$\Rightarrow c^2 = a^2m^2 + b^2 \Rightarrow \boxed{c = \pm\sqrt{a^2m^2+b^2}}$



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Condition of Tangency:

$$y = mx \pm \sqrt{a^2m^2 + b^2}$$

Slope form of Tangent

Various Forms of Tangents

(a) Cartesian Form:

Equation of the tangent at (x_1, y_1) is given by

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

(b) Slope Form:

The equation of the tangent with slope m is given by

$$y = mx \pm \sqrt{a^2m^2 + b^2}$$

(c) Parametric Form:

Equation of the tangent $P(a\cos\theta, b\sin\theta)$, where θ is the eccentric angle of the point P, is

$x_1 = a\cos\theta \qquad y_1 = b\sin\theta$

$$\frac{x(a\cos\theta)}{a^2} + \frac{y(b\sin\theta)}{b^2} = 1$$

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

Ques. The equation of the tangent at the point $(1/4, 1/4)$ of the ellipse $\frac{x^2}{4} + \frac{y^2}{12} = 1$, is-

(a) $3x + y = 48$ (b) $3x + y = 3$

(c) $3x + y = 16$ (d) None of these

Ans. $\frac{xx_1}{4} + \frac{yy_1}{12} = 1$

$$\frac{x \cdot \frac{1}{4}}{4} + \frac{y \cdot \frac{1}{4}}{12} = 1$$

$$\frac{x}{16} + \frac{y}{48} = 1 \qquad \text{WRONG}$$

$3x + y = 48$

First, check the position of the point

$$\therefore S_1 = \frac{\left(\frac{1}{4}\right)^2}{4} + \frac{\left(\frac{1}{4}\right)^2}{12} - 1$$

$$= \frac{1}{64} + \frac{1}{16 \times 4 \times 3} - 1 = \frac{1}{64}\left[1 + \frac{1}{3}\right] - 1$$

$$= \frac{1}{64} \times \frac{4}{3} - 1 = -\text{ve}$$

Since the point $\left(\frac{1}{4}, \frac{1}{4}\right)$ lies inside

$\Rightarrow$ no tangent is possible

Ques. The equations of the tangent of the ellipse $9x^2 + 16y^2 = 144$ which passes through the point $(2, 3)$ is

(a) $y = 3, x + y = 5$ (b) $y = -3, x - y = 5$

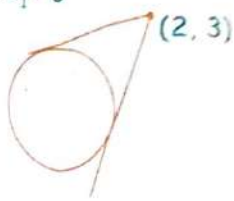
(c) $y = 4, x + y = 3$ (d) $y = -4, x - y = 3$

Ans. $S: 9x^2 + 16y^2 - 144 = 0$

$S_1: 9 \times 2^2 + 16 \times 3^2 - 144$

$= 36 + 144 - 144 = 36 \Rightarrow S_1 > 0$

$\Rightarrow$ P(2, 3) outside



$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

$a^2 = 16, \quad b^2 = 9$

$\therefore y = mx \pm \sqrt{a^2m^2 + b^2}$

$\Rightarrow y = mx \pm \sqrt{16m^2 + 9}$

$\because$ (2, 3) satisfies the equation

$\Rightarrow 3 - 2m = \pm\sqrt{16m^2 + 9}$,

$\Rightarrow (3 - 2m)^2 = 16m^2 + 9$

$\Rightarrow 9 + 4m^2 - 12m = 16m^2 + 9$

$\Rightarrow 12m^2 = -12m$

$\Rightarrow m^2 + m = 0$

$\Rightarrow m(m + 1) = 0 \Rightarrow m = 0, -1$

eqn of tangent: $y - 3 = m(x - 2)$

$m = 0 \Rightarrow y = 3$

at $m = -1$

$y - 3 = -x + 2, \Rightarrow y + x = 5$

Ques. The line $x\cos\alpha + y\sin\alpha = p$ will be a tangent to the conic $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. If

(a) $p^2 = a^2\sin^2\alpha + b^2\cos^2\alpha$

(b) $p^2 = a^2 + b^2$

(c) $p^2 = b^2\sin^2\alpha + a^2\cos^2\alpha$

(d) None of these

Ans. Let tangent be

$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$ and $x\cos\alpha + y\sin\alpha = p$ (given)

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Compare both the equations

$$\frac{\cos\theta}{a\cos\alpha} = \frac{\sin\theta}{b\sin\alpha} = \frac{1}{p}$$

$$\therefore \cos\theta = \frac{a\cos\alpha}{p} \text{ and } \sin\theta = \frac{b\sin\alpha}{p}$$

$$\because \cos^2\theta + \sin^2\theta = 1$$

$$\Rightarrow \frac{a^2\cos^2\alpha}{p^2} + \frac{b^2\sin^2\alpha}{p^2} = 1,$$

$$\Rightarrow a^2\cos^2\alpha + b^2\sin^2\alpha = p^2$$

Important Note

A variable tangent is drawn to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meeting the coordinate axes at points A and B respectively, then

(a) The minimum area of Triangle OAB is ab

(b) The minimum length of AB is (a + b)

Proof: **(a)** eqⁿ of tangent: $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$

For A, put y = 0

$$\frac{x\cos\theta}{a} + 0 = 1 \Rightarrow x = a\sec\theta$$

$$\text{Area} = \left|\frac{1}{2}a\sec\theta \cdot b\operatorname{cosec}\theta\right| = \left|\frac{1}{2}\frac{ab}{\sin\theta\cos\theta}\right|$$

$$= \frac{ab}{|\sin 2\theta|}$$

For Area (min):

$|\sin 2\theta|$ is max

$\Rightarrow |\sin 2\theta| = 1$

$\Rightarrow A_{min} = ab$

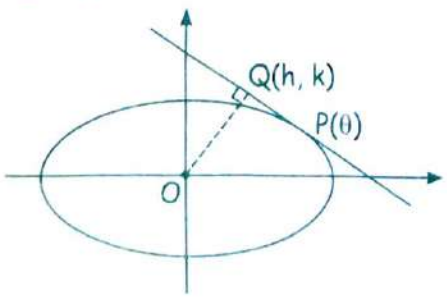
(b) $(AB)^2 = \underbrace{a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta}_{\text{min} \Rightarrow (a+b)^2}$

$\therefore (AB)^2_{min} = (a + b)^2$

$\Rightarrow (AB)_{min} = a + b$

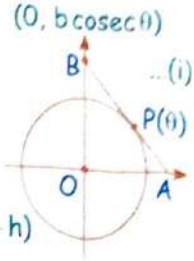
Ques. Find the locus of the foot of perpendicular drawn center to any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Ans.



eqⁿ of Tangent

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1 \quad \text{...(i)}$$



$$M_{OQ} = \frac{k}{h} \Rightarrow M_T = -\frac{h}{k}$$

eqⁿ of Tangent: $y - k = -\frac{h}{k}(x - h)$

$ky - k^2 = -hx + h^2$, $hx + ky = h^2 + k^2$

$hx + Ky = h^2 + k^2$ ...(ii)

By comparing eqn (i) and (ii)

$$\frac{\cos\theta}{ah} = \frac{\sin\theta}{bk} = \frac{1}{h^2 + k^2}$$

$$\therefore \cos\theta = \frac{ah}{h^2 + k^2} \text{ and } \sin\theta = \frac{bk}{h^2 + k^2}$$

$$\because \sin^2\theta + \cos^2\theta = 1$$

$$\Rightarrow \frac{a^2h^2}{(h^2 + k^2)^2} + \frac{b^2k^2}{(h^2 + k^2)^2} = 1$$

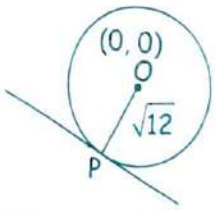
$$\Rightarrow a^2h^2 + b^2k^2 = (h^2 + k^2)^2$$

$\Rightarrow (h, k) \rightarrow (x, y)\ (x^2 + y^2)^2 = a^2x^2 + b^2y^2$

Ques. If m is the slope of a common tangent to the curves $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and $x^2 + y^2 = 12$, then $12m^2$ is equal to: **[JEE Main-2022 (June-11)]**

(a) 6 (b) 9 (c) 10 (d) 12

Ans. $\because OP = r = \sqrt{12}$



$$y = mx \pm \sqrt{16m^2 + 9}$$

$$\Rightarrow \left|\frac{\pm\sqrt{16m^2 + 9}}{\sqrt{1 + m^2}}\right| = \sqrt{12}$$

$\Rightarrow 16m^2 + 9 = 12(1 + m^2) \Rightarrow 4m^2 = 12 - 9$

$\Rightarrow \boxed{4m^2 = 3}, \Rightarrow 12m^2 = 9$

Ques. Let L be a tangent line to the parabola $y^2 = 4x - 20$ at (6, 2). If L is also a tangent to the ellipse $\frac{x^2}{2} + \frac{y^2}{b} = 1$, then the value of b is equal to

[JEE Main-2021 (March)]

(a) 11 (b) 14 (c) 16 (d) 20

Ans. $y = mx - \sqrt{2m^2 + b}$

$\because m = 1$

$\therefore y = x - \sqrt{2 + b}$,

put (6, 2) on it

$\Rightarrow \sqrt{2 + b} = 4 \Rightarrow 2 + b = 16 \Rightarrow b = 14$

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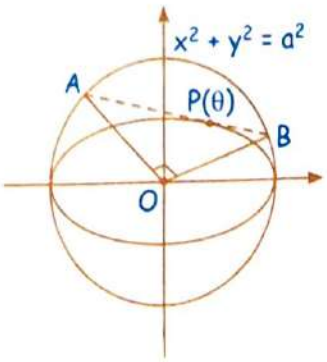
Brain Teaser

The tangent at the point $P(\theta)$ of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meet the auxiliary circle at A and B. Segment AB subtends a right at the origin. Eccentricity of the ellipse is

(a) $\frac{1}{\sqrt{1+\cos^2\theta}}$ (b) $\frac{1}{\sqrt{1+\operatorname{cosec}^2\theta}}$

(c) $\frac{1}{\sqrt{1+\sec^2\theta}}$ (d) $\frac{1}{\sqrt{1+\sin^2\theta}}$

Ans. By Homogenization equation of Tangent



Equation of tangent:

$$\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$$

Homogenization equation:

$$x^2+y^2-a^2\left(\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}\right)^2=0$$

$\therefore$ coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow 1-\frac{a^2\cos^2\theta}{a^2}+1-\frac{a^2\sin^2\theta}{b^2}=0$$

$$\Rightarrow 2=\cos^2\theta+\frac{a^2\sin^2\theta}{b^2}$$

$$\Rightarrow 2-\cos^2\theta=\frac{a^2\sin^2\theta}{a^2(1-e^2)}$$

$$\Rightarrow 1-e^2=\frac{\sin^2\theta}{2-\cos^2\theta} \Rightarrow 1-e^2=\frac{\sin^2\theta}{1+1-\cos^2\theta}$$

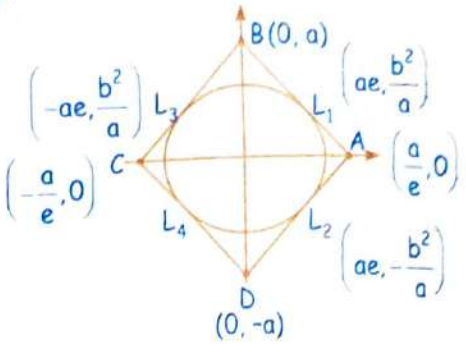
$$\Rightarrow 1-e^2=\frac{\sin^2\theta}{1+\sin^2\theta} \Rightarrow e^2=1-\frac{\sin^2\theta}{1+\sin^2\theta}$$

$$\Rightarrow e^2=\frac{1}{1+\sin^2\theta} \Rightarrow e=\frac{1}{\sqrt{1+\sin^2\theta}}$$

Important Note

Tangents drawn at the 4 ends of Latera Recta of an ellipse enclose a rhombus whose area is $2a^2/e$.

Proof:



$$T_1:\ \frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1, \Rightarrow \frac{xae}{a^2}+\frac{y\frac{b^2}{a}}{b^2}=1$$

$$\Rightarrow \frac{ex}{a}+\frac{y}{a}=1$$

$$\Rightarrow ex+y=a \quad \ldots(i)$$

$$P_2: ex-y=a \quad \ldots(ii)$$

$$T3: -ex+y=a \quad \ldots(iii)$$

By equation (i) and (ii) we get:

$$x=\frac{a}{e},\ y=0$$

$$\text{Area of ABCD} = \frac{1}{2}d_1d_2=\frac{1}{2}\cdot\frac{2a}{e}\cdot 2a=\frac{2a^2}{e}$$

Ques. Consider the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$. Let $H(\alpha, 0)$, $0<\alpha<2$, be a point. A straight line drawn through H parallel to the y-axis crosses the ellipse and its auxiliary circle at points E and F respectively, in the first quadrant. The tangent to the ellipse at the point E intersects the positive x-axis at a point G Suppose the straight line joining F and the origin an angle ϕ with the positive x-axis. **[JEE ADV-2022]**

List I		List II	
(I)	If $\phi=\frac{\pi}{4}$, then the area of the triangle FGH is	(P)	$\frac{(\sqrt{3}-1)^4}{8}$
(II)	If $\phi=\frac{\pi}{3}$, then the area of the triangle FGH is	(Q)	1

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(III)	If $\phi = \frac{\pi}{6}$, then the area of the triangle FGH is	(R)	$\frac{3}{4}$
(IV)	If $\phi = \frac{\pi}{12}$, then the area of the triangle FGH is	(S)	$\frac{1}{2\sqrt{3}}$
		(T)	$\frac{3\sqrt{3}}{2}$

The correct option is:

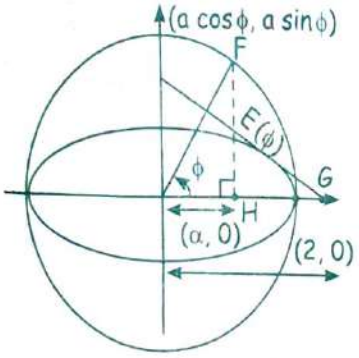
(a) (I) → (R); (II) → (S); (III) → (Q); (IV) → (P)

(b) (I) → (R); (II) → (T); (III) → (S); (IV) → (P)

(c) (I) → (Q); (II) → (T); (III) → (S); (IV) → (P)

(d) (I) → (Q); (II) → (S); (III) → (Q); (IV) → (P)

Ans. E[a cosϕ, b sinϕ], F[a cosϕ, a sinϕ]



equation of Tangent: $\frac{x\cos\phi}{a} + \frac{y\sin\phi}{b} = 1$

y = 0 for G,

$\frac{x\cos\phi}{a} + 0 = 1$

$\Rightarrow x = a\sec\phi$

$\therefore$ G[a secϕ, 0]

In ΔFGH

height (h) = a sinϕ

base = a secϕ - a cosϕ

$\therefore A = \frac{1}{2}(a\sec\phi - a\cos\phi)a\sin\phi$

$\Rightarrow A = \frac{1}{2}a^2[\sec\phi - \cos\phi]\sin\phi$

$\Rightarrow A = 2[\sec\phi - \cos\phi]\sin\phi \quad [\because a = 2]$

when, $\phi = \frac{\pi}{4} \Rightarrow A = 2\left[\sec\frac{\pi}{4} - \cos\frac{\pi}{4}\right]\sin\frac{\pi}{4}$

$= 2\left[\sqrt{2} - \frac{1}{\sqrt{2}}\right]\cdot\frac{1}{\sqrt{2}}$

(2 - 1) = (1)

$\therefore$ (I) → Q

when, $\phi = \frac{\pi}{3}$

$A = 2\left[2 - \frac{1}{2}\right]\frac{\sqrt{3}}{2} = \frac{3}{2}\cdot\sqrt{3} = \frac{3\sqrt{3}}{2}$

$\therefore$ (II) → T

Hence, option (c) is correct

Ques. The line x = 8 is the directrix of the ellipse E: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with the corresponding focus (2, 0). If the tangent to E at the point P in the first quadrant passes through the point $(0, 4\sqrt{3})$ and intersects the x-axis at Q, then $(3PQ)^2$ is equal to ___.

Ans. $\frac{a}{e} = 8$ ...(i)

and ae = 2 ...(ii)

By multiplying equation (i) and (ii)

$a^2 = 16 \Rightarrow a = 4$

$\therefore e = \frac{1}{2}$

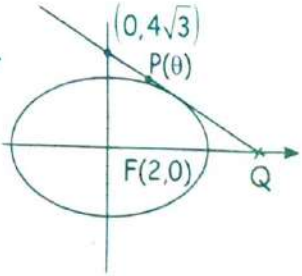
$\because b^2 = a^2(1 - e^2)$

$\Rightarrow b^2 = 16\left[1 - \frac{1}{4}\right] = 16 \times \frac{3}{4} = 12$

$\Rightarrow b = 2\sqrt{3}$

Equation of tangent

$\frac{x\cos\theta}{4} + \frac{y\sin\theta}{2\sqrt{3}} = 1$



Put $(0, 4\sqrt{3})$ on it

$\frac{4\sqrt{3}\sin\theta}{2\sqrt{3}} = 1$

$\Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$

$\therefore$ P[a cosθ, b sinθ]

$= P[4\cos30°, 7\sqrt{3}\sin30°] = [2\sqrt{3}, \sqrt{3}]$

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$= P[2\sqrt{3}, \sqrt{3}]$

and $Q[a\sec\theta, 0] = [4\sec 30°, 0]$

$= \left(4\cdot\frac{2}{\sqrt{3}}, 0\right) = \left(\frac{8}{\sqrt{3}}, 0\right)$

$\therefore (PQ)^2 = \left(2\sqrt{3} - \frac{8}{\sqrt{3}}\right)^2 + \left(\sqrt{3}\right)^2$

$= \left(\frac{6-8}{\sqrt{3}}\right)^2 + 3 = \frac{4}{3} + 3$

$\Rightarrow (PQ)^2 = \frac{4+9}{3}$

$\Rightarrow (PQ)^2 = \frac{13}{3}$

Therefore, $(3PQ)^2 = 9(PQ)^2 = 9\times\frac{13}{3} = 39$

Ques. Let a circle of radius 4 be concentric to the ellipse $15x^2 + 19y^2 = 285$. Then the common tangents are inclined to the minor axis of the ellipse at the angle.

[JEE Main-2023 (Apr,-11)]

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{12}$

Ans. Equation of circle: $x^2 + y^2 = 16$

$\frac{x^2}{19} + \frac{y^2}{15} = 1$

Eqⁿ of tangents: $y = mx \pm \sqrt{19m^2 + 15}$ (ellipse)

or $y = mx \pm 4\sqrt{1+m^2}$ (circle)

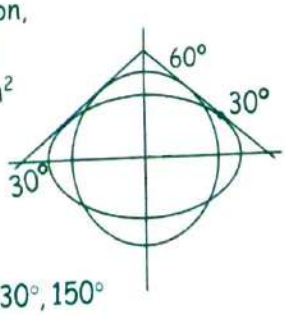
Comparing both equation,

$19m^2 + 15 = 16(1 + m^2)$

$\Rightarrow 19m^2 + 15 = 16 + 16m^2$

$\Rightarrow 3m^2 = 1$

$\Rightarrow m = \pm\frac{1}{\sqrt{3}}$



$\Rightarrow \tan\theta = \pm\frac{1}{\sqrt{3}} \quad \therefore \theta = 30°, 150°$

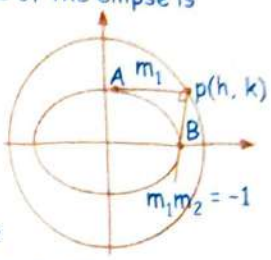
Angle to the minor axis = $90° - 30° = 60° = \frac{\pi}{3}$

DIRECTOR CIRCLE FOR ELLIPSE

The locus of the point of intersection of the perpendicular tangents to the ellipse is called the director circle of the ellipse.

Equation of the director circle of the ellipse is

$x^2 + y^2 = a^2 + b^2$



Proof: $y = mx \pm \sqrt{a^2m^2 + b^2}$

$(x, y) \to (h, k)$

$k - mh = \pm\sqrt{a^2m^2 + b^2}$

$\Rightarrow (k - mh)^2 = a^2m^2 + b^2$

$\Rightarrow k^2 + m^2h^2 - 2mhk = a^2m^2 + b^2$

$\Rightarrow m^2(h^2 - a^2) - 2mhk + k^2 - b^2 = 0$ (m_2, m_1)

$m_1m_2 = \frac{k^2 - b^2}{h^2 - a^2} = -1$

$\Rightarrow k^2 - b^2 = -h^2 + a^2$

$\Rightarrow h^2 + k^2 = a^2 + b^2$

$\therefore x^2 + y^2 = a^2 + b^2$

circle at centre (0, 0), $r = \sqrt{a^2 + b^2}$

Ques. Director circle of an ellipse is $x^2 + y^2 - 2x - 2y - 13 = 0$ and eccentricity is $\frac{\sqrt{3}}{2}$. Distance between the foci of the ellipse is

(a) 6 (b) 3 (c) 12 (d) $3\sqrt{2}$

Ans. centre [1, 1]

radius $= \sqrt{1^2 + 1^2 + 13} = \sqrt{15}$

$\therefore \sqrt{a^2 + b^2} = \sqrt{15}$

$\Rightarrow a^2 + b^2 = 15$ ...(i)

$\because b^2 = a^2(1 - e^2)$

$\Rightarrow b^2 = a^2\left(1 - \frac{3}{4}\right)$

$\Rightarrow b^2 = \frac{a^2}{4}$ ...(ii)

By equation (i) & (ii)

$a^2 + \frac{a^2}{4} = 15$

$\Rightarrow \frac{5a^2}{4} = 15 \Rightarrow a^2 = 12$ and $b^2 = 3$

Dis b/w Foci $= 2ae = 2\sqrt{12}\cdot\frac{\sqrt{3}}{2} = 6$

POINT OF INTERSECTION OF TANGENTS

Equation of trangetns from a point:

$\frac{x\cos\theta_1}{a} + \frac{y\sin\theta_1}{b} = 1$

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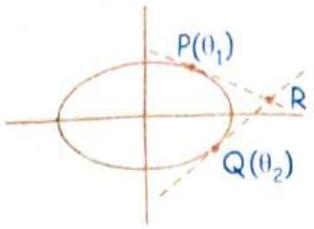
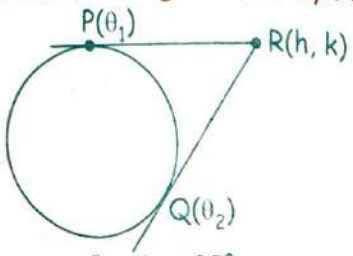
and $\frac{x\cos\theta_2}{a}+\frac{y\sin\theta_2}{b}=1$



$$\therefore R\left[\frac{a\cos\left(\frac{\theta_1+\theta_2}{2}\right)}{\cos\left(\frac{\theta_1-\theta_2}{2}\right)},\frac{b\sin\left(\frac{\theta_1+\theta_2}{2}\right)}{\cos\left(\frac{\theta_1-\theta_2}{2}\right)}\right]$$

Ques. Find the Locus of point of intersection of tangents drawn at the points on ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose eccentric angles differ by 90°.



$\theta_1-\theta_2=90°$

Ans. $h=\frac{a\cos\left(\frac{\theta_1+\theta_2}{2}\right)}{\cos\left(\frac{\theta_1-\theta_2}{2}\right)}$, $k=\frac{b\sin\left(\frac{\theta_1+\theta_2}{2}\right)}{\cos\left(\frac{\theta_1-\theta_2}{2}\right)}$

$\Rightarrow h=\sqrt{2}a\cos\left(\frac{\theta_1+\theta_2}{2}\right), k=\sqrt{2}b\sin\left(\frac{\theta_1+\theta_2}{2}\right)$

$\therefore \cos\left(\frac{\theta_1+\theta_2}{2}\right)=\frac{h}{\sqrt{2}a}$

and $\sin\left(\frac{\theta_1+\theta_2}{2}\right)=\frac{k}{\sqrt{2}b}$

$\because \sin^2\theta+\cos^2\theta=1$

$\Rightarrow \left(\frac{h}{\sqrt{2}a}\right)^2+\left(\frac{k}{\sqrt{2}b}\right)^2=1$

$\Rightarrow \frac{h^2}{a^2}+\frac{k^2}{b^2}=2 \Rightarrow \frac{x^2}{a^2}+\frac{y^2}{b^2}=2$

VARIOUS FORMS OF NORMALS

(a) Cartesian Form

$$\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2-b^2=a^2e^2$$

(b) Slope Form

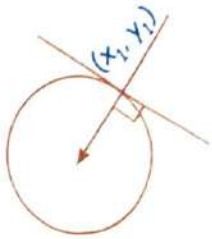
$$y=mx\mp\frac{(a^2-b^2)m}{\sqrt{a^2+b^2m^2}}$$

(c) Parametric Form

$$\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2=a^2e^2$$

Proof: $\because$ T: $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$

$m_T=-\frac{x_1\cdot b^2}{a^2\cdot y_1}=-\frac{b^2x_1}{a^2y_1}$



$\therefore m_N=\frac{a^2y_1}{b^2x_1}$

eqⁿ of Normal:

$y-y_1=\frac{a^2y_1}{b^2x_1}(x-x_1)$

$\Rightarrow \frac{y-y_1}{y_1}=\frac{a^2}{b^2}\left(\frac{x-x_1}{x_1}\right)$

$\Rightarrow b^2\left(\frac{y-y_1}{y_1}\right)=a^2\left[\frac{x-x_1}{x_1}\right]$

$\Rightarrow b^2\left[\frac{y}{y_1}-1\right]=a^2\left[\frac{x}{x_1}-1\right]$

$\Rightarrow \frac{b^2y}{y_1}-b^2=\frac{a^2x}{x_1}-a^2$

$\Rightarrow a^2-b^2=\frac{a^2x}{x_1}-\frac{b^2y}{y_1}$

$\Rightarrow \frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2e^2$ (Cartesian form)

Put, $x_1=a\cos\theta$, $y_1=b\sin\theta$

$\frac{a^2x}{a\cos\theta}-\frac{b^2y}{b\sin\theta}=a^2e^2$

$\Rightarrow \frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2e^2$

$=a^2-b^2$ (parametric form)

Ques. The equation of the normal at the point (2, 3) on the ellipse $9x^2+16y^2=180$, is

(a) $3y=8x-10$ (b) $3y-8x+7=0$

(c) $8y+3x+7=0$ (d) $3x+2y+7=0$

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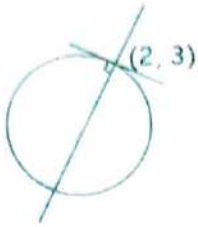
Ans. $S_1: 9(2)^2 + 16(3)^2 - 180$

$= 36 + 144 - 180 = 0$

$\Rightarrow (2, 3)$ lies on the ellipse



$\frac{x^2}{20} + \frac{y^2}{\frac{45}{4}} = 1$

$\therefore a^2 = 20,\ b^2 = \frac{45}{4}$

Equation of normal:

$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$

$\Rightarrow \frac{20x}{2} - \frac{45y}{4(3)} = 20 - \frac{45}{4}$

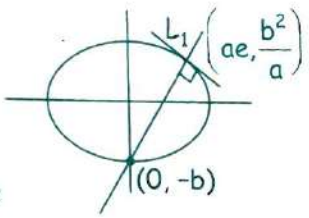
$\Rightarrow 10x - \frac{15y}{4} = \frac{80 - 45}{4}$

$\Rightarrow 40x - 15y = 35$

$\Rightarrow 8x - 3y = 7 \Rightarrow 3y - 8x + 7 = 0$

Ques. If the normal at an end of a latus-rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through one extremity of the minor axis, show that the eccentricity of the ellipse is given by $e = \sqrt{\frac{\sqrt{5} - 1}{2}}$.

Ans. $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$



$\because (x_1, y_1) = \left(ae, \frac{b^2}{a}\right)$

$\therefore \frac{a^2x}{ae} - \frac{b^2y}{\frac{b^2}{a}} = a^2 - b^2$

$\because b^2 = a^2(1 - e^2) \Rightarrow a^2 - b^2 = a^2e^2$

$\therefore \frac{ax}{e} - ay = a^2e^2$

$\Rightarrow \frac{x}{e} - y = ae^2$

$\because (0, -b)$ lies on the normal

$\therefore b = ae^2 \Rightarrow \frac{b}{a} = e^2$

$\Rightarrow \frac{b^2}{a^2} = e^4$ ...(i)

and $\frac{b^2}{a^2} = 1 - e^2$ ...(ii)

By comparing equation (i) & (ii)

$e^4 = 1 - e^2 \Rightarrow e^4 + e^2 - 1 = 0$

Let, $t = e^2 \Rightarrow t^2 + t - 1 = 0$

$\therefore t = \frac{-1 + \sqrt{5}}{2}$

$\Rightarrow e^2 = \frac{-1 + \sqrt{5}}{2} \Rightarrow e = \sqrt{\frac{\sqrt{5} - 1}{2}}$

Ques. Let T_1 and T_2 be two distinct common tangents to the ellipse E: $\frac{x^2}{6} + \frac{y^2}{3} = 1$ and the parabola P: $y^2 = 12x$. Suppose that the tangent T_1 touches P and E at the points A_1 and A_2 respectively and the tangent T_2 touches P and E at the points A_4 and A_3, respectively. Then which of the following statements is (are) true?

[JEE Adv, 2023]

(a) The area of the quadrilateral $A_1A_2A_3A_4$ is 35 square units

(b) The area of the quadrilateral $A_1A_2A_3A_4$ is 36 square units

(c) The tangents T_1 and T_2 meet x-axis at the point (-3, 0)

(d) The tangents T_1 and T_2 meet x-axis at the point (-6, 0)

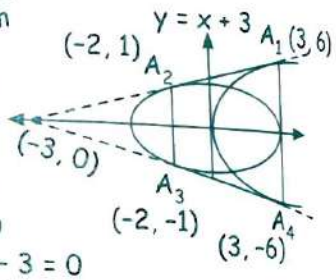
Ans. $y = mx \pm \sqrt{6m^2 + 3}$ (Tangent of ellipse)

$y = mx + \frac{3}{m}$ (Tangent of parabola)

$\because$ Both tangent are same

$\therefore \pm\sqrt{6m^2 + 3} = \frac{3}{m}$

$\Rightarrow 6m^2 + 3 = \frac{9}{m^2}$



Let $m^2 = z$

$\therefore (6z + 3)\, z = 9$

$\Rightarrow 2z^2 + z - 3 = 0$

$\Rightarrow 2z^2 + 3z - 2z - 3 = 0$

$\Rightarrow (z - 1)(2z + 3) = 0$

$z = 1,\ z = -\frac{3}{2}$ (m^2 cannot be -ve)

$\therefore m^2 = 1 \Rightarrow m = \pm 1$

$\therefore T_1: y = x + 3$ and $y = -x - 3$

Both meet at (-3, 0)

P $\rightarrow$ $m = \frac{1}{t} = 1 \Rightarrow t = 1$

$A_1(at^2, 2at) = (3, 6)$

$\therefore \frac{xx_1}{6} + \frac{yy_1}{3} = 1 \Rightarrow \frac{xx_1}{2} + yy_1 = 3$

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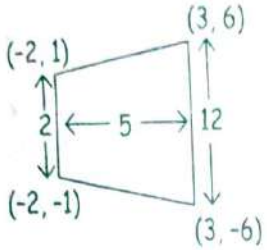
and $-x + y = 3$

By comparing both equation

$\frac{x_1}{2} = -1$ and $y_1 = 1$



Area of $A_1A_2A_3A_4 = \frac{1}{2}(12+2)\times 5 = 35$ sq.units

Ques. If the maximum distance of normal to the ellipse $\frac{x^2}{4}+\frac{y^2}{b^2}=1$, $b < 2$, from the origin is 1, then the eccentricity of the ellipse is:

[JEE Main-2023 (Jan,-1)]

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{4}$

Ans. $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2e^2$

$\because a = 2$

$\therefore \frac{2x}{\cos\theta} - \frac{by}{\sin\theta} = 4e^2 = a^2 - b^2$

$\because 1 = \left|\frac{4e^2}{\sqrt{a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta}}\right|$

$\because (a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta)_{min} = (a+b)^2$

$\Rightarrow 1 = \frac{a^2-b^2}{a+b}$

$\Rightarrow 1 = a - b$

$\Rightarrow 1 = 2 - b \Rightarrow b = 1$

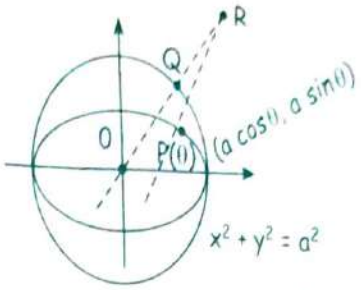
$\because b^2 = a^2(1-e^2)$

$\Rightarrow 1 = 4(1-e^2)$

$\Rightarrow 4e^2 = 3,\ e = \frac{\sqrt{3}}{2}$

Ques. P and Q are corresponding points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the auxilliary circle respectively. The normal at P to the ellipse and at Q to auxiliary circle meet at R. Prove that as P varies, locus of R is given by $x^2 + y^2 = (a+b)^2$.

Ans.



Equation of normal: $\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$ ...(i)

equation of OQ: $y - 0 = \frac{a\sin\theta - 0}{a\cos\theta - 0}(x - 0)$

$\Rightarrow \frac{x}{\cos\theta} = \frac{y}{\sin\theta}$ ...(ii)

By equation (i) & (ii)

$\frac{ax}{\cos\theta} - \frac{bx}{\cos\theta} = a^2 - b^2$

$\Rightarrow \frac{x}{\cos\theta}(a-b) = (a-b)(a+b)$

$\Rightarrow x = (a+b)\cos\theta$ ...(iii)

Put in equation (ii): $\frac{(a+b)\cos\theta}{\cos\theta} = \frac{y}{\sin\theta}$

$\Rightarrow y = (a+b)\sin\theta$ ...(iv)

[Equation (iii)]^2 + [Equation (iv)]^2

$x^2 + y^2 = (a+b)^2$

Important Note

Passing through a given point, a maximum of 4 normals can be drawn to an ellipse.

Also the sum of the eccentric angles of these 4 co-normal points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is an odd multiple of π.

Proof: equation of normal: $y = mx \pm \frac{(a^2-b^2)m}{\sqrt{a^2+b^2m^2}}$

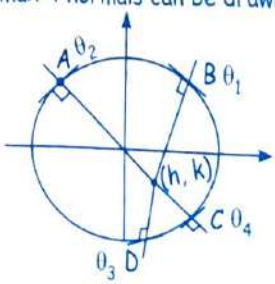
$(x, y) \to (h, k)$

$k - mh = \pm\frac{(a^2-b^2)m}{\sqrt{a^2+b^2m^2}}$

$\Rightarrow (k - mh)^2(a^2 + b^2m^2) = (a^2 - b^2)^2 m^2$

clearly it is 4° eq^n in m

$\Rightarrow$ max 4 normals can be drawn



$\theta_1 + \theta_2 + \theta_3 + \theta_4 = (2n+1)\pi$

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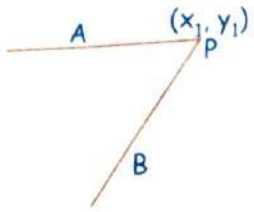
PAIR OF TANGENTS

Equation of pair of tangents of PA and PB is $\boxed{SS_1 = T^2}$ where

$$S_1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 \; ; \; T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$$

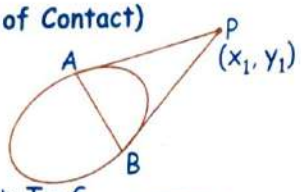
$$S: \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1$$

$$\Rightarrow \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1\right)\left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right) = \left(\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1\right)^2$$



Chords of Ellipse (Chord of Contact)

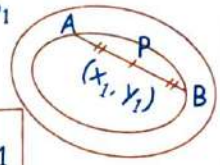
eqⁿ of AB: T = 0



$$\boxed{\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1}$$

Chord with a Given Mid Point: T = S_1

eqⁿ of AB: T = S_1



$$\boxed{\frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1}$$

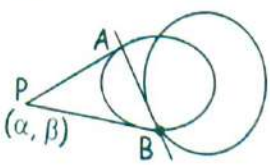
Ques. If chord of contact of the tangents drawn from the point (α, β) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, touches the circle $x^2 + y^2 = c^2$ then the locus of the point (α, β) is

(a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{c^2}$ (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{c^4}$

(c) ✓ $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{c^2}$ (d) None of these

Ans. $\frac{x\alpha}{a^2} + \frac{y\beta}{b^2} = 1 \rightarrow$ eqⁿ of AB

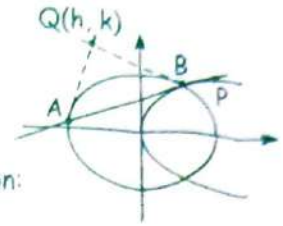
$$\because \left|\frac{-1}{\sqrt{\frac{\alpha^2}{a^4} + \frac{\beta^2}{b^4}}}\right| = c$$



$$\Rightarrow \frac{1}{c^2} = \frac{\alpha^2}{a^4} + \frac{\beta^2}{b^4} \Rightarrow \frac{1}{c^2} = \frac{x^2}{a^4} + \frac{y^2}{b^4}$$

Ques. If tangents to the parabola $y^2 = 4ax$ intersect the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at A and B, the find the locus of point of intersection of tangents at A and B.

Ans. AB: $\frac{hx}{a^2} + \frac{ky}{b^2} = 1$



AB: $y - mx = \frac{a}{m}$

Comparing both equation:

$$\frac{k/b^2}{1} = \frac{h/a^2}{-m} = \frac{1}{a/m}$$

$$\Rightarrow \frac{k}{b^2} = \frac{-h}{a^2 m}, \frac{h}{a^2} = \frac{-m^2}{a}$$

$$\Rightarrow m = \frac{-hb^2}{a^2 k}, m^2 = -\frac{h}{a}$$

$$\therefore \frac{b^4 h^2}{a^4 k^2} = \frac{-h}{a} \Rightarrow \frac{b^4 h}{a^3 k^2} = -1$$

$$\Rightarrow b^4 x = -a^3 y^2$$

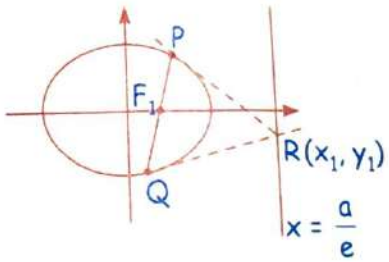
Important Note

For an Ellipse, Tangents drawn at the ends of any Focal chord meet on the corresponding directrix.

Proof:

Let R is (x_1, y_1)

PQ is chord of contact for Point R



PQ: $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

$\therefore$ (ae, 0) satisfies the equation

$$\therefore \frac{aex_1}{a^2} + 0 = 1$$

$$\Rightarrow \frac{ex_1}{a} = 1 \Rightarrow x_1 = \frac{a}{e}$$

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Ques. The equation of the chord of the ellipse $2x^2 + 5y^2 = 20$ which is bisected at the point $(2, 1)$ is

(a) $4x + 5y + 13 = 0$ (b) $4x + 5y = 13$

(c) $5x + 4y + 13 = 0$ (d) None of these

Ans. $T = S_1$

$T: 2xx_1 + 5yy_1 - 20$

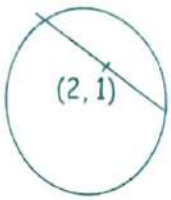
$= 4x + 5y - 20$

$S_1: 2(2)^2 + 5(1)^2 - 20$

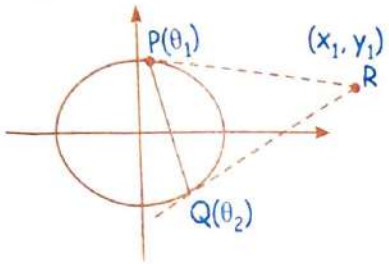
$= 8 + 5 - 20$

$\therefore 4x + 5y - 20 = -7$

$\Rightarrow 4x + 5y - 13 = 0$



Chords joining $P(\theta_1)$ and $Q(\theta_2)$



$$\frac{x}{a}\cos\left(\frac{\theta_1+\theta_2}{2}\right)+\frac{y}{b}\sin\left(\frac{\theta_1+\theta_2}{2}\right)=\cos\left(\frac{\theta_1-\theta_2}{2}\right)$$

Proof: eqⁿ of PQ: $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$

$$R\left[\frac{a\cos\left(\frac{\theta_1+\theta_2}{2}\right)}{\cos\left(\frac{\theta_1-\theta_2}{2}\right)}, \frac{b\sin\left(\frac{\theta_1+\theta_2}{2}\right)}{\cos\left(\frac{\theta_1-\theta_2}{2}\right)}\right]$$

$$\therefore \frac{xa\cos\left(\frac{\theta_1+\theta_2}{2}\right)}{a^2\cos\left(\frac{\theta_1-\theta_2}{2}\right)}+\frac{yb\sin\left(\frac{\theta_1+\theta_2}{2}\right)}{b^2\cos\left(\frac{\theta_1-\theta_2}{2}\right)}=1$$

$$\Rightarrow \frac{x}{a}\cos\left(\frac{\theta_1+\theta_2}{2}\right)+\frac{y}{b}\sin\left(\frac{\theta_1+\theta_2}{2}\right)=\cos\frac{(\theta_1-\theta_2)}{2}$$

Important Note

If α, β are eccentric angles of end points of a focal chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$,

then $\tan\alpha/2 \cdot \tan\beta/2 = \frac{e-1}{e+1}$

Proof:

eqⁿ of PQ:

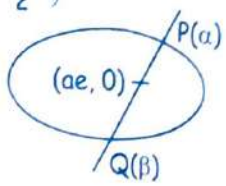
$$\frac{x\cos\left(\frac{\alpha+\beta}{2}\right)}{a}+\frac{y\sin\left(\frac{\alpha+\beta}{2}\right)}{b}=\cos\left(\frac{\alpha-\beta}{2}\right)$$

$\because (ae, 0)$ satisfies the equation

$$\therefore \frac{ae}{a}\cos\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\Rightarrow e\cos\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\Rightarrow \frac{\cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)}=\frac{e}{1}$$



By componendo and dividerdo:

$$\frac{\cos\left(\frac{\alpha-\beta}{2}\right)-\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)+\cos\left(\frac{\alpha+\beta}{2}\right)}=\frac{e-1}{e+1}$$

$$\Rightarrow \frac{2\sin\left(\frac{\alpha}{2}\right)\sin\left(\frac{\beta}{2}\right)}{2\cos\left(\frac{\alpha}{2}\right)\cos\left(\frac{\beta}{2}\right)}=\frac{e-1}{e+1}$$

$$\Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{e-1}{e+1}$$

Ques. If any two chords be drawn through two points on the major axis of an ellipse equidistant from the centre, show that $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}\cdot\tan\frac{\gamma}{2}\cdot\tan\frac{\delta}{2}=1$, where $\alpha, \beta, \lambda, \delta$ are the eccentric angles of the extremities of the chords.

Ans. $\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$

$\because (c, 0)$ satisfies the equation:

$$\therefore \frac{c\cos\left(\frac{\alpha+\beta}{2}\right)}{a}=\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\Rightarrow \frac{c}{a}=\frac{\cos\frac{\alpha-\beta}{2}}{\cos\left(\frac{\alpha+\beta}{2}\right)}$$

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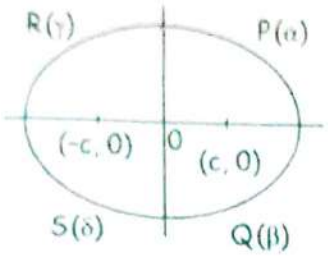


By componendo and dividerdo:

$$\Rightarrow \frac{c-a}{c+a} = \tan\frac{\alpha}{2}\tan\frac{\beta}{2} \quad ...(i)$$

Similarly,

$$\tan\frac{\gamma}{2}\tan\frac{\delta}{2} = \frac{-c-a}{-c+a} = \frac{c+a}{c-a} \quad ...(ii)$$

By multiplying equation (i) & (ii):

$$\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}\tan\frac{\delta}{2} = 1$$

Ques. Consider the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$. Let S(p, q) be a point in the first quadrant such that $\frac{p^2}{9}+\frac{q^2}{4}>1$. Two tangents are drawn from S to the ellipse of which one meets the ellipse at one end point of the minor axis and the other meets the ellipse at a point T in the fourth quadrant. Let R be the vertex of the ellipse with positive x-coordinate and O be the center of the ellipse. If the area of the triangle ΔORT is $\frac{3}{2}$, then which of the following options is correct? **[JEE Adv. 2024]**

(a) $q = 2, p = 3\sqrt{3}$

(b) $q = 1, p = 5\sqrt{3}$

(c) $q = 2, p = 4\sqrt{3}$

(d) $q = 1, p = 6\sqrt{3}$

Ans. Area of $\Delta ORT = \left|\frac{1}{2}\times 3\times y_1\right| = \frac{3}{2}$

$$\Rightarrow |y_1| = 1,\ y_1 = -1$$

$$\therefore \frac{x_1^2}{9}+\frac{1}{4}=1 \Rightarrow \frac{x_1^2}{9}=\frac{3}{4} \Rightarrow x_1^2=\frac{27}{4}$$

$$\Rightarrow x_1 = \frac{3\sqrt{3}}{2}$$

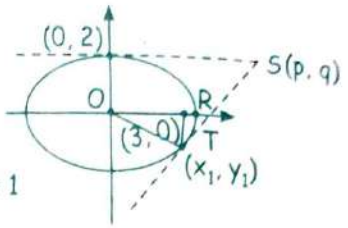
Tangent meet at the minor axis at the one end.

$\therefore q = 2$

equation of Tangent at T

$$\frac{xx_1}{9}+\frac{yy_1}{4}=1 \Rightarrow \frac{x3\sqrt{3}}{2x9}+\frac{y(-1)}{4}=1$$

$$\frac{\sqrt{3}x}{6}-\frac{y}{4}=1$$



(p, 2) satisfies

$$\therefore \frac{\sqrt{3}\times p}{6}-\frac{2}{4}=1$$

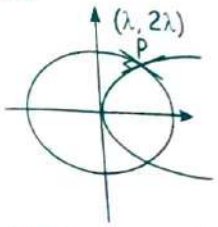
$$\Rightarrow \frac{\sqrt{3}p}{6}=1+\frac{1}{2} \Rightarrow \frac{p}{2\sqrt{3}}=\frac{3}{2} \Rightarrow p = 3\sqrt{3}$$

Ques. Let a, b and λ be positive real numbers. Suppose P is an end point of the latus rectum of the parabola $y^2 = 4\lambda x$, and suppose the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through the point P. If the tangents to the parabola and the ellipse at the point P are perpendicular to each other, then the eccentricity of the ellipse is **[JEE Adv., 2020 Paper-1]**

(a) $1/\sqrt{2}$ (b) 1/2

(c) 1/3 (d) 2/5

Ans. P[λ, 2λ]



$$\therefore \frac{\lambda^2}{a^2}+\frac{4\lambda^2}{b^2}=1$$

eqⁿ of Tangent to Parabola:

$$yy_1 = 2\lambda(x+x_1)$$

$$\Rightarrow y(2\lambda) = 2\lambda(x+\lambda) \Rightarrow y = x+\lambda \quad ...(i)$$

$$\rightarrow m = 1$$

eqⁿ to Tangent to ellipse: $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$

$$\Rightarrow \frac{x\lambda}{a^2}+\frac{y(2\lambda)}{b^2}=1 \rightarrow m=-1$$

$$\therefore \frac{-\lambda/a^2}{2\lambda/b^2}=-1 \Rightarrow \frac{b^2}{2a^2}=1$$

$$\Rightarrow b^2 = 2a^2 \Rightarrow b = \sqrt{2}a$$

$\because b > a$

Vertical ellipse: $a^2 = b^2(1-e^2)$

$$\Rightarrow a^2 = 2a^2(1-e^2)$$

$$\Rightarrow \frac{1}{2} = 1-e^2$$

$$\Rightarrow e^2 = \frac{1}{2} \Rightarrow e = \frac{1}{\sqrt{2}}$$

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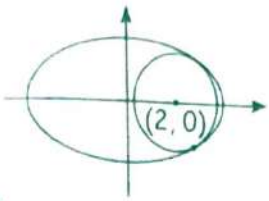
Ques. Let C be the largest circle centered at (2, 0) and inscribed in the ellipse $\frac{x^2}{36}+\frac{y^2}{16}=1$. If (1, α) lies on C, then $10\alpha^2$ is equal to

[JEE Main-2023 (Jan.-1)]

Ans. Equation of circle: $(x-2)^2+y^2=r^2$

$\Rightarrow y^2=r^2-(x-2)^2$

Put in ellipse



$$\frac{x^2}{36}+\frac{r^2-(x-2)^2}{16}=1$$

$$\Rightarrow \frac{4x^2+9(r^2-(x-2)^2)}{4\times4\times9}=1$$

$\Rightarrow 4x^2+9r^2-9x^2-36+36x=144$

$\Rightarrow -5x^2+36x+9r^2-180=0$

$D=0$

$\therefore (36)^2+4\times5\times9(r^2-20)=0$

$\Rightarrow 36+5(r^2-20)=0$

$\Rightarrow 36+5r^2-100=0 \Rightarrow 5r^2=64$

$\because (2-1)^2+(\alpha-0)^2=r^2 \Rightarrow 1+\alpha^2=\frac{64}{5}$

$\Rightarrow \alpha^2=\frac{64}{5}-1$

$\Rightarrow \alpha^2=\frac{59}{5} \Rightarrow 10\alpha^2=118$

DIAMETER OF AN ELLIPSE

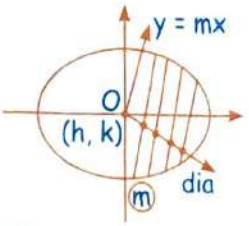
Locus of Mid point of System of parallel chords is known as diameter of Ellipse.

Proof:

$\because T=S_1$

$$\Rightarrow \frac{hx}{a^2}+\frac{ky}{b^2}-1=\frac{h^2}{a^2}+\frac{k^2}{b^2}-1$$

$$\Rightarrow \frac{hx}{a^2}+\frac{ky}{b^2}=\frac{h^2}{a^2}+\frac{k^2}{b^2}$$



$$\Rightarrow -\frac{h}{a^2}\frac{b^2}{k}=m$$

$$\Rightarrow -\frac{b^2}{a^2}h=mk \Rightarrow y=-\frac{b^2}{a^2}\frac{x}{m}$$

→ S.L. passing through (0, 0)

CONJUGATE DIAMETERS OF AN ELLIPSE

Two diameters are said to be conjugate if one bisects the chords parallel to the other and vice-versa.

Proof: If $y=m_1x$ and $y=\frac{-b^2}{a^2m}x$ are conjugate dia.

where $m_1=\frac{-b^2}{a^2m} \Rightarrow mm_1=\frac{b^2}{a^2}$

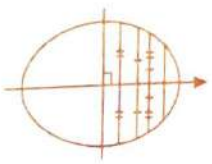
If $b=a \Rightarrow mm_1=-1$

$\Rightarrow$ Conj. dia. are $\perp$ also for circles.

If $m_1m_2=-1 \Rightarrow$ lines always $\perp$ always true

If two lines are $\perp \Rightarrow m_1m_2=-1$ or

$m=0$ and $m_2=\infty=$

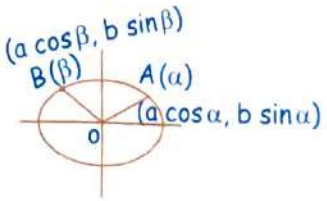


[only pair of conjugate $\perp$ dia is Major and Minor Axis]

Important Note

If α, β are eccentric angles of end points of two conjugate semi diameters of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then $|\alpha-\beta|=\frac{\pi}{2}$

Proof: $m_{oA}\cdot m_{oB}=\frac{-b^2}{a^2}$



$\because m_{OA}=\frac{b\sin\alpha-0}{a\cos\alpha-0}=\frac{b}{a}\tan\alpha$

$m_{OB}=\frac{b}{a}\tan\beta \Rightarrow \frac{b}{a}\tan\alpha\frac{b}{a}\tan\beta=\frac{-b^2}{a^2}$

$\Rightarrow \tan\alpha\tan\beta=-1 \Rightarrow \frac{\sin\alpha\sin\beta}{\cos\alpha\cos\beta}=-1$

$\Rightarrow \cos\alpha\cos\beta+\sin\alpha\sin\beta=0$

$\Rightarrow \cos(\alpha-\beta)=0 \Rightarrow |\alpha-\beta|=90^\circ$

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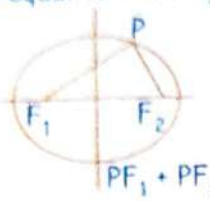
PROPERTIES OF AN ELLIPSE

Property-1

The sum of focal distances of any point P lying on the ellipse is always equal to the length of Major Axis.

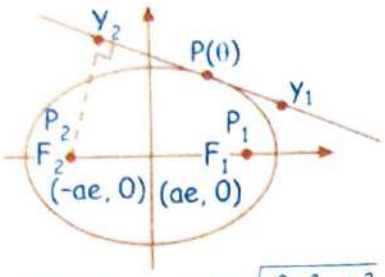


$PF_1 + PF_2 = 2a$

Property-2

Product of the lengths of the perpendiculars from either focus on a variable tangent to an Ellipse = (semi minor axis)^2 = b^2

Proof:



Let tangent be: $y = mx + \sqrt{a^2m^2 + b^2}$

$$P_1 = \left|\frac{aem + \sqrt{a^2m^2 + b^2}}{\sqrt{1+m^2}}\right| \quad ...(i)$$

$$P_2 = \left|\frac{-aem + \sqrt{a^2m^2 + b^2}}{\sqrt{1+m^2}}\right| \quad ...(ii)$$

By multiplying equation (i) & (ii)

$$P_1P_2 = \left|\frac{-a^2e^2m^2 + a^2m^2 + b^2}{(1+m^2)}\right|$$

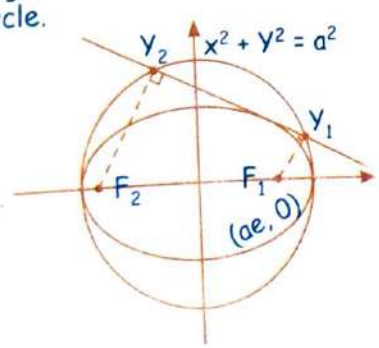
$$= \left|\frac{m^2a^2(1-e^2) + b^2}{1+m^2}\right| = \left|\frac{m^2b^2 + b^2}{1+m^2}\right|$$

$$= \left|\frac{b^2(1+m^2)}{(1+m^2)}\right| \Rightarrow P_1P_2 = b^2$$

Property-3

Feet of the perpendiculars from either foci on a variable tangent to an ellipse/hyperbola lies on its auxiliary circle.

Proof:



Equation of tangent $y = mx + \sqrt{a^2m^2 + b^2}$ ...(i)

equation of F_1Y_1: $y - 0 = -\frac{1}{m}(x - ae)$

$\Rightarrow my + x = ae$ ...(ii)

eliminating m from equation (i) and (ii)

must give locus of point Y_1

squaring eqn (i) & (ii)

$\Rightarrow y^2 + m^2x^2 - 2mxy = a^2m^2 + b^2$ ...(iii)

& $x^2 + m^2y^2 + 2mxy = a^2e^2$ ...(iv)

Adding both equation

$x^2(1 + m^2) + y^2(1 + m^2) = a^2m^2 + b^2 + a^2e^2$

$\Rightarrow (1 + m^2)(x^2 + y^2) = a^2(1 + m^2)$

$\Rightarrow x^2 + y^2 = a^2$

Ques. An ellipse, with foci (-1, 2) and (5, 6), touches the x-axis. Find its eccentricity.

Ans. $F_1F_2 = 2ae = \sqrt{6^2 + 4^2}$

$\Rightarrow 2ae = \sqrt{36 + 16}$

$\Rightarrow 2ae = 2\sqrt{13}$

$\Rightarrow ae = \sqrt{13}$ ...(i)

$\Rightarrow a^2e^2 = 13$

$\because b^2 = P_1P_2$

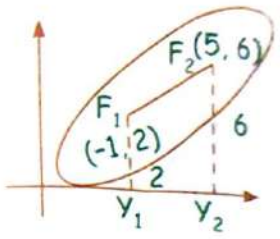
$\Rightarrow b^2 = 2 \times 6 = 12$

$\therefore a^2 - a^2e^2 = 12$

$\Rightarrow a^2 - 13 = 12$

$\Rightarrow a^2 = 25 \Rightarrow a = 5$

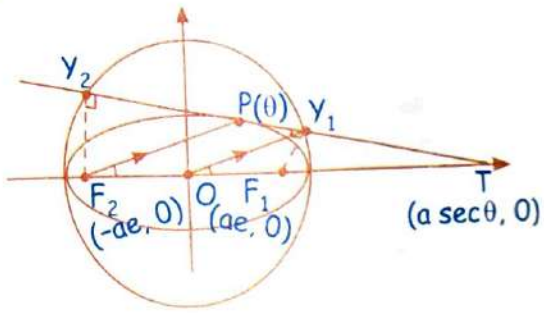
$\therefore 25e^2 = 13 \Rightarrow e = \frac{\sqrt{13}}{5}$



Property-4

Lines joining centre to the feet of perpendicular from a focus on any tangent at P and the line joining other focus to the point of contact 'P' are parallel.

Proof:



$OT = a\sec\theta$; $OY_1 = a$

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$F_2T = ae + a\sec\theta$; $PF_2 = a + ae\cos\theta$

$$\frac{OT}{F_2T} = \frac{a\sec\theta}{ae + a\sec\theta} = \frac{\sec\theta}{e + \sec\theta}$$

$$= \frac{1}{e\cos\theta + 1} \quad ...(i)$$

$$\frac{OY_1}{PF_2} = \frac{a}{a + ae\cos\theta} = \frac{1}{1 + e\cos\theta} \quad ...(ii)$$

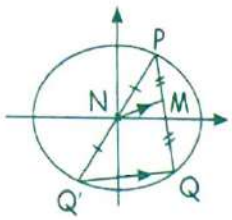
By equation (i) & (ii): $\frac{OT}{F_2T} = \frac{OY_1}{PF_2}$

$\therefore \Delta F_2PT \sim \Delta OY_1P$

Hence, $OY_1 \parallel F_2P$

Ques. Let E be the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. For any three distinct points P, Q and Q' on E, let M(P, Q) be the midpoint of the line segment joining P and Q, and M(P, Q') be the mid-point of the line segment joining P and Q'. Then the maximum possible value of the distance between M(P, Q) and M(P, Q'), as P, Q and Q' vary on E, is

Ans. $MN = \frac{1}{2}QQ'$



$\Rightarrow MN_{max} = \frac{1}{2} \cdot QQ'_{max}$

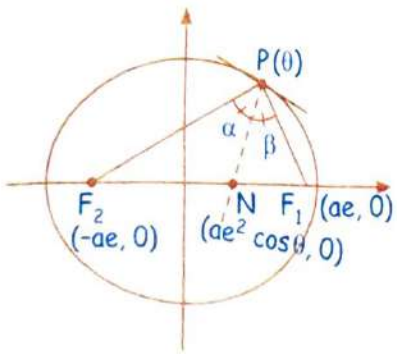
$\because QQ'_{max} = 2a$

$\therefore MN_{max} = \frac{1}{2}2a = a = 4$

Property-5: Reflection Property

The tangent and normal at a point P on the ellipse bisect the external and internal angles between the focal distances of P. This refers to the well known reflection property of the ellipse states that rays from one focus are reflected through other focus and vice-versa.

Proof:



Equation of normal

$$\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2e^2$$

$\because y = 0$

$\therefore x = ae^2\cos\theta$

$PF_1 = a - ae\cos\theta$; $F_2N = ae^2\cos\theta + ae$

$PF_2 = a + ae\cos\theta$; $F_1N = ae - ae^2\cos\theta$

$$\frac{PF_2}{PF_1} = \frac{a + ae\cos\theta}{a - ae\cos\theta} = \frac{1 + e\cos\theta}{1 - e\cos\theta} \quad ...(i)$$

$$\frac{F_2N}{F_1N} = \frac{ae^2\cos\theta + ae}{ae - ae^2\cos\theta} = \frac{e\cos\theta + 1}{1 - e\cos\theta} \quad ...(ii)$$

By equation (i) & (ii): $\frac{PF_2}{PF_1} = \frac{F_2N}{F_1N}$

Ques. A ray emanating from the point (-4, 0) is incident on the ellipse $9x^2 + 25y^2 = 225$ at the point P with abscissa 3. Equation of the reflected ray after first reflection is

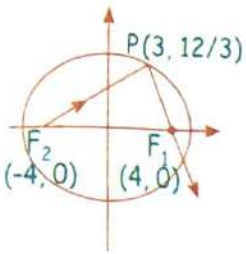
(a) $5x \pm 12y = 48$ (b) $5x \pm 12y + 48 = 0$

(c) $12x \pm 5y = 48$ (d) $12x \pm 5y + 48 = 0$

Ans. $a^2e^2 = a^2 - b^2$

$\Rightarrow a^2e^2 = 25 - 9$

$\Rightarrow (ae)^2 = 16 \Rightarrow ae = 4$



$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

$\because x = 3, \therefore \frac{3^2}{25} + \frac{y^2}{9} = 1$

$$\Rightarrow \frac{y^2}{9} = \frac{25 - 9}{25} \Rightarrow \frac{y^2}{9} = \frac{16}{25}$$

$$\Rightarrow \frac{y}{3} = \pm\frac{4}{5} \Rightarrow y = \pm\frac{12}{5}$$

$\therefore$ Eqn of reflection Ray: $y = \frac{\left(\pm\frac{12}{5} - 0\right)}{(3 - 4)}(x - 4)$

$\Rightarrow 12x \pm 5y = 48$

Property-6

If the normal at any point P on the ellipse with centre C meet the major and minor axes in G and g respectively, and if CF be perpendicular upon this normal, then

(i) $PF \cdot PG = b^2$ (ii) $PF \cdot Pg = a^2$

(iii) $PG \cdot Pg = SP \cdot S'P$

(iv) $CG \cdot CT = (CS)^2$

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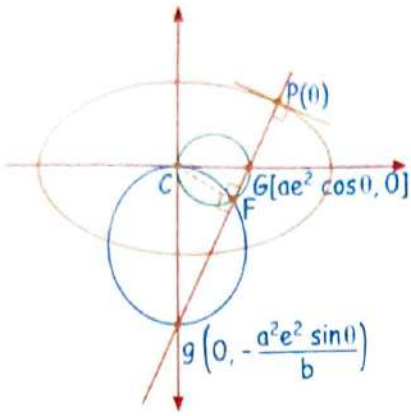


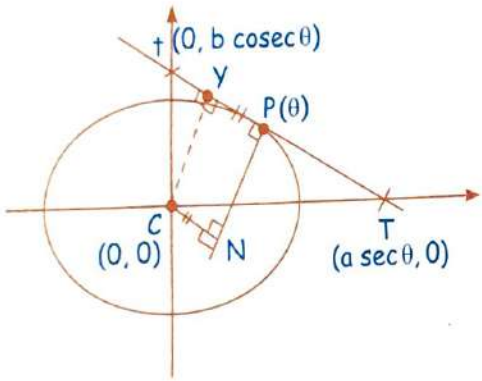
Property-7

If the tangent at the point P of a standard ellipse meets the axis in T and t and CY is the perpendicular on it from the centre then,

(i) Tt. PY = $a^2 - b^2$ and

(ii) Least value of Tt is a + b.

Proof:



PY = CN [Rectangle]

$\because$ $Tt = \sqrt{a^2 \sec^2\theta + b^2 \operatorname{cosec}^2\theta}$ ...(i)

equation of Normal

$$\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2e^2$$

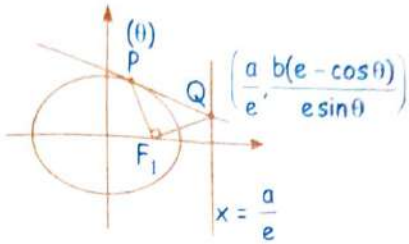
$$PY = CN = \left|\frac{a^2e^2}{\sqrt{a^2 \sec^2\theta + b^2 \operatorname{cosec}^2\theta}}\right| \quad ...(ii)$$

Tt. PY = $a^2e^2 = a^2 - b^2$

Property-8

The portion of the tangent to an ellipse between the point of contact and the directrix subtends a right at the corresponding focus.

Proof: $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$



Put $x = \frac{a}{e}$

$$\frac{a\cos\theta}{ea} + \frac{y\sin\theta}{b} = 1$$

$$\Rightarrow \frac{y\sin\theta}{b} = 1 - \frac{\cos\theta}{e}$$

$$\Rightarrow y = \frac{b(e - \cos\theta)}{e\sin\theta}$$

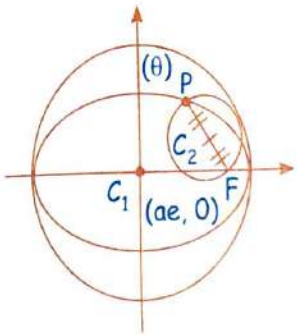
Property-9

The circle on any focal distance as diameter touches the auxiliary circle.

Proof:

$C_1 \rightarrow [0, 0]$

$C_2 \rightarrow \left[\frac{ae + a\cos\theta}{2}, \frac{b\sin\theta}{2}\right]$



$r_1 = a$,

$r_2 = \frac{a - ae\cos\theta}{2}$

$|r_1 - r_2|$

$= a - \left(\frac{a - ae\cos\theta}{2}\right)$

$= \left|\frac{a + ae\cos\theta}{2}\right|$

$$\therefore C_1C_2 = \sqrt{\left(\frac{ae + a\cos\theta}{2}\right)^2 + \left(\frac{b\sin\theta}{2}\right)^2}$$

$$= \frac{1}{2}\sqrt{(ae + a\cos\theta)^2 + b^2\sin^2\theta}$$

$$= \frac{1}{2}\sqrt{a^2e^2 + a^2\cos^2\theta + 2a^2e\cos\theta + b^2\sin^2\theta}$$

$$= \frac{1}{2}\sqrt{a^2e^2 + a^2\cos^2\theta + 2a^2e\cos\theta + (a^2 - a^2e^2)\sin^2\theta}$$

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$\sqrt{e^2 + a^2 + 2a^2e\cos\theta - a^2e^2\sin^2\theta}$

$\sqrt{a^2e^2(1-\sin^2\theta) + a^2 + 2a^2e\cos\theta}$

$\sqrt{a^2e^2\cos^2\theta + a^2 + 2a^2e\cos\theta}$

$\sqrt{(a + ae\cos\theta)^2}$

$r_2 = \left|\frac{a + ae\cos\theta}{2}\right| = |r_1 - r_2|$

y-10

licular from the centre upon all chords which ends of any perpendicular diameters of the are of constant length.

LE (POINT) & POLAR (LINE)

a fixed point $P(x_1, y_1)$ if chords are drawn to a meeting the Conic at points A and B the locus nt of intersection of tangents drawn at A and B traight line called polar of P.

F: equation of AB: (C.O.C) for (h, k)

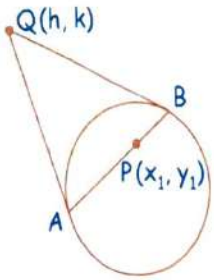
$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1$$

(x_1, y_1) satisfies

$$\frac{hx_1}{a^2} + \frac{ky_1}{b^2} = 1$$

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \rightarrow \text{Polar}$$

$\Rightarrow T = 0$



ues. The pole of the line $2x + y + 5 = 0$ w.r.t. the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is

(a) $\left(\frac{8}{5}, \frac{9}{5}\right)$ (b) $\left(-\frac{8}{5}, \frac{9}{5}\right)$

(c) $\left(\frac{8}{5}, -\frac{9}{5}\right)$ (d) $\left(\frac{-8}{5}, \frac{-9}{5}\right)$

Ans. Let Pole (x_1, y_1)

$\frac{xx_1}{4} + \frac{yy_1}{9} = 1$ and $2x + y = -5$

Compare: $\frac{x_1}{4 \times 2} = \frac{y_1}{9 \times 1} = \frac{1}{-5}$

$x_1 = \frac{-8}{5}$, $y_1 = \frac{-9}{5}$

Ques. The polar of (x_1, y_1), w.r.t. $x^2 + y^2 = a^2$, touches $xy = c^2$. Find the locus of (x_1, y_1)

Ans. Polar

$xx_1 + yy_1 = a^2$

$y = \frac{a^2 - xx_1}{y_1}$

and $xy = c^2$

$\therefore x\left[\frac{a^2 - xx_1}{y_1}\right] = c^2$

$\Rightarrow a^2x - x^2x_1 = c^2y_1 \Rightarrow x_1x^2 - a^2x + c^2y_1 = 0$

$D = 0$

$a^4 - 4x_1c^2y_1 = 0$

$x_1y_1 = \frac{a^4}{4c^2}$

CONJUGATE POINTS & CONJUGATE LINES

2 lines are said to be conjugate if pole of 1 line lies on the other line and vice verca.

2 points are said to be conjugate points if polar of one passes through other and vice verca.

Ques. The lines $2x + 3y + 1 = 0$, $x + y + k = 0$ are conjugate lines w.r.t. $3x^2 + 4y^2 = 12$. Then k =

(a) 7 (b) 17 (c) 27 (d) -7

Ans. Let Pole be (x_1, y_1)

$3xx_1 + 4yy_1 = 12 \rightarrow$ Polar

$2x + 3y = -1$

Compare

$\frac{3x_1}{2} = \frac{4y_1}{3} = \frac{12}{-1}$

$\frac{3x_1}{2} = -12$; $\frac{4y_1}{3} = -12 \Rightarrow x_1 = -8$; $y_1 = -9$

Pole $(-8, -9)$

$(-8, -9)$ must satisfy L_2

$\therefore -8 - 9 + k = 0 \Rightarrow k = 17$

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Short Notes

DEFINITION

- The general equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents an Ellipse if $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$ and $h^2 < ab$.
- The locus of a point such that the ratio of its distance from a fixed point to its distance from a fixed line is a constant, is called a conic section. The ratio e is called the eccentricity. A conic section, in which the eccentricity is less than unity, is called an Ellipse.

STANDARD EQUATIONS OF AN ELLIPSE

- STANDARD EQUATION-1 is given by $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ where $b^2 = a^2(1 - e^2)$, $(a > b)$.
- STANDARD EQUATION-2 It is given by $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ where $a^2 = b^2(1 - e^2)$, $(a < b)$. It is also known as conjugate ellipse.
- The parametric form of the ellipse is $x = a\cos\theta$, $y = b\sin\theta$.

FOCAL DISTANCE PROPERTY OF ELLIPSE

- The sum of the focal distances of any point on the ellipse is always a constant and is equal to the length of the major axis.
- Conversely, if F_1 and F_2 are given points and P is a variable point in the plane such that $|PF_1| + |PF_2| = 2a$, a constant then P describes an ellipse whose foci are F_1 and F_2 provided $2a > |F_1F_2|$.
- If $2a = |F_1F_2|$ then Locus is the line segment joining points F_1 & F_2.

EQUATION OF TANGENTS TO $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

- **Cartesian Form** Equation of the tangent at (x_1, y_1) is given by $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$
- **Slope Form** The equation of the tangent with slope m is given by $y = mx \pm\sqrt{a^2m^2 + b^2}$
- **Parametric Form** Equation of the tangent at $P(\theta)$, is $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$
- Tangents drawn at the 4 ends of Latera Recta of an ellipse enclose a rhombus whose area is $2a^2/e$.
- Point of intersection of the tangents at the point α & β is $\left(a\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, b\frac{\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$

DIRECTOR CIRCLE

- The locus of the point of intersection of the tangents to the ellipse which meet at right angle, is a circle. This circle is called the **director circle** of the ellipse.
- Equation of the director circle of the ellipse is $x^2 + y^2 = a^2 + b^2$

VARIOUS FORMS OF EQUATION OF NORMALS TO $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

- **Cartesian Form** The equation of the normal at $(x_1\ y_1)$ is $\frac{a^2x}{x_1}-\frac{b^2y}{y_1} = a^2 - b^2 = a^2e^2$
- **Slope Form** The equation of the normal with slope m is $y = mx \pm \frac{(a^2 - b^2)m}{\sqrt{a^2 + b^2m^2}}$
- **Parametric Form** Equation of the Normal at P (θ), is $\frac{ax}{\cos\theta}-\frac{by}{\sin\theta} = a^2 - b^2$

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CHORDS OF AN ELLIPSE

- **Chord joining P (θ_1) and Q (θ_2):** The equation of the chord PQ is

$$\frac{x}{a}\cos\left(\frac{\theta_1+\theta_2}{2}\right)+\frac{y}{b}\sin\left(\frac{\theta_1+\theta_2}{2}\right)=\cos\left(\frac{\theta_1-\theta_2}{2}\right)$$

- **Chord of Contact:** The equation of chord of contact of the tangents from P is given by $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$

- The equation of the chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, whose mid-point be (x_1, y_1) is $T = S_1$.

PROPERTIES OF ELLIPSE

- Product of the length's of the perpendiculars from either focus on a variable tangent to an Ellipse = (semi minor axis)^2 = b^2

- Feet of the perpendiculars from either foci on a variable tangent to an ellipse / hyperbola lies on its auxiliary circle.

- Lines joining centre to the feet of perpendicular from a focus on any tangent at P and the line joining other focus to the point of contact 'P' are parallel.

- The tangent & normal at a point P on the ellipse bisect the external & internal angles between the focal distances of P.

- This refers to the well known reflection property of the ellipse which states that rays from one focus are reflected through other focus & vice-versa.

- If the tangent at the point P of a standard ellipse meets the axis in T and t and CY is the perpendicular on it from the centre then Tt. PY = $a^2 - b^2$ and Least value of Tt is $a + b$.

"In an ellipse, two foci guide the path — in life, discipline and persistence guide success."

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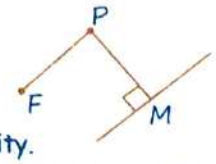
Chapter

14 Hyperbola

DEFINITION

The locus of a point such that the ratio of its distance from a fixed point to its distance from a fixed line is a constant, is called a conic section

$$\frac{PF}{PM} = e$$



The ratio e is called the eccentricity.

A conic section, in which the eccentricity is greater than unity, is called a hyperbola i.e., $e > 1$.

The fixed line is called the Directrix and the fixed point is called the Focus of hyperbola.

Equation $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represent a hyperbola, if $\Delta \neq 0$ and $h^2 > ab$

STANDARD EQUATION-1

Standard equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$,

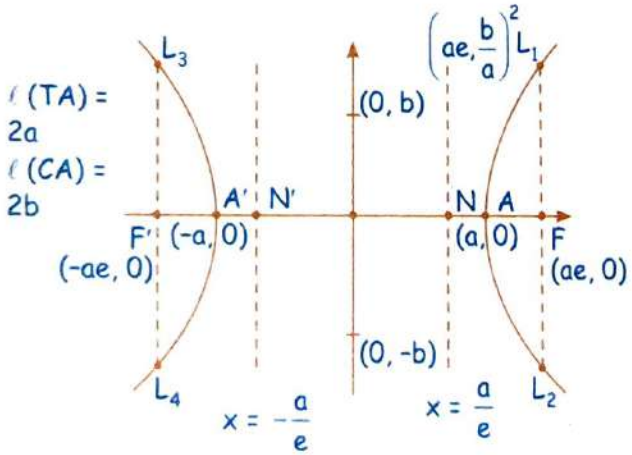
where $b^2 = a^2(e^2 - 1)$

If $a = b \Rightarrow$ Rectangular Hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{a^2} = 1 \Rightarrow x^2 - y^2 = a^2$$

And, $a^2 = a^2(e^2 - 1) \Rightarrow 1 = e^2 - 1$

$\Rightarrow e^2 = 2 \Rightarrow e = \sqrt{2}$



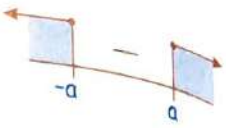
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{y^2}{b^2} = \frac{x^2}{a^2} - 1$$

$$\Rightarrow \frac{x^2}{a^2} - 1 \geq 0$$

$$\Rightarrow x^2 \geq a^2$$

$$(x - a)(x + a) \geq 0$$



Transverse Axis	y-axis (y = 0)
Ends of Transverse Axis/ Vertices	(± a, 0)
Length of Transverse Axis	2a
Conjugate Axis	y-axis (x = 0)
Ends of Conjugate Axis	(0, ± b)
Length of Conjugate Axis	2b
Centre	(0, 0)
Foci	(ae, 0) and (-ae, 0)
Directrices	$x = \pm\frac{a}{e}$
Feet of Directrices	$\left(\pm\frac{a}{e}, 0\right)$
Equation of Latus Rectum	x = ±ae
Ends of Latus Rectum	$\left(ae, \pm\frac{b^2}{a}\right)$ & $\left(-ae, \pm\frac{b^2}{a}\right)$
Length of Latus Rectum	$\frac{2b^2}{a}$
Relation (a, b & e)	$b^2 = a^2(e^2 - 1)$

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Important Note

Rectangular or Equilateral Hyperbola

The particular kind of hyperbola in which the lengths of the transverse and conjugate axis are equal is called an Equilateral Hyperbola ($a = b$)

In case of standard hyperbola if $a = b$, its equation will be $x^2 - y^2 = a^2$

The eccentricity of any rectangular hyperbola is $\sqrt{2}$ and the length of its latus rectum is equal to it's transverse or conjugate axis.

Ques. The eccentricity of the hyperbola whose length of the latus rectum is equal to 8 and the length of its conjugate axis is equal to half of distance between its foci, is **[2017 Main]**

(a) $\frac{4}{3}$ (b) $\frac{2}{\sqrt{3}}$

(c) $\frac{4}{\sqrt{3}}$ (d) $\sqrt{3}$

Ans. $\frac{2b^2}{a} = 8$ & $2b = \frac{1}{2}2ae$

$\Rightarrow 2b^2 = 8a$ $\Rightarrow 2b = ae$

$\Rightarrow 2b^2 = 8\sqrt{3}\,b$ Since, $b^2 = a^2(e^2 - 1)$

$\Rightarrow b = 4\sqrt{3}$ $\Rightarrow b^2 = a^2e^2 - a^2$

And, $a = \sqrt{3} \cdot 4\sqrt{3}$ $\Rightarrow b^2 = 4b^2 - a^2$

$a = 12$ $\Rightarrow 3b^2 = a^2$

$\because 2b = ae$ $\Rightarrow \sqrt{3}b = a$

$\therefore 2 \times 4\sqrt{3} = 12e$

$\Rightarrow 2\sqrt{3} = 3e$

$\Rightarrow e = \frac{2}{\sqrt{3}}$

Ques. If directrix of a hyperbola centered at the origin and passing through the point $(4, -2\sqrt{3})$ is $5x = 4\sqrt{5}$ and its eccentricity is e, then **[JEE Main 2019 (Apr.-1)]**

(a) $4e^4 - 12e^2 - 27 = 0$

(b) $4e^4 - 24e^2 + 27 = 0$

(c) $4e^4 + 8e^2 - 35 = 0$

(d) $4e^4 - 24e^2 + 35 = 0$

Ans. Given directrix, $x = \frac{4}{\sqrt{5}} = \frac{a}{e} \Rightarrow a = \frac{4}{\sqrt{5}}e$

Std. equation of hyperbola is, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\because$ hyperbola passing through the point $(4, -2\sqrt{3})$

$\therefore \frac{16}{a^2} - \frac{12}{b^2} = 1$

$\Rightarrow \frac{16}{a^2} - \frac{12}{a^2(e^2-1)} = 1$

$\Rightarrow 16 - \frac{12}{e^2-1} = a^2$

$\Rightarrow 16 - \frac{12}{e^2-1} = \frac{16e^2}{5}$

$\Rightarrow 4 - \frac{3}{e^2-1} = \frac{4e^2}{5}$

$\Rightarrow \frac{4e^2-4-3}{e^2-1} = \frac{4e^2}{5}$

$\Rightarrow (4e^2 - 7) \times 5 = 4e^2(e^2 - 1)$

$\Rightarrow 20e^2 - 35 = 4e^4 - 4e^2$

$\Rightarrow 4e^4 - 24e^2 + 35 = 0$

Ques. If the line $x - 1 = 0$, is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point. **[JEE Main (June-II)]**

(a) $(-2\sqrt{5}, 6)$ (b) $(-\sqrt{5}, 3)$

(c) $(\sqrt{5}, -2)$ (d) $(2\sqrt{5}, 3\sqrt{6})$

Ans. Directrix of the hyperbola is, $x = 1 = \frac{a}{e}$

$\Rightarrow e = a$

And given hyperbola is $\frac{x^2}{1/k} - \frac{y^2}{1} = 6$

$\Rightarrow \frac{x^2}{6/k} - \frac{y^2}{6} = 1$

$\Rightarrow a^2 = \frac{6}{k}, b^2 = 6$

$\Rightarrow b^2 = a^2(e^2 - 1)$

$\Rightarrow b^2 + a^2 = a^2e^2$

$\Rightarrow b^2 + a^2 = a^2 \cdot a^2$

$\Rightarrow 6 + \frac{6}{k} = \frac{6}{k} \cdot \frac{6}{k}$

$\Rightarrow 1 + \frac{1}{k} = \frac{6}{k^2}$

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$\Rightarrow \frac{k+1}{k} = \frac{6}{k^2}$

$\Rightarrow k(k+1) = 6$

$\Rightarrow k^2 + k - 6 = 0$

$\Rightarrow (k+3)(k-2) = 0$

$\Rightarrow k = -3$ · (X) as $a^2 = \frac{6}{k}$

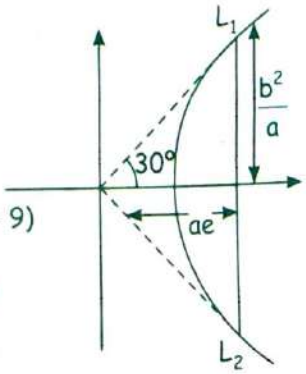
or $k = 2$

So, hyperbola is $2x^2 - y^2 = 6$.

point $(\sqrt{5}, -2)$ satisfy this hyperbola.

Ques. Let the latus rectum of the hyperbola $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$ subtend an angle of $\frac{\pi}{3}$ at the centre of the hyperbola. If b^2 is equal to $\frac{\ell}{m}(1+\sqrt{n})$, where ℓ and m are co-prime numbers, then $\ell^2 + m^2 + n^2$ is equal to **[JEE Main 2024 (30 Jan Shift-1)]**

Ans. $\tan 30° = \frac{b^2}{a \cdot ae}$



$\Rightarrow \frac{1}{\sqrt{3}} = \frac{b^2}{a^2 e}$

$\Rightarrow b^2 = \frac{9}{\sqrt{3}} \cdot e \ (\because a^2 = 9)$

Since, $b^2 = a^2(e^2 - 1)$

So, $\frac{9e}{\sqrt{3}} = 9(e^2 - 1)$

$\Rightarrow e = \sqrt{3}e^2 - \sqrt{3}$

$\Rightarrow \sqrt{3}e^2 - e - \sqrt{3} = 0$

$\Rightarrow e = \frac{1 \pm \sqrt{1 + 4 \times 3}}{2\sqrt{3}} = \frac{1 \pm \sqrt{13}}{2\sqrt{3}}$

So, $b^2 = \frac{9}{\sqrt{3}} \cdot \frac{1+\sqrt{13}}{2\sqrt{3}}$

$\Rightarrow b^2 = \frac{3}{2}(1+\sqrt{13})$

Given, $b^2 = \frac{l}{m}(1+\sqrt{n})$

So, $l = 3$, $m = 2$ $n = 13$

Then, $l^2 + m^2 + n^2 = 9 + 4 + 169$

$= 13 + 169 = 182$

Ques. Let the foci and length of the latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$ be $(\pm 5, 0)$ and $\sqrt{50}$, respectively. Then, the square of the eccentricity of the hyperbola $\frac{x^2}{b^2} - \frac{y^2}{a^2 b^2} = 1$ equals **[JEE Main 2024 (31 Jan Shift 1)]**

Ans. Given, $\sqrt{50} = \frac{2b^2}{a}$ and $ae = 5$

$\Rightarrow 2b^2 = 5\sqrt{2}a$ ...(i)

$\because b^2 = a^2(1 - e^2)$

$\Rightarrow a^2 - b^2 = a^2e^2$

$\Rightarrow a^2 - b^2 = 25$ ($\because$ Given $ae = 5$)

$\Rightarrow a^2 - \frac{5\sqrt{2}a}{2} = 25$ [From equation (i)]

$\Rightarrow 2a^2 - 5\sqrt{2}a - 50 = 0$

$\Rightarrow 2a^2 - 10\sqrt{2}a + 5\sqrt{2}a - 50 = 0$

$\Rightarrow (2a + 5\sqrt{2})(a - 5\sqrt{2}) = 0$

$\Rightarrow a = 5\sqrt{2}$

Let eccentricity of hyperbola $\frac{x^2}{b^2} - \frac{y^2}{a^2 b^2}$ is E.

Then,

$a^2b^2 = b^2[E^2 - 1]$

$\Rightarrow a^2 = E^2 - 1$

$\Rightarrow (5\sqrt{2})^2 = E^2 - 1$

$\Rightarrow 50 + 1 = E^2$

$\Rightarrow E^2 = 51$

Ques. If the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide, then find the value of b^2.

Ans. **Method - 1**

$\frac{x^2}{144/25} - \frac{y^2}{81/25} = 1$

$a^2 = \frac{144}{25}$, $b^2 = \frac{81}{25}$

$\because b^2 = a^2(E^2 - 1)$

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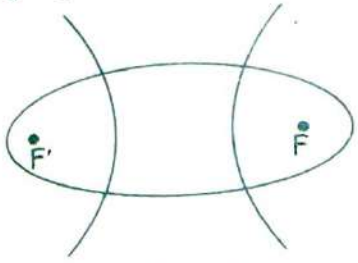
$\therefore \frac{81}{25} = \frac{144}{25}(E^2 - 1)$

$\Rightarrow 81 + 144 = 144E^2$

$\Rightarrow 225 = 144E^2$

$\Rightarrow E = \frac{15}{12} = \frac{5}{4}$

$F\left[\frac{12}{5} \times \frac{5}{4}, 0\right] = [3, 0]$



It is the foci of $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ also,

$\Rightarrow ae = 3$

$\Rightarrow 4e = 3 \qquad (\because a = 4)$

$\Rightarrow e = \frac{3}{4}$

Hence, $b^2 = 16(1 - e^2)$

$= 16\left[1 - \frac{9}{16}\right] = 16 - 9 = 7$

Method-2

$\Rightarrow ae = AE$

$\Rightarrow a^2e^2 = A^2E^2$

$\Rightarrow a^2 - b^2 = A^2 + B^2$

$\left(\because \quad b^2 = a^2(1 - e^2) \Rightarrow a^2 - b^2 = a^2e^2\right.$

$\left.\text{And, } B^2 = A^2(E - 1) \Rightarrow B^2 + A^2 = A^2E^2\right)$

$\Rightarrow 16 - b^2 = \frac{144}{25} + \frac{81}{25}$

$\Rightarrow 16 - b^2 = \frac{225}{25}$

$\Rightarrow 16 - b^2 = 9$

$\Rightarrow b^2 = 7$

Ques. Let H be the hyperbola, whose foci are $(1 \pm \sqrt{2}, 0)$ and eccentricity is $\sqrt{2}$. Then the length of its latus rectum is _____.

[JEE Main 2023 (Jan-II)]

(a) 2 (b) 3

(c) 5/2 (d) 3/2

Ans. Distance between foci = $2ae = 2\sqrt{2}$

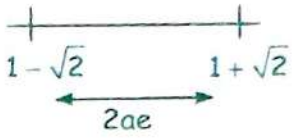
$\Rightarrow a = 1 \qquad (\because e = \sqrt{2})$

$e = \sqrt{2} \Rightarrow$ Rectangular Hyperbola

So, $a = b$

$LR = \frac{2b^2}{a}$

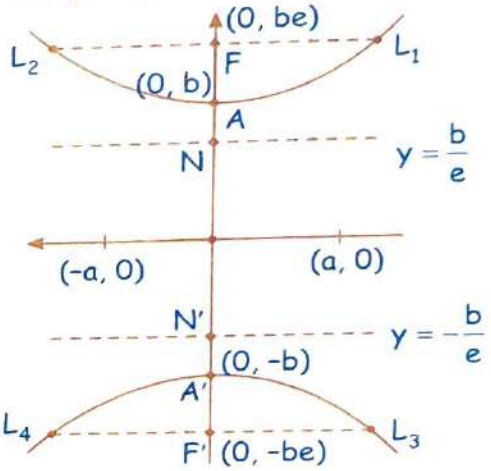
$= 2a = 2 \times 1 = 2$



STANDARD EQUATION-2

Standard equation of the hyperbola is $-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$,

where $a^2 = b^2(e^2 - 1)$



Transverse Axis	y-axis (x = 0)
Ends of T.A.	(0, b) & (0, -b)
Length of Transverse Axis	2b
Conjugate Axis	x-axis (y = 0)
Ends of C.A.	(± a, 0)
Length of Conjugate Axis	2a
Centre	(0, 0)
Foci	(0, be) and (0, -be)
Directrices	y = ±b/e
Feet of Directrices	(0, b/e) & (0, -b/e)
Equation of LR	y = ±be
Ends of L.R.	$\left(\pm\frac{a^2}{b}, \pm be\right)$
Length of LR	$2a^2/b$
Relation (a, b & e)	$a^2 = b^2(e^2 - 1)$

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Important Note

Conjugate Hyperbola

The conjugate hyperbola for $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=-1$. Both these hyperbolas are never intersect to each other.

If e_1 and e_2 are the eccentricities of a Hyperbola and its conjugate then $e_1^{-2}+e_2^{-2}=1$

Proof: $b^2=a^2(e_1^2-1)$ and $a^2=b^2(e_2^2-1)$

Multiply

$a^2b^2=a^2b^2(e_1^2-1)(e_2^2-1)$

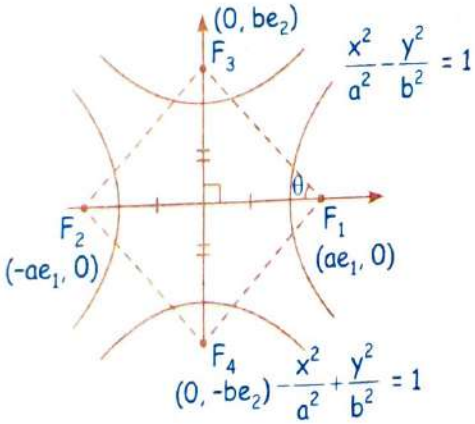
$1=(e_1^2-1)(e_2^2-1)$

$\Rightarrow 1=e_1^2e_2^2-e_1^2-e_2^2+1$

$\Rightarrow e_1^2+e_2^2=e_1^2e_2^2$

$\Rightarrow \frac{e_1^2+e_2^2}{e_1^2e_2^2}=1 \Rightarrow \frac{1}{e_1^2}+\frac{1}{e_2^2}=1$

Note: The figure formed by joining the foci of a hyperbola & its conjugate is a square.



Clearly diagonals are $\perp$ & bisect

$\Rightarrow$ $F_1F_3F_2F_4$ is a rhombus

To prove, $\theta = 45$

OR to prove, $be_2 = ae_1$

$a^2e_1^2=a^2+b^2$ ...(1)

Also

$b^2e_2^2=a^2+b^2$ ...(2)

so, $a^2e_1^2=b^2e_2^2$

$\Rightarrow ae_1=be_1$

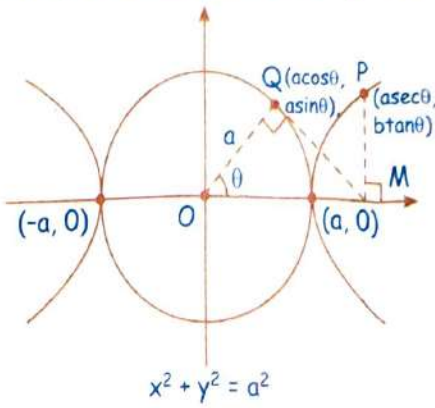
Here, F_1, F_3, F_2, F_4 is a square.

PARAMETRIC COORDINATES & AUXILIARY CIRCLE

The point $x = a\sec\theta$ and $y = b\tan\theta$ (where θ is a parameter) together represent the parametric co-ordinates of any general point lying on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.

A circle drawn with centre C and transverse axis as diameter is called the Auxiliary Circle of the hyperbola.

Equation of the auxiliary circle is $x^2+y^2=a^2$.



$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1 \Rightarrow x=a\sec\theta \,\&\, y=b\tan\theta,\ \theta\in R$

Q is called corresponding point of $P(\theta)$

In ΔOQM

$\cos\theta=\frac{OQ}{OM}$

$\Rightarrow \cos\theta=\frac{a}{OM} \Rightarrow OM=a\sec\theta$

and $PM=y=b\sqrt{\frac{a^2\sec^2\theta}{a^2}-1}=b\tan\theta$

Ques. AB is double ordinate of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ such that ΔAOB (where 'O' is the origin) is an equilateral triangle, then the eccentricity e of the hyperbola satisfies:

(a) $e>\sqrt{3}$ (b) $1<e<\frac{2}{\sqrt{3}}$

(c) $e=\frac{2}{\sqrt{3}}$ (d) $e>\frac{2}{\sqrt{3}}$

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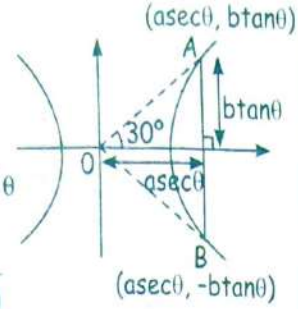
Ans. $\tan 30° = \dfrac{b\tan\theta}{a\sec\theta}$

$\Rightarrow \dfrac{1}{\sqrt{3}} = \dfrac{b}{a}\cdot\sin\theta$

$\Rightarrow \dfrac{1}{3} = \dfrac{b^2}{a^2}\sin^2\theta$

$\Rightarrow \dfrac{1}{3} = (e^2-1)\sin^2\theta$

$\Rightarrow \sin^2\theta = \dfrac{1}{3(e^2-1)}$



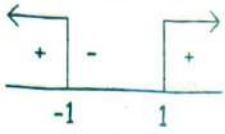
But $0 \le \sin^2\theta < 1$

$\Rightarrow 0 \le \dfrac{1}{3(e^2-1)} < 1$ and $\dfrac{1}{3(e^2-1)} < 1$

$\Rightarrow \dfrac{1}{3(e^2-1)} \ge 0$ $\Rightarrow 3(e^2-1) > 1$

$\Rightarrow 3e^2 > 4$

$\Rightarrow \dfrac{1}{3(e-1)(e+1)} \ge 0$ $\Rightarrow e > \dfrac{2}{\sqrt{3}}$



Hence, $e > \dfrac{2}{\sqrt{3}}$

Ques. Let P be a point on the hyperbola $H: \dfrac{x^2}{9} - \dfrac{y^2}{4} = 1$, in the first quadrant such that the area of triangle formed by P and the two foci of H is $2\sqrt{13}$. Then, the square of the distance of P from the origin is

[JEE Main 2024 (30 Jan Shift 2)]

(a) 18 (b) 26

(c) 22 (d) 20

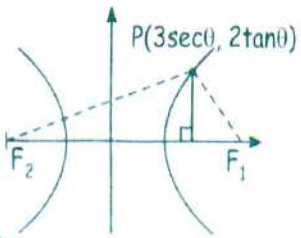
Ans. $a^2 = 9,\ b^2 = 4$

$\because a^2 + b^2 = a^2e^2$

$\Rightarrow 13 = a^2e^2$

$\Rightarrow ae = \sqrt{13}$

Base $= 2ae = 2\sqrt{13}$



Area of $\Delta PF_1F_2 = \dfrac{1}{2}(2\tan\theta)\cdot 2\sqrt{13} = 2\sqrt{13}$

$\Rightarrow \theta = 45°$

Then, P[3sec 45°, 2tan 45°]

$= P[3\sqrt{2}, 2]$

Hence, $(OP)^2 = 18 + 4 = 22$

FOCAL DISTANCE OF ANY POINT ON HYPERBOLA

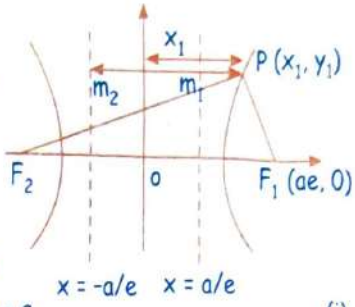
$\dfrac{PF_1}{PM_1} = e$

$\Rightarrow PF_1 = ePM_1$

$= e\left[x_1 - \dfrac{a}{e}\right]$

$= ex_1 - a$

$= ea\sec\theta - a$



$PF_1 = ae\sec\theta - a$ ...(i)

and $\dfrac{PF_2}{PM_2} = e \Rightarrow PF_2 = e\left(\dfrac{a}{e} + x_1\right)$

$\Rightarrow PF_2 = a + ex_1$ ...(ii)

$\Rightarrow PF_2 = a + ae\sec\theta$

From equation (i) & (ii)

$\boxed{|PF_2 - PF_1| = 2a}$

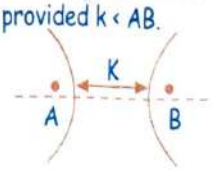
The difference of focal distances of any point P lying on the hyperbola is always equal to the length of Transverse Axis.

Conversely, if a point P moves such that difference of its distances from two fixed points is always a constant (k) then locus of P is a hyperbola provided $k < AB$.

$|PA - PB| = K$

Locus is hyperbola only if

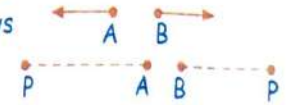
$K < AB$



If $K > AB \Rightarrow$ No locus

If $K = AB \Rightarrow$ Two rays

$\Rightarrow |PA - PB| = AB$



Ques. If the foci of a hyperbola are same as that of the ellipse $\dfrac{x^2}{9} + \dfrac{y^2}{25} = 1$ and the eccentricity of the hyperbola is $\dfrac{15}{8}$ times the eccentricity of the ellipse, then the smaller focal distance of the point $\left(\sqrt{2}, \dfrac{14}{3}\sqrt{\dfrac{2}{5}}\right)$ on the hyperbola, is equal to **[JEE Main 2024 (31 Jan Shift 2)]**

(a) $7\sqrt{\dfrac{2}{5}} - \dfrac{8}{3}$ (b) $14\sqrt{\dfrac{2}{5}} - \dfrac{4}{3}$

(c) $14\sqrt{\dfrac{2}{5}} - \dfrac{16}{3}$ (d) $7\sqrt{\dfrac{2}{5}} + \dfrac{8}{3}$

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Ans. $a^2 = 9, b^2 = 25$

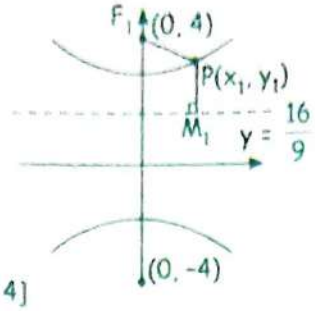
So, $9 = 25(1 - e^2)$

$\Rightarrow 16 = 25e^2$

$\Rightarrow e = \frac{4}{5}$

$F[0, be]$

$= F\left[0, 5 \times \frac{4}{5}\right] = [0, 4]$



Given, $E_H = \frac{15}{8}e \Rightarrow E_H = \frac{15}{8} \times \frac{4}{5} = \frac{3}{2}$

Now, $be = BE = 4 \Rightarrow B \times \frac{3}{2} = 4 \Rightarrow B = \frac{8}{3}$

Directrix of H:

$y = \frac{B}{E_H} = \frac{8}{3 \times 3} \times 2 = \frac{16}{9}$

Since, $PF_1 = EPM_1$

$\Rightarrow PF_1 = \frac{3}{2}\left[\frac{14}{3}\sqrt{\frac{2}{5}} - \frac{16}{9}\right] = \left[7\sqrt{\frac{2}{5}} - \frac{8}{3}\right]$

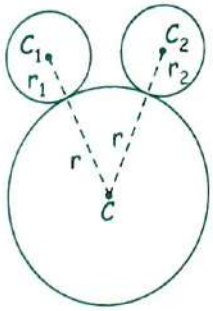
Ques. Show that the locus of the centre of a circle which touches externally two given circles is a hyperbola.

Ans. $CC_1 = r_1 + r$

$CC_2 = r + r_2$

$CC_1 - CC_2 = (r_1 - r_2) \rightarrow$ Const

Locus of C is hyperbola with C_1 & C_2 as foci and $2a = r_1 - r_2$



Ques. Consider a hyperbola H having centre at the origin and foci on the x-axis. Let C_1 be the circle touching the hyperbola H and having the centre at the origin. Let C_2 be the circle touching the hyperbola H at its vertex and having the centre at one of its foci. If areas (in sq units) of C_1 and C_2 are 36π and 4π, respectively, then the length (in units) of latus rectum of H is

[JEE Main 2024 (04 Apr Shift 2)]

(a) $\frac{14}{3}$ (b) $\frac{28}{3}$ ✓

(c) $\frac{11}{3}$ (d) $\frac{10}{3}$

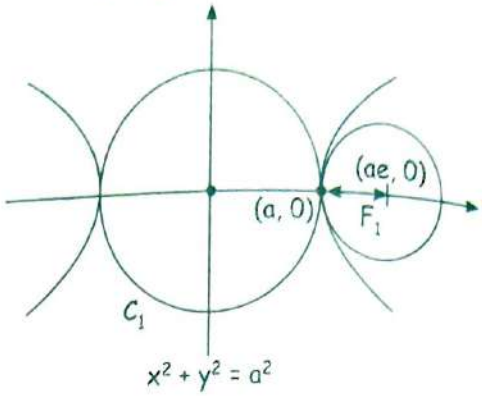
Ans. $\pi a^2 = 36\pi \Rightarrow a^2 = 36 \Rightarrow a = 6$

and $\pi(ae - a)^2 = 4\pi$

$\Rightarrow ae - a = 2$

$\Rightarrow 6e - 6 = 2$

$\Rightarrow e = \frac{8}{6} = \frac{4}{3}$



Now, $b^2 = a^2(e^2 - 1)$

$= 36\left[\frac{16}{9} - 1\right] = 4 \times 7 = 28$

Hence, $LR = \frac{2b^2}{a} = \frac{2 \times 28}{6} = \frac{28}{3}$

POSITION OF A POINT W.R.T. A HYPERBOLA

Let $S(x, y) = 0$ is equation of hyperbola and $P(x_1, y_1)$ is a point, if

(i) $S(x_1, y_1) < 0$: P is outside the hyperbola

(ii) $S(x_1, y_1) = 0$: is on the hyperbola

(iii) $S(x_1, y_1) > 0$: P is inside the hyperbola

EQUATION OF TANGENTS

1. Cartesian Form: $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1.$

2. Slope Form: $y = mx \pm \sqrt{a^2m^2 - b^2}$

For tangent to be real:

$a^2m^2 - b^2 \geq 0$

$\Rightarrow a^2m^2 \geq b^2 \Rightarrow m^2 \geq \frac{b^2}{a^2}$

3. Parametric Form: $\frac{x \sec\theta}{a} - \frac{y \tan\theta}{b} = 1.$

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EQUATION OF DIRECTOR CIRCLE

Director circle: $x^2 + y^2 = a^2 - b^2$, which is the locus of the point of intersection of perpendicular tangents.

It is real only if $a^2 \geq b^2$, i.e. only if $1 < e \leq \sqrt{2}$.

Proof: For real D.C.

$a^2 - b^2 \geq 0 \Rightarrow a^2 \geq b^2$

$\Rightarrow a^2 \geq a^2(e^2 - 1)$

$\Rightarrow 1 \geq e^2 - 1$

$\Rightarrow e^2 \leq 2$

$\Rightarrow 1 < e \leq \sqrt{2}$

Ques. The equation of a tangent to the hyperbola $4x^2 - 5y^2 = 20$ parallel to the line $x - y = 2$ is m = 1 **[JEE Main 2019, 10 Jan I]**

(a) $x - y - 3 = 0$ (b) $x - y + 9 = 0$

(c) $x - y + 1 = 0$ (d) $x - y + 7 = 0$

Ans. Equation of tangent:

$y = mx \pm \sqrt{a^2m^2 - b^2}$

$\Rightarrow y = mx \pm \sqrt{5m^2 - 4}$

$x - y = 2 \Rightarrow y = x - 2$

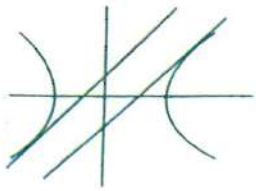
$\Rightarrow m = 1$

Hence, tangent of hyperbola parallel to $x - y = 2$ is,

$y = x \pm 1$

$\frac{x^2}{5} - \frac{y^2}{4} = 1$

$a^2 = 5, b^2 = 4$



Ques. If a hyperbola passes through the point $P(\sqrt{2}, \sqrt{3})$ and has foci at $(\pm 2, 0)$, then the tangent to this hyperbola at P also passes through the point **[2017 Main]**

(a) $(3\sqrt{2}, 2\sqrt{3})$ (b) $(2\sqrt{2}, 3\sqrt{3})$

(c) $(\sqrt{3}, \sqrt{2})$ (d) $(-\sqrt{2}, -\sqrt{3})$

Ans. Given, foci $(\pm 2, 0) = (\pm ae, 0) \Rightarrow ae = 2$

Let, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Hyperbola passes through the point $(\sqrt{2}, \sqrt{3})$

$\Rightarrow \frac{2}{a^2} - \frac{3}{b^2} = 1$

$\Rightarrow \frac{2}{a^2} - \frac{3}{4 - a^2} = 1$

$\Rightarrow \frac{2(4 - a^2) - 3a^2}{a^2(4 - a^2)} = 1$

$\Rightarrow 8 - 5a^2 = 4a^2 - a^4$

$\Rightarrow a^4 - 9a^2 + 8 = 0$

$\Rightarrow (a^2 - 1)(a^2 - 8) = 0$

$\Rightarrow a^2 = 1, a^2 = 8$ (✗)

And, $ae = 2$

So, $a^2 + b^2 = a^2e^2$

$\Rightarrow a^2 + b^2 = 4$

$\Rightarrow a = 1 \Rightarrow b^2 = 3$

Hyperbola is, $\frac{x^2}{1} - \frac{y^2}{3} = 1$

Then, equation of tangent, $\frac{x(\sqrt{2})}{1} - \frac{y(\sqrt{3})}{3} = 1$

$\Rightarrow 3\sqrt{2}x - \sqrt{3}y = 3$

put $(2\sqrt{2}, 3\sqrt{3})$ in it.

$3\sqrt{2}(2\sqrt{2}) - \sqrt{3}(3\sqrt{3}) = 12 - 9 = 3$

Hence, $(2\sqrt{2}, 3\sqrt{3})$ satisfy the tangent.

Ques. Tangents are drawn to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, parallel to the straight line $2x - y = 1$. The points of contacts of the tangents on the hyperbola are **[2012]**

(a) $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (b) $\left(-\frac{9}{2\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

(c) $(3\sqrt{3}, -2\sqrt{2})$ (d) $(-3\sqrt{3}, 2\sqrt{2})$

Ans. Equation of tangent, $y = mx \pm \sqrt{9m^2 - 4}$

$m = 2$ ($\because$ tangent is parallel to $y = 2x - 1$)

$\Rightarrow y = 2x \pm \sqrt{36 - 4}$

$\Rightarrow y = 2x \pm 4\sqrt{2}$

$\Rightarrow 2x - y = \pm 4\sqrt{2}$ ...(i)

Equation of tangent at point (x_1, y_1)

$\frac{xx_1}{9} - \frac{yy_1}{4} = 1$ ...(ii)

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From equation (i) and (ii)

$$\frac{x_1}{18} = \frac{y_1}{4} = \frac{1}{\pm 4\sqrt{2}}$$

Hence, point of contact $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ & $\left(\frac{-9}{2\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$

Ques. The foci of a hyperbola are (2, 0) and its eccentricity is 3/2. A tangent, perpendicular to the line $2x + 3y = 6$, is drawn at a point in the first quadrant on the hyperbola. If the intercepts made by the tangent on the x- and y-axis are a and b respectively, then $|6a| + |5b|$ is equal to **[JEE Main 2023 (April-II)]**

Ans. Given, $ae = 2$ and $e = \frac{3}{2}$

$$\Rightarrow a \times \frac{3}{2} = 2 \Rightarrow a = \frac{4}{3}$$

$m_T = \frac{3}{2}$ (Perpendicular to $2x + 3y = 6$)

$\because b^2 = a^2e^2 - a^2$

$$\therefore b^2 = 4 - \frac{16}{9}$$

$$\Rightarrow b^2 = \frac{36 - 16}{9} = \frac{20}{9}$$

Let equation of tangent,

$$y = mx \pm \sqrt{a^2m^2 - b^2}$$

$$\Rightarrow y = \frac{3}{2}x \pm \sqrt{\frac{16}{9} \times \frac{9}{4} - \frac{20}{9}}$$

$$\Rightarrow y = \frac{3x}{2} \pm \sqrt{\frac{36}{9} - \frac{20}{9}}$$

$$\Rightarrow y = \frac{3x}{2} \pm \frac{4}{3}$$

at $x = 0 \Rightarrow y_{in} = \pm\frac{4}{3}$

$$\Rightarrow |b| = \frac{4}{3}$$

and at $y = 0$, $\frac{3}{2}x_{in} = \pm\frac{4}{3}$

$$\Rightarrow x_{in} = \pm\frac{8}{9}$$

$$\Rightarrow |a| = \frac{8}{9}$$

Now, $6|a| + 5|b| = 6 \times \frac{8}{9} + 5 \times \frac{4}{3}$

$$= \frac{16}{3} + \frac{20}{3} = \frac{36}{3} = 12$$

Ques. A common tangent T to the curves $C_1: \frac{x^2}{4} + \frac{y^2}{9} = 1$ and $C_2: \frac{x^2}{42} - \frac{y^2}{143} = 1$ does not pass through the fourth quadrant. If T touches C_1 at (x_1, y_1) and C_2 at (x_2, y_2), then $|2x_1 + x_2|$ is equal to **[JEE Main 2022 (July-II)]**

Ans. Equation of tangent to $C_1 = mx \pm \sqrt{4m^2 + 9}$ ...(i)

Equation of tangent to $C_2 = mx \pm \sqrt{42m^2 - 143}$ ...(ii)

From equation (i) & (ii)

$\Rightarrow 4m^2 + 9 = 42m^2 - 143$

$\Rightarrow 38m^2 = 152$

$\Rightarrow 19m^2 = 76$

$$\Rightarrow m^2 = \frac{76}{19}$$

$\Rightarrow m^2 = 4$

$\Rightarrow m = \pm 2$

if, $m = 2$ then, the common tangent $y = 2x \pm 5$

if, $m = -2$ then, the common tangent $y = -2x \pm 5$

C.T.: $x_{in} < 0$ and $y_{in} > 0$

$\Rightarrow y = 2x + 5$

$$\Rightarrow \frac{-2x}{5} + \frac{y}{5} = 1$$

Equation of tangent at (x_1, y_1) on the ellipse C_1:

$$\frac{xx_1}{4} + \frac{yy_1}{9} = 1$$

Equation of tangent at (x_2, y_2) on the hyperbola C_2: $\frac{xx_2}{42} - \frac{yy_2}{143} = 1$

So, $\frac{x_1}{4} = -\frac{2}{5}$ and $\frac{y_1}{9} = \frac{1}{5}$

$$\Rightarrow x_1 = -\frac{8}{5}; y_1 = \frac{9}{5}$$

and $\frac{x_2}{42} = -\frac{2}{5}$

$$\Rightarrow x_2 = -\frac{84}{5}$$

Hence, $|2x_1 + x_2| = \left|\frac{16}{5} + \frac{84}{5}\right| = 20$

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EQUATION OF NORMALS

- Cartesian Form at the point (x_1, y_1) is

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2 = a^2e^2$$

- Parametric Form is $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2$.

- Slope Form is $y = mx \mp \frac{(a^2 + b^2)m}{\sqrt{a^2 - b^2m^2}}$

Ques. The length of the latus rectum and directrices of a hyperbola with eccentricity e are 9 and $x = \pm\frac{4}{\sqrt{13}}$, respectively. Let the line $y - \sqrt{3}x + \sqrt{3} = 0$ touch this hyperbola at (x_0, y_0). If m is the product of the focal distances of the point (x_0, y_0), then $4e^2 + m$ is equal to ____.

[JEE Main 2024 (06 Apr Shift 2)]

Ans. $\frac{2b^2}{a} = 9 \Rightarrow b^2 = \frac{9a}{2}$

$$\frac{a}{e} = \frac{4}{\sqrt{13}} \Rightarrow e = \frac{\sqrt{13}a}{4} \Rightarrow e^2 = \frac{13a^2}{16}$$

$\because b = a^2(e^2 - 1)$

$$\therefore \frac{9a}{2} = a^2\left[\frac{13a^2}{16} - 1\right]$$

$$\Rightarrow \frac{9}{2} = a\left[\frac{13a^2 - 16}{16}\right]$$

$\Rightarrow 72 = a[13a^2 - 16]$

$\Rightarrow a = 2$ (by HnT)

So, $b^2 = 9 \times \frac{2}{2} = 9$

Now, $e^2 = \frac{13a^2}{16} = \frac{13 \times 4}{16}$

$$\Rightarrow e^2 = \frac{13}{4}$$

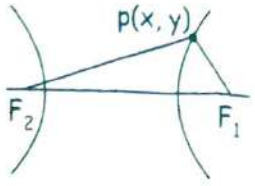
Then, hyperbola is $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Equation of tangent at point (x_0, y_0) on hyperbola, $x\frac{x_0}{4} - y\frac{y_0}{9} = 1$

and, $\sqrt{3}x - y = \sqrt{3}$ (given)

$$\Rightarrow x - \frac{y}{\sqrt{3}} = 1 \Rightarrow \frac{x_0}{4} = 1 \Rightarrow x_0 = 4$$

and, $\frac{y_0}{9} = \frac{1}{\sqrt{3}} \Rightarrow y_0 = 3\sqrt{3}$



Since, $PF_1 = ex_0 - a$

$PF_2 = ex_0 + a$

given, $m = PF_1PF_2$

$= (ex_0 - a)(ex_0 + a)$

$$\Rightarrow m = (e^2x_0^2 - a^2) = \left(\frac{13}{4} \times 16 - 4\right) = 52 - 4$$

$\Rightarrow m = 48$

Hence, $4e^2 + m = 13 + 48 = 61$

Brain Teaser

Ques. Show that for a Hyperbola, the portion of tangent between point of tangency (P) and Transverse Axis is the Harmonic Mean between the perpendiculars dropped from Foci upon the normal drawn at the same point P.

Ans. To prove

$$\frac{2}{\gamma} = \frac{1}{\alpha} + \frac{1}{\beta}$$

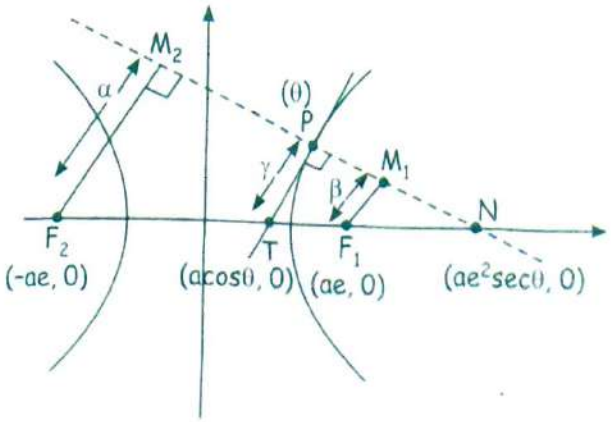
$$\Rightarrow \frac{\gamma}{\alpha} + \frac{\gamma}{\beta} = 2$$

Equation of tangent at P: $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$

For point T,

$$y = 0 \Rightarrow \frac{x\sec\theta}{a} = 1$$

$\Rightarrow x = a\cos\theta$



Equation of normal at point P,

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$$\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2e^2$$

For N, $y = 0$

$$\Rightarrow \frac{ax}{\sec\theta} = a^2e^2 \Rightarrow x = ae^2\sec\theta$$

Since $\Delta F_2M_2N \sim TPN \sim F_1M_1N$

$$\frac{\gamma}{\beta} = \frac{TN}{F_1N} = \frac{ae^2\sec\theta - a\cos\theta}{ae^2\sec\theta - ae} \quad ...(1)$$

$$\frac{\gamma}{\alpha} = \frac{TN}{F_2N} = \frac{ae^2\sec\theta - a\cos\theta}{ae^2\sec\theta + ae} \quad ...(2)$$

$$\frac{\gamma}{\beta} + \frac{\gamma}{\alpha} = \frac{a(e^2\sec\theta - \cos\theta)}{ae(e\sec\theta - 1)} + \frac{a(e^2\sec\theta - \cos\theta)}{ae(e\sec\theta + 1)}$$

$$= \left(\frac{e^2\sec\theta - \cos\theta}{e}\right)\left(\frac{1}{(e\sec\theta - 1)} + \frac{1}{(e\sec\theta + 1)}\right)$$

$$= \left(\frac{e^2\sec\theta - \frac{1}{\sec\theta}}{e}\right)\frac{2e\sec\theta}{(e^2\sec^2\theta - 1)}$$

$$= \frac{e^2\sec^2\theta - 1}{e\sec\theta} \cdot \frac{2e\sec\theta}{(e^2\sec^2\theta - 1)}$$

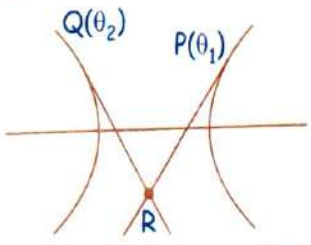
$$\Rightarrow \frac{\gamma}{\beta} + \frac{\gamma}{\alpha} = 2$$

$$\Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} = \frac{2}{\gamma}$$

POINT OF INTERSECTION OF TANGENTS

Point of intersection of the tangents at $P(\theta_1)$ and $Q(\theta_2)$ is

$$x = a\frac{\cos\left(\frac{\theta_1 - \theta_2}{2}\right)}{\cos\left(\frac{\theta_1 + \theta_2}{2}\right)}, \; y = b\tan\left(\frac{\theta_1 + \theta_2}{2}\right)$$



$$R\left[\frac{a\cos\left(\frac{\theta_1 - \theta_2}{2}\right)}{\cos\left(\frac{\theta_1 + \theta_2}{2}\right)}, \frac{b\sin\left(\frac{\theta_1 + \theta_2}{2}\right)}{\cos\left(\frac{\theta_1 + \theta_2}{2}\right)}\right]$$

EQUATION OF CHORDS

Chord of contact of the tangents is $T = 0$

where (x_1, y_1) is the point of intersection of tangents.

Chord with mid-point at (x_1, y_1) is $T = S_1$.

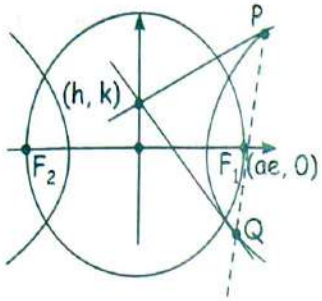
Chord joining two points $P(\alpha)$ and $Q(\beta)$ on the hyperbola is

$$\frac{x}{a}\cos\left(\frac{\alpha - \beta}{2}\right) - \frac{y}{b}\sin\left(\frac{\alpha + \beta}{2}\right) = \cos\left(\frac{\alpha + \beta}{2}\right)$$

Pair of tangents from (x_1, y_1) is $SS_1 = T^2$.

Ques. Prove that the locus of a point whose chord of contact touches the circle inscribed on the straight line joining the foci of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ as diameter is $x^2/a^4 + y^2/b^4 = 1/(a^2 + b^2)$.

Ans.



Chord of contact of hyperbola is $\frac{xh}{a^2} - \frac{ky}{b^2} = 1$

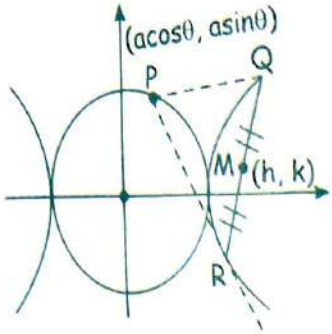
So the distance of (0, 0) from the chord of contact is equal to the radius of the circle.

$$\left|\frac{1}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}}\right| = ae$$

$$\Rightarrow \frac{1}{a^2e^2} = \frac{h^2}{a^4} + \frac{k^2}{b^4} \Rightarrow \frac{1}{a^2 + b^2} = \frac{x^2}{a^4} + \frac{y^2}{b^4}$$

Ques. From points on the circle $x^2 + y^2 = a^2$ tangents are drawn to the hyperbola $x^2 - y^2 = a^2$; prove that the locus of the middle points of the chords of contact is the curve $(x^2 - y^2)^2 = a^2(x^2 + y^2)$.

Ans.



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QR is C.O.C. of point P w.r.t. hyperbola

So, $xx_1 - yy_1 = a^2$

$\Rightarrow x(a\cos\theta) - y(a\sin\theta) = a^2$

$\Rightarrow x\cos\theta - y\sin\theta = a$ ...(i)

Also QR: $T = S_1$

$\Rightarrow xh - ky - a^2 = h^2 - k^2 - a^2$

$\Rightarrow hx - ky = h^2 - k^2$ ...(ii)

from equation (i) and (ii)

$$\frac{\cos\theta}{h} = \frac{\sin\theta}{k} = \frac{a}{h^2 - k^2}$$

$$\Rightarrow \cos\theta = \frac{ah}{h^2 - k^2}$$

$$\& \sin\theta = \frac{ak}{h^2 - k^2}$$

$\because \sin^2\theta + \cos^2\theta = 1$

$$\therefore 1 = \frac{a^2h^2 + a^2k^2}{(h^2 - k^2)^2}$$

$\Rightarrow (x^2 - y^2)^2 = a^2(x^2 + y^2)$ Hence proved

Ques. Show that the mid points of focal chords of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ lie on another similar hyperbola.

Ans. Equation of PQ: $T = S_1$

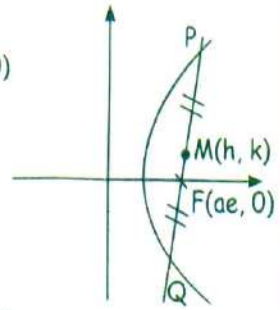
$$\Rightarrow \frac{hx}{a^2} - \frac{ky}{b^2} - 1 = \frac{h^2}{a^2} - \frac{k^2}{b^2} - 1$$

$$\Rightarrow \frac{hx}{a^2} - \frac{ky}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$

$\because$ PQ satisfy F (ae, 0)

$$\therefore \frac{hae}{a^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$

$$\Rightarrow \frac{eh}{a} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$



Replace $h \to x$ & $k \to y$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{ex}{a}$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{ex}{a} - \frac{y^2}{b^2} = 0$$

$$\Rightarrow \frac{x^2 - aex}{a^2} - \frac{y^2}{b^2} = 0$$

$$\Rightarrow \frac{\left(x - \frac{ae}{2}\right)^2}{a^2} - \frac{y^2}{b^2} = \frac{a^2e^2/4}{a^2}$$

$$\Rightarrow \frac{\left(x - \frac{ae}{2}\right)^2}{a^2} - \frac{y^2}{b^2} = \frac{e^2}{4}$$

$$\Rightarrow \frac{X^2}{a^2} - \frac{y^2}{b^2} = \frac{e^2}{4}$$

$$\Rightarrow \frac{x^2}{\frac{a^2e^2}{4}} - \frac{y^2}{\frac{b^2e^2}{4}} = 1$$

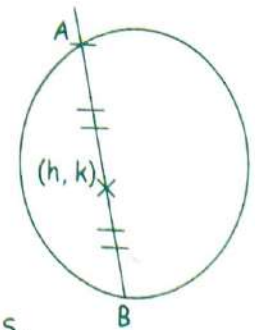
Let, $A^2 = \frac{a^2e^2}{4}$, $B^2 = \frac{b^2e^2}{4}$

$$\frac{B^2}{A^2} = \frac{b^2}{a^2}$$

Hence, the mid points of focal chords of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ lie on another similar hyperbola.

Ques. If a variable tangent of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, cuts the circle $x^2 + y^2 = 4$ at point A, B and locus of mid point of AB is $9x^2 - 4y^2 - \lambda(x^2 + y^2)^2 = 0$ then the value of λ is

Ans. AB: $\frac{x\sec\theta}{3} - \frac{y\tan\theta}{2} = 1$ ...(i)



AB: $T = S_1$

$\Rightarrow xh + ky - 4 = h^2 + k^2 - 4$

$\Rightarrow hx + ky = h^2 + k^2$ ...(ii)

From equation (i) & (ii)

$$\Rightarrow \frac{\sec\theta}{3h} = \frac{\tan\theta}{-2k} = \frac{1}{h^2 + k^2}$$

$$\Rightarrow \sec\theta = \frac{3h}{h^2 + k^2} \ \& \ \tan\theta = \frac{-2k}{h^2 + k^2}$$

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$\because \sec^2\theta - \tan^2\theta = 1$

$$\frac{9h^2}{(h^2+k^2)^2} - \frac{4k^2}{(h^2+k^2)^2} = 1$$

$$\Rightarrow 9h^2 - 4k^2 = (h^2+k^2)^2$$

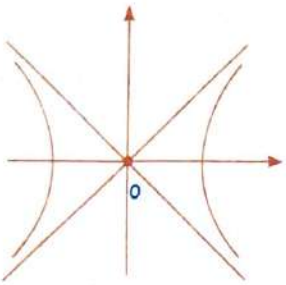
$$\Rightarrow 9x^2 - 4y^2 - (x^2+y^2)^2 = 0$$

Hence, $\lambda = 1$

ASYMPTOTES OF HYPERBOLA

The lines which appear to touch the curve at infinity are called Asymptotes $y = \frac{b}{a}x$ and $y = -\frac{b}{a}x$ are the 2 asymptotes for $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Asymptotes for a Rectangular Hyperbola are always perpendicular.



Equation of tangent

$$y = mx \pm \sqrt{a^2m^2 - b^2}$$

$$\Rightarrow -mx + y = \pm\sqrt{a^2m^2 - b^2}$$

and $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

Compare,

$$\frac{x_1}{a^2(-m)} = \frac{-y_1}{b^2} = \frac{1}{\pm\sqrt{a^2m^2 - b^2}}$$

$$\Rightarrow x_1 = \frac{-a^2m}{\pm\sqrt{a^2m^2 - b^2}} \text{ and } y_1 = \frac{-b^2}{\pm\sqrt{a^2m^2 - b^2}}$$

For asymptote

$(x, y_1) \to \pm\infty$

For asymptote

$a^2m^2 - b^2 = 0$

$$\Rightarrow m^2 = \frac{b^2}{a^2}$$

$$\Rightarrow m = \pm\frac{b}{a}$$

So, the asymptotes are $y = \pm\frac{b}{a}x$

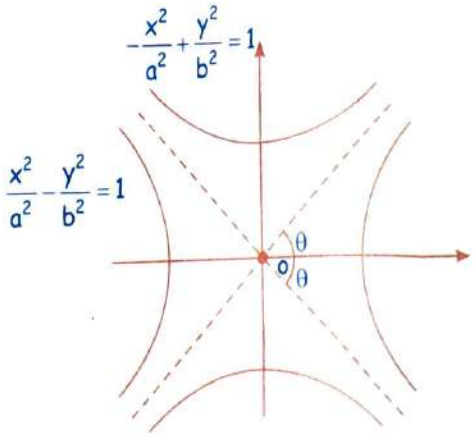
Ex: For hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$

the asymptote are, $y = \pm\frac{b}{a}x$

$$\Rightarrow y = \pm\frac{4}{3}x$$

Important Note

1. A hyperbola and its conjugate have the same asymptote.
2. The asymptotes pass through the centre of the hyperbola and the bisectors of the angles between the asymptotes are the axes of the hyperbola.
3. Asymptotes are the tangent to the hyperbola from the centre.
4. The equation of the pair of asymptotes differ from the equation of hyperbola by a constant.



Tangents:

$$\frac{x}{a} - \frac{y}{b} = 0 \quad \text{...(i)}$$

$$\frac{x}{a} + \frac{y}{b} = 0 \quad \text{...(ii)}$$

$$\text{POSL}: \left(\frac{x}{a} + \frac{y}{b}\right)\left(\frac{x}{a} - \frac{y}{b}\right) = 0$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

Note: $C_{POSL} - C_{CH} = C_H - C_{POSL}$

or $2C_{POSL} = C_{CH} + C_H$

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Ques. Find the hyperbola whose asymptotes are $2x - y = 3$ and $3x + y - 7 = 0$ and which passes through the point (1, 1). Also find equation of conjugate hyperbola

Ans. Equation of Hyperbola

$(2x - y - 3)(3x + y - 7) + k = 0$

Given, that hyperbola passes through the point (1, 1)

So, $(2 - 1 - 3)(3 + 1 - 7) + k = 0$

$\Rightarrow -2(-3) + k = 0 \Rightarrow k = -6$

Now, POSL: $(2x - y - 3)(3x + y - 7) = 0$

$\Rightarrow C_{POSL} = 21$

$C_H = 21 - 6 = 15$

$\because 2C_{POSL} = C_{CH} + C_H$

$\therefore 2 \times 21 = 15 + C_{CH}$

$\Rightarrow 42 - 15 = C_{CH}$

$\Rightarrow C_{CH} = 27$

POSL: $(2x - y - 3)(3x + y - 7) = 0$

$\Rightarrow 6x^2 + 2xy - 14x - 3xy - y^2 + 7y - 9x - 3y + 21 = 0$

To find conjugate hyperbola, replace 21 by 27 (C_{CH})

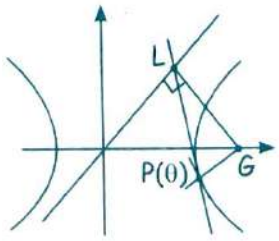
Hence, Conjugate Hyperbola:

$6x^2 - y^2 - xy - 23x + 4y + 27 = 0$

Ques. From a point G on the transverse axis of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, GL is drawn perpendicular to one of its asymptotes. Also GP is a normal to the curve at P. Prove that LP is parallel to the conjugate axis.

Ans. Equation of normal:

$\frac{ax}{\sec\theta} - \frac{by}{\tan\theta} = a^2e^2$



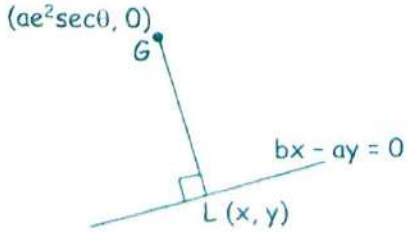
For point G, $y = 0$

$\Rightarrow \frac{ax}{\sec\theta} = a^2e^2$

$\Rightarrow x = ae^2 \sec\theta$

So, point $G(ae^2 \sec\theta, 0)$

To prove: $X_L = a \sec\theta$



$\frac{x - ae^2 \sec\theta}{b} = -\left[\frac{ae^2 \sec\theta \cdot b}{b^2 + a^2}\right]$

$\Rightarrow \frac{x - ae^2 \sec\theta}{b} = -\frac{ae^2 \sec\theta \cdot b}{a^2e^2}$

$\Rightarrow \frac{x - ae^2 \sec\theta}{b} = -\frac{b \sec\theta}{a}$

$\Rightarrow x - ae^2 \sec\theta = -\frac{a^2(e^2 - 1)\sec\theta}{a}$

$\Rightarrow x - ae^2 \sec\theta = -ae^2 \sec\theta + a \sec\theta$

$\Rightarrow x = a \sec\theta$

PROPERTIES OF ASYMPTOTES

Property-1

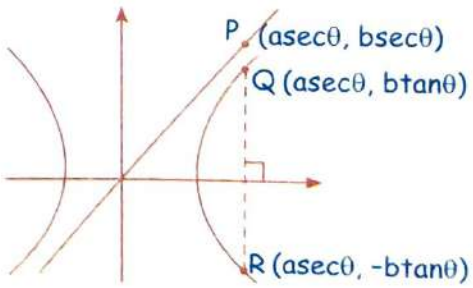
If from any point on the asymptote a straight line be drawn perpendicular to the transverse axis, the product of the segments of this line, intercepted between the point and the Hyperbola is always equal to the square of the semi conjugate axis.

Proof: To prove $PQ \cdot PR = b^2$

Let $Q[a \sec\theta, b \tan\theta]$ and $R[a \sec\theta, -b \tan\theta]$

Equation of asymptote is: $y = \frac{b}{a}x$

So, $y_p = \frac{b}{a} \cdot a \sec\theta = b \sec\theta$



Now, $PQ = b \sec\theta - b \tan\theta = b(\sec\theta - \tan\theta)$ ...(i)

and $PR = b \sec\theta + b \tan\theta = b(\sec\theta + \tan\theta)$ ...(ii)

Hence, $PQ \cdot PR = b^2(\sec^2\theta - \tan^2\theta) = b^2$

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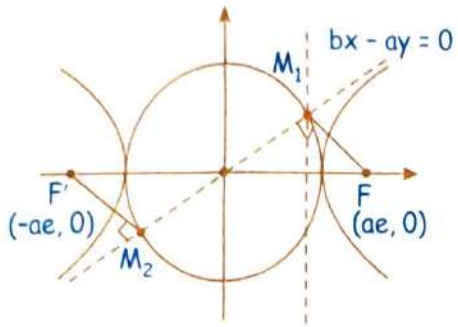
Property-2

Feet of perpendiculars from the foci on either asymptote meet on the corresponding directrix and as well as on the auxiliary circle.

Proof:



For co-ordinates of m_1,

$$\frac{x-ae}{b} = \frac{y-0}{-a} = -\left[\frac{abe-0}{a^2+b^2}\right]$$

$$\Rightarrow \frac{x-ae}{b} = \frac{y}{-a} = -\left[\frac{abe}{a^2e^2}\right]$$

$$\Rightarrow \frac{x-ae}{b} = \frac{y}{-a} = -\frac{b}{ae}$$

$$\Rightarrow \frac{x-ae}{b} = -\frac{b}{ae}$$

$$\Rightarrow x - ae = -\frac{a^2(e^2-1)}{ae}$$

$$\Rightarrow x - ae = \frac{-ae^2+a}{e}$$

$$\Rightarrow x - ae = -ae + \frac{a}{e}$$

$$\Rightarrow x = \frac{a}{e}$$

and, $\frac{y}{-a} = -\frac{b}{ae}$

$$\Rightarrow y = \frac{b}{e}$$

So, $M_1\left[\frac{a}{e}, \frac{b}{e}\right]$

$\because$ auxiliary circle is $x^2 + y^2 = a^2$

LHS: $\frac{a^2}{e^2} + \frac{b^2}{e^2} = \frac{a^2+b^2}{e^2}$

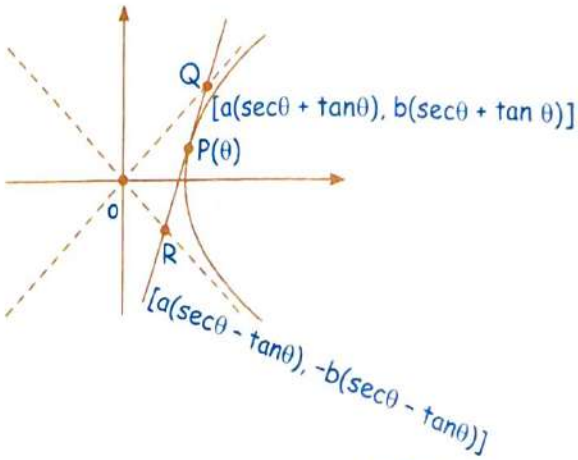
$= \frac{a^2e^2}{e^2} = a^2 = RHS$

Property-3

The portion of the tangent intercepted between the asymptotes is bisected at the point of contact.

For a rectangular hyperbola, the locus of the circum-centre of the triangle formed by pair of asymptotes and any tangent to the hyperbola is the same hyperbola.

Proof: P is mid point of QR



Equation of tangent at P(θ): $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$ ...(i)

Equation of asymptote, $y = \frac{b}{a}x$ ...(ii)

Form equation (i) & (ii)

$$\frac{x\sec\theta}{a} - \frac{bx}{a}\frac{\tan\theta}{b} = 1$$

$$\Rightarrow \frac{x}{a}(\sec\theta - \tan\theta) = 1$$

$$\Rightarrow x = \frac{a}{(\sec\theta-\tan\theta)}\frac{\sec\theta+\tan\theta}{(\sec\theta+\tan\theta)}$$

$$\Rightarrow x = a(\sec\theta + \tan\theta)$$

and $y = b(\sec\theta + \tan\theta)$ (from eq. (ii))

Equation of other asymptote,

$y = -\frac{b}{a}x$ ...(iii)

From equation (i) & (iii)

$$\frac{x\sec\theta}{a} + \frac{x\tan\theta}{a} = 1$$

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$\Rightarrow x = \dfrac{a}{\sec\theta + \tan\theta}$

$\Rightarrow x = a(\sec\theta - \tan\theta)$

and $y = -b(\sec\theta - \tan\theta)$ (from eq. (iii))

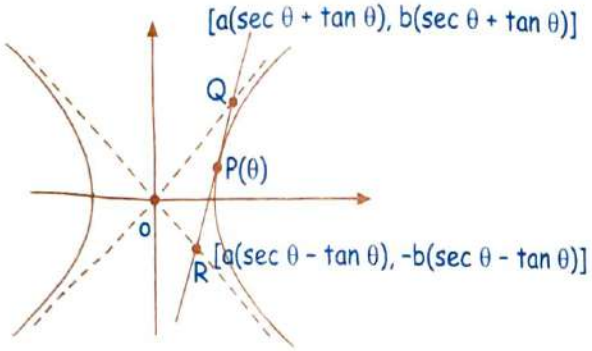
Now, mid point of QR

$x = \dfrac{2a\sec\theta}{2} = a\sec\theta,\ y = b\tan\theta$

Property-4

The area of the triangle formed by pair of asymptotes and any tangent to the hyperbola is always a constant equal to ab.

Proof:



$$\text{Area} = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ a(\sec\theta+\tan\theta) & b(\sec\theta+\tan\theta) & 1 \\ a(\sec\theta-\tan\theta) & -b(\sec\theta-\tan\theta) & 1 \end{vmatrix}$$

$= \dfrac{1}{2}|-ab - ab| = ab$

Ques. From any point on the hyperbola $H_1 : (x^2/a^2) - (y^2/b^2) = 1$ tangents are drawn to the hyperbola $H_2 : (x^2/a^2) - (y^2/b^2) = 2$. The area of the Δ cut-off by the chord of contact on the asymptotes of H_2 is equal to

(a) ab/2 (b) ab

(c) 2ab (d) 4ab

Ans. Let point be [a sec θ, b tan θ]

Chord of contact $\rightarrow \dfrac{xx_1}{a^2} - \dfrac{yy_1}{b^2} = 2$

$\Rightarrow \dfrac{x\,a\sec\theta}{a^2} - \dfrac{y\,b\tan\theta}{b^2} = 2$

$\Rightarrow \dfrac{x\sec\theta}{2a} - \dfrac{y\tan\theta}{2b} = 1$

Let A = 2a, B = 2b

Asymptotes for H_2 are:

$\Rightarrow y = \pm\dfrac{b}{a}x$

$\Rightarrow y = \pm\dfrac{2b}{2a}x$

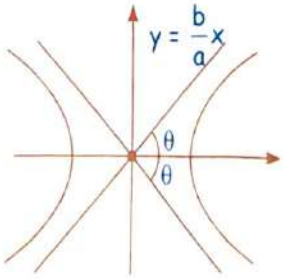
$\Rightarrow y = \pm\dfrac{B}{A}x$

Hence, the area of Δ = AB = 2a · 2b = 4ab

Property-5

If the angle between the asymptote of a hyperbola is 2θ then the eccentricity of the hyperbola is sec θ.

Proof:



$b^2 = a^2(e^2 - 1)$

$\Rightarrow \dfrac{b^2}{a^2} = e^2 - 1$

$\Rightarrow \tan^2\theta = e^2 - 1\ (\because \tan\theta = \dfrac{b}{a})$

$\Rightarrow e^2 = 1 + \tan^2\theta$

$\Rightarrow e^2 = \sec^2\theta \Rightarrow e = |\sec\theta|$

Ques. A hyperbola passing through origin has x − 2y = 1 and x + 3y = 6 as its asymptotes. Eccentricity of the hyperbola may be

(a) $\sqrt{4+2\sqrt{2}}$ (b) $\sqrt{2+2\sqrt{2}}$

(c) $\sqrt{4-2\sqrt{2}}$ (d) $\sqrt{4-\sqrt{2}}$

Ans. Let slopes of asymptotes x − 2y = 1 and x + 3y = 6 are m_1 and m_2.

then, $m_1 = \dfrac{1}{2}, m_2 = -\dfrac{1}{3}$

Now, $\tan 2\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right|$

$\tan 2\theta = \left|\dfrac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}}\right| = \dfrac{\frac{5}{6}}{\frac{5}{6}} = 1$

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$\Rightarrow$ Either $2\theta = 45°$

$\Rightarrow \theta = 22.5°$

$\because \tan 22.5° = \sqrt{2} - 1$

$e = \sec 22.5°$

$\Rightarrow e = \sqrt{1 + \tan^2 22.5°}$

$\Rightarrow e = \sqrt{1 + (\sqrt{2} - 1)^2}$

$\Rightarrow e = \sqrt{1 + 2 + 1 - 2\sqrt{2}}$

$\Rightarrow e = \sqrt{4 - 2\sqrt{2}}$

or $2\theta = 180° + 45°$

$\Rightarrow \theta = 90° + 225°$

$\because \tan(90° + 22.5°)$

$= \cot 22.5° = \sqrt{2} + 1$

$e' = \sec(90° + 22.5°)$

$= \operatorname{cosec} 22.5°$

$= \sqrt{1 + \cot^2 22.5°}$

$= \sqrt{1 + (\sqrt{2} + 1)^2}$

$= \sqrt{1 + 2 + 1 + 2\sqrt{2}}$

$= \sqrt{4 + 2\sqrt{2}}$

Ques. If the asymptotes of a hyperbola are given by the pair of lines $2x^2 - 7xy - 2y^2 = 0$. Find the eccentricity of the hyperbola.

Ans. For the given pair of line $a = 2, b = -2$

$\Rightarrow a + b = 0$

so, $\theta = 90°$

Then, hyperbola is Rectangular Hyperbola

Hence, the eccentricity, $e = \sqrt{2}$

Important Note

Confocal Ellipse & Hyperbola intersect orthogonally.

Ques. The curves $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$ and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with eccentricities e_1 and e_2 respectively, are confocal. The value of $e_1^2 + e_2^2$ is

Ans. Given ellipse is $\frac{x^2}{(2a)^2} + \frac{y^2}{(2b)^2} = 1$

Given that both ellipse and hyperbola are confocal

So, $4a^2 - 4b^2 = a^2 + b^2$

$\Rightarrow 3a^2 = 5b^2$ ...(i)

$\because 4b^2 = 4a^2(1 - e_1^2)$

$\Rightarrow b^2 = a^2(1 - e_1^2)$

from equation (i)

$\Rightarrow \frac{3a^2}{5} = a^2(1 - e_1^2)$

$\Rightarrow e_1^2 = \frac{2}{5}$

and $b^2 = a^2(e_2^2 - 1)$

$\Rightarrow \frac{3}{5}a^2 = a^2(e_2^2 - 1)$

$\Rightarrow e_2^2 = 1 + \frac{3}{5} = \frac{8}{5}$

Hence, $e_1^2 + e_2^2 = \frac{2}{5} + \frac{8}{5} = \frac{10}{5} = 2$

Ques. The ellipse $4x^2 + 9y^2 = 36$ and the hyperbola $4x^2 - y^2 = 4$ have the same foci and they intersect at right angles then the equation of the circle through the points of intersection of two conics is

(a) $x^2 + y^2 = 5$

(b) $\sqrt{5}(x^2 + y^2) - 3x - 4y = 0$

(c) $\sqrt{5}(x^2 + y^2) + 3x + 4y = 0$

(d) $x^2 + y^2 = 25$

Ans. $4x^2 + 9y^2 = 36$ ...(i)

$+ \quad 36x^2 - 9y^2 = 36$ ...(ii)

$40x^2 = 36 \times 2$

$\Rightarrow 5x^2 = 9$

$\Rightarrow x^2 = \frac{9}{5}$

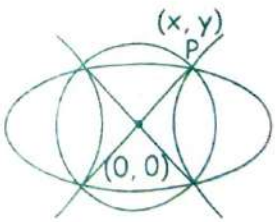
from equation (i)

$\Rightarrow 4 \times \frac{9}{5} + 9y^2 = 36$

$\Rightarrow \frac{4}{5} + y^2 = 4$

$\Rightarrow y^2 = 4 - \frac{4}{5}$

$\Rightarrow y^2 = \frac{16}{5}$



Now, $r^2 = x^2 + y^2$

$\Rightarrow r^2 = \frac{9}{5} + \frac{16}{5} \Rightarrow r^2 = \frac{25}{5}$

$\Rightarrow r^2 = 5$

Hence, the equation of circle is $x^2 + y^2 = 5$

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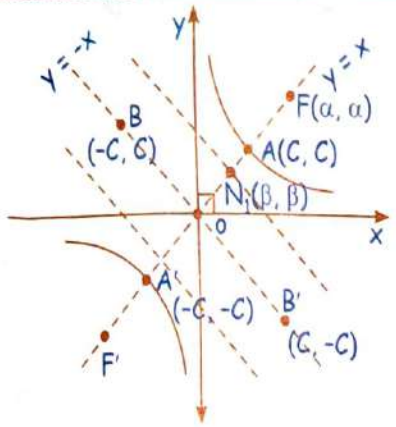
RECTANGULAR HYPERBOLA $xy = c^2$



Asymptotes are x-axis and y-axis

Transverse Axis	$y = x$
Ends of T.A.	(c, c) and $(-c, -c)$
Length of Transverse Axis	$2\sqrt{2}c$
Conjugate Axis	$y = -x$
Ends of C.A.	$(-c, c)$ and $(c, -c)$
Length of Conjugate Axis	$2\sqrt{2}c$
Centre	$(0, 0)$
Foci	$(\sqrt{2}c, \sqrt{2}c)$ and $(-\sqrt{2}c, -\sqrt{2}c)$
Directrices	$x + y = \sqrt{2}c$ and $x + y = -\sqrt{2}c$
Feet of Directrices	$\left[\frac{c}{\sqrt{2}}, \frac{c}{\sqrt{2}}\right]$ and $\left[-\frac{c}{\sqrt{2}}, -\frac{c}{\sqrt{2}}\right]$
Equation of LR	$x + y = 2\sqrt{2}c$ and $x + y = -2\sqrt{2}c$
Length of LR	$2\sqrt{2}c$
Asymptotes	$x = 0, y = 0$

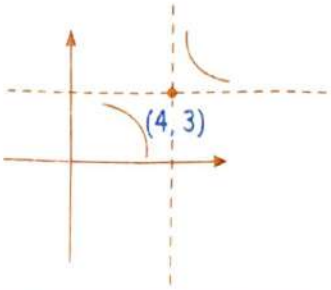
Shifted form of $xy = c^2$

$xy - xh - yk = c$

$\Rightarrow \underbrace{(x-k)}_{X}\underbrace{(y-h)}_{Y} = hk + c = hk + c$

Ex: $xy - 3x - 4y = 7 \Rightarrow (x - 4)(y - 3) = 12 + 7$

$\Rightarrow (x - 4)(y - 3) = 19 \Rightarrow XY = 19$



Parametric Coordinates on $xy = c^2$

Equation $xy = c^2$ has parametric co-ordinates $x = ct$, $y = c/t$, $t \in R - \{0\}$

Equation of Tangent & Normals

Equation of the tangent at $P(x_1, y_1)$ is $\frac{x}{x_1} + \frac{y}{y_1} = 2$ and at $P(t)$ is $\frac{x}{t} + ty = 2c$.

Equation of normal is $y - \frac{c}{t} = t^2(x - ct)$

Ques. The foot of the perpendicular from the origin, to any tangent to $xy = 1$, describes the curve

(a) $x^2 + y^2 = 1$

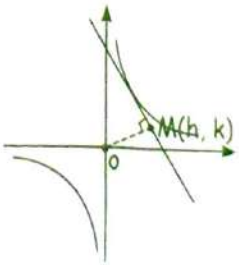
(b) $(x^2 + y^2)^2 = 4xy$

(c) $x^2 + y^2 = 4xy$

(d) None of these

Ans. Let (h, k) is the foot of perpendicular from origin to tangent.

Then, equation of tangent: $hx + ky = h^2 + k^2$



$\Rightarrow y = \frac{h^2 + k^2 - hx}{k}$

given, curve $xy = 1$

$\Rightarrow x\left[\frac{h^2 + k^2 - hx}{k}\right] = 1$

$\Rightarrow x(h^2 + k^2) - hx^2 = k$

$\Rightarrow hx^2 - x(h^2 + k^2) + k = 0$

$D = 0$

$\Rightarrow (h^2 + k^2)^2 - 4hk = 0$

$\Rightarrow (x^2 + y^2)^2 = 4xy$

Ques. The equation of a common tangent to the curves $y^2 = 16x$ and $xy = -4$, is **[JEE Main, 12 April 2019 II]**

(a) $x - y + 4 = 0$ (b) $x + y + 4 = 0$

(c) $x - 2y + 16 = 0$ (d) $2x - y + 2 = 0$

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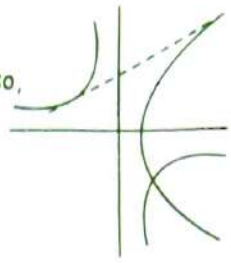
Ans. Let equation of tangent of parabola $y^2 = 16x$ is

$ty = x + 4t^2$

$\Rightarrow y = \dfrac{x + 4t^2}{t}$



It is tangent to $xy = -4$ also,

so, $x\left(\dfrac{x + 4t^2}{t}\right) = -4$

$\Rightarrow x^2 + 4t^2x + 4t = 0$

$D = 0$

$\Rightarrow 16t^4 - 4 \times 4t = 0$

$\Rightarrow t(t^3 - 1) = 0$

$\Rightarrow t = 0$ or $t = 1$

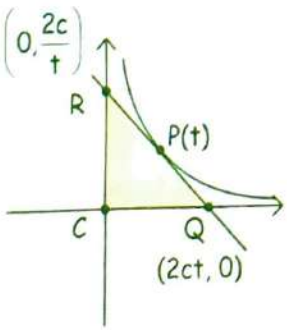
Hence, common tangent are, $x = 0$ and $y = x + 4$

Ques. For the hyperbola $xy = 8$ any tangent of it at P meets the coordinate axes at Q and R then area of the triangle CQR, where C is the centre of hyperbola.

Ans. Tangent of hyperbola $xy = c^2$ is $\dfrac{x}{t} + yt = 2c$.

Put $y = 0$, $x = 2ct$

Put $x = 0$, $y = \dfrac{2c}{t}$



Given hyperbola $xy = 8$

So, $c^2 = 8$

Area of triangle $= \dfrac{1}{2} \times 2ct \times \dfrac{2c}{t} = 2c^2 = 16$

Ques. The area of the triangle formed by any tangent of the rectangular hyperbola $12xy - 10x - 18y + 3 = 0$ with its asymptotes, is

Ans. $xy - \dfrac{5x}{6} - \dfrac{3y}{2} + \dfrac{1}{4} = 0$

$\Rightarrow \left(x - \dfrac{3}{2}\right)\left(y - \dfrac{5}{6}\right) = -\dfrac{1}{4} + \dfrac{15}{12}$

$\Rightarrow \left(x - \dfrac{3}{2}\right)\left(y - \dfrac{5}{6}\right) = -\dfrac{1}{4} + \dfrac{5}{4}$

$\Rightarrow \left(x - \dfrac{3}{2}\right)\left(y - \dfrac{5}{6}\right) = 1$

$XY = 1$

Area $= ab = \sqrt{2}c\,\sqrt{2}c = 2c^2 = 2(1)^2 = 2$

Equation of chords

Equation of a chord joining the points $P(t_1)$ and $Q(t_2)$ is $x + t_1t_2y = c(t_1 + t_2)$ with slope, $m = \dfrac{-1}{t_1t_2}$

Chord with a given middle point as (h, k) is $\dfrac{x}{h} + \dfrac{y}{k} = 2$

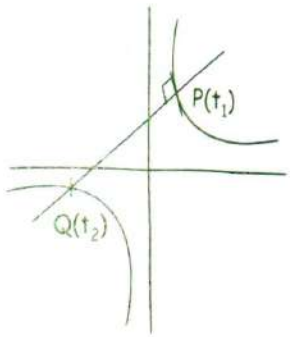
Ques. If normal drawn at point $P(t_1)$ to hyperbola $xy = c^2$ meets it again at $Q(t_2)$ then value of $t_1^3t_2$ =

Ans. $m_{Normal} = t_1^2$

$m_{Chord} = -\dfrac{1}{t_1t_2}$

so, $t_1^2 = -\dfrac{1}{t_1t_2}$

$\Rightarrow t_1^3t_2 = -1$



Important Note

The orthocenter of the triangle formed by 3 points lying on a rectangular Hyperbola always lies on the same Rectangular Hyperbola.

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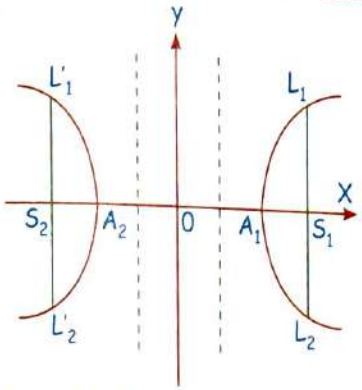
Short Notes

DEFINITION

- The general equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a hyperbola if $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$ and $h^2 > ab$ and a rectangular hyperbola if $\Delta \neq 0$ and $a + b = 0$.
- The locus of a point such that the ratio of its distance from a fixed point to its distance from a fixed line is a constant, is called a **conic section**. The ratio e is called the **eccentricity**. A conic section, in which the eccentricity is greater than unity, is called a **Hyperbola**.

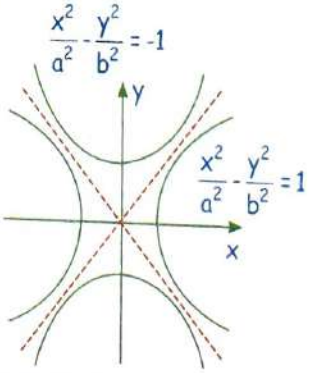
STANDARD EQUATIONS $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$:



- Vertices $A_1(a, 0)$, $A_2(-a, 0)$
- Transverse axis = 2a, Conjugate axis = 2b,
- Eccentricity $e = \sqrt{1 + \frac{b^2}{a^2}}$, $b^2 = a^2(e^2 - 1)$,
- Centre (0, 0), Foci $S_1(ae, 0)$ and $S_2(-ae, 0)$.
- Directrices $x = \frac{a}{e}$ and $x = -\frac{a}{e}$.
- Length of Latus rectum $= \frac{2b^2}{a} = \frac{(\text{Conjugate Axis})^2}{(\text{Transverse Axis})}$ $= 2a(e^2 - 1) = 2e$ (distance from focus to directrix)
- Ends of Latus Rectum $\left(ae, \frac{b^2}{a}\right), \left(ae, \frac{-b^2}{a}\right); \left(-ae, \frac{b^2}{a}\right), \left(-ae, \frac{-b^2}{a}\right)$
- Auxiliary circle: $x^2 + y^2 = a^2$
- Parametric co-ordinates: $x = a\sec\theta$, $y = b\tan\theta$

CONJUGATE HYPERBOLA

- $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is the conjugate hyperbola of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. So, their principal axes are interchanged.
- If their eccentricities be e_1 and e_2 respectively, then $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$
- The asymptotes of a hyperbola and its conjugate are common.



EQUATION OF TANGENTS TO: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Cartesian Form at point (x_1, y_1) is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$.

- Parametric Form is $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$.
- Slope form $y = mx \pm \sqrt{a^2m^2 - b^2}$, and point of contact is $\left(-\frac{a^2m}{c}, -\frac{b^2}{c}\right)$.
- Director circle: $x^2 + y^2 = a^2 - b^2$, which is the locus of the point of intersection of perpendicular tangents. It is real only if $a^2 \geq b^2$, i.e. only if $1 < e \leq \sqrt{2}$.

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- Point of intersection of the tangents at $P(\theta_1)$ & $Q(\theta_2)$ is $x = a\dfrac{\cos\left(\frac{\theta_1-\theta_2}{2}\right)}{\cos\left(\frac{\theta_1+\theta_2}{2}\right)}$, $y = b\tan\left(\frac{\theta_1+\theta_2}{2}\right)$
- Pair of tangents from (x_1, y_1) is $SS_1 = T^2$.

EQUATION OF NORMALS TO $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$:

- Cartesian Form at the point (x_1, y_1) is $\dfrac{a^2x}{x_1}+\dfrac{b^2y}{y_1} = a^2+b^2$
- Parametric Form is $\dfrac{ax}{\sec\theta}+\dfrac{by}{\tan\theta} = a^2+b^2$

EQUATION OF CHORDS

- Chord of contact of the tangents is $T = 0$ where (x_1, y_1) is the point of intersection of tangents.
- Chord with mid-point at (x_1, y_1) is $T = S_1$.
- Chord joining two points α and β on the hyperbola is $\dfrac{x}{a}\cos\left(\dfrac{\alpha-\beta}{2}\right)-\dfrac{y}{b}\sin\left(\dfrac{\alpha+\beta}{2}\right)=\cos\left(\dfrac{\alpha+\beta}{2}\right)$.

ASYMPTOTES

- The line which appears to touch the curve at infinity is known as an asymptote to that curve.
- The straight lines $\dfrac{x^2}{a^2}-\dfrac{y^2}{b^2}=0$ are asymptotes which pass through centre and are equally inclined to axes.
- When $b = a$, the asymptotes of the rectangular hyperbola $x^2 - y^2 = a^2$ are, $y = \pm x$.
- The asymptotes of a rectangular Hyperbola are always perpendicular.
- If the angle between the asymptotes is 2α then $e = \sec\alpha$.

RECTANGULAR HYPERBOLA $xy = c^2$

- Asymptotes are coordinates axes i.e. $xy = 0$.
- Foci $\left(\sqrt{2}c, \sqrt{2}c\right)$ and $\left(-\sqrt{2}c, -\sqrt{2}c\right)$, directrices $x + y = \pm\sqrt{2}\,c$, vertices (c, c), latus rectum $= 2\sqrt{2}\,c$.
- Parametric equation: $x = ct$, $y = \dfrac{c}{t}$, $t \neq 0$
- Cartesian Tangent at (x_1, y_1) is $\dfrac{x}{x_1}+\dfrac{y}{y_1}=2$
- Tangent at $P(t)$ is given by $x + yt^2 = 2ct$.
- Normal at $P(t)$ is $t^2x - y = c\left(t^3-\dfrac{1}{t}\right)$
- If a circle cuts $xy = c^2$ at four points (x_i, y_i), $i = 1, 2, 3, 4$, then $x_1x_2x_3x_4 = y_1y_2y_3y_4 = c^4$.
- The orthocentre of a triangle with vertices $\left(ct_i, \dfrac{c}{t_i}\right)$, $i = 1, 2, 3$ inscribed in $xy = c^2$ lies on the same curve, whose coordinates are $\left(-\dfrac{c}{t_1t_2t_3}, -ct_1t_2t_3\right)$.

"Even when branches diverge, they follow the same law — stay true to principles, even when paths differ."

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Chapter 15

Complex Numbers-II

n^{th} ROOTS OF UNITY

$z = (1)^{\frac{1}{n}} = (\cos 0 + i\sin 0)^{\frac{1}{n}}$

$= [\cos 2m\pi + i\sin 2m\pi]^{\frac{1}{n}}$

$z = \left(\cos\frac{2m\pi}{n} + i\sin\frac{2m\pi}{n}\right)$

$m = 0 \Rightarrow z_1 = \cos 0 + i\sin 0 = 1$

$m = 1 \Rightarrow z_2 = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n} = e^{i\frac{2\pi}{n}} \to \alpha_1$

$m = 2 \Rightarrow z_3 = \cos\frac{4\pi}{n} + i\sin\frac{4\pi}{n} = e^{i\frac{4\pi}{n}} \to \alpha_2$

$m = 3 \Rightarrow z_4 = \text{- - - - - - - - - - - - } = e^{i\frac{6\pi}{n}} \to \alpha_3$

$\vdots$

$m = n-1 \Rightarrow z_n = e^{i\frac{2(n-1)\pi}{n}} \to \alpha_{n-1}$

$1, \alpha_1, \alpha_2, \alpha_3 \text{ - - - } \alpha_{n-1}$ are called n roots of Unity.

where $\alpha_1 = e^{i\frac{2\pi}{n}}$

$1, \alpha_1, \alpha_2, \alpha_3 \text{ - - - } \alpha_{n-1}$ are in G.P.

So, $1, \alpha_1, \alpha_1^2, \alpha_1^3, \alpha_1^4, \text{ - - - - } (\alpha_1)^{n-1}$ are the n^{th} roots of unity.

PROPERTIES OF n^{th} ROOTS OF UNITY

(P-1) The n^{th} roots of unity are in G.P. with common ratio $e^{i(2\pi/n)} = \alpha_1$

(P-2) If α is any nth root of unity then $\alpha^n = 1$

(P-3) The points represented by n, n^{th} roots of unity are located at the vertices of a regular polygon of n sides inscribed in a unit circle having center at origin, one vertex being on positive real axis.

(P-4) The sum of n^{th} roots of unity is always zero.

(P-5) The product of the "n" n^{th} roots of unity is -1 if n is even and 1 if n is odd.

(P-6) $1^p + \alpha_1^p + \alpha_2^p + \cdots + \alpha_{n-1}^p = 0$; if p is not an integral multiple of n

$= n$; if p is an integral multiple of n

Ques. If α is the n^{th} root of unity, then $1 + 2\alpha + 3\alpha^2 + 4\alpha^3 + \cdots$ upto n terms is equal to

(a) $\frac{-n}{(1-\alpha)^2}$ (b) $\frac{-n}{1-\alpha}$

(c) $\frac{-2n}{1-\alpha}$ (d) $\frac{-2n}{(1-\alpha)^2}$

Ans. AGP

$S = 1 + 2\alpha + 3\alpha^2 + 4\alpha^3 + \text{- - - - - } n\alpha^{n-1}$

$\alpha S = \quad \alpha + 2\alpha^2 + 3\alpha^3 + \text{- - - - - } + n\alpha^n$

$S - \alpha S = 1 + \alpha + \alpha^2 + \alpha^3 + \text{- - - - } \alpha^{n-1} - n\alpha^n$

$\Rightarrow S(1-\alpha) = \frac{1(\alpha^n - 1)}{\alpha - 1} - n\alpha^n$

$\Rightarrow S(1-\alpha) = 0 - n.1 \qquad (\because \alpha^n = 1)$

$\Rightarrow S(1-\alpha) = -n$

$\Rightarrow S = \frac{-n}{1-\alpha}$

Ques. If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 + x^3 + x^2 + x + 1 = 0$, then $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021}$ is equal to:

(a) -4 (b) -1 (c) 1 (d) 4

Ans. Given, $1 + x + x^2 + x^3 + x^4 = 0$

$\Rightarrow \frac{x^5 - 1}{x - 1} = 0 \qquad (x \neq 1)$

$\Rightarrow x^5 - 1 = 0$

$\Rightarrow x^5 = 1 \Rightarrow x = (1)^{\frac{1}{5}} \to 5^{th}$ roots of unity

$\alpha, \beta, \gamma, \delta$ are the imaginary 5^{th} roots of unity

$\alpha^5 = \beta^5 = \gamma^5 = \delta^5 = 1$

$\alpha^5 = 1$

$\Rightarrow \alpha^{2020} = 1 \Rightarrow \alpha^{2021} = \alpha$

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Similarly, $\beta^{2021} = \beta, \gamma^{2021} = \gamma, \delta^{2021} = \delta$

So, $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021} = \alpha + \beta + \gamma + \delta$

$= \frac{-1}{1} = -1$

Ques. The value of $\sum_{k=1}^{10}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right)$ is

(a) i (b) 1 (c) -1 (d) -i

Ans. $\sum_{k=1}^{10}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right) = \sum\left(-i^2\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right)$

$= i\sum_{k=1}^{10}\left(\cos\frac{2k\pi}{11} - i\sin\frac{2k\pi}{11}\right)$

$= i\left[\left(\cos\frac{2\pi}{11} - i\sin\frac{2\pi}{11}\right) + \left(\cos\frac{4\pi}{11} - i\sin\frac{4\pi}{11}\right) + \ldots\right]$

$= i\left[\bar{\alpha}_1 + \bar{\alpha}_2 + \bar{\alpha}_3 + \cdots + \bar{\alpha}_{10}\right]$

where $1, \alpha_1, \alpha_2, \alpha_3, ---- \alpha_{10}$ are the 11^{th} roots of unity

$\because \alpha_1 + \alpha_2 + \alpha_3 + ---- \alpha_{10} + 1 = 0$

$\Rightarrow \alpha_1 + \alpha_2 + \alpha_3 + ---- \alpha_{10} = -1$

Conjugate both sides

$\Rightarrow \bar{\alpha}_1 + \bar{\alpha}_2 + \bar{\alpha}_3 + \cdots + \bar{\alpha}_{10} = -1$

Hence, $\sum_{k=1}^{b}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right) = -i$

Ques. The value of $\sum_{k=0}^{10}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right)$ is

(a) i (b) 1 (c) 0 (d) -i

Ans. $\sum_{k=0}^{10}\left(\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right)$

$= \sum\left(-i^2\sin\frac{2k\pi}{11} + i\cos\frac{2k\pi}{11}\right)$

$= i\sum_{k=0}^{10}\left(\cos\frac{2k\pi}{11} - i\sin\frac{2k\pi}{11}\right)$

$= i\left[1 + \left(\cos\frac{2\pi}{11} - i\sin\frac{2\pi}{11}\right) + \left(\cos\frac{4\pi}{11} - i\sin\frac{4\pi}{11}\right) + \ldots\right]$

$= i\left[1 + \bar{\alpha}_1 + \bar{\alpha}_2 + \bar{\alpha}_3 + \cdots + \bar{\alpha}_{10}\right]$

$= i \times 0 = 0 \quad (\because \alpha_1 + \alpha_2 + \alpha_3 + ---- \alpha_{10} + 1 = 0)$

Ques. If $a = \cos\left(\frac{2\pi}{7}\right) + i\sin\left(\frac{2\pi}{7}\right)$, then the quadratic equation whose roots are $\alpha = a + a^2 + a^4$ and $\beta = a^3 + a^5 + a^6$ is

(a) $x^2 - x + 2 = 0$ (b) $x^2 + x - 2 = 0$

(c) $x^2 - x - 2 = 0$ (d) $x^2 + x + 2 = 0$

Ans. $a \rightarrow 7^{th}$ roots $\Rightarrow a^7 = 1$

Let roots are $1, \alpha_1, \alpha_2, ---, \alpha_6$

$\alpha + \beta = \alpha_1 + \alpha_2 + \alpha_4 + \alpha_3 + \alpha_5 + \alpha_6$

Since, $1 + \alpha_1 + \alpha_2 + ---- + \alpha_6 = 0$

$\Rightarrow \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6 = -1$

So, $\alpha + \beta = -1$

$\alpha\beta = (a + a^2 + a^4)(a^3 + a^5 + a^6)$

$= a^4 + a^6 + 1 + a^5 + 1 + a + 1 + a^2 + a^3$

$= 3 + (a + a^2 + a^3 + a^4 + a^5 + a^6)$

$= 3 - 1 = 2$

So the equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$\Rightarrow x^2 - (-1)x + 2 = 0$

$\Rightarrow x^2 + x + 2 = 0$

(P-7) $(1 - \alpha_1)(1 - \alpha_2) \ldots (1 - \alpha_{n-1}) = n$

(P-8) $(1 + \alpha_1)(1 + \alpha_2) \ldots (1 + \alpha_{n-1}) = 0$; if n is even

$= 1$; if n is odd.

(P-9) $(\omega - \alpha_1)(\omega - \alpha_2)(\omega - \alpha_3) --- (\omega - \alpha_{n-1})$

$= 0$; if $n = 3k$

$= 1$; if $n = 3k + 1$

$= -\omega^2$; if $n = 3k + 2$

Brain Teaser

If $\alpha_1, \alpha_2, \alpha_3, \ldots \alpha_6$ are the 6 imaginary 7^{th} roots of -1 then find the value of

1. $(2 - \alpha_1)(2 - \alpha_2)(2 - \alpha_3) \ldots (2 - \alpha_6)$

2. $\frac{1}{2-\alpha_1} + \frac{1}{2-\alpha_2} + \frac{1}{2-\alpha_3} + \cdots + \frac{1}{2-\alpha_6}$

Ans. 1. $z = (-1)^{\frac{1}{7}}$

$\Rightarrow z^7 = -1 \Rightarrow z^7 + 1 = 0$

Let roots are $-1, \alpha_1, \alpha_2, \ldots\ldots \alpha_6$

Then, $z^7 + 1 = (z + 1)(z - \alpha_1)(z - \alpha_2) --- (z - \alpha_6)$

Put $z = 2$

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$\Rightarrow 2^7 + 1 = (2+1)(2-\alpha_1)(2-\alpha_2)\cdots(2-\alpha_6)$

$\Rightarrow (2-\alpha_1)(2-\alpha_2)(2-\alpha_3)\cdots(2-\alpha_6) = \frac{129}{3} = 43$

2. $\ln(z^7+1)$

$= \ln(z+1) + \ln(z-\alpha_1) + \ln(z-\alpha_2) \cdots + \ln(z-\alpha_6)$

diff wrt 'z'

$\Rightarrow \frac{1}{z^7+1}7z^6 = \frac{1\times1}{z+1} + \frac{1\times1}{z-\alpha_1} + \frac{1}{z-\alpha_2} + \cdots + \frac{1}{z-\alpha_6}$

$\Rightarrow \frac{7z^6}{z^7+1} = \frac{1}{z+1} + \frac{1}{z-\alpha_1} + \frac{1}{z-\alpha_2} + \cdots \frac{1}{z-\alpha_6}$

Put $z = 2$

$\Rightarrow \frac{7\cdot2^6}{2^7+1} = \frac{1}{3} + \frac{1}{2-\alpha_1} + \frac{1}{2-\alpha_2} + \frac{1}{2-\alpha_3} + \cdots \frac{1}{2-\alpha_6}$

$\Rightarrow \frac{1}{2-\alpha_1} + \frac{1}{2-\alpha_2} + \frac{1}{2-\alpha_3} + \cdots \frac{1}{2-\alpha_6}$

$= \left(\frac{7\times64}{129} - \frac{1}{3}\right) = \frac{405}{129}$

Brain Teaser

Factorize $z^7 + 1$ itno linear and quadratic factors having only real coefficients and hence deduce the value of $\cos\frac{\pi}{7}\cos\frac{3\pi}{7}\cos\frac{5\pi}{7}$.

Ans. $(z^7+1) = (z+1)(z-\alpha_1)(z-\alpha_2)\ldots(z-\alpha_6)$

$\because z = (-1)^{1/7}$

$\Rightarrow z = (\cos\pi + i\sin\pi)^{1/7}$

$= \cos\left(\frac{2m\pi+\pi}{7}\right) + i\sin\left(\frac{2m\pi+\pi}{7}\right)$

$m = 0 \Rightarrow z_1 = \cos\frac{\pi}{7} + i\sin\frac{\pi}{7} \to \alpha_1$

$m = 1 \Rightarrow z_2 = \cos\frac{3\pi}{7} + i\sin\frac{3\pi}{7} \to \alpha_2$

$m = 2 \Rightarrow z_3 = \cos\frac{5\pi}{7} + i\sin\frac{5\pi}{7} \to \alpha_3$

$m = 3 \Rightarrow z_4 = \cos\pi + i\sin\pi = -1$

$m = 4 \Rightarrow z_5 = \cos\frac{9\pi}{7} + i\sin\frac{9\pi}{7}$

$= \cos\left(2\pi - \frac{5\pi}{7}\right) + i\sin\left(2\pi - \frac{5\pi}{7}\right)$

$= \cos\frac{5\pi}{7} - i\sin\frac{5\pi}{7} = \bar{z}_3 \to \bar{\alpha}_3$

Similarly,

$m = 5 \Rightarrow z_6 = \bar{z}_2 \to \bar{\alpha}_2$

and $m = 6 \Rightarrow z_7 = -\bar{z}_1 \to \bar{\alpha}_1$

Therefore,

$z^7 + 1 = (z+1)(z-\alpha_1)(z-\bar{\alpha}_1)(z-\alpha_2)(z-\bar{\alpha}_2)(z-\alpha_3)(z-\bar{\alpha}_3)$

$\Rightarrow z^7 + 1 = (z+1)(z^2 - 2\cos\frac{\pi}{7}.z + 1)(z^2 - 2\cos\frac{3\pi}{7}.z + 1)(z^2 - 2\cos\frac{5\pi}{7}.z + 1)$

put $z = i$

$\Rightarrow i^7 + 1 = (i+1)(i^2 - 2i\cos\frac{\pi}{7} + 1)(i^2 - 2i\cos\frac{3\pi}{7} + 1)(i^2 - 2i\cos\frac{5\pi}{7} + 1)$

$\because i^7 = -i,\ i^2 = -1$

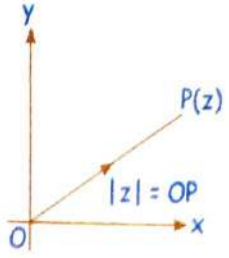
$\Rightarrow 1 - i = (1+i)\left[-2i\left(\cos\frac{\pi}{7}\cos\frac{3\pi}{7}\cos\frac{5\pi}{7}\right)\right]$

$\Rightarrow 1 - i = 2(-i - i^2)\left(\cos\frac{\pi}{7}\cos\frac{3\pi}{7}\cos\frac{5\pi}{7}\right)$

$\Rightarrow \cos\frac{\pi}{7}.\cos\frac{\pi}{7}.\cos\frac{5\pi}{7} = \frac{1}{2}$

VECTORIAL REPRESENTATION OF COMPLEX NUMBER

1. In complex number every point can be represented in terms of position vector. If the point P represents the complex number z then, $\overrightarrow{OP} = z$ and $|\overrightarrow{OP}| = |z|$



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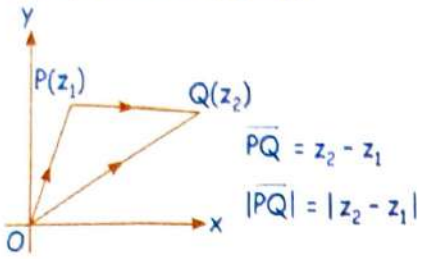
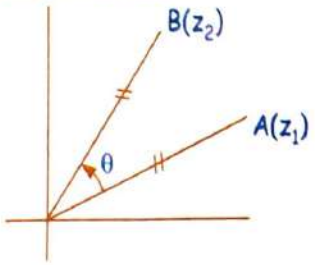
2. If $P(z_1)$ and $Q(z_2)$ be two complex numbers on argand plane then $\overline{PQ}$ represents complex number $(z_2 - z_1)$.



ROTATION OF COMPLEX NUMBER

Case-1: When Rotation is done about origin, keeping the magnitude same.



$z_2 = z_1 e^{i\theta}$

Important Note

1. (Final complex number) = (Initial complex number)$e^{i\theta}$; if rotation is anticlockwise.
2. (Final complex number) = (Initial complex number)$e^{-i\theta}$; if rotation is clockwise.
3. IF a complex number is multiplied with i then it gets rotated by 90° in anti clock wise direction.

$z_2 = iz_1 \rightarrow 90°$ rotation

Ques. Let w_1 be the point obtained by the rotation of $z_1 = 5 + 4i$ about the origin through a right angle in the anticlockwise direction, and w_2 be the point obtained by the rotation of $z_2 = 3 + 5i$ about the origin through a right angle in the clockwise direction. Then the principal argument $w_1 - w_2$ is equal to

(JEE Main 2023 (11 Apr Shift I))

(a) $\pi - \tan^{-1}\frac{8}{9}$ (b) $-\pi + \tan^{-1}\frac{33}{5}$

(c) $-\pi + \tan^{-1}\frac{8}{9}$ (d) $\pi - \tan^{-1}\frac{33}{5}$

Ans. $w_1 = iz_1$

$\Rightarrow w_1 = i(5 + 4i)$

$w_2 = -i(3 + 5i)$

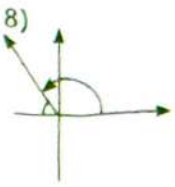
$\Rightarrow w_2 = 5 - 3i$

$\Rightarrow w_1 = 5i + 4i^2$

$\Rightarrow w_1 = -4 + 5i$

Now, $w_1 - w_2 = -9 + 8i$

$\arg(w_1 - w_2) = \pi - \tan^{-1}\left(\frac{8}{9}\right)$

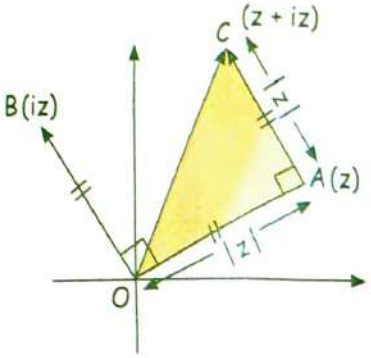


Ques. The area of the triangle with vertices A(z), B(iz) and C(z + iz) is:

(a) 1 (b) $\frac{1}{2}|z|^2$

(c) $\frac{1}{2}$ (d) $\frac{1}{2}|z + iz|^2$

Ans.



Area of triangles $= \frac{1}{2}|z| \cdot |z| = \frac{1}{2}|z|^2$

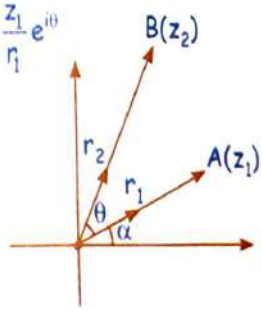
Case-2: When Rotation is done about origin and the magnitude is also changed.

M-1 Rotate unit vectors,

→ Final complex number = (Initial complex number) $e^{i\theta}$

$\Rightarrow \hat{z}_2 = \hat{z}_1 e^{i\theta} \Rightarrow \frac{z_2}{r_2} = \frac{z_1}{r_1} e^{i\theta}$

$\Rightarrow z_2 = \frac{r_2}{r_1} z_1 e^{i\theta}$



M-2

$z_1 = r_1 e^{i\alpha}$

$z_2 = r_2 e^{i(\theta + \alpha)} \Rightarrow z_2 = r_2 e^{i\alpha} e^{i\theta}$

$\Rightarrow z_2 = r_2 \cdot \frac{z_1}{r_1} e^{i\theta} \quad \left(\because e^{i\alpha} = \frac{z_1}{r_1}\right)$

$\Rightarrow z_2 = \frac{r_2}{r_1} z_1 e^{i\theta}$

Case-3: When rotation is done about any general point and magnitude is also changed. In this case also, we first find unit vectors along

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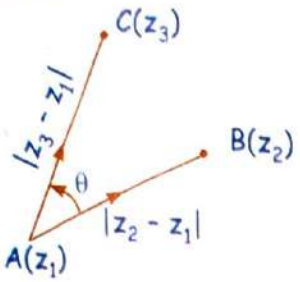
the two vectors AB and AC and then rotate them using Case 2.

$\overrightarrow{AC} = \overrightarrow{AB}e^{i\theta}$



$$\Rightarrow \frac{z_3 - z_1}{|z_3 - z_1|} = \frac{z_2 - z_1}{|z_2 - z_1|} e^{i\theta}$$

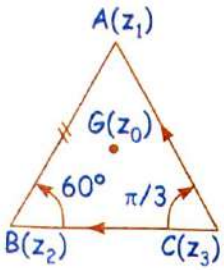
Important Note

If z_1, z_2, z_3 represent the vertices of an equilateral triangle where z_0 is its centre then

1. $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

   or $\sum z_1^2 = \sum z_1z_2$

   Proof:



$(z_1 - z_2) = (z_3 - z_2)\, e^{i\pi/3}$ ...(i)

$(z_1 - z_3) = (z_2 - z_3)\, e^{-i\pi/3}$ ...(ii)

By (i) × (ii)

$(z_1 - z_2)(z_1 - z_3) = (z_3 - z_2)(z_2 - z_3)$

$\Rightarrow z_1^2 - z_1z_3 - z_2z_1 + z_2z_3 = -[z_3^2 + z_2^2 - 2z_3z_2]$

$\Rightarrow z_1^2 + z_2^2 + z_3^2 = z_1z_3 + z_2z_1 + z_2z_3$

2. $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$

   Proof: $z_0 = \frac{z_1 + z_2 + z_3}{3}$

$\Rightarrow 3z_0 = z_1 + z_2 + z_3$

Square both side

$\Rightarrow 9z_0^2 = z_1^2 + z_2^2 + z_3^2 + 2(z_1z_2 + z_2z_3 + z_3z_1)$

$\Rightarrow 9z_0^2 = z_1^2 + z_2^2 + z_3^2 + 2(z_1^2 + z_2^2 + z_3^2)$

$\Rightarrow 9z_0^2 = 3(z_1^2 + z_2^2 + z_3^2)$

$\Rightarrow 3z_0^2 = z_1^2 + z_2^2 + z_3^2$

Ques. Let z_1, z_2 be the roots of the equation $z^2 + az + 12 = 0$ and z_1, z_2 form an equilateral triangle with origin. Then, the value of $|a|$ is

Ans. Given that z_1, z_2 are roots of equation,

$z^2 + az + 12 = 0$

So, $z_1 + z_2 = -a$ and $z_1z_2 = 12$

For equilateral triangle,

$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$

Put $z_3 = 0$

$\Rightarrow z_1^2 + z_2^2 = z_1z_2$

$\Rightarrow (z_1 + z_2)^2 - 2z_1z_2 = z_1z_2$

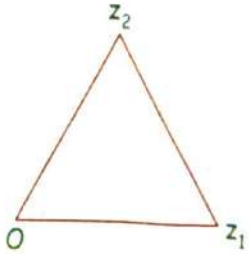
$\Rightarrow (z_1 + z_2)^2 = 3z_1z_2$

$\Rightarrow (-a)^2 = 3 \times 12$

$\Rightarrow a^2 = 36$

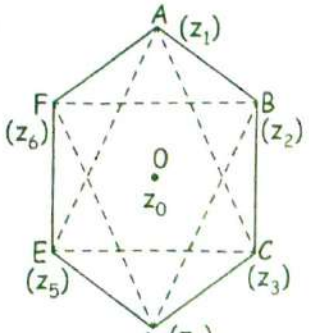
$\Rightarrow a = \pm 6$

Hence, $|a| = 6$



Ques. Let z_i (where $i = 1, 2, 3, 4, 5, 6$) represents the vertices of a regular hexagon in the complex plane and z_0 is its centre. If $\sum_{i=1}^{6} z_i^2 = k \cdot z_0^2$, find the value of k.

Ans.



Given, $z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2 + z_6^2 = kz_0^2$

$\because$ Δ BDF is equilateral triangle,

$\therefore z_2^2 + z_4^2 + z_6^2 = 3z_0^2$ ...(i)

Δ ACE is also equilateral triangle,

$\therefore z_1^2 + z_3^2 + z_5^2 = 3z_0^2$ ...(ii)

By equation (i) + (ii)

$z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2 + z_6^2 = 6z_0^2$

Hence, $k = 6$

Important Note

If Complex numbers z_1, z_2, z_3 respectively represent the vertices of a right angled isosceles triangle ABC with right angle at $B(z_2)$ then $z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$.

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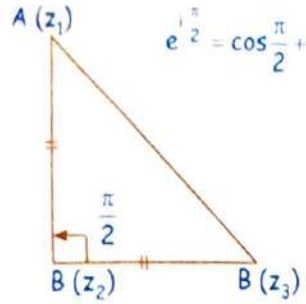
Proof:



$e^{i\frac{\pi}{2}} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$

Final CN = (Initial CN) $e^{i\frac{\pi}{2}}$

$\Rightarrow (z_1 - z_2) = (z_3 - z_2)e^{i\frac{\pi}{2}}$

$\Rightarrow (z_1 - z_2) = (z_3 - z_2)\, i$

Square both side

$\Rightarrow (z_1 - z_2)^2 = -(z_3 - z_2)^2$

$\Rightarrow z_1^2 + z_2^2 - 2z_1z_2 = -(z_3^2 + z_2^2 - 2z_2z_3)$

$\Rightarrow z_1^2 + 2z_2^2 + z_3^2 = 2z_1z_2 + 2z_2z_3 = 2z_2(z_1 + z_3)$

Ques. The complex numbers z_1, z_2 and z_3 satisfying $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2}$ are the vertices of a triangle which is **(JEE 2001)**

(a) of area zero

(b) right-angled isosceles

(c) equilateral

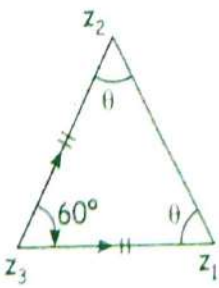
(d) obtuse- angled isosceles

Ans. $\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2} \Rightarrow z_1 - z_3 = \left(\frac{1 - i\sqrt{3}}{2}\right)(z_2 - z_3)$

$\Rightarrow (z_1 - z_3) = e^{-i\frac{\pi}{3}}(z_2 - z_3)$

$\left(\because \frac{1 - i\sqrt{3}}{2} = \cos\frac{\pi}{3} - i\sin\frac{\pi}{3} = e^{-i\frac{\pi}{3}}\right)$

$\Rightarrow$ Final CN = (Initial CN) $e^{-i\frac{\pi}{3}}$



Now, $\theta + \theta + 60° = 180°$

$\Rightarrow 2\theta = 120° \Rightarrow \theta = 60°$

Hence, triangle will be equilateral triangle

Use in Co-ordinate Geometry

Ques. Given ABCD is a square A (1, 3) & C(5, 9) find B & D.

Ans. PC → PD

Final CN = (Initial CN) i

$\Rightarrow (z_4 - z_0)$

$= (z_3 - z_0)\, i$

$\left[\because z_0 = \frac{6 + 12i}{2} = 3 + 6i\right]$

$\Rightarrow (z_4 - (3 + 6i))$

$= [(5 + 9i) - (3 + 6i)]\, i$

$\Rightarrow z_4 - (3 + 6i)$

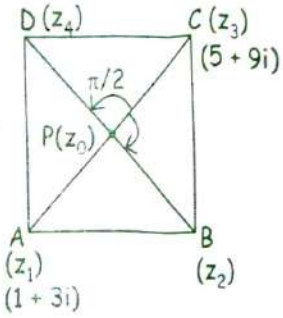
$= (2 + 3i)\, i$

$\Rightarrow z_4 - 3 - 6i$

$= 2i - 3$

$\Rightarrow z_4 = 8i$

Hence, D (0, 8)



Now, $\overrightarrow{PC} \rightarrow \overrightarrow{PB}$

$(z_2 - z_0) = (z_3 - z_0)(-i)$

$\Rightarrow z_2 - (3 + 6i) = (2 + 3i)(-i)$

$\Rightarrow z_2 - 3 - 6i = -2i + 3$

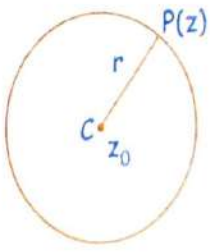
$\Rightarrow z_2 = 6 + 4i$

Hence, B (6, 4)

COMPLEX GEOMETRY INTRODUCTION

1. $|z - z_0| = r$ represents a circle with centre z_0 and radius = r.

PC = r



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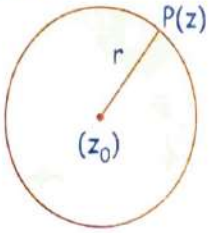
Ex→ $|z+3| = 10$

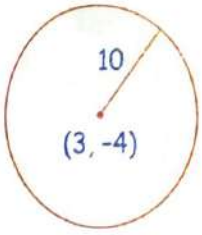
$|z-(-3)| = 10$

Centre $(-3, 0)$, $r = 10$

2. $|z - z_0| \le r$ represents a circular disc with centre $= z_0$ and radius $= r$.

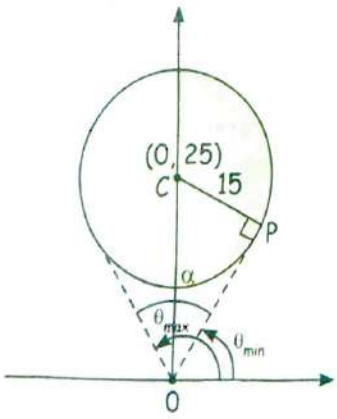


Example: $|z - 3 + 4i| \le 10$



Ques. If $|z - 25i| \le 15$ and $|\text{max. Arg}(z) - \text{min. Ang}(z)| = 2\cos^{-1}\left(\frac{a}{b}\right)$, where Arg (z) represents principal argument of z where a and b are co-prime to each other, then find the value of a + b.

Ans. Centre $(25\,i) = (0, 25)$



In ΔOPC, $\angle COP = \frac{\alpha}{2}$

$$\sin\frac{\alpha}{2} = \frac{15}{25} \Rightarrow \sin\frac{\alpha}{2} = \frac{3}{5}$$

$$\Rightarrow \cos\frac{\alpha}{2} = \frac{4}{5} \Rightarrow \frac{\alpha}{2} = \cos^{-1}\frac{4}{5}$$

$$\Rightarrow \alpha = 2\cos^{-1}\frac{4}{5} = 2\cos^{-1}\left(\frac{a}{b}\right)$$

Then, $a = 4, b = 5$

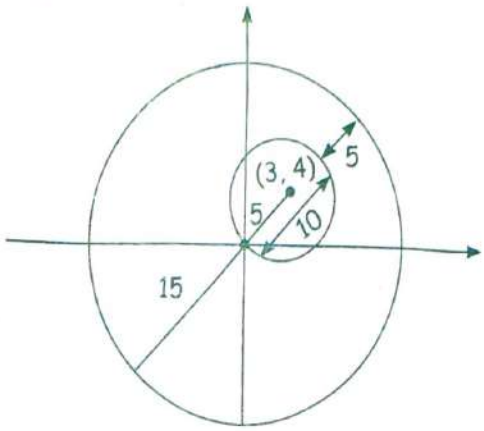
Hence, $a + b = 4 + 5 = 9$

Ques. If $|z_1| = 15$ and $|z_2 - 3 - 4i| = 5$, then

(a) $|z_1 - z_2|_{min} = 5$ (b) $|z_1 - z_2|_{min} = 10$

(c) $|z_1 - z_2|_{max} = 20$ (d) $|z_1 - z_2|_{max} = 25$

Ans. $|z_2 - (3 + 4i)| = 5$



$|z_1 - z_2|_{min} = 5$

$|z_1 - z_2|_{max} = 10 + 15 = 25$

Ques. If $\text{Re}\left(\frac{z-1}{2z+i}\right) = 1$, where $z = x + iy$, then the point (x, y) lies on a:

(JEE Main-2020 (January))

(a) straight line whose slope is -2/3

(b) straight line whose slope 3/2

(c) circle whose diameter is $\sqrt{5}/2$

(d) circle whose centre is at (-1/2, -3/2)

Ans. $\frac{z-1}{2z+i} = \frac{x+iy-1}{2(x+iy)+i} = \frac{[(x-1)+iy]\,[2x-i(2y+1)]}{2x+i(2y+1)\;\;2x-i(2y+1)}$

$$= \frac{2x(x-1)+y(2y+1)+i2xy-i(x-1)(2y+1)}{4x^2+(2y+1)^2}$$

Given, $\text{Re}\left(\frac{z-1}{2z+i}\right) = 1$

So, $\frac{2x(x-1)+y(2y+1)}{4x^2+(2y+1)^2} = 1$

$\Rightarrow 2x^2 - 2x + 2y^2 + y = 4x^2 + 4y^2 + 1 + 4y$

$\Rightarrow -2x + y = 2x^2 + 2y^2 + 1 + 4y$

$\Rightarrow 2x^2 + 2y^2 + 2x + 3y + 1 = 0$

$\Rightarrow x^2 + y^2 + x + \frac{3}{2}y + \frac{1}{2} = 0$

$\Rightarrow$ Centre $\left(-\frac{1}{2}, -\frac{3}{4}\right)$

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and, $r = \sqrt{g^2 + f^2 - c}$

$\Rightarrow r = \sqrt{\frac{1}{4} + \frac{9}{16} - \frac{1}{2}}$

$\Rightarrow r = \sqrt{\frac{4 + 9 - 8}{16}} \Rightarrow r = \frac{\sqrt{5}}{4}$

$d = 2r = \frac{\sqrt{5}}{2}$

Ques. Let z be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z - (6 + 8i)|$ is equal to

(2024 (09 Apr Shift 2))

(a) 12 (b) 10 (c) 8 (d) ∞

Ans. Let, $z = x + iy$

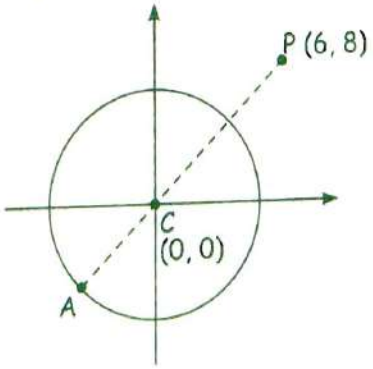
Then, $\frac{z-2i}{z+2i} = \frac{[x+i(y-2)]}{x+i(y+2)} \cdot \frac{[x-i(y+2)]}{x-i(y+2)}$

Given, $\text{Re}\left(\frac{z-2i}{z+2i}\right) = 0$

$\Rightarrow \frac{x^2 + (y-2)(y+2)}{x^2 + (y+2)^2} = 0$

$\Rightarrow x^2 + y^2 - 4 = 0$

$\Rightarrow x^2 + y^2 = 4$



$|z - z_1| \rightarrow$ represents distance b/w z & z_1

max value of $|z - (6 + 8i)| = PA$

$= PC + r = 10 + 2 = 12$

Ques. If z is a complex number such that $|z| \leq 1$, then the minimum value of $\left|z + \frac{1}{2}(3 + 4i)\right|$ is:

(JEE Main 01 Feb, 2024 Shift 2)

(a) $\frac{5}{2}$ (b) 2 (c) $\frac{3}{2}$ (d) 0

Ans. $\left|z - \left(-\frac{3}{2} - 2i\right)\right| = |z - z_1|$ (let)

Then, point P for z_1 is $\left(-\frac{3}{2}, -2\right)$

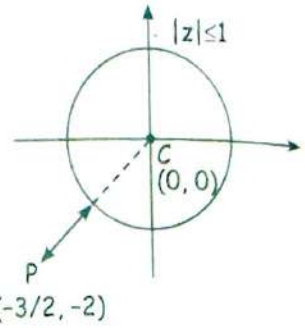
$(PC) = \sqrt{\left(\frac{3}{2}\right)^2 + 2^2}$

$= \sqrt{\frac{9}{4} + 4} = \sqrt{\frac{25}{4}}$

$\Rightarrow PC = \frac{5}{2}$

$d_{min} = PC - r$

$= \frac{5}{2} - 1 = \frac{3}{2}$



Ques. Let $S_1 = \{z \in C: |z| \leq 5\}$, $S_2 = \left\{z \in C: \text{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right) \geq 0\right\}$ and $S_3 = \{z \in C: \text{Re}(z) \geq 0\}$. Then the area of the region $S_1 \cap S_2 \cap S_3$ is: **(JEE Main 2024 (05 Apr Shift 2))**

(a) $\frac{125\pi}{12}$ (b) $\frac{125\pi}{4}$

(c) $\frac{125\pi}{24}$ (d) $\frac{125\pi}{6}$

Ans. Let, $z = x + iy$

Then, S_1 represent a circular disc with centre (0, 0) and radius 5 unit.

For S_2:

$\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right) = \frac{(x+iy+1-\sqrt{3}i)}{(1-\sqrt{3}i)} \frac{(1+\sqrt{3}i)}{(1+\sqrt{3}i)}$

$= \frac{\left[(x+1)+i\left(y-\sqrt{3}\right)\right]\left[1+\sqrt{3}i\right]}{1+3}$

Given, $\text{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right) \geq 0$

$\Rightarrow \frac{\sqrt{3}(x+1)+\left(y-\sqrt{3}\right)}{4} \geq 0$

$\Rightarrow \sqrt{3}x + \sqrt{3} + y - \sqrt{3} \geq 0$

S_2: $\sqrt{3}x + y \geq 0$

For S_3: $x \geq 0$

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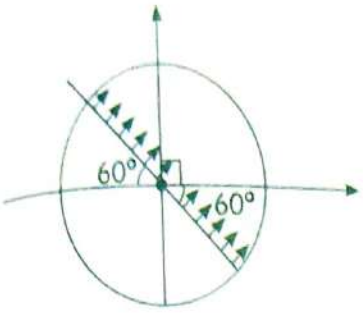


Now, $\theta = 60° + 90° = 150° = \frac{5\pi}{6}$

Hence, Area of sector $= \frac{1}{2}r^2\theta$

$= \frac{1}{2}(5)^2 \times \frac{5\pi}{6} = \frac{125\pi}{12}$

Ques. The sum of the square of the modulus of the elements in the set $\{z = a + ib;\ a, b \in Z,\ z \in C,\ |z - 1| \le 1,\ |z - 5| \le |z - 5i|\}$ is

Ans. Let, $z = a + ib$

Given, $|z - 1| \le 1$

$\Rightarrow (a - 1)^2 + b^2 \le 1$

& $|z - 5| \le |z - 5i|$

$\Rightarrow |(a - 5) + ib| \le |a + i(b - 5)|$ $(\because z = a + ib)$

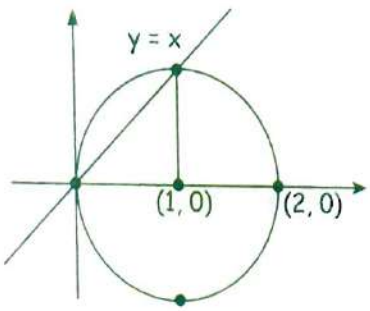
$\Rightarrow \sqrt{(a-5)^2 + b^2} \le \sqrt{a^2 + (b-5)^2}$

$\Rightarrow a^2 + 25 - 10a + b^2 \le a^2 + b^2 + 25 - 10b$

$\Rightarrow -10a \le -10b$

$\Rightarrow a \ge b$

$\Rightarrow a - b \ge 0$



a & b → Integers

a = 0, b = 0	$\|z\|^2 = 0$
a = 1, b = 0	$\|z\|^2 = 1^2$
a = 2, b = 0	$\|z\|^2 = 4$
a = 1, b = 1	$\|z\|^2 = 2$
a = 1, b = -1	$\|z\|^2 = 2$

Sum = 0 + 1 + 4 + 2 + 2 = 9

VARIOUS FORMS OF LINES

1. Canonical Form

Canonical form of a straight line is given by $\bar{a}z + a\bar{z} + b = 0$, where a is a complex constant, b is a real constant.

Proof:

$\bar{a}z + a\bar{z} + b = 0$

Let, $a = P + iq$ & $\bar{a} = P - iq$

$\Rightarrow (p - iq)z + (p + iq)\bar{z} + b = 0$

$\Rightarrow p(z + \bar{z}) + iq(-z + \bar{z}) + b = 0$

$\Rightarrow p(z + \bar{z}) - iq(z - \bar{z}) + b = 0$

$\Rightarrow p\ 2x - iq\ 2iy + b = 0$

$\Rightarrow 2xp + 2qy + b = 0 \rightarrow$ straight line

Ex: $(3 - 7i)z + (3 + 7i)\bar{z} = 12$

$\Rightarrow 3(z + \bar{z}) - 7i(z - \bar{z}) = 12$

$\Rightarrow 3(2x) - 7i\ 2iy = 12$

$\Rightarrow 6x + 14y = 12 \rightarrow$ straight line

2. Perpendicular Bisector Form

$|z - z_1| = |z - z_2|$ represents the perpendicular bisector of the line joining the points z_1 & z_2.

Ques. Let z be a complex number such that $|z - i| - |z + 5i| = 0$ and $|z| = 2$. If $z = x + iy$ then **(JEE Main-2022 (July))**

(a) $x + 2y = 4$ (b) $x^2 + y = 4$

(c) $x + 2y = -4$ (d) $x^2 - y + 3 = 0$

Ans. Given, $|z - i| = |z + 5i|$

Let, $z = x + iy$

Then, $|x + iy - i| = |x + iy + 5i|$

$\Rightarrow x^2 + (y - 1)^2 = x^2 + (y + 5)^2$

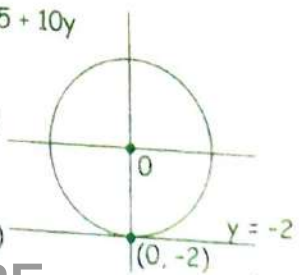
$\Rightarrow y^2 + 1 - 2y = y^2 + 25 + 10y$

$\Rightarrow 12y + 24 = 0$

$\Rightarrow y = -2$ ...(i)

and $|z| = 2$

$\Rightarrow x^2 + y^2 = 4$ ...(ii)



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From equation (i) & (ii)

$x = 0, y = -2$

Hence, equation $x + 2y = -4$ satisfy the point $(0, -2)$

3. Argument Form

$\arg(z - z_0) = \theta$ represents a 'ray' originating from point z_0 & inclined at an angle of θ with positive direction of x-axis. Locus will not include z_0 (hollow at z_0)

Ques. Find the point of intersection of the curves $\arg(z - 3i) = 3\pi/4$ and $\arg(2z + 1 - 2i) = \pi/4$

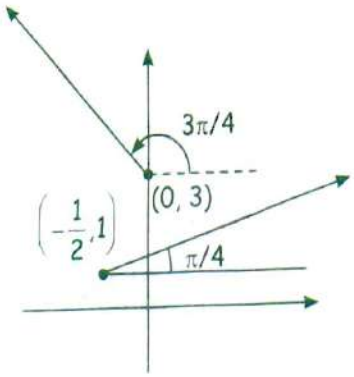
Ans. Given, $\arg(z - 3i) = \frac{3\pi}{4}$

and $\arg(2z + 1 - 2i) = \frac{\pi}{4}$

$\Rightarrow \arg\left(2\left(z + \frac{1}{2} - i\right)\right) = \frac{\pi}{4}$

$\arg 2 + \arg\left(z + \frac{1}{2} - i\right) = \frac{\pi}{4}$ $[\because \arg 2 = 0]$

$\Rightarrow \arg\left(z + \frac{1}{2} - i\right) = \frac{\pi}{4} \Rightarrow \arg\left(z - \left(-\frac{1}{2} + i\right)\right) = \frac{\pi}{4}$



No Point of Intersection

Ques. Let $S \{z \in \mathbb{C}: |z - 3| \le 1$ and $z(4 + 3i) + \bar{z}(4 - 3i) \le 24\}$. If $\alpha + i\beta$ is the point in S which is closest to 4i, then $25(\alpha + \beta)$ is equal to _____. **(JEE Main 24 June 2022-Shift II)**

Ans. Given, $z(4 + 3i) + \bar{z}(4 - 3i) \le 24$

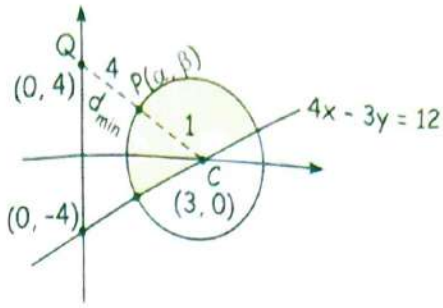
$\Rightarrow 4(z + \bar{z}) + 3i(z - \bar{z}) \le 24$

$\Rightarrow 4(2x) + 3i(2iy) \le 24$

$\Rightarrow 8x - 6y \le 24$

$\Rightarrow 4x - 3y \le 12$

and, $|z - 3| \le 1$



Now, $m_{QC} = \frac{4 - 0}{0 - 3} = \frac{-4}{3}$

$\Rightarrow \tan\theta = \frac{-4}{3}$

$\Rightarrow \sin\theta = \frac{4}{5}$ & $\cos\theta = \frac{-3}{5}$

Then, point P an line QC, $P[3 + 1\cos\theta, 0 + 1\sin\theta]$

$= P\left[3 - \frac{3}{5}, \frac{4}{5}\right] = P\left[\frac{12}{5}, \frac{4}{5}\right] = (\alpha, \beta)$

$\alpha = \frac{12}{5}, \beta = \frac{4}{5}$

Hence, $25(\alpha + \beta) = 25 \times \left(\frac{12}{5} + \frac{4}{5}\right) = \frac{25 \times 16}{5} = 80$

Ques. Let $S = \{z \in \mathbb{C}: |z - 2| \le 1$ and $z(1 + i) + \bar{z}(1 - i) \le 2\}$. Let: $|z - 4i|$ attains minimum and maximum values, respectively, at $z_1 \in S$ and $z_2 \in S$. IF $5(|z_1|^2 + |z_2|^2) = \alpha + \beta\sqrt{5}$, where α and β are integers, then the value of $\alpha + \beta$ is equal to _____. **(JEE Main 29 June, 2022 Shift-I)**

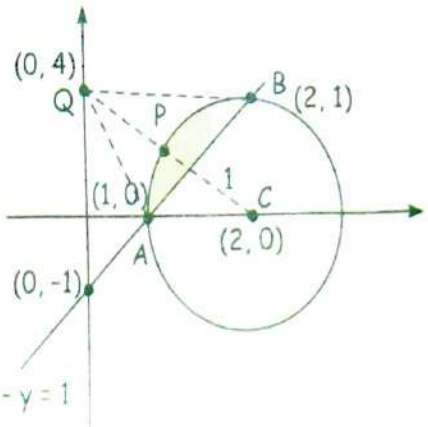
Ans. Given, $(z + \bar{z}) + i(z - \bar{z}) \le 2$

Let, $z = x + iy$

$\Rightarrow 2x + i(2iy) \le 2 \Rightarrow x - y \le 1$

And, $|z - 2| \le 1$

$\Rightarrow (x - 2)^2 + y^2 \le 1$



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Now, $m_{QC} = \frac{4-0}{0-2} = -2$

$\tan\theta = \frac{-2}{1}$

$\sin\theta = \frac{2}{\sqrt{5}}, \cos\theta = \frac{-1}{\sqrt{5}}$

So, point P on the circle, P [2 + 1cosθ, 0 + 1sinθ]

$\Rightarrow P\left[2-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right]$

Now, $QA = \sqrt{17}$

and $QB = \sqrt{2^2+3^2} = \sqrt{13}$

$d_{max} = \sqrt{17}$

$\Rightarrow z_2 = 1 + 0i = (1, 0) \Rightarrow |z_2|^2 = 1$

$d_{min} = PA$

$\Rightarrow z_1 = 2-\frac{1}{\sqrt{5}}+i\frac{2}{\sqrt{5}}$

$\Rightarrow |z_1|^2 = \left(2-\frac{1}{\sqrt{5}}\right)^2 + \frac{4}{5} = 4+\frac{1}{5}-\frac{4}{\sqrt{5}}+\frac{4}{5}$

$= 5-\frac{4}{\sqrt{5}}$

Hence, $5(|z_1|^2+|z_2|^2) = 5\left[1+5-\frac{4}{\sqrt{5}}\right] = 5\left[6-\frac{4}{\sqrt{5}}\right]$

$= 30-4\sqrt{5} = \alpha+\beta\sqrt{5}$ (given)

$\Rightarrow \alpha = 30, \beta = -4$

Hence, $\alpha + \beta = 26$

Complex Slope of a Line

For the line $\bar{a}z + a\bar{z} + b = 0$, Complex slope is $(C) = \frac{-a}{\bar{a}}$

Complex slope in 2 point form $(C) = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$

Ex → let the given line is

$(2 - 3i)z + (2 + 3i)\bar{z} + 10 = 0$, then

$a = 2 + 3i, \bar{a} = 2 - 3i$

So, the Complex slope $= -\frac{a}{\bar{a}} = -\frac{(2+3i)}{(2-3i)}$

Important Note

1. Complex slope of a line $= \frac{-a}{\bar{a}} = \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2}$
2. If real slope of a line is $\tan\alpha$, then its complex slope is $CiS(2\alpha) = \cos2\alpha + i\sin2\alpha$
3. If C_1 & C_2 are complex slopes of two lines L_1 & L_2 such that $L_1 \parallel L_2 \Rightarrow C_1 = C_2$
4. If C_1 & C_2 are complex slopes of two lines L_1 & L_2 such that $L_1 \perp L_2 \Rightarrow C_1 + C_2 = 0$

Ques. Find the complex slopes of the following straight lines:

(a) $y = x - 3$ (b) $3x - 4y = 7$

Ans. (a) Real slope $(m) = 1 = \tan45°$

$\Rightarrow \alpha = 45°$

Complex slope $(C) = \cos2\alpha + i\sin2\alpha$

$= \cos90° + i\sin90° = i$

(b) Real slope $(m) = \frac{-3}{-4} = \frac{3}{4} = \tan\alpha$

Complex slope $(C) = (\cos2\alpha) + i(\sin2\alpha)$

$\Rightarrow C = \left(\frac{1-\tan^2\alpha}{1+\tan^2\alpha}\right) + i\cdot\frac{2\tan\alpha}{1+\tan^2\alpha}$

$= \frac{1-\left(\frac{3}{4}\right)^2}{1+\left(\frac{3}{4}\right)^2} + i\cdot\frac{2\cdot\frac{3}{4}}{1+\left(\frac{3}{4}\right)^2} = \frac{7}{25} + i\frac{24}{25}$

Ques. Find the equation of a line for which foot of perpendicular from origin is given by the point $A(z_0)$.

Ans. $OA \perp PA$

$\Rightarrow C_{OA} + C_{PA} = 0$

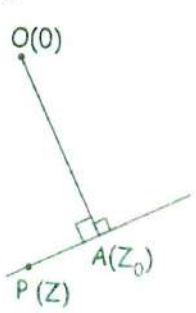
$\because C_{PA} = \frac{Z - Z_0}{\bar{Z} - \bar{Z}_0}$ & $C_{OA} = \frac{Z_0 - 0}{\bar{Z}_0 - 0}$

$\therefore \frac{Z_0}{\bar{Z}_0} + \frac{Z - Z_0}{\bar{Z} - \bar{Z}_0} = 0$

$\Rightarrow \frac{Z - Z_0}{\bar{Z} - \bar{Z}_0} = -\frac{Z_0}{\bar{Z}_0}$

$\Rightarrow Z\bar{Z}_0 - Z_0\bar{Z}_0 = -\bar{Z}Z_0 + Z_0\bar{Z}_0$

$\Rightarrow Z\bar{Z}_0 + \bar{Z}Z_0 = 2Z_0\bar{Z}_0$



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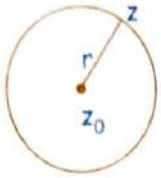
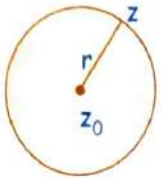
VARIOUS FORMS OF CIRCLE

1. Centre Radius Form

(i) $|z - z_0| = r$ represents a circle with centre z_0 and radius = r.



(ii) $|z - z_0| \le r$ represents a circular disc with centre = z_0 and radius = r.



2. Canonical Form

$z\bar{z} + \bar{a}z + a\bar{z} + b = 0$ represents a circle with centre (-a) & radius = $\sqrt{|a|^2 - b}$, where 'a' is a complex constant & 'b' is a real constant

Ex: let the circle is $z\bar{z} + (2 - 3i)\bar{z} + (2 + 3i)z + 5 = 0$

Then, $a = 2 - 3i$, $\bar{a} = 2 + 3i$, $b = 5$

So, the Centre = $-a = -2 + 3i$ or $(-2, 3)$

and radius $r = \sqrt{|a|^2 - b} = \sqrt{4 + 9 - 5} = \sqrt{8}$.

Ques. The line $z(1 - i) + (1 + i)\bar{z} - 2k = 0$ is secant to the circle $z\bar{z} - (1 - 2i)z - (1 + 2i)\bar{z} - 13 = 0$. If the set of all values of k is (a, b), then find the value of a + b.

Ans. Given line, $z(1 - i) + (1 + i)\bar{z} - 2k = 0$

$\Rightarrow z + \bar{z} - i(z - \bar{z}) = 2k$

$\Rightarrow 2x - i(2iy) = 2k$ (let $z = x + iy$)

$\Rightarrow 2x + 2y = 2k$

$\Rightarrow x + y = k$

And given Circle,

$z\bar{z} - (1 - 2i)z - (1 + 2i)\bar{z} - 13 = 0$

then, $a = -(1 + 2i)$, $b = -13$

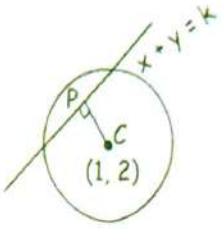
Centre = $-a = (1 + 2i)$ or $(1, 2)$

and radius, $r = \sqrt{|a|^2 - b} = \sqrt{1^2 + 2^2 - (-13)}$

$r = \sqrt{18} = 3\sqrt{2}$

Given that line is secant to the circle.

So, $CP < r$



$\Rightarrow \left|\frac{k - 3}{\sqrt{2}}\right| < 3\sqrt{2}$

$\Rightarrow |k - 3| < 6$

$\Rightarrow -6 < k - 3 < 6$

$\Rightarrow -3 < k < 9$

$\Rightarrow k \in (-3, 9) = (a, b)$ (given)

$\Rightarrow a = -3, b = 9$

Hence, $a + b = 6$

Ques. If the equation $a|z|^2 + \bar{\alpha}z + \alpha\bar{z} + d = 0$ represents a circle where a, d are real constants then which of the following condition is correct?

(18 Mar. 2021 (Shift-1))

(a) $|\alpha|^2 - ad \neq 0$

(b) $|\alpha|^2 - ad > 0$ and $a \in R - \{0\}$

(c) $|\alpha|^2 - ad \ge 0$ and $a \in R$

(d) $\alpha = 0$, $a, d \in R^+$

Ans. Given equation, $a|z|^2 + \bar{\alpha}z + \alpha\bar{z} + d = 0$

$\Rightarrow |z| + \frac{\bar{\alpha}z}{a} + \frac{\alpha\bar{z}}{a} + \frac{d}{a} = 0$

So, $a \to \frac{\alpha}{a}$, $b \to \frac{d}{a}$

This equation will be represents circle it, radius r is real.

So, $|a|^2 - b \ge 0$

$\Rightarrow \left|\frac{\alpha}{a}\right|^2 - \frac{d}{a} \ge 0$

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$$\Rightarrow \frac{|\alpha|^2}{a^2} - \frac{d}{a} \geq 0$$

$$\Rightarrow \frac{|\alpha|^2 - ad}{a^2} \geq 0$$

$$\Rightarrow |\alpha|^2 - ad \geq 0$$

Ques. If $\text{Re}\left(\frac{z-1}{2z+i}\right) = 1$, where $z = x + iy$, then the point (x, y) lies on a:

(a) straight line whose slope is $-\frac{2}{3}$

(b) straight line whose slope $\frac{3}{2}$

(c) circle whose diameter is $\frac{\sqrt{5}}{2}$

(d) circle whose centre is at $\left(-\frac{1}{2}, -\frac{3}{2}\right)$

Ans. Given, $\text{Re}\left(\frac{z-1}{2z+i}\right) = 1$ and $z = x + iy$

Let, $\alpha = \frac{z-1}{2z+i}$

$$\Rightarrow \text{Re}(\alpha) = \frac{\alpha + \bar{\alpha}}{2} = 1$$

$$\Rightarrow \alpha + \bar{\alpha} = 2$$

$$\Rightarrow \frac{z-1}{2z+i} + \frac{\bar{z}-1}{2\bar{z}-i} = 2$$

$$\Rightarrow \frac{(z-1)(2\bar{z}-i) + (\bar{z}-1)(2z+i)}{(2z+i)(2\bar{z}-i)} = 2$$

$$\Rightarrow \frac{2z\bar{z} - iz - 2\bar{z} + i + 2z\bar{z} + i\bar{z} - 2z - i}{4z\bar{z} - 2iz + 2i\bar{z} + 1} = 2$$

$$\Rightarrow 4z\bar{z} - iz + i\bar{z} - 2\bar{z} - 2z = 8z\bar{z} - 4iz + 4i\bar{z} + 2$$

$$\Rightarrow 4z\bar{z} - 3iz + 3i\bar{z} + 2\bar{z} + 2z + 2 = 0$$

$$\Rightarrow 4z\bar{z} + z(2 - 3i) + \bar{z}(2 + 3i) + 2 = 0.$$ It is a circle

$$\text{Centre} = -a = \frac{-2 - 3i}{4} = -\frac{1}{2} - \frac{3}{4}i$$

$$\text{and radius} = \sqrt{\frac{1}{4} + \frac{9}{16} - \frac{1}{2}} = \frac{1}{4}\sqrt{4 + 9 - 8} = \frac{\sqrt{5}}{4}$$

3. Argument Form

$\arg\left(\frac{z-z_1}{z-z_2}\right) = \theta$ (where z_1 & z_2 are complex constant & θ is a real constant) represents a circular arc subtending an angle of θ at the circumference.

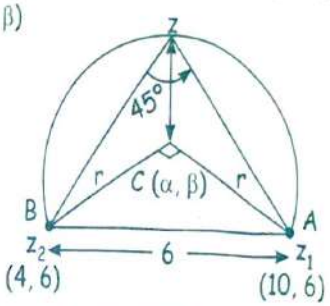
Ques. If $z_1 = 10 + 6i$, $z_2 = 4 + 6i$ and z is a complex number such that $\text{amp}\left(\frac{z-z_1}{z-z_2}\right) = \frac{\pi}{4}$, then the value to $|z - 7 - 9i|$ is equal to **(IIT 1990)**

(a) $\sqrt{2}$ (b) $2\sqrt{2}$ (c) $3\sqrt{2}$ (d) $2\sqrt{3}$

Ans. Given, $\arg\left(\frac{z-z_1}{z-z_2}\right) = \frac{\pi}{4}$

So, $\angle ACB = 90°$ (From Diagram)

Let centre (α, β)



In ΔCAB

$r^2 + r^2 = (AB)^2$

$\Rightarrow 2r^2 = 6^2$

$\Rightarrow r^2 = \frac{36}{2}$

$\Rightarrow r^2 = 18$

Now, $(\alpha - 4)^2 + (\beta - 6)^2 = (\alpha - 10)^2 + (\beta - 6)^2$

$\Rightarrow (\alpha - 4)^2 = (\alpha - 10)^2$

$\Rightarrow \alpha - 4 = 10 - \alpha$

$\Rightarrow \alpha = 7$

and $(\alpha - 4)^2 + (\beta - 6)^2 = r^2$

$\Rightarrow 3^2 + (\beta - 6)^2 = 18$

$\Rightarrow (\beta - 6)^2 = 9$

$\Rightarrow \beta - 6 = \pm 3 \Rightarrow \beta = 6 \pm 3 = 9, 3$(X)

$\Rightarrow \beta = 9$

So, the centre of circle = (7, 9)

Then, $|z - 7 - 9i| = r = 3\sqrt{2}$

Ques. Let arg (z) represent the principal argument of the complex number z. The $|z| = 3$ and $\arg(z - 1) - \arg(z + 1) = \frac{\pi}{4}$ intersect:

(29 June 2022-Shift II)

(a) Exactly at one point

(b) Exactly at two points

(c) Nowhere

(d) At infinitely many points

Ans. Given $\arg(z - 1) - \arg(z + 1) = \frac{\pi}{4}$

$$\Rightarrow \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$$

In ΔABC, $r^2 + r^2 = (2)^2$

$\Rightarrow 2r^2 = 4$

$\Rightarrow r = \sqrt{2}$

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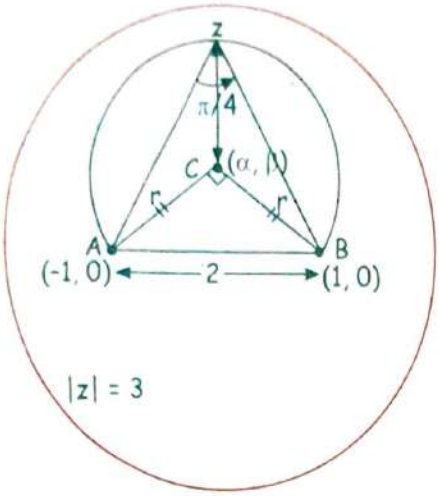


Now, $(\alpha + 1)^2 + \beta^2 = (\alpha - 1)^2 + \beta^2$

$\Rightarrow \alpha^2 + 1 + 2\alpha = \alpha^2 + 1 - 2\alpha \Rightarrow 4\alpha = 0 \Rightarrow \alpha = 0$

and $(\alpha - 1)^2 + (\beta - 0)^2 = r^2$

$\Rightarrow 1 + \beta^2 = 2 \Rightarrow \beta = 1$

So, centre = (0, 1)

$|z| = 3$ is a circle with centre (0, 0) and radius 3 and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circular arc with centre (0, 1) and radius $\sqrt{2}$. These are never intersect.

Ques. Locus of complex number satisfying $\arg[(z - 5 + 4i)/(z + 3 - 2i)] = -\pi/4$ is the arc of circle

(a) whose radius is $5\sqrt{2}$

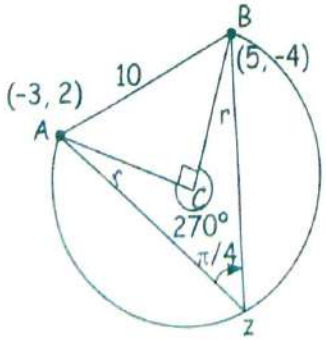
(b) whose radius is 5

(c) whose length (of arc) is $\frac{15\pi}{\sqrt{2}}$

(d) whose centre is -2 - 5i

Ans. Given, $\arg\left(\frac{z-5+4i}{z+3-2i}\right) = -\frac{\pi}{4}$

$\Rightarrow \arg\left(\frac{z-(5-4i)}{z-(-3+2i)}\right) = -\frac{\pi}{4}$



$(AB)^2 = 8^2 + 6^2$

$\Rightarrow AB = 10$

In ΔABC,

$r^2 + r^2 = 100 \Rightarrow r^2 = 50$

$\Rightarrow r = 5\sqrt{2}$

Length of arc $= \frac{\theta}{360^\circ}(2\pi r) = \frac{270^\circ}{360^\circ}(2\pi \times 5\sqrt{2})$

$= \frac{3}{4} \times 2\pi \cdot 5\sqrt{2} = \frac{15\pi}{\sqrt{2}}$

Check for centre, C [-2, -5]

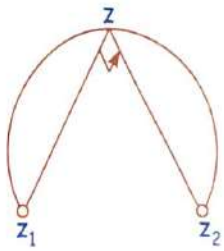
$\Rightarrow CA = \sqrt{1^2 + 7^2} = \sqrt{50}$

$\Rightarrow CB = \sqrt{7^2 + 1^2} = \sqrt{50}$

Hence, centre of circle is -2 -5i.

Special Cases

Case-I: If $\theta = 90^\circ \Rightarrow$ Semi Circle

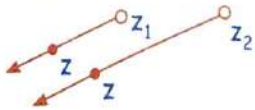


Case-II: If $\theta = 0^\circ \Rightarrow$ 2 parallel ray

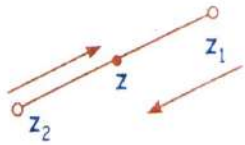
Then, $\arg\frac{(z-z_1)}{(z-z_2)} = \theta$

$\Rightarrow \arg(z - z_1) - \arg(z - z_2) = 0$

$\Rightarrow \arg(z - z_1) = \arg(z - z_2)$



Case-III: If $\theta = 180^\circ \Rightarrow$ anti parallel



4. Apollonius Circle

$\frac{|z-z_1|}{|z-z_2|} = k$ (k is not 1) represents a circle (where z_1, z_2 & k are constants)

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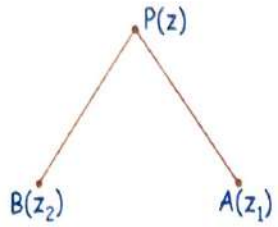
Proof:



$\frac{PA}{PB} = k, \quad k \neq 1$

$\Rightarrow \frac{|z - z_1|}{|z - z_2|} = k$

$\Rightarrow |z - z_1| = k\,|z - z_2|$

Square both sides

$\Rightarrow |z - z_1|^2 = k^2\,|z - z_2|^2$

$\Rightarrow (z - z_1)(\bar{z} - \bar{z}_1) - k^2(z - z_2)(\bar{z} - \bar{z}_2) = 0$

$\Rightarrow z\bar{z} - z\bar{z}_1 - z_1\bar{z} + |z_1|^2 - k^2\left(z\bar{z} - \bar{z}_2 z - z_2\bar{z} + |z_2|^2\right) = 0$

$\Rightarrow z\bar{z}(1 - k^2) + z\left(k^2\bar{z}_2 - \bar{z}_1\right) + \bar{z}\left(z_2 k^2 - z_1\right)$

$+ |z_1|^2 - k^2|z_2|^2 = 0$

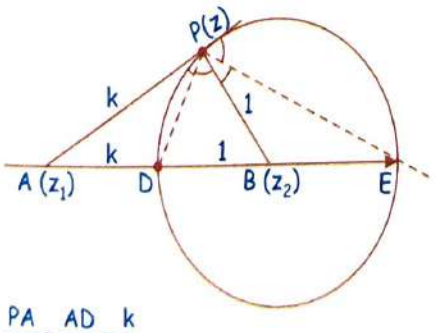
$\Rightarrow z\bar{z} + \underbrace{\frac{z(k^2\bar{z}_2 - \bar{z}_1)}{1 - k^2}}_{\bar{a}} + \underbrace{\frac{\bar{z}(z_2 k^2 - z_1)}{1 - k^2}}_{a} + \underbrace{\frac{|z_1|^2 - k^2|z_2|^2}{1 - k^2}}_{b} = 0$

Note: If $k = 1 \Rightarrow \frac{|z - z_1|}{|z - z_2|} = 1$

$\Rightarrow |z - z_1| = |z - z_2|$ → Represents a straight line

Geometrical Interpretation

$\frac{PA}{PB} = \frac{|z - z_1|}{|z - z_2|} = \frac{k}{1}$



$\frac{PA}{PB} = \frac{AD}{DB} = \frac{k}{1}$

Get D [Fixed pt] → Internal division

& Get E → External division

Circle with DE as diameter is the required locus.

Ques. Let z be a complex number such that $\left|\frac{z - 2i}{z + i}\right| = 2$, $z \neq -i$. Then z lies on the circle of radius 2 and centre **(JEE Main 2023 January)**

(a) (2, 0) (b) (0, 0)

(c) (0, 2) (d) (0, -2)

Ans. **Method-1**

$\left|\frac{z - 2i}{z + i}\right| = 2 \Rightarrow \frac{|z - 2i|}{|z + i|} = 2$

Let, $z = x + iy$

$\Rightarrow \frac{|x + i(y - 2)|}{|x + i(y + 1)|} = 2$

$\Rightarrow \frac{\sqrt{x^2 + (y - 2)^2}}{\sqrt{x^2 + (y + 1)^2}} = 2$

$\Rightarrow x^2 + (y - 2)^2 = 4(x^2 + (y + 1)^2)$

$\Rightarrow x^2 + y^2 + 4 - 4y = 4x^2 + 4y^2 + 4 + 8y$

$\Rightarrow 3x^2 + 3y^2 + 12y = 0$

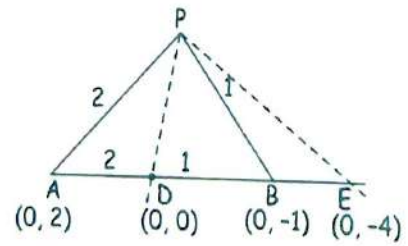
$\Rightarrow x^2 + y^2 + 4y = 0$

So, centre (-g, -f) = (0, -2)

Method-2

Given, $\frac{|z - 2i|}{|z + i|} = 2$

$\Rightarrow \frac{PA}{PB} = \frac{2}{1}$



$D = \left[\frac{2 \times 0 + 1 \times 0}{2 + 1}, \frac{2 \times (-1) + 1 \times (2)}{2 + 1}\right] = [0, 0]$

$E = \left[\frac{2 \times 0 - 1 \times 0}{2 - 1}, \frac{2(-1) - 1 \times 2}{2 - 1}\right] = [0, -4]$

So, eqⁿ of circle

$(x - 0)(x - 0) + (y - 0)(y + 4) = 0$

$\Rightarrow x^2 + y^2 + 4y = 0$

So, centre (-g, -f) = (0, -2)

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5. Diametric Form of a Circle

Circle with ends of diameter A (z_1) & B (z_2)

$PA^2 + PB^2 = AB^2$

$\Rightarrow |z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$

OR

$C_{AP} + C_{BP} = 0$

$\Rightarrow \frac{z - z_1}{\bar{z} - \bar{z}_1} + \frac{z - z_2}{\bar{z} - \bar{z}_2} = 0$

$\Rightarrow \frac{z - z_1}{\bar{z} - \bar{z}_1} = -\frac{z - z_2}{\bar{z} - \bar{z}_2}$

COMPLEX FORMS OF ELLIPSE

Let P is a variable point, $A(z_1)$ and $B(z_2)$ are 2 fixed points

$|z - z_1| + |z - z_2| = k$ represents an ellipse if $k > |z_1 - z_2|$.

It represents line segment joining A and B if $k = |z_1 - z_2|$.

It represents no real locus if $k < |z_1 - z_2|$

COMPLEX FORMS OF HYPERBOLA

Let P is a variable point, $A(z_1)$ and $B(z_2)$ are 2 fixed points,

$||z - z_1| - |z - z_2|| = k$ represents a hyperbola if $k < |z_1 - z_2|$.

It represents 2 rays if $k = |z_1 - z_2|$.

It represents no real locus if $k > |z_1 - z_2|$

Ques. Interpret the following Locus

1. $|z - 2| + |z + 2| = 5$
2. $|z - 3i| - |z + 3i| = 7$
3. $|z - 2| + |z| = 2$
4. $|z - 3 - 4i| - |z| = 5$
5. $|z - 2| - |z - 2i| = 6$

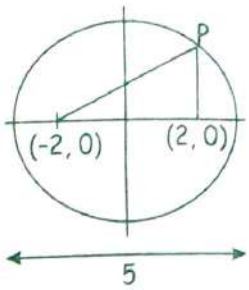
Ans. 1. Given, $|z - 2| + |z + 2| = 5$

Let, $PA + PB = 5$ where, A (2, 0) & B(-2, 0)

$\Rightarrow AB = 4$

and $k = 5$

$\Rightarrow k > AB \Rightarrow$ Locus is ellipse



$2a = 5$ &

$2ae = 4$

$e = \frac{4}{5}$

2. Given, $|z - 3i| - |z + 3i| = 7$

Let, $PA - PB = 7$

Where, A (0, 3) & B (0, -3)

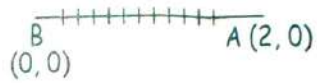
$\Rightarrow AB = 6 \Rightarrow k > AB$

But for Hyperbola, $k < AB$

$\Rightarrow$ It reprsents no real locus.

3. Given, $|z - 2| + |z| = 2$

Let, $PA + PB = 2$



where,

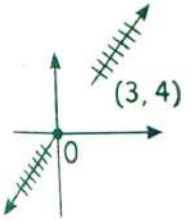
A (2, 0) & B (0, 0)

$\Rightarrow AB = 2 = k \Rightarrow$ straight line

4. Let, $PA - PB = 5$

where, A (3, 4) & B (0, 0)

$\Rightarrow AB = 5 = k \Rightarrow$ 2 rays



5. Given, $|z - 2| - |z - 2i| = 6$

Let, $PA - PB = 6$

where, A (2, 0) & B (0, 2)

$\Rightarrow AB = \sqrt{2^2 + 2^2} = 2\sqrt{2}$

$AB < k \Rightarrow$ non real locus.

Ques. Interprete the following locus and also find area: $|z - 3i| + |z + 3i| \leq 10$

Ans. Let, $PA + PB \leq 10$

where, A (0, 3), B (0, -3)

$\Rightarrow AB = 6$

and $k = 10$

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$\Rightarrow k > AB \rightarrow$ ellipse

Now, $2be = 6$, $2b = 10$

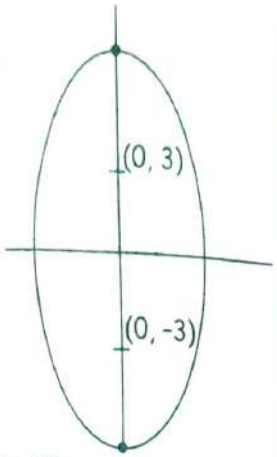
$\Rightarrow e = \frac{6}{10} = \frac{3}{5}$

$\because a^2 = b^2(1 - e^2)$

$\therefore a^2 = 5^2\left(1 - \frac{9}{25}\right)$

$\Rightarrow a^2 = 25 - 9$

$\Rightarrow a^2 = 16 \Rightarrow a = 4$



Area $= \pi ab = \pi \times 4 \times 5 = 20\pi$

Ques. For $z \in \mathbb{C}$ if the minimum value of $\left(|z - 3\sqrt{2}| + |z - p\sqrt{2}i|\right)$ is $5\sqrt{2}$, then a value of P is _____.

(a) 3 (b) $\frac{7}{2}$ (c) 4 (d) $\frac{9}{2}$

Ans. Let, $PA = |z - 3\sqrt{2}|$

$PB = |z - p\sqrt{2}i|$

Since, $PA + PB \geq AB$

min value of $(PA + PB) = AB = 5\sqrt{2}$

where, $A(3\sqrt{2}, 0)$, $B(0, p\sqrt{2})$

Now, $(AB)^2 = (3\sqrt{2})^2 + (p\sqrt{2})^2$

$\Rightarrow (AB)^2 = 18 + 2p^2$

$\Rightarrow (5\sqrt{2})^2 = 18 + 2p^2$

$\Rightarrow 50 - 18 = 2p^2$

$\Rightarrow 32 = 2p^2$

$\Rightarrow 16 = p^2$

$\Rightarrow p = \pm 4$

TRIANGLE INEQUALITY

$||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$

Proof: let, $z_1 = r_1 \text{cis}\theta_1$ & $z_2 = r_2 \text{cis}\theta_2$

Consider

$|z_1 + z_2| = |(r_1\cos\theta_1 + r_2\cos\theta_2) + i(r_1\sin\theta_1 + r_2\sin\theta_2)|$

$\Rightarrow |z_1 + z_2| = \sqrt{(r_1\cos\theta_1 + r_2\cos\theta_2)^2 + (r_1\sin\theta_1 + r_2\sin\theta_2)^2}$

$\Rightarrow |z_1 + z_2| = \sqrt{r_1^2 + r_2^2 + 2r_1r_2\cos(\theta_1 - \theta_2)}$

$\Rightarrow |z_1 + z_2|_{max} = \sqrt{r_1^2 + r_2^2 + 2r_1r_2} = \sqrt{(r_1 + r_2)^2} = r_1 + r_2$

when $\cos(\theta_1 - \theta_2) = 1$

$PA + PB \geq PB$

$|PA - PB| \leq AB$

$|z_1 + z_2|_{min} = \sqrt{r_1^2 + r_2^2 - 2r_1r_2} = \sqrt{(r_1 - r_2)^2} = |r_1 - r_2|$

when, $\cos(\theta_1 - \theta_2) = -1$

Hence, $||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$

Note: Equality Holds

(i) $|z_1 + z_2| = |z_1| + |z_2|$ If z_1 and z_2 are parallel.

(ii) $|z_1 + z_2| = ||z_1| - |z_2||$ If z_1 and z_2 are anti-parallel.

Ques. If $|z - 3i| = 1$ then show that $2 \leq |z| \leq 4$.

Ans. Consider

$||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$

Put, $z_1 = z$ and $z_2 = -3i$

$\Rightarrow ||z| - |-3i|| \leq |z - 3i| \leq |z| + |-3i|$

Let $|z| = r$

$\Rightarrow |r - 3| \leq 1 \leq r + 3$

$\Rightarrow |r - 3| \leq 1$ & $1 \leq r + 3$

$\Rightarrow -1 \leq r - 3 \leq 1$ $\Rightarrow r + 3 \geq 1$

$\Rightarrow 2 \leq r \leq 4$ $\Rightarrow r \geq -2$ (always true)

Hence, $2 \leq |z| \leq 4$

Ques. If $|z + 4| \leq 3$, then find the greatest value of $|z + 1|$.

Ans. **Method-1**

Given, $|z + 4| \leq 3$

$\Rightarrow |z - (-4)| \leq 3$

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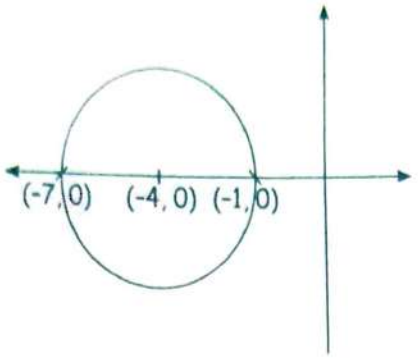

$|z+1| = |z-(-1)|$ = dis b/w z & (-1, 0)

$|z+1|_{min} = 0$

$|z+1|_{max} = 6$

Method-2

Consider

$||z_1| - |z_2|| \le |z_1 + z_2| \le |z_1| + |z_2|$

Put, $z_1 = z + 1$ and $z_2 = 3$

$\Rightarrow ||z+1| - |3|| \le |z+4| \le |z+1| + 3$

Let $|z+1| = r$

$\Rightarrow |r-3| \le |z+4| \le 3$

$\Rightarrow |r-3| \le 3 \Rightarrow -3 \le r-3 \le 3$

$\Rightarrow 0 \le r \le 6$

Hence, $|z+1|_{max} = 6$

Ques. If $\left|z+\frac{4}{z}\right| = 3$ then show that $|z|$ lies between 1 and 4.

Ans. Consider

$||z_1| - |z_2|| \le |z_1 + z_2| \le |z_1| + |z_2|$

Put, $z_1 = z$ & $z_2 = \frac{4}{z}$

$\Rightarrow \left||z| - \left|\frac{4}{z}\right|\right| \le \left|z+\frac{4}{z}\right| \le |z| + \left|\frac{4}{z}\right|$

$\Rightarrow \left||z| - \frac{4}{|z|}\right| \le 3 \le |z| + \frac{4}{|z|}$

Let $|z| = r$

$\Rightarrow \left|r - \frac{4}{r}\right| \le 3 \le r + \frac{4}{r}$

(i) $\left|r - \frac{4}{r}\right| \le 3$

$\Rightarrow -3 \le r - \frac{4}{r} \le 3$

$\Rightarrow -3 \le \frac{r^2-4}{r} \le 3$

$\Rightarrow -3r \le r^2 - 4 \le 3r$

$\Rightarrow r^2 - 4 \ge -3r$ & $r^2 - 4 \le 3r$

$\Rightarrow r^2 + 3r - 4 \ge 0$ $\Rightarrow r^2 - 3r - 4 \le 0$

$\Rightarrow (r+4)(r-1) \ge 0$ $\Rightarrow (r-4)(r+1) \le 0$

$\Rightarrow r \ge 1$ $\Rightarrow r \le 4$

So, $r \in [1, 4]$

(ii) $r + \frac{4}{r} \ge 3$

$\Rightarrow r^2 + 4 \ge 3r$

$\Rightarrow r^2 - 3r + 4 \ge 0$, D = -ve (always true)

Hence, $|z| \in [1, 4]$

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Short Notes

DE'MOIVRE'S THEOREM & CUBE ROOTS OF UNITY

- The value of $(\cos\theta + i\sin\theta)^n$ is $\cos(n\theta) + i\sin(n\theta)$ if 'n' is integer
- $\cos(n\theta) + i\sin(n\theta)$ is one of the values of $(\cos\theta + i\sin\theta)^n$ if n is a rational number of the form p/q, where p & q are co-prime.
- The 3 cube roots of unity are 1, $\frac{-1+i\sqrt{3}}{2}(\omega)$, $\frac{-1-i\sqrt{3}}{2}(\omega^2)$.
- $1 + \omega + \omega^2 = 0$
- $\omega^3 = 1$
- $1 + \omega^n + \omega^{2n} = 3$ (if n is multiple of 3)
- $1 + \omega^n + \omega^{2n} = 0$ (if n is not a multiple of 3).
- The three cube roots of unity when plotted on the argand plane constitute the vertices of an equilateral triangle.

n^{th} ROOTS OF UNITY

- The n n^{th} roots of unity are in G.P. with common ratio $e^{i(2\pi/n)}$.
- The points represented by n, n^{th} roots of unity are located at the vertices of a regular polygon of n sides inscribed in a unit circle having center at origin.
- The sum of the n, nth roots of unity is zero.
- The product of the n, nth roots of unity is -1 if n is even and 1 if n is odd.
- $(1-\alpha_1)(1-\alpha_2)\ldots\ldots(1-\alpha_{n-1}) = n$
- $(1+\alpha_1)(1+\alpha_2)\ldots\ldots(1+\alpha_{n-1}) = 0$ if n is even and it is equal to 1 if n is odd.

TRIANGLE INEQUALITY

- For any two complex numbers z_1 and z_2, $||z_1| - |z_2|| \le |z_1 + z_2| \le |z_1| + |z_2|$
- This result is called triangle inequality.
- If $|z_1 + z_2| = |z_1| + |z_2| \Rightarrow z_1$ & z_2 are parallel.
- If $|z_1 + z_2| = ||z_1| - |z_2|| \Rightarrow z_1$ & z_2 are anti-parallel.

VARIOUS FORMS OF STRAIGHT LINE

- **Canonical Form:** $\bar{a}z + a\bar{z} = b$ where a is a complex number & b is a real number.
- Quantity is called complex slope of this line and is defined as $\frac{-a}{\bar{a}}$.
- A line of real slope $\tan\alpha$ has complex slope $\text{cis}2\alpha$.
- **Perpendicular Bisector Form:** $|z - a| = |z - b|$ is the perpendicular bisector of the segment joining the points A (a) & B (b).
- **Argument Form:** $\text{amp}(z - \alpha) = \theta$ is a ray emanating from the complex number representing the point "α" and inclined at an angle θ to the positive x-axis.

VARIOUS FORMS OF CIRCLE

- **Centre Radius Form:** Equation of a circle with centre A(a) and radius r is $|z - a| = r$.

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- **Canonical Form:** $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$, where α is a complex constant and r is a real constant. Centre = $-\alpha$ & radius = $\sqrt{\alpha\bar{\alpha} - r}$.
- **Argument Form:** $\arg\left(\frac{z - z_1}{z - z_2}\right) = \alpha$, where $\alpha \in (0, \pi)$ represents a circular arc of segment of a circle on z_1z_2, containing angle α.

COMPLEX FORMS OF CONIC SECTION

- **Parabola:** $|z - z_0| = \frac{a\bar{z} + \bar{a}z + b}{2|a|}$
- **Ellipse:** $|z - z_1| + |z - z_2| = 2a$, where $2a > |z_1 - z_2|$
- **Hyperbola:** $||z - z_1| - |z - z_2|| = 2a$, where $2a < |z_1 - z_2|$

"Every point in complex plane is the proof that the beauty lies beyond the real axis."

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Chapter 16

Statistics

MEASURES OF CENTRAL TENDENCY

An average value or a central value of a distribution is the value of variable which is representative of the entire distribution, this representative value are called the measures of central tendency.

Generally the following five measures of central tendency.

(a) Mathematical Average
  (i) Arithmetic mean
  (ii) Geometric mean
  (iii) Harmonic mean

(b) Positional average
  (i) Median
  (ii) Mode

WAYS TO REPRESENT DATA

1) **Ungrouped data:** In this case the value of variate are written in the form of a list.
   Ex: 1, 2, 3, 5, 6, 7, 10, 12,...

2) **Ungrouped Frequency distribution:** In this case data is represented in tabular form where one column consists of different values of variate whereas the other column consists of their respective frequencies.

3) **Grouped Frequency distribution:** In this case data is grouped into intervals. These groupings are called class intervals and their size is called class width or class size. In each of these classes the least number is called lower class limit while the greatest number is called upper class limit.

Ques. Marks obtained out of 100 by students in class of 30 students (Discrete Data)

10	20	36	92	95	40	50	56	60	70
92	88	80	70	72	70	36	40	36	40
92	40	50	50	56	60	70	60	60	88

Ans. Marks obtained out of 100 by students in class of 30 students

Grouped frequency data/distribution

Marks	Frequency
0-20	1
20-40	4
40-60	9
60-80	9
80-100	7
Total	30

(Ungrouped frequency data)

Marks	No. of students (i.e., the frequency)
10	1
22	1
36	3
40	4
50	3
56	2
60	4
70	4
72	1
80	1
88	2
92	3
95	1
Total	30

MEAN

For Ungrouped Distribution

If $x_1, x_2, \ldots, x_n$ are n values of variate x_i then their A.M. $\bar{x}$ is defined as

$$\bar{x} = \frac{x_1 + x_2 \ldots x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n} \Rightarrow \sum_{i=1}^{n} x_i = n\bar{x}$$

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For Ungrouped and Grouped Frequency Distribution

If $x_1, x_2, \ldots, x_n$ are values of variate with corresponding frequencies $f_1, f_2, \ldots, f_n$ then their A.M. is given by

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + \ldots + f_nx_n}{f_1 + f_2 + \ldots f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{N},$$

where $N = \sum_{i=1}^{n} f_i$

Ques. The mean of 7 observations is 8. If one observation 14 is wrongly written. Find the mean of

(a) Remaining 6 Observations.

Ans. $\because \quad \frac{x_1 + x_2 + \ldots x_7}{7} = 8$

$\Rightarrow \quad x_1 + x_2 + x_3 \ldots\ldots + x_7 = 56$

$\because$ one observation 14 is wrongly written

$\therefore \quad x_1 + x_2 + x_3 + \ldots\ldots x_6 = 56 - 14$

$\Rightarrow \quad x_1 + x_2 + \ldots\ldots x_6 = 42$

$\therefore$ new mean $= \frac{42}{6} = 7$

(b) If the wrong observation is replaced with the correct value of 7.

Ans. New mean of correct observations $= \frac{42 + 7}{7}$

$= \frac{49}{7} = 7$

Ques. The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If now the mean age of the teachers in this school is 39 years, then the age (in years) of the newly appointed teacher is:

[JEE Mains (April) 2017]

(a) 25 (b) 30 (c) 35 (d) 40

Ans. $\because \quad \frac{x_1 + x_2 + \ldots x_{25}}{25} = 40$

$\Rightarrow x_1 + x_2 + \ldots\ldots x_{25} = 1000$

Let new appointed teacher's age is y years

$\therefore$ new sum of ages $= 1000 - 60 + y = 940 + y$

$\therefore$ new mean $= \frac{940 + y}{25} = 39$

$\Rightarrow 940 + y = 975 \Rightarrow y = 35$

WEIGHTED MEANS (FOR FREQUENCY DATA)

Let $a_1, a_2, \ldots\ldots, a_n$, be n-positive real numbers and $x_1, x_2, \ldots\ldots, x_n$ be n-positive rational numbers,

then Weighted Arithmetic Mean (A) =

$$\frac{x_1a_1 + x_2a_2 + \ldots + x_na_n}{x_1 + x_2 + \ldots\ldots + x_n}$$

Weighted Geometric Mean (G) =

$$\left(a_1^{x_1}.a_2^{x_2}\ldots\ldots a_n^{x_n}\right)^{\frac{1}{x_1 + x_2 + \ldots\ldots + x_n}}$$

Weighted Harmonic Mean (H) = $\frac{x_1 + x_2 + \ldots\ldots + x_n}{\frac{x_1}{a_1} + \frac{x_2}{a_2} + \ldots\ldots + \frac{x_n}{a_n}}$

Ques. The weighted mean of first n natural number whose weights are equal to the squares of the corresponding numbers is-

(a) $\frac{n+1}{2}$ (b) $\frac{3n(n+1)}{2(2n+1)}$

(c) $\frac{(n+1)(2n+1)}{6}$ (d) $\frac{n(n+1)}{2}$

Ans.

x_i	f_i
1	1^2
2	2^2
3	3^2
4	4^2
.	.
.	.
.	.
n	n^2

$$\sum f_i = \frac{n(n+1)(2n+1)}{6}$$

$$\Sigma f_ix_i = \sum r^3 = \left(\frac{n(n+1)}{2}\right)^2$$

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$$\bar{x} = \frac{1^3 + 2^3 + 3^3 + ... + n^3}{\frac{n(n+1)(2n+1)}{6}}$$

$$= \frac{\left(\frac{n^2(n+1)^2}{4}\right)}{\left(\frac{n(n+1)(2n+1)}{6}\right)} = \frac{3n(n+1)}{2(2n+1)}$$

Ques. Consider the data on x taking the value 0, 2, 4, 8, ... 2^n with frequencies, ${}^nC_0$, ${}^nC_1$, ${}^nC_2$, ... ${}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$ then n is equal to

[JEE Mains Online-2020]

Ans.

X_i	f_i
0	${}^nC_0$
2	${}^nC_1$
2^2	${}^nC_2$
2^3	${}^nC_3$
.	.
.	.
.	.
2^n	${}^nC_n$
	$\sum fi = 2^n$

$$\bar{x} = \frac{0 \cdot {}^nC_0 + 2 \cdot {}^nC_1 + 2^2 \cdot {}^nC_2 + ... + 2^n \cdot {}^nC_n}{2^n}$$

Consider

$(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + ... + {}^nC_nx^n$

Put x = 2

$3^n = {}^nC_0 + [{}^nC_1.2 + {}^nC_2 2^2 + ...{}^nC_n 2^n]$

$\Rightarrow 3^n - 1 = [{}^nC_1.2 + {}^nC_2.2^2 + ... + {}^nC_n 2^n]$

$\therefore \bar{x} = \frac{3^n - 1}{2^n} = \frac{728}{2^n}$

$\Rightarrow 3^n = 729 = 3^6$

$\Rightarrow n = 6$

PROPERTIES OF AM

(a) If $\bar{x}$ is the mean of variate x_i then

(i) AM of $(x_i + \lambda) = (\bar{x} + \lambda)$

(ii) AM of $(\lambda x_i) = \lambda\bar{x}$

(iii) AM of $(ax_i + b) = a\bar{x} + b$

(where λ, a, b are constants)

(b) Sum of deviations of variate from their AM is always zero i.e. $\sum(x_i - \bar{x}) = 0$, $\sum f_i(x_i - \bar{x}) = 0$

(c) Sum of square of deviations of variate from their AM is minimum i.e. $\sum(x_i - \bar{x})^2$ is minimum

Ques. In the year 2025, The mean age of 25 teachers in a school is 40 years. 2 teachers each of whose current age is 63 years will retire in 2030. Find the new mean age of the teachers in this school in 2030. (assuming that no other teacher retires in this duration):

(a) 38 (b) 43

(c) 40 (d) None of these

Ans. Current sum of age = 40×25 = 1000

Current sum of remaining 23 teachers

= 1000 - (63 + 63)

= 1000 - 126 = 874

$\therefore$ Current mean = $\frac{874}{23}$

Sum of age in 2030 = 874 + 23 × 5

$\therefore$ New mean = $\frac{874}{23} + 5 = 38 + 5 = 43$

ASSUMED MEAN METHOD

(iii) **By short cut method OR Assumed Mean Method:** If the value of x_i are large, then the calculation of A.M. by using previous formula is quite tedious and time consuming. In such case we take deviation of variate from an arbitrary point a.

Let $d_i = x_i - a$

$\therefore \bar{x} = a + \frac{\sum fd}{N}$, where a is assumed mean.

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Ques. Find the mean of the following frequency distribution by assumed mean method.

Marks obtained	Number of students (f_i)	Mid-points (x_i)	$d_i = x_i - 125$	$d_i/50 = w_i$	$f_i w_i$
0-50	17	25	-100	-2	-34
50-100	35	75	-50	-1	-35
100-150	43	125	0	0	0
150-200	40	175	50	1	40
200-250	21	225	100	2	42
250-300	24	275	150	3	72
	$\sum f_i = 180$				$\sum f_i w_i = 85$

Ans. $\frac{\sum f_i w_i}{\sum f_i} = \bar{x}$

$\bar{x}_{new} = \frac{85}{180}$

$\bar{x}_{current} = \frac{85}{180} \times 50 + 125$

$= \frac{85 \times 5}{18} + 125 = 148.61$

Ques. Find the mean of the following frequency distribution by assumed mean method.

Income Per day	Number of person (f_i)	x_i	$w_i = \frac{x_i - 350}{100}$	$f_i w_i$
0-100	4	50	-3	-12
100-200	8	150	-2	-16
200-300	9	250	-1	-9
300-400	10	350	0	0
400-500	7	450	1	7
500-600	5	550	2	10
600-700	4	650	3	12
700-800	3	750	4	12
	$\Sigma f_i = 50$			$\Sigma f_i w_i = 4$

Ans. $\bar{x}_{new} = \frac{4}{50}$

$\bar{x}_{current} = 100\,\bar{x}_{new} + 350$

$= 100 \times \frac{4}{50} + 350 = 8 + 350 = 358$

COMBINED MEAN

If x_1 and x_2 be the means of two groups having n_1 and n_2 terms respectively then the mean (combined mean) of their composite groups is given by combined mean

$= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

If there are more than two groups then, combined mean $= \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3 + \ldots}{n_1 + n_2 + n_3 + \ldots}$

Ques. The mean income of a group of persons is Rs. 400 and another group of persons is Rs. 480. If the mean income of all the persons of these two groups is Rs. 430, then find the ratio of the number of persons in the groups.

Ans. $\because \frac{400 \times n_1 + 480 \times n_2}{n_1 + n_2} = 430$

$\Rightarrow 40n_1 + 48n_2 = 43(n_1 + n_2)$

$\Rightarrow 40n_1 + 48n_2 = 43n_1 + 43n_2$

$\Rightarrow 5n_2 = 3n_1$

$\Rightarrow \frac{n_1}{n_2} = \frac{5}{3}$

MEDIAN

Median For Ungrouped Data

(i) The median of a series is the value of middle term of the series when the values are written in ascending order.

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{term, (when n is odd)} \\ \text{Mean of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2}+1\right)^{th} \text{ terms, (when n is even)} \end{cases}$$

Example: 1, 2, 3, 4, 7, 8, 12, 15, 16, 20

$\therefore$ middle terms are 7 and 8

$\therefore$ median $= \frac{7+8}{2} = \frac{15}{2} = 7.5$

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Ques. The mean and the median of the following ten numbers in increasing order 10, 22, 26, 29, 34, x, 42, 67, 70, y are 42 and 35 respectively, then $\frac{y}{x}$ is equal to:

[JEE Mains - 2019 (April)]

(a) $\frac{7}{3}$ (b) $\frac{9}{4}$ (c) $\frac{7}{2}$ (d) $\frac{8}{3}$

Ans. $\text{Median} = \frac{34 + x}{2} = 35$

$\Rightarrow 34 + x = 70$

$\Rightarrow x = 36$

$\therefore \text{mean} = 42$

$= \frac{10 + 22 + 26 + 29 + 34 + 36 + 42 + 67 + 70 + y}{10}$

$\Rightarrow 420 = 336 + y \Rightarrow y = 84$

$\therefore \frac{y}{x} = \frac{84}{36} = \frac{7}{3}$

Median For Ungrouped Frequency Data

(ii) First we prepare the cumulative frequency (c.f.) column and find value of N then

$$\text{Median} = \begin{cases} \left(\frac{N+1}{2}\right)^{th} \text{term, (when N is odd)} \\ \text{Mean of } \left(\frac{N}{2}\right)^{th} \text{ and } \left(\frac{N}{2}+1\right)^{th} \text{ terms, (when N is even)} \end{cases}$$

Median For Grouped Frequency Data

(iii) Prepare c.f. column and find value of N then find the class which contain value of c.f. is equal or just greater to N/2, this is median class

$$\therefore \text{Median} = l + \frac{\left(\frac{N}{2} - F\right)}{f} h$$

where

l = lower limit of median class

f = freq. of median class

F = c.f. of the class preceding median class

h = class interval of median class

Ques. Find the median of following frequency distribution.

Class	0-10	10-20	20-30	30-40	40-50	50-60
f_i	6	9	14	12	19	10

Ans.

Class	f_i	C.F.
0-10	6	6
10-20	9	15
20-30	14	29
30-40	12	41
40-50	19	60
50-60	10	70
	N = 70	

$\therefore N = 70$

$\Rightarrow \frac{N}{2} = 35$

$\therefore$ Median class: 30-40

l = 30, f = 12

F = 29, h = 10

$$M = l + \frac{\left(\frac{N}{2} - F\right)h}{f}$$

$$= 30 + \frac{(35 - 29)}{12} \times 10$$

$$= 30 + \frac{6}{12} \times 10$$

$= 30 + 5 = 35$

Ques. Find the missing frequencies in the following frequency distribution table, if N = 100 and median is 32.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	Total
No. of students	10	f_1	25	30	f_2	10	100

Ans.

Class	f_i	C.F.
0-10	10	10
10-20	f_1	$10+f_1$
20-30	25	$35+f_1$
30-40	30	$65+f_1$
40-50	f_2	$65 + f_1 + f_2$
50-60	10	$75 + f_1 + f_2$
	$\Sigma f_i = 100$	

$\because 75 + f_1 + f_2 = 100$

$\Rightarrow f_1 + f_2 = 25$ ...(i)

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Median = 32

Median class: 30-40

So, $l = 30, f = 30, F = 35 + f_1, h = 10$

$$\therefore 32 = 30 + \left[\frac{50 - (35 + f_1)}{30}\right] \times 10$$

$$\Rightarrow 2 = \frac{50 - 35 - f_1}{3}$$

$$\Rightarrow 6 = 15 - f_1$$

$$\Rightarrow f_1 = 9$$

By eqⁿ. (i)

$$f_2 = 25 - 9 = 16$$

MODE

In a frequency distribution the mode is the value of that variate which has the maximum frequency.

Method for determining mode

For Ungrouped Distribution

The value of that variate which is repeated maximum number of times.

For Ungrouped Frequency Distribution

The value of that variate which have maximum frequency.

Mode For Grouped Frequency Distribution

For grouped frequency distribution: First we find the class which have maximum frequency, this is model class.

$$\therefore \text{Mode} = l + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} h$$

Where,

l = lower limit of model class

f_0 = frequency of the model class

f_1 = frequency of the class preceding model class

f_2 = frequency of the class succeeding model class

h = class interval of model class

Ques. Find the mode of the following frequency distribution:

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
f_i	5	8	7	12	28	20	10	10

Ans. Modal class: 40-50 (because it have maximum frequency)

$l = 40, f_0 = 28, f_1 = 12, f_2 = 20, h = 10$

$$\text{Mode} = 40 + \frac{[28 - 12]}{2 \times 28 - (12 + 20)} \times 10$$

$$= 40 + \frac{16}{56 - 32} \times 10 = 40 + \frac{16}{24} \times 10$$

$$= 40 + \frac{2}{3} \times 10 = 40 + 6.66 = 46.66$$

RELATION BETWEEN MEAN, MEDIAN AND MODE

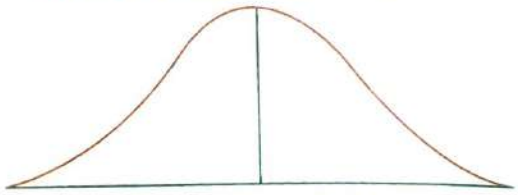
In a moderately asymmetric distribution, following is the relation between mean, median and mode of a distribution. It is known as Empirical formula.

Mode = 3 Median - 2 Mean.

Important Note

(i) Median always lies between mean and mode.

(ii) For a symmetric distribution the mean, median and mode are coinciding.



(Mean = Mode = Median)

Symmetrical Distribution/Bell Curve distribution

MEASURES OF DISPERSION

Generally the following measures of dispersion are commonly used,

(i) Range

(ii) Mean deviation

(iii) Variance and standard deviation

Range

The difference between the greatest and least values of variate of a distribution, are called the range of that distribution.

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If the distribution is grouped distribution, then its range is the difference between upper limit, of the maximum class and lower limit of the minimum class.

Mean Deviation

The mean deviation of a distribution is, the mean of absolute value of deviations of variate from their statistical average (Mean, Median, Mode).

If A is any statistical average of a distribution then mean deviation about A is defined as

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n}|x_i - A|}{n} \quad \text{(for ungrouped dist.)}$$

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} f_i|x_i - A|}{N} \quad \text{(for freq. dist.)}$$

Mean deviation is minimum when it is taken about the median

Example: 1, 2, 3, 5, 6, 7, 8, 9, 8, 1

$\bar{x} = \frac{50}{10} = 5$

$\bar{x} = 5$

$|x_i - \bar{x}| \rightarrow$ deviation

$d_i \rightarrow 4, 3, 2, 0, 1, 2, 3, 4, 3, 4$

$\sum|x_i - \bar{x}| = 4 + 3 + 2 + 0 + 1 + 2 + 3 + 4 + 3 + 4$

$= 26$

$\therefore$ M.D $= \frac{\sum|x_i - \bar{x}|}{N} = \frac{26}{10} = 2.6$

Ques. Let S be the set of all values of a_1 for which the mean deviation about the mean of 100 consecutive positive integers $a_1, a_2, a_3, \ldots, a_{100}$ is 25. Then, S is

[JEE Main-2023 (30 Jan.-II)]

(a) ϕ (b) {99} (c) N (d) {9}

Ans. $a_1, a_1 + 1, a_1 + 2, a_1 + 3, \ldots\ldots a_1 + 99$

$$\bar{x} = \frac{a_1 + a_1 + 1 + a_1 + 2 + \ldots\ldots + a_1 + 99}{100}$$

$$\frac{100a_1 + (1 + 2 + 3 + \ldots\ldots + 99)}{100}$$

$$= \frac{100a_1 + \frac{100 \times 99}{2}}{100} = a_1 + \frac{99}{2}$$

$$d_i = \left\{\frac{99}{2}, \frac{97}{2}, \frac{95}{2}, \ldots\ldots \frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \ldots\ldots \frac{99}{2}\right\}$$

$$\text{MD (about mean)} = \frac{\sum|x_i - \bar{x}|}{N}$$

$$= \frac{2\left[\frac{1}{2} + \frac{3}{2} + \frac{5}{2} + \ldots\ldots \frac{99}{2}\right]}{100}$$

$$= \frac{1 + 3 + 5 + \ldots\ldots + 99}{100}$$

$$= \frac{\frac{50}{2}[1 + 99]}{100} = 25$$

So, it is true for every natural number.

Ques. If the mean deviation about median for the number 3, 5, 7, 2k, 12, 16, 21, 24 arranged in the ascending order, is 6 then the median is

[JEE Main July 25, 2022 (II)]

(a) 11.5 (b) 10.5 (c) 12 (d) 11

Ans. $\because M = \frac{2k + 12}{2}$

$\Rightarrow M = k + 6$

M.D. about Median = 6

$d_i : |x_i - M|$

$= k + 3, k + 1, k - 1, 6 - k, 6 - k, 10 - k, 15 - k, 18 - k$

$$\text{M.D} = \frac{\sum|x_i - M|}{N}$$

$$\Rightarrow \frac{k + 3 + k + 1 + k - 1 + 6 - k + 6 - k + 10 - k + 15 - k + 18 - k}{8}$$

$= 6$

$\Rightarrow \frac{58 - 2k}{8} = 6$

$\Rightarrow 58 - 2k = 48 \Rightarrow 2k = 10 \Rightarrow k = 5$

$\therefore M = k + 6 = 11$

Ques. Let the median and the mean deviation about the median of 7 observation 170, 125, 230, 190, 210, a, b be 170 and $\frac{205}{7}$ respectively. Then the mean deviation about the mean of these 7 observations is:

[JEE Main-2024 (Feb.-1)]

(a) 31 (b) 28 (c) 30 (d) 32

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Ans. a, b, 125, 170, 190, 210, 230

deviations → 170 - a, 170 - b, 45, 0, 20, 40, 60

$\therefore$ M.D $= \dfrac{505 - a - b}{7} = \dfrac{205}{7}$

$\Rightarrow a + b = 300$

Mean $= \dfrac{a + b + 125 + 170 + 190 + 210 + 230}{7} = 175$

Mean deviation about the mean

$= \dfrac{175 - a + 175 - b + 50 + 5 + 15 + 35 + 55}{7}$

$= \dfrac{100 + 20 + 90}{7} = \dfrac{210}{7} = 30$

Ques. Consider the given data with frequency distribution **[JEE Adv.-2023]**

x_i	3	8	11	10	5	4
f_i	5	2	3	2	4	4

Match each entry in List-I to the correct entries in List-II.

List-I		List-II	
(P)	The mean of the above data is	(1)	2.5
(Q)	The median of the above data is	(2)	5
(R)	The mean deviation about the mean of the above data is	(3)	6
(S)	The mean deviation about the median of the above data is	(4)	2.7
		(5)	2.4

The correct option is:

(a) (P) → (3) (Q) → (2) (R) → (4) (S) → (5)

(b) (P) → (3) (Q) → (2) (R) → (1) (S) → (5)

(c) (P) → (2) (Q) → (3) (R) → (4) (S) → (1)

(d) (P) → (3) (Q) → (3) (R) → (5) (S) → (5)

Ans.

x_i	f_i	$x_i f_i$	$\lvert x_i - \bar{x}\rvert$	$f_i\lvert(x_i - \bar{x})\rvert$
3	5	15	3	15
8	2	16	2	4
11	3	33	5	15
10	2	20	4	8
5	4	20	1	4
4	4	16	2	8
	$\Sigma f_i = 20$	$\Sigma f_i x_i = 120$		$\Sigma f_i (x_i - \bar{x}) = 54$

$\bar{x} = \dfrac{120}{20} = 6$

M.D $= \dfrac{54}{20} = 2.7$

Which satisfy only option (a)

Variance and Standard Deviation

The variance of a distribution is, the mean of squares of deviation of variate from their mean. It is denoted by σ^2 or var (x).

Hence standard deviation, $\sigma = +\sqrt{\text{variance}}$

(i) For Ungrouped Distribution

$$\sigma_x^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$\sigma_x^2 = \frac{\sum x_i^2}{n} - \bar{x}^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$$

Ques. Find the variance and standard deviation of first n natural numbers.

Ans. {1, 2, 3, 4, 5, ..., n}

$$\bar{x} = \frac{1 + 2 + 3 + ... + n}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2}$$

$$\text{Var} = \frac{\sum x_i^2}{n} - \left(\bar{x}\right)^2$$

$$= \frac{1^2 + 2^2 + 3^2 + ... + n^2}{n} - \left(\frac{n+1}{2}\right)^2$$

$$= \frac{n(n+1)(2n+1)}{6n} - \frac{(n+1)^2}{4}$$

$$= \frac{2(n+1)(2n+1) - 3(n+1)^2}{12}$$

$$= (n+1)\left[\frac{2(2n+1) - 3(n+1)}{12}\right]$$

$$= (n+1)\left[\frac{4n + 2 - 3n - 3}{12}\right]$$

$$= \frac{(n+1)(n-1)}{12} \Rightarrow \text{Var} = \frac{n^2 - 1}{12}$$

and standard deviation $(\sigma) = \sqrt{\dfrac{n^2 - 1}{12}}$

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Ques. The S.D of first n odd natural numbers is

(a) $\sqrt{\frac{n^2-1}{2}}$ (b) $\sqrt{\frac{n^2-1}{3}}$

(c) $\sqrt{\frac{n^2-1}{6}}$ (d) $\sqrt{\frac{n^2-1}{12}}$

Ans. 1, 3, 5, 7, 9, ...(2n-1)

$$\bar{x} = \frac{1+3+5+....(2n-1)}{n} = \frac{n^2}{n} = n$$

$$\sum x_i^2 = 1^2 + 3^2 + 5^2 + 7^2 + ...(2n-1)^2$$

$$= [1^2 + 2^2 + 3^2 + 4^2 + 5^2 + ...(2n-1)^2 + (2n)^2] - [2^2 + 4^2 + 6^2 + ...(2n)^2]$$

$$= \frac{2n(2n+1)(4n+1)}{6} - 2^2\left[1^2 + 2^2 + ...n^2\right]$$

$$= \frac{n(2n+1)(4n+1)}{3} - 4\frac{n(n+1)(2n+1)}{6}$$

$$= \frac{n(2n+1)}{3}\left[(4n+1) - 2(n+1)\right]$$

$$= \frac{n(2n+1)(2n-1)}{3}$$

$$= \frac{n(4n^2-1)}{3}$$

$$\therefore \text{Var} = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$\therefore \sigma^2 = \frac{n(4n^2-1)}{3n} - n^2$$

$$= \frac{4n^2-1-3n^2}{3} = \frac{n^2-1}{3}$$

and $\sigma = \sqrt{\frac{n^2-1}{3}}$

Important Note

Variance of first n Natural Numbers is $\boxed{\frac{n^2-1}{12}}$

Variance of first n Odd Natural Numbers is $\boxed{\frac{n^2-1}{3}}$

Variance of first n Even Natural Numbers is $\boxed{\frac{n^2-1}{3}}$

Ques. If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16, then (m + n) is equal to

[JEE Main-2020 (January)]

Ans. $\frac{n^2-1}{12} = 10$

$\Rightarrow n^2 - 1 = 120$

$\Rightarrow n^2 = 121$

$\Rightarrow n = 11$

$\frac{m^2-1}{3} = 16$

$\Rightarrow m^2 - 1 = 48$

$\Rightarrow m^2 = 49$

$\Rightarrow m = 7$

$\therefore m + n = 18$

Ques. Let the mean and variance of 8 numbers x, y, 10, 12, 6, 12, 4, 8 be 9 and 9.25 respectively. If x > y then 3x - 2y is equal to

[JEE Main-2023 (08 Apr.-1)]

Ans. $\bar{x} = 9$ & Var = 9.25

$$\therefore 9 = \frac{x+y+10+12+6+12+12}{8}$$

$\Rightarrow 8 \times 9 = x + y + 52$

$\Rightarrow x + y = 72 - 52$

$\Rightarrow x + y = 20$ ...(i)

and $\frac{\sum x_i^2}{8} - 9^2 = 9.25$

$\Rightarrow \frac{\sum x_i^2}{8} = 81 + 9.25$

$\Rightarrow \frac{x^2+y^2+100+144+36+144+16+64}{8} = 90.25$

$\Rightarrow x^2 + y^2 + 504 = 722$

$\Rightarrow x^2 + y^2 = 218$ ...(ii)

By eqⁿ (i) and (ii)

x = 13, y = 7

$\therefore 3x - 2y = 39 - 14 = 25$

Ques. 5 students of a class have an average height 150 cm and variance 18 cm^2. A new student, whose height is 156 cm, joined them. The variance (in cm^2) of the height of these six student is:

[JEE Main-2019 (January)]

(a) 16 (b) 22 (c) 20 (d) 18

Ans. $\frac{\sum x_i}{5} = 150$

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$\sum x_i = 750$

New Mean $= \frac{750 + 156}{6} = 151$

$\because \frac{\sum x_i^2}{5} - (150)^2 = 18 \Rightarrow \sum_{i=1}^{5} x_i^2 = 5 \times 22518$

$\Rightarrow \frac{\sum x_i^2}{5} = 22500 + 18$

New Var $= \frac{\sum_{i=1}^{6} x_i^2}{6} - (151)^2$

$= \frac{5 \times 22518 + (156)^2}{6} - (151)^2$

$= \frac{5(22500 + 18) + (156)^2 - 6(150 + 1)^2}{6}$

$= \frac{5 \times 22500 + 90 + (156)^2 - 6[22500 + 1 + 300]}{6}$

$= \frac{90 + (156)^2 - 22500 - 301 \times 6}{6}$

$= \frac{90 + (156)^2 - (150)^2 - 301 \times 6}{6}$

$= \frac{90 + 306 \times 6 - 301 \times 6}{6}$

$= 15 + 306 - 301 = 15 + 5 = 20$

Ques. If the mean and variance of five observations are $\frac{24}{5}$ and $\frac{194}{25}$ and the mean of first four observations is $\frac{7}{2}$, then the variance of the first four observations is equal to

[JEE Main-2024 (Jan.-II)]

(a) $\frac{4}{5}$ (b) $\frac{77}{12}$ (c) $\frac{5}{4}$ (d) $\frac{105}{4}$

Ans. $\frac{x_1 + x_2 + ...x_5}{5} = \frac{24}{5}$ ...(i)

and $\frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{7}{2}$

$\Rightarrow x_1 + x_2 + x_3 + x_4 = 14$ ...(ii)

By eqⁿ (i) & (ii)

$x_5 = 10$

$\because \frac{\sum_{i=1}^{5} x_i^2}{5} - \left(\frac{24}{5}\right)^2 = \frac{194}{25}$

$\Rightarrow 5\sum x_i^2 - (24)^2 = 194$

$\Rightarrow \sum_{i=1}^{5} x_i^2 = \frac{(24)^2 + 194}{5}$

$\because x_5 = 10$

$\therefore \sum_{i=1}^{4} x_i^2 + (10)^2 = \frac{(24)^2 + 194}{5}$

$\Rightarrow \sum_{i=1}^{4} x_i^2 = \frac{(24)^2 + 194}{5} - 100$

$\Rightarrow \sum_{i=1}^{4} x_i^2 = \frac{76 + 194}{5} = \frac{270}{5} = 54$

New variance $= \frac{\sum x_i^2}{4} - (\bar{x})^2$

$= \frac{54}{4} - \frac{49}{4}$

$\therefore$ New variance $= \frac{5}{4}$

Ques. The mean and variance of the marks obtained by the students in a test are 10 and 4 respectively. Later, the marks of one of the students is increased from 8 to 12. If the new mean of the marks is 10.2. then their new variance is equal to:

[JEE Main-2023 (25 Jan.-1)]

(a) 4.04 (b) 4.08 (c) 3.96 (d) 3.92

Ans. $\bar{x} = 10$, Variance = 4

$\therefore \frac{x_1 + x_2 + ... + x_n}{n} = 10$

$\Rightarrow \sum x_i = 10n$

$\because$ New mean = 10.2

$\therefore 10.2 = \frac{10n - 8 + 12}{n}$

$\Rightarrow 10.2n = 10n + 4$

$\Rightarrow 0.2n = 4$

$\Rightarrow \frac{1}{5}n = 4 \Rightarrow n = 20$

$\because \frac{\sum x_i^2}{20} - (10)^2 = 4$

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$$\Rightarrow \frac{\sum x_i^2}{20} = 104 \Rightarrow \sum x_i^2 = 104 \times 20$$

$$\sum x_{i\ new}^2 = 104 \times 20 - 8^2 + 12^2$$

$$= 2080 + (12 - 8)(12 + 8)$$

$$= 2080 + 4 \times 20 = 2080 + 80$$

$$Var_{new} = \frac{2080 + 80}{20} - (10.2)^2$$

$$= 104 + 4 - (10.2)^2$$

$$= 108 - (10 + 0.2)^2$$

$$= 108 - [100 + 0.04 + 2 \times 10 \times 0.2]$$

$$= 8 - 0.04 - 4 \quad = 3.96$$

Ques. The mean and variance of 10 observations were calculated as 15 and 15 respectively by a student who took by mistake 25 instead of 15 for one observation. Then, the correct standard deviation is ____

Ans. $\bar{x} = 15$

Variance = 15

$$\therefore \frac{x_1 + x_2 + ... + x_{10}}{10} = 15$$

$$\Rightarrow \sum x_i = 150$$

$$\sum x_{i\ correct} = 150 - 25 + 15 = 140$$

$$\text{Correct Mean} = \frac{140}{10} = 14$$

$$\because \frac{\sum x_i^2}{10} - (15)^2 = 15$$

$$\Rightarrow \frac{\sum x_i^2}{10} = 15^2 + 15$$

$$\Rightarrow \sum x_i^2 = 15(16) \times 10 = 60 \times 40 = 2400$$

$$\sum x_{i\ correct}^2 = 2400 - 25^2 + 15^2$$

$$= 2400 - 400 = 2000$$

$$Var_{correct} = \frac{2000}{10} - (14)^2 = 200 - 196 = 4 = \sigma^2$$

Hence, standard deviation (σ) = 2

Ques. The mean and standard deviation of the marks of 10 students were found to be 50 and 12 respectively. Later, it was observed that two marks 20 and 25 were wrongly read as 45 and 50 respectively. Then the correct variance is

Ans. $\bar{x} = 50$

Variance = $(12)^2$

$$\therefore \frac{\sum x_i}{10} = 50$$

$$\Rightarrow \sum x_i = 500$$

$$\therefore \sum x_{i\ correct} = 500 - 45 - 50 + 20 + 25 = 450$$

$$\therefore \text{Correct Mean} = \frac{450}{10} = 45$$

$$\because (12)^2 = \frac{\sum x_i^2}{10} - (50)^2$$

$$\Rightarrow \frac{\sum x_i^2}{10} = (12)^2 + (50)^2$$

$$\Rightarrow \sum x_i^2 = 10[144 + 2500] = 26440$$

$$\therefore \sum x_{i\ correct}^2 = 26440 - 45^2 - 50^2 + 20^2 + 25^2$$

$$\sum x_i^2 = 26440 + (25 - 45)(25 + 45) + (20 + 50)(20 - 50)$$

$$= 26440 - (20 \times 70) + (70 \times (-30))$$

$$= 26440 - 70\ (50)$$

$$= 26440 - 3500$$

$$= 22940$$

$$Var_{correct} = \frac{22940}{10} - (45)^2$$

$$= 2294 - (45)^2 = 2294 - 2025 = 269$$

VARIANCE AND SD FOR FREQUENCY DATA

The variance of a distribution is, the mean of squares of deviation of variate from their mean. It is denoted by σ^2 or var (x).

Hence standard deviation, $\sigma = +\sqrt{\text{variance}}$

(ii) For frequency Distribution

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$$\sigma_x^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N}$$

$$\sigma_x^2 = \frac{\sum f_i x_i^2}{N} - (\bar{x})^2 = \frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N}\right)^2$$

Ques. Find the variance σ^2 of the data

[JEE Main-2024 (Jan.-II)]

x_i	0	1	5	6	10	12	17
f_i	3	2	3	2	6	3	3

Ans.

x_i	f_i	$x_i f_i$	$f_i x_i^2$
0	3	0	0
1	2	2	2
5	3	15	75
6	2	12	72
10	6	60	600
12	3	36	432
17	3	51	867
	$\Sigma f_i = 22$	$\Sigma x_i f_i = 176$	$\Sigma f_i x_i^2 = 2048$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{176}{22} = \frac{88}{11} = 8$$

$$\text{Variance} = \frac{\sum f_i x_i^2}{N} - (\bar{x})^2$$

$$\Rightarrow \text{Variance} = \frac{2048}{22} - (8)^2$$

$$= \frac{1024}{11} - 64 = 93 - 64 = 29$$

Mathematical Properties of variance

(1) Variance $(x_i + \lambda)$ = Variance (x_i)

(2) Variance $(\lambda x_i) = \lambda^2$ Variance (x_i)

(3) Variance $(ax_i + b) = a^2$. Variance (x_i)

where λ, a, b, are constant

Important Note

$1, 3, 5, 7, \ldots\ldots 2n-1\}$ Var $= \frac{n^2 - 1}{3}$

add 1 $\{2, 4, 6, 8, 10, \ldots\ldots 2n\} \rightarrow$ Var $= \frac{n^2 - 1}{3}$

Ques. Find the Variance of the 2025, 2026, 2027,.....100 numbers?

Ans. Subtract 2024

2025, 2026, ... = 1, 2, 3, 4,.....100

$$\sigma^2 = \frac{n^2 - 1}{12}$$

Put n = 100

$$= \frac{(100)^2 - 1}{12} = \frac{9999}{12}$$

Ques. The mean and the standard deviation (s.d.) of 10 observations are 20 and 2 respectively. Each of these 10 observations is multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new (s.d.) become half of their original values, then q is equal to

[JEE Main-2020 (January)]

(a) -20 (b) 10 (c) -10 (d) -5

Ans. $x_i \rightarrow p x_i - q$

new mean = $(P\bar{x} - q)$

$\Rightarrow 10 = 20p - q$

$\Rightarrow 20p - q = 10$ ...(i)

also, $\sigma_2^2 = p^2 \sigma_1^2$

$\Rightarrow (1)^2 = p^2 (2)^2$

$\Rightarrow P^2 = \frac{1}{4} \Rightarrow P = \pm \frac{1}{2} \Rightarrow P = \frac{1}{2}$

By eq^n (i):

$20 \times \frac{1}{2} - q = 10$

$\Rightarrow 10 - q = 10 \Rightarrow q = 0$

$\therefore P = \frac{1}{2}$ (reject because $q \neq 0$)

$\therefore P = \frac{-1}{2}$

By eq^n (i):

$20\left(\frac{-1}{2}\right) - q = 10$

$\Rightarrow -10 - q = 10 \Rightarrow q = -20$

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Brain Teaser

Consider 10 observation $x_1, x_2, ...x_{10}$, such that $\sum_{i=1}^{10}(x_i - \alpha) = 2$ and $\sum_{i=1}^{10}(x_i - \beta)^2 = 40$, where α, β are positive integers. let the mean and the variance of the observation be $\frac{6}{5}$ and $\frac{84}{25}$ respectively. the $\frac{\beta}{\alpha}$ is equal to:

[JEE Main-2024 (Feb-II)]

(a) 2 (b) $\frac{3}{2}$ (c) $\frac{5}{2}$ (d) 1

Ans. $\frac{\sum x_i}{10} = \frac{6}{5} \Rightarrow \sum x_i = 12$

$X_i \to X_i - \beta$

Var $= \frac{84}{25}$, $\bar{x} = \frac{6}{5} - \beta$

$\because \sum_{i=1}^{10}(x_i - \alpha) = 2$

$\Rightarrow \sum x_i - \sum \alpha = \sum x_i - 10\alpha = 2$

$\Rightarrow 12 - 10\alpha = 2 \Rightarrow 10 = 10\alpha$

$\Rightarrow \alpha = 1$

$\because \frac{84}{25} = \frac{\sum(x_i - \beta)^2}{10} - \left(\frac{6}{5} - \beta\right)^2$

$\therefore \frac{84}{25} = \frac{40}{10} - \left(\frac{6}{5} - \beta\right)^2$

$\Rightarrow \frac{84}{25} = 4 - \left(\frac{6}{5} - \beta\right)^2 \Rightarrow \left(\frac{6}{5} - \beta\right)^2 = 4 - \frac{84}{25}$

$\Rightarrow \left(\frac{6}{5} - \beta\right)^2 = \frac{16}{25} \Rightarrow \frac{6}{5} - \beta = +\frac{4}{5}$

Either $\frac{6}{5} - \beta = \frac{4}{5}$

$\Rightarrow \beta = \frac{2}{5}$ [Reject because β is an integer]

or $\frac{6}{5} - \beta = -\frac{4}{5}$

$\Rightarrow \beta = \frac{10}{5} = 2$

Hence, $\frac{\beta}{\alpha} = 2$

Ques. The mean square deviation of a set of n observations $x_1, x_2, ..., x_n$ about a point c is defined as $\frac{1}{n}\sum_{i=1}^{n}(x_i - c)^2$. The mean square deviation about -2 and 2 are 18 and 10 respectively, then standard deviation of this set of observaions is

(a) 3 (b) 2

(c) 1 (d) None of these

Ans. $\frac{1}{n}\sum_{i=1}^{n}(x_i + 2)^2 = 18$

$\Rightarrow \frac{1}{n}\sum\left(x_i^2 + 4 + 4x_i\right) = 18$

$\Rightarrow \frac{\sum x_i^2}{n} + \frac{\sum 4}{n} + 4\frac{\sum x_i}{n} = 18$

$\because \alpha + \frac{4n}{n} + 4\beta = 18$

$\Rightarrow \alpha + 4\beta = 14$ ...(i)

$\because \frac{1}{n}\sum_{i=1}^{n}(x_1 - 2)^2 = 10$

$\Rightarrow \frac{1}{n}\sum\left(x_i^2 + 4 - 4x_i\right) = 10$

$\Rightarrow \frac{\sum x_i^2}{n} + \frac{\sum 4}{n} - 4\frac{\sum x_i}{n} = 10$

$\Rightarrow \alpha + 4 - 4\beta = 10$

$\Rightarrow \alpha - 4\beta = 6$ ...(ii)

Adding eq^n (i) & (ii)

$2\alpha = 20$

$\alpha = 10$ and $\beta = 1$

$\therefore \frac{\sum x_i^2}{n} = 10$

$\bar{x} = \frac{\sum x_i}{n} = 1$

Therefore, $\sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$

$\Rightarrow \sigma^2 = 10 - 1^2 = 9 \Rightarrow \sigma = +3$

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COMBINED VARIANCE

If means of two series containing n_1, n_2 terms are $\bar{x}_1$, $\bar{x}_2$ and their variance's are σ_1^2, σ_2^2 respectively and their combined mean is $\bar{x}$ then the variance σ^2 of their combined series is given by following formula,

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)}$$

where $d_1 = \bar{x}_1 - \bar{x}$, $d_2 = \bar{x}_2 - \bar{x}$

(i) If $n_1 = n_2$

$$\sigma^2 = \frac{\sigma_1^2 + \sigma_2^2}{2} + \frac{1}{4}(\bar{x}_1 - \bar{x}_2)^2$$

(ii) If $\bar{x}_1 = \bar{x}_2$

$$\Rightarrow \sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2}$$

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

$$= \left(\frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2}\right) + \left(\frac{n_1d_1^2 + n_2d_2^2}{n_1 + n_2}\right)$$

$$\because \bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

$$\therefore \bar{x}_1 - \bar{x} = \bar{x}_1 - \frac{(n_1\bar{x}_1 + n_2\bar{x}_2)}{n_1 + n_2} = \frac{n_2\bar{x}_1 - n_2\bar{x}_2}{n_1 + n_2}$$

$$d_1 = \frac{n_2(\bar{x}_1 - \bar{x}_2)}{n_1 + n_2}$$

Similarly $d_2 = \frac{n_1(\bar{x}_2 - \bar{x}_1)}{n_1 + n_2}$

Consider:

$$\frac{n_1d_1^2 + n_2d_2^2}{n_1 + n_2} = \frac{\frac{n_1n_2^2(\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2} + \frac{n_2n_1^2(\bar{x}_2 - \bar{x}_1)^2}{(n_1 + n_2)^2}}{n_1 + n_2}$$

$$= n_1n_2(\bar{x}_1 - \bar{x}_2)^2\left[\frac{n_2 + n_1}{(n_1 + n_2)^2(n_1 + n_2)}\right]$$

$$= \frac{n_1n_2(\bar{x}_1 - \bar{x}_2)^2}{(n_1 + n_2)^2}$$

Hence, also $\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$

Ques. A group of 10 observation has mean 5 and S.D. $2\sqrt{6}$. Another group of 20 observation has mean 5 and S.D. $3\sqrt{2}$, then the S.D. of combined group of 30 observation is:

(a) $\sqrt{5}$ (b) $2\sqrt{5}$

(c) $3\sqrt{5}$ (d) None of these

Ans.

$n_1 = 10$	$n_2 = 20$
$\bar{x}_1 = 5$	$\bar{x}_2 = 5$
$\sigma_1 = 2\sqrt{6}$	$\sigma_2 = 3\sqrt{2}$
$\sigma_1^2 = 24$	$\sigma_2^2 = 18$

$$\because \sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2}$$

$$\Rightarrow \sigma^2 = \frac{10 \times 24 + 20 \times 18}{10 + 20} = \frac{24 + 36}{3} = 20$$

$$\Rightarrow \sigma^2 = 20 \Rightarrow \sigma = 2\sqrt{5}$$

Ques. The mean and variance of a set of 15 numbers are 12 and 14 respectively. The mean and variance of another set of 15 number are 14 and σ^2 respectively. If the variance of all the 30 numbers in the two sets is 13, then σ^2 is equal to:

(a) 10 (b) 11 (c) 9 (d) 12

Ans.

$n_1 = 15$	$n_2 = 15$
$\bar{x}_1 = 12$	$\bar{x}_2 = 14$
$\sigma_1^2 = 14$	σ_2^2

$\sigma^2 = 13$

$$\because \sigma^2 = \frac{15(\sigma_1^2 + \sigma_2^2)}{15 + 15} + \frac{15 \times 15}{30^2}(\bar{x}_1 - \bar{x}_2)^2$$

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$$= \frac{\sigma_1^2 + \sigma_2^2}{2} + \frac{1}{4}\left(\bar{x}_1 - \bar{x}_2\right)^2$$

$\bar{x}_1 = 12, \bar{x}_2 = 14$

$$\Rightarrow 13 = \frac{14 + \sigma_2^2}{2} + \frac{1}{4}(2)^2$$

$$\Rightarrow 12 = \frac{14 + \sigma_2^2}{2}$$

$$\Rightarrow 24 = 14 + \sigma_2^2 \quad \Rightarrow \sigma_2^2 = 10$$

Ques. An online exam is attempted by 50 candidates out of which 20 are boys. The average marks obtained by boys is 12 with a variance 2. The variance of marks obtained by 30 girls is also 2. The average marks of all 50 candidates is 15. If μ is the average marks of girls and σ^2 is the variance of marks of 50 candidates, then $\mu + \sigma^2$ is equal to:

[JEE Main Aug. 27, 2021(II)]

Ans.

B	G
$n_1 = 20$	$n_2 = 30$
$\bar{x}_1 = 12$	$\bar{x}_2 = \mu$
$\sigma_1^2 = 2$	$\sigma_2^2 = 2$

$\bar{x} = 15, \sigma^2 = ?$

$$\bar{x} = 15 = \frac{2 \times 12 + 3 \times \mu}{2 + 3}$$

$$\Rightarrow 75 = 24 + 3\mu$$

$$\Rightarrow 25 = 8 + \mu$$

$$\Rightarrow \mu = 17$$

$$\therefore \sigma^2 = \frac{20 \times 2 + 30 \times 2}{50} + \frac{20 \times 30}{50^2}\left[12 - 17\right]^2$$

$$\Rightarrow \sigma^2 = 2 + \frac{6}{25} \times 25$$

$$\Rightarrow \sigma^2 = 8$$

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Short Notes

WAYS TO REPRESENT DATA

- **Ungrouped data:** In this case the value of variate are written in the form of a list.
- **Ungrouped Frequency distribution:** In this case data is represented in tabular form where one column consists of different values of variate whereas the other column consists of their respective frequencies.
- **Grouped Frequency distribution:** In this case data is grouped into intervals. These groupings are called class intervals and their size is called class width or class size. In each of these classes the least number is called lower class limit while the greatest number is called upper class limit.

ARITHMETIC MEAN

- **For ungrouped distribution:** If $x_1, x_2, ... x_n$ are n values of variate x_i then their A.M. $\bar{x}$ is

defined as $\bar{x} = \frac{x_1 + x_2 + ... + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$

$\Rightarrow \Sigma x_i = n\bar{x}$

- **For ungrouped and grouped frequency distribution:** If $x_1, x_2, ... x_n$ are values of variate with corresponding frequencies $f_1, f_2, ... f_n$ then their A.M. is given by

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + ... + f_nx_n}{f_1 + f_2 + ... + f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{N}, \text{ where } N = \sum_{i=1}^{n} f_i$$

- **Weighted Means:** Let $a_1, a_2, ... a_n$ be n-positive real numbers and $x_1, x_2, ..., x_n$ be n-positive rational numbers, then

Weighted Arithmetic Mean

$$A = \frac{x_1a_1 + x_2a_2 + ... + x_na_n}{x_1 + x_2 + ... + x_n}$$

Weighted Geometyric Mean

$$G = (a_1^{x_1}.a_2^{x_2}...a_n^{x_n})^{\frac{1}{x_1+x_2+...+x_n}}$$

Weighted Harmonic Mean

$$H = \frac{x_1 + x_2 + ... + x_n}{\frac{x_1}{a_1} + \frac{x_2}{a_2} + ... + \frac{x_n}{a_n}}$$

PROPERTIES OF ARITHMETIC MEAN

- If $\bar{x}$ is the mean of variate x_i then

AM of $(x_i + \lambda) = (\bar{x} + \lambda)$
AM of $(\lambda x_i) = \lambda \bar{x}$
AM of $(ax_i + b) = a\bar{x} + b$
(where λ, a, b are constants)

- Sum of deviations of variate from their AM is always zero

i.e. $\Sigma(x_i - \bar{x}) = 0, \Sigma f_i(x_i - \bar{x}) = 0$

- Sum of square of deviations of variate from their AM is minimum i.e. $\Sigma(x_i - \bar{x})^2$ is minimum

BY SHORT CUT METHOD OR ASSUMED MEAN METHOD

If the value of x_i are large, then the calculation of A.M. by using previous formula is quite tedious and time consuming. In such case we take deviation of variate from an arbitrary point a.

Let $d_i = x_i - a$, where a is assumed mean.

MEDIAN

- The median of a series is the value of middle term of the series when the values are written in ascending order.

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{ term, (when n is odd)} \\ \text{Mean of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2}+1\right)^{th} \text{ terms, (when n is even)} \end{cases}$$

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- **For Ungrouped frequency distribution:** First we prepare the cumulative frequency (c.f.) column and find value of N then

$$\text{Median} = \begin{cases} \left(\frac{N+1}{2}\right)^{th} \text{ term, (when N is odd)} \\ \text{Mean of } \left(\frac{N}{2}\right)^{th} \text{ and } \left(\frac{N}{2}+1\right)^{th} \text{ terms, (when N is even)} \end{cases}$$

- **For Grouped frequency distribution:** Prepare c.f. column and find value of N then find the class which contain value of c.f. is equal or just greater to N/2, this is median class.

$$\therefore \text{Median} = l + \frac{\left(\frac{N}{2} - F\right)}{f} h$$

where

l = lower limit of median class

f = freq. of median class

F = c.f. of the class preceding median class

h = class interval of median class

MODE

In a frequency distribution the mode is the value of that variate which has the maximum frequency.

Method for determining mode

- **For ungrouped distribution:** The value of that variate which is repeated maximum number of times.
- **For ungrouped frequency distribution:** The value of that variate which have maximum frequency.
- **For grouped frequency distribution:** First we find the class which have maximum frequency, this is model class.

$$\therefore \text{Mode} = l + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} h$$

Where,

l - lower limit of model class

f_0 - frequency of the model class

f_1 - frequency of the class preceding model class

f_2 - frequency of the class succeeding model class

h - class interval of model class

RELATION BETWEEN MEAN, MEDIAN AND MODE

In a moderately asymmetric distribution, following is the relation between mean, median and mode of a distribution. It is known as Empirical formula.

Mode = 3 Median - 2 Mean.

- Median always lies between mean and mode.
- For a symmetric distribution the mean, median and mode are coinciding.

MEASURES OF DISPERSION

Generally the following measures of dispersion are commonly used,

- Range
- Mean deviation
- Variance and standard deviation

RANGE

The difference between the greatest and least values of variate of a distribution, are called the range of that distribution.

If the distribution is grouped distribution, then its range is the difference between upper limit of the maximum class and lower limit of the minimum class.

Mean deviation

The mean deviation of a distribution is, the mean of absolute value of deviations of variate from their statistical average (Mean, Median, Mode).

If A is any statistical average of a distribution then mean deviation about A is defined as

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} |x_i - A|}{n} \text{ (for ungrouped dist.)}$$

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} f_i |x_i - A|}{N} \text{ (for freq. dist.)}$$

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Variance or Standard Deviation

The variance of a distribution is, the mean of squares of deviation of variate from their mean. It is denoted by σ^2 or var (x).

Hence standard deviation, $\sigma = +\sqrt{\text{variance}}$

- For Ungrouped Distribution

$$\sigma_x^2 = \frac{\Sigma (x_i - \bar{x})^2}{n}$$

$$\sigma_x^2 = \frac{\Sigma x_i^2}{n} - \bar{x}^2 - = \frac{\Sigma x_i^2}{n} - \left(\frac{\Sigma x_i}{n}\right)^2$$

- For Frequency Distribution

$$\sigma_x^2 = \frac{\Sigma f_i (x_i - \bar{x})^2}{N}$$

$$\sigma_x^2 = \frac{\Sigma f_i x_i^2}{N} - (\bar{x})^2 = \frac{\Sigma f_i x_i^2}{N} - \left(\frac{\Sigma f_i x_i}{N}\right)^2$$

- Variance of first n Natural Numbers is $\frac{n^2 - 1}{12}$
- Variance of first n Odd Natural Numbers is $\frac{n^2 - 1}{3}$
- Variance of first n Even Natural Numbers is $\frac{n^2 - 1}{3}$

Mathematical Properties of variance

- Var. $(x_i + \lambda)$ = Var. (x_i)
- Var.$(\lambda x_i) = \lambda^2$ Var. (x_i)
- Var$(ax_i + b) = a^2$. Var. (x_i)

where λ, a, b are constant

COMBINED MEAN

If $\bar{x}_1$ and $\bar{x}_2$ be the means of two groups having n_1 and n_2 terms respectively then the mean (combined mean) of their composite groups is given by,

$$\text{Combined mean} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

If there are more than two groups then,

$$\text{Combined mean} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3 + \ldots}{n_1 + n_2 + n_3 + \ldots}$$

COMBINED VARIANCE

If means of two series containing n_1, n_2 terms are $\bar{x}_1, \bar{x}_2$ and their variance's are σ_1^2, σ_2^2 respectively and their combined mean is $\bar{x}$ then the variance σ^2 of their combined series is given by following formula

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)},$$

where $d_1 = \bar{x}_1 - \bar{x}$, $d_2 = \bar{x}_2 - \bar{x}$

$$\text{Also } \sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$$

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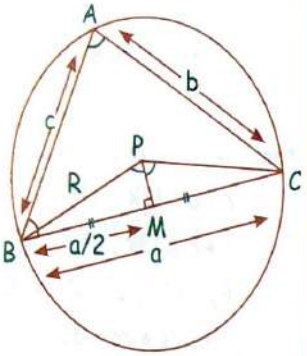
Chapter

17 Solution of Triangle

SINE LAW



In ΔPBM

$\because \angle BPC = 2A$

and $\Delta BMP \cong \Delta CMP$

$\therefore \angle BPM = \angle CPM = \frac{2A}{2}$

$\Rightarrow \angle BPM = A$

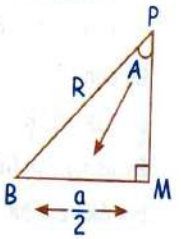
In ΔPBM

$\sin A = \frac{BM}{BP}$

$\Rightarrow \sin A = \frac{a/2}{R}$

$\Rightarrow \sin A = \frac{a}{2R}$

$\Rightarrow \frac{a}{\sin A} = 2R$

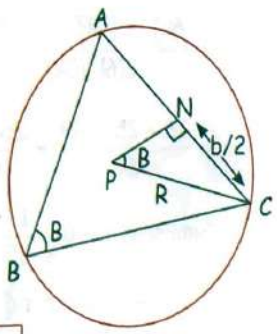


In ΔPNC

$\sin B = \frac{\left(\frac{b}{2}\right)}{R}$

$\Rightarrow \frac{b}{\sin B} = 2R$

Similarly $\frac{c}{\sin C} = 2R$



$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R}$$

where, R is the Radius of circum-circle.

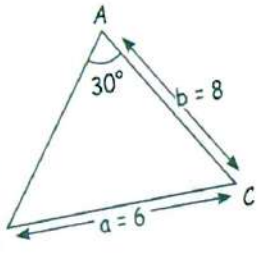
Important Note

Sine law relates 2 angles and sides.

Ques. In triangle ABC, $A = 30°$, $b = 8$, $a = 6$, then $B = \sin^{-1}x$, where $x =$

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) 1

Ans. Sine Law



$\frac{a}{\sin A} = \frac{b}{\sin B}$

$\Rightarrow \frac{6}{\sin 30°} = \frac{8}{\sin B}$

$\Rightarrow 6 \times 2 = \frac{8}{\sin B}$

$\Rightarrow \sin B = \frac{8}{12} = \frac{2}{3}$

$\Rightarrow B = \sin^{-1}\frac{2}{3} = \sin^{-1}x$

Hence, $x = \frac{2}{3}$

Ques. In a triangle ABC, if $a = 2$, $B = 60°$ and $C = 75°$, then $b =$

(a) $\sqrt{3}$ (b) $\sqrt{6}$ (c) $\sqrt{9}$ (d) $1 + \sqrt{2}$

Ans. $\because A + 60° + 75° = 180°$

$A = 105° - 60°$

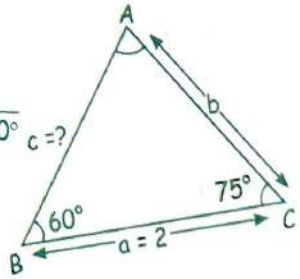
$A = 45°$

$\frac{2}{\sin A} = \frac{b}{\sin 60°}$

$\Rightarrow \frac{2}{\sin 45°} = \frac{b}{\sin 60°}$

$\Rightarrow \frac{2}{\frac{1}{\sqrt{2}}} = \frac{b}{\frac{\sqrt{3}}{2}}$

$\Rightarrow b = \sqrt{6}$



COSINE LAW

$BD = x$

$(AD)^2 = c^2 - x^2$ ...(i)

$\because DC = a - x$

In ΔADC

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$(AD)^2 = b^2 - (a - x)^2$ ...(ii)

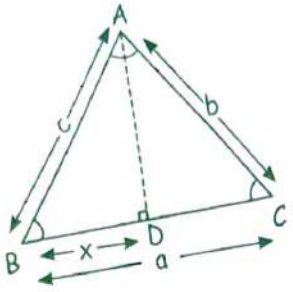
By eqn. (i) and (ii):

$c^2 - x^2 = b^2 - (a - x)^2$

$\Rightarrow c^2 - x^2 = b^2 - a^2 - x^2 + 2ax$

$\Rightarrow a^2 + c^2 - b^2 = 2ax$

$\Rightarrow x = \dfrac{a^2 + c^2 - b^2}{2a}$



ΔABD

$\cos B = \dfrac{x}{c}$

$$\boxed{\cos B = \frac{a^2 + c^2 - b^2}{2ac}}$$

Similarly,

$$\boxed{\cos C = \frac{b^2 + a^2 - c^2}{2ab}}$$

Important Note

Cosine law relates 1 angle & 3 sides.

Ques. If the sides of a triangle are p, q and $\sqrt{p^2 + pq + q^2}$, then the biggest angle is

(a) $\dfrac{\pi}{2}$ (b) $\dfrac{2\pi}{3}$ (c) $\dfrac{3\pi}{4}$ (d) $\dfrac{4\pi}{3}$

Ans. $\because a = p, b = q$

$\therefore c = \sqrt{p^2 + q^2 + pq}$

Clearly c is largest $\Rightarrow \angle C$ is largest

$\therefore \cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$

$= \dfrac{p^2 + q^2 - (p^2 + q^2 + pq)}{2pq} = -\dfrac{pq}{2pq}$

$\Rightarrow \cos C = \dfrac{-1}{2}$

$\Rightarrow C = 120° = \dfrac{2\pi}{3}$

Ques. In ΔABC, If $(a + b + c)(a - b + c) = 3ac$, then $\angle B =$

Ans. $(a + c + b)(a + c - b) = 3ac$

$\Rightarrow (a + c)^2 - b^2 = 3ac$

$\Rightarrow a^2 + c^2 + 2ac - b^2 = 3ac$

$\Rightarrow a^2 + c^2 - b^2 = ac$

$\Rightarrow \dfrac{a^2 + c^2 - b^2}{2ac} = \dfrac{1}{2}$

$\Rightarrow \cos B = \dfrac{1}{2}$

$\Rightarrow \angle B = 60°$

Ques. With usual notations, if in a triangle ABC, if $\dfrac{b + c}{11} = \dfrac{c + a}{12} = \dfrac{a + b}{13} = k$ then prove that $\dfrac{\cos A}{7} = \dfrac{\cos B}{19} = \dfrac{\cos C}{25}$

Ans. $b + c = 11k$ ...(i)

$c + a = 12k$ ...(ii)

$a + b = 13k$ ...(iii)

By adding eq^n (i), (ii) and (iii)

$2b + 2c + 2a = 36k$

$\Rightarrow a + b + c = 18k$

By eqn (i)

$a + 11k = 18k \Rightarrow a = 7k$

By using eqn (ii) and (iii), we get.

$b = 6k, \ c = 5k$

$\because \cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$,

$= \dfrac{(6k)^2 + (5k)^2 - (7k)^2}{2(6k)(5k)}$

$= \dfrac{36 + 25 - 49}{60} = \dfrac{25 - 13}{60} = \dfrac{12}{60} = \dfrac{1}{5} = \dfrac{7}{35}$

Similarly: $\cos B = \dfrac{19}{35}$, $\cos C = \dfrac{5}{7} = \dfrac{25}{35}$

$\therefore \cos A : \cos B : \cos C = 7 : 19 : 25.$

Hence, $\dfrac{\cos A}{7} = \dfrac{\cos B}{19} = \dfrac{\cos C}{25}$

Ques. In ΔABC, $\dfrac{\sin(A - B)}{\sin(A + B)} =$

(a) $\dfrac{a^2 - b^2}{c^2}$ (b) $\dfrac{a^2 + b^2}{c^2}$

(c) $\dfrac{c^2}{a^2 - b^2}$ (d) $\dfrac{c^2}{a^2 + b^2}$

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Ans. $$\frac{\sin A\cos B - \cos A\sin B}{\sin(180^\circ - C)}$$

$$= \frac{\sin A\cos B - \cos A\sin B}{\sin C}$$

$$\because \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} = k$$

$$\therefore \sin A = ak$$

$$\sin B = bk$$

$$\sin C = ck$$

$$= \frac{ka\cos B - \cos A.kb}{kc}$$

$$= \frac{a\cos B - b\cos A}{c}$$

$$= \frac{a\left(\frac{a^2 + c^2 - b^2}{2ac}\right) - b\left(\frac{b^2 + c^2 - a^2}{2bc}\right)}{c}$$

$$= \frac{\left(a^2 + c^2 - b^2\right) - \left(b^2 + c^2 - a^2\right)}{2c^2} = \frac{2a^2 - 2b^2}{2c^2}$$

$$= \frac{a^2 - b^2}{c^2}$$

Ques. The smallest of the triangle whose sides are $6 + \sqrt{12}$, $\sqrt{24}$, $\sqrt{48}$, is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

Ans. $\because$ b is the smallest side.

$\therefore \angle B$ will be the smallest angle.

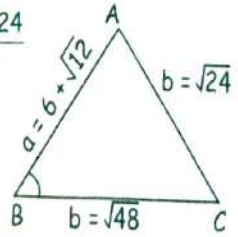
$$\therefore \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$= \frac{\left(6 + \sqrt{12}\right)^2 + 48 - 24}{2\left(6 + \sqrt{12}\right)\sqrt{48}}$$

$$= \frac{36 + 12 + 12\sqrt{12} + 24}{2\left(6 + \sqrt{12}\right)2\sqrt{12}}$$

$$= \frac{72 + 12\sqrt{12}}{4\sqrt{12}\left(6 + \sqrt{12}\right)}$$

$$= \frac{12\left(6 + \sqrt{12}\right)}{4\sqrt{12}\left(6 + \sqrt{12}\right)}$$



$$= \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \cos B = \frac{\sqrt{3}}{2} \Rightarrow B = 30^\circ = \frac{\pi}{6}$$

Ques. In ΔABC, $a\sin(B - C) + b\sin(C - A) + c\sin(A - B) =$

(a) 0 (b) $a + b + c$

(c) $a^2 + b^2 + c^2$ (d) $2(a^2 + b^2 + c^2)$

Ans. $\sin A = \sin(180^\circ - (B + C)) = \sin(B + C)$

$$\because \frac{a}{\sin A} = k$$

$$\therefore a = k\sin A$$

$$b = k\sin B,\ c = k\sin C$$

$$\therefore a\sin(B-C) + b\sin(C-A) + c\sin(A - B)$$
$$= k\sin A\sin(B - C) + k\sin B\sin(C - A) + k\sin C\sin(A - B)$$

$$= k\{\sin(B + C)\sin(B - C) + \sin(C + A)\sin(C - A) + \sin(A + B)\sin(A - B)\}$$

$$= k\{\sin^2 B - \sin^2 C + \sin^2 C - \sin^2 A + \sin^2 A - \sin^2 B\}$$
$$= k \times 0 = 0$$

$\sin(A + B)\sin(A - B) = \sin^2 A - \sin^2 B$

$\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B$

$2\sin A\sin B = \cos(A - B) - \cos(A + B)$

Ques. In ΔABC, if $b^2 + c^2 = 3a^2$, then $\cot B + \cot C - \cot A =$

(a) 1 (b) $\frac{ab}{4\Delta}$ (c) 0 (d) $\frac{ac}{4\Delta}$

Ans. $\because b^2 + c^2 = 3a^2$

$$\therefore \frac{b^2 + c^2 - a^2}{2bc} = \frac{2a^2}{2bc}$$

$$\Rightarrow \cos A = \frac{(k\sin A)^2}{k\sin B \cdot k\sin C} = \frac{\sin^2 A}{\sin B\sin C}$$

$$\Rightarrow \cot A = \frac{\sin A}{\sin B\sin C} \quad \ldots(i)$$

$$\because \cot B + \cot C - \cot A = \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C} - \frac{\cos A}{\sin A}$$

$$= \frac{\cos B\sin C + \sin B\cos C}{\sin B\sin C} - \frac{\cos A}{\sin A}$$

$$= \frac{\sin(B + C)}{\sin B\sin C} - \cot A$$

$\sin(B + C) = \sin(180^\circ - A) = \sin A$

$$= \frac{\sin A}{\sin B\sin C} - \cot A$$

By equation (i):

$= 0$

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Ques. In ΔABC, a^2, b^2, c^2 be in A.P., then cot A, cot B, cot C, are in:

(a) H. P. (b) G. P.

(c) A. P. (d) None of these

Ans. Given: $b^2 - a^2 = c^2 - b^2$

$\because a = k\sin A, b = k\sin B, c = k\sin C$

$\Rightarrow k^2\sin^2B - k^2\sin^2A = k^2\sin^2C - k^2\sin^2B$

$\Rightarrow \sin^2B - \sin^2A = \sin^2C - \sin^2B$

$\Rightarrow \sin(B + A)\sin(B - A) = \sin(C + B)\sin(C - B)$

$\Rightarrow \sin(180° - C)\sin(B - A) = \sin(180° - A)\sin(C - B)$

$\Rightarrow \sin C\sin(B - A) = \sin A\sin(C - B)$

$\Rightarrow \sin C(\sin B\cos A - \cos B\sin A) = \sin A(\sin C\cos B - \cos C\sin B)$

Divide both sides by $\sin A\sin B\sin C$

$$\frac{\sin B\sin C\cos A}{\sin A\sin B\sin C} + \frac{\cos B\sin A\sin C}{\sin A\sin B\sin C} = \frac{\sin A\sin C\cos B - \sin A\sin B\cos C}{\sin A\sin B\sin C}$$

$\Rightarrow \cot A - \cot B = \cot B - \cot C$

$\Rightarrow \cot A + \cot C = 2\cot B$

$\therefore \cot A, \cot B, \cot C \rightarrow AP$

Ques. The angles A, B and C of a triangle ABC are in A.P. and $a : b = 1 : \sqrt{3}$. if c = 4cm, then the area (in sq. cm) of this triangle is :

(JEE Mains, 10 Apr 2019-II)

(a) $\frac{2}{\sqrt{3}}$ (b) $4\sqrt{3}$ (c) $2\sqrt{3}$ (d) $\frac{4}{\sqrt{3}}$

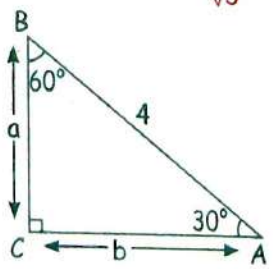
Ans. $\because 2B = A + C$

and $A + B + C = 180°$

$\therefore B + 2B = 180°$

$\Rightarrow 3B = 180°$

$\Rightarrow B = 60°$



$\because \frac{a}{\sin A} = \frac{b}{\sin B}$

$\Rightarrow \frac{a}{b} = \frac{\sin A}{\sin B}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{\sin A}{\sin 60°} \Rightarrow \frac{1}{\sqrt{3}} = \frac{2\sin A}{\sqrt{3}} \Rightarrow \sin A = \frac{1}{2}$

$\therefore A = 30°$

$\Rightarrow C = 90°$

$\therefore b = 4\cos 30°$

$\Rightarrow b = \frac{4\sqrt{3}}{2} = 2\sqrt{3}$

and $a = 4\sin 30° \Rightarrow a = 2$

$\therefore$ Area $= \frac{1}{2} \times 2\sqrt{3} \times 2$

$= 2\sqrt{3}$ sq. cm

Ques. In a triangle ABC, given $a^4 + b^4 + c^4 = 2c^2a^2 + 2b^2a^2$. Then find angle A.

Ans. $\because a^4 + b^4 + c^4 - 2a^2c^2 - 2b^2a^2 = 0$

$\Rightarrow (a^2 - b^2 - c^2)^2 - 2b^2c^2 = 0$

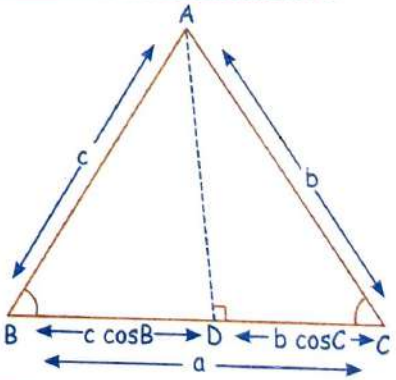
$\Rightarrow (a^2 - b^2 - c^2)^2 = 2b^2c^2$

$\Rightarrow (b^2 + c^2 - a^2)^2 = 2b^2c^2 \Rightarrow b^2 + c^2 - a^2 = \pm\sqrt{2}bc$

$$\Rightarrow \frac{b^2 + c^2 - a^2}{2bc} = \frac{\pm\sqrt{2}bc}{2bc}$$

$\Rightarrow \cos A = \frac{\pm 1}{\sqrt{2}} \Rightarrow A = \frac{\pi}{4}$ or $\frac{3\pi}{4}$

PROJECTION LAW



In any ΔABC;

(i) $a = b\cos C + c\cos B$

(ii) $b = c\cos A + a\cos C$

(iii) $c = a\cos B + b\cos A$

Ques. In any ΔABC, prove that

$$2\left[b\cos^2\left(\frac{C}{2}\right) + c\cos^2\left(\frac{B}{2}\right)\right] = a + b + c$$

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Ans. LHS: $b\,2\cos^2\frac{C}{2} + c.\,2\cos^2\left(\frac{B}{2}\right)$

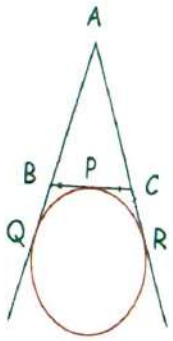
$= b(1 + \cos C) + c\,(1 + \cos B)$

$= b + b\cos C + c + c\cos B$

$= b + c + a = \text{RHS}$

Ques. In a triangle ABC, if a circle touches the side BC, AB produced and AC produced at P, Q & R respectively. Given, $(b + c)\cos A + (a + b)\cos C + (a + c)\cos B = 10$. Find the value of $(AQ)^2$

Ans. $b\cos A + c\cos A + a\cos C + b\cos C + a\cos B + c\cos B = 10$



$\Rightarrow c + b + a = 10$

$\because AQ = AR$

$\Rightarrow 2AQ = AQ + AR$

$= AB + BQ + AC + CR$

$= AB + AC + BP + CP$

$\Rightarrow 2AQ = AB + BC + AC = a + b + c$

$\Rightarrow 2AQ = 10 \Rightarrow AQ = 5$

$\Rightarrow (AQ)^2 = 25$

NAPIER'S ANALOGY (LAW OF TANGENTS)

In any ΔABC:

(i) $\tan\left(\frac{B-C}{2}\right) = \left(\frac{b-c}{b+c}\right)\cot\left(\frac{A}{2}\right)$

(ii) $\tan\left(\frac{A-B}{2}\right) = \left(\frac{a-b}{a+b}\right)\cot\left(\frac{C}{2}\right)$

(iii) $\tan\left(\frac{C-A}{2}\right) = \left(\frac{c-a}{c+a}\right)\cot\left(\frac{B}{2}\right)$

Ques. In a ΔABC, if $A = 30°$, $b = 2$, $c = \sqrt{3} + 1$, then $\frac{C-B}{2} =$

(a) 15° (b) 30°

(c) 45° (d) None of these

Ans. $\because \tan\left(\frac{C-B}{2}\right) = \frac{c-b}{c+b}\cot\left(\frac{A}{2}\right)$

$= \frac{\sqrt{3}+1-2}{\sqrt{3}+1+2}\cot 15°$

$= \frac{\sqrt{3}-1}{\sqrt{3}+3}\cot 15°$

$= \frac{\sqrt{3}-1}{\sqrt{3}(\sqrt{3}+1)} \times \frac{\sqrt{3}+1}{\sqrt{3}-1}$

$\Rightarrow \tan\left(\frac{C-B}{2}\right) = \frac{1}{\sqrt{3}} = \tan 30°$

$\Rightarrow \left(\frac{C-B}{2}\right) = 30°$

Ques. With the usual notation in ΔABC, if $\angle A + \angle B = 120°$, $a = \sqrt{3} + 1$ and $b = \sqrt{3} - 1$, then the ratio $\angle A : \angle B$, is: **(JEE Mains, 10 Jan 2019-II)**

(a) 7 : 1 (b) 5 : 3 (c) 9 : 7 (d) 3 : 1

Ans. $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{C}{2}$

$\because \angle A + \angle B = 120°$

$\therefore \angle C = 60°$

$\Rightarrow \tan\left(\frac{A-B}{2}\right) = \frac{\left(\sqrt{3}+1\right)-\left(\sqrt{3}-1\right)}{\sqrt{3}+1+\sqrt{3}-1}\cot 30°$

$= \frac{2}{2\sqrt{3}} \times \sqrt{3} \Rightarrow \tan\left(\frac{A-B}{2}\right) = 1$

$\therefore \frac{A-B}{2} = 45°$

$\Rightarrow A - B = 90°$ ...(i)

and $A + B = 120°$ ...(ii)

Adding eqn (i) & (ii)

$2A = 210°$

$\Rightarrow A = 105°$

$\therefore B = 15°$

Hence, $\frac{A}{B} = \frac{105°}{15°} = \frac{7}{1}$

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Ques. In a triangle ABC, let $AB = \sqrt{23}$, $BC = 3$ and $CA = 4$. Then the value of $\frac{\cot A + \cot C}{\cot B}$ is _____

Ans. $c = \sqrt{23}$, $a = 3$, $b = 4$

$$\therefore \frac{\cot A + \cot C}{\cot B} = \frac{\left(\frac{\cos A}{\sin A} + \frac{\cos C}{\sin C}\right)}{\left(\frac{\cos B}{\sin B}\right)}$$

$$= \frac{\left(\frac{\sin(A+C)}{\sin A \sin C}\right)}{\left(\frac{\cos B}{\sin B}\right)} = \frac{\sin B}{\sin A \cdot \sin C} \times \frac{\sin B}{\cos B}$$

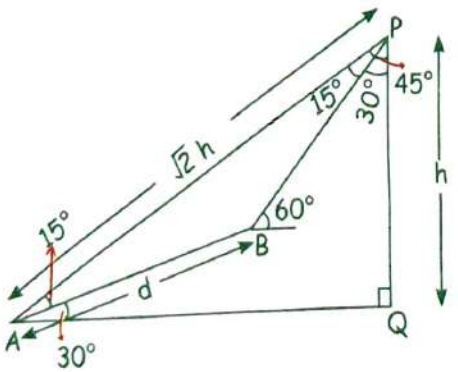
$$= \frac{\sin^2 B}{\sin A \sin C \cos B}$$

$$= \frac{(kb)^2}{ka \times kc} \times \frac{2ac}{a^2 + c^2 - b^2} = \frac{2b^2}{a^2 + c^2 - b^2}$$

$$= \frac{2(4)^2}{9 + 23 - 16} = \frac{32}{16} = 2$$

Ques. The angle of elevation of the top point P of the vertical tower PQ of height h from a point A is 45° and from a point B, the angles of elevation is 60°, where B is a point at a distance d from the point A measured along the line AB which makes an angle 30° with AQ. Prove that $d = h(\sqrt{3} - 1)$.

Ans.



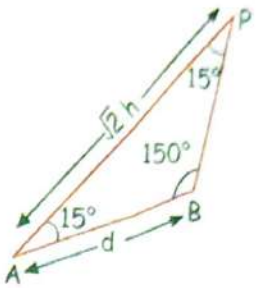
In $\triangle APQ$

$$\tan 45^\circ = \frac{PQ}{AQ}$$

$$\Rightarrow 1 = \frac{h}{AQ}$$

$$\Rightarrow AQ = h$$

In $\triangle ABP$: $\angle APB = 15^\circ$



$$\frac{d}{\sin 15^\circ} = \frac{\sqrt{2}h}{\sin 150^\circ}$$

$$\Rightarrow \frac{d2\sqrt{2}}{\sqrt{3} - 1} = \frac{\sqrt{2}h}{\frac{1}{2}}$$

$$\Rightarrow d = (\sqrt{3} - 1)h$$

LENGTH OF MEDIANS

$$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

$$m_b = \frac{1}{2}\sqrt{2c^2 + 2a^2 - b^2}$$

$$m_c = \frac{1}{2}\sqrt{2a^2 + 2b^2 - c^2}$$

By adding m_a^2, m_b^2, and m_c^2

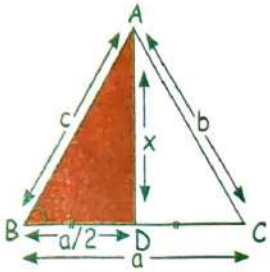
$$m_a^2 + m_b^2 + m_c^2 = \frac{1}{4}\left[3b^2 + 3c^2 + 3a^2\right]$$

$$\Rightarrow m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}\left[a^2 + b^2 + c^2\right]$$

Proof: In $\triangle ABC$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

In $\triangle ABD$



$$\cos B = \frac{c^2 + \left(\frac{a}{2}\right)^2 - x^2}{2 \times c \times \frac{a}{2}}$$

$$\text{equate : } \frac{a^2 + c^2 - b^2}{2ac} = \frac{c^2 + \frac{a^2}{4} - x^2}{ac}$$

$$\Rightarrow x = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

Length of Median: If m_a denotes the length of median from vertex A then $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

Ques. A lamp post is situated at the middle point M of the side AC of a triangular plot ABC with BC = 7m, CA = 8m and AB = 9m. Lamp post subtends an angle 15° at the point B. Determine the height of the lamp post.

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Ans. Let PM = h

$$BM = \frac{1}{2}\sqrt{2a^2 + 2c^2 - b^2}$$

$$= \frac{1}{2}\sqrt{2(7)^2 + 2(9)^2 - 8^2}$$

$$= \frac{1}{2}\sqrt{98 + 162 - 64}$$

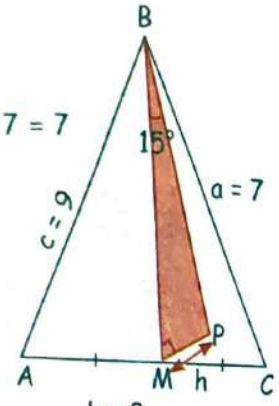
$$\frac{1}{2}\sqrt{98 + 98} = \frac{1}{2} \times 2 \times 7 = 7$$

In ΔPBM

$$\tan 15^\circ = \frac{h}{BM}$$

$$\Rightarrow 2 - \sqrt{3} = \frac{h}{7}$$

$$\Rightarrow h = 7(2 - \sqrt{3})$$



HALF ANGLE FORMULAS

$$\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}, \sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}},$$

$$\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$$

$$\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}, \cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$$

$$\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$$

$$\tan\frac{A}{2} = \frac{\Delta}{s(s-a)}, \tan\frac{B}{2} = \frac{\Delta}{s(s-b)}, \tan\frac{C}{2} = \frac{\Delta}{s(s-c)}$$

where, Δ Area of the triangle =

$$s = \text{Semi perimeter} = \frac{a+b+c}{2}$$

Proof:

$$\because \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow 2\cos^2\frac{A}{2} - 1 = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow 2\cos^2\frac{A}{2} = \frac{b^2 + c^2 - a^2}{2bc} + 1$$

$$\Rightarrow 2\cos^2\frac{A}{2} = \frac{b^2 + c^2 - a^2 + 2bc}{2bc} = \frac{(b+c)^2 - a^2}{2bc}$$

$$\Rightarrow 2\cos^2\frac{A}{2} = \frac{(b+c+a)(b+c-a)}{2bc}$$

$$\because a + b + c = 2s$$

$$\Rightarrow b + c = 2s - a$$

$$\Rightarrow 2\cos^2\frac{A}{2} = \frac{2s[2s - a - a]}{2bc}$$

$$\Rightarrow \cos^2\frac{A}{2} = \frac{s[2s - 2a]}{2bc}$$

$$\Rightarrow \cos^2\frac{A}{2} = \frac{s[s-a]}{bc} \Rightarrow \cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$$

$$\because \sin^2\frac{A}{2} = \sqrt{\frac{1 - \cos A}{2}}$$

Prove sin A/2 formula using the above formula

$$\because \tan\frac{A}{2} = \frac{\sin\frac{A}{2}}{\cos\frac{A}{2}}$$

$$= \sqrt{\frac{\frac{(s-b)(s-c)}{bc}}{\frac{s(s-a)}{bc}}}$$

$$= \sqrt{\frac{(s-b)(s-c)s(s-a)}{s(s-a)s(s-a)}}$$

$$= \sqrt{\frac{\Delta^2}{s^2(s-a)^2}} = \frac{\Delta}{s(s-a)}$$

Ques. In any triangle ABC, prove that

$$b\cos^2\left(\frac{C}{2}\right) + c\cos^2\left(\frac{B}{2}\right) = s$$

Ans. LHS: $b \times \frac{s(s-c)}{ab} + c \times \frac{s(s-b)}{ac}$

$$= \frac{s(s-c)}{a} + \frac{s(s-b)}{a}$$

$$= \frac{s}{a}[s - c + s - b] = \frac{s}{a}[2s - b - c]$$

$$= \frac{s}{a}[a + b + c - b - c] = \frac{s}{a} \times a = s$$

Ques. In a ΔABC,

Prove that $(b + c - a)\tan\left(\frac{A}{2}\right) = (c + a - b)\tan\left(\frac{B}{2}\right)$

$= (a + b - c)\tan\left(\frac{C}{2}\right)$

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Ans. $a + b + c = 2s$

$b + c = 2s - a$

$\because (b + c - a)\tan\left(\frac{A}{2}\right) = (b + c - a)\frac{\Delta}{s(s-a)}$

$= (2s - a - a)\frac{\Delta}{s(s-a)}$

$= \frac{2(s-a)\Delta}{s(s-a)} = \frac{2\Delta}{s}$

$\because (c + a - b)\tan\left(\frac{B}{2}\right) = (2s - b - b)\frac{\Delta}{s(s-b)}$

$= \frac{2(s-b)\Delta}{s(s-b)} = \frac{2\Delta}{s}$

Similarly, $(a + b - c)\tan\left(\frac{C}{2}\right) = \frac{2\Delta}{s}$

Hence, $(b + c - a)\tan\left(\frac{A}{2}\right) = (c + a - b)\tan\left(\frac{B}{2}\right)$

$= (a + b - c)\tan\left(\frac{C}{2}\right)$

Ques. In ΔABC, if $\tan\frac{A}{2}\tan\frac{C}{2} = \frac{1}{2}$, then a, b, c are in.

(a) A.P. (b) G.P.

(c) H.P. (d) None of these ✓

Ans. $\because \frac{\Delta}{s(s-a)}\frac{\Delta}{s(s-c)} = \frac{1}{2}$

$\Rightarrow \frac{\Delta^2}{s(s-a)s(s-c)} = \frac{1}{2}$

$\Rightarrow \frac{s(s-a)(s-b)(s-c)}{s(s-a)s(s-c)} = \frac{1}{2}$

$\Rightarrow \frac{s-b}{s} = \frac{1}{2} \Rightarrow 2s - 2b = s \Rightarrow s = 2b$

$\Rightarrow \frac{a+b+c}{2} = 2b$

$\Rightarrow a + b + c = 4b \Rightarrow a + c = 3b$

Ques. In ΔABC, BC is fixed and $\frac{\tan\frac{B}{2}}{\tan\frac{C}{2}} = k \neq 1$ is a constant. Show that vertex A lies on one or the other of the two branches of a hyperbola.

Ans. $a \to$ const

$\because \frac{\left(\frac{\Delta}{s(s-b)}\right)}{\left(\frac{\Delta}{s(s-c)}\right)} = k$

$\Rightarrow \frac{s-c}{s-b} = \frac{k}{1}$

By componendo and dividendo

$\frac{s-c+s-b}{s-c-s+b} = \frac{k+1}{k-1}$

$\Rightarrow \frac{2s-b-c}{b-c} = \frac{k+1}{k-1}$

$\Rightarrow \frac{a}{b-c} = \frac{k+1}{k-1}$

$\Rightarrow b - c = \frac{a(k-1)}{k+1} \longrightarrow$ const

$b - c =$ const

$\Rightarrow AC - AB =$ const

Hence, vertex A lies on a hyperbola.

Ques. If the length of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then the ratio of lengths of the sides of this triangle is:

(a) 5 : 9 : 13 (b) 4 : 5 : 6 ✓

(c) 3 : 4 : 5 (d) 5 : 6 : 7

Ans. $2b = a + c \Rightarrow s = \frac{a+b+c}{2} = \frac{3b}{2}$

also, $C = 2A$

$\Rightarrow \frac{C}{2} = A$

$\Rightarrow \sin\frac{C}{2} = 2\sin\frac{A}{2}\cos\frac{A}{2}$

$\Rightarrow \sqrt{\frac{(s-a)(s-b)}{ab}} = 2\sqrt{\frac{(s-b)(s-c)}{bc}} \cdot \sqrt{\frac{s(s-a)}{bc}}$

$\Rightarrow \frac{1}{a} = \frac{4s(s-c)}{bc^2}$

$\Rightarrow \frac{1}{a} = \frac{4(s-c)}{bc^2} \times \frac{3b}{2}$

$\Rightarrow \frac{1}{a} = \frac{6(s-c)}{c^2} \Rightarrow c^2 = 6a\left(\frac{3b}{2} - c\right)$

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$\because a = 2b - c$

$\Rightarrow c^2 = 6(2b - c)\left(\frac{3b}{2} - c\right)$

$\Rightarrow c^2 = 6\left[3b^2 - 2bc - \frac{3bc}{2} + c^2\right]$

$\Rightarrow c^2 = 6\left[3b^2 - \frac{7bc}{2} + c^2\right]$

$\Rightarrow c^2 = 18b^2 - 21bc + 6c^2$

$\Rightarrow 18b^2 - 21bc + 5c^2 = 0$

$\Rightarrow \frac{18b^2}{c^2} - \frac{21b}{c} + 5 = 0$

Let, $\frac{b}{c} = x$

$\therefore 18x^2 - 21x + 5 = 0$

$\Rightarrow 18x^2 - 15x - 6x + 5 = 0$

$\Rightarrow 3x[6x - 5] - 1(6x - 5) = 0$

$\therefore x = \frac{1}{3}$ (Reject because it gives the value of 'a' negative)

or $x = \frac{5}{6}$

$\Rightarrow \frac{b}{c} = \frac{5}{6}$ and $a = 2b - c = 4$

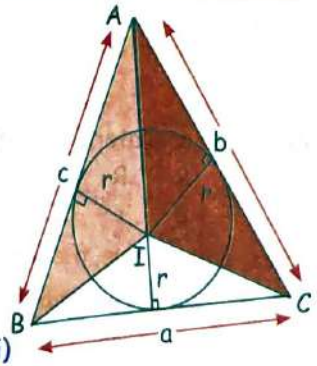
Hence, $a : b : c = 4 : 5 : 6$

FORMULA FOR IN-RADIUS

$r \rightarrow$ In Radius

$$r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2}$$

Proof:



$\Delta_{IBC} = \frac{1}{2}ra$ ...(i)

$\Delta_{IAC} = \frac{1}{2}rb$ ...(ii)

$\Delta_{IBA} = \frac{1}{2}rc$ ...(iii)

By adding eqn (i), (ii) & (iii)

$\Delta = \frac{1}{2}r(a + b + c) \Rightarrow \Delta = \frac{1}{2}r(2s) \Rightarrow r = \frac{\Delta}{s}$

$\because \tan\frac{A}{2} = \frac{\Delta}{s(s-a)}$

$\Rightarrow (s-a)\tan\frac{A}{2} = \frac{\Delta}{s}$

$\therefore r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2}$

Ques. The inradius of the triangle whose sides are 3, 5, 6, is-

(a) $\sqrt{\frac{8}{7}}$ (b) $\sqrt{8}$ (c) $\sqrt{7}$ (d) $\sqrt{\frac{7}{8}}$

Ans. $r = \frac{\Delta}{s}$

$\because s = \frac{3 + 5 + 6}{2} = 7$

and $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{7(7-3)(7-5)(7-6)}$

$= \sqrt{7 \times 4 \times 2 \times 1} = 2\sqrt{14}$

$\because r = \frac{\Delta}{s} = \frac{2\sqrt{14}}{7} = \frac{2\sqrt{2}}{\sqrt{7}} = \sqrt{\frac{8}{7}}$

FORMULA FOR CIRCUMRADIUS

$$R = \frac{abc}{4\Delta}$$

Proof: $\because \frac{a}{\sin A} = 2R$ [By Sine Rule]

$\Rightarrow R = \frac{a}{2\sin A}$

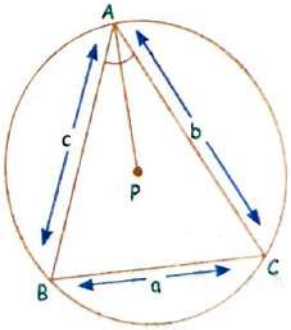
$\because$ Area of $\Delta = \frac{1}{2}ac\sin B = \frac{1}{2}bc\sin A = \frac{1}{2}ab\sin C$

$\Rightarrow \Delta = \frac{1}{2}bc\sin A$

$\Rightarrow \sin A = \frac{2\Delta}{bc}$

$\therefore R = \frac{a}{2 \times \frac{2\Delta}{bc}}$

$\Rightarrow R = \frac{abc}{4\Delta}$



Ques. Prove that in a triangle ABC, $\frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab} = \frac{1}{2Rr}$

Ans. LHS: $\frac{1}{bc} + \frac{1}{ac} + \frac{1}{ab}$

$= \frac{a + b + c}{abc}$

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$$= \left(\frac{2s}{\frac{abc}{4\Delta}} \cdot 4\Delta\right)$$

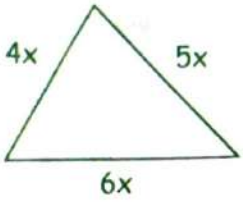
$$= \frac{2s}{R} \cdot 4\Delta = \frac{s}{2R\Delta}$$

$$= \frac{1}{2R\frac{\Delta}{s}} = \frac{1}{2Rr} = \text{RHS}$$

Ques. In a Δ ABC, the sides are in ratio 4 : 5 : 6. The ratio of the circum-radius and the in-radius is-

(a) 8 : 7 (b) 3 : 2 (c) 7 : 3 (d) 16 : 7

Ans. $a = 4x, b = 5x, c = 6x$



$\therefore a + b + c = 15x$

$$\Rightarrow s = \frac{15x}{2}$$

$$\Delta = \sqrt{\frac{15x}{2}\left(\frac{15x}{2} - 4x\right)\left(\frac{15x}{2} - 5x\right)\left(\frac{15x}{2} - 6x\right)}$$

$$\Rightarrow \Delta = \sqrt{\frac{15x}{2} \times \frac{7x}{2} \times \frac{5x}{2} \times \frac{3x}{2}}$$

$$\Rightarrow \Delta^2 = \frac{15 \times 7 \times 15x^4}{16}$$

$$\because \frac{R}{r} = \frac{abc}{4\Delta \times \frac{\Delta}{s}} = \frac{abc \times s}{4\Delta^2}$$

$$\Rightarrow \frac{R}{r} = \frac{4x \times 5x \times 6x \times \frac{15x}{2}}{\left(\frac{4 \times 15 \times 7 \times 15x^4}{16}\right)} \Rightarrow \frac{R}{r} = \frac{16}{7}$$

Ques. In ΔABC, the lengths of side AC and AB are 12cm and 5cm, respectively. If the area of ΔABC is $30cm^2$ and R and r are respectively the radii of circumcircle and incircle of ΔABC then the value of 2R + r (in cm) is equal to

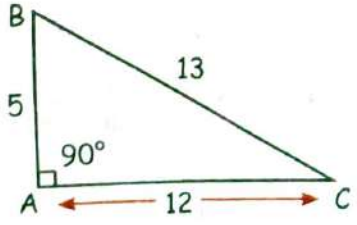
Ans. $\Delta = 30$

$$\Rightarrow 30 = \frac{1}{2} \times 5 \times 12 \sin A$$

$$\Rightarrow 30 = 30 \sin A$$

$$\Rightarrow \sin A = 1$$

$$\Rightarrow A = 90°$$



$$\therefore R = \frac{13}{2} \Rightarrow 2R = 13$$

and $r = \frac{\Delta}{s}$

$$\because s = \frac{5 + 12 + 13}{2} = \frac{30}{2}$$

$$\Rightarrow r = \frac{30}{15} \Rightarrow r = 2$$

Hence, $2R + r = 13 + 2 = 15$ cm

RELATION BETWEEN r AND R

$$r = 4R \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2}$$

Proof: $\text{RHS} = 4R \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2}$

$$= \frac{4abc}{4\Delta} \sqrt{\frac{(s-b)(s-c)}{bc} \times \frac{(s-a)(s-c)}{ac} \times \frac{(s-a)(s-b)}{ab}}$$

$$= \frac{abc}{\Delta} \frac{s(s-a)(s-b)(s-c)}{s \quad abc} = \frac{\Delta^2}{\Delta s} = \frac{\Delta}{s} = r = \text{LHS}$$

Important Note

$\because \cos A + \cos B + \cos C \le \frac{3}{2}$ for any Δ

$$\Rightarrow 1 + 4\sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2} \le \frac{3}{2}$$

$$\Rightarrow 1 + \frac{r}{R} \le \frac{3}{2} \Rightarrow \frac{r}{R} \le \frac{1}{2} \Rightarrow r \le \frac{R}{2}$$

when triangle is equilateral:

$$\Rightarrow r = \frac{R}{2}$$

Ques. The ratio of the circumradius and in radius of an equilateral triangle is:

(a) 3 : 1 (b) 1 : 2 (c) $2 : \sqrt{3}$ (d) 2 : 1

Ans. For equilateral triangle:

$$r = \frac{R}{2} \Rightarrow \frac{R}{r} = \frac{2}{1}$$

Ques. In a triangle, the sum of lengths of two sides is x and the product of the lengths of the same two sides is y. if $x^2 - c^2 = y$, where c is the length of the third side of the triangle, then the circumradius of the triangle is: **(JEE Mains 11 Jan 2019-I)**

(a) $\frac{3}{2}y$ (b) $\frac{c}{\sqrt{3}}$ (c) $\frac{c}{3}$ (d) $\frac{y}{\sqrt{3}}$

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Ans. $a + b = x$ & $ab = y$

$\because x^2 - c^2 = y$

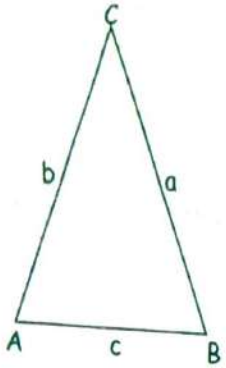
$\Rightarrow (a+b)^2 - c^2 = ab$

$\Rightarrow a^2 + b^2 + 2ab - c^2 = ab$

$\Rightarrow \dfrac{a^2+b^2-c^2}{2ab} = \dfrac{-ab}{2ab}$

$\Rightarrow \cos C = -\dfrac{1}{2}$

$\therefore \angle c = 120°$



$\Rightarrow \dfrac{c}{\sin C} = 2R \Rightarrow \dfrac{c}{\sin 120°} = 2R$

$\Rightarrow \dfrac{2c}{\sqrt{3}} = 2R \Rightarrow R = \dfrac{c}{\sqrt{3}}$

Ques. In ΔABC the value of $\dfrac{a\cos A + b\cos B + c\cos C}{a+b+c}$ is equal to

(a) $\dfrac{R}{r}$ (b) $\dfrac{R}{2r}$ (c) $\dfrac{r}{R}$ (d) $\dfrac{2r}{R}$

Ans. $\because \dfrac{a}{\sin A} = 2R \Rightarrow a = 2R\sin A$

$\therefore \dfrac{a\cos A + b\cos B + c\cos C}{a+b+c}$

$= \dfrac{2R\sin A\cos A + 2R\sin B\cos B + 2R\sin C\cos C}{2s}$

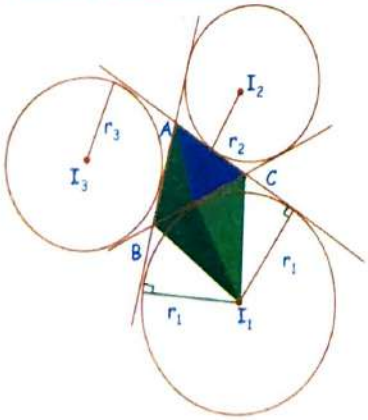
$= R\left[\dfrac{\sin 2A + \sin 2B + \sin 2C}{2s}\right]$

$= \dfrac{R\left[4\sin A\sin B\sin C\right]}{2s}$

$= \dfrac{2R}{s}\sin A\sin B\sin C = \dfrac{2R}{s}\cdot\dfrac{a}{2R}\cdot\dfrac{b}{2R}\cdot\dfrac{c}{2R}$

$= \dfrac{\Delta abc}{sR^2 4\Delta} = \dfrac{rR}{R^2} = \dfrac{r}{R}$

EX-RADII



$\Delta A_{I_1C} = \dfrac{1}{2}r_1 b$ ...(i)

$\Delta_{AI_1B} = \dfrac{1}{2}r_1 c$ ...(ii)

$\Delta_{BI_1C} = \dfrac{1}{2}r_1 a$ ...(iii)

Eqn (i) + Eqn (ii) - Eqn (iii)

$\Delta = \dfrac{1}{2}r_1\left[b + c - a\right]$

$= \dfrac{1}{2}r_1\left[2s - a - a\right] = \dfrac{1}{2}r_1(2s - 2a)$

$\Rightarrow \Delta = r_1(s-a) \Rightarrow r_1 = \dfrac{\Delta}{s-a} = s\tan\dfrac{A}{2}$

$r_1 = \dfrac{\Delta}{(s-a)} = s\tan\dfrac{A}{2}$

$r_2 = \dfrac{\Delta}{(s-b)} = s\tan\dfrac{B}{2}$

$r_3 = \dfrac{\Delta}{(s-c)} = s\tan\dfrac{C}{2}$

IN-RADIUS, CIRCUM-RADIUS AND EX-RADIUS

- $R = \dfrac{abc}{4\Delta}$
- $r = \dfrac{\Delta}{s} = (s-a)\tan\dfrac{A}{2} = (s-b)\tan\dfrac{B}{2}$
  $= (s-c)\tan\dfrac{C}{2} = 4R\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2}$
- $r_1 = \dfrac{\Delta}{s-a} = s\tan\dfrac{A}{2} = 4R\sin\dfrac{A}{2}\cos\dfrac{B}{2}\cos\dfrac{C}{2}$

  $r_2 = \dfrac{\Delta}{s-b} = s\tan\dfrac{B}{2} = 4R\sin\dfrac{B}{2}\cos\dfrac{C}{2}\cos\dfrac{A}{2}$

  $r_3 = \dfrac{\Delta}{s-c} = s\tan\dfrac{C}{2} = 4R\sin\dfrac{C}{2}\cos\dfrac{A}{2}\cos\dfrac{B}{2}$

Ques. Prove that in a Triangle ABC,

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2+b^2+c^2}{\Delta^2}$$

Ans. $\dfrac{s^2}{\Delta^2} + \dfrac{(s-a)^2}{\Delta^2} + \dfrac{(s-b)^2}{\Delta^2} + \dfrac{(s-c)^2}{\Delta^2}$

$= \dfrac{1}{\Delta^2}\left[s^2 + (s-a)^2 + (s-b)^2 + (s-c)^2\right]$

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$$= \frac{1}{\Delta^2}\left[s^2 + s^2 + a^2 - 2as + s^2 + b^2 - 2bs + s^2 + c^2 - 2cs\right]$$

$$= \frac{1}{\Delta^2}\left[4s^2 - 2s(a+b+c) + a^2 + b^2 + c^2\right]$$

$$= \frac{1}{\Delta^2}\left[4s^2 - 2s(2s) + a^2 + b^2 + c^2\right] = \frac{a^2+b^2+c^2}{\Delta^2}$$

Ques. If the ex-radii of a triangle are in H.P. then corresponding sides are in-

(a) A.P. (b) G.P.

(c) H.P. (d) None of these

Ans. $\frac{1}{r_1}, \frac{1}{r_2}, \frac{1}{r_3} \rightarrow$ A.P.

$\Rightarrow \frac{s-a}{\Delta}, \frac{s-b}{\Delta}, \frac{s-c}{\Delta} \rightarrow$ A.P.

$\Rightarrow s-a, s-b, s-c \rightarrow$ A.P.

$\Rightarrow -a, -b, -c \rightarrow$ A.P.

$\Rightarrow a, b, c \rightarrow$ A.P.

Ques. In a Triangle ABC, $r_2r_3 + r_3r_1 + r_1r_2 =$

(a) s^2 (b) r^2 (c) $\frac{s}{r^3}$ (d) R^2

Ans. $\frac{\Delta^2}{(s-b)(s-c)} + \frac{\Delta^2}{(s-c)(s-a)} + \frac{\Delta^2}{(s-a)(s-b)}$

$$= \Delta^2 \frac{s-a+s-b+s-c}{(s-a)(s-b)(s-c)}$$

$$= \frac{s(s-a)(s-b)(s-c)[3s-2s]}{(s-a)(s-b)(s-c)}$$

$$= s \cdot s = s^2$$

Ques. In a Triangle ABC, $(r_1 + r_2)(r_2 + r_3)(r_3 + r_1) =$

(a) Rs^2 (b) $2Rs^2$ (c) $3Rr^2$ (d) $4Rs^2$

Ans. $\Delta^3\left[\frac{1}{s-a}+\frac{1}{s-b}\right]\left[\frac{1}{s-b}+\frac{1}{s-c}\right]\left[\frac{1}{s-c}+\frac{1}{s-a}\right]$

$$= \Delta^3\left[\frac{2s-a-b}{(s-a)(s-b)}\right]\left[\frac{2s-b-c}{(s-b)(s-c)}\right]\left[\frac{2s-c-a}{(s-c)(s-a)}\right]$$

$$= s^2\Delta^3 \frac{abc}{s^2(s-a)^2(s-b)^2(s-c)^2}$$

$$= \frac{s^2\Delta^3 abc}{\Delta^4} = \frac{4s^2abc}{4\Delta} = 4s^2R$$

Ques. In an equilateral triangle, the in-radius, circumradius and one of the ex-radii are in the ratio-

(a) 2 : 3 : 5 (b) 1 : 2 : 3

(c) 1 : 3 : 7 (d) 3 : 7 : 9

Ans. In an equilateral triangle:

$s = \frac{3a}{2}$ and $\Delta = \frac{\sqrt{3}}{4}a^2$

$r_1 = \frac{\Delta}{s-a}$

$\Rightarrow r_1 = \frac{\sqrt{3}}{4}\cdot\frac{a^2}{\frac{3a}{2}-a} = \frac{\sqrt{3}}{4}\cdot\frac{2a^2}{a} \Rightarrow r_1 = \frac{\sqrt{3}a}{2}$

$\because r = \frac{\Delta}{s} = \frac{\sqrt{3}}{4}\frac{a^2}{\frac{3a}{2}}$

$= \frac{\sqrt{3}}{2}\cdot\frac{a}{3} = \frac{a}{2\sqrt{3}}$

$\therefore \frac{r}{r_1} = \frac{a \times 2}{2\sqrt{3} \times \sqrt{3}a} = \frac{1}{3} \Rightarrow r : r_1 = 1 : 3$

$\Rightarrow r : R = 1 : 2$ (For an equilateral triangle)

and $R = \frac{abc}{4\Delta} = \frac{a^3}{4.\left(\frac{\sqrt{3}}{4}a^2\right)} = \frac{a}{\sqrt{3}}$

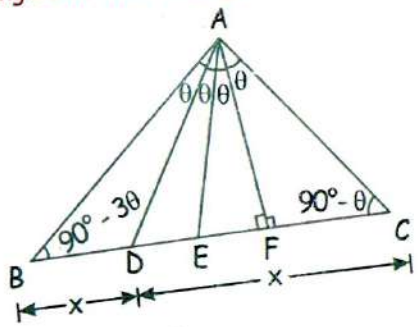
$R : r_1 = 2 : 3$

Hence, $r : R : r_1 = 1:2:3$

Brain Teaser

In a ΔABC, if the median, Angle bisector and Altitude respectively drawn from vertex A divides the angle A into 4 equal parts. Then find angles of this Triangle.

Ans.



AD is the median.

BD = DC = x (let)

In Δ ABD : $\frac{AD}{\sin(90^\circ - 3\theta)} = \frac{x}{\sin\theta}$

$\Rightarrow \frac{AD}{x} = \frac{\cos 3\theta}{\sin\theta}$ ...(i)

In Δ ADC : $\frac{AD}{\sin(90^\circ - \theta)} = \frac{x}{\sin 3\theta}$

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$$\frac{AD}{x} = \frac{\cos\theta}{\sin 3\theta} \quad \ldots(ii)$$

By eqn (i) and (ii)

$$\frac{\cos 3\theta}{\sin\theta} = \frac{\cos\theta}{\sin 3\theta} \Rightarrow 2\sin 3\theta \cos 3\theta = 2\sin\theta \cos\theta$$

$\Rightarrow \sin 6\theta = \sin 2\theta \Rightarrow 6\theta = 180° - 2\theta$.

$$\Rightarrow 8\theta = 180° \Rightarrow \theta = \frac{45°}{2} \Rightarrow \theta = \frac{\pi}{8}$$

$$\because \angle A = 4\theta = \frac{\pi}{2}$$

$$\angle B = \frac{\pi}{2} - \frac{3\pi}{8} = \frac{\pi}{8}$$

and $\angle C = \frac{\pi}{2} - \frac{\pi}{8} = \frac{3\pi}{8}$

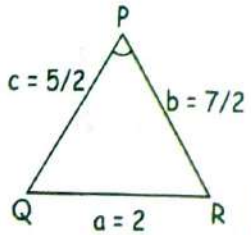
Ques. Let PQR be a triangle of area Δ with $a = 2$, $b = \frac{7}{2}$ and $c = \frac{5}{2}$; where a, b, and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals.

(a) $\frac{3}{4\Delta}$ (b) $\frac{45}{4\Delta}$ (c) $\left(\frac{3}{4\Delta}\right)^2$ (d) $\left(\frac{45}{4\Delta}\right)^2$

Ans. $$\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P} = \frac{2\sin P - 2\sin P\cos P}{2\sin P + 2\sin P\cos P}$$

$$= \frac{2\sin P(1 - \cos P)}{2\sin P(1 + \cos P)}$$

$$= \frac{\sin^2\frac{P}{2}}{\cos^2\frac{P}{2}}$$



$$= \tan^2\frac{P}{2} = \frac{\Delta^2}{s^2(s-a)^2}$$

$$= \frac{s(s-a)(s-b)(s-c)}{s^2(s-a)^2} = \frac{(s-b)(s-c)}{s(s-a)}$$

$$\because s = \frac{2 + \frac{5}{2} + \frac{7}{2}}{2} = \frac{8}{2} = 4$$

$$\therefore \frac{(s-b)(s-c)}{s(s-a)} = \frac{\left(4 - \frac{5}{2}\right)\left(4 - \frac{7}{2}\right)}{4(4-2)}$$

$$= \frac{\frac{3}{2} \times \frac{1}{2}}{4 \times 2} = \frac{3}{32} = \frac{3 \times 3}{32 \times 3} = \frac{3 \times 3}{16 \times 2 \times 3}$$

$\because \Delta^2 = s(s-a)(s-b)(s-c)$

$$= 4 \times 2 \times \frac{3}{2} \times \frac{1}{2} \Rightarrow \Delta^2 = 6$$

$$\therefore \frac{3 \times 3}{16 \times 2 \times 3} = \frac{3 \times 3}{16\Delta^2} = \frac{9}{16\Delta^2} = \left(\frac{3}{4\Delta}\right)^2$$

Ques. In a triangle PQR let $\angle PQR = 30°$ and the sides PQ and QR have length $10\sqrt{3}$ and 10, respectively. Then which of the following statement(S) is (are) TRUE? **(JEE Adv. 2018)**

(a) $\angle QPR = 45°$

(b) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120°$

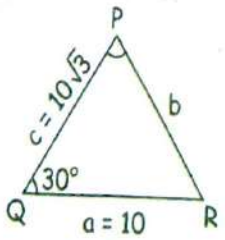
(c) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$

(d) The area of the circumcircle of the triangle PQR is 100π.

Ans. $$\because \cos 30° = \frac{10^2 + \left(10\sqrt{3}\right)^2 - b^2}{2 \times 10 \times 10\sqrt{3}}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{100 + 300 - b^2}{2 \times 100\sqrt{3}}$$

$\Rightarrow 300 = 100 + 300 - b^2 \Rightarrow b^2 = 100 \Rightarrow b = 10$



Isosceles $\Delta \Rightarrow \angle Q = \angle P = 30° \Rightarrow \angle R = 120°$

$$\therefore \Delta = \frac{1}{2} \times 10 \times 10\sqrt{3} \times \sin 30° = 25\sqrt{3}$$

$$\because r = \frac{\Delta}{s}$$

$$= \frac{25\sqrt{3}}{\left(\frac{20 + 10\sqrt{3}}{2}\right)} = \frac{25\sqrt{3}}{10 + 5\sqrt{3}}$$

$$= \frac{5\sqrt{3}}{\left(2 + \sqrt{3}\right)} \times \frac{2 - \sqrt{3}}{\left(2 - \sqrt{3}\right)} = 5\sqrt{3}\left(2 - \sqrt{3}\right)$$

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$= 10\sqrt{3} - 15$

$\because R = \frac{abc}{4\Delta}$ or $\frac{a}{\sin P} = 2R \Rightarrow \frac{10}{\sin 30^\circ} = 2R$

$10 \times 2 = 2R \Rightarrow R = 10$

Area of the circumcircle of the triangle $= \pi R^2$

$= 100\pi$

Ques. For a triangle ABC, the value of $\cos 2A + \cos 2B + \cos 2C$ is least. If its inradius is 3 and incentre is M, Then which of the following is NOT correct?

(a) Perimeter of ΔABC is $18\sqrt{3}$

(b) $\sin 2A + \sin 2B + \sin 2C = \sin A + \sin B + \sin C$

(c) $\overrightarrow{MA}.\overrightarrow{MB} = -18$

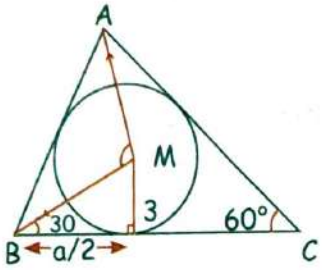
(d) area of ΔABC is $\frac{27\sqrt{3}}{2}$

Ans. $\because \tan 30^\circ = \frac{3}{\left(\frac{a}{2}\right)} \Rightarrow \frac{1}{\sqrt{3}} = \frac{6}{a} \Rightarrow a = 6\sqrt{3}$

$\therefore \Delta = \frac{\sqrt{3}}{4} \times \left(6\sqrt{3}\right)^2$

$= \frac{\sqrt{3}}{4} \times 36 \times 3$

$= 27\sqrt{3} \neq \frac{27\sqrt{3}}{2}$



PARAGRAPH

Consider an obtuse angled triangle ABC in which the difference between the largest and the smallest angle is $\frac{\pi}{2}$ and whose sides are in arithmetic progression.

Suppose that the vertices of this triangle lie on a circle of radius 1. (There are two questions based on PARAGRAPH "I", the question given below is one of them)

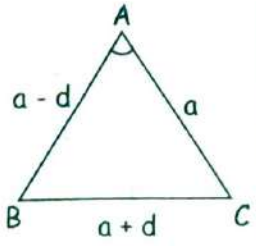
Ques. Let a be the area of the triangle ABC. Then the value of $(64a)^2$ is **(JEE Adv. 2023)**

Ans. $\because A - C = \frac{\pi}{2}$

$A + B + C = 180^\circ$

$\Rightarrow 90^\circ + C + B + C = 180^\circ$

$\Rightarrow B = 90^\circ - 2C$



and $R = 1$

$\therefore \frac{a+d}{\sin A} = \frac{a}{\sin B} = \frac{a-d}{\sin C} = 2$

$\Rightarrow \frac{a+d}{\cos C} = \frac{a}{\sin B} = \frac{a-d}{\sin C} = 2$

$\Rightarrow \frac{a+d}{\cos C} = \frac{a}{\cos 2C} = \frac{a-d}{\sin C} = 2$

$\Rightarrow a + d = 2\cos C$ ...(i)

$a - d = 2\sin C$ ...(ii)

$[\text{eqn(i)}]^2 + [\text{eqn(ii)}]^2$

$(a+d)^2 + (a-d)^2 = 4$

$\Rightarrow 2(a^2 + d^2) = 4$

$\Rightarrow a^2 + d^2 = 2$ ...(iii)

$[\text{eqn(i)}]^2 - [\text{eqn(ii)}]^2$

$(a+d)^2 - (a-d)^2 = 4\cos^2 C - 4\sin^2 C$

$\Rightarrow 4ad = 4\cos 2C$

$\Rightarrow \cos 2C = ad$ ...(iv)

$\because \frac{a}{\cos 2C} = 2$

$\Rightarrow \frac{a}{ad} = 2$ [By eqn (iv)]

$\Rightarrow \frac{1}{d} = 2 \Rightarrow d = \frac{1}{2}$

$\therefore a^2 + \frac{1}{4} = 2$ [By eqn (iii)]

$\Rightarrow a^2 = 2 - \frac{1}{4} = \frac{7}{4} \Rightarrow a = \frac{\sqrt{7}}{2}$

PEDAL TRIANGLE

Distance of Orthocentre from Vertices

(i) $PA = 2R\cos A$

(ii) $PB = 2R\cos B$

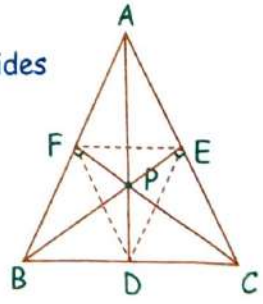
(iii) $PC = 2R\cos C$

Distance of Orthocentre from Sides

(i) $PD = 2R\cos B\cos C$

(ii) $PE = 2R\cos A\cos C$

(iii) $PF = 2R\cos A\cos B$



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Short Notes

IMPORTANT THEOREMS

- In a triangle ABC, the angles are denoted by capital letters A, B and C and the lengths of the sides opposite to these angles are denoted by small letters a, b and c respectively.
- Semi-perimeter of the triangle is written as $s = \frac{a+b+c}{2}$, and its area denoted by Δ.
- **Sine Rule:** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \lambda = \frac{abc}{2\Delta} = 2R$

  where R is the radius of the circum-circle of the ΔABC.
- **Cosine Rule:**

$$\cos A = \frac{b^2+c^2-a^2}{2bc}, \cos B = \frac{a^2+c^2-b^2}{2ac},$$

$$\cos C = \frac{a^2+b^2-c^2}{2ab}.$$

- **Length of Median:** If m_a denotes the length of median from vertex A then,

$$m_a = \frac{1}{2}\sqrt{2b^2+2c^2-a^2}$$

$$\text{Also } m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2+b^2+c^2)$$

- **Projection Rule:** $b\cos C + c\cos B = a$, $c\cos A + a\cos C = b$, $a\cos B + b\cos A = c$.
- **Napier's Analogy:**

$$\tan\frac{B-C}{2} = \frac{b-c}{b+c}\cot\frac{A}{2}, \tan\frac{C-A}{2} = \frac{c-a}{c+a}\cot\frac{B}{2},$$

$$\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2}.$$

TRIGONOMETRIC RATIOS OF HALF-ANGLES

- $\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}, \sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}},$

  $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}},$
- $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}, \cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}},$

  $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}},$
- $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}, \tan\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}},$

  $\tan\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}.$

AREA OF A TRIANGLE

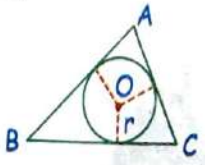
- $\Delta = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \frac{1}{2}ab\sin C$

  $= \sqrt{s(s-a)(s-b)(s-c)} = \frac{abc}{4R} = rs$

  where R and r are the radii of the circum-circle and the in-circle of the ΔABC respectively.
- The area of the cyclic quadrilateral with sides a, b, c and d is $\sqrt{(s-a)(s-b)(s-c)(s-d)}$

IN-CIRCLE

- The circle touching the three sides of the triangle internally is called the inscribed or the in-circle of the triangle. Its radius r is called the in-radius of the circle. In the triangle ABC,



- $r = \frac{\Delta}{s} = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2}$

  $= (s-c)\tan\frac{C}{2} = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$
- $2r \le R$. Here equality holds for the equilateral triangle.

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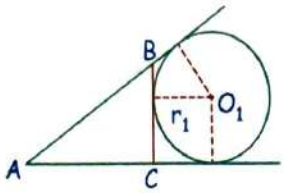
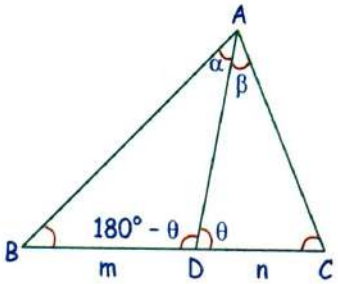
ESCRIBED CIRCLES

- The circle touching BC and the two sides AB and AC produced of ΔABC externally is called the escribed circle opposite A.
- Its radius is denoted by r_1. Similarly r_2 and r_3 denote the radii of the escribed circles opposite to angles B and C respectively. r_1, r_2, r_3 are called the ex-radii of ΔABC.



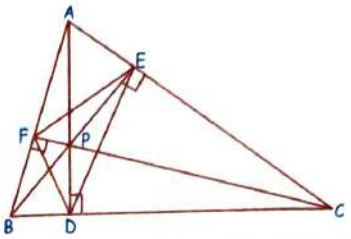
- $r_1 = \frac{\Delta}{s-a} = s\tan\frac{A}{2} = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$
- $r_2 = \frac{\Delta}{s-b} = s\tan\frac{B}{2} = 4R\sin\frac{B}{2}\cos\frac{C}{2}\cos\frac{A}{2}$,
- $r_3 = \frac{\Delta}{s-c} = s\tan\frac{C}{2} = 4R\sin\frac{C}{2}\cos\frac{A}{2}\cos\frac{B}{2}$

m-n THEOREM



- $(m+n)\cot\theta = m\cot\alpha - n\cot\beta$
- $(m+n)\cot\theta = n\cot B - m\cot C$

PEDAL TRIANGLE



- Point of intersection of altitudes is the orthocentre P & the triangle DEF which is formed by joining the feet of the altitudes is called the pedal triangle of triangle ABC.
- The distances of the ortho-centre from the angular points of the ΔABC are given by $PA = 2R\cos A$, $PB = 2R\cos B$ & $PC = 2R\cos C$
- The distance of P from sides are given by $PD = 2R\cos B\cos C$, $PE = 2R\cos C\cos A$ and $PF = 2R\cos A\cos B$.
- Side length of pedal triangle are given by $DE = R\sin 2C$, $EF = R\sin 2A$ & $DF = R\sin 2B$.
- Angles of Pedal Triangle $\angle D = \pi - 2A$, $\angle E = \pi - 2B$ & $\angle F = \pi - 2C$.
- The point which is orthocentre for original triangle ABC acts as the in-centre for the pedal triangle.

"Every triangle stands firm because of its three sides — just like your effort, patience, and focus."

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Chapter 9

Sequence & Series

INTRODUCTION

Sequence or Progression

(1) A.P. (Arithmetic Progression)

Example: → 1, 3, 5, 7, 9, _ _ _ _ _ _

(2) G.P. (Geometric Progression)

Example: 1, 4, 16, 64, - - -

(3) H.P (Harmonic Progression)

(4) A.G.P. (Arithmetic-Geometric. Progression)

Series

Sum of a sequence is called series

Example: 1 + 3 + 5 + 7 + 9 + ...

ARITHMETIC PROGRESSION (A.P.)

General Term of A.P.

$T_n = a + [n - 1]d$,

where a = First terms, d = common difference

If a is First term and d is common difference
Then AP: a, , a + b, a + 2b , a + 3b , a + 4b, ------

Example: - 7, - 4, - 1, 2, 5, 8, ... ⇒ a = - 7, d = 3

$T_{20} = a + 19d = -7 + 19 \times 3 = -7 + 57 = 50$

Ques. If m^{th} terms of an AP is n & n^{th} terms is m. then show that its $(m + n)^{th}$ term is Zero

Ans. $T_m = n = a + (m - 1)\, d$ ....(i)

$T_n = m = a + (n - 1)\, d$ ....(ii)

By eq (i), (ii)

$n - m = (m - 1)d - (n - 1)d.$

$\Rightarrow (n - m) = d\,[m - 1 - n + 1]$

$\Rightarrow -(m - n) = d\,(m - n)$

$\Rightarrow d = -1$

By eqⁿ (i)

$n = a + (m - 1)(-1)$

$\Rightarrow a = m + n - 1$

Now, $T_{m+n} = a + (m + n - 1)d$

$= m + n - 1 + (m + n - 1)(-1)$

$= m + n - 1 - m - n + 1 = 0$

Hence, $T_{m+n} = 0$.

Ques. Let $a_1 = 8, a_2, a_3, \ldots a_n$ be an A.P. If the sum of its first four terms is 50 and the sum of its last four terms is 170, then the product of its middle two terms is .....

[JEE Main Jan. 2023 Shift I]

Ans. Let AP is, a, a + d, a + 2d, a + 3d, ...

Given, a + a + d + a + 2d + a + 3d = 50

$\Rightarrow 4a + 6d = 50$

$\because a = 8$

$\Rightarrow 32 + 6d = 50$

$\Rightarrow 6d = 18$

$\Rightarrow d = 3$

Last four terms:

$T_{n-3}, T_{n-2}, T_{n-1}, T_n$

$\therefore 170 = a + (n - 1)\, d + a + (n - 2)d + a + (n - 3)d + a + (n - 4)d$

$\Rightarrow 170 = 4a + d\,[n - 1 + n - 2 + n - 3 + n - 4]$

$\Rightarrow 170 = 4a + d\,[4n - 10]$

$\Rightarrow 170 = 32 + 3\,[4n - 10]$ ($\because a = 8, d = 3$)

$\Rightarrow 138 = 3\,[4n - 10] \Rightarrow 46 = 4n - 10$

$\Rightarrow 4n = 56 \Rightarrow n = 14$

So, AP will be:

$T_1, T_2, T_3, T_4, __ T_6\ T_7\ T_8\ T_9 ___ T_{14}$

Middle two terms of AP are T_7 and T_8.

$T_7 = a + 6d = 8 + 6 \times 3 = 26$

$T_8 = a + 7d = 8 + 7 \times 3 = 29$

Hence, $T_7 T_8 = 26 \times 29 = 754$.

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Sum of n Terms of A.P.

$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}[a + L]$

Where a = first term, d = common difference and L = last term

Ex: 1, 3, 5, 7, 9, ... upto n terms

$a = 1, d = 2$

$S_{10} = \frac{n}{2}[2a + (n-1)d]$

$= \frac{10}{2}[2 \times 1 + 9 \times 2]$

$= 10[1 + 9]$

$= 10 \times 10 = 100$

$S_n = \frac{n}{2}[2a + (n-1)d]$

$= \frac{n}{2}[2 \times 1 + (n-1) \times 2] = n[1 + n - 1] = n^2$

Important Note

1. Sum of First n Natural Numbers is

   $1 + 2 + 3 + 4 + 5 ___ + n = \frac{n}{2}[1 + n] = \frac{n(n+1)}{2}$

2. Sum of First (n - 1) Natural Numbers is

   $1 + 2 + 3 + 4 + ___ + (n-1) = \frac{(n-1)}{2}[1 + n - 1]$

   $= \frac{(n-1)n}{2} = \frac{n(n-1)}{2}$

3. Sum of First n odd Natural Numbers is

   $1 + 3 + 5 + 7 + 9 + ____ = n^2$

Ques. Find the sum of all natural Numbers upto 200 which are either divisible by 2 or by 3 but not by both.

Ans. A: div by 2: $2 + 4 + 6 + 8 + ____ + 200$,

$n = 100$; $A = \frac{100}{2}[2 + 200] = 10100$

B: div by 3: $3 + 6 + 9 + 12 + - - - + 198$

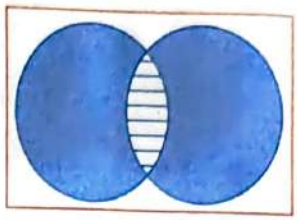
$n = 66$

$B = \frac{66}{2}[3 + 198] = 33 \times 201$

C: div by 6: $6 + 12 + 18 + - - - - - + 198$

$n = 33$

$C = \frac{33}{2}[6 + 198] = \frac{33}{2} \times 204 = 102 \times 33$



$A + B - 2C = 10100 + 33 \times 201 - 2 \times 33 \times 102$

$= 10100 + 33[201 - 204]$

$= 10100 + 33 \times (-3) = 10100 - 99 = 10001$

Ques. For $A = \{n \in [100, 700] \cap \mathbb{N}\}$, n is neither a multiple of 3 nor a multiple of 4}. Then the number of elements in A is

[JEE Main 6 Apr. 2024 Shift I]

(a) 290 (b) 280 (c) 300 (d) 310

Ans. p: multiple of 3: 102, 105, 108, ... , 699

$T_n = 699 = 102 + (n-1) \times 3$

$\Rightarrow \frac{597}{3} = n - 1 \Rightarrow 199 = n - 1 \Rightarrow n = 200$

q: multiple of 4: 100, 104, 108, ... 700

$\because 700 = 100 + (n-1) \times 4$

$\Rightarrow 600 = (n-1).4 \Rightarrow n - 1 = 150 \Rightarrow n = 151$

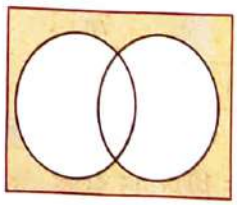
$p \cap q$: multiple of 12: 108, 120, ... 696

$\because 696 = 108 + (n-1) \times 12$

$\Rightarrow \frac{588}{12} = n - 1 \Rightarrow n = 50$

$\because n(A \cup B) = n(p) + n(q) - n(p \cap q)$

$\Rightarrow n(A \cup B) = 200 + 151 - 50 = 301$



Total Numbers: 100, 101, 102, 103, ___ 700

$\because 700 = a + (n-1) \times 1$

$\Rightarrow 700 = 100 + (n-1) \Rightarrow n = 601$

$n(A)$ = Total numbers - $n(A \cup B)$

$= 601 - n(A \cup B) = 601 - 301 = 300$

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Ques. If for an AP, $T_n = (4n - 7)$ Find S_n

Ans. If $n = 1$

$\Rightarrow T_1 = 4 \times 1 - 7$

$\Rightarrow T_1 = -3 = a$

If $n = 2$

$\Rightarrow T_2 = (4 \times 2 - 7)$

$\Rightarrow 1 = a + d$

$\Rightarrow 1 = -3 + d$

$\Rightarrow d = 4$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$= \frac{n}{2}[2 \times (-3) + (n-1) \times 4]$$

$$= n[-3 + 2(n-1)]$$

$$= n(2n - 5) = 2n^2 - 5n$$

Ques. If for an AP, $S_n = (4n^2 + 7n)$, Find T_n

Ans. Put $n = 1$, $S_1 = 4(1)^2 + 7(1) = 11$

$\Rightarrow a = 11$

$\because S_2 = T_1 + T_2$

Put $n = 2$

$\Rightarrow S_2 = 4(2)^2 + 7(2)$

$\Rightarrow T_1 + T_2 = 16 + 14 = 30$

$\Rightarrow a + a + d = 30$

$\Rightarrow 2 \times 11 + d = 30$

$\Rightarrow d = 8$

Now, $T_n = a + (n-1)d = 11 + (n-1)8 = 8n + 3$

Ques. The sums of first n terms of two A.P.'s are in the ratio $(7n + 1) : (4n + 27)$ for all n. If the ratio of their 11^{th} terms by p : q where p and q are co-primes then find the value of [p + q].

Ans. Given, $\frac{Sn}{Sn'} = \frac{7n+1}{4n+27}$

$$\Rightarrow \frac{\frac{n}{2}[2a + (n-1)d]}{\frac{n}{2}[2a_1 + (n-1)d_1]} = \frac{7n+1}{4n+27}$$

$$\Rightarrow \frac{2a + (n-1)d}{2a_1 + (n-1)d_1} = \frac{7n+1}{4n+27} \quad \forall\, n \in N$$

$$\Rightarrow \frac{a + \frac{(n-1)}{2}d}{a_1 + \frac{(n-1)}{2}d_1} = \frac{7n+1}{4n+27}$$

Put $\frac{n-1}{2} = 10$ or $n = 21$

$$\Rightarrow \frac{T_{11}}{T_{11}} = \frac{a + 10d}{a' + 10d'} = \frac{7 \times 21 + 1}{4 \times 21 + 27}$$

$$= \frac{147 + 1}{84 + 27} = \frac{148}{111} = \frac{37 \times 4}{37 \times 3} = \frac{4}{3}$$

Hence, $p + q = 4 + 3 = 7$

Common Terms of 2 A.P.s

Example: 1, 3, 5, 7, 9, 11, - - - $d_1 = 2$

1, 4, 7, 10, 13, - - - - $d_2 = 3$

common AP: 1, 7, 13, - - - -

First terms → Hit & trial, $d = LCM(2, 3) = 6$

$d_{\text{common AP}} = LCM\{d_1 \& d_2\}$

Ques. Let 3, 7, 11, 15, ... 403 and 2, 5, 8, 11 ...,404 be two arithmetic progressions. Then the sum, of the common terms in them, is equal to

[JEE Main 1 Feb 2024 Shift I]

Ans. $d_1 = 4$, $d_2 = 3$

$\Rightarrow LCM\{d_1, d_2\} = 12 = d$

First common terms = 11

Common AP: 11, 23, 35, 47, - - -.

Smaller of the both AP last term

$T_n < 403$

$\Rightarrow 11 + (n-1)12 < 403$

$\Rightarrow 12n - 1 < 403$

$\Rightarrow 12n < 404$

$\Rightarrow n < \frac{404}{12}$

$\Rightarrow n < \frac{101}{3}$

$\Rightarrow n_{max} = 33$

$$\text{Sum} = \frac{33}{2}[2 \times 11 + 32 \times 12]$$

$$= 33[11 + 32 \times 6]$$

$$= 33[11 + 192]$$

$$= 33 \times 203 = 6699$$

Ques. The sum of the common terms of the following three arithmetic progressions.

[JEE Main Feb. 2023 Shift I]

3, 7, 11, 15,.....399,

2, 5, 8, 11.....359 and

2, 7, 12, 17.....197, is equal to .....

Ans.

3, 7, 11, 15,.....399, $d_1 = 4$

2, 5, 8, 11.....359 and $d_2 = 3$

$\Big\}$ → 11, 23, 35, (47), ---

2, 7, 12, 17.....197, is equal to ..... $d_3 = 5$

$\Big\}$ → 2, 17, 32, (47)

$d = LCM\{d_1, d_2, d_3\} = 4 \times 3 \times 5 = 60$

$a = 47$

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common AP: 47, 107, 167, 167 + 60

Sum = 47 + 107 + 167 = 321

Important Note

If sum of numbers in A.P. is given then always assume the numbers to be.

3 numbers in A.P. : $a - d, a, a + d$

4 numbers in A.P. : $a - 3d, a - d, a + d, a + 3d$

5 numbers in A.P. : $a - 2d, a - d, a, a + d, a + 2d$

6 numbers in A.P. : $a - 5d, a - 3d, a - d, a + d, a + 3d, a + 5d$

Ques. An AP has an even number of terms. The sum of the odd numbered terms is 24 and that of even numbered terms is 30. The last terms exceeds the 1st by $\frac{21}{2}$. Find the numbers of terms.

Ans. Let, no of terms terms = n (even)

Given, $T_1 + T_3 + T_5 + \text{------} + T_{n-1} = 24$ ...(i)

& $T_2 + T_4 + T_6 + \text{------} + T_n = 30$ ...(ii)

& $T_n = a + \frac{21}{2}$

$\Rightarrow a + (n - 1)\, d = a + \frac{21}{2}$

$\Rightarrow (n - 1)\, d = \frac{21}{2}$ ...(iii)

equation (ii) - (i)

$\underbrace{(T_2 - T_1)}_{d} + \underbrace{(T_4 - T_3)}_{d} + \underbrace{(T_6 - T_5)}_{d} + \text{---} \underbrace{(T_n - T_{n-1})}_{d} = 30 - 24$

$\Rightarrow \frac{n}{2} \times d = 6$

$\Rightarrow nd = 12$ ...(iv)

Divide eq. (iii) by (iv)

$\Rightarrow \frac{(n - 1)\, d}{nd} = \frac{\frac{21}{2}}{12} \Rightarrow \frac{n - 1}{n} = \frac{7}{8} \Rightarrow n = 8$

Ques. Let AP (a; d) denote the set of all the terms of an infinite arithmetic progression with first terms a and common difference $d > 0$.
if $AP[1; 3] \cap AP[2; 5] \cap AP[3; 7] = AP[a; d]$
then $a + d$ equals ____ **[JEE Main 2019]**

Ans. $a_1 = 1$ $a_2 = 2$ $a_3 = 3$

$d_1 = 3$ $d_2 = 5$ $d_3 = 7$

$d = \text{lcm}(3, 5, 7) = 105$

AP - 1: 1, 4, 7, 10, 13, - - - - → 7, 22, 37, 52, - - -

AP - 2: 2, 7, 12, 17, 22, - - → 17, 52, - - -

AP - 3: 3, 10, 17, 24, 31, - - -

$a = 52, d = 105$

Hence, $a + d = 157$

Ques. Four numbers are in A.P. If their sum is 20 and the sum of their squares is 120, then find the numbers.

Ans. Let numbers be $a - 3d, a - d, a + d, a + 3d$

Add, $4a = 20 \Rightarrow a = 5$

$\therefore$ no's are $5 - 3d, 5 - d, 5 + d, 5 + 3d$

$\Rightarrow (5 - 3d)^2 + (5 - d)^2 + (5 + d)^2 + (5 + 3d)^2 = 120$

$\Rightarrow 2(25 + 9d^2) + 2[25 + d^2] = 120$

$\Rightarrow 25 + 9d^2 + 25 + d^2 = 60$

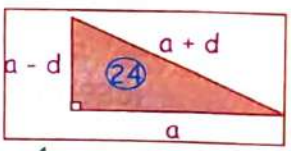
$\Rightarrow 10d^2 = 10$

$\Rightarrow d^2 = 1 \Rightarrow d = \pm 1$

Hence, numbers are 2, 4, 6, 8

Ques. The side of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side? **[JEE Main 2018]**

Ans.



$\therefore \frac{1}{2}\, a\, (a - d) = 24$..(i)

and $(a + d)^2 = (a - d)^2 + a^2$...(ii)

By eq. (ii)

$(a - d)^2 + 4ad = (a - d)^2 + a^2$

$\Rightarrow a = 4d$

put the value in eq. (i)

$\frac{1}{2} \times 4d \times 3d = 24$

$\Rightarrow d = 2$ and $a = 8$

Hence, smallest side = $a - d = 6$

Properties of an A.P.

(i) If a fixed number is added (or subtracted) to each of a given A.P. then the resulting sequence is also on A.P. with the same common difference as that of the given A.P.

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Example: 1, 5, 9, 13, 17, 21, ...

add k: $1+k, 5+k, 9+k, 13+k, 17+k, \ldots \to AP$

If $a+e, b+e, c+e, \to AP \Rightarrow a, b, c, \to AP$

(ii) If each term of an A.P. is multiplied by a fixed number (say k) (or divided by a non-zero fixed number), the resulting sequence is also an A.P. with the common difference multiplied by k.

Example: 1, 5, 9, 13

multiply by 2: 2, 10, 18, 26 → AP

(iii) In an A.P. the sum of terms equidistant from the beginning and end is constant and equal to sum of first and last terms.

Example: $a_1 + a_{100} = a_3 + a_{98}$

For example, If $a_1, a_2, a_3, \ldots$ an are in A.P.,

then $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \ldots$ and so on.

(iv) If a, b, c are three numbers in A.P. then $2b = a + c$

Proof: a, b, c are in AP

$\Rightarrow b - a = c - b$

$\Rightarrow 2b = a + c$

Ques. If $a_1, a_2, a_3, \ldots$, an are in A.P and $a_1 + a_4 + a_7 + \ldots \ldots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to:

[JEE Main 2019]

Ans. $a_1 + a_4 + a_7 + a_{10} + a_{13} + a_{16} = 114$

$\Rightarrow 3(a_1 + a_{16}) = 114$

$\Rightarrow a_1 + a_{16} = 38$

$\therefore a_1 + a_6 + a_{11} + a_{16} = 2(a_1 + a_{16})$

$= 2 \times 38 = 76$

Ques. Let $a_1, a_2, a_3, \ldots$, an be in A.P. If $a_3 + a_7 + a_{11} + a_{15} = 72$, then the sum of its first 17 terms is equal to:

[JEE Main 2016]

(a) 306 (b) 204 (c) 153 (d) 612

Ans. $\because a_3 + a_7 + a_{11} + a_{15} = 2(a_3 + a_{15}) = 72$

$\Rightarrow a_3 + a_{15} = 36 = a_1 + a_{17}$

$\therefore S_{17} = \frac{17}{2}(a_1 + a_{17})$

$= \frac{17}{2} \times 36$

$= 17 \times 18$

$= 17(17 + 1)$

$= 189 + 17 = 306$

Ques. For $x \geq 0$, the least value of k, for which $4^{1+x} + 4^{1-x}, \frac{k}{2}, 16^x + 16^{-x}$ are three consecutive terms of an A.P., is equal to:

[JEE Main 5 Apr. 2024 Shift II]

(a) 8 (b) 4 (c) 10 (d) 16

Ans. $2\frac{k}{2} = 4^{1+x} + 4^{1-x} + 16^x + 16^{-x}$

$\Rightarrow k = 4 \cdot 4^x + \frac{4}{4^x} + 16^x + \frac{1}{16^x}$

$k = 4\underbrace{\left(4^x + \frac{1}{4^x}\right)} + \underbrace{\left(16^x + \frac{1}{16^x}\right)}$

$\downarrow \qquad \downarrow$

$\left(t + \frac{1}{t}\right) \quad \left(t^2 + \frac{1}{t^2}\right) \qquad (\text{let } t = 4^x)$

$\because t + \frac{1}{t} \geq 2$ for $t > 0$ & $t + \frac{1}{t} \leq 2$ for $t < 0$

$k_{min} = 4 \times 2 + 2 \qquad [\text{when } x = 0]$

$= 10$

GEOMETRIC PROGRESSION INTRODUCTION

A sequence in which sub-sequent terms have common ratio is called G.P.

Example 1: 2, 6, 18, 54, ... $a = 2, r = 3$

Example 2: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \ldots\ldots$ $a = 1, r = \frac{1}{2}$

Example 3: $1, \frac{-1}{3}, \frac{1}{9}, \frac{-1}{27}, \frac{1}{8}, \ldots\ldots$ $a = 1, r = \frac{-1}{3}$

Example 4: $-2, 4, -8, 16, -32, \ldots\ldots$ $a = -2, r = -2$

Example 5: $1, \omega, \omega^2 \to GP$

4^{th} roots of unity $(i, -1, -i, 1) \to GP$

or $i, i^2, i^3, i^4, \to GP$

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General Term of G.P.

$T_n = a r^{n-1}$

Where a = first term & r = common ratio

If a is first term and r is common ratio

then GP: a, ar, ar^2, ar^3, ...

Example: 1, $\sin\theta$, $\sin^2\theta$, $\sin^3\theta$, ... GP

Ques. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is

(a) $\frac{8}{5}$ (b) $\frac{4}{3}$ (c) 1 (d) $\frac{7}{4}$

Ans. $T_2, T_5, T_9 \rightarrow GP$

$\Rightarrow a + b, a + 4d, a + 8d \rightarrow GP$

$\therefore (a + 4d)^2 = (a + d)(a + 8d)$

$\Rightarrow a^2 + 16d^2 + 8ad = a^2 + 8ad + ad + 8d^2$

$\Rightarrow 8d^2 = ad$

$\Rightarrow 8d = a$

$\therefore r = \frac{a + 4d}{a + d} = \frac{8d + 4d}{8d + d} = \frac{12d}{9d} = \frac{4}{3}$

Ques. Let 2nd, 8th and 44th, terms of a non-constant A.P. be respectively the 1st, 2nd and 3rd terms of G.P. If the first terms of A.P. is 1 then the sum of first 20 terms of this AP is equal to-

[JEE Mains 31 Jan. 2024 Shift II]

(a) 980 (b) 960 (c) 990 (d) 970

Ans. $T_2, T_8, T_{44} \rightarrow GP$

$\Rightarrow a + d, a + 7d, a + 43d \rightarrow GP$

$\Rightarrow 1 + d, 1 + 7d, 1 + 43d \rightarrow GP$

$\therefore (1 + 7d)^2 = (1 + d)(1 + 43d)$

$\Rightarrow 1 + 49d^2 + 14d = 1 + 44d + 43d^2$

$\Rightarrow 6d^2 = 30d$

$\Rightarrow 6d = 30$

$\Rightarrow d = 5$

$\therefore S_{20} = \frac{20}{2}[2a + 19d]$

$= 10[2 \times 1 + 95]$

$= 10 \times 97 = 970$

Ques. Let the first three terms 2, p and q, with $q \neq 2$, of G. P. be respectively the 7th, 8th and 13th terms of an A.P. the 5th term of the G.P. is the nth terms of the A.P., then n is equal to:

[JEE Main 4 Apr. 2024 Shift I]

(a) 163 (b) 151 (c) 177 (d) 169

Ans. Given $2 = T_7 = a + 6d$, ...(i)

$p = T_8 = a + 7d$,

$q = T_{13} = a + 12d$

$\therefore$ 2, p, q are in G.P.

$\therefore (a + 7d)^2 = (a + 6d)(a + 12d)$

$\Rightarrow 49d^2 + 14ad = 72d^2 + 18ad$

$\Rightarrow 23d^2 + 4ad = 0$

$\Rightarrow 23d + 4a = 0$ ...(ii)

multiply eq. (i) by 4:

$4a + 24d = 8$

$\Rightarrow 4a + 23d + d = 8$

By eq. (ii):

$d = 8$

By eqn (i)

a for A.P. = -46

$\therefore P = 2 + 8 = 10$

In G.P.

$a = 2, ar = 10$

$\Rightarrow r = 5$

$\therefore ar^4 = 2 \times 5^4 = 625 \times 2 = 1250$

Given $T_n = ar^4$

$\therefore T_n = 1250$

$\Rightarrow a + (n - 1)d = 1250$

$\Rightarrow 1250 = -46 + (n - 1)8$

$\Rightarrow n - 1 = 162$

$\Rightarrow n = 163$

Ques. If three sides of a triangle are in G.P. find range of common ratio.

Ans. $a + ar > ar^2 \Rightarrow r + 1 > r^2 \Rightarrow r^2 - r - 1 < 0$

$\therefore r = \frac{1 \pm \sqrt{5}}{2}$

& $a + ar^2 > ar \Rightarrow r^2 + 1 > r$

$\Rightarrow r^2 - r + 1 > 0$ (D < 0) always true

& $ar^2 + ar > a$

$\Rightarrow r^2 + r > 1$

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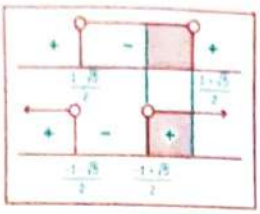
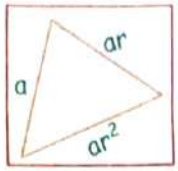
$\Rightarrow r^2 + r - 1 > 0$

$\therefore r = \dfrac{-1 \pm \sqrt{5}}{2}$





$r \in \left(\dfrac{\sqrt{5}-1}{2}, \dfrac{\sqrt{5}+1}{2}\right)$

Ques. In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the 4^{th}, 6^{th} and 8^{th} terms is equal to :

[JEE Main 8 Apr. 2024 Shift II]

(a) 96 (b) 91 (c) 84 (d) 78

Ans. Given $ar + ar^5 = \dfrac{70}{3}$ ...(i)

& $ar^2 . ar^4 = 49$

$\Rightarrow a^2 . r^6 = 49$

$\Rightarrow (ar^3)^2 = 49$

$\Rightarrow ar^3 = +7 \Rightarrow ar = \dfrac{7}{r^2}$

By eqⁿ (i)

$\dfrac{7}{r^2}[1 + r^4] = \dfrac{70}{3}$

$\Rightarrow \dfrac{1 + r^4}{r^2} = \dfrac{10}{3}$

$\Rightarrow r^2 + \dfrac{1}{r^2} = \dfrac{10}{3}$

$\Rightarrow r^2 = 3$

Hence, $ar^3 + ar^5 + ar^7$

$= ar^3 [1 + r^2 + r^4]$

$= 7 [1 + 3 + 9]$

$= 7 \times 13 = 91$

Ques. If a, b, c, are real numbers satisfying the equation $25(9a^2 + b^2) + 9c^2 - 15(5ab + bc + 3ca) = 0$, then prove that a, b, c are in A.P.

Ans. Given $225a^2 + 25b^2 + 9c^2 - 75ab - 15bc - 45ac = 0$

Multiply by 2

$450a^2 + 50b^2 + 18c^2 - 150ab - 30bc - 90ac = 0$

$\Rightarrow (15a - 5b)^2 + (5b - 3c)^2 + (3c - 15a)^2 = 0$

$\therefore 15a - 5b = 0 \Rightarrow 15a = 5b$

and $5b - 3c = 0 \Rightarrow 5b = 3c$

and $3c - 15a = 0 \Rightarrow 3c = 15a$

$\therefore 15a = 5b = 3c$

$\Rightarrow \dfrac{a}{1} = \dfrac{b}{3} = \dfrac{c}{5}$

$\Rightarrow a : b = c = 1 : 3 : 5$

$k : 3k : 5k$

Hence, a, b, c are in AP

Ques. If the sum of 3 numbers in GP is 14 and the sum of their squares is 84. Find the numbers.

Ans. Let a, ar ar^2 are 3 numbers in G.P.

then $a + ar + ar^2 = 14$

$\Rightarrow a(1 + r + r^2) = 14$ ...(i)

Also, $a^2 + (ar)^2 + (ar^2)^2 = 84$

$\Rightarrow a^2(1 + r^2 + r^4) = 84$ ...(ii)

Divide square of eq. (i) by (ii)

$\dfrac{a^2(1 + r + r^2)^2}{a^2(1 + r^2 + r^4)} = \dfrac{14^2}{84}$

$\Rightarrow \dfrac{(r^2 + r + 1)^2}{r^4 + r^2 + 1} = \dfrac{14 \times 14}{84}$

$\Rightarrow \dfrac{(r^2 + r + 1)^2}{r^4 + r^2 + 1} = \dfrac{7}{3}$

$\Rightarrow \dfrac{(r^2 + r + 1)^2}{(r^2 + r + 1)(r^2 - r + 1)} = \dfrac{7}{3}$

$\Rightarrow \dfrac{r^2 + r + 1}{r^2 - r + 1} = \dfrac{7}{3}$

$\Rightarrow r = 2, \dfrac{1}{2}$

By equation (i)

a = 2 or 8

Hence, numbers are 2, 4, 8

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Important Note

$r^4 + r^2 + 1 = (r^2 + r + 1)(r^2 - r + 1)$

Proof: $r^4 + 2r^2 + 1 - r^2$

$= (r^2 + 1)^2 - r^2$

$= (r^2 + 1 + r)(r^2 + 1 - r)$

$=$ RHS

Sum of n Terms of G.P.

$S_n = \frac{a(r^n - 1)}{(r - 1)}$ $(r > 1)$ OR $S_n = \frac{a(1 - r^n)}{(1 - r)}$ $(r < 1)$

Where a is first term, & r is common ratio

Proof: $S = a + ar + ar^2 + ar^3 + - - - - - + ar^{n-1}$ ...(i)

Multiply with 'r'

$rS = ar + ar^2 + ar^3 + - - - - - - - - + ar^n$ ...(ii)

By subtracting equation (ii) from (i)

$\Rightarrow S - rs = a - ar^n$

$\Rightarrow S(1 - r) = a(1 - r^n)$

$s = \frac{a(1 - r^n)}{(1 - r)}$ $r \neq 1$

If, $r = 1$

$S = a + a + a + \ldots + a$

$S = na$

Example: $1 + 3 + 3^2 + 3^3 + \ldots\ldots$ up to n terms

$a = 1$ $r = 3$ $n \to n$

$S_n = \frac{a(r^n - 1)}{(r - 1)}$

$= \frac{1(3^n - 1)}{(3 - 1)} = \frac{3^n - 1}{2}$

Ques. $3, -1, \frac{1}{3}, \frac{-1}{9}, \frac{1}{27}, \ldots$ sum of 10 terms

Ans. $\therefore a = 3$

$r = -\frac{1}{3}$

$S_{10} = \frac{a(1 - r^{10})}{1 - r}$

$= 3\frac{\left[1 - \left(-\frac{1}{3}\right)^{10}\right]}{1 - \left(-\frac{1}{3}\right)}$

$= 3 \times \frac{\left[1 - \left(-\frac{1}{3}\right)^{10}\right]}{\frac{4}{3}} = \frac{9}{4}\left(1 - \left(\frac{1}{3}\right)^{10}\right)$

Ques. Find the sum $1 - 2 + 2^2 - 2^3 + 2^4 \ldots\ldots\ldots$ upto n terms

Ans. $a = 1$

$r = -2$

$\therefore S_n = \frac{a(1 - r^n)}{1 - r}$

$= 1\frac{[1 - (-2)^n]}{1 - (-2)}$

$= \frac{1 - (-2)^n}{3}$

Ques. The sum of 10 terms of the series $.7 + .77 + .777 + \ldots\ldots\ldots$ is

(a) $\frac{7}{9}\left(89 + \frac{1}{10^{10}}\right)$ (b) $\frac{7}{81}\left(89 + \frac{1}{10^{10}}\right)$

(c) $\frac{7}{9}\left(89 + \frac{1}{10^9}\right)$ (d) None of these

Ans. $S = 0.7 + 0.77 + 0.777 + \ldots\ldots$

$\Rightarrow S = 7[0.1 + 0.11 + 0.111 + \ldots\ldots]$

$\Rightarrow S = \frac{7}{9}[0.9 + 0.99 + 0.999 + \ldots\ldots]$

$\Rightarrow S = \frac{7}{9}[1 - (0.1) + 1 - (0.01) + 1 - (0.001) \ldots\ldots]$

$\Rightarrow S = \frac{7}{9}[10 - ((0.1) + (0.1)^2 + (0.1)^3 + \ldots\ldots (0.1)^{10})]$

$= \frac{7}{9}\left[10 - \frac{a(1 - r^{10})}{1 - r}\right]$

$= \frac{7}{9}\left[10 - (0.1)\frac{1 - (0.1)^{10}}{0.9}\right]$

$= \frac{7}{9}\left[10 - \frac{\left(1 - \left(\frac{1}{10}\right)^{10}\right)}{9}\right]$

$= \frac{7}{9}\left[\frac{90 - 1 + \left(\frac{1}{10}\right)^{10}}{9}\right]$

$= \frac{7}{81}\left[89 + \left(\frac{1}{10}\right)^{10}\right]$

Properties of a G.P.

(i) If each terms of a G.P. be multiplied or divided by the same non-zero quantity, then the resulting sequence is also a G.P.

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(ii) In a G.P., the product of two terms which are equidistant from the first and the last terms, is constant and is equal to the product of first and last terms.

i.e. $T_1 \, T_n = P_2 \cdot P_{n-1} = T_3 . T_{n-1} = ... = a.l$

(iii) If a, b, c are in G.P. then $\frac{a}{b} = \frac{b}{c} \Rightarrow b^2 = ac$

(iv) If each terms of a G.P. be reaised to the same power, then resulting sequence is also a G.P.

Important Note

If Product of some in G.P. is given then also assume the numbers to be:

3 numbers in G.P.: $\frac{a}{r}, a, ar$

4 numbers in G.P. $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$

5 numbers in G.P.: $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$

6 numbers in G.P.: $\frac{a}{r^5}, \frac{a}{r^3}, \frac{a}{r}, ar, ar^3, ar^5$

Ques. The dimension of a Cuboid are $a > b > c$. The volume = 216 and the total outer surface area = 252. If a,b,c, are in G.P., then c =

(a) 3 (b) 1 (c) 5 (d) 2

Ans. Let side be $\frac{a}{r}, a, ar$

$\because abc = 216$

$\Rightarrow \frac{a}{r} . a . ar = 216$

$\Rightarrow a^3 = 216 = 6^3$

$\Rightarrow a = 6$

$\therefore \frac{6}{r}, 6, 6r$ are sides

$\because 2(lb + bh + hl) = 252$

$\Rightarrow 2\left[\frac{a}{r}a + a\,ar + \frac{a}{r}ar\right] = 252$

$\Rightarrow 2a^2\left[\frac{1}{r} + r + 1\right] = 252$

$\Rightarrow \left[\frac{1}{r} + r + 1\right] = \frac{252}{2 \times 36}$

$\Rightarrow \frac{1}{r} + r + 1 = \frac{7}{2}$

$\Rightarrow r + \frac{1}{r} = \frac{5}{2}$

$\Rightarrow r + \frac{1}{r} = \frac{5}{2}$

$\Rightarrow r = 2, \frac{1}{2}$

If, r = 2

Sides, are 3,6,12

or if, $r = \frac{1}{2}$

Sides, are 12,6,3

$\therefore c = 3$

Ques. If the sum and product of four positive consecutive terms of a G.P., are 126 and 1296 respectively, then the sum of square of common ratio of all such G.P.s is

[JEE Advance 2023]

(a) 9/2 (b) 3 (c) 7 (d) 14

Ans. Given sum = 126

product = 1296

Let no's be $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$

$\therefore a^4 = 1296 = (36)^2$

$\Rightarrow a^2 = 36$

$\Rightarrow a = \pm 6$

$\Rightarrow a = 6$

& $r > 0$

also, $\frac{6}{r^3} + \frac{6}{r} + 6r + 6r^3 = 126$

$\Rightarrow 6\left[r + \frac{1}{r} + r^3 + \frac{1}{r^3}\right] = 126$

$\Rightarrow \left(r + \frac{1}{r}\right) + r^3 + \frac{1}{r^3} = 21$

$\because r^3 + \frac{1}{r^3} = \left(r + \frac{1}{r}\right)^3 - 3\left(r + \frac{1}{r}\right)$

$\therefore \left(r + \frac{1}{r}\right) + \left(r + \frac{1}{r}\right)^3 - 3\left(r + \frac{1}{r}\right) = 21$

Let, $r + \frac{1}{r} = t$

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$\Rightarrow t + t^3 - 3t = 21$

$\Rightarrow t^3 - 2t - 21 = 0$

$\Rightarrow t^3 - 2t - 21 = 0$

$t^2(t - 3) + 3t(t - 3) + 7(t - 3) = 0$

$(t^2 + 3t + 7)(t - 3) = 0$

$\Rightarrow t = 3$ or $t^2 + 3t + 7 = 0$ ($D < 0$, no sol^n)

$\therefore$ sides are $t = 3$

$\Rightarrow r + \frac{1}{r} = 3$

$\Rightarrow \frac{r^2 + 1}{r} = 3 \Rightarrow r^2 - 3r + 1 = 0$ $\begin{matrix} \nearrow r_1 \\ \searrow r_2 \end{matrix}$

$\therefore r_1 + r_2 = 3, r_1 r_2 = 1$

Hence, $r_1^2 + r_2^2 = (r_1 + r_2)^2 - 2r_1 r_2$

$= 3^2 - 2 \times 1 = 9 - 2 = 7$

Brain Teaser

Show that $\underbrace{444\ldots}_{n}\underbrace{4888\ldots}_{n-1}89$ is always a perfect square.

Ans. $\underbrace{9 + [8 \times 10^1 + 8 \times 10^2 + 8 \times 10^3 + \ldots 8.10^{n-1}]}_{\text{GP } (n-1 \text{ term})} + \underbrace{[4 \times 10^n + 4 \times 10^{n+1} 4 \times 10^{n+2} + \ldots]}_{\text{GP no. of terms} = n}$

$= 9 + 80\frac{(10^{n-1} - 1)}{10 - 1} + 4.10^n\frac{[10^n - 1]}{10 - 1}$

$= 9 + 80\frac{(10^{n-1} - 1)}{9} + 4.\frac{10^n(10^n - 1)}{9}$

$= \frac{81 + 80.10^{n-1} - 80 + 4.10^{2n} - 4.10^n}{9}$

$= \frac{1 + 8 \times 10^n + 4.10^{2n} - 4.10^n}{9}$

$= \frac{1 + 4.10^{2n} + 10^n(8 - 4)}{9}$

$= \frac{1 + 4.10^{2n} + 4.10^n}{9} = \left(\frac{1 + 2.10^n}{3}\right)^2$

Which is a perfect square.

INFINITE G.P.

$S_\infty = \frac{a}{1 - r}, (-1 < r < 1)$

Important Note

The sum of Infinite GP is finite only when $|r| < 1$ or $-1 < r < 1$ and is given by $S_\infty = \frac{a}{1 - r}$

Proof:

$S_n = \frac{a(1 - r^n)}{(1 - r)}$

as $n \to \infty$

$\Rightarrow r^n \to 0$

$S_\infty = \frac{a(1 - 0)}{(1 - r)}$

Hence, $S_\infty = \frac{a}{1 - r}$

Ques. $S = 3.3^{\frac{1}{3}}.3^{\frac{1}{9}}.3^{\frac{1}{27}} \ldots\ldots\infty$

Ans. $3^{\left(1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots\ldots\infty\right)}$

$a = 1, r = \frac{1}{3}$

$\therefore \frac{a}{1 - r} = \frac{1}{1 - \frac{1}{3}} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$

Hence, $S = 3^{\frac{3}{2}} = 3.3^{\frac{1}{2}} = 3\sqrt{3} = \sqrt{27}$

Ques. The product $2^{\frac{1}{4}}.4^{\frac{1}{16}}16^{\frac{1}{128}}$ .....to ∞ is equal to

[JEE Mains Jan. 2020]

(a) $2^{\frac{1}{4}}$ (b) $2^{\frac{1}{2}}$ (c) 1 (d) 2

Ans. $2^{\frac{1}{4}} \cdot 2^{\frac{2}{16}} 2^{\frac{3}{48}} \cdot 2^{\frac{4}{128}} \ldots\ldots \infty$

$= 2^{\left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \ldots\ldots \infty\right)}$

$\therefore a = \frac{1}{4}, r = \frac{1}{2}$

$\therefore S_\infty = \frac{\frac{1}{4}}{1 - \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{2}{4} = \frac{1}{2}$

Hence, product $= 2^{\left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16}\frac{1}{2}\frac{1}{32} + \ldots\ldots \infty\right)} = 2$

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Ques. Let $a, ar, ar^2 \ldots$ be an infinite G.P. If $\sum_{n=0}^{\infty} ar^n = 57$ and $\sum_{n=0}^{\infty} a^3 r^{3n} = 9747$, then $a + 18r$ is equal to **[JEE Mains 9 Apr. 2024 Shift II]**

(a) 46 (b) 38 (c) 31 (d) 27

Ans. $\sum_{n=0}^{\infty} ar^n = ar^0 + ar^1 + ar^2 + ar^3 + \ldots\ldots \infty$

$\Rightarrow S_\infty = \frac{a}{1-r} = 57$

$\Rightarrow a = 57(1-r)$ ....(i)

$\Rightarrow a^3 = (57)^3 (1-r)^3$

Also, $a^3r^0 + a^3r^3 + a^3r^6 + \ldots\ldots \infty$

$\Rightarrow \frac{a^3}{1-r^3} = 9747$

By equation (i),

$\frac{(57)^3 (1-r)^3}{1-r^3} = 9747$

$\Rightarrow \frac{(1-r)^3}{(1-r)(1+r^2+r)} = \frac{9747}{57 \times 57 \times 57} = \frac{1}{19}$

$\Rightarrow \frac{(1-r)^2}{1+r^2+r} = \frac{1}{19}$

$\Rightarrow \frac{r^2+1-2r}{r^2+r+1} = \frac{1}{19}$

$\Rightarrow 19r^2 + 19 - 38r = r^2 + r + 1$

$\Rightarrow 18r^2 - 39r + 18 = 0$

$\Rightarrow 6r^2 - 13r + 6 = 0$

$\Rightarrow 6r^2 - 9r - 4r + 6 = 0$

$\Rightarrow 3r(2r-3) - 2(2r-3) = 0$

$\Rightarrow (2r-3)(3r-2) = 0$

$\Rightarrow r = \frac{3}{2}$ or $r = \frac{2}{3}$ (Accept) $\boxed{-1 < r < 1}$

$\therefore a = 57(1-r) = 57 \times \frac{1}{3} = 19$

Hence, $a + 18r = 19 + 18 \times \frac{2}{3} = 31$

Ques. If $x = \sum_{n=0}^{\infty} (-1)^n \tan^{2n}\theta$ and $y = \sum_{n=0}^{\infty} \cos^{2n}\theta$, for $0 < \theta < \frac{\pi}{4}$, then:

[JEE Mains Jan. 2020 Shift II]

(a) $x[1+y] = 1$ (b) $y[1-x] = 1$

(c) $y[1+x] = 1$ (d) $x[1-y] = 1$

Ans. $y = 1 + \cos^2\theta + \cos^4\theta + \cos^6\theta + \ldots\ldots \infty$

$\Rightarrow y = \frac{1}{1-\cos^2\theta}$ ...(i)

$x = 1 - \tan^2\theta + \tan^4\theta - \tan^6\theta + \tan^8\theta \ldots\ldots \infty$

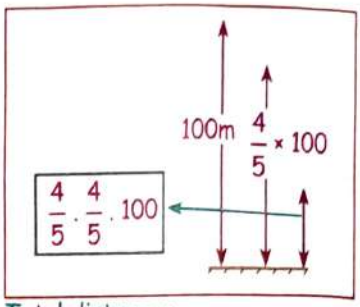
$\Rightarrow x = \frac{1}{1+\tan^2\theta} = \frac{1}{\sec^2\theta} = \cos^2\theta$

By equation (i),

$y = \frac{1}{1-x} \Rightarrow y(1-x) = 1$

Ques. A ball falls from a height of the 100 mts on a floor. If in each rebound, it describes $(4/5)^{th}$ height of the previous falling height, then the total distance travelled by the ball before it come to rest is ?

Ans.



Total distance =

$100 + 2 \cdot \frac{4}{5} \cdot 100 + 2 \cdot \left(\frac{4}{5}\right)^2 \cdot 100 + 2\left(\frac{4}{3}\right)^3 \cdot 100 + \ldots\ldots \infty$

$= 2\left[100 + \frac{4}{5} \cdot 100 + \left(\frac{4}{5}\right)^2 \cdot 100 + \ldots\ldots \infty\right] - 100$

G.P. $\Rightarrow S_\infty = \frac{100}{1-\frac{4}{5}} = 500$

Hence, total distance $= 2 \times 500 - 100 = 900$ m

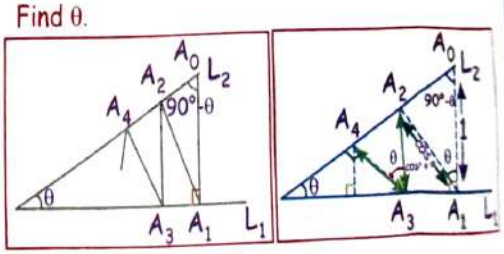
Brain Teaser

In the figure, $A_0A_1, A_2A_3, A_4A_5, \ldots\ldots$ are all perpendicular to L_1; $A_1A_2, A_3A_4, A_5A_6, \ldots\ldots$ are all perpendicular to L_2.

If $A_0A_1 = 1$ and $A_0A_1 + A_1A_2 + A_2A_3 + \ldots \infty = 2(2+\sqrt{2})$

Find θ.

Ans.



$\because A_0A_1 + A_1A_2 + A_2A_3 + \ldots \infty =$

$1 + \cos\theta + \cos^2\theta + \cos^3\theta + \ldots\ldots \infty = 2(2+\sqrt{3})$

$\Rightarrow \frac{1}{1-\cos\theta} = \frac{2(2+\sqrt{3})(2-\sqrt{3})}{(2-\sqrt{3})}$

$\Rightarrow \frac{1}{1-\cos\theta} = \frac{2}{2-\sqrt{3}}$

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$$\Rightarrow \frac{1}{1-\cos\theta} = \frac{1}{1-\frac{\sqrt{3}}{2}}$$

$$\Rightarrow 1-\cos\theta = 1-\frac{\sqrt{3}}{2}$$

$$\Rightarrow \cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30°$$

HARMONIC PROGRESSION (H.P)

A Sequence whose reciprocals from an AP is called H.P.

Example: $\frac{1}{2}, \frac{1}{5}, \frac{1}{8}, \frac{1}{11}, \ldots$ HP

Example: $\frac{2}{3}, \frac{2}{5}, \frac{2}{7}, \frac{2}{9}, \ldots$ HP

General Term of H.P.

$$T_n = \frac{1}{a+(n-1)d}$$

$$S_n = \frac{1}{\frac{n}{2}-(2a+(n-1)d}$$

$S_n \rightarrow$ HP $\rightarrow$ not defined

Important Note

(i) The general form of a harmonic progression is

$$\frac{1}{a}, \frac{1}{a+b}, \frac{1}{a+2b}, \ldots \frac{1}{a+(n-1)d}$$

(ii) No term of any H.P. can be zero.

(iii) If a, b, c are in HP, then $b = \frac{2ac}{a+c}$.

Proof: a, b, c $\rightarrow$ HP

$$\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \rightarrow AP$$

$$\Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c} \Rightarrow \frac{2}{b} = \frac{a+c}{ac} \Rightarrow b = \frac{2ac}{a+c}$$

(iv) We do not have general formula for the sum of the n terms of an HP.

Ques. If the 3rd, 6th and last terms of a H.P. are $\frac{1}{5}, \frac{3}{203}$ find the number of terms

Ans. $\underset{\downarrow T_3}{3}, \underset{\downarrow T_6}{5}, \underset{\downarrow T_n}{\frac{203}{3}} \rightarrow$ (A.P)

$\therefore T_3 = 3, T_6 = 5$

$\therefore a + 2d = 3$ ....(i)

and $a + 5d = 5$ ....(ii)

Equation (i) - (ii),

$-3d = -2$

$\Rightarrow d = \frac{2}{3}$

By equation (i),

$a = 3 - \frac{4}{3} \Rightarrow a = \frac{5}{3}$

$T_n = \frac{203}{3}$

$\Rightarrow a + (n-1)d = \frac{203}{3}$

$\Rightarrow \frac{5}{3} + (n-1)\frac{2}{3} = \frac{203}{3}$

$\Rightarrow 2n = 200$

$\Rightarrow n = 100$

Ques. If a,b,c is in AP, b C, d is in GP and c,d,e is in HP then show that a, c, e is in GP.

Ans. Given $2b = a + c$, $c^2 = bd$, $d = \frac{2ce}{c+e}$

$\therefore c^2 = bd$

$c^2 = \frac{a+c}{2} \times \frac{2ce}{(c+e)}$

$\Rightarrow c^2 = \frac{(a+c)ce}{(c+e)}$

$\Rightarrow c(c+ce) = (a+c)e$

$\Rightarrow c^2 + ce = ae + ce$

$\Rightarrow c^2 = ae$

Hence, a, c, e, is in G.P.

GP

a, b, c, d, e

AP HP

Arithmetic Geometric Progression (A.G.P.)

A series whose each terms is formed by multiplying corresponding terms of a A.P. and a G.P. is called an Arithmetic -geometric series.

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Example: $\frac{1}{3}+\frac{3}{32}+\frac{5}{33}\frac{7}{34}+\ldots$

Example: $1, 2x, 3x^2, 4x^3, 5x^4, \ldots$

General Term of A.G.P.

$$T_n = (a + (n-1)d)\, r^{n-1}$$

Sum of First n Term of A.G.P.

$$S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]}{1-r} r^n$$

Steps to find S_n for A.G.P.

Step I: Write the given series upto T_n.

Step II: Multiply the whole series with common ratio (r) and write the series again by shifting one term to right.

Step III: Subtract the two equations to obtain S_n term.

Ques. $S = 1+\frac{3}{5}+\frac{5}{5^2}+\frac{7}{5^3}+\frac{9}{5^4}+\ldots$ Find S_n and S_∞

Ans. $S_n = 1+\frac{3}{5}+\frac{5}{5^2}+\frac{7}{5^3}+\frac{9}{5^4}+\ldots+\frac{(2n-1)}{5^{n-1}}$

$$\frac{S_n}{5} = \quad \frac{1}{5}+\frac{3}{5^2}+\frac{5}{5^3}+\frac{7}{5^4}+\ldots\ldots+\frac{2n-1}{5^n}$$

$\ominus$ ----------

$$S_n - \frac{S_n}{5} = 1+\left(\frac{2}{5}+\frac{2}{5^2}+\frac{2}{5^3}+\ldots\ldots\right)-\frac{(2n-1)}{5^n}$$

$a = \frac{2}{5}$, $r = \frac{1}{5}$, no of terms $= n-1$

$$\Rightarrow S_n\left(1-\frac{1}{5}\right) = 1+\frac{2}{5}\,\frac{\left[1-\left(\frac{1}{5}\right)^{n-1}\right]}{1-\frac{1}{5}}-\frac{2n-1}{5^n}$$

$$\Rightarrow 4\frac{S_n}{5} = 1+\frac{1}{2}\left[1-\left(\frac{1}{5}\right)^{n-1}\right]-\left(\frac{2n-1}{5^n}\right)$$

$$\Rightarrow S_n = \frac{5}{4}+\frac{5}{8}\left[1-\left(\frac{1}{5}\right)^{n-1}\right]-\frac{5}{8}\left(\frac{2n-1}{5^n}\right)$$

$$S_\infty = 1+\frac{3}{5}+\frac{5}{5^2}+\frac{7}{5^3}+\ldots\ldots\infty$$

$$\frac{S_\infty}{5} = \quad \frac{1}{5}+\frac{3}{5^2}+\frac{5}{5^3}+\ldots\ldots\infty$$

$\ominus$ ----------

$$S_\infty\left[1-\frac{1}{5}\right] = 1+\left(\underbrace{\frac{2}{5}+\frac{2}{5^2}+\frac{2}{5^3}+\ldots\ldots\infty}_{\text{Infinite GP}}\right)$$

$$\frac{4}{5}\cdot S_\infty = 1+\frac{\frac{2}{5}}{1-\frac{1}{5}}$$

$$\frac{4}{5}\cdot S_\infty = 1+\frac{\frac{2}{5}}{\frac{4}{5}}$$

$$4\frac{S_\infty}{5} = \frac{3}{2} \Rightarrow S_\infty = \frac{15}{8}$$

Ques. $S = 3+(3+d)\frac{1}{4}+(3+2d)\frac{1}{4^2}+\ldots\ldots\infty = \frac{44}{9}$, then $d = ?$ **[JEE Mains 2019]**

Ans. $S = 3+(3+d)\frac{1}{4}+(3+2d)\frac{1}{4^2}+(3+3d)\frac{1}{4^3}+\ldots\ldots\infty$ ....(i)

$$\frac{S}{4} = \frac{3}{4}+(3+d)\frac{1}{4^2}+(3+2d)\frac{1}{4^3}+\ldots\ldots\infty \quad \ldots(ii)$$

Equation (i) - (ii)

$$\Rightarrow \frac{3S}{4} = 3+\frac{d}{4}+\frac{d}{4^2}+\frac{d}{4^3}+\ldots\ldots\infty$$

$$\Rightarrow \frac{3S}{4} = 3+\frac{\frac{d}{4}}{1-\frac{1}{4}}$$

$$\Rightarrow \frac{3S}{4} = 3+\frac{\frac{d}{4}}{\frac{3}{4}}$$

$$\Rightarrow \frac{3S}{4} = 3+\frac{d}{3}$$

$\because S = \frac{44}{9}$

$$\therefore \frac{44}{3\times 4} = 3+\frac{d}{3}$$

$$\Rightarrow \frac{11}{3}-3 = \frac{d}{3}$$

$$\Rightarrow \frac{11-9}{3} = \frac{d}{3}$$

$$\Rightarrow 2 = d$$

Ques. The Sum $\sum_{k=1}^{20} k\frac{1}{2^k}$ is equal to **[JEE Mains 2019]**

(a) $2-\frac{3}{2^{17}}$ (b) $1-\frac{11}{2^{20}}$

(c) $2-\frac{21}{2^{20}}$ (d) $2-\frac{11}{2^{19}}$

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Ans. $S = \frac{1}{2^1} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \ldots\ldots\ldots + \frac{20}{2^{20}}$

$\frac{S}{2} = \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \ldots\ldots\ldots + \frac{20}{2^{21}}$

$\ominus$ ----------

$\Rightarrow S - \frac{S}{2} = \left(\frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \ldots\ldots + \frac{1}{2^{20}}\right) - \frac{20}{2^{21}}$

$\Rightarrow \frac{S}{2} = \frac{\frac{1}{2}\left(1 - \left(\frac{1}{2}\right)^{20}\right)}{\left(1 - \frac{1}{2}\right)} - \frac{20}{2^{21}}$

$\Rightarrow \frac{S}{2} = 1 - \left(\frac{1}{2}\right)^{20} - \frac{20}{2^{21}}$

$\Rightarrow S = 2 - \frac{2}{2^{20}} - \frac{2\times 20}{2^{21}}$

$\Rightarrow S = 2 - \frac{1}{2^{19}} - \frac{20}{2^{20}}$

$\Rightarrow S = 2 - \frac{1}{2^{19}} - \frac{10}{2^{19}}$

$\Rightarrow S = 2 - \frac{11}{2^{19}}$

Ques. If $(10)^9 + 2(11)^1(10)^8 + 3(11)^2(10)^7 + \ldots\ldots\ldots + 10(11)^9 = k(10)^9$, then k is equal to

(a) 100 (b) 110 (c) $\frac{121}{10}$ (d) $\frac{441}{100}$

Ans. $S = (10)^9 + 2(11)^1(10)^8 + 3(11)^2(10)^7 + \ldots + 10(11)^9$

$\frac{11S}{10} = (11)(10)^8 + 2(11)^2(10)^7 + \ldots\ldots + \frac{10(11)^{10}}{10}$

$\ominus$ ----------

$\therefore S - \frac{11S}{10} = \underbrace{\left(10^9 + (11)(10)^8 + (11)^2(10)^7 + \ldots\right)}_{n = 10} - \frac{10(11)^{10}}{10}$

$\Rightarrow S\left(1 - \frac{11}{10}\right) = \frac{10^9\left[\left(\frac{11}{10}\right)^{10} - 1\right]}{\frac{11}{10} - 1} - (11)^{10}$

$\Rightarrow \frac{-S}{10} = \frac{10^9\left[\left(\frac{11}{10}\right)^{10} - 1\right]}{\frac{1}{10}} - (11)^{10}$

$\Rightarrow \frac{-S}{10} = 10^{10}\left[\left(\frac{11}{10}\right)^{10} - 1\right] - (11)^{10}$

$\Rightarrow \frac{-S}{10} = (11)^{10} - (10)^{10} - (11)^{10}$

$\Rightarrow \frac{-S}{10} = -(10)^{10}$

$\Rightarrow S = (10)^{11} = k(10)^9$

$\Rightarrow (10)^2 = k$

$\Rightarrow k = 100$

Brain Teaser

The sum of infinite terms of the series $1^2 + \frac{3^2}{2} + \frac{5^2}{2^2} + \frac{7^2}{2^3} + \ldots\ldots\ldots\infty$ is equal to

Ans. $S = 1^2 + \frac{3^2}{2} + \frac{5^2}{2^2} + \frac{7^2}{2^3} + \ldots\ldots\ldots \infty$

$\frac{S}{2} = \frac{1^2}{2} + \frac{3^2}{2} + \frac{5^2}{2} + \ldots\ldots\ldots\infty$

$\ominus$ ----------

$\frac{S}{2} = 1^2 + \frac{(3^2 - 1^2)}{2} + \frac{(5^2 - 3^2)}{2^2} + \frac{(7^2 - 5^2)}{2^3} + \frac{(9^2 - 7^2)}{2^4} + \ldots \infty$

$\Rightarrow \frac{S}{2} = 1 + \frac{4.2}{2} + \frac{8.2}{2^2} + \frac{12.2}{2^3} + \frac{16.2}{2^4} + \ldots \infty$

$\Rightarrow \frac{S}{2} = 1 + 4 + \frac{8}{2} + \frac{12}{2^2} + \frac{16}{2^3} + \ldots\ldots\ldots\infty \rightarrow S'$

Let, $S' = 4 + \frac{8}{2} + \frac{12}{2^2} + \frac{16}{2^3} + \ldots\ldots\ldots\infty$

$\frac{S'}{2} = \frac{4}{2} + \frac{8}{2^2} + \frac{12}{2^3} + \ldots\ldots\ldots\infty$

$\ominus$ ----------

$S' - \frac{S'}{2} = \frac{S'}{2} = \left(4 + \frac{4}{2} + \frac{4}{2^2} + \frac{4}{2^3} + \ldots\ldots\ldots \infty\right)$

$\Rightarrow \frac{S'}{2} = \frac{4}{1 - \frac{1}{2}}$

$\Rightarrow \frac{S'}{2} = 8 \Rightarrow S' = 16$

Hence, $\frac{S}{2} = 1 + 16 = 17 \Rightarrow S = 34$

INSERTION OF MEANS

Arithmetic Mean

Case-1: Inserting Single AM between 2 given numbers a & b.

a, A, b $\rightarrow$ A.P.

$A - a = b - A \Rightarrow 2A = a + b$

$\Rightarrow A = \frac{a + b}{2}$

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Case-2: Inserting "n" AMs between 2 given number a & b.

$a, A_1, A_2, A_3, A_4, \ldots\ldots A_n, b \rightarrow$ A.P.

This AP has $(n+2)$ terms

$b = T_{n+2}$

$b = a + (n+1)d$

$d = \dfrac{b-a}{n+1}$

$A_1 = a + d = a + \dfrac{b-a}{n+1}$

$A_2 = a + 2d = a + 2\dfrac{b-a}{n+1}$

$A_n = a + nd = a + n\dfrac{b-a}{n+1}$

Ques. Insert 12 AMs between the numbers 1 to 99. Also find their sum.

Ans. $1, A_1, A_2, A_3, \ldots\ldots A_{12}, 99 \rightarrow$ A.P.

This AP has 14 terms

$99 = T_{14}$

$\Rightarrow 99 = a + 13d$

$\Rightarrow 99 = 1 + 13d$

$\Rightarrow d = \dfrac{98}{13}$

$A_1 = a + d = 1 + \dfrac{98}{13}$

$A_2 = a + 2d = 1 + \dfrac{98 \times 2}{13}$

$A_3 = a + 3d = 1 + \dfrac{98 \times 3}{13}$

$A_{12} = a + 12d = 1 + \dfrac{98 \times 12}{13}$

$A_1 + A_2 + \ldots\ldots A_n = n\left(\dfrac{a+b}{2}\right)$

$\Rightarrow A_1 + A_2 + \ldots\ldots A_{12} = 12\left(\dfrac{1+99}{2}\right) = 600$

OR

$S_{14} = \dfrac{14}{2}[1 + 99] = 700$

$\Rightarrow A_1 + A_2 + A_3 \ldots\ldots A_{12} = 700 - (1 + 99) = 600$

Important Note

The sum of n AMs inserted between 2 given number is always equal to n times the single AM.

Proof:

$A_1 + A_2 + A_3 + \ldots\ldots A_n = \dfrac{n}{2}[A_1 + A_n]$

$= \dfrac{n}{2}[a + b]$

$= n\left(\dfrac{a+b}{2}\right) = \text{RHS}$

Ques. Find the sum of 100 AMs inserted between the numbers 1 to 2021.

(a) 5050 (b) 202200

(c) 50550 (d) None

Ans. $A_1 + A_2 + \ldots\ldots A_n = n\left(\dfrac{a+b}{2}\right)$

$\text{Sum} = 100\left(\dfrac{1+2021}{2}\right) = \dfrac{100}{2} \times 2022$

$= 100 \times 1011$

$= 101100$

Geometric Mean

Case-1: Inserting Single GM between 2 given numbers a & b.

$a, G, b \rightarrow$ G.P.

$\dfrac{G}{a} = \dfrac{b}{G} \Rightarrow G^2 = ab \Rightarrow G = \sqrt{ab}$

GM is +ve (a & b are +ve)

Case-2: Inserting "n" GMs between 2 given number a & b.

$a, G_1, G_2, G_3, G_4, \ldots\ldots G_n, b \rightarrow$ GP

$T_{n+2} = b = ar^{n+1}$

$\dfrac{b}{a} = (r)^{n+1}$

$r = \left(\dfrac{b}{a}\right)^{\frac{1}{n+1}}$

$G_1 = ar$

$G_2 = ar^2$

$G_3 = ar^3$

$G_n = ar^n$

Product: $G_1.G_2.G_3.G_4 \ldots\ldots G_n = a^n r^{1+2+3+\ldots+n}$

$= a^n r^{\frac{n(n+1)}{2}} = a^n\left(\dfrac{b}{a}\right)^{\frac{1}{n+1}\frac{n(n+1)}{2}} = a^n\left(\dfrac{b}{a}\right)^{\frac{n}{2}} = \dfrac{a^n.b^{\frac{n}{2}}}{a^{\frac{n}{2}}} = a^{\frac{n}{2}}.b^{\frac{n}{2}} = (\sqrt{ab})^n$

Ques. Insert 7 GMs between the numbers 2 and 32. Also find their product.

Ans. $2, G_1, G_2, G_3 \ldots\ldots G_7, 32 \rightarrow$ GP

$T_9 = 32 = ar^8$

$\Rightarrow 32 = 2r^8$

$\Rightarrow r^8 = 16$

$\Rightarrow r^2 = 2$

$\Rightarrow r = +\sqrt{2}$

$G_1 = ar = 2\sqrt{2}$

$G_2 = ar^2 = 2(\sqrt{2})^2 = 4$

$G_3 = ar^3 = 4\sqrt{2}$

$G_7 = ar^7 = 16\sqrt{2}$

Product: $G_1, G_2 \ldots\ldots G_7 = (\sqrt{ab})^7$

$= (\sqrt{2 \times 32})^7 = (\sqrt{64})^7 = 8^7$

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Important Note

The product of n GMs inserted between 2 given number is always equal to nth power of the single GM.

$$G_1, G_2, G_3 \ldots\ldots G_n = (\sqrt{ab})^n$$

Ques. If n AMs are inserted between two numbers "a" and 100. Such that the ratio of first mean to last mean is 1 : 7 and a + n = 33, then find the value of n? **[JEE Mains 2022]**

Ans. $a + n = 33$ ....(i)

$\Rightarrow a = 33 - n$

$a, A_1, A_2, A_3 \ldots\ldots A_n, 100$

$$\frac{A_1}{A_n} = \frac{1}{7}$$

$$\Rightarrow \frac{a+d}{a+nd} = \frac{1}{7}$$

$\Rightarrow 7a + 7d = a + nd$

$\Rightarrow 6a = nd - 7d$

$\Rightarrow 6a = d(n - 7)$ ....(ii)

$$\Rightarrow d = \frac{6a}{n-7}$$

$\because 100 = T_{n+2}$

$\Rightarrow 100 = a + (n + 1)\, d$ ....(iii)

$$\Rightarrow 100 = a + \frac{(n+1)\,6a}{n-7}$$

$$\Rightarrow 100 = \frac{(n-7)\,a + (n+1)\,6a}{n-7}$$

$\Rightarrow 100(n - 7) = a[n - 7 + 6n + 6]$

$\Rightarrow 100(n - 7) = a[7n - 1]$

$\Rightarrow 100(n - 7) = (33 - n)[7n - 1]$

$\Rightarrow 100n - 700 = 231n - 33 - 7n^2 + n$

$\Rightarrow -7n^2 + 132n + 667 = 0$

$\Rightarrow 7n^2 - 161n + 29n - 667 = 0$

$\Rightarrow 7n^2 - [n - 23] + 23(n - 23) = 0$

$\Rightarrow (7n + 29)(n - 23) = 0$

$\Rightarrow n = 23$

Harmonic Mean

Case-1: Inserting Single HM between 2 Given numbers a & b.

$a, H, b \rightarrow$ H.P.

$$\frac{2}{H} = \frac{1}{a} + \frac{1}{b}$$

$$\Rightarrow \frac{1}{a}, \frac{1}{H}, \frac{a}{b} \rightarrow \text{A.P.}$$

$$\Rightarrow \frac{2}{H} = \frac{a+b}{ab} \Rightarrow H = \frac{2ab}{a+b}$$

Case-2: Inserting "n" HMs between 2 Given numbers a & b.

$a, H_1, H_2, H_3 \ldots\ldots H_n, b$

$$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3} \ldots\ldots \frac{1}{H_n}, \frac{1}{b} \Rightarrow \text{A.P.}$$

$$T_{n+2} = \frac{1}{b} = \frac{1}{a} + (n+1)d$$

$$d = \frac{1}{(n+1)}\left(\frac{1}{b} - \frac{1}{a}\right)$$

$$\frac{1}{H_1} = \frac{1}{a} + d$$

$$\frac{1}{H_2} = \frac{1}{a} + 2d$$

.

.

.

$$\frac{1}{H_n} = \frac{1}{a} + nd$$

Ques. If 4 HM's be inserted between 2/3 and 2/13 then the second HM is

(a) 2/5 (b) 2/7 (c) 2/11 (d) 2/17

Ans. $\frac{2}{3}, H_1, H_2, H_3, H_4, \frac{2}{13} \rightarrow$ HP

$$\Rightarrow \frac{3}{2}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \frac{1}{H_4}, \frac{13}{2} \rightarrow \text{AP}$$

$$\Rightarrow \frac{13}{2} = \frac{3}{2} + 5d$$

$$\Rightarrow \frac{10}{2} = 5d \Rightarrow d = 1$$

$$\because \frac{1}{H_2} = \frac{3}{2} + 2d$$

$$\Rightarrow \frac{1}{H_2} = \frac{3}{2} + 2$$

$$\Rightarrow \frac{1}{H_2} = \frac{7}{2} \Rightarrow H_2 = \frac{2}{7}$$

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Important Note

Note 1: The Sum of Reciprocals of n HMs inserted between 2 given number is always equal to n times the reciprocal of the single HM.

Let $a, H_1, H_2, H_3 \ldots\ldots\ldots H_n, b \to HP$

$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3} \ldots\ldots\ldots \frac{1}{H_n}, \frac{1}{b} \to AP$

Then, $\frac{1}{a} + \frac{1}{H_1} + \frac{1}{H_2} + \frac{1}{H_3} + \ldots\ldots\ldots + \frac{1}{H_n} = \frac{n}{H}$

Proof: $LHS = \frac{n}{2}\left[\frac{1}{H_1} + \frac{1}{H_n}\right]$

$= \frac{n}{2}\left[\frac{1}{a} + \frac{1}{b}\right]$

$= \frac{n(a+b)}{2ab} = \frac{n}{\frac{2ab}{a+b}} = \frac{n}{H} = RHS.$

Note 2: A, G & H are always in GP.

Proof: Let a, b are two positive number.

$A = \frac{a+b}{2}$,

$G = \sqrt{ab}$,

$H = \frac{2ab}{a+b}$

$\therefore AH = \frac{a+b}{2} \cdot \frac{2ab}{a+b}$

$\Rightarrow AH = ab$

$\Rightarrow AH = G^2$

Example: 2, 32

$A = \frac{2+32}{2} = 17$

$G = \sqrt{2 \times 32} = 8$

$H = \frac{2 \times 2 \times 32}{32+2} = \frac{2 \times 2 \times 32}{34} = \frac{64}{17}$

$G^2 = AH \Rightarrow A, G, H \to G.P.$

SPECIAL SERIES - 1

1. Sum of the first n natural numbers

$\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$

Proof: $\sum_{r=1}^{n} r = 1 + 2 + 3 + 4 + \ldots\ldots + n \Rightarrow A.P.$

$= \frac{n(n+1)}{2}$

2. Sum of the squares of the first n natural numbers

$\sum_{r=1}^{n} r^2 = (1)^2 + (2)^2 + (3)^2 + \ldots\ldots\ldots + (n)^2$

$= \frac{n(n+1)(2n+1)}{6}$

Proof: $1^2 + 2^2 + 3^2 + \ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$

consider $n^3 - (n-1)^3 = 3n^3 - 3n + 1$

n = 1	$1^3 - 0^3 = 3(1)^2 - 3(1) + 1$
n = 2	$2^3 - 1^3 = 3(2)^2 - 3(2) + 1$
n = 3	$3^3 - 2^3 = 3(3)^2 - 3(3) + 1$
.	
.	
n = n	$n^3 - (n-1)^3 = 3(n)^2 - 3(n) + 1$

$\Rightarrow n^3 - 0^3 = 3(1^2 + 2^2 + 3^2 + \ldots + n^2) - 3(1 + 2 + 3 + \ldots + n) + n$

Let, $1^2 + 2^2 + 3^2 + \ldots + n^2 = x$

$\Rightarrow n^3 = 3x - \frac{3n(n+1)}{2} + n$

$\Rightarrow 3x = n^3 + \frac{3n(n+1)}{2} - n$

$\Rightarrow 3x = n\left[n^2 + \frac{3(n+1)}{2} - n\right]$

$\Rightarrow 3x = n\left[\frac{2n^2 + 3n + 3 - 2}{2}\right]$

$\Rightarrow 3x = \frac{n}{2}[2n^2 + 3n + 1]$

$\Rightarrow 3x = \frac{n}{2}(2n+1)(n+1)$

$\Rightarrow x = \frac{n}{6}(2n+1)(n+1)$

3. Sum of the cubes of the first n natural numbers

$\sum_{r=1}^{n} r^3 = 1^3 + 2^3 + 3^3 + \ldots\ldots\ldots + n^3 = \frac{n^2(n+1)^2}{4}$

$= \left[\sum_{r=1}^{n} r\right]^2 = \left[\frac{n(n+1)}{2}\right]^2$

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Proof: $1^3 + 2^3 + 3^3 + ... + n^3 = \left(\frac{n(n+1)}{2}\right)^2$

consider $n^4 - (n-1)^4 = 4n^3 - 6n^2 + 4n - 1$

$n = 1$	$1^4 - 0^4 = 4(1)^3 - 6(1)^2 + 4(1) - 1$
$n = 2$	$2^4 - 1^4 = 4(2)^3 - 6(2)^2 + 4(2) - 1$
$n = 3$	$3^4 - 2^4 = 4(3)^3 - 6(3)^2 + 4(3) - 1$
$n = n$	$n^4 - (n-1)^4 = 4(n)^3 - 6(n)^2 + 4n - 1$

$\Rightarrow n^4 - 0^4 = 4(1^3 + 2^3 + ... + n^3) - 6(1^2 + 2^2 + ... + n^2) + 4(1 + 2 + ... + n) - n$

Let, $y = 1^3 + 2^3 + ... + n^3$

$\Rightarrow 4y = n^4 + 6.\frac{n}{6}(n+1)(2n+1) - 4.\frac{n(n+1)}{2} - n$

$\Rightarrow y = \left[\frac{n(n+1)}{2}\right]^2$

Ques. The sum of the series $1^3 - 2^3 + 3^3 - ..... + 9^3 =$

(a) 300 (b) 125 (c) 425 (d) None

Ans. $1^3 - 2^3 + 3^3 - ..... + 9^3 = [1^3 + 2^3 + 3^3 + 4^3 + ... + 9^3] - 2[2^3 + 4^3 + 6^3 + 8^3]$

$= \left(\frac{9 \times 10}{2}\right)^2 - 2.2^3 [1^3 + 2^3 + 3^3 + 4^3]$

$= (45)^2 - 16 \times \left(\frac{4 \times 5}{2}\right)^2$

$= (45)^2 - 16 \times (10)^2$

$= (45)^2 - (40)^2$

$= (45 + 40)(45 - 40)$

$= 85 \times 5 = \frac{850}{2} = 425$

Ques. Let $S_k = \frac{1 + 2 + 3 + ... + k}{k}$. If $S_1^2 + S_2^2 + ... + S_{10}^2 = \frac{5}{12} A$, then A is equal to :

(a) 283 (b) 301 (c) 303 (d) 156

Ans. $S_k = \frac{k(k+1)}{2.k}$

$\Rightarrow S_k = \frac{k+1}{2}$

$\Rightarrow (S_k)^2 = \frac{(k+1)^2}{4}$

$\therefore (S_1)^2 = \frac{2^2}{4}$,

$(S_2)^2 = \frac{3^2}{4}$,

$(S_3)^2 = \frac{4^2}{4}$,

.

.

Hence, $S_1^2 + S_2^2 + ... + S_{10}^2 = \frac{2^2}{4} + \frac{3^2}{4} + \frac{4^2}{4} + ... + \frac{11^2}{4}$

$= \frac{1}{4}[1^2 + 2^2 + 3^2 + 4^2 + ... + 11^2 - 1^2]$

$= \frac{1}{4}\left[\frac{n(n+1)(2n+1)}{4} - 1\right]$

$= \frac{1}{4}\left[\frac{11 \times 12 \times 23}{6} - 1\right]$

$= \frac{1}{4}[22 \times 23 - 1]$

$= \frac{1}{4} \times 505 = \frac{5A}{12}$

$A = \frac{505 \times 3}{5} = 101 \times 3 = 303$

Ques. If gcd (m, n) = 1 and $1^2 - 2^2 + 3^2 - 4^2 + ... + (2021)^2 - (2022)^2 + (2023)^2 = 1012m^2n$, then $m^2 - n^2$ is equal to

Ans. $1^2 - 2^2 + 3^2 - 4^2 + ... + (2021)^2 - (2022)^2 + (2023)^2$

$= (1 + 2)(1 - 2) + (3 + 4)(3 - 4) + (5 + 6)(5 - 6) .... + (2021 + 2022)(2021 - 2022) + (2023)^2$

$= -[1 + 2 + 3 + 4 + 5 + 6 + ... 2021 + 2022] + (2023)^2$

$= -\left[\frac{2022 \times 2023}{2}\right] + (2023)^2$

$= 2023[-1011 + 2022] = 2023 \times (1012) = 1012\, m^2n$

$\therefore m^2n = 2023 = 17^2 \times 7$

$\Rightarrow m = 17, n = 7$

Hence, $m^2 - n^2 = 240$

PROPERTIES OF SIGMA

1. $\sum_{r=1}^{n}(a_r \pm b_r) = \sum_{r=1}^{n} a_r \pm \sum_{r=1}^{n} b_r$

2. $\sum a_r b_r \neq \sum a_r \sum b_r$

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3. $\sum\left(\frac{a_r}{b_r}\right) \neq \frac{\sum a_r}{\sum b_r}$

4. $\sum_{r=1}^{n} ka_r = k\sum_{r=1}^{n} a_r$

Ques. $\sum_{r=1}^{10}(2r^2 - 3r) = ?$

Ans. $\sum_{r=1}^{10}(2r^2 - 3r) = 2\sum_{r=1}^{10} r^2 - 3\sum_{r=1}^{10} r$

$= \frac{2n(n+1)(2n+1)}{6} - \frac{3n(n+1)}{2}$

$= \frac{n(n+1)(2n+1)}{3} - \frac{3n(n+1)}{2}$

$= n(n+1)\left[\frac{2n+1}{3} - \frac{3}{2}\right]$

$= 10 \times 11\left(7 - \frac{3}{2}\right)$ $(\because n = 10)$

$= 10 \times 11 \times \frac{11}{2} = 121 \times 5 = 605$

Ques. Find S_n for the following series having General Term T_n as follows:

(i) $T_n = n^2 - 3n + 2$

Ans. $T_r = r^2 - 3r + 2$

$S_n = \Sigma T_r = \Sigma(r^2 - 3r + 2)$

$= \Sigma r^2 - 3\Sigma r + 2\Sigma 1$

$= \frac{n(n+1)(2n+1)}{6} - \frac{3n(n+1)}{2} + 2n$

(ii) $T_n = 3n^3 - 1$

Ans. $T_r = 3r^3 - 1 \Rightarrow S_n = \sum_{r=1}^{n}(3r^3 - 1) = 3\Sigma r^3 - \Sigma 1$

$= 3\left[\frac{n(n+1)}{2}\right]^2 - n$

(iii) $T_n = 2n^2 - 3^n + 4$

Ans. $T_r = 2r^2 - 3^r + 4$

$S_n = \sum_{r=1}^{n}(2r^2 - 3^r + 4)$

$= 2\Sigma r^2 - \Sigma 3^r + 4\Sigma 1$

$\because \sum_{r=1}^{n} 3^r = 3^1 + 3^2 + 3^3 + \ldots\ldots + 3^n$

$a = 3, r = 3$

$= \frac{3(3^n - 1)}{3 - 1}$

$= \frac{3}{2}(3^n - 1)$

$\therefore S_n = \frac{2n(n+1)(2n+1)}{6} - \frac{3}{2}(3^n - 1) + 4n$

Ques. The sum $1.3^2 + 2.5^2 + 3.7^2 + \ldots\ldots$ up to 20 terms

(a) 188090 (b) 94045

(c) 325178 (d) 812715

Ans. $T_r = r(2r+1)^2 = r[4r^2 + 1 + 4r]$

$= (4r^3 + r + 4r^2)$

$\therefore S_n = \Sigma T_r$

$= \Sigma(4r^3 + 4r^2 + r)$

$= 4\Sigma r^3 + 4\Sigma r^2 + \Sigma r$

$S_n = 4\left[\frac{n(n+1)}{2}\right]^2 + \frac{4n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$

$\because n = 20$

$\therefore S_{20} = 4 \times (210)^2 + \frac{2}{3} \times 20 \times 21 \times 41 + 210$

$\Rightarrow S_{20} = 210\left[4 \times 210 + \frac{4}{3} \times 41 + 1\right]$

$= 210\left[4 \times \left(210 + \frac{41}{3}\right) + 1\right]$

$= 210\left[4 \times \left(\frac{630 + 41}{3}\right) + 1\right]$

$= 210\left[\frac{4 \times 671}{3} + 1\right] = 188090$

Ques. The sum $\frac{3 \times 1^3}{1} + \frac{5 \times (1^3 + 2^3)}{1^2 + 2^2} + \frac{7 \times (1^3 + 2^3 + 3^3)}{1^2 + 2^2 + 3^2}$ + ... up to 10 term.

(a) 660 (b) 620 (c) 680 (d) 600

Ans. $T_r = (2r+1)\frac{(1^3 + 2^3 + 3^3 + \ldots\ldots + r^3)}{(1^2 + 2^2 + 3^2 + \ldots\ldots + r^2)}$

$= \frac{(2r+1)\left(\frac{r(r+1)}{2}\right)^2}{\frac{r(r+1)(2r+1)}{6}} = \frac{3r(r+1)}{2}$

$S_{10} = \sum_{r=1}^{10} T_r$

$= \frac{3}{2}\Sigma r(r+1)$

$= \frac{3}{2}[\Sigma r^2 + \Sigma r)$

$= \frac{3}{2}\left[\frac{10 \times 11 \times 21}{6} + \frac{10 \times 11}{2}\right]$

$= \frac{3}{2}[5 \times 11 \times 7 + 5 \times 11]$

$= \frac{3}{2} \times 5 \times 11 \times 8 = 3 \times 5 \times 11 \times 4$

$= 33 \times 20 = 660$

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SPECIAL SERIES - 2

Series in which the Difference of Consecutive terms are in A.P or G.P.

Method of difference to find sum

Step 1: Write the given series.

Step 2: Write the series again by shifting one term to right.

Step 3: Subtract the two equations to obtain n^{th} term (T_n) of the series.

Step 4: Find the sum of the n terms n of the sequence as $S_n = \sum_{n=1}^{n} T_n$.

Ques. Find S_n for the series $3 + 7 + 14 + 24 + 37 + ...$ up to n terms.

Ans. $S = 3 + 7 + 14 + 24 + 37 + ... + T_n$

$S = \quad 3 + 7 + 14 + 24 + ... + T_n$

$-$ ---------- $-$ $-$ $-$ $-$ $-$ ----------

$0 = 3 + (4 + 7 + 10 + 13 + ...+) - T_n$

$T_n = 3 + (4 + 7 + 10 + 13 + ...+)$

$= 3 + \frac{(n-1)}{2}[2 \times 4 + (n-2) \times 3]$

$= 3 + \frac{n-1}{2}[3n + 2]$

$T_n = \frac{6 + (n-1)(3n+2)}{2}$

$T_n = \frac{6 + 3n^2 - n - 2}{2}$

$T_n = \frac{3n^2 - n + 4}{2}$

$T_r = \frac{3r^2 - r + 4}{2}$

$S_n = \Sigma T_r = \sum \frac{3r^2 - r + 4}{2}$

$= \frac{1}{2}[3\Sigma r^2 - \Sigma r + 4\Sigma 1]$

$= \frac{1}{2}\left[3.\frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n\right]$

Ques. Find the sum of n-terms of the series $1 + 4 + 10 + 22 + 43 + ...$

Ans. $S = 1 + 4 + 10 + 22 + 46 + ... + T_n$

$S = \quad 1 + 4 + 10 + 22 + ... + T_n$

$-$ $-$ $-$ $-$ $-$ $-$

$0 = 1 + [3 + 6 + 12 + 24 + ...] - T_n$

$\Rightarrow T_n = 1 + [3 + 6 + 12 + 24 + ...]$

$\therefore T_n = 1 + \frac{3(2^{n-1} - 1)}{2 - 1}$

$\Rightarrow T_n = 1 + 3(2^{n-1} - 1)$

$\Rightarrow T_r = 3.2^{r-1} - 2$

$S_n = \Sigma T_r$

$= \Sigma(3.2^{r-1} - 2)$

$= 3\sum_{r=1}^{n} 2^{r-1} - 2\sum_{r=1}^{n} 1$

$= 3[2^0 + 2^1 + ... + 2^{n-1}] - 2n$

$= \frac{3(2^n - 1)}{(2 - 1)} - 2n$

Hence, $S_n = 3(2^n - 1) - 2n$

Brain Teaser

Natural numbers ware written as follows :

1

2 3

4 5 6

7 8 9 10

T_n

Find sum of numbers in n^{th} row.

Ans. 1, 2, 4, 7, 11, ... T_n

$S = 1 + 2 + 4 + 7 + 11 + ... + T_n$

$S = \quad 1 + 2 + 4 + 7 + ... + T_n$

$-$ ---------- $-$ ----------

$0 = 1 + (1 + 2 + 3 + ...) - T_n$

$T_n = 1 + (1 + 2 + 3 + ...)$

$= 1 + \frac{n(n-1)}{2}$

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$\frac{n^2 - n + 2}{2} = T_n$

Now, sum of numbers in n^{th} row.

First term = $\frac{n^2 - n + 2}{2}$, d = 1

Number of term = n

Sum = $\frac{n}{2}[n^2 - n + 2 + (n - 1) \times 1]$

$\therefore S = \frac{n}{2}[n^2 + 1]$

Ques. An arithmetic progression is written in the following way

		2	
	5		8
11			17
20	23	26	29

The sum of all the terms of the 10^{th} row is .....

Ans. $S = 2 + 5 + 11 + 20 + 32 + ... + a_n$

$S = \quad 2 + 5 + 11 + 20 + ... + a_n$

$a_n = 2 + (3 + 6 + 9 + 12 + ...)$

$a_n = 2 + \frac{n - 1}{2}[2 \times 3 + (n - 2) \times 3]$

$= 2 + \left(\frac{(n - 1)}{2}\right).3n$

$a_n = 2 + \frac{3n(n - 1)}{2}$

Put n = 10

$\Rightarrow a = 2 + \frac{3 \times 10 \times 9}{2} = 137$

d = 3, no. of terms = 10

The sum of all the terms of the 10^{th} row is

$S = \frac{10}{2}[2 \times 137 + 9 \times 3]$

$\Rightarrow S = 5[274 + 27]$

$= 5 \times [301] = 1505$

Important Note

The general term of a series in which difference of 2 consecutive terms are in AP is always of the form $T_n = an^2 + bn + c$

TELESCOPING SERIES

A series in which the general term can be written as a difference of two symmetric terms is called Telescoping Series.

Ques. If $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + ... S_n$

Ans. $T_r = \frac{1}{r(r + 1)}$

$= \frac{(r + 1) - r}{r(r + 1)}$

$= \frac{r + 1}{r(r + 1)} - \frac{r}{r(r + 1)}$

$\Rightarrow T_r = \frac{1}{r} - \frac{1}{r + 1}$

$S_n = \sum T_r$

$= \sum \frac{1}{(r^2 + r)}$

$= \frac{1}{\sum r^2 + \sum r}$ → wrong

$\sum \frac{ar}{br} \neq \frac{\sum ar}{\sum br}$

$S_n = \frac{1}{1} - \frac{1}{n + 1}$

$S_n = 1 - \frac{1}{n + 1}$

$= \frac{n + 1 - 1}{n + 1} = \frac{n}{n + 1}$

and $S_\infty = 1 - 0 = 1$

$T_1 = \frac{1}{1} - \frac{1}{2}$

$T_2 = \frac{1}{2} - \frac{1}{3}$

$T_3 = \frac{1}{3} - \frac{1}{4}$

.

.

$T_n = \frac{1}{n} - \frac{1}{n + 1}$

Ques. $S = \frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + ...$ upto n terms.

Ans. $T_r = \frac{(r + 1) - r}{r(r + 1)(r + 2)}$

WRONG APPROACH

$T_r = \frac{1}{r(r + 2)} - \frac{1}{(r + 1)(r + 2)}$

$T_1 = \frac{1}{1.3} - \frac{1}{2.3}$

$T_2 = \frac{1}{2.4} - \frac{1}{3.4}$

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$$T_r = \frac{1}{2}\left[\frac{(r+2)-r}{r(r+1)(r+2)}\right]$$

$$T_r = \frac{1}{2}\left[\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)}\right]$$

$$T_1 = \frac{1}{2}\left[\frac{1}{1.2} - \frac{1}{2.3}\right]$$

$$T_2 = \frac{1}{2}\left[\frac{1}{2.3} - \frac{1}{3.4}\right]$$

$$T_3 = \frac{1}{2}\left[\frac{1}{3.4} - \frac{1}{4.5}\right]$$

.

.

$$T_n = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

$$S_n = \frac{1}{2}\left[\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right]$$

and $S_\infty = \frac{1}{2}\left[\frac{1}{2} - 0\right] = \frac{1}{4}$

Ques. Find the sum to n terms and S_∞ for the series

$$S = \frac{1}{1.2.3.4} + \frac{1}{2.3.4.5} + \frac{1}{3.4.5.6} + \ldots\ldots$$

Ans. $$T_r = \frac{1}{3}\left[\frac{(r+3)-r}{r(r+1)(r+2)(r+3)}\right]$$

$$T_r = \frac{1}{3}\left[\frac{1}{r(r+1)(r+2)} - \frac{1}{(r+1)(r+2)(r+3)}\right]$$

$$T_1 = \frac{1}{3}\left[\frac{1}{1.2.3} - \frac{1}{2.3.4}\right]$$

$$T_r = \frac{1}{3}\left[\frac{1}{2.3.4} - \frac{1}{3.4.5}\right]$$

.

.

$$T_n = \frac{1}{3}\left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)}\right]$$

$$S_n = \frac{1}{3}\left[\frac{1}{6} - \frac{1}{(n+1)(n+2)(n+3)}\right]$$

and $S_\infty = \frac{1}{3} \times \frac{1}{6} = \frac{1}{18}$

Ques. $S = 1.2 + 2.3 + 3.4 + 4.5$ ..... upto n terms.

Ans. $$T_r = \frac{1}{3} r(r+1)[(r+2) - (r-1)]$$

$$T_r = \frac{1}{3}[r(r+1)(r+2) - (r-1)r(r+1)]$$

$$T_1 = \frac{1}{3}[1.2.3 - 0.1.2]$$

$$T_2 = \frac{1}{3}[2.3.4 - 1.2.3]$$

$$T_3 = \frac{1}{3}[3.4.5 - 2.3.4]$$

.

.

$$T_n = \frac{1}{3}[n(n+1)(n+2) - (n-1)n(n+1)]$$

$$S_n = \frac{1}{3}[n(n+1)(n+2) - 0]$$

$$\Rightarrow S_n = \frac{1}{3} n(n+1)(n+2)$$

Ques. $S = 1.2.3 + 2.3.4 + 3.4.5 + 4.5.6 +$ ..... upto n terms.

Ans. $$T_r = \frac{1}{4} r(r+1)(r+2)[(r+3) - (r-1)]$$

$$T_r = \frac{1}{4}\Big[r(r+1)(r+2)(r+3) - (r-1)r(r+1)(r+2)\Big]$$

$$T_1 = \frac{1}{4}[1.2.3.4 - 0]$$

$$T_2 = \frac{1}{4}[2.3.4.5 - 1.2.3.4]$$

.

.

$$T_n = \frac{1}{4}\Big[n(n+1)(n+2)(n+3) - (n-1)n(n+1)(n+2)\Big]$$

$$S_n = \frac{1}{4}\Big[n(n+1)(n+2)(n+3)\Big]$$

Ques. $S = 1.2.3.4 + 2.3.4.5 + 3.4.5.6 + 4.5.6.7 +$ ..... upto n terms.

Ans. $$S_n = \frac{1}{5} n(n+1)(n+2)(n+3)(n+4)$$

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Ques. If the sum of the series $\frac{1}{1(1+d)} + \frac{1}{(1+d)(1+2d)} + \ldots + \frac{1}{(1+9d)(1+10d)}$ is equal to 5, then 50d is equal to : **[JEE Mains Apr. 2024 Shift I]**

(a) 10 (b) 5 (c) 15 (d) 20

Ans. $T_1 = \frac{1}{d}\left[\frac{1}{1} - \frac{1}{1+d}\right]$

$T_2 = \frac{1}{d}\left[\frac{1}{1+d} - \frac{1}{1+2d}\right]$

.

.

$T_{10} = \frac{1}{d}\left[\frac{1}{1+9d} - \frac{1}{1+10d}\right]$

$S_{10} = \frac{1}{d}\left[\frac{1}{1} - \frac{1}{1+10d}\right]$

$= \frac{1}{d}\left[\frac{1+10d-1}{(1+10d)}\right]$

$= \frac{1}{d} \cdot \frac{10d}{1+10d}$

$= \frac{10}{1+10d} = 5$

$\Rightarrow 2 = 1 + 10d$

$\Rightarrow 1 = 10d$

Hence, $50d = 5$

Ques. Find the sum to n terms and S_∞ for the series $\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^2} + \frac{3}{1+3^2+3^4} + \ldots\ldots$

Ans. $T_r = \frac{r}{1+r^2+r^4}$

$\because 1 + r^4 + r^2 = (r^2+r+1)(r^2-r+1)$

$= \frac{1}{2}\left[\frac{(r^2+r+1)-(r^2-r+1)}{(r^2+r+1)(r^2-r+1)}\right]$

$T_r = \frac{1}{2}\left[\frac{1}{r^2-r+1} - \frac{1}{r^2+r+1}\right]$

$T_1 = \frac{1}{2}\left[\frac{1}{1} - \frac{1}{3}\right]$

$T_2 = \frac{1}{2}\left[\frac{1}{3} - \frac{1}{7}\right]$

$T_3 = \frac{1}{2}\left[\frac{1}{7} - \frac{1}{13}\right]$

.

$T_n = \frac{1}{2}\left[\frac{1}{n^2-n+1} - \frac{1}{n^2+n+1}\right]$

$S_n = \frac{1}{2}\left[\frac{1}{1} - \frac{1}{n^2+n+1}\right]$

Hence, $S_\infty = \frac{1}{2}$

Ques. If $a_n = \frac{-2}{4n^2-16n+15}$ then $a_1 + a_2 + \ldots\ldots + a_{25}$ is equal to : **[JEE Mains Jan. 2023 Shift I]**

(a) $\frac{51}{144}$ (b) $\frac{49}{138}$ (c) $\frac{50}{141}$ (d) $\frac{52}{147}$

Ans. $a_n = \frac{-2}{4n^2-10n-6n+15}$

$a_n = \frac{-2}{(2n-3)(2n-5)}$

$a_n = \frac{(2n-5)-(2n-3)}{(2n-3)(2n-5)}$

$a_n = \frac{1}{2n-3} - \frac{1}{2n-5}$

$a_1 = \frac{1}{-1} - \frac{1}{-3}$

$a_2 = \frac{1}{1} - \frac{1}{-1}$

$a_3 = \frac{1}{3} - \frac{1}{1}$

.

$a_{25} = \frac{1}{47} - \frac{1}{45}$

$S_{25} = \frac{1}{47} + \frac{1}{3}$

$= \frac{3+47}{47\times3} = \frac{50}{47\times3} = \frac{50}{141}$

Brain Teaser

Evaluate $\sum_{k=1}^{360} \frac{1}{k\sqrt{k+1} + (k+1)\sqrt{k}}$

Ans. $\sum_{k=1}^{360} \frac{1}{\sqrt{k}\sqrt{k+1}\,[\sqrt{k}+\sqrt{k+1}]} \cdot \frac{\sqrt{k+1}-\sqrt{k}}{(\sqrt{k+1}-\sqrt{k})}$

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$$= \sum_{k=1}^{360} \frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k}\sqrt{k+1}\,[(k+1) - k]}$$

$$= \sum_{k=1}^{360} \frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k}\sqrt{k+1}}$$

$$T_k = \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}}\right)$$

$$T_1 = \frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}}$$

$$T_2 = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}}$$

$$T_3 = \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}}$$

.

$$T_{360} = \frac{1}{\sqrt{360}} - \frac{1}{\sqrt{361}}$$

$$S_{360} = \frac{1}{\sqrt{1}} - \frac{1}{\sqrt{361}} = 1 - \frac{1}{19} = \frac{18}{19}$$

Brain Teaser

Evaluate $\sum_{r=1}^{n} \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right)$

Ans. Let $\frac{x}{2^r} = \theta$

$$\Rightarrow \left(\frac{x}{2^{r-1}}\right) = \frac{x}{2^r/2} = \frac{2x}{2^r} = 2\theta$$

$$\therefore \sum_{r=1}^{n} \tan\left(\frac{x}{2^r}\right) \sin\left(\frac{x}{2^{r-1}}\right) = \sum \tan\theta \sec 2\theta$$

$$= \sum \frac{\sin\theta}{\cos\theta \cos 2\theta} = \sum \frac{\sin(2\theta - \theta)}{\cos\theta \cos 2\theta}$$

$$= \sum \frac{\sin 2\theta \cos\theta - \cos 2\theta \sin\theta}{\cos\theta \cos 2\theta}$$

$$= \sum \tan 2\theta - \tan\theta$$

$$T_r = \tan\left(\frac{x}{2^{r-1}}\right) - \tan\left(\frac{x}{2^r}\right)$$

$$T_1 = \tan\frac{x}{2^0} - \tan\frac{x}{2^1}$$

$$T_2 = \tan\frac{x}{2^1} - \tan\frac{x}{2^2}$$

$$T_3 = \tan\frac{x}{2^2} - \tan\frac{x}{2^3}$$

.

$$T_n = \tan\frac{x}{2^{n-1}} - \tan\frac{x}{2^n}$$

$$S_n = \tan x - \tan\frac{x}{2^n}$$

AM-GM-HM INEQUALITY

1. **For 2 positive numbers**

$$AM = \frac{a+b}{2},\ GM = \sqrt{ab},\ HM = \frac{2ab}{a+b} = \frac{2}{\frac{1}{a} + \frac{1}{b}}$$

Equality holds (AM = GM = HM), where given numbers are equal.

2. **For 3 positive numbers**

a, b, c

$$AM = \frac{a+b+c}{3},\ GM = (abc)^{1/3}$$

$$HM = \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

3. **For n positive numbers**

Let $a_1, a_2, \ldots\ldots, a_n$ be n positive real numbers, then we define their arithmetic mean (A), geometric mean (G) and harmonic mean (H) as $A = \frac{a_1 + a_2 + \ldots\ldots + a_n}{n}$,

$G = (a_1, a_2, \ldots. a_n)^{1/n}$ and $H = \frac{n}{\left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \ldots\ldots + \frac{1}{a_n}\right)}$

$A \geq G \geq H$

Important Note

1. Minimum value of $a^2\tan^2\theta + b^2\cot^2\theta = 2ab$ where a, b > 0.
2. Minimum value of $a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta = (a+b)^2$ where a, b > 0.
3. $x + \frac{1}{x} \geq 2$ for $x > 0$

   $x + \frac{1}{x} \leq -2$ for $x < 0$.

Ques. If $x \in (0, 90°)$ then find the minimum value of $3\tan^2x + 4\cot^2x$.

Ans. $\because AM \geq GM$

$$\Rightarrow \frac{3\tan^2x + 4\cot^2x}{2} \geq \sqrt{3\tan^2x . 4\cot^2x}$$

$$\Rightarrow 3\tan^2x + 4\cot^2x \geq 2\sqrt{12}$$

$$\Rightarrow (3\tan^2x + 4\cot^2x)_{min} = 2\sqrt{12} = 4\sqrt{3}$$

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Ques. If all roots of equation $x^3 - 6x^2 + px - 8 = 0$ are positive and real then find value of p.

Ans. $x^3 - 6x^2 + px - 8 = 0 \Leftarrow \begin{matrix}\alpha\\ \beta\\ \gamma\end{matrix}$

$\alpha + \beta + \gamma = 6 \quad \Rightarrow \dfrac{\alpha+\beta+\gamma}{3} = 2 = AM$

$\alpha\beta + \beta\gamma + \gamma\alpha = p \;\&\; \alpha\beta\gamma = 8$

$\Rightarrow (\alpha\beta\gamma)^{1/3} = 2 = GM$

$\because AM = GM$

$\therefore \alpha = \beta = \gamma = 2$

Now, $\alpha\beta + \beta\gamma + \gamma\alpha = p$

$\Rightarrow 4 + 4 + 4 = p \Rightarrow p = 12$

Ques. Let $f : R \to R$ such that $\forall x \in R$, $(2^{1+x} + 2^{1-x})$, $f(x)$ and $(3^x + 3^{-x})$ are in A.P., then find the minimum value of $f(x)$. **[JEE Mains 2022]**

Ans. $2f(x) = 2^{1+x} + 2^{1-x} + 3^x + 3^{-x}$

$\Rightarrow 2f(x) = 2.2^x + \dfrac{2}{2^x} + 3^x + \dfrac{1}{3^x}$

$\Rightarrow 2f(x) = 2\left(2^x + \dfrac{1}{2^x}\right) + \left(3^x + \dfrac{1}{3^x}\right)$

Let, $t = 2^x$, $z = 3^x$

$2f(x) = 2\underbrace{\left(t + \dfrac{1}{t}\right)}_{\geq 2} + \underbrace{\left(z + \dfrac{1}{z}\right)}_{\geq 2}$

$\Rightarrow 2f(x)_{min} = 2 \times 2 + 2$ (occurs at same $x = 0$)

$\Rightarrow 2f(x)_{min} = 6 \Rightarrow f(x)_{min} = 3$

Ques. Find the maximum value of $x^3 y^4$ where x & y are positive real numbers satisfying $3x + 4y = 5$

Ans. $\underbrace{x, x, x}_{③}, \underbrace{y, y, y, y}_{④}$

$\because AM \geq GM$

$\therefore \dfrac{3x + 4y}{7} \geq (x^3 y^4)^{\frac{1}{7}}$

$\Rightarrow \dfrac{5}{7} \geq (x^3 y^4)^{\frac{1}{7}}$

$\Rightarrow \left(\dfrac{5}{7}\right)^7 \geq x^3 y^4 \Rightarrow x^3 y^4 \leq \left(\dfrac{5}{7}\right)^7$

Hence, maximum value of $x^3 y^4 = \left(\dfrac{5}{7}\right)^7$

Ques. Find the maximum value of $x^2 y^3$ where (x, y) is point in first Quadrant lying on the line $3x + 4y = 5$ Also find the value of x and y for which this maxima occurs.

Ans. $3x + 4y = 5$ ...(i)

$\underbrace{\dfrac{3x}{2}, \dfrac{3x}{2}}_{②}, \underbrace{\dfrac{4y}{3}, \dfrac{4y}{3}, \dfrac{4y}{3}}_{③}$

$\because AM \geq GM$

$\therefore \dfrac{\frac{3x}{2} + \frac{3x}{2} + \frac{4y}{3} + \frac{4y}{3} + \frac{4y}{3}}{5} \geq \left[\left(\dfrac{3x}{2}\right)^2 \left(\dfrac{4y}{3}\right)^3\right]^{\frac{1}{5}}$

$\Rightarrow \dfrac{3x + 4y}{3} \geq \left[\dfrac{3^2}{2^2} x^2 \dfrac{4^3}{3^3} y^3\right]^{\frac{1}{5}}$

$\Rightarrow 1 \geq \left[\dfrac{x^2 4^2 y^3}{3}\right]^{\frac{1}{5}} \Rightarrow 1^5 \geq \dfrac{x^2 \times 16\, y^3}{3}$

$\Rightarrow \dfrac{3}{16} \geq x^2 y^3 \Rightarrow x^2 y^3 \leq \dfrac{3}{16}$

$\Rightarrow x^2 y^3{}_{max} = \dfrac{3}{16}$

at max value AM = GM

Given no's are equal

For maxima

$\dfrac{3x}{2} = \dfrac{4y}{3} \Rightarrow 9x = 8y$ ...(ii)

By eqⁿ (i)

$8y + 12y = 15 \Rightarrow 20y = 15 \Rightarrow y = \dfrac{3}{4}$

By eqⁿ (ii)

$9x = 8 \times \dfrac{3}{4} \Rightarrow 9x = 6 \Rightarrow x = \dfrac{2}{3}$

Ques. Let a, b, c and d be positive real numbers such that $a + b + c + d = 11$. If maximum value of $a^5 b^3 c^2 d$ is 3750β, then the value of β is **[JEE Mains Apr. 2023 Shift II]**

(a) 90 (b) 110 (c) 55 (d) 108

Ans. $\dfrac{a}{5} \to 5$ times, $\dfrac{b}{3} \to 3$ times, $\dfrac{c}{2} \to 2$ times, $d \to 1$ times — 11 no's

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AM ≥ GM

$$\frac{a+b+c+d}{11} \geq \left(\frac{a^5 b^3 c^2 d}{5^5 \times 3^3 \times 2^2 \times 1}\right)^{\frac{1}{11}}$$

$$\Rightarrow a^5b^3c^2d^1 \leq 3,37,500 = 3750 \times 90$$

$$\therefore \beta = 90$$

WEIGHTED MEAN

Let $a_1, a_2, \ldots a_n$ be n-positive ral numbers and $x_1, x_2, \ldots x_n$ be n-positive rational numbers, then

Weighted Arithmetic Mean:

$$A = \frac{x_1 a_1 + x_2 a_2 + \ldots + x_n a_n}{x_1 + x_2 + \ldots + x_n}$$

Weighted Geometric Mean:

$$G = (a_1^{x_1} . a_2^{x_2} \ldots a_n^{x_n})^{\frac{1}{x_1 + x_2 + \ldots + x_n}}$$

Weighted Harmonic Mean:

$$= H = \frac{x_1 + x_2 + \ldots + x_n}{\frac{x_1}{a_1} + \frac{x_2}{a_2} + \ldots \frac{x_n}{a_n}}$$

and satisfy $A \geq G \geq H$ (equality holds if and only if $a_1 = a_2 \ldots a_n$)

EXPONENTIAL & LOGARITHMIC SERIES

1. $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$

put $x = 1$:

$$e^1 = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \ldots \infty = \sum_{r=0}^{\infty} \frac{1}{r!} \quad \ldots(i)$$

put $x = -1$:

$$e^{-1} = \frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \ldots \infty \quad \ldots(ii)$$

Adding eqⁿ (i) and (ii)

$$e + e^{-1} = 2\left[\frac{1}{0!} + \frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \ldots \infty\right]$$

$$\Rightarrow \frac{e + \frac{1}{e}}{2} = \frac{1}{0!} + \frac{1}{2!} + \frac{1}{4!} + \ldots \infty = \sum_{r=0}^{\infty} \frac{1}{(2!)!}$$

Subtract eqⁿ (i) and (ii)

$$\frac{e - \frac{1}{e}}{2} = \left[\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \ldots \infty\right] = \sum_{r=0}^{\infty} \frac{1}{(2r+1)}$$

2. $a^x = 1 + \frac{x \ln a}{1!} + \frac{x^2 (\ln a)^2}{2!} + \frac{x^3 (\ln a)^3}{3!} + \ldots \infty$

3. $\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \ldots \infty$

Put, $x = 1$

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \ldots \infty$$

4. $\ln(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \ldots \infty$

$$= -\left[x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \ldots \infty\right]$$

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \ldots \infty$$

Ques. If α and β are the roots of the equations $3x^2 - 6x + 2 = 0$ the find the value of $\left(1 + \frac{\alpha}{2} + \frac{\alpha^2}{2!4} + \frac{\alpha^3}{3!8} + \ldots \infty\right) \cdot \left(1 + \frac{\beta}{2} + \frac{\beta^2}{2!4} + \frac{\beta^3}{3!8} + \ldots \infty\right) = ?$

(a) $2e$ (b) e^2 (c) $e^{2/3}$ (d) e

Ans. $\left(1 + \frac{\alpha}{2} + \frac{\alpha^2}{2!4} + \frac{\alpha^3}{3!8} + \ldots \infty\right)\left(1 + \frac{\beta}{2} + \frac{\beta^2}{2!4} + \frac{\beta^3}{3!8} + \ldots \infty\right)$

$= e^{\alpha/2} . e^{\beta/2} = e^{\frac{\alpha + \beta}{2}}$

$= e^{2/2} \quad \left(\because \alpha + \beta = \frac{6}{3} = 2\right)$

$= e^1 = e$

Ques. Simplify: $S = \frac{2}{1!} + \frac{4}{3!} + \frac{6}{5!} + \frac{8}{7!} + \ldots \infty$

Ans. $S = \sum_{r=1}^{\infty} \frac{(2r-1)+1}{(2r-1)!}$

$$= \sum_{r=1}^{\infty} \left[\frac{2r-1}{(2r-1)!} + \frac{1}{(2r-1)!}\right]$$

$$= \sum_{r=1}^{\infty} \frac{2r-1}{(2r-1)(2r-1)!} + \sum_{r=1}^{\infty} \frac{1}{(2r-1)!}$$

$$= \sum_{r=1}^{\infty} \frac{2r-1}{(2r-2)!} + \sum_{r=1}^{\infty} \frac{1}{(2r-1)!}$$

(even) (odd)

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$$= \frac{1}{0!} + \frac{1}{2!} + \frac{1}{4!} + \ldots + \frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \ldots = e$$

Ques. Find sum: $\sum_{r=0}^{\infty} \frac{n^2}{2!} = ?$

Ans. $$\sum_{r=0}^{\infty} \frac{n^2}{n!} = \sum_{r=1}^{\infty} \frac{n^2}{n\,(n-1)!} = \sum_{r=1}^{\infty} \frac{(n-1)+1}{(n-1)!}$$

$$= \sum_{r=1}^{\infty} \left[\frac{n-1}{(n-1)!} + \frac{1}{(n-1)!}\right]$$

$$= \sum_{r=2}^{\infty} \frac{(n-1)}{(n-1)\,(n-2)!} + \sum_{r=1}^{\infty} \frac{1}{(n-1)!}$$

$$= \sum_{r=2}^{\infty} \frac{1}{(n-2)!} + \sum_{r=1}^{\infty} \frac{1}{(n-1)!}$$

$$= \underbrace{\left(\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \ldots \infty\right)}_{e} + \underbrace{\left(\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \ldots \infty\right)}_{e} = 2e$$

Ques. Find sum: $\sum_{r=0}^{\infty} \frac{n^3}{n!} = ?$

Ans. $$\sum_{r=1}^{\infty} \frac{n^2 - 1 + 1}{(n-1)!} = \sum_{r=1}^{\infty} \frac{(n-1)\,(n+1)}{(n-1)!} + \sum_{r=1}^{\infty} \frac{1}{(n-1)!}$$

$$= \sum_{r=2}^{\infty} \frac{n+1}{(n-2)!} + \sum_{r=1}^{\infty} \frac{1}{(n-1)!}$$

$$= \sum_{r=2}^{\infty} \frac{(n-2)+3}{(n-2)!} + \sum_{r=1}^{\infty} \frac{1}{(n-1)!}$$

$$= \sum_{r=3}^{\infty} \frac{1}{(n-3)!} + 3 \sum_{r=2}^{\infty} \frac{1}{(n-2)!} + \sum_{r=1}^{\infty} \frac{1}{(n-1)!}$$

$$= e + 3e + e = 5e$$

Ques. $S = 1 + \frac{1+2}{2!} + \frac{1+2+2^2}{3!} + \frac{1+2+2^2+2^3+}{4!} \ldots \infty$

Ans. $$\sum_{r=1}^{\infty} \frac{1 + 2 + 2^2 + 2^3 + \ldots + 2^{r-1}}{r!}.$$

$$= \sum_{r=1}^{\infty} \left(1.\frac{(2^r - 1)}{2 - 1}\right) \frac{1}{r!} = \sum_{r=1}^{\infty} \frac{2^r - 1}{r!}$$

$$= \sum_{r=1}^{\infty} \frac{2^r}{r!} - \sum_{r=1}^{\infty} \frac{1}{r!}$$

$$\therefore \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \ldots \infty = e$$

$$\& \frac{2^1}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \ldots \infty = e^2 - 1$$

Hence, $S = e^2 - 1 - (e - 1) = e^2 - e$

Ques. If $S = \frac{1}{1.2} - \frac{1}{2.3} + \frac{1}{3.4} - \frac{1}{4.5} \ldots\ldots \infty$, then find e^S

(a) $\ln \frac{e}{4}$ (b) $\frac{4}{e}$ (c) $\frac{e}{4}$ (d) $\ln \frac{4}{e}$

Ans. $$S = \left(\frac{2-1}{1.2}\right) - \left(\frac{3-1}{2.3}\right) + \left(\frac{4-3}{3.4}\right) - \left(\frac{5-4}{4.5}\right) \ldots$$

$$= \left(\frac{1}{1} - \frac{1}{2}\right) - \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) - \left(\frac{1}{4} - \frac{1}{5}\right)$$

$$= 1 - \frac{1}{2} - \frac{1}{2} + \frac{1}{3} + \frac{1}{3} - \frac{1}{4} - \frac{1}{4} + \frac{1}{5} \ldots$$

$$= 1 - 2.\frac{1}{2} + 2.\frac{1}{3} - 2.\frac{1}{4} + 2.\frac{1}{5} \ldots$$

$$= -1 + \left[2.1 - 2.\frac{1}{2} + 2.\frac{1}{3} - 2.\frac{1}{4} + 2.\frac{1}{5} \ldots\right]$$

$$= -1 + 2\left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} \ldots\right]$$

$$= -1 + 2 \ln 2 = -1 + \ln 4 = -\ln e + \ln 4 = \ln \frac{4}{e}$$

$$e^S = e^{\ln\left(\frac{4}{e}\right)} = \frac{4}{e}$$

Ques. The sum of $\frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2^3} + \frac{1}{5} + \frac{1}{2^5} + \ldots \infty$ is

(a) $\log_e \sqrt{\frac{3}{2}}$ (b) $\log_e \sqrt{2}$

(c) $\log_e \sqrt{\frac{1}{2}}$ (d) $\log_e \sqrt{3}$

Ans. $$\ln(1+x) = \frac{x}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \ldots \infty$$

$$\ln(1-x) = -\frac{x}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} \ldots \infty$$

$$- \quad + \quad + \quad + \quad +$$

$$\ln(1+x) - \ln(1-x) = 2\left[x + \frac{x^3}{3} + \frac{x^5}{5} + \ldots \infty\right]$$

$$\Rightarrow \ln \frac{(1+x)}{(1-x)} = 2\left[x + \frac{x^3}{3} + \frac{x^5}{5} + \ldots \infty\right]$$

put $x = \frac{1}{2}$, $\ln\left(\frac{1 + \frac{1}{2}}{1 - \frac{1}{2}}\right) = 2(s)$

$$\Rightarrow \ln\left(\frac{\frac{3}{2}}{\frac{1}{2}}\right) = 2s \Rightarrow \frac{1}{2} \ln 3 = s \Rightarrow s = \ln\sqrt{3}$$

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Short Notes

ARITHMETIC PROGRESSION (A.P.)

- n^{th} term a_n is given by $a_n = a + (n-1)d$.
- The sum S_n of the first n terms of an A.P. is given by $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$ where l is the last term.
- If n^{th} term of any sequence is a linear expression in n, then the sequence is an AP, whose common difference is the coefficient of n.
- If sum of n terms of any sequence is a quadratic in n, then the sequence is an AP, whose common difference is twice the coefficient of n^2.
- If three terms are in A.P., then the middle term is called the arithmetic mean (A.M.) between the other two i.e. if a, b, c are in A.P. then $b = \frac{a+c}{2}$ is the A.M. of a and c.
- The n arithmetic means $a_1, a_2 ..... a_n$ are inserted between the numbers a and b then $a_1 + a_2 + a_3 + ... + a_n = n\frac{(a+b)}{2}$

GEOMETRIC PROGRESSION (G.P.)

- If 'a' is the first term and 'r' the common ratio of a G.P. can be written as a, ar, ar^2, ...
- The n^{th} term 'a_n' is given by $a_n = ar^{n-1}$.
- The sum S_n of the first n terms is
  $S_n = \frac{a(r^n - 1)}{r-1}, \quad r \neq 1$
  $= na, \quad r = 1$
- If $-1 < r < 1$, then the sum of the infinite G.P. $a + ar + ar^2 + ... = \frac{a}{1-r}$.
- If three terms are in G.P., then the middle term is called the geometric mean (G.M.) between the two. So if a, b, c are in G.P. then $b = \sqrt{ac}$ is the geometric mean of a and c.
- If $G_1, G_2, ... G_n$ are n geometric means between a and b then $G_1 G_2 .... G_n = (ab)^{n/2}$

HARMONIC PROGRESSION (H.P.)

- The n^{th} term a_n of the H.P. is $a_n = \frac{1}{a + (n-1)d}$, where $a = \frac{1}{a_1}$, and $d = \frac{1}{a_2} - \frac{1}{a_1}$.
- There is no formula for S_n of H.P.
- If a and b are two non-zero numbers, then the harmonic mean of a and b is a number H such that the numbers a, H, b are in H.P. We have $H = \frac{2ab}{a+b}$.
- Sum of reciprocals of n H.Ms inserted between two given numbers = $n\left(\frac{a+b}{2ab}\right)$

ARITHMETIC-GEOMETRIC PROGRESSION

Steps to find S_n or S_∞ for A.G.P

- Write the given series upto T_n.
- Multiply the whole series with common ratio (r) and write the series again by shifting one term to right.
- Subtract the two equations to obtain S_n or S_∞.

SPECIAL SERIES

- Sum of the first n natural numbers $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$

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- Sum of the first (n-1) natural numbers $\sum_{r=1}^{n-1} r = \frac{n(n-1)}{2}$
- Sum of the squares of the first n natural numbers $\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$
- Sum of the cubes of the first n natural number.

$$\sum_{r=1}^{n} r^3 = \frac{n^2(n+1)^2}{4} = \left[\sum_{r=1}^{n} r\right]^2$$

Method of Difference

- Used to find T_n of a series in which the difference of consecutive terms are in A.P. or G.P.
- **Steps to apply Method of Difference:**

Step - (i) Write the given series.

Step - (ii) Write the series again by shifting one term to right.

Step - (iii) Subtract the two equations to obtain n^{th} term (T_n) of the series.

Step - (iv) Find the sum of the n terms of the sequence as $S_n = \sum_{n=1}^{n} T_n$.

Telescoping Series

- A series in which the general term can be expressed as a difference two symmetric terms is called a Telescoping series.
- The expression $\sum_{r=1}^{n-1}(a_r - a_{r+1}) = a_1 - a_n$ is called a Telescoping series

INEQUALITIES

- Let $a_1, a_2, \ldots a_n$ be n positive real numbers, then their arithmetic mean (A), geometric mean (G) and harmonic mean (H) as $A = \frac{a_1 + a_2 + \ldots + a_n}{n}$, $G = (a_1\ a_2 \ldots a_n)^{1/n}$ and $H = \frac{n}{\left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \ldots \frac{1}{a_n}\right)}$.
- For positive numbers A.M. $\geq$ G. M. $\geq$ H. M.
- Equality holds at either place (A.M. = G. M. = H. M) if and only if $a_1 = a_2 = \ldots = a_n$.

WEIGHTED MEAN

- Let $a_1, a_2, \ldots, a_n$ be n positive real numbers and $m_1, m_2, \ldots, m_n$ be n positive rational numbers. Then we define weighted Arithmetic mean (A), weighted Geometric mean (G) and weighted harmonic mean (H) as
- $A = \frac{m_1a_1 + \ldots + m_na_n}{m_1 + m_2 + \ldots + m_n}$,
- $G = \left(a_1^{m_1}.a_2^{m_2}\ldots\ldots a_n^{m_n}\right)^{\frac{1}{(m_1+m_2+\ldots+m_n)}}$
- $H = \frac{m_1 + m_2 + \ldots + m_n}{\left(\frac{m_1}{a_1} + \frac{m_2}{a_2} + \ldots + \frac{m_n}{a_n}\right)}$.
- It can be shown that $A \geq G \geq H$. Equality holds at either place if and only if $a_1 = a_2 = \ldots = a_n$.

"Success is also a series – One positive step repeated makes the biggest difference."

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